# Six attempts at quasi-polynomial unknot recognition: synthesis, cross-validation, and a faster exact recognizer

Synthesis report prepared with Claude Code

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**Abstract**

Six independently produced archives each tried to implement the quasi-polynomial unknot recognition algorithm announced in Marc Lackenby's February 2021 talk. This report extracts them into numbered directories, reads their reports, reviews their code, re-runs their test suites, and cross-validates them against each other on shared inputs. All six reach the same honest conclusion: the quasi-polynomial algorithm was not implemented, because the accelerated hierarchy operations named on the last slide have no published algorithmic specification with cost bounds. What they deliver instead is consistent and correct as far as we can test: five reduced-Khovanov recognizers agree on 44 shared inputs, six ball-pattern testers agree on 2000 random spherical patterns, and the grid-diagram search of archive 01 agrees with Khovanov homology on every rectangular diagram we checked. Building on their ideas we add a seventh recognizer, `fastunknot`, whose exact part is a scanning computation of Khovanov homology (Bar-Natan's local algorithm with delooping and Gaussian elimination) preceded by an Alexander-polynomial filter. It is only justified by an exponential general bound; its practical cost depends on scan width and intermediate complex size. For example, the 80-crossing unknot family on which every cube implementation needs $2^{80}$ states is decided in eleven milliseconds. Nine further runs then proposed accelerations of that recognizer. We extracted and measured all nine on one machine, integrated every idea with a measurable effect into version 0.2 (the input that timed out at 600 seconds now takes under a second, and recognition of the Conway knot went from 59 ms to 1.3 ms), recorded a per-idea ablation, and wrote a Rust implementation that is faster again by an order of magnitude. The $n^{O(\log n)}$ target remains open; Section 5 states precisely what is missing. The October continuation integrates report 07: a linear structural certificate and exact component sharing, including a conditional finite-type complexity bound. Section 12 separates this bound from the unresolved general target.

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# 1 Task, sources, and outcome

## 1.1 What was asked

The original request to each of the six runs was an implementation of the main theorem of the supplied talk [23]: an algorithm that decides whether an $n$-crossing knot diagram represents the unknot in time $n^{O(\log n)}$. The present task was to unpack the six resulting archives, read their reports, review their code, and synthesize what was achieved and what obstacles remain.

A second task was added mid-way: write the asymptotically fastest algorithm possible, trying to reach $n^{O(\log n)}$. A third task followed: unpack nine proposals for accelerating that algorithm, compare their ideas, incorporate those that give a measurable speedup, and write an optimized Rust implementation.

### 1.2 The talk

The supplied file `docs/quasipolynomial-talk.pdf` has 109 pages including progressive-reveal duplicates. Its logical content is short:

- pp. 11–14: the theorem, $n^{O(\log n)}$ for an $n$-crossing diagram;
- pp. 15–30: hierarchies, boundary patterns, essential patterns (a proper disc meeting the pattern at most three times must cut off an empty disc, an arc, or a tripod), and the Waldhausen–Johannson characterization: the knot is nontrivial iff the exterior admits an essential hierarchy ending in balls;
- pp. 38–53: simplification of an inessential hierarchy by a violating disc (compress or pattern-compress the last surface the disc runs over, discard later surfaces), and the basic algorithm as a flowchart;
- pp. 54–65: termination by a lexicographic complexity $(g(S_1), \dots, g(S_\ell))$, with the warning that "genus" may mean $\chi_-$ or a pattern-sensitive variant, and the estimate "running time at most $Lg^L$" if every surface has complexity at most $g$ and hierarchies have length at most $L$;
- pp. 66–74: the four speed-ups needed for $n^{O(\log n)}$: (1) surfaces with $g \le O(n^2)$, (2) compressed hierarchy encodings (Agol–Hass–Thurston), (3) multi-surfaces, (4) generalised Heegaard splittings and Cheeger regions, "the effect of this is to ensure that $L \le O(\log n)$";
- pp. 75–108: sketches of each speed-up, including the Cheeger-region condition $g(\partial M_i) \le \frac{1}{3}\min\{g(H_i), g(H) - g(H_i)\}$;
- p. 109: the final flowchart with the boxes Build hierarchical multi-surface, Cut along surface, Use Cheeger region, Haken's lemma, Weakly reduce, Simplify multi-surface, and the terminal test "does the graph dual to the boundary pattern have a violating cycle?".

The slides describe the strategy; they contain no data structures, no algorithms for the boxes, and no proof of the depth bound.

### 1.3 The 2026 preprint

All six archives located Lackenby's July 2026 preprint [24] ("Incompressible surfaces, hierarchies and unknot recognition", 54 pages). We downloaded it and read Section 9 ("The number of steps") ourselves. It opens: "There are various steps in the algorithm from Theorem 1.3 that are not described with enough precision to be able to estimate their running time. So, we will initially content ourselves with an analysis of the number of times that the algorithm goes through any of the steps", in terms of the maximal hierarchy length $L$ and the maximal pattern-complexity $g$, and "At present, we will not attempt to bound these two quantities." Proposition 9.1 then gives the iteration bound $L(g+1)^L$ using exactly the base-$(g+1)$ digit argument that the talk sketches in base $g$. Section 10 refines the algorithm to obtain a bound on $L$ that is *linear* in the initial $q$-complexity of the handle structure, not logarithmic in $n$. The preprint also supplies two facts the archives use: Proposition 2.6 (a pattern on a 3-ball is essential iff it is empty, a circle, or a connected graph whose planar dual is simple and 4-connected or $K_3$ or $K_4$) and Definition 3.4 (pattern-complexity $\chi_P(S) = -4\chi(S) + |S \cap P|$).

So the accelerated, logarithmic-depth algorithm of the talk remains announced but not specified in any source available to us on rechecking the author's page and the preprint on 8 October 2026. This is the single obstacle behind every "partial" verdict below, and it is mathematical, not an engineering shortfall.

**Plausibility and the precise target.** Later seminar abstracts, including the November 2024 talk [15], state $2^{O((\log n)^3)} = n^{O((\log n)^2)}$. The earlier supplied slides state the stronger $n^{O(\log n)}$ target used elsewhere in this report; these are distinct formulations. Both are quasi-polynomial and hence $2^{o(n)}$. The hierarchy strategy is a credible route to such a bound: it replaces exhaustive global witness search with decomposition, compression and controlled corrections. The preprint supplies substantial topological machinery for that route. This is our assessment of plausibility, not a verification of its announced asymptotics. Even given the iteration estimate $L(g+1)^L$, an implementation must bound the bit cost $C(n)$ of each operation on its largest intermediate encoding. A sufficient accounting condition is

$$\log C(n) + \log(L+1) + L\log(g+1) = O((\log n)^3).$$

Efficiently handling one compressed surface, or proving termination, does not establish this condition. No general sub-exponential bound for our recognizer follows from the current report or from adaptive switching among its present backends.

For intuition, $L = O(\log n)$ and $g = n^{O(1)}$ would make the iteration count $2^{O((\log n)^2)}$; allowing suitably bounded quasi-polynomial operation cost could still fit the later cubic-logarithm exponent. The difficulty is proving compatible bounds throughout all corrections, including the size of intermediate compressed descriptions. A short certificate or a terminating hierarchy does not itself give a fast way to find one. Conversely, the growth of our explicit Khovanov complexes is not evidence against Lackenby's different hierarchy algorithm. Our assessment is that the claim is mathematically plausible and worth pursuing, while the available specification is insufficient to verify or implement its announced asymptotics. Section 36 proves a general subexponential scalar computation, and Section 38 extends this to full Jones polynomial recovery. Section 56 sharpens the implemented full-polynomial bound to $2^{O(\sqrt{n})}$. Neither computation supplies the missing complete recognition bound.

### 1.4 Outcome in one paragraph

None of the six archives, and not the new package either, implements a recognizer with a proved $n^{O(\log n)}$ bound. Each archive contains a complete exact recognizer with an exponential bound, plus genuinely implemented polynomial pieces of the hierarchy strategy (the terminal ball-pattern test, cocycle-dual normal coordinates, the potential arithmetic), and each report says so without hedging. We verified every test suite, cross-validated the archives against each other and against independent computations, found no incorrect verdict, and identified one gap in a correctness argument (Section 3.4). The new package `fastunknot` improves the exact part substantially in practice (Section 6); its worst case is still $2^{O(n)}$.

## 2 The six archives

### 2.1 Extraction

The six ZIP files in `reports/` were extracted into `reports/01` through `reports/06`, each with the single wrapper directory removed, in the order of the archive modification times:

| Dir | archive | package | Python lines | tests |
|---|---|---|---|---|
| 01 | `unknot_recognition_implementation.zip` | `unknot` (grid search) | 1753 | 40 |
| 02 | `unknot_recognition_partial_implementation.zip` | `unknotlab` | 1939 | 64 |
| 03 | `unknot_workbench.zip` | `unknot_lab` | 1431 | 49 |
| 04 | `unknot_recognition_partial.zip` | `unknot` | 1422 | 62 |
| 05 | `unknot_recognition_partial_implementation (1).zip` | `unknot_recognition` | 1882 | 58 |
| 06 | `unknot-recognition-implementation.zip` | `unknot` | 1278 | 45 |

Every archive contains a README, a LaTeX report with PDF, examples, recorded results, and a standard-library-only Python package. All six test suites were re-run here on CPython 3.14.4 under Windows (the archives recorded CPython 3.13.5 on Linux) and pass with exactly the claimed number of tests.

### 2.2 What each archive delivers

**Archive 01: rectangular diagrams and Dynnikov moves.** The only archive that does not use Khovanov homology. It represents knots as grid diagrams, searches the finite space of grids reachable by non-increasing moves (exchanges and destabilizations, modulo cyclic shifts) and relies on Dynnikov's monotonic simplification theorem [37]: an unknot reaches the $2 \times 2$ grid, a knot is certified by a state set closed under all moves. Certificates (positive path, non-unit determinant, closed set) are replayed by a verifier that enumerates moves independently. Bound $2^{O(g \log g)}$ in the grid size $g$; the report is careful that $g$ is not the crossing number. It validated all 44 719 connected grids of size $\leq 6$ (1285 shift orbits). It also contains the pattern test, exact rational $H^1$ of a simplicial complex, cocycle-dual normal coordinates, and the potential arithmetic.

**Archive 02: Khovanov cube with fast certificates.** Descending-diagram test (a diagram traversed meeting every crossing first on its over-strand is trivial), Fox determinant, then the full reduced Khovanov complex over $\mathbb{F}_2$. Its normal-surface module accepts genuine face-pairing triangulations, checks orientability and sphere/disc vertex links, and computes cocycle-dual normal coordinates without expanding discs. Six braid/determinant pairs were checked against Wolfram `KnotData`.

**Archive 03: workbench with an explicit exponential lower bound.** Khovanov cube with default resource ceilings (65 536 states), a bond-enumeration pattern test with checkable witnesses, and a `bound` command for the conditional $L(g+1)^L$ arithmetic. Its report proves that its own implementation is not quasi-polynomial: on the unknot $\sigma_1\widehat{\sigma_2 \cdots}\sigma_n$ the cube has $2^n$ states and $3^n$ generators, and the determinant filter cannot help since the determinant is 1.

**Archive 04: Khovanov cube with Knot Atlas fixtures.** Khovanov cube, dual-4-connectivity pattern test, potential arithmetic. It checks $10_{124} = T(3,5)$ from an Atlas PD code and contains a clean chain-level proof that the unreduced $\mathbb{F}_2$ complex is two copies of the reduced one, which justifies reading reduced ranks off published unreduced tables.

**Archive 05: Khovanov cube with Reidemeister preprocessing.** Crossing-decreasing Reidemeister I/II moves with replayable traces, determinant filter, Khovanov cube; bond-enumeration pattern test; simplicial normal coordinates with manifold validation; the pattern-complexity $-4\chi + |S \cap P|$ of the preprint. It exhausts all 682 two-strand words of odd length $\leq 9$ and gives an independent braid-theoretic proof that its eight-crossing "hard unknot" is trivial.

**Archive 06: Khovanov cube with 11-crossing tests.** Khovanov cube, bond-enumeration pattern test, a `verify` command that recomputes a report with $d^2 = 0$ checks. Its fixtures include the Kinoshita–Terasaka and Conway knots (Alexander polynomial 1, reduced rank 33). Its source audit notes that a 34-page Oxford version of the slides uses the factor $1/10$ instead of $1/3$ in the Cheeger condition and displays $O((\log n)^2)$ for the hierarchy length; substituting that depth into the digit count would give $n^{O((\log n)^2)}$.

### 2.3 Common structure of the six reports

Read side by side, the reports agree on every substantive point, which is itself evidence since they were produced independently:

1. The requested bound is not established; the delivered recognizer is a different, exponential algorithm; resource limits return `UNKNOWN` and never a verdict.
2. The Khovanov route is justified by Kronheimer–Mrowka [35] plus a universal-coefficient argument: $\dim_{\mathbb{F}_2} \widetilde{\mathrm{Kh}}(K;\mathbb{F}_2) \geq \operatorname{rank} \widetilde{\mathrm{Kh}}(K;\mathbb{Z})$, and the rational reduced homology is nonzero because its Euler characteristic is the reduced Jones polynomial, so $\mathbb{F}_2$-rank one forces rational rank one.
3. Determinant one is never a positive certificate ($T(3,5)$ and the Kinoshita–Terasaka knot are the standard witnesses).
4. The talk's "base $g$" digit argument needs base $g+1$ when the digit $g$ is allowed, since $(1,0)$ and $(0,g)$ have the same base-$g$ value; all six use base $g+1$, as does Proposition 9.1 of the preprint.
5. The terminal pattern test is implemented only under an explicit "already a 3-ball" hypothesis, and $b_1 = 0$ is not ball recognition ($S^2 \times I$ is the standard counterexample).
6. The missing operations are listed by name: initial layered handle structure of the exterior, compressed cutting with provenance, hierarchical multi-surfaces with the pattern-sensitive quadratic bound, lifting a violating dual cycle to a disc in an earlier stage, Cheeger-region detection and Heegaard modification, Haken's lemma and weak reduction, and an end-to-end bit-cost proof including restarts.

## 3 Code review

### 3.1 The Khovanov recognizers (02–06)

All five have the same architecture, evidently converged upon rather than copied (variable names, data layouts and gradings differ): enumerate the $2^n$ resolutions with a disjoint-set structure, encode a reduced generator as a bit mask over the unmarked circles with the marked circle fixed to $x$, build each differential column as a Python integer (XOR arithmetic over $\mathbb{F}_2$), and compute ranks by pivot elimination. Points checked in the review:

- Input validation is uniformly careful: one component, spherical rotation system via $V-E+F=2$ on the ribbon graph (so virtual diagrams are rejected, not misclassified), labels normalized, `{"pd": []}` defined as one circle.
- The Frobenius algebra tables ($m(x\otimes x)=0$, $\Delta(1)=1\otimes x + x\otimes 1$, $\Delta(x)=x\otimes x$) are identical everywhere, and every implementation checks that the marked circle keeps its $x$ label.
- $d^2=0$ is available as an exhaustive optional check in all five and is exercised by their tests.
- Gradings are deliberately unnormalized; only the total rank is used, so writhe shifts are irrelevant. This is correct for recognition.
- Resource semantics are consistent (`UNKNOWN`, exit code 2 or 3; the codes differ between archives, which is a minor interoperability wart).
- Archive 03 imposes default ceilings, so its default invocation is incomplete above 16 crossings; the README says so.

We found no correctness problem. The determinant filters (Fox colouring matrix, Bareiss elimination with exact-division checks) are also equivalent.

### 3.2 The pattern testers (all six)

Three different criteria are implemented for "is this cubic spherical pattern essential on a ball": dual simple and 4-connected with the $K_3$, $K_4$ exceptions (01, 04), no dual loop, parallel pair or

non-facial dual triangle (02), and no bond of size 1 or 2 and no three-edge bond with more than one vertex on each side (03, 05, 06). Each report proves its own criterion from the definition. The proofs are correct; the bond argument in archive 06 (the cycle rank of a side with $k$ vertices and $r$ cut edges is $(k - r + 2)/2$) is the cleanest. Input conventions differ (darts $2e, 2e + 1$; signed edge ends; positional darts; edge identifiers occurring twice), which is why a converter was needed for cross-validation.

### 3.3 Normal coordinates and potentials

Archives 01, 02 and 05 compute integral cocycles spanning $H^1(\,\cdot\,;\mathbb{Q})$ in a spanning-tree gauge and convert a cocycle to normal coordinates by half-integral level sets of local vertex heights. The constructions agree (seven coordinates per tetrahedron, matching equations checked, Euler characteristic without disc expansion). All are honest that this is not a saturated $\mathbb{Z}$-basis, not a connected surface, and not a cut. The four potential modules implement the same base-$(g + 1)$ lemma and differ only in interface.

### 3.4 The grid search of archive 01

This is the one place where the review found a gap in an argument. The recognizer's negative certificate is a set of grids closed under all implemented moves; soundness needs the implemented move set to be exactly a set for which monotonic simplification holds. The report states that "rows sharing an endpoint are not accepted by this implementation's exchange rule" and cites Dynnikov's Proposition 6, but does not argue that Dynnikov's theorem still holds when exchanges of two adjacent rows (or columns) with a common corner column are excluded. Depending on the reading of Dynnikov's definition of interleaving, that exclusion may remove legal moves, in which case a closed set could exist for an unknot. We could not settle this from the sources at hand, so we tested it: Section 4.3 compares the search verdict with Khovanov homology on every grid of size $\leq 6$ and on random larger grids, and found no disagreement. The gap is in the proof, not (as far as tested) in the answers.

Everything else in archive 01 checked out: the move implementation, the Booth canonicalization, the independent verifier, and the exhaustive counts ($g!(g-1)!/2$ connected grids per size).

## 4 Cross-validation

None of the archives compared itself with another; each ran its own tests. We wrote three scripts (in `synthesis/data/`) that load the packages side by side.

### 4.1 Five Khovanov implementations on shared inputs

The same braid words and Knot Atlas PD codes were passed to all five packages (after converting label conventions). Reduced $\mathbb{F}_2$ ranks:

| Input | 02 | 03 | 04 | 05 | 06 | agree |
|---|---|---|---|---|---|---|
| unknot circle | 1 | 1 | 1 | 1 | 1 | yes |
| trefoil | 3 | 3 | 3 | 3 | 3 | yes |
| figure-eight | 5 | 5 | 5 | 5 | 5 | yes |
| T(2,5) | 5 | 5 | 5 | 5 | 5 | yes |
| T(2,7) | 7 | 7 | 7 | 7 | 7 | yes |
| T(3,4) | 5 | 5 | 5 | 5 | 5 | yes |
| T(3,5) det 1 | 7 | 7 | 7 | 7 | 7 | yes |
| unknot s1 s2 | 1 | 1 | 1 | 1 | 1 | yes |
| unknot s1s2s3s4 | 1 | 1 | 1 | 1 | 1 | yes |
| conjugated unknot | 1 | 1 | 1 | 1 | 1 | yes |
| hard unknot 05 | 1 | 1 | 1 | 1 | 1 | yes |
| KT 11n42 (Atlas PD) | 33 | 33 | 33 | 33 | 33 | yes |
| Conway 11n34 (Atlas PD) | 33 | 33 | 33 | 33 | 33 | yes |
| 10_124 (Atlas PD) | 7 | 7 | 7 | 7 | 7 | yes |
| 30 random braid closures (3–4 strands, 6–10 letters) | | | ranks 1–13 | | | 30/30 |

All 44 inputs agree, including the two Alexander-polynomial-one 11-crossing knots and 30 random closures whose ranks range from 1 to 13.

### 4.2 Six pattern testers on random spherical patterns

Random perfect matchings of the darts of $V$ trivalent vertices were kept when every component's rotation system is spherical, converted to the six input formats, and classified by all six testers (400 distinct systems per vertex count) together with six named patterns:

| Vertices | spherical systems tested | random matchings drawn | essential | disagreements |
|---|---|---|---|---|
| 2 | 400 | 883 | 102 | 0 |
| 4 | 400 | 737 | 12 | 0 |
| 6 | 400 | 1179 | 0 | 0 |
| 8 | 400 | 2603 | 0 | 0 |
| 10 | 400 | 6144 | 0 | 0 |

| Pattern | verdict | all six agree |
|---|---|---|
| theta | essential | yes |
| tetrahedron | essential | yes |
| cube | essential | yes |
| prism | inessential | yes |
| dodecahedron | essential | yes |
| dumbbell | inessential | yes |

No disagreement among 2006 patterns and three different criteria. (Random matchings on many vertices are almost always inessential because short bonds are common; the named cube and dodecahedron are the essential large examples.)

### 4.3 Grid search versus Khovanov homology

We wrote an independent grid-to-PD converter (traversal of the grid, crossing signs from the directions of the vertical and horizontal strands) and compared archive 01's search (determinant filter disabled) with archive 04's Khovanov rank:

| Grid size | sample | diagrams | unknots | knots | disagreements |
|---|---|---|---|---|---|
| 2 | exhaustive (all shift orbits) | 1 | 1 | 0 | 0 |
| 3 | exhaustive (all shift orbits) | 2 | 2 | 0 | 0 |
| 4 | exhaustive (all shift orbits) | 6 | 6 | 0 | 0 |
| 5 | exhaustive (all shift orbits) | 64 | 62 | 2 | 0 |
| 6 | exhaustive (all shift orbits) | 1212 | 1185 | 27 | 0 |
| 7 | random, $\leq 14$ crossings | 300 | 286 | 14 | 0 |
| 8 | random, $\leq 14$ crossings | 200 | 188 | 12 | 0 |
| 9 | random, $\leq 14$ crossings | 100 | 82 | 18 | 0 |

The knotted orbits of size 5 are the two trefoils (grid number 5) and those of size 6 are 26 trefoil presentations and one figure-eight, exactly as the grid-number theory predicts. No disagreement was found, which supports but does not prove the completeness of the restricted move set. The sample stops at size 9 because the search itself becomes expensive there (one size-9 grid needed 900 863 states, and the 100 grids took 1.9 hours), while the Khovanov side stayed below 14 crossings by construction.

### 4.4 What the cross-validation does and does not show

Agreement of five independently written cubes is strong evidence that all five compute the same invariant, and agreement with the published ranks (trefoil 3, figure-eight 5, $T(3,5)$ 7, $K11n34$ and $K11n42$ 33) ties that invariant to reduced Khovanov homology. It says nothing about the quasi-polynomial question, and all runs are on diagrams of at most 14 crossings because the cubes cannot go further.

## 5 What is missing for $n^{O(\log n)}$

Consolidating the six audits with our own reading of the sources, an implementation with a proved $n^{O(\log n)}$ bound would require the following assembled executable obligations. Published algorithms supply some components, including cutting and parallelity bundles discussed in Section 156; the complete source-authenticated integration and uniform complexity bounds below are not implemented here.

1. **A representation.** A layered handle structure of the knot exterior with a Morse function, boundary-pattern labels, the ambient embedding in $S^3$ (used by the generalised Seifert construction of pp. 75–78), compressed normal-surface data, and regluing maps, with a proved bound on the bit size of all of this after arbitrarily many operations. The archives' normal-coordinate modules are local ingredients only.
2. **Bounded multi-surfaces.** An algorithm producing independent relative homology classes with embedded representatives whose *pattern-sensitive* complexity is $O(n^2)$, and a proof that compression and pattern-compression preserve the invariants that allow later surfaces to be kept. A rational $H^1$ basis is not this.
3. **Compressed cutting.** Cutting, component extraction, Euler characteristic, orientation and pattern transport on compressed data without expanding parallel discs, with provenance sufficient to lift a violating dual cycle in a terminal ball to a disc in the correct earlier stage. Agol–Hass–Thurston orbit counting is a tool, not this procedure.
4. **Cheeger regions and weak reduction.** Recognition of the Heegaard–Morse hypotheses, construction of the modified generalised splitting, Haken's lemma and weak reduction, and a global progress measure bounding restarts. The inequality $3g(\partial M_i) \leq \min\{g(H_i), g(H) - g(H_i)\}$ is a test on three integers, not the operation.
5. **The depth bound.** A proof that the executable process, after every rebuild and restart, has hierarchy length $O(\log n)$ (the talk) or $O((\log n)^2)$ (the Oxford version cited by archive 06),

together with polynomial digit bounds. The preprint's Section 10 gives a linear bound in the initial $q$-complexity, and its Section 9 explicitly declines to bound $g$ and $L$.

6. **The cost accounting.** Only after all of the above does the conditional lemma proved in every archive, $L(g+1)^L$ iterations times a polynomial cost per iteration times polynomially many restarts, become a running-time theorem: $\log((g+1)^L) = O((\log n)^2)$.

The archives also list, and we agree with, the substitutions that do not work: an exponential search with a time cap is not a total algorithm; a polynomial iteration count does not bound work per iteration; expanding a compressed normal surface loses the compression; $b_1 = 0$ is not ball recognition; a Jones or Alexander test does not detect the unknot.

# 6 A faster exact recognizer: `fastunknot` 0.1

## 6.1 Design

Since the geometric route cannot be completed from the available sources, the second task was addressed by making the *exact* route as fast as we could while keeping every verdict provable. The package `fast/fastunknot` (standard library only, Python $\geq$ 3.10) runs:

1. validation (PD codes, braid closures, or grid diagrams in the archive-01 convention, converted by the traversal of Section 4.3);
2. crossing-decreasing Reidemeister I/II moves, as in archive 05;
3. the descending-diagram test of archive 02;
4. the *Alexander polynomial*, computed exactly by fraction-free Bareiss elimination over $\mathbb{Z}[t]$ on a first minor of the Fox matrix of the Wirtinger presentation (rows $(1-t)x_o + tx_u - x_v$ at positive and $(1-t)x_o - x_u + tx_v$ at negative crossings, with signs from the traversal); $\Delta \neq 1$ certifies knottedness and strictly dominates the determinant filter of the archives ($T(3,5)$ and all torus knots are now rejected in polynomial time; the Kinoshita–Terasaka and Conway knots are not);
5. reduced Khovanov rank over $\mathbb{F}_2$ by *scanning*.

## 6.2 The scanning computation

This is Bar-Natan's local algorithm [36] specialised to $\mathbb{F}_2$ and implemented directly in terms of the arc-algebra description of morphisms. The crossings are processed one at a time in a greedy order chosen (from up to twelve starting crossings) to keep the boundary of the processed region small. After $k$ crossings the partial complex has:

- objects: perfect matchings of the boundary edges, each with a homological degree; closed circles are delooped, one object with a circle becoming two objects (labels 1 and $x$);
- morphisms $a \to b$: $\mathbb{F}_2$-combinations of dot-subsets of the circles of $a \cup \bar{b}$, i.e. elements of $A^{\otimes \text{circles}}$ with $A = \mathbb{F}_2[x]/(x^2)$.

Composition is the two-dimensional TQFT of the glued surface: a component of genus $\geq 1$ or with two dots is zero (because the handle operator $m \circ \Delta = 2x$ and $x^2$ vanish), a closed component is 1 exactly when it carries one dot, and a planar component with $\beta$ boundary circles and $d \in \{0,1\}$ dots is $\Delta^{(\beta-1)}(x^d)$, which over $\mathbb{F}_2$ is $x \otimes \cdots \otimes x$ for $d = 1$ and the sum over one undotted position for $d = 0$. Delooping is the same computation with a cup on the source circle (dotted for label $x$) and a cap on the target circle (dotted to extract the coefficient of 1, since the dual basis of $\{1, x\}$ under the counit is $\{\varepsilon(x\,\cdot\,), \varepsilon\}$). After every crossing all invertible entries are cancelled: an entry $b \to c$ between equal matchings is a unit of the local ring $\mathrm{End}(a) \cong A^{\otimes k}$ iff its expansion contains the undotted term, its inverse is the finite geometric series of the nilpotent part, and Bar-Natan's cancellation lemma replaces $\varepsilon$ by $\varepsilon - \gamma\varphi^{-1}\delta$. When the last crossing is added the boundary is empty, every morphism is a scalar, so a minimal complex has zero differential and its number of

objects is the unreduced total rank; the reduced rank is half of it (archive 04's two-copies lemma) and equals 1 exactly for the unknot.

**Proposition 6.1** (Correctness)**.** *On a validated one-component diagram,* `fastunknot` *returns* `UNKNOT` *iff the knot is trivial, unless a user-selected ceiling stops it, in which case it returns* `UNKNOWN`*.*

*Proof.* Steps 2–3 are isotopies and a sufficient triviality criterion; step 4 uses $\Delta_{\text{unknot}} = 1$. Step 5 computes the homology of the Khovanov complex: the tensor product of crossing complexes in Bar-Natan's category, delooping, and Gaussian elimination are chain homotopy equivalences, and the closed complex is the standard one. The decision rule is the detection theorem with the universal-coefficient argument of the archives. □

**Proposition 6.2** (Cost)**.** *Steps 1–4 are polynomial in $n$. Step 5 is $2^{O(n)}$ in the worst case.*

*Proof.* The Alexander determinant is $O(n^3)$ polynomial operations of degree $\leq n$ with polynomially bounded coefficients. In step 5 the objects before elimination after crossing $k$ number at most $2 \cdot 2^c$ times the objects after crossing $k-1$, where $c \leq 2$ closed circles can appear, so the whole computation is dominated by the cube; compositions cost polynomial time in the boundary size. □

The interesting quantity is the size after elimination. Empirically it stays close to (number of crossingless matchings of the boundary) $\times$ (a multiplicity that grows with the tangle's homology), i.e. roughly $2^{O(w)} \cdot \text{poly}$ for scan width $w$. **We have no proof of this**, and there cannot be one in general for the multiplicity: the reduced rank of $\#_k\, K11n34$ is $33^k$ with Alexander polynomial 1, so any method that outputs the full homology is exponential on such inputs, whatever the scan width. A subexponential unknot recognizer along these lines would have to detect "rank $> 1$" without finishing the computation; we do not know how.

### 6.3 Validation

The package has 19 unit tests (polynomial arithmetic; Alexander polynomials of the trefoil, figure-eight, $5_2$, $T(3,5)$, $\Delta = 1$ for the two 11-crossing knots and two hard unknots; symmetry $\Delta(t) = \Delta(t^{-1})$, $|\Delta(1)| = 1$, mirror invariance and odd determinant on 40 random braids; Reidemeister moves; the local endomorphism ring; $d^2 = 0$ after every crossing; independence of the processing order; resource-limit semantics; known ranks; an independent dense reduced-cube computation on 25 random braids; the command line). Beyond the tests, the scanner agrees with archive 04's cube on 120 random braid closures (3–5 strands, 6–13 letters, with random processing orders), and reproduces the Atlas ranks 33, 33 and 7.

### 6.4 Measurements

Timings are on the same Windows machine as everything else in this report; they illustrate the dependence on scan width and are not complexity evidence.

| Family | $n$ | reduced rank | max. boundary | objects before | objects after | seconds |
|---|---|---|---|---|---|---|
| unknot sigma_1..sigma_n | 8 | 1 | 2 | 6 | 2 | 0.001 |
| unknot sigma_1..sigma_n | 16 | 1 | 2 | 6 | 2 | 0.004 |
| unknot sigma_1..sigma_n | 32 | 1 | 2 | 6 | 2 | 0.013 |
| unknot sigma_1..sigma_n | 64 | 1 | 2 | 6 | 2 | 0.036 |
| unknot sigma_1..sigma_n | 128 | 1 | 2 | 6 | 2 | 0.133 |
| unknot sigma_1..sigma_n | 256 | 1 | 2 | 6 | 2 | 0.475 |
| T(2,n) | 5 | 5 | 4 | 30 | 10 | 0.003 |
| T(2,n) | 11 | 11 | 4 | 66 | 22 | 0.011 |
| T(2,n) | 21 | 21 | 4 | 126 | 42 | 0.043 |
| T(2,n) | 41 | 41 | 4 | 246 | 82 | 0.153 |
| T(3,n) | 8 | 5 | 6 | 42 | 10 | 0.010 |
| T(3,n) | 10 | 7 | 6 | 54 | 14 | 0.022 |
| T(3,n) | 14 | 9 | 6 | 78 | 18 | 0.032 |
| T(3,n) | 16 | 11 | 6 | 90 | 22 | 0.045 |
| T(3,n) | 20 | 13 | 6 | 114 | 26 | 0.092 |
| T(3,n) | 22 | 15 | 6 | 126 | 30 | 0.089 |
| random 3-braid | 20 | 75 | 6 | 762 | 150 | 0.217 |
| random 3-braid | 40 | 321 | 6 | 3354 | 642 | 1.035 |
| random 4-braid | 15 | 5 | 6 | 30 | 10 | 0.015 |
| random 4-braid | 21 | 25 | 8 | 438 | 146 | 0.576 |
| random 4-braid | 31 | 45 | 8 | 604 | 216 | 0.947 |
| random 4-braid | 41 | 109 | 8 | 654 | 218 | 2.683 |
| random 5-braid | 24 | 13 | 6 | 109 | 42 | 0.099 |
| random 5-braid | 36 | timeout | timeout | timeout | timeout | 600.147 |
| random 6-braid | 31 | 85 | 8 | 548 | 216 | 1.194 |

| Example | $n$ | verdict | deciding step | seconds |
|---|---|---|---|---|
| conway.json | 11 | KNOTTED | reduced-khovanov-F2-scan | 0.0528 |
| figure_eight.json | 4 | KNOTTED | alexander-polynomial | 0.0001 |
| grid_determinant_one_knot.json | 10 | KNOTTED | alexander-polynomial | 0.0010 |
| grid_scrambled_unknot.json | 6 | UNKNOT | reidemeister-reduction | 0.0004 |
| hard_unknot_8.json | 8 | UNKNOT | reduced-khovanov-F2-scan | 0.0022 |
| kinoshita_terasaka.json | 11 | KNOTTED | reduced-khovanov-F2-scan | 0.0593 |
| torus_3_5.json | 10 | KNOTTED | alexander-polynomial | 0.0009 |
| trefoil.json | 3 | KNOTTED | alexander-polynomial | 0.0001 |
| unknot.json | 0 | UNKNOT | reidemeister-reduction | 0.0000 |
| unknot_braid40.json | 40 | UNKNOT | reidemeister-reduction | 0.0090 |

One case timed out: a random 36-letter closure on five strands exceeded the 600-second budget, while a 31-letter closure on six strands took about a second. When this section was first written we attributed the difference to the size of the minimal complexes of the intermediate tangles. **That explanation was wrong.** Section 7 shows that the minimal complexes of this input never exceed 5898 objects; the time went into fill-in caused by cancelling pivots in last-in-first-out order, and with min-fill pivots the same scan takes under a second.

Two comparisons with the archives' cubes on identical inputs: the Kinoshita–Terasaka knot (11 crossings, 2048 states, 11 559 generators) takes 0.24–0.31 s in the cubes and 0.07 s in the scan with a boundary of six edges and at most 66 objects after elimination; the unknot $\widehat{\sigma_1 \cdots \sigma_n}$, archive 03's explicit exponential family, needs $2^n$ states in every cube and is decided by the scan in 0.011 s for $n = 80$ with two objects after every elimination.

### 6.5 Limits of the new package

It is a reference implementation in pure Python; a compiled implementation of the same algorithm (as in Bar-Natan's or the KnotJob programs) would be far faster. Its default scan-order heuristic is greedy; the optional certified separator order added in Section 36 guarantees $O(\sqrt{n})$ width. Grid-like projections can still require $\Omega(\sqrt{n})$ raw width, and minimal complexes can be large. Homology verdicts do not produce a spanning disc or a Reidemeister sequence. No general quasi-polynomial recognition bound is proved.

## 7 Nine acceleration proposals

### 7.1 What was delivered

After version 0.1 of `fastunknot` (Section 6) was written, nine further runs were each asked to accelerate it. Their archives were extracted into `proposals/01`–`09` (wrapper directories removed, ZIP files deleted, archive order recorded in `proposals/ORDER.txt`). Every proposal is a complete replacement package with its own report, tests, an unchanged copy of the 0.1 baseline and same-machine benchmarks, and every one of them states that the general worst case remains exponential. As with the six original archives, the independent runs converged on the same ideas, which is evidence that the ideas are the natural ones.

### 7.2 The ideas

1. **Units are involutions.** In $\mathrm{End}(m) \cong \mathbb{F}_2[x_1, \ldots, x_k]/(x_i^2)$ a unit is $1+\nu$ with $\nu$ in the maximal ideal, and $\nu^2 = 0$ in characteristic two (squares of monomials vanish, cross terms occur twice). So $(1+\nu)^{-1} = 1+\nu$ and the geometric series of 0.1 is unnecessary. All nine.
2. **Min-fill pivots.** Version 0.1 cancelled invertible entries in last-in-first-out order. Choosing the pivot $b \to c$ that minimizes the Markowitz count $(\mathrm{out}(b)-1)(\mathrm{in}(c)-1)$ through a lazy heap limits fill-in [38]. Seven of nine (not 05, 06).
3. **Cheaper cobordism algebra.** The topology of a gluing (which discs merge, Euler characteristics, boundary circles) does not depend on the dots, so it can be computed once per triple of matchings and reused; identity and square shortcuts; memoized results. All nine; proposal 03 goes furthest and packs a morphism into one integer whose bit number is a dot mask.
4. $O(n \log n)$ **scan order and** $O(n)$ **descending test.** The greedy score of a crossing changes only when a neighbour is processed, so a lazy heap reproduces the 0.1 order exactly; the descending test becomes a sweep over a difference array of bad cyclic intervals. All nine.
5. **Modular Alexander test.** Evaluate the first Alexander minor in a prime field; for the unknot it is $\pm t^k$, $0 \le k < n$, so any other value certifies knottedness in $O(n^3)$ field operations. Six of nine, three of them only at $t = -1$ (the determinant).
6. **Modular Jones test by a frontier scan.** Evaluate the Kauffman bracket at a fixed $A \in \mathbb{F}_p$ while scanning the diagram, adding the weights of partial states as soon as they induce the same boundary matching. A value different from $\delta(-A^3)^w$ certifies knottedness. All nine. This is the most valuable single idea for recognition: it rejects the Kinoshita–Terasaka and Conway knots, which have trivial Alexander polynomial, in about a millisecond, and its cost is *provably* $(w-1)!! \cdot \mathrm{poly}(n)$ for scan width $w$, because states with equal matchings are merged; there is no multiplicity factor.
7. **Visible connected sums.** Two edges bordering the same two faces are a two-edge cut of the projection, so the diagram is a connected sum of the two sides. Reduced Khovanov homology over a field is multiplicative, so ranks multiply and degree counts convolve; for recognition the summands are examined separately. Seven of nine.
8. **Crossing signs from both strands.** Version 0.1 assumed that a PD row starts at the

incoming under-port (true for its own converters); seven proposals compute the sign from both incoming ports, which the Jones normalization needs.

9. **Incremental Reidemeister simplification** (02 only), **no cancellation on the last crossings followed by linear algebra** (06 only), **an exact integer bracket after the modular one** (05 only), and **witness replay commands** (six proposals).

### 7.3 Same-machine comparison

The harness `synthesis/data/proposals_bench.py` runs every engine on the same inputs, each measurement in a fresh process with a 120-second limit; timers exclude start-up, imports and input construction; medians of three runs (one run above five seconds). "scan" is the raw Khovanov scan with factorization disabled, "rank" the rank of a connected sum of eight Conway knots through each engine's factored API when it has one, "recognize" the default pipeline. Milliseconds; 0.1 is the baseline, 0.2 the integrated version of Section 8, and Rust the port of Section 9.

| Task | 0.1 | 01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 0.2 | Rust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| scan: conway | 34.4 | 11.6 | 10.4 | 9.86 | 11.3 | 10.2 | 10.4 | 9.80 | 10.0 | 12.5 | 8.44 | 0.59 |
| scan: kinoshita_terasaka | 41.5 | 12.4 | 11.2 | 10.5 | 12.8 | 12.3 | 13.3 | 10.8 | 11.3 | 13.0 | 9.05 | 0.62 |
| scan: T(3,11) | 51.8 | 29.9 | 29.4 | 27.9 | 31.2 | 28.8 | 31.7 | 26.9 | 30.3 | 30.8 | 26.8 | 1.22 |
| scan: braid3_40 | 611 | 103 | 115 | 94.0 | 108 | 133 | 140 | 143 | 164 | 201 | 84.1 | 6.88 |
| scan: braid4_41 | 984 | 263 | 274 | 212 | 350 | 520 | 321 | 273 | 268 | 284 | 160 | 10.2 |
| scan: braid6_31 | 495 | 96.7 | 108 | 88.2 | 115 | 95.1 | 93.6 | 94.6 | 104 | 114 | 67.7 | 5.25 |
| scan: braid5_36 | >120 s | 1064 | 1208 | 887 | 1153 | 44995 | 7708 | 1080 | 1163 | 1491 | 691 | 68.1 |
| scan: chain256 | 312 | 28.8 | 24.9 | 29.7 | 26.9 | 24.8 | 335 | 24.1 | 24.2 | 25.3 | 19.3 | 0.97 |
| scan: conway_sum_2 | 1481 | 239 | 241 | 173 | 204 | 220 | 205 | 213 | 207 | 261 | 136 | 11.9 |
| rank: conway_sum_8 | >120 s | 31.5 | >120 s | 20.2 | 31.4 | >120 s | 16.0 | 29.3 | 51.7 | 18.6 | 20.7 | 1.90 |
| recognize: trefoil | 0.13 | 0.19 | 0.19 | 0.13 | 1.67 | 0.15 | 0.15 | 0.28 | 0.16 | 0.15 | 0.27 | 0.008 |
| recognize: torus_3_5 | 0.98 | 0.34 | 1.03 | 0.32 | 2.24 | 1.06 | 0.80 | 1.04 | 0.32 | 0.66 | 0.56 | 0.027 |
| recognize: hard_unknot_8 | 2.84 | 5.05 | 2.40 | 4.02 | 3.86 | 2.45 | 3.07 | 2.18 | 2.16 | 3.03 | 2.78 | 0.056 |
| recognize: unknot_braid40 | 11.2 | 8.00 | 0.41 | 7.82 | 8.31 | 7.64 | 7.50 | 7.93 | 8.64 | 8.42 | 0.17 | 0.005 |
| recognize: conway | 58.8 | 1.53 | 0.89 | 1.79 | 4.01 | 3.50 | 4.05 | 1.57 | 3.95 | 1.20 | 1.27 | 0.055 |
| recognize: kinoshita_terasaka | 65.6 | 2.12 | 1.02 | 1.49 | 3.43 | 1.55 | 3.31 | 1.08 | 1.81 | 0.72 | 1.25 | 0.030 |
| recognize: conway_sum_2 | 1817 | 3.54 | 2.05 | 3.47 | 2.75 | 2.32 | 12.6 | 2.07 | 2.77 | 2.17 | 1.56 | 0.059 |
| recognize: conway_sum_3 | 58090 | 6.77 | 2.96 | 5.15 | 3.60 | 3.43 | 35.3 | 2.10 | 2.78 | 2.51 | 2.47 | 0.099 |
| recognize: T(3,61) | 4747 | 3.71 | 12.5 | 5780 | 13.8 | 19.0 | 5811 | 22.3 | 5.91 | 8.80 | 8.84 | 0.19 |
| recognize: braid5_36 | 9.89 | 2.53 | 0.69 | 2.38 | 8.63 | 2.63 | 11.2 | 6.20 | 2.40 | 2.37 | 0.90 | 0.023 |

All eleven Python engines and the Rust binary return identical ranks and verdicts on every input. Observations:

- The stress case `braid5_36`, which 0.1 abandoned after 600 seconds, takes about a second wherever min-fill pivots are used, 45 seconds in proposal 05 (LIFO kept) and 7.7 seconds in proposal 06 (LIFO plus the tail trick). Its reduced rank is 2949. **This corrects our own 0.1 analysis**: we had attributed the timeout to large minimal complexes of the intermediate tangles. The minimal complexes are small (at most 5898 objects); what exploded was fill-in caused by the pivot order.
- Proposal 03 has the fastest raw scanner, which is why its representation was adopted.
- On $T(3,61)$ proposals 03 and 06 take 5.8 seconds. Proposal 06 runs the symbolic Alexander polynomial before anything modular. Proposal 03 is more instructive: it evaluates the minor at $t = 2$ modulo $2^{31} - 1$, where 2 has multiplicative order 31, so the set of "units" $\{\pm 2^k\}$ has only 62 elements and is closed under the cyclotomic arithmetic of torus knots; $\Delta_{T(3,61)}(2) \equiv -2^{29}$ and the filter misses. Our own first implementation had the same structural weakness (2 has order 61 modulo $2^{61} - 1$) even though it happened to fire. Version 0.2 and the Rust port evaluate at fixed generic field elements instead, for both the Alexander and the Jones test. The filters are one-sided, so this was never a soundness issue, only a lost speedup.
- Tiny inputs get slower: the trefoil goes from 0.1 ms to 0.2–0.3 ms in most engines, because more stages run before one fires.

# 8 `fastunknot` 0.2

## 8.1 What was adopted

Version 0.2 (`fast/`) integrates every proposed idea that produced a measurable speedup, re-implemented in the structure of the existing package; the 0.1 scanner is kept unchanged as `scan_reference` so that each idea can be switched off and so that the new code has an independent oracle.

The pipeline is now: validation; incremental R1/R2 reduction; linear descending test; visible connected-sum factorization (summands examined smallest first, the first knotted one decides, and a sum is trivial exactly when all summands are); then per summand the Alexander minor at $t = -1$ and at a generic field element modulo $p = 2^{61} - 1$, the Kauffman bracket at a generic $A$ modulo $p$ (skipped when it exceeds 4096 frontier states), the exact Alexander polynomial over $\mathbb{Z}[t]$, and finally the Khovanov scan with bit-packed morphisms, compiled plans and min-fill pivots. Every filter is one-sided: only a mismatch with the unknot's value gives a verdict, and that verdict is KNOTTED.

Not adopted: the exact integer bracket (equality with the unknot's value is inconclusive whether it is computed modulo $p$ or over $\mathbb{Z}$, so it cannot change a verdict, and it showed no measurable effect); replay commands (useful, but not a speedup); tails longer than one crossing (slower, see below).

## 8.2 Ablation

`fast/ablation.py` measures each idea in isolation, again in fresh processes (limit 300 s). First the scanner, in milliseconds:

| Scanner configuration | conway | braid3_40 | braid4_41 | braid6_31 | braid5_36 |
|---|---|---|---|---|---|
| 0.1 reference scanner | 42.9 | 1115 | 1264 | 612 | >300 s |
| sets, LIFO, series inverse | 44.1 | 1044 | 1206 | 618 | >300 s |
| sets, LIFO, self-inverse | 43.0 | 703 | 1113 | 618 | >300 s |
| sets, min-fill | 43.0 | 594 | 845 | 553 | 10376 |
| bits, LIFO | 11.4 | 106 | 203 | 73.4 | 35622 |
| bits, min-fill (default) | 11.1 | 103 | 224 | 78.6 | 670 |
| bits, min-fill, tail 1 | 10.3 | 97.0 | 232 | 81.8 | 566 |
| bits, min-fill, tail 2 | 13.7 | 105 | 225 | 84.6 | 815 |

Then the polynomial-time stages and the filters:

| Task | new (ms) | old (ms) | ratio |
|---|---|---|---|
| greedy order, 2048 crossings | heap O(n log n): 32.6 | 0.1 rescoring O($n^2$): 26685 | 818× |
| descending test, 2048 crossings | linear sweep: 3.72 | 0.1 quadratic: 314 | 84× |
| R1/R2 reduction, 2048 crossings | incremental: 3.77 | 0.1 rebuild per move: 28089 | 7458× |
| T(3,61) rejection, 122 crossings | modular minor: 8.07 | 0.1 exact Z[t]: 5829 | 722× |
| recognize Conway | with Jones filter: 0.81 | without Jones filter: 10.0 | 12× |
| recognize Conway#Conway#Conway, no Jones | with factorization: 12.6 | without factorization: 6781 | 539× |
| rank of Conway#Conway#Conway | factored: 20.7 | one scan: 6972 | 337× |

What the ablation shows:

- **The bit-packed algebra** is the across-the-board gain: 4 to 10 times on every scan, with either pivot rule.

- **Min-fill pivots** matter on exactly one kind of input, and there they are decisive. On the stress case the bit-packed scanner needs 35.6 s with LIFO pivots and 0.67 s with min-fill; the set-based scanner goes from more than 300 s to 10.4 s. On the other four inputs the pivot rule is neutral to slightly negative (the heap costs a few percent). The final complex is the same in all configurations, so the difference is fill-in during cancellation, not the size of the homology.
- **Self-inverse units** save between nothing and 30%, depending on how many cancellations involve a non-identity unit.
- **The tail trick** is mixed: one crossing without cancellation is 16% faster on the stress case and 6% faster on one other input, a few percent slower on the rest; two crossings are slower almost everywhere because the uncancelled complex quadruples. It is kept as an option (`tail`) with default zero.
- **Heap ordering, incremental reduction, modular Alexander test**: three to four orders of magnitude at 2048 crossings and at $T(3, 61)$. These are asymptotic improvements of polynomial stages: the old code spent 27 s on ordering and 28 s on R1 moves for a diagram that reduces to nothing.
- **The linear descending test** needs an adversarial input to show (a 2048-crossing chain with one crossing flipped: 3.7 ms against 314 ms); on typical inputs the old test exits early and the two are equal.
- **The Jones filter** turns recognition of the Conway knot from a 10 ms Khovanov scan into a 0.8 ms bracket evaluation, and **factorization** turns a 6.8 s scan of three Conway summands into 12.6 ms even with the Jones filter disabled.

### 8.3 Complexity: what changed and what did not

The polynomial stages improved asymptotically: scan ordering from $O(n^2)$ to $O(n \log n)$ per start, the descending test from $O(n^2)$ to $O(n)$, R1/R2 reduction from one $O(n)$ revalidation per move to $O(1)$ per move, and the Alexander rejection from fraction-free elimination over $\mathbb{Z}[t]$ to $O(n^3)$ field operations.

Two statements with an exponential flavour are now *proved* rather than observed:

**Proposition 8.1.** *The Jones filter costs $O(n \cdot (w-1)!! \cdot \mathrm{poly}(w))$ field operations for a scan of width $w$.*

*Proof.* After each crossing the state is a map from perfect matchings of at most $w$ boundary points to $\mathbb{F}_p$; each entry spawns two successors. The earlier Catalan claim required an unproved common-disk hypothesis. Report 09 supplies explicit counterexamples; the October integration corrects this statement without changing the exact bracket evaluation. □

**Proposition 8.2.** *If the visible factors of the reduced diagram have $n_1, \ldots, n_k$ crossings, recognition and the factored rank cost $\mathrm{poly}(n) + \sum_i 2^{O(n_i)}$. In particular the cost is polynomial when every $n_i = O(\log n)$ and quasi-polynomial when every $n_i = O(\log^2 n)$.*

This is a statement about a restricted class of diagrams (several proposals make it too), not about unknot recognition: a composite knot can be drawn without any two-edge cut, and prime knots gain nothing. The family $\#_k\, K11n34$, which Section 6 used to show that a scan cannot be subexponential if it must output the whole homology, is exactly the family that factorization now handles in polynomial time; the obstruction moves to diagrams in which the summands are hidden.

The general worst case of the exact fallback is unchanged, $2^{O(n)}$, and the $n^{O(\log n)}$ target of Section 5 is no closer: nothing here touches the hierarchy machinery.

### 8.4 Verification

The package now has 33 unit tests. The 14 new ones compare bit-packed with set-based composition on random planar matchings, check every unit of a three-circle endomorphism ring, run all scanner configurations against the 0.1 scanner, check that heap orders equal the 0.1 orders from every start, check sign and filter invariance under half-turns of PD rows, check that no filter fires on any unknot diagram and that every filter hit has reduced Khovanov rank above one, check $33^k$ for $k = 2, 3, 8$ Conway summands and equality of factored and unfactored degree counts, replay simplifier traces, and recompute the stress case (rank 2949, with and without the tail).

## 9 A Rust implementation

### 9.1 Design

`rust/` is a dependency-free crate (about 2500 lines) implementing the 0.2 pipeline except the exact Alexander polynomial over $\mathbb{Z}[t]$, which would need big integers and on which exactness never depended. It was designed for Rust rather than transliterated:

- matchings are interned, so objects, plans and caches hold 32-bit ids;
- a morphism is a bitset over dot masks, a single inline `u64` when the Hom space has at most six circles (every scan of width at most twelve) and boxed words otherwise;
- composition and transfer plans are a few integer masks per surface component, evaluated with popcounts;
- memo tables use an Fx-style multiplicative hasher; measured against the standard SipHash it is worth 22% on the stress case (65 ms against 86 ms) and 18% on two Conway summands;
- objects live in flat vectors, the min-fill queue is a binary heap, and field arithmetic modulo $2^{61} - 1$ uses 128-bit products;
- factored ranks such as $33^{16}$ use a small base-$10^9$ integer.

### 9.2 Verification

Four unit tests (known ranks under both pivot rules, tails 0–2 and mirrors; input validation; one-sidedness of the filters; the pipeline), and `rust/cross_check.py`, which compares ranks by cube degree in three configurations and verdicts in two with the Python package on random braid closures: 200 closures, no disagreement.

### 9.3 Profile

`rust/profile.py` records the binary's own timers (medians of five runs inside one process, after parsing). The comparison columns of the table in Section 7 show the Rust binary against the Python engines on identical inputs. Larger random braid closures, raw scan, one run each with a 300-second budget:

| Input | reduced rank | boundary | objects | total s | transfer s | eliminate s |
|---|---|---|---|---|---|---|
| random 3-braid, 80 crossings | 5189 | 6 | 12240 | 0.235 | 0.101 | 0.122 |
| random 3-braid, 160 crossings | budget of 300 s exhausted | | | | | |
| random 4-braid, 61 crossings | 643 | 8 | 3310 | 0.113 | 0.055 | 0.055 |
| random 4-braid, 81 crossings | budget of 300 s exhausted | | | | | |
| random 5-braid, 48 crossings | 39 | 10 | 450 | 0.015 | 0.011 | 0.004 |
| random 5-braid, 60 crossings | 481 | 10 | 7398 | 4.663 | 0.525 | 4.108 |
| random 6-braid, 43 crossings | 177 | 10 | 530 | 0.023 | 0.014 | 0.008 |
| random 6-braid, 55 crossings | 1171 | 10 | 13294 | 1.484 | 0.434 | 1.021 |
| random 7-braid, 48 crossings | 29 | 10 | 1838 | 0.070 | 0.032 | 0.036 |
| random 8-braid, 49 crossings | 1709 | 12 | 3418 | 0.309 | 0.153 | 0.149 |

What the profile shows:

- On identical inputs the Rust scanner is 10 to 17 times faster than Python 0.2 (Conway 0.59 ms against 8.4 ms, the 40-crossing 3-braid 6.9 ms against 84 ms, the stress case 68 ms against 691 ms) and recognition takes between 5 and 200 microseconds. Against version 0.1 the stress case went from more than 600 s to 0.068 s.
- Time splits roughly evenly between adding crossings (gluing, delooping, transfer plans) and cancellation, except when fill-in dominates: with LIFO pivots the stress case takes 2.1 s in Rust, 2.03 s of it in cancellation, against 0.068 s with min-fill. The pivot rule matters as much in Rust as in Python; on the other inputs it is again neutral.
- One uncancelled tail crossing saves 5 to 20% in Rust as well.
- **The exponential wall is still there.** A random 3-strand closure with 80 crossings has reduced rank 5189 and takes 0.23 s; with 160 crossings the scan exhausted 300 s. A 4-strand closure with 61 crossings (rank 643) takes 0.11 s; with 81 crossings it exhausted 300 s. The scan width is 6 or 8 throughout, so this is the multiplicity factor of Section 6 and not the boundary: the rank of the homology itself grows exponentially with the length of a random braid word, and any algorithm that writes the homology down inherits that. (Both runs overshot their cooperative deadline by 40 to 70 seconds, because the deadline is checked between cancellations.) For *recognition* these inputs are easy, since the Alexander test rejects them in microseconds; the hard inputs for recognition are diagrams of the unknot and of knots with trivial Alexander and Jones values at our evaluation points, where the scan must run to completion.

### 9.4 Where the three versions stand

| Task | Python 0.1 | Python 0.2 | Rust |
|---|---|---|---|
| Scan, stress closure (36 crossings, rank 2949) | $> 600$ s | 0.69 s | 0.068 s |
| Scan, Conway knot | 34 ms | 8.4 ms | 0.59 ms |
| Recognize Conway knot | 59 ms | 1.3 ms | 0.055 ms |
| Recognize Conway#Conway#Conway | 58 s | 2.5 ms | 0.099 ms |
| Recognize $T(3, 61)$ | 4.7 s | 8.8 ms | 0.19 ms |
| Rank of eight Conway summands ($33^8$) | $> 120$ s | 21 ms | 1.9 ms |
| Recognize the 8-crossing hard unknot | 2.8 ms | 2.8 ms | 0.056 ms |

The last row is the honest one for the original question: on an unknot diagram that no filter can decide, the accelerations of the Python version change nothing, because the Khovanov scan still has to run; only the constant factor of the Rust port helps. (This describes 0.1 against 0.2. The later work of Section 10 does speed up the scan itself, and this case with it, by a factor of 1.4.)

# 10 A third round: constant factors in Python, and how to measure them

After version 0.2 the Rust work was paused and the Python package was optimized further, with no change to the mathematics, the verdicts or the asymptotic behaviour. This section records what was gained, what was tried and rejected, and an episode about measurement that changed how every number below was obtained. All verdicts and ranks are exact and identical to those of 0.2 (the deciding step can differ: the Reidemeister III help below settles some unknots before the scan): 41 unit tests pass, and `crosscheck.py` compares the ranks by homological degree of the new default scanner with the generic 0.2 scanner and the 0.1 set algebra on random braid closures (400 closures with 2 to 5 strands, no mismatch).

## 10.1 What changed

A profile of 0.2 on the stress closure showed that the cobordism algebra was no longer the bottleneck: of 1.6 s under the profiler, the scanner's own bookkeeping (method calls per differential entry, tuple-keyed dictionaries, hashing of nested frozensets, a heap of pivot candidates) took most of it.

- **A scanner written for CPython** (`scan_fast.py`). Objects of a stage are numbered $0, \dots, N-1$ and live in lists; the objects created from one old object form a contiguous block, so a delooping label is an offset; matchings are interned as integers, so every cache key is a tuple of small integers; transfers of an edge are cached per (source matching, target matching, value) for both smoothings at once; cancellation is one inlined loop. The generic scanner of 0.2 is kept for the ablation configurations.
- **A bucket queue for pivots.** On the stress closure 114 197 unit entries are candidates but only 16 884 are used; the rest die when a neighbour is cancelled. Markowitz costs are tiny, so one list per small cost plus a heap for the rare large ones replaces the heap.
- **Integer planar geometry** (`planar.py`). Circles of $a \cup \bar{b}$ are found by walking partners alternately; a crossing is glued by following the at most four strands through it; a transfer plan is compiled only over the crossing and the circles it touches, the other circles being renumbered; and since monomials of different components live on disjoint circles, multiplying a bit-packed sum by a monomial $c$ is the shift `<< c`.
- **Outside the scanner.** The modular Alexander filter builds the matrix by sparse rows (at most three nonzero entries per row) and evaluates each distinct entry once; the boundary profile of a greedy scan order is tracked inside the greedy in $O(1)$ per step and a losing start is abandoned early; the Jones frontier scan uses the interned matchings and the strand-following gluing. Each was checked to be output-preserving against the previous implementation on 500 to 600 random diagrams, including diagrams with kinks.

## 10.2 Results

Same harness on both sides (`perf.py`: in one process, best of $N$, timers around the call only), so the columns are comparable with each other but *not* with the fresh-process medians of the previous section.

| Task (milliseconds) | 0.2 | after the scanner work | ratio |
|---|---|---|---|
| Scan, stress closure (36 crossings, rank 2949) | 575.4 | 216.8 | 2.65 |
| Scan, 41-crossing 4-strand closure | 135.3 | 51.1 | 2.65 |
| Scan, 40-crossing 3-strand closure | 71.0 | 26.2 | 2.71 |
| Scan, 31-crossing 6-strand closure | 59.3 | 21.4 | 2.77 |
| Scan, Conway#Conway (unfactored) | 114.9 | 41.7 | 2.75 |
| Scan, $T(3,11)$ | 22.8 | 8.7 | 2.61 |
| Scan, Conway knot | 7.7 | 3.0 | 2.54 |
| Scan, chain of 256 kinks | 15.9 | 10.9 | 1.47 |
| Recognize the 8-crossing hard unknot | 1.26 | 0.90 | 1.41 |
| Recognize Conway knot, exact steps only | 7.8 | 3.2 | 2.43 |
| Total of the 14 cases of `perf.py` | 1022.2 | 394.6 | 2.59 |

The right-hand column is a run whose control loop (see below) spread by 4.8%. The changes outside the scanner came later and were measured as paired ratios new/old with 95% confidence intervals, which do not need a quiet machine: recognition of $T(3,61)$ 0.60 $[0.53, 0.67]$ and then 0.69 $[0.62, 0.72]$, together about 0.41; the chain of 256 kinks 0.78 $[0.73, 0.82]$; the Jones scan of the stress closure 0.67 $[0.46, 0.72]$; recognition of the Conway knot 0.92 $[0.89, 0.93]$. Four later sequential runs were all rejected by the control rule and are not used; they are flagged in `perf_log.json`.

**A cache that began as a trade-off.** The last change was at first not a free win; how it became nearly one is told in order. Every crossing brings new edge labels, so plans cached per stage are never reused across stages, although a long braid compiles the same few pictures at every crossing: at boundary size 6 there are five matchings, and about 35 transfer plans were compiled per crossing of $T(3,n)$. Transfer plans are now also cached under a label-independent key, the matchings and the crossing relabelled by rank. This is sound because the relabelling is monotone and circles are numbered by their minimal label whichever matching is walked first (tested on random matchings). The reuse is 80% of the plans for $T(3,11)$, 94% for $T(3,31)$, 99% for the chain of kinks, 38% and 19% for the random 3- and 4-strand closures, but 1% for the stress closure and none for the Conway knot. With the cache always on, three agreeing A/B runs: scan of $T(3,11)$ 0.61 $[0.59, 0.66]$, chain of kinks 0.78, 40-crossing 3-strand closure 0.97; *scan of the Conway knot* 1.06 $[1.04, 1.07]$ *and recognition of the hard unknot* 1.07*, slower*; no claim on the large scans. Timing partial variants against the old code (A/A ratio 0.999) splits the cost into ranking the boundary at each stage (+1.5%), building and looking up the key (+3.0%) and storing (+1.3%). Two explanations offered first were wrong: interning the shapes as integers, against the hashing of nested tuples, changed nothing, and the ratio was the same with the cyclic garbage collector disabled; `shape_cache=False` restores the previous speed exactly (1.003 $[0.999, 1.007]$).

A first attempt at an automatic switch failed, and I concluded too early that none was possible: an early hit count predicts the wrong way, since the inputs that profit get their first hit only after 35 to 150 lookups, while the stress closure gets 12 hits in its first 64 lookups and then none. A different predictor works. A plan can be reused only in a stage whose picture, the position of the four crossing slots among the boundary points up to monotone relabelling, has occurred before, and that is known from the scan order alone, before the scan. The number of such stages is 0 for the Conway knot, the Kinoshita–Terasaka knot and Conway#Conway, 1 of 36 for the stress closure, 4 of 31 and 11 of 41 for the 6- and 4-strand closures, 16 of 40 for the 3-strand closure and 16 of 22 for $T(3,11)$. The default now turns the cache on when the diagram has at least 16 crossings and an eighth of its stages repeat. Transfer and composition *results* are shared across stages too, but only after the first reused plan of a scan, because before that no shared result can exist; without that gate they cost a further 2.6% on the small irregular scans. Against the code before any of this, with 301 interleaved pairs: scan of $T(3,11)$ 0.505 $[0.486, 0.508]$, Conway knot 1.008 $[1.005, 1.010]$, hard unknot 1.007; on the corpus, $T(3,11)$ 0.51, chain of kinks 0.81, and nothing slower.

**A stage that was not needed.** The benchmark corpus contained one input that no filter decides, the hard unknot with 8 crossings, and that hid the cost of a whole stage. A generator of scrambled unknot braids (`hard_unknots.py`: the closure of $v\,\sigma_1 \cdots \sigma_{n-1} v^{-1}$ rewritten by random braid relations) gives diagrams of which about half survive the Reidemeister I/II reduction with 15 to 27 crossings and pass every filter. On 26 of them the exact Alexander polynomial over $\mathbb{Z}[t]$ took a median 24%, and up to 63%, of the recognition. That stage can only answer KNOTTED, and only for a polynomial that is not a unit, which the modular test has just failed to see at $t = -1$ and at a generic point of $\mathbb{F}_p$; and whatever it might still catch, the exact scan decides anyway, so verdicts cannot change. It now runs only when the modular test did not. Over 720 diagrams the two modes never disagree, and the exact stage decided nothing once the modular test had passed. Recognition of the 55 scan-decided scrambled unknots is faster on all 55: median ratio 0.81, range 0.53 to 0.93.

**Reidemeister III, the one algorithmic gain of this round.** The scan is exponential in what the reduction leaves, and the reduction used to stop wherever a Reidemeister III move was needed. On the dart structure of the reduction a triangular face is a 3-cycle of the face walk through three distinct crossings; it can be inverted if and only if some side passes over at both ends; and inverting it only reverses the order in which each of the three strands meets the other two, a relinking of at most twelve darts that keeps every crossing's slots and the direction of both strands through it, hence its sign. When no I/II move is left, each such triangle is inverted on trial, kept if that creates a I/II move and undone otherwise (the move is an involution). The move was tested on its own before it was used: 150 random moves give valid spherical diagrams with the same Alexander polynomial, writhe and Khovanov ranks by degree, and a second move restores the structure exactly (300 of 300). On 1500 random closures the helped reduction never leaves more crossings than I/II alone, leaves no I/II move, replays from its trace, and preserves the Alexander polynomial and the Khovanov rank; 26% of the diagrams shrink further and 0.80 of the crossings remain. The 8-crossing "hard unknot" used throughout this report goes from 6 crossings to none, and so do all 120 scrambled unknots above, which is no surprise, their scrambling being made of such moves. Honest hard inputs had to be found by search: 117 closures among 40 000 random words that survive I, II, the III search and the Alexander filter, and are unknots by reduced Khovanov rank 1, with up to 21 crossings. Verdicts with and without the search agree on 900 diagrams.

Two lessons came with it. *Placement:* run together with the first reduction, the search cost 58% on $T(3, 61)$, 20% on a random 5-strand closure and 4 to 6% on the Conway knot, inputs that a filter decides a moment later. It now runs only after every filter has failed, immediately before the scan; then no slowdown can be claimed on those inputs, while recognition of the hard unknot takes 0.48 and of a scrambled unknot of 27 crossings 0.18 of the time. *Fewer crossings is not always less work:* one survivor was flagged 1.43 times slower, correctly, since the help takes it from 22 to 20 crossings and its scan work from 2081 to 3277. Over the 54 survivors where crossings are removed (3.4 on average) the scan work is 0.43 of what it was, cheaper on 44 and dearer on 10, with per-diagram ratios from 0.03 to 7.3; over all 117 it is 0.69.

*Depth, and a budget.* One move of lookahead left those 117 unknots untouched. Sequences of two moves, the second next to the first, reduce 97 of them to nothing, three moves 109 and four moves 110, without a violation at any depth. What costs is a search that fails: on $T(3, 61)$, full of triangles that lead nowhere, depths 1 to 4 take 2, 10, 44 and 189 ms. Useful sequences are found early, so the search watches its own cost and deepens only while fewer than a fixed number of trial moves per crossing have been made; budgets of 5, 15, 40 and none give the same 97, 109 and 110, and a budget of 5 caps the $T(3, 61)$ case at 10 ms. With the defaults (depth 4, budget 10) only 15 of 60 000 random closures are unknots that survive the reduction, the largest with 16 crossings.

**Introspective selection and racing (two suggestions of the user).** The first suggestion was to choose algorithms from a runtime profile, as introsort does. It works where a cheap signal separates the populations and switching is free, and three parts of the package are now of that kind: the shape cache chosen from the scan order, result sharing enabled by the first reused plan, and the budgeted deepening just described. For scan *orders* it fails with the rules at hand, in both forms tried. Restarting with another tie-break rule when a growing work budget is overrun was simulated exactly on the saved work of three rules on 81 diagrams: all 27 policies cost 1.4 to 4.8 times the default. Introsort can switch because heapsort carries a guarantee and recursion depth signals degeneracy; here no rule has a guarantee, expensive is not degenerate, and an abandoned scan is lost. Without restarts (the previous stage is never mutated, so tied candidates can each be tried from the same state and the smallest real complex kept) the cost with all trial work counted is 0.41 on the scrambled unknots but 1.44 to 1.52 on random closures and about 1.4 overall: the size of the complex now does not predict the cost of the choice later.

The second suggestion, racing two algorithms on separate cores, is where the same data does promise something, because by wall clock a race costs only the winner's time: the better of two rules needs 0.75 of the default on the expensive inputs, and 0.07, 0.24 and 0.30 on single ones. In Python a competitor has to be a process, and starting one costs 30 ms bare, 76 ms with the package imported and 111 ms for a complete scan of the trefoil from the command line, against scans that mostly finish within 300 ms. Following the user's instruction for that case, the race was built in the Rust version, where a thread costs microseconds, together with a port of the Reidemeister III search: up to three greedy orders (ties by crossing index, by longest wait, by most recent touch) are scanned on separate threads and the first to finish stops the others through a flag; the rank does not depend on the order, so the result is the same whoever wins. On 40 random closures of 34 to 55 crossings whose single scan takes 30 ms to 8 s, with the settings interleaved over five rounds, racing two orders needs 0.650 of the total time and racing three 0.689. The gain is insurance against the heavy tail and not uniform: per input the ratio runs from 0.05 to 1.68; 6.8 s become 0.34 s and 890 ms become 140 ms, but on the half of the inputs where the default order was the best anyway the race costs 10 to 60%. I first blamed the shared heap for that premium and was wrong: two or three separate processes running the same single scan at once slow each other down exactly as much as the racing threads (1.15 against 1.18, 2.11 against 2.06), so it is contention in the hardware of this machine and will differ elsewhere. The race is therefore an option (`-race 2`), not the default.

Dismissing a race in Python was, on reflection, a wrong comparison: the cost of starting a process was set against the typical scan, while a race only matters on the heavy tail, where Python scans take seconds to minutes. The Python package therefore has it too, in the form the two suggestions combine to: the default order runs alone for a head start of one second, and only then are other greedy orders started as separate processes, which the scanner polls where it looks at the clock; the first to finish wins. On 26 random closures with single scans of 0.15 to 3.6 s the total is 0.910 of single with two orders and 0.925 with three. That total consists of two inputs on which a competitor wins in every round (3.22 s to 1.74 s, 3.55 s to 1.56 s) and of a cost of 10 to 20% on scans of 1 to 3 s that the default order wins. The other per-input ratios of that experiment are noise, and the data says so itself: 18 inputs finish inside the head start, where nothing is started and the ratio must be 1, yet they read 0.80 to 1.43 with three rounds each. Measured with 61 interleaved pairs, an armed race that starts nothing costs 1.010 $[0.984, 1.041]$ against an A/A control of 1.042, that is, nothing.

**Start-up, the largest gain for a command-line user.** Everything above saves well under a millisecond on a small diagram, while a whole `python -m fastunknot recognize` process took 115 ms, 29.5 of them for the bare interpreter. Importing the package cost 56 ms: 25 for `dataclasses`, which loads `inspect`, `re`, `dis`, `ast` and `tokenize`, used for five simple records (now written out by hand with the same constructors, equality, hashing, immutability and `repr`); 9 for

`subprocess` and `json`, needed only when a race starts; 5 for `typing`, used only in annotations that are never evaluated. Then `argparse`: 55 to 59 ms in Python 3.14, where it loads `_colorize`, `dataclasses` again and `shutil`. The options are now one table feeding a small direct parser for the well-formed common case and, on demand, the real `argparse` parser for everything else, so help and error messages are unchanged; tests check that the two agree wherever the fast one accepts. With 41 interleaved pairs of whole processes a run takes 0.52 of the time (0.510 to 0.541; 115.5 to 60.9 ms for the Conway knot) on every short invocation and 0.86 on the half-second stress scan. My first two such comparisons showed no difference and were invalid: `python -m` puts the current directory first on the module path and both arms ran from the working tree, which was thus compared with itself.

Profiling `recognize` by cumulative time, rather than the scanner by own time, then found four more things of the same unglamorous kind. The Jones filter paid for the bookkeeping of a shape cache it never uses (9% of a recognition of the Conway knot); the reduction rebuilt and revalidated the diagram even when no move applied (12%); the dart involution of an immutable diagram was recomputed seven times per recognition; and on the commonest input of all, a reducible closure that the Alexander filter decides, a quarter of the time went into validating the reduced diagram from scratch, although the moves can only break its topology, which is now checked directly on the darts (one traversal through every crossing twice, $n + 2$ faces; of 282 deliberately corrupted structures 273 are rejected and the other 9 also pass the full validation). Against the code before: recognition of the Conway knot 0.765 $[0.733, 0.794]$, of $T(3, 61)$ 0.759, of Conway#Conway#Conway 0.731, and of the random 5-strand closure 0.865 and then 0.796 $[0.772, 0.831]$, with status, deciding step, reduced size and rank identical on 700 diagrams and reduced diagrams and traces identical on 1500. The `tail` option was re-tested on the present scanner and stays at 0: one crossing of tail is a wash (0.94 to 1.01), two or three are worse, up to three times.

Two remarks. First, the closing sentence of the previous section, that on an unknot diagram no filter can decide the Python accelerations change nothing, was true of 0.1 against 0.2 and is no longer true: that case is now 1.4 times faster, because the scan itself is. Second, the gain is a constant factor: between 2.5 and 2.8 on every scan in the table except the chain of kinks (1.5), where fixed work per crossing dominates. Nothing here touches the exponential growth, and the Rust port remains several times faster still.

### 10.3 Measured and not adopted

All of these were decided with deterministic counters (compositions performed during cancellation, differential entries written while adding crossings), and the ranks were identical in every variant.

- *One queued pivot candidate per source object.* Fewer queue operations, but 42 197 instead of 29 347 compositions on the stress closure: the queued candidate goes stale. Rejected.
- *Other pivot rules.* First-in-first-out inside a cost bucket needs 64% more compositions on the stress closure; the cost $(\text{out} - 1) + (\text{in} - 1)$ is worse on all four inputs tried. The product cost with last-in-first-out stays.
- *Another score for scan orders.* Scoring greedy orders by $\sum 2^{|\partial|/2}$ selects the same start as the current rule (maximal, then total boundary size) on all four inputs.
- *Scoring all greedy starts instead of twelve.* On 40 random closures the order changed in 18: equal work in 8, less in 6, more in 4, with ratios from 0.29 to 6.69. The total, 0.952, is carried by three inputs and changes sign without one of them. Not adopted.
- *Cancelling saddle isomorphisms before writing their entries.* Between 74 and 83% of the entries written are incident to a cancelled pair when it is cancelled, but only 30 to 53% of the cancellations are between the two smoothings of one old object, the predictable kind, and they account for 24 to 36% of the entries. That bounds the gain near 15% on the stress closure before the cost of the extra code. Not pursued.
- *A fast path for single-entry transfers.* It would apply to 13% of the edges of the stress closure:

too little to measure. Not written.

The experiments on scan orders have a positive content. The score is the weak point, not the number of candidates: on the stress closure the greedy starts 14 to 16 need 23 710 compositions and have boundary profile $(8, 226)$, while start 0, with profile $(8, 230)$, needs 895 182, that is 38 times more, with 2.6 times the peak size; 20 of the 36 starts do not finish in 4 s. A matchings-only simulation cannot replace the score, because the gap occurs at equal boundary size, where there are at most 14 matchings: it is all multiplicity, i.e. homology of the partial tangle, which is what the computation computes. A cheap predictor of that multiplicity would be worth more than everything in this section.

A larger experiment, on 81 diagrams (36 scrambled unknots of the kind described above and 45 random closures with 20 to 35 crossings), confirms this and closes one more door. The greedy rule breaks ties by crossing index, which is arbitrary. Breaking them towards the most recently touched crossing costs 1.09 and 1.05 of the present total work on the two families, and towards the longest-waiting crossing 1.52 and 1.57; per diagram the ratios run from 0.01 to 28, better on about as many inputs as worse, so the rules are incomparable rather than ranked. A portfolio does not rescue this. The best of the three rules on each diagram would need 0.31 of the work on the unknots and 0.74 overall, but running them interleaved until the first one finishes costs three times the winner: 0.94 on the unknots and 2.2 overall, because 8 of the 81 inputs carry 86% of the work and there the rules lie within a factor two of each other.

### 10.4 An episode about measurement

Midway, a benchmark run showed everything 40% slower, including code that had not been touched. No process of ours was running; the desktop's own load had changed. A fixed pure-Python control loop, added to the harness, went from 5.3 ms to 12 ms *inside one run*. Sequential before/after timing was therefore abandoned for decisions: `perf.py` now times the control next to every case and rejects a run whose control spreads by more than 10% (eight of the fifteen logged runs are rejected), and `ab.py` times the old and the new code alternately in one process.

The first criterion of `ab.py`, "the interquartile range of the paired ratios lies below 1", was then caught making false claims in both directions. A change confined to order selection, which is 0.2 to 1.8% of the scans concerned and was measured on its own as 32 to 58% faster with identical output, was reported as 14% slower on one input and 4% faster on another. The bound came from asking what share of the run time the change could touch at all. The tool now times the old code twice per round; on mid-sized scans this A/A control shows identical code differing by 20 to 30% between adjacent runs. A difference is claimed only if the order-statistic 95% confidence interval of the median ratio excludes 1 *and* the median lies outside the interquartile range of the A/A ratios. The one claim that had been published under the lenient criterion (7 to 13% from the refinements of the geometry on small scans, no claim on large ones) was re-tested between the two revisions under the strict criterion and holds: 0.87 to 0.92 with intervals inside $[0.87, 0.95]$, no claim on the large scans.

A smaller finding of the same kind: the new scanner at first overshot a time budget of 1 s by up to 0.24 s (the generic scanner: 0.01 s), because a single crossing was added without looking at the clock. With a check every 512 objects the overshoot is at most 0.03 s on the same runs.

## 11 A port to Bend 2, and what its parallelism delivered

Bend 2 (version 2.0.10) is a pure, total, affine functional language that compiles through C to native code and promises parallelism for free: a parallel let `a b = f(x) g(y)` forks, the type checker guarantees that the two calls share nothing, and the result cannot depend on the number of threads. The algorithm of `fastunknot` was ported to it (directory `bend/` of the standalone Knots

repository; the port was not carried into ProveIt when that repository was merged on 4 October 2026) to find out what that promise is worth for this problem. Nothing in this section changes the complexity discussion: the scan is exponential in the boundary width, in every language.

### 11.1 Setting

Bend runs on Linux, macOS and WSL only, so all Bend work ran over SSH on an Ubuntu virtual machine with 6 virtual CPUs of an i7-3770. That CPU has no AVX2 and the distributed binary died with an illegal instruction; Bend was run from source under the baseline build of its JavaScript runtime instead. All times below are from that host, and the Python package was copied there for every comparison, so the two sides of a ratio always share the hardware. Timings are interleaved with an A/A control as in Section 10; the control's ratio stayed within 0.96 to 1.05.

### 11.2 What was ported, and how it was checked

- the modular Alexander filter (Wirtinger matrix, determinant modulo 65521, because `U32` is the only machine integer);
- the planar geometry, the dotted cobordism algebra over $\mathbb{F}_2$ and Bar-Natan's scan with delooping and Gaussian elimination, that is `scan_fast.py` with, of its caches, only the transfer plans;
- a greedy scan order from eight starting crossings;
- a standalone recognizer that chains the three.

Not ported: diagram simplification, the Jones filter, factorization, and the cache of composition plans. Each layer was compared with its Python counterpart before the next was written: 200 Alexander minors, 1920 gluings and circle numberings, 2736 compositions and transfers, the Khovanov ranks of 160 random closures and of the timing inputs (up to rank 5898), and the verdicts of the recognizer on 60 diagrams with up to 40 crossings, 40 of them unknots, among the knots Conway's and the Kinoshita–Terasaka knot: no mismatch anywhere. Eighteen laws about the control logic (for instance: of two scan orders with equal profile the left one wins, so the work does not depend on scheduling) are stated in `LAWS.bend` and proved in `PROOF.bend`; they were drafted by the AI as proposals, and they do not say that a rank is right, which Bend cannot express about `U32` arithmetic.

### 11.3 What the fork delivers

Measured on kernels first: a deep, balanced, uniform recursion reaches 5.0 on 6 threads; 32 coarse leaves reach 2.1; every switch from sequential to parallel execution costs about 80 microseconds; forks inside the sparse elimination of the Alexander determinant made it 8 to 15 times *slower*; one fork over four independent determinants gained nothing. Tasks are dealt once and never moved, and there is no cancellation, so the race of scan orders, the best parallel gain of the Rust port (0.65 of the time in total), cannot be written.

In the scan, the transfer of all objects of a stage through a crossing is a map that shares nothing, and it was written as a fork over the tree of the array. Its first effect was negative: with six threads the scan took 1.28 to 1.85 s instead of 1.04 s. The cause was found in the runtime source and the emitted C, and then confirmed by measurement: generated code makes a non-tail call either by pushing a stack frame or, in "parallel mode", by allocating the continuation as a task on the heap; a program runs in parallel mode wherever a fork is reachable, which for a program with one fork is everywhere. With the forking transfer *switched off* and six threads the scan took 1.85 s: a factor 1.8 for the mere presence of a fork. The remedy is to make every long sequential phase a task of its own, a fork next to a trivial second child, since a task whose function reaches no fork runs in the fast mode. Three natural ways of writing that child were silently compiled away (inlined, constant-folded, dropped as unused), leaving no fork; it has to be recursive, depend on

run-time data, and its result has to be used. After that: 1.02 s on one thread, 0.81 s on six; the transfer alone became 2.7 times faster.

### 11.4 Where the time goes

| | braid closure, rank 218 | | stress closure, rank 5898 | |
|---|---|---|---|---|
| version | 1 thread | 6 threads | 1 thread | 6 threads |
| first working scan, first-found pivots | 1.04 s | 1.28–1.85 s | 63 s | 58 s |
| sequential phases as tasks | 1.05 s | 0.89 s | 63 s | 58 s |
| least-fill pivots in batches, work lists | 0.65 s | 0.50 s | 7.0 s | 4.9 s |
| one transfer plan per pair of matchings | 0.36 s | 0.39 s | 4.07 s | 3.9 s |
| composition plans once per source matching | 0.32 s | | 3.59 s | |
| one alternative evaluated instead of all | 0.24 s | | 2.65 s | |
| number of sources kept, not counted | 0.23 s | 0.25 s | 2.20 s | 1.99 s |
| Python package, same host | 0.134 s | | 0.542 s | |

The largest gain was algorithmic and sequential: cancelling only pivots whose fill-in is within a limit, and raising the limit to the least fill-in that was skipped, brought the stress closure from 58 to 5 s. A sampling profile of the emitted C then showed 43% of the time in reference counting: a value used more than once is shared and counted, and every step through a shared list bumps and drops counts. Matchings are shared by design, and the linear lookups in them were most of those steps; a differential profile put the transfers at 48 to 61% of the single-thread time.

Python avoids that work by interning matchings and caching every plan (2817 plans serve 251 406 transferred entries on the stress closure). In Bend a cache is a value every call must thread, the only map has string keys, and an array cannot be shared between the branches of a fork. The port therefore splits a transfer into its plan, which depends on the two matchings only, and its application to a morphism, and before the forked map of a stage a sequential pass builds a table with one entry per pair of matchings that occurs: a binary trie over 20 bits of a hash of the pair, which is plain data and may be shared, read-only, by all branches. A leaf remembers its pair, and a lookup that finds another pair or nothing computes the plan on the spot, so a collision costs time and never correctness (two of the laws). That halved the single-thread time (0.53 and 0.56, A/A control within 0.98 to 1.01).

Two further findings were about the language rather than the algorithm. Bend is strict and has no `if`: `Bool.pick` evaluates both alternatives and `&&` both sides. The port had used them as conditionals, and so ran twelve rounds of list work per surface component whether needed or not, compared matchings for every entry of every visited row, and iterated a connectivity closure sixteen times after it had converged; evaluating one alternative takes a separate function that matches on a Boolean parameter. Five such repairs gave 0.75 of the time. And a third of all reference-count drops then came from counting a list of sources at every pivot candidate; the list now carries its length.

The table of plans also ended the parallel gain. It took the work out of the one part that forks: the map fell to 22 to 25% of the time, and with six threads the two inputs took 0.39 and 3.9 s against 0.36 and 4.07 s with one (at the end: 0.25 and 1.99 s against 0.23 and 2.20 s). What had been running in parallel was the redundant work.

As a process the native recognizer (one thread) beats the Python command line on every input tried up to 30 crossings, because Python needs 28 ms to start (2 ms against 27 ms on an 8-crossing unknot, 34 against 44 ms at 30 crossings); in-process, Python is on a par up to 25 crossings on the same steps, and 1.7 to 4.1 times faster on the two large scans of the table.

### 11.5 Verdict

The promise holds in the narrow sense and not in the useful one. Writing a fork is trivial and safe. But the scan alternates a parallel map with a Gaussian elimination in which each cancellation changes the rows the next one reads. While the map recomputed every plan for every entry it was 20 to 60% of the time and six threads gave 1.2 to 1.4 overall; once the plans were computed once per pair of matchings, which is the first thing a sequential programmer does, one thread was as fast as six. The cost model (a fork anywhere taxes all sequential code by 1.8 unless it is wrapped as a task) is invisible in the language, and the compiler removes the wrapper unless it is written just so. For this algorithm the measured ranking stays: Rust, then Python, then Bend, the last 1.7 to 4.1 times behind the Python package on large scans, after becoming 4.5 to 29 times faster than its own first version by sequential changes alone.

## 12 Structural certificates and compressed complexes: October continuation

### 12.1 Intake, implementation, and scope

Report 07 [21] extends the Python recognizer at a pinned repository snapshot. Its patch was reviewed and integrated on 7 October 2026 as version 0.3.0. The delivered archive stays unchanged; this section is the maintained account of the implementation. The Rust implementation still uses the earlier pipeline. The complete integrated Python suite passes all 69 tests on CPython 3.13.14, including randomized comparisons with the inherited scanner, generic set algebra, independent cube calculations in the inherited suite, certificate replay, and stagewise checks that $d^2 = 0$. These checks support the implementation; they are not formal proofs of its algebra or of the topological theorems it imports.

The new default first tries a signed Seifert-graph certificate. If it is inconclusive, the previous pipeline runs. The ordinary scanner remains the default backend, because compression adds overhead on complexes that do not split. Three optional backends provide exact component sharing, saturated decision weights, and early Euler obstructions. They preserve the distinction between a proved decision and resource exhaustion. No universal quasi-polynomial running-time guarantee follows from this integration.

### 12.2 A linear structural decision

Let $D$ be a validated, oriented, one-component classical diagram, with $n$ crossings, writhe $r$, and $s$ Seifert circles. The Seifert graph has one vertex per circle and one signed edge per crossing. Let $c_+$ and $c_-$ count the connected components of its positive and negative spanning subgraphs, including isolated vertices in both counts. Define

$$g_D = (n - s + 1)/2, \qquad \delta = s + 1 - c_+ - c_-.$$

The Seifert surface consists of $s$ discs and $n$ bands and has one boundary component; its Euler characteristic is $s - n = 1 - 2g_D$. Abe's criterion identifies $\delta = 0$ with homogeneity. For a homogeneous diagram, the canonical Seifert surface realizes knot genus [19, Theorems 2.3 and 3.4]. The Kawamura–Lobb bounds give

$$r - s + 2c_+ - 1 \;\le\; s(K) \;\le\; r + s - 2c_- + 1.$$

Here $s(K)$ is Rasmussen's invariant, distinct from the circle count $s$; the upper bound follows by applying the lower bound to the mirror [20, 19]. The interval width is $2\delta$.

Consequently the implementation certifies the unknot when $g_D = 0$, a nontrivial knot when $\delta = 0$ and $g_D > 0$, and a nontrivial knot when the interval excludes zero. The first condition gives a spanning disc. The second gives positive knot genus. The third uses $s(U) = 0$. In every other case the stage is inconclusive. In particular, the interval $[0, 0]$ alone does not certify triviality: the figure-eight knot is a homogeneous positive-genus example with that interval.

The implementation follows the oriented smoothing and edge-pairing involutions to label circles, then traverses each signed subgraph. This costs $O(n)$ word operations after input validation and normalization. Each word holds an $O(\log n)$-bit index; this statement does not include reading arbitrarily long external edge labels. The separate certificate verifier rebuilds orientation from the PD code and uses union by size with path compression, costing $O(n\alpha(n))$ word operations. It checks all counts and the decision criterion rather than trusting cached production data. Both implementations share the classical one-component input promise and the cited topology.

### 12.3 What exact sharing represents

At each scan frontier the inherited Bar–Natan complex has matching objects and differential entries in the finite dot-mask algebra over $\mathbb{F}_2$. Connect two objects whenever a differential entry between them is nonzero. The connected components of this graph are literal direct summands. The shared scanner stores

$$C \simeq \bigoplus_a \bigoplus_h T_a[h]^{\oplus p_{a,h}}, \qquad p_a(z) = \sum_h p_{a,h} z^h \in \mathbb{N}[z].$$

A template $T_a$ records a complete component with its minimum homological degree subtracted. Its key includes the ordered matching identifiers, relative degrees, and every differential target and coefficient. Equality of complete keys exhibits a chain isomorphism. A hash collision cannot identify unequal keys; a missed isomorphism merely loses a saving. Multiplicity coefficients are nonnegative integers, never scalars in $\mathbb{F}_2$. Two identical isolated generators have rank two, not rank zero.

Crossing extension is additive and commutes with shifts. Unit elimination preserves direct sums: the Schur-complement correction $\gamma\phi^{-1}\eta$ remains in the pivot's parent block. Thus it suffices to extend and contract one representative, split its resulting differential graph, and attach each child's normalization shift to its parent's weight. Merging equal child keys adds their shifted weight polynomials. Induction proves preservation of the represented chain-homotopy type. At closure all nonzero differential entries are units and have been eliminated; summing weights gives the exact unreduced rank and its raw homological-degree counts. The inherited characteristic-two splitting yields the reduced rank by dividing by two. The method does not output a homology basis, quantum grading, or integral torsion.

### 12.4 Safe decision compression and early obstructions

The map $q(m) = \min(3, m)$ is a semiring homomorphism from nonnegative integer bookkeeping to saturated addition and multiplication. Evaluating each $p_a$ at one and applying $q$ therefore preserves the final total rank capped at three. It discards absolute grading but does not modify the differential algebra. For a validated knot, unreduced $\mathbb{F}_2$ rank two identifies the unknot, while a capped rank of three identifies a nontrivial knot, using the inherited rank criterion and [35]. No decision is made from large multiplicities at an incomplete scan stage.

An early decision needs information about the actual remaining crossings. Let $F_i$ extend through that suffix and close the diagram. The completed component bound is

$$B_i = \sum_a p_a(1)\,|\chi(F_i(T_a))| \le \dim_{\mathbb{F}_2} H(C).$$

This follows by additivity of homology and $|\chi(X)| \le \dim H(X)$ for each completed summand. When a component splits, the triangle inequality implies $B_i \le B_{i+1}$; at complete closure the bound equals the rank. The multiplicity uses $p_a(1)$, not $p_a(-1)$, since shifts do not change the absolute Euler characteristic of an individual copy.

The memoized suffix functional on a matching is

$$E_i(m) = 2^{\ell_0} E_{i+1}(m_0) - 2^{\ell_1} E_{i+1}(m_1), \qquad E_n(\varnothing) = 1,$$

where smoothing $j$ creates $\ell_j$ closed circles and matching $m_j$. The terminal empty object here is the monoidal unit; the separate empty input-PD convention returns one unknot of rank two. A component's completed Euler value is $\sum_v (-1)^{h_v} E_i(m_v)$. Signed intermediate values must remain exact: saturating them before subtraction would invalidate the bound. Only the final nonnegative bound and multiplicities may be capped. The implementation uses an explicit dependency stack, avoiding recursion depth failures on long narrow diagrams.

The default Euler budget registers at most 4096 suffix states. Exhaustion disables this optional inference and continues the saturated scanner. Object or time exhaustion instead propagates to an `UNKNOWN` result. An early result exposes `rank_lower_bound_capped`; a completed decision exposes `rank_capped`. Neither is reported as an exact rank when it equals three.

### 12.5 The conditional complexity statement

Fix the complete scan order. Let $w$ bound every frontier, including those in an unvisited suffix after an early decision, and let $b$ bound the size of every differential component after elimination and before sharing. These are execution parameters, not knot invariants or conjecturally optimal decompositions. Write $M(0) = 1$ and $M(w) = (w-1)!!$ for positive even $w$. The number of complete template keys of size at most $b$ is at most

$$K(b, w) = \sum_{t=1}^{b} t^t M(w)^t 2^{t^2 2^{w/2}}.$$

For a numbered connected component with $t$ vertices, relative degrees lie in $\{0, \dots, t-1\}$, giving $t^t$ assignments. There are at most $M(w)$ boundary matchings per vertex. A morphism has at most $2^{w/2}$ coefficient bits, so it has up to $2^{2^{w/2}}$ possible values. Allowing all $t^2$ entries gives the last factor. Confusing coefficient bits with the number of morphisms would incorrectly remove an exponential here.

At a compressed stage there are at most $bK(b, w)$ physical objects. A crossing creates at most eight objects per parent before elimination: each of its two smoothings can create at most two circles. Matrix elimination is polynomial in physical size; coefficient operations cost $2^{O(w)}$ times polynomial factors. After $j$ crossings the virtual expanded state has at most $8^j$ objects, so each multiplicity coefficient has $O(j)$ bits and each exact weight has at most $j+1$ degree positions. The conservative bit bound, including exact multiplicity arithmetic, is

$$T(n; b, w) \le n^{O(1)} (bK(b, w))^{O(1)} 2^{O(w)}.$$

The suffix Euler table has at most $(n+1)M(w)$ states with $O(n)$-bit values and fits within the same conservative bound.

In particular $b^2 2^{w/2} = O(\log^2(n+2))$ suffices for $n^{O(\log n)}$ time. This includes fixed $w$ with $b = O(\log n)$, or fixed $b$ with $w \le 4 \log_2 \log(n+2) + O(1)$. The order must be supplied or efficiently constructed. Neither the existence of such orders for every knot nor the required component-size bound has been established. Small experimental values do not prove either statement. Moreover, report 07 supplies RI/RII-reduced unknot diagrams of homogeneity defect one with arbitrarily large canonical genus; defect alone cannot close the missing bound. The hierarchy obligations in Section 5 remain a separate unresolved route.

## 12.6 Local measurements and reproducibility

The integrated benchmark `fast/benchmark_structural.py` records seven randomized paired rounds, with five fresh Diagram records per arm and an identical-baseline A/A control. Parsing, validation, imports, and serialization are excluded from every arm. The baseline disables the Seifert stage; the treatment enables it. These are recognition-function timings, not end-to-end command-line times. The raw record is `fast/results/structural_integration_20261007.json`.

On this host the median paired baseline/new ratios were 27.10 for a 1000-crossing weaving braid, 26.34 for a 1000-crossing positive three-braid, and 50.56 for a 1004-crossing positive five-braid. Their median A/A ratios were 0.993, 1.002, and 1.008. Inconclusive controls show the added cost: Conway 0.97, hard unknot 0.95, and the tiny scrambled-grid unknot 0.60. Ratios below one mean the added stage slowed the recognition call. These are descriptive measurements with host-dependent variability, not fitted complexity exponents or hardware-independent speed guarantees.

Separate raw-scanner checks bypass all geometric factorization and recognition filters. Exact sharing agrees with the standard rank and degree counts on one and two Conway summands; eight summands give $2 \cdot 33^8 = 2{,}812{,}817{,}236{,}482$ without expanding that many generators. This demonstrates representation compression. It does not claim a corresponding recognition speedup: the existing recognition pipeline already factors visible connected sums. The archive's original timing observations remain separate from these local checks.

## 12.7 Rechecking structure after crossing reduction

The first integration checks the original diagram only. This misses an inexpensive opportunity: cancelling opposite-sign crossing pairs may turn an inconclusive signed graph into a homogeneous one. The maintained pipeline therefore repeats the Seifert check after the initial RI/RII pass, provided that the crossing count strictly decreased and is nonzero. An unchanged diagram incurs no second graph traversal. Disabling Seifert certificates or reduction preserves the corresponding opt-out behavior.

Correctness is a composition of two certificates: the recorded legal Reidemeister moves preserve the knot, and the signed-graph certificate decides the reduced diagram. The evidence key `seifert_after_reduction` explicitly identifies this target. To verify it, replay `reidemeister_trace` on the original diagram, then call the independent Seifert verifier on the resulting diagram. The original-input key `seifert_certificate` is not populated with a certificate for a different diagram. This distinction prevents a numerically correct verdict from carrying misleading replay evidence.

For example, append the cancelling braid word $\sigma_2\sigma_2^{-1}\sigma_1^{-1}\sigma_1$ to $(\sigma_1\sigma_2^{-1})^m$, with $3 \nmid m$ and $m \geq 2$. The resulting knot diagram has both signs on both braid generators and a Rasmussen interval containing zero, so the original Seifert test is inconclusive. Reduction exposes the homogeneous weaving diagram of positive genus. The recheck certifies nontriviality before factorization or matrix filters. The added work is one $O(n)$ traversal; combined with the incremental RI/RII reducer it gives linear word-operation recognition on this family after validation. This is a family result, not a global complexity bound.

The reproduction script `fast/benchmark_reduction.py` loads only the baseline recognition module from commit `1323898c3`, sharing the other implementation modules to isolate this change. Nine paired rounds with fresh diagrams, five calls per arm, and A/A controls measured speedups 1.99, 2.80, and 8.24 for reduced weaving sizes 10, 100, and 1000 (input sizes 14, 104, and 1004). The corresponding A/A ratios were 1.011, 0.993, and 1.023. Control ratios for Conway, the hard unknot, and the scrambled grid were 0.99, 0.95, and 0.95; no improvement is claimed for them. Raw samples are in `fast/results/reduction_structural_20261007.json`. All 71 integrated tests pass, including mirrors, independent certificate replay on the reduced diagram, and a regression

asserting that the matrix filter is not entered on this family.

## 12.8 Preparing suffix geometry within the inference budget

The delivered Euler engine eagerly constructed a separate geometry object for every crossing, retaining all intermediate frontier sets. This cost was incurred even with a zero-state budget, or if scanning finished without attempting an Euler inference. The maintained implementation instead constructs a stage's geometry only after registering a state within the budget. The complete matching of that state supplies the frontier. All matchings at a fixed suffix index have the same boundary labels, so a single cached geometry object serves every visit to that stage. This changes preparation, not the Euler recurrence or its values.

Construction still checks closure in $O(n)$ word operations by toggling each of the four label occurrences at every crossing. Occurrences must be toggled individually: a loop label appearing twice in one crossing does not remain on the frontier. The pass keeps only the current boundary set and checks the time deadline throughout. The crossing list occupies $O(n)$ references. For a budget of $B$ registered states, at most $\min(n, B)$ stage geometries are prepared; their frontier storage is $O(\min(n, B)w)$. Thus the geometry preparation storage is $O(n + \min(n, B)w)$ rather than $O(nw)$, before accounting for the separately budgeted memo and gluing entries. This is especially useful when the optional inference is disabled or rejected immediately. It does not change the worst-case complexity theorem.

The additional statistic `prepared_stages` reports actual geometry construction. Tests check zero- and one-state budgets on a 1201-crossing diagram, deadline exhaustion during construction, the existing exact suffix comparisons against independent generic-algebra continuation, and the long-order case. All 73 tests pass on CPython 3.13.14.

`fast/benchmark_euler_setup.py` isolates construction and the first budget-exhausted evaluation against commit `9765612b8`, with seven paired rounds and A/A controls. On 1201 crossings, budgets zero and one reduced traced allocation peaks from about 2.87 MB to 12.0 and 16.5 kB; on 4801 crossings they reduced approximately 11.02 MB to 43.7 and 48.4 kB. Setup time improved by 33–43 times, with A/A ratios 0.979–0.994. Allocation peaks were measured separately from timing, using Python's `tracemalloc`; they are not process resident memory. The raw data are in `fast/results/lazy_euler_setup_20261007.json`. These measurements concern optional inference setup only, and make no claim of an equivalent speedup for full recognition.

## 12.9 Replacing Euler state enumeration by link connectivity

The suffix dynamic program computes more intermediate information than the ungraded Euler functional needs. A surviving matching comes from a partial resolution of the original classical diagram. Closing it through the unprocessed crossings therefore gives a classical link $L_m$, possibly with several components. Gaussian elimination removes objects and changes differentials but does not introduce unrealizable matching objects. Let $c(L_m)$ be the component count and let $n_-(L_m)$ count negative crossings after choosing an orientation on each completed component. Then, in the scanner's raw homological convention,

$$E_i(m) = (-1)^{n_-(L_m)} 2^{c(L_m)}.$$

This formula concerns each completed matching separately; using the original knot's writhe or component count would be incorrect.

To justify it, the normalized graded Euler characteristic of Khovanov homology is the Jones polynomial in the normalization with unknot value $q + q^{-1}$ [39]. At $q = 1$, a classical $c$-component link has value $2^c$; equivalently the specialized skein relation makes the normalized value unchanged by crossing changes, which reduce the link to the unlink. The homological normalization shifts

raw cube degree by $-n_-$, accounting for the sign. Reversing a component changes signs only at its intersections with other components. The total number of such crossings is even for classical plane curves, so the parity of $n_-$ is independent of this choice. The empty monoidal closure has $c = 0$ and value one. Removed crossingless circles are already represented by the scanner's delooping multiplicities and are not counted twice.

`ClosureEuler` constructs the remaining crossings' dart pairing: internal labels pair their two occurrences, and the supplied matching pairs the remaining boundary occurrences. Alternating straight-through crossing passages with edge pairings orients and counts all components. The incoming under/over slots determine each sign using the same PD convention as the rest of the package. Each evaluation takes $O(n - i + w)$ word operations, plus linear-bit integer construction, instead of enumerating up to $M(w)$ smoothing states per suffix stage. At most one stage's base graph is retained; complete query values are memoized. Thus a budget of $B$ queries gives polynomial work $O(n + B(n + w))$ at this level, and removes the exponential dependence on frontier size from the Euler inference itself. The shared complex can still be exponentially large. This does not settle the general target.

The default `euler` backend now uses this evaluator; the exact `SuffixEuler` recurrence remains available as an independent oracle. The state budget now counts requested completed matchings, not recursive smoothing states. The zero-budget fallback is unchanged. This formula is used only for matchings arising from validated classical diagrams; the low-level method is not an evaluator for arbitrary virtual closures.

A separate exploratory comparison checked 34,447 live matching queries on 500 seeded random classical knot diagrams with shuffled orders, with no differences. The maintained tests include 100 random diagrams across all live suffix matchings, exact component-bound equality, mirrors, the monoidal empty closure, malformed frontier rejection, and a 1201-crossing suffix evaluated using a single query. All 76 tests pass. Seven paired raw Euler-assisted scanner runs against commit `1fe98cb7e` reduced the eight-Conway-summand example's Euler states from 286 to one and measured a 1.326 baseline/new time ratio (A/A 1.013), with the same early decision at stage 11. Conway and the hard-unknot controls performed no Euler inference and provide no evidence of speedup. The script `fast/benchmark_euler_connectivity.py` and `fast/results/euler_connectivity_20261007.json` preserve the measurement boundaries and raw samples.

### 12.10 Source screening: the OpenAI mathematics collection

At the user's suggestion, we screened the manuscript catalogue at `https://github.com/openai/math`, pinned to commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. The catalogue contained no directly named unknot-recognition or Khovanov algorithm. This is a catalogue screening result, not an assertion that every manuscript was audited. The collection itself distinguishes different verification stages. No theorem from it is imported as a correctness dependency of the recognizer.

The September 25 manuscript *Deterministic quasipolynomial-time mean-payoff games*, family 104, asks for forced boundary labels rather than the entire numerical fixed point. Its complexity section separately counts recursion nodes and the bit cost of local operations. The useful methodological lesson here is to compute the required observable directly when possible: the preceding connectivity evaluator calculates Euler values without constructing the suffix smoothing state space. Its proof uses the classical Jones specialization, independently of the game's claimed result.

One recursion-counting lemma from that manuscript is directly elementary. If every path has length at most $H$, has at most $B$ exceptional edges, and each node has at most one ordinary child

and at most $s$ exceptional children, the node count is bounded by

$$\sum_{\ell=0}^{H} \sum_{j=0}^{\min(\ell,B)} \binom{\ell}{j} s^j \le (H+1)(B+1)(1+Hs)^B.$$

Choose the exceptional positions and their child indices; every other position is forced. Thus polynomial $H, s$ and logarithmic $B$ imply a quasi-polynomial call count, provided local bit cost is also polynomial. For an unknot hierarchy, constructing such a budget and proving that all but one child spend it would be an alternative precise research target. The current scanner has no such proved budget; relabelling its branches does not supply one.

Family 089's planar $L_1$ embedding manuscript controls metric distortion and charging across scales. Even its asserted constant distortion would not bound the number of edges in our frontier or the size of connected differential components. Family 306's cosmetic-surgery manuscript has a topological conclusion about filled manifolds and uses instanton and spinor counts; its introductory construction supplies no bounded-bit algorithm for these counts on a PD input. Neither is presently an algorithmic replacement for the missing width or hierarchy operations. Their introductions and relevant source sections were inspected for applicability; this is not a full verification or rejection of those manuscripts' main mathematical claims.

# 13 Further incoming reports and checked integrations

## 13.1 Intake status and provenance

Reports 08–12 arrived during the October continuation and were merged from the shared main branch before further integration. Their delivered contents remain unchanged. Local reruns passed 59 test methods for report 08, 69 for report 09, 15 for report 10's numerical kernels, and 62 for report 11. These are separate suites and their counts must not be added as though they were independent mathematical cases. Logs and local cross-checks are retained under `synthesis/data/`. The current production suite passes 209 tests. Report 12's 29 archive test methods also pass locally. The braid specialization of report 08, selected preprocessing of report 09, component contraction from report 12, and the twist backend from report 11 are integrated. Report 10's parallel kernel was cross-checked; report 09's pointed backend and report 12's succinct block layer remain under integration review. Section 14 gives the restricted-family complexity proof and degree-sensitive refinement.

## 13.2 Complete recognition for checked short-strand braids

Report 08 contributes a complete decision procedure for the closure of an explicit braid on at most three strands. This is a restricted input class, not a method for reducing every diagram to three strands. The public raw-word entry point is `braid_certificate(m, word)`. It checks integer generators, strand bounds, and that the strand permutation is one cycle before issuing a knot verdict. If the word has $n$ letters, one-component closure requires $m \le n+1$, so even permutation validation uses linear space and word operations.

Write $e$ for the exponent sum. Bennequin's inequality applied to a braid and its mirror gives $|e| \le m-1$ for an unknot closure. A violation therefore proves knottedness on any number of strands. For one or two strands, this test leaves only the trivial closure classes, so it already gives a complete answer. Its converse is not asserted for arbitrary $m$.

For three strands put $a = \sigma_1$, $b = \sigma_2$, $x = aba$, $y = ab$, and $z = x^2 = y^3$. The central quotient is

$$B_3/\langle z \rangle \cong C_2 * C_3 = \langle x, y \mid x^2 = y^3 = 1 \rangle.$$

The Birman–Menasco classification says that the three conjugacy classes with unknot closure are represented by $ab$, $(ab)^{-1}$, and $ab^{-1}$; see Theorem 4.1(i) of Datta's account of the classification. Only this specific classification result is needed here. The implementation maps letters to at most two quotient tokens:

$$a \mapsto y^2x, \quad a^{-1} \mapsto xy, \quad b \mapsto xy^2, \quad b^{-1} \mapsto yx.$$

A stack multiplies adjacent tokens in the same cyclic factor. A deque then removes or combines equal-factor end tokens until the word is cyclically reduced. Each push, pop, and end operation can be charged to an input token, so the total is $O(n)$ symbol operations. Cyclic words are compared up to rotation, not by an expensive conjugator search.

The exact test accepts the following exponent/quotient pairs:

$$(e, \text{cyclic class}) = (2, [y]), \quad (-2, [y^2]), \quad (0, [xyxy^2]).$$

For the last case the implementation counts two $x$ tokens, one $y$, and one $y^2$ in a reduced word of length four. Alternation forces precisely the desired cyclic class. To justify lifting the quotient test, lift a quotient conjugator to $B_3$. The remaining difference from the target is $z^k$; since $e(z) = 6$, equality of exponent sums forces $k = 0$. Thus forgetting the center has not lost information needed by the classifier. A conservative bit bound, including counters and indices, is $O((n+m)\log(n+m+2))$.

An optional independent backend multiplies the integral matrices

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

The map $B_3 \to \mathrm{SL}_2(\mathbb{Z})$ has kernel $\langle z^2 \rangle$, whose exponent sum is a multiple of twelve. Trace one has two conjugacy classes, lifted from the order-three elements of $C_2 * C_3$; exponent residue distinguishes them. Trace three has the single class of $AB^{-1}$. One way to see the latter claim is to express a hyperbolic projective conjugacy class as a positive word in $L = A$ and $R = B^{-1}$. A mixed word can be rotated to start with $L^r R^s$, $r, s \geq 1$, of trace $2 + rs$. Each additional letter increases the trace of this strictly positive matrix. Trace three forces $r = s = 1$ and no further letters. Lifting and matching exponent sums again eliminates the central ambiguity.

Consequently the exact trace/exponent pairs for the unknot are $(2,1), (-2,1), (0,3)$. After the writhe bound and knot-permutation validation, this is equivalently $|2 - \mathrm{tr}(M)| = 1$. Without the writhe condition the determinant test is unsound as a recognizer: $(ab)^5$ has determinant one but is the nontrivial torus knot $T(3,5)$. Matrix entries can have $\Theta(n)$ bits and the addition-only implementation uses $O(n^2)$ bit time. The symbolic backend is therefore the default; matrix witnesses use hexadecimal strings to avoid decimal serialization limits.

For more strands, an optional sufficient simplification freely and cyclically cancels inverse letters, then looks for an endpoint generator $\sigma_1^{\pm 1}$ or $\sigma_{m-1}^{\pm 1}$ occurring exactly once. Cyclic rotation exposes an elementary Markov destabilization. Deleting that letter and relabeling the remaining strands preserves the closure. There are at most $m - 3$ steps, each using $O(n)$ work. The accompanying verifier replays the supplied choices and validates every singleton, rather than trusting the asserted final word. This $O(nm)$ procedure does not find every destabilization: Morton's four-strand unknot remains inconclusive and exercises the existing exact diagram fallback.

**Integration and provenance.** Only `Diagram.from_braid` attaches checked braid provenance. PD and grid input acquire none; a reduced factor also does not inherit the original word. Mirrors negate the word's letters, while serialized provenance is reconstructed and compared to the stored PD, allowing a half-turn of an individual crossing row. Equality and hashing continue to depend only on the PD. Standard JSON output remains PD; the explicit `preserve_braid=True` option retains the source word.

Recognition now tries the linear braid test before the Seifert stage. For $m > 3$, the potentially quadratic endpoint descent waits until the Seifert, Reidemeister I/II, repeated Seifert, and descending tests have failed. Resource handling surrounds both braid stages. The evidence records which representation made the decision, and any Markov trace. `-no-braid`, `-braid-backend matrix`, and `-no-braid-reduction` expose ablations. Existing structural and scanner tests explicitly disable earlier braid dispatch where needed to keep testing their intended code paths.

**Validation and measurement scope.** The integrated tests exhaust all 46,376 one-component three-braid words of lengths two, four, six, and eight against both representations. For the 2,856 words through length six they also compare the verdict to an independent dense Khovanov cube. Separate checks cover Alexander determinants, Artin relations, conjugation, mirror and pickle behavior, invalid inputs, endpoint replay, resource limits, and the four-strand fallback. The raw symbolic counter records exactly two token operations per input letter; timings do not substitute for its amortized proof. `fast/benchmark_braid.py` compares fresh diagram construction plus recognition to the current `use_braid=False` ablation, including the already-integrated Seifert shortcut. Python startup and JSON parsing are excluded. Its raw-word scaling experiment separately avoids PD construction. The raw samples are `fast/results/braid_integration_20261007.json`.

**Local braid results.** Nine paired rounds, rerun after correctness tests completed, measured baseline/new ratios 3.30–4.89 on the five scrambled three-braid unknots with default Reidemeister III enabled, and 9.73–41.12 when that optional search was disabled in both arms. Positive three-braid cases improved 1.23–1.35 times, and the hard eight-crossing braid fixture 6.14 times. Conway, Kinoshita–Terasaka, grid input, and the four-strand fallback support no timing claim under the recorded paired/A–A rule. The 256-strand stabilized unknot was about 2.8% slower: the additional linear provenance check has a real constant cost when a different cheap certificate would already finish. These are finite local results, not a uniform speedup claim. The symbolic raw-word test took 0.112 seconds at 128,000 letters in the committed run, with exactly 256,000 token operations.

### 13.3 Canonical visible factors from interlacement

Traverse the validated spherical knot diagram and record the crossing number at each visit, obtaining a double-occurrence word $W$ of length $2n$. Join two crossing labels if their occurrences alternate cyclically. Report 09 identifies the connected components of this interlacement graph with the atomic crossing sets of visible connected-sum factorization. This is a statement about the given diagram; it does not find hidden prime factors of an arbitrary knot presentation.

For completeness, the central argument is short. A chord not adjacent to any chord of a connected component has both endpoints in one gap between successive endpoints of that component. Connectedness forces every chord of another component into the same gap. Consequently the component intervals are nested or disjoint. An innermost component is a contiguous closed interval: every crossing it visits is visited twice. Only the two edges at the ends of that interval connect its projection subgraph to the rest. In the projection sphere these form a visible splitting curve. Removing the interval preserves interlacements among the remaining labels, so induction splits all graph components. Conversely, a visible two-edge split separates whole intervals and cannot divide a connected interlacement component. The induced traversal, with original crossing rotations and over/under data, determines each factor diagram.

The implementation never builds the potentially quadratic graph. A stack tracks active components, union–find merges them when a closing chord crosses later-open components, and circular linked lists maintain an open representative of each component. Every successful merge decreases the number of components. Each chord is opened and closed once, giving $O(n\alpha(n))$ combinatorial

work and $O(n)$ storage. Checked PD output construction sorts labels, for $O(n \log(n+1))$ total interface work.

The certificate contains a partition of crossing labels and a forest of actual interlaced pairs. A separate verifier checks every forest edge, connectivity within each proposed group, and proper component nesting. It does not rerun the discovery algorithm. Production recognition now uses this factorizer and exposes `connected_sum_factorization` evidence. The historical API and cut-trace schema remain available through `factor_backend="legacy"` or `-legacy-factor`.

On a cyclic sum of $k$ standard trefoil diagrams, the old recursive factorizer peels off one factor per internal call. Its internal crossing visits total

$$3\sum_{j=2}^{k} j = 3\left(k(k+1)/2 - 1\right),$$

which is quadratic in input size. The new factorizer reads the word once and constructs constant-size factors directly. Seven local paired kernel runs measured baseline/new ratios 14.20, 56.59, and 117.20 for $k = 32, 128, 256$; A/A ratios were 1.007, 1.005, and 0.985. These are factorizer timings, not default recognition speedups: the Seifert stage already decides this homogeneous family before factorization.

### 13.4 Sparse modular Alexander elimination

The Alexander minor starts with at most three nonzero entries per row. Report 09's sparse determinant keeps row dictionaries and column incidence sets. It selects a shortest active row, then a least-populated column in that row, and records row and column permutation signs. Removing a pivot updates only its incident Schur-complement positions. The determinant equals the signed product of pivots, by the same block-elimination identity as dense Gaussian elimination; a zero active row proves singularity. All arithmetic is exact in the fixed prime field.

If $m$ is the minor dimension, $E$ its initial nonzero count, and

$$F = \sum_{\text{pivots}} (\text{row degree} - 1)(\text{column degree} - 1),$$

the arithmetic work is $O(m + E + F)$ plus at most $m$ inversions. Heap maintenance and stable neighbor order add logarithmic bookkeeping. The bound is conditional on actual fill; $F$ may still be cubic. The logical number of live entries also differs from Python container capacity and resident memory. The implementation reports these work counters rather than claiming linear complexity for every Alexander matrix.

The sparse path is selected automatically at 256 crossings; dense and sparse modes remain explicit low-level options. On positive three-braid examples with 62, 262, and 514 crossings, local paired sparse-kernel speedups were 0.67, 1.10, and 1.98 (A/A ratios 0.999, 0.999, 1.002). The small-case regression motivates retaining the dense path below the threshold. These kernel measurements also bypass the Seifert stage. The exact witness dictionaries agree, including determinant signs. The reproduction script is `fast/benchmark_factor_sparse.py`; raw samples are in `fast/results/factor_sparse_20261007.json`.

### 13.5 Resource handling and the corrected frontier bound

Factorization and preprocessing now execute inside the same resource handler as the exact backend. Cooperative deadline checks occur before and after coarse operations and between factors; sparse elimination receives the callback at each pivot. An exhausted deadline or a memory allocation failure yields `UNKNOWN`. Tests inject these failures and advance a simulated clock during preprocessing, ensuring an expired operation does not leak an exception or produce a late verdict.

Report 09 also corrects an older theoretical statement. The scan stores endpoint pairings but supplies no certificate that all frontier points lie on one disk boundary in a common cyclic order. On arbitrary frontiers the safe scalar-state bound is

$$s_i \le \min\{2^i, (w_i - 1)!!\}, \qquad T \le \mathrm{poly}(n, w, \log p)(w - 1)!!.$$

The Catalan count is valid only with that additional disk condition. We reproduced three counterexamples by independent prefix-resolution enumeration: an explicit six-crossing input, a twelve-crossing greedy order, and the existing double-Conway fixture under its default best order. Each has a six-endpoint frontier with six nonzero Laurent and modular matching states, exceeding the third Catalan number five. The data are in `synthesis/data/frontier-counterexamples-integration.json`. The earlier proposition and user documentation have been corrected; the exact bracket calculation and all verdicts are unchanged.

### 13.6 Later algebra and twist reports: production cross-checks

Report 10's component-contraction kernel completed 453,611 matching comparisons against the current production algebra, with all completed comparisons equal. Both inspected source hashes matched at the preintegration checkpoint; Conway and Kinoshita–Terasaka scans agree in every raw homological degree. Their adaptive runs selected no compressed calls, so those checks establish compatibility but no ordinary-diagram speedup. Report 11's twist backend agreed on 200 production reduced-rank comparisons. Report 12's preintegration source hash checks also passed; baseline, adaptive, and forced contraction agreed on each degree for six production fixtures, with $d^2 = 0$ checked after every crossing. The 29 standalone tests pass. These results are saved as the corresponding `report10`, `report11`, and `report12-production-crosscheck.json` files under `synthesis/data/`; the delivered reports themselves are unchanged.

The scope of the later claims matters. A faster dense cobordism product reduces one algebra operation, while the number of objects and calls can still be exponential. Twist compression requires a supplied braid-block presentation and measures size in expanded crossings. Report 12's succinct Schur cancellation assumes that tensor factors for a same-matching block and its interfaces have already been supplied. Its trillion-generator example verifies an algebraic identity and is not a knot decision. General extraction of these descriptions and control of their size under all crossing and cancellation operations remain unresolved. In particular, a scalar unit in a square-free characteristic-two algebra is its own inverse, but a matrix unit need not be: radical corrections require a finite matrix geometric series. That distinction must survive any future block integration.

### 13.7 Typed scalar shortcuts in the default planar algebra

The report 12 adapter drew attention to a small mismatch between the two production algebra engines: the older bit engine shortcuts identities and scalar squares, whereas the default integer-matching engine compiled them as general cobordisms. The default `Planar.compose` now returns zero immediately for a zero factor, and recognizes the identity only when its source and target matching agree. A polynomial encoded by the integer one in a different morphism space is not an identity.

For an endomorphism of one matching, the coefficient ring is $R_b = \mathbb{F}_2[x_1, \ldots, x_b]/(x_i^2)$. If $F = c + \sum_{S \neq \varnothing} f_S x_S$, then

$$F^2 = c^2 + \sum_{S \neq \varnothing} f_S^2 x_S^2 = c,$$

because cross terms cancel in characteristic two and every nonconstant square-free monomial has square zero. Thus equal scalar endomorphisms compose to their constant coefficient. This avoids

the old Cartesian loop over $|\operatorname{supp} F|^2$ monomial pairs. Comparing packed integers may still read the coefficient vector; it is not constant bit time for arbitrary dense inputs. Equal coefficients in differently typed morphism spaces do not satisfy this identity, and matrix-valued endomorphisms are explicitly outside its scope.

The new tests compare every three-arc scalar and typed identity against the original compiled evaluator, enumerate equal coefficients across all small matching triples to catch a missing type guard, and check 1,000 random four-arc products. The suite at this scalar-only checkpoint passed 116 tests. Nine paired raw-scan runs are recorded in `fast/results/planar_shortcuts_20261007.json`, with a pinned pre-change implementation, fresh algebra objects, and separate untimed call profiling. Ordinary scans support no general timing gain: the scanner already shortcuts most identities before calling this layer. For example, Conway calls the general algebra only 32 times, three of which are scalar squares; its exact square-zero debug checks call the algebra much more frequently. The separate dense scalar microcases measure the algebra operation alone and must not be advertised as full recognition speedups. The reproduction command is `python benchmark_planar_shortcuts.py -output RESULTS.json`.

### 13.8 Opt-in component contraction in the production scanner

Reports 10 and 12 share an exact algebraic improvement. We have integrated report 12's contraction kernel and adapter, preserving its attribution to the established fast subset-convolution algorithm. The independent matrix-block and tensor prototypes remain in the research archive. The production option is `composition="component"`; the explicit `component-dense` mode selects the dense asymptotic algorithm on every nontrivial composition. The standard scalar engine remains default.

Let the middle matching have $b$ arcs and hence $w = 2b$ boundary points. A compiled gluing plan decomposes into connected surfaces. For component $j$, write $L_j, Q_j$ for its input-circle sets, $B_j$ for its output-circle set, and $e_j$ for its fixed dots. Sending every left input variable in $L_j$ to $z_j$, and every right input variable in $Q_j$ to the same $z_j$, defines algebra maps into $R_c = \mathbb{F}_2[z_1, \dots, z_c]/(z_j^2)$, where $c$ counts components. Two dots entering one component vanish. Distinct input monomials with the same projected support cancel by XOR; keeping only the number of dots would lose this information.

Define a linear output map $\Lambda$ componentwise. Total dot occupancy at least two vanishes. An open component with one dot gives the monomial on all its output circles; with zero dots it gives the sum of monomials omitting exactly one output circle. A closed component gives one exactly when its total dot occupancy is one, and otherwise zero. Multiplication across disjoint output blocks defines the complete map. The composition identity is

$$\mathcal{B}(F, G) = \Lambda\big(\pi_L(F)\pi_R(G)\big).$$

Checking input monomials proves it: the projections and multiplication kill precisely the repeated-dot cases, while the output rule is the original Frobenius evaluation. Bilinearity extends the equality to arbitrary coefficient vectors. A positive-genus component vanishes because a handle acts by $2x = 0$ over $\mathbb{F}_2$.

The ring product is disjoint subset convolution,

$$(FG)_S = \sum_{T \subseteq S} F_T G_{S \setminus T}.$$

We use the ranked transform of Björklund, Husfeldt, Kaski, and Koivisto, *Fourier meets Möbius*. Group coefficients by subset cardinality, take the subset zeta transform at each rank, convolve the rank polynomials, and invert the transform. At output subset $S$, selecting rank $|S|$ retains exactly disjoint pairs whose union is $S$. In characteristic two the inverse transform uses the same

XOR butterfly. The cost is $O((c+1)^2 2^c)$ field operations and $O((c+1)2^c)$ coefficient storage; this algorithm is not a new invention.

If the left, right, and output spaces have $a, d, h$ circles, the complete factorized operation has cost

$$O\big((a+1)2^a + (d+1)2^d + (c+1)^2 2^c + (h+1)2^h\big)$$

in field and subset-mask operations. For the output step, component $j$ with $r_j$ output circles contributes at most $r_j + 1$ alternatives, and $\prod_j (r_j + 1) \le 2^{\sum_j r_j} = 2^h$. Each surviving component contains a middle arc, so $a, d, h, c \le b$ in ordinary composition. Consequently the conservative bit cost is $\mathrm{poly}(w)2^{w/2}$, improving the dense Cartesian-pair baseline, which can visit $4^b = 2^w$ pairs. Python coefficient integers contain up to $2^b$ bits and are not treated as unit-cost words. The implementation unpacks their bytes once when traversing support and uses rank polynomials with only $O(b)$ bits in the dense transform.

Sparse input requires a different choice. The adaptive mode delegates small support products to the existing engine. After projection, it also uses the characteristic-two identities

$$FG = F(F+G) + F(0) = G(F+G) + G(0)$$

to choose the cheapest of three sparse pair enumerations. A ranked transform is selected only when its estimated work is lower. The adaptive threshold is a performance heuristic; the explicit dense mode is the implementation to which the displayed asymptotic bound applies.

**Implementation boundaries.** The adapter retains the current `Planar` matching IDs and circle numbering. It strips only the redundant fifth field of each compiled component. Its added plan cache is cleared at each stage, and installation is permitted only on a fresh scanner before any crossing. It does not replace the specialized crossing-transfer operation. The options are available through both `recognize` and `khovanov_rank`, and through `-composition standard|component|component-dense` in the CLI. The current integration requires the standard scanner, minimum-fill pivots, bit coefficients, and no racing; incompatible combinations are rejected before an early certificate can hide them. Earlier recognition filters still run normally, and explicit rank computation bypasses them.

`composition_max_variables`, or the CLI flag of the same name with hyphens, defaults to eighteen. It bounds the component and output variables on selected factored allocations. This is an allocation guard, not a process-memory bound or a theorem for arbitrarily large uncapped inputs. Exceeding it produces `UNKNOWN` in recognition and a resource exception in the raw Python rank API; the CLI reports resource exhaustion with exit code three. Kernel deadline callbacks share the scanner's deadline. Counts of delegated, factored, and scalar-shortcut calls are reported under `composition_stats`. Neither a declined dense call nor an allocation estimate is evidence of knottedness.

The dense improvement leaves the number and size of live chain objects uncontrolled. The finite-type sharing theorem and the need for a uniform topological progress bound therefore remain unchanged. Faster local algebra alone does not prove quasi-polynomial unknot recognition.

**Integrated validation and measurements.** At the component-contraction checkpoint, all 134 production tests passed. The contraction-specific tests exhaust 65,536 three-variable ring products, compare 2,245 small matching/monomial compositions to both the production evaluator and an independent set-based engine, and check all 2,744 four-arc matching triples with seeded arbitrary polynomials. Complete scans, per-stage $d^2 = 0$, and exact homological-degree counts agree on supplied fixtures and forty additional random braid closures. Pipeline tests exercise allocation failure, option rejection, evidence fields, and both CLI resource paths. They also exposed an existing unclosed input file in the CLI loader, which now uses a context manager.

The seven-round production benchmark includes separate standard/standard controls. Adaptive isolated dense products improved 2.87, 10.25, and 37.90 times at six, eight, and ten variables; nearly equal dense products improved 6.51, 35.10, and 182.84 times through polarization. The ten-variable sparse control was about 1.13 times slower adaptively and 43.86 times slower under forced dense contraction. No adaptive factored calls were selected on the six ordinary scanner fixtures. Their observed adaptive baseline/new ratios were 0.94–1.04, with A/A variation, and establish no broad scan speedup. The forced path was slower on several ordinary scans. Raw samples, exact inputs, counters, and timing boundaries are retained in `fast/results/component_algebra_20261007.json`; reproduce with `fast/benchmark_component_algebra.py`. The dense-kernel improvement is real for the supplied synthetic morphisms, but their density is not claimed to arise on every difficult knot input.

# 14 Twist compression and a restricted quasi-polynomial family

Report 11 supplies a different way to reduce the exponential dependence: simplify an entire two-strand twist before constructing the closed chain complex. We have integrated this backend as an explicit option, after checking its local model, degree convention, and production comparisons. Unlike component contraction, which changes one morphism product, this replacement reduces the number of terms in the selected complex. The parameter is the number of signed runs in a supplied braid word. It is neither the minimal braid index nor an invariant of an arbitrary PD presentation.

## 14.1 Local replacement and exact assembly

Write the braid on $b$ strands as

$$\beta = \sigma_{i_1}^{\epsilon_1 m_1} \cdots \sigma_{i_t}^{\epsilon_t m_t}, \qquad m_j > 0, \quad \epsilon_j \in \{-1, 1\}, \quad n = \sum_j m_j.$$

The production adapter groups maximal equal signed letters. It does not perform a hidden search for a shorter braid presentation. The standalone run API also accepts binary-encoded exponents, but every complexity bound below is in the expanded crossing count $n$, not their encoding length.

Let $I$ be the identity smoothing and $E$ the turnback. A positive $m$-crossing twist has the local representative

$$I \xrightarrow{s} E \xrightarrow{D} E \xrightarrow{D} \cdots \xrightarrow{D} E, \qquad D = X_{\text{top}} + X_{\text{bottom}},$$

with terms in degrees $0, \ldots, m$. The negative twist is its dual, in degrees $-m, \ldots, 0$, with the saddle entering the final $I$. This is the characteristic-two, quantum-ungraded specialization of Thompson's Proposition 4.1 (version 2, December 2018), using Bar-Natan's local delooping and cancellation. It is a homotopy equivalence relative to the boundary, so inserting it into the surrounding braid preserves homology. The integer dot sums and differences become the same map over $\mathbb{F}_2$. This locality theorem, not merely a $d^2 = 0$ check, justifies replacing the crossing cube.

A macro state chooses $k_j \in \{0, \ldots, m_j\}$ in each block. Its homological degree is $\sum_j \epsilon_j k_j$, and its geometry depends only on the support $S = \{j : k_j > 0\}$. The implementation closes the corresponding $I/E$ picture on a $(t+1)$-by-$b$ grid and caches its circle partition by $S$. A marked closing arc fixes one circle's label to $X$ in $\mathbb{F}_2[X]/(X^2)$; the other circles carry binary labels. A state with $c(S)$ circles therefore contributes $2^{c(S)-1}$ reduced basis elements.

Transitions at the end of a run use the multiplication or comultiplication saddle. Internal transitions apply the two dot maps. If the top and bottom arcs lie on one circle their dot contributions cancel; a dot on the marked circle vanishes. Different blocks commute, so cross terms cancel in

characteristic two. Exact bit-column elimination computes each boundary rank and hence each homology degree. The existing rank-one reduced Khovanov detector then decides the knot.

## 14.2 Counting the selected complex

For each support $S$, precisely $\prod_{j\in S} m_j$ states have that support. Thus the full reduced chain dimension is exactly

$$N = \sum_{S\subseteq[t]} 2^{c(S)-1} \prod_{j\in S} m_j. \tag{1}$$

This is a chain dimension, not a Betti number. The same formula with $m_j$ replaced by $z^{\epsilon_j} + \cdots + z^{m_j\epsilon_j}$ gives the dimension in every degree. The Temperley–Lieb trace of $\prod_j (I + m_j E_{i_j})$ at loop value two, divided by two for the mark, computes $N$ without expanding the run lengths. This provides the adapter's exact size preflight. A separate polynomial transfer computes the degree profile using sliding windows for these consecutive powers.

Here is the useful structural bound from report 11. If $d(S)$ counts distinct generator indices active in $S$, then

$$c(S) \le b + |S| - 2d(S).$$

Begin with $b$ vertical circles and activate turnbacks in word order. At the first use of index $i$, no prior connection crosses the cut between positions $i$ and $i+1$. Its two arcs are therefore in different circles, and that saddle must merge them. Every other activation can increase the circle count by at most one. Counting the $d(S)$ forced merges proves the inequality; its slack is an even nonnegative integer.

Set $Q_i = \prod_{j:i_j=i}(1+2m_j)$. Inserting the bound into (1) and grouping subsets by their generator gives

$$N \le 2^{b-1} \prod_{i=1}^{b-1} \left(1 + \frac{Q_i - 1}{4}\right) = U := \prod_{i=1}^{b-1} \frac{Q_i + 3}{2}.$$

A knot closure uses every generator index: otherwise its strand permutation preserves the two nonempty sides of an unused cut. Consequently $Q_i \ge 3$, $b \le t+1$, and, for $t > 0$,

$$N \le U \le \prod_j (1+2m_j) \le \left(1 + \frac{2n}{t}\right)^t. \tag{2}$$

The last step is arithmetic–geometric mean. The bound is sharp for two-strand presentations and when each generator appears in just one block. It is not a bound on the smallest possible presentation of a given knot.

There are at most $N$ macro states, each with polynomial bookkeeping in $n+b$. Differential matrices have at most $N$ rows and columns, and their bit elimination costs at most a fixed polynomial times $N^3$, with storage at most a fixed polynomial times $N^2$. Therefore the uncapped algorithm satisfies

$$T \le \mathrm{poly}(n+b)N^3 = \mathrm{poly}(n)2^{O(t\log(2+n/t))}.$$

In particular, for every fixed $c$, supplied knot braids with $t \le c\log(n+2)$ admit a uniform $2^{O(\log^2(n+2))}$ bound. Fixed $t$ gives a polynomial whose exponent may depend on $t$; this is not an FPT claim. Finite allocation and time limits can still cause an invocation to return `UNKNOWN`. No algorithm here constructs such a short-run braid for every knot diagram.

## 14.3 A sharper degree-sensitive bound

The actual implementation eliminates each homological degree separately. Writing $d_h = \dim C^h$, define

$$M = \sum_h d_h d_{h+1}, \qquad W = \sum_h d_h d_{h+1} \min(d_h, d_{h+1}).$$

There are at most $d_h \min(d_h, d_{h+1})$ column reductions for the $h$th differential, each touching at most $d_{h+1}$ bits. Thus $W$ bounds their total XOR bit work; reading columns and bookkeeping add at least $O(N)$ work and polynomial factors in index bit lengths. $M$ is the dense coefficient-bit storage bound. These more informative quantities are reported by both the assembly and degree-profile preflight as `matrix_bit_upper_bound` and `rank_xor_bit_upper_bound`. They are analytical bounds, not sampled CPU times or process-memory measurements.

This gives an additional consequence for a single lengthening block. Hold the strand count, block positions, signs, and all $m_j$ except $m_r$ fixed. For any assignment of the other local indices and any degree $h$, the equation $\sum_j \epsilon_j k_j = h$ determines at most one value of $k_r$. Every state has dimension at most $2^{b+t-1}$. Hence

$$d_h \le 2^{b+t-1} \prod_{j \ne r} (m_j + 1),$$

independent of $m_r$. The number of degrees and macro states is $O(m_r + 1)$, while the finite list of support geometries and local maps is unchanged. It follows that $N, M, W = O(m_r + 1)$, with constants depending on the fixed surrounding word. The code therefore has linear word-level work in this family, with logarithmic factors for index arithmetic in a conservative bit model. This conclusion is substantially sharper than substituting a linear $N$ into the generic cubic bound.

For a single positive two-strand twist, explicitly $d_0 = 2$ and $d_1 = \cdots = d_m = 1$. Thus $N = m + 2$, the peak degree dimension is two, and $M = W = m + 1$. Tests compare these closed formulas to independently computed profiles and assembled complexes at lengths 3, 31, and 301. This is a homology-computation improvement; recognition already has the faster elementary two-strand certificate.

### 14.4 Integration, provenance, and failure semantics

`backend="twist"` or `-backend twist` selects this backend after the existing reduction, factorization, and invariant stages. The direct rank CLI uses `khovanov -twist`. The adapter accepts only retained, checked source-braid provenance. If the entire diagram was simplified, its original braid still represents the same knot. If factorization produced several diagrams, that braid is never passed to an individual factor. Cheap tests may still decide a factor and thus the sum. Remaining factors are recorded as inconclusive with `twist_deferred`; if necessary the backend then decides the original whole braid once, with its rank recorded only at the top level. This avoids losing the restricted run-count bound by sending factors without braid provenance to an unrestricted scanner. Ordinary PD or grid input has no source word at all and uses the standard scanner, recording `twist_skipped`. The raw twist-rank API instead rejects input without provenance.

The macro degree convention differs from the scanner's cube degree:

$$h_{\mathrm{macro}} = n_+ - h_{\mathrm{cube}}, \qquad \dim H^{h_{\mathrm{cube}}}_{\mathrm{unred}} = 2 \dim H^{h_{\mathrm{macro}}}_{\mathrm{red}}.$$

Here $n_+$ counts positive letters of the *original source braid*. The adapter converts its output to that original cube grading and labels the evidence explicitly, even when the current PD has fewer crossings. A guessed shift or a factor's crossing count would be wrong. Normalized macro degrees are also retained for inspection.

The adapter first rejects a macro-state count exceeding its budget, then computes the exact reduced basis count by Temperley–Lieb transfer. Only a successful count within the basis ceiling permits matrix assembly. Preflight and assembly consume one common time budget. The preflight has separate limits of 4096 matchings and 256 strands; exhausting a guard raises a resource limit, not a verdict. The all-$I$ state's dimension also provides an immediate lower bound before any large geometry is allocated. The state-count guard uses bounded multiplication, avoiding construction and decimal formatting of enormous products for succinct run inputs.

The production recognition option `twist_max_basis` defaults to one million, independently of the standard scanner's object ceiling. The raw `Budget` also controls macro states, matrix bits, XORs, and time. Matrix-bit limits concern coefficient capacity rather than Python resident memory. Allocation/time exhaustion yields `UNKNOWN` in recognition and the CLI, and `ScanLimit` or `MemoryError` in the raw production adapter. The separate `fastunknot.twist` module retains the advanced signed-run, homology, and preflight interfaces from the report.

For context, Kelomäki and Schütz (2026) give a polynomial homology algorithm for all three-braids and show exponential behavior of ordinary scanning on simple three-braid families. The run-count bound above is a different restriction and does not subsume their result. Our existing three-braid recognition shortcut already decides that knot class without computing its homology; it remains ahead of this optional backend.

### 14.5 Production checks and observed scaling

The full production suite passes 209 tests, including the report's 62 local algebra/topology tests and new production integration regressions. A seeded corpus of 100 knot braid closures on two through five strands agrees with the existing scanner in every raw cube degree, with $d^2 = 0$ checked in both engines. The earlier 200-case production cross-check remains recorded separately. These are comparisons of presentations, not counts of distinct knots. New tests also exercise mirrors, whole-diagram simplification, proper-factor deferral, cheap factor obstructions, missing provenance, huge encoded counts, exact preflight rejection, shared time accounting, and CLI resource handling.

Seven randomized paired rounds compare fresh raw homology computations, with a second standard-scanner arm as an A/A control. Diagram creation and input loading are outside the timed interval; scanner ordering and twist size preflight are inside. Every timed computation must agree in rank and every original cube degree. The results are:

| Input family | Size parameter | Scanner/twist | A/A |
|---|---|---|---|
| Single two-strand twist | 51 | 31.61 | 1.013 |
| | 201 | 134.74 | 0.973 |
| | 801 | 574.45 | 1.004 |
| Fixed four-strand context | 3 | 0.19 | 1.012 |
| | 15 | 0.30 | 1.002 |
| | 51 | 0.48 | 0.995 |
| | 201 | 1.26 | 1.016 |
| | 801 | 4.29 | 1.024 |
| Alternating three-braid, no run compression | 8 crossings | 0.22 | 0.995 |
| Morton's four-strand unknot | 11 crossings | 0.06 | 0.972 |

The four-strand word is $\sigma_1^m \sigma_2 \sigma_3^{-1} \sigma_2 \sigma_3 \sigma_1 \sigma_2^{-1}$ with odd $m$. Its measured chain dimension is $N = 177m + 234$, verified by exact assembly and the independent preflight in each sample. Its peak degree dimension is 177 for the listed $m \geq 15$, while the small $m = 3$ example peaks at 160. Thus the code exhibits the fixed-context bound above; this finite sample is not a proof of the general bound. The larger context measurements were run separately after the first corpus, with the same paired protocol.

The raw scanner took approximately 2.302 seconds on the 801-crossing two-strand case, compared with 0.004 seconds for the macro adapter. These are *homology* timings: the default recognition pipeline already decides all the long-twist examples by its braid obstruction. The alternating control is about 4.5 times slower under the macro backend, and the Morton example about 15.8 times slower. These regressions preclude an unconditional default switch. In particular, a small

run count gives a useful asymptotic theorem but need not beat the scanner's cancellations at a particular finite size.

Reproduce with `fast/benchmark_twist.py`; add `-context-only` for the larger four-strand cases. Raw timings, inputs, degree ranks, work bounds, and the method chosen by default recognition are saved in `fast/results/twist_integration_20261007.json` and `fast/results/twist_context_scaling_20261007.json`. No archive files were changed to obtain these results.

# 15 Survivor prediction and adaptive cancellation

The incoming package `unknot_radical_transfer_20261007.zip` is now preserved as report 15. It analyzes minimal complexes before performing cobordism-valued elimination. We reviewed an isolated extraction of the delivered archive, reran its checks, and integrated a narrower operation into the production scanner: exact prediction of the surviving matching multiplicities, followed by a certified shortcut when no differential can remain. An adaptive variant uses this operation only after ordinary cancellation becomes expensive. The delivered report and the production measurements are separate artifacts.

## 15.1 What binary residue homology determines

At a fixed frontier with $2k$ endpoints, write $c(a, b)$ for the number of circles in the overlay of two matching objects. A dotted basis morphism $x_S : a \to b$ has nonnegative intrinsic weight

$$\omega(x_S) = k - c(a, b) + 2|S|.$$

Weight zero means exactly the undotted identity of one matching. The Euler-characteristic calculation in the composition engine makes weights additive on every nonzero term of a valid cobordism composition. More explicitly, for input dot counts $u, v$ and output dot count $q$, the nonzero genus-zero terms obey

$$2q = 2u + 2v + k - c(a, b) - c(b, c) + c(a, c),$$

which gives the stated additivity after substitution. Consequently the operation $\rho$ that retains only the scalar identity coefficient between equal matchings is multiplicative. An undotted coefficient between *different* matchings is discarded. Forgetting this typing would give an incorrect binary complex.

Let $m_{a,h}$ count copies of matching $a$ in degree $h$ of a valid partial complex $C$. The residue differential $d_0 = \rho(d)$ is a direct sum, over $a$, of ordinary binary complexes. Put

$$r_{a,h} = \operatorname{rank}_{\mathbb{F}_2}(d_0 : C_{a,h} \to C_{a,h+1}), \qquad \mu_{a,h} = m_{a,h} - r_{a,h} - r_{a,h-1}.$$

**Proposition 15.1** (Exact survivor profile)**.** *Complete scalar-unit cancellation leaves exactly $\mu_{a,h}$ copies of each matching in each degree. This profile is independent of pivot order at the fixed processed tangle.*

*Proof.* An allowed pivot is an endomorphism $\phi$ with scalar coefficient one, so $\rho(\phi) = \rho(\phi^{-1}) = 1$. Applying $\rho$ to its cobordism Schur update gives the ordinary binary pivot update. Hence each cancellation preserves residue homology. Cancellation terminates, removing two objects at each step. Its final residue differential is zero: a remaining nonzero scalar coefficient would be another allowed pivot. The final multiplicities are therefore the dimensions of the original residue homology, namely $\mu_{a,h}$. □

This argument uses only the valid compositions already required by the scanner. It does not require a universal Catalan count, the full classification of minimal complexes, or the incoming report's sharp nilpotence exponent. In particular, pivot heuristics can improve fill, composition work, and intermediate memory, but cannot decrease the final matching multiplicities of a fixed prefix. Changing the scan order changes the prefix and is a different operation.

## 15.2 An exact shortcut, and an essential non-shortcut

The profile is not itself the surviving differential. For example, $a \xrightarrow{x} a$ in degrees zero and one has zero residue differential, yet its dotted map is nonzero. Cancelling other scalar pivots can also create such maps between surviving objects.

**Proposition 15.2** (Degree-gap shortcut)**.** *Suppose the occupied degrees of $\mu$ contain no consecutive integers. Then $C$ is homotopy equivalent, relative to its frontier, to the direct sum specified by $\mu$, with zero differential. The same conclusion holds at a closed frontier without the degree-gap hypothesis.*

*Proof.* Perform unit cancellation conceptually and apply the survivor proposition. A differential increases homological degree by one, so the degree-gap hypothesis forbids every remaining entry, even between unequal matchings. At a closed frontier $k = 0$, all coefficients are scalars and a fully cancelled differential is zero. The usual local cancellation equivalences therefore identify $C$ with exactly the stated zero-differential complex. □

The implementation need not carry out that conceptual sequence: it computes binary ranks and constructs the resulting objects directly. There are no cobordism compositions in the shortcut. Because the equivalence is relative to the frontier, subsequent crossing gluing preserves it. Zero survivors are allowed. Nonconsecutive integer degrees and previously cancelled object slots are handled explicitly. The raw helper assumes a valid complex, including $d^2 = 0$; a nonnegative profile alone is not a validity certificate.

## 15.3 Switching implementations in response to progress

Eager residue prediction can be slower than the existing sparse cancellation loop. The adaptive mode follows an introsort-style policy, with a mathematical distinction: the alternative operation is a conditional shortcut, not a universal asymptotic fallback.

At the beginning of each crossing stage let $N$ be the live object count. Start the existing minimum-fill cancellation with an allowance

$$B = \max(256, 4N)$$

for Schur update pairs. Before cancelling a pivot from $b$ to $c$, its candidate work is

$$(|\operatorname{out}(b)| - 1)(|\operatorname{in}(c)| - 1).$$

These are the Markowitz costs already available to the queue. They bound touched pairs, not actual expensive compositions: scalar identities, zero products, and caches may make a pair cheaper. When the next pivot would exceed $B$, pause before that pivot, finish the counters, and discard the candidate queues. All completed pivots, live maps, and composition caches are retained.

Compute the residue profile of this partially reduced complex. If the degree-gap or closed-boundary certificate applies, finish the stage using the shortcut. Otherwise resume ordinary cancellation on the same partial result. There is at most one prediction attempt per crossing stage, so a

failed attempt cannot cause oscillation between implementations. A later crossing starts a fresh allowance. The constants are a transparent cost heuristic, not a proved optimal threshold. Deadline and object limits keep their existing meaning; neither slow progress nor a failed shortcut decides whether a knot is trivial.

The shared state is what makes this switch inexpensive and sound. Every pause lies between complete homotopy equivalences; a partially applied Schur update is never exposed to another engine. Rebuilding the queue on fallback costs a fresh pass over its remaining entries, while the completed algebra is reused. The additional binary rank work is not bounded by elapsed sparse work, so this policy does not claim the constant-factor competitive guarantee of a time-sliced portfolio. In particular, it does not acquire introsort's global worst-case theorem merely by adopting its switching principle.

**Interfaces and evidence.** Both recognition and raw homology accept the Python option `reduction="adaptive"` or `reduction="residue"`; the latter predicts at every reduced stage and is an ablation/control. The CLI uses `-reduction adaptive|residue|standard`. This controls chain cancellation and is distinct from the existing `-no-reduction` switch for Reidemeister simplification. The default remains `standard`. The new modes require the standard backend, bit coefficients, minimum-fill pivots, and no race; they compose with component contraction and the existing tail option. Unsupported combinations are rejected before early recognition filters. Counters expose prediction attempts, successful shortcuts, switches, fallbacks, and Schur update pairs. Earlier recognition certificates remain enabled, so raw homology measurements are not pipeline speedups.

### 15.4 The stronger report and its boundary assumptions

For noncrossing matchings in a *prescribed common cyclic order*, the incoming report proves the sharp arc-algebra radical exponent $J^{2k} = 0 \neq J^{2k-1}$ for $k \geq 1$. Its dot derivative and central marked dot also yield a two-pass transfer formula. Given a scalar contraction $(i, p, h)$ and $\delta = d - d_0$, the induced differential is

$$D = \sum_{r=0}^{2k-2} p\delta(h\delta)^r i.$$

This is a finite specialization of the homological perturbation lemma, whose inverse formula and special contraction conditions are stated in Crainic's Section 2. The report supplies sharp-depth examples, so truncating this general expression after the first correction would be unsound.

Those stronger formulas are not installed into the arbitrary PD scanner here. Its frontier pairing representation has no global common-disk certificate. As a simple warning, the three pairings $(01)(23), (02)(13), (03)(12)$ are not simultaneously noncrossing in one cyclic order; blindly composing the associated abstract triple in the planar gluing engine yields an impossible surface component. The narrow production proof above needs only its existing valid geometric compositions. A full transfer adapter must establish its category and geometry assumptions separately. The report's provided profile adapter targets the older dictionary-based scanner, not the current interned-list implementation; our adapter is new.

The survivor profile also exposes the main asymptotic obstruction. A small number of matching *types* does not bound their *multiplicities*. Even fixed-width scans can have exponentially many surviving copies. The report's parameterized transfer estimate still depends on both boundary half-width and the largest minimal survivor count. No bound forcing these parameters into a quasi-polynomial regime for every input has been established.

### 15.5 Validation and measured limits

The isolated report rerun passed 18 unit tests, 2,244 exhaustive composition/derivation comparisons, 500 full transfer certificates, 500 minimal-profile comparisons, 308 marked-transfer certificates, 306 braid closures against an independent crossing cube, and 1,365 intermediate-stage contraction certificates. Sharp radical witnesses through $k = 12$ and transfer-depth witnesses through $k = 8$ also passed. These checks cover the report's model; they do not prove a global runtime bound or establish compatibility with arbitrary scanner geometry.

Production tests compare each predicted profile to actual complete cancellation at every stage of 120 seeded knot closures on two to five strands, using both greedy and shuffled crossing orders. Every final homological degree agrees for eager, adaptive, and ordinary scans; twenty cases also use the independent older set-based engine. Additional tests check a nontrivial unit, nonzero maps after a declined shortcut, pause/resume validity, both adaptive outcomes, budget exhaustion, CLI and pipeline evidence, and incompatible options. All 222 integrated tests passed. The integrated run and the report validation summaries are retained under `synthesis/data/`.

| Input | Eager ratio | Adaptive ratio | A/A |
|---|---|---|---|
| two_strand_201 | 0.699 | 0.983 | 0.998 |
| torus_3_10 | 0.897 | 1.004 | 0.997 |
| alternating_3_4 | 0.977 | 0.995 | 1.021 |
| morton | 0.897 | 0.986 | 1.020 |
| conway | 0.922 | 0.998 | 1.006 |
| hard_unknot_8 | 0.934 | 0.986 | 1.006 |
| Synthetic, 33 objects | 87.29 | 4.91 | 1.001 |
| Synthetic, 65 objects | 272.23 | 132.33 | 0.995 |
| Synthetic, 129 objects | 409.70 | 203.12 | 1.002 |

Ratios are median paired standard/new times; values above one are faster.

The benchmark includes seven shuffled paired trials with a second ordinary-scanner control. Diagram construction is excluded and scan order selection is included. Synthetic two-term complexes are timed only during elimination, after deterministic construction: they have a dense dotted differential, a full-rank binary residue, and a single survivor. Their density is not asserted to occur on a measured knot scan. The adaptive mode is useful on this kernel family, but the ordinary diagram results do not establish a general recognition gain. Raw inputs, all timings, counters, and controls are stored in `fast/results/residue_integration_20261007.json`; reproduce with `fast/benchmark_residue.py`. These measurements keep both new modes opt-in while giving subsequent adaptive policies a tested, interruptible cancellation interface.

## 16 Recovered grading, homogeneous algebra, and sharper type bounds

The incoming package `unknot_recognition_dense_algebra_20261008.zip` contains another dense coefficient engine, a recovered-grading argument, and a bounded finite-quotient certificate search. It is now preserved as report 16. Its component contraction overlaps the implementation already integrated from reports 10 and 12. We therefore retain the current adapter and incorporate the additional homogeneous multiplication case, rather than replacing the production tree with the archive's older baseline. We also derive a packed shortcut for products in the top degree and audit the recovered grading against the current scanner.

## 16.1 The grading that survives after its labels are erased

Start the empty object with an auxiliary quantum shift $q = 0$. When smoothing $i \in \{0, 1\}$ of one crossing creates $c$ closed circles, its delooping label $\ell$ gives the child shift

$$q' = q + i + c - 2\,\mathrm{popcount}(\ell).$$

For matching objects $a_u, a_v$ at a frontier of $2m$ endpoints, every monomial with $d$ dots in a nonzero differential entry $u \to v$ obeys

$$c(a_u, a_v) - m - 2d + q_v - q_u = 0. \tag{3}$$

Here $c(a_u, a_v)$ counts the circles of their overlay, and the normalization is that of the raw scanner, before oriented link shifts. This is the dotted grading of Bar–Natan's tangle formalism, specialized to the implementation's smoothing and delooping order.

**Proposition 16.1.** *Every actual differential entry in the ordinary scanner is homogeneous in dot degree. Every cancellable entry between equal matchings is the scalar identity* 1.

*Proof.* A $k$-disc cobordism carrying $d$ dots has normalized degree $k - m - 2d$. A crossing saddle contributes $-1$, compensated by the smoothing shift. Delooping a circle contributes shifts $+1$ and $-1$, giving the displayed recurrence and establishing (3) after crossing extension.

The intrinsic weights from Section 15 add under nonzero composition; their negatives are these normalized degrees. If a map between equal matchings has scalar coefficient one, homogeneity forces its dot degree to be zero. There is only one zero-dot monomial, so the map is exactly 1 and the two shifts agree. The source and target shifts telescope in a Schur correction through this identity. That correction has the same degree as the existing entry, and their sum remains homogeneous. Induction over the cancellation sequence proves the claim. □

The auxiliary labels are a proof and diagnostic device; the normal scanner still does not store quantum grading. Component sharing preserves the conclusion: identical connected keys have identical relative shift differences along their nonzero entries, and an overall shift of a disconnected summand is irrelevant to homogeneity. The adaptive degree-gap shortcut produces a zero differential, which also admits such a grading before subsequent crossing extension. This does not turn its output into a computation of quantum-graded homology; those discarded absolute labels are not reconstructed.

The production audit explicitly propagates $q$ and checks every term before and after elimination, for ordinary and forced component composition. It completed 520 scans of 87 diagrams, using input, greedy, and seeded shuffled orders where specified: 196,928 entries and 220,736 supported terms satisfy the equation. The largest dot degree observed was two. This finite observation is not a universal bound on dot degree. Inputs, orders, exact ranks, and counts are in `fast/results/grading_audit_20261008.json`; reproduce with `fast/audit_grading.py`.

## 16.2 Cheaper multiplication with an explicit applicability check

In the component ring $R_r = \mathbb{F}_2[x_1, \ldots, x_r]/(x_1^2, \ldots, x_r^2)$, suppose $F, G$ have homogeneous dot degrees $a, b$. Apply the ordinary subset zeta transform to their binary coefficient arrays, multiply pointwise, and apply the inverse transform, also XOR over $\mathbb{F}_2$. The resulting coefficient at $S$ is

$$U_S = \sum_{A \cup B = S} F_A G_B.$$

Retain only subsets of size $a + b$. Each supported pair has $|A| + |B| = a + b$, so this restriction is equivalent to $A \cap B = \varnothing$. The surviving array is exactly the square-free product $FG$. If $a + b > r$, the product is zero.

This branch uses $O((r+1)2^r)$ binary field operations. The generic ranked transform, retained for mixed-degree inputs, has the usual $O((r+1)^2 2^r)$ ring-operation bound of Bjoerklund–Husfeldt–Kaski–Koivisto. These are arithmetic bounds; array indices, monomial masks, and coefficient conversions still carry their bit costs. Both remain within the earlier $\mathrm{poly}(r)2^r$ coefficient envelope. The ranked implementation itself is cheaper on many special inputs, so the two displayed bounds are not a prediction of a factor-$r$ measured speedup on every homogeneous product.

The final size restriction is essential. For example, $x_1(x_1 + x_2) = x_1x_2$, whereas unfiltered union convolution would incorrectly keep $x_1$. The software verifies the dot degree of each actual operand and falls back to ranked convolution for mixed degrees. It never infers this property merely from a caller selecting the scanner API. Projection to component variables preserves degree for every surviving monomial; a repeated component annihilates a term and collisions may cancel terms, but cannot create another degree. Thus the optimization applies after the existing component projection.

**Complementary subsets in the top degree.** We add a further specialization when $a + b = r$. Only the full monomial $x_{[r]}$ can survive, and its coefficient is

$$[x_{[r]}]FG = \sum_{A \subseteq [r]} F_A G_{[r]\setminus A}.$$

In the packed coefficient convention, complementing the subset index is exactly reversing its $2^r$ coefficient bits. Reverse the bytes and the bits within each byte, remove padding when $2^r < 8$, intersect with the other polynomial, and take the parity of the population count. This computes the coefficient using linear work in the packed bit length after the degrees have been checked, with no zeta transform. The homogeneity check still incurs its own support and index-processing cost. Generic mixed-degree inputs cannot use this shortcut to compute their other output degrees.

The new cases are inside the existing opt-in component-contraction backend. No additional recognition flag is required; the ordinary sparse engine remains the default. The old ranked function remains available for independent comparisons. Existing allocation limits, deadline callbacks, scalar identities, and polarization are retained.

## 16.3 A sharper sufficient regime for component sharing

Let $B_m = \binom{m}{\lfloor m/2 \rfloor}$ and let $M(2m) = (2m-1)!!$, with $M(0) = 1$. The latter is the safe unrestricted pairing count; this argument does not restore a Catalan bound without a common-disk certificate.

A homogeneous coefficient with at most $m$ circle variables lies in one of at most $m + 1$ degree layers, each of dimension at most $B_m$. There are therefore at most

$$H_m = (m+1)2^{B_m}$$

possibilities, intentionally counting zero more than once. A connected component on $s$ objects has normalized homological degrees between zero and $s - 1$: each edge changes degree by one, and a simple path from a minimum-degree object has at most $s - 1$ edges. Consequently the number of ordered serialized keys with at most $b$ objects is bounded by

$$K(m, b) = \sum_{s=1}^{b} s^s M(2m)^s H_m^{s^2}.$$

We have overcounted zero entries, invalid degree differences, disconnected graphs, and failures of $d^2 = 0$, so no unproved realizability assertion enters this upper bound. Taking logarithms,

$$\log K = O\big(b^2 B_m + bm \log(m+1) + b \log(b+1)\big) = O(b^2 B_m).$$

The existing exact-sharing argument now gives the stronger sufficient condition

$$b^2\binom{m}{\lfloor m/2\rfloor} = O(\log^2(n+2))$$

for quasi-polynomial cost, provided it bounds every reduced component and frontier actually encountered. Extending a representative through one crossing only multiplies its object count by a constant; different parent components stay independent. Shift weights have polynomial degree and polynomial bit length, as in Section 12. For bounded $b$, writing $L = \log_2(n+2)$ and $\ell = \log_2 L$, the sufficient frontier scale improves from $2m \le 4\ell + O(1)$ to

$$2m \le 4\ell + \log_2 \ell + O(1).$$

This is a refinement of a conditional parameter regime, not a proof that arbitrary diagrams admit such scans. Actual component size is still uncontrolled, and large multiplicities still defeat an explicit-object implementation.

### 16.4 Adaptive controls that satisfy the grading constraint

The mixed-dot two-term matrices used in the previous adaptive benchmark are valid ungraded complexes, but generally cannot be genuine scan differentials: their entries mix dot degrees. Their large speedups therefore do not establish the behavior of a graded scan. We retain those samples and add a separate graded family.

For size $s$, begin with an invertible binary $s$-by-$s$ matrix, constructed by reversible row additions, and add one source column. Interpret each nonzero entry as the identity of a fixed matching. The resulting two-term complex has $s+1$ sources, $s$ targets, quantum shifts all zero, and exactly one surviving source object. It satisfies both $d^2 = 0$ and the recovered grading equation. Dense scalar Schur updates can still be expensive even though they require no cobordism products. The existing adaptive allowance detects that update work and switches to binary residue homology.

The new graded controls are still synthetic: we do not claim that our crossing order naturally generates their density. This makes them a stronger applicability check for the switch, not evidence of a broad recognition improvement.

| Kernel | Product ratio | Full component ratio |
|---|---:|---:|
| homogeneous_8_2_3 | 1.14 | 1.08 |
| homogeneous_10_3_4 | 1.08 | 1.07 |
| homogeneous_12_4_5 | 1.19 | 1.13 |
| homogeneous_14_5_6 | 1.20 | 1.13 |
| homogeneous_8_4_4 | 16.25 | 4.69 |
| homogeneous_10_5_5 | 22.62 | 5.21 |
| homogeneous_12_6_6 | 28.86 | 5.97 |
| homogeneous_14_7_7 | 34.21 | 6.54 |
| mixed_12 | 1.01 | 1.00 |
| sparse_12 | 1.10 | 1.11 |

| Graded two-term objects | Adaptive ratio | A/A |
|---|---:|---:|
| 65 | 2.72 | 1.026 |
| 129 | 8.91 | 1.005 |
| 257 | 20.74 | 1.001 |
| 513 | 58.52 | 0.987 |

| Forced component scan | Ranked/new ratio | A/A |
|---|---|---|
| trefoil | 0.803 | 1.058 |
| conway | 0.991 | 1.015 |
| kinoshita_terasaka | 1.092 | 1.060 |
| torus_3_5 | 0.987 | 0.990 |
| hard_unknot_8 | 1.006 | 1.002 |

All speedups are median paired old/new times; values above one are faster.

All measurements use seven shuffled paired trials and an identical old/old control. Product timings compare the prior ranked transform to the checked dispatch; full-component timings include projection and output expansion. Ordinary scan measurements force component contraction to exercise the changed path and include order selection. The small trefoil scan regressed in this sample; other scan differences are modest and include control variation. The default sparse engine does not use this new dispatch, and no general recognition speedup is claimed. Reproduction scripts are `fast/benchmark_homogeneous.py` and `fast/benchmark_adaptive_graded.py`; their complete samples are in `fast/results/homogeneous_integration_20261008.json` and `fast/results/adaptive_graded_20261008.json`.

The full suite passes 229 tests. New tests exhaust every pair of four-variable homogeneous polynomials, test overlap rejection and byte padding, compare mixed inputs with the ranked fallback, verify projection behavior, and combine forced component algebra with adaptive cancellation. The isolated incoming archive passed its 74 tests, the independent eleven-test algebra suite, 2,100 associativity checks, 8,768 transfer comparisons, and replay of ten finite-quotient certificates. Local logs are retained under `synthesis/data/`.

### 16.5 A limit of fixed finite-quotient portfolios

The archive's optional $A_5$ search has not been merged into the current pipeline. Its witnesses are useful nontriviality certificates; exhaustion must remain inconclusive. The report also supplies a structural limitation worth retaining in the research plan.

For prime $p > 3$, the torus-knot group of $T(3, p)$ has presentation $\langle x, y \mid x^3 = y^p \rangle$, with $x^3 = y^p$ central. In any finite image in which the order of the image of $y$ is coprime to $p$, that image is a power of its central $p$th power. It is therefore central, so the image generated by $x, y$ is abelian. In particular this holds for $A_5$ when $p \geq 7$.

For a coloring by a finite quandle $Q$ with $|Q| < p$, its inner action lies in the symmetric group on fewer than $p$ points, so the same argument makes the induced group image abelian. All meridian images then agree. The common right translation fixes each color by idempotence, making every crossing preserve its under-arc color; following the one-component knot forces the coloring to be constant. Thus this family needs palettes of at least $p$ for nonconstant colorings, although its usual braid diagram has only $2p$ crossings. These knots are already decided by our three-braid or homogeneous diagram stages. The obstruction rules out a universal fixed-palette strategy; it does not rule out larger adaptive targets, bounded-seed searches, or a quasi-polynomial algorithm by another method.

## 17 Exact homological windows and adaptive widening

Report 13 supplies an exact way to compute a prescribed interval of unreduced Khovanov homology over $\mathbb{F}_2$ without constructing every chain group. We integrate its window scanner and explicit mirror bound into the current implementation, then add a bounded adaptive recognition stage. This stage first queries degree zero, widens only while its budget permits, and falls back to the

complete selected backend if the answer is inconclusive. The earlier adaptive cancellation strategy can also run inside each window. Both mechanisms are optional.

## 17.1 Why deleting chain groups can be exact

Raw cube degree counts 1-resolutions. Each remaining crossing contributes homological support $\{0, 1\}$, and delooping preserves homological degree. Suppose the requested final raw interval is $[a, b]$ and $r$ crossings remain. Retain the current complex only in

$$[a - r - 1, b + 1]. \tag{4}$$

This is brutal truncation: groups outside the interval and maps incident to them are deleted. The two extra degrees are guards.

**Lemma 17.1.** *Truncation to* (4) *preserves the final homology in* $[a, b]$ *after the actual remaining crossing extensions and closure.*

*Proof.* A current object of degree $p$ can contribute only to final degrees $p + q$ with $0 \le q \le r$. For a final degree $j \in [a - 1, b + 1]$, every contributing $p = j - q$ belongs to $[a - r - 1, b + 1]$. Thus all chain groups and maps between them in this three-degree neighborhood survive unchanged. Homology in degree $j$ uses precisely its group and the incoming and outgoing differential. Additive planar extension, closure, and the Khovanov functor preserve these equalities. □

Neither guard can be dropped in general. The complex $\mathbb{F}_2 \xrightarrow{1} \mathbb{F}_2$ in degrees $a - 1, a$ has zero $H^a$; deleting the first group invents a class. The same example in degrees $b, b + 1$ proves the need for the other guard. Homology in a guard degree can itself be artificial and is never returned as part of the answer.

After processing $i$ of $n$ crossings, the lower bound is $a - (n - i) - 1$ and the upper bound remains $b + 1$. At each step the code allocates only the delooped children in that band, then constructs crossing saddles and transferred old maps only when both endpoints survive. This is exactly the truncated extension, so the lemma applies before allocation as well as after it. An object ceiling therefore limits retained allocation, rather than an unnecessarily constructed full extension. The frontier still advances when the retained complex is empty.

Unit cancellations may be interleaved arbitrarily: they are chain homotopy equivalences, and the remaining additive extension and closure operations preserve their homotopy equations. Applying the lemma after each extension and this invariance after each cancellation proves correctness inductively. The proof compares each intermediate state after its own actual suffix; it does not require the truncated and full scanners to choose the same pivots. An isolated object in degree $h$ may additionally be removed when $[h, h + r] \cap [a, b] = \varnothing$, since its independent continued summand has no homology in the target. Incident nonzero maps forbid this shortcut.

## 17.2 Mirror choice and a parameterized complexity bound

Let $s_0, s_1$ count circles in the all-zero and all-one complete smoothings. Intersect the target with $[0, n]$ first; an empty intersection has zero homology. Define

$$Q(n, k) = \sum_{j=0}^{k} 2^j \binom{n}{j}, \qquad B_0 = 2^{s_0} Q(n, \min(n, b + 1)).$$

A prefix state with $j$ one-resolutions has at most $s_0 + j$ closed circles: complete it with zero-resolutions, then compare the resulting state with the all-zero smoothing using $j$ surgeries, each changing the circle count by at most one. Thus its delooping gives at most $2^{s_0+j}$ objects. Summing

over at most $\binom{n}{j}$ states bounds every retained allocation by $B_0$. Cancellation and truncation remove physical objects and cannot increase this bound; surviving labels extend injectively to cube labels.

If $w$ is the largest frontier size in the selected order, the ordinary explicit coefficient representation has cost

$$2^{O(w+1)}(n+1)^{O(1)}(B_0+1)^{O(1)}.$$

There are polynomially many object pairs and pivot updates per stage, and at most $2^{w/2}$ dotted basis elements in a morphism space. This argument counts objects actually present, so it requires no common boundary disk or Catalan matching bound.

Mirroring reverses raw degree: the rank in original degree $h$ equals the mirror rank in degree $n-h$. It preserves the crossing graph and the width of the same crossing order, while interchanging $s_0$ and $s_1$. The corresponding bound is

$$B_1 = 2^{s_1}Q(n, \min(n, n-a+1)).$$

The implementation compares these exact integers and selects the smaller bound, mapping returned degrees back to the original diagram. This is a proved size-bound selection, not a prediction of observed running time. The integers have $O(n)$ bits since $Q(n,k) \le 3^n$. The recorded per-stage profile explicitly states whether its internal degrees belong to the original or mirrored diagram.

For normalization let $\nu$ be the number of negative crossings returned by `Diagram.signs()`; normalized degree is raw degree minus $\nu$. The source braid word uses a different sign convention: the closure of `[1]` on two strands has $\nu = 1$ and raw homology in degree one. Counting negative braid letters would select the wrong window. If $s$ is the Seifert-circle count and $\mu = \min(\nu, n-\nu)$, surgery comparison with the oriented smoothing gives $s_0 \le s+\nu$ and $s_1 \le s+n-\nu$. For the normalized band $[-d,d]$, the resulting bound is

$$2^{O(w+s+\mu+d)}(n+1)^{O(\mu+d+1)}. \tag{5}$$

It is quasi-polynomial when $w+s = O(\log^2(n+2))$ and $\mu+d = O(\log(n+2))$. For a $t$-strand braid scanned in braid order, $s=t$ and $w \le 2t$. These are bounds for the prescribed query. No theorem here bounds a detecting radius for every nontrivial knot, finds such a low-width presentation for every input, or turns an agreeing partial query into a complete unknot recognizer.

### 17.3 An introspective recognition policy

The new recognition options are `window_radius`, `window_max_objects`, and `window_seconds`; the corresponding CLI flags use hyphens. A missing radius disables the stage. An enabled stage runs after the existing inexpensive certificates and simplification, before the selected complete backend. Each actual factor is queried independently and its PD is included in the evidence.

The requested radius is a maximum. The stage tries normalized bands $[-d,d]$ at radii $0,1,2,4,\ldots$, including the requested endpoint once and stopping when every raw degree is covered. Each wider query restarts: discarded chain groups cannot be recovered from a narrow scan. The last successful crossing order is reused, including when the selected mirror changes. All queries for one factor share the same time allowance (default 0.1 seconds) and retained-object ceiling (default 20 000), both bounded by the enclosing recognition limits. Before another query, the policy stops if the remaining local time is less than twice the last query's time. This is an empirical progress rule, not a competitive bound.

An exact queried profile differing from $\{0:2\}$ proves knottedness, since every queried band includes normalized degree zero. Agreement is inconclusive unless the query covers all raw degrees $0,\ldots,n$. Only in that complete case does rank two certify the unknot using the existing Khovanov detection theorem. For example, a trefoil agrees in normalized degree zero and is detected after widening. A

local time, object, or memory limit abandons the probe and continues the full backend; exhausting the enclosing deadline returns `UNKNOWN`.

This differs operationally from adaptive cancellation: that method pauses between complete pivots and resumes the same complex when its residue shortcut fails. Window widening must restart. Both fit the same design principle of measuring progress, switching only at mathematically valid boundaries, and retaining an exact fallback. Their combination does not improve the general exponential worst-case guarantee.

The standalone `window` CLI and the Python functions in `fastunknot/window_scan.py` and `fastunknot/window_bounds.py` expose exact partial ranks independently of recognition. CLI `-normalized` interprets input endpoints as normalized degrees and adds a separate normalized profile; `by_degree` always uses original raw degrees. The API result distinguishes `window_rank` from `complete_rank`. A partial rank of two alone is never a verdict.

### 17.4 Validation and measured tradeoffs

The integrated suite passes 245 tests. New checks compare 100 deterministic random knot braids on two through five strands with full homology, using shuffled crossing orders and arbitrary intervals. Every interval is checked with ordinary, residue, and adaptive cancellation; the first 20 also force the dense component kernel. Intermediate differentials are checked to square to zero. Other regressions cover both guards, mirror degree mapping, wrong-sign hazards, factor evidence, partial agreement, complete coverage, local versus global limits, and budget-sensitive widening. Separately, report 13 passes its 64 scanner tests and 10 hierarchy tests; report 14 passes 94 tests. These later review runs do not alter the as-delivered archive trees. Logs are retained under `synthesis/data/`.

| Input | Full/window | Probe 0 | Probe 4 | A/A |
|---|---|---|---|---|
| conway | 0.973 | 0.988 | 0.959 | 0.950 |
| kinoshita_terasaka | 1.142 | 1.030 | 1.030 | 0.984 |
| hard_unknot_8 | 0.983 | 0.514 | 0.207 | 1.032 |
| stress_braid5_36 | 1.374 | 1.478 | 1.469 | 0.988 |
| torus_4_7 | 8.460 | 0.885 | 1.230 | 1.009 |
| two_strand_201 | 2.733 | 0.741 | 0.946 | 1.019 |

Ratios are median paired baseline/new times; values above one are faster.

The table uses seven shuffled paired rounds on CPython 3.13.14. A/A compares identical full-decision arms. Preparation and imports are excluded; order selection and mirror selection are included. The raw-window ratio compares *different output contracts*: degree zero versus all homology. The two probe columns compare complete decision procedures with fallback, using a two-second shared local window budget and maximum radii zero and four. Earlier certificates are disabled in those timed arms to isolate this stage. With default certificates enabled, Jones decides Conway and Kinoshita–Terasaka, the short-braid engine decides the hard unknot, Alexander decides the stress braid, and Bennequin decides both torus examples. Thus these are fallback measurements, not claimed default-pipeline speedups.

The stress input reduces peak retained allocation from 17 694 to 7 692 objects and its degree-zero rank is 698, already an obstruction. At a 10 000-object cap, the full baseline returns `UNKNOWN`; the window stage returns `KNOTTED`. Conversely, the hard unknot gains no allocation reduction and maximum-radius-four probing is about 4.8 times slower than the complete baseline. The large raw-window gain on the four-strand torus example does not directly translate to a decision gain: degree zero agrees with the unknot and requires widening or fallback. These negative results justify keeping the policy optional.

### 17.5 Review boundaries for the adjacent reports

Report 13's finite nilpotent transfer and terminal patterned-ball routine are separate proposals; their passing tests do not supply the missing complete hierarchy construction. They are not substituted for a knot recognition backend here. Report 14's interval normalization, decision quotients, and scalar Fitting split were initially held for grading review. Section 18 records their subsequent optional integration, including the additional restriction of scalar basis changes to recovered quantum-shift blocks. These operations remain separate from the exact window scanner; an agreeing partial window may fall back to the new complete decision backend, but its shortened model is never used to answer an exact window query.

## 18 Interval normalization, continuation quotients, and graded splitting

Report 14 contributes a different compression mechanism: change bases to expose direct summands, then shorten certain entire summands when only a final unknot verdict is required. The production implementation now includes optional `barcode` and `fitting` recognition backends and exact-rank interfaces. We retain the standard backend as default. The scalar splitter is strengthened to respect recovered quantum shifts by default, allowing its representation to retain the homogeneous coefficient invariant from Section 16.

### 18.1 Exact decomposition of a pure component

Let $m$ be one matching on $2p$ boundary endpoints. Its endomorphism ring is

$$R_m = \mathbb{F}_2[x_1, \dots, x_p]/(x_1^2, \dots, x_p^2).$$

Every element $\theta$ with zero constant term satisfies $\theta^2 = 0$: squares of nonconstant monomials vanish, and cross terms cancel in characteristic two. Call a connected differential component pure if all its objects are $m$ and every nonzero entry is one fixed nonzero $\theta$ of this form. Singletons are also considered pure.

Write a pure component's degreewise maps as $\theta B_h$, where $B_h$ is binary. Crucially $B_{h+1}B_h$ need not vanish: it is the factor $\theta^2$ that makes the actual differential square to zero. The $B_h$ form a representation of a finite linearly oriented quiver, whose interval decomposition is a simultaneous scalar change of basis. Define

$$\rho(a, b) = \operatorname{rank}(B_{b-1} \cdots B_a), \qquad \rho(a, a) = \dim V_a.$$

The number of intervals supported on $[a, b]$ is

$$\rho(a, b) - \rho(a - 1, b) - \rho(a, b + 1) + \rho(a - 1, b + 1),$$

with unavailable ranks interpreted as zero. Indeed $\rho(a, b)$ counts intervals covering $[a, b]$, and this finite difference selects exactly the intervals born at $a$ and ending at $b$. Successive image-space bases compute these ranks without materializing all matrix products.

Reinserting $\theta$ gives a chain isomorphism to a direct sum of intervals $I_l(m, \theta)$, each with $l$ consecutive copies of $m$ and maps $\theta$. The implementation reconstructs intervals in degree order and then shares identical templates using the existing nonnegative multiplicity polynomials. This decomposition preserves full raw homology and every later additive continuation. It is applied to whole differential components, never to a submatrix that still has unaccounted attachments.

In exact mode, lengths and birth shifts are retained. In decision mode, birth shifts may be discarded and multiplicities saturated at three. These are two distinct interfaces. In particular, scalar coefficients remain in $\mathbb{F}_2$, whereas multiplicities are counts and must not be reduced modulo two.

## 18.2 Why length two is enough for the decision observation

Put $A = \mathbb{F}_2[\epsilon]/(\epsilon^2)$ and let $J_l$ have $l$ free rank-one terms with differential $\epsilon$. For an actual remaining crossing sequence and ordinary closure $F$, put $D = F(m)$ and let $\epsilon$ act by $F(\theta)$. This action commutes with the suffix differential and squares to zero. In the strict raw complex, additive planar gluing gives

$$F(I_l(m, \theta)) \cong J_l \otimes_A D.$$

This module structure, not linearity alone, justifies the replacement.

For a bounded complex $D$ of finite-dimensional $A$-modules, the finite dual-number derived classification gives

$$r_l(D) = \dim H(J_l \otimes_A D) = al + 2 \sum_{j \geq 1} b_j \min(l, j), \tag{6}$$

where $a, b_j$ are nonnegative and finitely supported. Here $a$ counts shifted simple modules and $b_j$ counts shifted free intervals $J_j$. Report 14 supplies an elementary proof: take a bounded-above degreewise finite free resolution and cancel its units downward. The remaining matrices are $\epsilon B_h$. In an exact far-left tail, $2r_h - \operatorname{rank} B_{h-1} - \operatorname{rank} B_h = 0$ forces constant rank and invertible $B_h$. Choose identity bases on that tail, cut it once, and interval-decompose the finite remainder. Intervals born after the cut stay finite; those born at the cut extend left and represent shifted simple modules. Thus only finitely many summands are needed.

For completeness, the tensor calculation behind (6) is

$$\dim H(J_l \otimes_A J_j) = 2 \min(l, j), \qquad \dim H(J_l \otimes_A \mathbb{F}_2) = l.$$

The first differential is $\epsilon(u + v)$ on $A \otimes \mathbb{F}_2[u, v]/(u^l, v^j)$. Its homology dimension is twice the cokernel dimension of multiplication by $u + v$, namely $2 \min(l, j)$. The second differential vanishes. Since $J_l$ is bounded and free, tensoring preserves the quasi-isomorphisms used above.

A nonzero radical endomorphism requires a nonempty matching. Every complete smoothing of an actual suffix glued to it contains at least one circle, so its unreduced chain-space dimension is even. Homology has the same parity, implying $r_1(D) = a + 2 \sum b_j$ is even and therefore $a$ is even. For $l \geq 2$, if $a > 0$ or any $b_j > 0$ with $j \geq 2$, then already $r_2(D) \geq 4$. Otherwise $r_l(D) = 2b_1$ is constant. Consequently

$$\min(3, r_l(D)) = \min(3, r_{\min(l,2)}(D)). \tag{7}$$

Length one cannot replace length two uniformly: close a two-arc matching to one circle and take $\theta = x_1 + x_2$. The two dots become the same circle operator and cancel, so the interval's closed rank is $2l$. For arbitrary algebraic suffixes without the evenness premise, the simple module gives $r_l = l$, and cutoff two fails. The optional algebraic cutoff three remains valid in that broader setting.

Shortening preserves an observation, not the homotopy type or the exact homology. Repeated replacements are sound because every later continuation is also a continuation before the replacement. Direct sums and copies preserve capped-rank equality, and saturation commutes with addition and multiplication of nonnegative counts. At final closure, elimination leaves isolated generators; their saturated total is exactly the original rank capped at three. Rank two certifies the unknot, and a capped value three means at least three, hence at least four for a classical knot. The code labels shortened-model object statistics accordingly and never exports their degree counts as exact knot homology.

## 18.3 Certified scalar splitting with recovered grading

Purity need not be visible in the current basis. The optional Fitting search looks for scalar endomorphisms $Q$ satisfying $Qd = dQ$, solving the binary equations coefficient by coefficient.

The archive permits variables between objects with equal matching and homological degree. We additionally require equality of recovered quantum shift by default.

For a nonzero homogeneous entry $u \to v$, let $d_{uv}$ be its dot degree and $c_{uv}$ the overlay-circle count. The required relation is

$$q_v - q_u = p - c_{uv} + 2d_{uv}.$$

Propagating these differences along the undirected differential graph recovers all $q_v$ up to one additive constant per component. The code checks homogeneity of every inspected entry and rejects inconsistent cycle sums. Scalar variables exist only within blocks $(m, h, q)$. Both solving and basis-change verification use these blocks. Cache keys include the relative shifts, so a cached ungraded search cannot silently stand in for a graded search or a geometrically different shift constraint.

For a candidate $Q$, take a power $Q^N$ with $N$ a power of two at least the largest block dimension. Then $\ker Q^N \oplus \operatorname{im} Q^N$ is the stable Fitting decomposition. Because $Q$ commutes with $d$, these spaces form subcomplexes. The code builds bases for both parts, checks invertibility, verifies the commutator independently, transforms every differential coefficient, and requires every entry across the proposed split to vanish. It accepts only nontrivial splits. Optional witnesses record the original and transformed matrices, matching and degree labels, relative quantum shifts, and the change of basis. A separate dense test verifier checks both directions of the basis transformation and rejects corrupted witnesses.

The search is deliberately incomplete: defaults limit each considered component to 48 objects, the scalar system to 1024 variables, accepted splits to 16 per crossing, and candidates to a bounded deterministic set. A local size or search limit leaves the exact component intact. Accepted splits are followed by interval normalization and sharing. Exhausting the global scanner budget gives `UNKNOWN`. The global time allowance includes order selection, and retained-object ceilings are checked before crossing allocation. The low-level `preserve_grading=False` option retains the archive's exact ungraded search for controlled comparisons; it does not justify the homogeneous bounds below.

For a connected pure component with homogeneous $\theta$ of dot degree $d$, $q_v - 2dh_v$ is constant throughout the component. Thus all objects in a given homological layer have the same recovered quantum shift, and its interval basis changes are automatically graded. Sharing identical connected templates preserves the existence of a graded lift up to componentwise shifts. Shortening keeps a graded interval representative, although it does not preserve the original bigraded homology. These facts justify using homogeneous coefficient counts after our restricted scalar splits; no claim to output the erased absolute quantum grading is made.

## 18.4 A sharper checkpoint bound

A pure checkpoint is a stage where, after the implemented splitting and normalization, every component is pure. Such a stage is recognizable; a small maximum gap between such stages is an additional family hypothesis. Let $w = 2p$ bound the frontier throughout the chosen order, and let $g$ bound the number of crossings between successive pure checkpoints, including the initial and final stages.

Without grading, there are $2^{2^p-1} - 1$ possible nonzero radical coefficients on a fixed matching. With our homogeneous invariant there are at most

$$S(p) = \sum_{d=1}^{p} \left(2^{\binom{p}{d}} - 1\right) \le p\, 2^{\binom{p}{\lfloor p/2 \rfloor}} \quad (p \ge 1), \qquad S(0) = 0.$$

At a checkpoint, a shortened interval type is determined by its matching and coefficient; singletons need no coefficient. With $M(w) = (w-1)!!$ for $w > 0$ and $M(0) = 1$, there are at most

$$M(w)(1 + S(p)) \text{ templates}, \qquad M(w)(1 + 2S(p)) \text{ stored objects}.$$

No common boundary disk is assumed. This bound is independent of the original connected component size, the number of hidden copies, and the original interval lengths. Exact mode retains lengths through $n + 1$ and pays only an additional polynomial factor in $n$.

One crossing makes at most eight delooped children per object. Hence an impure stretch costs at most a factor $8^g$ in stored objects. Gaussian elimination, interval ranks, the bounded scalar search, and coefficient operations have polynomial cost in this physical size and $2^{O(w)}$. Multiplicity polynomials have polynomially many degree positions and $O(n)$-bit coefficients. The resulting conservative bound is

$$T(n; w, g) \leq n^{O(1)} \exp\left(O\left(g + w \log(w + 1) + \binom{w/2}{\lfloor w/4 \rfloor}\right)\right). \tag{8}$$

This sharpens the report's $2^{w/2}$ term by incorporating the recovered grading. It gives a complete quasi-polynomial recognizer on families with constructible suitable orders for which the displayed exponent is $O(\log^2(n + 2))$. It does not prove that arbitrary knot diagrams have such orders or frequent pure checkpoints. A single gap $g = n$ leaves an exponential bound. A normalization size cap that skips a nontrivial component prevents that stage from counting as a pure checkpoint in the implemented statistics.

### 18.5 Validation and practical scope

All 267 production tests pass. The imported interval tests independently change quiver bases, reconstruct interval multiplicities, and compare dense suffix closures. The scalar tests replay witnesses, check corruption, and compare 80 seeded knot braids with exact full ranks and capped decisions; another 60 seeded braid cases exercise interval normalization. New production regressions cover grading constraints, inconsistent cycles, cache separation, order-selection deadlines, input validation, CLI incompatibilities, and recognition fallback after an agreeing window query. Recovered shifts also agree up to componentwise constants with independently propagated shadow labels on Conway, Kinoshita–Terasaka, and the hard unknot.

Report 14's independent continuation verifier imports no production code. It enumerates 861 finite complexes of dual-number modules, including nonfree modules and unit differentials, and passes 10 206 threshold comparisons through interval length six. This supports the implementation and finite algebra examples; the all-suffix statement rests on the proof. Its output and the full integrated test log are retained under `synthesis/data/`.

| Input | Interval exact | Fitting exact | Fitting decision | A/A |
|---|---|---|---|---|
| conway | 0.821 | 0.460 | 0.469 | 1.063 |
| kinoshita_terasaka | 0.841 | 0.487 | 0.481 | 0.992 |
| hard_unknot_8 | 0.819 | 0.777 | 0.795 | 1.028 |
| stress_braid5_36 | 0.747 | 0.776 | 0.766 | 0.988 |
| torus_3_5 | 0.901 | 0.889 | 0.871 | 1.003 |

Ratios are median paired full-scanner/new times; values below one are slower.

These are seven shuffled paired raw-backend rounds on CPython 3.13.14, including order selection and excluding imports and diagram construction. Every exact result is checked degree by degree against the full scanner; every decision result is checked against its rank capped at three. A/A compares identical full-scanner arms. Witness recording is performed separately, outside the timed arms. Earlier certificates are bypassed, so these measurements describe backend costs rather than end-to-end default recognition. Default front-end methods and all inputs are recorded.

The data include overhead and unsuccessful normalization attempts. None of these five actual diagrams produces an eligible nonsingleton pure component, so they demonstrate no real-diagram

interval-shortening gain. The synthetic ladder regression does exhibit the intended structural saving: 31 hidden copies of a seven-term interval require 217 objects under literal sharing, seven after exact interval normalization, and two after decision shortening. This is a graded algebraic fixture, not a claim that it appears in the knot corpus. Scalar splitting produces two verified splits on both Conway and Kinoshita–Terasaka, but their existence alone does not imply faster recognition. On this corpus the new modes are slower than the full baseline. The quantum restriction also costs time: the median ungraded/graded Fitting ratio is about 0.85 on those two diagrams and near one on the other three. The measured tradeoffs support keeping both backends optional.

Report 14's bounded crossing-order optimizer is a separate proposal and is not part of this integration. Its objective is a frontier proxy rather than elapsed scanner time. The general quasi-polynomial recognition goal still requires a structural theorem or additional machinery beyond these conditional compression bounds.

# 19 Verified rank-two braid preprocessing

The incoming archive `unknot_rank_two_kernels_20261007.zip` is now preserved as report 19. It contributes an exact braid-word compressor. It finds an optimal set of disjoint two-index subwords to replace by the empty word or a single Artin letter. We integrate the search and its independent replay verifier as an optional bounded preprocessing stage, after the inexpensive existing certificates and before the more expensive braid descent and diagram fallbacks. Unlike the closed-three-braid gateway, this procedure proves braid equality inside the surrounding word.

## 19.1 Faithful local states without copying prefixes

For adjacent indices write $a = \sigma_i$, $b = \sigma_{i+1}$. The local group is

$$B_3 = \langle a, b \mid aba = bab \rangle = \langle x, y \mid x^2 = y^3 \rangle, \qquad x = aba, \quad y = ab.$$

The common element $z = x^2 = y^3$ is central and has Artin exponent sum six. The quotient by $z$ is $C_2 * C_3$. Thus a local braid is uniquely determined by its reduced quotient word together with its exponent sum: equal quotients differ by $z^k$, and equal exponents force $6k = 0$.

The quotient normal form must not be cyclically reduced. The conjugate letters $a$ and $b$ are not equal inside an arbitrary context. Nor may the central exponent guard be omitted: $(aba)^2$ has empty quotient word but is nontrivial. For distant indices the local subgroup is $\mathbb{Z}^2$, so two exponent counts suffice. A single index uses one count.

In the quotient, the four signed letters have images $y^2x, xy, xy^2, yx$. Each appends two tokens to an alternating normal-form stack. The search stores these stacks persistently in a trie with three child slots per node. Push creates or reuses a child, cancellation follows a parent pointer, and combination of order-three tokens uses a child of the parent. Node identity is an exact canonical word identity; no hash fingerprint substitutes for group equality. The prefix tables themselves use deterministic AVL trees in the integrated recognition stage.

## 19.2 An optimal pass over bounded-overlap intervals

Compress maximal runs of a common absolute generator index. Label each edge between consecutive runs by its unordered pair of indices. A maximal string of equal edge labels specifies a maximal two-index interval, called a corridor in the source. Consecutive corridors overlap in exactly one run; nonneighbors are disjoint. Every permitted subinterval lies in a corridor, each letter belongs to at most two, and the sum of their lengths is at most $2n$.

Let $F[j]$ be the maximum number of letters removable from the first $j$ letters by disjoint permitted replacements. Inside one active corridor, write $p_j$ for its local prefix product. Maintain an exact table

$$H(q) = \max_{i:p_i=q}(F[i] - i),$$

with an attaining start position. For a target $v$ of length zero or one,

$$w[i:j] = v \quad \Longleftrightarrow \quad p_i = p_j v^{-1}.$$

Therefore the next optimum is

$$F[j] = \max\left(F[j-1], \max_{\text{active corridors, } v}\{j - |v| + H(p_j v^{-1})\}\right).$$

Queries precede insertion of the current endpoint. Backtracking the attaining starts recovers disjoint replacements. The last replacement of any optimal prefix plan is represented in this recurrence, proving optimality by induction. This is optimality for one pass and this explicit rule set, not global minimality among braid presentations or Markov moves.

There are at most five targets per corridor and at most two active corridors. Each adjacent-group update takes two trie-token operations. There are $O(n)$ exact states, queries, and backpointers. AVL table access costs $O(\log(n+2))$ comparisons, giving

$$O(n\log(n+2)) \text{ word-RAM time}, \qquad O(n) \text{ words}.$$

State identifiers, exponent counts, and indices have $O(\log(n+s+2))$ bits for $s$ strands, giving the conservative bit bound $O(n\log^2(n+s+2))$. The low-level package also retains an optional hash backend whose dictionary-time bound is expected rather than deterministic. Recognition uses one AVL pass. Repeated passes remain available through the low-level API, but their conservative total bound is quadratic up to logarithmic factors.

### 19.3 Independent replay and evidence association

The verifier does not import the optimizer or its quotient trie. It uses a central normal form $z^k f_1 \cdots f_r$ with alternating $x, y, y^2$ factors. Each signed Artin letter expands into $z^{-1}$ times the same two positive factors used above; reductions $xx = z$ and $y^3 = z$ update the central integer explicitly. A replacement is accepted only if its source and target differ by the identity in this form, or by equal exponent counts in the commuting case.

Stable linked-list node identifiers describe interval endpoints across passes. Inserted target letters get fresh identifiers. Replay checks that endpoints remain live and correctly ordered, that the interval uses at most two indices, that the target is permitted, that equality holds, and that the word strictly shortens. The final output is reconstructed and compared exactly with the certificate. Its input digest binds the transcript to the supplied word; it is not an authenticity signature.

Braid equality preserves exponent parity, so each successful replacement saves at least two letters. Across all passes there are at most $n/2$ insertions and $3n/2$ removed nodes. Thus valid-certificate replay takes linear word-RAM work and memory, even when search used multiple passes. The verifier establishes soundness of the replacements; it does not certify the optimizer's separate optimality or saturation claims.

The pipeline requires checked source-braid provenance. It compresses that source, independently replays the transcript, and constructs a new validated PD only if the word is shorter than the current simplified diagram. An accepted replacement restarts recognition on the new source with compression disabled, preserving all other options and the remaining global time. Earlier PD reductions and later reductions lie on distinct branches of the evidence, named `before_ranktwo` and `after_ranktwo`; the source word and braid-equality certificate connect the original knot to

the restarted input. They must not be concatenated as one PD move trace. The existing factor and twist provenance rules then apply to the restarted recognition without modification.

The stage is enabled by `use_ranktwo=True` or `-ranktwo`. It has one shared local allowance for search and replay, defaulting to 0.1 seconds, configurable through `ranktwo_seconds` or `-ranktwo-seconds`. Budgets are cooperative. Local time or allocation failure skips the probe and continues the existing complete route; global exhaustion returns `UNKNOWN`. A failed certificate is an error, never a knot verdict. Already decisive cheap certificates retain priority. Inputs without checked braid provenance bypass this stage.

## 19.4 A constructive quasi-polynomial input class and its barrier

Let $\kappa_1(w)$ be the smallest output length under the one-pass rules. The optimizer finds this number without receiving a decomposition. A *flat rank-two inflation* of a $k$-letter core replaces disjoint letters by equal rank-two words and inserts disjoint rank-two identity blocks. Those blocks themselves define an admissible plan, so $\kappa_1(w) \le k$. Following the verified pass by an uncapped exact exponential homology computation takes

$$O(n \log^2(n+s+2)) + \mathrm{poly}(\kappa_1(w)) 2^{O(\kappa_1(w))}.$$

For a knot closure the reduced strand count satisfies $s \le \kappa_1(w) + 1$: a word on $s$ strands whose permutation is one cycle needs at least $s-1$ adjacent transpositions. This removes a hidden strand factor. Flat inflations of $O(\log^2 n)$-letter cores consequently admit quasi-polynomial recognition without a supplied inflation decomposition. The statement concerns the uncapped route, including `ranktwo_seconds=None` in the integrated API. A local cutoff can miss the compression and invoke an exponential fallback on the larger input even when that fallback eventually completes; a finite global budget can return `UNKNOWN`. The integrated earlier simplifications have polynomial overhead and do not invalidate this conditional bound.

No universal small-core theorem follows. An explicit obstruction is

$$U = abcaba, \quad V = cbacbc, \quad q_h = (UV^{-1})^h abc \quad \text{in } B_4.$$

The positive words $U, V$ are equal half-twists, so $q_h$ closes to the unknot and has length $12h+3$. Within a uniform-sign block, exponent sum proves every subword geodesic. Any two-index subword crossing a sign boundary is contained in $ac^{-1}$ or $c^{-1}a$, flanked by index $b$; its commuting exponent vector has norm two, proving geodesicity there as well. Thus no strictly shortening rank-two substitution of any target length can begin on this family. Escaping it requires, for example, length-preserving relations, a larger support, or separately certified closed-knot operations. Increasing the number of identical shortening passes cannot solve that obstruction.

## 19.5 Production validation and end-to-end measurements

The archive's 33 tests pass locally. Its previously unrun real-upstream smoke test now passes all 16 sleeve cases against the current braid gateway: the gateway alone is inconclusive before compression and certifies the unknot after replay. This gateway result does not by itself establish an end-to-end speedup; the current diagram simplifier may already decide them.

All 272 integrated tests pass. New tests compare 200 random words with an independent exhaustive interval oracle, both dictionaries, both target radii, and the Artin action on a free group. Sixty knot inputs compare full normalized homological profiles before and after shortening, checking $d^2 = 0$. Other regressions cover central and conjugacy guards, certificate corruption, the higher-rank barrier, original input counts, evidence scopes, local and global budgets, absent provenance, preservation of cheap verdicts, and restart into the standard, shared, twist, barcode, and Fitting backends. Logs and the actual gateway smoke-test output are retained in `synthesis/data/`.

| Input | Disabled/enabled | A/A |
|---|---|---|
| sleeve_1 | 3.397 | 1.019 |
| sleeve_16 | 12.471 | 1.007 |
| sleeve_64 | 33.777 | 1.005 |
| barrier_8 | 0.994 | 1.011 |
| cheap_braid | 0.979 | 1.014 |
| stress_braid5_36 | 0.559 | 0.969 |
| conway | 1.012 | 1.014 |
| kinoshita_terasaka | 1.021 | 1.027 |

Ratios are median paired end-to-end times; values above one are faster.

The benchmark uses seven shuffled paired rounds with an identical A/A control. It includes PD construction, all default certificates and simplifications, compression, replay, and the complete fallback; imports are excluded. It uses a five-second global budget and the default 0.1-second compression budget. The source inputs, verdicts, methods, transcripts, and timings are recorded. Separate traced-allocation runs record both their peak and the path actually taken, since instrumentation can affect a time-limited branch. These are measurements of this machine, not asymptotic guarantees or a broadly sampled knot distribution.

The sleeve cases with parameters $1, 16, 64$ improve by approximately 3.4, 12.5, and 33.8 times respectively. Their baseline already certifies the unknot through diagram reduction; compression reaches the short source braid and its genus-zero certificate sooner. The stress five-braid instead runs about 1.8 times slower: its probe cannot supply a shorter input than the current simplified diagram, and Alexander still provides the verdict. The barrier fixture is settled by the descending diagram certificate before compression, and the positive braid by Bennequin's obstruction; neither timing measures a compressor success. The two PD-only named knots bypass the stage. These contrasting outcomes support keeping the new stage optional.

In the separate traced-allocation run, the largest sleeve baseline hit the five-second global deadline. Its recorded peak is censored and must not be compared as a peak of a completed recognition. All ordinary timed arms completed with agreeing verdicts.

# 20 Minimal extremal windows with certified binomial bounds

The incoming archive `ProveIt_Unknot_Extremal_Windows_2026-10-07.zip` is now preserved as report 20. It adds a stronger size analysis for an alternative window schedule. The schedule is now available in recognition through the `-window-strategy` option with value `minimal`, and through `window -minimal` for exact partial homology. The existing allocation-pruned strategy remains the default. Both compute exact requested ranks, but they construct different guard complexes and have different size guarantees.

## 20.1 Cancellation and truncation do not commute

Let $K = k + 1$ when the desired raw degrees are $0, \dots, k$. The new strategy extends the whole retained complex by one crossing, deloops, exhaustively cancels units, and only then discards degrees above $K$. In particular it constructs temporary degree $K + 1 = k + 2$. The contractible complex $X \xrightarrow{1} X$ in degrees $K, K + 1$ explains the order: cancellation removes it, while early truncation leaves a nonzero stalk in degree $K$. The latter is artificial guard homology.

This does not contradict the exactness of the earlier support-window algorithm. That algorithm returns degrees below the guard and has a separate bounded-suffix support proof. But that proof does not establish the binomial size bound derived here. The two implementations are kept

explicitly separate; the new strategy never suppresses temporary objects before cancellation. Its object ceiling therefore includes their allocation.

## 20.2 Minimality controls all exhaustive pivot choices

In the finite dotted matching category over $\mathbb{F}_2$, let $J$ consist of all morphisms between distinct matchings and all equal-matching morphisms with zero constant coefficient. This is the nilpotent radical; the semisimple quotient has one scalar factor per matching. The positive weight $m - c(a, b) + 2|S|$ from the radical analysis gives a finite nilpotence bound. An entry outside $J$ is precisely a unit between equal matchings. Exhaustive scalar-unit cancellation therefore leaves a minimal complex, meaning one whose differential lies in $J$.

Every bounded finite complex splits, by the invertible changes of basis from Gaussian cancellation, into a minimal complex and contractible disks. If minimal complexes $M, N$ are homotopy equivalent, their homotopy inverse equations reduce modulo $J$ to degreewise inverse maps. Lift these inverses using the finite geometric series in the nilpotent matrix ideal. The chain maps are thus degreewise invertible and give a chain isomorphism. Consequently the minimal matching multiplicity in every homological degree is unique, although matrix entries and computation times can depend on the chosen pivots.

Write $\tau_K$ for brutal upper truncation. Consider additive extension functors $F_t$ that preserve chain homotopies and have degree amplitude $[0, 1]$. Starting from a common initial complex, compare

$$A_t = \tau_K \operatorname{Min}(F_t A_{t-1}), \qquad B_t = \tau_K R_t(F_t B_{t-1}),$$

where $R_t$ performs any collection of valid unit cancellations. Then

$$B_t \simeq A_t \oplus E_t[K] \tag{9}$$

for a stalk object $E_t$ in degree $K$. In particular $A_t$ has no larger matching multiplicities than $B_t$ in any degree.

To prove this, extend the inductive decomposition. Its extra stalk extends only into degrees $K, K + 1$. Uniqueness of minimal complexes identifies the minimal part of the extension with the sum of the desired minimal extension and a minimal complex supported in those two degrees. The partially reduced comparison complex is *chain-isomorphic* to that sum plus contractible disks. Truncate this chain isomorphism: disks below $K$ stay contractible, disks above $K$ disappear, and disks crossing the cutoff become stalks in $K$. The extra minimal two-degree piece also becomes a stalk. This proves (9). It never assumes that brutal truncation preserves an arbitrary homotopy equivalence.

## 20.3 The geometric certificate and inherited object count

The external ingredient is Kelomäki–Schütz, Lemma 5.5, checked directly against the primary paper during this review. For their comparison scan on a nice disc order, its degree-$j$ object count after $t$ crossings is bounded by $\binom{t}{j}$ before final closure and by $2\binom{n}{j}$ at closure. Reducing its coefficients modulo two preserves the allowed cancellations. Applying (9) transfers these object bounds to our exhaustive cancellation policy.

The production certificate verifies an actual crossing permutation, absence of projection loop edges, connected processed and remaining vertex sets, consecutive attaching slots at every intermediate crossing, and the unique final four-slot closure. For a validated plane projection, a cut with connected sides is a bond; its dual cycle supplies a separating curve and a disc presentation. Consecutive attachments are checked explicitly. The certificate records the measured maximum boundary size $W$, not an estimate of the best possible girth.

When no order is supplied, the implementation first tries the ordinary fast greedy order. If its certificate fails, a polynomial search attempts a connected-prefix, connected-suffix order on the loopless two-connected projection. It repeatedly chooses a vertex attached to the prefix whose removal leaves the suffix connected, breaking ties by the next cut size. The block decomposition of the connected suffix guarantees a permissible extension under the stated two-connectedness hypothesis. Direct candidate connectivity checks give a conservative $O(n^3)$ construction cost. The resulting order is reverified. If construction fails, the original order remains valid for exact computation and only the stronger complexity certificate is withheld. Supplied orders are verified without replacement. All planning and verification is charged to the query deadline.

At each certified stage the implementation checks the actual retained degree counts against the binomial bound. This is a diagnostic consequence of the proof, not a substitute for the geometric hypothesis. Tests also compare matching-by-degree profiles across standard, residue, and adaptive exhaustive cancellation. Optional dense coefficient operations preserve the same exact category and can be used as well.

## 20.4 A uniform bound without the all-state-circle factor

Put

$$B(n,K)=\sum_{j=0}^{\min(n,K)}\binom{n}{j}.$$

Retained population is at most $2B(n,K)$. One crossing creates at most eight delooped children per object, so even the fully allocated temporary extension has population $O(B(n,K))$. There are quadratically many possible entries and linearly many pivots; a pivot changes quadratically many entries. Each coefficient computation costs $\mathrm{poly}(n,W)2^{O(W)}$ bit operations. Thus the new strategy has the bounds

$$T\le \mathrm{poly}(n,W)2^{O(W)}B(n,k+1)^3,\qquad S\le \mathrm{poly}(n,W)2^{O(W)}B(n,k+1)^2.$$

The implementation disables the cross-stage shape cache for this strategy and caps its per-stage coefficient-composition memo at 4096 entries. This avoids silently charging cubic memo storage as quadratic matrix storage. Disc matching and plan metadata has $2^{O(W)}$ possible types per stage; accumulated stage labels add polynomial factors.

For $k=O(\log(n+2))$ and $W=O(\log^2(n+2))$, these bounds are quasi-polynomial. Unlike the previous elementary support-window estimate, this bound has no all-zero smoothing circle factor. It requires a certified nice order and this particular cancellation schedule. It is not asserted for early-allocation pruning or arbitrary uncertified frontiers.

The API `khovanov_minimal_window_auto` accepts an arbitrary raw interval $[a,b]$. On one side it computes through upper degree $b$ and returns only the requested interval. Mirroring instead computes through $n-a$ and maps degrees back by $h\mapsto n-h$. The default mirror choice minimizes this raw upper depth; it does not estimate wall time. The complexity parameter is the selected upper depth, not the width $b-a$. The `binomial_bound_certified` field records whether the nice-order premise was established. A failed certificate does not weaken the exactness of the returned queried ranks.

Recognition can use this strategy inside the existing bounded widening portfolio. Its same normalized-band comparison, local allowance, global limits, restart rule, and complete fallback remain in force. Agreement with the unknot on a partial band is still inconclusive. Widening restarts; it does not attempt to reconstruct discarded chain groups. No new default policy or general quasi-polynomial verdict guarantee is introduced.

## 20.5 Raw endpoint depth has an exact padding obstruction

Insert $s$ cancelling positive-negative crossing pairs into a diagram. Its knot type and normalized Khovanov homology do not change, while its crossing count rises by $2s$ and its negative PD crossing count by $s$. Therefore the raw profile obeys

$$\beta_{D_s}(r) = \beta_D(r - s).$$

Every discrepant degree relative to the unknot profile shifts by $s$, and its distance from either raw endpoint increases by $s$. Such padding can be inserted with constant additional frontier width. Consequently bounded width alone does not force a logarithmic detecting endpoint depth. Ordinary diagram simplification removes this particular padding, so this is an obstruction to an unqualified raw-window theorem, not a lower bound for the whole preprocessing pipeline. A theorem covering all reduced inputs would need additional information. Even shallow agreement on an unknot needs a positive certificate or eventual full coverage.

## 20.6 Validation and comparison with the current window engine

The integrated suite passes 277 tests. New checks compare 100 random knot braids and arbitrary intervals with complete production homology, using three exhaustive reducers and forced dense composition on the first 15 inputs. They check intermediate $d^2 = 0$, certified binomial inequalities, mirror labels, invalid orders, padding shifts, time and allocation limits, and the guard-disk counterexample. The CLI and recognition portfolio are tested separately, including an agreeing partial query followed by exact fallback. The actual stage profiles are also compared across reducers.

The archive's 14 tests pass locally. Its separately rerun independent cube audit compares 393 windows on 100 diagrams, checks 1772 certified stages and 15 padding cases, and reports no failures. Its archived adapter is not applied: the production implementation here targets the current array-based `FastScan`, with its existing coefficient engines and adaptive cancellation. Source archive files remain unchanged; review runs use a scratch extraction. Local logs are retained under `synthesis/data/`.

| Input | Full/support | Full/minimal | Support/minimal | A/A |
|---|---:|---:|---:|---:|
| weaving_14 | 4.52 | 4.09 | 0.895 | 1.002 |
| torus_3_4 | 1.67 | 1.29 | 0.777 | 0.996 |
| padded_trefoil | 1.74 | 1.20 | 0.708 | 1.009 |
| conway | 1.63 | 1.35 | 0.834 | 0.994 |
| kinoshita_terasaka | 1.72 | 1.43 | 0.831 | 1.017 |
| stress_braid5_36 | 425.47 | 323.49 | 0.792 | 0.999 |

Ratios are medians of paired times; larger than one favors the denominator.

These seven-round paired measurements use common supplied crossing orders and depth-two raw queries. Full homology and a window have different output contracts; both window strategies answer exactly the same interval. A/A compares identical full-homology calls. Order construction and imports are excluded, while the minimal strategy's certificate verification is included. These are backend query measurements, not default recognition speedups. In particular, a supplied nice order can differ from the ordinary recognizer's preferred order.

The new strategy is slower than allocation pruning on all six cases. It sometimes saves retained guard objects nevertheless: on the 14-crossing weaving case, peak pre-cancellation allocation is 492 rather than 516, versus 5046 for full homology in that order. On Conway it instead allocates 84 objects versus 42 for the support strategy, because it constructs the temporary degree needed for

cancellation. Its practical tradeoff is an additional proved size guarantee under explicit hypotheses, not a measured replacement for the faster default on this corpus.

# 21 Exact Potts filters and independent component tensors

Report 17 contributes an optional scalar obstruction before the complete homology fallback. The integrated selectors are `potts5`, `potts-exact`, and `potts-exact-factorized`; the existing `matching` Jones backend remains the default. These selectors are independent of the Khovanov backend and of adaptive cancellation. Potts representations of Jones evaluations and width-sensitive contraction are established techniques, including the tensor implementation of Meichanetzidis–Kourtis [40] and Maria's parameterized analysis of quantum invariants [41]. The contribution here is the audited implementation, exact coefficient representation, component factoring, and explicit resource and fallback contracts.

## 21.1 Normalization without approximate algebraic numbers

Let $D$ be a nonempty validated plane knot diagram, $G$ either checkerboard Tait multigraph, $v$ its number of vertices, and $r$ the diagram writhe. Loops and parallel edges are retained. At crossing $e$, let $\epsilon_e \in \{1, -1\}$ be the bracket exponent of the smoothing that omits its Tait edge; put $s = \sum_e \epsilon_e$. A subset $S$ of included edges produces

$$c(S) = 2k(S) - v + |S|$$

smoothing circles. Indeed a regular neighborhood of the spanning plane subgraph has Euler characteristic $v - |S| = 2k(S) - c(S)$, including its isolated vertices. With $\delta = -A^2 - A^{-2}$ and the full bracket of one circle equal to $\delta$, expansion of the smoothing sum gives

$$B_D(A) = \delta^{-v} A^s \sum_{S \subseteq E} (\delta^2)^{k(S)} \prod_{e \in S} \delta A^{-2\epsilon_e}.$$

At the integer specialization $q = \delta^2$, expanding a spin product shows that the sum is a partition function: each selected edge requires equal endpoint colors, so the assignments constant on $(V, S)$ number $q^{k(S)}$. Writing $x = A^4$ gives

$$q = x + 2 + x^{-1}, \qquad Z_G = \sum_{\sigma : V \to [q]} \prod_{e=uv} \begin{cases} -x^{-\epsilon_e}, & \sigma(u) = \sigma(v), \\ 1, & \sigma(u) \neq \sigma(v). \end{cases}$$

The normalized Jones polynomial is $V_D(A^{-4}) = B_D(A)/(\delta(-A^3)^r)$. Substitution of $\delta = -(x+1)A^{-2}$ yields

$$V_D(x^{-1}) = (-1)^{r+v+1} x^h \frac{Z_G}{(x+1)^{v+1}}, \qquad h = \frac{s - 3r + 2(v+1)}{4}. \tag{10}$$

For a knot, $h$ is an integer. The residue modulo four of $s_\sigma - 3r + 2(c_\sigma - 1)$ is unchanged by a smoothing flip. In the Seifert state $s_\sigma = r$ and $c_\sigma - 1 \equiv r \pmod 2$; in the all-omitted state $c_\sigma = v$, giving the assertion. Thus the exact obstruction compares $Z_G$ with

$$Z_U = (-1)^{r+v+1} x^{-h} (x+1)^{v+1}.$$

No fourth root or division by $x + 1$ is computed. The crossing-free knot is handled separately with $Z_G = Z_U = q$.

For every integer $q \geq 5$, work in $R_q = \mathbb{Z}[x]/(x^2 - (q-2)x + 1)$. The discriminant lies strictly between $(q-3)^2$ and $(q-2)^2$, so this polynomial is irreducible and each element has a unique integer pair $(a, b)$ representing $a + bx$. Here $x^{-1} = q - 2 - x$ and

$$(a, b)(c, d) = (ac - bd,\ ad + bc + (q-2)bd).$$

Multiplication by an equal-spin edge uses just additions and multiplication by $q - 2$:

$$-x^{-1}(a, b) = (-(q-2)a - b, a), \qquad -x(a, b) = (b, -a - (q-2)b).$$

Partition inequality therefore certifies a nontrivial Jones value exactly. Partition equality is inconclusive. The exact implementations default to $q = 6$, whose roots are $2 \pm \sqrt{3}$; no numerical embedding is chosen. Witness pairs are serialized as signed hexadecimal strings, avoiding Python's limit on conversion of very large integers to decimal strings.

## 21.2 A half-frontier bound for every supplied crossing order

At a cut in the crossing order, let $w_i$ count shadow edges joining a processed crossing to an unprocessed one. A Tait vertex is active when it is incident to both processed and unprocessed edges; write $f_i$ for their number. Traverse the cyclic boundary walk of an active selected-color face. The processed/unprocessed marks change at least twice. Each change traverses a cut shadow edge. A cut edge has just one side of the selected color and can be charged to just that face. Hence

$$2f_i \leq w_i. \tag{11}$$

This argument allows repeated vertices and edges in face walks, and does not require a common-disk ordering or a Catalan matching hypothesis.

The single-table algorithm introduces a spin at its first incident edge, applies the edge weight, and forgets spins at their last incident edge. Its key records the equality pattern of active colors, canonically naming classes in first-occurrence order. Coefficients aggregate *all* labeled assignments in an orbit. An introduced spin can join any existing class with multiplicity one, or create a new class with multiplicity $q - k$ when $k$ classes already occur. Restriction and recanonicalization implement forgetting. There are at most $q^{f_i}$ keys; an edge introduces at most two extra vertices. At fixed $q$, the work is

$$\mathrm{poly}(n, w)\, q^{w/2}, \qquad w = \max_i w_i,$$

including polynomial bit arithmetic. To justify the latter, each coefficient after $i$ edges and $I$ introduced vertices is a sum of at most $q^I$ assignments with pair norm bounded by $(q-1)^i$. Therefore its bit length is $O(n \log q)$, since $v \leq n + 1$. The normalization pair also has that bit bound. This is a scalar-query bound, not a bound on full homology.

## 21.3 Independent processed components and exact orbit division

The factored implementation keeps one tensor for each connected component of processed Tait edges. A vertex forgotten from the frontier does not erase interactions through it. Consequently this decomposition follows processed connectivity, not connectivity of just the active vertices. Relative color equalities between two components become relevant only when an edge joins them. A closed component contributes a scalar factor.

For a pattern using $k$ colors, write $(q)_k = q(q-1)\cdots(q-k+1)$. Its aggregate coefficient $C$ is $(q)_k$ times its per-assignment value, because simultaneous color permutation preserves the partial partition sum. When joining patterns with $k$ and $\ell$ classes, enumerate partial bijections between their classes. An alignment with $K \leq q$ total classes has pre-edge aggregate coefficient

$$\frac{C_L}{(q)_k}\frac{C_R}{(q)_\ell}(q)_K. \tag{12}$$

The integer coordinates of each division must have zero remainder; the code checks this before multiplying pairs. The operation is exact integer division, not division in a finite field or a numerical approximation. Each combined equality pattern determines its two restrictions and alignment uniquely, proving that the merge neither loses nor duplicates assignments.

There are at most $q!$ relative alignments per pair of keys. Let $g$ be the largest number of active vertices in any one processed component after an edge. Only the current edge's endpoints can be forgotten at that step, so a merged component has at most $g+2$ vertices before forgetting. Thus the Cartesian product of its two input key sets is bounded by $q^{g+2}$, not merely by the weaker $q^{2g}$. The resulting bound is

$$\operatorname{poly}(n, g)\, q!\, q^{g+2}$$

coefficient and key work, with $O(nq^g)$ logical keys and polynomial coefficient lengths. Here $g \le \max_i f_i \le w/2$. The extra alignment work can dominate when factoring saves few states.

## 21.4 A specialization collision that changing primes cannot repair

Report 17 analyzes $W_m = \widehat{(\sigma_1\sigma_2^{-1})^m}$. It is a knot exactly when $3 \nmid m$. Its three-braid Jones trace identity gives, at $t + t^{-1} = q - 2$,

$$V_{W_m}(t) = q - 2 + T_m, \quad T_0 = 2, \quad T_1 = 3 - q, \quad T_{m+1} = (3-q)T_m - T_{m-1}.$$

At $q = 5$ this is $3 + 2(-1)^m$, equal to one for every odd $m$. In particular the nontrivial members with odd $m \ge 5$ are missed even in characteristic zero. Changing primes or using the conjugate root preserves this equality. For $q \ge 6$, set $a = q - 3 \ge 3$ and $U_m = (-1)^m T_m$. Then $U_0 = 2$, $U_1 = a$, $U_2 = a^2 - 2 > a$, and the recurrence proves strict increase thereafter. Thus $a + 1 + (-1)^m U_m \ne 1$ for every $m \ge 2$. For example, $V_{W_5} = -119$ at $q = 6$. This proves detection for this family only. The existing short-braid stage already decides these inputs when their source braid is available.

## 21.5 Integration, budgets, and validation

The optional filter runs in the same place as the existing matching Jones filter and reuses its crossing order. A mismatch returns `KNOTTED`; equality or local state/transition exhaustion continues the existing Alexander, simplification, window, and complete homology stages. The global cooperative deadline still returns `UNKNOWN`. `-jones-backend` selects the filter in recognition, while `jones -backend` selects a standalone query. The standalone query now exits 3 with an inconclusive result when its resource limit is reached: a nonempty error dictionary is never interpreted as an obstruction. Invalid options exit 2.

State limits count represented keys. Inputs and a replacement table can coexist during a merge, and normalized lists and alignment caches consume additional memory, so this is not a resident-memory limit. Transition limits count actual orbit updates, including relative alignments. At fixed $q \le 6$ the alignment cache has finitely many bounded entries; larger color counts use lazy alignment generation. The factored wrapper additionally checks the global deadline after final normalization. The archived source remains unchanged.

The archived suite passed 172 tests; the first integrated checkpoint passed 315 tests. The retained independent full-smoothing oracle constructs Laurent Jones polynomials without Tait graphs or production ring multiplication. It checked 248 diagrams and 30,116 cube states, followed by 4,464 comparisons per exact backend over three color counts, two shadings, and three orders. Another 45 weaving checks reached 98 crossings. These are correctness observations, not complexity proofs. The report's certificate replayer, which recomputes the same backend, is not presented as an independent oracle. The ring-valued Fitting proposals in report 17 are not integrated by this change.

### 21.6 Measurements and the quasi-polynomial objective

The reproducible driver `fast/benchmark_potts.py` measures seven shuffled paired rounds with an exact-$q = 6$ A/A control. Kernel timings exclude construction of their common supplied order; the fragmented-order case is deliberately adverse. A separate normal-pipeline comparison includes diagram construction, ordering, cheap certificates, and fallbacks. Matching and modular $q = 5$ arms evaluate different specializations, so their timing ratios alone do not establish equivalent detection power. Detailed measurements are recorded in `fast/results/potts_20261008.json`.

| Input | Exact keys | Factored keys | Exact/factored | Pipeline gain |
|---|---|---|---|---|
| Weaving, 10 crossings | 5 | 6 | 0.799 | bypassed |
| Weaving, 98 crossings | 5 | 6 | 0.934 | bypassed |
| Fragmented order, 16 crossings | 4111 | 205 | 23.612 | bypassed |
| Conway | 5 | 6 | 0.803 | 1.143 |
| Kinoshita–Terasaka | 5 | 6 | 0.734 | 1.152 |
| Stress braid, 36 crossings | 15 | 16 | 0.787 | bypassed |
| Hard unknot, 8 crossings | 2 | 3 | 0.749 | bypassed |

Here "Exact/factored" is the median paired kernel-time ratio; a value above one favors factoring. "Pipeline gain" compares default matching recognition with the single-table exact-$q = 6$ option, including construction and all earlier stages. On Conway and Kinoshita–Terasaka, every compared pipeline reached its Jones filter and returned `KNOTTED`. The factored option's corresponding pipeline ratios were only 1.069 and 1.053. The other five cases were decided earlier, so their pipeline timings measure noise and common stages rather than a benefit from Potts evaluation.

Factoring reduced transitions from 30,197 to 1,558 on the fixed fragmented order but was slower in all six other kernel cases. The exact A/A ratios ranged from 0.973 to 1.159, making small timing changes unsuitable for a broad speedup claim. The robust large saving on the adverse order supports keeping the factored implementation available; these data do not support making it the default. Nor does the modest normal-pipeline improvement on two fixtures establish superiority of the six-color specialization on unseen diagrams.

At fixed $q$, a supplied order with $w = O(\log^2 n)$ makes this filter quasi-polynomial. The stronger component parameter $g$ can make a poor global frontier affordable. Neither guarantees a suitable order for every diagram, nor that the scalar differs from the unknot value. The fallback therefore retains its exponential worst case. This integration adds options for a future adaptive scalar-query policy; it does not change the existing progress-based adaptive Khovanov cancellation policy or promote an unmeasured automatic Potts switch.

## 22 Disk-certified transfer after an adaptive pause

Report 21 supplies the geometric certificate needed to apply a finite radical transfer to actual scan prefixes. The integrated option `-reduction disk-adaptive` extends the existing adaptive policy: ordinary sparse cancellation first, the cheap binary survivor test second, and a bounded full-transfer attempt only if adjacent surviving degrees prevent that shortcut. Successful transfer retains the radical differential between those degrees. Failure of the geometric or local-work gate resumes the same partially reduced complex. The default remains `standard`. This is an implementation of another exact cancellation strategy within the local-complex framework of Bar-Natan [36], not a new recognition criterion or a general quasi-polynomial algorithm.

## 22.1 Certifying the boundary from the processed rotation system

For a processed prefix $P$, pair the darts of every internal edge and fix each frontier dart. Call this permutation $\alpha$; let $\sigma$ rotate the four darts at each crossing. The cycles of $\sigma\alpha$ are boundary walks of the ribbon neighborhood of $P$. The implementation checks that this prefix is connected, that

$$|V(P)| - |E_{\text{internal}}(P)| + \#\text{boundary walks} = 2,$$

and that all frontier darts occur on one boundary walk. These conditions certify a genus-zero neighborhood with one marked boundary component. Filling its unmarked holes gives a disk. In a validated connected full diagram, no unprocessed crossing component can hide in such a hole: a connection to the rest would produce a frontier dart on that boundary.

Reading the marked boundary gives the common cyclic endpoint order. Every smoothing is embedded in this disk, so its matching is noncrossing in that order. Delooping closed circles leaves endpoint pairings unchanged. The code also checks all distinct live matching types against the order, using the usual stack test for nested arcs. Numeric edge labels do not themselves specify an order. In particular the three pairings of four points cannot all be noncrossing in one cyclic order; applying a disk-algebra theorem to them without this check would be invalid.

The certificate records the boundary walks, vertex and internal-edge counts, cyclic order, and a digest of the actual prefix. The digest binds data but does not authenticate a source. Replay recomputes all the fields. Certificate construction is linear in indexed prefix darts; checking every prefix independently costs $O(n^2)$ such graph operations. The adapter records the actual supplied scan order and verifies its current frontier against the certificate. It does not replace a good greedy order automatically.

The report also supplies a polynomial ear-insertion construction of connected prefix-and-suffix orders for biconnected crossing graphs. Such a cut admits the disk certificate, but the construction need not preserve a small width. The helper is retained for experiments and independently checked on smoothing states; no claim of a width approximation or a universally good order follows.

## 22.2 Retaining the maps that the binary profile cannot determine

For a frontier of $2b$ points, let $J$ be the positive-weight radical in its disk matching category. A dotted basis map from matching $a$ to matching $c$ has weight $b - k(a,c) + 2|S|$, where $k(a,c)$ is the number of overlay circles. Composition adds weights; the only weight-zero basis maps are identity maps of equal matchings. The earlier radical analysis gives $J^{2b} = 0$ for $b \geq 1$. Briefly, the weight bound first gives $J^{2b+1} = 0$; a potentially nonzero product of $2b$ positive factors would consist entirely of undotted weight-one surfaces and produce a fully dotted endomorphism. Over $\mathbb{F}_2$, an undotted positive-genus component vanishes and every surviving genus-zero component leaves an output circle undotted. The fully dotted term is therefore impossible. If all positive maps preserve their matching, the square-free endomorphism ring gives the stronger exponent $b + 1$.

Split the actual differential as $d = d_0 + \delta$, retaining in $d_0$ only the constant coefficient of maps between equal matchings. Undotted maps between distinct matchings belong to $\delta$; bit zero alone is not an identity test. Binary elimination constructs a contraction onto the residue homology with inclusion $i$, projection $p$, and degree-minus-one homotopy $h$:

$$pi = 1, \quad d_0h + hd_0 = 1 + ip, \quad hi = ph = h^2 = 0, \quad d_0i = pd_0 = 0.$$

All three maps have scalar identity coefficients. Hence nilpotence makes the following inverse series finite:

$$I = (1 + h\delta)^{-1}i, \qquad D = p\delta(1 + h\delta)^{-1}i,$$
$$H' = (1 + h\delta)^{-1}h, \qquad P = p(1 + \delta h)^{-1}.$$

These are the characteristic-two perturbation formulas; the standard homological perturbation lemma supplies the deformation retract [43]. In particular

$$D = \sum_{j=0}^{2b-2} p\delta(h\delta)^j i$$

for $b \geq 1$, with at most $b$ positive factors needed in the matching-preserving case. At a closed frontier $D = 0$ and binary homology already suffices. The output has exactly the previously predicted minimal matching multiplicities, with its nonzero adjacent-degree maps intact.

The kernel streams one residual column at a time: apply $\delta$, accumulate its projection by $p$, and replace the current vector by its image under $h\delta$. It computes an additional final zero check rather than silently cutting off a nonzero term. Unexpected failure of nilpotence raises an algebraic error; it does not become a local resource decline or a knot verdict. Optional full certificates include $I, P, H'$ and are verified by separate typed matrix multiplication. The checks include both square-zero equations, both chain-map equations, $PI = 1$, $dH' + H'd = 1 + IP$, and the three side conditions $H'I = PH' = (H')^2 = 0$.

## 22.3 The adaptive implementation and its transactional boundary

As before, a stage permits $\max(256, 4N)$ ordinary Schur-update pairs before pausing between complete pivots. A successful cheap residue/degree-gap test finishes the stage immediately. Otherwise the disk option takes a compact snapshot and attempts the geometric certificate and finite transfer. The new differential and reverse incidences are built separately, and all five object arrays are installed together only after the last budget check. The old coefficient-result cache is cleared at installation; typed algebra plans remain valid. A declined attempt leaves the old differential and its completed sparse cancellations intact.

The default experimental allowance is $\max(4096, 64N)$ cooperative polls for one attempt. Geometry traversal, contraction columns, series steps, coefficient products, and scalar incidences poll this guard. The global deadline is checked first at each poll; global exhaustion still raises `ScanLimit` and becomes `UNKNOWN` in recognition. Local exhaustion only declines the attempt. This poll allowance is neither a bit-operation budget nor a memory ceiling: binary vector work and a coefficient composition can cost more than one elementary operation. There is no competitive-runtime theorem for this scheduler. Direct kernel calls can perform the complete finite transfer without that experimental allowance.

The new policy is available to raw homology, recognition, allocation-pruned windows, and minimal windows. The window adapter records each processed crossing before either its ordinary or pruned extension calls cancellation, so the geometric certificate remains bound to the correct prefix. Existing component coefficient engines are compatible; component-owner arrays, saturated weights, races, and alternative scanner representations are not silently reinterpreted. The same standard-scanner restrictions as the earlier adaptive policy apply. Evidence includes attempts, completed transfers, geometry declines, local-budget fallbacks, transfer compositions, and maximum radical depth. Full replay certificates are an explicit kernel diagnostic, not an extra default allocation during recognition.

## 22.4 Cost separation and the missing global parameter bound

Let $N$ be the pre-reduction object count, $r$ the residue survivor count, $E_\delta$ the number of nonzero radical entries, and $S_h, S_p$ the numbers of scalar incidences in the homotopy and projection. Write $A_b, M_b$ for coefficient addition and correctly typed composition costs. The column algorithm has the conservative mixed-cost bound

$$O(N^3) + O\big(N + E_\delta + \ell r\big(E_\delta M_b + (S_h + S_p)A_b\big)\big), \quad \ell \leq 2b.$$

The first term is binary contraction. Dense coefficient work is thus $O(brN^2M_b)$, instead of a generic $O(N^3M_b)$ scalar-pivot bound. It can help when $r \ll N$ and sparse elimination would create expensive fill. It need not help sparse matrices. The product cache is bounded at 4096 entries; optional full certificates require additional storage and work.

For a fixed disk-certified full scan, let $B$ be its maximum boundary half-width and $R$ its largest minimal explicit object count. Attaching one crossing creates at most eight objects per old object, so stage input size is at most $8R$. Coefficients occupy at most $2^B$ bits and even elementary composition has cost $\mathrm{poly}(B)2^{O(B)}$. Noncrossing matching types number at most $4^B$. Charging algebra caches and geometry as well as matrices yields

$$\mathrm{poly}(n, 1+R)2^{O(B)}.$$

This conditional explicit-scan bound is compatible with ordinary exhaustive pivots too; the new transfer improves how expensive coefficient work is organized. It does not prove smaller intrinsic minimal multiplicities. The condition $B + \log(1+R) = O(\log^2 n)$ is sufficient for quasi-polynomial time, but no universal extraction theorem for those parameters is supplied.

## 22.5 Why changing crossing order alone cannot suffice

Report 21 constructs an unknot diagram on an $m \times m$ crossing grid, with $n = m^2$. Pair exterior endpoints adjacent in their remaining cyclic order, always joining distinct open paths until only one closed strand remains. This adds no crossings and preserves the internal grid as a spanning subgraph. Make the first visit to each crossing an overpass; the result is descending and hence an unknot.

Every ordering has a prefix containing between $m^2/3$ and $2m^2/3$ vertices. If at least $m/3$ rows are mixed across the prefix cut, each contributes a horizontal cut edge. Otherwise there is both a full and an empty row: mixed rows alone cannot contain either side's required size. Each column then contributes a vertical cut edge between those rows. Thus every ordering has at least $m/3$ raw frontier edges, or $B \geq m/6 = \Omega(\sqrt{n})$. This refutes a universal polylogarithmic *raw-order-width* strategy. It is not a recognition lower bound: the descending certificate and ordinary Reidemeister simplification bypass these diagrams. In the production audit, all generated inputs through 64 crossings were recognized before homology; independent cube ranks were checked only through nine crossings.

## 22.6 Validation and measured scope

The report's 52 tests pass unchanged. Its source-derived fixture was audited against the pinned Git blobs for `scan_fast.py`, `planar.py`, and the scoped geometry definitions; executable ASTs agree after stripping documentation. The production suite now passes 344 tests, including 100 seeded small diagram comparisons with an independent cube, intermediate minimal-profile comparisons, adjacent-map synthetic checks, failure atomicity, both window implementations, coefficient composition, and global limits. An eager test control deliberately exercises transfer on actual prefixes; this is distinct from observing a production adaptive switch.

The separately rerun geometric audit accepted 71 of 80 order candidates, checked 14,658 complete smoothing states and 71 independent cube ranks, and generated and verified seven typed full chain-homotopy certificates. Data and the source audit are retained under `synthesis/data/`; the reproduction driver is `fast/check_disk_geometry.py`.

Seven shuffled paired rounds in `fast/benchmark_disk_transfer.py` compare fresh states under ordinary, A/A, residue-adaptive, and disk-adaptive policies. Dense synthetic two-term blocks with nonzero survivor maps gained 1.126, 2.948, 4.078, and 3.307 times for the four retained cases; every disk run made one successful full-transfer call. Independent binary regular- representation

homology agrees before and after reduction. These synthetic blocks, including their coefficient-degree mixtures, are not asserted to be actual knot-prefix complexes.

The sparse diagonal control stayed on ordinary pivots: no transfer attempt was made, and its ratio was 0.959 rather than the eager prototype's large regression. The four full diagram scans (Conway, Kinoshita–Terasaka, the 36-crossing stress braid, and the eight-crossing hard unknot) made zero full transfer calls. Their ratios ranged from 0.960 to 1.010, while A/A controls ranged from 0.992 to 1.012. Normal recognition decided all four in earlier certificate or invariant stages. These measurements support the overhead guard on these controls; they establish no recognition speedup from full transfer. Defaults remain unchanged and the goal of general quasi-polynomial recognition remains open.

# 23 Exact long-run recurrences and streamed macro homology

Report 18 extends the existing twist macro complex in two directions: an exact compressed homological profile for one sufficiently long run, and a separate backend that streams degree layers to reduce retained storage. The module `fastunknot.twist.continuation` exposes both on signed run-encoded braid input, with structural recognition before homology in its recognition mode. Its homology mode always computes the requested invariant. The existing production defaults and `backend="twist"` adapter retain their original macro implementation. Ordinary recognition additionally offers the opt-in `-braid-profile` structural evaluation on checked source words.

## 23.1 The repeated differential, not just repeated dimensions

Select a run $\sigma_i^{\epsilon m}$ and freeze the remaining context. Let its macro homological support be $[a, b]$, where $a$ is the sum of negative context exponents and $b$ the sum of positive ones. Write $W = b - a$ and $m_0 = W + 2$. When the selected run is in its turnback sector $E$, the direct sum of all context groups is a fixed vector space $C$ of dimension $M$. Let $D$ be the context differential and $L$ the sum of the dots on the cap and cup. The repeated interior differential is the *same* square-zero operator

$$F = D + L, \qquad \eta = M - 2\,\mathrm{rank}\,F.$$

Indeed $D^2 = 0$, each dot squares to zero, and distinct dot actions commute. The Frobenius identities make exterior dots commute with every context saddle, giving $DL = LD$ and $F^2 = 0$ over $\mathbb{F}_2$. The argument restricts to the reduced marked-circle subcomplex. This is an ungraded endomorphism statement about $C$; $D$ and $L$ need not have the same original bidegree.

For a positive run, complete copies of $C$ occur in total macro degrees $b + 1 \le h \le a + m$. At $m = m_0$ there are exactly two adjacent complete copies, in degrees $b + 1, b + 2$, with boundary map $F$ between them. For a negative run the complete copies lie in $b - m \le h \le a - 1$, and the reference map is the boundary in degree $a - 2$. Geometry depends on whether the selected sector is $I$ or $E$, not on its positive internal index, which identifies these interior maps exactly.

Let $P_{m_0}(z)$ be the reference reduced homological polynomial and $\Delta = m - m_0 \ge 0$. Split its positive-run profile into $A_+$ in degrees at most $b + 1$ and $B_+$ in degrees at least $b + 2$. For the negative run split into $A_-$ at most $a - 2$ and $B_-$ at least $a - 1$. Then

$$P_m(z) = A_+(z) + z^{\Delta} B_+(z) + \eta \sum_{h=b+2}^{a+m-1} z^h,$$

$$P_{-m}(z) = z^{-\Delta} A_-(z) + B_-(z) + \eta \sum_{h=b-m+1}^{a-2} z^h.$$

The lower and upper end pieces retain both incident boundary maps, with the displayed shift where needed. Every inserted degree has $F$ as both incoming and outgoing differential and therefore has dimension $\eta$. There are exactly $\Delta$ inserted degrees, so

$$R(m) = R(m_0) + \Delta\eta.$$

The implementation extracts $\eta$ from the reference *boundary rank*, not from either end-piece Betti number. Those Betti numbers need not equal the interior dimension. These formulas are proved identities of the macro complex; they are not extrapolations fitted to sampled ranks.

## 23.2 Why the slope is positive

Let $J_E$ be the link obtained by replacing the selected run with its cap and cup. Coherent braid orientation forces the two outside paths at that box to join top ends to bottom ends. Adding the cap and cup therefore puts both marked arcs on the same actual component, regardless of the other components of the closure.

Jaeger's weighted-dot sliding theorem states that the characteristic-two twisted complex depends on the sum of weights on each actual component [44]. Two unit weights on that component cancel. In the report's explicit specialization, sliding across one crossing uses a reverse saddle $H$ with $H^2 = 0$ and $DH + HD = X_p + X_{p'}$; the change of basis $1 + H$ conjugates the corresponding dot-deformed differentials. Thus the ordinary reduced cube with the two-dot deformation has the homology of $J_E$. Relative local reductions in the other run boxes commute with the exterior dots, so this identification passes to the macro context as well. Consequently

$$\eta = \dim_{\mathbb{F}_2} \widetilde{Kh}(J_E; \mathbb{F}_2) > 0.$$

Positivity follows from the nonzero reduced Jones Euler polynomial of a nonempty link: its value at one has absolute value $2^{\ell-1}$. A free integral homology summand therefore survives over $\mathbb{F}_2$. This argument does not use the generally false assertion that adding an arbitrary null-homotopic operator preserves homology.

The dot argument identifies a total dimension after collapsing grading. The original run's homological profile follows instead from the separate identification of the complete groups and both incident maps above. Quantum grading is not returned. The raw continuation API retains macro degrees. Converting to the original production cube convention requires $h_{\mathrm{cube}} = n_+^{\mathrm{original}} - h_{\mathrm{macro}}$ and, for a knot, doubling reduced dimensions. The shortened reference's positive crossing count is not the original shift. Also, shortening can change component parity: recognition checks the original run list, and the reference may be a link.

## 23.3 Compressed output and a fixed-context bound

The tail algorithm keeps two finite sets of exceptional degree–dimension pairs and one constant interval, with inclusive endpoints. Exceptions and the interval are disjoint. Queries and total rank use integer arithmetic; explicit expansion is optional and capped. The tests include exponents with 100,001 bits while the two-strand reference still has three macro states and four reduced basis elements. This is an in-memory API test; the JSON frontend retains Python's decimal-integer conversion limit.

The reference has $2W + 2$ expanded crossings. At $s$ strands a conservative basis bound is $2^{s-1}3^{2W+2}$, so even elementary assembly and elimination give $2^{O(s+W)}$ reference work, with polynomial dependence on the binary length of $m$ for reconstruction. For fixed context and strand count this removes enumeration of the long exponent. For growing context the reference can still be exponential. On a knot closure the context must touch all other strand separators, so

$s \le W + 2$; a context of $W = O(\log^2 n)$ yields a quasi-polynomial bound in the expanded crossing count $n$. This remains a restricted homology statement.

Since $\eta > 0$, an explicit list of the inserted nonzero degrees has at least $\Delta$ entries. No algorithm returning that expanded list can be polynomial in $\log m$ for fixed context. The compact output contract is essential, rather than an optional presentation improvement. The largest run is the default candidate. Two runs cannot both strictly dominate their complement, so this one-run recurrence cannot simply be iterated to compress every large exponent independently.

### 23.4 Direct structural evaluation on run-encoded braids

For each generator index let $p_i, n_i$ count its positive and negative occurrences, and set $P = \sum p_i$, $N = \sum n_i$, $e = P - N$, $u_+ = |\{i : p_i > 0\}|$, $u_- = |\{i : n_i > 0\}|$. The braid's signed Seifert graph is a path multigraph on $s$ vertices. On a one-component closure, the existing structural formulas become

$$e + s - 2u_+ - 1 \le s(K) \le e - s + 2u_- + 1, \qquad 2g = P + N - s + 1.$$

Its homogeneity defect is the number of indices occurring in both signs. Genus zero certifies the unknot; positive homogeneous genus or an interval excluding zero certifies knottedness. A zero-containing interval alone is inconclusive. Sparse permutation tracking uses each exponent's parity to check the original component count without expanding the word or allocating an array for untouched strands. Missing a strand separator rejects a knot candidate immediately; otherwise $s - 1$ is bounded by the number of runs. These checks are polynomial in the encoded input length.

The production PD convention reverses the Artin crossing sign. Public Seifert fields therefore use writhe $-e$ and interval $[-U, -L]$ when the Artin interval is $[L, U]$. Separate Artin fields make that conversion explicit. Tests compare the entire dictionary with the independent PD Seifert graph, rather than comparing only verdicts.

Every one-component dominant-run input is already structurally knotted. For a positive run of magnitude $m \ge W + 2$, the lower bound equals $2(P - u_+) - d$, with $d = m + W - s + 1$, and

$$2(P - u_+) - d \ge 2(m - 1) - (m + W - s + 1) = m - W + s - 3 \ge s - 1 > 0.$$

The negative case is symmetric. Thus tail timings do not represent a newly solved recognition family. The report's signature-packing diagnostic also adds no cases: its selected definite rank is at most $P - u_+$ or $N - u_-$, so its successful inertia bound is dominated by the same interval. Its diagnostic and verifier are retained for reproducibility, but recognition uses the cheaper dominating calculation.

### 23.5 Streaming changes storage, with an explicit time tradeoff

The separate streamed backend enumerates bounded compositions of degree totals. Each macro state is visited once, without repeatedly scanning the whole Cartesian product for each degree. A depth-first enumerator prunes prefixes using suffix capacities and avoids recursion limits. Its total enumeration work is $O(t \prod_j (|e_j| + 1))$ for $t$ runs.

Only two adjacent state indexes are live. Local Frobenius maps retain circle indexes instead of exponentially large label tables; generated columns feed directly into binary elimination. Geometry and map caches have bounded entry and weighted-slot capacities. If square-zero checking is requested, two adjacent matrices are additionally retained. The limits distinguish total enumerated states and basis from live states, layer basis, matrix bit payload, column references, geometry vertices, and map slots. They are mathematical storage envelopes, not allocator or RSS reservations. The returned degree, chain-dimension, and boundary-rank dictionaries still grow with the number of degrees; two live state layers do not imply constant total memory.

After computing ranks of $d_{h-1}$ and $d_h$, the value $\beta_h = \dim C^h - \operatorname{rank} d_{h-1} - \operatorname{rank} d_h$ is final. On a knot, a cumulative finalized reduced rank greater than one certifies `KNOTTED` and permits early exit. Such a result explicitly has `reduced_rank=null`, a lower bound, and `homology_complete=false`. Agreement with rank one before completion cannot certify the unknot. Full homology mode never uses this early exit.

The combined RLE frontend checks structural certificates first, then uses the selected `tail`, `streaming`, or `macro` backend. It is an explicit interface, not an unmeasured automatic storage switch. Resource exhaustion returns `UNKNOWN`; CLI exit codes are 0 for completion, 2 for invalid input or configuration, and 3 for exhaustion. Tail's finite reference and reconstruction share one cooperative deadline in the production integration, and mutable streaming budgets are copied and revalidated. Tail's square-zero flag refers to the finite reference, not to an unbuilt enormous matrix. The ordinary source-word profile stage is opt-in and never assigns the whole source braid to an individual connected-sum factor.

### 23.6 Validation and measurements

The archive passed 254 tests unchanged. The production suite now passes 391, including 400 tail/macro profile comparisons, 80 independent cube comparisons, both signs and boundary thresholds, original component parity, huge binary exponents, streamed enumeration/maps/storage limits, partial-rank semantics, and deadline/budget regressions. Separate reruns produced 832 independently formulated recurrence comparisons, 220 cube comparisons, and 500 checks identifying the slope with a separately extracted ordinary turnback subcomplex. Their records are in `synthesis/data/twist-*-audit.json`.

Seven paired A/B/A rounds compare complete homology contracts, with tail's compact profile expanded only for validation outside timing. At exponent 1025, the two-strand, figure-eight-context, and four-strand-context cases gained 38.70, 58.59, and 46.84 times respectively. Their reduced reference bases were 4, 93, and 126, versus 1027, 15393, and 18486 for full macro assembly. The no-compression control ratio was 0.974. Raw samples and the separate A/A control are retained in `fast/results/twist_tail_20261008.json`.

The separate eight-case streaming benchmark includes fresh A/A controls, seven shuffled rounds, and independent traced-allocation runs. Streaming was slower in every full-homology case: macro/stream ratios ranged from 0.229 to 0.694. Peak traced allocation ratios ranged from 1.020 to 3.064 in streaming's favor. These are Python allocation peaks, not process RSS; early-decision arms have a different output contract and are recorded separately. Data are in `fast/results/twist_streaming_20261008.json`. This supports streaming as a storage option, with no default change. The structural benchmark separately measures encoded input versus expansion and recognition on fresh prebuilt diagrams; these scopes must not be combined into a general recognition-speedup claim. For the four-strand family with runs $(1, m), (2, -(2m-1)), (3, m)$ and odd $m = 17, 129, 1025, 4097$, the source-profile option gained 2.296, 2.824, 2.905, and 3.334 times on fresh prebuilt diagrams. Its A/A ratios ranged from 0.997 to 1.022. These inputs have homogeneous positive genus despite an Artin Rasmussen interval $[0, 0]$, so the homogeneous criterion, not interval equality, supplies the verdict. The separate encoded-input comparison includes expansion, PD construction, and Seifert evaluation in its baseline; its ratios range from 94.4 to 29687.0. These large numbers primarily measure the benefit of preserving compressed input. Morton's 11-crossing case remains a structural-only inconclusive control, with no full-pipeline timing in this experiment. All raw data are in `fast/results/braid_profile_20261008.json`.

# 24 Arithmetic continuations and checked Montesinos sources

Report 22 supplies a useful change of representation: some prefixes whose explicit homotopy complexes are necessarily large have short arithmetic descriptions sufficient for the actual closure decision. Production now implements its supplied-Montesinos classifier and a separate optional local obstruction. This is a complete polynomial decision on a specified source grammar, not an algorithm discovering that grammar in every PD. The incoming archive remains unchanged; this section describes the maintained integration.

## 24.1 What the source describes

An input consists of an integer $e$ and separate finite rational tangles with ordinary continued fractions. Its knot is

$$N([e] + T(p_1/q_1) + \cdots + T(p_s/q_s)), \qquad \gcd(p_i, q_i) = 1, \quad q_i > 0.$$

The summands remain separate throughout. Reducing their numerical sum would lose topological information: two $1/2$ summands give uncancelled numerator 4, whereas the numerical sum is 1. A quarter rotation sends a slope to $-1/x$; a rotation followed by crossing reversal gives $1/x$. These conventions agree with the rational-tangle operations in [45]. The compiler performs those operations on actual four-terminal crossing graphs, joins adjacent terminal pairs, and adds the numerator caps. A lazy crossing-switch bit avoids repeatedly mirroring earlier crossings during continued-fraction construction.

For each continued fraction, read coefficients backwards from $(p, q) = (1, 0)$:

$$(p, q) \longmapsto (ap + q, p).$$

Every update matrix is unimodular. Hence no gcd computation is required, intermediate infinity is harmless, and the final pair is primitive. A final infinite slope is rejected. Write $p_i = a_i q_i + \beta_i$ with $0 \le \beta_i < q_i$, add all $a_i$ to $e$, and discard integral summands. The remaining $r$ pairs $(\beta_i, \alpha_i)$ have $0 < \beta_i < \alpha_i$ and $\alpha_i \ge 2$. Integral-twist transfer through rational summands preserves the numerator closure. Set

$$Q = \prod_{i=1}^{r} \alpha_i, \qquad D = eQ + \sum_{i=1}^{r} \beta_i Q/\alpha_i. \tag{13}$$

The implemented accumulator starts at $(D, Q) = (e, 1)$ and updates $(D, Q) \mapsto (D\alpha + \beta Q, Q\alpha)$, without cancellation.

**Lemma 24.1** (Component gate)**.** *The numerator closure in this grammar is a knot exactly when $D$ is odd.*

*Proof.* A primitive rational column modulo two specifies one of the horizontal, vertical, and crossed endpoint pairings, respectively $(0, 1)$, $(1, 0)$, and $(1, 1)$. Twisting and reciprocal operations preserve this correspondence. If $k > 0$ denominators are even, the open horizontal sum has $k - 1$ internal closed strings and its numerator closure has $k$ components. If $k = 0$, the number of components is one or two according as $e + \sum_i \beta_i$ is odd or even. Formula (13) has odd parity exactly for $k = 1$, or for $k = 0$ with the stated odd sum. For $k \ge 2$ every term is even. This also covers the empty exceptional list. □

**Proposition 24.2** (Decision on supplied Montesinos knots)**.** *After the component gate, the answer is UNKNOT precisely when $r \le 2$ and $|D| = 1$; otherwise it is KNOTTED.*

*Proof.* Use the classical double-cover construction: the cover is a Seifert manifold whose group, in this convention, is

$$\langle h, x_1, \dots, x_r \mid [h, x_i] = 1, x_i^{\alpha_i} h^{\beta_i} = 1,\ x_1 \cdots x_r = h^e \rangle.$$

Rational tangle balls lift to solid tori; their meridians have slopes $\alpha_i x_i + \beta_i h$. Abelianizing this presentation gives homology order $|D|$ when $D \neq 0$. If $r \geq 3$, kill $h$ and all but three of the $x_i$. The resulting triangle group with orders $a, b, c \geq 2$ is nontrivial. Here is an explicit verification avoiding an algorithm for triangle groups. Put $\lambda = e^{\pi i/a}$, $t = 2\cos(\pi/b)$, $s = 2\cos(\pi/c)$,

$$u = \frac{s - \lambda^{-1} t}{\lambda - \lambda^{-1}}, \quad v = \frac{\lambda t - s}{\lambda - \lambda^{-1}}, \quad X = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix}, \quad Y = \begin{pmatrix} u & 1 \\ uv - 1 & v \end{pmatrix}.$$

Both determinants are one and the traces of $Y, XY$ are $t, s$. Their distinct eigenvalues give $X^a = Y^b = (XY)^c = -I$. Thus $X, Y, (XY)^{-1}$ define a nontrivial projective representation; $X$ has projective order $a$. The cover cannot be $S^3$ and the knot cannot be the unknot. This is a symbolic proof, not a numerical subroutine.

For at most two exceptional summands the closure is two-bridge, even though the open sum need not be rational. The rational-link classification gives the unknot exactly when its numerator parameter $|D|$ equals one. In the two-summand case this is also the rational gluing criterion of [46, Theorem 5]. The Seifert-cover and local-closure foundations are discussed in [47, Theorem 4.9]. □

This detects determinant-one nontrivial knots. For example, for every odd $a \geq 3$ take slopes $-1/a, 1/(a+2), 1/r$ with $r = (a^2 + 2a - 1)/2$. The signed pretzel expression is $-a(a+2) + 2r = -1$, so the normalized positive-denominator determinant has absolute value one. All three denominators exceed one, proving nontriviality. The usual genus-one Seifert matrix for odd pretzel parameters $(p, q, r)$ is

$$\frac{1}{2} \begin{pmatrix} p + q & q + 1 \\ q - 1 & q + r \end{pmatrix}.$$

Its determinant is $(pq + pr + qr + 1)/4 = 0$ here; its Alexander polynomial is a unit and its symmetrization has signature zero. Thus these easy arithmetic decisions include a family on which those two obstructions are inconclusive.

**Bit cost and output.** Let $B$ include all binary coefficients, signs, and list delimiters. Each continued-fraction column has bit length bounded by the sum of $1 + \lceil \log_2(1 + |a|) \rceil$ along its block. All final columns together have $O(B)$ bits. Multiplying an $s$-bit partial entry by a $b$-bit coefficient costs $O(sb + s)$ with schoolbook arithmetic; summing over the input costs $O(B^2)$. The divisions producing $(a_i, \beta_i)$ and the uncancelled accumulator have the same bound, by charging products to both operand lengths. The classifier therefore uses $O(B^2)$ bit operations and $O(B)$ bits of integer/source data. This statement does not count Python object overhead as a single bit. The certificate has $O(B)$ encoded bits; integers exceeding 1024 bits use exact hexadecimal records. The independent replay uses forward matrix products and full denominator-product sums, with a conservative $O(B^3)$ bound because it also replays prefix size statistics. The reported maximum entry size concerns tracked states, not every temporary or peak memory.

## 24.2 Provenance, budgets, and integration

The compressed API is `montesinos_certificate`, taking $e$ and the tangle list. The Diagram constructor `from_rational` (also `from_montesinos`) builds all crossings and validates the resulting spherical, one-component PD. Its default expansion ceiling is 100000 crossings, checked before

allocating crossings. Input parsing and construction precede the ordinary recognition timer. The ordinary JSON grammar uses a `montesinos` object with `e` and `tangles`; its CLI expands that input, rather than silently switching to the compressed interface.

Only the checked constructor attaches a detached immutable source. Bare PD input has no such assertion. Mirroring negates all coefficients; serialized sources are reconstructed and must match the stored PD up to half-turns of individual rows. Multiple serialized source types are rejected. An initialized Diagram cannot be reinitialized with a different PD while retaining old provenance. Equality and hashing still depend solely on PD. These are supported API invariants, not protection against arbitrary mutation of interpreter internals.

The recognizer checks this source before other certificates; `use_rational=False` or `-no-rational` disables that stage. Arithmetic polls the same cooperative deadline and the recognizer checks again after certificate construction. The source is used only for the original whole diagram, before factorization. The rank-two restart option continues to carry all recognition settings. PDs without this new source retain the previous pipeline. The general quasi-polynomial flag stays false.

### 24.3 Literal local obstructions with a checked disk

For a genuine two-string tangle with $r \geq 2$ finite nonintegral summands, Nogueira–Salgueiro's criterion says it embeds in some unknot exactly when $r = 2$ and

$$p_1 q_2 + p_2 q_1 \equiv \pm 1 \pmod{q_1 q_2}.$$

We use only its negative direction. A verified occurrence failing this criterion proves the entire host knot nontrivial, regardless of the exterior. An allowed complement gives no positive verdict for the actual host. The implemented templates require initial $e = 0$, connected crossing shadow, four distinct ports, and no closed strand component.

The source compiler recomputes fractions and builds the literal crossing pattern. Its rotation system must be spherical with every port on one facial boundary in the declared order. Opposite-dart tracing must produce exactly two open strings. An occurrence maps crossings injectively into a freshly validated host PD. Offsets zero and two preserve the cyclic order and under-pair. Every internal edge must match and every other dart must lead outside the selected set, yielding four distinct cut edges.

These checks prove the needed disk condition. Take a regular neighborhood of the connected selected graph, ending halfway along its outgoing edges. Every remaining graph component attaches at a port, since the entire knot shadow is connected. All ports lie in one face, so the remaining graph lies there too. Fill the other neighborhood boundary components with their empty complementary disks. The result is a disk meeting precisely the four cut edges; thickening it gives a tangle ball. An abstract four-edge cut alone would not justify this argument.

Fixing one root image and its half-turn forces the image and rotation of every neighbor through internal edges. Connectedness leaves only $2n$ root trials, each costing $O(k)$ for a $k$-crossing template. The matching core thus costs $O(nk)$ word operations. Public host validation also sorts labels, adding $O(n \log(n+2))$; source reading and compilation must be charged separately. A huge zero-coefficient description is not free just because it expands to few crossings. Python dictionaries use the usual table-cost model in this count; indexed arrays give the corresponding deterministic algorithmic bound. The six default patterns have slopes $(1/3, 1/3)$, $(1/2, 1/5)$, $(1/3, 1/4)$ and their mirrors.

`recognize_with_subtangles` is a separate opt-in wrapper. No match continues ordinary recognition with the remaining time; exhaustion yields UNKNOWN. The production integration also checks time after successful final verification, closing a gap in the delivered wrapper. Time values must be finite and nonnegative. Template compilation and PD validation are cooperative operations, not interruptible real-time allocations.

An abstract extension enumerating at most $h$ coefficients and all separators, with coefficients in $[-n, n]$ and total expanded size at most $n$, has at most $\sum_{\ell \le h} (2n+1)^{\ell} 2^{\ell-1} = (n+2)^{O(h)}$ candidates. Compiling and checking them gives quasi-polynomial discovery when $h = O(\log n)$. This is a restricted literal-witness theorem from report 22, not the implemented fixed catalogue and not a bound for every knot diagram.

### 24.4 Why arithmetic helps when a small linear summary cannot

Let

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \quad v_w = G_w(1,1)^t, \quad x_w = G_w(3,2)^t$$

for words $w \in \{A, B\}^m$ and their literal matrix products $G_w$. Two twists realize each letter on rational tangles. The associated prefix and mirrored continuation have at most $2m+1$ and $2m+3$ crossings. All cross closures are knots: the two columns have parity $(1,1)$ and $(1,0)$. The rational gluing test is $|\det(v_u, x_w)| = 1$.

For $u = w$ the determinant is $-1$. For distinct words, their positive-cone images are disjoint: $A$ sends a positive column into $x > 2y$, while $B$ sends it into $y > 2x$; remove the common prefix to prove the assertion inductively. Consequently $G_w^{-1} v_u = (a, b)^t$ has one nonpositive and one positive integer coordinate. The determinant $2a - 3b$ has absolute value at least two, and odd parity raises this to at least three. Thus the Boolean continuation matrix is the $2^m$-dimensional identity over every field. Distinct exact continuation states require at least $m$ bits; a linear representation of all Boolean answers needs dimension at least $2^m$. Neither statement is a general running-time lower bound. The signed determinant matrix has rank at most two, and its nonlinear absolute-value-one readout decides the same family using $O(m)$ bits and $O(m^2)$ bit operations.

There is also an explicit full-complex obstruction inside unknots. Put $G_k = (AB)^k$, $v_k = (p_k, q_k)^t = G_k(1,1)^t$, and $x_k = G_k(3,2)^t$. The whole closure is an unknot with $N_k = 8k+4$ displayed crossings. Its designated rational prefix has $4k+1$ crossings and

$$p_0 = 1, \quad p_1 = 7, \quad p_{k+2} = 6p_{k+1} - p_k, \quad p_k \ge (3 + 2\sqrt{2})^k.$$

Any full, explicitly delooped four-ended complex homotopy equivalent to that prefix needs at least $p_k/2$ summands. Indeed, closing each matching and reducing on a fixed exterior marked arc yields at most two scalar generators per summand. Closing and applying the marked TQFT is additive and preserves chain-homotopy equations. Homology dimension is therefore at most twice the summand count, whereas its graded Euler characteristic at quantum parameter $i$ gives $\dim \widetilde{\mathrm{Kh}} \ge \det = p_k$.

If the quantum grading is preserved, the stronger bound is $p_k + q_k$. At $i$, a reduced closure with more than one circle contributes zero because $i + i^{-1} = 0$; a one-circle closure contributes absolute value one. Numerator closure bounds one matching's multiplicity below by $p_k$ and denominator closure bounds the other by $q_k$. Both are knots since $p_k, q_k$ are odd. Orientations may be chosen separately: the orientation-free bracket complex changes only by global grading shifts, which do not change these absolute-value estimates. This obstruction concerns that specified prefix, full homotopy type, and expanded summands. It does not cover every order, binary multiplicity encodings, or the actual arithmetic decision procedure.

For adaptive recognition the consequence is concrete: bounding geometric frontier alone does not bound full surviving object count. When checked arithmetic structure is available, its decision stage should precede a full complex. For general PDs, discovering and composing such structures without expansion remains a research obligation; switching between existing complex representations alone does not prove the missing bound.

### 24.5 Production validation and measured scope

All 152 delivered tests pass unchanged. The maintained suite has 411 tests, including two additional resource/option regressions. The patch hash and all 81 delivered patched-file hashes agree with the patch manifest; all 82 source Git blobs agree with the pinned commit. The package-wide checksum launcher cannot run because the delivered tree lacks its `SHA256SUMS` file; the narrower checks are recorded explicitly in `rational-source-audit.json`. Independent reruns against production check 1200 sources: 718 knot-PD Alexander comparisons, 482 link component comparisons, and 1436 mirrored pickle checks. The local audit rejects 12 forged certificates and a crossing switch, checks 40 reorder/half-turn/relabel/mirror instances, and finds no false rejection on 31 known unknots. The identity diagnostic checks 1398101 matrix entries through $m = 10$. The checkpoint audit constructs 21 whole diagrams and adds 14 Alexander and six independent cube comparisons. These are finite implementation checks; the proofs above supply asymptotic conclusions.

`fast/rational_research/benchmark.py` retains seven paired rounds with fresh prebuilt Diagrams, one-second recognition budgets, and a 20000-object cap. Each round measures baseline, five source-enabled samples, then baseline. The after/before ratio is a drift control, not a separate randomized A/A trial. Construction plus new recognition is recorded separately; recognition-only speedups must not be read as construction-inclusive speedups. Local patterns use the same before/new/after protocol, with the template cache warm. Raw data and exact source hashes are in `fast/results/rational_20261008.json`.

| Input | Crossings | Old ms | New ms | Old/new or censoring |
|---|---|---|---|---|
| Diagonal $m = 2$ | 12 | 2.953 | 0.0300 | 96.8× |
| Diagonal $m = 4$ | 20 | 6.157 | 0.0302 | 203.2× |
| Diagonal $m = 6$ | 28 | 15.494 | 0.0333 | 454.3× |
| Diagonal $m = 8$ | 36 | 121.454 | 0.0658 | 1896.4× |
| Diagonal $m = 10$ | 44 | 133.061 | 0.0673 | object limit, no ratio |
| Diagonal $m = 12$ | 52 | 94.483 | 0.0394 | object limit, no ratio |
| Off-diagonal $m = 8$ | 36 | 0.102 | 0.0282 | 3.64× |
| Pretzel $a = 3$ | 15 | 1.154 | 0.0286 | 41.4× |
| Pretzel $a = 7$ | 47 | 3.446 | 0.0317 | 109.0× |
| Pretzel $a = 31$ | 575 | 52.195 | 0.0527 | 992.6× |

Ratios are medians of paired ratios, not ratios of displayed medians. The 44- and 52-crossing baselines stop at the object cap; they are not completed solves. Source-enabled construction plus recognition ranges from 0.175 to 0.700 ms on these diagonal cases, and is 5.37 ms for the 575-crossing pretzel. The largest compressed pretzel input uses a 65537-bit parameter and takes 1.46 ms median for arithmetic only; no PD is built. These compressed times describe a different input contract.

Local search loses on all five measured cases: old/new ratios are 0.682 (Conway), 0.634 (eight-crossing unknot), 0.256 (trefoil), 0.876 (planted two-summand pattern), and 0.160 (planted three-summand pattern). In the planted cases the old modular Alexander or Rasmussen stage is already cheaper. This is why local search remains explicit. Its proven rejection family permits arbitrarily complicated exteriors, but these measurements do not establish an end-to-end gain on that family.

## 25 Full transfer ordered by quantum grading

The graded kernel from report 23 supplies another exact response to slow sparse cancellation. It computes all maps between scalar survivors, including adjacent-degree maps, without the common-disk certificate required by Section 22. Production exposes `graded` and `graded-adaptive` as

optional reduction policies. This integration covers the graded kernel; Section 26 reviews the report's twist-threshold, succinct interface and geometric-dependency proposals.

### 25.1 Retaining the grading rather than reconstructing it afterwards

At a frontier of $2k$ points let $c(a, b)$ be the number of circles in the overlay of matchings $a, b$. A basis morphism dotted on a set $S$ of those circles has intrinsic weight

$$\omega(a, b, S) = k - c(a, b) + 2|S|.$$

The existing coefficient engine and grading audit use the convention $q_b - q_a = \omega$ for every nonzero monomial in a homogeneous differential entry. The new scanner keeps these object shifts during allocation:

$$q' = q + i + c - 2\,\mathrm{popcount}(\ell),$$

where $i \in \{0, 1\}$ is the smoothing, $c$ is the number of delooped circles, and $\ell$ is their binary label. The array follows the exact object order used by `FastScan.add_crossing`. Scalar cancellation preserves the shifts of all remaining objects. This is the ordinary dotted grading and cancellation calculus of [36], used here as an algorithmic order.

Weights are nonnegative because $c(a, b) \le k$. Equality requires equal matchings and no dots: a circle in the overlay then consists of the two copies of the same arc. Composition adds weights on every nonzero term. In the engine's dot-count convention, for a composition through $b$ with input dot counts $u, v$ and output count $r$,

$$2r = 2u + 2v + k - c(a, b) - c(b, c) + c(a, c),$$

which proves additivity by substitution. Thus writing $d = d_0 + \delta$, the zero-weight part $d_0$ is a direct sum of binary complexes at fixed matching and $q$, and $\delta$ strictly raises $q$. An undotted morphism between distinct matchings is not a scalar identity. Taking weight zero in $d^2 = 0$ proves $d_0^2 = 0$.

### 25.2 A contraction with representatives and homotopies

For each fixed matching and quantum shift, binary elimination splits each chain group as

$$C^h = B^h \oplus R^h \oplus L^h,$$

where $B^h = \operatorname{im} d_0^{h-1}$, $R^h$ represents its quotient in $\ker d_0^h$, and $d_0 : L^h \to B^{h+1}$ is an isomorphism. Let $I, P$ include/project $R$ and let $H$ invert the last isomorphism, vanishing on the other summands. Then

$$PI = 1, \quad d_0 I = P d_0 = 0, \quad d_0 H + H d_0 = 1 + IP, \quad H^2 = HI = PH = 0.$$

The implementation constructs bases, lifts, and inverse coordinates; a list of ranks would not supply the maps needed for this step. Each scalar map preserves matching and $q$, and $H$ lowers homological degree by one.

There are finitely many object shifts, so $H\delta$ and $\delta H$ are nilpotent. In characteristic two form the finite inverses

$$U = (1 + H\delta)^{-1}, \qquad V = (1 + \delta H)^{-1}.$$

The finite perturbation identity [43] gives

$$D = P\delta U I, \quad I' = UI, \quad P' = PV, \quad H' = UH = HV. \tag{14}$$

These are a special chain contraction from $(C, d)$ to $(R, D)$:

$$dI' = I'D, \quad P'd = DP', \quad P'I' = 1, \quad dH' + H'd = 1 + I'P', \quad H'^2 = H'I' = P'H' = 0, \quad D^2 = 0.$$

For an explicit algebraic check, $Z = UI$ solves $Z = I + H\delta Z$. Using $d_0\delta + \delta d_0 + \delta^2 = 0$ and the scalar contraction gives $dZ = ID + H\delta dZ$, hence $dZ = ZD$. The right-hand calculation gives the projection identity. Expanding the finite inverses with $H^2 = HI = PH = 0$ proves the side identities. Multiplying $dH' + H'd + 1 + I'P'$ on the left by $1 + H\delta$ and on the right by $1 + \delta H$ reduces it to zero using $d^2 = 0$ and the scalar homotopy equation. The multipliers are invertible. Finally $D^2 = P'd^2I' = 0$. Every term of $D$ contains $\delta$ and has positive weight, so the returned complex has no scalar unit left to cancel.

This termination argument uses the actual finite grading and typed composition law. It does not invoke a uniform nilpotence claim for arbitrary ungraded formal matching tables. Both the disk-certified and graded algorithms remain applicable only to valid complexes in their respective contracts.

## 25.3 Computing a path sum without enumerating its paths

For each scalar survivor column $e$, evaluate

$$Z = Ie + HW, \qquad W = \delta Z, \qquad De = PW$$

one quantum block at a time in increasing order. All contributions to a block arrive from lower blocks before it is processed; binary $P, H$ then mix within that block. Each radical edge is propagated at most once for this source, combining equal destinations by XOR. This computes the finite sum in (14) without constructing matrix powers or enumerating exponentially many individual paths.

There is also a source-dependent band:

$$0 \le \omega = k - c(a,b) + 2|S| \le k + c(a,b) \le 2k.$$

A source of shift $q_0$ cannot reach a nonzero coefficient beyond $q_0 + 2k$. The kernel drops such destinations before composition. It still retains higher-order corrections within the band. For example, two radical steps with a scalar contracting arrow between them can yield $x_1x_2$ even when $P\delta I = 0$. On a single arc the analogous $x^2$ term is zero. Both cases are explicit regression tests, as is the nonzero minimal map $a \xrightarrow{x} a$. Replacing the latter by two isolated survivors would change later gluing.

Let $N$ count allocated object slots including holes left by completed sparse pivots, $S$ count scalar survivors, and $E_\delta$ count positive-weight entries. There are at most $SE_\delta$ coefficient-composition attempts. If $A_H, A_P$ count nonzero scalar incidences, the remaining transfer work is bounded by $O(S(N + E_\delta + A_H + A_P))$ coefficient additions and control operations, after polynomial binary preprocessing. Packed indices and integer costs must also be charged. The implementation's global bit columns and arrays can depend on $N$, even if few slots remain live. Thus a live-only bound must first account for compaction or the size of the allocated arrays. The component coefficient engine costs $2^{O(k)}\,\mathrm{poly}(k)$ per operation, giving an explicit bound polynomial in $N, S$ and exponential in $k$. None of these statements bounds necessary object multiplicity.

Optional diagnostic mode additionally computes all of $I', P', H'$ and checks the eight displayed identities in the full cobordism algebra. It is excluded from timed production reductions. If the whole differential is scalar and diagnostics are disabled, the kernel obtains quantum-resolved homology multiplicities directly and avoids constructing unused contraction maps.

## 25.4 Adaptive scheduling and installation

The eager policy transfers each stage. The adaptive policy first allows $\max(256, 4M)$ candidate Schur update pairs, where $M$ is the live count at stage entry. The existing sparse eliminator

pauses only between complete pivots and clears its candidate queues. Full transfer then consumes that partially reduced complex, retaining completed cancellations. Counters distinguish stages, actual switches, coefficient attempts, identity shortcuts, and scalar-only reductions.

The maintained adapter prepares the new incoming-edge index before replacing any differential arrays, checks the shared deadline again, and installs the matching, degree, quantum, outgoing and incoming arrays together. A failure while preparing the replacement preserves the current differential; composition caches may have acquired reusable entries. Tests force both a final-boundary deadline expiration and an incoming-index allocation failure, then resume the same object successfully. The kernel also polls during long composition sequences. A global resource failure propagates from the raw API and becomes UNKNOWN in recognition; no rate of progress is a topological verdict.

Direct constructors have an optional transfer-object ceiling; exceeding it resumes ordinary cancellation. The public reduction modes retain the existing object and time ceilings. Their policy is a heuristic, not a competitive running-time guarantee or a byte-memory budget. In particular, a dense binary basis change can be more expensive than the sparse pivots it replaces and can change future fill. The default remains `standard`.

Both modes require minimum-fill pivots, bit coefficients, self-inverse cancellation, and one scan order. Standard and component coefficient composition and tail finishing are supported. Homological windows are rejected before early recognition certificates: their pruned allocation needs a separate quantum-label adapter. Shared, saturated, twist and other alternative scanners are also rejected. Existing disk-adaptive window modes remain available.

### 25.5 Validation and measurement scope

The delivered report 23 snapshot passes all 281 tests unchanged and its 114 source Git blobs match the declared baseline. Those full archive tests also cover additions outside this integration; passing them does not constitute integration of those additions. Production adds 13 graded integration tests and two transactional-resource tests to the previous 411-test suite. The transfer audit uses 86 diagrams in 258 order presentations and checks 1812 full stage contractions, with eight identities each. Four scanner variants agree on raw ranks and three quantum-tracking variants agree on quantum-resolved ranks. Synthetic controls retain nonzero minimal maps, test higher corrections, exercise capacity fallback and resume after sparse progress. Independent set-coefficient scans provide an additional oracle on 18 small random diagrams.

The maintained drivers are in `fast/graded_research`; raw audit and source-provenance evidence is in `synthesis/data/graded-*`. Measurements use seven shuffled paired rounds, an identical standard A/A arm, fresh states and batched calls. Whole-diagram timings cover complete raw scans including reduction setup, but exclude construction and order selection; they bypass recognition filters. Synthetic timings cover the entire reduction stage and exclude cloning. Synthetic controls are valid graded algebraic complexes, not proved realizations of classical knot prefixes.

| Case | Old/eager | Old/adaptive | A/A | Adaptive switches |
|---|---|---|---|---|
| hard_unknot_8 | 0.689 | 0.957 | 0.998 | 0 |
| conway | 0.557 | 0.920 | 0.929 | 0 |
| kinoshita_terasaka | 0.616 | 0.980 | 1.019 | 0 |
| unknot_braid40 | 0.633 | 0.886 | 0.984 | 0 |
| stress_braid5_36 | 0.260 | 0.942 | 1.021 | 0 |
| synthetic 16 | 0.888 | 0.741 | 0.999 | 1 |
| synthetic 32 | 1.909 | 1.582 | 0.992 | 1 |
| synthetic 64 | 4.968 | 4.174 | 1.008 | 1 |
| synthetic 128 | 13.180 | 11.266 | 1.015 | 1 |

Ratios above one favor the new policy. Each is a median of paired ratios. The five actual diagrams invoke zero adaptive transfers. Eager transfer slows all five raw scans; tracking the grading also adds overhead to adaptive scans. The largest synthetic case gains 13.18× eagerly and 11.27× adaptively, retaining a nonzero dotted minimal differential. The smallest synthetic case regresses. A/A ratios range from 0.929 to 1.021 across all cases, so small changes should not be treated as reliable improvements. No end-to-end recognition gain on the actual-diagram controls is established. The raw measurements are in `fast/results/graded_transfer_20261008.json`.

The conditional complexity target remains explicit. If all allocated intermediate state counts satisfy $N = n^{O(\log n)}$, all frontiers have $k = O(\log^2 n)$, and there are polynomially many stages, these explicit operations fit a quasi-polynomial bound. The integration proves neither universal hypothesis. Section 24 shows why even a four-ended frontier can have large necessary expanded multiplicity. The next algebraic refinement in report 24 prunes paths irrelevant to survivors and chooses propagation direction; its performance and topology obligations remain separate from this kernel's correctness.

# 26 Succinct transport, the earlier rank law, and repair dependencies

The remaining parts of report 23 overlap substantially with the integrated one-run recurrence of Section 23. We retain that backend and its established profile schema. The useful additions are exact transport for enormous integers, an independent structural replay, a sharper statement of the total-rank threshold, and a precisely conditional repair bound. None requires a second competing implementation of the same tail.

## 26.1 An earlier total-rank threshold does not give a smaller cap for free

Fix a selected signed run of magnitude $m$, and let the remaining expanded context have degree range $[a, b]$ and length $L = b - a$. For a positive run, the interior starts at degree $b + 1$. At $m = L + 1$ there is exactly one full chain slice $V$. At $m = L + 2$ there are two, with the canonically identified square-zero matrix $A : V \to V$ between them. Write $v = \dim V$, $\rho = \operatorname{rank} A$, and $\nu = v - 2\rho$.

All lower boundary matrices are fixed; all upper boundary matrices agree after translating degrees. Increasing $m$ from $L + 1$ inserts one more full slice and one matrix $A$, without changing any other matrix rank. The field identity

$$\dim H(C) = \sum_h \dim C^h - 2 \sum_h \operatorname{rank} d^h$$

therefore proves

$$R(m) = R(L + 1) + (m - L - 1)\nu \qquad (m \geq L + 1).$$

For a negative run use the reversed coordinate $q = -h$; the original cochain matrix still goes from $h$ to $h + 1$, so its central source is $a - 2$, as in the existing implementation. The formula is about total rank, not a decomposition of the single full slice's two boundary contributions. Exact degree profiles still use the $L + 2$ cap.

The integrated recurrence now reports $L + 1$ and $R(L + 1) = R(L + 2) - \nu$ in its compressed-result certificate. It still builds the $L + 2$ reference: the earlier equality by itself does not construct the missing square matrix or compute its rank from a single slice. No reference cost reduction is claimed. An independent referee run checked 392 explicit complexes: 96 signed contexts at four magnitudes plus eight sharp boundary examples. It checks canonical interior matrix equality, $d^2 = 0$, Euler characteristics, the rank law and full profiles against the ordinary macro builder.

The boundary examples $\sigma_1^{L+1}\sigma_1^{-L}$ are unknots, so a dominance rejection threshold cannot simply be lowered to $L+1$.

The cap also has a sharper elementary size estimate. A support with $r$ active turnbacks has at most $s+r$ closure circles on $s$ strands, since adding one turnback changes the count by at most one. Summing reduced dimensions over active-run choices gives, at selected magnitude $L+2$,

$$K \le 2^{s-1}(2L+5)\prod_{i\neq j}(1+2|e_i|) \le 2^{s-1}(2L+5)3^L.$$

For an original knot, every strand separator occurs. Apart from the selected generator at least $s-2$ distinct generators occur in the context, so $s \le L+2$. Finite-reference geometry, rank computations and compressed output consequently cost $2^{O(L)}\,\mathrm{poly}(N)$ for succinct input bit length $N$. This becomes quasi-polynomial when $L = O(\log^2(N+2))$. Reading an explicit word has its additional linear input cost. Discovering a small context for arbitrary diagrams is still unproved, and dominant-run knot decisions were already covered by the existing structural criteria.

## 26.2 JSON is part of the succinct representation contract

The previous Python API could compute exponents with tens of thousands of bits, but ordinary JSON decimal conversion could reject their input or output. This included degree *keys* of the finite exceptional profile, not just rank values. The maintained run frontend now accepts literal integers or explicitly signed hexadecimal strings. Above 4096 bits it emits hexadecimal strings for both integer values and integer dictionary keys; smaller values retain the ordinary JSON representation. No global decimal conversion limit is relaxed. Booleans, floats and ambiguous numeric strings are rejected. Distinct dictionary keys must retain distinct encodings.

The frontend accepts exactly one of `runs` and `word`; the latter is already an explicit list of signed generators. Neither path constructs a PD, and run input is never expanded. Recognition, homology and structural profile modes retain their existing meanings and resource outcomes. A final deadline check now covers the arithmetic after source counting, including successful structural answers.

`decode_degree_profile` reconstructs integer points and interval fields without enumerating the plateau. It rejects duplicate degrees after decoding. This function validates a transport schema, not a homology proof; the normal profile-query functions operate on its decoded Python result. The structural certificate verifier decodes only numeric fields prescribed by its exact schema, rejecting additional fields and boolean/float impostors. Its replay uses a separately written dense permutation and generator-support calculation, independent of the sparse producer. Before allocating the permutation it checks $s \le \#\mathrm{runs}+1$, which every knot source must satisfy. It recomputes the PD/Artin sign conversion and both Rasmussen intervals. This checks arithmetic premises, not the imported topology.

New regressions send $2^{20000}+1$ through the actual CLI with both signs, reconstruct the degree profile, and verify the exact rank while constructing only four reference basis vectors. A five-strand example with four balanced huge signed runs has no dominant exponent; its homogeneous source certificate round-trips and verifies while the producer is disabled by the test. Tampered fields, duplicate degree encodings, original component parity, and unchanged global decimal limits are checked. These changes enable an already-proved succinct computation to cross the public interface; they do not provide a new homology runtime improvement.

## 26.3 What an acyclic repair certificate would need to establish

The report's geometric proposal is an accounting theorem for a supplied fixed DAG, not a constructed hierarchy of a knot exterior. Suppose module $i$ has an integer counter $0 \le x_i \le g_i$. Every charged event strictly decreases its distinguished counter and can increase only counters at

its out-neighbors. Only these charged events may increase counters. Partition time at events of predecessors of $i$; within each interval $i$ has at most $g_i$ events. Thus the event counts obey

$$r_i \le g_i \left(1 + \sum_{j \to i} r_j\right).$$

Induction in topological order gives the weighted-path bound

$$\sum_i r_i \le \sum_{\pi \neq \varnothing} \prod_{v \in \pi} g_v \le \ell g \sum_{j=0}^{B} (dg)^j,$$

where $\ell$ is the number of vertices, $d$ bounds outdegree, $g$ bounds the numerical counter values, and $B$ bounds path length in edges. The exact path sum is computable by dynamic programming without enumerating paths. Polynomial $\ell, d, g$ and logarithmic $B$ give a quasi-polynomial number of charged events. A short binary encoding of an enormous $g$ is insufficient; the event count depends on $g$, not merely its bit length.

The maintained research diagnostic agrees with independent path enumeration on 380 fixtures and rejects cycles and invalid bounds. A complete DAG has bound $(g+1)^\ell - 1$, while a chain has $\sum_{j=1}^{\ell} (\ell - j + 1) g^j$; sparse edges alone do not control depth. For a knot algorithm one must still construct and validate the modules, fixed dependencies, counter bounds, and legal update rules. Disjoint local supports do not by themselves imply independence of global nonseparatingness, homology predicates or attachment certificates. Total work must also charge uncharged steps, event cost, and representation size. We retain this as a research target rather than installing an unsupported geometric scheduler.

# 27 Survivor corridors and adaptive transfer direction

Report 24 refines the full transfer of Section 25. It removes algebraic paths that cannot contribute to a surviving map and chooses which end of each remaining component to propagate from. The maintained implementation exposes this as `corridor` and `corridor-adaptive`; neither changes the default sparse reducer. The latter first permits $\max(256, 4N_{\mathrm{live}})$ ordinary Schur-update pairs in a stage. Completed pivots are retained when that allowance is exhausted, and the remaining complex is transferred exactly. The policy therefore responds to observed elimination work in the spirit of introsort. Unlike introsort's fallback theorem, it supplies neither a competitive runtime guarantee nor a bound on the number of objects generated by a scan.

## 27.1 From perturbation transfer to a directed graph

Write $d = d_0 + \delta$, with $d_0$ the weight-zero scalar differential, and let $(I, P, H)$ be its typed binary special contraction. Here $H$ preserves matching type and quantum shift and lowers homological degree by one; $\delta$ raises homological degree by one and strictly raises quantum shift. For the degree-$h$ part of

$$D = P\delta(1 + H\delta)^{-1} I = \sum_{r \ge 0} P\delta(H\delta)^r I,$$

make one vertex $z_a$ for each original degree-$h$ object and one $w_b$ for each degree-$(h+1)$ object. Include terminal vertices $s, t$ for the survivors in those degrees. Its edges are precisely

$$s \xrightarrow{I_{as}} z_a, \qquad z_a \xrightarrow{\delta_{ba}} w_b, \qquad w_b \xrightarrow{H_{ab}} z_a, \qquad w_b \xrightarrow{P_{tb}} t,$$

omitting zero entries. The $I, H, P$ edges carry typed identities, whereas the radical edges retain their actual cobordism coefficients. Ordering by quantum shift, and at equal shift by phase $w, s, z, t$, makes every edge go forward. Thus this graph is acyclic even when the homological degrees of the survivors are consecutive.

**Proposition 27.1** (Exact survivor restriction)**.** *The matrix entry $D_{ts}$ is the sum of the ordered coefficient products over all directed paths from $s$ to $t$. Deleting every vertex not both reachable from an input and able to reach an output preserves all entries of $D$.*

*Proof.* Expansion of $P\delta(H\delta)^r I$ gives exactly the paths with $r$ homotopy edges. Quantum acyclicity makes the expansion finite. A path contributing to any matrix entry has both required reachability properties at each vertex, so deleting the other vertices removes none of its terms. Conversely every retained path was already present. An edge between two retained vertices lies on a complete path by concatenation, so retaining all such edges is safe. □

Reachability is computed by Boolean union, not by coefficient XOR. It is an overestimate of algebraic support: equal path products can cancel and nonzero factors can have zero composite. In particular, neither a nonempty corridor nor the scalar survivor multiplicities decide a homology rank. The implementation evaluates the entire remaining differential.

The weak components of the retained graph include its terminal vertices. They give disjoint blocks of this adjacent-degree map. Within a block, forward propagation accumulates each source's coefficients in topological order; reverse propagation accumulates each target's suffix coefficients in reverse order. For an edge $u \xrightarrow{f} v$, the latter forms $g \circ f$ when the current suffix is $g : v \to t$. It does not reverse the order of multiplication or assume that differently typed coefficients commute. Distributivity and associativity show that both evaluations give the same finite path sum.

## 27.2 What the direction heuristic bounds

Let $L(u)$ be the set of sources reaching $u$ and $R(v)$ the set of targets reachable from $v$. For a retained component $G_j$, define

$$F_j = \sum_{(u,v)\in E_j} |L(u)|, \qquad B_j = \sum_{(u,v)\in E_j} |R(v)|.$$

The `reach` policy constructs these endpoint sets as packed integers and chooses forward when $F_j \le B_j$, reverse otherwise.

**Proposition 27.2** (Propagation bound)**.** *The number of coefficient edge propagations under this policy is at most $\sum_j \min(F_j, B_j)$. If component $j$ has $a_j$ inputs and $b_j$ outputs, the alternative* `ports` *policy uses only Boolean reachability flags and bounds the propagation count by $\sum_j |E_j| \min(a_j, b_j)$.*

*Proof.* Forward propagation from one source processes an edge at most once, and only if that source reaches its initial vertex. Summing gives $F_j$. The suffix argument gives $B_j$. The implemented choice takes their minimum. Replacing the two reach-set sizes by the total numbers of ports gives the second bound, choosing the side with fewer ports. □

These bounds count attempted edge propagations, not cache misses or seconds. The code bypasses typed identities and caches equal products. It also scans every ordered vertex of a component once per chosen root, including vertices whose current coefficient is zero. If $r_j$ is the number of roots on the chosen side, that costs $O(\sum_j |V_j| r_j)$ dictionary visits. The `reach` choice need not minimize this term or choose the side with fewer ports. Nor does its bound include scalar setup and endpoint masks.

## 27.3 Sparse scalar components and complete costs

The scalar graph has only equal-matching, equal-shift identity arrows. The `components` engine finds its weak components and contracts each separately. Isolated vertices are survivor identities; a

component consisting of one scalar arrow has only its reverse homotopy entry. Other components use the existing binary contraction on local indices. The result stores sorted tuples of nonzero indices for $I, P, H$, including empty tuples at dead slots, instead of integers indexed by all old objects.

This changes representation without changing the established scalar basis. Within a matching/shift/degree group, elimination sees source and target indices in their original order. Operations in disjoint scalar components cannot interact. Each selected kernel representative has a distinct largest original source index, its dependent-column anchor. Sorting group keys and restoring anchor order therefore reproduces the original representative order. The same permutation of projection columns, and unchanged local homotopy coefficients, recover all three original maps entrywise.

Let $N$ count allocated slots, including dead slots; let $E_{\mathrm{in}}$ count input differential entries, and $M_{\mathrm{sc}}$ count nonzero scalar-map indices. If scalar components have sizes $s_i$, a conservative binary setup bound is, up to polynomial factors in index and label bit lengths,

$$A_{\mathrm{sc}} = O\left(N + E_{\mathrm{in}} + M_{\mathrm{sc}} + \sum_{i:s_i>2} s_i^3\right).$$

The scan of $E_{\mathrm{in}}$ is necessary to discover scalar entries. Across adjacent degrees, graph sizes satisfy $V = O(N+S)$ and $E = O(E_\delta + M_{\mathrm{sc}})$, where $S \le N$ is the number of survivors. Graph construction, sorting, Boolean reachability, and union by size with path compression cost near-linear work in this explicit representation, apart from index and label arithmetic.

For the `ports` policy the remaining bound has the form

$$T \le A_{\mathrm{sc}} + A_{\mathrm{graph}} + \Lambda \sum_j |V_j| r_j + \Lambda(1 + C(k)) \sum_j |E_j| r_j, \qquad r_j = \min(a_j, b_j),$$

where $\Lambda$ charges index/dictionary arithmetic and $C(k)$ bounds coefficient work at $2k$ boundary points. A conservative coefficient envelope is $\mathrm{poly}(k)2^{O(k)}$. The `reach` version replaces the edge term by $\sum_j \min(F_j, B_j)$, but uses its actually chosen $r_j$ in the vertex term and additionally pays up to $O((V+E)S)$ bit work for endpoint masks and population counts. Python hash-table timings are empirical; ordered dictionaries give a deterministic accounting with polynomial index overhead.

Sparse indices matter even for a singleton column: an integer $2^j$ uses $j+1$ bits of payload. Many such global columns can have quadratic total bit length while their sparse representation has only linearly many indices. Conversely, a dense connected scalar block can still require large local binary matrices. This optimization does not remove all fill.

## 27.4 A strict separation, and its scope

Fix the three-arc dot algebra $\mathbb{F}_2[x, y, z]/(x^2, y^2, z^2)$. Take $R$ degree-zero survivors $s_j$ at shift zero and one degree-one survivor $t$ at shift six. Add identity pairs $L_0 \to B_0$ at shift two and $L_i \to B_i$ at shift four for $1 \le i \le M$, with $L$ in degree zero and $B$ in degree one. The only radical maps are

$$s_j \to B_0 : x, \qquad L_0 \to B_i : y, \qquad L_i \to t : z.$$

This is a homogeneous two-term complex, hence $d^2 = 0$. For odd $M$ its transferred entry is $D_{ts_j} = Mxyz = xyz \neq 0$. Forward propagation visits $R(1 + 2M)$ radical edges, while reverse propagation visits $R + 2M$. The latter accumulates the common suffix once before distributing it to the $R$ sources. For even $M$ the output vanishes by parity despite unchanged structural reachability, as a regression test checks.

There are $R + 2M + 3$ original objects, $R + 1$ isolated scalar survivors, and $M + 1$ identity pairs. Sparse scalar setup has no nontrivial binary blocks. Using `components` and `ports`, the single-target

graph is evaluated once in reverse; all coefficients lie in a fixed eight-dimensional algebra. Including input reading, scalar setup, sorting and output writing, the stage takes $O((R+M)\log(R+M+2))$ index-word operations and $O(R+M)$ words. The previous forward evaluation alone takes $\Omega(RM)$ propagations. At $R = M = 255$, the radical counts are 130305 and 765; the contraction uses 768 sparse indices versus 164735 packed payload bits. These are representation counts, not heap measurements.

This is a separation between complete transfer implementations on an explicit algebraic family. Its realization by knot prefixes is unproved, and minimum-fill sparse cancellation is already efficient on the family. The default public corridor policy still uses `reach` and the global scalar engine; the near-linear theorem refers specifically to the direct `ports/components` configuration.

## 27.5 Integration, validation, and measured limits

The new reducers support the standard scanner, minimum-fill pivots, bit coefficients, the existing component coefficient engines, and tail contraction. Races, other scanner backends, and window truncation are rejected before early certificates. Window integration needs its own quantum-shift and truncation argument. Global resource exhaustion raises the ordinary scan limit in the raw API and yields `UNKNOWN` in recognition. Full replacement arrays, reverse indices, and an empty cache are allocated before a final deadline check and installation. Tests inject both allocation failure and expiry after transfer, then resume successfully on the preserved complex. Completed earlier sparse pivots remain valid.

The maintained independent audit compares seven configurations entrywise against the frozen forward reference on 555 prefixes from 80 diagrams: 3885 comparisons, including partially cancelled inputs. All eight special contraction identities are checked at each prefix. Separate tests compare graded homology, continued scans and independent set coefficients. The full maintained suite passes 458 tests. The stronger graded support-only certificate uses the condition

$$\Delta > 0, \quad k - c(a,b) \le \Delta \le k + c(a,b), \quad \Delta \equiv k - c(a,b) \pmod 2$$

for a possible radical map. Its audit accepts 196 of 639 stages in 72 diagrams, exactly the same stages as the older degree-gap shortcut. The experimental support reducer is retained for comparison, not substituted for an established production policy.

The delivered package lacks its advertised full SHA-256 manifest. The maintained reproduction wrapper verifies the separate 20-file integration manifest and runs the archive in a temporary copy: 268 scanner tests, the transfer and support audits, and the independent surface-cover suite all pass. This is a narrower provenance claim than verifying every file. The surface-cover theorem, reviewed and extended in Section 29, has a separate presentation contract and is not used to make a knot verdict in this integration.

| Raw scan | Std/corridor | Std/adapt. | Std/sparse | A/A | Switches |
|---|---|---|---|---|---|
| two_strand_201 | 0.105 | 0.808 | 0.084 | 1.019 | 0 |
| torus_3_10 | 0.327 | 0.932 | 0.305 | 0.987 | 0 |
| alternating_3_4 | 0.348 | 0.938 | 0.302 | 1.017 | 0 |
| morton | 0.422 | 0.927 | 0.411 | 1.013 | 0 |
| conway | 0.341 | 0.923 | 0.309 | 0.991 | 0 |
| hard_unknot_8 | 0.500 | 0.918 | 0.490 | 0.986 | 0 |

Sparse denotes the component-scalar/Boolean-port configuration. Ratios above one favor the denominator.

| Constructed stage | Fwd/corridor | Fwd/sparse | Std/sparse | A/A |
|---|---|---|---|---|
| source_heavy | 2.390 | 2.214 | 0.340 | 0.988 |
| sink_heavy | 0.461 | 0.426 | 0.336 | 1.016 |
| balanced | 0.623 | 0.592 | 0.299 | 1.014 |
| dead_branches | 6.234 | 5.457 | 1.188 | 0.993 |
| shared_suffix_15 | 1.303 | 1.558 | 0.204 | 1.017 |
| shared_suffix_63 | 4.194 | 4.859 | 0.172 | 0.981 |
| shared_suffix_255 | 15.661 | 18.141 | 0.154 | 1.041 |
| dead_leaves_255 | 0.677 | 0.860 | 0.187 | 0.993 |

Fwd denotes the maintained quantum-ordered forward transfer. These timings include scalar and graph setup.

All six actual scans favor ordinary sparse cancellation: eager corridor transfer is approximately 2.0–9.5 times slower. The adaptive policy switches zero times, with standard/adaptive ratios 0.808–0.938; its quantum bookkeeping therefore buys no transfer benefit in these cases. The sparse-scalar/port refinement also loses on every actual scan. Actual standard/control ratios span 0.986–1.019. On the constructed size-255 shared-suffix case, the sparse refinement is 18.14 times faster than the maintained forward transfer, but ordinary sparse cancellation is still approximately 6.49 times faster than that refinement. The constructed dense dead-branch case gives a different result: eager corridor and adaptive corridor beat ordinary cancellation by factors 1.406 and 1.380, respectively. Other constructed cases include regressions. These observations support retaining optional policies and publishing their negative results, rather than changing the default or claiming a uniform timing improvement.

The measurements use fresh states, seven shuffled paired warm rounds, and identical standard/control arms. Actual raw scans include construction of the scanner, quantum tracking and all reduction work; diagram construction and shared crossing-order choice are outside timing. Synthetic timings include the complete reduction stage, with fixture construction outside timing. Early recognition filters are bypassed, so these are not end-to-end recognition speedups. Raw samples, repetition counts, exact checks and source hashes are retained with the result file.

Finally, a scan with every allocated $N_i \leq \exp(O(\log^2(n+2)))$ and every $k_i = O(\log^2(n+2))$, suitably bounded labels and an affordable order, would meet the target by the displayed bounds and ordinary extension costs. No such uniform bounds have been proved for arbitrary diagrams. Corridor size also depends on the scalar basis; it is not a knot invariant. Faster evaluation of an explicit stage leaves the central multiplicity and representation problems unresolved.

The weaker general sub-exponential target is also unproved for this implementation. Here sub-exponential means $2^{o(n)}$ in the crossing count, not merely a speedup over the previous program. Under the same polynomial label-size assumptions, the stage accounting would suffice if both $\max_i k_i = o(n)$ and $\max_i \log(N_i + 2) = o(n)$, with an order found in sub-exponential time. No such simultaneous guarantee has been established. Adaptive switching among the current representations does not supply it. This implementation status must also be distinguished from Lackenby's announced quasi-polynomial result: the July 2026 hierarchy paper explicitly leaves some step runtimes unspecified in its Section 9 [24].

## 28 Certified cyclic Garside preprocessing

Report 25 extends the earlier rank-two interval optimizer to arbitrary supports in an Artin braid group. The maintained implementation shares exact prefix arithmetic across cyclic cuts and target words. It supplies an optional source-braid probe, not a new knot invariant. A successful proposal must shorten the current simplified diagram, must pass independent replay, and restarts

recognition under the remaining global budget. The probe follows the existing braid, Seifert, descending, and modular Alexander obstructions; its default setting is disabled.

## 28.1 Exact states and a separately checked equality witness

Let $b$ be the strand count and $\Delta$ the positive half-twist of length $B = \binom{b}{2}$. Classical Garside theory gives a unique left-greedy form

$$g = \Delta^z s_1 \cdots s_r,$$

where $z \in \mathbb{Z}$ and the $s_i$ are proper nonidentity simple braids. A simple braid crosses each pair at most once and is determined by its permutation. These are classical facts, not new compression theorems; see Dehornoy [42]. The implementation stores complete permutation arrays and the integer power. Sorting and binary search compare these exact states; neither a permutation quotient nor a hash fingerprint is used to decide braid equality.

Our convention is $p[i]$ = bottom position of top strand $i$. For two adjacent simple factors $A, C$, a positive atom $\sigma_{i+1}$ can move from the start of $C$ to the end of $A$ exactly when

$$C[i] > C[i+1], \qquad A^{-1}[i] < A^{-1}[i+1].$$

The move swaps values $i, i+1$ in $A$ and positions $i, i+1$ in $C$. The first condition factors $C = \sigma_{i+1}C'$ and reduces simple length by one; the second makes $A\sigma_{i+1}$ simple and increases its length by one. Thus the local product is unchanged. An identity right factor is deleted. Positive input letters append an atom; negative letters use

$$\sigma_i^{-1} = \Delta^{-1}(\Delta\sigma_i^{-1}), \qquad S\Delta^{-1} = \Delta^{-1}\tau(S), \qquad \tau(p)[i] = b - 1 - p[b-1-i].$$

After transfers make all pairs left weighted, leading half-twists are extracted into $z$. Any half-twist factor must then be initial, since a left-weighted factor immediately before $\Delta$ must itself be $\Delta$.

**Proposition 28.1** (Elementary normalization bound)**.** *Normalizing an $m$-letter signed word uses at most $Bm(m-1)/2$ atom transfers, at most $m$ simultaneous simple factors, and $O(m^2b^3)$ index-word operations. Appending one atom to a normal state with $r$ factors costs $O((r+1)^2b^3)$ conservatively.*

*Proof.* For a temporary factor list indexed from zero, use $\Phi = \sum_j j\,\ell(s_j)$. Each transfer decreases it by one before any deletion. Deletion and leading half-twist extraction cannot increase it, and $\tau$ preserves simple length. The $k$th append raises it by at most $B(k-1)$ and creates at most one factor. Summing proves the transfer bound. Each pair test or transfer scans $O(b)$ permutation positions; failed tests advance the scan index, and a successful transfer backs it up by at most one. List copying, twisting and deletion fit the stated quadratic bound. Starting with $r$ factors instead gives initial potential $O(B(r+1)^2)$ and the single-append bound. □

The certificate records local transfer positions, generator indices, and half-twist extraction counts. A separate verifier imports neither the normalizer nor the optimizer. It applies the elementary permutation operations only after checking their inequalities and indices. Therefore it proves equality of the represented braid throughout the trace, even without trusting normal-form uniqueness or the optimizer's choices. Version one checks equality of a source interval to the identity or a signed generator. Version two checks that source times inverse target replays to the identity. It also checks source binding, rotation, disjoint intervals, strict shortening, target radius, and reconstructed output. The digest is an input-binding aid; replay on the actual source supplies the equality proof. It certifies neither kernel optimality nor a knot verdict. A cyclic cut preserves oriented closure by conjugacy, while uncut substitutions preserve the braid element itself.

## 28.2 Sharing the algebra across cuts and targets

Write $w$ for the input of length $n$, and $P_j$ for the exact prefix products of the doubled word $ww$. A source interval $[i, j)$ equals a target word $v$ precisely when

$$P_i^{-1} P_j = v \quad \Longleftrightarrow \quad P_i = P_j v^{-1}.$$

This ordering matters in a noncommutative group. The doubled table lets all $n$ cyclic cuts query the same prefix states. The radius-$\rho$ dictionary contains every literal signed word of length at most $\rho$, including the empty word; its size is

$$M = \sum_{d=0}^{\rho} (2b-2)^d.$$

Different words may represent the same braid, which causes no correctness problem. A trie of reversed inverse target words shares the appends needed to evaluate $P_j v^{-1}$ for one endpoint. Canonical query states are looked up in the sorted prefix table.

For a fixed cut $s$, let $F[j]$ be the maximum saving on the first $j$ letters. Immediately before processing $j$, maintain

$$H(q) = \max_{0 \le i < j:\ P_{s+i} = q} (F[i] - i).$$

The recurrence is

$$F[j] = \max\left(F[j-1],\ \max_{v \in D}\{j - |v| + H(P_{s+j} v^{-1})\}\right).$$

Only afterwards insert $F[j] - j$ under $P_{s+j}$, excluding empty source intervals. Backpointers are updated only on a strict improvement.

**Proposition 28.2** (Optimal disjoint one-pass kernel)**.** *The recurrence returns the shortest word obtainable by disjoint interval substitutions to the chosen dictionary at that cut. Minimizing across all cyclic cuts gives the corresponding cyclic optimum.*

*Proof.* An optimal prefix plan either retains its last letter or ends with a replacement $[i, j) \mapsto v$. The latter saves $F[i] + j - i - |v|$ and is permitted exactly by the prefix-state equality. Conversely every term extends a valid optimal prefix plan. Induction proves the recurrence. A nonshortening substitution has saving at most $F[i] \le F[j-1]$ and cannot win strictly, so recorded intervals strictly shorten. Backtracking to their starts makes them disjoint. Any circular disjoint plan admits a cut at a block boundary or an unused boundary, including a block covering the entire circle, and is therefore among the tested cuts. □

This is not optimization over arbitrary sequences of substitutions, conjugations or Markov moves. It extends the earlier interval dynamic program by supplying exact unrestricted group queries. The lower bound

$$k \ge \max\left(|\exp(w)|, b - \#\text{cycles of the strand permutation}\right)$$

allows an exact early exit: each letter changes exponent by one and changes permutation cycle count by at most one. Both quantities are unchanged by conjugacy. In particular, a knot closure needs at least $b - 1$ letters.

### 28.3 Height-sensitive costs and the small-core theorem

Let $R = \max_{j \le 2n} \ell_c(P_j)$, where $\ell_c$ counts proper simple factors. Inversion preserves this length, and multiplication is subadditive: move half-twist powers left, concatenate factor lists, and normalize without increasing their number. Thus interval prefixes have height at most $2R$, and target queries at most $R + \rho$; always $R \le 2n$. The following conservative word bound includes a winning proof and replay:

$$T = O\big(nM(R+\rho+1)^2 b^3 + nM(R+\rho+1)b\log(n+2) \\ + n^2 M + M\rho\log(M+2) + n(R+\rho+1)b^3\big).$$

The first term charges the $O(nM)$ exact append queries, the second their full-state comparisons, and the third all per-cut scalar dynamic programs. The next term builds the target dictionary and trie. In the last term, disjoint winning source intervals total at most $n$ letters and their strictly shorter targets total at most $n$; the height bound and transfer potential therefore charge proof generation and replay to $O(n(R+\rho+1)b^3)$. Proof size is $O(n(R+\rho+1)b^2)$ elementary records. Search workspace, excluding that output, is

$$O\big(n(R+1)b + nM + M\rho + \rho(R+\rho+1)b\big)$$

words. Each index word occupies $O(\log(n+b+\rho+M+2))$ bits, giving the corresponding bit-storage bound. Elementary index arithmetic adds at most a polynomial in this logarithm to the word-operation cost in bit time. With fixed $\rho$, the computation is polynomial in $n, b$; leading conservative terms are $O(n^3b^4)$ for radius one and $O(n^3b^5)$ for radius two. An arbitrary binary-encoded radius is not a fixed parameter: its dictionary is exponential in $\rho\log(b+1)$.

A cheaper portfolio stage first expands the whole-word normal form only if its calculated Artin length is smaller than the input. It independently proves this equality. If the candidate attains the universal lower bound, it returns immediately; otherwise it compares that candidate with the cyclic kernel. The returned word is no longer than the kernel optimum, but need not itself be an optimal radius-specific interval plan. Evidence distinguishes `whole-normal-form` from `cyclic-kernel`.

**Proposition 28.3** (Constructive restricted recognition bound)**.** *For fixed radius $\rho$, uncapped certified preprocessing followed by an uncapped conventional exact recognizer has cost*

$$\mathrm{poly}_\rho(n,b) + \mathrm{poly}(k)2^{O(k)},$$

*where $k$ is the resulting word length. It is quasi-polynomial on families whose cyclic radius-$\rho$ optimum is $O(\log^2(n+2))$.*

*Proof.* The optimizer and verifier have the polynomial bounds just given. Closure is preserved and remains one component, so $b \le k+1$. A conventional Khovanov cube on the resulting $k$ crossings has $2^k$ resolutions with at most $b+k$ circles each. Hence its enhanced-state dimension is $2^{O(k)}$, and exact matrix ranks over $\mathbb{F}_2$ can be obtained in polynomial work in that dimension. Reduced rank one detects the unknot: the rational unknot-detection theorem and the universal coefficient theorem imply the same rank-one criterion over $\mathbb{F}_2$. The maintained exact fallback supplies this decision. Substitution of the small-core bound gives $2^{O(\log^2(n+2))}$. Local probe cutoffs are disabled in this theorem. □

There is a proved family for which the decomposition need not be supplied. Partition a short core into blocks of length at most $\rho$, replace each by a no-shorter equal braid word, and insert disjoint identity blocks, forming $y$. Surround it by $u, v$ with $vu = 1$, forming $uyv$. A cyclic cut exposes $yvu$; deleting the sleeve and identity blocks and shortening the inflated core blocks is an admissible

disjoint plan. The optimizer discovers a result no longer than the original core without being told this decomposition. This generalizes rank-two inflation to arbitrary supports and multi-letter targets. It does not establish a small core for every unknot or a general sub-exponential recognition algorithm.

## 28.4 Two limitations of short-word approaches

First, increasing target radius can be essential even for an unknot. In $B_4$ write $a = \sigma_1$, $b = \sigma_2$, $c = \sigma_3$ and, for $m \geq 2$,

$$X_m = a^{-m}c^m a^{m+1}c^{-(m-1)}, \qquad W_m = X_m b.$$

Since $ac = ca$, $X_m = ac$ and the closure of $W_m$ is the unknot. Nevertheless, the cyclic radius-one optimum is $4m + 1$, while radius two returns three letters. In $\langle a, c\rangle \cong \mathbb{Z}^2$, the prefix path visits successively $(0, 0), (-m, 0), (-m, m), (1, m), (1, 1)$. It is simple and has no nonconsecutive lattice-neighbor vertices. Thus no interval in $X_m$ is an identity or a long signed-letter replacement. An interval containing the sole $b$ has form $pbq$ with $p, q$ in this commuting subgroup. The permutation double cosets force both $p, q$ to be pure if this equals a signed $b$. Write $p = a^{2r}c^{2s}$ and $q = a^{2u}c^{2v}$. Signed crossings between original strand labels occur on pairs $12, 34, 13, 24$ with counts $2r, 2s, 2u, 2v$ respectively, and pair 23 has count $+1$. These equality invariants force $p = q = 1$ and the positive sign. The simple prefix path then permits only the original single letter, which does not shorten. This covers intervals at every cyclic cut. Radius two replaces $X_m$ by $ac$; the four-cycle permutation requires at least three output letters. A whole-word normal form also exposes the short simple braid, so the practical portfolio avoids the larger dictionary on this family. The theorem separates interval rules, not all braid algorithms.

Second, preserving the braid element itself cannot universally give a polylogarithmic core, even at three strands. In $B_3$, the unknot braid

$$G_m = a^{m+1}ba^{-m} = a^m(ab)a^{-m}$$

has minimum equal-braid Artin length exactly $2m + 2$. To see this, quotient $B_3$ by its center as $C_2 * C_3$, with $x = aba$, $y = ab$. The image of $G_m$ is $(y^2x)^m y(xy)^m$, of reduced syllable length $4m + 1$. Each signed Artin letter contributes at most two syllables, so an equal word has length at least $2m + 1$. Exponent sum two forces even length, giving $2m + 2$, attained by $G_m$. A shortening subinterval would contradict this geodesicity. Cyclic rotation, however, exposes the sleeve $a^{-m}a^m$ and leaves $ab$ of length two. Existing free cyclic cleanup already handles this example. The obstruction applies to equality-only explicit words, not to compressed representations or closure-preserving transformations in general.

## 28.5 Production contract and empirical results

The `-garside` option retains the original checked source braid through PD simplifications. The proposal compares its verified output length with the current PD's crossing count, not merely with source length. An accepted source branch and the earlier PD move trace are recorded separately; they are not concatenated into a fictitious move sequence. Restart preserves backend, coefficient, reduction, window, and other user options, subtracts elapsed time from the global allowance, and disables this probe to prevent recursive reapplication.

Search, dictionary construction, and independent replay share a cooperative local deadline and operation allowance. The host's global check always runs first. A local limit or allocation failure skips the probe; global exhaustion yields `UNKNOWN`. Invalid certificates propagate as errors. Even the lower-bound shortcut checks resources. These are cooperative limits, not hard process-memory or interrupt-latency guarantees; sorting and individual array operations can be noninterruptible.

The existing modular Alexander obstruction runs before the optional probe. When it is inconclusive and the probe declines, this result is reused on the unchanged single-factor diagram. Individual factors receive their own checks. The selected Jones and requested exact Alexander stages follow the probe: successful compression can avoid their work on the larger PD. This ordering avoids spending a search allowance on a knot already detected by the inexpensive modular test, while retaining the opportunity to shorten before a larger invariant computation. Source-free PD input bypasses the probe. No production default changes.

The delivered suite passes 47 tests, including 7016 short-word normal forms checked against a faithful Artin action and 1512 independent interval optimization comparisons. All 45 archive hashes and six inspected source Git blobs were checked. Three blobs are tied to the named source commit and path; three were recorded only by exact blob. The previously unexecuted adapter smoke now passes. Production adds normalized-homology comparisons on 60 small knot closures, independent replay without the normalizer, backend/reduction restarts, corrupted certificates, shared resource hooks, and filter-priority/cache-reuse tests. All 517 maintained tests pass. These are executable checks, not formal verification of Python semantics or exhaustive knot classification.

| Recognition input | $n$ | Off/radius 1 | Off/radius 2 | A/A |
|---|---|---|---|---|
| barrier_h4 | 51 | 0.979 | 0.971 | 1.003 |
| rectangle_m16 | 65 | 0.977 | 1.007 | 1.032 |
| geodesic_m12 | 26 | 0.865 | 0.888 | 0.981 |
| hidden_sleeve_m8 | 24 | 0.841 | 0.823 | 1.008 |
| unknot_h2_u8 | 43 | 1.018 | 1.037 | 0.989 |
| unknot_h2_u16 | 59 | 0.811 | 0.814 | 0.997 |
| figure8_h2_u8 | 45 | 1.099 | 1.073 | 0.998 |
| figure8_h2_u16 | 61 | 1.102 | 1.107 | 1.014 |
| positive_h2_u8 | 49 | 0.775 | 0.763 | 0.961 |
| positive_h2_u16 | 65 | 0.834 | 0.827 | 1.018 |
| unknot_h8_u8 | 115 | 2.000 | 1.974 | 1.000 |
| unknot_h8_u16 | 131 | 2.053 | 2.042 | 0.997 |
| figure8_h8_u8 | 117 | 1.094 | 1.099 | 1.005 |
| figure8_h8_u16 | 133 | 1.096 | 1.092 | 1.006 |
| positive_h8_u8 | 121 | 0.870 | 0.876 | 1.000 |
| positive_h8_u16 | 137 | 0.882 | 0.869 | 1.000 |
| unknot_h16_u8 | 211 | 1.529 | 1.523 | 1.004 |
| unknot_h16_u16 | 227 | 1.340 | 1.327 | 0.988 |
| figure8_h16_u8 | 213 | 1.077 | 1.098 | 1.004 |
| figure8_h16_u16 | 229 | 1.099 | 1.096 | 0.996 |
| positive_h16_u8 | 217 | 0.895 | 0.902 | 0.975 |
| positive_h16_u16 | 233 | 0.911 | 0.911 | 0.997 |
| conway | 11 | 0.960 | 0.982 | 0.993 |
| hard_unknot_8 | 8 | 0.750 | 0.749 | 0.996 |
| stress_braid5_36 | 36 | 1.182 | 1.196 | 1.046 |

Ratios are medians of paired complete-recognition times; values above one favor the enabled policy.

| Isolated compressor | $n$ | Repeated/shared | A/A |
|---|---|---|---|
| single_crossing | 1 | 0.891 | 1.004 |
| positive_4_braid | 9 | 3.901 | 0.983 |
| ranktwo_barrier_h1 | 15 | 7.769 | 1.016 |
| ranktwo_barrier_h4 | 51 | 22.510 | 1.000 |
| geodesic_m12 | 26 | 6.231 | 1.009 |
| hidden_sleeve_m8 | 24 | 6.441 | 1.000 |
| seeded_5_braid | 22 | 9.832 | 0.979 |

Both compressor arms search all cuts without lower-bound pruning and generate and replay only the winning certificate.

The complete-recognition experiment keeps the normal filters enabled and uses fresh validated diagrams, with their construction outside timing. Probe search, replay, candidate construction, and restarted recognition are included. Seven shuffled paired warm rounds compare disabled, identical control, and radius-one/radius-two enabled policies. The local allowance is 0.1 seconds and 100000 ticks, with 100000 target words; all arms have the same two-second global and 50000-object ceilings. An exhausted local probe is not censored if the remaining exact pipeline completes; a global `UNKNOWN` would receive no speedup ratio. Raw samples, repeats, verdicts, stages and source hashes are retained. Isolated compressor measurements are reported separately and do not imply the same gain in recognition.

All 25 inputs complete with agreeing verdicts in every measured arm. The four larger inflated-unknot fixtures have radius-one gains of 1.34–2.05 relative to the disabled pipeline. The two smaller ones have ratios 1.018 and 0.811, so successful shortening need not reduce total work. Several inputs are already decided before any probe; enabled-option overhead makes some very short certificates slower, with a ratio as low as 0.750. The identical-control ratios range from 0.961 to 1.046. These observations support an optional policy, not a universal or default performance win. On modular-Alexander-detected examples the enabled path returns before factorization, which also accounts for modest gains unrelated to Garside arithmetic. Source-free PD inputs do not run the probe.

Two retained diagnostic experiments explain the placement choice. Putting compression before modular Alexander made some already-detectable knots roughly 140 times slower. Moving all invariant filters before it removed that regression but lost the useful inflated-unknot gains, because Jones work on the larger PD had already been spent. The final placement lies between modular Alexander and Jones. These diagnostics retain their own source hashes and are not pooled with the final timings.

The isolated shared-prefix compressor improves by 3.90–22.51 on the six multi-crossing fixtures, but loses on the one-letter control. At the 51-letter barrier, sharing uses 714 algebra appends instead of 18207. That mechanism result is distinct from the full recognition table: the ordinary pipeline already decides the displayed barrier without invoking the compressor. Radius two does not deliver a consistent recognition gain on this sample, since the inexpensive whole-normal-form proposal often already attains the lower bound.

## 29 Compressed surface covers and ordered attachment points

Report 24's second contribution concerns geometric compression rather than chain cancellation. Its dihedral-cover classifier is now maintained in `surface_cover.py`. We also extend it with exact comparison of components carrying an ordered list of points over a fixed basepoint. This is a possible interface for part of the attachment bookkeeping in a future hierarchy algorithm. It currently receives a covering presentation; it does not extract one from a knot exterior and

supplies no knot verdict.

The general correspondence between connected covers and transitive fundamental-group actions, with equivariant bijections representing cover isomorphisms, is classical [17, Section 1.3]. The formulas below specialize it to binary affine data. They are not a replacement for general normal-surface orbit algorithms: Agol–Hass–Thurston already give polynomial algorithms for orbit counting and associated normal-surface topology [18]. Our new marked signatures have the narrower global dihedral input promise stated here.

## 29.1 A checked presentation and three unmarked families

Let $F$ be a connected compact surface with $b \geq 1$ boundary circles and free fundamental group of rank $r = 1 - \chi(F)$. The input specifies its orientability, genus (crosscap number when nonorientable), and $b$, together with $r$ monodromies on a fibre of binary size $W$:

$$g_i(x) = \epsilon_i x + a_i \pmod{W}, \qquad \epsilon_i \in \{1, -1\}.$$

Each assignment defines a cover because this fundamental group is free. Peripheral words are generated from the canonical surface schema: the last is the inverse of the product of handle commutators, or crosscap squares, followed by the first $b - 1$ boundary generators. Arbitrary supplied words are not accepted as a peripheral-system certificate. Closed bases are outside this interface. The disk is included.

If all signs are positive, put $d = \gcd(W, a_1, \ldots, a_r)$ and $m = W/d$. Otherwise fix $R(x) = -x + a_0$ and take the gcd of $W$, all positive shifts, and all differences $a_i - a_0$ for negative generators. In either case the translation subgroup is generated by $T(x) = x + d$. Products of two reflections give the differences, and conjugating a translation by a reflection negates its shift; Bezout supplies the converse containment.

Without reflections there are $d$ components of degree $m$, all isomorphic over $F$. With a reflection, components are the orbits of $r_0 \mapsto a_0 - r_0$ on residues modulo $d$. Let $f$ be the number of solutions of $2r_0 = a_0 \pmod{d}$. There are $(d - f)/2$ paired-residue components of degree $2m$ and $f \leq 2$ fixed-residue components of degree $m$. All paired components are equivariantly isomorphic: for two nonfixed residues $r_0, s_0$, send

$$r_0 + jd \mapsto s_0 + jd, \qquad a_0 - r_0 + jd \mapsto a_0 - s_0 + jd.$$

These maps commute with $T, R$, hence with every monodromy generator. Thus at most three records, with binary multiplicities, suffice.

When two fixed residues occur, their covers are isomorphic exactly when $m$ is odd. An equivariant map between their transitive translation orbits must be a translation by $A = d/2 + dt$. Commutation with $R$ requires $2A = 0 \pmod{W}$, or $2t = -1 \pmod{m}$. This is solvable exactly for odd $m$. The records retain distinct component labels even when their cover types coincide. Type equality never means that external attachments can be discarded. The three-type bound is sharp: for a genus-one base with two boundaries and maps $x + d, -x, -x$, even $d \geq 4, m \geq 2$ give component genera $m + 1, m/2, m/2 + 1$ and boundary counts $2m, m + 2, m$.

## 29.2 Orientation and peripheral information are retained

An affine sign is not the orientation character of a surface loop. Attach the actual character $w_i \in \mathbb{Z}/2$ to every generator. For the translation increments $u_j$, retain bits $v_j = w_j$ for positive generators and $v_j = w_j + w_0$ for reflection differences. An extended gcd identity gives $(d, \tau)$ as a combination of $(W, 0)$ and $(u_j, v_j)$. The signed translation action is consistent precisely when

$$m\tau = 0 \pmod{2}, \qquad v_j = (u_j/d)\tau \pmod{2} \quad \text{for every } j.$$

If a relation fails, subtracting multiples of $(d, \tau)$ produces $(0, 1)$, an orientation-reversing loop fixing every sheet. Otherwise the signed translation group is $(jd, j\tau)$, with no orientation-reversing translation stabilizer. A reflection stabilizes a fixed-residue component at $r_0$ with orientation bit $w_0 + ((2r_0 - a_0)/d)\tau$. Thus all paired components are orientable in the consistent case, and this last bit decides each fixed component. Pure translation covers use the same consistency test. In particular, an even-degree connected cyclic cover of a Möbius band is an annulus; an odd-degree one remains a Möbius band.

For boundary monodromy $x + c$, put $q = \gcd(m, c/d)$. There are $q$ lifts of degree $m/q$ on a single-residue component, or $2q$ on a paired one. A boundary reflection exchanges the two residues of a paired component, giving $m$ lifts of degree two. On a fixed residue $r_0$, its fixed points solve $2y = (c - 2r_0)/d \pmod m$. If this has $\nu$ solutions, the profile is $\nu$ degree-one and $(m - \nu)/2$ degree-two lifts, omitting zero multiplicities. Together with

$$\chi(C) = \deg(C)\chi(F), \qquad 2 - \#\partial C - \chi(C) = \begin{cases} 2g(C), & C \text{ orientable}, \\ g(C), & C \text{ nonorientable}, \end{cases}$$

these profiles determine the topology. The code checks the resulting integer identities. They describe the orientation interval bundle with orientable total space; a different bundle character would be extra input.

The component of sheet $x$ is labeled by $x \bmod d$, together with its reflection partner when distinct. A boundary translation cycle has label $x \bmod \gcd(W, c)$; a reflection cycle has label $\{x, c - x \pmod W\}$. These give individual component and boundary-lift queries without enumerating other sheets or lifts.

### 29.3 An exact extension for ordered marked points

Fix the normalized input presentation. A marked component consists of one of its components and an ordered list $(x_1, \dots, x_q)$ of fibre points in that component. Repeated points are allowed. Equivalence means an isomorphism over the identity of $F$ taking each $x_i$ to its corresponding point. The implementation's component-selector sheet identifies the component and is not an additional mark. This definition does not include unparameterized boundary-circle markings or arbitrary attaching maps.

**Proposition 29.1** (Rooted equivariant transport)**.** *An equivariant map between connected components is uniquely determined by the image $y$ of one point $x$. For two translation components any $x, y$ are admissible and the map is $z \mapsto z - x + y$. For two paired components, any $x, y$ are admissible and the map is*

$$f_{x,y}(z) = \begin{cases} z - x + y, & z \equiv x \pmod d, \\ z + x - y, & z \equiv a_0 - x \pmod d, \end{cases} \pmod W.$$

*For two fixed components it exists precisely when $2(y-x) = 0 \pmod W$, and is then the translation $z \mapsto z - x + y$. A paired and a fixed component cannot be isomorphic over the base.*

*Proof.* Transitivity forces $f(gx) = gy$ for every monodromy element $g$. Translations act regularly on a single residue class. On a paired component, the points have unique expressions $T^j x$ or $T^j Rx$, for $j \in \mathbb{Z}/m$; the two displayed affine formulas follow. They are bijective and commute with $T, R$. On a fixed component the translation formula is forced, and commuting with $R$ is equivalent to $2(y - x) = 0 \pmod W$. The degree difference excludes the remaining case. These proofs include $W = 1, 2$ and $m = 1$ even when formal affine signs describe the same permutation. □

For $q > 0$, take $x = x_1$. A complete marked signature is obtained as follows. For translation components record $j_i = (x_i - x)/d \pmod m$. For paired components record $(e_i, j_i)$ according as

$x_i = T^{j_i}x$ or $T^{j_i}Rx$, with $e_i = 0$ or 1. For fixed components record all $j_i = (x_i - x)/d \pmod m$ and also

$$k_x = (2x - a_0)/d \pmod m.$$

The extra value records the reflection stabilizer of the root. Equality of $k_x, k_y$ is exactly the rooted transport condition. In all cases include the unmarked cover-type identifier. For $q = 0$, that identifier alone suffices. The API also includes the entire normalized base and monodromy in its signature, so comparisons between different presentations cannot silently become a claim of equivalence under a change of base.

**Corollary 29.2** (Exact marked-component comparison)**.** *Within one checked presentation, two signatures are equal if and only if there is an equivariant isomorphism respecting every ordered mark. After preparing the cover, a signature costs $O(q)$ arithmetic operations on binary integers. The unique map determined by two admissible anchors can be evaluated at any requested sheet in a constant number of such operations.*

*Proof.* An equivariant map preserves the root stabilizer and the relative $T, R$ coordinates. Conversely, equal signatures satisfy the rooted transport criterion; the resulting map takes all coordinates, hence all marks, to their prescribed targets. The empty-mark case is the previously proved unmarked type classification. No search through sheet bijections occurs. □

**Proposition 29.3** (Two marks defeat a constant type bound)**.** *There can be $W$ inequivalent ordered two-point markings of one connected component, even though its unmarked cover has just one type. If the points must be distinct, there can be $W - 1$ types.*

*Proof.* Take the annulus cover with monodromy $T(x) = x + 1 \pmod W$. Any equivariant automorphism is a translation, since the image of zero determines it. The pairs $(0, j)$ are equivalent precisely when their second coordinates agree. Exclude $j = 0$ for the distinct-point version. □

Thus polynomial bit complexity for each state and each equivalence query does not bound the number of possible marked states by a polynomial in $\log W$. Nor does this example show that an unknot hierarchy must visit all these states. A quasi-polynomial search bound needs an additional restriction on reachable attachment data or a way of skipping states.

## 29.4 Implementation bounds, verification and measurements

Let $L$ be the total peripheral-word length and let $B$ bound input integer bit lengths and $\log(r + L + b + q + 2)$. The classifier uses $O((r + L + b)B^3)$ elementary bit operations conservatively; here $L = O(r + b)$ for the canonical schema. It outputs $O((r + L + b)B)$ bits. Prepared marking queries use $O(qB^2)$ bit work and $O(qB)$ extra storage using elementary division bounds. The normalized model header is shared inside a prepared index; comparing or serializing complete signatures includes its size. No loop depends on $W$, its divisors, the component count, or the number of unmarked boundary lifts.

`CoverIndex` builds its own validated classification and returns a defensive copy of its summary. A serialized summary is not accepted as a proof of a covering presentation. Integer fields support the existing literal-integer or signed-hexadecimal transport, including values beyond Python's decimal conversion limit. The standalone command supports a cooperative deadline; cancellation propagates and never becomes a topology answer. Checks occur in schema construction, gcd iteration, boundary processing and marked queries. Individual integer operations and bulk serialization are not hard-interruptible.

The ten maintained tests derived from the archive repeat 13846 comparisons against explicit lifted-spine graph search, orientation propagation, and boundary permutation cycles. The new

independent audit exhausts all two-generator affine actions through six sheets. Across 364 presentations it performs 196712 marked and 1061 unmarked equivalence comparisons, 9100 rooted comparisons and 38746 transported-sheet checks against literal equivariant bijections. Additional tests cover binary sheet counts beyond $2^{24000}$, invalid marks, ordered repetitions, normalized input binding, defensive copies, cancellation, and hexadecimal CLI transport. The complete maintained suite passes 534 tests.

| Supplied cover | Sheets | Records | Expanded/compressed | A/A |
|---|---|---|---|---|
| annulus | 16 | 1 | 4.06 | 1.034 |
| mobius_reflection | 16 | 3 | 4.15 | 0.991 |
| three_types | 16 | 3 | 3.52 | 0.990 |
| annulus | 256 | 1 | 15.65 | 1.016 |
| mobius_reflection | 256 | 3 | 25.41 | 1.156 |
| three_types | 256 | 3 | 17.58 | 0.988 |
| annulus | 4096 | 1 | 244.61 | 1.072 |
| mobius_reflection | 4096 | 3 | 396.16 | 0.956 |
| three_types | 4096 | 3 | 318.54 | 1.046 |
| annulus | 65536 | 1 | 4621.74 | 1.045 |
| mobius_reflection | 65536 | 3 | 8290.72 | 0.989 |
| three_types | 65536 | 3 | 7002.16 | 0.970 |

Ratios are median paired times for complete cover topology; sheet expansion is an independent oracle, not a competing compressed algorithm.

| Sheet exponent | Marks | Query (ms) | JSON bytes | A/A |
|---|---|---|---|---|
| 32 | 2 | 0.010 | 100 | 1.001 |
| 32 | 32 | 0.077 | 456 | 0.999 |
| 32 | 512 | 1.187 | 6135 | 1.008 |
| 1024 | 2 | 0.014 | 698 | 1.013 |
| 1024 | 32 | 0.192 | 5534 | 1.009 |
| 1024 | 512 | 3.059 | 82897 | 1.001 |
| 16384 | 2 | 0.412 | 8286 | 0.998 |
| 16384 | 32 | 12.330 | 70055 | 1.000 |
| 16384 | 512 | 148.544 | 1058344 | 1.004 |

Marked queries reuse a prepared index, with sheet count $W = 2^{\text{exponent}}$. Preparation and serialization are outside query timing.

The topology benchmark keeps all input validation, peripheral evaluation, orientation and genus classification inside the compressed timer. Its expanded arm independently constructs the lifted spine and boundary permutations. Seven shuffled paired warm rounds retain all samples and A/A controls, with input generation and result comparisons outside timing. On 65536 sheets the observed expanded/compressed ratios range from 4622 to 8291; these measure avoidance of explicit expansion, not superiority to another compressed topology algorithm. The topology controls range from 0.956 to 1.156, so small constant-factor differences should not be overinterpreted.

The separate prepared-query experiment uses $W = 2^{32}, 2^{1024}, 2^{16384}$ and $2, 32, 512$ ordered points spread across a paired component. It includes mark validation and signature construction, but excludes the shared preparation and serialization. At $W = 2^{16384}$, 512 marks take about 149 milliseconds and the JSON signature has 1058344 bytes. The latter illustrates that mark output size is charged explicitly. Query A/A controls range from 0.998 to 1.013. These finite timings support the implementation checks; the polynomial bit-complexity claim rests on the arithmetic proof.

These measurements concern a geometric kernel with a supplied presentation. They are not timings of unknot recognition. To use it in a knot-exterior hierarchy we still need a checked extraction of the base and common fibre coordinate, an exact representation of all relevant attachments, and a bound on the hierarchy's search and repair states. Ordered fibre points are only one part of that attachment interface. The general recognizer does not dispatch to this module or claim a sub-exponential bound from it.

# 30 Causal RIII search: review and unambiguous certificates

The next intake placed three archives in arrival order as reports 26–28: determinant continuations, causal RIII search, and classical closure resets. The second archive calls itself "Report 25" internally; our collection number is 27 to avoid colliding with the earlier Garside archive. Original checksum manifests were verified in scratch before placement, then omitted under the repository intake rules. Delivered source and results remain unchanged apart from the mandated line-ending normalization of report 28's benchmark CSV. The retained intake audit records all original hashes, normalized hashes and source pins. Local unit reruns pass 35, 7 and 29 tests respectively; these counts do not establish mathematical soundness or integration with the current recognizer.

## 30.1 What the causal argument can give

Report 27 considers bounded-local reversible moves on a fixed set of sites. Each move has a footprint containing everything it reads and changes. Actions on disjoint footprints commute in both directions and preserve each other's legality. Terminal reductions also have local footprints. A move is a *birth* when its footprint is disjoint from the union of all preceding footprints. That union is accumulated support, not merely the footprint of the last move.

Every birth can commute to the front of the trace: it is disjoint from every earlier action. The births are pairwise disjoint and can be ordered canonically. Remaining actions retain an intersection with the preceding support. A reduction may become available sooner after this rearrangement, in which case the trace is truncated. Thus one can search a normal form with a canonical sequence of births followed by connected extensions. Connectedness of the full dependency graph does not imply that every chronological prefix was connected.

For RIII on a fixed input diagram, let $D(T)$ be the twelve darts at the three crossings of a triangular face $T$, and let $\alpha$ be the edge pairing. The conservative footprint

$$F(T) = D(T) \cup \alpha(D(T)), \qquad |F(T)| \le 24$$

contains every modified pairing. Inverting the face leaves this footprint unchanged. Globally there are $O(N)$ triangular faces in a connected $N$-crossing spherical diagram. The number whose footprint meets a set of $O(k)$ darts is $O(k)$, since only incident crossings and their partners need inspection. The report consequently obtains a conditional search bound

$$N^{b+O(1)}(Ck)^k$$

for traces of length at most $k$ and birth number at most $b$, with an absolute constant $C$. A depth-first implementation without a global memo table uses polynomial workspace in $N, k$. The delivered abstract prototype does retain a memo table, so its memory usage should not be identified with the latter claim.

For example, fixed $b$ and $k = O(\log N/\log\log N)$ give polynomial time; $b, k = O(\log N)$ give quasi-polynomial time. These are parameterized statements. No argument bounds the parameters uniformly for arbitrary unknots. Repeated preprocessing also requires the bound at each episode selected by the actual deterministic reduction policy; existence of a different successful sequence of choices is insufficient.

We reran the report's independent breadth-first comparisons in scratch. The generated audit covers 5,000 six-site systems, with 1,660 witnessed instances, 3,209 already reducible and 131 without a depth-six witness. The exhaustive three-site catalog covers 608,400 instances through depth four, with 171,968 witnessed, 163,800 already reducible and 272,632 without a witness. Both reproduce zero discrepancies in the recorded normal-form checks. These are abstract Boolean systems, not knot-diagram benchmarks; they exercise the prototype, not the proposed production adapter.

### 30.2 Why immediate inverse pruning needs care

The proposed RIII adapter suppresses the inverse of the preceding move. The abstract prototype deliberately does not. For unrestricted shortest paths an adjacent inverse pair is removable, but this need not preserve the birth bound. Here is an explicit four-site example under precisely the abstract locality assumptions. Start with all bits zero:

| Move | Footprint | Toggled bits | Guard |
|---|---|---|---|
| $A$ | $\{0,1,2,3\}$ | $\{0,1\}$ | $x_2 = x_3 = 0$ |
| $B$ | $\{0,2\}$ | $\{2\}$ | $x_0 = 0$ |
| $C$ | $\{1,3\}$ | $\{3\}$ | $x_1 = 0$ |

Each guard avoids the toggled bits, so every legal move is an involution. The terminal predicate is $x_2 = x_3 = 1$. The trace $BC$ succeeds but has two births. The trace $AABC$ succeeds with one birth: after $AA$ the physical state is again zero, while accumulated support contains all four sites. After the first $A$, both $B$ and $C$ are disabled, so the inverse $A$ is essential for this one-birth trace.

Exhaustion of all words through length four gives exactly $AABC$ and $AACB$ as first-unlocking traces with at most one birth. Suppressing adjacent inverses loses both. The script `data/causal_inverse_audit.py` compares this independent enumeration with the delivered prototype and retains the source hashes and results. This is an abstract counterexample to the pruning justification; it is not an actual RIII counterexample. It shows that locality alone cannot justify transplanting that extra pruning rule into the claimed complete parameterized search. Physical-state memoization likewise cannot ignore support and search phase. The existing bounded local simplifier remains a heuristic; its failure is not a knottedness certificate.

### 30.3 Integrated prerequisite: identify the actual triangular face

Review also exposed a separate defect in the existing production trace format. Recording only the three crossing indices does not uniquely identify an RIII move. The two-strand braid $(1, 1, -1)$ has the PD code

$$((5,4,0,1),(1,0,2,3),(2,4,5,3)).$$

Both $(11,4,0)$ and $(9,2,6)$ are legal triangular faces on crossings $\{0,1,2\}$. Their inversions give distinct labeled PD codes:

$$((0,1,2,0),(3,4,1,3),(2,4,5,5)),$$
$$((0,1,1,2),(3,4,4,0),(5,5,3,2)).$$

The old replay routine selected a face according to traversal order and therefore could reconstruct the wrong intermediate diagram. Both moves preserve knot type; the defect concerns reproducibility of the recorded sequence, including the legality of its subsequent moves.

New RIII records include a `triangle` field containing the face's three input dart indices. These indices remain fixed after earlier crossing deletions. Replay checks distinct, present crossing indices,

literal integer types, range, face membership and RIII legality. It recovers cyclic order from the current pairing, so reordering the three supplied dart indices cannot accidentally request a different reconnection. Legacy crossing-only RIII records are accepted when they identify a unique face, and rejected with an ambiguity error otherwise. R1/R2 serialization is unchanged. This is a certificate correctness fix, not a new recognition criterion or an asymptotic improvement.

Five new regression tests cover the explicit two-face example with all six dart orders, invalid certificates, every candidate RIII face in deterministically generated knot diagrams, legacy unambiguous replay, and JSON round trips after crossing deletions. The full integrated suite passes 539 tests in 109.018 seconds at that checkpoint; the log is `data/reidemeister-trace-integrated-tests.txt`.

### 30.4 Integrated search and adaptive dispatch

The production module `fastunknot/causal_r3.py` now implements the support-aware normal-form search. It uses an explicit depth-first stack with no global memo table. Each frame owns a journal of at most 24 pairing entries. Failure, local trial exhaustion and exceptions restore those entries and the path; successful first-unlocking traces remain applied. The retained footprint unions, per-frame candidate sets and pairing journals use polynomial workspace in $N, k$. Candidate enumeration polls an optional resource callback, as do terminal tests and the main search loop.

An additional pruning rule is justified under a narrower hypothesis than in the delivered adapter. Suppose an action with footprint $F$ is applied when the earlier accumulated support is $S$. Its immediate inverse can be skipped if $F \subseteq S$. In that case deleting the pair preserves the physical state, accumulated support, birth count and last birth key. It can only delay the start of the connected phase; all subsequent connected actions remain legal in the normal form. The resulting trace is shorter. If $F \not\subseteq S$, the inverse is retained, including inverses of initial births. This distinction preserves the preceding counterexample and the parameterized completeness argument.

The `r3_search` option accepts `last`, `clustered`, and `adaptive`; the last-move neighborhood implementation remains the default. The clustered implementation searches with a caller-specified `r3_births` bound. Adaptive dispatch first gives the last-move search half the remaining trial allowance. If it fails to make progress, the clustered search uses the remaining shared allowance. When the total trial limit is disabled, the first phase is still limited to $5N$ trials before switching. Each phase uses iterative deepening up to `r3_depth`. At a depth of at least two, the first move is not filtered by the last-step face shortcut. If a completed search at that depth tried zero nondegenerate moves, there is no legal first RIII move at all. Both search implementations stop deepening in that case, and adaptive dispatch also skips its second phase. This is a proved dead end for RIII-only search; it says nothing about crossing-increasing moves or knot type. In particular, failure of the face-filtered depth-one search alone does not justify stopping.

Successful reductions trigger another episode, with the same accumulated trial counter. Only paths ending in a legal RI/RII reduction are committed. The independent starting moves and retained support are local to each search episode, not retained across changes of crossing count.

The API and CLI expose the depth, births and trial allowance. The existing `use_r3=False` switch still disables the whole stage. Global time exhaustion interrupts either implementation and yields `UNKNOWN`; local trial exhaustion is inconclusive and continues the exact recognition pipeline. These options are preserved through certified source-reduction restarts and passed to each summand. Recognition records per-implementation trial counts and switch counts for the new modes.

Eleven new production tests compare bounded searches against unrestricted all-face enumeration on actual diagrams, verify disjoint births commute, check exact trace replay, retain inverse moves that add support, and exercise nested cancellation and injected mutation failures. They also check the no-first-move shortcut and deeper recognition of a stored hard unknot. The full integrated suite passes 550 tests in 98.849 seconds. Its log is `data/causal-r3-integrated-tests.txt`; the

separate focused log is `data/causal-r3-production-tests.txt`.

### 30.5 Actual-diagram evidence and limits

The reproducible discovery run in `benchmark_causal_r3.py` uses seed 2703 and tests 12,000 random braid words on four to six strands. Of these, 2,296 close to knots and 1,537 remain above six crossings after the historical search at depth four and ten trials per input crossing. On those 1,537, each new mode ties the resulting crossing count in 1,529 cases and leaves more crossings in eight; neither improves it. This is a negative practical result for the proposed dispatch at those settings, even though the unlimited clustered search has a stronger parameterized completeness claim. Finite budgets and the choice of committed reductions matter.

Another 1,200 generated unknots use the existing conjugation-and-scrambling constructor with 800 rewriting steps. All already reduce below seven crossings with the historical simplifier; none enters the stalled-diagram comparison. These are easy examples for existing RIII preprocessing and supply no evidence of a gain from clustered search. The data and exact source hashes are in `results/causal_r3_discovery_20261008.json`. For all 3,496 validated diagrams, the current default simplifier's complete trace and output PD code also agree exactly with pinned revision `28c974650`; the new no-first-move shortcut changes work, not choices.

The stored hard unknot corpus gives a separate lead: two examples admit further reductions when the historical search depth is raised from four to six, using fewer than forty trials. This benefit comes from greater depth, not from the new support representation. Full-pipeline paired measurements are maintained separately from the discovery counts. The final run covers 21 inputs, seven shuffled paired rounds and three fresh constructions per arm, after one untimed warmup. The baseline loads both the simplifier and recognizer from revision `28c974650`, sharing all other modules; an identical control arm measures variation. All recognition results finish and agree, with every default invariant and structural filter enabled and a five-second global limit.

At depth six, the historical search improves full recognition time by 1.71× on stored survivor 0 (sixteen crossings reduced to twelve) and 3.61× on survivor 1 (fifteen crossings reduced to zero). The adaptive depth-six arm gives 1.48× and 3.65× respectively; the second example finishes in its historical phase without switching. Across all inputs, the depth-six historical arm ranges from 0.907× to 3.605×, with median 0.957×. Consequently the default depth remains four; selective examples do not justify a universal increase.

The final default arm, including stricter option validation, cooperative deadline checks and the no-first-move shortcut, ranges from 0.945× to 1.061× in full recognition, with median 0.989×. The recognition control ratios range from 0.983× to 1.054×. For survivor 8 the no-first-move shortcut reduces isolated simplification from about 391 to 325 microseconds (paired ratio 1.20×), with little effect on full recognition. On the empty diagram, validation adds about 1.34 microseconds to isolated simplification and 0.56 microseconds to recognition. These costs are retained in the reported measurements.

At depth four, the clustered and adaptive full-recognition median ratios are 0.942× and 0.931×; neither supplies evidence for a default switch. Raw simplifier ratios must not be confused with end-to-end speedups: for example, spending more on simplification can save a larger subsequent homology computation on the depth-six witnesses. Inputs, individual samples, verdicts, methods, per-mode trial counts and exact source hashes are retained in `results/causal_r3_integration_20261008.json`. These are finite measurements, not asymptotic evidence.

A local search budget may stop preprocessing inconclusively; it cannot replace the exact fallback. The conditional bound still requires bounded unlocking parameters at each episode selected by the actual reduction policy. Neither this integration nor adaptive switching proves a general sub-exponential or quasi-polynomial recognition theorem.

# 31 Marked determinant continuations and an adaptive exact fallback

Report 26 supplies a stronger observation of a partially scanned complex: complete each genuine relative direct summand with the actual remaining diagram, reduce at a marked point, and retain its Euler characteristic in each quantum degree modulo four. Its norm can detect knottedness before the whole scan closes. The maintained implementation is `fastunknot/shadow_scan.py`, selected by `backend="shadow"`. It evaluates actual completed diagrams by exact integer determinants. The report's reusable terminal kernel is verified separately; a new geometric producer for that kernel is still a research prototype. Neither observation gives a general bound on when a useful summand appears.

## 31.1 Why the marked completion is necessary

At a proper prefix of a crossing order, choose a point on an edge incident to the last unprocessed crossing. The remaining tangle retains this point through every prefix observation. The marked circle carries the action of $A = \mathbb{F}_2[x]/(x^2)$. The relative cobordism maps, cancellations, and their explicit chain homotopies commute with this action. Completing the relative complex and applying the marked quotient therefore preserves these chain homotopies: apply the quotient to each identity $fg - 1 = dh + hd$. No assertion that the quotient functor is exact or flat is needed.

Suppose the maintained relative complex splits as whole differential components $C_a$, with multiplicities $m_a$. After the actual suffix is attached and the marked quotient is applied, it follows that

$$\widetilde{\mathrm{Kh}}(K;\mathbb{F}_2) \cong \bigoplus_a H\big(\widetilde{\mathrm{cl}}(C_a)\big)^{\oplus m_a},$$

up to the recorded grading shifts. This is a statement about components of a genuine relative knot complex. An arbitrary graded matrix satisfying $d^2 = 0$ is insufficient evidence of this provenance.

There is also a crucial endpoint restriction. After the final crossing is processed, individual unmarked scalar generators need not be free marked $A$-summands. Dividing the Euler value of each such generator by $q + q^{-1}$ would be invalid. The observer rejects the final stage, and the complete fallback instead uses the established unreduced rank test.

## 31.2 A four-coordinate lower bound

For a finite bigraded completed complex $X$, write

$$\sigma_r(X) = \sum_{h,\, j\equiv r \ (4)} (-1)^h \dim_{\mathbb{F}_2} X^{h,j}, \qquad r = 0, 1, 2, 3.$$

This vector is unchanged by a graded chain homotopy equivalence. A uniform homological shift changes its overall sign and a uniform quantum shift cyclically permutes its coordinates. Thus neither changes its $\ell^1$ norm. Taking Euler characteristics after passing to homology proves

$$\|\sigma(X)\|_1 \le \dim_{\mathbb{F}_2} H(X), \qquad B_4 = \sum_a m_a \, \|\sigma(\widetilde{\mathrm{cl}}(C_a))\|_1 \le \dim_{\mathbb{F}_2} \widetilde{\mathrm{Kh}}(K;\mathbb{F}_2).$$

More precisely, if $E_r, O_r$ are the even and odd homological dimensions in residue $r$, the deficit is $2\sum_r \min(E_r, O_r)$. Since reduced Khovanov homology detects the unknot, $B_4 > 1$ certifies knottedness. A value of zero or one is inconclusive.

**Aggregation order matters.** For each whole differential component the implementation recovers relative quantum shifts using `recover_shifts`. A nonzero coefficient of dot degree $d$ between matchings with $m$ arcs and $c$ glued circles requires $q_{\text{target}} - q_{\text{source}} = m - c + 2d$. Consistency is checked on all edges and cycles. The completed object vectors are shifted, multiplied by the object's homological sign, and *added before taking a norm.* Taking separate object norms would discard legitimate differential cancellations and could produce an unsound rejection.

Exact refinement of a direct-sum decomposition cannot decrease $B_4$: this is the triangle inequality, together with additivity of the Euler vector. Shared isomorphic components add their multiplicities. The production scanner caps those multiplicities at three; it preserves the threshold $\min(2, B_4)$, rather than the full uncapped bound. Tests compare both exact multiplicities and this saturated threshold.

### 31.3 Computing the vector without enumerating the cube

For a nonempty closed diagram use the raw reduced convention

$$J(q) = \sum_{s \in \{0,1\}^n} (-q)^{|s|} (q + q^{-1})^{c(s)-1}.$$

Its quantum degrees have one parity $\epsilon = (c(0) - 1) \bmod 2$. Put $a = J(1)$ and $d = i^{-\epsilon} J(i)$, which is an integer. Only residues $\epsilon$ and $\epsilon + 2$ can occur, and

$$\sigma_\epsilon = (a + d)/2, \qquad \sigma_{\epsilon+2} = (a - d)/2.$$

The engine verifies the parity and the exact divisions. The existing orientation-based closure calculation supplies the unreduced value $2a$ at $q = 1$; no exponential state sum is involved.

For the other specialization, construct the actual completed dart involution and follow the face permutation $d \mapsto \mathrm{rot}(\alpha(d))$. Check that its Euler characteristic is that of a disjoint union of spheres, then checkerboard color its faces. A disconnected projection has $J(i) = 0$. For a connected projection choose either color as black, with $v$ black regions. At crossing $e$, let $b_e$ be the smoothing bit which does *not* join its black sectors, put $w_e = (-1)^{b_e}$, and set $B = \sum_e b_e$. With $L_G^{(0)}$ a grounded cofactor of the signed Tait Laplacian,

$$J(i) = (-i)^{B+v-1} \det L_G^{(0)}.$$

One-circle smoothings correspond to spanning trees of the Tait graph; expanding their signed weights gives this formula by the matrix-tree identity. The identity is polynomial in the edge weights, so negative weights do not require a positivity assumption. Tait loops contribute to $B$ even though they make no Laplacian contribution. Dropping this phase, or replacing the determinant by its absolute value, would corrupt cancellation within a component.

Either checkerboard color is valid if its phase is computed consistently. The maintained engine chooses the smaller color class to reduce the dense cofactor dimension. Fraction-free Bareiss elimination, with row pivoting and exact integer division, computes the determinant. Its integer intermediate sizes are polynomially bounded for these integer Laplacians; the determinant calculation for one completion has polynomial bit complexity in its crossing count.

For a single parity vector the norm is $\max(|a|, |d|)$, using $|a + d| + |a - d| = 2\max(|a|, |d|)$. Thus the observation retains both Euler and determinant information. The significant new possibility is their application to separate genuine completed components, whose contributions may cancel in the global polynomial. Abstract examples of stronger component bounds do not by themselves establish that a knot scan realizes those examples.

### 31.4 A budgeted observer on the same exact scan

The driver first tries the existing component Euler obstruction, then the four-coordinate bound. Stage zero is always eligible. Later observations are skipped while there is just one component copy, since its completed Euler vector is the unchanged global vector up to shifts. Observation resumes when a nontrivial direct-sum decomposition appears. The suffix preparation and completed values are cached by stage and boundary matching. Euler and determinant observations share the `euler_max_states` allowance, default 4096 distinct queries.

An additional `shadow_max_work` allowance, default one million, charges geometry visits, dense matrix allocation and arithmetic updates. It is an operation allowance, not a count of bit operations or a hard wall-clock guarantee. The cooperative global deadline is polled during geometry and between determinant elimination rows, including on cache hits. An incomplete determinant is never cached as a completed value.

At this integration checkpoint, exhaustion of either local allowance disabled further observations and continued the *same* capped exact scan. Section 35 subsequently refines this policy by retaining a separately query-capped Euler reserve after marked-work exhaustion. No restart, discarded completed pivot, or speculative verdict is needed. Global resource exhaustion still returns `UNKNOWN` through the recognition API. An unknot verdict is available only when the complete rank computation has closed. The empty diagram is handled by its known unreduced rank two.

This is an adaptive policy in the sense suggested by introsort: expensive optional work has a budget and a complete fallback. Its guarantee is that inference cannot continue consuming unbounded local work after that allowance. Unlike introsort's worst-case sorting theorem, it does not improve the fallback's asymptotic bound. The default recognition backend remains unchanged because the measurements below do not show a broad end-to-end benefit.

### 31.5 A reusable terminal kernel, including singular interiors

Report 26 also proposes factoring a common signed Laplacian into interior vertices $I$ and boundary terminals. A completion identifies terminals; it must leave interior vertices distinct. A nonsingular interior block could be eliminated by an ordinary Schur complement, but signed weights allow a singular interior even when some completed cofactor is nonzero. The archive uses symmetric one- or two-coordinate pivots to extract a nonsingular principal block $P$. The remaining interior block is zero, of dimension $r$. With residual coupling $R$, terminal matrix $S$, and grounded partition matrix $Q$ for a query with $p$ terminal blocks, its cofactor is

$$\det P \, \det \begin{pmatrix} 0 & RQ \\ Q^{\mathsf{T}}R^{\mathsf{T}} & Q^{\mathsf{T}}SQ \end{pmatrix}, \qquad Q \text{ has } q = p - 1 \text{ columns.}$$

If $r > q$, or $RQ$ lacks row rank $r$, the answer vanishes. In particular, $r > b - 1$ makes every query vanish when there are $b$ terminals. At the critical size $q = r$ the answer becomes

$$(-1)^r \det P \, \det(RQ)^2,$$

independent of $S$. Consequently two nonzero critical answers have the same sign and rational square class; their product is a positive integer square. These are identities for a verified common graph and partition, not a license to identify arbitrary regions of unrelated completions.

**An explicit geometric producer.** The new research program `fast/determinant_research/terminal_geometry_audit.py` retains the original black regions incident to the suffix crossings, and retains exactly those crossings as signed edges. Original black regions touching a frontier edge become terminals. For each completed matching it verifies that completed faces retain the

original colors, that each retained original black region maps to one completed face, and that every fiber containing more than one old region consists only of terminals. These conditions establish the graph quotient by checking each retained edge's endpoints; no dense matrix comparison is required. The phase uses the *completed* black vertex count.

The program constructs one archive terminal kernel per suffix and verifies its algebraic certificate. Disconnected completed projections have zero determinant specialization and are handled separately. Any other failed geometry check declines the query; it cannot certify a knot. Across 300 random braid attempts, 78 validated knots and all encountered prefix matchings, 3186 connected queries passed the checks and agreed with direct completed determinants. A further 1317 queries had disconnected projections. There were no color, region-splitting, or interior-identification failures; the largest terminal set had size ten. The input words, crossing orders, and source hashes accompany these 4503 queries in `data/determinant-terminal-geometry-audit.json`. This is finite evidence for a useful construction, not a proof that every realizable matching passes its checks. The recognizer currently uses direct completed determinants; kernel reuse needs a maintained implementation, interruptible exact arithmetic, and measurements of setup amortization before production dispatch is justified.

## 31.6 Independent checks and actual performance

The unchanged delivered package passes its 35 tests. Its full validation rerun covers 701 diagrams, 133923 direct cube states, 83 homology calculations, 62904 columns checked for $d^2 = 0$, and 4000 quotient queries on 160 signed graphs, including 57 singular interiors. The upstream probe performs 176 scans of 44 diagrams, using two crossing orders and both physical and shared components. All 820 proper-prefix observations obey the exact-rank bound and the appropriate monotonicity condition.

Ten maintained tests independently compare Bareiss elimination with permutation expansion, completed vectors with direct cube polynomials, and component bounds with reduced homology. They check ordinary and capped multiplicities, mirror diagrams, global deadlines, partial determinant interruption, shared cache limits, exact fallback with zero inference budget, invalid inputs, and actual CLI dispatch. They also verify that observing a component does not mutate its differential. The full integrated suite passes all 560 tests.

One concrete strict improvement is the natural crossing order of $T(3, 5)$. Its initial four-coordinate norm is one. After nine of ten crossings, the completed components have total norm seven, while their reduced Euler bound is only one. The marked backend therefore rejects at stage nine; the Euler backend reaches the final stage ten. Similar one-stage improvements occur on the measured $T(3, 7)$ and $T(3, 11)$ inputs. An earlier certified stage does not automatically imply a faster runtime.

| Input | Full/shadow | A/A | Raw/shadow | Raw/Euler | A/A |
|---|---|---|---|---|---|
| unknot | 0.985 | 0.997 | 0.079 | 0.969 | 1.020 |
| trefoil | 1.014 | 1.028 | 3.015 | 0.980 | 0.974 |
| conway | 0.979 | 1.015 | 0.948 | 0.990 | 0.993 |
| kinoshita_terasaka | 1.007 | 1.013 | 0.951 | 0.992 | 0.987 |
| hard_unknot_8 | 0.991 | 1.002 | 0.854 | 1.019 | 1.002 |
| grid_scrambled_unknot | 0.986 | 0.982 | 0.803 | 0.980 | 0.979 |
| stress_braid5_36 | 1.000 | 0.993 | 360.190 | 1.032 | 0.987 |
| conway_sum_2 | 1.004 | 1.004 | 1.475 | 1.539 | 1.007 |
| conway_sum_8 | 1.012 | 0.969 | 4.144 | 5.603 | 1.003 |
| torus_3_5 | 1.007 | 1.002 | 1.023 | 1.002 | 1.003 |
| torus_3_7 | 1.002 | 1.017 | 1.000 | 0.983 | 0.986 |
| torus_3_11 | 0.994 | 1.000 | 0.972 | 0.976 | 0.997 |

Ratios are median paired times; values above one favor the denominator. Full uses the previous default recognizer; raw uses the saturated scanner. A/A compares identical implementations in each scope.

The complete recognizer shows no material benefit from the new backend on these inputs: old-default/marked ratios range from 0.979 to 1.014, while the identical-control range is 0.969–1.028. The changed default entry point has old/new ratios 0.962–1.019, with median 0.998. In contrast, the raw marked scanner is about 3.02 times faster on the trefoil and 360 times faster on the 36-crossing stress braid, rejecting both at stage zero. Existing recognition filters already decide those inputs. The Conway sums gain from component Euler information available to the old Euler backend too; that backend is faster than the marked observer on both measured sums. The raw marked scanner is slower on Conway, Kinoshita–Terasaka, both nonempty unknot fixtures, and the empty diagram. The one-stage torus improvements give ratios of 1.023, 1.000, and 0.972, respectively, against saturated scanning. These results support optional use and further work on selective observation, rather than enabling the backend by default. For scale, the empty raw scan rises from about 2.56 to 34.90 microseconds, while the stress braid falls from about 635 to 1.72 milliseconds; these are medians of the individual arm times.

The reproducer is `fast/benchmark_shadow.py`. It compares twelve inputs over seven shuffled paired rounds, with a duplicate control arm. Recognition measurements keep every default filter enabled and include fresh validated diagram construction. A separate raw-decision scope bypasses those filters and compares the saturated, Euler and marked scanners. Each call has a five-second and 50000-object allowance; any timed repetition returning `UNKNOWN` suppresses ratios for that case. The historical recognition control loads only `recognize.py` from commit `3a6d80ed6`, sharing its unchanged dependencies. Imports and one warmup are excluded. Complete samples and source hashes are retained in `fast/results/marked_shadow_20261008.json`.

### 31.7 What is still missing for a general complexity theorem

Each determinant query is polynomial, but the number and size of physical relative components need not be small. The report's stopping-prefix bound has the conditional form $\mathrm{poly}(n, R)2^{O(W)}$, where $R$ bounds the actual maintained physical data and $W$ bounds the frontier through the first decisive observation. Neither parameter has a universal bound sufficient for quasi-polynomial recognition. A planar separator bound on a diagram alone does not bound the surviving algebraic multiplicities or ensure an early decisive decomposition.

The new observer is useful precisely because it may stop before computing all homology. That possibility does not establish that it stops early on every nontrivial knot, nor that the complete fallback is subexponential on all unknots. The earlier assessment of Lackenby's hierarchy claim remains unchanged: it is a plausible geometric route, while this implementation still lacks a general quasi-polynomial or subexponential running-time proof.

## 32 Classical closures, first jets, and bounded diagram resets

Report 28 studies a different way to avoid carrying a large complex through the entire crossing order. When a whole relative summand has one matching, its *actual classical completion* imposes restrictions absent from an arbitrary algebraic continuation. In certain cases the entire current scan has the same total homology rank as that completed matching alone. We can then splice the matching into the remaining diagram and start a fresh scan with fewer crossings. The maintained decision-only backend `closure`, implemented in `fastunknot/closure_scan.py`, combines these resets with closure-rank obstructions and bounded optional work. The standard backend remains the default.

### 32.1 Source audit and the exact scope of the rerun

The delivered package distinguishes its shortened reference fixture from the production checkout. We ran its AST checker against the current checkout. The scanner agrees after docstrings are removed, but the geometry fixture's `compose` omits four maintained fast paths: zero factors, the two identity cases, and equal endomorphism factors. The latter uses the square-zero radical identity. Moreover, the geometry blob listed in the provenance file is not the blob at the stated inspected revision `13f6dbfdd`: the actual blob there starts `92135876`, whereas the supplied pin starts `1078526e`. The other three pins match that revision. The earlier intake ledger already flagged this mismatch, but its summary and catalog wording overstated the verification. Those summaries are now corrected. The detailed follow-up is retained in `data/closure-reset-source-audit.json`.

The 29 delivered tests pass with imports redirected to the actual production package. We also reran the independent cube audit through `fast/closure_research/audit_upstream.py`. This adapter changes only the import paths, output destinations, and scope description; the delivered files remain unchanged. All 480 driver comparisons on 160 diagrams, 480 interval-rank checks, and 130 quiver-rank checks pass. Eighty diagrams reset and two have early nontriviality certificates. No nonsingleton eligible block occurs in this corpus. The independent cube does not import scanner code.

### 32.2 Why a first jet controls a knot completion

Let $m$ be a nonempty geometric matching with $p$ arcs, and let a whole differential component supported on $m$ have coefficient matrix

$$Q(\boldsymbol{x}) = A + \sum_{i=1}^{p} x_i B_i + \text{higher dot terms}, \quad Q^2 = 0, \qquad B = \sum_i B_i,$$

over $\mathbb{F}_2[x_1, \dots, x_p]/(x_i^2)$. The matrix raises homological degree by one. Gluing the actual suffix gives a link $L_m$, whose Khovanov complex $C$ carries commuting square-zero dot operators $X_i$. The completed block has differential $D = d_C + Q(X)$ on $C \otimes V$. The component must have no incoming or outgoing attachments; a selected submatrix does not meet this premise.

**The chain-level calculation.** Moving a marked point across a crossing has reverse-saddle homotopy $H$ with $d_C H + H d_C = X_a + X_b = \delta$, $H^2 = 0$, and commutation with all fixed dot operators. It also satisfies $H\delta = 0$. These are strict chain identities, recorded in Nahm's Lemma 2.4 and its cited predecessors [16]; equality of induced dot maps on homology would not suffice here. For $P = \partial Q/\partial x_i$, differentiation of $Q^2 = 0$ gives $PQ + QP = 0$. Thus $G = 1 + HP(X)$ is self-inverse and

$$GDG = D + \delta P(X).$$

Indeed the linear error is $\delta P$, while the quadratic error is $H d_C H P^2 = H\delta P^2 = 0$. There is no requirement that $P^2 = 0$. This conjugation moves one variable's dot action to the new marked edge.

If $L_m$ is a knot, all marks can consequently be moved to one edge. All higher monomials vanish under $x_i \mapsto X$, leaving $A + XB$. The characteristic-two vertical operator $\nu$, which changes one circle label $x$ to 1 and sums over circles, satisfies

$$\nu^2 = 0, \qquad \nu d_C = d_C \nu, \qquad \nu X + X\nu = 1.$$

Hence $C = E \oplus XE \cong \mathbb{F}_2[x]/(x^2) \otimes E$, where $E = \operatorname{im} \nu$ is a subcomplex. Contracting $(V, A)$ onto its homology transfers $A + xB$ to $xB_*$: two perturbation factors vanish because $x^2 = 0$. If

$t = \dim H(V, A)$ and $b = \operatorname{rank} B_*$, the transferred complex has dimension $2t$, differential rank $b$, and homology dimension $2(t - b)$. Therefore

$$\dim H(\mathrm{cl}(Q)) = \kappa(Q)\, r(L_m), \qquad \kappa(Q) = \dim H(V, A) - \operatorname{rank} B_*,$$

where $r$ denotes total unreduced characteristic-two Khovanov rank. The formula preserves total rank; it does not assert equality of the original open complexes or their bigraded profiles.

After exact scalar-unit cancellation, $A = 0$, so

$$\kappa = N - \sum_h \operatorname{rank} B_h \geq 1.$$

The implementation reads only the parity of singleton-monomial coefficients in each entry, then computes adjacent binary ranks. Counting all monomials modulo two would give the wrong matrix. Neither barcode reconstruction nor ranks of products of adjacent matrices are needed. The maintained reader declines blocks containing scalar units; the general formula above explains the theory but is not an additional production scalar eliminator.

### 32.3 Pure blocks and the indispensable link distinction

For a pure block $Q = \theta T$ with one nonzero radical coefficient, put $\beta = N - \sum_h \operatorname{rank} T_h$ and let $\sigma = \sum_i [x_i]\theta$ be its total-linear parity. The scalar quiver has $\beta$ intervals, although $T^2$ need not be zero. When $\sigma = 1$, the derivation identity $\nu\theta + \theta\nu = U$ has $U^2 = 1$ and $\nu U = U\nu$. Then $U\nu$ splits the suffix complex into a free dual-number factor for the action of $\theta$. This proves completed rank $\beta r(L_m)$ for *every* classical link completion.

If $\sigma = 0$ and the completion is a knot, the first-jet matrix vanishes, giving completed rank $Nr(L_m) \geq 2N$. If the completion has at least two components, each quiver interval contributes at least $r(L_m) \geq 4$. The first inequality follows from the monotone interval profile already proved in the continuation analysis; the second follows from the link's Euler specialization. Consequently a pure block has lower bound

$$\begin{cases} 2\beta, & \sigma = 1, \\ 2\min(N, 2\beta), & \sigma = 0. \end{cases}$$

Any even pure block with an edge therefore certifies rank at least four. The implementation uses this threshold directly instead of computing unnecessary exact interval data.

A nonpure block requires the knot-completion premise. For example, on the two-component unlink, $x_1 + x_2$ acts nontrivially, and a three-term interval joined by this coefficient has rank four. Applying the knot formula would incorrectly give twelve. The maintained tests include this counterexample; no completion with several components is silently treated as a knot.

### 32.4 Recognition control, splicing, and fallback

The backend scans with exact component sharing and multiplicities capped at three. At each proper prefix it inspects whole differential components. Pure components give the closure-uniform lower bounds above. Otherwise the driver constructs the actual completion for each eligible matching, checks its spherical rotation system, and counts link components. A knot completion supplies $2\kappa$ as a lower bound. Contributions are multiplied by their nonnegative integer multiplicities and added; rank at least four rejects the original knot.

When the entire scan has one component copy, one matching, and $\kappa = 1$ with a knot completion, that residual knot has the same total rank as the current computation. Each matching arc joins two boundary labels of the unprocessed suffix. The splice merges these labels, canonicalizes the remaining PD code, and retains the selected crossing order. Removed smoothing circles have

already been handled by delooping; they are not reintroduced. A fresh scanner starts from empty boundary on the strictly shorter residual diagram. No claim of a knot isotopy between the original and residual diagrams is needed or made.

The local `closure_max_work` allowance, default one million, charges observer traversal, polynomial extraction, binary elimination, and completion geometry. It is an operation allowance, not a bit-cost or strict wall-clock bound. The global cooperative deadline has priority. Local exhaustion disables observation and continues the current exact scan, preserving all completed algebraic work. A partially computed bound or residual is never committed as a reset. Global exhaustion returns `UNKNOWN` through recognition. Only final closed rank two certifies the unknot; a proper-prefix failure to reject is inconclusive.

The decision interface exposes capped ranks and reset events. It is not connected to the exact homology or window APIs, because resets discard grading information. The optional backend retains the same alternate backend restrictions as the existing shared scanners.

### 32.5 Actual reset gaps and what they do not prove

Let $g$ be the maximum number of crossings processed between resets or the final decision, including the initial and terminal segments. After $j$ crossings in one segment, there are at most $4j$ frontier labels and, using a deliberately loose delooping bound, at most $8^j$ physical objects. The coefficient basis has size $2^{O(j)}$. Exact arithmetic, elimination, first-jet inspection, and geometry checks cost $\mathrm{poly}(n)2^{O(j)}$ in bit operations. Saturated sharing does not increase the literal object bound. The maintained ordering heuristic has polynomial overhead, and preserving the suffix order means each original crossing is processed only once across all segments. Thus this implementation satisfies the parameterized bound

$$T(n,g),\ S(n,g)\ \leq \mathrm{poly}(n)2^{O(g)}.$$

The recorded `max_gap` is this execution parameter, including scanning after observation has been disabled. A polylogarithmic bound on *actual* gaps would give quasi-polynomial recognition. We do not know such a bound for arbitrary knots or unknots, or an efficiently constructible ordering that guarantees it. Adaptive fallback does not supply that missing theorem.

### 32.6 Expanded discovery, validation, and paired measurements

The maintained discovery program `fast/closure_research/audit_blocks.py` samples 1200 braid attempts with seed 2801, accepting 312 knots. Each is scanned in a random order and the width heuristic's order, with a quarter-second and 3000-object allowance per scan. All 624 scans complete: 4972 crossing stages produce 785 singleton observations and no nonsingleton pure or first-jet block. This larger negative result supports no claim of actual-diagram gains from the mixed-coefficient rule. It does not prove that such blocks cannot occur. Inputs, orders, limits and source hashes are retained in `data/closure-reset-block-discovery.json`.

The maintained unit checks directly complete forty mixed polynomial matrices against the independent trefoil cube, verify $d^2 = 0$, and compare their ranks with $\kappa r(L_m)$. They also compare sixty random knots against the independent cube under resets, no resets, and zero observation budget; compare actual splices with the independent implementation; check malformed blocks and external attachments; and test global deadlines, CLI dispatch, and local exhaustion after successful resets. A seven-crossing Coxeter unknot has six resets and maximum gap one. This is an illustrative family already easy for earlier structural recognizers. The complete maintained suite passes all 567 tests. Shape caching is tested both enabled and disabled; automatic mode applies the established repeated-stage heuristic to each residual diagram, so the raw comparison uses the same cache policy as the existing saturated scanner.

| Input | Full/reset | Raw/reset | Raw/no reset | Raw/Euler | Raw A/A |
|---|---|---|---|---|---|
| unknot | 0.996 | 0.104 | 0.108 | 0.984 | 0.965 |
| trefoil | 1.028 | 0.953 | 1.126 | 0.991 | 1.180 |
| conway | 0.999 | 0.938 | 0.942 | 0.996 | 1.005 |
| kinoshita_terasaka | 1.001 | 0.942 | 0.939 | 0.997 | 0.998 |
| hard_unknot_8 | 1.022 | 0.889 | 0.923 | 0.961 | 1.001 |
| grid_scrambled_unknot | 0.986 | 0.646 | 0.719 | 0.978 | 0.993 |
| stress_braid5_36 | 1.001 | 0.927 | 0.949 | 1.014 | 0.990 |
| conway_sum_2 | 0.994 | 1.420 | 1.427 | 1.538 | 1.003 |
| conway_sum_8 | 1.012 | 5.283 | 5.241 | 5.576 | 1.005 |
| R28 cancel_pairs_26 | 1.003 | 0.465 | 0.589 | 0.973 | 1.000 |
| R28 coxeter_24 | 0.978 | 0.297 | 0.383 | 0.974 | 1.006 |
| R28 figure_eight | 1.001 | 0.951 | 0.900 | 0.962 | 0.996 |
| R28 positive_2braid_11 | 1.015 | 0.940 | 0.941 | 0.993 | 1.004 |
| R28 trefoil | 1.009 | 0.959 | 0.962 | 0.994 | 0.996 |
| R28 trefoil_then_curls_12 | 1.007 | 2.034 | 1.967 | 1.996 | 0.980 |
| R28 trefoil_then_curls_24 | 1.004 | 2.401 | 2.371 | 2.307 | 1.016 |
| R28 trefoil_then_curls_6 | 1.003 | 1.627 | 1.617 | 1.633 | 0.990 |
| torus_3_5 | 0.995 | 0.965 | 0.961 | 0.988 | 1.007 |
| torus_3_7 | 0.998 | 0.968 | 0.969 | 0.994 | 0.998 |
| torus_3_11 | 1.000 | 0.929 | 0.922 | 0.987 | 1.002 |

Ratios are median paired times; values above one favor the denominator. Full uses the previous default recognizer; raw uses the saturated scanner. The no-reset arm retains closure observations but does not restart.

The full-recognition ratios lie between 0.978 and 1.028, with median 1.001, while the identical full-control ratios span 0.976–1.081. All twenty inputs are decided by earlier enabled filters, so this corpus shows no complete-recognition gain from closure dispatch. The unchanged default path has median old/new ratio 1.008, with variation consistent with the controls. One raw trefoil control reaches 1.180; its small scanner differences should not be interpreted as reliable gains.

The raw observer improves the Conway sums by about 1.42 and 5.28 times, and the trefoil-with-curls examples by 1.63, 2.03, and 2.40 times. These are early obstructions with *zero resets*; the existing Euler backend already provides comparable gains. The actual reset cases are slower: saturated/reset ratios are 0.889 for the hard unknot, 0.646 for the scrambled grid unknot, 0.465 for cancellation pairs, and 0.297 for the Coxeter example. Their reset counts are 1, 4, 13, and 23. The no-reset control is faster on each of these four inputs, showing that splicing and scanner reinitialization add overhead here. There is no observed nonsingleton first-jet compression in this benchmark either. These measurements justify retaining the optional backend for further research and leaving automatic recognition dispatch unchanged.

`fast/benchmark_closure.py` measures full recognition separately from raw saturated, Euler, reset, and no-reset scanners. It includes the eight delivered examples and twelve maintained examples, uses seven shuffled paired rounds, includes validated diagram construction, and retains an identical control arm in each scope. Imports and one warmup are excluded. Calls have five-second and 50000-object limits; any timed repetition returning `UNKNOWN` suppresses that case's timing ratios. The old recognition arm loads only `recognize.py` from `e6f975521`, sharing unchanged dependencies. All raw samples and source hashes are retained in `fast/results/closure_reset_20261008.json`.

## 33 Adaptive boundary summaries for repeated completion queries

Repeated observations of one suffix should share its topology. The maintained Euler backend now starts with direct closure walks and switches to a pair of boundary path summaries when enough distinct queries reach the same stage. A separate research implementation extends this

idea to report 26's signed Tait kernel. Its common vertices are *cut-face fragments*; this choice makes both the quotient construction and its geometric checks depend on the boundary rather than on another traversal of the entire suffix. Neither improvement bounds the number of relative generators or proves a general subexponential recognition algorithm.

### 33.1 Two path summaries determine the raw Euler value

Fix a suffix with $n$ crossings and $b$ boundary darts. Its partial edge involution is $\alpha$; at each crossing the opposite-dart pairing describes link strands, and the zero-smoothing pairing describes the all-zero state. Combining either local pairing with the internal edges gives disjoint closed circles and paths whose endpoints lie on the boundary. Store the number of closed circles and the induced boundary matching for each of these two graphs. Preparation takes $O(n + b)$ combinatorial work and temporary space, while the retained summaries have size $O(b)$ apart from the integer counts.

For a proposed boundary matching $m$, the circles of the completed graph are exactly the stored closed circles plus the alternating cycles of the stored boundary matching with $m$. Thus two boundary walks compute the link component count $c$ and the all-zero circle count $c_0$. Coverage is checked: every boundary label must occur once, with no missing, repeated, or unknown endpoint. No suffix dart need be revisited by these queries.

Here is why these counts suffice even for the sign. Choose any orientation of the classical completion, let $h$ be the number of one-smoothings in its oriented Seifert state, and let $s$ be its number of Seifert circles. Changing one smoothing changes circle parity, giving $s \equiv c_0 + h \pmod 2$. The orientable Seifert surface has Euler characteristic $s - n$ and $c$ boundary components, hence $s - n \equiv c \pmod 2$. This also holds for a disconnected surface. Therefore

$$h \equiv n + c + c_0 \pmod 2, \qquad \chi_{\mathrm{raw,unred}}(m) = (-1)^{n+c+c_0} 2^c.$$

This is the unnormalized Euler value needed by the existing relative scan. The production caller supplies classical completions from a valid scan; these two path summaries alone are not a test for virtual planarity. At the empty scalar terminal the convention is $n = c = c_0 = 0$, with value one; it does not denote an implicitly added crossingless unknot.

Each subsequent query uses $O(b)$ combinatorial operations. This is not an $O(b)$ bound on bit complexity independent of $n$: the integer $2^c$ itself has $c + 1 = O(n + b)$ bits. The implementation retains exact integers and the existing coefficient-size statistics.

### 33.2 When the maintained backend switches

The summaries are implemented in `fast/fastunknot/boundary_connectivity.py`.

`AdaptiveClosureEuler` in `fast/fastunknot/euler_scan.py` uses the independent orientation-based `ClosureEuler` for the first four distinct, uncached queries at a stage. Later queries prepare the summaries if $b \leq n$; a wide boundary on a short suffix keeps direct evaluation. This is an empirical setup policy, not an optimal stopping theorem. A low-level `compression_after=0` option forces eager preparation for diagnostics. The public Euler backend uses the adaptive policy; the standard recognizer backend remains unchanged.

Changing the stage discards the active summaries. Cached exact values do not consume another query allowance. The global deadline is checked even on a cache hit, and the local query cap is checked before preparation. A summary is published only after both path traversals finish, so an interrupted setup cannot leave a usable half-summary. Local inference exhaustion continues the same exact saturated scan, while global exhaustion returns `UNKNOWN`, as before. The direct closure evaluator and the older suffix-smoothing recurrence remain independent references.

Five new tests cover random classical completions and their mirrors, independent recurrence values and explicit circle counts, switching and stage changes, the wide-boundary guard, cache and budget

behavior, malformed coverage, and interruption after the first of the two path summaries. The full maintained suite passes all 572 tests; logs are retained in `data/boundary-connectivity-tests.txt` and `data/boundary-connectivity-integrated-tests.txt`.

### 33.3 A common Tait graph made from cut-face fragments

The research implementation `fast/determinant_research/boundary_tait.py` starts with a checkerboard coloring of the original classical diagram. Write $\rho$ for cyclic rotation of a crossing's darts. A completed face follows the permutation $\rho\alpha$. Retain only its edges arising from internal suffix edges. Each resulting component is either a closed face cycle or an open face path. These *fragments*, rather than entire original regions, are the common graph vertices. In particular, a region that splits when the prefix is removed is represented by separate fragments.

For a boundary label $a$ with dart $d_a$, store the fragment containing $d_a$ as its outgoing endpoint and the fragment containing $\rho(d_a)$ as its incoming endpoint. Pairing labels $a, b$ inserts exactly the two face connections

$$d_a \longrightarrow \rho(d_b), \qquad d_b \longrightarrow \rho(d_a).$$

Consequently every completed face is obtained by identifying these open fragments, while every closed fragment remains untouched. This proves the face reconstruction directly from the partial permutation; it does not assume that the original region partition survives a new closure.

There are at most $2b$ open fragments. A query runs union-find only on them and on the suffix projection components incident to the boundary. Counts of closed fragments and closed projection components are stored once. Every inserted face connection must preserve the inherited checkerboard color. If $F$ is the resulting face count and $k$ the resulting projection component count, the query also requires

$$F = n + 2k.$$

The completed rotation system has $V = n$ and $E = 2n$. Its orientable ribbon components have total Euler characteristic $F - n = 2k - 2\sum g_i$; the displayed check is therefore equivalent to every $g_i = 0$. It certifies spherical completion for accepted queries. Color incompatibility is a decline of this particular representation, even when a completion is classical; it is never evidence that a knot is nontrivial.

Choose the smaller shading by fragment count. Each crossing contributes one signed edge between its two black fragments, with the same local weight as in Section 31. The black open fragments are terminals. All identifications of the common graph occur among these terminals; black closed fragments are fixed interior vertices. Thus an accepted connected completion has exactly the quotient Tait graph required by report 26's terminal kernel. Its phase is recomputed from the resulting black vertex count $v$:

$$J_{\mathrm{raw}}(i) = (-i)^{B+v-1}\tau(G/m).$$

The crossing sum $B$ includes loop edges even though loops contribute zero to the Laplacian. If the completed projection is disconnected, the specialization is zero and no kernel is needed. If there is no boundary, one black vertex serves as the artificial grounding terminal.

The common Laplacian is prepared once; the delivered exact rational kernel is built lazily. Canonical terminal partitions cache completed integer cofactors, so distinct matchings inducing the same partition share a value. Phase and topology counts are still computed for each matching. The two boundary path summaries give the raw value at one and the parity $\epsilon = c_0 - 1 \pmod 2$. Together with the value at $i$, they reconstruct the four marked residues by the existing $(a + d)/2, (a - d)/2$ formula.

The geometric query costs $O(b\,\alpha(b))$ combinatorial operations. For the determinant, the singular-safe elimination described earlier leaves a border of dimension $r + q$, where $r$ is interior nullity

and $q$ is one less than the number of terminal blocks. If $r > q$, the cofactor vanishes; otherwise $r + q = O(b)$. This controls matrix dimension, not coefficient bit length or the number of scan queries. Rational arithmetic can still depend polynomially on the original suffix size. Production integration also requires an interruptible maintained kernel and useful amortization on actual observer traces. The present prototype imports report 26's kernel and has no production deadline or local-work interface; production shadow observations continue to compute their closures directly.

### 33.4 Independent geometry audit

`fast/determinant_research/audit_boundary_tait.py` repeats all 4,503 actual completion queries on 582 proper-prefix stages of 78 retained input diagrams. Every accepted value agrees with an independently rebuilt completed diagram, including component counts, zero circles, raw Euler values, and the Gaussian value at $i$. There are 1,317 disconnected projections, 15 singular stage kernels, maximum terminal count ten, and maximum interior nullity two. Kernel certificates are checked independently.

A further exhaustive pairing audit covers boundary sizes four through eight in the first 20 retained diagrams: among 3,468 pairings, 745 are accepted, 472 classical completions are declined because the inherited coloring does not fit, and 2,251 nonclassical completions are rejected. Every accepted pairing passes the independent classical completion check and both value comparisons. The scope, exact inputs, and source hashes are recorded in `data/boundary-tait-audit.json`. These finite checks support the implementation; the fragment and Euler-characteristic arguments above supply the general accepted-query contract.

### 33.5 Measured reuse and its limits

`fast/benchmark_boundary.py` uses seven shuffled paired rounds with identical control arms and one excluded warmup. Every timed sample includes fresh engine setup. The isolated workload consists of 200 distinct actual matchings from one nine-crossing prefix, all with 18 boundary darts. Appended cancelling braid pairs increase the suffix from 5 to 37, 133, and 517 crossings without changing the knot. Matching-stream discovery is outside the primitive timings. These deliberately easy added crossings isolate reuse; they are not a hard family for complete recognition.

| $n$ | Euler adaptive | Euler eager | A/A | Tait boundary | A/A |
|---|---|---|---|---|---|
| 5 | 0.873 | 0.906 | 1.006 | 1.152 | 1.004 |
| 37 | 2.589 | 2.622 | 1.002 | 5.829 | 0.993 |
| 133 | 7.273 | 8.332 | 1.001 | 19.476 | 1.008 |
| 517 | 15.828 | 23.640 | 0.997 | 53.905 | 1.002 |

Ratios are median paired direct/alternative times; values above one favor the alternative. A/A compares identical direct implementations.

The adaptive Euler policy achieves up to 15.8 times speedup, including its first four direct queries and summary setup. Eager setup reaches 23.6 times on the longest suffix. On the shortest suffix, the guard prevents preparation but the adaptive wrapper still costs about 15 percent in this isolated 200-query workload. This overhead is retained in the table.

The research determinant path reaches 53.9 times speedup, including coloring, fragment geometry, kernel construction, and partition-cache setup. However, all four of these common graphs have nine vertices, all nine are terminals, and interior nullity is zero. These gains therefore demonstrate boundary geometry and partition reuse, not amortization of a nontrivial interior elimination or a singular-kernel runtime advantage. Only 61 or 96 distinct cofactors are computed for each 200-matching stream; disconnected queries also avoid determinants. Independent kernel-certificate checking is performed in the separate audit, outside these timings.

| Input | Raw adaptive | A/A | Full adaptive | A/A |
|---|---|---|---|---|
| conway | 0.997 | 1.003 | 1.001 | 0.958 |
| kinoshita_terasaka | 0.996 | 1.000 | 0.990 | 0.989 |
| conway_sum_2 | 1.005 | 1.001 | 1.010 | 1.015 |
| conway_sum_8 | 1.008 | 1.003 | 0.987 | 1.000 |
| hard_unknot_8 | 0.997 | 0.999 | 0.982 | 1.002 |
| grid_scrambled_unknot | 1.010 | 1.005 | 0.990 | 0.999 |
| stress_braid5_36 | 0.961 | 0.985 | 0.994 | 0.997 |

Ratios are median paired direct/alternative times; values above one favor the alternative. A/A compares identical direct implementations.

For complete raw Euler scans, the baseline loads the earlier `euler_scan.py` from revision `f0179ab50`; the other dependencies are shared. Status, method, and observation stage agree in every paired run. None of these seven raw scans prepares a boundary summary. They show no consistent gain, and the stress case is about four percent slower. For complete recognition with the Euler backend requested, earlier filters decide every example, so no Euler evaluator runs. No whole-recognizer speedup is established. The raw samples, per-arm evidence, inputs, and source hashes are retained in `fast/results/boundary_reuse_20261008.json`. Future scheduling decisions must distinguish frequent cheap boundary queries from early-stopping scans; neither a favorable microbenchmark nor an adaptive switch supplies the missing universal width bound.

# 34 Signed Tait block products for actual observer workloads

The preceding boundary experiment isolates a useful primitive, but actual recognition decides when that primitive runs. We therefore traced the unmodified marked observer on every retained example before choosing its next optimization. Most examples ask for just the initial determinant. The determinant-one torus and grid examples ask for two additional distinct closures at a later stage. No example in this collection reaches the repeated-query threshold of four at any stage. This evidence favors reducing the cost of a single completed Tait graph before enabling terminal-kernel setup in the production observer.

The maintained shadow backend now factors larger signed Tait graphs at their articulation vertices. For example, the initial eightfold Conway connected sum has a $41 \times 41$ full cofactor, but its block factors require matrices of size at most $6 \times 6$. This optimization applies directly to each actual completion, with the original signed phase retained. It makes no assumption that a partial complex splits as a knot connected sum.

## 34.1 Why the signed product identity is exact

Let $G$ be the black Tait multigraph of a connected classical projection, with integer edge weights $w_e \in \{1, -1\}$. Its weighted tree sum is

$$\tau(G) = \sum_{T \text{ spanning tree of } G} \prod_{e \in T} w_e.$$

Loops never occur in a tree and may be discarded for this computation. For parallel edges between $u$ and $v$, a tree can use at most one, so replacing them by the single weight $w_{uv} = \sum_{e:u\leftrightarrow v} w_e$ preserves the sum. This is a polynomial identity and does not require positive weights. A zero combined weight may be deleted. If these deletions disconnect the graph, the sum is zero even though the original knot projection remains connected.

Suppose the resulting graph is connected, and let $G_1, \ldots, G_s$ be its edge blocks, with each bridge

treated as a two-vertex block. Then

$$\boxed{\tau(G) = \prod_{j=1}^{s} \tau(G_j).}$$

To prove this, first split at an articulation vertex into two graphs meeting only at that vertex. Restricting a spanning tree to either graph gives a spanning tree there: a disconnected restriction cannot be reconnected by leaving and returning through the same articulation vertex. Conversely, the union of two such trees is connected and has no cycle. The bijection preserves the product of edge weights. Induction on the block-cut tree gives the formula. Thus the identity also holds when factors are negative or vanish by cancellation. A bridge contributes its combined weight; the one-vertex graph has tree sum one.

For a nontrivial block, the integer matrix-tree theorem identifies its tree sum with any grounded Laplacian cofactor. The implementation uses the existing fraction-free Bareiss determinant with exact-division checks and row-level interruption points. It stops once a zero factor is found. No positive-definite elimination or nonsingular-interior premise is used.

The full specialization is still

$$J_{\mathrm{raw}}(i) = (-i)^{B+v-1} \prod_{j} \tau(G_j),$$

where $v$ is the black vertex count of the *full completed graph* and $B$ is the original crossing sum. In particular, loops and cancelled parallel edges continue to contribute to the crossing phase. Multiplying independently normalized block Jones polynomials without these conventions would not justify the implementation.

This factorization is performed after the actual closure has been checked. It cannot automatically be applied to the fixed common graph before arbitrary terminal identifications: those identifications can create cycles crossing old articulation blocks. Terminal-kernel reuse and completed-graph block factorization therefore have different geometric premises.

### 34.2 Implementation, resources, and complexity

`fast/fastunknot/tait_blocks.py` combines parallel weights and uses an iterative low-link depth-first search to produce edge blocks. A stack of DFS frames replaces recursion, so a long chain does not impose Python's recursion-depth limit. When a child's low-link index is at least its parent's discovery index, the edge stack is popped through that tree edge to emit a block. A visit count detects disconnection of the nonzero graph.

The observer chooses the smaller checkerboard shading as before. For a cofactor of size at most sixteen it retains the direct dense path; larger cofactors use block decomposition before any dense matrix is allocated. This cutoff avoids adding graph bookkeeping to the smallest queries. It is an implementation policy, not a claim that sixteen is universally optimal. The observer exposes block, bridge, cancellation, and block-determinant counts. `max_unsplit_cofactor_size` records the full graph's size, while `max_cofactor_size` now records the largest matrix actually allocated. The preexisting `determinants` field continues to count connected closure specializations, including interrupted attempts.

Write $b_j = |V(G_j)| - 1$. Apart from integer bit costs, the determinant path uses

$$O\left(|V| + |E| + \sum_j b_j^3\right) \quad \text{work and} \quad O\left(|V| + |E| + \max_j b_j^2\right) \text{ auxiliary space,}$$

with bridge factors evaluated directly. This replaces the full cofactor's cubic arithmetic and quadratic allocation by the corresponding block costs. The graph decomposition itself is linear.

Integer sizes still grow with the input; the displayed arithmetic count is not a unit-cost claim for arbitrarily large numbers. These polynomial savings do not bound the Khovanov scanner's exponential state growth or establish a general subexponential recognition algorithm.

All graph traversal, edge aggregation, matrix allocation, and elimination paths charge the existing local work allowance and poll the global deadline. Only a complete four-residue result enters the closure cache. A partially multiplied block product is local state and is discarded on interruption. The implementation measured in this section disables observations on local exhaustion and continues the same exact capped scan. The later scheduler in Section 35 retains a query-capped Euler phase; a global deadline still yields `UNKNOWN`. The bounded-work policy counts combinatorial and arithmetic updates, not integer bit operations. The change remains inside the optional `shadow` backend; default recognition still uses `standard`.

### 34.3 Validation and workload evidence

Five new tests compare 245 random small weighted multigraphs with independent spanning-tree enumeration and 192 larger graphs with independent dense cofactors. They include parallel edges, loops, negative weights, deletion of cancelled edges, and connected graphs whose signed tree sum vanishes. A 5,000-vertex bridge chain verifies the iterative traversal and avoids any dense allocation. Four larger classical diagrams and their mirrors match independently reconstructed values at both one and $i$; this checks that the block product retains the global phase.

An injected interruption after one completed block verifies that no partial result is cached. Resumption recomputes the exact value, global deadlines win even on a cache hit, and a real 2,000-unit optional allowance on the Conway sum exhausts locally while exact scanning still returns knotted. All 577 maintained tests pass after the final cutoff change. The test records are `data/tait-block-tests.txt` and `data/tait-block-integrated-tests.txt`.

`fast/benchmark_tait_blocks.py` keeps three scopes separate: the initial closure query, the complete raw marked scan, and complete recognition with the shadow backend requested. Each sample includes fresh diagram and observer setup. Two additional inputs concatenate 16 and 32 oriented Conway traversals to form larger connected sums; their validated PD codes are retained in the benchmark data. Seven paired rounds shuffle direct, identical control, and block-factor arms; imports and one warmup are excluded. The direct baseline loads `shadow_scan.py` from revision `2c7c8df3f`, sharing other dependencies. Actual observer traces are collected separately outside the timed calls. Inputs, source hashes, work counts, and raw samples accompany the resulting tables.

| Input | Initial | A/A | Raw | A/A | Full | A/A |
|---|---|---|---|---|---|---|
| Conway | 0.983 | 0.994 | 1.004 | 0.999 | 0.999 | 0.999 |
| Conway double three | 0.999 | 0.996 | 1.007 | 1.005 | 0.998 | 0.998 |
| Conway pretzel | 1.043 | 1.061 | 0.972 | 0.995 | 0.967 | 0.992 |
| Conway sum 2 | 0.995 | 1.000 | 1.015 | 0.998 | 0.995 | 0.993 |
| Conway sum 3 | 1.008 | 1.009 | 0.984 | 0.987 | 1.048 | 1.001 |
| Conway sum 8 | 2.152 | 0.999 | 1.186 | 1.009 | 1.030 | 1.027 |
| Figure eight | 1.001 | 1.026 | 1.009 | 1.013 | 1.003 | 0.994 |
| Grid determinant one | 0.991 | 0.982 | 0.999 | 0.993 | 1.005 | 0.989 |
| Scrambled unknot | 0.984 | 1.006 | 0.986 | 0.998 | 0.996 | 0.997 |
| Hard unknot 8 | 0.994 | 0.997 | 1.018 | 1.013 | 1.003 | 0.994 |
| Kinoshita–Terasaka | 0.996 | 1.004 | 0.987 | 0.991 | 1.001 | 1.004 |
| Stress braid 36 | 1.004 | 1.000 | 0.998 | 0.986 | 0.986 | 0.992 |
| $T(3,5)$ | 0.994 | 1.007 | 0.999 | 0.997 | 0.999 | 0.990 |
| Trefoil | 0.989 | 0.996 | 0.982 | 0.995 | 0.992 | 1.007 |
| Crossingless unknot | — | — | 0.986 | 1.002 | 0.983 | 1.003 |
| Braid unknot 40 | 1.302 | 0.989 | 1.082 | 1.009 | 1.000 | 1.002 |
| Conway sum 16 | 5.236 | 1.007 | 1.623 | 1.008 | 1.000 | 0.977 |
| Conway sum 32 | 17.418 | 0.995 | 21.547 | 1.024 | 1.001 | 1.000 |

Ratios are median paired direct/block times; values above one favor block evaluation. A/A compares identical direct implementations. Every scope includes fresh setup.

**Choosing the direct-path cutoff from measured overhead.** The first run, preserved in `fast/results/tait_blocks_threshold8_20261008.json`, used cutoff eight and the implementation at `07c02e10c`. It improved the eightfold Conway sum's initial query by 2.10 times, but made the irreducible size-16 stress cofactor about nine percent slower. The size-11 twofold sum also lost about five percent. Raising the cutoff to sixteen removes those decomposition costs, at the price of foregoing the initial run's modest size-16 threefold sum gain. The displayed final run uses this revised policy. Both source versions and all measurements are retained; the comparison does not replace the earlier unfavorable samples with the later ones.

**Single queries and complete raw scans.** The final initial-query speedups are 2.15, 5.24, and 17.42 times for 8, 16, and 32 Conway summands. The 40-crossing braid unknot improves by 1.30 times: all twenty effective blocks are bridges, so it allocates no dense determinant matrix. The size-16 irreducible stress case now keeps its direct path and has ratios near one. The corresponding complete raw scan gains are 1.19, 1.62, and 21.55 times for the three larger sums, and 1.08 times for the braid unknot.

The 32-fold sum illustrates why resource behavior matters. Its initial unrestricted direct query needs 1,398,785 work units, compared with 13,252 for block products. With the normal million-unit optional allowance, the old driver stops observing during this query and scans all 352 crossings, eventually certifying knottedness from closed rank. The new driver uses only 14,619 units including its later Euler query and obtains a component-Euler certificate after 11 crossings. Its largest determinant is $5 \times 5$, compared with the original $160 \times 160$ cofactor. The raw benchmark therefore requires agreement of verdicts while recording method and stage differences explicitly. The large gain here includes preserving the opportunity for later inference, not just faster arithmetic. Both runs complete; no timeout is assigned a speedup ratio.

**Complete recognition and the remaining limit.** Earlier recognizer filters decide all 18 measured inputs before the shadow backend runs. Full-recognition ratios range from 0.967 to 1.048, with no demonstrated overall benefit from the new determinant path. For the 32-fold sum the existing factorization and Jones filter already return in about 15 milliseconds, ahead of

either raw scanner. The results support the optimization inside the optional observer; they do not justify changing the default backend or claiming a general asymptotic improvement in unknot recognition. Final inputs, traces, source hashes, and measurements are in `fast/results/tait_blocks_20261008.json`.

# 35 Preserving a cheap observer after marked-work exhaustion

Block products reduce the cost of many determinant queries, but a large irreducible cofactor can still exhaust its optional allowance. The earlier scheduler then stopped *all* observations, including cheap component-Euler checks. We now distinguish those resource limits. This is analogous to introsort at the scheduling level: an expensive optional method has a bounded attempt, then gives way to a cheaper exact method, then to the complete scanner when its own allowance is exhausted.

The marked backend has the one-way progression

$$\text{marked residues and Euler} \longrightarrow \text{Euler only} \longrightarrow \text{unobserved capped scan.}$$

The first transition occurs on marked-work exhaustion; shared query exhaustion can take either observer state directly to the unobserved scan. The crossing order, simplified relative complex, completed pivots, exact cached values, and global deadline survive every transition. There is no restart and no new query allowance.

## 35.1 Two limits with separate meanings

Let $W$ be `shadow_max_work` and $K$ be `euler_max_states`. While marked observations are active, $W$ pays for their geometry, matrix allocation, arithmetic updates, and accompanying Euler queries, as before. The exception `ShadowWorkBudget`, a subtype of `EulerBudget`, identifies exhaustion of this particular allowance. The driver then stops requesting marked observations permanently.

If query capacity remains, later Euler queries use the direct closure connectivity evaluator with the *same* exact cache and state counter. Its new geometric work is separately measured by `euler_fallback_traversed_darts`; it is not charged to, or hidden by resetting, the already spent marked-work allowance. The corresponding `euler_fallback_evaluations` and `euler_fallback_states` fields distinguish all completed fallback requests from newly computed values. Cached values do not consume another query, but still check the global deadline. Result fields `shadow_exhausted` and `euler_exhausted`, together with `observer_mode`, distinguish which methods remain available.

This is an explicit resource-semantics change: setting $W = 0$ disables marked work but permits the query-capped Euler reserve. Setting $K = 0$ disables all new completion queries. Either limit may be explicitly unbounded through the existing API option `None`; an unbounded $K$ does not give a finite fallback-query guarantee. No additional public switch is required.

If $k$ distinct Euler values have already been computed, the fallback can compute at most $K - k$ more. Each direct completion walk uses $O(n)$ combinatorial work, so additional *new closure geometry* is $O(n(K - k))$. This bound does not include every observation cost: aggregating whole relative components still visits the current generators, and cache keys encode boundary matchings. Integer bit costs also depend on the suffix size and coefficient growth. In particular, this is not a polynomial bound on the complete Khovanov scan, whose number of live generators remains uncontrolled in general.

## 35.2 Why partially completed observations are safe to abandon

For a whole relative component, completion gives a genuine chain complex and its signed Euler characteristic $\chi$. Therefore $|\chi| \le \dim H$. Summing these inequalities with the already certified nonnegative component multiplicities gives the existing Euler rank lower bound. Capping the final nonnegative bound preserves its decision meaning; the individual signed Euler values themselves remain exact integers. Thus an Euler certificate does not depend on finishing a marked determinant.

The implementation first completes the component-Euler bound for a stage. If marked work fails afterward, the incomplete marked bound is discarded. That stage's already completed Euler bound remains valid; later stages may request further Euler values. If the marked allowance instead runs out while preparing the Euler part, its adapter switches before completing the outstanding query. Both paths use the same budget-checked evaluator and cache. A generic query-cap exception stops all further observations. Global `ScanLimit` exceptions are never converted into local declines: they propagate to `UNKNOWN` at the recognition boundary.

As before, an early lower bound can certify knottedness only. An unknot verdict requires the completed rank calculation. These invariants hold for every allowed transition regardless of how favorable the timing turns out to be. The policy provides bounded optional work and exact continuation; it supplies no general quasi-polynomial recognition theorem.

## 35.3 Transition tests and benchmark design

Five new tests exercise zero marked allowance, exhaustion during a block determinant, one versus two available queries, preserved cached Euler values, and a global deadline expiring during the transition. Two hundred budgeted scans of twenty small random diagrams and their mirrors agree with independent reduced homology. Zero-cap and low-cap paths are checked for both knotted and unknotted outcomes, and no partial observation returns an unknot verdict. Existing fallback expectations now reflect the retained Euler phase. All 582 maintained tests pass; logs are `data/shadow-fallback-tests.txt` and `data/shadow-fallback-integrated-tests.txt`.

`fast/benchmark_shadow_fallback.py` compares the earlier scheduler from revision `82fdfe15b` with the new one, sharing all other dependencies. It times seven shuffled paired rounds with an identical baseline control, fresh input and observer setup, and one excluded warmup. Every successful arm must agree on the verdict. Method and stage differences are retained as evidence, and any `UNKNOWN` suppresses all ratios for that case. The raw backend and full recognition are separate scopes.

Alongside all sixteen existing examples under default limits, the benchmark uses reduced-work and single-query controls. A larger default-limit example joins the Conway knot to $T(3, 151)$, giving a validated 313-crossing PD code. It is tested with the ordinary heuristic order and with a supplied order that processes the Conway summand first. The latter case is explicitly raw-only; its favorable ordering is not attributed to the default heuristic. Complete input codes, actual scan orders, budgets, source hashes, and raw samples are retained in `fast/results/shadow_fallback_20261008.json`.

| Input/order | $W$ | $K$ | Old stop | New stop | Speedup | A/A |
|---|---|---|---|---|---|---|
| Conway sum 8 | $10^6$ | 4096 | 11 | 11 | 0.997 | 0.996 |
| Conway sum 8 | 0 | 4096 | 88 | 11 | 5.530 | 1.028 |
| Conway sum 8 | 2800 | 4096 | 88 | 11 | 5.154 | 1.008 |
| Conway sum 8 | 2800 | 1 | 88 | 88 | 1.010 | 1.003 |
| $T(3, 5)$ | 0 | 4096 | 10 | 10 | 0.982 | 0.997 |
| Hard unknot 8 | 0 | 4096 | 8 | 8 | 0.984 | 0.998 |
| 313 crossings, heuristic | $10^6$ | 4096 | 313 | 302 | 1.016 | 1.009 |
| 313 crossings, supplied | $10^6$ | 4096 | 313 | 11 | 4.786 | 1.007 |

Ratios are median paired baseline/fallback times for complete raw scans. Stop columns give the processed crossing count. A/A compares identical baselines.

**Retained query capacity produces the gain.** On the eightfold Conway sum, $W = 0$ improves the raw scan by 5.53 times, and $W = 2800$ by 5.15 times. Both schedulers respect the same marked-work cap. In the latter case each spends all 2,800 units and has computed one Euler value; the new scheduler uses one further query to obtain the component-Euler certificate at crossing 11. The baseline completes all 88 crossings. With $K = 1$, neither can compute that further value and both finish the full scan, with a timing ratio of 1.01. This negative control checks that the improvement does not come from silently resetting the query budget. With default marked work, this example never switches and its ratio is 0.997.

The retained reserve can also cost time without strengthening a bound. With $W = 0$, $T(3, 5)$ uses three fallback values but still requires closed rank, with ratio 0.982. The hard eight-crossing unknot uses one value and still finishes the entire scan, with ratio 0.984. These small regressions are part of the comparison, not excluded as inconclusive examples.

**Ordering determines when the reserve can help.** For the 313-crossing connected sum, both policies stop marked work at 999,932 units under the default million-unit allowance. The heuristic order first processes the torus part. Fallback gives a certificate at crossing 302 rather than 313, but the measured speedup is only 1.016, with an A/A ratio of 1.009. This does not demonstrate a substantial timing gain. With the supplied order, fallback needs only one new query and certifies knottedness at crossing 11. Its measured raw-scan speedup is 4.79 times, with an A/A ratio of 1.007. The respective median times are about 170 and 821 milliseconds. This is a scheduler gain for the supplied order, not evidence that the ordinary order selector discovers it.

All 45 case/scope comparisons complete and agree on verdict. None of the sixteen default-limit examples switches modes; their raw ratios range from 0.971 to 1.031. Earlier recognition filters decide every full-recognition case before this observer runs, giving ratios from 0.982 to 1.019 and no demonstrated full-recognizer gain. In particular, the larger connected sum is already detected by the Rasmussen interval filter. The new policy therefore improves robustness of an optional backend under its resource limits while leaving the universal complexity question unresolved.

# 36 Certified square-root frontiers and a subexponential scalar filter

The previous integration reduced the cost of observing a fixed scan. We now address its order. The maintained Python implementation constructs a crossing order of width $O(\sqrt{n})$ for every closed classical projection with $n$ crossings. It supplies a separately checkable hierarchy and compares the result with the existing greedy order, retaining the smaller measured frontier. Combined with the exact fixed-color Potts evaluator, this gives a general $\mathrm{poly}(n)2^{O(\sqrt{n})}$ bit bound for *one scalar Jones specialization*, with optional resource caps disabled. It does not give that bound for unknot recognition: equality with the unknot's scalar is inconclusive.

This is an implementation of classical planar-separator ideas [14], not a new planar-separator theorem. The contribution here is a constructive order, a graph-only certificate verifier, its integration with the existing exact scalar backend, and explicit accounting of which recognition bottleneck remains. Nothing in this section requires connected prefixes, a common boundary disk, or a bound on surviving Khovanov objects.

## 36.1 A bounded-radius middle band

Represent the projection by a plane multigraph $G$: crossings are vertices, diagram arcs are edges, and the PD rows specify the cyclic orders of half-edges. The degree is four, with loops counted twice. The constructor checks that each edge label occurs twice and that the rotation system is spherical on every connected component. Loops may then be deleted for separator construction: they neither connect distinct components nor cross a cut of crossing vertices. Parallel edges remain in the embedded map. Disconnected projections are handled component by component.

Consider one connected recursive subproblem with $s$ vertices. Take breadth first search levels $V_0, \dots, V_d$ from a root, and put $r = \lceil \sqrt{s} \rceil$. Let $a$ be the first level for which $\sum_{i \le a} |V_i| \ge s/2$. If $a \ge r$, choose a smallest level in $[a - r, a - 1]$ as the lower boundary $\ell$; otherwise use the empty virtual level $\ell = -1$. If $a + r \le d$, choose a smallest level in $[a + 1, a + r]$ as the upper boundary $u$; otherwise put $u = d + 1$. Each real chosen level has size at most $s/r \le r$.

The vertices strictly below $\ell$ and strictly above $u$ number at most $s/2$ on either side. An edge changes its BFS distance by at most one, so removing the two boundary levels separates these outside vertices from the middle band. Retain levels below $u$, and, when $\ell \ge 0$, contract the connected inner BFS tree on levels at most $\ell$ to the root. Delete the resulting loops. The resulting plane multigraph $H$ has radius at most $2r$. Give weight one to each original middle-band vertex and weight zero to the contracted root. The total weight is at most $s$.

The ribbon-map implementation contracts a tree by following its boundary through the removed half-edges. It does not repeatedly rebuild the graph after individual edge contractions. This matters for long bands and large inner balls. An independent check after construction verifies the actual separator size and every remaining component's size in the original graph.

## 36.2 Three short paths from a dual centroid

For each face of $H$, add a center and a spoke to every corner occurrence. Keep the old edges. A bridge or a repeated vertex on a face walk therefore causes repeated spokes, not an assumption that the face boundary is a simple cycle. The resulting map is triangulated; every new center has distance at most $2r + 1$ from the root. The one-vertex case is handled directly.

Choose a BFS spanning tree $T$ of this triangulation. The dual edges of primal edges outside $T$ form a dual spanning tree $T^*$. Assign each weighted original vertex to one incident triangular face. Choose a weighted centroid face $f$ of $T^*$: every component of $T^* - f$ has at most half the total assigned weight. Remove the union $P$ of the three primal tree paths from the root to the vertices of $f$.

To see balance, consider a dual edge incident to $f$. Its primal edge and the tree path between its endpoints bound a fundamental cycle. The cycle's vertices lie in $P$, since both endpoints belong to $f$. It separates the corresponding branch of the dual cotree from the other branches. Consequently each surviving primal component has its assigned faces in one branch of $T^* - f$. Any weight assigned to $f$ itself belongs to a vertex of $f$, hence to $P$. Every remaining component therefore has at most half the middle-band weight. This argument permits arbitrary nonnegative face weights; it does not require each individual assigned weight to be small.

Return only the *original middle-band vertices* of $P$, together with the two boundary levels. Discarding artificial centers from this separator is safe: paths in the original graph use old vertices and old edges only. The old edges were retained during triangulation, so any original path avoiding the returned vertices is also a path in the triangulation avoiding $P$. The contracted root represents vertices already separated from the middle by the lower level. Thus the original graph has a separator $S$ with

$$|S| \le 2r + 3(2r + 2) = 8\lceil \sqrt{s} \rceil + 6, \qquad |C| \le s/2 \quad (C \text{ a component of } G - S).$$

The constants are deliberately loose. Small separators on practical inputs matter more than optimizing this universal numerical bound.

## 36.3 From the hierarchy to a certified order

Recursively order each component of $G - S$, then put $S$ last. Stop at a fixed leaf size, eight by default; the API permits integer cutoffs from one through 64. Process distinct initial connected components consecutively. At a prefix cut, any crossing arc is incident to a separator on the current recursion path, or to a vertex of the current leaf. Completed sibling components have no edges to unvisited siblings. Since the original degree is four, the frontier is at most

$$4\left(\sum_{S \text{ on the path}} |S| + |\text{leaf}|\right).$$

Sizes halve along a path. The square-root terms form a convergent geometric sum, while the additive constants contribute $O(\log n)$, giving width $O(\sqrt{n})$. This holds for every prefix, including prefixes inside a separator's own vertex list.

The JSON certificate records the input digest, leaf cutoff, separator and child lists, flattened order, actual maximum and total frontier, and the recursively computed bound

$$b(\text{leaf}) = 4|\text{leaf}|, \qquad b(\text{node}) = 4|S| + \max_C b(C).$$

Each crossing occurs once in the hierarchy. With normalized crossing indices the hierarchy uses $O(n \log n)$ bits; arbitrary input-label bit length also counts toward input processing. Construction and verification take polynomial time. The verifier rebuilds adjacency directly from edge owners and checks coverage, disjointness, the exact remaining components, half-balance, separator sizes, order, actual profile and the bound. It calls no planar-map or separator constructor. Thus even a mistake in the embedded construction cannot silently publish an incorrect certified frontier bound.

The practical wrapper also computes or accepts a candidate order. It chooses the lexicographically smaller pair (maximum frontier, total frontier) between that order and the separator order. If greedy wins, the separator witness is retained: explicit profile comparison proves that the selected order has the same asymptotic upper bound. This is a bounded-choice ordering policy, not a progress-based restart of a scan. It complements the earlier adaptive elimination and observer fallbacks.

## 36.4 Exactly which subexponential computation follows

For a fixed checkerboard shading, let $f_i$ count Tait vertices incident to both processed and unprocessed crossings, and let $w_i$ be the crossing frontier. The corner argument in Section 21 gives $2f_i \le w_i$, including repeated face walks. A fixed $q \ge 5$ Potts query uses at most $q^{f_i+2}$ temporary assignments and exact pairs in

$$R_q = \mathbb{Z}[x]/(x^2 - (q-2)x + 1).$$

Coefficient bit lengths are $O(n \log q)$; arithmetic and boundary-key work add polynomial factors. Substituting the certified width yields

$$T_{\text{scalar}}(n) = \text{poly}(n)\, 2^{O(\sqrt{n})} \quad \text{for fixed } q.$$

This is a bit-complexity statement, not a count of unit-cost operations on unbounded integers. For the existing modular matching filter, allowing all boundary pairings gives the weaker but still subexponential bound $\text{poly}(n)2^{O(\sqrt{n}\log(n+1))}$ on the same order. These are universal upper bounds for scalar evaluation, not claims that our implementation or constants are optimal.

The optional `potts-separator` Jones backend constructs the witness and then calls the existing exact evaluator. With the Python API, setting both `max_states=None` and `max_transitions=None` disables the local caps for the theoretical complete query. Production defaults retain the 4096-state and 200 000-transition caps. Exhaustion is inconclusive, and the recognition deadline remains global. A zero local budget declines before constructing a hierarchy. Once a complete certificate has been published, scalar exhaustion retains its order for the homology fallback. If subsequent Reidemeister reductions change crossing indices, the witness is rebuilt on the reduced diagram before supplying an order to a window query or the scanner. A global interruption during preparation publishes no partial certificate.

Three obstacles prevent promoting this to a general recognition theorem. First, agreement at a scalar specialization is not an unknot certificate. Second, a frontier bound alone does not bound the number of surviving relative Khovanov objects or intermediate matrix size. Third, this hierarchy does not certify common-disk prefixes, so the stronger conditional bounds of the disk-transfer and nice-window backends cannot simply be substituted. The descending grid family already described in this report has raw width $\Omega(\sqrt{n})$ for every order. That obstructs a universal polylogarithmic-width strategy on a fixed diagram, but it is no lower bound for recognition: a descending certificate immediately handles the family.

### 36.5 Validation and measurements

All 589 integrated tests pass. The new tests compare acceptance of 500 random ribbon maps with an independent spherical-map implementation, verify certificate corruption is rejected, and exercise loops, disconnected projections, long bands and grids. Sixteen small knot diagrams are checked against independent cube homology using actual separator orders; both Potts shadings agree with the previous exact evaluator. Resource tests distinguish partial preparation from a completed certificate followed by scalar exhaustion, and check that a changed diagram receives a fresh certificate.

The separate reproducible driver `fast/audit_separator_orders.py` audits 114 inputs, including 100 randomly generated, crossing-permuted and edge-relabelled braid closures. Its independent hierarchy checker uses union-find over edge-owner pairs rather than the production component traversal. It independently counts cut edges and checks balance, coverage, components and the stored bound. The result archive retains all inputs, certificates, source hashes, five shuffled ordering rounds and three shuffled kernel rounds, with warm-ups excluded. Table 1 separates measured frontiers from preparation costs.

| Family | $n$ | Greedy | Separator | Selected | Greedy ms | Bounded ms |
|---|---|---|---|---|---|---|
| Grid $8 \times 8$ | 64 | 10 | 46 | 10 | 0.626 | 4.524 |
| Grid $32 \times 32$ | 1024 | 34 | 214 | 34 | 6.082 | 142.782 |
| Tree medial, height 5 | 62 | 16 | 12 | 12 | 1.118 | 4.492 |
| Tree medial, height 7 | 254 | 64 | 14 | 14 | 5.218 | 26.624 |
| Tree medial, height 9 | 1022 | 256 | 18 | 18 | 23.653 | 140.075 |

Table 1: Widths and median order-preparation times. Bounded preparation includes the separator witness, its verification and the greedy comparison; these are additional costs, not speedups.

The tree-medial family consists of medial diagrams of embedded complete binary trees, with one crossing per tree edge. These are simple nugatory unknot diagrams used to stress the greedy order; they are not hard recognition instances. In isolated modular matching queries, height five falls from a median 47.665 ms to 1.941 ms (24.6 times), excluding order preparation. At heights six and seven, all greedy runs exhaust the state cap while the selected orders finish in 4.126 and 8.788 ms. Those censored runs have no completion-time speedup ratio. All completed scalar answers are inconclusive for recognition.

The exact Potts queries on these same trees differ little between orders: the advantageous checkerboard shading already has trivial frontier. At height seven the medians are 2.572 and 2.585 ms, far below the extra order preparation cost. The grids also retain their better greedy order. These results support keeping the new option explicit: the certified asymptotic bound and a reduction in matching-filter work do not establish a general practical speedup, a full-recognition speedup, or a smaller homology complex.

# 37 Deferring separator work until the scalar query needs it

The separator backend supplies a useful universal bound, but eagerly building its certificate slows many easy scalar queries. The new optional `potts-adaptive` backend starts with the ordinary order and watches *actual completed transitions*. It postpones separator construction until that work becomes large, while retaining the preceding section's subexponential fixed-color scalar bound. This is the introsort principle applied to one exact invariant query: limit speculative work, then use a method with a proved bound. It is not yet an analogous worst-case bound for the complete unknot recognizer.

## 37.1 The policy and its persistent state

For an $n$-crossing input, let the initial work threshold be $B = 64n$ and the short-tail allowance be $r_0 = 2\lceil \log_2(n+1) \rceil$. The evaluator begins on a supplied order, or the existing greedy order when none is supplied. At each completed crossing before the last, it observes the cumulative transition count. The actions are as follows.

1. Below $B$, continue normally. A completed scalar answer needs no separator certificate.
2. On first reaching $B$, if at most $r_0$ crossings remain, finish the current table. This enters `polynomial-tail` mode; the argument below supplies a bound without requiring a width certificate.
3. Otherwise construct and independently verify a separator hierarchy. Compare its order with the better of the current order and a freshly computed greedy order, using (maximum frontier, total frontier).
4. If the selected order is unchanged, retain the *same* Potts table and continue. No processed crossing or transition is repeated. This is `certified-continue` mode.
5. A changed order enters `certified-restart` mode. Discard the unfinished scalar table and restart once. The old exception handler is exited before allocating the replacement table, releasing the abandoned evaluator's traceback and local references.

A local state-cap failure can also trigger the preparation when it has not already happened and transition allowance remains. If the selected order is unchanged, repeating the query cannot repair that failure: publish the completed order certificate and decline immediately. A spent transition allowance never triggers preparation. There is at most one separator preparation and at most one restart. A fixed state cap may still interrupt the polynomial tail, in which case the same rules apply.

The `ordering_policy` evidence records the mode, threshold, short-tail allowance, preparation/restart counts, abandoned crossings and transitions, and total spent transitions. `PottsLimit` carries work counters only; it never exposes a partial partition function as a completed value. The public exact evaluator's existing complete-result behavior is unchanged.

## 37.2 A single transition budget, including discarded work

Suppose the user supplies transition allowance $T$, and the first attempt spends $a$ transitions. A restarted evaluator receives exactly $T - a$. If it spends another $b$, the returned count and any

exhaustion exception report $a + b \le T$. There is no fresh allowance for the second implementation. State caps keep their existing meaning as limits on represented table keys; the reported peak is the larger attempt peak, not a byte-memory estimate. Other scalar geometry and coefficient statistics describe the final attempt.

Order preparation is charged to the shared global deadline, not to the scalar-transition counter. The deadline callback propagates through the policy and verifier. A preparation interrupted globally publishes no partial certificate. After a complete witness is available, a later local scalar failure retains it for the recognition fallback. Subsequent crossing-changing Reidemeister simplification rebuilds it on the reduced diagram, as in the preceding section. When the scalar query finishes in ordinary or polynomial-tail mode, no separator witness has been computed; those modes make no width promise for the later homology scan.

## 37.3 Why the deferral preserves the asymptotic scalar bound

Fix the number of colors $q \ge 5$. One crossing introduces at most two Tait vertices. From one stored spin-equality state there are at most $q^2$ extensions, because every canonical extension represents at least one labeled choice for those two vertices. Aggregation and cancellation only decrease the number of states. Moreover, after at least one crossing the number of stored states is at most the number of transitions spent so far.

Immediately before the first completed stage reaching $B$, at most $B + 1$ states are available. The crossing that first passes the threshold therefore takes at most $(B + 1)q^2$ further transitions. Stopping only at a crossing boundary causes polynomial, rather than unbounded, overshoot. The same observation bounds a final crossing that completes the entire query before the policy is invoked. Canonical-key work and exact coefficient arithmetic have polynomial bit cost at fixed $q$.

If the policy enters a short tail of $r \le r_0$ crossings, the number of additional transitions is bounded by

$$O\left((B+1)q^2 \sum_{j=1}^{r} q^{2j}\right) = O\Big(nq^{2r+O(1)}\Big) = n^{O(1)} \quad \text{for fixed } q.$$

The degree of this polynomial can be large. This is an asymptotic safeguard, not a promise that every permitted tail is practically cheap. Existing state, transition and deadline limits remain active. If a state-cap failure interrupts the tail and requests a new order, the abandoned tail work is still bounded by this polynomial.

In every other uncapped case the policy eventually obtains a certified $O(\sqrt{n})$-frontier order. Continuing an unchanged order or restarting on the selected order costs $\mathrm{poly}(n)2^{O(\sqrt{n})}$ bit operations. A continuing run does not repeat its earlier work; a restarted run adds only the polynomially bounded first attempt. Construction of the hierarchy is polynomial. Hence the complete adaptive *scalar* query has the same general subexponential upper bound as eager separator evaluation.

This theorem does not assert competitiveness with the fastest order on each input. Under a finite allowance, discarded transitions may prevent completion where some single-order query would have completed. Frontier improvement also does not force fewer spin states after exact cancellations. These are reasons to expose the option and its policy evidence while retaining the existing default, not reasons to reset the budget silently.

## 37.4 Independent checks and measured scope

The six new test cases cover actual measured-work restarts, state-cap restarts, unchanged-order continuation, a completed short tail, exact shared budget boundaries, globally interrupted preparation, zero budgets, invalid options, CLI routing and the complete recognition fallback.

Twenty small diagrams, three color counts and both shadings give 120 comparisons with an independent whole-cube Laurent-Jones oracle. That oracle uses neither Tait graphs nor production ring arithmetic. Its cases exercise policy switches as well as ordinary completion. The five-color blind weaving example still falls through to Khovanov recognition rather than receiving an unknot verdict.

The benchmark uses seven shuffled rounds after an excluded warm-up, fresh diagram construction and all order preparation inside every timer. It separates 15 scalar scopes from 11 full-recognition scopes. Ordinary exact Potts has an identical A/A control; eager separator, final adaptive and the initial adaptive policy are compared on the same scalar queries. Every completed scalar comparison checks cross-multiplied normalized values, because changing shading changes both the partition function and its normalizing factor. Limits remain censored outcomes.

The initial policy lacked the short-tail rule. Its source snapshot matches the hash in the retained initial measurements, and the final driver also reruns it as a paired control. On the $10 \times 10$ descending grid, the threshold is crossed after crossing 90, at 6555 transitions; the whole query requires 7109. With ten crossings left and $r_0 = 14$, the final policy avoids the otherwise unnecessary certificate. Its median time is 21.457 ms versus 30.556 ms for the initial policy, a median paired improvement of 1.415 times. On a shuffled 16-crossing grid, however, the tail rule forgoes a useful restart and is slower than the initial policy (paired ratio 0.913). Both results are retained; the rule is not claimed to optimize every instance.

| Scalar input | Ordinary | A/A | Eager | Initial | Adaptive |
|---|---|---|---|---|---|
| Conway | 0.499 | 0.514 | 1.134 | 0.534 | 0.547 |
| Tree medial 254 | 9.617 | 9.636 | 34.412 | 9.705 | 9.703 |
| Grid 16, shuffled | 5.875 | 5.670 | 1.390 | 5.236 | 5.692 |
| Grid 64, shuffled | limit | limit | limit | 23.877 | 23.893 |
| Grid 100 | 21.241 | 21.129 | 29.452 | 30.556 | 21.457 |
| Grid 144 | 135.286 | 135.237 | 146.759 | 149.998 | 153.190 |

Median milliseconds including all setup. A/A repeats ordinary exact Potts. A limit is a censored query, not a completed scalar value.

On the 254-crossing tree-medial diagram, ordinary and adaptive scalar times are 9.617 and 9.703 ms, versus 34.412 ms for eager separator preparation. On the shuffled 36-, 64- and 100-crossing grids, both adaptive policies finish within the default budgets while ordinary and eager supplied-order queries hit their state caps. The adaptive fallback's fresh greedy comparison is part of this benefit; the eager backend compares only the supplied order with its separator order. These capped runs have no completion-time speedup ratio. On the ordinary 144-crossing grid, the final policy continues the same table after certification and takes 153.190 ms versus 135.286 ms for ordinary exact Potts: the guarantee still has a preparation cost.

Full recognition on the measured inputs gains no consistent advantage over ordinary exact Potts. Earlier certificates bypass the scalar query on many cases. Avoiding eager preparation helps where it would have been used, but this is not a general improvement over the historical matching default. The new backend remains optional.

## 37.5 A relevant geometric result, with a different scope

The research refresh found no new delivered unknot report in the drop zone. Lackenby's revised *Some fast algorithms for curves in surfaces* [13] supplies a useful related primitive: Theorem 1.1 computes geometric intersection numbers of normal 1-manifolds in time polynomial in triangulation size and the logarithms of their weights. Theorem 1.2 gives a corresponding isotopy test. The normalization theorem also depends on the input's simplification number; that extra parameter must not be dropped from a complexity claim. These results strengthen the case that very large

geometric objects can admit efficient compressed operations. They neither bound the depth of an accelerated 3-manifold hierarchy nor the number of surviving chain objects in our implementation. We therefore record them as candidate geometric primitives, without treating them as a missing recognition theorem or adding an unused surface API to the scalar scanner.

# 38 A collision-free Jones query and full polynomial recovery

The fixed-color queries above can certify a nontrivial Jones value, but equality can hide a nonconstant polynomial. The optional `potts-faithful` backend removes this specialization ambiguity. It chooses an input-dependent quadratic integer and either decides whether the *whole* normalized Jones polynomial is one, or recovers that polynomial. With resource caps disabled, both queries have the general bit bound

$$\mathrm{poly}(n)\, 2^{O(\sqrt{n}\log(n+1))}.$$

This is subexponential in the crossing count. It is a bound for Jones computation, not for complete unknot recognition. The implementation does not assume that the Jones polynomial detects the unknot: a completed identity result still proceeds to an independent recognition certificate.

This section combines elementary coefficient bounds and integer encoding with the preceding maintained separator and Potts algorithms. It makes no priority claim for subexponential Jones computation. In particular, efficient Jones computation under graph-width restrictions predates this implementation; see the literature review in `reports/17/research/literature_landscape.md`.

## 38.1 Uniform coefficient and degree bounds

Write $V_D(t)$ for the normalized Jones polynomial of an $n$-crossing knot, with $V_{\bigcirc} = 1$, and put $W(x) = V_D(x^{-1})$ for $x = A^4$. The bracket state sum has $2^n$ states. Each state has at most $n+1$ circles: changing one smoothing changes the circle count by one, and smoothing a connected $n$-crossing projection cannot produce more than $n+1$ components. The normalized circle factor is $(-A^2 - A^{-2})^{c-1}$, whose coefficient $\ell^1$ norm is $2^{c-1}$. Writhe normalization only changes signs and shifts degrees. Thus

$$\|W\|_1 = \|V_D\|_1 \le 2^n 2^n = 4^n =: C.$$

The smoothing monomial has $A$-exponent of absolute value at most $n$; circle factors contribute at most $2n$; writhe contributes at most $3n$. Every resulting exponent is divisible by four for a knot, giving $|\deg_x| \le 3n/2$. We use the simpler integral bound $D = 2n$. The crossing-free knot is handled separately with polynomial one.

## 38.2 Choosing a specialization that cannot hide nontriviality

Choose

$$M = 2^{4n+2} = 4 \cdot 16^n, \qquad q = M + 2, \qquad x^2 - Mx + 1 = 0,$$

and consider the larger real root $x$. Since $M \ge 4$, this root satisfies $x > M - 1 > C$. Suppose that $W - 1$ is a nonzero Laurent polynomial. Multiplying by a monomial produces an ordinary integer polynomial $P$ with nonzero leading coefficient of absolute value at least one. Its norm is at most $C + 1$, so the sum of absolute values of all its other coefficients is at most $C$. At a real argument $x > C$, the leading term has absolute value greater than the sum of the lower terms. Thus $P(x) \neq 0$, and hence $W(x) \neq 1$.

Consequently equality of the exact quadratic-ring pairs for the partition function $Z$ and the unknot normalization $U$ is equivalent to $V_D(t) = 1$ as a polynomial. The normalization is nonzero: its formula is a signed monomial times a power of $x + 1$. No floating-point approximation or

probabilistic identity test is used. The same leading-term argument applied to a difference of two bounded polynomials proves injectivity, because $M - 1 > 2C$.

The root argument alone would permit a smaller base for identity testing. We deliberately use a single larger base for both identity testing and the coefficient recovery below, keeping their result semantics consistent.

## 38.3 Recovering all coefficients from two integers

The existing evaluator supplies $Z$ and $U$ as pairs representing elements of $\mathbb{Z}[x]/(x^2 - Mx + 1)$. First recover the normalized pair exactly. If $U = c + dx$, its conjugate is $(c + Md) - dx$ and its norm is

$$N(U) = c^2 + Mcd + d^2.$$

Multiply $Z$ by this conjugate and divide both coordinates by $N(U)$. The divisions have zero remainder because $Z/U = W(x)$ is a Laurent polynomial with integer coefficients and $x^{-1} = M - x$ is integral. The code checks the remainders, rather than rounding them.

To understand the coordinate encoding, introduce formal polynomials $U_{-1}(M) = 0$, $U_0(M) = 1$, $U_1(M) = M$, and $U_j(M) = MU_{j-1}(M) - U_{j-2}(M)$. For $k \geq 1$,

$$x^k = -U_{k-2}(M) + U_{k-1}(M)x, \qquad x^{-k} = U_k(M) - U_{k-1}(M)x.$$

The first formula for $k = 1$ uses $U_{-1} = 0$. Each $U_j$ has coefficient norm at most $2^j$, by induction on the recurrence. Therefore reduction of $W$ yields coordinate polynomials $a(M), b(M)$ of degree at most $D$, each with every coefficient bounded by

$$C2^D \leq 4^n 2^{2n} = 16^n = M/4.$$

The computed integers $a(M), b(M)$ consequently determine these coordinate polynomials uniquely by balanced base-$M$ expansion. Repeatedly take the residue modulo $M$, subtract $M$ when it is at least $M/2$, and divide the remaining difference by $M$. This handles negative coordinates as well as positive ones. Actual coefficients lie strictly inside the balanced digit interval; there is no carry ambiguity.

Finally substitute the formal identity $M = x + x^{-1}$. A decoded term $cM^j x^e$, where $e \in \{0, 1\}$, contributes

$$c \sum_{k=0}^{j} \binom{j}{k} x^{j-2k+e}.$$

Aggregate equal exponents, remove zero coefficients, and negate the exponents to return the standard variable $t = A^{-4}$. The implementation independently checks the final degree and coefficient bounds, and consistency with the pair-equality identity result. These checks detect inconsistent input to the decoder; they do not replace the state-sum correctness argument. There are $O(n)$ output coefficients of $O(n)$ bits each. Balanced decoding, exact normalization division and binomial expansion take polynomial bit time.

## 38.4 Why exponentially many colors do not cause exponential work

Applying the fixed-$q$ bound $q^f$ directly would lose the desired result. The implementation instead stores *canonical equality partitions* of the $f$ active Tait vertices. There are at most the Bell number $B_f$ such partitions, independent of $q$. Its extension loop considers each existing color class and at most one new class, weighted by the number of unused colors $q - k$. It never enumerates $q$ colors.

The certified crossing order has frontier $w = O(\sqrt{n})$, hence at most $f \leq w/2$ active faces of either chosen checkerboard shade. As $B_f \leq \max(1, f^f)$, the number of represented states is

$2^{O(\sqrt{n}\log(n+1))}$. Each crossing introduces at most two vertices; the number of extensions from one state is at most $(n+3)^2$. The original scalar coefficient bound is $O(n \log q)$ bits, now $O(n^2)$. Key manipulation, multiplicities, arithmetic and the number of stages therefore contribute only polynomial factors.

The adaptive preliminary scan needs separate accounting because $q$ is no longer fixed. Before its first threshold crossing it has at most $64n+1$ states. One completed-stage overshoot costs at most $(64n+1)(n+3)^2$ transitions. If it takes the short-tail option with at most $2\lceil \log_2(n+1) \rceil$ crossings remaining, a conservative work bound is

$$\mathrm{poly}(n)(n+3)^{O(\log(n+1))} = 2^{O(\log^2(n+1))}.$$

Thus the mode called `polynomial-tail` in the shared fixed-color policy has only a *quasi-polynomial* bound here. That cost is still dominated by $2^{O(\sqrt{n}\log(n+1))}$. All other uncapped runs obtain the separator certificate and continue or restart once. Discarded work and a possible state-cap interruption obey the same bounds. The polynomial construction and verification costs do not change the final bound.

## 38.5 Interface, evidence and resource semantics

`faithful_potts_exact` in `fastunknot/faithful_jones.py` returns the existing exact query fields together with `polynomial_identity`: the algorithm version, crossing count, whether the polynomial is one, and the exponents defining the coefficient and encoding bounds. Its optional `include_polynomial=True` returns sorted Laurent coefficients in `jones_polynomial`, with variable `t` and signed hexadecimal coefficient strings. Raw quadratic pairs and the raw color count remain Python integers.

The command `python -B -m fastunknot jones FILE -backend potts-faithful` requests full recovery. Recognition through `-jones-backend potts-faithful` requests identity testing only. The color count is derived from the input; `-potts-colors` does not override it. The Python API and recognition defaults remain 4096 states and 200000 transitions; the standalone `jones` command has no local caps unless supplied. An exhausted allowance remains inconclusive. The uncapped complexity statement requires `max_states=None` and `max_transitions=None` and no external deadline. A restarted evaluation shares the original transition allowance. Global checks run during decoding too. No incomplete identity or polynomial is published if the query fails before completion.

Exact witnesses encode $q$ in `q_hex` above 256 bits and retain the existing integer `q` field for smaller values. Large ring descriptions refer symbolically to $q$. This avoids Python's decimal-conversion limit without changing process-global settings. The factorized exact backend uses the same ring description and witness conversion.

Identity equality has two deliberately separate consequences: it settles the algebraic question $V_D = 1$, but yields `INCONCLUSIVE` for the Jones recognition filter. The complete recognizer retains this stronger evidence and continues to its existing exact fallback. Neither this backend nor the adaptive scheduler establishes a general subexponential recognition bound.

## 38.6 Validation and measurement

The initial integration passed all 602 tests in 111.769 seconds. That run is preserved in `data/faithful-jones-integrated-tests.txt`.

Seven new maintained tests compare 156 full polynomials against the independent whole-cube Laurent oracle: both checkerboard shades and both mirrors of 35 random diagrams, three named fixtures, and the crossing-free knot. Another 187 synthetic bounded Laurent polynomials check recovery at extreme signed degrees and coefficient sizes, including arbitrary exact normalization.

Tests exercise a real certified restart, the precise shared transition limit, the known $q = 5$ weaving collision, recognition fallback after polynomial identity, cancellation during decoding, zero budgets, CLI output, and 20001-bit color counts through both exact witness backends.

`benchmark_faithful_jones.py` measures fixed $q = 6$ twice as an A/A control, faithful identity, and full recovery in seven shuffled rounds after an excluded warmup. Every timing includes construction of a fresh diagram and all order preparation. Small inputs have independent cube checks; completed larger recovered polynomials are checked by specialization back to the fixed-$q$ result. These arms deliver different guarantees, so their timings describe the price of stronger information rather than speedups for an identical task. Capped results are retained as such. The backend remains optional, and these invariant timings do not claim faster full recognition.

| Input | Fixed $q = 6$ | A/A | Identity | Full polynomial |
|---|---|---|---|---|
| Conway | 0.563 | 0.568 | 0.608 | 0.718 |
| Hard unknot 8 | 0.588 | 0.588 | 0.647 | 0.673 |
| Weaving 10 | 0.553 | 0.557 | 0.598 | 0.648 |
| Tree medial 254 | 17.981 | 18.105 | 125.268 | 239.190 |
| Grid 64, shuffled | 21.704 | 21.792 | 25.720 | 25.703 |
| Grid 100 | 21.873 | 21.874 | 38.744 | 38.943 |

Median milliseconds including setup; distinct output guarantees. A limit is a censored query.

In the initial implementation, all 13 measured inputs completed in every arm and round. The Conway knot costs 0.563 ms at fixed $q = 6$, 0.608 ms for collision-free identity, and 0.718 ms for full recovery. The larger integer arithmetic is more costly on the 254-crossing tree medial: 17.981 ms, 125.268 ms, and 239.190 ms, respectively. Grid 100 rises from 21.873 ms to 38.744 ms for identity and 38.943 ms for recovery. This cost is a reason to retain the inexpensive fixed specializations and the current default filter; stronger asymptotic information is not a measured per-input runtime improvement.

## 38.7 Using proved identity to avoid redundant reconstruction

The first decoder always formed the norm of the unknot normalization, divided both coordinates and performed balanced decoding, even when $Z = U$ had already proved $V_D = 1$. These potentially large integer operations are unnecessary in that case. The revised decoder validates the input-sized color count and nonzero normalization, checks the two exact pairs for equality, and returns the complete polynomial $\{0 \mapsto 1\}$ immediately when they agree. For every other input it follows the original recovery path.

This shortcut uses the collision-free theorem above; applying it to an arbitrary fixed specialization would be incorrect. The regression check therefore rejects a fixed-$q$ result, rejects zero normalization even when both pairs are zero, and exercises cancellation immediately before returning the identity. Global checks still run before validation and before this early publication. The complete maintained suite now passes 603 tests in 110.500 seconds, recorded in `data/jones-identity-shortcut-integrated-tests.txt`.

The added benchmark preserves the exact earlier decoder from commit `58006240a`, records its source hash, and compares complete polynomial queries in seven shuffled paired rounds after an excluded warmup. Every arm constructs a fresh diagram and includes all ordering and scalar work. A repeated old-decoder arm supplies an A/A control. The full returned results, including all exact pairs, coefficients and work counters, agree in every arm. The retained initial benchmark above and this new paired measurement are separate runs; their absolute times should not be compared across runs.

| Full polynomial query | Before | A/A | After | Paired speedup |
|---|---|---|---|---|
| Trefoil | 0.161 | 0.165 | 0.167 | 0.966 |
| Conway | 0.605 | 0.579 | 0.594 | 1.038 |
| Hard unknot 8 | 0.342 | 0.341 | 0.327 | 1.045 |
| Tree medial 126 | 17.718 | 17.727 | 10.135 | 1.742 |
| Tree medial 254 | 142.257 | 141.964 | 71.874 | 1.959 |
| Grid 100 | 38.266 | 38.046 | 38.644 | 0.998 |
| Grid 64, shuffled | 24.363 | 24.570 | 25.246 | 0.999 |

Median milliseconds including setup. Ratios are medians of paired before/after times for identical complete results.

On the 126-crossing tree medial, the median paired speedup is 1.742; on the 254-crossing example it is 1.959, with median times falling from 142.257 ms to 71.874 ms. Grid and small-knot timings show little change; the trefoil's paired ratio is 0.966, so no broad improvement is claimed. The large tree inputs benefit because redundant normalization arithmetic is a substantial part of full recovery. The scalar evaluation is still required, and the general asymptotic bound is unchanged. Recognition already requests identity without coefficient recovery, so these are full-Jones-query gains, not gains in the complete recognition pipeline.

# 39 Auditing a claimed route from Jones identity to recognition

A June 2026 preprint by Carmi and Cohen claims that the Jones polynomial detects the unknot [12]. We checked a specific prerequisite: Lemma 5.8 bounds residual Jones–Vassiliev degree by $\widehat{B} + 2$, under its machine hypotheses. Section 5.1.1 sets $\widehat{B} = 3n + \ell - 1 + 11$ from the pre-planting host, and Proposition 5.12 claims the required local construction at a seed crossing, recovering an ordinary odd anti-parallel twist. Together these statements would bound every such residual twist on a three-crossing knot host by $3 \cdot 3 + 1 - 1 + 11 + 2 = 22$, independently of its length.

The explicit computation below contradicts this combined consequence. It does not distinguish an error in the conditional algebraic lemma from an error in the claimed geometric realization, and it does not provide a counterexample to Jones unknot detection itself. It is sufficient to reject this proof as a dependency of the maintained recognition algorithm. The stronger Jones identity backend therefore continues to require an independent unknot certificate.

## 39.1 An explicit host and a twist with fixed exterior

Use the oriented trefoil with PD rows

$$H = ((5, 1, 2, 0), (1, 4, 3, 2), (0, 3, 4, 5)).$$

Keep its first two rows fixed. Replace the last row by a linear twist of odd length $L$, preserving its four exterior labels $0, 3, 4, 5$. Start with $(a, b) = (0, 3)$. For $i = 0, \ldots, L-2$, insert $(a, b, 6+2i, 7+2i)$ and replace $(a, b)$ by $(7 + 2i, 6 + 2i)$. Finish with $(a, b, 4, 5)$. Call the resulting knot diagram $D_L$. For $L = 1$ this is precisely $H$, and $D_L$ has $L + 2$ crossings. All of its crossings are positive in the orientation given by traversing opposite ports and equal-label arc endpoints.

The two strands in the inserted band are anti-parallel: at every inserted row, exactly one of ports $0, 1$ is incoming and exactly one of ports $2, 3$ is incoming. The oriented smoothing pairs ports $(0, 1)$ and $(2, 3)$. After smoothing any one band crossing, the others successively become adjacent-port monogons. Deleting these by Reidemeister I moves gives exactly the two exterior rows obtained by smoothing the seed of $H$. All moves take place in the inserted band. This shadow calculation is independent of the signs assigned to the remaining crossings.

There is also an explicit host anchor in every odd sign cube: assign the first $L-1$ crossings alternating signs $+,-$, and the last sign $+$. Cancel each consecutive opposite-sign pair by Reidemeister II. The surviving row and its exterior are exactly $H$, up to arc relabeling. Thus the audit does not infer applicability merely from a knot-family name or from a numerical resemblance of polynomials; it checks the specified local anti-parallel and anchor behavior.

## 39.2 An independent polynomial calculation

Use $t$ for the standard Jones variable and $z = t^{1/2}$ for the variable called $x$ in the audited preprint. This $z$ is different from the $x = A^4 = t^{-1}$ used in our numerical Potts ring. The seed polynomial is

$$V_1(t) = t + t^3 - t^4.$$

Oriented smoothing of a band crossing gives the positive Hopf link, with $V_0(z) = -z - z^5$. The four bracket states of the two-crossing smoothed host independently check this value. Switching one band crossing allows a Reidemeister II cancellation and produces $D_{L-2}$. The usual oriented Jones skein relation consequently gives

$$t^{-1}V_L(t) - tV_{L-2}(t) = (t^{1/2} - t^{-1/2})(-t^{1/2} - t^{5/2}),$$

or

$$V_L(t) = t^2V_{L-2}(t) + t - t^2 + t^3 - t^4 \qquad (L \geq 3 \text{ odd}).$$

In particular, the leading term is $-t^{L+3}$. For the smallest positive example exceeding the claimed cap, $L = 9$,

$$V_9(t) = t - t^2 + 2\sum_{j=3}^{9}(-1)^{j+1}t^j - t^{10} + t^{11} - t^{12}.$$

An independent whole-cube bracket computation checks this eleven-crossing diagram using only $2^{11}$ states. It uses neither a Tait graph nor the production quadratic evaluator. The exact faithful backend independently recovers the same complete polynomial.

Now write $p = z - z^{-1}$, so $z^2 = pz + 1$. Reducing $V_L(z^2)$ into $a(p) + b(p)z$ is a formal operation in a free rank-two $\mathbb{Z}[p]$-module. Multiplication by $z$ sends the pair $(a, b)$ to $(b, a + pb)$. Thus for positive exponent $k$, the coefficient of $z$ in $z^k$ has degree $k-1$ and leading coefficient one. The term $-z^{2L+6}$ therefore contributes a unique leading term $-p^{2L+5}$ to $b(p)$; every other Jones monomial has exponent at most $2L+4$ and cannot cancel it. Consequently

$$\deg_p V_L = 2L + 5, \qquad f_{23}(D_9) = (a_{23}, b_{23}) = (0, -1) \neq (0, 0).$$

This contradicts the degree-22 consequence above. There is no numerical approximation or degree estimation involved. Reflection provides another check: the mirrored family has degree $2L+6$, so the mirror of $D_9$ has degree 24. For any fixed proposed cap, the same recurrence supplies larger counterexamples to a uniform residual-twist bound.

## 39.3 Reproducibility and the scope of the finding

The maintained research audit is `fast/jones_research/audit_residual_twists.py`, with results in `fast/results/residual_twist_audit_20261008.json`.

It checks lengths $1, 3, 5, 7, 9, 23, 31$, both mirrors, and stores each PD, full Jones polynomial, both formal JVP coordinates and their actual degrees. Ten small cases have whole-cube comparisons; all fourteen have faithful recovery and skein-recurrence comparisons. Formal substitution $p = z - z^{-1}$ reconstructs every original Laurent polynomial, checking the coordinate conversion independently of the numerical quadratic ring. The longer cases include lengths beyond 22, so the discrepancy is not resolved by requiring a twist longer than that threshold.

The geometry record contains 79 individual smoothings and 1576 explicit Reidemeister I deletions with the fixed exterior retained, together with the alternating-sign Reidemeister II anchor traces. Another 42 crossing-sign assignments, exhausting sign cubes of lengths one, three and five, compare whole-cube polynomials with their signed-sum twist reductions. For each assignment, the independent bracket computation sums all smoothing states.

The audit does not construct the four-clasp machine or identify the precise step of its claimed collapse that fails. Its conclusion is the narrower, fully reproducible one: the uniform residual cap and universal geometric realization asserted by the audited statements cannot both be correct. In particular, the preprint does not close our gap between fast Jones computation and fast complete recognition.

# 40 A complete polynomial test on certified treewidth-two projections

## 40.1 The structural restriction that makes determinant one sufficient

The general determinant-one implication is false: the Conway and Kinoshita–Terasaka knots both survive that scalar test. Here we add an explicit graph hypothesis to make it a complete decision rule. Let $G_D$ be the projection graph of a validated classical one-component diagram $D$: vertices are crossings and edges are the arcs between crossings. This is *not* a Tait graph or a knot-complement triangulation graph. Loops and multiple edges do not affect the presence of a $K_4$ minor.

The structural theorem of Rué, Thilikos and Velona [10] classifies links admitting a $K_4$-minor-free projection as disjoint and connected sums of two-strand torus links. For a knot, this reduces to connected sums of $T(2, q_i)$ with odd $q_i$, with trivial factors allowed. Indeed a connected sum of links with $c_1, c_2$ components has $c_1 + c_2 - 1$ components, so a two-component torus link cannot occur in a decomposition resulting in one component.

**Proposition 40.1.** *For a classical knot diagram $D$ with* $\mathrm{tw}(G_D) \le 2$*,*

$$D \text{ represents the unknot} \quad \Longleftrightarrow \quad \det(D) = 1.$$

*Proof.* Treewidth at most two is equivalent to excluding a $K_4$ minor. Apply the classification above. For odd $q$, the Alexander polynomial of $T(2, q)$, up to a Laurent unit and mirroring, is $(t^{|q|} + 1)/(t + 1)$, whose value at $t = -1$ has absolute value $|q|$. The Alexander polynomial multiplies under connected sum, hence $\det(D) = \prod_i |q_i|$. This product is one exactly when all factors are trivial. Conversely every unknot has determinant one. □

This uses the given projection, with no supplied braid or Montesinos presentation. A rejected projection does not imply that its knot type has no other diagram in the class. In particular, it is not a knottedness certificate.

Bodlaender, Burton, Fomin and Grigoriev [11] give a stronger, linear-time recognition algorithm on the same class, using generalized diagrams with twist-labelled double edges. Our implementation uses the classification and an exact determinant instead; it does not implement their linear-time algorithm or inherit its time bound. This separates a directly checkable implementation from a future optimization of its arithmetic.

## 40.2 A graph certificate before any positive invariant verdict

The producer in `fastunknot/treewidth_two.py` builds the simple projection adjacency sets. It queues vertices of degree at most two. Deleting a degree-two vertex joins its two remaining

neighbours; existing parallel edges coalesce. It records the deletion order. Every operation preserves the property of having treewidth at most two in both directions. For degree two, suppression is an edge contraction followed by removal of redundant parallel edges, so it cannot increase treewidth. Conversely, a width-two decomposition of the remaining graph has a bag containing the new neighbour edge; attach the deleted vertex in a new bag of size three. The cases of degree zero or one are immediate.

Every nonempty graph of treewidth at most two has a vertex of degree at most two. Consequently elimination either deletes everything and supplies a certificate, or stops at a graph of minimum degree at least three and rejects the input projection. Deletion never increases a remaining vertex's degree: a possible fill edge replaces the incident edge just removed. This justifies the queue implementation. Each deletion examines at most two neighbours and creates at most one fill edge. The graph phase uses $O(n)$ adjacency operations and $O(n)$ words; the recorded order uses $O(n \log(n+1))$ bits.

A verifier does not trust the producer's queue or adjacency state. It reconstructs projection edges from PD labels, checks that the order is a permutation of all crossings, and replays deletion and fill on an edge set. It rejects any step with more than two remaining neighbours. The simple independent replay is polynomial rather than linear. Crucially, it must succeed *before* an exact determinant of one is allowed to certify an unknot. Certificates also bind the crossing count, schema version, method and exact verdict. The width witness is a sufficient structural certificate, not an implementation of a formal proof of the imported classification.

## 40.3 Exact integer arithmetic and independent verification

The producer builds the signed checkerboard graph using the existing PD corner convention. For either shading, the absolute signed spanning-tree sum is the knot determinant. We choose the shading with fewer vertices, discard loops, sum parallel signed weights and use the articulation-block product from Section 34. The reduced signed Laplacian of every nontrivial block is evaluated by fraction-free Bareiss elimination with exact division and row exchanges when a leading pivot is zero. Bridge blocks contribute just their integer weight. The shared determinant routine was moved to `integer_determinant.py`; the marked-continuation backend retains its existing interface and algorithm.

No modular residue is promoted to an exact determinant. A cancelled parallel sum and a zero leading pivot require exact handling even though a validated classical knot ultimately has an odd, nonzero determinant. The stored determinant is hexadecimal, so large composite knots do not hit Python's decimal-conversion ceiling when their certificates are serialized as JSON.

The independent certificate verifier avoids the checkerboard construction and Bareiss entirely. It constructs Wirtinger arcs by joining only the overpassing arc labels at each crossing. At $t = -1$, every Fox row is

$$2x_{\text{over}} - x_{\text{under},1} - x_{\text{under},2},$$

independently of crossing sign and orientation. It removes one row and one column and evaluates the resulting cofactor using exact rational Gaussian elimination. Its absolute value must equal the reported integer, and the verdict must be precisely the determinant-one decision. This provides a separate graph representation, knot-matrix construction and elimination algorithm for replay.

With $n$ crossings, checkerboard matrices have $O(n)$ rows and integer entries of size $O(n)$. Hadamard bounds give $O(n \log(n+1))$ bits for all relevant minors, hence for Bareiss entries; numerator products and exact divisors remain polynomial in bit length. Summing cubic arithmetic cost over blocks gives at most $O(n^3)$ integer arithmetic operations. Thus the uncapped certificate producer has a polynomial bit bound on this class, with a conservative dense memory bound also polynomial. This is weaker than the cited linear algorithm, but independent of the size of any Khovanov complex.

### 40.4 Adaptive integration and the exact scope of the guarantee

The CLI option `-treewidth-two`, or Python option `use_treewidth_two=True`, enables this stage after checked source certificates and before the usual Seifert and simplification stages. The local allowance defaults to 0.1 seconds and includes lazy module loading, graph certification and determinant work. Its callback first checks the global deadline, then the local one; graph loops and determinant rows poll cooperatively. A local timeout or allocation failure records a skipped probe and continues the existing recognizer. A projection outside the class also continues. A global timeout retains `UNKNOWN`. No interrupted order or partially multiplied determinant is published as a verdict. The local interval and discarded work remain part of the complete query's global elapsed time.

The standalone producer has no local cap. An uncapped local query in the Python recognizer sets `treewidth_two_seconds` to `None`. The polynomial claim belongs to the uncapped restricted-class procedure. A capped probe that falls back to Khovanov does not give the whole pipeline a polynomial guarantee, even on inputs whose projection lies in the class. The optional stage also does not change the general exponential upper bound or establish a general subexponential recognizer.

### 40.5 Validation and whole-query measurements

Eight added tests cover: an independent branch-set search for $K_4$ minors on all 64 four-vertex graphs and 25 larger graphs; all 170 signed one-component two-strand braids through seven crossings against full cube-of-resolutions recognition; random braid and named-diagram determinants against exact Alexander polynomials; mirrored and relabelled diagrams; forged certificates; CLI dispatch; early and late cancellation; and local fallback versus global exhaustion. The Conway, Kinoshita–Terasaka and stored determinant-one grid knots explicitly test that the graph restriction cannot be omitted. A connected sum of 10,000 trefoils produces determinant $3^{10000}$ and a JSON-safe certificate on 30,000 crossings. The full suite result is recorded in `data/treewidth-two-integrated-tests.txt`.

`benchmark_treewidth_two.py` measures validation plus full recognition on fresh, source-free PD inputs. Fourteen inputs each receive one excluded warm-up and seven shuffled paired rounds: default, an identical default control, and the optional stage. The interpreter is warm; imports are not part of the measured rounds. All 294 measured queries finish with matching verdicts. Median paired speedups on mixed-sign two-strand unknot diagrams of 101, 501 and 2001 crossings are respectively 1.085, 1.142 and 1.126. The default/control ratios range from 0.954 to 1.038.

Other inputs show why the stage remains optional. The trefoil has a 0.606 speedup ratio, tree medials range from 0.756 to 0.838, and the two 120-crossing trefoil sums have ratios near 0.67: existing Seifert certificates already decide these cheaply. Outside-class inputs range from 0.862 to 1.018. These measurements support a modest improvement on a specific family and a new independently certified input-class bound, not a default-wide performance improvement. Exact inputs, execution order, individual timings, methods and certificates are retained in `results/treewidth_two_20261008.json`.

## 41 An adaptive normal-surface fallback in an isolated worker

### 41.1 A complete geometric decision with a different bottleneck

The built-in complete backend computes Khovanov homology. We now add an optional alternative based on the topology of the knot exterior. For a classical knot $K \subset S^3$, its exterior is a solid torus exactly when $K$ is the unknot. Thus a complete solid-torus recognition algorithm on an

exact triangulation of the exterior decides both outcomes; a failed attempt to find a compressing disc is not by itself a negative verdict.

Normal-surface theory provides an algorithmic framework for that decision. A normal surface in a triangulation with $t$ tetrahedra has coordinates for four triangle types and three quadrilateral types per tetrahedron. The coordinates are nonnegative integers satisfying face-matching equations; at most one quadrilateral type is permitted in each tetrahedron. These constraints encode properly embedded surfaces, including candidates for compressing discs. The local quadrilateral alternatives are one source of exponential search even though the matching equations are linear. Successful recognition implementations combine such searches with simplification and crushing operations.

Burton and Ozlen [7] develop a complete branching algorithm using normal-surface methods and report experimental polynomial behaviour, while retaining exponential worst-case complexity. This is motivation for trying a different complete algorithm when Khovanov work is expensive, not evidence of a polynomial or subexponential theorem for our inputs.

We use Regina's documented `isSolidTorus` decision routine [8]. The wrapper first applies the library's fast link simplification, constructs an ideal triangulation of the complement, and calls that routine. It never substitutes `knowsSolidTorus`, whose Boolean value says whether the property is already known or cheap to compute, rather than whether the manifold is a solid torus. The implementation trusts Regina's conversion, simplification and exact recognition. It does not yet export and independently check a sequence of normal surfaces or crushing operations.

## 41.2 Convention checks and the trust boundary

Our PD row $(a, b, c, d)$ lists ports counterclockwise, with under-strand ports zero and two. Regina's PD input starts at the incoming underpass. The worker obtains that port $u \in \{0, 2\}$ from the validated orientation and rotates each row by $u$. Rotation by two preserves the crossing; rotation by one would interchange over and under and is forbidden. Normalized arc labels increase by one for the external API. The crossing-free input is explicitly one trivial link component, rather than an empty link.

The worker independently revalidates the supplied PD with `Diagram.from_pd`. It checks that the imported link has one component, is classical and has the same writhe. After simplification and complement construction, it checks that the triangulation is valid, orientable and has exactly one boundary component. These checks guard conversion mistakes but are not a replacement for the imported topology.

Every request and result carries a SHA-256 digest of the normalized PD. The digest binds the response to its input; it is not a mathematical certificate. Evidence includes the Regina engine version and installed distribution version (respectively 7.4 and 7.4.1 in this experiment), initial and simplified crossing counts, tetrahedron count, elapsed time and an explicit external-engine trust label. A result with a wrong digest, invalid schema, unsupported status, nonzero worker exit or unparsable output is inconclusive. Only a completed, well-formed `UNKNOT` or `KNOTTED` response can decide the query. Serialized results involving such a verdict, including inside connected-sum factors or restarted queries, explicitly exclude the external wrapper from the built-in Khovanov running-time statement.

## 41.3 Switching after cheap filters, with bounded discarded work

The optional `-regina` flag enables the probe only for an undecided diagram with at least 32 crossings. It runs after the existing one-sided invariant filters and before RIII search and Khovanov construction. Small inputs avoid interpreter and native-library startup. Visible factors are processed as their own diagrams: an external query never receives the whole source braid as if it presented a proper factor.

The local allowance is two seconds by default and is configurable with `-regina-seconds`. Python also permits `regina_seconds=None` for an uncapped local query. The standalone `regina_decide` routine has the same local allowance semantics and accepts diagrams of any size. Installing `fastunknot[normal]` supplies the optional dependency; the default package remains dependency-free and the default recognizer does not dispatch to Regina.

The parent launches a fresh worker using the current Python interpreter. It never imports the native Regina module itself. Native import, request conversion, process startup, simplification and recognition all consume the local query interval. The parent polls communication in slices of at most 50 milliseconds, checking the global callback before each slice and before accepting a result. Retried communication retains partially sent input and already received output. Every exit path kills a live child, waits for it to finish, and closes all three pipes; this includes local expiry, global cancellation and exceptions. Process creation and cleanup remain operating-system operations, so this is not a hard real-time latency guarantee.

A missing dependency, local expiry or failed worker records an inconclusive probe and leaves the original diagram available to the existing RIII and Khovanov stages. A global deadline propagates to `UNKNOWN` after cleanup. Time spent on unsuccessful native work is not reset or refunded. The `max_objects` parameter still bounds the Khovanov scanner; it is not a native-memory ceiling. This worker uses a time allowance rather than an independent bound on Regina's allocations.

Regina's simplifier makes random choices. Each worker resets its random engine to the library default, making these calls reproducible on the same build and platform. This does not promise identical simplification paths across machines or versions, and it does not make correctness probabilistic: the simplification moves preserve the knot and the terminal recognizer is an exact decision routine. Exhaustion remains inconclusive.

This is an adaptive portfolio based on failure of the cheaper stages, with a local cap before the built-in complete fallback. It is not simultaneous racing or a claim of constant-factor optimality relative to the best possible backend. A useful next step is to extract independently checkable geometric witnesses. No general subexponential or quasi-polynomial bound follows from the current integration.

## 41.4 Measurements with explicit censoring

The new fixture directory stores PD data generated by Regina 7.4.1 for its `ExampleLink.monster`, `gordian` and `gst` examples [9]. Their crossing counts are 10, 141 and 48. The last is the Gompf–Scharlemann–Thompson knot from their slice-ribbon investigation, not an unknot fixture. The Conway diagram is also included in the benchmark from the existing corpus.

`normal_research/benchmark.py` runs five measured shuffled rounds plus an excluded warm-up for each of four inputs. Its three arms are the default pipeline, an identical default control, and the optional portfolio. Each arm receives a fresh validated PD, a four-second global allowance and a 50,000-object Khovanov ceiling. Native attempts always start a new interpreter; the portfolio's local native allowance is two seconds.

| Input | Default median (s) | Portfolio median (s) | Completed default / portfolio |
|---|---|---|---|
| Monster unknot | 0.000965 | 0.000968 | 5/5 / 5/5 |
| GST knot | 0.001993 | 0.001983 | 5/5 / 5/5 |
| Gordian unknot | 4.043 (censored) | 1.788 | 0/5 / 5/5 |
| Conway knot | 0.001041 | 0.001043 | 5/5 / 5/5 |

All conclusive verdicts agree. Both default arms exhaust their time allowance on every measured Gordian query, whereas the portfolio recognizes it in all five. The number 4.043 includes validation and is an observed resource-limited duration, *not* a completion time or a denominator for a

reported completion-speedup factor. Native evidence records simplification from 141 to 111 crossings followed by recognition of a 119-tetrahedron complement.

The dispatch policy matters. Standalone native queries, with a three-second allowance, take about 0.278 seconds for Monster and 1.402 seconds for Gordian; GST exhausts that allowance. The ordinary Alexander filter rejects GST in about two milliseconds, so the integrated portfolio avoids the slower native call. Monster and Conway remain below the crossing threshold. These are four concrete examples and one successful hard-unknot improvement, not a representative average-case study or an asymptotic conclusion. Raw timing samples and verdict evidence are stored in `results/normal_surface_20261008.json`.

Eight new tests check per-crossing conversion, communication across repeated polls, missing dependencies, malformed responses, input binding, pipeline dispatch, fallback and actual worker cleanup on local and global cancellation. With Regina installed they also check six exact native verdicts, global exhaustion and an actual CLI recognition of the Gordian unknot. The full integration log and the dependency-free focused log are retained alongside the article. The optional tests are skipped explicitly when Regina is not installed; the failure-handling tests still run in that configuration.

# 42 Cyclic knot-group certificates and adaptive Whitehead descent

## 42.1 A positive certificate from the full fundamental group

The optional `-group` stage provides a new sufficient test for unknottedness. It simplifies a Wirtinger presentation by exact changes of presentation, records every nontrivial step, and independently replays that record. A successful trace ends with precisely one generator and no nonempty relators. It therefore proves that the knot group is isomorphic to $\mathbb{Z}$. For classical knots in $S^3$, this implies that the knot is the unknot, by the classical consequence of the Loop Theorem [25]. This criterion concerns the whole fundamental group: its abelianization alone is $\mathbb{Z}$ for every knot and cannot supply this conclusion.

Here is the topological implication explicitly. Write $X$ for the compact knot exterior. If $\pi_1(X) = \mathbb{Z}$, the map from the boundary torus group $\mathbb{Z}^2$ cannot be injective. The Loop Theorem supplies an embedded compressing disc. The exterior is irreducible: an embedded sphere bounds two balls in $S^3$, and the ball on the side away from the connected knot lies in the exterior. Compressing its torus boundary along the disc gives a sphere; irreducibility then identifies the exterior with a solid torus. Equivalently, the compression slope is the longitude, which bounds a disc and gives a spanning disc for the knot after adjoining the collar annulus. The input contract is essential: our PD loader validates a classical, single-component knot diagram. No conclusion for virtual diagrams or arbitrary finitely presented groups is asserted.

## 42.2 A deterministic presentation and reversible eliminations

For each PD row $(a, b, c, d)$, join the labels $b, d$ along the overpass. The connected classes of arc labels are the Wirtinger generators. Name these classes by increasing minimum arc label; this makes certificate indices independent of the producer's union–find root choices. Orient the knot by following its component from dart zero. At a crossing let $u, v$ denote the incoming and outgoing underpass generators and let $o$ be its overpass generator. Our relator is

$$ouo^{-1}v^{-1} \quad \text{at a positive crossing,} \qquad ovo^{-1}u^{-1} \quad \text{at a negative crossing.}$$

These choices give the same signed Fox rows as the established Alexander convention. We retain all crossing relators, including the redundant Wirtinger relator, and freely and cyclically reduce

each one. Free reduction changes no word in the free group; cyclic reduction conjugates the relator and hence preserves its normal closure.

Suppose a relator contains a generator $g$ exactly once, with sign $\epsilon \in \{1, -1\}$. Write it as $Pg^{\epsilon}S$ where neither $P$ nor $S$ contains $g^{\pm 1}$. Solve the defining relation:

$$g = P^{-1}S^{-1} \quad (\epsilon = 1), \qquad g = SP \quad (\epsilon = -1).$$

Substitute this word, and its inverse for $g^{-1}$, into every other relator. Delete this one defining relation and the generator. This is a Tietze elimination, giving an isomorphic presented group. In particular, it is not the operation of imposing an extra relation. The trace records only the original relation index and generator; replay derives the replacement itself. Empty slots retain their original indices.

The producer prefers an eligible pair minimizing $(|R| - 2)\,\mathrm{occ}(g)$, followed by relator length, index and generator. This is a deterministic estimate of expansion cost, not a completeness argument. Renumbering or mirroring a diagram can change the greedy path. Indeed, two of the twelve tested mirror diagrams stall even though every original succeeds. This is a concrete reason to retain other decision procedures in the portfolio.

## 42.3 Cut-based changes of free basis

When no generator can be eliminated, try to shorten the whole relator list by a Whitehead automorphism. Fix a signed letter $a$ and a subset $A$ of the signed alphabet with $a \in A$ and $a^{-1} \notin A$. Fix $a^{\pm 1}$, and for every other letter $x$ put

$$\tau_{A,a}(x) = (a^{-1} \text{ if } x^{-1} \in A)\; x\; (a \text{ if } x \in A),$$

where an absent factor means the empty word. This preserves inversion and defines a free-group automorphism. Its inverse uses multiplier $a^{-1}$ and subset $(A \setminus \{a\}) \cup \{a^{-1}\}$. Applying it to every relator therefore preserves the isomorphism type of the quotient group.

For a cyclically reduced list of words, add a weighted undirected edge $\{x, y^{-1}\}$ for each cyclic adjacent pair $xy$, including the last–first pair of each word. Add weights across all relators. For this convention, the exact change in total cyclic length is

$$|\tau_{A,a}(R)|_{\mathrm{cyc}} - |R|_{\mathrm{cyc}} = \mathrm{cut}(A, A^c) - \deg(a).$$

The cut-capacity approach to Whitehead minimization is developed by Roig–Ventura–Weil [26]; their polynomial algorithm handles varying free rank, rather than enumerating exponentially many subsets. We implement the relevant minimization directly on the sum of the relators' weighted graphs. For each signed multiplier, an exact integer-capacity Edmonds–Karp flow finds a minimum cut separating $a$ from $a^{-1}$. The best strictly negative length change selects the next move. A nonnegative minimum ends this search attempt.

The producer explicitly substitutes the chosen automorphism and checks that the observed length change equals the cut prediction. The verifier needs neither this formula nor the minimum-cut computation: it checks that the supplied pair is a valid Whitehead automorphism and directly applies it. Soundness does not require the proposed move to shorten the words. Thus an error in search strategy cannot by itself create a false positive through the certificate interface.

## 42.4 A reproducible convention audit of a new preprint

Kapovich's July 2026 preprint on compressed primitivity [27] motivates keeping group words compressed. It states polynomial compressed Whitehead-minimality tests at fixed rank, with further results for compressed primitivity. It also cautions that fast compressed operations alone

do not bound the number of descent steps: elementary descent on $ab^{2^m}$ can require exponentially many steps in the compressed input length. Fixed-rank hypotheses also need attention when the rank grows with a knot diagram.

Our audit found a convention mismatch in version 1's equations (10)–(12), as printed together. Its automorphism (10) is the one above, but its graph (11) uses edges $\{x^{-1}, y\}$. Take

$$w = abc, \qquad A = \{a, b^{-1}\}, \qquad \text{multiplier } a.$$

Literal substitution gives $aa^{-1}bc$, hence cyclic length change $-1$. The printed graph has edges $\{a^{-1}, b\}$, $\{b^{-1}, c\}$, $\{c^{-1}, a\}$; the stated cut has weight two and $\deg(a) = 1$, predicting $+1$ via (12). With our edges $\{x, y^{-1}\}$ the cut has weight zero, so the predicted change is correctly $-1$.

This local inconsistency is repaired by inverting the graph's vertex labels, or by transforming the associated characteristic pair consistently. It is not a disproof of the paper's compressed-primitivity claims: minimization over all pairs can survive such a relabelling. Our regression test records the three-letter example and checks 15,552 short-word/characteristic-pair combinations against an independent four-case substitution implementation. The production code currently uses explicit word lists. It does not implement compressed free-group operations or inherit a compressed-input complexity guarantee.

## 42.5 Independent replay and local resource accounting

The certificate contains its version, method, positive verdict, exact normalized input PD, move list, and surviving generator. The checker reconstructs the presentation by graph traversal rather than union–find, reconstructs orientation from paired darts rather than cached Diagram orientation, and performs word cancellation with a separate deque-based implementation. It checks unique occurrence before an elimination and valid alphabet membership and multiplier conditions before a Whitehead move. After every substitution it reduces every remaining relator. Acceptance requires exactly the declared surviving generator and an entirely empty relator list. All relators are accounted for; neither the stored verdict nor a claim of rank one is accepted on its own.

The search and checker share a default wall allowance of 0.05 seconds per attempt. Each separately has a two-million-unit cooperative work budget and a 200,000-letter expansion ceiling. Work units conservatively charge letter processing in blocks, graph scans and flow exploration; they are not a hardware-independent instruction count. Substitution sizes are checked before allocating expanded words. The Whitehead producer uses a conservative bound of three times the current length. A trace that is mathematically valid can therefore be rejected by a resource ceiling. Only a fully completed and checked trace supplies a verdict.

Local exhaustion or stalled search returns `INCONCLUSIVE`; global cancellation propagates and the surrounding recognizer returns `UNKNOWN`. The optional stage runs after cheap invariant obstructions and before the optional normal-surface worker and late Reidemeister III/Khovanov stages. A local failure consumes only its allowance and resumes the portfolio. Verification is inside the local allowance, including the check immediately before acceptance.

## 42.6 What can and cannot be bounded

Let $n$ be the crossing number, and suppose every expanded relator list in one completed search has total length at most $B$. There are at most $n - 1$ generator eliminations. Between eliminations each chosen Whitehead step decreases a nonnegative integer length, so there are at most $B$ such steps in each of at most $n$ phases. The graph has $O(n)$ vertices and $O(n^2)$ edges. Edmonds–Karp takes $O(n^5)$ graph operations per multiplier and we try $O(n)$ multipliers. Thus, conservatively, the search cost is polynomial in $n$ and $B$, bounded by

$$O\big(n(B + 1)(B + n^6)\,\mathrm{polylog}(nB + 2)\big)$$

with straightforward bit accounting for integer capacities. Replay is polynomial in explicit trace size and expanded word volume. This is our implementation-level conditional bound, not a compressed-word theorem.

Two essential gaps prevent a general fast recognition theorem. First, eliminating a generator can substantially expand the other relators; no polynomial or quasi-polynomial bound on $B$ in terms of $n$ is established. Second, even unlimited length-decreasing Whitehead search followed by this greedy elimination rule can stall on an unknot. A larger Tietze search would require its own completeness and quantitative bounds. Compressed words could address representation growth, but would not alone supply those missing search bounds. The bounded optional stage preserves the existing complete fallback and does not improve the general exponential upper bound proved for our built-in decision pipeline.

### 42.7 Whole-query measurements and regression tests

`fast/benchmark_group.py` measures 21 inputs in seven randomized A/A/B rounds after one excluded warmup: 441 timed whole queries. Every query starts with fresh validated PD data and carries no source-braid provenance. The two control arms use the default pipeline. The third enables the group stage, including independent replay. Every arm has a 0.25-second global allowance and a 50,000-object Khovanov ceiling. Raw samples, resulting certificates and environment metadata are recorded in `fast/results/group_20261008.json`.

The group stage certifies all twelve maintained Khovanov-only survivor inputs in every measured round. Ten show lower median query costs. The two largest gains are 10.90 to 4.66 ms and 12.86 to 3.86 ms. Two regress, from 4.62 to 4.89 ms and from 3.67 to 4.69 ms. The sum of the twelve median costs is 60.85 ms for the default control, 60.56 ms for the repeated control, and 39.19 ms with group certificates. This aggregate describes these twelve selected inputs, not a distribution of arbitrary knots or an asymptotic law.

An eight-crossing hard unknot and Monster already have cheap default Reidemeister paths; inserting group search increases their median costs from 0.66 to 1.01 ms and 0.95 to 1.82 ms respectively. Easy structural and invariant obstructions generally bypass the new stage. All arms time out on Gordian under this benchmark's short global cap; these censored observations are not completion times or speedups. The standalone group attempt also exhausts its local work allowance on Gordian. These results support retaining `-group` as opt-in.

Tests cover exhaustive short-word cut identities, minimum-cut versus exhaustive characteristic-pair optimization, the printed-convention counterexample, all twelve survivor certificates, mirror behaviour, 70 random small knots against an independent full Khovanov cube, known nontrivial determinant-one knots, malformed and input-mismatched traces, independent replay with producer helpers disabled, local exhaustion, global and verification-stage cancellation, and actual CLI routing.

The integrated suite passes all 626 tests with the optional Regina backend installed (126.707 seconds); see `synthesis/data/group-integrated-tests.txt`. The rebuilt article contains 154 pages.

## 43 Relator overlaps: a native certificate for Gordian

### 43.1 Preserving a normal closure while shortening a relator

The first group-certificate stage could stall after exhausting generator elimination and length-reducing Whitehead moves. An additional exact move now changes the relators themselves.

Suppose two distinct relator slots, after cyclic rotation and optional inversion of the donor, read

$$R_i = UW, \qquad R_j^{\epsilon} \sim UV, \qquad i \neq j, \quad \epsilon \in \{1, -1\}.$$

Replace the target by $V^{-1}W$ and retain the donor. This preserves the normal closure of the relator list, because

$$(UV)^{-1}(UW) = V^{-1}W, \qquad (UV)(V^{-1}W) = UW.$$

Both the old and new target therefore lie in the normal closure of the other presentation. Cyclic conjugation and inversion preserve a relator's normal closure, so undoing the displayed rotations does not change the argument. The quotient group, and hence the positive unknot conclusion at terminal rank one, is unchanged.

Requiring different target and donor slots is essential. A relator cannot be deleted by using itself as the donor: that would remove a relation rather than replace it by an equivalent one. Distinct slots containing equal relators can legitimately reduce to one copy. The independent checker enforces this distinction.

If $u = |U|$ and $d = |R_j|$, the literal target length decreases by $2u - d$. The producer proposes only moves with $2u > d$; free and cyclic cancellation can shorten further. Among these moves it maximizes the guaranteed decrease, not the decrease after all incidental cancellations. This is a Dehn-style presentation simplification. Regina documents a related use of relator substitutions and generator elimination [28]. We implement and replay our own individual moves; a declared external isomorphism is not used as proof evidence.

No small-cancellation hypothesis is required for the correctness of an individual substitution. Such hypotheses enter completeness theorems for Dehn algorithms. We do not assume that arbitrary knot presentations satisfy them, and failure to find an overlap has no negative meaning.

## 43.2 Finding the best cyclic overlap without listing rotations

A direct search that builds a long tuple key for each rotation can spend quadratic space or work on a single long relator. Small fixed-length seeds avoid that copying but can still produce many candidates on repetitive words. The implementation instead uses an exact suffix automaton, following the classical linear-size subword-automaton framework [29].

For each nonempty donor $v$, and separately for $v^{-1}$, build the suffix automaton of $v$ followed by all but its last letter. This word of length $2|v| - 1$ contains every cyclic rotation and every cyclic segment of length at most $|v|$. An automaton state stores its maximum represented length, suffix link, exact letter transitions, and the end position of one occurrence. Online extension creates a new state, follows suffix links to add missing transitions, and splits a state when a transition would otherwise identify incompatible maximum lengths. The clone copies transitions and the occurrence position, while redirecting the necessary suffix-link transitions.

Scan each other doubled target word through this automaton. On a missing transition, follow suffix links and shorten the currently matched suffix. For a match of length $\ell$ against target $w$, use

$$u = \min(\ell, |v|, |w|).$$

The caps forbid a match from wrapping around either cyclic word more than once. The target scan position and stored donor occurrence position give the two rotation offsets modulo their respective lengths. Compare $2u - |v|$ with the best positive decrease found so far. A target whose upper bound $2|w| - |v|$ cannot improve the current best can be skipped. No probabilistic fingerprints or collision assumptions enter equality: letters are signed generator integers, and a literal prefix comparison checks the selected witness before application.

Let $m$ be the number of nonempty relators and $L$ their total explicit length. Building each automaton and scanning every target costs $O(mL)$ dictionary operations overall, plus traversal of the original relator slots. The auxiliary automaton occupies $O(L)$ space, since only one donor orientation is indexed at a time. With the usual amortized hash-table costs this is the corresponding expected word-operation bound; the correctness is deterministic. Even replacing dictionary lookup by a linear scan over the at most $2n$ letters gives a polynomial worst-case bound. The implementation charges construction, clone copies, suffix links and target scanning against its cooperative work allowance.

The five new tests compare the optimized search with an independent exhaustive search over all rotations and both donor signs on 2,141 word lists. These include every pair of cyclically reduced rank-two words of length at most three, random signed lists, empty slots and periodic words. The test checks the maximum guaranteed decrease and the literal selected substitution, not only whether some move was returned.

## 43.3 Certificate format and independent verification

A trace containing an overlap uses certificate version 2. The previous elimination/Whitehead-only traces remain version 1 and are still accepted. The new move records target and donor indices, their cyclic offsets, the donor inversion flag, and overlap length. The checker validates types, indices, distinct slots, offsets, overlap bounds and letter equality. It reconstructs the two cyclic words by modular indexing rather than by the producer's suffix automaton or rotation helper, then directly replaces the target by the inverse donor remainder followed by the target remainder. The unchanged donor remains available throughout.

The checker does not require a maximum overlap or even positive length gain: the normal-closure identity is valid without those search-policy properties. Its expansion ceiling and operation allowance still apply. Forged self-donors, false matching prefixes, invalid offsets, incorrect versions and malformed fields are rejected. Acceptance still requires one surviving generator and every remaining relator empty. Tests replay the Gordian trace with the search and application helpers disabled, and cross-check another 80 random small knot inputs against the full Khovanov cube whenever a positive certificate is produced.

## 43.4 What the larger search establishes

On the supplied 141-crossing Gordian PD, the native producer finds a trace with 140 generator eliminations and one relator-overlap move. Independent replay proves its full knot group infinite cyclic. This replaces the previously missing native positive certificate for this example; it does not need Regina at runtime. The two mirrored survivor diagrams that stalled under the original group search also acquire positive certificates. The input itself remains the numeric PD fixture documented with the normal-surface experiments; no claim based on its filename or provenance is used by the verifier.

The same conditional complexity obstacle remains. Between generator eliminations, both overlap moves and chosen Whitehead moves strictly reduce total relator length. If every expanded list has length at most $B$, there are at most $O(n(B+1))$ moves. Adding a polynomial-time overlap search to each step preserves a polynomial bound in $n$ and $B$. Neither a quasi-polynomial bound on $B$ nor completeness of these greedy moves is proved. The existing work, wall and expansion caps yield safe fallback rather than a new general asymptotic guarantee.

Kapovich's August 2026 preprint [30] studies generic small-cancellation stability and isomorphism rigidity for fixed generator and relator counts. Its stated generic quadratic isomorphism algorithm is for its random-presentation model, not for all finitely presented groups. Our Wirtinger generators and relations grow with the crossing number and are strongly constrained by topology. Thus its

generic conclusions do not establish an average-case or worst-case bound for this knot recognizer. We use exact normal-closure identities for soundness and measured search behaviour for scheduling, rather than importing that unverified distributional transfer.

### 43.5 Adaptive scheduling rather than unconditional overlap search

The first implementation tried overlaps whenever direct generator elimination stalled, before trying Whitehead moves. A paired 29-input experiment showed why that order should not be universal. Although it certified Gordian, the sum of median query costs on the twelve survivors and twelve mirrors increased from 113.14 ms with the previous group stage to 191.77 ms with overlaps first. These are sums of individual medians, not end-to-end batch timings. The code and all 580 measured queries at this checkpoint remain in commit `772b24bca` and `results/relator_overlap_20261008.json`.

The revised policy observes the surviving generator count after each elimination. At rank at most six, it tries a shortening Whitehead move first and searches for relator overlaps only if none exists. Above six, it tries overlaps first, then Whitehead moves. It reuses the result of an already completed cut search, rather than performing it twice. Six is an empirical dispatch threshold: it reflects the different costs of small graphs and many signed-multiplier flow problems, not a theorem that low-rank presentations are easy. The regression test verifies that all twelve original survivors retain exactly their prior successful certificate traces, without invoking overlap search. The stalled mirrors still gain the new move when they need it.

The new mode is selected by `-group-relators`, which also enables the optional group stage. Python callers use `use_group=True, group_relators=True`. The previous `-group` behaviour remains available. Time and work caps are explicit: the Gordian experiment uses a two-second shared search-and-replay allowance and ten million work units per producer/checker, while the default group settings remain 0.05 seconds and two million units. Python callers set `group_seconds` and `group_max_work`; the standalone producer and checker expose `max_work`. Raising the time allowance alone need not overcome the separate work ceiling.

The benchmark uses fresh validated PD inputs, strips braid-source provenance, randomizes the order of default, previous group, overlap and Regina arms, and excludes one warmup before five measured rounds. All arms share a four-second global cap and a 50,000-object Khovanov ceiling; both group arms use the same two-second and ten-million-work allowances. Regina also has a two-second local allowance including process startup. Certificates are independently replayed inside the measured query. The initial overlap-first checkpoint completed Gordian in all five rounds at median 1.452 seconds, versus 1.957 seconds for Regina. Both the default and previous group arms timed out in every round. No ratio against these censored control times is reported as a completion speedup.

A fresh 580-query experiment of the revised dispatcher is stored in `results/relator_adaptive_20261008.json`. Its 24 survivor/mirror inputs all finish with independently checked group certificates in every round. Their summed median query costs are 120.28 ms for the default, 84.17 ms for the previous group search, 84.86 ms for adaptive overlap search, and 120.00 ms for the Regina arm, which bypasses native startup on these small inputs. Thus the additional reach costs less than one percent against the previous group-stage aggregate in this experiment. Absolute timings vary between experiments; compare arms within the same randomized run rather than attributing the entire change between the two archived runs to scheduling.

Gordian completes in all five adaptive runs at median 1.212 seconds, including validation, all prior pipeline work, search and independent replay. Regina completes all five at median 1.708 seconds in the same run. The default and previous group arms remain censored at four seconds. The two formerly stalled mirrors receive group certificates, but are not faster than the default Khovanov route; this is greater certificate reach, not universal acceleration. These observations keep the

group extensions opt-in. They do not show that relator-overlap search dominates normal surfaces or Khovanov on other knot families.

# 44 Exact compressed-word replay without expanding relators

## 44.1 Separating word representation from proof search

The group-search implementations above store their relators as explicit letter lists. This makes free reduction straightforward, but ties their operation counts and storage ceilings to expanded length. The new `compressed_words.py` implements a deterministic compressed-word kernel, and `compressed_group.py` uses it to replay the same group certificates. The kernel represents finite words by binary straight-line programs (SLPs), retaining concatenation and repeated subwords as a DAG. It supports exact equality, substrings, inversion, free reduction, cyclic reduction, simultaneous letter substitution and occurrence queries.

The equality and reduction algorithms follow the constructive account of Plandowski's and Lohrey's results in Schleimer [34], especially Sections 3, 7 and 8. These are existing polynomial algorithms; the contribution here is an exact, resource-bounded implementation and its integration into certificate replay. No probabilistic fingerprint or assumed hash-collision bound decides word equality. Python dictionaries index exact rule tuples and assertions, with ordinary collision-resolving equality checks.

The initial integration was a representation improvement for verification. The default producer still uses explicit relators. The optional compressed search developed below extends the representation to discovery. The compressed checker does not by itself construct a short proof, control all grammar growth over a long search, or prove general quasi-polynomial recognition. Its present practical cost is also higher on most measured knot queries, so it is separately opt-in.

## 44.2 The immutable word arena

Node zero represents the empty word. Other nodes either contain one signed generator or concatenate two earlier nonempty nodes. Identical rules share a node. Each node caches its length and endpoint letters; lengths and substring coordinates use exact integers. A binary program with $N$ nonempty nodes represents words of length at most $2^N$, so these coordinates have $O(N)$ bits, apart from the separate bit cost of generator names. Repeated squaring constructs powers using logarithmically many concatenations.

A substring operation descends only at the two interval boundaries, reusing complete covered subtrees. It creates $O(h)$ nodes for input height $h$ without increasing that height. Explicit stacks avoid Python recursion limits. Inversion reverses concatenation order and negates terminal letters, sharing memoized inverse nodes. Occurrence counting uses a topological pass: a generator count at a concatenation is the sum of its child counts. Following the child with a positive count locates the first occurrence and its sign without reading preceding letters.

The arena has separate limits on work and allocated nodes. The node allowance also caps live equality assertions and the equality-result cache. Exceeding any limit raises an inconclusive resource exception, never a guessed equality result. Cached nodes are immutable; an equality result cannot become stale after a later operation. Diagnostic expansion is explicitly bounded and is not called by compressed equality, reduction or certificate replay.

## 44.3 Equality by splitting and periodicity compaction

For clarity, treat the two compared programs as disjoint tagged copies, even when they share arena nodes. An assertion $(U, V, i)$ aligns $V$ at offset $i$ in $U$ and requires equality on the intersecting

interval. If $|U| \le i + |V|$, it compares a suffix of $U$ with a prefix of $V$; otherwise it asserts that $V$ occurs inside $U$. A proper prefix inclusion at offset zero is written with the two symbols interchanged. Root length and endpoint mismatches are immediate rejections. Equal root identities are immediate exact acceptances.

Start with $(A, B, 0)$. At each round choose a mentioned nonterminal of maximum expanded length, and replace its assertions by assertions involving its two children. The new assertions are exactly the intersections of the old comparison interval with the child intervals. Terminal comparisons are checked directly. Every split preserves the conjunction of all assertions. Each selected nonterminal disappears permanently, because nonempty children have strictly smaller length.

Splitting alone can create too many overlapping comparisons. For an ordered pair $(U, V)$, consider three suffix–prefix assertions at offsets $i < j < k$. If

$$(j - i) + (k - i) \le |U| - i,$$

replace those three assertions by the assertions at offsets

$$i, \qquad i + \gcd(j - i, k - i).$$

Two overlapping copies imply a period of the relevant suffix. The displayed inequality allows the periodicity lemma to replace the two periods by their greatest common divisor. Thus this compaction is logically equivalent, including when the compared words are unequal. It is not merely a way of sampling a few positions.

The code groups assertions by ordered symbol pair and repeatedly checks consecutive triples of sorted offsets. This suffices: if any triple starting at $i$ meets the inequality, the next two offsets after $i$ are no larger and also meet it. When no triple can be compacted, every two successive gaps advance more than half the remaining suffix length. There are consequently $O(\log |U|) = O(N)$ overlap assertions per pair. Across $O(N^2)$ symbol pairs there are $O(N^3)$ overlap assertions; at most $O(N)$ split rounds accumulate $O(N^4)$ subword assertions. Our elementary sorting-and-compaction implementation therefore has a polynomial bound in the grammar size and coordinate bit lengths. Its high worst-case degree is not claimed to match the best compressed-string algorithms.

Tests compare the implementation with literal strings on 400 random 30-rule grammars. They also force compaction on distinct parses of periodic words and on a same-length interior mutation with identical endpoints. The latter exercises rejection through the actual equality procedure, rather than through an endpoint or length shortcut.

## 44.4 Indexed assertions and a bounded adaptive equality probe

The initial checker rebuilt the entire assertion set after every split, including comparisons that did not mention the chosen symbol. Profiling compressed discovery identified these repeated scans as a major practical cost. The revised implementation retains the same logical assertions and split rule, but stores four associated structures: the assertion set; incidence sets indexed by each mentioned tagged symbol; overlap offsets indexed by ordered symbol pair; and a priority queue of nonterminals by decreasing expanded length. This adds only a constant number of stored references per assertion, plus $O(N)$ symbol-queue metadata.

After choosing a symbol, remove its incident assertions and insert their split consequences. All other assertions remain untouched. A selected symbol cannot return: subsequent splits introduce only strictly shorter children, and periodicity compaction introduces no symbols. Each tagged nonterminal therefore needs at most one queue entry. A stale entry whose assertions have already been discharged can be skipped. Every remaining assertion mentions a nonterminal, since comparisons between terminals are checked immediately, so the queue cannot run out before the conjunction has been resolved.

Only overlap buckets that receive an insertion need another compaction pass. Deleting offsets cannot create a triple satisfying the compaction inequality: any such triple would already have been present. Buckets untouched since their last compaction are therefore still fully compact. The incidence and overlap indices are updated on both insertion and removal. Thus the implementation preserves exactly the same conjunction and the same polynomial assertion-count argument as before; it avoids repeated work on unrelated comparisons. The maximum-length queue adds a logarithmic scheduling cost per entry. The cooperative work counter is an implementation allowance, not a count of machine instructions or integer bit operations.

Before allocating these structures, the current checker tries a direct synchronized walk of the two grammar derivations for at most $B = 64$ steps. Each side is a stack of unconsumed chunks. If the leading chunks are the same interned node, remove both, skipping their entire represented word. If their first letters differ, reject. Otherwise split the longer leading chunk into its two children. A terminal cannot require splitting: two terminals with the same letter have the same interned node.

The invariant is that equal-length prefixes have been consumed on both sides, with every consumed pair proved identical. Hence a differing next letter is an exact mismatch. If one stack empties, equal original lengths and positive chunk lengths force the other to empty too, proving equality. If the step cap is reached, the walk returns no answer and invokes the indexed polynomial checker on the original roots. It neither guesses nor treats the compared prefix as a certificate for the unvisited suffix. No expanded word list is allocated.

This is an introspective choice of implementation: inexpensive direct progress is attempted first, with a bounded cost before switching to the algorithm with the proven polynomial bound. For default fixed $B$, an unresolved probe adds $O(B)$ abstract work to the fallback. It cannot turn the compressed equality procedure into a scan of an exponentially long expanded word. The low-level option `equality_probe_steps=0` disables this probe for comparison and testing; a custom nonnegative cap is charged to the same work allowance. This is adaptivity inside equality, not a new worst-case theorem for the surrounding group search.

Two hundred additional random pairs of differently parsed words exercise the indexed engine with the probe disabled, including interior mutations that preserve both endpoints. Separate tests force compaction, verify both truth values after a full 64-step probe on words longer than $2^{100}$, and interrupt a comparison by work exhaustion and caller cancellation. Repeating an interrupted comparison succeeds: no partial answer is placed in the equality cache. The finite tests supplement the invariants above; they do not replace them.

## 44.5 Free reduction from exact longest-prefix queries

An exact bounded grammar walk first attempts to compute the common prefix directly. If it remains unresolved, binary search beyond the prefix already proved equal, using SLP substring construction and exact equality, finishes the computation. This fallback uses at most $O(\log L)$ equality calls for expanded length $L$; since the SLP length is binary-encoded, this number is polynomial in the grammar size.

If $u$ and $v$ are freely reduced, all cancellations in their product occur at their common boundary. Let

$$c = \operatorname{lcp}(u^{-1}, v).$$

Then its freely reduced form is

$$u[0 : |u| - c]\; v[c : |v|].$$

A bottom-up traversal applies this formula at every concatenation node. When the boundary letters are not inverse, no prefix query is necessary. The reduced children and their slices remain compressed and shared. A second prefix comparison against the inverse word, capped at half its length, removes the conjugating prefix and suffix for cyclic reduction.

For a single input SLP, reduction does not increase expanded length or parse height. Boundary slicing and the logarithmic prefix search create only polynomially many additional rules over the original nodes. Inversion, occurrence counting and one simultaneous substitution followed by reduction are likewise polynomial operations in their encoded inputs. However, repeatedly applying polynomial transformations need not maintain polynomial-size intermediate grammars in the original knot size. A bound on the total constructed grammar size over the search remains a separate obligation.

One exact test compares the two differently parsed words

$$(ab)^{2^{100}}a = a(ba)^{2^{100}}$$

and freely reduces their quotient to the empty word. Neither word is expanded. Another test uses a 1,500-level repeated-power grammar to check substring and inversion operations without recursive-stack overflow.

### 44.6 Adaptive cancellation and certified-prefix reuse

The initial prefix implementation tested candidate lengths by comparing both words from position zero. Although this has a polynomial bound, it repeated comparisons over prefixes already known equal. In addition, many cancellation boundaries in group traces are easily resolved by skipping shared grammar nodes. The current kernel therefore uses the synchronized walk for prefix queries as well as full equality.

Let $H = \min\{|a|, |b|, c\}$ when a cap $c$ is supplied, or $H = \min\{|a|, |b|\}$ otherwise. The desired answer is $\min\{\mathrm{lcp}(a, b), H\}$. Maintain two stacks of unconsumed chunks and the length $p$ of a proved common prefix. Matching node identities allow an entire chunk to be skipped and its length added to $p$. If this reaches $H$, return $H$, even when the cap lies inside the skipped chunk. If the next letters differ, return $p$. Otherwise split the longer leading chunk and continue. The same invariant as for equality proves these answers: both stacks begin at the same original position, and the consumed prefixes are exactly equal. Reaching the end of the shorter word necessarily reaches $H$, so neither stack is accessed after exhaustion. Empty inputs and identical roots are handled before the walk.

The low-level option `prefix_probe_steps`, independent of `equality_probe_steps`, bounds this traversal and defaults to 64. If the walk exhausts its allowance, $p$ is only a lower bound on the answer. The checker retains that certified progress and starts bisection with $\ell = p$ and $h = H$. Its invariant is

$$a[0 : \ell] = b[0 : \ell], \qquad \ell \le \min\{\mathrm{lcp}(a, b), H\} \le h.$$

For $m = \lceil(\ell + h)/2\rceil$, compare only

$$a[\ell : m] \quad \text{and} \quad b[\ell : m].$$

An equality proves the larger prefix and permits $\ell \leftarrow m$. A mismatch proves the answer is below $m$ and permits $h \leftarrow m - 1$. Previously proved letters are not included in either substring query. This remains valid when $\ell$ was obtained from earlier positive bisection queries rather than the direct walk. Setting the probe allowance to zero therefore still uses the improved bisection rule.

Each iteration strictly shrinks the integer interval. There are at most $O(\log(H - p + 1))$ exact equality queries after the bounded walk, and substring construction retains its polynomial grammar-size bound. The direct walk adds only its fixed allowance of abstract steps, with the usual bit cost for encoded lengths. Thus this adaptation preserves the previous polynomial bound for each compressed word operation. It does not bound how often the recognizer needs such an operation.

For a concrete capacity example, let $P = (ab)^{2^{500}}$ and compare $Pc$ with $Pd$, where $c \neq d$. Their shared root can be skipped in one step after exposing it; the answer $|P| = 2^{501}$ requires no equality

query or expanded allocation. Caps strictly inside $P$ are equally exact. Tests also attach differently parsed exponential periodic tails to a shared prefix, force the direct walk to exhaust all 64 steps, and check that fallback slices never start before the proved prefix. An interior mutation far beyond that prefix tests a negative comparison without an endpoint shortcut. Eight hundred random word pairs check capped prefix results against literal strings using probe allowances 0, 1, 7 and 64. Work exhaustion and caller cancellation raise rather than returning the current lower bound as a completed answer.

## 44.7 Replaying the existing knot certificates

The public checker accepts `compressed=True`; the optional pipeline flag is `-group-compressed`, and Python recognition uses `use_group=True, group_compressed=True`. Existing certificate versions 1 and 2 are unchanged. The independent PD reconstruction remains the same as in the explicit checker, and only its small initial relators are materialized. The compressed engine then replays every move.

For a generator elimination, an exact DAG occurrence count must equal one. Its position splits the defining relator into prefix, signed generator and suffix; the derived replacement and its inverse are substituted into all other relators with shared reduction. For a Whitehead move, validated letter images are substituted simultaneously. For a relator-overlap move, compressed rotations and inversions are formed, the stated prefix equality is checked exactly, and the target is replaced using the same normal-closure identity as before. A distinct donor must remain present. Terminal acceptance still requires one generator and no nonempty relators.

The original `max_letters` bound applies to the reconstructed initial presentation in compressed mode. Intermediate expanded lengths are not subject to that limit; `max_nodes` and the work allowance instead bound compressed replay. The default node/assertion allowance is 100,000. Local exhaustion propagates as `GroupLimit` and resumes recognition when called through the optional stage. Global cancellation propagates unchanged. Search and replay still share the same local wall allowance. Evidence records whether replay used explicit letters or compressed SLPs.

The initial presentation construction and classical cyclic-knot-group criterion are shared by the two checkers. The compressed and explicit word algorithms are distinct, but this is not two entirely unrelated proof systems. The regression suite checks every maintained survivor, the two overlap-dependent mirrors and Gordian, along with forged moves, resource exhaustion and actual CLI dispatch.

## 44.8 Capacity gains, measured overhead, and their scope

`benchmark_compressed_words.py` records whole-query, replay-only, constructed-trace and word-kernel measurements separately. Each query starts from fresh validated PD data. The 15 knot cases comprise twelve original survivors, the two previously stalled mirrors and Gordian. There are five measured randomized A/A/B rounds after one excluded warmup, with explicit verification, a repeated explicit control, and compressed verification. Search settings are identical: relator moves enabled, two local seconds, ten million work units and four global seconds.

On the fourteen small cases, the sums of median whole-query costs are 51.02 ms, 50.89 ms and 137.53 ms respectively. On Gordian the medians are 1.236, 1.235 and 1.417 seconds. All queries complete, but the compressed verifier adds overhead on these inputs. The default stays explicit. We do not present compression as a current general speedup.

To test representation capacity separately, construct valid but deliberately inflated proof traces for a three-crossing unknot. Apply alternating Whitehead shears repeatedly, undo them, and append an ordinary certificate. These are artificial verification stresses, not hard unknot-recognition instances; a trace simplifier could cancel inverse moves. At eight rounds, both checkers finish, at

medians 8.84 ms explicit and 2.12 ms compressed. At sixteen, thirty-two and sixty-four rounds, the explicit verifier exhausts its 200,000-letter allowance. The compressed verifier finishes all three; its medians are 5.69, 18.28 and 64.06 ms. The sixty-four-round trace uses 904 SLP nodes and represents an intermediate node length with 90 bits. The explicit limit observations are not completed verification times.

Finally, the noncanonical equality-and-cancellation kernel above, with exponent $2^{100}$, completes in a median 0.676 seconds using 10,614 allocated nodes and 835,846 charged work units. This includes grammar construction, equality and reduction. Raw samples, actual synthetic certificates, environment metadata and separate timing scopes are in `fast/results/compressed_words_20261008.json`.

The resulting advance is exact access to words far larger than available expanded storage. The following extension adds compressed search operations; a quantitative bound on the grammars and moves needed for arbitrary knot diagrams remains open. The complete recognition fallback remains exponential; this implementation does not establish general sub-exponential or quasi-polynomial unknot recognition.

## 44.9 Compressed discovery: multiplicities and cyclic adjacency

The separately selectable `compressed_search.py` now keeps relators as SLP roots throughout elimination and Whitehead search. The key additional operation extracts the weighted Whitehead graph without examining expanded letters. Let $N$ be the number of reachable grammar rules, $m$ the number of relator slots and $r$ the surviving generator rank. Seed each nonempty root with multiplicity one, adding multiplicities when several relator slots share that root. Visit rules in reverse topological order. At $X = UV$, add the multiplicity of $X$ to both children, even when $U = V$. At a terminal, add its multiplicity to the absolute generator's total occurrence count.

This computes exact counts: the coefficient of a node is the number of its occurrences in the fully unfolded derivation forest. Distributing a coefficient through a concatenation partitions these occurrences into its two child occurrences. Induction down the DAG proves the terminal counts. It also proves the following adjacency construction. At each $X = UV$ with multiplicity $c_X$, insert weight $c_X$ on the undirected edge

$$\{\mathrm{last}(U), -\mathrm{first}(V)\}.$$

For each nonempty relator root, insert its closing edge with weight one. Every internal adjacent pair in an unfolded word has exactly one lowest concatenation separating its letters, so this counts each pair exactly once. Singleton relators receive just their closing edge. Empty relators contribute nothing. As with the explicit implementation, both signed vertices are retained, and the edge convention is $\{x, -y\}$.

The traversal takes $O(N \log N + m)$ abstract operations, including the sort used to obtain a topological order, with $O(N + m + r^2)$ storage as a loose bound. These are integer-operation counts, not constant-time claims for unbounded integers: multiplicities have $O(N + \log(m + 1))$ bits. In particular, a power containing $2^{100}$ copies of a subword contributes a 101-bit weight, not $2^{100}$ edge visits. The existing exact integer minimum-cut routine can therefore operate on the compressed graph. Its Edmonds–Karp augmentation bound is independent of the numeric capacity values; arithmetic still pays their bit cost.

## 44.10 Unique occurrences and elimination without expansion

Elimination needs to know which generators occur exactly once in each individual relator. Repeating full occurrence counting independently for every generator and every relator wastes sharing.

Instead store two masks for every reachable node: $P_X$ records presence and $R_X$ records at least two occurrences. For a terminal $g$, set $P_X = \{|g|\}$ and $R_X = \varnothing$. For a concatenation,

$$P_{UV} = P_U \cup P_V, \qquad R_{UV} = R_U \cup R_V \cup (P_U \cap P_V).$$

Thus the unique generators at a root are precisely $P_X \setminus R_X$. The implementation uses dense generator indices and integer bit masks; the cost of a mask operation grows with its $r$ bits. The transient mask storage is $O(Nr)$ bits. This is polynomial in grammar size and rank, with no dependence on expanded word length except its encoded bit length.

Combine these unique-generator sets with aggregate counts to retain the explicit producer's elimination priority

$$((|w| - 2)\ \mathrm{count}(g),\ |w|,\ \text{relation index},\ g).$$

For the chosen occurrence $w = ug^{\epsilon}v$, obtain its position using the existing compressed occurrence query. Form $vu$ by two slices and a concatenation. Substitute $(vu)^{-1}$ when $\epsilon = 1$, or $vu$ when $\epsilon = -1$, into all other relators; retain the eliminated relator's empty slot. Exact compressed free and cyclic reduction finishes the move. The usual Tietze transformation proves preservation of the group.

When no generator can be eliminated, a negative Whitehead cut supplies the same length-reducing automorphism used by the explicit producer. Its generator images are short words, but substitution and cancellation remain compressed. The implementation checks that the resulting total length change equals the cut prediction. All lengths and heuristic scores use exact integers. A zero or positive best cut is only a stalled search, not a nontrivial-knot certificate.

### 44.11 The bounded overlap bridge and verification boundary

The cyclic overlap matcher is still a suffix automaton on explicit letters. The new producer retains the rank-adaptive policy from the previous section, but enters this matcher only when the total current expanded length is at most `max_letters`. It charges expansion and matching to the same search work allowance. If the length exceeds the cap, it skips overlap matching and continues to the compressed Whitehead test. It does not report that no overlap exists. Consequently the new backend is a hybrid, and its search paths need not cover all paths available with a larger explicit overlap allowance. Section 47 adds a later exact whole-donor fallback when this initial policy would stall above the cap; the explicit suffix-automaton path remains unchanged.

The CLI flag `-group-compressed-search` enables compressed discovery and compressed replay; Python callers combine `group_compressed_search=True` with `use_group=True`. The earlier `-group-compressed` flag continues to select replay only. Standalone `compressed_certificate` returns an untrusted candidate trace. Before a recognition verdict, the checker reconstructs the full Wirtinger presentation independently from the input PD and replays the trace. It shares the deterministic word kernel, but does not call the new summary, eligibility, cut-selection or producer move routines. On the finite knot corpus, the independently implemented literal checker also accepts all the produced traces. This corroborates these examples; it is not a proof that two implementations sharing a kernel cannot share a bug.

`max_nodes` bounds allocated grammar nodes and equality assertions; `max_work` bounds cooperative search operations. The independent replay has its own work allowance, and both share the local wall deadline. The explicit-letter cap bounds initial PD reconstruction and optional overlap expansion, rather than all intermediate represented lengths. Exhaustion remains inconclusive and caller cancellation propagates.

## 44.12 What the extension does and does not bound

For a supplied grammar, counts, unique-occurrence masks and Whitehead graphs are polynomial-time constructions. So are the individual compressed word operations established above. These statements do not bound the number of Whitehead moves in terms of the original crossing count, or the accumulated grammar size after repeated substitutions. Even a strictly length-decreasing search may have exponentially many steps when its initial length is exponentially compressed. Generator elimination can also increase subsequent relator lengths. A proof of a general quasi-polynomial recognizer must address these global quantities.

The practical policy therefore keeps explicit search as the default. Representation adaptivity should be justified by measured progress and resource use, with independent verification retained after any transition. Switching implementations alone supplies no improved worst-case exponent.

## 44.13 Discovery audit and measured limitations

The 8 October 2026 discovery audit repeats the same fifteen input diagrams: twelve original Khovanov survivors, their two previously stalled mirrors, and Gordian. For each input it executes a randomized A/A/B ordering of explicit search, an identical explicit control, and compressed search. Every arm includes fresh PD validation, the complete recognition query and independent certificate replay. One warm-up round is excluded and five rounds are measured, giving 225 measured queries. The local group allowance is ten seconds, the global allowance twelve seconds, and each search/replay receives ten million abstract work units. The larger time allowance deliberately permits comparison of completed compressed searches; it differs from the earlier two-second replay-only experiment.

All 225 queries return UNKNOT with group evidence. The sum of median times over the fourteen small cases is 73.52 ms for explicit search, 74.00 ms for its control and 391.33 ms for compressed search. Gordian's respective medians are 1.247, 1.252 and 7.076 seconds. Thus this implementation increases representation capacity but substantially slows the present corpus. A profile of Gordian identifies exact compressed equality inside cancellation as the main cost; the shared eligibility masks do not remove that bottleneck. Its producer allocates 11,529 grammar nodes and uses one bounded overlap expansion. These are observed values, not worst-case complexity bounds.

A separate synthetic audit builds $(ab)^{2^k}$, computes counts and the cyclic graph for two copies, checks its unique-occurrence masks, then constructs the abstract presentation $\langle a, b \mid ab^{2^k} \rangle$ and eliminates $a$ without expanding. This presentation is explicitly known to be infinite cyclic; it is not a new family of hard knot diagrams. The table includes grammar construction and all these operations, with five measured repetitions after a warm-up:

| $k$ | allocated nodes | charged work | median seconds |
|---:|---:|---:|---:|
| 16 | 53 | 511 | 0.000267 |
| 64 | 197 | 1855 | 0.000836 |
| 256 | 773 | 7231 | 0.003128 |
| 1024 | 3077 | 28735 | 0.015473 |

The represented lengths exceed $2^{1024}$ in the last row, while no expanded-word scan occurs. Unit tests additionally compare 250 random grammars' counts, singleton sets, weighted graphs and cut predictions with literal computations, check oversized-overlap skipping, and replay all fifteen real certificates through both implementations. Resource limits, global cancellation, nontrivial knots and CLI routing are checked too. The full integrated suite passes all 642 tests with the optional Regina dependency installed in 166.446 seconds; its log is `data/compressed-search-integrated-tests.txt`. The raw experiment is `fast/results/compressed_search_20261008.json`; its script is `fast/benchmark_compressed_search.py`.

The incoming-report directory still contains only its README at this checkpoint. The upstream `openai/math` head remains `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`; its latest listed changes update manuscripts and Lean formalizations and provide no new unknot-specific bound for this integration. No general sub-exponential or quasi-polynomial recognition theorem is claimed.

### 44.14 Controlled measurements of indexed and adaptive equality

The follow-up audit loads the previous word kernel directly from the recorded repository commit `e4f471c971359b74d30d5f85284850d462a74dfd`, without changing the checkout. It patches only the arena constructors used by compressed search and compressed replay. This permits five interleaved arms on the same fifteen input diagrams: previous kernel, identical previous-kernel control, indexed assertions with the probe disabled, indexed assertions with the 64-step probe, and the established explicit search/replay. Each arm includes fresh PD validation and the complete recognition query. Five measured rounds follow one excluded warm-up, giving 375 measured queries. The local group allowance is fifteen seconds, the global allowance eighteen seconds, and search and replay each receive twenty million abstract work units. The reported repeat ran without concurrent local tests.

All 375 queries finish with UNKNOT and group evidence. The fourteen small cases and Gordian give the following results:

| Equality/search arm | Small cases (ms) | Gordian (s) |
|---|---:|---:|
| Previous compressed kernel | 265.32 | 6.764 |
| Previous-kernel control | 266.16 | 6.760 |
| Indexed assertions | 265.05 | 2.514 |
| Indexed plus bounded probe | 248.40 | 2.403 |
| Explicit search and replay | 51.46 | 1.241 |

The adaptive compressed arm improves Gordian by about 2.81× relative to the previous compressed kernel. The bounded probe additionally reduces the small-case aggregate by about 6.3% relative to that baseline. The explicit backend remains faster on this corpus and stays the default; these improvements do not justify automatically moving small presentations to compressed search.

Gordian's previous producer charges 8,804,494 abstract work units. Indexing reduces this to 2,200,397 while preserving its 31,088 split rounds; the bounded probe lowers the counts to 2,165,522 and 25,130 respectively. The latter resolves 394 uncached comparisons directly and falls back on 203, after 22,060 total probe steps. All variants allocate 11,529 grammar nodes and perform one explicit overlap expansion. These figures explain the reduction in repeated assertion processing; work units remain specific to the implementation and are not used as portable CPU timing predictions.

The separate noncanonical periodic kernel builds $(ab)^{2^k}a$ and $a(ba)^{2^k}$, checks equality, and freely cancels their quotient. Five measurements after warm-up give the following median seconds:

| $k$ | previous | control | indexed | adaptive |
|---:|---:|---:|---:|---:|
| 8 | 0.005065 | 0.005044 | 0.005205 | 0.005497 |
| 32 | 0.071208 | 0.072552 | 0.068555 | 0.070064 |
| 100 | 0.656635 | 0.675381 | 0.650547 | 0.650003 |

These examples have only six peak assertions, so indexing brings little timing benefit. Every probe here exhausts its 64 steps and falls back; at $k = 8$ the adaptive version is slower. Thus the experiment does not support a uniform speedup for all compressed words. It supports removing

unrelated-assertion scans when a comparison has a substantial live set, and bounding the cost of an unsuccessful direct walk.

The reproducible script is `fast/benchmark_indexed_equality.py`; raw samples, execution orders, caps, version identifiers and per-query statistics are in `fast/results/indexed_equality_20261008.json`. All 645 integrated tests pass with the optional Regina dependency installed in 123.169 seconds; `data/indexed-equality-integrated-tests.txt` preserves the output. All finite examples continue to rely on the same exact mathematical decision criteria. Neither faster word equality nor the adaptive probe bounds the length or grammar growth of the full knot-group search.

### 44.15 Measured effects of adaptive prefix cancellation

The prefix follow-up loads the preceding word kernel from repository commit `97587b366` (the full identifier is recorded in the raw artifact). Its five randomized arms are that baseline, an identical control, the new bisection rule with direct prefix walks disabled, the complete adaptive prefix implementation, and explicit search/replay. The previous 64-step equality probe remains enabled in every compressed arm, so the experiment isolates the prefix changes. Each whole query includes fresh PD validation, discovery and independent replay. There are five measured rounds after a warm-up on each of the same fifteen diagrams: 375 measured queries. Caps remain fifteen local seconds, eighteen global seconds, twenty million search/replay work units each, 100,000 grammar nodes and 200,000 explicit letters. No local tests run concurrently with these measurements.

All queries complete with UNKNOT and group evidence. In the small-case column below, the number is the sum of medians over fourteen diagrams:

| Prefix/search arm | Small cases (ms) | Gordian (s) |
|---|---:|---:|
| Previous compressed kernel | 245.18 | 2.461 |
| Previous-kernel control | 246.33 | 2.450 |
| Suffix-only bisection | 248.09 | 2.110 |
| Adaptive prefix walk | 185.27 | 1.922 |
| Explicit search and replay | 51.50 | 1.259 |

The adaptive prefix implementation reduces Gordian's observed median by about 21.9% and the small-case aggregate by about 24.4% relative to the immediately preceding compressed kernel. Explicit search still wins on this corpus and remains the default. The suffix-only arm by itself does not improve the small-case aggregate.

On Gordian, the producer's equality calls fall from 3,035 to 271 and its allocated nodes from 11,529 to 9,025. Of 971 prefix queries, 895 finish through the direct walk, 41 enter bisection, and the remaining 35 finish through identity, empty-length or first-letter shortcuts. The walks take 11,248 total steps. The suffix-only arm still makes 3,035 equality calls, but reduces split rounds from 25,130 to 4,264; the adaptive arm uses 3,113 split rounds. These counters support the distinction between rechecking fewer letters and avoiding an equality query altogether.

Two synthetic kernel families separate these mechanisms. The first builds $P = (ab)^{2^k}$ and computes the common prefix of $Pc$ and $Pd$ for distinct letters $c, d$. The second checks equality of $(ab)^{2^k}a$ and $a(ba)^{2^k}$ and freely cancels their quotient. Each measurement includes fresh grammar construction, with five measured repetitions after a warm-up:

| Kernel | $k$ | Previous (s) | Suffix-only (s) | Adaptive (s) |
|---|---|---|---|---|
| Shared prefix | 16 | 0.001902 | 0.002300 | 0.000054 |
| Shared prefix | 64 | 0.026006 | 0.033483 | 0.000111 |
| Shared prefix | 128 | 0.103473 | 0.134563 | 0.000637 |
| Periodic cancellation | 16 | 0.019466 | 0.009441 | 0.011670 |
| Periodic cancellation | 64 | 0.273239 | 0.143347 | 0.143935 |
| Periodic cancellation | 128 | 1.091430 | 0.578284 | 0.595711 |

In the shared-prefix family the adaptive walk uses four steps and zero equality queries for every tested $k$, excluding grammar construction and the bit cost of length arithmetic. At $k = 128$ it allocates 135 nodes, compared with 8,391 for the previous kernel. The suffix-only method allocates fewer nodes too, but takes longer because its nonzero-offset slices require more traversal work. In the periodic family the walk exhausts its 64 steps; bisection still runs. At $k = 128$, both new arms allocate 779 nodes rather than 17,166. The probe does not beat the suffix-only arm on these examples. At $k = 16$ its partial progress also changes the later slice boundaries and increases allocation from 107 to 224 nodes. Retaining a proved prefix therefore preserves correctness and the polynomial bound, but does not guarantee a monotone timing or storage improvement on every input.

Raw samples and controls are in `fast/results/compressed_prefix_20261008.json`; the reproducible driver is `fast/benchmark_compressed_prefix.py`. These are improvements to word operations and finite measured recognition queries. The global search-length and grammar-growth obligations remain open. The incoming directory still contains no new report, and rechecking upstream `openai/math` again returns the previously reviewed head `fd4aeeb2e`.

All 648 integrated tests pass with the optional Regina dependency installed in 124.837 seconds. The output is preserved in `data/compressed-prefix-integrated-tests.txt`.

### 44.16 A further literature route for cancellation queries

Tomohiro I's deterministic recompression result [31] builds a longest-common-extension index from a supplied SLP. For grammar size $s$ and represented length $M$, Theorem 2 gives $O(s \log(M/s))$ construction time and $O(\log M)$ query time, with logarithms bounded below by one, in its word-RAM model, whose words have $\Omega(\log M)$ bits and whose alphabet is integer encoded. The introduction explicitly leaves efficient dynamization unresolved. This is a promising route to replacing repeated substring-equality searches, but construction and updates must be charged when our substitutions change the strings. We have not implemented that index, and its query bound supplies no bound on the number of group moves. The current incidence index stores equality assertions, not a recompressed text index. This source review therefore identifies a further concrete research direction without attributing its stronger query bound to the present code.

## 45 Continuing group search across word representations

### 45.1 Why the representation should depend on progress

The preceding measurements show two distinct regimes. Explicit relators are faster on the current finite knot corpus, while compressed words can represent intermediate strings beyond an explicit-letter allowance. The new optional adaptive stage starts with the explicit producer and changes representation once when the next substitution would exceed a chosen threshold. It continues the existing search state, preserving the certificate prefix and remaining resources. This implements a form of introspective search without requiring compressed arithmetic on every small initial relation.

Let $C$ be the hard explicit-letter allowance and $T \le C$ the switching threshold. By default $T = C$. A caller may request an earlier nonnegative threshold; values above $C$ are clamped to $C$. Initial presentation construction must still satisfy the hard allowance, including the $4n$-letter bound before crossing relations are simplified. Adaptivity does not bypass that check.

## 45.2 Predicting an allocation before changing state

Suppose the selected defining relator $w$ has length $\ell$ and contains the eliminated generator $g$ exactly once. Let $L$ be the total relator length and $c_g$ the total number of occurrences of $g^{\pm 1}$. The replacement for $g$ has length $\ell - 1$. Removing $w$ and substituting into the remaining words would allocate

$$E = L - \ell + (c_g - 1)(\ell - 2)$$

letters before free reduction. This is an exact raw substitution length, including the case $\ell = 1$, where all remaining occurrences of $g$ are deleted. If $E > T$, the producer switches before deleting $w$, constructing the replacement, changing the generator set or recording the pending move. Otherwise the existing explicit substitution and reduction proceed.

A Whitehead image has length at most three, so $3L$ is a conservative allocation bound. A selected shortening Whitehead move triggers conversion if $3L > T$, again before applying or recording it. Its final reduced length may be much smaller: the trigger controls the raw allocation allowance, not a theorem that the move necessarily expands the presentation. Relator overlap moves strictly decrease total length and retain their existing explicit implementation during this phase.

The producer converts each current relator to an SLP root while preserving its slot, including empty slots. Generator names and the surviving set are unchanged. It passes the same move list to the compressed continuation; there is no second reconstruction of the initial presentation and no replay of the producer's completed moves. The continuation may recompute the pending search choice, which is charged normally. This duplication can occur at only one handoff. Once compressed, the producer never returns to explicit search, avoiding repeated conversion near the threshold.

## 45.3 Soundness and the resource boundary

Conversion changes only representation, so the group presented by the current relators is unchanged. Every subsequent move uses the same Tietze, Whitehead or relator-overlap justification as before. The resulting certificate format is unchanged and remains bound to the original input PD. Acceptance still uses the cyclic-knot-group criterion of Section 42.

If no conversion occurred, independent replay remains explicit unless the caller separately requested compressed verification. If conversion did occur, compressed replay is selected automatically: literal replay might exhaust the very allowance that prompted the producer to switch. In both cases the checker independently reconstructs the full original Wirtinger presentation and verifies every recorded move from the beginning. It does not accept the handoff state as evidence. The two implementations still share the underlying compressed-word kernel when compressed replay is selected, as described in the previous section.

The explicit prefix consumes part of the original work allowance. The SLP arena receives exactly the remainder, and conversion itself is charged to it. The continuation does not obtain a fresh work budget or a new wall deadline. Its node allowance is the usual compressed node/assertion cap. Independent replay retains its separately specified work allowance, while search and verification share the local wall deadline. A depleted work allowance, node-limited conversion, local timeout or stalled continuation remains inconclusive; global caller cancellation propagates.

Evidence reports whether conversion occurred and records its trigger, the number of preceding moves, surviving rank, current and projected lengths, threshold, work already spent and work

remaining. These are diagnostics, not additional certificate axioms. A regression test leaves exactly one work unit at handoff and checks that conversion exhausts it rather than silently refilling the budget.

## 45.4 A bound on the explicit prelude

Although no quasi-polynomial bound for the full recognizer is established, the explicit portion has a useful conditional bound. Let $r$ be the initial generator rank, let $L_0$ be its initial reduced total relator length, and put $M = \max\{L_0, T\}$. There are at most $r - 1$ generator eliminations. Between eliminations, each accepted Whitehead or overlap move decreases the nonnegative integer total length by at least one. An elimination performed explicitly has raw output length at most $T$, and free reduction cannot increase that length. Every interval between eliminations therefore starts at length at most $M$ and contains at most $M$ shortening moves. Including eliminations and the possible final switch decision gives

$$O\big(r(M + 1)\big)$$

explicit move iterations. Counts, suffix-automaton overlap searches, minimum cuts and literal substitutions are polynomial in the current explicit input size and rank. Conversion is polynomial in the current length too, and occurs once. Thus a polynomially bounded threshold gives a polynomially bounded explicit prelude on inputs admitted by its initial resource checks.

This does not establish a polynomial or quasi-polynomial recognizer. The compressed continuation still needs a completeness argument and bounds on both grammar growth and the number of moves for the reachable knot presentations. Moreover, finding positive cyclic-group certificates quickly would not itself supply a fast conclusive route on nontrivial knots. The surrounding recognizer retains its previous complete exponential fallback. The policy preserves progress and limits representation overhead; it does not manufacture a stronger worst-case guarantee from its current backends.

## 45.5 Interface and validation

The CLI flag `-group-adaptive` enables this group stage with $T = C$. `-group-switch-letters T` also enables it and selects an earlier threshold. Python recognition accepts:

```
recognize(diagram, use_group=True, group_adaptive=True,
          group_switch_letters=T)
```

For standalone group calls, enable `adaptive_search`. Always-compressed search and adaptive search are mutually exclusive. Replay-only compression may be combined with either policy. The default recognition pipeline is unchanged.

Tests show that the default adaptive policy preserves all fifteen corpus certificates exactly and never constructs a search arena on those paths. Earlier-switch traces pass both independent replayers. Tests force both elimination and Whitehead handoffs, verify the preserved trace prefix, prohibit reconstruction at conversion, check resource and cancellation behavior, reject invalid option combinations and exercise CLI routing. Known nontrivial knots remain inconclusive at this positive-certificate stage. These finite checks support the invariants above without implying completeness of the group heuristic.

The full integrated suite passes all 653 tests with the optional Regina dependency installed in 126.251 seconds. Its output is preserved in `data/adaptive-group-integrated-tests.txt`.

### 45.6 Whole-query and matched-cap measurements

The 8 October 2026 audit runs five interleaved whole-query arms on the same fifteen real diagrams: explicit search, an identical explicit control, adaptive search with $T = C = 200{,}000$, adaptive search with $T = 4{,}096$ and $C = 200{,}000$, and always-compressed search. Every arm includes fresh PD validation, all recognition stages and independent group-certificate replay. There are five measured rounds after a warm-up, giving 375 whole queries. Search and replay each have twenty million work units; the local group allowance is fifteen seconds, the global allowance eighteen seconds, and the node allowance 100,000. Local tests do not run concurrently.

All queries finish with UNKNOT and group evidence. The default adaptive policy does not switch on any of these inputs. The earlier threshold switches only on Gordian; its fourteen small-case paths stay explicit. The small-case column is the sum of the fourteen per-input medians:

| Search policy | Small cases (ms) | Gordian (s) |
|---|---:|---:|
| Explicit | 52.54 | 1.291 |
| Explicit control | 52.18 | 1.251 |
| Adaptive, $T = C$ | 52.00 | 1.242 |
| Adaptive, $T = 4{,}096$ | 52.05 | 1.641 |
| Always compressed | 190.37 | 1.970 |

The no-switch adaptive results are near the explicit controls; these small differences are not presented as an algorithmic speedup. Early conversion costs more on Gordian than letting its explicit path finish. Accordingly, the default trigger remains the hard letter ceiling, not the earlier threshold tested here. The optional policy is primarily a capacity extension that retains the faster representation while it fits.

A second experiment lowers the hard letter cap on four real-input cases. It measures fresh PD validation and the group stage alone, including independent replay, without the surrounding recognition filters or fallback. The four arms are explicit, its control, adaptive with $T = C$, and always compressed. Five measured rounds after a warm-up give 80 probes. The explicit arms exhaust the letter allowance in all forty measured probes. Both adaptive and always-compressed search produce independently verified certificates in all forty of their probes:

| Input | $C$ | Adaptive median (s) | Compressed median (s) |
|---|---:|---:|---:|
| Survivor 00 | 64 | 0.01093 | 0.01380 |
| Survivor 02 | 60 | 0.01102 | 0.01285 |
| Gordian | 4,096 | 3.72104 | 3.41667 |
| Gordian | 8,192 | 1.15899 | 1.23270 |

The explicit exhaustion times are recorded in the raw artifact, but are not completed recognition times and are not used in speedup ratios. This experiment establishes additional certificate capacity under matched caps, not newly decidable knots: the overall recognizer already has its complete fallback, and sufficiently generous explicit limits certify these examples.

Hard and soft thresholds have different effects. With $T = 4{,}096$ but $C = 200{,}000$, Gordian switches after 127 completed moves from 3,228 current letters before an 8,098-letter raw substitution. The continuation still has the large overlap allowance and finishes with 141 total moves and 7,270 allocated nodes. When the hard cap is also 4,096, the switch point is the same, but the compressed continuation skips 75 oversized overlap probes. It follows a longer path of 220 moves and allocates 76,758 nodes. At hard cap 8,192, conversion occurs after 128 moves; the continuation skips five overlap probes and finishes with 149 moves and 21,279 nodes. The cap therefore changes more than memory use: it can change the chosen sequence of valid transformations and its cost. No threshold is claimed uniformly optimal.

The reproducible driver is `fast/benchmark_adaptive_group.py` and the raw artifact is `fast/results/adaptive_group_20261008.json`. The evidence supports a correct one-way continuation mechanism and measured capacity gains. It leaves the general sub-exponential recognition question open.

# 46 Batching powers of a Whitehead automorphism

## 46.1 The remaining cost of elementary descent

Compressed words remove the obligation to materialize every letter, but not the obligation to make every search move. For example, the abstract presentation

$$\langle a, b \mid (ab^N)^2, (ab^N)^3 \rangle, \qquad N = 2^h,$$

has an $O(h)$-size straight-line program (SLP). Neither generator occurs once in either relator. A unit Whitehead substitution $a \mapsto ab^{-1}$, $b \mapsto b$ reduces total cyclic length by five, and the same direction remains available for $N$ iterations. Our recorded producer takes these unit steps before attempting an overlap in rank two. The presentation is infinite cyclic: writing $u = ab^N$, the two relations imply $u = u^3(u^2)^{-1} = 1$, so $a = b^{-N}$. This example is a supplied group presentation, not a family of hard knot diagrams. It nevertheless isolates an exponential iteration cost in the previous compressed search.

The implementation now minimizes cyclic length along a selected Whitehead power in one exact operation when a cheap adaptive trigger justifies it. The minimization below is elementary and is proved here; it is not an assumption imported from an unverified complexity claim in a research report.

## 46.2 An exact convex length formula

Use our established convention for a type-II Whitehead automorphism $\alpha = \alpha_{a,A}$, where $a \in A$ and $-a \notin A$. Signed letters $x$ with $|x| \neq |a|$ satisfy

$$\alpha^k(x) = a^{-k[-x \in A]} x a^{k[x \in A]}, \qquad \alpha^k(a) = a.$$

Here brackets denote indicators, and $k$ is a nonnegative integer. For a cyclically freely reduced relator, retain its cyclic sequence of letters other than $a^{\pm 1}$. Between successive retained letters $x, y$ there is a possibly empty run $a^e$, where the sign of $e$ is measured relative to the signed multiplier $a$. Its transformed exponent is

$$e + kd, \qquad d = [x \in A] - [-y \in A] \in \{-1, 0, 1\}.$$

**Lemma 46.1** (The retained letters do not cancel)**.** *For every nonnegative integer $k$, the retained cyclic sequence is the same before and after applying $\alpha^k$ and cyclic free reduction.*

*Proof.* Only a gap that becomes zero can bring two retained letters together. If these letters are inverses, $y = -x$, then $d = 0$. Their initial intervening run cannot be empty because the input is cyclically freely reduced, so that run remains nonempty for every $k$. Thus no first cancellation of retained letters is possible, including at the cyclic seam. Words with no retained letters are powers of $a$ and are fixed. □

**Proposition 46.2** (Weighted-median line minimization)**.** *For a tuple of cyclically freely reduced relators, total cyclic length under $\alpha^k$ has the form*

$$F(k) = C + \sum_t w_t |k - t|, \qquad w_t \in \mathbb{Z}_{>0}.$$

*Each gap with $d \neq 0$ contributes one occurrence of $t = -ed$; all other contributions belong to $C$. If the histogram is nonempty, the smallest nonnegative minimizer is $k_* = \max(0, t_*)$, where $t_*$ is the lower weighted median: the first sorted key whose cumulative weight is at least half the total. For an empty histogram, $k_* = 0$.*

*Proof.* The lemma gives the exact contribution $|e + kd|$ from each gap. For $d = \pm 1$ this is $|k - (-ed)|$. The integer forward difference of a term $|k - t|$ is $-1$ before $t$ and $+1$ at and after $t$. Summing these nondecreasing differences gives a convex function. The first nonnegative forward difference occurs at the lower weighted median; restricting to $k \geq 0$ clips that point at zero. □

Consequently, the exact unit change and optimal change are

$$F(1) - F(0) = \sum_t w_t \begin{cases} 1 & t \leq 0, \\ -1 & t > 0, \end{cases} \qquad F(k_*) - F(0) = \sum_t w_t(|k_* - t| - |t|).$$

When the selected min-cut has negative unit change, $k_* \geq 1$ and every unit step before the first minimum strictly decreases length. The macro therefore never passes through a length-increasing intermediate cyclic word. It may choose a different overall search path from recomputing the best cut after every unit step; the procedure remains a bounded heuristic for finding a certificate.

## 46.3 Computing the histogram on a shared grammar

For each reachable SLP node store four values: first and last retained letters (zero if absent), and signed lengths of its prefix and suffix $a$-runs. A terminal supplies these immediately. For a concatenation $UV$, the first retained letter is that of $U$, if present, and otherwise that of $V$; the last letter is determined symmetrically. A prefix run extends into $V$ exactly when $U$ has no retained letter, and similarly for a suffix. If both children contain retained letters, their concatenation introduces one internal gap with exponent

$$e = \operatorname{suffix}(U) + \operatorname{prefix}(V).$$

Initialize root multiplicities from the entire relator list and propagate them downward in reverse topological order. An internal gap contributes its node's multiplicity to the histogram. If both child identifiers are the same, both occurrences are still counted. Finally, add one seam gap per root having a retained letter. Duplicate relators have separate seams. Every internal gap is created at exactly one concatenation occurrence in the expanded parse, so the weighted count is exact.

Let $s$ be the number of reachable nonempty grammar rules and $m$ the number of roots, counting duplicates. There are at most $s+m$ gap events; seam gaps for duplicate roots have the same key, so there are at most $2s$ distinct histogram keys. The metadata, multiplicities and histogram use $O(s)$ integer records plus the root list. Traversal, sorting and median selection use $O(s \log(s+1) + m)$ arithmetic operations. Rule lengths are at most $2^s$, multiplicities at most $m2^s$, and the run-length and count integers have $O(s + \log(m+1))$ bits. Thus this is polynomial bit complexity in the compressed input size, with no expanded-word traversal.

The powered images share a binary construction of $a^{k_*}$ and its inverse. Before reduction, substitution creates only polynomially many new grammar nodes. Exact compressed free and cyclic reduction use the previously established word kernel. The producer checks both that the profile's unit change equals the independent min-cut prediction and that the actual reduced output length equals the profile's predicted gain. These checks supplement the proof; the certificate verifier does not trust any optimization assertion.

## 46.4 Adaptive triggering and independently checked certificates

Computing the full profile is unnecessary when the chosen power is one. The producer profiles when either the immediately preceding move used the same multiplier and subset, or current expanded total length exceeds four times the number of allocated arena rules (including the empty rule and unreachable historical rules). Both triggers are cheap and exact; the factor four is a policy choice, not a theorem about optimal scheduling. When neither trigger fires, it performs the original unit move. Thus the large-compression example above is batched immediately, while an ordinary run of repeated moves triggers profiling on its second step. Counters record profile calls, actual power moves, and maximum exponent bit length. The explicit producer remains unchanged except that a compressed handoff can return the new certificate format.

Version 3 adds a `whitehead_power` move with the existing multiplier and signed subset plus an integer exponent at least two. Booleans, floats, missing or extra fields, and use in older certificate versions are rejected. A power of one retains the old `whitehead` representation. Versions 1 and 2 remain supported; version 3 also permits version-2 relator moves. The producer selects the lowest version needed by its trace.

The compressed verifier reconstructs the Wirtinger presentation independently, validates the automorphism, and independently builds its powered substitution. It never calls the gap profiler or the producer's image builder. The literal verifier instead repeats the elementary automorphism, checking resources on every iteration. A gigantic exponent is rejected by the literal work budget before attempting that many iterations. Such a resource failure is inconclusive, not a proof of knottedness or invalidity. Both verifiers check all surviving relators and the final single generator. Powers can represent length-increasing moves in externally supplied certificates; the optimizer's shortening condition is not a prerequisite for a valid automorphism.

## 46.5 Validation and the remaining global problem

The profile is checked against literal elementary substitutions for every cyclically reduced word of length at most three over three generators, for all 96 valid multiplier/subset choices. Duplicate roots, empty roots, all-multiplier words, signed multipliers and the seam are included. A further 400 random tuples check mixed gap distributions. The abstract presentation above completes in three moves for $h = 10, 100, 500$: one power, one relator overlap and one generator elimination. Expansion is permitted only after the relators have total length five. A separate artificially inflated trace on a small genuine unknot is replayed independently with exponent $2^{100}$; this tests the verifier and is not evidence of a hard knot family.

These results eliminate one exponential elementary-descent mechanism. They do not bound the number of changes of Whitehead direction, grammar growth across such changes, the number of relator moves, or the completeness of presentation simplification. In particular, the implementation still has no general subexponential or quasi-polynomial recognition guarantee. The next useful theoretical target is a global potential controlling alternating directions and representation growth.

The incoming-report directory contained no new delivery on 8 October 2026. The checked head of the suggested `openai/math` repository remained `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`; its latest manuscript and Lean updates did not supply a new knot-recognition algorithm. The new power optimizer follows from the algebraic analysis above. It does not assume the announced quasi-polynomial bound for Lackenby's geometric algorithm.

## 46.6 Paired measurements and an unchanged knot-corpus regime

The reproducible driver `benchmark_whitehead_power.py` loads the trusted historical producer and verifier from commit `6773a9b6e2ab50cdae07a627397054ce4a522b6d` without changing the

checkout. It compares that baseline, an identical A/A control, and the power-enabled implementation in shuffled order with five measured repetitions following one excluded warmup. Each of the fifteen knot inputs is reconstructed from PD and passed through complete recognition with compressed group search and independent replay: 225 measured queries, all successful. A separate thirty measured group probes use adaptive search on Gordian with letter caps 4096 and 8192; all complete. Group time is capped at 15 seconds, whole recognition at 18 seconds, search work at twenty million units and arena storage at 100000 nodes.

| Scope | Before | A/A | Powers |
|---|---|---|---|
| Small 14, summed medians | 220.779 | 216.227 | 232.865 |
| Gordian, full query | 1959.344 | 1940.392 | 1950.736 |
| Gordian group, cap 4096 | 3681.056 | 3637.441 | 3619.508 |
| Gordian group, cap 8192 | 2081.825 | 2112.004 | 2107.494 |

Median milliseconds; all queries and group probes complete.

No knot query or capped probe uses a power greater than one. The trigger requests at most one profile per such run. Gordian's compressed recognition remains about 1.95 seconds; its tight-cap continuation remains about 3.6 seconds. The first run's sum of fourteen small-case medians is worse with powers, 232.86 versus 220.78 and 216.23 milliseconds. In particular, `survivor-02` varies from 16 to 57 milliseconds despite identical search work, nodes and prefix counters in all three arms. We retain these raw observations and do not claim a knot-corpus speedup. A separately retained follow-up uses 21 measured rounds on `survivor-00` and `survivor-02`, or 126 additional full queries. Before/control/powers medians are 16.919/17.069/16.904 milliseconds and 15.753/15.685/15.733 milliseconds respectively; process CPU medians are within 0.002 milliseconds of these wall-time medians. The large slowdown from the first run does not reproduce. These observations support no material difference on those inputs, rather than a general speed claim. The follow-up is saved separately in `results/whitehead_power_followup_20261008.json`.

The synthetic family uses an independent one-million-unit work cap, 100000 arena nodes and an explicit overlap allowance of five letters. There are 75 measured kernel runs. The old producer completes for $h = 4, 8$ using $2^h + 2$ moves, but exhausts its work budget for $h = 12, 100, 500$. All new runs complete in three moves. The $h = 500$ result allocates 1011 nonempty nodes and consumes 17115 charged work units including construction. The old failed runs are censored observations; their elapsed times are not completion times and no speedup ratio is formed against them.

| $h$ in $N = 2^h$ | Before | A/A | Powers | Moves | Nodes |
|---|---|---|---|---|---|
| 4 | 5.169 | 5.054 | 0.736 | 3 | 19 |
| 8 | 101.550 | 101.625 | 1.031 | 3 | 27 |
| 12 | limit | limit | 1.859 | 3 | 35 |
| 100 | limit | limit | 4.270 | 3 | 211 |
| 500 | limit | limit | 10.899 | 3 | 1011 |

Median milliseconds including grammar construction. Limits are censored; moves and nodes describe the new search.

The full maintained suite passes 658 tests with Regina 7.4.1 in 130.885 seconds. A targeted rerun additionally checks that historical unreachable grammar allocations can suppress the size trigger while the second identical direction still activates batching. Raw results and execution orders are retained in `results/whitehead_power_20261008.json`; test and benchmark logs are retained under `synthesis/data/whitehead-power-*`.

# 47 Exact substring matching without expanding relators

## 47.1 What the new stage needs to decide

Whitehead powers address repeated elementary descent, but not the earlier backend's inability to compare overlaps once the explicit-letter allowance is exceeded. The new fallback first handles a precise subproblem: find a complete occurrence of a donor's recorded spelling, or its inverse, inside a cyclic target. The substring primitive is complete for its specified pattern and text. Section 48 extends the one-copy deletion described here to a checked quotient of consecutive donor copies. The resulting relator search is restricted: it does not try every cyclic conjugate of the donor or find the longest partial overlap.

The implemented occurrence table follows the fully compressed pattern-matching framework of Lifshits [32]. It represents occurrences touching a grammar cut by a single arithmetic progression, using dynamic programming on pattern and text rules. The paper's analysis counts operations on string positions; bit costs require a further factor depending on their binary length. Independently, Matsubara et al. [33] give a polynomial algorithm for longest common substrings of two SLP strings. That stronger primitive supports complete cyclic-overlap search on doubled relators. It was not implemented at this checkpoint; the subsequent implementation and local completeness proof appear in Section 49.

## 47.2 The implemented occurrence-table invariant

For a nonempty pattern node $P$ and text node $T = UV$, define

$$\mathcal{A}(P,T) = \{i : 0 \le i \le |T| - |P|, \quad T[i : i + |P|] = P, \quad i \le |U| \le i + |P|\}.$$

Positions are zero based, substrings are half open, and touching either endpoint of the cut is allowed. For a terminal text the cut is zero. The cut-occurrence lemma states that this set is an arithmetic progression, including the empty and singleton cases. We store a progression as $(s, d, c)$: first position, step and cardinality. A singleton has $d = 0$. The table includes empty entries for sufficiently long text nodes; shorter text nodes are omitted because they cannot contain the pattern.

For a one-letter pattern, a concatenation's only possible occurrences at its cut are the last letter of its left child and the first letter of its right child. Both are available from the word arena. A terminal text requires only one exact letter comparison. The remaining cells are computed in ascending pattern-node order, so both children of the pattern have complete rows available. No occurrence list is expanded.

## 47.3 An iterative local-window query

The internal query $\mathrm{Local}(P, T, [\alpha, \beta))$ returns all occurrences lying entirely within the specified interval, provided that its width is at most $3|P|$. Bounds are first clipped to the text. The implementation visits the parse of $T$ using an explicit stack. It obtains central occurrences from the stored cell and descends into a child only when the child's intersection with the interval has length at least $|P|$. Left-child, central and right-child results are emitted in that order. Occurrences shared between adjacent pieces can only coincide at an endpoint. The stack avoids dependence on Python's recursion depth.

Write $q = |P|$. Partition occurrence starts into

$$i \le \alpha + q \qquad \text{and} \qquad i > \alpha + q.$$

Every occurrence in the first part touches $\alpha + q$, and every occurrence in the second touches $\alpha + 2q$. Indeed, starts lie between $\alpha$ and $\beta - q \le \alpha + 2q$. Each part is therefore one cut-occurrence

progression restricted to an interval of starts. This gives at most two output progressions without a greedy merge heuristic.

To combine the ordered pieces of one part, the code computes their first and last positions, the greatest common divisor of their nonzero steps and offsets from the first position, and their total cardinality after removing duplicate endpoints. It then checks that this cardinality equals the number of points in the proposed progression. All supplied points have the same residue modulo the resulting step. Consequently equality of the cardinalities certifies that no hole was filled accidentally. A violated invariant raises an arithmetic error instead of manufacturing a match.

The local traversal has at most a constant number of branches at any level: its live intervals have disjoint interiors, each has length at least $q$, and together they fit in an interval of width at most $3q$. Its depth is at most the number of text rules. Repeated grammar occurrences therefore do not cause an exponential traversal.

## 47.4 Extending a larger pattern child

Suppose $P = AB$ and $|A| \geq |B|$. If a full occurrence touches text cut $c$, its $A$ part is contained in

$$[c - |P|,\ c + |A|),$$

a window of width at most $3|A|$. Query that row locally, then restrict candidate starts so the full occurrence fits in the text and touches $c$. Each resulting progression describes overlapping copies of the same word $A$ that share a position. Their union is periodic with the progression's step, unless it is a singleton.

Let $e$ be the last ending position of those $A$ occurrences. Candidate endings at most $e - |B|$ leave their entire prospective $B$ block inside that periodic union. These blocks begin at congruent positions and are identical. One exact local query therefore either accepts or rejects this entire first range of candidates. The remaining endings are within $|B|$ of $e$. A local $B$ query on $[e-|B|, e+|B|)$ gives at most two progressions; intersect them with the candidate endings and shift back by $|A|$.

When the right child is larger, the implementation uses the symmetric construction on starting positions: prospective left blocks inside the periodic union are checked together, and the first $|A|$-wide region is checked locally. Both orientations are exercised by randomized unbalanced parses. Candidate progression intersections use exact congruences and the extended-Euclidean modular inverse. Inconsistent residues yield the empty set. The final pieces describe exactly the full-pattern occurrences that touch the text cut and are merged with the checked progression operation.

This establishes the cell invariant inductively. To recover the globally first occurrence, propagate the earliest position of every text node down the DAG, in reverse topological order. Every nonempty occurrence touches a cut of some containing parse node (or belongs to a terminal). Taking the minimum of a node's earliest position plus its first cut-occurrence position therefore finds the first occurrence in the entire text. Failure to find any such cell is an exact negative substring answer.

## 47.5 Complexity, fast positive queries, and resource semantics

Let $m, n$ be the numbers of pattern and text rules. There are at most $mn$ cells. Each nontrivial cell uses a constant number of local traversals, each visiting $O(n)$ parse pieces and performing exact progression arithmetic. Thus there are $O(mn^2)$ such visits and integer operations, including gcd and modular-inverse calls. This is a polynomial bit-time algorithm: positions and counts have $O(n)$ bits when the text has $n$ binary rules, and gcd and modular inverse are polynomial in that bit length. We do not identify the Python implementation's charged work counter with a unit-cost machine model for these big-integer operations. Space is $O(mn + n)$ integer records, in addition

to the input arena. The arena's node allowance also independently caps the number of live table cells.

Before constructing a full table, the public primitive checks whether the pattern is a prefix using exact compressed LCP. If not, a bottom-up DAG pass finds the earliest occurrence of its first signed letter. If there is no such letter, or insufficient remaining text even at that first position, absence is proved. Otherwise one exact substring LCP tests that candidate. Success is necessarily the first match; failure invokes the complete table. This is an exact fast path with a complete fallback, not a sample of possible occurrences. Its positive benefit can be large when the first candidate begins after an exponentially long block.

Table construction, local traversal, grammar operations and the fast path all share the caller's remaining work and cancellation allowance. Resource exhaustion raises `CompressedLimit`; the group wrapper translates this to an inconclusive group probe. Partial tables never escape as a negative substring result. Global cancellation propagates unchanged. The matcher uses signed letters and exact integer arithmetic, without hashes deciding word equality or substring occurrence.

## 47.6 Integrating whole-donor deletion

The fallback runs only with relator moves enabled, when current total expanded length exceeds `max_letters` and the Whitehead test offers no shortening. This preserves the ordinary compressed descent paths. Nonempty donors are visited in descending length, with stable slot order for ties. For each donor, the search tries its spelling and inverse against distinct target slots at least as long. Per-relator absolute-generator counts supply a necessary-condition filter: a donor cannot fit in one cyclic turn of a target with fewer occurrences of any generator.

For donor $D$ and target $R$, searching $D$ in $RR$ finds every complete occurrence of this particular spelling across a cyclic seam. Since $|D| \le |R|$, an occurrence found in the second copy has an equivalent start in the first; reducing its position modulo $|R|$ is valid. The search does not turn a multiple-wrap match into a legal overlap. If the resulting cyclic target is $DW$, keeping $D$ and replacing $DW$ by $W$ preserves normal closure:

$$\langle\langle D, DW \rangle\rangle = \langle\langle D, W \rangle\rangle.$$

Inversion and cyclic conjugation of a relator also preserve normal closure. The producer checks the entire proposed prefix by an exact LCP before slicing it away and cyclically reducing the remainder. It records the existing `relator` move, with donor rotation zero and overlap $|D|$. The independent literal and compressed verifiers already support these moves and never call the matcher or donor-search helper. A standalone presentation simplification result is not a knot verdict; production still reconstructs the full Wirtinger presentation from the original valid PD.

A useful stalled example is

$$\langle a, b \mid a^N,\ ba^N b^{-1} a \rangle, \qquad N = 2^h.$$

For the tested $h = 10, 100, 500$, the previous capped producer cannot proceed. The new producer finds the donor beginning at target position one. Deletion leaves a cyclic conjugate of $a$, which eliminates $a$ and empties both relations. The two-move trace leaves the free generator $b$, as expected from the presentation itself. No expanded relator is allocated, including for $N = 2^{500}$. This is again an abstract supplied presentation family; it is not asserted to be a hard knot-diagram family.

**Proposition 47.1** (A uniform stalled family resolved without expansion)**.** *For every $N \ge 2$, the presentation above has no eligible generator elimination and no shortening Whitehead cut in its initial state. With relator moves enabled and an explicit overlap allowance below $2N + 3$, the preceding internal producer stalls. Given sufficient compressed work and storage allowances, the new fallback completes in two moves. For the binary-power grammar at $N = 2^h$, both moves and their discovery use polynomial bit complexity in $h$.*

*Proof.* Neither generator occurs once in either relator. The weighted Whitehead graph has an edge between $a$ and $a^{-1}$ of weight $2N-1$, and four weight-one edges joining each of $b, b^{-1}$ to each of $a, a^{-1}$. An $a$–$a^{-1}$ cut has minimum weight $2N+1$, equal to the degree of either endpoint: the large edge is mandatory and each of $b, b^{-1}$ contributes at least one more edge. A $b$–$b^{-1}$ cut has minimum weight two, also equal to the endpoint degree, because the two length-two paths through $a$ and $a^{-1}$ are edge-disjoint. Thus every optimal Whitehead change is zero. The old overlap bridge is unavailable above its cap. The new donor search finds $a^N$ at target position one and leaves the cyclic word $b^{-1}ab$, which reduces to $a$. One generator elimination then removes both relators. The initial grammar has $O(h)$ rules. The first-letter probe finds the exact donor node, and the fixed-rank graph, compressed slices, cyclic reduction and substitution all operate on a polynomial-size grammar. No loop iterates over the $N$ letters. □

## 47.7 Validation and remaining work

The maintained tests compare 7340 exhaustive binary pattern/text pairs, including all text lengths through seven and pattern lengths through four. Three hundred random signed-letter instances check every stored table cell against literal cut occurrences and also check local-window results. A separate 3000-case progression audit checks intersections, including negative starts and singleton cases. Large tests represent an occurrence progression with $2^{60}+1$ elements, force a failed first candidate before an exponential later match, and use an unbalanced 1500-rule grammar. Table, work and global cancellation limits are exercised. Three hundred fifty random relator tuples compare the complete recorded-donor search against literal cyclic scans. Finally, genuine small-unknot traces inflated by a Whitehead power through $2^{100}$ receive a new donor-deletion move and pass independent compressed replay; a smaller exponent also passes literal replay.

The implementation removes an expansion barrier for exact substring queries and enables additional certified search paths. Completing arbitrary cyclic partial overlaps, reusing occurrence tables across changing presentations, and bounding the total number of presentation moves remain separate tasks. In particular, polynomial cost for an occurrence query does not give a polynomial, quasi-polynomial or general subexponential bound for recognition.

## 47.8 Controlled measurements

The driver is `benchmark_compressed_match.py`. Its historical baseline is commit `59054ec761d1`; the full revision is retained in the raw artifact. It measures the historical search, an identical A/A control and the new search in shuffled order, with five measured rounds after one excluded warmup. Full queries include fresh PD validation, recognition and independently reconstructed certificate replay. There are 225 measured queries on the fifteen-case knot corpus and thirty additional adaptive group probes on Gordian at letter caps 4096 and 8192. Every query and probe completes. Full-query certificate digests agree across the three arms. None of these runs reaches the new stalled-state fallback, so the following data check preservation of existing paths and do not demonstrate a knot-corpus speedup.

| Scope | Before | A/A | Matching |
|---|---|---|---|
| Small 14, summed medians | 348.682 | 344.734 | 351.222 |
| Gordian, full query | 1976.739 | 1976.758 | 1985.460 |
| Gordian group, cap 4096 | 3682.891 | 3718.799 | 3706.699 |
| Gordian group, cap 8192 | 1146.458 | 1165.369 | 1158.745 |

Median milliseconds; all knot queries and group probes complete.

Sixty additional measurements construct and simplify the abstract presentation family above. The old search and its control stall for every tested exponent; the new search completes in two

moves in every run. At $h = 500$, median construction plus search time is 7.712 milliseconds, with 1014 allocated nonempty grammar nodes and 17325 charged work units. The earliest-letter probe finds the useful occurrence, so this family does not require the full occurrence table. These are capacity results on supplied presentations, not ratios against completed knot recognition.

| $h$ in $N = 2^h$ | Before/A/A | New (ms) | Moves | Nodes |
|---:|---|---:|---:|---:|
| 8 | stalled | 0.385 | 2 | 30 |
| 32 | stalled | 0.862 | 2 | 78 |
| 100 | stalled | 1.897 | 2 | 214 |
| 500 | stalled | 7.712 | 2 | 1014 |

Full construction and search on abstract presentations; stalls are incomplete searches, not completed recognition times.

A separate 135-query audit tests the complete matcher. For $N = 2^h$ let $P = (ab)^N$. The first-hit text is $cPc$; the later-hit text is $ac\,c^N P$, so its first $a$ is a false candidate; the absent text is $ac(ab)^{N-1}ac$. The expected first positions are respectively 1, $N + 2$ and absent. Each timing includes grammar construction. The full-table arm bypasses the fast path; the adaptive and A/A arms both use the first-letter probe with table fallback. All 135 results complete and agree with the constructed expected answers.

| Substring query | $h$ | Full table | Adaptive | A/A |
|---|---|---:|---:|---:|
| First candidate matches | 8 | 4.779 | 0.055 | 0.053 |
| Later candidate matches | 8 | 5.267 | 5.290 | 5.278 |
| No occurrence | 8 | 7.111 | 7.067 | 7.089 |
| First candidate matches | 32 | 135.163 | 0.127 | 0.100 |
| Later candidate matches | 32 | 167.487 | 163.197 | 160.450 |
| No occurrence | 32 | 247.027 | 246.467 | 244.871 |
| First candidate matches | 64 | 946.014 | 0.216 | 0.203 |
| Later candidate matches | 64 | 1164.627 | 1193.875 | 1174.301 |
| No occurrence | 64 | 3274.404 | 3301.257 | 3236.010 |

Median milliseconds including grammar construction. A/A repeats the adaptive query; both positive and negative results are exact.

At $h = 64$, the first-hit probe takes about 0.216 milliseconds and avoids all table cells; direct table construction takes about 946 milliseconds and stores 2415 cells. The later-hit and absent cases still require full tables: 4626 and 4556 cells respectively. Their adaptive medians are about 1.194 and 3.301 seconds. Thus the implementation supports exact queries on exponential strings, but the polynomial fallback has significant constants. These measurements motivate future table reuse and better string indexes; they do not justify moving this fallback onto already successful small-word search paths.

All runs use twenty million charged work units and 100000 arena nodes/table cells; whole recognition allows 18 seconds, and its group stage 15 seconds. The explicit overlap allowance for the abstract presentations is four letters. The raw artifact `results/compressed_match_20261008.json` retains the inputs or exact construction, execution order, statuses, timings and per-query counters. The integrated suite passes 666 tests with Regina 7.4.1 in 140.313 seconds; its log is `synthesis/data/compressed-match-integrated-tests.txt`.

No new delivery was present in `docs/incoming` during this iteration. The suggested `openai/math` repository remained at `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. The newly integrated research is the exact compressed-pattern-matching method cited above; the fuller longest-common-substring algorithm was the next step at this checkpoint and is implemented in Section 49.

# 48 Quotient deletion and exact uniform-word summaries

## 48.1 From repeated subtraction to a compressed quotient

The occurrence matcher of Section 47 supplies an exact place to delete a whole donor, but one deletion can make very little progress. Consider the supplied presentation

$$\langle a, b \mid a^N,\ a^{N^2+1} \rangle, \qquad N = 2^h.$$

Both relators have binary straight-line programs of size $O(h)$. There is no singleton generator until one exponent becomes one, and the cyclic Whitehead graph has only the edge $\{a, a^{-1}\}$, with weight $N^2 + N + 1$. A cut separating its endpoints has weight equal to the multiplier degree, so it gives no strict shortening. With the explicit overlap bridge disabled, the previous whole-donor fallback subtracts $N$ from $N^2 + 1$ once per move: it needs $N$ deletions followed by an elimination. Compression alone has not removed this exponential number of iterations.

The new move performs the quotient in one step. At the occurrence selected by the existing matcher, rotate the target to a word $R$ beginning with $D$, where $D$ is the donor's recorded spelling or its inverse. Put

$$Q = \left\lfloor \frac{|R|}{|D|} \right\rfloor, \qquad \ell = \mathrm{LCP}(R, D^Q), \qquad q = \left\lfloor \frac{\ell}{|D|} \right\rfloor.$$

Binary powering constructs $D^Q$ without expanding it. Since the occurrence was checked, $q \geq 1$. This is the maximal number of consecutive complete copies at that chosen occurrence, restricted to one turn of the target. It is not the maximum gain over every occurrence or cyclic donor spelling. If $q = 1$ the producer emits the existing `relator` move. Otherwise it emits `relator_power`, replaces $R = D^q W$ by the cyclic reduction of $W$, and retains the donor. An incomplete final copy is retained.

**Proposition 48.1** (Soundness of quotient deletion)**.** *Replacing a target $D^q W$ by $W$ while retaining the donor relation $D$ preserves the normal closure of the relators, for every positive integer $q$. Inverting or cyclically conjugating either relator before applying this operation does not change this conclusion.*

*Proof.* Modulo the relation $D = 1$, the equations $D^q W = 1$ and $W = 1$ are equivalent. More explicitly, $W = D^{-q}(D^q W)$ lies in the normal closure of the old pair, and $D^q W$ lies in the normal closure of the new pair. The two normal closures therefore coincide. Inversion and conjugation preserve the normal closure generated by a word. Free and cyclic reduction preserve its relation as well. □

## 48.2 A versioned certificate with independent replay

Version 4 adds the move fields

| Field | Meaning |
|---|---|
| `kind` | `relator_power` |
| `target`, `donor` | Distinct nonempty relator slots |
| `target_rotation`, `donor_rotation` | Valid cyclic offsets |
| `inverse` | Whether to invert the donor before rotation |
| `overlap` | The entire donor length |
| `copies` | Integer $q \geq 2$ |

The record must have exactly these fields. Booleans are not accepted as integer copy counts, and the count must not exceed the target length divided by the donor length. The producer currently

uses donor offset zero; replay supports all valid donor offsets. The verifier does not require the count to be maximal: any fully checked positive quotient is sound.

The literal verifier compares each letter of the removed target prefix with the corresponding donor letter modulo $|D|$. It does not allocate $q$ explicit copies of the donor. It then normalizes the existing target suffix. The compressed verifier independently constructs a binary power $D^q$ and checks exact equality with the target prefix of length $q|D|$, then cyclically reduces the remaining suffix. Neither verifier calls the producer's occurrence or application helper. A public knot verdict still requires reconstructing the presentation from the original planar diagram and checking the complete trace. Internal success on supplied relators is not by itself a knot-recognition verdict.

Version 4 retains the earlier elimination, elementary Whitehead, ordinary relator and Whitehead-power moves. The latter is valid in versions 3 and 4; older certificate formats remain accepted. New quotient moves require version 4, and unknown versions are rejected. Invalid claims return false; work, time or storage exhaustion propagates to an inconclusive search, not a negative knot verdict.

## 48.3 Why batching also needs a better local query

Initially, batching alone still exhausted the twenty-million-work allowance at $h = 500$. The target prefix and binary donor power describe the same long constant word with different parses. A general compressed equality algorithm can prove that fact, but invoking it repeatedly during LCP search wastes substantial work. The arena now stores an exact summary $u(X)$:

$$u(\epsilon) = 0, \qquad u(x) = x, \qquad u(XY) = \begin{cases} u(X), & u(X) = u(Y) \neq 0, \\ 0, & \text{otherwise.} \end{cases}$$

Here $x$ is a signed generator, and zero denotes either the empty word or a mixed word. Concatenation already removes empty children, so each new binary rule combines two nonempty words. Induction on the grammar proves that $u(X) = x \neq 0$ exactly when $X$ consists entirely of copies of $x$. This costs constant additional work and one stored label per new rule. It is an exact summary, not a hash or a probabilistic fingerprint.

If two nonempty nodes have the same nonzero summary, their LCP is the minimum of their lengths and the optional cap. They are equal precisely when their lengths agree. These queries require constant charged work and create no grammar nodes, even for unrelated parses of words with exponential length. Their integer operations still have the bit cost of the stored lengths. Cap validation and budget checks precede the shortcut. Words with mixed interiors continue through the general deterministic engine; equal endpoints alone do not justify the shortcut.

There is also a separate allocation improvement. Before constructing the inverse of a candidate donor, the search filters its possible targets by distinct slot, length and absolute-generator multiplicities. If no target could contain the donor, neither spelling is searched and no inverse is built. For the family $a^N, ba^Nb^{-1}a$, the largest candidate cannot fit in the shorter relator; previously its unused inverse consumed roughly another copy of the large grammar. The new order succeeds with a 600-node allowance at $N = 2^{500}$. This pruning changes allocation costs, not which exact occurrences are eligible.

## 48.4 A polynomial bound for the pure-power subproblem

**Proposition 48.2** (Restricted Euclidean descent)**.** *Let $m, n \geq 1$ be supplied in binary, and initialize the internal compressed search with the standard binary-power grammars for $\langle a, b \mid a^m, a^n \rangle$. Enable relator moves, disable explicit overlap expansion by setting its allowance to zero, and allow sufficient compressed resources. The search succeeds exactly when* $\gcd(m, n) = 1$*. It takes $O(\log \max(m, n))$*

*quotient or ordinary donor steps, plus at most one elimination, and has polynomial bit complexity in the binary input length. On $m = N, n = N^2 + 1$ with $N \geq 2$, it takes exactly two moves.*

*Proof.* If either exponent is one, eliminating $a$ removes both relators and leaves $b$ free. Otherwise there are no singleton generators, and the Whitehead graph described above has no shortening cut. A larger donor cannot fit the smaller target. The smaller donor matches the larger target at position zero, and the LCP quotient removes exactly $\lfloor \max(m, n)/\min(m, n) \rfloor$ copies. The surviving exponent is the Euclidean remainder. Equal exponents leave one empty relator. Continuing therefore reaches the gcd; gcd one permits elimination, whereas a remaining single power $a^g$ with $g > 1$ has no eligible donor target or shortening move and the search stalls. Stalling does not emit a general nontriviality certificate.

Every two nonterminal Euclidean divisions reduce the larger positive exponent by at least a factor of two. Thus there are $O(H)$ divisions, where $H$ is the binary length of the larger initial exponent. The initial power grammars have height and size $O(H)$. A suffix operation in the implemented grammar rebuilds only boundary paths, preserving or reducing the height: this follows by induction at each retained binary rule. All residual relators are suffixes of earlier positive powers, so their height remains $O(H)$. Each binary quotient power and each slice adds $O(H)$ rules. Positive words have no free or cyclic cancellation, so their reduction does not increase height or allocate a larger normal form. Doubled targets contribute only constantly many further rules per step. Consequently the accumulated arena has $O(H^2)$ rules. The uniform summary answers every needed positive-power equality and LCP query directly; the first occurrence is a prefix, so no general occurrence table is built. Summary traversals, sorting, cuts on this fixed-rank graph and terminal elimination all require polynomial work on this arena. Stored lengths and multiplicities have $O(H)$ bits. Including integer arithmetic gives a polynomial bit bound, rather than merely a bound on move count.

For $(N, N^2 + 1)$ the first quotient is $N$, the remainder is one, and the second move eliminates $a$. The old one-copy search needed $N + 1$ moves. □

This proposition concerns an explicitly specified internal presentation family and search policy. It does not assert that arbitrary knot diagrams reach such a family cheaply. Nor does it bound the number of changing nonuniform donor moves, general grammar growth, or the cost of finding all partial cyclic overlaps. The proof therefore removes one concrete exponential descent example without establishing a general subexponential or quasi-polynomial recognition algorithm.

## 48.5 Validation and controlled measurements

The six new tests include 400 randomly parsed words for the uniform invariant, 300 cyclic donor-copy examples checked against literal deletion, 100 small random Euclidean pairs, a long Fibonacci pair and its noncoprime multiple, and quadratic exponent families through $N = 2^{500}$. Expansion is patched to fail in the compressed search tests. The uniform tests use distinct parses, capped queries, opposite signs, a changed interior letter, large cancellation and exhausted budgets. The allocation test forbids inverse construction before the positive useful match.

A 142-move alternative trace for the genuine Gordian diagram includes a checked two-copy donor move after 132 prior moves. Both public replayers accept it with producer helpers disabled. Mutations of its count, offsets, overlap, fields and version are rejected. This is a legal alternative trace, not evidence that the default search discovers that macro or speeds up this knot. A separate internal trace starts with small supplied relators, uses a Whitehead power to create an exponential word, and checks deletion of $2^{500} + 1$ copies without expansion. It tests compressed replay, not a public knot verdict on an invented presentation.

The integrated suite passes 672 tests with Regina 7.4.1 in 146.146 seconds. The benchmark compares archived word-arena and donor-helper components from commit `10bc9bb754` with the

new components inside a common current recognition and replay harness. Thus it is a component-controlled historical comparison, not a run of the entire old executable. The A/A arm repeats the archived components. Each case has one excluded warmup and five measured rounds, with randomized arm order using seed 2709. All full knot queries include fresh PD construction and independent certificate replay.

| Scope | Before | A/A | Both changes |
|---|---|---|---|
| Small 14, summed medians | 191.178 | 192.193 | 195.016 |
| Gordian, full query | 1950.123 | 1946.408 | 1953.006 |
| Gordian group, cap 4096 | 3640.016 | 3572.292 | 3651.645 |
| Gordian group, cap 8192 | 1135.538 | 1136.436 | 1147.343 |

Median milliseconds; all knot queries and group probes complete.

The small-fourteen sum of medians is 191.178 milliseconds for the archived components, 192.193 for A/A and 195.016 for both changes: about two percent above the baseline. Gordian takes 1.950, 1.946 and 1.953 seconds respectively. The capped probes also remain close, with the new arm slightly slower. No full knot query reaches the compressed donor fallback, and every trace digest is unchanged. These data show a small overhead on existing paths, not a demonstrated knot-corpus speedup.

The audit retains 225 full queries on fifteen knots and thirty adaptive group probes at the two Gordian letter caps. Full-query certificate digests must agree across arms. The kernel audit adds 150 measurements across six supplied-presentation cases and five arms: archived components, their A/A control, quotient/pruning alone, uniform summaries alone, and both changes. Each kernel timing includes construction of the entire initial grammar. The quotient cases use $a^N, a^{N^2+1}$; the allocation case uses $a^N, ba^N b^{-1} a$ with a 600-node cap. The explicit overlap allowance is four letters in these measurements, as distinct from zero in the proposition and its randomized test.

| $h$ | Before | A/A | Quotient | Uniform | Both | Moves | Nodes |
|---|---|---|---|---|---|---|---|
| 4 | 3.161 | 3.125 | 2.269 | 2.419 | 0.248 | 2 | 11 |
| 8 | 331.534 | 335.181 | 12.077 | 57.182 | 0.442 | 2 | 19 |
| 32 | limit/stall | limit/stall | 203.623 | limit/stall | 1.264 | 2 | 67 |
| 100 | limit/stall | limit/stall | limit/stall | limit/stall | 2.909 | 2 | 203 |
| 500 | limit/stall | limit/stall | limit/stall | limit/stall | 14.960 | 2 | 1003 |
| Allocation, 500 | limit/stall | limit/stall | 5.978 | limit/stall | 6.139 | 2 | 512 |

Median milliseconds including grammar construction; Quotient also includes pair pruning. Moves and nodes describe both changes. Limits/stalls are incomplete searches.

At $h = 8$, the archived search takes 257 moves and 331.534 milliseconds, whereas both changes take two moves and 0.442 milliseconds. Quotient/pruning alone takes 12.077 milliseconds; uniform summaries alone still take 257 moves and 57.182 milliseconds. Thus batching removes repeated subtraction, while uniform summaries remove the expensive local comparisons. At $h = 500$, the combined search takes two moves and 14.960 milliseconds, allocates 1003 nonempty grammar nodes and charges 27624 work units. Each isolated arm exhausts the one-million-work cap. These are restricted-presentation results, not a speedup ratio against a completed general recognition algorithm.

In the separate allocation case, pair pruning is sufficient: quotient/pruning alone and the combined arm both finish with 512 nodes, while archived and uniform-only arms exhaust the 600-node allowance before completing a move. The combined median is 6.139 milliseconds. This isolates a capacity benefit from avoiding an unused inverse; it is not evidence of a faster completed baseline query.

Full queries and group probes have twenty million charged work units, with 18 seconds for whole recognition and 15 for the group stage. Kernels have one million work units, with 100000 nodes except in the allocation case. A resource limit or stall is an incomplete search: its elapsed time is never used as a completed-recognition denominator. The JSON artifact `results/relator_power_20261008.json` retains raw orders, timings, statuses, move counts and counters. An earlier aborted benchmark log is retained separately: its harness did not catch the archived module's distinct resource-exception class. The reported JSON comes from a complete rerun after correcting that catch; no aborted samples enter its tables. Test and benchmark logs are in `synthesis/data/relator-power-*`. No incoming report was present during this iteration, and the previously recorded `openai/math` head was unchanged. The quotient and uniform-summary arguments above are local extensions of the implemented compressed search.

# 49 Complete cyclic overlap queries on compressed relators

## 49.1 The local completeness gap being removed

The preceding whole-donor fallback leaves a concrete gap: it can miss an overlap longer than half the donor if that overlap uses only part of the donor or starts at another cyclic position. The new fallback computes a longest common substring of doubled relators, capped at one donor turn. It therefore searches all cyclic rotations and both donor signs, without expanding either word. It uses the existing ordinary relator certificate; no new certificate format or trusted verifier operation is introduced.

The cut-pair and periodic-extension strategy follows Matsubara et al. [33]. Their implementation uses a particular overlap/FM data structure and proves an $O(s^4 \log s)$ arithmetic-operation bound. Our implementation combines the existing deterministic SLP equality engine with a dyadic construction of overlap progressions using the Lifshits occurrence table. These substituted primitives have different costs. We prove a polynomial local bound below; we do not assign the published degree to this implementation. The later refinement in Section 50 adds stronger inexpensive bounds and incremental witness processing; the measurements below describe the initial complete-query checkpoint.

## 49.2 Dyadic construction of suffix-prefix overlaps

Write $X[i : j]$ for a zero-based half-open substring and define

$$\mathrm{OL}(X, Y) = \{k : 1 \le k \le \min(|X|, |Y|), \quad X[|X| - k : |X|] = Y[0 : k]\}.$$

We represent this set as disjoint arithmetic progressions of lengths. For each power of two $l$, consider the band $[l, h]$, where $h = \min(2l - 1, |X|, |Y|)$. An overlap of length in this band must begin with the seed $P = Y[0 : l]$, occurring in $X[|X| - h : |X|]$. The window has length less than $2l$; all seed occurrences share a cut and are represented by an arithmetic progression. Their starts $p$ correspond to candidate overlap lengths $|X| - p$.

A singleton candidate is checked by exact LCP. For a progression $p_i = p_0 + id$ with at least two members, $0 < d < l$. Adjacent overlapping occurrences imply that the seed has period $d$, and their union is one $d$-periodic run. Compute its maximal right endpoint in $X$ and the maximal $d$-periodic prefix length of $Y$:

$$E_X = p_0 + d + \mathrm{LCP}(X[p_0 : |X|], X[p_0 + d : |X|]), \qquad E_Y = d + \mathrm{LCP}(Y, Y[d : |Y|]).$$

Every candidate starts in the same periodic phase as $Y$.

**Lemma 49.1** (Filtering a seed progression)**.** *If $E_X = |X|$, exactly the candidates with $p_i \geq |X| - E_Y$ are suffix-prefix overlaps. If $E_X < |X|$, the only possible overlap candidate is $p_i = E_X - E_Y$; when this value belongs to the progression, an exact LCP query determines whether its entire suffix matches.*

*Proof.* Compare the periodic run beginning at $p_i$ with the prefix of $Y$. They agree until the smaller of the two periodic extents $E_X - p_i$ and $E_Y$. If these extents differ and the shorter string has not ended, the first departure from the periodic word causes a mismatch: the longer side still has the periodic letter. Thus a match extending through a nonperiodic tail of $X$ requires equal departure positions. That equality determines at most one progression member. If $X$ ends while still periodic, the whole suffix matches precisely when its length is no greater than the periodic extent of $Y$. A possible end of $Y$ causes no exception, since every candidate length in the band is at most $|Y|$. □

Clipping and reversing the surviving progression of starts gives a progression of overlap lengths. Distinct bands are disjoint and cover all positive lengths, so the resulting list is exact and has $O(\log \min(|X|, |Y|))$ progressions. Uniform equal-letter words have the direct answer: every positive length up to the shorter word length. Different uniform letters have no overlaps. Neither case enumerates that set.

## 49.3 From overlaps to longest common substrings

Consider grammar rules $U = AB$ and $V = CD$. For each $k \in \mathrm{OL}(A, D)$, let

$$L(k) = \mathrm{LCSuf}(A[0 : |A| - k], C), \qquad R(k) = \mathrm{LCP}(B, D[k : |D|]).$$

Then $k + L(k) + R(k)$ is a common substring length, with start positions $|A| - k - L(k)$ in $U$ and $|C| - L(k)$ in $V$. Inversion reverses and negates both words, so the existing LCP primitive on their inverses computes LCSuf exactly. The alignment $k = 0$ is also valid and uses $\mathrm{LCSuf}(A, C) + \mathrm{LCP}(B, D)$.

**Lemma 49.2** (Coverage by pairs of grammar cuts)**.** *Every common substring of length at least two is covered by one of these alignments for some pair of binary rules, allowing $U$ and $V$ to be interchanged. Single-letter matches can be handled separately.*

*Proof.* In each occurrence's derivation tree, choose the lowest node containing the whole occurrence. Since the occurrence has length at least two, it crosses that node's binary cut. Place the two equal substrings on the same coordinate line. If the first cut is to the right of the second by $k > 0$, the intervening word is both a suffix of $A$ and a prefix of $D$, and is therefore in $\mathrm{OL}(A, D)$. Extending in both directions covers the original common substring. Reverse the roles if the cuts are ordered the other way; coincident cuts give $k = 0$. □

Although the derivation trees may be exponential, each rule has a fixed spelling. We examine rule pairs, not tree-node occurrences. A top-down pass on the grammar DAG records the earliest position of each rule in the root word. Adding that offset to a local witness gives a valid root witness. We examine longer rules first and stop examining rules whose length cannot improve the current best result.

## 49.4 Only six members of a progression need extension

Let $k = a + id$, $0 \leq i < t$, be an overlap progression with $t \geq 2$. The two largest overlaps imply that the common suffix of $A$ and prefix of $D$ of length $a + (t - 1)d$ have period $d$. Let $e_2$ be the maximal $d$-periodic suffix length of $A$ and $e_3$ the maximal $d$-periodic prefix length of $D$. Let $e_1$ be

the extent to which $B$ continues the periodic suffix of $A$, and $e_4$ the extent to which $C$ continues the periodic prefix of $D$ backwards. All four values are obtained with compressed suffix/prefix queries. For example,

$$e_2 = d + \mathrm{LCSuf}(A[0 : |A| - d], A[d : |A|]), \qquad e_3 = d + \mathrm{LCP}(D, D[d : |D|]).$$

For $e_1$, compare $B$ with enough binary-powered copies of the last $d$ letters of $A$. For $e_4$, compare $C$ backwards with enough copies of the first $d$ letters of $D$. Only a logarithmic number of new power rules is required.

Except when periodic departure positions coincide, the extension length is

$$F(k) = k + \min(e_2 - k, e_4) + \min(e_1, e_3 - k).$$

Its two breakpoints are $e_2 - e_4$ and $e_3 - e_1$. Between breakpoints it is affine: first nondecreasing, then constant, then nonincreasing. At a breakpoint, equal departures may match for longer; the true extension must therefore be checked there. It follows that the maximum over the progression is attained among its endpoints and the progression members immediately below and above the two breakpoints, at most six values. We evaluate those candidates by the exact extension formula, not by assuming that the periodic approximation remains valid at a breakpoint. Singleton progressions require one exact extension. This argument permits arbitrary starting residues and does not require the first overlap to be less than the step.

## 49.5 Cyclic search, early bounds and certificate replay

For relators $R, D$, every cyclic substring of length at most $\min(|R|, |D|)$ appears in $RR$ and $DD$. Conversely, any such common substring of the doubled words determines valid cyclic starting positions by reducing its offsets modulo the original lengths. Thus the longest cyclic overlap is the longest linear common substring of the doubled words, capped at one shorter relator turn. Repeat with $D^{-1}$ to cover the inverse sign.

For each unordered pair, it suffices to choose the shorter relation as donor. If an overlap of length $k$ gives gain $2k - |D|$ with the longer donor, it gives at least that gain with the shorter one. Hence this policy is complete for the existence of an overlap satisfying $2k > |D|$ across all distinct pairs and rotations. It returns the first pair and sign with such an overlap, not the globally largest gain.

Absolute-generator counts give the additional bound

$$k \le M(R, D) = \sum_g \min(\#_g R, \#_g D).$$

If $2M \le |D|$, the pair cannot help. Otherwise the LCS query may stop at $M$: no one-turn cyclic overlap can exceed that value. This is a proved upper bound, not a timeout heuristic. A direct prefix LCP sometimes attains it before any overlap table is built. The generic LCS API likewise accepts an explicit cap and returns the smaller of that cap and the exact longest common substring length, together with a witness.

After rotating the relators to $R = PS$ and $D = PQ$, the producer checks the prefix equality and replaces the target by the cyclic reduction of $Q^{-1}S$. The donor is retained. This preserves normal closure, and before further cancellation the target loses $2|P| - |D| > 0$ letters. Existing literal and compressed verifiers independently check the ordinary `relator` move from the original knot presentation. The new matcher is only a search mechanism, never a trusted shortcut in recognition.

The integration retains the fast paths: singleton elimination and Whitehead shortening, the bounded explicit overlap bridge, then whole-donor matching and quotient deletion. Only an oversized state that would otherwise stall reaches the complete cyclic fallback. Every primitive

shares the arena's work allowance and cancellation callback. Node allocation, occurrence table cells and cached overlap entries are bounded; resource exhaustion is inconclusive and must not be mistaken for proof that no overlap exists.

## 49.6 What complexity has actually been established

Let $s$ be the input grammar size and $b$ the largest binary length of a represented word length. There are $O(s^2)$ rule pairs and $O(b)$ dyadic bands per required overlap set. Each band builds one polynomial-sized occurrence table and makes a bounded number of exact compressed queries. Each overlap progression needs at most six extensions. Every substring used here is a slice of an input rule, its inverse, or a bounded binary power of a slice. Slice height does not increase; powering adds $O(b)$ height and rules. The number and size of all intermediate grammars are therefore polynomial in $s$ and $b$. Applying the established polynomial bounds of the occurrence and equality primitives gives polynomial bit complexity for one uncapped-resource LCS or cyclic-overlap query.

This is deliberately a local statement. It does not bound the number of relator moves, the size of intermediate grammars across the entire search, or whether a trivial knot group can always be simplified using strictly shortening moves. The general subexponential and quasi-polynomial recognition targets remain unproved. What has changed is that partial cyclic overlap search is now implemented, rather than merely cited as a possible future primitive.

## 49.7 Validation and measurements

The tests compare overlap sets for all 3844 ordered pairs of nonempty binary words of length at most five, plus 300 signed random pairs with independently randomized parses. Longest-common-substring tests cover 961 ordered binary pairs including the empty word, 300 further signed random pairs and five cap choices per pair, checking both lengths and witness positions. A separate periodic test compares the critical-candidate maximum against every member of the small progression, using independent noise on all four sides. There are 350 randomized cyclic-pair trials checked against all literal rotations and signs, including exact application and strict length decrease.

An exponential test represents $2^{500}$ overlap lengths by one progression and finds a witness in $a^N b$ and $ca^N$ without expansion. Another test starts with $a^N b^2, a^N c^2$: neither a Whitehead shortening nor a whole donor deletion exists, but the new stage replaces the second word by $b^{-2}c^2$. Total expanded relator length drops from $2N + 4$ to $N + 6$. The supplied group remains inconclusive afterward; no knot verdict or recognition-time speedup is inferred from this example. Invalid caps and exhausted work and cache allowances are also checked.

The integrated suite passes 678 tests with Regina 7.4.1 in 163.634 seconds; the targeted six-test run passes in 28.136 seconds.

| Scope | Before | A/A | Complete overlap |
|---|---|---|---|
| Small 14, summed medians | 277.600 | 278.931 | 280.322 |
| Gordian, full query | 3425.051 | 3218.354 | 3264.758 |

Median milliseconds; all full knot queries complete.

All 225 full knot queries complete with identical certificate digests; none reaches the new fallback. The small-fourteen sum of medians is 277.600, 278.931 and 280.322 milliseconds for archived, A/A and new search. Gordian takes 3.425, 3.218 and 3.265 seconds respectively. The variation between the historical controls prevents interpreting the lower Gordian median as a demonstrated speedup. The audit supports preservation of the existing successful paths.

| LCS family | $h$ | Full query | Count bound | A/A |
|---|---|---|---|---|
| Cyclic | 8 | 201.535 | 0.108 | 0.072 |
| Cyclic | 16 | 1969.533 | 0.164 | 0.105 |
| Cyclic | 32 | limit | 0.126 | 0.143 |
| Cyclic | 100 | limit | 0.462 | 0.320 |
| Cyclic | 500 | limit | 0.802 | 0.865 |
| Shifted | 8 | 0.671 | — | 0.675 |
| Shifted | 16 | 1.252 | — | 1.247 |
| Shifted | 32 | 2.359 | — | 2.371 |
| Shifted | 100 | 7.285 | — | 7.241 |
| Shifted | 500 | 37.239 | — | 36.928 |

Median milliseconds including grammar construction. A/A repeats the bounded query for Cyclic and the full query for Shifted. Limits are censored queries.

| Partial move, $h$ | Milliseconds | Nodes | Charged work |
|---|---|---|---|
| 8 | 0.302 | 20 | 246 |
| 32 | 0.615 | 44 | 606 |
| 100 | 1.438 | 112 | 1626 |
| 500 | 6.301 | 512 | 7626 |

Includes grammar construction, failed whole-donor search, complete cyclic search and checked application. This is not a knot verdict.

There are 125 measured LCS queries and twenty separate partial-move runs. At $h = 16$, the full cyclic-family query takes about 1.970 seconds and 1421272 work units; the count-bound query takes 0.164 milliseconds and 75 work units. At $h = 32, 100, 500$, the full query exhausts its two-million-work allowance. The bounded query still completes at $h = 500$ in 0.802 milliseconds, with 509 nonempty grammar nodes and 1527 work units. Submillisecond A/A measurements are visibly noisy, so the stable work counts matter more than precise ratios. Exhausted full queries are not completed-time denominators.

The shifted family tests a successful uncapped query that actually uses periodic extension. At $h = 500$, it takes 37.239 milliseconds versus 36.928 for A/A, with 3003 nodes and 27039 work units. The complete partial move at the same exponent takes 6.301 milliseconds, 512 nodes and 7626 work units, including construction and checked application. These local results do not demonstrate completion of the supplied group search or a speedup on genuine knot recognition.

The audit uses archived search-loop code from `032ca00758` in the common current recognition and replay harness, with an identical A/A arm. There is one excluded warmup and five measured rounds per case, with randomized arm order using seed 2810. Full queries include fresh planar-diagram construction and independent replay. Their certificate digests are checked outside the timer. All kernel timings include grammar construction; exhausted queries remain censored. The cyclic query family compares a full LCS query with the proved shared-count cap on $(a^N b^2)^2, (a^N c^2)^2$, whose exact LCS is $N$. A second family uses $a^N b, ca^N$ and compares two identical uncapped queries. A separate move audit includes failed whole-donor search, complete cyclic search and checked application.

Whole recognition allows 18 seconds, with 15 seconds and twenty million work units for its group stage. Kernels allow two million work units and 100000 nodes/table cells. Raw results are in `results/compressed_lcs_20261008.json`. Test and build logs are in `synthesis/data/compressed-lcs-*`.

No new incoming report was present; the suggested `openai/math` head remained at the previously recorded `fd4aeeb2ee`. This iteration implements and checks the previously identified compressed-LCS research direction.

# 50 Alphabet barriers and incremental optimality checks

## 50.1 The next bottleneck is unnecessary exact work

The complete overlap implementation has a polynomial local bound, but some compressed substring queries still exhaust millions of work units on simple repeated words. For example, the longest common substring of $(a^N b^2)^2$ and $(a^N c^2)^2$ has length $N$. A total multiplicity bound on these doubled strings permits $2N$, so it does not prove that the first match of length $N$ is optimal. Continuing with exact occurrence tables can be much more expensive than discovering the match itself.

There is a stronger inexpensive obstruction: a common substring cannot cross a signed letter that is absent from the other word. Here $b$ and $c$ are separators, and the longest possible common-alphabet run has length $N$. The new implementation computes this bound on the grammar DAG and uses it at both the root and individual grammar-rule pairs. It also processes witnesses incrementally. Once a verified witness attains the relevant upper bound, all remaining candidates for that scope are unnecessary. This is exact optimality detection, not speculative termination.

## 50.2 An exact summary of allowed-letter runs

Let $C$ be the intersection of the two words' signed alphabets. For a nonempty grammar word $W$, store three lengths: $p(W)$, its longest prefix over $C$; $q(W)$, its longest suffix over $C$; and $r(W)$, its longest contiguous substring over $C$. A terminal in $C$ has summary $(1, 1, 1)$, and any other terminal has $(0, 0, 0)$. For a binary rule $W = UV$,

$$\begin{aligned} p(W) &= p(U) + \mathbf{1}_{p(U)=|U|}\, p(V), \\ q(W) &= q(V) + \mathbf{1}_{q(V)=|V|}\, q(U), \\ r(W) &= \max\{r(U), r(V), q(U) + p(V)\}. \end{aligned}$$

The third formula includes the only possible run crossing the rule's cut. All quantities are computed bottom up without expanding any word.

**Proposition 50.1** (Shared-alphabet upper bound)**.** *For any two reachable grammar words $U, V$ in the query,*

$$\mathrm{LCS}(U, V) \le \min(r(U), r(V)),$$

*where the summaries use the intersection $C$ of the original root alphabets. The summaries take linear arithmetic work and storage in the reachable grammar size, with the bit cost of stored lengths.*

*Proof.* Induction on the grammar gives the stated prefix, suffix and maximum-run meanings: a run either lies in one child or crosses the cut as a suffix followed by a prefix. Every letter in a common substring of $U, V$ occurs in both root words, so it lies in $C$. The substring is therefore contained in an allowed-letter run on each side. The recurrence uses a constant number of arithmetic operations per rule. □

The intersection is signed: a letter and its inverse are different string symbols. Terminal node identifiers are unique for each signed letter in one arena, so intersecting the reachable terminal sets computes $C$ exactly. If one root's entire alphabet is shared, every substring of that root is over $C$; the implementation uses ordinary node lengths on that side and skips its summary pass. Disjoint alphabets give an immediate zero answer. Bounds are local to the query, since changing the other word can change $C$; they are not permanent word-arena metadata.

This bound can be stronger than shared total counts. Counts constrain how many matching letters exist, while the run bound also constrains their contiguity. Neither changes the equality test or accepts a witness that has not been checked.

## 50.3 Using progress as soon as it is proved

The complete search initially tries the existing exact prefix query. If that does not finish the query, it builds root offsets and alphabet sets, computes any needed run summaries, and reduces its global upper bound to

$$U = \min(\text{requested cap}, r(X), r(Y)),$$

with the requested cap omitted when absent. An already found prefix or single-letter witness of length $U$ settles the query immediately.

Binary rules are examined in descending run-bound order. A rule with bound no greater than the best known common substring length cannot improve the answer, nor can any subsequent rule with a smaller bound. For a selected pair $A, B$, its local upper bound is $U_{A,B} = \min(r(A), r(B), U)$. The matcher now yields the zero-offset extension immediately, followed by the critical progression extensions as they are computed. After every witness it updates the best answer. Reaching $U$ finishes the whole query; reaching $U_{A,B}$ stops that pair and continues only where a larger answer is still possible.

Previously the matcher collected all extension witnesses for a pair before considering any of them. Even a first witness attaining a known upper bound could be followed by expensive occurrence tables and further extensions. The new loop does not request those computations. A local stop is never treated as global completion unless the global bound is also attained. The original exact fallback remains available whenever the inexpensive bounds are loose, including queries with identical signed alphabets but different letter order.

## 50.4 Restricted families with inexpensive exact completion

The first family is $(a^N b^2)^2, (a^N c^2)^2$, with $N = 2^h$. The common alphabet is $\{a\}$, the maximum allowed run is $N$ on both sides, and the initial prefix witness has length $N$. No cut-pair search is needed.

A second family separates discovery from stopping. Represent $ba^N b$ and $ca^N c$ with root cuts in the middle of the long run:

$$(ba^{N/2})(a^{N/2}b), \qquad (ca^{N/2})(a^{N/2}c).$$

There is no common prefix. The first aligned-cut extension matches $N/2$ letters backwards and $N/2$ forwards, reaching the run bound $N$. Thus the new loop finishes after one extension, without requesting the suffix-prefix overlap sets. Computing bounds while retaining eager witness collection does not obtain this same early stop.

On these particular binary-power parses, construction, offsets, summaries and the required prefix/suffix comparisons use $O(h \log h)$ traversal and arithmetic operations in the constant-time-dictionary model, on $O(h)$-bit lengths. This includes grammar traversal and sorting. Allowing also a conservative linear worst-case dictionary lookup overhead gives a polynomial bit bound of $O(h^3 \log(h+1))$. This statement concerns these explicit families and parse shapes; the general LCS primitive retains its polynomial fallback with larger costs.

There is also a direct relator-search example with a loose total-count bound. Let $M = N/4$ and use

$$D = a^N b^2 a^M b^2, \qquad R = a^N c^2 a^M c^2.$$

The shared count is $N + M$, but the maximum shared-alphabet run, including cyclic seams, is $N$. The checked prefix attains it. For $N > M + 4$, the overlap gives positive raw gain $N - M - 4$ and replaces the target by

$$b^{-2} a^{-M} b^{-2} c^2 a^M c^2.$$

The total relator length drops from $2N + 2M + 8$ to $N + 3M + 12$. The earlier whole-donor stage cannot match either relation because each has a generator absent from the other. The new bound therefore accelerates an actual partial-move query whose count cap alone is insufficient. This is a supplied-presentation operation, not a completed knot verdict.

### 50.5 Validation and controlled ablation

Four new tests check run bounds against literal expansion at every reachable node of 400 randomly parsed signed words; exponential uncapped queries through $N = 2^{500}$ with occurrence tables forbidden; an interior witness with further overlaps forbidden; local-pair stopping against a literal LCS oracle; a same-alphabet negative control; exhausted work; and the large partial-move family above without expansion. The existing exhaustive and randomized LCS, cyclic-overlap and witness tests are retained.

All 682 tests pass with Regina 7.4.1 in 182.222 seconds. The preceding ten-test targeted run passed in 27.990 seconds; the integrated run also checks the interior example with both root cuts inside the long run.

| Scope | Before | A/A | Both changes |
|---|---|---|---|
| Small 14, summed medians | 280.857 | 280.086 | 282.904 |
| Gordian, full query | 1952.276 | 1939.691 | 1940.255 |

Median milliseconds; all whole-knot queries complete.

All 225 whole-knot queries retain identical certificate digests, and none reaches the changed matcher. The small-fourteen sum of medians is 280.857, 280.086 and 282.904 milliseconds for baseline, A/A and both changes. Gordian takes 1.952, 1.940 and 1.940 seconds respectively. These data check preservation of existing paths; they do not demonstrate a knot-corpus speedup.

| LCS family | $h$ | Before | A/A | Bounds | Incr. | Both |
|---|---|---|---|---|---|---|
| Repeated | 8 | 109.458 | 106.942 | 0.135 | 107.604 | 0.142 |
| Repeated | 32 | limit | limit | 0.245 | limit | 0.319 |
| Repeated | 500 | limit | limit | 3.078 | limit | 3.028 |
| Interior | 8 | 60.409 | 59.386 | 58.950 | 59.761 | 0.177 |
| Interior | 32 | limit | limit | limit | limit | 0.414 |
| Interior | 500 | limit | limit | limit | limit | 4.912 |
| Same alphabet | 8 | 1.124 | 1.116 | 1.111 | 1.127 | 1.117 |
| Same alphabet | 32 | 8.550 | 8.543 | 8.647 | 8.598 | 8.595 |
| Same alphabet | 500 | limit | limit | limit | limit | limit |

Median milliseconds including construction. Bounds retains eager witness collection; Incr. retains ordinary length bounds. Limits are censored queries.

| Partial move, $h$ | Before | A/A | Both | Nodes | Work |
|---|---|---|---|---|---|
| 8 | 287.444 | 286.253 | 0.409 | 35 | 527 |
| 32 | limit | limit | 0.875 | 83 | 1439 |
| 500 | limit | limit | 8.418 | 1019 | 19223 |

Median milliseconds including construction, failed whole-donor search, cyclic query and checked application. Nodes/work describe both changes.

The kernel audit contains 225 uncapped LCS measurements. At $h = 8$, the repeated family falls from 109.458 milliseconds to 0.142 milliseconds. Bounds alone obtain this improvement; incremental witnesses alone do not. At $h = 500$, both changes complete in 3.028 milliseconds,

with 509 nonempty nodes and 5562 work units, while the original query exhausts its two-million-work allowance.

The interior family separates the two effects. At $h = 8$, archived search takes 60.409 milliseconds, bounds alone 58.950, incremental witnesses alone 59.761, and both changes 0.177. At $h = 500$, both changes finish in 4.912 milliseconds with 1012 nodes and 8562 work units; every isolated or archived arm exhausts the cap. A bound does not save the later exact work unless the checked witness is considered before that work is requested.

The same-alphabet control completes with identical work counts through $h = 32$: 16187 units and approximately 8.6 milliseconds in each arm. At $h = 500$, every arm still exhausts two million work units. Thus the new bound leaves a measurable class of expensive queries unresolved at this checkpoint. Section 51 adds directed-pair constraints that resolve this family, while retaining a stronger negative control.

Forty-five further measurements include the complete partial-move operation. At $h = 8$, construction plus search and application falls from 287.444 to 0.409 milliseconds. At $h = 500$, the combined path completes in 8.418 milliseconds with 1019 nodes and 19223 work units; both archived arms hit the cap. These are exact local operations on supplied relators, not completed recognition of those groups. No speedup ratio is assigned to an exhausted baseline.

The audit compares the archived matcher from `e678ef097` against two isolated components and their combination. The bounds-only arm uses run summaries and bound-based ordering but eagerly collects pair witnesses. The incremental-only arm uses the new witness loop with the previous ordinary-length bounds. An A/A arm repeats the archived class. These classes run in the common current word arena, recognition and replay harness, so this is a controlled component comparison.

One warmup is excluded before five measured rounds per case; arm order is shuffled with seed 2910. Full knot timings include fresh planar-diagram construction, recognition and independent replay; certificate digests are checked outside the timer. Kernel timings include grammar construction and uncapped exact LCS. A same-alphabet family, $(ab)^N ac, (ab)^N ca$, has exact LCS length $2N$ and provides a control where run summaries are skipped. The partial-move timings additionally include the unsuccessful whole-donor query, cyclic search and checked application. Resource exhaustion is censored, never a negative substring answer or a completed-time denominator.

Whole recognition allows 18 seconds, with 15 seconds and twenty million work units for the group stage. Kernels allow two million work units and 100000 nodes/table cells. Raw results are in `results/lcs_bounds_20261008.json`; logs are in `synthesis/data/lcs-bounds-*`. No new incoming report was present, and the suggested `openai/math` head was unchanged at the previously recorded `fd4aeeb2ee`. These are local refinements of the complete compressed-overlap implementation. They do not settle global move count, grammar growth or completeness, so the general subexponential recognition bound remains unproved.

# 51 Directed transition bounds for compressed overlap search

## 51.1 Equal alphabets can still have different local order

The same-alphabet control of Section 50, $(ab)^N ac, (ab)^N ca$, exhausted the work allowance at $N = 2^{500}$. An alphabet-only bound cannot distinguish these words, but their adjacent letter pairs differ. Any common substring must use pairs that occur in both words. The new stage applies this stronger necessary condition after the alphabet-run stage has failed to finish the query.

It is also necessary to improve the preceding whole-donor stage. A fast partial-overlap query is of little use if an impossible full-donor query has already consumed its budget. We therefore use directed pair sets to reject impossible whole-donor orientations before building inverse words

or invoking the occurrence matcher. Both uses are exact pruning; the existing complete string algorithms remain available when the conditions are insufficient.

## 51.2 Exact directed pairs from grammar cuts

For a word $W$, let $E(W)$ be its set of ordered adjacent signed-letter pairs. This is a linear-word definition; no cyclic closing edge is added unless explicitly requested by a caller. For each reachable binary rule $U = AB$, record the pair $(\text{last}(A), \text{first}(B))$.

**Lemma 51.1** (Cut-pair extraction)**.** *The union of these pairs over reachable binary rules is exactly $E(W)$. It has at most as many entries as there are reachable binary rules and can be collected without expanding $W$.*

*Proof.* A rule's cut contributes an actual adjacent pair whenever that rule occurs in the derivation tree. Conversely, any adjacent positions in the expanded word have a lowest common ancestor in that tree. They lie on opposite sides of its cut, so their pair is contributed by the corresponding grammar rule. Repeated occurrences of a rule contribute the same pair and need not be visited separately. □

Let $C$ be the shared signed alphabet and let $E = E(X) \cap E(Y)$ for the two query roots. A common substring is a word over $C$ whose every adjacent pair lies in $E$. It is a walk in the directed graph of allowed transitions, although the implementation only needs a set of pairs and three lengths per grammar rule. Signs and order are retained; inverse words transform an original pair $(u, v)$ into $(-v, -u)$.

## 51.3 Longest runs respecting the common transitions

As before, summarize each grammar word by its longest valid prefix $p$, suffix $q$ and contiguous run $r$. A terminal in $C$ has all three values one, and any other terminal has all three zero. At a binary rule $W = UV$, let $\delta = 1$ when $(\text{last}(U), \text{first}(V)) \in E$, and zero otherwise. The recurrence is

$$\begin{aligned} p(W) &= p(U) + \delta\, \mathbf{1}_{p(U)=|U|}\, p(V), \\ q(W) &= q(V) + \delta\, \mathbf{1}_{q(V)=|V|}\, q(U), \\ r(W) &= \max\{r(U), r(V), \delta\,(q(U) + p(V))\}. \end{aligned}$$

A run crossing the cut must have an allowed transition there. If either side contributes no valid letters, the child maxima already cover the remaining run. Induction therefore gives exact run lengths, and

$$\text{LCS}(U, V) \le \min(r(U), r(V))$$

for any pair of reachable words in the query. The summaries use the shared root constraints; they remain upper bounds for every smaller rule pair.

The implementation first uses prefix and alphabet-run checks. Only when those fail to finish does it collect directed pairs. A side whose entire pair set is shared needs no new transition-summary pass; its existing alphabet bound remains valid. Otherwise the stronger summaries replace its bounds. The existing incremental witness loop then uses the improved global and local bounds, returning only after an exact witness reaches one of them. This preserves the complete fallback and its polynomial local bound. The extra extraction and summary work is linear in reachable rules in the usual arithmetic/dictionary model; integer and dictionary costs remain polynomial in the encoded input size.

For $X = (ab)^N ac$ and $Y = (ab)^N ca$, the shared pairs are $ab$ and $ba$. The maximum valid run has length $2N + 1$ in $X$ and $2N$ in $Y$. The already checked prefix of length $2N$ therefore

proves the exact answer without any occurrence table. There is also an interior-match family $c(ab)^N b, b(ab)^N c$. Cutting both grammars halfway through their long periodic blocks lets the first aligned-cut extension find $2N$ letters. The transition bound is $2N$, so no further extensions are needed. On the specified binary-power parses, these tests use polynomial work in $\log N$ and no expansion.

### 51.4 Filtering whole donors before matching

An occurrence of a donor spelling $D$ inside a cyclic target $R$ requires

$$E(D) \subseteq E(R) \cup \{(\mathrm{last}(R), \mathrm{first}(R))\}.$$

The target's closing pair is essential: a genuine match may cross its seam. For an inverse donor, transform every internal source pair as described above. The test is only necessary; passing it still requires the exact compressed occurrence query. A one-letter donor has no internal pairs and is not rejected by this condition.

After the existing length and absolute-count filters, the implementation collects each needed root's pairs lazily and caches them for that call. It rejects incompatible orientations before allocating an inverse donor. The total cached entries are limited by the arena's node allowance; exhaustion raises the existing resource exception and remains inconclusive. All traversals and candidate checks consume the shared work allowance.

For the cyclic relators

$$D = (ab)^N ac, \qquad R = (ab)^N ca,$$

the absolute counts agree. However $ac$ is absent from the target's cyclic pair set, and $bc$ is absent from the donor's cyclic pair set. The reversed negative pairs cannot occur in either positive word. Thus every whole-donor orientation is rejected without an occurrence table or inverse construction. The complete cyclic stage then finds the common prefix $(ab)^N$ using transition bounds on the doubled words. The ordinary relator move replaces $R$ by $c^{-1}a^{-1}ca$, preserving the donor and normal closure. Total expanded relator length falls from $4N + 4$ to $2N + 6$.

This measures a local search-and-application operation on supplied relators. It does not establish that the optional group search reaches this state before its other moves, nor does it certify an unknot from this presentation. The public recognizer still reconstructs and independently checks a complete trace from the original planar diagram.

### 51.5 A control that the new condition does not resolve

Let $T =$ `aabbccacbabc`. Its internal pairs include all nine ordered pairs over $\{a, b, c\}$. The words

$$X = (ab)^N cT, \qquad Y = (ab)^N aT$$

therefore have identical alphabets and identical pair sets. Their exact LCS is $2N$ for $N \geq 16$, but neither cheap bound proves this. To see the exact value, the common prefix gives the lower bound. A common substring longer than $2N$ would have to start within the first thirteen positions of each word, since each tail has length thirteen. Unequal starting parities mismatch immediately. With equal parity, if $X$ reaches its tail first, its $c$ disagrees with the other alternating prefix. If $Y$ reaches its tail first, its first $a$ can agree but its next $a$ disagrees with the following $b$. Equal starts disagree at $c$ versus $a$. None yields a match longer than $2N$.

The complete fallback is still required on this control. At this checkpoint, its retained resource limits demonstrate that a stronger necessary condition does not solve general compressed substring search by itself. More importantly, none of these local bounds proves a bound on total relator moves, grammar growth or recognition completeness. The general subexponential target remains

unresolved. Section 52 subsequently resolves this particular resource limit by restricting the fallback's occurrence tables to their query windows.

### 51.6 Validation and controlled measurements

Four new tests compare directed pair sets and transition-run summaries with every reachable literal word in 400 random signed grammars. They check same-alphabet queries through $N = 2^{500}$ with occurrence tables forbidden, an interior witness with later overlaps forbidden, an equal-pair control requiring the exact fallback, cyclic-seam matches, the sign/order change under inversion, impossible orientations with matching and inverse allocation forbidden, an independent cache limit, exhausted work, and the complete local partial-move operation without expansion. The integrated suite passes 686 tests with Regina 7.4.1 in 155.970 seconds; the fourteen-test targeted suite passes in 20.006 seconds. A fifth new test forbids adjacency traversal when uniform metadata already determines the pair set, including one-letter and inverse-only matches. After that shortcut, the final integrated suite passes 687 tests in 162.201 seconds.

| Scope | Before | A/A | Transitions |
|---|---|---|---|
| Small 14, summed medians | 197.913 | 196.984 | 199.021 |
| Gordian, full query | 1947.100 | 1951.987 | 1941.596 |

Median milliseconds; all full knot queries complete.

All 225 full knot queries complete with identical certificate digests, and none reaches the changed fallback. The small-fourteen sum of medians is 197.913 milliseconds before, 196.984 for A/A and 199.021 after. Gordian medians are 1.947, 1.952 and 1.942 seconds. These data check existing paths; they do not demonstrate a knot-corpus speedup.

| LCS family | $h$ | Before | A/A | Transitions |
|---|---|---|---|---|
| Prefix | 8 | 1.089 | 1.099 | 0.122 |
| Prefix | 32 | 8.395 | 8.376 | 0.283 |
| Prefix | 500 | limit | limit | 3.612 |
| Interior | 8 | 85.855 | 87.077 | 0.197 |
| Interior | 32 | limit | limit | 0.463 |
| Interior | 500 | limit | limit | 5.120 |
| Equal bigrams | 8 | 7.066 | 7.085 | 7.235 |
| Equal bigrams | 32 | 71.284 | 71.929 | 71.226 |
| Equal bigrams | 500 | limit | limit | limit |

Median milliseconds including construction; limits are censored.

| Operation | $h$ | Before | A/A | Matcher | Pruning | Both |
|---|---|---|---|---|---|---|
| Partial | 8 | 176.782 | 174.430 | 12.035 | 165.455 | 0.356 |
| Partial | 32 | limit | limit | 285.692 | limit | 0.762 |
| Partial | 500 | limit | limit | limit | limit | 7.078 |
| Quotient | 8 | 0.330 | 0.325 | — | — | 0.363 |
| Quotient | 32 | 0.909 | 0.930 | — | — | 1.025 |
| Quotient | 500 | 14.130 | 14.105 | — | — | 15.340 |

Median milliseconds including construction. Both denotes the initial filter before the uniform-word shortcut. Partial operations do not produce knot verdicts.

The primary audit contains 135 substring queries, 75 local partial-move operations and 45 successful abstract quotient searches. At $h = 500$, the prefix family finishes in 3.612 milliseconds with 508 nodes and 6576 work units; the interior family takes 5.120 milliseconds with 1014 nodes and 9596

work units. Both archived queries exhaust the two-million-work cap. On the equal-bigram control the new stage adds little charged work but cannot improve the bound: all arms still exhaust the cap at $h = 500$.

For the full partial-move operation, at $h = 8$ the baseline takes 176.782 milliseconds, the new matcher alone 12.035, pruning alone 165.455 and both 0.356. At $h = 500$, only both changes complete: 7.078 milliseconds, 514 nodes and 15775 work units. Both isolated components exhaust the cap. This demonstrates why progress in a fallback must be accompanied by a bounded route through the preceding stages on this family.

The successful quotient control exposes a regression in the initial filter. At $h = 500$, its median rises from 14.130 milliseconds (14.105 A/A) to 15.340, and work increases from 27624 to 32135. The pair sets of these uniform words were being recomputed by grammar traversal even though the arena already stores their exact signed letter. The final implementation uses that metadata: a one-letter word has no internal pair, and a longer uniform word with letter $a$ has exactly the pair $(a, a)$. This takes constant work and avoids the redundant traversal. Mixed words retain the normal cut-based extraction, and cache accounting is unchanged.

| Operation | $h$ | Before | A/A | Filter | Shortcut |
|---|---|---|---|---|---|
| Quotient | 8 | 0.337 | 0.336 | 0.365 | 0.331 |
| Quotient | 32 | 0.950 | 0.952 | 1.062 | 0.969 |
| Quotient | 500 | 13.601 | 13.549 | 15.112 | 13.545 |
| Partial | 500 | — | 7.329 | 7.445 | 7.346 |

Median milliseconds over 21 measured rounds. A/A repeats Before for Quotient and Filter for Partial. All operations complete with identical traces.

At $h = 500$, the follow-up medians are 13.601 milliseconds before, 13.549 for its A/A control, 15.112 for the initial filter and 13.545 for the final shortcut. The extra charged work falls from 4511 units to six; the node count remains 1003. The nonuniform partial-operation medians are 7.445 milliseconds for the initial filter, 7.329 for its A/A control and 7.346 for the final code, all with the same trace, 514 nodes and 15775 work units. Thus reuse of exact uniform metadata removes the measured quotient regression without changing the mixed-word filtering behavior.

The follow-up compares the original donor helper, the initial filtered helper archived at `3e9afde43`, and the final uniform shortcut in the same current kernel and search harness. It uses 21 measured rounds after one warmup, with shuffled arm order and seed 3011. There are 252 quotient measurements and 63 partial-operation controls. The partial controls repeat the initial filter as A/A and verify that the shortcut leaves nonuniform traces unchanged. Every follow-up operation completes, and traces agree across its arms. Raw follow-up samples are in `results/donor_pair_shortcut_20261008.json`.

The baseline uses the matcher class and whole-donor helper archived at `87e9585ec`, inside the common current word arena, recognition and replay harness. A/A repeats those components. Local partial-move tests also isolate the new matcher with the old whole-donor stage and the new whole-donor filter with the old matcher. A successful pure-power quotient family is retained to expose any overhead added to an already successful whole-donor path.

Every case has one excluded warmup and five measured rounds with shuffled arm order, using seed 3010. Full knot timings include fresh PD construction, recognition and independent certificate replay; certificate digests are checked outside the timer. Kernel and operation timings include grammar construction. Resource-limited runs are censored, not negative answers or completed-time denominators. Whole recognition allows 18 seconds, with 15 seconds and twenty million work units in its group stage. Local queries allow two million work units and 100000 nodes/table cells. Raw results are in `results/lcs_transitions_20261008.json`, with logs in `synthesis/data/lcs-transitions-*`. There was no new incoming report, and the suggested `openai/math` head

remained at the previously recorded `fd4aeeb2ee`.

# 52 Local occurrence tables for overlap windows

## 52.1 A short query inside a long compressed word

The equal-bigram control in Section 51 exposes a different source of unnecessary work. For

$$T = \texttt{aabbccacbabc}, \qquad X = (ab)^N cT, \quad Y = (ab)^N aT,$$

the exact longest common substring has length $2N$, but shared alphabets and directed pairs do not prove optimality. The fallback must examine grammar cuts. Some of its overlap queries compare a very long child against a short tail. The previous implementation built an occurrence table over the entire long child even though the query used only a short suffix of it. A profile at $N = 2^{500}$ places almost all charged work in these table constructions and their local occurrence queries. Profiling times are diagnostic only and are excluded from the timing comparison.

The correction is to localize the table's text operand as well as the query interval. This retains the exact algorithm and its complete fallback; it changes the grammar presented to each occurrence table.

## 52.2 Two exact restrictions and one coordinate translation

Write $\mathrm{OL}(U, V)$ for the positive lengths $k$ for which the suffix of $U$ of length $k$ equals the prefix of $V$ of length $k$. Set $m = \min(|U|, |V|)$. Then

$$\mathrm{OL}(U, V) = \mathrm{OL}(U[|U| - m : |U|], V[0 : m]).$$

Every feasible overlap is at most $m$, so its two compared substrings are unchanged by these restrictions. The implementation first replaces the operands by these equal-length words. Cache keys retain the original root identities, while cached values remain overlap lengths and require no translation. If the restricted words are uniform, the existing exact uniform-word shortcut now applies even when the original roots were mixed.

For a dyadic band $\ell \le k \le h$ with $h = \min(2\ell - 1, m)$, let $P = V[0 : \ell]$ and $Z = U[m - h : m]$ in the restricted coordinates. A candidate overlap of length $k$ has its seed starting at $m - k$ in $U$, or at $h - k$ in $Z$. Its entire seed lies in $Z$:

$$0 \le h - k \le h - \ell, \qquad (h - k) + \ell \le h.$$

Consequently the occurrences of $P$ in $Z$ are in bijection with precisely the seed occurrences needed for this band. No occurrence from outside the suffix window can contribute an overlap in the band.

The matcher therefore builds its arithmetic-progression table for $(P, Z)$ and queries the interval $[0, h]$. For every returned progression of starts $(p, d, t)$ it uses

$$(p + m - h, d, t)$$

in the existing periodic-extension test. Only the first start changes; step and multiplicity do not. The extension test still operates on the equal-length restricted operands and returns lengths $m - p'$. Thus both singleton verification and periodic-run departure comparisons have exactly the same coordinate meaning as before. The dyadic bands partition positive overlap lengths, so the resulting progressions remain disjoint and complete.

## 52.3 Cost and limits of the improvement

Occurrence-table construction now depends on the reachable grammars of the prefix seed and suffix window, rather than on all rules reachable from the original text. Slices reuse complete interior subtrees and create only boundary nodes. With grammar height $H$, the two initial restrictions and $O(\log m)$ dyadic windows create at most $O(H \log m)$ such nodes before accounting for reuse. This remains a polynomial local operation on a binary straight-line program. The usual word-operation statement must still be supplemented by polynomial costs for encoded integers.

There is no universal claim that the new reachable grammar is smaller: boundary nodes and repeated slices can add work, and the largest band can contain the whole text. The full-overlap control below measures that case. All slices, tables, extension checks and cached progressions retain the shared work and memory limits. A resource exception is still an inconclusive search, never a negative knot verdict.

On the specified equal-bigram binary-power family, the difficult overlap queries concern only constant-length tails. The new tables therefore have constant size even as the long prefix grows. Root-bound preparation and grammar navigation still depend on $\log N$. This family-specific improvement does not prove a bound on the number of relator moves, intermediate grammar size, or completeness of the group search. In particular it supplies neither a general subexponential recognizer nor a quasi-polynomial recognition theorem.

## 52.4 Validation and controlled measurements

The existing literal oracles cover 3844 binary overlap pairs, random signed parses, exact substring witnesses, periodic critical alignments, and complete cyclic overlap searches. Three additional tests check asymmetric periodic operands and empty boundaries, caller-root cache reuse, uniform overlaps surrounded by exponentially long irrelevant context, and the exact equal-bigram answer at $N = 2^{500}$ under a 20000-work allowance with expansion forbidden. The complete suite passes 690 tests with Regina 7.4.1 in 153.084 seconds.

| Scope | Before | A/A | Windows |
|---|---|---|---|
| Small 14, summed medians | 215.475 | 215.870 | 215.012 |
| Gordian, full query | 1906.227 | 1982.828 | 1917.508 |

Median milliseconds; all full knot queries complete.

| Family | $h$ | Before | A/A | Windows |
|---|---|---|---|---|
| Prefix | 8 | 0.118 | 0.110 | 0.110 |
| Prefix | 32 | 0.265 | 0.274 | 0.274 |
| Prefix | 500 | 3.376 | 3.385 | 3.376 |
| Interior | 8 | 0.161 | 0.159 | 0.159 |
| Interior | 32 | 0.405 | 0.397 | 0.405 |
| Interior | 500 | 4.944 | 4.946 | 4.955 |
| Equal pairs | 8 | 7.249 | 7.053 | 2.174 |
| Equal pairs | 32 | 70.293 | 70.937 | 2.487 |
| Equal pairs | 500 | limit | limit | 7.893 |
| Full overlap | 8 | 33.549 | 33.367 | 29.741 |
| Full overlap | 32 | limit | limit | limit |
| Full overlap | 500 | limit | limit | limit |

Median milliseconds including construction; limits are censored. Full overlap enumerates all lengths in compressed progressions, not all occurrences.

| Family | $h$ | Before | A/A | Windows |
|---|---|---|---|---|
| Partial | 8 | 0.319 | 0.322 | 0.297 |
| Partial | 32 | 0.650 | 0.637 | 0.630 |
| Partial | 500 | 6.775 | 6.915 | 6.936 |
| Equal-pair cyclic | 8 | 0.514 | 0.496 | 0.501 |
| Equal-pair cyclic | 32 | 0.805 | 0.808 | 0.804 |
| Equal-pair cyclic | 500 | 6.664 | 6.586 | 6.629 |
| Quotient | 8 | 0.333 | 0.339 | 0.333 |
| Quotient | 32 | 0.983 | 0.953 | 0.942 |
| Quotient | 500 | 13.486 | 13.291 | 13.532 |

Median milliseconds including construction; limits are censored. Local operations do not produce knot verdicts.

For the equal-pair family, medians at $h = 8$ fall from 7.249 milliseconds (7.053 A/A) to 2.174; at $h = 32$ they fall from 70.293 (70.937 A/A) to 2.487. At $h = 500$ both archived controls exhaust the work allowance, whereas localization finishes in 7.893 milliseconds with 1034 arena nodes and 12609 charged work units. All three localized sizes use eight window queries and 162 cumulative table cells. These are actual completed exact LCS queries, not merely longer partial searches.

The full-overlap control compares $(ab)^N$ with itself and asks for all positive overlap lengths, represented by arithmetic progressions. At $h = 8$, localization takes 29.741 milliseconds versus 33.549 before and 33.367 A/A, with work falling from 42506 to 37641 and nodes rising from 103 to 108. At $h = 32$ and $h = 500$, every arm still exhausts the allowance. The localized $h = 500$ run spends 2.639 seconds before exhaustion versus 1.177 before, and creates 2120 nodes versus 503. This is a material limit of localization: it reaches expensive equality/periodic-extension work that the archived table construction never reaches at the same charged budget. The changed mix of operations has a higher wall-clock cost per charged unit. Neither run completes, so their times do not measure a completed-query slowdown or speedup. Future work must address that remaining bottleneck rather than extrapolate from short-tail queries. Section 53 subsequently supplies a separate certified short-period algorithm for this control; the general fallback remains needed.

All full knot queries complete with identical certificate digests and zero calls to the changed fallback. The small-fourteen sum of medians is 215.475 milliseconds before, 215.870 A/A and 215.012 after; Gordian medians are 1.906, 1.983 and 1.918 seconds. The full relator operations also have identical traces between arms. At $h = 500$, the equal-pair cyclic medians are 6.664, 6.586 and 6.629 milliseconds; successful quotient-search medians are 13.486, 13.291 and 13.532. These controls show no corresponding knot-corpus or relator-pipeline speedup. Timing variation is visible in A/A and should not be promoted to an algorithmic effect.

The comparison uses the LCS class archived at `34f2689e9` inside the common current word arena, donor helpers, recognizer and certificate replayer. A/A repeats that archived class. Seed 3012 shuffles arm order in each round; each case has one excluded warmup and five measured rounds. The audit includes 225 full knot queries, 180 string queries and 135 local relator operations. Full query timings include fresh PD construction and independent replay; certificate digests are compared outside the timer. String and relator timings include grammar construction. Local queries allow two million work units and 100000 nodes/table cells. Whole recognition allows 18 seconds, with 15 seconds and twenty million work units in its group stage. Censored runs are never used as completed runtime denominators.

The cyclic equal-bigram operation is deliberately a separate control. Its doubled words have a common substring $T(ab)^N$ of length $2N + 12$, so the existing cyclic algorithm already finds a useful checked move quickly. Applying it reduces the target to $c^{-1}a$, with total relator length $2N + 15$. Faster exact linear queries should not be reported as a speedup for this already fast cyclic operation. All completed relator traces are checked for agreement between arms; these supplied abstract presentations are not knot-recognition verdicts.

Raw samples are in `results/lcs_windows_20261008.json`, with the profile, test and build logs under `synthesis/data/lcs-window-*`. The synchronization brought in fifteen research archives. Static relevance triage inspected their overview material and scanned textual members, with archive hashes, member names and matches retained in `data/lcs-window-incoming-scan.json`. The new material concerns recurrence avoidance, special functions, affine partitions and probabilistic asymptotics. The affine-partition article uses perfect hash families to coordinate rainbow zero sets, but its parameter regime is fixed field size and growing numbers of coordinate arms; its scope statement explicitly excludes a faster unknot recognizer. Sparse-graph moment and tree-broadcast results concern probabilistic limits rather than the deterministic planar diagrams or compressed group words in this implementation. No reduction connecting these results to a useful exact recognition step was identified, so none was incorporated as a complexity premise. This was a relevance review, not an independent validation of those manuscripts' theorems. The suggested `openai/math` head remained `fd4aeeb2ee`.

# 53 Exact phase certificates for short-period overlaps

## 53.1 Recognizing a structured case before general matching

The full-overlap control of Section 52 asks for all positive $k$ satisfying

$$\operatorname{suffix}_k((ab)^N) = \operatorname{prefix}_k((ab)^N).$$

Its answer is simply the progression $2, 4, \ldots, 2N$, but the general periodic-extension machinery still exhausts its allowance on the larger test sizes. A short known repeating block gives a more direct route. It can be verified throughout a compressed grammar by a finite-state computation, without occurrence tables or long substring equality queries.

The new stage attempts this route only when both feasible overlap operands have length at least 128. It reads a prefix of exactly 64 letters from the right operand and proposes its least period $p \le 32$, if one exists. These fixed constants bound the speculative work and the size of phase masks; they are policy choices, not mathematical thresholds for general recognition. The proposed block is $B = b_0 \cdots b_{p-1}$. The sample does not prove that the remaining word is periodic. Both entire operands must pass the grammar certificate below. Otherwise the existing complete matcher runs unchanged.

This is another instance of adaptive implementation selection: a small structural test can choose a cheaper exact algorithm. Failure of that test is not failure of the underlying search, and exhaustion remains inconclusive.

## 53.2 A phase mask at every grammar rule

For a grammar word $W$, define

$$M(W) = \{r \in \mathbb{Z}/p\mathbb{Z} : W_i = b_{(r+i) \bmod p} \text{ for every } 0 \le i < |W|\}.$$

This set is represented by a $p$-bit integer. A terminal $a$ has precisely the phases at which $b_r = a$. At a binary rule $W = UV$,

$$M(UV) = M(U) \cap (M(V) - |U|).$$

Equivalently, rotate the right child's bit mask to the right by $|U| \bmod p$, then intersect it with the left mask. This condition is necessary and sufficient: the first child starts at phase $r$ and the second at phase $r + |U|$. Induction over the grammar proves that every mask describes the entire

represented word, including positions far beyond the sampled prefix. Signed letters are compared exactly, with no hashing.

The certificate succeeds only if $0 \in M(Y)$ and $M(X)$ is nonempty. Select any certified starting phase $r \in M(X)$ and set $e = (r + |X|) \bmod p$. The right operand is then a prefix of $B^\infty$, and every suffix of $X$ starts at its certified phase $e - k$.

## 53.3 Recovering every overlap length

The candidate block is cyclically primitive. Indeed, if a nonzero cyclic shift fixed $B$, it would be a repetition of a block of length $\gcd(p, r) < p$. The sampled prefix, being a prefix of $B^\infty$, would then have that smaller period, contradicting the chosen least period. Consequently a full block matches $B$ only at phase zero.

For $k \geq p$ an overlap therefore exists exactly when

$$k \equiv e \pmod{p}.$$

The same congruence is sufficient for shorter lengths as well. It produces one arithmetic progression with first element $e$ if $e > 0$, or $p$ if $e = 0$, and step $p$, clipped at $\min(|X|, |Y|)$. Its multiplicity is stored as an integer even when exponentially large.

Shorter overlaps outside this congruence class are possible because a proper fragment of a primitive block may agree at several phases. The implementation checks each $1 \leq k < p$ outside the main progression by comparing

$$b_{(e-k+i) \bmod p} = b_i \qquad (0 \leq i < k).$$

Each successful check contributes a singleton progression. This gives at most $p$ disjoint progressions, all exact. Their union is the complete overlap set, rather than a selected useful witness. The existing critical alignment routine can therefore consume them without changes to its proof or the ordinary relator certificate format.

For $X = Y = (ab)^N$, $p = 2$ and $e = 0$, so the output is $(2, 2, N)$. For a block of 32 distinct letters, with $X$ starting at phase seven and both words containing $N$ full blocks, the answer is $(7, 32, N)$. The latter case checks a nonzero rotation and the maximal supported period.

## 53.4 Work, memory and the scope of the bound

The prefix sample contains at most 64 letters; its explicit local traversal has bounded size, independent of the represented word lengths. Creating that prefix slice costs $O(H)$ for grammar height $H$. Period proposal and short-exception checks take bounded work because $p \leq 32$. The mask pass uses one bounded-width integer per reachable rule and one transition per binary rule. Its work is $O(s)$ after a topological order is available. The current shared reachability helper sorts node identifiers, so an implementation-level bound is $O(s \log s + H)$ word operations, plus polynomial costs for the binary-encoded lengths. No loop counts to $N$ or enumerates the members of a returned progression.

The arena's shared allowance charges sample traversal, candidate-period checks, reachability, mask evaluation and short-overlap checks. Existing node and overlap-cache limits remain in force. At most one mask is retained per reachable node, and each mask has at most 32 bits. A failed proposal adds bounded sample work; a plausible sample that fails globally adds one grammar pass before the fallback. The negative controls measure this speculation cost.

This stage establishes an efficient exact local algorithm for two words certified to follow a common short periodic block. It does not bound the periods arising throughout unknot recognition, guarantee that relator search reaches such states, or bound the number of moves and intermediate grammar sizes. The general subexponential and quasi-polynomial targets remain open for this implementation.

### 53.5 Validation and matched measurements

Literal-oracle tests cover signed blocks, nonprimitive supplied spellings, random grammar parses, phase rotations, truncated final blocks, and exceptional short overlaps. Mutations after position 64 and a 33-letter period verify that a plausible prefix or an unsupported period does not bypass the complete matcher. A 32-letter shifted block repeated $2^{500}$ times must finish under 20000 work units with whole-word expansion and occurrence tables forbidden. A separate oracle checks that the returned progressions feed the exact critical-extension calculation correctly. The integration run passes 693 tests in 183.274 seconds. The four-test periodic suite, including the additional extension oracle, passes in 0.786 seconds; together the runs cover all 694 tests currently present.

| Scope | Before | A/A | Phases |
|---|---|---|---|
| Small 14, summed medians | 300.269 | 309.512 | 299.232 |
| Gordian, full query | 2113.390 | 2228.724 | 2140.719 |

Median milliseconds; all full knot queries complete.

| Family | $h$ | Before | A/A | Phases |
|---|---|---|---|---|
| Prefix | 8 | 0.111 | 0.112 | 0.111 |
| Prefix | 32 | 0.277 | 0.277 | 0.283 |
| Prefix | 500 | 3.609 | 3.483 | 3.577 |
| Interior | 8 | 0.164 | 0.166 | 0.163 |
| Interior | 32 | 0.406 | 0.404 | 0.406 |
| Interior | 500 | 5.254 | 5.227 | 5.184 |
| Equal pairs | 8 | 2.230 | 2.253 | 2.250 |
| Equal pairs | 32 | 2.595 | 2.562 | 2.586 |
| Equal pairs | 500 | 8.021 | 7.997 | 7.980 |
| Full overlap | 8 | 31.393 | 31.449 | 0.124 |
| Full overlap | 32 | limit | limit | 0.347 |
| Full overlap | 500 | limit | limit | 2.406 |
| Shifted 32 | 8 | 242.843 | 243.305 | 0.473 |
| Shifted 32 | 32 | limit | limit | 0.741 |
| Shifted 32 | 500 | limit | limit | 4.582 |
| Late defect | 8 | 33.184 | 33.148 | 32.865 |
| Late defect | 32 | limit | limit | limit |
| Late defect | 500 | limit | limit | limit |

Median milliseconds including construction; limits are censored. Full overlap enumerates all lengths in compressed progressions, not all occurrences.

| Family | $h$ | Before | A/A | Phases |
|---|---|---|---|---|
| Partial | 8 | 0.581 | 0.586 | 0.585 |
| Partial | 32 | 1.165 | 1.200 | 1.193 |
| Partial | 500 | 7.128 | 7.086 | 7.116 |
| Equal-pair cyclic | 8 | 0.495 | 0.523 | 0.501 |
| Equal-pair cyclic | 32 | 0.832 | 0.846 | 0.838 |
| Equal-pair cyclic | 500 | 6.776 | 6.792 | 6.851 |
| Quotient | 8 | 0.348 | 0.351 | 0.340 |
| Quotient | 32 | 0.966 | 0.965 | 0.980 |
| Quotient | 500 | 13.381 | 13.702 | 13.413 |

Median milliseconds including construction; limits are censored. Local operations do not produce knot verdicts.

For the full-overlap family at $h = 8$, median time falls from 31.393 milliseconds (31.449 A/A) to 0.124; charged work falls from 37641 to 328 and arena nodes from 108 to 11. At $h = 32$ and $h = 500$, both archived controls exhaust two million work units, while phase certification completes in 0.347 and 2.406 milliseconds respectively. The $h = 500$ run uses 503 nodes, 4264 work units and no occurrence-table cells, returning one progression representing $2^{500}$ overlap lengths.

For the shifted 32-letter block, at $h = 8$ the medians are 242.843 milliseconds before, 243.305 A/A and 0.473 after. At $h = 500$, only the new stage completes: 4.582 milliseconds, 1094 nodes and 10110 charged work units, again with one progression and no occurrence tables. This confirms that the method handles a nonzero phase and is not restricted to identical binary-power roots or a two-letter alphabet.

The late-defect control records the cost of an unsuccessful proposal. At $h = 8$, medians are 33.184, 33.148 and 32.865 milliseconds; the exact work increase is 301 units, from 40113 to 40414. Nodes and cumulative table cells remain 121 and 541 respectively. At $h = 32$ and $h = 500$, all arms still exhaust the allowance. For $h = 500$ the recorded times to exhaustion are 2.717, 2.759 and 2.803 seconds. These are censored searches, not completed-query timing ratios. The failed certificate correctly leaves this more general obstacle unresolved. Section 54 subsequently resolves the late-defect family by counting endpoint-forced candidate lengths, without certifying a short period.

All full knot queries produce identical certificate digests, and none attempts the new phase stage. The small-fourteen sums of medians are 300.269 milliseconds before, 309.512 A/A and 299.232 after; Gordian medians are 2.113, 2.229 and 2.141 seconds. Every local relator operation also completes with the same trace across arms. Successful $h = 500$ quotient-search medians are 13.381, 13.702 and 13.413 milliseconds. These controls provide no evidence of a knot-corpus speedup, and their A/A variation limits interpretation of small timing differences.

The archived baseline is the window-localized matcher at `ca4681932`; A/A repeats it. Every arm runs inside the same current arena, donor helpers, recognizer and independent certificate replayer. Seed 3013 shuffles the three arms in one excluded warmup and five measured rounds per case. The audit includes 225 complete knot queries, 270 string queries and 135 local relator operations. String and relator timings include grammar construction. Whole-query timings include fresh PD construction and replay, with certificate digest comparison outside the clock. Local operations have the unchanged two-million-work and 100000-node/table-cell allowances; whole recognition has 18 seconds, including a 15-second, twenty-million-work group stage. Limited runs are censored, not negative answers or completed runtime denominators.

The late-defect control uses $X = Y = (ab)^N c$. Its sampled prefix suggests period two, but the final $c$ invalidates every phase mask. Its only overlap is the full word; the complete fallback must establish this. This is a correctness and speculation-cost control, not an expected shortcut success.

Raw samples are in `results/lcs_periodic_20261008.json`, and logs are under `synthesis/data/lcs-periodic-*`. The incoming archives and their relevance review are unchanged from Section 52; the suggested `openai/math` head remains `fd4aeeb2ee`. The phase certificate is the elementary evaluation of a fixed periodic language on a straight-line grammar, not a claim of a new general recognition theorem.

# 54 Sparse endpoint certificates for overlap lengths

## 54.1 A defective periodic word can still have few candidates

The late-defect control of Section 53, $X = Y = (ab)^N c$, is not periodic with a short block. Nevertheless its overlap set is immediate from the final letter: every matching prefix of $Y$ must end in $c$, and $Y$ contains that letter only at its end. Hence the only candidate is the full length,

whose equality is easy to check.

This suggests a general exact filter that does not assume periodicity. For any nonempty words $X, Y$, a positive suffix/prefix overlap of length $k$ necessarily satisfies both

$$Y_{k-1} = \mathrm{last}(X), \qquad X_{|X|-k} = \mathrm{first}(Y).$$

If either required letter is rare in the corresponding word, these conditions give a small complete list of candidate lengths. The letters are signed free-group letters; $a$ and $a^{-1}$ are distinct here.

## 54.2 Saturated counts preserve precisely the needed information

After the existing feasible suffix/prefix restriction, the implementation tries this stage on operands of length at least 128, following the uniform and certified short-period cases. It uses a fixed candidate allowance $K = 64$. Over the union of the two reachable grammars it computes counts of $\mathrm{first}(Y)$ and $\mathrm{last}(X)$, each saturated at $K + 1 = 65$. A terminal contributes zero or one; concatenation contributes

$$\widehat{c}(UV) = \min(K + 1, \widehat{c}(U) + \widehat{c}(V)).$$

Induction gives $\widehat{c}(W) = \min(K + 1, c(W))$ for the true signed-letter count $c(W)$. Thus a value at most $K$ is exact, zero proves absence, and $K + 1$ means only that enumeration would exceed the policy allowance. No exponentially large frequency integer is needed.

Compare the count of $\mathrm{first}(Y)$ in $X$ with the count of $\mathrm{last}(X)$ in $Y$. Choose the smaller; ties use $Y$. If both counts exceed $K$, the stage declines and the complete occurrence matcher runs. Otherwise a grammar traversal enumerates the chosen letter's positions, pruning every subtree whose count is zero. Shared grammar nodes may have several distinct occurrence offsets; these are intentionally kept distinct. There are at most $K$ retained terminal occurrences, even if the other letters occupy exponentially many positions.

An occurrence at offset $i$ in $X$ proposes $k = |X| - i$; one at offset $j$ in $Y$ proposes $k = j + 1$. Candidates outside the feasible length bound are discarded. Each remaining candidate is checked by deterministic compressed equality of

$$X[|X| - k : |X|] \quad \text{and} \quad Y[0 : k].$$

A successful check contributes the singleton progression $(k, 0, 1)$. The stage uses an equality query because only this fixed-length predicate is needed; it does not compute a longest prefix by bisection.

## 54.3 Soundness, completeness and cost

Every returned length is supported by exact equality. Conversely, every actual overlap satisfies the chosen endpoint condition, hence occurs in the enumerated candidate list and passes its equality check. The conversion from a terminal offset to a length is injective, so the output progressions are disjoint. If a required endpoint letter is absent, the complete overlap set is empty and no equality query is necessary. Counts above the threshold produce no negative conclusion; they simply select the fallback.

For $s$ reachable grammar rules, the two count summaries use $O(s)$ bounded-size storage and transitions. The current reachability helper's sorting gives $O(s \log s)$ word operations overall for that pass. If grammar height is $H$, traversal of at most $K$ retained terminal paths uses $O(KH + 1)$ work, including discarded siblings. Candidate slicing adds at most $O(KH)$ grammar nodes. If $E(t)$ bounds an exact compressed equality query on a grammar of size $t$, a conservative local bound is

$$O\big(s \log s + KH + KE(s + KH)\big),$$

with the binary-length arithmetic costs included in the underlying primitive's bit-complexity accounting. The candidate limit is a fixed implementation parameter. The potentially expensive equality term must not be omitted merely because candidate enumeration is cheap.

All traversals and checks consume the existing shared allowance. Node allocation, equality assertions and overlap-cache storage retain their existing limits. A candidate check that exhausts its resources remains inconclusive. The result is cached under the original caller roots, as for the other exact overlap implementations.

This route also handles arbitrarily long repeating blocks when their marker count is small. Set

$$B = (ab)^N c, \qquad X = Y = B^q.$$

The endpoint $c$ occurs exactly $q$ times, so for $q \leq 64$ the candidate lengths are precisely $|B|, 2|B|, \ldots, q|B|$. All pass equality. The block length can be exponential in its grammar size and need not fit the earlier 32-letter phase model. For $q = 65$, the sparse stage declines. This is a policy boundary, not a mathematical hardness transition; the benchmark retains it to expose the limitation directly.

No bound has been proved here for the number of recognition moves or the sizes and marker distributions of all intermediate relators. These exact local improvements therefore do not establish general subexponential or quasi-polynomial unknot recognition.

## 54.4 Validation and controlled comparison

New literal-oracle tests use 160 random signed parses with both endpoint directions, deliberately invalid candidates, and at least one genuine overlap. They distinguish opposite signs, count multiple occurrences of shared grammar nodes, check absent endpoints without equality calls, retain the dense complete fallback, and exercise the work limit. Compressed regressions use $N = 2^{500}$ with $q = 1, 17, 64$, under 30000 work units and with whole-word expansion and occurrence tables forbidden. The $q = 65$ case checks the saturated-count refusal. The periodic-rejection oracle explicitly bypasses the independent sparse stage so that it continues to exercise the general occurrence table. All 697 tests pass with Regina 7.4.1 in 155.753 seconds. The benchmark writes $N = 2^h$.

| Scope | Before | A/A | Endpoints |
|---|---|---|---|
| Small 14, summed medians | 187.459 | 188.118 | 188.697 |
| Gordian, full query | 1901.241 | 1895.539 | 1893.101 |

Median milliseconds; all full knot queries complete.

| Family | $h$ | Before | A/A | Endpoints |
|---|---|---|---|---|
| Prefix | 8 | 0.107 | 0.109 | 0.107 |
| Prefix | 32 | 0.271 | 0.278 | 0.272 |
| Prefix | 500 | 3.432 | 3.471 | 3.446 |
| Interior | 8 | 0.160 | 0.159 | 0.170 |
| Interior | 32 | 0.410 | 0.406 | 0.405 |
| Interior | 500 | 5.218 | 5.084 | 5.205 |
| Equal pairs | 8 | 2.167 | 2.168 | 2.149 |
| Equal pairs | 32 | 2.464 | 2.434 | 2.461 |
| Equal pairs | 500 | 7.900 | 7.879 | 7.889 |
| Full overlap | 8 | 0.097 | 0.095 | 0.097 |
| Full overlap | 32 | 0.198 | 0.200 | 0.199 |
| Full overlap | 500 | 2.344 | 2.335 | 2.319 |
| Shifted 32 | 8 | 0.399 | 0.424 | 0.414 |
| Shifted 32 | 32 | 0.604 | 0.586 | 0.602 |
| Shifted 32 | 500 | 4.054 | 4.009 | 4.014 |
| Late defect | 8 | 32.704 | 33.066 | 0.142 |
| Late defect | 32 | limit | limit | 0.354 |
| Late defect | 500 | limit | limit | 2.909 |
| Markers 17 | 8 | 121.041 | 121.152 | 0.834 |
| Markers 17 | 32 | limit | limit | 1.105 |
| Markers 17 | 500 | limit | limit | 3.682 |
| Markers 64 | 8 | 202.026 | 201.761 | 11.055 |
| Markers 64 | 32 | limit | limit | 11.628 |
| Markers 64 | 500 | limit | limit | 13.985 |
| Markers 65 | 8 | 204.693 | 202.723 | 202.383 |
| Markers 65 | 32 | limit | limit | limit |
| Markers 65 | 500 | limit | limit | limit |

Median milliseconds including construction; limits are censored. Full overlap enumerates all lengths in compressed progressions, not all occurrences.

| Family | $h$ | Before | A/A | Endpoints |
|---|---|---|---|---|
| Partial | 8 | 0.314 | 0.302 | 0.305 |
| Partial | 32 | 0.641 | 0.627 | 0.641 |
| Partial | 500 | 6.934 | 6.845 | 6.953 |
| Equal-pair cyclic | 8 | 0.496 | 0.476 | 0.497 |
| Equal-pair cyclic | 32 | 0.815 | 0.805 | 0.827 |
| Equal-pair cyclic | 500 | 6.739 | 6.798 | 6.700 |
| Quotient | 8 | 0.345 | 0.333 | 0.319 |
| Quotient | 32 | 1.102 | 1.025 | 1.050 |
| Quotient | 500 | 13.067 | 13.081 | 13.144 |

Median milliseconds including construction; limits are censored. Local operations do not produce knot verdicts.

For the late-defect family at $h = 8$, median time falls from 32.704 milliseconds (33.066 A/A) to 0.142; charged work falls from 40414 to 377. At $h = 500$, both archived controls exhaust two million work units, whereas the endpoint route completes in 2.909 milliseconds with 505 nodes, 5789 work units and one checked candidate. No occurrence table is built.

At $h = 500$, the 17-marker and 64-marker queries complete in 3.682 and 13.985 milliseconds, with 6621 and 15658 work units respectively. Both archived controls are limited on each family. The 64-marker run stores 625 arena nodes and checks exactly 64 candidates; equality work is visible in its larger runtime. At $h = 8$, the 17-marker median falls from 121.041 milliseconds (121.152 A/A) to 0.834, and the 64-marker median falls from 202.026 (201.761 A/A) to 11.055.

The 65-marker control shows the cost of declining this route. At $h = 8$, all arms complete, with

medians 204.693 milliseconds before, 202.723 A/A and 202.383 after. The new stage adds exactly 56 work units, from 259264 to 259320, while nodes and cumulative table cells remain 425 and 2673. At $h = 32$ and $h = 500$, all arms still exhaust the allowance. At $h = 500$, times to exhaustion are 2.627, 2.591 and 2.602 seconds. These are censored searches, not completed runtime comparisons. The unresolved dense case is recorded explicitly rather than hidden by the enumeration threshold.

The short-period families retain identical charged statistics across arms, because their earlier phase certificate still answers before the new stage. Every full knot query has the same certificate digest in all arms, and none invokes the sparse stage. The small-fourteen sums of medians are 187.459 milliseconds before, 188.118 A/A and 188.697 after; Gordian medians are 1.901, 1.896 and 1.893 seconds. All local relator operations also retain identical traces. At $h = 500$, successful quotient-search medians are 13.067, 13.081 and 13.144 milliseconds. These controls support preservation of existing paths, not a demonstrated knot-corpus speedup.

The archived baseline is the phase-certified matcher at `0c9ac4f1b`; A/A repeats that class inside the common current arena, donor helpers, recognizer and independent certificate replayer. Each case has one excluded warmup and five measured rounds, with shuffled arm order and seed 3014. The audit contains 225 full knot queries, 405 string queries and 135 local relator operations. Grammar construction and local answer checks are inside the string and relator timers. Full query timings include fresh PD creation and replay, with certificate digest comparison outside the timer. Local queries retain the two-million-work and 100000-node/table-cell allowances. Whole recognition allows 18 seconds, including a 15-second, twenty-million-work group stage. Resource-limited runs remain censored.

The dense 65-marker control is included alongside the successful sparse families and the previously optimized short-period families. Full knot queries and abstract relator operations are reported separately from the string primitives; a faster overlap query alone is not a faster knot verdict.

Raw measurements are in `results/lcs_sparse_20261008.json`, with test and build logs under `synthesis/data/lcs-sparse-*`. The incoming-archive review remains the one in Section 52, and the suggested `openai/math` head is unchanged at `fd4aeeb2ee`. The endpoint condition is elementary; its role here is to select a bounded exact verification path inside the compressed implementation.

# 55 Dense endpoint progressions certified by two overlaps

The 65-marker limitation of Section 54 suggests retaining an entire arithmetic progression of candidate lengths. This removes both the fixed marker-count limit and the short-period limit for a useful family, while preserving the general exact occurrence matcher as fallback.

## 55.1 Certifying an occurrence grid without enumeration

For a signed letter $a$, summarize its positions in a grammar word by $(f, l, d, c)$: first position, last position, gcd of successive gaps, and exact cardinality. Empty sets have $c = 0$; singletons have $d = 0$. At a concatenation $UV$, shift the right positions by $|U|$. If both sets are nonempty, the new gap gcd is

$$\gcd(d_U, d_V, |U| + f_V - l_U).$$

The first and last positions and count combine directly. Every position is congruent to $f$ modulo $d$. Since positions are distinct, the set fills its entire grid if and only if

$$l - f = d(c - 1).$$

This criterion also covers a singleton. Absence, a certified progression, and a nonprogression are distinct outcomes. The traversal uses the reachable grammar, including binary arithmetic on exponentially large counts and gaps; it never expands those counts into lists.

Every suffix/prefix overlap of $X, Y$ has a length in both necessary sets

$$\{|X| - i : X_i = \text{first}(Y)\}, \qquad \{j + 1 : Y_j = \text{last}(X)\}.$$

If both sets are progressions, intersect them using exact CRT arithmetic. If only one is a progression, it still supplies a necessary candidate set. Clip to $1 \le k \le \min(|X|, |Y|)$. An absent endpoint or empty intersection proves that there are no overlaps. If neither set is a progression, decline this stage. No conclusion is drawn from failure to find a grid.

### 55.2 The two-largest-overlaps certificate

Suppose the resulting candidate progression contains at least two lengths, with maximum $L$ and step $d > 0$. Check exactly, using compressed equality, that both $L$ and $L - d$ are overlaps. If so, the common word

$$W = \text{suffix}_L(X) = \text{prefix}_L(Y)$$

has a border of length $L-d$, hence period $d$. Repeated application of that period implies that every positive $L - jd$ is also a border of $W$. It follows that every member of the candidate progression is an overlap. Conversely, endpoint necessity already ensures that every overlap belongs to that progression. Thus two exact equalities certify the entire answer. The progression need not begin at $d$, and its cardinality need not fit a machine integer or any fixed enumeration allowance.

For a singleton, one equality decides it. If either of the two checks for a nonsingleton fails, the stage declines: smaller candidates may still be valid. The complete matcher then runs. This distinction is essential; a failed largest overlap is not a certificate of an empty overlap set.

For $s$ reachable rules, grammar height $H$, and maximum integer bit length $b$, the two grid passes cost $O(s \log s)$ word operations for reachability and a polynomial amount of $b$-bit arithmetic per rule. The CRT intersection has polynomial bit cost. At most four slices and two equality queries remain, adding $O(H)$ grammar nodes. If $E(t, b)$ bounds the deterministic equality primitive, the equality term is at most $2E(s + O(H), b)$. These costs remain charged to the common work and node allowances; a limit is inconclusive. The implementation tries this route after the cheaper uniform, short-period, and sparse-endpoint routes, for operands of length at least 128.

Set $B = (ab)^N c$ and $X = Y = B^q$. The occurrences of $c$ give precisely the length progression $(|B|, |B|, q)$. For $q \ge 2$, the two largest candidates are true overlaps, so the complete answer is certified without enumerating $q$. In particular both $N$ and $q$ may equal $2^{500}$. This is a local compressed-string theorem. It does not bound the number of relator moves, intermediate grammar growth, or total unknot-recognition time.

### 55.3 Validation and measurements

The four new tests compare occurrence grids with literal sets on all binary words of length at most seven and 300 random signed words; compare complete overlap sets and critical-extension maxima on 40 randomized repeated blocks; exercise disjoint grids and both failed-check fallbacks; and certify the $N = q = 2^{500}$ family under a 100000-work allowance with expansion and generic occurrence tables forbidden. Existing generic-matcher coverage explicitly bypasses the independent fast stages. The full suite passes all 701 tests with Regina 7.4.1 in 166.342 seconds.

| Scope | Before | A/A | Grid |
|---|---|---|---|
| Small 14, summed medians | 197.850 | 198.286 | 198.954 |
| Gordian, full query | 2278.670 | 1997.217 | 3209.060 |

Median milliseconds; all full knot queries complete.

| Family | $h$ | Before | A/A | Grid |
|---|---|---|---|---|
| Prefix | 8 | 0.103 | 0.104 | 0.105 |
| Prefix | 32 | 0.272 | 0.271 | 0.271 |
| Prefix | 500 | 3.714 | 3.708 | 3.664 |
| Interior | 8 | 0.181 | 0.158 | 0.158 |
| Interior | 32 | 0.434 | 0.405 | 0.431 |
| Interior | 500 | 5.345 | 5.375 | 5.359 |
| Equal pairs | 8 | 2.301 | 2.303 | 2.311 |
| Equal pairs | 32 | 2.608 | 2.635 | 2.612 |
| Equal pairs | 500 | 8.469 | 8.472 | 8.490 |
| Full overlap | 8 | 0.097 | 0.095 | 0.094 |
| Full overlap | 32 | 0.226 | 0.200 | 0.199 |
| Full overlap | 500 | 2.506 | 2.467 | 2.485 |
| Shifted 32 | 8 | 0.459 | 0.441 | 0.440 |
| Shifted 32 | 32 | 0.619 | 0.621 | 0.633 |
| Shifted 32 | 500 | 4.279 | 4.310 | 4.319 |
| Late defect | 8 | 0.119 | 0.119 | 0.120 |
| Late defect | 32 | 0.250 | 0.275 | 0.252 |
| Late defect | 500 | 3.076 | 3.097 | 3.105 |
| Markers 17 | 8 | 0.871 | 0.841 | 0.833 |
| Markers 17 | 32 | 1.059 | 1.033 | 1.066 |
| Markers 17 | 500 | 3.759 | 3.871 | 3.877 |
| Markers 64 | 8 | 11.050 | 11.055 | 11.092 |
| Markers 64 | 32 | 11.775 | 11.798 | 11.789 |
| Markers 64 | 500 | 14.938 | 15.058 | 15.126 |
| Markers 65 | 8 | 213.510 | 209.198 | 0.675 |
| Markers 65 | 32 | limit | limit | 0.942 |
| Markers 65 | 500 | limit | limit | 4.810 |
| Markers power | 8 | 314.478 | 321.601 | 0.917 |
| Markers power | 32 | limit | limit | 3.399 |
| Markers power | 500 | limit | limit | 52.894 |
| Broken grid | 8 | 255.670 | 271.733 | 259.146 |
| Broken grid | 32 | limit | limit | limit |
| Broken grid | 500 | limit | limit | limit |

Median milliseconds including construction; limits are censored. Full overlap enumerates all lengths in compressed progressions, not all occurrences.

| Family | $h$ | Before | A/A | Grid |
|---|---|---|---|---|
| Partial | 8 | 0.311 | 0.325 | 0.314 |
| Partial | 32 | 0.665 | 0.650 | 0.662 |
| Partial | 500 | 7.000 | 7.059 | 7.310 |
| Equal-pair cyclic | 8 | 0.510 | 0.505 | 0.508 |
| Equal-pair cyclic | 32 | 0.835 | 0.837 | 0.842 |
| Equal-pair cyclic | 500 | 7.158 | 7.248 | 7.164 |
| Quotient | 8 | 0.336 | 0.335 | 0.339 |
| Quotient | 32 | 0.979 | 0.978 | 0.963 |
| Quotient | 500 | 13.937 | 13.636 | 13.726 |

Median milliseconds including construction; limits are censored. Local operations do not produce knot verdicts.

For $q = 65$ and $N = 2^8$, median time falls from 213.510 milliseconds (209.198 A/A) to 0.675; charged work falls from 259320 to 860 and arena nodes from 425 to 26. At $N = 2^{500}$ both old arms exhaust two million work units; the new query completes in 4.810 milliseconds, with 518 nodes, 9224 work units, two exact equalities and no occurrence table. For $q = N = 2^{500}$ the complete answer is one progression containing $2^{500}$ lengths. Its median is 52.894 milliseconds, using 2003

nodes, 48292 work units and two equalities. Both controls are limited. For $q = N = 2^8$, completed medians are 314.478, 321.601 and 0.917 milliseconds.

The negative control changes the first block to $(ba)^N c$ while retaining 64 ordinary blocks, and compares that word with itself. Its endpoint grid is regular, but the second-largest proposed overlap fails. Only the full word is an overlap. At $N = 2^8$, the fallback completes in 255.670 milliseconds before, 271.733 A/A and 259.146 after. The unsuccessful certificate adds 210 work units and six nodes. At $N = 2^{32}$ and $N = 2^{500}$ every arm is still limited. A regular endpoint grid alone is therefore insufficient.

All 225 whole-knot queries preserve their certificate digests and none activates this stage. Small-fourteen sums of medians are 197.850, 198.286 and 198.954 milliseconds. Gordian medians are 2.279, 1.997 and 3.209 seconds; this batch is noisy and does not demonstrate a recognition speedup. The 135 local relator operations retain identical traces; the $N = 2^{500}$ quotient medians are 13.937, 13.636 and 13.726 milliseconds. Earlier phase and sparse routes likewise retain their counters on the corresponding controls.

The archived baseline is `8170a64c7`. A/A repeats its matcher inside the common current arena and recognition/replay pipeline. Seed 3015 shuffles three arms in each of five measured rounds, after one excluded warmup. The 495 string queries include construction and local answer validation; no progression with exponentially many entries is enumerated for validation. Local allowances are two million work units and 100000 arena/table cells. Whole recognition allows 18 seconds, with a 15-second, twenty-million-work group stage. Limited samples remain censored, with no completed-time ratio. Raw measurements are in `results/lcs_endpoint_progressions_20261008.json`; the test and benchmark logs are under `synthesis/data/lcs-endpoint-*`.

# 56 Binary boundary tensors for faithful Jones computation

## 56.1 The two incoming reports and the integration decision

Commit `1be2abc8c` supplies two research archives dated 8 October 2026. The binary-tensor report develops a complete Jones evaluator, a near-prefix compressed matcher, and dihedral boundary transport. The modular-boundary report develops finite-field reuse of the marked Euler observer. Both target upstream `8a9583494`; neither snapshot includes our subsequent window, phase, sparse-endpoint, or endpoint-progression refinements. Their patches must therefore be integrated by component, preserving those later changes. The supplied manifests verify all 480 binary-package files and all 380 modular-package files. These integrity checks establish source identity, not mathematical correctness.

This section integrates the binary Jones component, its three test files, benchmark driver and historical policy sources. The other components remain under review. In particular, the modular report reports improved repeated boundary queries but slower whole raw scans on all seven measured examples; its fixed prime palette can be inconclusive. The new near-prefix branch must be compared with the current matcher, since its archived baseline predates our window optimization. No speedup from either unintegrated component is attributed to the maintained recognizer here.

The new optional `spin-faithful` backend improves the maintained full Jones bound from the faithful Potts implementation's $2^{O(\sqrt{n}\log n)}$ to $2^{O(\sqrt{n})}$. Comparable general bounds are known in the literature, including Maria [41]; this is an implementation improvement, not a claim of a new complexity class result. Jones polynomial one remains inconclusive for unknot recognition.

## 56.2 An integral turning certificate from the PD rotation system

For a validated nonempty classical knot diagram, let $\alpha$ exchange the two darts on an edge and let $\sigma$ advance one slot counterclockwise at a crossing. The cycles of $\phi = \sigma\alpha$ are the $n + 2$ faces. Pick an outer face $f_\infty$. An antisymmetric integral cochain $b$ on directed edges must satisfy

$$b_{\alpha(d)} = -b_d, \qquad \sum_{d \in f} b_d = |f| - 4 + 8\mathbf{1}_{f=f_\infty}.$$

A face corner turns by $-1$ quarter-turn, so these are the edge contributions to the total turning $-4$ of a bounded face and $+4$ of the outer face. The demands sum to zero, since the face lengths sum to $4n$.

Root a dual spanning tree at the outer face. On each child face's parent edge, assign the total demand of its subtree, with opposite sign on the reverse dart; assign zero to the other edges. At each face, the parent subtree total minus its children's totals equals its demand. The root follows from total demand zero. Each edge value has magnitude at most $4n + 4$, and the sum $L_b = \sum_e |b_e|$ is at most $4(n + 1)^2$. The certificate therefore has $O(n)$ integers of $O(\log n)$ bits. An independent verifier reconstructs the PD faces and checks the displayed equations directly, without repeating the tree solver.

Why do these equations suffice for all smoothing circles? In any smooth planar realization with cardinal crossing tangents, geometric edge turns satisfy the same equations. The difference from $b$ integrates to zero around every face. On the sphere, face boundaries span the cycle space, so that difference integrates to zero around every closed graph walk. Replacing geometric edge turns by $b$ therefore leaves each oriented smoothing circle's total turning unchanged: it is $+4$ or $-4$. This proof permits loops, repeated face vertices, and a smoothing circle visiting the same crossing twice. It does not apply unchanged to virtual diagrams on surfaces of positive genus.

## 56.3 Two orientations replace boundary connectivity

Write the unnormalized bracket as

$$\mathcal{B}_D(A) = \sum_{s \in \{0,1\}^n} A^{n-2|s|} \delta^{c(s)}, \qquad \delta = -A^2 - A^{-2}.$$

Introduce $\rho$ with $\rho^4 = -A^2$. Orient every smoothing circle. Its two orientations contribute $\rho^4 + \rho^{-4} = \delta$ by the turning certificate. Thus the same state sum can be expressed with one binary arrow variable on each diagram edge. At a crossing, retain only smoothings whose pairs have one incoming and one outgoing arrow; multiply by their usual $A$ weight and by $\rho$ to the sum of their local turns. On an edge, multiply by $\rho^{\epsilon b_e}$, where $\epsilon$ is its chosen arrow. Expanding this tensor product gives precisely all oriented smoothing states and recovers the bracket.

A crossing order with $w$ cut edges now has at most $2^w$ tensor keys. There is no connected-prefix or common-disk assumption: a key records arrows, not a pairing of boundary points. The local tensor has bounded arity, and self-loop arrows are summed at their crossing. This exact summary preserves the bracket under every remaining tensor contraction. It does not preserve the relative Khovanov complex; cancellations in a polynomial may erase data needed for homology.

## 56.4 One phase and one integer per key

Set $A = B^2$ and $\rho = \zeta B$, where $\zeta^4 = -1$. Initially this suggests four integer coordinates per tensor key. In fact the phase modulo four is determined by the boundary arrows. To see this, for an edge from slot $j$ at $v$ to slot $k$ at $u$, the cochain $b_d + j - k - 2$ has zero face integrals modulo four. It is therefore a vertex potential difference $\nu(u) - \nu(v)$. At a crossing the local

turning is $\sum_d \epsilon_d j$ modulo four, and $\sum_d \epsilon_d = 0$. After adding the vanishing potential terms, each internal edge contributes $2\epsilon_d = 2$ modulo four, independent of its arrow. The remaining terms are determined by the boundary. Hence one phase $r \in \{0, 1, 2, 3\}$ and one signed integer suffice. The implementation checks phase agreement whenever terms merge, and checks phase zero for the final scalar.

Multiply each crossing tensor by $B^4$ and each edge tensor by $B^{|b_e|}$. Every local exponent is then nonnegative. Local multiplication requires only shifts and additions of signed integers, with phase/sign updates. If $K_0 = 4n + L_b$ and $K = \max(K_0, 6n + 4)$, the final padded scalar is

$$S = B^K \mathcal{B}_D(B^2).$$

All expanded shifted monomials have $B$-degree at most $8n + 2L_b$ before final padding. There are at most $2^{3n}$ smoothing–arrow assignments. The triangle inequality consequently bounds every intermediate integer by $O(n^3)$ bits for the base chosen next; large cancellation does not invalidate this upper bound.

## 56.5 Faithful evaluation and coefficient recovery

For a knot with $n$ crossings, the normalized Jones polynomial $V_D(t) = \sum_j c_j t^j$ obeys

$$\sum_j |c_j| \le 4^n, \qquad |j| \le 2n \quad (c_j \neq 0).$$

The first follows from $2^n$ states, each with at most $n + 1$ circles and coefficient norm at most $2^n$ after reduction. Smoothing degrees and the writhe correction give the second, deliberately loose, degree bound. Choose

$$\beta = \lceil (2n + 2)/8 \rceil, \quad B = 2^\beta, \quad X = B^8.$$

Then $X \ge 4 \cdot 4^n$. Balanced base-$X$ digits uniquely recover every coefficient after multiplying the Laurent polynomial by a known monomial. This is an injective exact encoding, not a random evaluation.

If $a$ is the writhe, the expected scaled unknot bracket is

$$U = (-1)^{a+1}(B^8 + 1)B^{K+6a-4}.$$

Its exponent is nonnegative. Equality $S = U$ proves $V_D = 1$; inequality proves knottedness. For the full polynomial, form the exact integer

$$X^{2n} V_D(X^{-1}) = \frac{(-1)^{a+1} S B^{16n+4-K-6a}}{X + 1},$$

moving a negative power to the denominator. Check divisibility, then recover balanced digits $d_i$ and set $c_j = d_{2n-j}$. The decoder checks normalization, degree and coefficient bounds, including before its identity shortcut. These checks validate the decoding interface; they are not an independent certificate of a supplied scalar's contraction. The empty PD represents one unknot circle and receives its explicit special case.

## 56.6 Complexity and recognition semantics

Each crossing has a bounded number of transitions per binary frontier key, and all keys and integer coefficients have polynomial bit length. A fixed width-$w$ order therefore costs $\mathrm{poly}(n)2^{O(w)}$ deterministic bit operations. Even charging linear scans for worst-case Python dictionary collisions only squares the exponential factor and preserves this bound. Our existing verified separator order has $w = O(\sqrt{n})$. The default spin query considers a greedy candidate even when the caller

supplies an order, and retains an order within the separator certificate's bound. With local caps and deadlines disabled, full Jones computation thus has the stated square-root exponential bound. The polynomial prefactor is absorbed here: for every fixed $c$,

$$n^c 2^{C\sqrt{n}} = 2^{C\sqrt{n}+c\log_2 n} = 2^{O(\sqrt{n})}.$$

The factor $\mathrm{poly}(n)$ remains explicit in the width-parameterized bound because $w$ may be constant even when the diagram is large. For supplied orders of width $O(\log^2 n)$ the corresponding fixed-order query is quasi-polynomial.

The production default remains unchanged: the report exhibits regressions as well as improvements. Users can select `spin-faithful` in either the Jones command or the recognizer's Jones-filter option. Local state or transition exhaustion publishes no unfinished scalar or identity claim; a completed separator certificate may still be reused. Caller cancellation propagates. Large evidence integers are serialized in hexadecimal without changing global decimal conversion settings. A polynomial-one result continues to the independent exact recognizer. Consequently this backend does not establish a general subexponential unknot-recognition algorithm.

## 56.7 Independent validation and controlled experiments

The imported tests perform 140 full-polynomial comparisons with the independent whole-cube Laurent oracle, covering mirrors, crossing orders and outer faces. They verify oriented-circle turning directly, test 20 vertex gauges, exercise 180 bounded signed polynomial decodes, reject forged normalization, check budget/cancellation behavior, and verify CLI and recognition fallback behavior. A shuffled 64-crossing grid guards against the report's earlier supplied-order policy defect.

All 714 maintained tests pass in 164.627 seconds with Regina 7.4.1. The integrated benchmark uses the delivered final source, including the normalization check added after the report's historical timing runs. It records 490 measurements and 70 excluded warmups over fourteen inputs, five arms and seven shuffled rounds (seed 26100860). There are 469 complete measurements and 21 local-cap outcomes. Default queries include fresh PD validation and order preparation; the fixed-order scope uses the same externally prepared certified order and includes all tensor and cochain setup. Each query requests the full polynomial, with 4096 states, 200000 transitions and a three-second cooperative deadline. Completed arms agree; small inputs use an independent cube oracle, and grids have checked descending certificates.

| Input | $n$ | Potts | A/A | Spin | Fixed P | Fixed S |
|---|---|---|---|---|---|---|
| Trefoil | 3 | 0.190 | 0.187 | 0.544 | 0.158 | 0.416 |
| Conway | 11 | 0.638 | 0.655 | 2.314 | 0.445 | 1.568 |
| KT | 11 | 0.667 | 0.647 | 2.348 | 0.491 | 1.576 |
| Hard 8 | 8 | 0.381 | 0.372 | 1.287 | 0.283 | 0.993 |
| Braid5 36 | 36 | 3.104 | 3.112 | 11.495 | 2.755 | 8.455 |
| Weaving 10 | 10 | 0.698 | 0.647 | 2.515 | 0.522 | 1.784 |
| Torus 101 | 101 | 37.243 | 37.376 | 39.350 | 34.529 | 32.174 |
| Tree 6 | 126 | 10.277 | 10.341 | 62.948 | 7.905 | 48.508 |
| Tree 7 | 254 | 73.257 | 73.270 | 394.441 | 68.170 | 352.804 |
| Grid 8 | 64 | 6.791 | 6.747 | 23.368 | 6.119 | 17.737 |
| Grid 10 | 100 | 39.177 | 39.008 | 67.193 | 39.033 | 57.578 |
| Grid 12 | 144 | 380.542 | 459.847 | 306.520 | 369.836 | 224.954 |
| Grid 14 | 196 | limit | limit | 732.041 | limit | 717.790 |
| Grid 8 shuffled | 64 | 29.152 | 29.297 | 22.882 | 6.043 | 17.528 |

Median milliseconds for full Jones queries. Fixed P/S share an externally prepared order; other columns include ordering. Limits are censored, not completed-query times.

On the 144-crossing grid, ordinary Potts/A/A/spin medians are 380.542/459.847/306.520 milliseconds. On the common order they are 369.836/224.954 milliseconds for Potts/spin. Median within-round ratios are 1.638 and 1.648 respectively; they need not equal ratios of the displayed medians. Potts stores 877 peak states and performs 48386 transitions; spin stores 240 and performs 32098. On the 196-crossing grid, every Potts arm reaches the state cap, while all spin queries complete with 480 peak states and 89122 transitions. This is a capacity result, with no completed-time ratio against the capped arms.

Regressions remain substantial. The 254-crossing tree medial has ordinary Potts/spin medians 73.257/394.441 milliseconds and fixed-order medians 68.170/352.804. The Potts cancellation pattern leaves only one state on that input, while spin reaches 272. On the shuffled 64-crossing grid, ordinary medians favor spin, 29.152/22.882 milliseconds, but fixed-order medians favor Potts, 6.043/17.528. The ordinary gain on this case comes from ordering rather than a faster contraction. These full invariant queries do not measure completed recognition on a hard-knot corpus.

Raw maintained measurements are in `results/spin_jones_integrated_20261008.json`, with per-file source hashes. Its inherited `baseline_commit` field identifies the report's integration base `8a9583494`; the actual compared source is identified by the hashes. The original `spin_jones_20261008.json`, initial-policy results and ordering follow-up remain separate historical records. The report's previous performance numbers are not relabeled as results of this integrated run.

## 57 Review queue for the six new October research packages

The user identified commits `1be2abc8c` and `bc5b91385` on `origin/main`. Both have been merged. They add six research archives, whose supplied manifests all verify. The first two contain 480 and 380 manifested files; the later braid, causal, certified-primitives and graded-torus packages contain 47, 38, 507 and 42 respectively. Archive and manifest hashes are recorded in `data/incoming-20261008-review.json`. The suggested `openai/math` head remains `fd4aeeb2ee`. No result below is inferred merely from integrity checks or a supplied test log.

| Package | Current assessment and next integration check |
|---|---|
| Binary tensors | Jones component integrated in Section 56. Near-prefix matching and boundary transport require separate comparisons and review against the updated code. |
| Modular boundary | The reviewed modular norm and completion arguments are consistent with a sound one-sided observer. Production geometry, singular response handling, replay and fallback remain to be audited before promotion. |
| Braid kernel | Claims a certified minority-sign kernel and linear endpoint Markov descent. Review the connected-sum topology and independent replay; compare producer plus replay with the current braid prefilter. |
| Causal kernels | Claims complete fixed-parameter bounded RIII unlocking. Review the initial connected-core and full-footprint arguments and adapt transactional replay. Its own genuine-diagram timings show no speedup. |
| Certified primitives | Supplies a second binary Jones implementation, valuation arithmetic, minimum-disc cocycle seeds and an optional majority word-overlap policy. Valuation arithmetic is transferred in Section 58, and the bounded Potts prelude in Section 59; audit each geometric or group primitive independently. |
| Graded torus | Claims graded occupancy bounds and a polynomial four-strand torus scanner using an external Morse-model theorem. Review that dependency and transfer the grading conventions to the maintained representation. |

These are component-level integration decisions, not blanket acceptance or rejection of the reports. Their prototypes target different snapshots and internal representations. The current work preserves the later compressed matcher refinements rather than replacing them with an older bundled tree.

The second tensor implementation is particularly relevant to adaptive selection. It represents a scalar as a signed mantissa with a separate valuation, removing large powers from the integer payload. Its report gives a tree254 comparison of 130177 bits for the shifted scalar versus 1536 bits for the mantissa. This is supplied experimental evidence, not yet a measurement of the maintained spin backend. It motivates a direct comparison and possibly transferring that arithmetic, rather than adding an unexamined second default path.

The braid and causal reports provide parameterized complete subproblems. The graded-torus report provides a claimed complete recognition bound for a syntactically restricted class. None proves that arbitrary diagrams enter those favorable parameter regimes. The modular observer can miss existing obstructions with a finite prime palette, and even its completed threshold comparison does not detect every knot. The complete general recognition bound therefore remains open after these deliveries.

## 57.1 A direct comparison of the two tensor implementations

The maintained `benchmark_tensor_candidates.py` loads the second report's `tensor_jones.py` directly from its original ZIP member, recording both archive and source hashes. This is a research comparison, not a second production backend. Four cases use one common certified order per case, prepared outside timing. Each arm includes fresh PD construction, tensor preparation, contraction and full polynomial decoding. The four arms are maintained spin, an identical A/A spin control, and the report's valuation and global-shift arithmetic. Seed 26100862 shuffles five measured rounds after one excluded warmup. All 80 measured queries and 16 warmups complete under the common 4096-state, 200000-transition and three-second allowances. Every completed polynomial agrees. The weaving case uses the independent whole-cube oracle; grids have independently checked descending certificates; the tree case uses agreement between completed arms.

| Input | Spin | A/A | Valuation | Global shift |
|---|---|---|---|---|
| Weaving 10 | 1.650 | 1.651 | 1.232 | 1.146 |
| Tree 7 | 365.409 | 365.028 | 68.833 | 278.227 |
| Grid 12 | 209.727 | 209.850 | 118.829 | 160.606 |
| Grid 14 | 714.712 | 717.063 | 338.190 | 569.544 |

Median milliseconds on a common prepared order. The last two arms load the delivered research module directly.

On the 254-crossing tree, the four arm medians are 365.409, 365.028, 68.833 and 278.227 milliseconds respectively. The median paired spin/valuation ratio is 5.134; within the report's own implementation, the global-shift/ valuation ratio is 4.039. On the 144-crossing grid, spin/valuation medians are 209.727/118.829 milliseconds; on the 196-crossing grid they are 714.712/338.190, with paired ratios 1.765 and 2.113. The ten-crossing weaving case slightly favors the report's global-shift mode over its valuation mode, 1.146 versus 1.232 milliseconds, retaining a useful small-input control.

The tree's maintained spin counter includes its large final normalization integer and reaches 195585 bits. The report's global-shift contraction counter reaches 130177 bits; its valuation mantissa reaches 1536 bits. Those are differently represented quantities and not total memory usage. The two report modes have identical transition counts, so their direct comparison isolates the arithmetic choice. Spin and the report use different local tensor aggregation and turn/sign conventions: for example the tree has 12589 spin transitions versus 15931 in either report mode. Their timing ratio must not be attributed solely to valuation normalization.

These results motivated the arithmetic transfer in Section 58. The measurements in this subsection precede that transfer. The exact representation is $z^v m$, with nonzero $m$ not divisible by the

power-of-two base $z$. Addition aligns the smaller valuation, adds signed mantissas, and removes all common factors of $z$; exact cancellation has its own zero representation. This can remove large intermediate monomial padding while preserving the exact evaluated number, including cancellation and carries. A production transfer must retain the spin backend's phase convention, normalization checks, budgets and independently tested polynomial output. Raw measurements are in `results/tensor_candidates_20261008.json`.

# 58 Valuation arithmetic in the maintained binary Jones backend

## 58.1 Integrating the second report's arithmetic idea

The common-order comparison in Section 57 identified large intermediate monomial padding as a practical cost of exact tensor contraction. The certified-primitives report supplied by commit `bc5b91385` removes such padding with valuation-normalized arithmetic. We transfer that idea into the maintained spin backend while retaining its own turning certificate, cyclotomic phases, local tensor aggregation, crossing-order policy and faithful Jones decoder.

The new default within `spin-faithful` stores a phase, a binary valuation and a signed odd mantissa. The overall recognizer's default Jones backend remains unchanged. Python callers can select `arithmetic='shifted'` to use the previous integer representation; `arithmetic='valuation'` selects the new one. Both return the same materialized exact scalars and polynomial encoding. This makes the arithmetic change independently comparable without changing any topological decision.

## 58.2 Canonical exact integer arithmetic

For a nonzero integer $x$, write

$$x = 2^v m, \qquad v \geq 0, \quad m \text{ odd},$$

with $m$ signed. Zero has the unique representation $(0, 0)$. We remove all binary factors, rather than only whole powers of the faithful base $B$. The representation therefore concerns the evaluated integer, including arithmetic carries, and makes no assertion about a formal polynomial's lowest degree.

For addition of $(v, m)$ and $(u, a)$, let $q = \min(v, u)$ and compute

$$t = (m \ll (v - q)) + (a \ll (u - q)).$$

If $t = 0$, return the canonical zero. Otherwise let $r$ be the number of trailing zero bits of $|t|$ and return $(q + r, t/2^r)$. The equality $2^q t = 2^v m + 2^u a$ proves exactness. The signed division is exact, so Python's arithmetic right shift produces the required signed quotient even for negative integers. Cancellation can increase the valuation by more than either input's trailing zeros, which is why normalization is performed after addition.

A compiled local tensor has one phase $\zeta^s$ and a bounded sum of monomials $\sum_j a_j 2^{e_j}$. Its common offset $e = \min_j e_j$ is factored out once during local preparation. Multiplication by a stored coefficient $\zeta^r 2^v m$ therefore computes

$$t = \sum_j a_j (m \ll (e_j - e)), \qquad \zeta^{r+s} 2^{v+e} t.$$

Normalize $t$ as above and reduce $r + s$ modulo four, negating the mantissa when $r + s \geq 4$, since $\zeta^4 = -1$. The per-transition work still uses shifts and additions with bounded small multipliers,

without multiplying two arbitrary large coefficients. Phase agreement is checked before every merge.

Induction on the crossing scan proves that every stored entry represents exactly the integer and phase stored by the previous implementation. A key is removed if and only if its exact coefficient is zero in either mode. Thus all frontier keys, transition counts, peak state counts and cap boundaries are identical. At the end, materialize $m \ll v$ once and apply the existing padding and normalization. The independent turning verifier and faithful polynomial decoder retain their original contracts.

## 58.3 Bit complexity and resource interpretation

Every shifted integer represented by a frontier entry has the polynomial bit bound of Section 56. Its odd mantissa cannot be longer. The valuation is at most the represented integer's bit length and itself requires only logarithmically many bits in that bound. Aligning unequal valuations never creates a larger integer than the old arithmetic would have materialized for the same terms, apart from a constant carry allowance. The new representation therefore preserves the deterministic $\mathrm{poly}(n)2^{O(w)}$ bound for a width-$w$ order and the $2^{O(\sqrt{n})}$ general Jones bound for the certified order.

The polynomial prefactor is intentionally retained only in the parameterized bound. For the general bound, $n^c = 2^{O(\log n)}$ is absorbed by $2^{O(\sqrt{n})}$ for every fixed $c$. Keeping it visible can explain the per-entry arithmetic cost, but it does not define a stronger asymptotic class. As usual, arbitrary noncanonical input label bit lengths also have their own reading and validation cost; the crossing-count bound uses a normalized combinatorial diagram encoding.

The established `max_coefficient_bits` counter still reports the bit length of represented full integers, including final scalar and normalization values. Two new counters distinguish the largest stored frontier mantissa and largest frontier valuation. Neither is a total memory measurement: keys, tuples, Python object overhead, both active dictionaries, preparation and the final materialized scalars also consume memory. The state and transition allowances retain their meanings and do not become byte budgets. Caller cancellation and inconclusive local exhaustion keep their existing behavior. No unfinished polynomial is published.

## 58.4 Validation and controlled comparison

Four new tests compare 2000 randomized signed additions and local tensor products with literal integer and four-coordinate cyclotomic arithmetic, including exact cancellation and carry normalization. They test a common valuation of $2^{500}$ without allocating its expanded integer; compare all historical scalar, certificate and work fields across 70 mirrored ordered knot queries; validate arithmetic choices before preparation; and verify a large reduction in stored tree-frontier mantissas with identical exact results. The existing 140 independent whole-cube polynomial comparisons, turning and gauge tests, decoder tests, CLI/fallback tests and cancellation checks now run through valuation arithmetic by default. All 17 focused spin tests pass. The full integration suite passes all 718 tests with Regina 7.4.1 in 158.291 seconds.

| Input | Before | A/A | Shifted | Valuation | Potts |
|---|---|---|---|---|---|
| Trefoil | 0.535 | 0.553 | 0.584 | 0.618 | 0.196 |
| Conway | 2.338 | 2.335 | 2.371 | 2.574 | 0.655 |
| KT | 2.411 | 2.418 | 2.421 | 2.638 | 0.683 |
| Hard 8 | 1.282 | 1.268 | 1.309 | 1.379 | 0.395 |
| Braid5 36 | 21.104 | 21.700 | 24.396 | 23.757 | 5.586 |
| Weaving 10 | 4.171 | 4.145 | 4.187 | 4.575 | 1.086 |
| Torus 101 | 39.956 | 39.709 | 39.855 | 40.577 | 37.326 |
| Tree 6 | 64.914 | 58.730 | 60.099 | 42.778 | 10.667 |
| Tree 7 | 387.402 | 390.981 | 394.490 | 112.079 | 77.981 |
| Grid 8 | 22.359 | 22.487 | 23.525 | 24.025 | 6.592 |
| Grid 10 | 66.397 | 64.900 | 68.029 | 62.089 | 38.484 |
| Grid 12 | 218.472 | 216.920 | 255.359 | 176.728 | 361.450 |
| Grid 14 | 752.957 | 745.677 | 785.157 | 480.545 | limit |
| Grid 8 shuffled | 22.533 | 22.503 | 23.160 | 23.859 | 29.374 |

Median milliseconds for fresh full Jones queries, including ordering. Current shifted arithmetic controls for the new dispatch and counters. Limits are censored.

The benchmark compares the archived spin implementation at `ef374a365`, an identical A/A repeat, current shifted arithmetic, current valuation arithmetic, and faithful Potts. Each timer includes fresh PD validation, ordering, turning preparation, contraction, normalization and full polynomial recovery. Seed 26100865 shuffles five measured rounds after one excluded warmup on each of fourteen inputs. Of 350 measurements, 345 complete and five Potts queries reach the state cap on the 196-crossing grid. The 70 warmups are excluded. Every completed polynomial agrees, and every completed new spin query reproduces all historical scalar, certificate and work fields, including the old full-integer bit counter. Source hashes are checked before and after measurement. Limits remain incomplete queries, with no runtime ratio against a completed answer.

On the 254-crossing tree, archived/A/A/shifted/valuation medians are 387.402, 390.981, 394.490 and 112.079 milliseconds. The median paired archived/valuation ratio is 3.479; shifted/valuation is 3.482. The largest frontier mantissa falls from 195585 to 1536 bits, while peak states remain 272 and transitions remain 12589. The largest valuation is 195584 bits of binary shift, stored as a small integer; this is not a 195584-bit allocation. The materialized final scalars still account for the unchanged full-integer bit counter.

On the 144-crossing grid, archived/valuation medians are 218.472 and 176.728 milliseconds, with paired ratio 1.236. Faithful Potts takes 361.450 milliseconds, giving a paired Potts/valuation ratio of 2.045. The 196-crossing grid improves from 752.957 milliseconds (745.677 A/A) to 480.545, with paired ratio 1.588. Its frontier mantissa maximum falls from 42701 to 1600 bits; the full-integer counter, which also includes normalization, remains 59201. Every spin arm completes all five queries, whereas Potts is capped in all five. That last comparison is a capacity result, not a completed-runtime speedup.

The small-input controls expose normalization overhead. Trefoil medians increase from 0.535 to 0.618 milliseconds, Conway from 2.338 to 2.574, and the 64-crossing grid from 22.359 to 24.025. The current shifted arm separates the new counter/dispatch overhead from normalization itself. Faithful Potts also remains faster on the tree, with median 77.981 milliseconds, and on most small inputs. Thus neither valuation arithmetic nor the spin representation dominates all cases. The overall default Jones backend remains unchanged, and the shifted mode remains available.

These are full invariant queries, not whole-recognition measurements. The new arithmetic does not reduce frontier width, the number of represented states, or the number of Khovanov generators. It improves the actual cost of exact scalar arithmetic on padded coefficients. A next adaptive experiment can use measured coefficient size or padding to decide when to switch representations, while charging conversion work and preserving one exact contract. A cross-backend policy must additionally account for discarded Potts work and its total transition allowance.

Raw evidence is in `results/spin_valuation_20261008.json`; test, benchmark and source-review records are under `synthesis/data/spin-valuation-*`. All 21 incoming ZIP archives match their preceding recorded hashes, and the suggested `openai/math` head is unchanged at `fd4aeeb2ee`. The remaining report components retain the review queue of Section 57; this transfer is not blanket approval of either complete supplied patch.

# 59 Adaptive faithful Jones computation with a charged fallback

## 59.1 A portfolio with an asymptotic guarantee

The earlier measurements favor faithful Potts on small diagrams and tree medials, and binary spin on larger grid diagrams. Neither implementation dominates. The user's introsort suggestion and the certified-primitives report's bounded-Potts-prelude proposal suggest a one-way adaptive policy: try the practically favorable method, but bound its work before invoking the method with the stronger general complexity guarantee.

The optional `faithful-adaptive` backend implements this policy for full Jones computation and exact polynomial-identity queries. For an input with $n$ crossings, it gives faithful Potts a default trial of $128n$ transitions. The constant is a scheduling choice, not a topological theorem. The trial includes any internal Potts ordering restart. If Potts completes, its faithful answer is final for this invariant query. If it hits its trial allowance or frontier cap, the wrapper abandons the table and starts spin. A zero trial selects spin directly. The crossing-free unknot is handled by the existing exact empty-diagram case even with zero local caps.

This uses measured progress: the transition counter and actual state population determine whether the prelude completes. It does not choose a backend solely from the input crossing count or from a predicted width. Unlike introsort, the two representations do not currently share partially contracted coefficients. Switching restarts the invariant calculation, and all completed work in the discarded representation is explicitly charged.

## 59.2 One resource ledger across both implementations

Let $C$ be the caller's transition allowance, possibly infinite, and $T = 128n$. The Potts attempt receives $\min(T, C)$. If it fails after $p$ transitions, spin receives exactly $C - p$. Its completed work $s$ produces total $p + s$; its local exhaustion reports that same cumulative count. No fresh full allowance is allocated to the replacement. If $C - p = 0$, the wrapper returns an inconclusive limit before starting spin. An explicitly configured trial has its own cost and does not inherit the default linear-prelude guarantee.

Both stages use the same state cap. The reported peak state count is the maximum across attempts, including a rejected insertion that first exceeds a state cap. It is not a byte-memory limit, nor the sum of simultaneous populations. The failed Potts exception handler is exited before allocating the spin table, releasing the abandoned traceback and its evaluator locals. Only progress counters and completed order information are retained.

Potts may already have constructed a checked separator order while adapting its own scan. The wrapper passes that order to spin without reconstructing its certificate. Otherwise spin constructs its usual certified order. Thus a failed prelude does not require a second certificate construction. The retained certificate's order still has the square-root frontier bound. Caller cancellation exceptions propagate. A local limit can retain a completed order certificate and policy counters, but never an unfinished polynomial, identity assertion or scalar value.

The public result preserves the winning backend's scalar format and names its selected backend. The obstruction wrapper chooses the corresponding hexadecimal serialization, including large

quadratic coordinates and color counts for Potts or bracket scalars for spin. CLI output includes the policy record. Recognition preserves that record even when the Jones result is inconclusive and the independent Khovanov calculation continues.

### 59.3 Why the prelude preserves the stronger complexity bound

With external caps and deadlines disabled, the default Potts attempt still has only $128n$ transitions. Each transition changes the represented table by at most one key. Thus it retains polynomially many keys; keys have at most $O(n)$ entries, and faithful quadratic coefficients have $O(n^2)$ bits. Even pessimistic dictionary lookup costs remain polynomial in this bounded attempt. Input validation, ordering, at most one separator preparation, normalization and optional coefficient recovery likewise have polynomial bit cost. Internal reordering does not multiply the work allowance: both attempts contribute to the same Potts transition counter.

If the prelude completes, the whole query has polynomial cost. Otherwise the remaining spin computation has the certified $2^{O(\sqrt{n})}$ bound, including exact arithmetic and full coefficient recovery. Hence

$$\mathrm{poly}(n) + 2^{O(\sqrt{n})} = 2^{O(\sqrt{n})}.$$

This guarantee is independent of whether the prelude proves useful on the input. It applies to the default bounded trial; an arbitrarily large user-supplied trial contributes its own additional work. Fixed local caps can make the query inconclusive and do not turn a partial result into a proof.

Both backends compute the same faithful invariant. They cannot disagree about a completed polynomial-one result. However, that result still does not certify an unknot. The general subexponential bound established here is for Jones computation, not for complete unknot recognition. The policy offers no theorem that spin is within a constant factor of the faster backend on every input, and the chosen transition constant is not a runtime optimum.

### 59.4 Validation and measurements

The focused tests compare both winners and direct-spin mode against an independent whole-cube polynomial oracle on 36 diagram/mirror inputs, giving 108 full-polynomial comparisons. They verify that easy Potts completions avoid spin and separator preparation; that a measured-work switch reuses its certificate; that a state-limited Potts query can finish in spin; and that exact shared-budget boundaries give completion or inconclusive exhaustion. A shuffled grid forces a Potts ordering restart before the cross-backend switch and verifies that both discarded stages are charged. Zero budgets, empty input, malformed options, cancellation during either decoder, large JSON evidence, CLI dispatch and Khovanov fallback after Jones identity are also covered.

The integrated suite passed 725 tests in 165.183 seconds. A final refinement counts a switch only when spin actually starts, and extends the nested ordering-restart assertion. After that refinement, all seven focused test methods passed again in 1.637 seconds. The broad suite predates that small counter clarification; the final focused gate covers it explicitly.

The maintained `benchmark_adaptive_jones.py` compares historical Potts at commit `153bc2362`, an identical A/A control, current valuation spin, and the new adaptive wrapper. Fourteen inputs use seed 26100867 to shuffle five measured rounds after one excluded warmup. Each query includes fresh PD validation, ordering, tensor preparation, contraction, normalization and full coefficient recovery. The common allowances are three cooperative seconds, 4096 states and 200000 transitions. Source hashes are frozen across the run. All 56 warmup attempts are excluded from timings. Of 280 measured attempts, 270 complete; the ten incomplete attempts are the two Potts arms on the 196-crossing grid. They are never treated as completed-query times. Small diagrams use an independent whole-cube oracle, grids have verified descending certificates, and

the other cases compare completed polynomials across arms. These are full invariant queries, not full recognition timings.

| Input | Potts | A/A | Spin | Adaptive |
|---|---|---|---|---|
| Trefoil | 0.185 | 0.194 | 0.589 | 0.193 |
| Conway | 0.646 | 0.656 | 2.555 | 0.671 |
| KT | 0.695 | 0.714 | 2.675 | 0.721 |
| Hard 8 | 0.360 | 0.378 | 1.382 | 0.393 |
| Braid5 36 | 3.127 | 3.141 | 13.151 | 3.149 |
| Weaving 10 | 0.631 | 0.629 | 2.508 | 0.644 |
| Torus 101 | 37.607 | 37.543 | 41.431 | 37.330 |
| Tree 6 | 10.586 | 10.680 | 43.437 | 10.660 |
| Tree 7 | 73.716 | 122.667 | 116.558 | 84.612 |
| Grid 8 | 6.740 | 6.799 | 24.626 | 6.726 |
| Grid 10 | 38.881 | 38.092 | 67.187 | 38.037 |
| Grid 12 | 367.153 | 362.538 | 176.774 | 279.949 |
| Grid 14 | limit | limit | 480.226 | 507.184 |
| Grid 8 shuffled | 28.519 | 28.516 | 24.128 | 28.537 |

Median milliseconds for fresh full Jones queries, including ordering. Limits are censored, not completed-query times.

On the 144-crossing grid, adaptive computation uses 18432 discarded Potts transitions and 32098 spin transitions, totaling 50530. It reuses the checked order. The adaptive median is 279.949 milliseconds, compared with 367.153 for Potts and 176.774 for direct spin; the median paired Potts/adaptive ratio is 1.330. Thus the policy improves on Potts here, but the unsuccessful prelude still has a measurable cost relative to direct spin.

On the 196-crossing grid, Potts exceeds its state cap after 6405 transitions. Adaptive computation then spends 89122 spin transitions and completes in 507.184 milliseconds, compared with 480.226 for direct spin. The ten capped Potts attempts establish a completion difference under the same resource limits, not a finite completed-query speedup. On the 144-crossing case, the focused test confirms completion at the exact combined work allowance; one transition less gives an inconclusive result with the charged cumulative count.

Most small cases retain Potts's timing advantage over spin, with some wrapper overhead. For example, the trefoil medians are 0.185 milliseconds for Potts, 0.589 for spin and 0.193 for adaptive computation. The 126-crossing tree has medians 10.586, 43.437 and 10.660 respectively. The 254-crossing tree has substantial timing variation, including different medians for the identical Potts arms, so it does not support a precise small percentage comparison. The shuffled 64-crossing grid finishes within the Potts prelude after an internal ordering restart; adaptive takes 28.537 milliseconds versus 24.128 for direct spin. These counterexamples rule out interpreting the policy as an oracle for the fastest representation. It remains optional, with the existing overall backend default unchanged.

Raw measurements are in `results/adaptive_jones_20261008.json`; test logs and PDF validation are retained under `synthesis/data/`. The research refresh finds unchanged hashes for all 21 incoming archives and unchanged `openai/math` head `fd4aeeb2ee`. This checkpoint promotes the bounded-prelude proposal; the other component reviews in Section 57 remain separate work. The subsequent intake commit `b71e64c3b` places those six deliveries in `reports/29/` through `reports/34/`. Comparison against the original archives finds only omitted checksum manifests and CRLF normalization in two CSV files; all other placed bytes match. This placement adds no new research claim to the current review queue.

# 60 Polynomial projectors, certified local algebras, and static reuse

## 60.1 What the new algebraic path changes

The projector-and-seam delivery extends the existing scalar Fitting search. For a component $C$ of the live cobordism complex, let $R$ be its complete algebra of scalar chain endomorphisms over $\mathbb{F}_2$. Scalar blocks mix only objects with the same matching, homological degree and recovered quantum shift. The solver imposes $Qd = dQ$ separately for every morphism coefficient, including every attachment in the component. It does not discard the non-scalar coefficients of $d$ when constructing those equations.

The old search tries a bounded deterministic list of elements $Q \in R$. A sufficiently large power gives the Fitting decomposition $C = \ker Q^k \oplus \operatorname{im} Q^k$. This discovers a proper summand when both parts are nonzero, but every invertible candidate has trivial kernel. An invertible candidate can nevertheless have several distinct irreducible primary factors. The optional `primary` backend tests for the associated polynomial projectors after the ordinary Fitting test fails. A nilpotent or identity candidate cannot supply such a projector and skips this extra work. The component, variable and candidate caps remain unchanged.

## 60.2 A complete polynomial test for one candidate

Let $D$ be the number of scalar coordinates and $\mu_Q$ the minimal polynomial of the direct sum of the typed blocks. The first linear dependence among $I, Q, \ldots, Q^D$ determines $\mu_Q$ exactly. Flattening the blocks preserves all matrix entries, so an earlier annihilating polynomial would have caused an earlier dependence. The evaluation map

$$\mathbb{F}_2[t]/(\mu_Q) \longrightarrow R, \qquad p \longmapsto p(Q)$$

is injective. Write $\mu_Q = \prod_{i=1}^{s} f_i^{e_i}$ with distinct irreducible $f_i$. Each Chinese-remainder factor $\mathbb{F}_2[t]/(f_i^{e_i})$ is local. Its only idempotents are zero and one: for an idempotent $e$, either $e$ or $1 - e$ is a unit, giving $e = 1$ or $e = 0$ respectively. Therefore

$$\mathcal{B}_Q = \{p \bmod \mu_Q : p^2 = p\} \cong \mathbb{F}_2^s.$$

This is Berlekamp's classical fixed-algebra construction [48], including repeated irreducible factors. In characteristic two, squaring is linear in this commutative quotient. The implementation finds the nullspace of the columns $t^{2j} + t^j \bmod \mu_Q$. A basis element other than $0, 1$ gives an actual nonzero, nonidentity idempotent $p(Q)$.

Discovery uses retained packed powers. A separate positive checker uses Horner evaluation, checks an annihilator, verifies the displayed projector polynomial and squares the actual matrices. A positive witness does not need to establish minimality of the displayed annihilator. The scanner additionally checks the full commutator, the invertibility of the splitting bases and every transformed cross term. Thus the accepted decomposition preserves all typed attachments. Elementary methods take $O(D^4)$ binary operations and $O(D^3)$ bits for one polynomial-projector query; building the complete commutant and trying multiple candidates are separate costs.

For example, $f = t^3 + t + 1$ and $g = t^3 + t^2 + 1$ are irreducible. The invertible matrix $C_f \oplus C_g$ has no ordinary stable-kernel split, while $p = t^4 + t^2 + t$ satisfies $p \equiv 0 \pmod f$, $p \equiv 1 \pmod g$ and projects onto the second summand. This example explains the stronger test without assuming that such a matrix occurs in every difficult knot scan.

## 60.3 A new stopping certificate for the entire scalar algebra

A failed polynomial test usually says only that this particular $Q$ offers no polynomial projector. The maintained continuation adds an exception that can be proved complete for the current scalar

algebra. Let $a = \dim_{\mathbb{F}_2} R$. If

$$\deg \mu_Q = a \quad \text{and} \quad \dim \mathcal{B}_Q = 1,$$

then the injective evaluation map above is surjective by dimension equality. Consequently $R = \mathbb{F}_2[Q]$ is a local algebra. It contains no nontrivial idempotents, so no subsequent scalar candidate can split this component. The remaining candidate trials can be skipped exactly.

The implementation uses the complete commutant dimension, not the number of candidates sampled. It rechecks that the generator commutes with the complete differential, and records the minimal polynomial, degree, candidate matrices and algebra dimension in the search metrics. A repeated irreducible factor is allowed: the algebra can have nilpotents and still be local. Conversely, two copies of a field admit a primary candidate of small degree while their full commutant is a matrix algebra containing projections. The degree equality is essential and is tested explicitly.

This certificate concerns the specified scalar endomorphism algebra of the current complex. It is not a claim that no more general cobordism operation, homotopy reduction or future crossing extension can simplify the complex. Unlike the positive projector checker, replay of this negative certificate must justify completeness of the commutant and minimality of $\mu_Q$. The current code obtains both from exact linear solves; the saved numerical dimensions alone would not suffice as an external proof.

## 60.4 Transporting complete child spaces without changing the search

The delivery proposes, but does not implement, static reuse of corner algebras. We implement it behind `fitting_reuse_commutant=True`. Suppose a verified split gives $C = \bigoplus_i C_i$, with inclusion $u_i$, projection $v_i$, and $e_i = u_i v_i$. Every scalar endomorphism of $C_i$ extends by zero to $C$; conversely $r \in R$ restricts to $v_i r u_i$. Hence

$$\mathrm{End}_{\mathrm{scalar}}(C_i) \cong e_i R e_i, \qquad \sum_i \dim(e_i R e_i) \le \dim R.$$

Conjugating a complete parent basis by the accepted change of coordinates and restricting it to a child therefore spans the entire child commutant. No new equations against the differential need be solved for that child.

A basis change in the commutant could otherwise change the first sampled candidates. To avoid this confounder, the transported spanning set is reduced to the unique basis with increasing least-bit pivots and zero coefficients at the other pivot positions. That is precisely the existing nullspace solver's basis in the same variable ordering. Direct tests compare the whole basis, candidate count, every split witness and final complex against fresh solving. The two implementations consequently follow the same search, apart from the common certified stopping rule.

Transport is lazy: size skips, exhausted split allowances and cache hits do not construct a child space. A cached positive witness does not contain a complete parent basis, so its children use fresh solves. Basis reuse stays inside one static compression call; adding a crossing or changing the differential starts from fresh spaces. Failed transport leaves the current exact differential intact. The option remains off by default because avoiding a solve is not itself a runtime advantage: transporting and canonicalizing a large basis can cost more than solving a small child's equations.

The sum-of-dimensions bound gives at most $\min(a, D)$ nonzero leaves in a proper static refinement tree. A straightforward simultaneous transport has $O(aD^3)$ cost per level and $O(aD^4)$ over such a tree, before candidate search, dependency elimination and differential transport. These bounds are in the explicit live dimension, which remains uncontrolled in terms of the original crossing count. They provide no general quasi-polynomial theorem.

## 60.5 Validation and measurement protocol

The transferred kernel passes exhaustive binary-polynomial tests through degree six, independently evaluated small matrix minimal polynomials, repeated-primary-factor cases, corrupted-witness rejection and full-differential conjugation checks. Public API and CLI tests exercise exact ranks, capped recognition, incompatible options and fallback. Additional maintained tests compare 200 canonical spans against fresh nullspaces and audit transported child spaces on multiple algebraic families and genuine knot scans. They cover cache hits, split-budget ordering and interruption inside transport. An exhaustive small-commutant test enumerates every endomorphism for each accepted local-algebra stop and confirms that its only idempotents are zero and identity. A repeated-field control has a primary nongenerator but still splits, excluding the unsound shortcut of stopping after any primary candidate.

The maintained `benchmark_primary_continuation.py` pins the original Fitting module at `437b48c7f` and the transferred primary module at `74446633a`. The five arms are original Fitting, an identical A/A control, delivered primary search, maintained primary with the local-algebra stop, and maintained primary with both the stop and static corner reuse. Every source hash is checked before and after timing. Five shuffled rounds with two calls per batch follow one excluded warmup per arm; the seed is 26100879. Raw knot scans use one common prepared order and compare complete graded ranks with the ordinary Khovanov scanner. Ordering and diagram construction are outside those timers. Algebraic search includes the commutant solve, all candidate work and the verified basis change; complete compression also includes recursive splitting and sharing. Fixture construction is outside both timers. The latter inputs are graded algebraic complexes, not established classical-knot prefixes. Identical final primary witnesses and representations are checked across the delivered and two maintained primary arms.

The final integrated gate passes all 748 tests in 179.037 seconds with Regina available. The benchmark completes 950 measured calls and excludes 95 warmup calls. There are 250 raw knot scans and 350 calls each for algebraic search and recursive compression. The supplementary field audit checks 320 exact candidate pairs and 240 field elements by an independent Frobenius-orbit calculation. Its uniform-sampling probabilities are not attributed to the deterministic production candidate policy.

| Input | Fitting | A/A | Delivered | Local stop | Corner |
|---|---|---|---|---|---|
| Conway | 11.874 | 11.934 | 11.998 | 12.000 | 12.239 |
| KT | 13.362 | 13.728 | 13.462 | 13.608 | 14.097 |
| Hard 8 | 1.404 | 1.406 | 1.419 | 1.414 | 1.424 |
| Braid5 36 | 623.583 | 629.583 | 617.867 | 623.707 | 646.495 |
| Torus (3,5) | 5.993 | 7.494 | 5.941 | 6.012 | 6.264 |

Median milliseconds; all calls complete. Full raw knot scans at a common prepared order.

All five knot cases have zero primary candidates, primary splits and local-algebra stops. Corner reuse occurs three times each on Conway and Kinoshita–Terasaka, but supplies no measured improvement. The other knot cases do not invoke it. These timings give no evidence of a natural-knot speedup or a reason to change the overall default scanner. They are raw complete homology scans, not the full recognition portfolio with its earlier filters.

| Input | Fitting | A/A | Delivered | Local stop | Corner |
|---|---|---|---|---|---|
| 3 dense | 1.883 | 1.885 | 2.159 | 1.583 | 1.900 |
| 4 dense | 2.354 | 2.362 | 3.013 | 1.950 | 2.600 |
| 6 dense | 7.731 | 7.747 | 8.758 | 4.815 | 6.954 |
| 8 dense | 19.198 | 18.928 | 18.685 | 9.836 | 15.048 |
| 10 dense | 37.043 | 54.307 | 40.615 | 24.584 | 51.019 |
| 12 dense | 65.092 | 60.793 | 66.129 | 32.529 | 50.975 |
| 8 bridge | 8.355 | 8.436 | 13.306 | 4.656 | 8.739 |

Median milliseconds; all calls complete. Complete recursive compression of graded algebraic fixtures; construction excluded.

On the dense degree-eight family, original Fitting exhausts 32 candidates without splitting. The delivered primary kernel finds a parent split after two candidates, but its recursive pass still examines 46 candidates in total. The maintained local-algebra certificate reduces that total to six, including three polynomial-projector calls and two certified stops. Complete-compression medians fall from 18.685 milliseconds for delivered primary to 9.836 for the maintained continuation, with median paired ratio 1.905. Both retain exactly the same final complex with 75 differential entries, compared with 142 before the parent split. This is a gain in complete recursive compression, beyond merely the already faster search for its first split.

The degree-ten family likewise reduces 52 total candidates to six and retains 102 entries instead of the original 221. The degree-twelve family reduces 58 candidates to six and has delivered/maintained medians 66.129/32.529 milliseconds, with median paired ratio 2.033. Degree twelve raises the scalar variable allowance above the production default, which must remain explicit when interpreting the result. The degree-eight bridge fixture has paired ratio 2.849. Timing variation is substantial in the degree-ten samples, including the identical Fitting control, so its raw samples and paired statistics are retained rather than turning a ratio of medians into a precision claim.

Corner transport lowers fresh commutant builds from three to one on each split two-field fixture, but slows every measured complete-compression family. For dense degree eight it takes 15.048 milliseconds rather than 9.836; its median paired primary/corner ratio is 0.655. The canonical-basis checks show that this is transport cost, not a different search outcome. Thus the polynomial-projector backend enables the local-algebra stop, while corner reuse remains a separately selectable experimental option. Raw measurements and the exact source pins are in `results/primary_continuation_20261008.json`.

# 61 Four further October deliveries and their integration boundaries

The user pointed out four newly arrived `docs/incoming/unknot_*.zip` archives. Their hashes and all supplied manifest entries verify: 705 for projectors and seams, 482 for presentations and early certificates, 393 for compressed certificates, and 40 for separator transfer. These checks establish integrity, not mathematical correctness.

The later intake places these deliveries, followed by the SU(2) and port register reports, in `reports/35/`–`reports/40/`. A separate byte comparison verifies all retained files against the original ZIPs; only the separator benchmark CSV has CRLF-to-LF normalization, and checksum-only manifests are omitted under the existing intake policy. The placement record is `data/incoming-35-40-placement-audit.json`.

The audit record is:

`data/incoming-four-20261008-review.json`

The suggested `openai/math` head remains `fd4aeeb2ee` at this review.

**Projectors and seams.** The reviewed polynomial-projector construction is integrated and extended in Section 60. Its separate geometry kernel assembles complete boundary covers with dihedral sheet maps, retaining original ports and marks. It does not extract such pieces from an arbitrary knot exterior or verify their embedding in that exterior. Geometry and the independently proposed worker-communication repair remain separate integration items.

**Presentations and early certificates.** The report claims exact real-feasibility bounds controlled by generator count, explicit relator length and retained word-circuit cuts. Its quaternion compiler uses a noncommuting-generator test, rather than assuming arbitrary generator images must differ for a nonabelian representation. It distinguishes an arbitrary presentation from certified meridional generators, charges dense polynomial construction, and identifies the published traceless-meridian specialization as not yet implemented in its benchmarked solver. The required small presentation or small circuit cut is not proved for every residual knot. The delivered Z3 path is an optional exact external solver, not an independent UNSAT proof checker. Those provenance and solver boundaries require review before a production verdict path is added.

The same report supplies a first-jet observer and reverse-suffix modular responses. The response proof retains a singular interior radical instead of silently applying an invertible Schur-complement formula. Its claimed $O(n(W+1)^2)$ field-operation bound concerns shared preparation of all suffixes under a valid reverse filtration. It does not bound the number of live Khovanov components or automatically establish a full-recognition speedup. The next useful check is a maintained consumer and complete accounting for response preparation, query reuse and exact fallback. The first-jet report retains its observed regressions, so automatic dispatch is not justified by the delivered timing evidence alone.

**Compressed certificates.** The endpoint matcher certifies an arithmetic progression using short endpoint anchors, proves a period on the largest candidate window, and gallops and bisects in candidate rank to locate the true overlap cutoff. This is stronger than merely accepting an entire candidate progression when its two largest members match. Section 63 now integrates it after the existing shortcuts and compares it against the current maintained matcher, including partial cutoffs and failed period certificates. The delivered whole-knot runs have zero endpoint activations and establish no end-to-end speedup; primitive capacity gains must be assessed separately.

The regular-cover kernel represents compatible isomorphisms by at most two congruence families per connected base graph, but fast equivalence checking does not bound the number of distinct states. The disk checker validates a finite triangulation, normal coordinates, primitivity, extreme-ray rank, Euler characteristic and essential boundary parity. Even a correct accepted disk still needs a verified link from that triangulation to the input knot diagram. Neither geometry module is promoted to a knot-verdict path at this checkpoint.

**Separator transfer.** The first-hit identity factors a typed DAG transfer through a vertex cut. Prefixes must stop at their first cut vertex; unrestricted prefixes would count a path multiple times when the cut is not an antichain. The numerical rank budget then maximizes $\sum_h r_h$ subject to $r_{h-1} + r_h \le c_h$ and supplied upper bounds on $r_h$. Saturating the leftmost available path edge is optimal: any deficit can be moved from the next edge to it without decreasing the objective, then the argument repeats. Thus $\sum_h c_h - 2\max\sum_h r_h$ is the sharp lower bound supplied by those numbers. For a verified reduced knot complex, a bound greater than one is a one-sided obstruction; a smaller bound does not certify an unknot.

All 36 delivered separator tests pass in the current environment. This is a prototype gate, not a

production Khovanov integration gate. The report supplies neither a maintained cobordism-closure evaluator nor a cheap general exporter of a suitable state; its braid utility first constructs an exponential cube. Its actual-knot timings favor ordinary propagation. A future transfer must validate the closure and grading conventions, compare expanded maps with the current corridor implementation, and charge construction and cut discovery. These obligations remain outside the current integration.

## 61.1 The subsequent degree-budgeted SU(2) delivery

Commit `ea9cf443c` adds the degree-budgeted SU(2) archive:

`unknot_su2_degree_budget_20261008.zip`

All 50 checksum entries verify, and all 30 delivered test methods pass in 1.049 seconds. The report has a particularly concrete restricted recognition route: two *certified* meridian generators permit a complete compressed integer calculation. The underlying topological input is established: Kronheimer–Mrowka, Corollary 7.17 of *Knots, sutures and excision*, gives an irreducible SU(2) representation sending a chosen meridian to $i$ for every nontrivial knot [49]. The primary source was checked directly during this review.

For noncommuting pure imaginary generator images $A, B$, set $T = AB$. Since $A^2 = B^2 = -1$ and $ATA^{-1} = T^{-1}$, every word has the form $(-1)^e T^k A^s$ with $e, s \in \{0, 1\}$. Multiplication changes only one integer and two parity bits:

$$(e, k, s)(f, \ell, t) = (e + f + st \bmod 2,\ k + (-1)^s \ell,\ s + t \bmod 2).$$

Thus a binary straight-line word circuit can be evaluated without expansion. For comparison relators with triples $(e_j, k_j, s_j)$, odd $s_j$ is impossible. Otherwise the equations are $k_j \phi \equiv e_j \pi \pmod{2\pi}$ with $0 < \phi < \pi$. After ruling out $k_j = 0, e_j = 1$, let $d = \gcd_j |k_j|$. If $d = 0$, every interior phase works. If $d > 0$, an odd $k_j/d$ fixes a parity $e$; all equations are compatible exactly when $e_j \equiv (k_j/d)e \pmod 2$. An interior phase then exists exactly when $\lfloor (d - 1 + e)/2 \rfloor > 0$. A compact witness is $\phi = \pi/d$ for odd parity or $2\pi/d$ for even parity, subject to that strict interior condition. These normal-form and phase arguments check out at the algebraic interface.

The report's bit bound is polynomial in compressed input size: intermediate exponent bit lengths are linear in the number of word-circuit nodes, and the remaining operations are gcd, quotient and parity calculations. It also gives an exhaustive two-seed Wirtinger propagation class, with a claimed polynomial complete recognition bound on diagrams admitting such a derivation. The unrestricted multi-generator path is a formula compiler; the invoked general existential-real algorithm is not implemented in this package.

The maintained adapter in Section 62 now constructs and replays the seed derivation from the validated PD, keeps every original crossing relation and carries meridian provenance to the two surviving generators. An arbitrary two-generator endpoint of a Whitehead search is insufficient: exponent sum one does not prove that a generator is a meridian. The delivered Boolean declaration is explicitly a precondition, not a topology certificate. The production path checks that precondition from the diagram itself and exposes a bounded optional recognizer with verified verdicts on the resulting restricted class. The general formula compiler remains a separate research interface, with no all-knot small-seed or degree bound established.

# 62 Complete recognition from a checked two-meridian derivation

## 62.1 The delivered idea and the maintained topology adapter

Commit `ea9cf443c` supplied a particularly useful consequence of the traceless-representation theorem: recognition is polynomial when every Wirtinger generator can be derived from two original meridians by crossing conjugations. The maintained implementation now checks this condition directly on a validated planar diagram. This produces actual two-sided knot verdicts on a restricted class. Exhaustion of the search says nothing about knot type.

The new module is `fastunknot.two_meridian`.

Enable the stage with `--two-meridian` or Python's `use_two_meridian=True`. It runs after the existing invariant filters and before optional group simplification or the remaining exact fallback. The ordinary defaults are unchanged. A successful negative test of representation existence returns `UNKNOT`; a successful existence test returns `KNOTTED`. Failure to obtain the required derivation, a local timeout or local work exhaustion returns `INCONCLUSIVE` and resumes the pipeline.

The adapter revalidates the spherical, one-component PD instead of trusting a presentation-level declaration of meridian provenance. At each crossing, union the two overpassing edge ports. The resulting classes are the actual Wirtinger arcs. The oriented underpassing ports and crossing sign give

$$x_v = x_o^{\epsilon} x_u x_o^{-\epsilon}, \qquad \epsilon \in \{1, -1\}.$$

Both directions are retained: knowing $x_o, x_u$ derives $x_v$, and knowing $x_o, x_v$ derives $x_u$. The adapter hashes the canonical validated PD and binds every certificate to that input. A connected-sum factor uses its own PD and its own meridians; the original source braid is not substituted for a factor presentation.

## 62.2 A seed derivation is an elimination certificate

A seed set consists of one or two *original* Wirtinger arcs. A queue propagates the crossing rules whose dependencies have become known. A rule with identical over and source arcs has one distinct dependency, not two. Every accepted step records a previously unknown target, the original crossing index and the direction. Its two dependencies have already been derived, so the trace is acyclic and has one step per nonseed arc.

For a fixed seed set this computes the least closed set of known arcs. Induction shows that every queued step belongs to every rule-closed superset of the seeds. Conversely, when the queue is empty, every rule with known dependencies has already fired. Therefore queue failure cannot be repaired by choosing a different order of rules with those same seeds. The initial search tried all single seeds and then all unordered pairs, subject to its caps. Section 77 now removes proved unsuccessful pairs under a checked structural condition while preserving the first successful trace. This recognizes the diagram-specific two-seed class exactly when uncapped. It does not search all diagrams of the same knot.

Replay uses the recorded rules to define a word $w_a$ in the seed meridians for every arc $a$. The chosen crossing relation is a valid Tietze elimination of its new target. Substituting the definitions into *every original crossing relation*, including unused ones, preserves the original group presentation. Equivalently, assigning values to the seeds extends uniquely along the trace, and is a representation of the knot group exactly when all substituted original relations hold. No omitted residual relation can create a spurious representation. An arbitrary two-generator presentation obtained by a separate group search does not suffice: its generators need not be meridians, even if their abelianized exponents are $\pm 1$.

The delivered prototype also minimizes formal degrees for fixed seeds. The maintained two-seed path needs only a valid derivation and uses ordinary queue closure. It does not implement that

degree optimization or the report's general multivariate real-feasibility compiler.

## 62.3 Why two traceless meridians reduce to integer arithmetic

Kronheimer–Mrowka's Corollary 7.17 states that every nontrivial knot admits an irreducible SU(2) representation sending a chosen meridian to $i$ [49]. All positively oriented Wirtinger meridians are conjugate; their images therefore have trace zero. Conversely, the infinite cyclic group of the unknot has only abelian images. For a checked two-seed presentation, nontriviality is thus equivalent to existence of noncommuting unit pure imaginary quaternion values $A, B$ satisfying the substituted relations. If the seeds commute, every derived value commutes, so such a representation is not irreducible.

Set $T = AB$. Since $A^2 = B^2 = -1$ and $ATA^{-1} = T^{-1}$, all word values have normal form

$$(-1)^e T^k A^s, \qquad e, s \in \{0, 1\}, \quad k \in \mathbb{Z}.$$

The seed triples are $A = (0, 0, 1)$ and $B = (1, -1, 1)$. Direct multiplication gives

$$\begin{aligned}(e, k, s)(f, \ell, t) &= (e + f + st \bmod 2,\ k + (-1)^s \ell,\ s + t \bmod 2),\\ (e, k, s)^{-1} &= (e + s \bmod 2,\ (-1)^{s+1} k,\ s).\end{aligned}$$

These operations evaluate all derivation words without expanding them. The checker forms each original relation as the predicted outgoing meridian times the inverse of the actual outgoing value.

For noncommuting $A, B$, write $T = \cos\phi + J \sin\phi$ with $0 < \phi < \pi$. Every such phase occurs, for example with $A = i$ and $B = -\cos\phi\, i + \sin\phi\, j$. Terms with $s = 1$ lie in the orthogonal quaternion plane and cannot equal the identity. For a relation with $s = 0$, the remaining condition is

$$k_j \phi \equiv e_j \pi \pmod{2\pi}.$$

A relation with $k_j = 0, e_j = 1$ is immediately inconsistent. Otherwise put $d = \gcd_j |k_j|$. If $d = 0$, all relations are identities and any interior phase, such as $\pi/2$, supplies a noncommuting representation.

If $d > 0$, at least one $k_j/d$ is odd. Its central bit fixes a parity $e$. The equations are compatible exactly when

$$e_j \equiv (k_j/d)e \pmod 2 \quad \text{for every } j.$$

To see necessity, a Bézout combination of the integers $k_j/d$ shows that $d\phi/\pi$ is an integer $q$, and each equation fixes $(k_j/d)q \equiv e_j \pmod 2$. An odd quotient determines $q \bmod 2$. Conversely, the displayed compatibility makes every integer $q \equiv e \pmod 2$ a simultaneous solution with $\phi = q\pi/d$. The strict interior condition requires $1 \le q \le d - 1$, so there are

$$\left\lfloor \frac{d - 1 + e}{2} \right\rfloor$$

allowed phases. A positive count supplies the compact witness $q = 1$ for odd parity or $q = 2$ for even parity. A zero count leaves only commuting endpoint phases and certifies the unknot. Floating-point trigonometry and a general real-algebraic solver are unnecessary.

For one seed, every crossing derivation is formally the same meridian: a conjugate of $A$ by $A^{\pm 1}$ is $A$. All original relations become identities, the group is cyclic, and the diagram is an unknot. The zero-crossing diagram is handled directly.

## 62.4 Completeness, complexity and certificate scope

The uncapped standalone algorithm is complete on diagrams admitting a derivation from at most two seeds. There are $O(n^2)$ seed sets; each closure uses $O(n)$ incidence and queue operations.

Indices have $O(\log(n+2))$ bits. Hence seed search costs $O(n^3 \log(n+2))$ bit operations with an elementary model.

For a successful derivation, the maximum absolute exponent grows by at most a fixed factor plus a constant at each step, so it is at most $3^{O(n)}$. Every stored integer consequently has $O(n)$ bits. There are $O(n)$ word operations and relations. Even allowing $O(n)$ long-division steps at $O(n^2)$ cost for each of the $O(n)$ gcd computations gives the conservative complete bound

$$O(n^4 \log(n+2))$$

on this class, including PD preparation, substitution, arithmetic and replay. This deliberately loose bound does not depend on expanding a word of length $3^{O(n)}$. The trace and relation triples occupy $O(n^2)$ bits. More efficient integer arithmetic can improve this accounting but is not needed for polynomiality. This bound is for the standalone stage. The optional pipeline placement retains earlier invariant filters, whose costs must be added; an expensive Jones computation before this stage is not covered by this restricted-class polynomial bound.

This theorem does not cover arbitrary input knots, nor prove a cheap conversion of every diagram into the two-seed class. In particular, seed failure is not evidence of nontriviality. The general recognition fallback still has an exponential worst-case bound. The maintained faithful Jones query has a general $2^{O(\sqrt{n})}$ bound, with fixed polynomial factors absorbed, but equality of Jones polynomials does not finish recognition. The new two-seed theorem supplies another complete restricted class, not the missing general quasi-polynomial theorem.

A certificate records the version, input hash, arc count, seed indices, derivation trace, every relation triple, the gcd/parity calculation and the verdict. The public verifier reconstructs the signed presentation from the PD and replays the trace without rerunning seed search. It recomputes all relation triples and the final answer and compares the complete strict schema, including integer types. Producer and verifier share the PD parser and integer arithmetic; this is search-independent replay, not an independent implementation of topology or quaternion multiplication. Independent exact quaternion and whole-cube comparisons test that shared arithmetic separately.

The local defaults are 50 milliseconds, two million weighted work units and 10,000 attempted seed sets. Preparation, failed attempts, successful arithmetic and the mandatory replay share the same allowance; replay does not receive a fresh budget. The counter charges integer bit lengths but is a cooperative work metric, not an exact bit-operation count or a hard memory bound. Deadline and caller cancellation checks occur throughout the search and replay. Exhaustion before verified publication emits no certificate. A global pipeline limit retains the existing `UNKNOWN` semantics. Python callers can set all three local caps to `None` for exhaustive restricted-class recognition; the default finite caps are a practical probe.

## 62.5 Validation and full-pipeline measurements

Eight focused tests cover 300 rational-quaternion product/inverse comparisons, 120 small phase systems compared with explicit rational phases, signed PD relations compared with the existing full Wirtinger builder, and 80 seeded small classical diagrams compared with an independent expanded Khovanov cube. For every seed set of those small inputs, a separate fixed-point closure agrees with the queue algorithm. Mirrors, rotated underpassing ports and permuted crossing rows exercise orientation handling. Replay is tested with seed search disabled, forged schema fields and derivations, a mismatched PD, large integer JSON encoding, exact work boundaries, caller cancellation, local fallback and CLI dispatch. The connected-sum test checks the actual interlacement factor and rejects replay against the whole input. All 756 integrated tests pass with Regina installed in 174.738 seconds; the eight focused methods pass in 1.131 seconds.

The benchmark fixes the previous recognizer at `7c45c41bc`, loads it under the unchanged package dependencies, and runs it twice as an A/A control. The other arms use the maintained recognizer

with the new stage disabled or enabled. Source hashes are frozen before timing and verified afterward. All four arms revalidate a fresh PD and run the whole pipeline, including filters, certificate verification and fallback. Import time and construction of the previously reduced hard-input corpus are excluded. The corpus consists of the 15 existing compressed-group hard-unknot cases and five named controls: trefoil, figure-eight, Hard 8, Conway and Kinoshita–Terasaka.

Each input is tested under the ordinary pipeline, with a three-second global allowance, and the stronger compressed-group configuration, with 15 seconds for group search, 20 million group work units and an 18-second global allowance. Both configurations retain all existing filters and allow 50,000 Khovanov objects. There is one excluded warmup per input, configuration and arm, then five measured rounds with shuffled arm order. The optional stage uses its ordinary 50-millisecond local allowance. This gives 800 measured whole-recognition calls and 160 excluded warmup attempts. A separate one-shot standalone probe on each input omits the local time cap while retaining the work and attempt caps; those are discovery observations, not repeated timing comparisons. All known labels predate this recognizer.

Raw samples, completion status, local work counts and deduplicated certificate contents are retained in:

`fast/results/two_meridian_20261008.json`

Completed-time medians exclude censored runs. A paired ratio is formed only when both queries complete; it is the median of per-round ratios, not a ratio of medians. The following table displays timings only for arms completing all five measured queries and ratios only for five complete pairs.

| Input | $n$ | Ord. off | Ord. on | Ratio | Group off | Group on | Ratio |
|---|---|---|---|---|---|---|---|
| Survivor 00 | 16 | 12.45 | 3.14 | 3.84 | 18.05 | 3.09 | 5.85 |
| Survivor 01 | 15 | 13.90 | 15.40 | 0.90 | 16.49 | 17.71 | 0.91 |
| Survivor 02 | 15 | 5.05 | 2.19 | 2.26 | 16.46 | 2.02 | 8.18 |
| Survivor 03 | 13 | 3.97 | 1.79 | 2.22 | 11.83 | 1.84 | 6.45 |
| Mirror 03 | 13 | 4.05 | 1.83 | 2.33 | 25.49 | 1.80 | 14.15 |
| Survivor 04 | 13 | 3.49 | 1.76 | 1.97 | 10.37 | 1.84 | 5.55 |
| Survivor 05 | 13 | 4.38 | 1.84 | 2.40 | 19.91 | 1.86 | 10.79 |
| Survivor 06 | 12 | 3.98 | 1.84 | 2.16 | 7.89 | 1.85 | 4.27 |
| Survivor 07 | 12 | 3.48 | 1.88 | 1.85 | 22.84 | 1.94 | 11.55 |
| Survivor 08 | 11 | 3.13 | 1.51 | 2.06 | 7.79 | 1.52 | 5.14 |
| Mirror 08 | 11 | 3.16 | 1.53 | 2.04 | 20.88 | 1.61 | 13.08 |
| Survivor 09 | 11 | 3.08 | 1.52 | 2.00 | 7.80 | 1.53 | 5.17 |
| Survivor 10 | 11 | 2.94 | 1.53 | 1.94 | 7.10 | 1.60 | 4.45 |
| Survivor 11 | 11 | 3.37 | 1.63 | 2.07 | 7.68 | 1.66 | 4.61 |
| Gordian | 141 | limit | limit | – | 2091.80 | 2113.05 | 0.99 |
| Trefoil | 3 | 0.06 | 0.06 | 1.00 | 0.06 | 0.06 | 1.00 |
| Figure eight | 4 | 0.06 | 0.06 | 1.00 | 0.06 | 0.06 | 1.01 |
| Hard 8 | 8 | 0.68 | 1.02 | 0.67 | 2.56 | 1.05 | 2.41 |
| Conway | 11 | 1.04 | 1.01 | 1.03 | 1.02 | 1.01 | 1.00 |
| KT | 11 | 1.05 | 1.06 | 1.00 | 1.05 | 1.05 | 1.00 |

Median milliseconds for whole recognition, including PD validation and replay. Ratios are medians of paired off/on times; limits are censored. The standalone stage resolves 13 of the 15 established hard-unknot inputs. It exhausts all 120 seed sets on survivor 01, reaching at most seven of its 15 arcs. Gordian reaches the two-million-unit work cap after 4,609 attempts; that capped failure does not determine whether its diagram belongs to the class. Trefoil, figure-eight and Hard 8 are also conclusive; Conway and Kinoshita–Terasaka exhaust their 66 seed sets and remain inconclusive. Their ordinary pipeline verdicts come from earlier Jones obstructions.

For the 13 successful hard-corpus cases, ordinary full-recognition paired speedups range from 1.85 to 3.84, with a median of the 13 per-case ratios of 2.07. Against the compressed-group

configuration, the corresponding range is 4.27–14.15 and median 5.85. For example, mirror 03 changes from 25.49 to 1.80 milliseconds in the group configuration; the ordinary configuration changes from 4.05 to 1.83 milliseconds. These are measured whole-query improvements with replay included, on an existing finite corpus, not an asymptotic claim or an independent random-knot sample.

There are material counterexamples to unconditional dispatch. Hard 8 changes from 0.68 to 1.02 milliseconds in the ordinary pipeline: the original late path is cheaper than constructing this certificate. Its paired off/on ratio is 0.666, about a 50% slowdown, although the same probe improves the optional group configuration by a factor of 2.41. The unsuccessful search on survivor 01 gives ordinary/group ratios 0.903 and 0.914. Gordian's 50-millisecond probe times out in both configurations. All 20 measured ordinary Gordian queries exhaust the global allowance and remain censored; all group-configured Gordian queries complete using the original group certificate, with disabled/enabled medians 2.092/2.113 seconds and paired ratio 0.987. No timeout is reported as a successful three-second decision.

Overall, 780 of the 800 measured queries complete, every completed verdict agrees with the prior label, and 20 are censored. Ordinary A/A ratios range from 0.965 to 1.027 on completed pairs; group A/A ratios range from 0.946 to 1.052. Comparing the old recognizer to the current disabled arm also retains an ordinary survivor-00 outlier of 0.882, so small timing changes are not interpreted as improvements from argument validation or other incidental work. The large successful-stage improvements coexist with the retained regressions and limited corpus size. The implementation therefore remains opt-in. Choosing when to attempt it earlier, or when to skip it in favor of an already cheap path, requires a separately measured adaptive policy.

# 63 Endpoint patterns and certified overlap cutoffs

## 63.1 Extending the existing compressed-word path

The compressed-certificates delivery adds a useful continuation of Section 55: short endpoint patterns can have regularly spaced occurrences even when neither endpoint letter does, and a progression of candidate overlap lengths can have a proper initial segment of successes. The maintained implementation keeps the earlier uniform, short-period, sparse-letter and two-largest-overlap shortcuts first. Only their unresolved queries of length at least 128 reach the new continuation in `fastunknot/compressed_endpoint.py`. A failed structural certificate still invokes the complete occurrence-table matcher.

For words $x, y$ represented by binary straight-line programs, put

$$\mathcal{O}(x, y) = \{k : 1 \le k \le \min(|x|, |y|), \quad x[|x| - k : |x|] = y[0 : k]\}.$$

The output must describe *all* these lengths by disjoint arithmetic progressions, not merely certify one good overlap. Both operands may first be cropped to length $L = \min(|x|, |y|)$, retaining the suffix of $x$ and prefix of $y$. This does not change $\mathcal{O}$.

## 63.2 Summarizing every occurrence of a short pattern

Fix an explicit pattern $a$ of length $q$. For all its occurrence starts in a grammar word, retain the count, first and last start, and minimum and maximum successive gap. Empty and singleton sets have separate exact conventions. At a concatenation $uv$, occurrences fall into three disjoint, ordered groups: wholly in $u$, crossing the cut, and wholly in $v$. The middle group uses at most $q - 1$ possible shares from the suffix of $u$; retaining only the first and last $q - 1$ letters of each grammar word suffices to check every such occurrence. The last group is translated by $|u|$.

Concatenating two nonempty occurrence lists adds their counts and introduces exactly one new gap between the first list's last start and the second list's first start. Updating both gap extrema therefore gives an exact summary by induction. A set with at least two positions is an arithmetic progression precisely when its two gap extrema agree. Counts and gaps are arbitrary-size integers, not saturated counters or numerical iteration bounds.

For $g$ reachable rules, one pattern costs $O(q^2 g + g \log g)$ elementary operations and $O(qg)$ stored symbols and integer fields. The sorting term comes from the existing reachable-rule ordering. Coordinate and label bit lengths must still be charged in a bit model. The production trial lengths are $q \in \{1, 2, 4, 8\}$, in both endpoint directions; a bounded pattern prefix is a proposal, never evidence that the rest of a word is periodic.

## 63.3 The period makes candidate rank monotone

Let $a$ be the length-$q$ suffix of $x$. Every overlap of length at least $q$ ends at an occurrence of $a$ in $y$. Suppose these necessary candidate lengths form

$$K = \{b, b + d, \ldots, b + (r - 1)d\}, \qquad M = \max K,$$

with $r > 1$. Check by exact compressed equality that $P = y[0 : M]$ has period $d$, meaning $P[0 : M - d] = P[d : M]$. For every $k \in K$, the difference $M - k$ is a multiple of $d$, so

$$y[0 : k] = P[0 : k] = P[M - k : M].$$

Thus the candidate is a true overlap exactly when the length-$k$ suffixes of $x$ and $P$ agree. Equivalently,

$$\mathcal{O}(x, y) \cap [q, \infty) = K \cap [\, q, \mathrm{lcsuffix}(x, P)\,].$$

The dual argument starts with the length-$q$ prefix of $y$, takes candidate lengths $|x| - s$ from its starts $s$ in $x$, and checks period $d$ on $P = x[|x| - M : |x|]$. The cutoff is then $\mathrm{lcp}(P, y)$. The first candidate need not equal the period. Material outside the maximum candidate window is irrelevant to the period certificate.

Consequently exact overlap equality is true on an initial segment of the candidate ranks. The implementation first tests the largest rank. If it passes, every candidate passes. Otherwise it tests the smallest rank; failure there rules out all candidates. In the remaining case, probe backwards from the maximum at distances $1, 2, 4, \ldots$, stopping at a success or the known minimum, then bisect the bracket. With $b_{\mathrm{bad}}$ rejected top candidates, this takes

$$O(1 + \log(1 + b_{\mathrm{bad}})) \le O(\log(r + 1))$$

exact candidate comparisons. All-pass and all-fail cases use one and at most two comparisons, respectively. Exactly one rejected top candidate needs three comparisons, even when both the period and the candidate count have hundreds of bits. Computing a full LCP on a much larger numerical range is unnecessary.

An empty occurrence set excludes all lengths at least $q$. A singleton candidate is checked directly, without a period assumption. Every shorter length $1, \ldots, q - 1$ is checked separately. A successful trial therefore returns the complete overlap set: at most one progression of long overlaps and at most $q - 1$ short singletons. A nonprogression or failed exact period check gives no conclusion and tries another bounded anchor, then falls back.

## 63.4 Costs, integration and examples

If $H$ bounds the input grammar height, $c$ is the rank-comparison count and $E(G)$ bounds exact equality at the actual intermediate grammar size, a conservative successful-trial cost is

$$O(q^2 g + g \log g + H(q + c)) + O(q + c + 1)E(G), \qquad G = O(g + H(q + c))$$

for a fresh arena containing the input and query artifacts. Earlier failed trials have their own bounded preparation and equality costs. This is an operation bound, not a unit-cost assumption for binary positions or letter labels. In particular, a middle cutoff can still require many slices and exhaust the node allowance. No general linear-time claim is made.

All preparation and exact checks use the caller's existing word-arena budget. Local resource exceptions and global cancellation propagate. Neither becomes an empty overlap answer; incomplete queries are not inserted in the overlap cache. The old endpoint counters are preserved, while new preparation, period, cutoff and allocation counters have the `lcs_anchor_` prefix. The existing longest-common-substring critical-extension calculation receives the same complete progression interface. Forty separately checked clipped families verify that its finite critical candidates still attain the maximum over every explicit overlap.

For $n, m \geq 2$, the words

$$((21)^n 1)^m, \qquad ((12)^n 11)^m$$

have dense single-letter endpoints. A suffix bigram in the first case gives all positive multiples of $2n+1$ through $m(2n+1)$. A four-letter anchor in the second gives all positive multiples of $2n+2$ through $m(2n+2)$, together with the short overlap one. Their grammars have $O(\log n + \log m)$ rules. For $P_n = (12)^n 3$, the pair

$$x = 4^{2n+1} P_n^{m-1}, \qquad y = P_n^m$$

has exactly one rejected top candidate, so the three-comparison cutoff applies. The older all-success marker family remains on the original two-largest path. A damaged word whose relevant windows have no certified period still needs the general matcher; the extension is not a universal pattern shortcut.

### 63.5 Validation and comparison with the current baseline

The 12 new test methods cover exact signed occurrence summaries, exhaustive short-word overlaps, 360 random parse pairs, eight-letter anchors, a depth-4000 grammar, 500-bit periods and counts, interior and near-end cutoffs, allocation and work limits, and cancellation without partial cache entries. The four previous endpoint-progression tests still pass, including a trap proving that their successful old shortcut never invokes the new module. Existing general fallback tests explicitly disable the additional optional shortcut so that they continue exercising the table path. All 768 maintained tests pass in 167.676 seconds with Regina installed.

The benchmark fixes the old matcher at `8a9c08bcf` inside otherwise common current word, relator and recognition code. It runs old, identical A/A control and maintained clipping arms, with one excluded warmup followed by five measured rounds in shuffled order. Source hashes are frozen before measurement and checked afterward. Each compressed operation uses fresh state, two million shared work units and at most 100,000 grammar nodes. Grammar construction is included in timings; output containment and cardinality checks are outside the overlap-query clock. Numerical word lengths are never used to enumerate the large expected answers.

Eight overlap families use parameters $n = 2^b$ at $b = 8, 64, 500$, covering new dense endpoints and cutoff classes, successful old shortcuts, and uncertifiable periods. Whole phased LCS and cyclic relator searches use the dense bigram family at $b = 8, 32$; the relator scope includes failed whole-donor search, the complete cyclic overlap query and checked application. A pure-power quotient control retains the established path. These are compressed algebraic inputs, not diagrams with $2^b$ crossings.

| Family | Bits | Before | A/A | Clipping | Ratio |
|---|---|---|---|---|---|
| Marker control | 8 | 0.821 | 0.862 | 0.870 | 0.94 |
| Marker control | 64 | 6.120 | 6.187 | 6.046 | 1.00 |
| Marker control | 500 | 51.187 | 50.593 | 52.121 | 1.00 |
| Dense bigram | 8 | 307.677 | 310.654 | 1.259 | 244.48 |
| Dense bigram | 64 | limit | limit | 8.145 | – |
| Dense bigram | 500 | limit | limit | 65.936 | – |
| Dense fourgram | 8 | 308.246 | 313.962 | 1.530 | 201.36 |
| Dense fourgram | 64 | limit | limit | 10.193 | – |
| Dense fourgram | 500 | limit | limit | 85.070 | – |
| One rejected | 8 | 329.133 | 344.696 | 1.198 | 279.34 |
| One rejected | 64 | limit | limit | 7.965 | – |
| One rejected | 500 | limit | limit | 62.965 | – |
| Half rejected | 8 | 308.250 | 303.671 | 2.076 | 150.16 |
| Half rejected | 64 | limit | limit | 62.613 | – |
| Half rejected | 500 | limit | limit | limit | – |
| Period rejected | 8 | 384.168 | 378.957 | 378.362 | 1.02 |
| Period rejected | 64 | limit | limit | limit | – |
| Period rejected | 500 | limit | limit | limit | – |
| Short-period control | 8 | 0.090 | 0.091 | 0.099 | 0.91 |
| Short-period control | 64 | 0.353 | 0.331 | 0.330 | 1.07 |
| Short-period control | 500 | 2.243 | 2.281 | 2.243 | 1.00 |
| Sparse control | 8 | 0.798 | 0.800 | 0.796 | 1.00 |
| Sparse control | 64 | 1.139 | 1.122 | 1.126 | 1.01 |
| Sparse control | 500 | 3.737 | 3.701 | 3.753 | 0.99 |

Median milliseconds; ratios are medians of complete paired before/clipping times. Censored runs are shown as limits and excluded from ratios.

| Family | Bits | Before | A/A | Clipping | Ratio |
|---|---|---|---|---|---|
| Phased LCS | 8 | 940.259 | 934.837 | 59.307 | 15.88 |
| Phased LCS | 32 | limit | limit | 1053.355 | – |
| Phased relator | 8 | limit | limit | 155.734 | – |
| Phased relator | 32 | limit | limit | limit | – |
| Quotient control | 8 | 0.344 | 0.340 | 0.335 | 1.02 |
| Quotient control | 32 | 0.963 | 0.962 | 0.962 | 1.00 |

Median milliseconds; ratios are medians of complete paired before/clipping times. Censored runs are shown as limits and excluded from ratios.

The complete-query comparison retains all 15 established hard-unknot inputs, all existing filters, compressed group search and certificate replay. The new two-meridian stage is disabled so that it does not bypass the matcher being tested. Every query starts from a freshly validated PD. Group work is capped at 20 million units and 15 seconds, with an 18-second global allowance and 50,000 Khovanov objects. Certificate digests and complete search counters are retained outside the timed region. Timeout records never enter completed-time medians or paired speedup denominators.

| Input | Before | A/A | Clipping | Ratio |
|---|---|---|---|---|
| Survivor 00 | 17.824 | 17.633 | 17.360 | 1.00 |
| Survivor 01 | 14.852 | 14.786 | 14.795 | 1.00 |
| Survivor 02 | 16.129 | 16.147 | 16.122 | 1.00 |
| Survivor 03 | 11.309 | 11.290 | 11.266 | 1.00 |
| Mirror 03 | 24.843 | 24.946 | 25.149 | 0.99 |
| Survivor 04 | 9.927 | 10.000 | 9.954 | 1.00 |
| Survivor 05 | 19.091 | 19.155 | 19.104 | 1.00 |
| Survivor 06 | 7.834 | 7.662 | 7.626 | 1.00 |
| Survivor 07 | 19.815 | 19.931 | 19.789 | 1.00 |
| Survivor 08 | 7.694 | 7.674 | 7.687 | 1.00 |
| Mirror 08 | 20.825 | 20.633 | 20.695 | 1.02 |
| Survivor 09 | 7.600 | 7.528 | 7.502 | 1.00 |
| Survivor 10 | 6.518 | 6.513 | 6.650 | 0.97 |
| Survivor 11 | 7.210 | 7.122 | 7.142 | 1.01 |
| Gordian | 1968.756 | 1988.703 | 1972.944 | 1.00 |

Median milliseconds; ratios are medians of complete paired before/clipping times. Censored runs are shown as limits and excluded from ratios.

All raw samples and source hashes are retained in:

`fast/results/endpoint_clipping_20261008.json`

The record contains 675 measured calls: 360 complete-overlap attempts, 90 whole algebraic operations and 225 whole-knot queries, plus 135 excluded warmups. Of the measured calls, 525 complete and 150 exhaust an allowance. All completed answers agree with their structural oracle or prior knot label; every completed knot arm produces the same certificate digest as its paired competitors.

At the eight-bit parameter, dense bigram and fourgram overlap queries improve by paired factors 244.48 and 201.36, respectively. The one-rejected family improves by 279.34 and the half-rejected family by 150.16. At 500 bits, dense bigram, dense fourgram and one-rejected queries complete in median 65.94, 85.07 and 62.96 milliseconds; both old arms exhaust the work cap. Those capped comparisons demonstrate increased capacity, not numerical speedup ratios. The one-rejected query uses exactly three rank comparisons at all three parameter sizes. The established all-success marker, short-period and sparse controls never invoke the new continuation.

The finite comparison also exhibits its limits. At 64 bits the half-rejected family completes with 128 rank comparisons; at 500 bits it exhausts the 100,000-node allowance after 72 comparisons, before finding its cutoff. The failed-period family remains on the complete fallback and still exhausts the work cap at 64 and 500 bits. These are unresolved matching computations, not certificates that the overlap set is empty.

The whole phased LCS improves from 940.26 to 59.31 milliseconds at eight bits, a paired factor 15.88, and completes at 32 bits where the old arms exhaust the cap. The full phased cyclic-relator query and checked application complete in 155.73 milliseconds at eight bits where both old arms are capped. At 32 bits that operation remains capped even with eight successful anchor hits: other matching and extension work still dominates the shared allowance. The quotient control remains on its old path.

All 225 measured whole-knot queries complete, but their new anchor activation counts are zero. Their before/clipping ratios range from 0.967 to 1.018, while their A/A ratios range from 0.993 to 1.025. There is consequently no measured recognition speedup on this corpus. For completed primitive pairs, A/A ratios range from 0.961 to 1.067; small changes on unaffected controls are retained as timing variation. The supported conclusion is broader exact compressed-word capacity and faster selected complete algebraic queries. Neither a general relator-search move bound nor a quasi-polynomial knot recognizer follows from these results.

# 64 Port-register multiplicities: a checked consumer and the missing producer

## 64.1 The new delivery and what it changes

Commit `820c8b3f4` supplies:

`docs/incoming/unknot_port_register_20261008.zip`

All 53 manifest entries verify. The later intake commit `4c4c2254c` places its non-manifest files in `reports/40/`; the original ZIP remains recoverable from its arrival commit. The report proposes a more expressive multiplicity representation than literal direct-sum sharing: small template complexes repeated over large finite-state domains, with nonzero coupling between copies through finitely many scalar rank-one channels, called ports. A differential support graph can be connected even with one such port. These algebraic ports are not topological boundary circles.

This is a relevant response to the explicit-multiplicity obstruction. The consumer computes exact homology in polynomial time in its supplied compact description. It does not construct that description from an arbitrary diagram or retain the relative chain maps required at an open tangle frontier. The report's large virtual complexes are abstract algebraic examples; its small knot examples first expand their cubes. Accordingly, this review adds an actual maintained-scanner compatibility audit, but no production recognition branch or new general asymptotic guarantee.

## 64.2 Why the small core remains valid when singular

Work over $\mathbb{F}_2$ and write

$$A = \bigoplus_{\nu} A_\nu \otimes I_{\Omega_\nu}, \qquad D = A + UV,$$

where every template $A_\nu$ is square-zero and homogeneous, and $U, V$ have $r$ columns and rows, respectively. Before homology is computed, the full $D$ must also be proved homogeneous and square-zero.

For any matrix $A$, rectangular if necessary, choose a reflexive generalized inverse $h$ with $AhA = A$ and $hAh = h$. An invertible rank minor constructs one by ordinary elimination. Put

$$U_0 = (I - Ah)U, \qquad V_0 = V(I - hA), \qquad E = I_r + VhU.$$

Let $\overline{U}$ be a row basis of $U_0$ and $\overline{V}$ a column basis of $V_0$, with dimensions $s, t \le r$. The retained core is

$$K = \begin{pmatrix} 0_{s\times t} & \overline{U} \\ \overline{V} & E \end{pmatrix}, \qquad \operatorname{rank}(D) = \operatorname{rank}(A) + \operatorname{rank}(K) - r.$$

Here is the rank argument. The bordered matrix $\left(\begin{smallmatrix} A & U \\ V & I_r \end{smallmatrix}\right)$ has rank $r + \operatorname{rank}(A+UV)$ by eliminating its bottom identity block. Alternatively, put $A, h$ into compatible identity-plus-zero bases and eliminate the rank-rank$(A)$ block. What remains is $\left(\begin{smallmatrix} 0 & U_0 \\ V_0 & E \end{smallmatrix}\right)$ in quotient coordinates. Independent operations on its zero-corner rows and columns retain precisely the displayed row and column bases. This contributes $\operatorname{rank}(K)$ to the rank. Equating the two expressions proves the formula without inverting $E$.

For a valid differential, $\beta(D) = \dim C - 2\operatorname{rank}(D)$, giving

$$\beta(D) = \beta(A) + 2r - 2\operatorname{rank}(K).$$

Singularity is essential: even the scalar update $A = U = V = (1)$ has $A + UV = 0$ and $E = 0$. An inverse-based update would fail on this elementary case, while the bordered formula gives the correct rank zero.

## 64.3 Finite-state coordinates supply ranks and contractions

Each block uses a layered weighted register

$$F(x_1, \dots, x_m) = \alpha T_1^{x_1} \cdots T_m^{x_m} O$$

whose simultaneous outputs are all local $U, V$ coordinates. All $m$ layers are explicitly listed; the register length is not an exponentially large repetition count supplied in binary. At each layer retain an independent subset of the actual rows obtained by multiplying the previous basis by the two transition matrices. Linearity proves inductively that these at-most-$B$ rows span all reachable states, where $B$ is the maximum layer width. Store a predecessor and a bit for every retained witness. This constructs actual word witnesses in $O(mB)$ records instead of copying all prefixes. The underlying forward-space method is classical weighted-automaton linear algebra [50].

These evaluation rows, followed by template actions, span all quotient rows and columns needed in $K$. Pairings such as $VhU$ require a separate parity contraction. For terminal state rows $w(x)$, define

$$W_0 = \alpha^T \alpha, \qquad W_j = \sum_{b \in \{0,1\}} (T_j^b)^T W_{j-1} T_j^b.$$

Then $W_m = \sum_x w(x)^T w(x)$ by induction. Terminal coefficient vectors $a, b$ give

$$\sum_x (w(x)a)(w(x)b) = a^T W_m b.$$

This is not a rank test: a nonzero constant function on a positive-length Boolean cube has evaluation rank one and zero self-pairing in characteristic two. Reachability determines ranks; the matrices $W_j$ compute contractions only.

The full square-zero condition is likewise finite-state:

$$D^2 = [\, AU + U(VU) \quad U \,] \begin{bmatrix} V \\ VA \end{bmatrix}.$$

After computing $VU$, both factors are linear transforms of the register outputs. Every pairing between a row-span basis of the first and a column-span basis of the second must vanish. Bilinearity makes this equivalent to vanishing of every entry of $D^2$, rather than a sample of entries. Forbidden homological-degree coordinates are checked on spanning evaluations in the same way.

A supplied deterministic language can restrict the multiplicity domain. Tensoring its state transitions with the weighted register masks rejected words to zero, with effective width at most the product of the two widths. Integer dynamic programming counts accepted words exactly; counting paths in an arbitrary nondeterministic automaton would not suffice. If masking is implemented by padding to the full Boolean domain, subtract the known uncoupled complement *before* saturating any homology count. Cardinalities are integers, not parities.

## 64.4 Fixed-base minimization and a necessary cancellation budget

For a fixed base $A$, the fewest scalar ports is $r_{\min} = \operatorname{rank}(D - A)$. The report obtains it without enumerating the domain. Within one homogeneous degree group, take row and column bases $P, Q$ for $U, V$ and suitable one-sided inverses. Then $U = LP$, $V = QR$, with $L$ full column rank and $R$ full row rank, so $\operatorname{rank}(UV) = \operatorname{rank}(PQ)$. A rank factorization of this small product transforms only the terminal register outputs. Transition matrices, widths and register lengths are unchanged. Summing over disjoint degree blocks preserves homogeneity and gives the global minimum for this base. Choosing a better base is a separate problem.

Rank subadditivity gives the sharp budget

$$|\beta(D) - \beta(A)| \le 2 \operatorname{rank}(D - A) \le 2r.$$

Thus $\beta(A) > 2r + 2$ already rules out rank two, and the sufficient base cap for this decision is $\min(\beta(A), 2r + 3)$, rather than a fixed cap of three. For example, one arrow on four zero-base generators leaves homology two; one arrow on six leaves homology four. The nontrivial port core is identical, and capping both base dimensions at three would lose the distinction. This does not invalidate saturation of genuinely isolated direct sums in the maintained scanner: those summands have no coupling to cancel their homology.

A useful description of an unknot must therefore have a mostly contractible base, unless its port count is large. Starting with an enormous zero base cannot evade this inequality. Moreover, passive multiplicity extension itself can defeat a fixed-base port bound:

$$\operatorname{rank}((D - A) \otimes I_q) = q \operatorname{rank}(D - A).$$

This follows by tensoring a rank-normal form with $I_q$ and is independent of register width. Appending $t$ passive Boolean factors can force $2^t$ times as many scalar ports. A native construction must absorb such replicated updates into new templates, cancel them with valid relative operations, or use a richer structured coupling representation with its own size theorem.

## 64.5 The conditional bound and the next constructive obligation

For block size $d_\nu$, explicitly listed register length $m_\nu$, effective width $B_\nu$ and total port count $r$, the report supplies the elementary field-operation bound

$$O\left(\sum_{\nu} \left[d_\nu^3 + (m_\nu + 1)B_\nu^3 + d_\nu^2 B_\nu r + d_\nu B_\nu^2 r + d_\nu B_\nu r^2\right] + r^3\right).$$

Dimensions and language counts also have polynomial bit cost in the explicit descriptor. Hence a proved quasi-polynomial-time producer and verifier of a quasi-polynomial-size description would give quasi-polynomial recognition. A succinct description found only after an exponential scan does not meet that premise. Nor does a good description of a closed scalar complex supply chain-homotopy data for arbitrary open-frontier cobordism gluing.

The next useful target is therefore a producer that selects near-contractible templates *before* explicit copy allocation, preserves typed relative maps until closure, and bounds effective widths and minimum port ranks after every operation. Its measurements must include failed admission attempts, conversion, exact validity checks and fallback. The tensor rank identity is an early rejection criterion for a proposed fixed-base construction, not a reason to assume that all bases are impossible.

## 64.6 Reproduced evidence and actual maintained-scanner compatibility

The delivered 48 test methods pass locally in 1.517 seconds. Its separate validation script reproduces 4,352 exhaustive rank updates, including 1,728 singular middle matrices, and 404 independent signed braid-cube comparisons. Those are 210 knot and 194 link presentations, not 404 distinct knot types. The recorded validation run takes 4.070 seconds; these are correctness-audit times, not competitive benchmark results.

The maintained audit adds 30 nonempty, validated one-component diagrams: trefoil, figure-eight and Hard 8 with their mirrors, plus seeded small braids. For each diagram it checks original, reversed and shuffled crossing orders, with full cancellation or the final two crossings left uncancelled. All 180 closed scalar complexes agree with an independent expanded reduced cube, doubled to the unreduced rank, and with ordinary subsequent cancellation. The adapter rejects all 144 encountered nonclosed first-crossing frontiers. Port minimization preserves every answer, and all prototype topological verdict fields remain unset.

This audit includes actual maintained list-based scanner states, rather than only API-shaped fixtures. Nevertheless, all converted registers have length zero. The largest input has 354 already allocated scalar generators and needs 209 raw ports; the adapter has not compressed an exponentially large state. The delivered benchmark also reports slowdowns on all four already-expanded small knot cubes. No automatic production dispatch follows from these checks. The singular-safe consumer and the sharp constraints on a future producer are the useful retained results at this checkpoint.

The reproducible audit and detailed records are:

`data/audit_port_register.py`
`data/port-register-maintained-audit.json`
`data/port-register-delivery-validation.json`

The audit recovers the original archive from its arrival commit if intake has retired the drop-zone ZIP. It extracts into a temporary directory, checks its hash and all manifest entries, then uses the prototype without installing it into the production package. Its source hashes bind the maintained parser, scanner, independent cube oracle and audit script. The prototype's own replay checker recomputes through its solver; it is not an independently implemented rank proof checker. The new compatibility checks do not change that boundary.

# 65 Integral shears: exact local optimization and a phase-sensitive bound

Commit `2b6068c604` supplies `docs/incoming/unknot_integral_shears_20261008.zip`. We verified all 42 entries of its checksum manifest before executing its code. Intake commit `7ad289e2b` subsequently placed the delivery in `reports/41`; all 42 retained files are byte-identical to the archive. Only the checksum manifest was omitted. The native audit recovers the original ZIP from its arrival commit, so it remains reproducible after intake removal. The full archive SHA256 and the reproduced results are recorded in `data/integral-shears-review.json`. Its useful contribution is an exact joint optimization of commuting automorphisms, together with a direct compressed output construction and a separately checked optimality witness. This is stronger than optimizing the power along one selected Whitehead direction. The distinction between this local result and a complete recognition algorithm is essential: neither a general phase bound nor a complete terminal-state theorem is supplied.

## 65.1 From powered Whitehead moves to integer tensions

Fix a positive alive generator $a$. For each signed letter $x$ with $|x| \neq a$, choose an integer $z_x$ and define

$$\Phi_{a,z}(a) = a, \qquad \Phi_{a,z}(x) = a^{-z_{-x}} x a^{z_x}.$$

The inverse-letter formula is consistent, and substitution gives $\Phi_{a,z}\Phi_{a,t} = \Phi_{a,z+t}$. Thus every potential vector defines an automorphism, with inverse $\Phi_{a,-z}$. A powered Whitehead direction is the special case $z_x = k[x \in A]$. Conversely, the integral shear factors into at most $2(r-1)$ powered singleton directions at rank $r$. Negative potentials use the inverse multiplier. These factors commute, although individual factors can temporarily increase length.

For a literally cyclically reduced relator, retain the cyclic sequence of non-$a^{\pm 1}$ letters. If consecutive retained letters are $u, y$, with intervening run $a^e$, write $v = -y$. The transformed gap exponent is

$$e + z_u - z_v.$$

No retained letters cancel. Indeed, a first such cancellation would require $y = -u$, hence $v = u$; its exponent remains $e$, which was nonzero by the input reduction contract. This argument includes the cyclic seam. Pure multiplier powers are fixed. Consequently the exact total cyclic length is

$$L_a(z) = C_a + \sum_{(u,v,e)} w_{uve}|e + z_u - z_v|, \tag{15}$$

where $C_a$ counts retained letters and pure-root lengths. Duplicate relator slots contribute with multiplicity. Self-loop records contribute constant costs and must be retained.

The gap histogram can be computed on a repetition grammar without expanding its words. Each rule stores its first and last retained letters and its signed outer multiplier runs. A concatenation creates at most one new gap; a repetition creates a copy-seam gap with multiplicity $k - 1$. Reverse propagation of root multiplicities counts all rule occurrences, and a separate root seam closes each non-pure relator. There are $O(s + m)$ gap records for $s$ rules and $m$ roots. Large multiplicities and exponents are binary integers, not iteration counts. The verifier instead builds bottom-up histograms, providing a distinct polynomial recurrence.

## 65.2 An exact certificate and a polynomial solver

For each gap choose an integer $f_{uve}$ satisfying

$$|f_{uve}| \le w_{uve}, \qquad \operatorname{div} f = 0.$$

For any potentials $z$, termwise comparison and cancellation of the vertex potentials give

$$\sum w_{uve}|e + z_u - z_v| \;\ge\; \sum f_{uve}(e + z_u - z_v) = \sum e f_{uve}.$$

Equality certifies a global integer minimum. The checker needs only exact support, integer types, capacities, conservation and equality of these two objectives. It does not rerun the optimizer or rely on hashes for mathematical word equality.

The delivered optimizer replaces a gap by opposite arcs of capacity $w$ and costs $-e, e$. A circulation on these arcs yields $f = g^+ - g^-$. Residual-feasible vertex potentials provide the primal solution through $z = -\pi$. Its capacity-scaling implementation processes bits from most to least significant: it doubles old capacities and flows, installs the next capacity bits, saturates new negative-reduced-cost arcs, then repairs the resulting imbalances along shortest residual paths. Each newly opened negative arc has capacity at most one, so the total imbalance is at most the number of arcs. Each integral repair reduces this total. There are at most $A$ repairs per capacity bit, not one repair per unit of the original large capacities.

Truncated shortest-path distance updates preserve nonnegative residual reduced costs, including unreachable vertices. Bellman–Ford recalibration after repairs bounds potential bit lengths by simple-path costs. This gives a polynomial bit-time algorithm in graph size and the binary input lengths. The zero-imbalance self-loop case is included. Balanced offset graphs can solve every gap exactly by graph traversal. Forests of parallel-edge bundles admit weighted-median differences on each bundle; acyclicity makes those independent choices compatible. The same primal–dual checker validates the shortcut results. These observations explain why forcing the general flow solver on every structured input is unnecessary.

The archive's work budget charges circulation operations. It is not a global service allowance: histogram construction, balanced/forest shortcuts, certificate checking and grammar allocation also need host accounting before any bounded production probe is introduced.

## 65.3 Why the retained representation matters

For each non-pure rule $U$, construct a transformed core $M_U$ between its first and last retained letters. Concatenation joins child cores with the one transformed multiplier run. If $U = V^k$ is

non-pure and $h$ is the transformed copy-seam exponent, use

$$M_U = (M_V a^h)^{k-1} M_V.$$

Append the transformed cyclic seam at each root. Skeleton invariance proves that the output is already freely and cyclically reduced. This construction uses no generic free-reduction or equality queries. Each old rule introduces a bounded number of new rules, provided repetition rules with binary exponents remain part of the representation.

Writing $G = s + m + r + 2$, the report's conservative bound is $G' \le 8G$. A different verifier recurrence stores decorated images with outer multiplier runs and associates the repeated core in the opposite order. It has the same constant-factor growth property. During strictly shortening phases, expanded length is bounded by its initial value and integer bit lengths grow only with the initial encoding size and the logarithm of the rule count. Accordingly $q$ such phases, their local optimality certificates and their independent replay cost

$$\mathrm{poly}(\sigma)\, 2^{O(q)}$$

in the initial binary encoding size $\sigma$. Polynomial factors in $q$ are absorbed into the exponential; the zero-phase case remains polynomial. For $\sigma \le n^{O(1)}$ and $q = O(\log^2(n+2))$, this is $2^{O(\log^2(n+2))}$, a quasi-polynomial bound. With logarithmically many phases the same computation is polynomial.

This is a conditional class bound, not a general unknot theorem. A knot algorithm must also provide checked full-presentation provenance, a phase bound, a sufficient terminal predicate, and bounds for interleaved relator operations. In particular, the maintained `WordArena` represents powers by binary concatenation. Exporting every new repetition rule to that arena adds a bit-length factor per phase and does not inherit the displayed constant-factor growth theorem. Legacy substitution and reduction have their own costs as well.

## 65.4 A concrete limit of automorphic descent

The pure-power terminal predicate has a simple complete algebraic meaning. If each nonempty relator is a power of a single alive generator, take the gcd $d_j$ of its nonzero exponents, with $d_j = 0$ when that generator has no relation. The group is the free product of the cyclic factors $C_{d_j}$, where $C_0 = \mathbb{Z}$ and $C_1$ is trivial. It is infinite cyclic precisely when exactly one $d_j$ is zero and all others are one. Absent generators must therefore remain in the alive set.

Even unlimited strictly shortening automorphisms do not always reach this terminal class. For $N \ge 2$, consider

$$\langle a, b \mid a^N, a^{N+1}, [a, b] \rangle.$$

The first two relators kill $a$, so this is an infinite-cyclic group, but it has no singleton relation and the commutator is not a pure-power root. Under any free-group automorphism the first two relators have combined cyclic length at least $2N + 1$. The commutator image is a nontrivial element with zero abelianization, hence cyclic length at least four. The original total $2N + 5$ already attains that lower bound. This is a plateau under all free-group automorphisms, not merely a poor choice of multiplier.

Relator combinations or other Tietze operations can escape this example; our maintained overlap and relator-power machinery is relevant to that separate task. The example is an abstract presentation, not a family of hard knot diagrams. It establishes that a phase-count result by itself would still leave a terminal-completeness obligation.

## 65.5 Reproduced delivery and maintained-source experiments

All 51 delivered unit tests passed in 3.325 seconds. The deterministic audit also reproduced 1,200 circulation instances, 360 stages from 180 connected braid closures, 1,012 individual relator-image

comparisons and three large serialized presentation certificates. It completed in 5.220 seconds. These are correctness checks, not comparative performance measurements.

The delivery's source-pinned gateway passed all 160 comparisons against the actual maintained `WordArena`, selected-power formulas and legacy replay. All four pinned Git blobs match; no unpinned override was required. The independently implemented Artin-presentation corpus in the delivery is useful, but it does not exercise our actual PD presentation constructor or singleton scheduler. We therefore added `data/audit_integral_shears.py`.

This audit reconstructs full native Wirtinger presentations from 30 validated one-component diagrams already recorded in the port-register audit. For each diagram it checks the raw state and the state after the maintained `_search` performs its singleton prelude. The exact selected-power comparator minimizes the power along the maintained cut direction, even where the production heuristic would skip that extra query. It is a local length comparator, not a timing of the production policy.

On all 60 states the delivered legacy macro replay agrees, up to cyclic rotation, with both the direct constructor and explicit letter substitution. This gives 234 relator-image checks. The joint query strictly improves the selected-power length on 12 of the 30 raw states and on none of the 30 post-singleton states. These are paired stages of the same diagrams, not 60 independent knot samples.

We then continue the maintained search from the legacy representatives and the corresponding recorded moves. Forty-six traces finish and pass both explicit and compressed independent full-source certificate verification; 14 stall and none hits the continuation cap. A stall remains inconclusive. These checks establish compatibility of successful composed traces, not a speedup or a larger recognized class.

The experiment also finds 19 literal representative differences between direct construction and legacy replay. Their cyclic words agree, but later overlap records refer to positions in a particular representative. Thus installing direct roots while recording only legacy moves can invalidate subsequent positional replay. An atomic-shear certificate would need its own specified representative convention and independently tested verifier. Our audit deliberately continues from the actual legacy roots.

The delivery likewise reports 21 improved raw Artin presentations and zero improvements after its singleton elimination. Its supplied small-stage measurements show overhead; we have not rerun those comparative timings. The reported gains on large synthetic presentations concern an independently implemented selected-line baseline, not our maintained recognizer. Combined with the native-stage experiment, this is insufficient evidence to change runtime defaults or add an unconditional joint query.

## 65.6 Integration decision and next proof obligation

We retain the source-pinned gateway result, reproducible native adapter audit, mathematical construction and explicit counterexample in the maintained synthesis. The producer remains unchanged. A future runtime experiment should first identify highly compressed post-singleton states where joint optimization changes the search, then compare complete verified runs with the same resource allowances. If direct construction is needed for asymptotics, that experiment also requires a retained repetition representation and an atomic certificate protocol. A bound on multiplier changes and on intervening relator operations would be the next substantive step toward the quasi-polynomial target.

# 66 Linear endpoint descent with original-position certificates

The endpoint kernel of report 34 removes a repeated whole-word scan from the maintained braid recognizer. Previously, every elementary destabilization ran cyclic free reduction, counted both endpoint generators, copied a cyclic rotation, and potentially renumbered every surviving letter. A chain on $m$ strands and $n$ letters could therefore use $O(nm)$ word operations. The replacement retains the same sufficient endpoint criterion and the same right-before-left preference. It does not find general Markov moves, internal connected-sum cuts, or a minimum braid index.

## 66.1 Representation and local updates

Keep the original signed generators $w_i$, with predecessor and successor arrays forming a cyclic list of live positions. For each original generator index, maintain its live multiplicity and the xor of its live positions. The xor is used only when the multiplicity is one, when it identifies that unique position exactly. A live interval of strand labels is represented by $\ell < h$: the current strand count is $h - \ell$, and an original signed letter becomes

$$\operatorname{sgn}(w_i)(|w_i| - \ell).$$

A left destabilization increments $\ell$; a right destabilization decrements $h$. No surviving word is renumbered until output.

Initially queue every original position as a possible left endpoint of an inverse seam. Remove a live adjacent inverse pair when found, then enqueue only the predecessor of the newly exposed seam. After the queue drains, the cyclic word has no inverse adjacency. If the current right endpoint $h - 1$ occurs once, remove it; otherwise test the left endpoint $\ell + 1$. Set the head to the successor of the removed letter, implementing precisely the cyclic conjugation required to put that crossing last before Markov destabilization. Enqueue the newly exposed seam and repeat. Stop at three strands or when neither endpoint is a singleton.

Every inverse cancellation is a braid-group equality, including a conjugation for a wrapped cyclic cancellation. An endpoint generator occurring exactly once can be moved to the last position by conjugation and destabilized; left renumbering merely identifies the surviving consecutive strand labels. Thus every recorded move preserves closure isotopy. Reduction order can change the chosen cyclic representative; comparison with the earlier algorithm is therefore up to cyclic rotation.

**Proposition 66.1** (Amortized endpoint cost)**.** *On an explicit braid with $n$ letters and $m$ strands, the linked kernel and its original-position replay use $O(n + m)$ word operations and space. Their conservative bit cost, including the certificate, is $O((n + m)\log(n + m + 2))$.*

*Proof.* Each position is deleted at most once. A cancellation deletes two positions and a destabilization deletes one; each deletion event adds at most one queue entry. Including the initial $n$ entries gives at most $2n$ queue pops and at most $n$ trace records. Each pop and deletion performs a bounded number of array, counter and link operations. Every successful strand iteration removes a letter, and there is at most one final unsuccessful iteration. Initialization costs $O(n + m)$ and final output costs $O(n)$. Replay checks only the supplied local adjacencies and singleton counts, with the same initialization and deletion bounds. Indices and signed generator labels have $O(\log(n + m + 2))$ bits, giving the conservative bit bound. □

The maintained dispatcher selects the linked path for at least 64 strands and 128 letters, provided $m \le n + 1$. Otherwise it retains the simpler version-1 implementation. A one-component closure necessarily satisfies $m \le n + 1$, since each adjacent transposition can reduce the number of permutation cycles by at most one. On this domain the remaining branch has bounded rank or

bounded word length, so its whole-word passes also have a linear bound with fixed constants. Sparse standalone link inputs retain the earlier $O(nm)$ upper bound and avoid an allocation indexed by an arbitrarily large strand count. These statements concern explicit input; they are not complexity claims for a binary-compressed braid word.

## 66.2 Replay and resource boundaries

The new `singleton-markov-descent-v2` certificate records original positions for every cancellation and destabilization, plus the final live order and signed word. The verifier constructs its own lists and multiplicities; it checks live distinct positions, directed adjacency, inverse generators, endpoint identity and singleton count. It does not use the producer's xor lookup or queue, and it does not search for a reducible endpoint. Exact field, integer and trace-length checks reject malformed records. The queue counter is a diagnostic, never a mathematical witness for an isotopy. A valid prematurely stopped trace proves only its recorded equivalence.

Version-1 evidence remains replayable by the old verifier. Its historical replay complexity is unchanged; the new linear replay bound applies to the new certificates. Both new production and replay paths call the cooperative cancellation callback during their traversals. Resource interruption raises without returning a partial recognition result.

The complete free-product decision for the resulting at-most-three-strand braid remains linear on explicit input, so this endpoint-recognized class now has a linear symbolic path. The independent matrix backend retains its quadratic bit-arithmetic term. A supplied PD still needs checked braid provenance; neither this improvement nor a stalled endpoint search makes a new general unknot decision.

## 66.3 Validation and measurement contract

Seven added test methods compare exhaustive signed words through three strands, additional four-strand cases, and 120 seeded conjugated mixed stabilization chains with a separate explicit whole-word algorithm. A deliberately slow list-based original-position oracle checks the local trace independently of both linked implementations. Additional tests cover both signs, both recognition backends, chains through 2048 strands, malformed and reused positions, false inverse cancellations, input binding, small-path dispatch and resource interruption. Small stabilizations agree with independently computed reduced Khovanov rank. The complete maintained suite passes all 775 tests in 181.567 seconds with Regina installed.

`fast/benchmark_braid_descent.py` pins the previous reducer and verifier to commit `c4c23b61d`. It compares baseline, identical A/A control, and the adaptive implementation in shuffled order, using one excluded warmup and five measured rounds. Kernel measurements include certificate generation and independent replay. Raw braid measurements also include closure validation and symbolic recognition; fresh-PD measurements include diagram construction and the ordinary recognition pipeline. Input construction for the kernel is excluded; imports and JSON serialization are excluded in all arms. Source hashes are checked before and after the experiment. Any unknown result suppresses a timing ratio for that case.

The experiment contains 300 completed measured calls and 60 excluded warmups. At 4096 strands, right-chain generation plus replay falls from 4300.024 ms to 14.142 ms, with median paired ratio 305.29; the left chain falls from 5953.078 ms to 14.304 ms, with paired ratio 380.36. At 256 strands the corresponding ratios are 19.10 and 24.27. Complete raw-braid recognition plus replay at 2048 strands falls from 1042.500 ms to 7.582 ms, with paired ratio 138.34. These are large separated effects, supported independently by the deletion and queue bounds rather than extrapolated from timings.

| Case | Before | A/A | Adaptive | Ratio |
|---|---|---|---|---|
| right-16 | 0.127 | 0.128 | 0.128 | 0.99 |
| left-16 | 0.202 | 0.175 | 0.172 | 1.20 |
| right-64 | 1.248 | 1.187 | 1.196 | 1.05 |
| left-64 | 1.587 | 1.570 | 1.579 | 1.01 |
| right-128 | 4.280 | 4.216 | 4.259 | 0.99 |
| left-128 | 5.388 | 5.394 | 0.446 | 12.35 |
| right-256 | 15.470 | 15.670 | 0.810 | 19.10 |
| left-256 | 19.456 | 21.660 | 0.822 | 24.27 |
| right-1024 | 267.443 | 270.631 | 3.470 | 77.70 |
| left-1024 | 334.841 | 347.082 | 3.489 | 93.84 |
| right-4096 | 4300.024 | 4480.186 | 14.142 | 305.29 |
| left-4096 | 5953.078 | 5419.346 | 14.304 | 380.36 |
| small-rank-stall | 0.093 | 0.093 | 0.094 | 1.00 |
| raw-recognition-16 | 0.147 | 0.150 | 0.143 | 1.01 |
| raw-recognition-128 | 4.080 | 4.187 | 4.105 | 0.99 |
| raw-recognition-512 | 61.126 | 61.666 | 1.857 | 33.97 |
| raw-recognition-2048 | 1042.500 | 1051.843 | 7.582 | 138.34 |
| pd-recognition-16 | 0.187 | 0.197 | 0.188 | 1.02 |
| pd-recognition-128 | 1.314 | 1.296 | 1.292 | 1.02 |
| pd-recognition-512 | 5.473 | 5.559 | 5.567 | 1.01 |

Median milliseconds; ratios are medians of paired before/adaptive times.

The ordinary PD examples invoke the linked kernel zero times: earlier simplification already settles those stabilized unknots. Their paired ratios are 1.005–1.021 and give no meaningful complete-pipeline improvement. A/A ratios range from 0.940 to 1.194 across this short experiment; small apparent gains in unchanged paths fall within that variability. In particular, the 128-strand right chain has only 127 letters and deliberately stays on the old branch, while the slightly longer left chain crosses the threshold. These results support a practical large-chain optimization, not a universal recognition speedup or a quasi-polynomial general algorithm.

# 67 Affine families: tractable global queries and a geometric extraction step

Commit `58ee11a97d` supplies `docs/incoming/unknot_affine_families_20261008.zip`. All 466 manifest entries match the original archive. The delivery combines three separate interfaces: optimizing a supplied family of cyclic translations, extracting partial interval maps from actual normal coordinates, and inheriting scalar endomorphism algebras through a verified direct-sum decomposition. The first two are complementary but are not yet connected by a complete geometric adapter. The third overlaps our earlier scalar-locality and static-corner work, with additional transport and one-generator optimizations worth measuring separately. This review preserves these distinctions.

## 67.1 A factor-free optimum over an affine modular family

Let $R = \mathbb{Z}/m\mathbb{Z}$, and let feasible parameters satisfy $Az = b$. For a connected base whose monodromies are translations by the entries of $h(z) = c + Hz$, the number of covering components is

$$\kappa(z) = \gcd(m, h_1(z), \ldots, h_\beta(z)).$$

Solving the seam congruences gives the full affine coset $z = z_0 + Bt$. Its holonomy image is $v + Ut$, with $v = c + Hz_0$ and $U = HB$. The minimum component count is exactly the gcd of $m$ and all entries of $v$ and $U$. This is stronger than a lower bound: some assignment attains that gcd.

The constructive step merges two vectors $a, b$. Divide them and $m$ by their common content $g$, and put $M = m/g$ and $d = \gcd(M, a_1/g, \ldots, a_\beta/g)$. If $d = 1$, retain $a$. Otherwise iterate $D \leftarrow \gcd(M, D^2)$ from $D = d$ until stable, and use $a + (M/D)b$. At termination $D$ contains the full prime-power parts of $M$ supported on $d$. On those primes $a/g$ vanishes and $(M/D)b/g$ has a nonzero coordinate; on the other primes the added term vanishes and $a/g$ has a nonzero coordinate. The new vector has content $g$. This proof uses prime valuations to justify the algorithm, while the algorithm itself only computes gcds. Saturation doubles positive valuations, giving $O(\log\log(M + 2))$ iterations. For example $m = 12$, $a = 2$, $b = 1$ needs the saturated coefficient three; the unsaturated coefficient six would give content four rather than one.

A row-by-row modular solver maintains the complete affine coset using unimodular two-column gcd transformations and a single congruence restriction per row. All stored residues are reduced modulo $m$. The parameterization may be redundant, but uniform parameters have equal-size fibres, so dividing the assignment count by the size of each new scalar image counts distinct feasible assignments correctly. Explicit matrix dimensions and witness lengths are charged; a binary declaration of an enormous number of unused variables is not treated as a constant-size explicit-variable input.

For a proposed minimum $\mu \mid m$, set $s = m/\mu$. A feasible witness $z_*$ with $\gcd(m, c + Hz_*) = \mu$, together with

$$YA = sH, \qquad Yb + sc = 0 \pmod{m},$$

proves optimality without a kernel solver: every feasible holonomy is annihilated by $s$ and hence divisible by $\mu$, while $z_*$ attains it. An infeasibility witness is a row $y$ with $yA = 0$ and $yb \neq 0$ modulo $m$. The finite-module annihilator identity guarantees existence also for composite moduli. The implemented verifier checks these elementary identities and does not invoke the optimizer. Its minimal certificate contract deliberately excludes the separately computed feasible-assignment count.

## 67.2 Where the family query stops

The surface-family compiler permits disconnected local cyclic covers, full-boundary seams, shared affine parameters and a connected patch graph. A spanning tree transports sheet labels; local generators and non-tree seam voltages generate the global translation action. Seam compatibility is the equality of peripheral translations with the specified base-circle sign. A negative base-circle degree is not a reflection of the cyclic fibre. The compiler recomputes the holonomy data from the original surface schema before checking an arithmetic certificate.

The polynomial optimum does not enumerate every marked covering or preserve all information needed for future attachments. Even connected covers can have different genus: over a five-sheet pair of pants, peripheral translations $(1, 1, -2)$ give three boundary components and genus two, whereas $(1, -1, 0)$ give seven boundary components and genus zero. Both covers are connected and have Euler characteristic $-5$.

The report gives a sharp warning about correlated peripheral queries. For a graph, assign one parameter $z_v$ modulo three to each vertex and one boundary translation $z_u - z_v$ to each edge, on a planar surface with one additional boundary. Requiring every designated edge-boundary preimage to be connected is exactly proper three-colouring. This proves NP-completeness for that unrestricted affine peripheral query, not for unknot recognition, and does not unconditionally exclude quasi-polynomial time. Similarly, counting connected assignments in the family $h(z) = z$ includes computing $\varphi(m)$; for a semiprime this recovers its factors. Neither issue contradicts the polynomial minimum-and-witness theorem.

## 67.3 From actual normal coordinates to local cut data

The normal extractor accepts reciprocal tetrahedral face pairings and an admissible seven-coordinate normal vector. It checks nonnegative coordinates, quadrilateral compatibility and matching equations. The ambient manifold condition remains a precondition; no PD-to-exterior conversion is supplied. Each tetrahedron has at most four triangle stacks and one quadrilateral stack. Face arcs are ordered from the cut-off vertex, with triangles before quadrilaterals. Quadrilateral order reverses on the opposite side of its chosen vertex partition. Equal face ranks therefore give affine maps $i \mapsto \varepsilon i + c$ between intervals of disc indices.

There are at most five such surface bands per paired face: for $k \leq 3$ nonempty arc types, either side has at most $k+1$ blocks, and overlaying the two partitions gives at most $2(k+1) - k = k + 2 \leq 5$ intersections. Consecutive-disc prism gaps require both endpoint discs to lie in the band. Their map is $i \mapsto i + c$ for a translation, but $i \mapsto -i + c - 1$ for a reflection. The minus one is necessary because a gap is indexed by its smaller disc index.

If a tetrahedron has $D$ normal discs and $k$ nonempty types, it has $D - k$ between-disc prisms and $k + 1 \leq 6$ remaining local cells. Across a paired face at most two exceptional prism sides meet residual cells: one in each direction where an arc block crosses a triangle–quadrilateral transition. These are singleton gap contacts. The output size and arithmetic work are linear in tetrahedra and polynomial in coordinate bit length, with no multiplicity expansion.

Concatenating disc-stack intervals produces an input for interval-orbit counting. A signed orientation double cover similarly distinguishes orientable and nonorientable component counts once those orbits are known. The delivery prepares these inputs; it does not implement unrestricted Agol–Hass–Thurston reduction, all weighted component data, residual edge attachments, or a full cut-manifold handle structure. Its partial signed interval maps are not automatically total translations on one cyclic fibre. A sound extraction and coverage theorem is still needed to connect them to the affine-family optimizer. The later integration in Section 72 supplies unweighted AHT counting and aggregate topology of supplied normal surfaces; weighted extraction and attachment-aware cutting remain open implementation obligations.

## 67.4 Scalar inheritance relative to the maintained implementation

For a complete scalar commutant $A$ and an actual typed direct-sum child with coordinate selector $e$, restriction identifies its full algebra with $e(S^{-1}AS)e$: every child endomorphism extends by zero. The map from the whole algebra to the corner is only linear in general, not multiplicative. Canonical least-pivot reduction recovers the same ordered basis as a fresh commutator solve, preserving bounded candidate order. Our maintained `corner_split.py` already implements this exact static principle; its reuse option remains separate from runtime defaults.

The delivery additionally selects direct transport or a precomputed packed linear map on matrix variables. When a generator $Q$ saturates the full commutant dimension, $A = \mathbb{F}_2[Q]$. A child of this commutative algebra is generated by the restricted $Q$, so its full algebra can be recovered from one transported generator. A primary minimal polynomial then proves scalar locality. Our earlier dimension-saturation stopping already checks the locality implication; the new contribution to evaluate is propagation through one generator and its transport cost.

The delivery reports fewer equations and solves on attached-copy fixtures, and about twofold improvements on selected saturated two-field fixtures. Those are graded algebraic complexes without a claimed realization as knot-scan prefixes. Its six natural-knot timings range from 0.961 to 1.022 in paired baseline/new ratios, providing no convincing recognition gain. These supplied timings are not measurements of our current implementation; its baseline predates several maintained changes. A blanket patch would also overwrite our independently integrated scalar options. We therefore retain the proof and reproduction evidence, and defer runtime promotion

until a direct comparison with the maintained backend establishes a benefit.

### 67.5 Reproduction and integration status

We reran all 65 targeted methods in the distributed source tree: 13 arithmetic, 19 family-compiler, eight normal-extractor, 13 scalar-inheritance and 12 primary-split tests. All passed. The rational oracle reproduced 1,537 geometric cases, 9,771 disc identifications and 4,850 prism identifications, including 591 reflected prism bands. The separate Regina audit reproduced 807 inputs and 1,537 component records, including 92 nonorientable and 274 closed components; it compared full coordinate multisets, Euler characteristic, orientability and boundary counts. Family example and peripheral-reduction checks also passed. Logs and JSON evidence are stored under `data/affine-families-*`.

These are reproduced delivery tests, not a claim that the entire distributed source tree replaces our maintained implementation. We did not rerun its 790-test historical suite or its comparative timing experiments. The current 775-test maintained suite validates the separate braid change in Section 66. The archive also repeats the earlier single-submission worker-communication repair; it is distinct from all three new mathematical interfaces and should be assessed as a focused resource handling change.

The most consequential new geometric result is the actual normal-coordinate extractor. The next integration target is a checked native consumer of those intervals, with general weighted orbit reduction and full residual attachments. A polynomial family query can remove enumeration only when its invariant is all that the caller needs. Neither component minimization nor inherited scalar locality supplies the missing global state-count, reset or completeness bound for the unknot algorithm.

## 68 Whitehead exposure: rank drops beyond strict length descent

Commit `613bec60c3` supplies `docs/incoming/unknot_whitehead_exposure_20261008.zip`; all 44 manifest entries verify. Its main idea addresses a different objective from integral shear length minimization: deliberately expose a defining occurrence, then eliminate that generator, even if the intervening automorphism increases length. The delivered adapter now runs against our maintained code in an offline audit. No runtime policy is changed by this review.

### 68.1 An exact one-step exposure oracle

For a cyclically reduced relator $w$, put one unit on the undirected edge $\{x, -y\}$ for every cyclic adjacent pair $xy$. Parallel edges aggregate as positive capacities. For an admissible Whitehead pair $(a, S)$ with $a \in S$ and $-a \notin S$, the transformed cyclic word has exactly $c_w(S)$ occurrences of $a^{\pm 1}$, where $c_w$ is the graph cut capacity. Every other unsigned generator count is unchanged. The same noncancelling skeleton argument used for integral shears proves this stronger occurrence identity, and subtracting the original multiplier count gives the classical length-change formula.

Consequently a one-step multiplier singleton exists exactly when the graph has a capacity-one support bridge. A unit cut clearly isolates such a bridge. Conversely, one component after deleting the bridge has odd weighted degree sum. Signed degrees occur in equal inverse pairs, so that shore cannot be closed under inversion: it separates some $a, -a$ and defines an admissible Whitehead move. A bridge of capacity greater than one does not suffice.

To optimize a target relation's exposure, choose its unit bridge and contract all its other support edges. A cut with that bridge as its only target edge is constant on every contracted component. Pin the two bridge sides and the multiplier pair in opposite shores, then solve a minimum cut in

the quotient of the *total* presentation graph. Enumerating target bridges and signed multipliers covers all one-step exposures. Absent signed letters must remain as isolated vertices: assigning them can change the costs of other relators. Equal pin patterns can share a flow computation. A feasible flow whose value equals the cut certifies the selected pinned optimum; the checker does not certify that every other bridge was searched. Topological validity only requires replaying the chosen automorphism and elimination.

There is also an exact allocation objective. Let $L$ be total initial length, $d(a)$ the total multiplier count, $c$ the total selected cut capacity, and $t = |w_j| - d_j(a)$ for the target. Its transformed length is $t + 1$ and its solved replacement has length $t$. Removing it and substituting into the other $c - 1$ multiplier occurrences allocates exactly

$$U = L - d(a) + t(c - 2)$$

letters before the final reductions. Since $t \geq 0$, the same minimum-cut query optimizes $U$ for fixed target and multiplier. This is a bound on raw substitution, not an estimate of final reduced length or future difficulty.

## 68.2 A separation from strict automorphism descent

Let $u = a^2bab$, $t = (ab^{-1})^3$, and let $v$ be the cyclically reduced commutator $[u, t]$, of length 20. The presentation

$$P_m = \langle a, b \mid u, v^m \rangle$$

presents $\mathbb{Z}$: the Whitehead map fixing $a$ and sending $b \mapsto a^{-1}b$ takes $u$ to $ab^2$, so $u$ is primitive, and the commutator relator already lies in its normal closure. After this exposure, substituting $a = b^{-2}$ kills every relation.

The complete rank-two cut table gives the non-inner elementary length changes on $(u, v^m)$ as $2m-2$, $2m-1$, 2, and 3, with complement symmetries. Thus for $m \geq 2$ every non-inner elementary Whitehead move increases length. Signed permutations and inner moves preserve this obstruction and the absence of a singleton. The least one-step exposing increase is $2m - 2$, yet the following elimination leaves length zero. For the selected exposure, $L = 20m + 5$, the intermediate length is $22m + 3$, and the raw bound is $U = 34m$. This explains why optimizing only immediate length can miss an affordable algebraic simplification.

It is a separation for the restricted automorphism policy. It does not establish failure of our overlap-enabled group search, and these abstract presentations are not a family of difficult knot diagrams. Nor is the one-step increase a lower bound on every longer increasing path.

The fused verifier composes the automorphism with its solved quotient map and evaluates that map directly on all original grammar roots. On $P_{2^k}$, the base commutator maps to the empty word before repeated copies are evaluated. A grammar with $k + 51$ nodes therefore supports exact replay with a literal image cap of 100. The cap is enforced at every needed grammar node; this is a bounded small-image evaluator, not a general compressed free-reduction algorithm. The supplied $k = 1000$ certificate was successfully replayed without expanding its enormous relator.

## 68.3 Rank bounds do not establish completeness

The explicit rank-first stage performs at most $r - 1$ rank drops and $2(r - 1)$ elementary moves. With stored-letter cap $M$, Whitehead workspace is at most $3M$. Choosing a quasi-polynomial cap bounds this particular possibly inconclusive positive search, including replay, by a quasi-polynomial function on native Wirtinger inputs. It does not prove that every unknot passes the stage, that the greedy choice finds every affordable path, or that an exponential complete fallback inherits this bound.

The report gives two distinct obstructions. A primitive word can have no unit bridge until a preparatory move is made, so absence of a bridge is only a negative answer to the depth-one query. More permanently, $\langle a, b \mid a^2, a^3 \rangle$ is infinite cyclic, while an automorphism keeps both relators proper powers; neither can acquire a singleton. Relator interaction is indispensable. Exposure should therefore cooperate with certified relator replacement rather than replace it.

### 68.4 Reproduction and actual maintained inputs

All 49 delivered unit tests passed. The deterministic audit reproduced 17,664 occurrence and length checks on 1,104 cyclic words, 1,104 bridge existence comparisons, 640 selector comparisons against literal enumeration, 468 checked witnesses, 500 independent flow comparisons, 500 rejected capacity mutations, and 32 barrier presentations. All three supplied positive certificates passed their respective replay checks. The five-case maintained adapter smoke test also passed; this closes the delivery's previously unexecuted compatibility check, without implying a new production integration.

Our reproducible `data/audit_whitehead_exposure.py` recovers the original archive from its arrival commit and reconstructs actual maintained PD presentations. On the 30 small diagrams from the independent port-register Khovanov corpus, it returns 23 positives and seven inconclusive results. Every positive has independent explicit and compressed full-source replay and agrees with the stored Khovanov oracle. None uses an exposure: existing singletons suffice. External cancellation propagates and a zero time budget returns inconclusive.

We additionally ran the native adapter on the maintained 141-crossing Gordian fixture. It uses one exposure, then stops at rank 11 with 97,684 letters. Reconstructing every recorded move from the native full presentation and scanning the resulting relator graphs finds zero unit bridges. Thus the final failure is not just an allocation cap. This reproduces the delivery's negative residual through the actual maintained input boundary. Across all 31 diagrams there are 23 positives and eight inconclusive results, with one exposure and no demonstrated new recognition coverage.

The delivery's reported selector speedups compare against exhaustive shore enumeration, not our existing minimum-cut producer. We did not rerun those timings or treat them as pipeline gains. A further useful experiment is a bounded handoff from an intact maintained residual after ordinary group search stalls, preserving relation slots, names, prior moves and input binding. The supplied adapter restarts from the original PD and does not implement that handoff. Its search and verification have separate work allowances but a shared wall-clock callback; a production integration must account for both. Section 69 now resolves the narrower Gordian residual question: an overlap creates unit bridges and permits a fully verified continuation. It also compares that handoff against the already successful maintained search.

## 69 Escaping the exposure residual and pruning overlap queries

The stalled native presentation in Section 68 does admit an inexpensive algebraic next step, although the resulting search policy is slower than our incumbent on this example. This distinction matters: escaping a restricted policy's obstruction is evidence about which operations are needed, not evidence of an improved recognition complexity bound.

### 69.1 A full-source certified continuation

The offline driver `../fast/exposure_research/residual_bridge.py` loads the immutable exposure delivery from its arrival commit, checks its archive digest and all 44 manifest entries, and constructs the maintained Wirtinger presentation of the 141-crossing Gordian fixture. It reconstructs all 131 recorded prefix moves, preserving generator names and relation slots. The

result has rank 11 and 97,684 letters, with neither a singleton occurrence nor a unit bridge in any relator graph.

One maintained overlap query selects donor slot 32 and target slot 54, with donor rotation 294, target rotation 2, positive donor sign and overlap length 14,722. Applying its normal-closure-preserving replacement leaves 84,400 letters: a decrease of 13,284. There are now two singleton occurrences and four unit bridges. The maintained compressed continuation reaches rank one; the full certificate contains 142 moves. Independent explicit and compressed verifiers both replay it from the original PD. The certificate is saved in `../fast/results/gordian_exposure_overlap_certificate.json`. The bridges are diagnostics of the new opportunity; the proof of the unknot is the independently replayed complete group certificate.

This experiment answers the concrete residual question left open by the delivery. It does not establish that a rank-first prefix is necessary or beneficial. Existing overlap-enabled group search already solves the fixture, and uses 141 moves. Nor is the offline reconstruction a hardened public interface: it accepts the fixed, checked delivery's move format and relies on final independent replay. It is not used by runtime dispatch.

## 69.2 The policy comparison includes reconstruction and replay

At checkpoint `875c8565d`, before the pruning below, we compared five arms on fresh PD input: explicit search, an identical explicit control, compressed search, adaptive search with a 20,000-letter switch, and the exposure–overlap handoff. Each search has 20 million charged work units; each of the two independent verifiers has its own 20 million. One shared 15-second wall allowance covers input construction, search and both replays. All arms allow 600,000 replay letters. The exposure producer permits its specified $3M$ temporary Whitehead workspace, while the incumbent explicit producer retains its existing $M = 200,000$ search-letter limit. Thus this is a comparison of the stated policies, not identical intermediate storage. Archive import and result serialization are excluded. Reconstruction and the before/after bridge audits are included in the exposure arm.

One excluded warmup per arm and five shuffled measured rounds, with seed 261008113, give 25 completed positives and no capped observations. Median times in seconds are 0.866 for explicit, 0.851 for its identical control, 1.326 for compressed, 1.031 for adaptive, and 4.478 for exposure–overlap. The median paired explicit/control ratio is 1.018; the paired explicit/handoff ratio is 0.193. The handoff is approximately 5.2 times slower and gains no recognition coverage here. Its charged search work is 8,857,800, including prefix reconstruction and graph audits. These results support retaining the existing dispatch. Raw samples and frozen source hashes are in `../fast/results/exposure_residual_20261008.json`; the production source hashes refer to that checkpoint, not to the subsequent optimization.

## 69.3 An exact upper bound on further overlap improvements

For a donor of length $d$ and a cyclic match of length $u$, the overlap producer guarantees shortening

$$g = 2u - d, \qquad u \le d, \qquad g \le d.$$

If the best already selected gain is $G$, a donor with $d \le G$ cannot improve it. The producer replaces its current witness only on a strict improvement, so such a donor cannot change tie breaking either. If a scan attains $g = d$, its remaining positions, targets and inverse sign can all be skipped. A later longer donor is still searched. These bounds are applied before constructing a suffix automaton and inside the existing loops; donor, sign, target and scan order are otherwise unchanged.

Consequently, with adequate resources the complete selected witness, including rotations, is identical to the earlier producer's. Independent certificate replay is unchanged. Under a work cap

the new producer may finish when the old one exhausts its allowance. Each skipped donor still charges a work unit and polls cancellation. The general bound remains $O(n + mL)$ dictionary operations for $n$ slots, $m$ nonempty relators and total stored length $L$. For $m$ equal length-$d$ donors whose first pair fully matches, this particular query now uses $O(n + d)$ operations on the already stored words: one automaton, one successful scan, and slot traversal. The earlier implementation still built both signed automata for every donor, costing $\Omega(md)$ even after the optimum was known. No general quasi-polynomial recognition bound follows from this local pruning.

| Query or complete search | Before | A/A | Pruned | Ratio |
|---|---|---|---|---|
| Repeated 16 by 128 | 6.203 | 6.134 | 0.279 | 22.15 |
| Repeated 128 by 128 | 51.202 | 52.065 | 0.319 | 159.06 |
| Disjoint 16 by 128 | 47.409 | 47.035 | 49.360 | 0.99 |
| Later longer donor | 0.086 | 0.083 | 0.023 | 3.85 |
| Gordian explicit | 855.693 | 849.902 | 572.097 | 1.51 |
| Gordian exposure handoff | 4338.314 | 4393.262 | 2892.362 | 1.51 |

Median milliseconds; ratios are medians of paired before/pruned times. Native rows include two independent full certificate replays.

The pruning audit swaps only the overlap query against its source pinned at `875c8565d`; its identical old-function control detects measurement drift. It first compares 2,000 complete randomly generated witnesses, including signs, tie choices and cyclic positions. Six workloads then use one excluded warmup and five measured shuffled rounds per arm (seed 261008114). The repeated-donor cases exercise the new early stop, disjoint alphabets exercise an unsuccessful full search, and a short mixed-length case ensures that an early winner does not suppress a later longer donor. The two native workloads include fresh PD construction and both independent replay modes; their full certificate hashes must agree across all arms and rounds. Their resource limits match the earlier policy comparison. The raw record is `../fast/results/overlap_bounds_20261008.json`.

All 90 measured operations and 18 warmups complete, with identical witnesses or full certificate hashes across arms. For 128 repeated 128-letter donors, charged query work falls from 179,840 to 1,022 and the median paired speedup is 159.06. The 16-donor version improves by 22.15. The disjoint control has unchanged charged work and a paired ratio of 0.986; it provides no evidence of a speedup on unsuccessful scans. On the actual Gordian explicit search, including both replayers, median elapsed time falls from 0.856 to 0.572 seconds, with paired speedup 1.51. The exposure handoff falls from 4.338 to 2.892 seconds, also 1.51 times faster, and charged search work decreases from 8,857,800 to 6,455,975. It remains substantially slower than the incumbent. Identical-control paired ratios range from 0.991 to 1.040. These are local same-machine observations on the stated workloads, not a universal speedup.

All 778 maintained tests pass with Regina enabled in 185.977 seconds. New regressions require the repeated-donor query to fit a linear work allowance, preserve a later longer winner, and honor cancellation and work limits while skipping donors. Existing overlap tests independently check maximum gains and forged certificates; the pinned audit additionally checks the entire deterministic witness. The resulting native improvement changes neither the recognized input class nor the general worst-case guarantee.

# 70 Dynamic terminal contractions: algebra, native replay and zero cones

Commit `49a8af358` supplies `docs/incoming/unknot_dynamic_terminal_20261008.zip`. Its 51-file manifest verifies. The report concerns repeated signed Tait-graph quotient observations at a fixed suffix, extending the boundary geometry of Section 33. It supplies a quadratic field-update

method for a given terminal merge, including singular intermediate matrices. It does not supply a complete unknot criterion or bound how many observations a complete recognition procedure must generate. We reproduced its algebra checks, ran the previously unexecuted native adapter, and implemented its proposed zero-descendant-cone optimization as a research primitive.

## 70.1 Keep singular interior directions at each prime

Let a signed graph have $v$ vertices and $b$ terminals, with terminal zero grounded. Write its grounded Laplacian as

$$A = \begin{pmatrix} D & B \\ E & F \end{pmatrix},$$

where $D$ indexes the interior and there are $t = b - 1$ non-root terminal coordinates. Over a chosen prime field, find invertible left and right matrices $P, Q$ such that $PDQ = \operatorname{diag}(I_\rho, 0_r)$. Separate the active and null rows of $PB$ and columns of $EQ$, and eliminate the active identity block. The surviving matrix is

$$K = \begin{pmatrix} 0_r & C \\ G & S \end{pmatrix}, \qquad S = F - (EQ)_{\text{active}}(PB)_{\text{active}}.$$

The determinant multiplier is $\gamma = (\det P \det Q)^{-1}$. For a partition with $q$ non-root blocks and terminal incidence matrix $R$, its quotient cofactor is

$$\gamma \det \begin{pmatrix} 0_r & CR \\ R^T G & R^T S R \end{pmatrix}.$$

In particular, $r > q$ forces zero. Different primes can have different $r$; a singular interior is retained rather than discarded as a bad prime. If $r > t$, all terminal partitions are zero at that prime. Otherwise $\dim K = r + t \le 2(b-1)$. Separate left and right elimination need not preserve symmetry, so $G$ cannot silently be replaced by $C^T$.

## 70.2 A fixed matrix and a rank-two merge

Give each terminal block a retained anchor, with root block anchored at zero. Let $e_i$ be the terminal unit column of $K$ and put $e_0 = 0$. Keep the null-direction rows. Replace a non-root anchor row by the sum of the *original* terminal rows of $K$ in its block, and replace each non-anchor row $i$ in anchor block $a$ by $e_i^T - e_a^T$. Denote the resulting fixed-size matrix by $M_\pi$.

Its determinant equals the smaller quotient border above, with no extra partition sign. Indeed set $x_i = z_a + z_i$ at non-anchor members of a non-root block, and keep its anchor coordinate $x_a = z_a$. This column change is $I + N$ with $N^2 = 0$ and determinant one. Constraint rows become unit rows. Clear their columns, then reorder rows and columns by the same permutation; the result is the small border direct-summed with an identity. The two permutation signs cancel. The identity works over any commutative ring.

Now merge source block $B$ with anchor $s \neq 0$ into target anchor $a$. Define $e = e_s - e_a$, $\chi_B = \sum_{i \in B} e_i$, and let $w_B^T$ be the sum of the original base rows indexed by $B$. Direct row comparison gives

$$M_{\pi'} - M_\pi = -e w_B^T + \chi_B e^T.$$

The target anchor gains $w_B^T$; the old source anchor becomes a constraint; every other source member changes its constraint anchor from $s$ to $a$. The root case follows from $e_0 = 0$. Using current constraint rows to compute $w_B$ would be wrong. Anchors need not be minimum labels, but a certificate must retain their actual identities to identify the represented matrix.

## 70.3 Rank updates without a nonsingularity assumption

Maintain $LMU = \operatorname{diag}(I_k, 0)$ and $\eta \det L \det U = 1$. A rank-one update transforms to $xy^T$, where $x = Lu$ and $y^T = v^T U$. If $x$ has a nonzero null coordinate, pivot that coordinate and use its original zero row to clear the other entries of $x$. Clear the active part of the new row using active unit rows. A remaining null component of $y$ adds a pivot; otherwise rank stays unchanged. The case with a null component only in $y$ follows by transposition.

When both tails vanish, set $\alpha = 1 + y^T x$. For $\alpha \neq 0$ the inverse rank-one correction restores the active identity, and $\eta$ is multiplied by $\alpha$. If $\alpha = 0$, conjugate within the active block to send $x$ to its final coordinate; its diagonal entry becomes zero and elimination removes one pivot. All branches use $O(b^2)$ field operations and never invert zero. The determinant is $\eta$ at full rank and zero otherwise. Applying this routine twice realizes a terminal merge even if the first update temporarily changes rank.

Persistent cursors clone the two normalizing matrices before either update. The parent therefore survives an interrupted child construction. This transactional property belongs to the cursor API; the low-level mutable rank-update object does not promise it.

## 70.4 What the local complexity bound includes

For a supplied merge tree with $J$ edges and $Q$ observations, preparation and evaluation take $O(v^3 + Jb^2 + Q)$ field and indexed operations per prime. The cubic term includes interior preparation and the root normalization. Depth is at most $b - 1$, so depth-first traversal retains $O(v^2 + b^3)$ field elements, plus the explicit tree and results. Query generation is separate. A static competitor should eliminate the smaller $(r + q)$ border, not the original graph at every query. Heavy coarsening can make that static border much cheaper than retaining a large dynamic matrix.

Signed reconstruction uses the graph-derived bound $B = \prod_{e \text{ non-loop}} (1 + |w_e|)$: every quotient spanning tree uses a subset of the original edges. Select distinct primes with product greater than $2B$ and reconstruct the unique signed integer in $[-B, B]$. Independent choices of signs modulo each prime do not prove an integer is a unit: 41 is $-1, +1, -1$ modulo $3, 5, 7$, although their product exceeds $2(41 + 1)$. For $H = \lceil \log_2(B + 1) \rceil$, the report gives elementary polynomial bounds on the number and size of trial-generated primes, so scalar bit costs do not invalidate the stated explicit-graph polynomial bound.

The delivered exact tree walker reconstructs at every visited node, including unobserved intermediate nodes. For that implementation, signed reconstruction must be charged at $J + 1$ nodes, not merely $Q$ requested observations. Our per-prime driver reconstructs only requested outputs. This distinction does not change the per-prime theorem, but matters for its exact-integer wrapper. Neither implementation includes independent endpoint verification in the quadratic update bound. Direct verification of the normal forms is cubic; the delivered endpoint schema also repeats interior certificates.

All partitions can share a merge tree with Bell-number many nodes. This removes repeated intermediate work but leaves $\exp(\Theta(b \log(b + 2)))$ possible partitions. Thus a suitable polylogarithmic boundary bound can make this observer quasi-polynomial; it does not supply such a bound for general diagrams or control the other recognition work.

## 70.5 Implemented zero cones and independent native validation

The maintained research helper `../fast/determinant_research/terminal_plan.py` evaluates a supplied prime-specific merge plan. Once $q < r$, every descendant has still fewer blocks, so its residue is zero. The helper skips matrix cloning, updates and algebraic traversal of that entire

cone. Before pruning, it validates *every* tree edge with a depth-first live-anchor set, undoing anchor deletions on return. An impossible merge hidden below a zero node is still rejected. This costs $O(J + Q + b)$ indexed work even when all algebra vanishes. The remaining algebra uses $O(v^3 + J_{\text{live}}b^2)$ field work, where $J_{\text{live}}$ counts actually updated edges. Requested callbacks propagate through validation and the delivered cursor operations; unrequested work statistics are not collected. Prime preparation and the full exact adapter remain research interfaces without a complete production deadline contract.

The 39 delivered unit tests pass. Its deterministic audit reproduces 19,200 rank-one updates, 600 graph kernels with 193 singular interiors, 1,772 graph quotient comparisons, all 877 partitions of seven terminals, and the three exported certificates. Our new traversal additionally agrees with direct integer quotient determinants on 15,840 values across 600 randomized field plans, with nonminimum anchors and branching. Invalid merges beneath a zero cone are rejected, and cancellation propagates.

The native driver `data/audit_dynamic_terminal.py` recovers the immutable archive, rechecks its manifest and exports actual maintained scanner query streams. On 78 diagrams and 582 stages, all 4,503 queries agree in value *and metadata* among the maintained rational observer, delivered static and dynamic modular adapters, and independently rebuilt completed diagrams. There are 1,317 disconnected projection queries. Connected observations also agree with the new per-prime traversal and signed reconstruction. All 24 selected nontrivial native endpoint certificates pass independent algebraic replay; that graph-level check is supplemented by the separate geometry comparison, not used as a substitute for it.

Across 1,819 prepared prime kernels, 132 interiors are singular and 24 are universally zero. The connected query plans contain 4,535 merge edges. Beyond universal zeros, the new pruning reduces field-edge updates from 16,862 to 16,785: only 77 are avoided. This is a real activation on native inputs, but a small reduction, not evidence for a broadly effective shortcut. These stages have at most ten graph vertices, ten terminals and 200 queries; their field-interior nullities are at most two. They therefore test actual integration but do not probe the large-boundary asymptotic regime of the report's graph microbenchmarks.

## 70.6 Controlled native-stage measurements

| Observer | All stages | Widest | Most queries | Singular | Ratio |
|---|---|---|---|---|---|
| Rational | 351.585 | 1.366 | 10.829 | 1.007 | 1.00 |
| Identical control | 360.448 | 1.323 | 9.815 | 0.999 | 0.97 |
| Integer quotient | 293.190 | 1.299 | 7.278 | 0.663 | 1.24 |
| Static modular | 506.603 | 1.770 | 12.928 | 1.731 | 0.69 |
| Cached static modular | 514.974 | 1.969 | 13.275 | 1.635 | 0.70 |
| Delivered dynamic | 1276.307 | 7.055 | 35.974 | 2.638 | 0.28 |
| Pruned field traversal | 1406.933 | 6.993 | 37.377 | 2.837 | 0.27 |

Median milliseconds per fresh replay; the final column is the median paired rational/arm ratio for all 582 stages. These are observer timings.

The final benchmark includes fresh PD validation, boundary geometry, kernel preparation, matching validation, plan compilation and exact reconstruction. It excludes archive import, output hashing and the separate independent audits. The rational reference and its identical control retain their normalized partition cache. The integer arm constructs full integer quotient cofactors and uses fraction-free elimination with the same cache. We measure both delivered static modular evaluation and a cached version, delivered dynamic evaluation, and the new per-prime pruned traversal. The last arm also avoids reconstruction at unrequested tree nodes, so its timing is not an isolated measurement of the zero-cone test.

One excluded warmup batch and five shuffled measured rounds use seed 261008116. The complete 582-stage corpus uses one replay per sample; widest, most-query and most-singular stages use respectively 40, 10 and 50 fresh replays per batch. All repeated results must have identical hashes including metadata. The initial unbatched record is retained separately: its smallest case had a 1.78 identical-control ratio, and its new traversal unnecessarily collected work counters. The final record uses optional counters and batches to address those concrete measurement problems. Frozen initial source and audit snapshots permit verification of both records.

All 140 measured batches and 28 warmup batches complete with matching output hashes. Across the full corpus the rational median is 351.6 ms, its identical control is 360.4 ms, and the integer quotient median is 293.2 ms; the paired rational/integer ratio is 1.245. Delivered dynamic evaluation takes 1,276.3 ms with paired rational/dynamic ratio 0.278, roughly a 3.6-fold slowdown. The new traversal takes 1,406.9 ms and supplies no aggregate timing gain. Static modular evaluation is also slower, with or without a partition cache. The final identical-control paired ratios range from 0.970 to 1.033 over the four workloads, much closer than in the initial small-case measurement. These results motivate testing integer/rational adaptation across larger native stages before choosing a new policy; they do not establish a competitive adaptive bound or justify a cutoff fitted to this small corpus.

These are boundary-observer measurements, not full-recognizer speedups. The native adapter and zero-cone helper remain research-only. The report's general local theorem is useful, but the native evidence does not justify replacing maintained dispatch with an eager dynamic modular observer.

# 71 Fraction-free terminal kernels and the cost of switching

Section 70 found that direct integer quotient elimination beat the rational observer on small native graphs. This does not mean every query should repeat the interior elimination. We now retain the terminal-kernel structure while preparing and querying it with integers. The validated fraction-free kernel becomes the boundary research observer's default; `arithmetic='rational'` retains the earlier reference. Neither observer is dispatched by ordinary knot recognition. The improvement changes exact arithmetic cost, not the meaning or sufficiency of a boundary observation.

## 71.1 An integer border with a determinant scale

Ground terminal zero, and order the remaining coordinates as $h$ interior vertices followed by $t = b - 1$ terminals. Perform fraction-free elimination while a nonzero pivot remains in the interior block. Row and column swaps are allowed independently within that block. With previous pivot $\Delta_{k-1}$, initially one, and current pivot $p$, update trailing entries by

$$a'_{ij} = \frac{p a_{ij} - a_{ik} a_{kj}}{\Delta_{k-1}}.$$

Every division is checked for zero remainder. The bordered-minor identity underlying Bareiss elimination guarantees exactness [5]. A zero diagonal is insufficient to stop: off-diagonal interior pivots are searched too, and both permutation signs are recorded. Only a wholly zero remaining interior block ends preparation.

After $\rho$ pivots put $r = h - \rho$, let $\Delta \neq 0$ be the last pivot (one if none were taken), and let $\epsilon$ be the accumulated swap sign. The retained integer matrix $K$ equals $\Delta$ times the remaining rational Schur complement. It has a zero $r \times r$ interior corner. Independent row and column pivots can destroy symmetry, so its two coupling blocks are not assumed to be transposes.

For a partition with $q$ non-root terminal blocks, sum rows and columns of $K$ within those blocks and discard terminals in the root block, retaining the null directions. Write the resulting integer

border as $K_\pi$ and set $d = r + q$. Its exact quotient cofactor is

$$\tau_\pi = \begin{cases} 0, & r > q, \\ \epsilon\Delta, & d = 0, \\ \epsilon\dfrac{\det K_\pi}{\Delta^{d-1}}, & d > 0. \end{cases}$$

The final division is checked as well. The empty-border case is essential: eliminating every coordinate leaves the eliminated determinant, not an empty determinant divided by an integer power with negative exponent.

*Proof.* The selected interior rows and columns form a nonsingular $\rho\times\rho$ block of the permuted source, of determinant $\Delta$. Terminal identification does not affect these coordinates. Schur elimination therefore factors each quotient determinant as $\epsilon\Delta$ times its remaining Schur determinant. The integer border scales all $d$ remaining rows by $\Delta$, so its determinant acquires a factor $\Delta^d$. This proves the formula for $d > 0$; the empty Schur determinant gives the $d = 0$ case. If $r > q$, the $r$ null rows couple to only $q$ terminal columns, forcing zero. A failed exact division is consequently an arithmetic failure, never evidence about knot type. □

## 71.2 Independent replay and resource bounds

`../fast/fastunknot/terminal_determinant.py` stores immutable border rows, the scale, sign, source labels and interior pivot positions. Its independent diagnostic verifier replays those positions with rational Schur subtraction, checks nonzero pivots, multiplies the rational pivots, and compares the scaled residual entry by entry. It checks that the claimed null interior is zero. It never calls the constructor or its fraction-free recurrence. This binds the kernel to its supplied matrix; no new serialized knot-certificate protocol is claimed.

Preparation costs $O(v^3)$ integer operations and $O(v^2)$ space. If $r \le b-1$, a query uses $O(b^2+(r+q)^3)$ integer operations on a border of size at most $2(b-1)$; if $r > b-1$, every partition returns zero immediately. This matches the earlier static kernel's arithmetic bound while avoiding repeated fraction normalization. It is not a uniform bit-complexity speedup. For source entries of at most $\tau$ bits, bordered minors have $O(v(\tau + \log(v+1)))$ bits. Aggregation adds logarithmic factors, and the final numerator and denominator can acquire another factor $O(b)$ in their bit lengths before exact cancellation. These bounds are polynomial for explicit input, but individually reduced fractions can be more economical than a common scale on some matrices.

Construction, query elimination and diagnostic replay accept a cooperative callback. Interrupted construction publishes no kernel. Queries use fresh mutable border storage and preserve the prepared kernel. The boundary adapter keeps its geometry checks, phases, component counts and normalized partition cache. Its existing whole-adapter deadline limitations remain; an algebra callback alone does not establish a production resource contract.

Four new tests compare 8,000 random signed-graph quotients with independently built full integer cofactors, and replay each kernel independently. They exercise off-diagonal pivots, singular interiors, empty quotients, nonunit scales, malformed input, corrupted kernels and interruption. Twenty-five actual random knot diagrams compare all boundary values and metadata against explicit rational mode. Before the default change, all 782 maintained tests passed with Regina in 167.115 seconds; the new default is checked separately against the full suite and native completion audit. The final default passes all 782 tests with Regina in 182.267 seconds. Across 78 diagrams and 582 stages, all 4,503 native queries agree with independently rebuilt completions, including 15 singular rational-interior cases. Of 3,468 additional arbitrary pairings, 745 are accepted and independently agree, 472 classical pairings are declined by the inherited-color contract, and 2,251 nonclassical pairings are rejected. These totals match the earlier geometry audit. Every prepared kernel in this audit also passes independent rational source replay.

### 71.3 Larger native stages and a switching experiment

The benchmark pins the rational observer at `7951ae852` and includes an identical control. Other arms use direct full integer quotient elimination, the reusable integer kernel, or a candidate policy that handles four distinct connected partitions directly and prepares the integer kernel on the fifth. All share the same normalized partition cache. Cache hits do not consume another distinct-query allowance. The four-query threshold is an experiment, not an optimal stopping theorem.

In addition to the previous 4,503 native observations, we extend an actual five-strand scanner prefix by copies of the identity braid word $\sigma_1\sigma_2\sigma_2^{-1}\sigma_1^{-1}$. This preserves its closure while enlarging both face shadings, producing selected graphs with 17, 41 and 73 vertices and nine terminals. Repeating only one generator's cancelling pair would keep the selected graph at nine vertices. We measure one connected query, the full 200-query stream, and 200 repeats of one connected query. The completions are actual maintained prefix states; these constructed workloads separate setup, reuse and caching, rather than represent hard knots.

| Stage | Rational | A/A | Direct | Kernel | Switch | Ratio |
|---|---|---|---|---|---|---|
| Native corpus | 340.179 | 339.129 | 273.497 | 271.764 | 275.203 | 1.23 |
| 17 vertices, one | 4.386 | 4.303 | 0.739 | 0.857 | 0.736 | 5.17 |
| 17 vertices, stream | 11.104 | 10.856 | 8.396 | 6.779 | 6.875 | 1.65 |
| 41 vertices, one | 72.510 | 71.096 | 3.674 | 4.333 | 3.734 | 16.55 |
| 41 vertices, stream | 79.756 | 79.575 | 56.839 | 9.949 | 17.897 | 8.08 |
| 73 vertices, one | 428.080 | 432.740 | 16.765 | 16.580 | 17.009 | 25.79 |
| 73 vertices, stream | 438.952 | 437.307 | 355.343 | 22.835 | 77.097 | 19.22 |
| 73 vertices, repeats | 447.043 | 433.479 | 22.467 | 22.039 | 22.355 | 19.97 |

Median milliseconds; final column is the median paired rational/integer-kernel ratio. These are complete observer-stage timings.

Fresh PD validation, geometry, preparation, queries and cache activity are timed. Input generation, output hashing and independent verification are excluded. Seed 261008119 shuffles one warmup round and five measured rounds for five arms and eight workloads: 40 warmups and 200 measured samples. The smallest one-query case uses ten fresh replays per sample. All complete values and metadata agree across every arm and round. Raw samples are in `../fast/results/integer_terminal_20261008.json`. The measured integer implementation is frozen at `fc8b7da1c`, before changing the research constructor's default; the benchmark selected integer mode explicitly. The later default change and explicit rational-reference test are covered by the separate final validation above.

Across the original native corpus, the reusable integer kernel has median 271.8 ms against 340.2 ms for the rational observer, with paired speedup 1.229. On the full 73-vertex stage it takes 22.8 ms against 439.0 ms, with paired speedup 19.223. Direct integer evaluation still takes 355.3 ms there because it repeats the interior work on 25 distinct partitions. The four-query policy takes 77.1 ms: paying for four large full quotients before setup costs more than preparing immediately. With only one distinct query, the adaptive arm correctly avoids setup even when that query is repeated 200 times. On the small native corpus it reduces kernel preparations from 582 to 128, but its 275.2 ms median does not beat the reusable kernel's 271.8 ms. Identical-control paired ratios range from 0.985 to 1.019 across the workloads.

These results justify the integer kernel as the research observer's default, but not the candidate switching rule. A future adaptive policy should account for the actual cost of discarded direct-query work, especially as the interior grows. The measured improvement is at the observer boundary, not a full-knot recognition speedup, and leaves the global quasi-polynomial problem open.

# 72 Binary interval orbits and supplied normal-surface topology

The quotient-kernels delivery in `6cfd26c80` supplies an implementation of the classical Agol–Hass–Thurston orbit algorithm, with a sharper periodic merger. We integrate its interval kernel, normal-coordinate adapter, fixtures and tests. This closes the unweighted orbit-counting implementation gap identified in the earlier normal-surface reviews. It does not close the surface-search, weighted component-extraction, cutting, attachment or diagram-to-exterior provenance gaps. Ordinary recognition does not yet dispatch this supplied-surface interface.

The integration provenance records the archive's full SHA-256. All 84 manifest entries verify. Seven integrated source/test files match the reviewed delivery exactly, as recorded in `data/quotient-orbit-integration.json`. The coefficient-span and joint cyclic-overlap changes in the same delivery are separate, pending work. This section reports our reproduction, not the delivery's historical full-suite timings or its measurements against an older checkout.

## 72.1 The encoded relation and a sharp local merger

An interval pairing is an isometry from $[a, b]$ to $[c, d]$, with inclusive integer endpoints, equal positive widths, and $a \le c$. Its map is either $x \mapsto x + c - a$ or $x \mapsto a + d - x$. Endpoints have $O(\log(N + 1))$ bits; the interface stores $k$ such records on $[0, N - 1]$, never $N$ vertices. The requested answer is the number of equivalence classes generated by these identifications, including uncovered singleton points.

A positive translation with $a < c \le b + 1$ has period $p = c - a$ and support $[a, d]$ of size at least $2p$. Its classes are precisely the residues modulo $p$ within this support. Consider two such supports, of periods $p, q$, with $h$ points of intersection, and put $g = \gcd(p, q)$.

**Proposition 72.1.** *The relation generated by the two pairings equals the residue relation modulo $g$ on their union if and only if $h \ge p + q - g$.*

*Proof.* Every identification preserves residue modulo $g$. Color each point of the intersection by its generated orbit. This sequence has periods $p$ and $q$. The Fine–Wilf theorem gives period $g$ at length $p + q - g$ [4]. This length is at least $\max(p, q)$, so every residue class in either support meets the intersection. Thus every point reaches the appropriate common $g$-class, proving sufficiency.

For necessity, first replace each support separately by its residue classes: there are $p + q$ vertices. Each shared point supplies one edge between these two sets of vertices. These $h$ edges can reduce the number of components by at most $h$, leaving at least $p + q - h$. Below the proposed threshold this exceeds $g$. Both supports contain every residue modulo $g$, so equality with the $g$-residue relation is impossible. □

The condition is exact for these two generators alone. Additional pairings can identify further classes; failure of this test is not a disconnectedness certificate. A justified replacement remains valid when any common set of other generators is adjoined. The maintained default uses this threshold; `periodic_rule='aht'` retains $p + q$ as a comparison mode.

## 72.2 Counting without expanding integer points

`../fast/fastunknot/interval_orbits.py` repeatedly deletes identities, contracts uncovered intervals while adding their singleton counts, trims duplicate reflection arrows, merges qualifying periodic supports, transmits pairings through a maximal rightmost carrier, and truncates its exclusive suffix. Trimming a reflection leaves its fixed midpoint unpaired; the next static contraction counts it if no other generator meets it. Truncation removes points already identified with surviving representatives and adds no singleton count. Transmission computes maximal translation powers

by integer division, rather than executing one shift per point. These are the classical AHT operations [18]; their order and maximality conditions are part of the implementation contract.

For the sharper merger, let $w_1, w_2$ be the old domain widths. The merged width is

$$w' = w_1 + w_2 + p + q - h - g \leq w_1 + w_2 \leq 2w_1w_2.$$

Consequently the AHT potential $4^k \prod_i w_i$ decreases by at least a factor two at each such merge. When no sharp merge is available, no classical merge is available either. The remaining maximal-transmission progress argument is therefore unchanged. There are polynomially many cycles in $k$ and $\log(N+1)$; sorting, scanning all candidate pairs and binary integer arithmetic within each cycle preserve a polynomial bit bound. We do not infer an optimal polynomial degree from these simple list-based data structures.

For $r$ adjacent period-$p$ pairings on $(r+1)p$ points,

$$[ip, (i+1)p-1] \longrightarrow [(i+1)p, (i+2)p-1], \qquad 0 \leq i < r,$$

the count is $p$. The classical threshold misses every neighboring overlap of size $p$, and the rightmost suffix is removed one block per cycle: there are $r+1$ cycles including the final static contraction. The sharp rule merges the entire chain in one cycle and finishes in two. Candidate-pair scanning still has a cost; cycle savings alone are not timing evidence.

`max_cycles` limits begun cycles. It is neither a bit-operation allowance nor a wall-clock deadline. Exhaustion returns an incomplete result with no count. Cooperative callbacks propagate cancellation. The signed wrapper constructs $2k$ pairings on a two-sheeted interval of size $2N$, with an independent parity bit on each identification. If its orbit count is $C_2$ and the base count is $C$, then $C_2 - C$ components have a consistent parity labeling and $2C - C_2$ do not. The two counts share one cycle allowance. Order reversal of an interval does not by itself specify this parity character. A same-orbit query compares counts before and after adjoining a singleton identification; it does not extract a component.

## 72.3 From normal coordinates to three orbit queries

`../fast/fastunknot/normal_surface_orbits.py` accepts a finite connected compact orientable tetrahedral manifold with exactly one torus boundary, and seven nonnegative binary coordinates per tetrahedron. It reconstructs reciprocal face identifications, edge orientations and links, vertex links, ambient orientation and boundary topology. It rejects reversed edge self-identifications, invalid links and unsupported ambient manifolds. Coordinates must satisfy matching equations and the quadrilateral constraint. They need not be primitive, fundamental or extremal.

Allocate one interval for the intersection points on each global edge. The ordered points nearest a face corner are determined by its triangle coordinate plus the appropriate quadrilateral coordinate. Each face has at most three such arc families, represented by interval isometries. Local edge directions are transported through the actual face gluings; global vertex labels cannot determine direction in a one-vertex manifold. One copy of each paired face is used. This gives $O(t)$ pairings, with at most $4W$ points when the surface has $W$ normal discs. The normal surface has the same components as this graph: every attached disc has a connected boundary already in the graph. A second graph uses only boundary edges and faces, avoiding spurious isolated interior points.

The three queries count the surface, the surface with all coordinates doubled, and its boundary. In an orientable ambient manifold the doubled surface is the horizontal boundary of its normal interval bundle, hence its orientation double. If the first two counts are $C, C_2$, the numbers of orientable and nonorientable components are $C_2 - C$ and $2C - C_2$. This geometric doubling handles one-sided components correctly: two copies of the coordinate vector of a Möbius band describe an annulus. Total Euler characteristic is computed directly as $V - E + F$ from global

edge intersections, normal arcs and discs. For connected output, boundary count and Euler characteristic determine genus or crosscap number. For disconnected output these are aggregate statistics; no distribution of genera among components is asserted.

A connected surface with $\chi = 1$ and one boundary curve is a disc. To test whether that disc compresses the torus, the adapter checks the boundary intersection cocycle over $\mathbb{F}_2$. Solving for vertex potentials tests whether this cocycle is a coboundary. An inconsistency means nonzero boundary homology. A simple closed curve on a torus is essential exactly when this mod-two class is nonzero: an essential curve has a primitive integral slope. This last assertion uses simplicity and the torus hypothesis, and is not a general multicurve criterion.

All three orbit calls share one cycle allowance. If any stops early, the adapter returns `INCONCLUSIVE` without a disc assertion. Completed topology has polynomial cost in $t$ and $\log(W+1)$. A returned compressing disc assertion concerns the supplied manifold only. It does not establish that this manifold is the exterior of the user's knot, and it is not a new independent serialized knot certificate.

### 72.4 Validation and controlled measurements

The 22 focused tests pass with Regina. They include exhaustive small pairing systems, random literal union-find comparisons, both merger thresholds, binary endpoint inflation, signed parity graphs, cancellation and shared budgets. The geometric tests compare layered meridians with literal arc graphs, compare admissible normal-surface sums and component types with Regina, exhaust one-tetrahedron one-face-pairing manifold validation, and randomize tetrahedron and vertex labels. They also check huge parallel discs, odd and even multiples of a one-sided surface, boundary-parallel discs, and a 128-tetrahedron meridian with more than $10^{26}$ normal discs. Detailed logs are retained under `data/quotient-orbit-*`. The complete maintained suite passes all 805 tests with Regina in 191.413 seconds.

| Query | Classical | Sharp | A/A | Ratio | Cycles |
|---|---|---|---|---|---|
| Chain, 8 pairings | 0.414 | 0.085 | 0.086 | 4.871 | 9/2 |
| Chain, 16 pairings | 2.212 | 0.162 | 0.162 | 13.066 | 17/2 |
| Chain, 32 pairings | 12.904 | 0.317 | 0.315 | 41.464 | 33/2 |
| Chain, 64 pairings | 89.518 | 0.602 | 0.609 | 147.258 | 65/2 |
| Chain, 128 pairings | 759.598 | 1.195 | 1.199 | 649.887 | 129/2 |
| Meridian, 4 tetrahedra | 1.907 | 1.971 | 1.954 | 0.971 | 23/25 |
| Meridian, 32 tetrahedra | 73.945 | 70.930 | 71.129 | 1.038 | 107/109 |
| Meridian, 128 tetrahedra | 2854.560 | 2862.672 | 2822.536 | 0.982 | 395/397 |
| $2^{2000}$ parallel discs | 1.956 | 2.059 | 2.061 | 0.952 | 20/22 |

Median milliseconds; ratio is the median paired classical/sharp time. Cycles are classical/sharp. Surface queries include validation and all three orbit counts; chain queries count interval orbits only.

The maintained driver `../fast/benchmark_orbits.py` uses seed 261008171, one warmup and five measured rounds with shuffled classical, sharp and identical-sharp control arms. There are 29 interval cases and 13 surface cases, giving 126 warmup calls and 630 measured calls. Source hashes agree before and after the run, pinned to the integrated code at `ac0dfceb9`. Raw samples, arm orders, operation counts and complete last outputs are in `../fast/results/orbits_20261008.json`.

At 128 adjacent pairings the paired classical/sharp ratio is 649.887, with 129 versus two cycles. This is a constructed interval relation, not a knot-diagram workload. Six independently generated tiny parity graphs also determine expected counts after replacing each vertex by $2^b + 1$ ordered points, for $b = 16, 256, 4096, 16384$. A component with consistent reversal parity lifts to $2^b + 1$

components; an inconsistent one lifts to $2^{b-1}+1$. All expanded counts agree without materializing the large graph. Five templates have essentially unchanged cycle counts; one drops from nine to seven, with measured ratios from 1.126 to 1.211.

For actual layered-torus meridians with 4–128 tetrahedra, complete topology takes about 2 ms to 2.9 seconds. The sharper rule has paired ratios 0.971–1.038, and takes two more total cycles on these examples. Earlier valid merges can change later scheduling, so a sharper local threshold does not monotonically reduce the total cycle count. At 128 tetrahedra both modes validate the surface with more than $10^{26}$ discs. On three tetrahedra with $2^{2000}$ parallel meridians, the sharp query takes about 2.06 ms and 22 cycles. Across these parallel-copy cases it is roughly 4–7% slower than the classical mode. Identical-control paired ratios range from 0.982 to 1.030 across all cases.

Explicit connectivity checks run only below a declared 100,000-point cap; larger cases are omitted, not described as timeouts. They perform less work than the encoded surface query, which includes manifold validation and three counts, so their single samples are diagnostic rather than a matched speedup comparison. The evidence establishes useful encoded local queries and a sharp-merger gain on its target family. It establishes neither a uniform local speedup nor a whole-recognizer speedup.

## 72.5 Repairing worker communication

The integration also repairs Regina worker communication. A retry with `communicate(None)` after timing out on a large request can leave input unsent on supported Python implementations. The worker then waits for data while its parent waits for completion. The revised code stages UTF-8 input, output and diagnostics in temporary files, waits cooperatively, and reads the completed streams. An exit stack closes every stream; cancellation kills and reaps a live child. Startup, staging, waiting and reading share the allowance through checks before and after blocking operations. Filesystem calls are not made preemptible by these checks.

All nine focused worker tests pass, including delayed reading of a large request, simultaneous large UTF-8 output and diagnostics, and local/global cancellation cleanup. This is a reliability repair to an existing complete backend, separate from the new supplied-surface API.

## 72.6 Christoffel delivery: initial review before the native contract

The adjacent Christoffel archive has SHA-256 prefix `9dc35414f591da0b`; its full hash and reproduction status are recorded in `data/quotient-orbit-review.json`. All 32 manifest entries verify; its 22 tests and both saved example replays pass locally. The report supplies a compressed primitive-power test for freely and cyclically reduced rank-two words, with independent local replay. For coherently signed exponent counts $(a, b)$, divide by $d = \gcd(|a|, |b|)$ to obtain primitive counts $(u, v)$ after sign normalization. Prefix weights $(v, -u)$ characterize Christoffel powers by width $u + v - 1$, including the pure-letter case. Concatenation summaries store total displacement and minimum/maximum prefix displacement, so the exponentially long word need not be materialized.

The useful group-theoretic consequence is conditional on provenance: in a torsion-free group a relation $P^d = 1$ with $P$ primitive kills $P$. For a two-generator knot group this leaves a cyclic group, necessarily $\mathbb{Z}$ by knot-group abelianization. The resulting unknot conclusion requires a verified presentation trace from the input knot. The delivery's small braid demonstration has an independent literal, capped trace replay; its higher-rank projection prototype lacks an independent whole-round checker and assumes torsion-freeness. Its WordArena adapter is tested against a schema stub, not our maintained search.

At this initial checkpoint the compressed verifier required its existing rank-one/empty-relator terminal. The later integration in Section 84 adds a separate version-five terminal and independent source-bound replay, while preserving the old contract. The report's conditional depth estimate,

polynomial initial cost times $2^{O(d)}$, still needs a controlled presentation producer, a depth bound and a guarantee of reaching a decisive endpoint. Neither this estimate nor the polynomial orbit kernel proves unrestricted quasi-polynomial recognition.

During this review, the original deliveries were archived unchanged as reports 45 (Christoffel) and 46 (quotient kernels); the incoming ZIP paths in the provenance record refer to their arrival commit. The newer report 47 supplies a separate orbit implementation with independent local-rule replay, marked-component incidence and a two-meridian search proposal. Its 512-entry manifest verifies. Reconciliation of that certificate API with the sharper maintained kernel is subsequent work, not validation claimed for this checkpoint. Section 73 subsequently integrates independent orbit replay and marked-union incidence, while retaining the geometric and recognition boundaries above.

# 73 Independent orbit proofs and reusable marked-union queries

Report 47 supplies an independent verifier for the six AHT reduction rules and a reduction from marked-component incidence to ordinary orbit counts. We reconcile these with the sharper kernel of Section 72, instead of installing a second production orbit scheduler. The maintained extension records optional local proofs, replays both merger conventions, computes complete incidence histograms, and shares repeated marked-union queries. These are local components of a possible compressed hierarchy, not a recognition procedure.

## 73.1 What the verifier establishes

`../fast/fastunknot/interval_orbit_verify.py` imports neither the orbit producer nor its scheduling helpers. It checks the universe size and the ordered, canonical input pairing list, then applies six independently implemented local rules. A certificate must finish with an empty universe and no pairings. Its claimed count must equal the number of singleton orbits removed by static contractions. A truncated valid prefix proves no count. Boolean values are rejected in integer fields, and exact hexadecimal transport is accepted without changing Python's global decimal limit.

Identity deletion removes only reflexive relations. Static contraction recomputes the entire complement of all pairing domains and ranges, compares the supplied gaps, and shifts remaining endpoints by their exact deleted rank. Reflection trimming retains one arrow for each two-point orbit and leaves a fixed midpoint available for subsequent counting. A periodic merger checks both periods and the union hull. Version one requires overlap $p+q$; version two requires $p+q-\gcd(p,q)$, using the proved criterion in the preceding section. Changing a sharp-only trace to version one therefore makes it fail replay.

Transmission records the powers applied to the source and image intervals of a target pairing. The verifier checks that these partial inverses are defined throughout each interval. For a positive translation of distance $p$, power $s$ is legal only when the original interval is in the range and its last intermediate left endpoint remains in that range. Monotonicity then covers every intermediate power with one inequality; the cost depends on the bit length of $s$, not on $s$. A trimmed reflection permits one inverse step. The verifier composes the two affine maps independently and normalizes their resulting intervals.

Suffix truncation requires the carrier to end at the universe boundary, the removed suffix to lie in its range, and every other pairing to avoid that suffix. A positive carrier has strictly positive displacement, so iteration reaches a surviving earlier representative. A reflection must already have disjoint domain and range. The verifier does not require the producer's preferred scheduling choices: any sequence satisfying these local conditions is sound. Scheduling is needed for the producer's polynomial running-time theorem, not for the validity of a supplied trace.

Replay is polynomial in the explicit trace length, pairing-list size and integer bit lengths. The producer's polynomial cycle bound gives a polynomial-size trace: a cycle emits polynomially many deletion, gap, merge, transmission and truncation records, each with polynomially many binary integers. This is not a hostile-input memory cap; callers can limit serialized size and use cooperative cancellation. Production recording is optional. Incomplete runs expose neither a partial count nor a certificate.

## 73.2 Coning a marked union

Let $A$ be a nonempty union of half-open marked intervals in the orbit universe. For each interval $[a, b)$ of length at least two, adjoin the translation $[a, b-2] \to [a+1, b-1]$. Its points become one class. Join one point of each interval to a common marked anchor by singleton pairings. There are at most twice as many new pairings as marked intervals, with no expansion into individual points.

If $C$ is the original orbit count and $C_A$ is the count after these identifications, then the number of original orbits meeting $A$ is

$$H(A) = C - C_A + 1.$$

Indeed, exactly those original orbits meeting $A$ become one class; all others retain their identities. The empty union has $H(\varnothing) = 0$ and requires no coned query. This argument allows partial translations, reflections, intersecting marks, static points and disconnected supports.

For ordered ports $A_1, \ldots, A_r$, let $h(T)$ count orbits meeting exactly the ports indexed by $T \subseteq [r]$. Define

$$F(S) = H\left(\bigcup_{i \in S} A_i\right), \qquad G(U) = C - F([r] \setminus U).$$

An orbit contributes to $G(U)$ precisely when its signature is a subset of $U$, so $G(U) = \sum_{T \subseteq U} h(T)$. The producer recovers $h$ by Boolean Möbius inversion in $O(r2^r)$ integer subtractions. It retains $h(\varnothing)$, the completely unmarked component count.

## 73.3 Sharing queries and proofs without changing the output

Different port subsets can describe the same marked set. The implementation sorts intervals, merges overlap and adjacency, and uses the resulting ordered disjoint union as its cache key. Equal unions introduce exactly the same coning pairings in the same order. Consequently one result and one proof can serve every mask with that union; neither an approximate shape signature nor equality of counts is used to justify reuse.

If all $r$ nonempty ports coincide, there is one distinct nonempty union: only two orbit queries are needed, including the original count. If the ports form a strict nonempty nested chain, there are $r$ distinct unions and $r + 1$ queries. Disjoint nonempty ports can have $2^r - 1$ distinct unions, so no uniform query reduction follows. The output remains a dense array of $2^r$ counts, even when query reuse is substantial.

The new incidence certificate has version two. It contains one baseline orbit proof, one proof per distinct nonempty union, and an index array binding every subset mask to its proof. Empty unions have no proof index. The verifier reconstructs the marked unions from the supplied ports, rejects rebinding one proof to a different union, checks every distinct orbit proof, and rejects unused or duplicate proof assignments. To check the histogram it computes its *forward* subset sums, rather than repeating the producer's inverse transform. This catches incorrect histograms independently of inversion code. Input normalization and the explicit coning construction are shared; their correctness is also checked against literal marked graphs in the small-instance tests.

Let $M$ be the total number of marked intervals and $U$ the number of distinct nonempty unions. If

$P(k, B)$ bounds the orbit kernel for $k$ pairings and $B$-bit endpoints, a coarse bound is

$$(U + 1)P(k + 2M, B) + O(2^r M \log(M + 1)B + r2^r B) .$$

The first term pays for the baseline and distinct coned queries; the second pays for union construction, cache keys and dense inversion. For hash-based lookups this uses ordinary expected dictionary costs; balanced ordered keys would add polynomial comparison factors without changing the $2^r \operatorname{poly}(k + M + r, B)$ bound. Certificate storage contains one trace per distinct union and $O(2^r \log(U + 2))$ bits of proof indices, in addition to the dense counts and input records.

Polynomially bounded $k, M, B$ give a polynomial query for $r = O(\log n)$, and a quasi-polynomial query for $r = O((\log n)^2)$. Neither condition has been established for a complete unknot hierarchy. The weighted AHT extension offers a different route to sparse component-weight information; this dense reduction is not a new general complexity bound for weighted orbits. It implements one useful aggregate query over the unweighted kernel.

Unlike the delivery's separate allowance for each coned query, the maintained interface shares one `max_cycles` budget across all orbit calls. Exhaustion returns `INCONCLUSIVE` with work statistics and no histogram or certificate. Port validation and Boolean inversion also poll the global callback. The default explicit port cap is 12. Raising it permits the corresponding exponential output cost; the cap is not a theorem about geometric input. A signature records set incidence only. It omits point multiplicity, cyclic order, slope and the action of future attachment maps.

### 73.4 Validation and measured costs

Nineteen new verifier tests adapt the report's adversarial replay suite and add sharp-threshold version checks, a forged merger below the exact threshold, 1,000 randomized comparisons with literal graphs, and a transmission of $2^{16000} - 1$ steps stored in a single record. They compare traced and untraced results and operation statistics, exercise exact JSON transport, and disable producer functions while replaying proofs. Malformed inputs, changed source bindings, invalid local operations and unfinished traces are rejected. Seven incidence tests compare 250 random marked graphs and 18 actual normal-arc systems against literal component signatures, including parallel meridians, overlapping and coincident ports, huge static universes, shared budget boundaries and certificate mutations. All 831 maintained tests pass with Regina in 179.802 seconds. A separate cross-replay audit imports the unchanged report-47 producer and verifier under another package namespace. For 1,000 interval systems, the maintained verifier accepts the delivery's classical proofs and the delivery's verifier accepts our classical proofs; our sharp proofs also pass independent replay. The audit exercises all six rule types. One hundred marked systems agree with the delivery's complete incidence histograms, with both proof formats checked by their respective verifiers. Evidence and source hashes are in `data/orbit-certificate-cross-replay.json`.

| Marked system | Fresh | Reuse | Certified | Ratio | Queries |
|---|---|---|---|---|---|
| Static, coincident | 8.13 | 1.41 | 3.16 | 5.76 | 256/2 |
| Static, nested | 8.95 | 1.41 | 3.00 | 6.09 | 256/9 |
| Static, disjoint | 36.78 | 38.81 | 67.63 | 0.95 | 256/256 |
| Meridian, coincident | 128.47 | 3.79 | 5.76 | 33.31 | 64/2 |
| Meridian, nested | 52.57 | 10.06 | 15.31 | 5.26 | 64/7 |
| Meridian, disjoint | 61.32 | 61.59 | 98.67 | 0.99 | 64/64 |
| Parallel, coincident | 2410.80 | 40.47 | 68.65 | 61.82 | 64/2 |
| Parallel, nested | 1828.16 | 137.31 | 242.90 | 13.24 | 64/7 |
| Parallel, disjoint | 800.09 | 782.48 | 1357.14 | 1.02 | 64/64 |

Median milliseconds per complete incidence query. Certified includes proof construction and independent replay; ratio is the median paired fresh/reuse time. Queries are fresh/reuse.

The incidence benchmark uses the same maintained orbit kernel for both the fresh-query ablation and the cached implementation, isolating reuse from the difference between the reports' two orbit schedulers. There are nine workloads, four shuffled arms, one warmup and five measured rounds: 36 warmup and 180 measured incidence calls. The fourth arm is an identical cached control. Seed 261008483 fixes the order. Each complete histogram is compared outside its timed region; the certified arm includes local proof construction and independent verification in its timing. The inputs use eight marks on a static $2^{4000}$-point universe, six marks on the normal arc graph of an eight-tetrahedron meridian, or six marks on $2^{500}$ parallel copies of that meridian. Marks are chosen coordinate intervals, not claimed outputs of a geometric attachment-port extractor.

Coincident marks reduce queries from 256 to two in the static case and from 64 to two in the native cases. The native unscaled query improves from 128.5 to 3.8 ms, with paired ratio 33.308. The large parallel-copy query improves from 2.411 seconds to 40.5 ms, with paired ratio 61.817; including proof generation and replay takes 68.6 ms. Ratios can exceed the ratio of total query counts because the unchanged baseline count is cheaper than a coned count. Nested marks also benefit, while disjoint marks leave every union distinct. The latter show no reliable improvement and can incur cache overhead. All timing arms and proof-size costs are retained in `../fast/results/orbit_certificates_20261008.json`; for example, the disjoint parallel-copy certificate is about 13.3 MB in the measured JSON transport. Binary input does not imply a tiny trace.

The same driver separately measures ordinary counts, optional recording, and recording plus replay against the pre-certificate kernel pinned at `8cd6dcfcd`. Its initial small-query samples had divergent identical controls, so a separate batched follow-up retains those results and measures fresh batches of the four workloads. It uses seed 261008485, seven measured rounds, and 1,000 repeats for each tiny translation/reflection query, 30 for the 64-pair chain and ten for the native 32-tetrahedron arc query. Its raw results are in `../fast/results/orbit_trace_cost_20261008.json`. On the native arc query, the old/current paired ratio is 0.976: the optional recording branches have a small default-path cost. Median times are 33.34 ms before the change, 33.92 ms without recording, 37.48 ms with recording and 42.88 ms with recording plus replay. The translation and chain ratios are 0.968 and 0.993. Reflection samples retain some variability, so no precise default-path gain is claimed there. Paired identical-current ratios in the follow-up range from 0.999 to 1.026. Proof production is optional precisely because its time and storage are real costs. These local costs do not alter ordinary knot dispatch, which does not currently call the orbit interface.

The remaining normal-disc certificate adapter and sparse two-meridian seed search in report 47 require separate review. The report's source snapshots are unchanged. The maintained coning and replay code establish assertions about supplied interval relations, including relations extracted by our validated normal-arc adapter; they do not establish diagram-to-exterior provenance or search completeness.

## 74 Native compressed braid recognition and singleton forests

The compressed-braid delivery arriving at commit `36f1dd8ef` provides a useful complete algorithm on a restricted input domain. We integrate its three-braid normal forms, singleton decompositions and local proofs with the maintained exact string kernel. The result decides a supplied binary grammar for a three-braid in polynomial bit time, given sufficient resources, even when its expanded word is exponentially long. It is a separate source-bound interface, not a conversion from arbitrary planar diagrams. The earlier explicit three-braid criterion is unchanged; the contribution is compressed discovery with bounded intermediate representation growth and replay.

## 74.1 Input and exact finite summaries

An input contains a strand count $m$, an ordered rule list and a root index. Rule zero is empty. Subsequent rules are either a signed Artin generator $\sigma_j^{\pm 1}$, $1 \le j < m$, or concatenation of two strictly earlier rules. The represented object is the ordinary closure of the root word. Builder operations for powering and inversion produce these same binary rules: binary powering needs $O(\log(q+1))$ rules for exponent $q$, and inversion reverses concatenations with memoization. Neither is an extra input operation.

Let $g$ denote input bit size, $s$ the number of nonempty rules, and

$$N_* = \max_i |W_i|, \qquad B = \lceil \log_2(N_* + 2) \rceil.$$

All supplied rules count, including unreachable ones, since validation examines them. Strict back references give $N_* \le 2^s$ and $B = O(s+1)$. Length and exponent sum are additive at concatenations; strand permutations compose. For three strands this costs $O(s)$ operations on $O(B)$-bit integers and constant-size permutations. The root permutation's cycles are exactly the closure components. A non-knot closure returns `LINK`, separately from `KNOTTED`. Boolean aliases for integer fields are rejected.

## 74.2 The complete three-braid test

Write $a = \sigma_1$, $b = \sigma_2$, $x = aba$, $y = ab$, and $z = y^2$. Then

$$B_3 = \langle x, y \mid x^2 = y^3 \rangle, \qquad B_3 / Z(B_3) = C_2 * C_3.$$

The central generator has exponent sum six. Thus quotient element and exact exponent sum together determine a braid: an element trivial in both is a central power with zero exponent. The same argument lifts quotient conjugacy when the exponent sums agree. In the quotient,

$$a \mapsto zx, \quad a^{-1} \mapsto xy, \quad b \mapsto xz, \quad b^{-1} \mapsto yx.$$

Reduced syllables alternate between the $x$ factor and the $\{y, z\}$ factor. For a one-component closure with exponent $e$ and cyclically reduced quotient core $C$, the existing complete criterion is

| $e$ | unknot core, up to cyclic rotation |
|---|---|
| 2 | $y$ |
| $-2$ | $z$ |
| 0 | four syllables: two $x$'s, one $y$ and one $z$. |

Only cores of length at most four need explicit inspection.

Here is the algebraic reason the criterion is complete. Bennequin's inequality applied to an unknot and its mirror gives $|e| \le 2$. For the reduced Burau matrix $M$ at $t = -1$, the three-braid Alexander formula gives $\det K = |2 - \operatorname{tr} M|$. An unknot therefore requires trace one or three. A cyclically reduced quotient word of length greater than one rotates to a product of $xy$ and $xz$. Up to overall sign and a fixed conjugation, its matrix is a positive product of

$$L = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad R = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

If it contains $p > 0$ copies of $L$ and $q > 0$ copies of $R$, expansion of these matrices as identity plus a nonnegative elementary matrix gives trace at least $2 + pq$: each choice of one upper and one lower elementary term contributes one to trace. Absolute trace three forces $p = q = 1$; a pure power has trace two. The remaining cores $1, x, y, z$ have absolute traces $2, 0, 1, 1$. Their exponent residues modulo six, together with $|e| \le 2$, give precisely the three rows above. Conversely these rows are quotient conjugate to $ab$, $b^{-1}a^{-1}$ and $ab^{-1}$ with the same exponent. The central ambiguity vanishes, and each canonical closure destabilizes to an unknot. This restricted argument does not make determinant one a general unknot certificate.

## 74.3 One prefix query replaces expanded reduction

The native `WordArena` supplies exact concatenation, substring, equality and longest common prefix (LCP) on straight-line programs. Syllable inversion here reverses strings, fixes $x$ and exchanges $y, z$; it does not use free-group integer sign negation. For reduced strings $U, V$, compute

$$k = \mathrm{LCP}(U^{-1}, V), \qquad U' = U[0 : |U| - k], \quad V' = V[k : |V|].$$

If either remainder is empty, the other is the product. If the surviving endpoints have different factors, concatenate them. Otherwise maximality excludes inverse endpoints, leaving only $yy$ or $zz$; replace these by $z$ or $y$. The neighbors of this new syllable, if present, are $x$, so reduction cannot cascade. Each input concatenation requires one LCP and at most one additional factor multiplication, regardless of expanded length.

For cyclic reduction of a reduced $W$, take

$$t = \mathrm{LCP}(W, W^{-1}, \lfloor |W|/2 \rfloor), \qquad P = W[0 : t], \quad R = W[t : |W| - t].$$

If $R$ has at most one syllable or endpoints in different factors, its core is $R$ and conjugator $P$. Otherwise maximality forces $R = fMf$ with $f \in \{y, z\}$. Set $C = Mf^2$ and $Q = Pf$. Then $QCQ^{-1} = PfMfP^{-1} = W$, and $C$ is cyclically reduced. All trimming positions are binary integers; the operation never iterates over removed syllables.

## 74.4 Representation size and bit complexity

A substring of a height-$h$ binary string grammar requires $O(h)$ new nodes: only its two boundary paths are partly retained, and complete subtrees are reused. Reassembling in the original tree shape does not increase height. Inversion preserves height. A reduced input rule at dependency depth $d$ therefore has height at most $2d + 2$: a terminal image has height two, and product reassembly adds at most two levels above sliced children.

The maintained LCP routine first tries a bounded direct DAG comparison, then uses exact equality and bisection. Its worst-case equality query count is $O(B + 1)$. There are $O(s)$ product/cyclic prefix queries, and every scratch substring adds $O(s)$ nodes. Memoized inversion adds only a constant number of rules per newly visited rule. Consequently the total string allocation is

$$S = O(s^2(B + 1)), \qquad \text{equality calls} = O(s(B + 1)).$$

If $E(S, B)$ is the deterministic polynomial bit bound for the maintained exact SLP equality procedure, discovery costs

$$O\Big(s(B + 1)E(Cs^2(B + 1), B + O(1))\Big) + \mathrm{poly}(g)$$

for a fixed constant $C$. Exact compressed equality is the essential primitive [34]; a bounded heuristic alone would not justify this theorem. Since $B = O(s + 1)$, this is polynomial in input bit size. The cubic specialization of the allocation bound is *not* a cubic running-time claim. Dictionary and integer costs also belong in the polynomial term. The finite default budgets are operational limits; increasing all allowances geometrically recovers the uncapped decision procedure.

## 74.5 Independent replay and the host contract

A proof binds the original grammar by exact canonical JSON, not merely a hash. It records finite summaries, the string-rule DAG, one reduced root and cancellation length per source rule, and the cyclic reduction witness. The verifier reconstructs the supplied string DAG in a fresh native arena. It checks reducedness, exact inverse cancellation equality and residual endpoints; these endpoints

establish maximality without searching for an LCP. It then checks the cyclic witness and finite unknot criterion. It imports neither the reducer nor its prefix-search procedure. The trusted base still includes exact string equality, integer arithmetic, input parsing and the topological criterion. Independent structure does not mean absence of shared primitives.

Certificates have $O(g + S\log(S+2) + sB)$ bits and replay uses $O(s)$ exact equality calls plus polynomial work. Arbitrary supplied proofs are verified in time polynomial in their combined input and proof sizes. The public `recognize` function publishes a proof-bearing result only after independent replay. Standalone `verify` performs the same source binding with fresh budgets.

The host shares one work allowance and one cumulative string-node allowance across metadata, every factor, discovery and replay. It also caps input rules, canonical input bytes, proof bytes and optional elapsed time. Defaults are 100,000 input rules, 100,000 allocated string nodes, ten million abstract work units, 16 MB of canonical input and 64 MB of proof. Exhaustion returns `INCONCLUSIVE` without a partial certificate. External cancellation and unexpected implementation errors propagate; they are not converted into mathematical verdicts. These are cooperative allowances, not hard CPU or resident-memory guarantees. CLI input-byte checking precedes JSON parsing; the API time allowance does not include that CLI parse.

## 74.6 Singleton cuts on a compressed braid

For general $m$, a reverse DAG pass propagates root multiplicities and sums unsigned generator occurrences. A repeated child receives both contributions. This counts occurrences in the root, rather than counting distinct terminal rules. A missing generator separates two nonempty strand blocks with no crossing between them, so the closure is a link. This test precedes allocating an $m$-entry permutation. Otherwise all $m-1$ generators occur and $m \le s+1$, bounding subsequent permutation work polynomially in $g$ even when the declared strand count was supplied in binary.

A generator $\sigma_j$ occurring exactly once gives a connected-sum cut. Rotate the braid to put that crossing at the closure seam. Every other crossing lies entirely within the first $j$ or last $m-j$ strands, and crossings in these two groups commute. A sphere through the single seam separates the two projected closures, joined by the one crossing, of either sign. The closure component count is the sum of the two projected component counts minus one; a one-component parent therefore has knot factors. Doing all such cuts yields ordered strand intervals. Projecting to an interval removes other generators, shifts retained indices, and suppresses empty concatenations in one pass through the input DAG. There are at most $m$ factors and $O(s)$ projected rules per factor.

The verifier recomputes occurrence counts, exact cut locations and projected grammars, then checks each leaf proof. Leaves with at most two strands use their complete exponent criterion; three-strand leaves use compressed quotient replay. Wider leaves may be rejected by $|e| > m-1$. In the initial integration an unresolved wider leaf stayed explicitly `INCONCLUSIVE`; Section 75 adds its complete bounded fallback. A replayed forest certificate can certify that residual description without certifying unknotness. For a forest of knots, all trivial factors imply an unknot, and a nontrivial factor implies a nontrivial connected sum.

## 74.7 A conditional exceptional-factor bound

After all singleton cuts, every retained generator in an unresolved factor occurs at least twice. For its expanded crossing count $n_i$, strand count $m_i$ and minority-sign count $\kappa_i$, this gives

$$m_i - 1 \le n_i/2, \qquad |e_i| = n_i - 2\kappa_i.$$

If its Bennequin obstruction does not already decide the factor, then $n_i - 2\kappa_i \le m_i - 1$, hence

$$n_i \le 4\kappa_i, \qquad m_i \le 2\kappa_i + 1.$$

A complete exponential decision fallback on just these expanded exceptional factors would therefore yield

$$\mathrm{poly}(g) + \sum_{i \text{ unresolved}} 2^{O(\kappa_i)}.$$

The solved compressed three-braid factors are excluded from the sum even when their minority counts are exponentially large. With polynomially many factors and $\max_i \kappa_i = O((\log g)^2)$, this bound is quasi-polynomial in $g$. This was a conditional composition theorem at the delivery's initial integration. Section 75 connects the complete fallback and its source-projection certificate contract, making the bound an implemented restricted-input guarantee. No theorem bounds the exceptional parameters for every knot diagram.

## 74.8 Why matrix evaluation can destroy compression

Let $u = ab^{-1}$ and $\beta_m = u^m(ab)u^{-m}$. Its grammar has $O(\log(m+1))$ rules and its closure is an unknot. Under Burau at $-1$, $u$ maps to $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Writing $b = F_{2m}$ and $d = F_{2m-1}$, one entry of the conjugated matrix is $3b^2 + 3bd + d^2$. It has $\Omega(m)$ bits despite trace one. Taking $m = 2^k$ therefore makes explicit integer matrix evaluation exponentially costly in $k$, while the string proof remains polynomial. This is an obstruction to materializing those integers, not to symbolic matrix circuits or modular methods.

## 74.9 Native validation and measurements

The delivery is now preserved as report 48. Its later numbered placement was checked against the original archive and the prior integration hashes: all 53 manifest files are identical, and the omitted manifest was restored verbatim. This relocation does not constitute a new algorithm or a new execution of the earlier audit. All 53 files in the delivery manifest match their hashes. The original 42 tests pass, and the adapted 42 tests plus 12 host tests pass with the actual maintained `WordArena`. The full maintained suite passes 885 tests with Regina in 204.549 seconds. Host tests cover shared budget boundaries across factor production and replay, cancellation during validation and replay, Boolean field substitutions, source rebinding, legacy proofs and CLI limits. Replay succeeds with producer reduction and LCP disabled. Huge inputs are checked with the debug expansion method disabled.

The broader native audit checks all 87,381 words through length eight against the maintained explicit gateway, an explicit free-product oracle and a separate Burau computation. It finds 41,005 links, 22,208 unknots and 24,168 nontrivial knots, with no disagreement. Four hundred shared random grammars agree with the explicit controls; another 400 quotient checks force the polynomial fallback by disabling fast prefix/equality probes. Capacity cases include conjugating powers $2^{4096}$, differently associated conjugators, and forests with 96 strands. Public discovery and standalone replay both pass. The 36.061-second audit and its exact source hashes are retained in `data/compressed-braid-native-audit.json`.

| Input | rules | explicit ms | native ms | paired ratio |
|---|---|---|---|---|
| sleeve_0_0 | 10 | 0.031 | 0.828 | 0.038 |
| sleeve_8_0 | 26 | 1.198 | 1.641 | 0.730 |
| sleeve_12_0 | 34 | 19.030 | 2.180 | 8.727 |
| sleeve_16_0 | 42 | 305.270 | 2.580 | 118.476 |
| sleeve_16_1 | 41 | 254.164 | 17.952 | 14.169 |
| sleeve_4096_0 | 8202 | – | 434.503 | – |
| explicit_random_500 | 435 | 1.651 | 48.793 | 0.034 |
| forest_16 | 4286 | – | 393.375 | – |

The paired benchmark starts each arm with the same grammar. Its explicit arm validates, expands under a $2^{20}$-letter cap, and calls the maintained explicit braid gateway. Its native arm includes input/proof limits, discovery and independent replay. Thus the native arm pays for a stronger proof output; we do not subtract replay time to improve its comparison. Each algorithm has an identical A/A control, one shuffled warmup and five shuffled measured rounds. Small cases are batched ten times. Oversized explicit inputs are omitted by preflight, not reported as timeouts or assigned speedup ratios. Forest cases have no matched complete wider explicit gateway and report only native timings. All raw samples, orders, budgets and source hashes are in `../fast/results/compressed_braid_20261008.json`.

For the unknot sleeve with conjugating exponent $2^{16}$ (262,146 expanded letters), the explicit median is 305.270 ms and native discovery plus replay is 2.580 ms, with paired ratio 118.476. The nontrivial-core sleeve at the same exponent improves from 254.164 to 17.952 ms, with paired ratio 14.169. At exponent $2^8$, the explicit unknot path is still faster (1.198 versus 1.641 ms). Random explicit conjugators of length 500 likewise favor the explicit gateway: 1.651 versus 48.793 ms. These regressions are retained. The $2^{4096}$ unknot sleeve takes 434.503 ms including replay and emits 349,628 canonical proof bytes. The 48-strand forest of 16 compressed factors takes 393.375 ms. These last two have no explicit comparison. Identical-arm ratios show a few percent variation, so the measurements support the large family-specific gains, not fine timing distinctions.

The ordinary explicit recognizer remains the default for explicit braid words. This separate native grammar interface removes an exponential expansion cost on a useful input class; it does not demonstrate an improvement for arbitrary planar diagrams. An adaptive composition should use the explicit path when expansion is cheap, retain compression for large shared words, and send only unsupported residual factors to a complete fallback. A crossover observed on one family is insufficient to install a universal timing threshold.

# 75 A complete fallback for exceptional compressed-braid factors

The preceding section left a precise missing implementation step: completing wider factors after singleton cuts without expanding the already solved three-braid factors. We now connect a bounded reduced Khovanov cube, with source-bound linear-algebra proofs, to the native grammar interface. Given sufficient allowances this makes recognition complete for every supplied braid grammar. Its running time is polynomial in grammar size plus an exponential term for each exceptional factor's minority-sign count. The latter parameters remain uncontrolled for general knots.

## 75.1 Dispatch and the source contract

The existing component check and exact singleton cuts run first. One- and two-strand factors use exponent sums; three-strand factors use compressed free-product normal forms. A wider factor with $|e| > m - 1$ is already nontrivial. For every remaining wider factor the host compares its exact binary expanded length with `fallback_max_crossings`, default 12. Only a factor that passes this comparison is expanded. All larger residuals remain explicitly inconclusive; no iteration up to their represented lengths occurs.

Expansion walks the projected grammar and skips zero-length children. This last detail is necessary: a depth-$s$ grammar of concatenated empty strings can have $2^s$ occurrences but length zero. With zero children skipped, every visited node lies on a path to one of the $n$ actual letters, giving an $O(s(n+1))$ traversal bound. The expansion is checked against its previously computed length. Three-braid factors never enter this routine, including factors with exponentially large minority-sign counts.

A forest proof containing an exceptional cube uses version two. Its original input, singleton

cuts, strand intervals and projected grammars are checked exactly as before. The child proof additionally records the expanded signed word and strand count, which replay recomputes from that projected grammar. Thus neither an independently chosen small braid nor a convenient PD diagram can substitute for the actual factor. The cube operates directly on the braid; a separate diagram-to-braid conversion is not part of its trusted input. Version-one proofs remain replayable with their original unsupported-factor semantics. Changing a version-two cube proof to version one is rejected.

## 75.2 The exact reduced cube

We adapt the explicit braid-resolution construction from report 34's MIT-0 oracle, retaining it as a separately loaded comparison implementation. For an $m$-strand word of length $n$, vertices comprise the $m$ top endpoints and two new bottom endpoints per crossing. In each state $\epsilon \in \{0,1\}^n$, smoothing joins either the two horizontal pairs or the two vertical pairs. The choice of state bit is reversed for a negative Artin generator. Joining final endpoints to their initial strand positions closes the resolution. A disjoint-set computation returns its circles. The circle containing vertex zero is the marked circle in every state.

Assign each circle a label in $\{1, x\}$ with $x^2 = 0$, and require the marked circle's label to be $x$. A state with $c$ circles therefore contributes $2^{c-1}$ basis elements to cube degree $|\epsilon|$. A cube edge changes one zero state bit to one and applies the ordinary Frobenius maps

$$\begin{aligned} &\mu(1,1) = 1, && \mu(1,x) = \mu(x,1) = x, \quad \mu(x,x) = 0, \\ &\Delta(1) = 1 \otimes x + x \otimes 1, && \Delta(x) = x \otimes x. \end{aligned}$$

Unchanged circles keep their labels. The marked-$x$ subcomplex is preserved. Over $\mathbb{F}_2$ the cube signs disappear. Homological normalization shifts and quantum gradings are unnecessary for the total rank used here.

Let $D_h$ be the dimension in raw cube degree $h$, and $r_h$ the rank of its differential to degree $h+1$. The implementation constructs each differential as integer bit-vector columns and checks $d^2 = 0$ before using

$$b_h = D_h - r_h - r_{h-1}, \qquad \dim \widetilde{Kh}(K; \mathbb{F}_2) = \sum_h b_h.$$

The same chain check runs during replay. Negative dimensions or zero total rank on a knot are implementation errors, not verdicts. The unknot test is total reduced rank one. Kronheimer–Mrowka gives the rational rank-one criterion [35]; the universal coefficient theorem bounds rational rank by mod-two dimension, while the unknot has a single torsion-free reduced generator. These facts give the same detection criterion over $\mathbb{F}_2$. They do not identify the Jones polynomial with a complete decision procedure.

## 75.3 A rank proof with no pivot search in replay

For each matrix column, discovery records a list of earlier basis-column indices whose XORs eliminate its leading bits, followed by the leading row of the residual, or $-1$ for zero. The proof stores indices, not copies of all intermediate bit vectors. Discovery uses a map from leading row to its basis column to select eliminations.

Replay reconstructs the differential from the source word and reads the proposed XOR list. Each reference must name a distinct preceding basis column. A zero residual must have pivot $-1$. A nonzero residual must have exactly the stated highest set bit, and that row must differ from all earlier basis pivots. Replay stores this residual as the new basis column. It does not call the rank producer or select a pivot to eliminate.

The rank assertion follows by two inductions. First, distinct highest set bits make the retained residual vectors linearly independent: among vectors in a putative dependence, the largest leading row cannot cancel. Second, each original column is the XOR of its residual and earlier basis vectors, while each new residual is in the span of original columns already processed. The two spans are consequently equal after every column. The number of retained residuals is the exact rank, including for zero and rectangular matrices. Both lower and upper rank bounds are proved; a list of independent columns alone would prove only the lower bound and would be insufficient for an unknot verdict.

The rank verifier is structurally independent of elimination discovery. It shares the explicit Frobenius complex construction, expansion helper, integer arithmetic and $d^2$ check with the producer. This trusted boundary is intentional and visible. Comparisons with the maintained planar-diagram scanner test a different geometric construction and reduction algorithm; comparisons with report 34 additionally check that the adaptation preserves its original ranks.

## 75.4 Budgets across discovery and replay

The public API adds `fallback_max_crossings=12` and `fallback_max_generators=200000`, also available as hyphenated CLI options. The latter is a cumulative allowance across *all* cube constructions, including reconstruction during replay. A resolution reserves its $2^{c-1}$ generators before allocating the differential matrices. The ordinary work callback is polled during expansion, resolution building, Frobenius edge assembly, the chain check, elimination and rank replay. String arenas, metadata, JSON transport and these cube operations retain the same shared work and time budgets.

A factor above the crossing threshold has not started a cube and may remain as a verified inconclusive residual in a forest proof. Exhaustion of shared work, generator, node, time or proof-size allowances returns inconclusive without a partial proof. External cancellation propagates. Setting the crossing threshold to zero restores the earlier unsupported-factor behavior; raising it permits a larger exponential computation. Proof byte limits are checked after construction, so they are not hard peak-memory guarantees. Neither a generator count nor a work counter claims to count individual machine instructions or bits allocated by Python.

## 75.5 The exceptional-factor theorem is now implemented

Let $g$ be the original grammar bit size and let $I$ index only wider factors not decided by the finite checks. For factor $i$, write $n_i, m_i, \kappa_i$ for its expanded length, strand count and minority-sign count. The previous singleton and Bennequin arguments give

$$n_i \le 4\kappa_i, \qquad m_i \le 2\kappa_i + 1.$$

The cube has $2^{n_i}$ states. Each resolution has at most $m_i + 2n_i$ circles, since that is its number of vertices, so a deliberately coarse bound for its total reduced basis size is

$$D_i \le 2^{m_i+3n_i-1} = 2^{O(\kappa_i)}.$$

Building all columns and checking $d^2$ use polynomially many bit operations in $D_i + n_i + m_i$. Elimination performs at most $D_i$ XORs per column on at most $D_i$ bits. A trace has at most $O(D_i^2)$ indices of $O(\log(D_i + 2))$ bits, and replay has the same exponential upper bound. The polynomial costs of source validation, projections and solved compressed factors are paid separately. Products of a source-polynomial term and an exponential factor can be absorbed into an additive bound using $ab \le a^2 + b^2$, with only a constant change in the exponential coefficient. With sufficiently large allowances, implemented discovery, certificate production and replay therefore satisfy

$$\mathrm{poly}(g) + \sum_{i \in I} 2^{O(\kappa_i)}.$$

All factors eventually decide as the finite allowances increase. The number of factors is polynomial in $g$. Thus the implementation has a restricted quasi-polynomial guarantee when $\max_{i \in I} \kappa_i = O((\log g)^2)$, and a polynomial guarantee when those counts are $O(\log g)$. There is no restriction on the expanded minority counts of the solved three-braid factors. This is a complete algorithm for supplied braid grammars, but an exponential worst case remains. We have not proved the exceptional-parameter condition or a favorable grammar-construction bound for arbitrary knot diagrams.

## 75.6 Validation and measured scope

Twelve new tests exercise both knot verdicts, exact source binding, Boolean field substitutions, altered dimensions and ranks, invalid column spans, replay without the rank producer, and shared budgets through multiple factors and replay. All 512 binary $3 \times 3$ matrices are checked against their explicitly enumerated column spans. Random braid cubes are compared with the maintained planar-diagram scanner. A mixed seven-strand forest contains a $2^{256}$ conjugation sleeve and a nine-crossing exceptional factor; only the latter is expanded. A separate binary-sized wider factor is declined before expansion. The empty-subgrammar test ensures that represented emptiness does not cause exponential traversal.

The complete maintained suite passes 897 tests with Regina in 226.425 seconds. A separate seeded audit constructs 160 knot closures on three through six strands with at most ten crossings. All reduced ranks agree with both the unchanged report-34 oracle and the maintained planar-diagram scanner, with $d^2$ checked in all three computations. Every public result also replays. There are 108 unknots and 52 nontrivial knots; 17 public calls require the new cube fallback, while the other cases are decided by the earlier exact stages. The audit takes 44.679 seconds and retains every signed input word, rank, verdict and source hash in `data/exceptional-cube-audit.json`.

| Input | whole ms | hybrid ms | paired ratio | generators |
|---|---|---|---|---|
| four_unknot | 159.49 | 161.83 | 1.012 | 4638 |
| four_knot | 139.22 | 137.07 | 1.016 | 3774 |
| small_sum_unknot | 7702.33 | 169.43 | 44.278 | 4638 |
| small_sum_knot | 5557.81 | 139.17 | 40.209 | 3774 |
| huge_sum_8 | – | 168.05 | – | 4638 |
| huge_sum_64 | – | 187.69 | – | 4638 |
| huge_sum_512 | – | 249.54 | – | 4638 |

The benchmark uses seed 261008493, one shuffled warmup and five measured rounds with identical controls. Its whole-cube ablation expands the entire supplied braid, builds a reduced complex, produces a rank trace and replays it. The hybrid uses the public source-bound forest interface, including its transport checks, decomposition, factor proofs and replay. The ablation also uses the host's transport and work-accounting conventions, with larger finite allowances for its larger complex. Neither timed arm exhausts its allowances on the compared cases. The explicit comparison is admitted only for inputs with at most 12 crossings; larger inputs are omitted by preflight.

The two four-strand controls separately time the maintained scanner on an already constructed PD diagram. That timing has no rank certificate, source conversion or independent replay. It measures the practical cost of choosing a straightforward proof-producing cube instead of the faster ordinary recognizer; it is not a comparison between equivalent proof interfaces. The mixed cases adjoin the trivial three-braid $ab$ through a singleton seam to either a nine-crossing trivial or nontrivial four-braid. Larger mixed cases replace $ab$ with its $u^{2^k}$ conjugate for $k = 8, 64, 512$; the exceptional factor remains nine crossings. No whole-input speedup ratio is inferred for these enormous

represented words. Raw samples and source hashes are in `../fast/results/exceptional_cube_20261008.json`.

For the small trivial connected sum, whole-cube production and replay take 7.702 seconds at the median, versus 169.427 ms for the hybrid; the median paired ratio is 44.278. For the nontrivial connected sum, the corresponding values are 5.558 seconds, 139.172 ms and ratio 40.209. Both comparisons include proof replay. The trivial-sum hybrid samples vary from 159 to 331 ms, and its identical control has a different median despite a paired A/A ratio near one. We therefore interpret these as roughly fortyfold family-specific gains, not precise crossover thresholds. All variation is retained.

The large mixed examples require exactly 4,638 cumulative cube generators for production and replay, independent of the compressed sleeve exponent. At $k = 512$ the complete hybrid median is 249.543 ms and its proof has 137,317 canonical bytes. The measured time grows with the compressed source and proof, not its expanded conjugating length.

On the standalone nine-crossing four-braids, the hybrid cube path takes 161.835 ms for the unknot and 137.068 ms for the nontrivial knot. The ordinary scanner takes 1.673 and 4.559 ms respectively, without independent proof output. The cube is consequently a conservative complete fallback, not a replacement for the faster scanner in ordinary diagram recognition. These measurements motivated two further optimizations: verified explicit reductions before a small factor's cube, and immediate termination on a proved nontrivial factor. Section 76 implements both with source-bound proofs. Neither optimization is assumed in the checkpoint measurements above.

# 76 Adaptive factor scheduling and checked pre-cube reduction

The complete exceptional-factor fallback exposes two avoidable costs. A small wider braid may simplify to three strands, yet a direct cube pays exponential cost before using that structure. A nontrivial connected-sum factor may already have an inexpensive proof, yet an earlier expensive factor can consume the shared allowance first. We address both costs while retaining the same complete fallback and exceptional-parameter bound. This section records the initial eager-preparation policy; Section 80 subsequently makes preparation lazy while retaining the proof formats and checked reductions below.

## 76.1 Prioritize decisions before exponential work

The producer validates the original grammar, computes exact singleton cuts, and prepares each projected leaf and its finite summaries. It then assigns three priority classes. The first contains factors with at most two strands and factors already excluded by the exponent obstruction $|e| > m - 1$. The second contains the remaining three-braid factors, which have a complete compressed polynomial algorithm. The third contains wider residual factors. Within a class, shorter projected grammars precede longer ones; original factor index breaks ties. Preparing these projections remains polynomial in the input size and does not expand any word.

This is a deterministic structural policy, not a timing threshold fitted to the benchmark. It promotes cases whose available decision method is cheap and complete. It does not promise to minimize running time on every input: a particularly simple wider factor might finish before an awkward compressed three-braid. All attempted operations still share the host's resource budget. The returned `factor_order` is execution metadata, not an extra mathematical assertion accepted by the verifier.

As soon as a factor is proved nontrivial, the producer stops. It need not finish unvisited factors or retain proofs for earlier trivial factors. This changes bounded behavior as well as runtime: an exact negative result can now survive a zero cube-generator allowance when a finite obstruction

appears later in strand order. Without priority, the first wider factor could exhaust that allowance and leave the whole input inconclusive.

An unknot verdict still requires every factor to be proved trivial. Complete split proofs retain all factor records in their original strand order, even when discovery visits them in a different order. An unsupported factor may remain inconclusive while later factors are examined; any subsequent exact negative result is sufficient to decide the whole sum.

## 76.2 A proof for one nontrivial summand

The new forest format is version three. Its `knotted-factor` phase contains the original input, the exact complete singleton-cut list, a factor index and precisely one leaf record. The verifier recomputes the source summaries and cut list, verifies that the original closure is a knot, and checks that the index names one of the resulting consecutive strand intervals. It reconstructs that interval's projected grammar and checks its metadata and child proof. It accepts this phase only when the child proves `KNOTTED` and the parent asserts `KNOTTED`.

Soundness uses the connected-sum decomposition from Section 74: an unknot has only trivial knot summands. Thus one nontrivial summand disproves unknotness of the original source. The unvisited factors require no certificate. The original one-component check matters, since the connected-sum-of-knots argument is not a statement about arbitrary links. A single positive factor cannot prove the whole input trivial; the verifier explicitly rejects that substitution. It also rejects an index rebound to a different interval, Boolean index aliases, an omitted leaf in a complete split proof, and downgrades of the selected-factor phase to an older version.

This selective proof can be much smaller than the former complete forest proof. It still includes the whole source and exact cut data, plus the selected factor's projection and certificate. Preparation currently constructs all projections before scheduling, so early termination saves decision and replay work, not that polynomial preparation cost. Versions one and two remain accepted with their original complete-list semantics.

## 76.3 A checked reduction before building a cube

A wider factor that passes its expanded-length preflight is converted to an explicit word under the existing cap. Before building its reduced Khovanov cube, the producer invokes the maintained singleton endpoint Markov descent from Section 66. This routine freely and cyclically cancels adjacent inverse letters, then deletes singleton endpoint generators by a cyclic rotation and destabilization, stopping on reaching three strands or when no endpoint move applies. Every operation preserves the isotopy type of the closure and never increases word length or strand count.

If both the final word and strand count are unchanged, the original direct cube certificate is used. No witness of an unsuccessful simplification search is required. If there is progress, an `exceptional-reduction-v1` child contains the checked Markov trace followed by a terminal proof. Three-strand residuals use the compressed quotient recognizer on a deterministically built balanced grammar for the explicit residual word. Wider residuals still use the complete cube, now on the shorter word. Reaching three strands, rather than guessing minimal braid index, is sufficient for the complete terminal criterion. Failure to reach three strands remains eligible for homology.

Replay starts from the original factor's projected grammar, expands it under the same cap, and invokes the independent maintained endpoint verifier on the supplied trace. It reconstructs the actual resulting strand count and word. The terminal grammar is rebuilt from that word, then the corresponding three-braid or cube proof is checked against it. The declared final word is never used as a substitute for replay. Version three is required for this nested reduction proof; an older forest cannot silently acquire new terminal semantics.

The endpoint verifier checks recorded choices without searching for an endpoint. Its legacy version-one path shares deterministic free/cyclic cancellation with discovery, while its version-two path replays local moves on original positions. That existing trusted boundary is retained; we do not claim two independent implementations of every simplification primitive. Both maintained Markov schemas contain no Boolean fields, so the adapter rejects Boolean values throughout the nested witness before legacy replay. This prevents Python integer equality from accepting an aliased signed generator or metadata field. External exceptions, including a user callback's `ValueError` subtype, propagate without reclassification as a knot verdict.

## 76.4 Resources and asymptotic accounting

The public producer enables this reduction by default. The control `use_fallback_reduction=False`, or CLI flag `-no-fallback-reduction`, provides an ablation of only the new pre-cube stage. It retains factor priority and selective negative proofs. Verification accepts every supported proof independently of the producer's choice of this control. Existing crossing preflight, input and proof bytes, shared work, time, string nodes and cumulative cube-generator allowances remain in force. A solved reduced three-braid consumes no cube generators, but its string production and replay still consume the shared string budget.

Let $f$ be the number of factors and $g$ the input bit size. The new ordering adds $O(f \log(f+1))$ comparisons after the existing polynomial projections. For an admitted wider factor, explicit endpoint descent costs $O(n+m)$ word operations on this one-component domain. The residual has no more letters or strands than its source. Constructing a balanced residual grammar and applying the complete three-braid procedure therefore adds polynomial work in that explicit factor size; a remaining cube retains the previous exponential upper bound. Consequently the implemented uncapped bound remains

$$\mathrm{poly}(g) + \sum_{i \in I} 2^{O(\kappa_i)},$$

where $I$ and $\kappa_i$ refer to the exceptional factors *before* this new elementary reduction. We do not replace them by residual minority counts without proving a new singleton bound for those residuals. Early negative termination can remove summands from the actual work, but a worst-case unknot still requires all factors. No new general quasi-polynomial claim follows.

## 76.5 Validation and measurements

Twelve new tests exercise a reducible wider unknot with zero cube allowance, a stalled nontrivial four-braid, and a shorter residual that still needs the cube. They verify that later finite and compressed negative factors run before an earlier wider factor; that selective proofs cannot prove unknotness or be rebound to another interval; and that all positive factors remain necessary. They mutate reduction output, terminal source grammars and versions, replay with reduction/rank discovery disabled, and exercise exact shared-work boundaries and external cancellation. The earlier direct-cube tests explicitly disable the new reduction, preserving coverage of the fallback rather than accidentally testing only a simplification shortcut.

All 78 focused tests pass, followed by all 909 maintained tests with Regina in 212.426 seconds. The paired experiments load the previous maintained package from commit `666a62f6a` under a separate package name, sharing the unchanged native string kernel. Baseline module hashes are retained, rather than approximating the old behavior by disabling one new option.

A 160-case cross-version audit reuses the preceding section's independently checked signed braid inputs. Old and new public verdicts agree in all cases: 108 unknots and 52 nontrivial knots. The new verifier accepts every old proof and every newly produced proof. Public cube use falls from 17 to 12 cases. The audit also fixes the operational difference at a zero generator allowance: a

wider trivial factor followed by a two-strand trefoil was previously inconclusive and now has a checked nontriviality proof for the later factor. The 5.581-second audit records source hashes and the prior input audit's hash in `data/adaptive-forest-audit.json`. An initial baseline-loader path failure is retained separately; it occurred before any algorithm comparison.

| Input | old ms | current ms | no reduction ms | paired ratio |
|---|---|---|---|---|
| reducible_four | 265.06 | 2.04 | 270.63 | 128.769 |
| irreducible_four | 134.80 | 135.77 | 135.70 | 0.995 |
| shorter_wider | 1203.18 | 138.64 | 1208.10 | 8.889 |
| finite_late | 163.02 | 0.70 | 0.73 | 234.638 |
| compressed_negative_late | 186.87 | 19.30 | 19.26 | 9.539 |
| positive_sum | 173.29 | 4.87 | 173.45 | 35.050 |
| huge_positive_sum | 238.69 | 79.08 | 239.35 | 2.980 |
| positive_forest | 91.40 | 93.27 | 92.60 | 0.989 |
| three_braid_control | 6.98 | 7.16 | 7.01 | 0.998 |

The performance experiment uses seed 261008494, one shuffled warmup and five measured rounds on nine inputs. It compares the actual previous maintained host, an identical old control, the current host, an identical current control, and the current host with only elementary fallback reduction disabled. Every arm includes its normal proof production and independent replay. The last arm retains priority and selective negative proofs, separating their benefit from pre-cube simplification. All signed inputs come from the checked fixture constructors, and all arms must agree on the verdict. Source hashes, raw samples, execution orders, proof sizes and resource counts are retained in `../fast/results/adaptive_forest_20261008.json`.

The reducible nine-crossing four-braid improves from a 265.060 ms median to 2.039 ms, with median paired ratio 128.769. Its cumulative cube-generator count falls from 4,638 to zero, and its proof shrinks from 68,785 to 1,395 canonical bytes. Disabling reduction restores the expensive cube. For the 11-crossing word whose shorter residual remains four-stranded, the median improves from 1,203.179 to 138.640 ms, with paired ratio 8.889; generator reservations fall from 28,470 to 3,774. This case checks that useful progress need not reach a polynomial terminal to reduce complete-fallback cost.

When the nine-crossing wider unknot precedes a two-strand trefoil, priority and the selected-factor proof reduce 163.016 ms to 0.696 ms, with paired ratio 234.638. The proof shrinks from 69,095 to 656 bytes. The no-reduction arm is similarly fast, isolating scheduling and selective replay as the mechanism. A later compressed nontrivial three-braid gives a paired ratio of 9.539 by the same mechanism. The original shared budget would have spent work on the irrelevant earlier cube in both cases.

For a positive mixed forest containing a $2^{512}$ conjugation sleeve, all factors remain necessary, but the wider factor no longer needs a cube. The median falls from 238.687 to 79.079 ms, with paired ratio 2.980. A smaller positive sum improves by paired ratio 35.050. On the stalled four-braid, the current 135.774 ms median is close to the old 134.800 ms; its complete cube remains necessary. The positive all-three-braid forest and direct three-braid control likewise show no meaningful gain. Their small differences are retained, not used to claim an improvement.

Identical-arm paired ratios range from 0.961 to 1.077. The short reducible case has the widest control variation, so the large ratios should be read as approximate family-specific improvements, not exact machine-independent constants. The experiment has 225 measured calls and 45 warmup calls. It does not measure arbitrary-diagram recognition or bound the general exceptional minority parameter. The next opportunities include avoiding unneeded projection preparation on negative forests and integrating the pending sparse two-meridian seed search from report 47; neither is included in these timings.

# 77 Sparse two-meridian seed search with complete candidate pruning

Report 47 contains a second useful native optimization besides its orbit certificates: a faster search for the two-meridian derivations of Section 62. We integrate its reusable closure state, structurally restricted candidate pairs and proper-closure pruning. The representation arithmetic, source-bound certificate format and search-independent verifier are unchanged. The delivered frozen baseline is byte-identical to the previously maintained module, so the comparison does not depend on reconstructing an approximate historical implementation.

## 77.1 The implication system and its checked sparse regime

A signed Wirtinger crossing $(o, u, v, \epsilon)$ supplies implications $\{o, u\} \Rightarrow v$ and $\{o, v\} \Rightarrow u$. Coincident dependencies count once. For seeds $S$, let $\mathrm{cl}(S)$ be the least set containing $S$ and closed under these rules. A successful one- or two-seed closure supplies an acyclic original-crossing derivation of all meridians. The unchanged exact arithmetic then imposes every original relation and returns an unknot or nontrivial-knot verdict. Exhaustive closure failure still says only that this diagram lacks the required derivation; it remains `INCONCLUSIVE` for recognition.

The sparse regime requires every singleton closure to be trivial. This is equivalent to absence of a productive unary rule: a rule with one distinct dependency and a different target. A productive unary rule immediately grows its singleton. Conversely, without one, no first step can grow any singleton. The implementation checks the original rule list before using sparse pairs. It does not assume that every knot diagram satisfies this condition.

Suppose there are at least three arcs and a pair $S$ generates all arcs. Some rule must derive the first arc outside $S$. Its prerequisites are in $S$. They cannot consist of one arc in the checked regime, so they are exactly the two elements of $S$, and its target is outside $S$. Consequently every generating pair occurs among the distinct prerequisite pairs of productive binary rules. With $r$ directed rules, at most $r$ pairs need examination. A knot diagram with $n$ crossings has $O(n)$ arcs and at most $2n$ such rules, in place of the former quadratic pair set. On two arcs, the sole pair is retained explicitly: it already contains the whole universe and requires no first rule. Zero and one arc are handled by the earlier empty/singleton logic.

The productive-unary guard is necessary. For example, crossings $(0, 0, 1)$ and $(1, 2, 3)$ on four abstract arcs imply $0 \Rightarrow 1$ and $\{1, 2\} \Rightarrow 3$. The pair $\{0, 2\}$ generates all four arcs, but it is not a productive binary prerequisite pair. The apparent candidates $\{1, 2\}$ and $\{1, 3\}$ cannot derive arc zero. The maintained algorithm therefore retains exhaustive pair enumeration whenever a productive unary rule exists. The example concerns the implication logic, not a claim that these two crossings are a valid four-arc knot diagram.

## 77.2 Reuse closure arrays without changing derivation order

The old engine allocates dependency counters for every attempted seed set, paying $\Theta(v + r)$ initialization even if the trial reaches only one or two of its $v$ arcs. The new engine allocates two arrays once. Advancing a generation number makes older entries logically absent. A known-generation stamp records enqueueing; a processed-generation stamp records removal from the queue. The reached list is rebuilt only from actually visited vertices.

A rule fires when all its distinct dependencies have processed stamps for the current generation and its target lacks a current known stamp. The processed distinction is essential for preserving the exact previous trace. In the old counter algorithm, the remaining counter is precisely the number of dependencies not yet removed from the queue. Thus the two firing conditions agree after every dequeue. Induction on queue events gives the same enqueues, rule choices and derivation order,

including coincident dependencies. Testing only the known stamps would still compute closure, but could change the timing of rule firings and the certificate trace.

One attempt pays for its dequeued vertices and their rule incidences, not for untouched vertices. Both stamps are needed because a vertex may have been enqueued but not yet processed when another dependency is examined. Generation counters have $O(\log(n+2))$ bits over the at-most-quadratic number of attempts; no array of all possible pairs is allocated.

### 77.3 Prune pairs inside failed closed sets

If a failed trial exhausts its queue in a proper closed set $C$, then every seed pair $S' \subseteq C$ also fails:

$$\mathrm{cl}(S') \subseteq \mathrm{cl}(C) = C \neq V.$$

The producer stores candidate pairs as a sparse undirected graph. After a failed trial, it examines candidate edges incident to reached vertices and marks later edges whose other endpoint is also reached. It skips those edges on subsequent enumeration. Only proper completed closures justify pruning; a resource-interrupted prefix proves no failed candidate.

Candidates remain in lexicographic order. Every omitted or skipped pair is proved unsuccessful, so the first successful pair remains the same as in the old exhaustive algorithm. Singleton attempts are also retained in their original order. Combined with identical queue traces, this preserves the entire successful certificate, including its exact arithmetic and source binding. The number of charged attempts now counts actually examined seeds; proved failures need not consume a full closure computation. Under a fixed budget more of the class may be exhausted, but no monotonic improvement for every tiny resource allowance is promised: the new graph has a preparation cost of its own.

### 77.4 Search bounds and their limits

Let $v$ be the number of arcs, $r$ the number of rules and $\mathcal{A}$ the seed sets actually examined. For a reached closure $C_S$, let $d_R(x)$ and $d_P(x)$ be its vertices' rule-incidence and candidate-graph degrees. A coarse output-sensitive bound is

$$O\left(v + r\log(r+1) + \sum_{S\in\mathcal{A}} \sum_{x\in C_S} (1 + d_R(x) + d_P(x))\right)$$

word operations, with $O(v+r)$ working words. The graph term is needed only for failed candidate-pair closures. This formula uses ordinary amortized set lookup for pair deduplication; sorting and adjacent deduplication offer a deterministic comparison alternative. Even quadratic worst-case hash preparation leaves the following quadratic Wirtinger bound intact.

There are $v$ singleton trials and at most $r$ candidate-pair trials, each visiting at most $O(v+r)$ incidences. For a validated knot diagram, $v, r = O(n)$, giving $O(n^2)$ search word operations or $O(n^2\log(n+2))$ elementary bit operations for indices. This replaces the old cubic closure-search bound on the checked regime. The unrestricted productive-unary fallback retains $O(v^2(v+r))$ search work, although its individual attempts still benefit from reused state. Memory remains linear in the rule system rather than in the number of potential pairs.

This theorem concerns finding the derivation. Successful representation arithmetic, gcd/parity checks and mandatory certificate replay remain as in Section 62; their deliberately conservative $O(n^4\log(n+2))$ complete-stage bound is not improved merely by changing closure enumeration. The public stage remains opt-in in the ordinary pipeline, after invariant filters, with the same time, work and attempt caps. Earlier filter costs are outside the standalone bound. Exhaustive failure proves no knot-type result, and no cheap conversion of every knot diagram into the two-seed class has been established.

## 77.5 Native validation

All six delivered source, test, corpus, benchmark and documentation files were verified against the archived report-47 manifest at intake. The maintained research README now distinguishes fresh native evidence from the delivery's historical measurements. The previous maintained module and the delivered frozen baseline both have SHA256 beginning `5eb36230e9b9`. The optimized module is integrated without changing the delivered search implementation. Fourteen focused tests pass in 2.309 seconds. They include all 4,096 four-arc systems of distinct crossing triples and all 512 three-arc systems allowing productive unary rules. For every one- and two-seed set, a separate fixed-point sweep agrees with the stamped engine, whose trace also agrees exactly with the old counter engine. The tests compare the first successful pair before and after pruning. Four hundred additional random degenerate systems cover coincident labels, empty rules and repeated generations.

The twenty frozen actual-diagram fixtures retain exactly the same successful certificates. The suite exercises proper-closure pruning, the planar productive-unary fallback, search-independent replay, exact shared-work boundaries and global cancellation. These checks preserve the distinction between exhausting a seed class and completing knot recognition.

The full maintained suite passes 915 tests with Regina in 209.658 seconds. A further seeded audit compares 240 held-out actual planar braid closures against the unchanged frozen module. Verdicts and successful certificates agree exactly: 102 unknots, 86 nontrivial knots and 52 exhaustive two-seed failures. There are 153 sparse-regime inputs and 87 productive-unary fallback inputs. Both old and new verifiers accept each conclusive certificate. This comparison checks the search integration; the pre-existing geometric and quaternion tests supply the independent arithmetic controls.

The same audit constructs disconnected three-vertex implication blocks to check scaling of unsuccessful closures. With 5,000 blocks there are 15,000 vertices and 10,000 directed rules. The candidate graph has 10,000 edges, and closed-set pruning skips 5,000 of them. There are 20,000 actual trials, 30,000 vertex dequeues, 40,000 visited rule incidences and 20,000 pruning incidences, versus 112,492,500 potential unordered pairs. The corresponding counts scale exactly with the number of blocks in four audited sizes. Preparation's charged sorting term is retained. These are synthetic implication systems, not asserted knot diagrams. The 0.508-second audit, all signed planar inputs, the unary guard counterexample and exact source hashes are in `data/sparse-seeds-native-audit.json`.

## 77.6 Standalone and whole-pipeline measurements

The benchmark retains the delivery's fixed twenty-diagram corpus and seed 2610081249, but reruns it against the current maintained pipeline. Standalone arms are the frozen baseline, an identical baseline control, stamped closure without candidate restriction, candidates without closed-set pruning, and the complete optimization. Each has one shuffled warmup and nine measured rounds. Every timed call includes fresh PD validation, seed search, successful representation arithmetic and replay. An exhaustive unsuccessful seed search counts as a completed class query but stays inconclusive for recognition.

| Input | old ms | stamped ms | pairs ms | full ms | ratio |
|---|---|---|---|---|---|
| survivor-00 | 1.303 | 1.007 | 0.935 | 0.923 | 1.410 |
| survivor-01 | 1.244 | 0.754 | 0.521 | 0.424 | 2.917 |
| survivor-04 | 0.734 | 0.678 | 0.718 | 0.703 | 1.056 |
| survivor-09 | 0.628 | 0.593 | 0.625 | 0.631 | 1.002 |
| gordian | 456.551 | 40.373 | 4.837 | 4.581 | 100.120 |
| trefoil | 0.237 | 0.243 | 0.251 | 0.249 | 0.951 |
| conway | 0.697 | 0.501 | 0.401 | 0.369 | 1.882 |
| kinoshita_terasaka | 0.710 | 0.508 | 0.409 | 0.435 | 1.791 |

Whole-pipeline measurements separately use the ordinary configuration with a three-second budget and the compressed-group configuration with an 18-second total budget and 15-second group allowance. Each compares the old stage, its identical control, the new stage and disabling the stage, with one shuffled warmup and five measured rounds. Local two-meridian caps remain 50 ms, two million work units and 10,000 attempted seeds. A pipeline `UNKNOWN` remains a censored sample; it is not assigned a completion time or used in paired completed-time ratios. All raw samples, actual stage invocations, proof hashes and loaded-dependency hashes are retained in `../fast/results/two_meridian_search_native_20261008.json`.

| Input | ordinary old/new ms | ratio | group old/new ms | ratio |
|---|---|---|---|---|
| survivor-00 | 3.003/2.541 | 1.184 | 3.004/2.624 | 1.158 |
| survivor-01 | 14.347/13.218 | 1.085 | 16.406/15.616 | 1.063 |
| survivor-04 | 1.742/1.731 | 1.005 | 1.749/1.728 | 1.010 |
| survivor-09 | 1.911/2.106 | 0.974 | 1.607/1.594 | 1.005 |
| gordian | –/– | – | 1683.188/1692.476 | 1.037 |
| trefoil | 0.055/0.056 | 0.998 | 0.054/0.054 | 1.004 |
| conway | 1.023/1.024 | 1.001 | 1.019/1.020 | 1.001 |
| kinoshita_terasaka | 1.052/1.047 | 1.005 | 1.048/1.058 | 0.999 |

All 900 measured standalone calls finish their class query. On Gordian, the old median is 456.551 ms and the optimized median is 4.581 ms; the median paired ratio is 100.120. Stamped state alone takes 40.373 ms and restricting to prerequisite pairs takes 4.837 ms. The number of attempted seed sets falls from 10,011 to 274, and charged work from 4,347,002 to 12,251. With the unchanged two-million-work cap, the old search stops before exhaustion whereas the new search completely excludes its seed class. Both answers remain inconclusive for knot recognition. This is a search improvement on a hard negative instance for the derivation class, not a new obstruction to unknottedness.

The smaller Conway and Kinoshita–Terasaka fixtures have paired standalone ratios 1.882 and 1.791. Other fixtures gain little: survivor-09 has ratio 1.002, and the trefoil has ratio 0.951. Metadata construction can cost more than it saves when a tiny successful derivation is found immediately. The identical-baseline controls range from 0.975 to 1.038 across the standalone corpus. These measurements support the large search improvement and its mechanism; they do not support a universal constant-factor gain.

There are 800 measured whole-pipeline calls, of which 780 complete. All 20 censored calls are ordinary-configuration Gordian runs, across the four arms. The group configuration completes Gordian through the existing cyclic-group backend. Its old/new medians are 1,683.188/1,692.476 ms, while its median of paired old/new ratios is 1.037. The latter statistic need not equal the ratio of separate medians. With only five rounds and identical-control ratios ranging from 0.909 to 1.035 across the group corpus, this is weak evidence for a whole-pipeline gain. Disabling the

seed stage has Gordian median 1,653.628 ms. The optimized stage quickly finishes its unsuccessful class query; most elapsed time belongs to other work.

Survivor-00 improves from 3.003 to 2.541 ms in the ordinary pipeline (paired ratio 1.184), while several invariant-filtered fixtures never reach the seed stage and consequently show no systematic change. All timings, including slower arms and censored samples, are retained. The measured loaded-source hashes are unchanged throughout the run. The public default therefore remains disabled: the evidence justifies integrating the faster search for callers that enable this class test, without yet establishing an automatic dispatch policy that improves the broader input distribution.

**Remaining theoretical gap.** The new bound is quadratic for closure discovery in its checked regime, with an exact fallback outside it. It neither bounds the number of Reidemeister moves required to enter the two-seed class nor supplies a quasi-polynomial complete recognizer for arbitrary planar diagrams. Future adaptive scheduling can use reached-closure sizes and charged work as progress signals, but any early departure must remain inconclusive and leave a complete fallback available. Such scheduling also requires measured stage costs rather than extrapolating the standalone 100-fold ratio to whole-pipeline performance.

# 78 Source-bound certificates for binary normal-surface topology

The next report-47 integration connects the local orbit proofs of Section 73 to the maintained normal-surface adapter. The report's polygon-stack adapter is useful as an independent reference, but leaves ambient-manifold validity as a precondition. The maintained adapter already checks the finite manifold itself. We retain its edge-point interval model and manifold validator, and add optional certificates to its three existing orbit queries. Section 79 subsequently reduces common coordinate multiplicity and introduces a compatible second proof version; the measurements here retain the direct version-one path. Thus this integration strengthens the checkable evidence without weakening the existing ambient input contract.

## 78.1 Exact scope and reconstructed geometry

The input is a reciprocal tetrahedral face-pairing array and seven nonnegative normal coordinates per tetrahedron. The native validator checks connectedness, ambient orientability, absence of reversed-edge identifications, circle or interval edge links, sphere or disc vertex links, and exactly one finite torus boundary. Matching equations, quadrilateral constraints and globally consistent edge-intersection counts are checked from the supplied coordinates. No recognition of a knot exterior, and no relation to an input planar diagram, is inferred from these checks.

For an ambient edge $e$, the normal intersection weight is $w_e$. The surface's one-skeleton is represented on the disjoint union of these edge-point intervals. At a face corner, the endpoints of its normal arcs are paired by their rank from the corner. A parity union–find determines which end of an edge interval is nearest the corner; vertex labels alone would be insufficient in a one-vertex triangulation. The already validated normal polygons attach to this one-skeleton, so its connected components are exactly those of the surface. Restricting to boundary faces gives its boundary circles. Neither construction enumerates individual discs, intersection points or sheets.

The finite geometry routines now live in `../fast/fastunknot/normal_surface_geometry.py`; their validation and interval construction are shared by producer and checker. This is an explicit common trust boundary. The checker does not invoke the topology producer, orbit scheduler or boundary-cohomology search. Independent polygon-stack extraction and Regina supply cross-checks of the shared geometry rather than being required production dependencies.

## 78.2 The three orbit proofs and orientation doubling

The optional `normal-surface-topology-v1` certificate contains the normalized source digest, the complete proofs for `surface`, `double` and `boundary`, a boundary-cohomology witness and a precisely typed table of topological claims. Integer coordinates and their hexadecimal encodings have the same normalized source digest. The exact face records remain bound; an isomorphic relabelling requires fresh reconstruction and evidence. The digest is a source consistency check, not a replacement for mathematical verification.

Replay revalidates the caller's manifold and coordinates and reconstructs all three ordered interval systems. It then invokes the independent local orbit verifier on those reconstructed inputs. Rewriting the outer digest cannot make a foreign trace acceptable. Each orbit trace must finish; an otherwise valid prefix supplies no count.

Let $C$, $C_2$ and $b$ be the verified counts for the normal vector $x$, its double $2x$, and its boundary. The doubled normal surface is the horizontal boundary of a regular normal interval neighbourhood of the original. In an orientable ambient manifold this is its orientation double cover. An orientable component contributes two lifted components and a nonorientable component contributes one. Hence

$$C_{\mathrm{or}} = C_2 - C, \qquad C_{\mathrm{nonor}} = 2C - C_2.$$

The checker rejects negative values. The finite reconstruction also computes $\chi = V - E + F$: $V = \sum_e w_e$, $E$ counts normal arcs after face identifications, and $F$ is the sum of normal coordinates. All these are binary arithmetic operations on a compressed description.

For a connected orientable surface, the checker verifies $g = (2 - b - \chi)/2$, including nonnegativity and integrality. For a connected nonorientable surface it verifies the positive crosscap number $2 - b - \chi$. Every supplied component count, Euler value, disc count, genus or crosscap claim must match the reconstructed calculation. Boolean values are not accepted as integer counts. The empty surface has zero components and no connected-surface genus claim.

## 78.3 A finite witness for essential boundary

Normal intersections with each ambient boundary edge define a mod-two cochain $z(e) = w_e \bmod 2$. The normal matching equations make it a cocycle: on each boundary triangle the endpoints of normal arcs occur in pairs. Its cohomology class is dual to the surface boundary's homology class. The graph used below has one edge for each ambient boundary edge, retaining loops and multiple edges, and is independent of sheet multiplicity.

The report supplies two complementary finite witnesses. To prove that the class vanishes, the producer supplies one bit $p(v)$ per graph vertex, and the checker verifies

$$z(uv) = p(u) + p(v) \pmod 2$$

for every edge. To prove nonvanishing, it supplies a set of graph edges having even incidence at every vertex and odd sum of $z(e)$. This is a closed mod-two chain with pairing one against the cocycle. Either witness has size linear in the finite boundary graph. The positive witness can be found by a breadth-first potential assignment: an inconsistent edge closes an odd cycle using the two parent paths. Replay checks the chain directly and does not repeat that search.

A connected normal surface with $\chi = 1$ and one boundary component is a disc. Its boundary is a simple closed curve in the validated boundary torus. Such a curve is essential exactly when its mod-two homology class is nonzero: every essential simple torus curve represents a primitive integral class. The checker therefore certifies a compressing disc exactly when $C = 1$, $\chi = 1$, $b = 1$, and the nonzero boundary witness verifies. A vertex-linking disc correctly fails the last condition. An aggregate zero boundary class on a disconnected surface says nothing about each individual boundary curve; the disc conclusion requires connectedness.

## 78.4 Complexity, budgets and API

For $t$ tetrahedra and largest coordinate bit length $B$, each interval system has $O(t)$ pairings and $O(B + \log(t + 1))$-bit endpoints. The ordinary normal-disc count may be exponential in $B$, but neither the finite validation nor any of the three orbit queries expands it. The polynomial AHT production and trace-size bounds of Section 72 therefore yield polynomial-size normal-topology certificates and polynomial verification in the supplied encoded geometry and proof length. This is a theorem for a supplied normal vector, not a polynomial algorithm for finding a compressing disc.

`normal_surface_topology(..., record_certificate=True)` records the proofs; the default remains count-only. A single `max_cycles` allowance is shared across all three production queries. An incomplete query returns `INCONCLUSIVE` without any topology certificate or compressing-disc answer. Recording preserves cycle counts and all previous result fields. Proof production and independent verification are separate API calls; requesting a proof does not implicitly replay it.

`verify_normal_surface_certificate` optionally checks a shared `max_operations` cap on the sum of recorded orbit events before replay. Exceeding it means unverified, not a negative topological claim. Caller cancellation exceptions propagate, including `ValueError`. Invalid supplied manifold data raises the same `NormalOrbitError` as production; malformed proof data returns false. Callers can additionally bound serialized size or provide cooperative deadlines. These controls do not assert that the entire untrusted input was bounded before allocation.

## 78.5 Native validation and performance evidence

Four delivered source and test files were checked against report 47's manifest. The native adapter retains the prior finite geometry, whose count-only results are compared against the actual maintained implementation at commit `98f21ce333bf`. Twenty-eight focused tests pass in 11.167 seconds. New proof tests disable all producer search functions during replay, alter every topological claim, substitute foreign queries, rewrite source digests, corrupt both boundary-witness forms, exhaust shared caps, and inject cancellation at early, middle and late checkpoints. Empty, one-sided, vertex-linking and parallel surfaces include coordinates with more than 20,000 bits and hexadecimal JSON round trips.

The seeded native audit starts with 252 actual normal vectors in layered solid tori with one through eight tetrahedra, including compatible sums and empty vectors. Every result agrees with both Regina and the report's separate polygon-stack extraction; both native and report certificates replay. Eighteen further cases attach tetrahedral balls along boundary faces, preserving the manifold while adding boundary vertices; meridian extensions and each vertex-linking disc exercise the larger boundary graph. There are twelve compressing-disc cases, 34 cases with a nonorientable component and 94 disconnected cases among the 270 total comparisons. The old native count-only result also agrees exactly in every case. Raw vectors, proof digests and source hashes are retained in `data/normal-certificate-native-audit.json`.

The full maintained suite passes 924 tests with Regina in 246.517 seconds. The final focused run includes additional odd-cycle and exact-potential checks on a three-vertex graph. The ten moved geometry definitions are AST-identical to their baseline versions; the refactoring adds no alternate manifold-validation path.

The benchmark uses the actual baseline module from that commit with unchanged orbit-kernel dependencies. Its six arms are the old count-only call, the new count-only call, an identical new control, optional proof recording, production followed by independent replay, and replay of an already supplied proof. There is one shuffled warmup and five shuffled measured rounds per case. Small fixtures use batches to reduce timer overhead; each call starts with fresh source validation. Proof transport and JSON serialization are outside the timed region. All results and produced

certificates are compared outside timing; the combined production/replay arm checks successful replay inside its call. Source hashes are checked before and after the experiment.

| Input | count ms | record ms | both ms | replay ms | ratio |
|---|---|---|---|---|---|
| meridian1 | 0.928 | 1.120 | 1.890 | 0.750 | 1.270 |
| meridian8 | 5.043 | 5.662 | 8.596 | 2.453 | 2.079 |
| meridian32 | 70.736 | 82.329 | 95.175 | 16.466 | 4.364 |
| meridian96 | 1267.563 | 1688.191 | 1514.220 | 125.874 | 9.852 |
| parallel5000 | 16.594 | 19.522 | 27.104 | 7.130 | 2.449 |
| one_sided20000 | 0.740 | 1.164 | 1.624 | 0.728 | 1.059 |

Here "both" means production plus independent verification, and the ratio is the median paired count-only/replay time. Replay therefore has a different input contract: a complete proof is already available. The table does not claim that the full task of finding and verifying a normal surface has been accelerated by the same ratio. Raw shuffled samples, exact representative certificates and their encoded sizes are in `../fast/results/normal_certificates_20261008.json`.

The 840 measured calls and 168 excluded warmup calls finish in a 56.791-second run. For the 96-tetrahedron meridian, fresh topology computation has median 1,267.563 ms and source-bound replay 125.874 ms, with median paired ratio 9.852. Its proof has 24,038 recorded orbit events and occupies 2,018,995 bytes in compact JSON. This establishes substantially cheaper checking when a proof is already supplied. Production plus replay takes 1,514.220 ms and is more expensive than count-only discovery.

At 32 tetrahedra, count-only/replay medians are 70.736/16.466 ms (paired ratio 4.364); at eight tetrahedra they are 5.043/2.453 ms (ratio 2.079). The $2^{5000}$ parallel multiplicity has ratio 2.450, whereas the 20,001-bit one-sided example has ratio 1.059. In the latter case finite reconstruction and binary arithmetic leave little orbit search to avoid. Large coordinate bit length by itself does not imply a large replay advantage.

Identical current controls have paired ratios from 0.983 to 1.011. Nevertheless the large fixture's recording-only median, 1,688.191 ms, exceeds its combined production/replay median. The raw shuffled samples retain this timing fluctuation; subtracting those separate medians would not estimate the cost of verification. Nor do the old/new count-only measurements establish an optimization of discovery: their geometry and orbit algorithms are unchanged. The useful performance result here is proof reuse, with independently checked source reconstruction, and its benefit depends on how much orbit search the source would otherwise need.

**What remains before a diagram recognizer can use this.** A checked compressing disc implies unknottedness only after the supplied manifold has been established as the original knot exterior. This adapter validates a finite manifold with torus boundary but supplies no such correspondence. It also does not search for a normal vector, bound the cost of a hierarchy that finds one, identify slopes of general components, or distinguish every component in an aggregate summary. Report 47's boundary-incidence extensions and the maintained marked-union proofs can support further work, but neither replaces those missing source and complexity arguments. General quasi-polynomial recognition is still open for the maintained implementation.

## 79 Remove common normal multiplicity before orbit reduction

The previous section provides independently replayable normal-topology proofs. Its measurements also expose a remaining avoidable cost: scaling a fixed normal vector enlarges every endpoint in the three interval queries and every recorded endpoint in the proofs. We now divide out common

coordinate multiplicity before those queries, then recover the original surface's topology by an exact lifting formula. This improves computation and proof size on repeated surfaces. The source triangulation and full input vector are still validated before reduction.

## 79.1 Why scaling is not always a disjoint union

Let $y$ be an admissible integral normal vector in the supported orientable three-manifold, and let $x = ky$ for a positive integer $k$. The normal surface for $x$ can be placed in the normal interval neighbourhood of the surface for $y$. Over each normal disc it has $k$ ordered local sheets. Across a face the sheet order is either preserved or reversed, so the monodromy on sheet indices is generated by

$$j \longmapsto k - 1 - j, \qquad 0 \le j < k.$$

There are no arbitrary permutations of this ordered normal fibre. In the maintained interval model the same fact follows directly by writing an edge-point index as $kq + j$: a scaled translation preserves $j$ and a scaled reflection replaces it by $k - 1 - j$.

A two-sided component has trivial normal monodromy. It therefore contributes $k$ disjoint copies of itself. A one-sided component has nontrivial monodromy: its sheet pairs $\{j, k-1-j\}$ each form a connected normal orientation double cover, with a central original component only when $k$ is odd. Since the ambient manifold is orientable, two-sided and orientable components coincide, and the paired double covers are orientable. If $O$ and $N$ are the numbers of orientable and nonorientable components of $y$, the scaled surface has

$$O_k = kO + \lfloor k/2 \rfloor N, \qquad N_k = (k \bmod 2)N, \qquad C_k = O_k + N_k.$$

Multiplying the total component count by $k$ would be wrong when $N > 0$. For example, twice a normal Möbius band is one annulus, not two Möbius bands: it has one orientable component, two boundary circles, Euler characteristic zero and genus zero.

Every boundary circle has an orientable collar in the surface. The normal orientation cover consequently restricts to two separate copies of each boundary circle. Thus the total number of boundary circles scales by $k$, including for one-sided components. Euler characteristic and normal-disc count also scale by $k$, both from the covering argument and directly from the linear coordinate formulas. These statements apply to disconnected surfaces and mixtures of orientable and nonorientable components; no component-by-component extraction is needed.

## 79.2 The producer and the original source

After full native validation, the producer computes the gcd $k$ of all $7t$ coordinates. It stops the gcd scan as soon as the value is one. The empty vector uses the identity case $k = 1$. When $k > 1$, it divides each coordinate exactly. Admissibility is preserved: matching equations are homogeneous, nonnegativity is unchanged, and no new quadrilateral type appears. Global edge weights, Euler characteristic, normal-disc count and boundary-arc counts are linear, so their quotient values are obtained by exact division from the already validated source. There is no second manifold-validation pass.

The three orbit queries now concern $y$, $2y$ and $\partial y$. Their counts give $O = C_2 - C$ and $N = 2C - C_2$ as before. The lifting formula then returns $O_k$, $N_k$, $C_k$ and $kb$ for the original input $x$. Connected-surface genus and crosscaps are computed from these lifted counts and the original Euler characteristic. The original normal-disc count, maximum coordinate bit length, boundary-arc count and boundary-cohomology witness are retained. In particular, even $k$ can make the original boundary class zero although the quotient has nonzero class; using the quotient witness in its place would be incorrect.

The public `reduce_multiplicity` option defaults to true. Setting it to false retains direct unreduced queries. Primitive and empty inputs have exactly the previous result structure. A reduced result additionally reports `coordinate_divisor`; its query statistics describe the quotient, while every topological claim describes the original surface. The original shared `max_cycles` limit applies to the three actual queries. Cancellation propagates during validation, gcd scanning, division and orbit work. Incomplete queries remain inconclusive and emit no topology proof or component count.

## 79.3 Versioned proofs without gcd search in replay

Version-one normal-topology proofs continue to bind and verify the original three interval systems. The producer retains that format for primitive and empty inputs and whenever reduction is disabled. A reduced proof uses `normal-surface-topology-v2` and supplies its `coordinate_divisor`. Its source digest still binds the full input $x$, not merely the quotient. The three contained local orbit proofs bind the reconstructed quotient systems, and its claims concern $x$.

The verifier checks that the supplied divisor is an integer at least two, that the source is nonempty, and that every source coordinate is divisible by it. It then reconstructs the quotient geometry and independently checks all three local traces. Finally it applies the lifting formula and checks every original topological claim, together with the original boundary witness. A different divisor with unchanged query traces is rejected because the reconstructed interval inputs differ.

Replay does not need to prove that the divisor is the greatest common divisor. Any exact common divisor at least two gives a sound quotient certificate. For example, a source of multiplicity twelve may use divisor six and a complete proof for the twice-primitive quotient. This avoids rerunning producer gcd discovery and supports independently constructed proofs. The producer chooses the full gcd to minimize the remaining coordinate sizes. A Boolean, fractional, zero, unit, negative or nondividing witness is rejected; arbitrary-size hexadecimal integers remain supported. An empty source cannot masquerade as a version-two reduction.

The event cap remains shared across the three supplied traces. Source validation and binary division remain necessary work even for very short traces; a short certificate is not an assertion that the original input was cheap to read. Version-one proofs from the actual prior implementation continue to replay with the new checker.

## 79.4 Bit costs and proof-size improvement

Let $B$ bound the original coordinate bit lengths, $B_0$ those after division, and $t$ the tetrahedron count. Write $P(t, b)$ for the maintained three-query orbit cost with $O(t)$ pairings and $O(b + \log(t+1))$-bit endpoints. Validation still processes the full source. In addition, the reduction uses $O(t)$ gcd and division operations on integers of at most $B$ bits, followed by

$$P(t, B_0)$$

rather than $P(t, B)$ orbit work. Lifting the fixed number of summary counts uses ordinary binary multiplication and parity. With schoolbook arithmetic, a coarse bound for the extra gcd/division work is polynomial in $t$ and $B$; fast integer arithmetic can reduce that front-end cost. No unit-cost assumption is made for the supplied large coordinates.

The reduced proof stores the divisor and a fixed number of original summary counts, together using $O(B + \log(t+1))$ bits, plus the three quotient traces. Consequently common multiplicity contributes only to the outer arithmetic fields, rather than to every endpoint in those traces. For a fixed primitive surface scaled through arbitrarily large integers, the orbit-search portion and its local-event structure depend only on the primitive surface. Source decoding, validation, division, output arithmetic and the outer proof fields continue to depend on the encoded multiplicity. When

the source gcd is one there is no such reduction; the algorithm retains its previous worst-case polynomial dependence on the input bit length. This is a parameter-sensitive improvement of a supplied-vector subroutine, not a new bound on complete normal-vector search.

### 79.5 Validation and paired measurements

The first mathematical check compared the scaling identities with Regina on 1,080 multiples of the preceding section's 270 actual normal vectors. These include mixed orientability, disconnected vectors and triangulations with multiple boundary vertices. The integrated audit additionally compares complete old and new topology results and both proof formats against the actual prior producer and checker at commit `a3f10a6637bb`.

Thirty-six focused tests pass in 7.619 seconds. They cover even and odd one-sided scaling, connected annuli produced by doubling, mixed components, 20,001-bit multiplicities, original-versus-quotient boundary parity, nonmaximal but valid divisors, foreign quotient traces, empty and primitive inputs, strict witness types, exact shared resource boundaries, cancellation during quotient construction, and replay with gcd and orbit search disabled. The historical version-one research driver explicitly disables multiplicity reduction so its original comparison remains reproducible.

All 932 maintained tests pass with Regina in 202.074 seconds. The integrated native audit completes 1,350 comparisons in 17.493 seconds: 1,090 inputs use the reduction and 102 resulting surfaces retain a nonorientable component. Every old proof replays with the current checker, every new proof verifies, and every topology agrees with Regina and the pinned producer. Raw scale factors, input-manifest indices, topology, cycle counts and proof sizes are in `data/normal-multiplicity-audit.json`.

The paired benchmark loads the actual prior producer and verifier in separate module namespaces, retaining unchanged finite geometry and orbit kernel dependencies. It compares prior count-only calls, current calls with reduction disabled, current defaults and an identical current control. It also separately compares old/new production plus verification and old/new replay of already supplied proofs. Each fixture has one shuffled warmup and five measured rounds, with small inputs batched. Every computation includes fresh source validation. Serialization is outside timing; final topology and certificates are checked against the reference after the measured calls. The combined proof arms include verification inside timing.

| Input | old ms | new ms | count ratio | proof ratio | proof bytes old/new |
|---|---:|---:|---:|---:|---:|
| primitive1 | 0.505 | 0.514 | 0.983 | 0.994 | 2820/2820 |
| primitive8 | 4.948 | 4.976 | 0.995 | 1.017 | 20387/20387 |
| primitive32 | 70.897 | 71.281 | 0.979 | 0.990 | 234409/234409 |
| primitive96 | 1218.134 | 1217.641 | 1.000 | 0.989 | 2018995/2018995 |
| parallel5000 | 9.821 | 5.109 | 1.914 | 1.692 | 434035/27931 |
| parallel20000 | 21.757 | 5.671 | 3.889 | 2.987 | 1671535/50431 |
| one_sided20000 | 0.738 | 0.352 | 2.102 | 1.779 | 231732/26445 |
| mixed20000 | 3.152 | 0.968 | 3.283 | 2.434 | 620065/34638 |

Count and certified ratios are medians of paired old/new times. The former compares the full supplied-vector topology call; the latter includes proof production and replay on both sides. Neither includes finding a normal vector or constructing its source manifold from a knot diagram. Exact representative old/new proofs, all samples, query statistics and loaded source hashes are retained in `../fast/results/normal_multiplicity_20261008.json`.

The principal run contains 1,560 measured calls and 312 excluded warmup calls and takes 59.287 seconds. For $2^{20000}$ parallel copies of the eight-tetrahedron meridian, the complete count-only median falls from 21.757 to 5.671 ms, with median paired ratio 3.889. Production plus replay

falls from 33.796 to 11.313 ms (ratio 2.987), and compact proof transport falls from 1,671,535 to 50,431 bytes, about a 33-fold reduction. For $2^{5000}$ copies, count and certified ratios are 1.914 and 1.692, with proof size 434,035 versus 27,931 bytes.

The odd one-sided multiplicity $2^{20000} + 1$ has count and certified ratios 2.102 and 1.779; its proof shrinks from 231,732 to 26,445 bytes. A compatible mixture of a vertex-linking disc and a Möbius band at the same scale has ratios 3.283 and 2.434, with 620,065 versus 34,638 proof bytes. These are complete supplied-vector computations, rather than timings that exclude source validation or correctness checks from one side of the comparison.

Primitive controls have paired count ratios from 0.979 to 1.000 and retain identical proof sizes. The identical current controls range from 0.974 to 1.019 across the principal run. No benefit is claimed for primitive inputs; the gcd scan adds a small amount of work and quickly stops when it reaches one. The timing differences on those controls are much smaller than the large-multiplicity effects.

A separate small-multiplicity follow-up uses four families—one- and eight-tetrahedron meridians, the one-sided core, and a mixed disc/band surface—at scales 2, 3, 8, 64 and $2^{32}$. Seven shuffled measured rounds of ten calls per arm produce 5,600 timed calls, with 800 excluded warmups. Old versus identical-old control ratios range from 0.969 to 1.021. Count-only old/new ratios range from 0.995 to 1.179. The twice-meridian at one tetrahedron is essentially unchanged (0.488/0.492 ms), while the eight-tetrahedron $2^{32}$ case improves from 5.477 to 4.817 ms (paired ratio 1.123). Every one of these twenty cases finishes with the previous cycle allowance. Raw samples and cap probes are in `../fast/results/normal_multiplicity_small_20261008.json`.

Neither runtime nor proof size is asserted to improve on every possible input. In the 1,350-case native audit, 58 small certificates grow rather than shrink; the largest growth is from 12,883 to 13,405 bytes. The divisor field and differences in local reduction traces can outweigh endpoint savings at small scale. No audited case uses more orbit cycles after reduction, but that finite comparison is not a theorem of cycle-count monotonicity for the scheduler. The unchanged direct path remains available for callers and future adaptive scheduling experiments.

**Complexity target.** This change isolates a large numeric factor and preserves a proof for the original object. It removes common multiplicity from the expensive local orbit stage, while charging full-source arithmetic. It does not supply a normal vector, certify correspondence with the input knot exterior, or bound a complete hierarchy search. Those obligations remain necessary for a general quasi-polynomial unknot recognizer.

# 80 Prepare compressed braid factors only when they are visited

Section 76 stops recognition after a complete nontrivial-factor proof, but still constructs and summarizes every projected factor before choosing the first one. Thus the earlier scheduler saves leaf recognition work while paying all factor-preparation costs. The maintained recognizer now constructs a cheaper schedule directly from the input grammar, and prepares factors on demand. The topology theorem and certificate formats are unchanged.

## 80.1 Keep global component validation before scheduling

The input grammar is validated in full, including unreachable rules. Its root occurrence counts determine every singleton-generator cut. Its exact root permutation must describe a knot before any factor can prove a nontrivial-knot verdict. A link containing a nontrivial component remains a link; one obstructing factor does not replace this global check.

As before, deleting singleton generators partitions consecutive strand intervals and expresses

the knot as a connected sum of their projected closures. One nontrivial summand proves the whole knot nontrivial, whereas an unknot verdict requires every summand to be an unknot. A selected-factor proof therefore remains the existing version-three certificate with one source-bound projected grammar. Positive certificates retain every factor in original strand order, regardless of discovery order. The checker still reconstructs the selected projection, or all projections for a complete positive proof, without consulting the producer's schedule.

## 80.2 A schedule without constructing the projected grammars

Let $s$ be the number of supplied grammar rules, $m$ the number of strands, and $f$ the number of factor intervals. Complete root support implies $m \leq s + 1$, so a strand-label-to-factor array is safe after global knot validation. Cut labels map to no factor. Summing root occurrence counts inside each interval gives its exact projected word length.

A reverse DAG pass propagates root weights to children, adding both contributions when a concatenation repeats the same child. Each terminal contributes its weight with its sign to the exponent of its factor. Unreachable terminals have weight zero. Consequently these projected lengths and exponents are exact without expanding any word or building any factor grammar.

To estimate construction cost, a forward pass assigns each node either an empty span or the hull of the factor indices touched by its terminal descendants. A terminal at a cut has empty span. A concatenation takes the hull of its nonempty child spans. The hull may contain factors that the node does not actually touch; this only makes the estimate larger.

A terminal contributes a possible new rule to its single factor. A concatenation contributes only where both children can survive: the intersection of the two child hulls. If one child is empty in a factor, projection aliases the surviving child and creates no rule. An empty child span or a disjoint pair of child spans therefore contributes nothing.

For each nonempty contribution interval $[a, b]$, add one to a difference array at $a$ and subtract one at $b + 1$. Prefix summation gives the possible rule creations per factor. Its value plus one bounds the number of rules in the canonical projection. Every created terminal has the corresponding factor label; every created concatenation has two nonempty projected children, so that factor belongs to both child hulls. The extra rule is the empty node. Interning and false-positive hull intersections can only make the estimate larger than the actual count. All supplied nodes are considered, including unreachable ones, because the established canonical projection includes their surviving rules. This differs deliberately from root-weighted length and exponent.

The schedule keeps the previous three priority classes: at-most-two-strand or exponent-obstructed factors first, other three-strand factors next, and wider residual factors last. Within a class it sorts by this rule-count upper bound, then original index. The old tie-breaker was the actual projected rule count, which required eager construction. The new estimate is a scheduling heuristic, not a topological certificate or a guarantee of optimal discovery order.

## 80.3 Lazy preparation and unchanged replay

Only when a factor is visited does the producer construct its canonical projection and run the full leaf summary. It checks the resulting length and exponent against the schedule and checks that the leaf is a knot. The existing finite, compressed three-braid, elementary-reduction and exceptional-cube backends then produce the same kinds of leaf proofs. A negative verdict immediately discards the remaining schedule; unvisited factor grammars are never constructed. An inconclusive wider factor does not justify a negative verdict, and a positive result still requires all factors.

A one-strand projection has a further exact shortcut: every generator is erased, so its canonical grammar is just the empty rule with root zero. This agrees byte-for-byte with the old projection, including in the presence of unreachable source rules. Producer and verifier can construct it

directly after global source validation. Invalid unreachable rules still fail the earlier validation.

The public option `use_lazy_factors=False`, or CLI flag `-eager-factors`, retains eager preparation and its former exact rule-count ordering. The one-strand shortcut applies to both policies. No certificate version change is necessary: scheduling choices are not trusted proof fields, and every published verdict still undergoes fresh independent replay. Old proofs replay with the current checker and new proofs replay with the pinned previous checker.

## 80.4 Preparation bound and its limits

Let $q$ be the number of factors visited before completion, and let $s_i, m_i$ be the rule and strand counts of a visited projection. The plan uses $O(s + m + f \log f)$ structural operations and $O(s + m + f)$ working entries, with binary root weights charged according to their bit lengths. A single-factor input bypasses the scheduling pass entirely. Ordinary projection scans the $s$ supplied rules; a one-strand projection is constant size. Apart from leaf recognition and binary arithmetic on lengths and weights, preparation now costs

$$O\left(s + m + f \log f + \sum_{i \text{ visited}} (s + s_i m_i)\right),$$

where the second term inside the sum bounds the unchanged full leaf permutation summary. Previously every factor paid its projection and summary cost before recognition could stop. The new producer retains only visited projections and their required evidence, rather than materializing all factors first.

At the checkpoint measured in this section, the global root's dense-permutation pass costs $O(sm)$ word operations and may dominate a large forest. That work is intentionally included in every complete-call measurement. This optimization therefore does not claim to remove the root permutation cost or improve the overall worst-case asymptotic bound merely by reducing factor preparation. If all factors are needed, $q = f$, and planning adds work. The exceptional-factor exponential bound and the restricted quasi-polynomial regimes of the prior sections remain unchanged.

All planning, preparation, leaf solving, proof transport accounting and replay share the existing public work/time allowances. The estimate affects only which work is tried first. A limit still returns inconclusive with no partial certificate; external cancellation, including `ValueError`, is not converted to an ordinary resource result.

## 80.5 Native validation

Seventy-eight existing compressed-braid tests pass in 20.043 seconds, and eight final focused tests pass in 0.244 seconds. The new tests compare exact projected metadata and structural bounds on 107 grammars, including nonconvex support spans, dead rules and repeated child references. They check that a thirteen-factor finite-obstruction example makes exactly two projection calls: one in production and one in mandatory replay, both for the selected factor. Positive forests retain identical complete certificates and include every factor. Link inputs are rejected before planning; one-strand shortcuts preserve exact canonical output; replay works with the planner disabled; and exact work limits and cancellation retain their prior semantics.

A further native audit compares 160 prior explicit braid cases and 80 seeded forests against the actual maintained implementation at commit `586bfc8d0ffc`. There are 110 unknot and 130 nontrivial-knot cases. Every old and new certificate replays in both implementations; every positive certificate is identical. Scheduling metadata and rule bounds are also checked against actual projections. The 4.424-second final audit keeps all source grammars, work counts and factor orders in `data/lazy-forest-audit.json`. Its bounded-work example completes a finite

negative proof within an allowance at which the old implementation is inconclusive, with both string-node and cube-generator allowances zero.

The final full maintained suite passes 940 tests with Regina in 208.823 seconds. The benchmark compares the pinned old implementation, an identical old control, current defaults, an identical current control and current eager preparation. Each of ten fixtures has one shuffled warmup and five measured rounds. Every timed call is the complete public recognizer: source checks, root summaries, factor discovery, proof construction, transport allowance accounting and mandatory independent replay. Projection-call counts come from separate untimed instrumented calls. No precomputed projections are supplied to a timed arm.

| Input | old ms | lazy ms | eager ms | ratio | projections old/new |
|---|---|---|---|---|---|
| late_finite_8 | 14.092 | 11.469 | 14.063 | 1.255 | 10/2 |
| late_finite_24 | 64.446 | 42.462 | 62.005 | 1.465 | 26/2 |
| late_finite_64 | 328.629 | 184.098 | 327.261 | 1.785 | 66/2 |
| late_compressed | 127.182 | 112.913 | 128.522 | 1.161 | 25/2 |
| early_compressed | 127.902 | 117.397 | 127.392 | 1.071 | 25/2 |
| positive_forest | 115.400 | 115.676 | 112.968 | 1.025 | 32/32 |
| all_singleton | 36.872 | 12.712 | 12.484 | 2.902 | 256/256 |
| positive_mixed | 7.053 | 7.205 | 7.203 | 0.976 | 4/4 |
| single_wider | 1.171 | 1.184 | 1.167 | 1.006 | 2/2 |
| direct_three | 4.020 | 4.026 | 4.024 | 1.000 | 0/0 |

The ratio is the median paired old/current elapsed time. All source grammars, samples, factor orders, proof sizes, projection calls and shared resource counts are in `../fast/results/lazy_forest_20261008.json`. The experiment includes positive forests, a negative factor early in strand order, a single wider factor and a direct three-braid control; the selected examples are not restricted to favorable early exits.

The first tested estimate counted every nonempty node hull. A complete-call benchmark exposed a scheduling defect: concatenations between disjoint factors inflated the estimate for early factors, even though they alias away under projection. On the early-compressed negative fixture, the producer visited 22 factors instead of the old policy's one, with paired old/new ratio 0.849. The first measurements are preserved in `../fast/results/lazy_forest_initial_20261008.json`.

The final bound counts possible rule creations, using child-hull intersections as proved above. It is exact on the tested block-separated forest grammars, regardless of negative-factor placement. A dedicated regression checks that both early and late compressed negative factors are visited first. Nonconvex supports still give a conservative bound, so no claim of optimal scheduling is made for arbitrary input DAGs.

The final run has 250 measured calls and 50 excluded warmups, taking 24.451 seconds. For 64 positive three-braid factors followed by a finite negative factor, complete recognition falls from 328.629 to 184.098 ms, with median paired ratio 1.785. Production plus replay makes two projection calls instead of 66, and charged work falls from 414,154 to 98,011. The eight- and 24-positive-factor versions have paired ratios 1.255 and 1.465. The current eager arm remains close to the old implementation on these cases, isolating preparation as the mechanism.

For the 24-factor compressed-negative fixtures, both early and late placements now visit exactly one factor and make two projection calls instead of 25. Charged work is 81,033 instead of 124,353 in both placements. The paired time ratios are 1.071 for early placement and 1.161 for late placement. These smaller elapsed-time differences should be read with the controls: identical-old ratios across the run range from 0.935 to 1.031, and identical-current ratios from 0.966 to 1.060. The structural correction is exact even where timing evidence for a small gain is noisy.

The 128-strand chain consisting entirely of singleton generators improves from 36.872 to 12.712 ms (paired ratio 2.902). It still needs all 128 factors and 256 producer/replay projection calls. Here

the improvement comes from the canonical one-strand shortcut, not early stopping; current eager preparation also benefits, with median 12.484 ms. Its charged work falls from 77,209 to 13,205.

The sixteen-factor positive compressed forest needs every projection. Its old/new medians are 115.400/115.676 ms, with no systematic gain asserted. The smaller mixed positive case has paired ratio 0.976; planning costs can outweigh savings when all factors are needed. Single wider-factor and direct three-braid controls remain close to unchanged. The implementation therefore retains an eager policy for comparison and makes no universal claim about the best discovery order.

**Remaining complexity work.** The complete recognizer now avoids unused factor projections. Its input is still a supplied grammar. It neither constructs such a grammar of controlled size from every knot diagram nor bounds the exceptional parameter for every input. Section 81 subsequently reduces permutation storage and evaluation work on local forest nodes while preserving the full component check. Section 82 subsequently reconciles the incoming shared overlap index with the maintained group backend. The general source and complete-search bounds remain open.

# 81 Compact permutations for braid component checks

The lazy factor preparation of Section 80 exposes a remaining cost: computing a strand permutation for every supplied grammar rule. A local three-strand subword in a large forest previously occupied a full $m$-entry tuple. This cost was paid in production and again in independent replay, even when one finite factor settled the knot verdict. The maintained implementation now stores local permutations compactly and evaluates only root-reachable permutation nodes. It still validates every supplied rule.

## 81.1 Exact composition with adaptive storage

Use zero-based strand indices. A generator of either sign exchanges two adjacent indices. If a concatenation has child permutations $p, q$, the established convention is

$$(p \circ q)(i) = p(q(i)).$$

A sparse representation stores exactly the pairs $i \mapsto p(i)$ for which $p(i) \neq i$. An absent key means a fixed point, and the empty map represents the identity. To compose two sparse operands, start with a copy of $p$. For every stored pair $i \mapsto j$ of $q$, set its image to $p(j)$, using $j$ when that key is absent in $p$. Delete the result entry if its image is $i$. For indices outside the support of $q$, $q(i) = i$, so the initial copy already has the correct image. This proves the formula for every strand, including overlapping supports and products that cancel to the identity.

When the resulting map moves more than half the strands, it is converted to a dense tuple. Dense–dense composition indexes the first tuple by the second. Dense–sparse composition copies the dense tuple and updates just the sparse operand's domain. Sparse–dense composition performs one sparse lookup for each image in the dense tuple. These are the same formula in three storage cases. A dense descendant remains dense even if its permutation later becomes sparse or the identity; there is no claim that storage always tracks minimal support.

All operands are treated as immutable. An identity product can therefore reuse its nonidentity operand without copying, including when a grammar node has several parents. Strands counts at most eight retain the simple dense path to avoid representation dispatch overhead. The half-density and small-strand thresholds are implementation heuristics, not topological facts or claims of an optimal policy. The internal dense ablation retains the new reachability check while disabling compact storage.

## 81.2 Reachability, source validation and proof compatibility

The existing reverse root-weight pass already assigns zero to unreachable nodes. A live concatenation has two live children because both child weights receive a positive contribution, including twice the contribution if the children coincide. Thus a forward pass can skip every zero-weight permutation without losing any dependency of the root. Unreachable rules still undergo full syntax, index, generator and metadata validation. They remain part of the exact source grammar and serialized-byte accounting; this is not source pruning or a change in certificate identity.

Before allocating any strand-sized representation, the validator checks for a missing generator label in the root's unsigned occurrence counts. A missing label separates the strand intervals and proves that the closure is a link. This guard works with a binary strand count far larger than the supplied rule count. If there is no gap, every one of the $m - 1$ labels occurs in some terminal, so $m \le s + 1$ for $s$ supplied rules. The strand-sized allocations are made only after this bound is established.

Finally the root is materialized as the same full permutation list as before. Its orbit from strand zero must contain all $m$ strands to establish a knot. This check remains before factor scheduling or any selected negative-factor proof. A link containing a nontrivial component cannot acquire a nontrivial *knot* verdict from that component. Producer and replay use the same validated permutation arithmetic but replay still reconstructs it afresh from the exact source, without trusting a claimed component count.

No public option or certificate schema changes. Exact summary fields, canonical proofs, factor ordering, verdicts and charged abstract work remain unchanged on the differential audit. The callback still runs at each supplied rule in the permutation pass, even when its permutation is skipped. The work counter counts cooperative operations, not every tuple entry or hash probe; a faster or smaller representation need not reduce that counter. Work and time limits still yield no partial certificate, and external cancellation propagates.

## 81.3 Cost and remaining asymptotic limitations

Let $s$ be the supplied rule count and $m$ the number of strands. For a live node $v$, write $r_v$ for its representation size: the moved-point count for a sparse map, and $m$ for a dense tuple. A sparse–sparse composition performs $O(r_u + r_w)$ dictionary operations before any dense conversion. A conversion costs $O(m)$ operations but occurs only after more than $m/2$ output entries have been constructed. A composition with a dense operand costs $O(m)$, which is also $O(r_u + r_w)$. Live terminals use two entries in the compact path; identity aliases allocate no permutation entries.

Thus, with constant-cost dictionary operations, the permutation pass uses

$$O\left(s + m + \sum_{v=(u,w)\ \text{live}} (r_u + r_w)\right)$$

word operations and at most $O(s + m + \sum_{v\ \text{live}} r_v)$ stored entries. The sums bound storage even when aliases share objects. The ordinary dense bound $O(sm)$ remains the worst case in this model, while local forest nodes can have representation size independent of the total strand count. The current implementation retains node representations until the root has been evaluated; it does not claim optimal lifetime management.

These are representation and dictionary-operation bounds, not an unconditional bit-cost theorem for Python hash tables. Python's conventional amortized hash-table cost model gives the stated elapsed-work interpretation. Charging up to $O(m)$ probes per lookup instead gives a conservative polynomial $O(sm^2)$ probe bound. Strand keys and images have $O(\log m)$ bits; the separate validated lengths, exponents and root weights must still be charged at their own binary lengths.

No deterministic subexponential or quasi-polynomial theorem is inferred from a hash-table timing result.

In particular, a large dense descendant need not become compact again after cancellation. All-factor positive recognition still pays every required projection and leaf proof. The input remains a supplied braid grammar; this change neither constructs one of controlled size from an arbitrary knot diagram nor bounds the exceptional-factor parameter on every diagram. The restricted recognition theorems of the preceding sections are unchanged.

## 81.4 Native differential validation and measurements

Eight focused tests pass in 0.196 seconds. They exhaust all 576 ordered pairs of four-strand permutations in each of the four sparse/dense representation combinations, checking results and operand immutability. A separate oracle expands 400 bounded random shared grammars and applies adjacent swaps directly. Other cases check repeated child references, a $2^{256}$-power grammar, cancellation, dead-rule validation, a strand count of $10^{100}$, complete component checks on links, exact public work ceilings and proof equality with the dense ablation. The small-strand path is tested with compact composition disabled.

The native compatibility audit loads the actual previous compressed-braid package at commit `d0f0e37761b5`. It combines 240 prior cases, twelve benchmark sources and 150 additional seeded sources with shared reachable and unreachable powers. With the exponential fallback disabled equally in both versions, the 402 cases give 103 unknot, 132 nontrivial-knot, 148 link and nineteen inconclusive results. All public result fields and root summaries agree exactly, and every old and new certificate replays in both versions. The 4.117-second audit retains its complete sources and hashes in `data/compact-permutations-audit.json`.

The full maintained suite passes 948 tests with Regina in 222.385 seconds.

The benchmark has twelve fixtures, five arms, one shuffled excluded warmup and seven shuffled measured rounds per fixture: 420 measured calls and sixty warmups. The arms are the pinned old package, an identical old control, current defaults, an identical current control, and current code with dense permutation storage forced internally. Policy binding for this last arm is outside the timer; all source validation, permutation construction, factor recognition, proof transport accounting and mandatory replay are inside. Every arm produces the same complete public result. Separate untimed `tracemalloc` calls measure peak traced allocations of a complete root summary, including its validation and metadata. These are not whole-process resident-memory measurements or enforced memory ceilings.

| Input | old ms | new ms | dense ms | ratio | peak KiB old/new |
|---|---|---|---|---|---|
| late_finite_8 | 11.967 | 11.155 | 12.085 | 1.072 | 204.0/184.9 |
| late_finite_24 | 43.256 | 33.085 | 42.199 | 1.310 | 1292.0/578.3 |
| late_finite_64 | 188.472 | 89.181 | 178.327 | 2.103 | 8002.7/1674.1 |
| early_compressed | 109.791 | 99.178 | 108.404 | 1.115 | 1257.1/574.8 |
| positive_forest | 118.311 | 115.619 | 117.032 | 1.023 | 609.1/374.2 |
| all_singleton | 13.562 | 12.305 | 13.787 | 1.103 | 288.4/187.6 |
| single_wider | 1.212 | 1.235 | 1.242 | 0.977 | 2.5/2.6 |
| direct_three | 4.102 | 4.125 | 4.120 | 0.994 | 11.6/11.7 |
| dense_random_link | 49.501 | 34.507 | 48.795 | 1.432 | 2492.5/1578.6 |
| dense_power_link | 19.512 | 16.999 | 18.060 | 1.139 | 968.4/874.8 |
| unreachable_rules | 43.521 | 32.854 | 33.371 | 1.302 | 1236.1/137.1 |
| huge_strand_gap | 0.088 | 0.088 | 0.090 | 1.003 | 0.5/0.5 |

The ratio is the median paired old/new time. The dense column still skips unreachable permutation nodes, so it separates compact storage from that optimization. Raw source grammars, sample orders and times, resource counts, proof sizes and source hashes are retained in

`../fast/results/compact_permutations_20261008.json`. The complete run, including separate memory measurements, takes 27.469 seconds. Identical-old paired ratios range from 0.972 to 1.017; identical-current ratios range from 0.989 to 1.021. Small changes near this range are not evidence of a gain.

On the 64-positive-factor forest followed by a finite negative factor, complete recognition falls from 188.472 to 89.181 ms, with paired ratio 2.103. The dense ablation takes 178.327 ms, locating most of the gain in compact representations. Root-summary peak traced allocation falls from 8,194,752 to 1,714,252 bytes, a factor of 4.780. Both public calls still charge 98,011 abstract work steps and produce the same proof. The eight- and 24-positive-factor cases have paired ratios 1.072 and 1.310 respectively; the smaller case also has a smaller memory reduction.

The positive sixteen-factor forest needs all leaf work and has paired ratio 1.023, close to the control range; no systematic whole-call improvement is claimed. Its root-summary allocation nevertheless falls from 623,728 to 383,220 bytes. The early compressed-negative case has paired ratio 1.115; the singleton chain has ratio 1.103. The small wider-factor and direct three-braid controls have ratios 0.977 and 0.994, so the compact policy is not presented as universally faster.

The dense random-link control improves from 49.501 to 34.507 ms, with ratio 1.432. Although long prefixes become dense, their terminal operands remain sparse, permitting copy-and-update composition. The repeated dense-power control has ratio 1.139 but only a 1.107-fold memory reduction, illustrating that dense descendants remain dense after cancellation. In the unreachable rule control, old/new medians are 43.521/32.854 ms and the dense ablation is 33.371 ms: here skipping dead permutations explains most of the gain. Its root-summary peak allocation falls from 1,265,744 to 140,368 bytes. The enormous strand-count gap control remains constant-size and has no timing gain.

**Scope of the advance.** Local permutation support now reduces a measured preparation bottleneck in both proof production and replay. The finite obstruction, three-braid and exceptional-cube mathematics is unchanged. Establishing a complete source-bound conversion and controlling its size for arbitrary knot diagrams remains necessary before these restricted algorithms can imply a general quasi-polynomial recognition theorem.

# 82 Shared overlap indexing with bounded adaptive search

Report 46 supplies an exact shared index for cyclic relator overlap. Its operation bound removes the number of nonempty relators from the leading pairwise scan cost. The report was based on commit `58ee11a97d5f`, before the maintained donor-pruning improvement. The native integration therefore preserves those later cutoffs and compares against the actual maintained implementation at `bec1afd07cd2`. It also retains the as-delivered report as a separate measured arm. No group-theoretic certificate rule changes.

## 82.1 The exact local query and its topological scope

Let $s$ be the number of supplied relator slots, $m$ the number of nonempty slots, and $L$ their total explicit length. Empty slots are retained because certificate moves refer to original slot indices. For a target and a distinct donor, choose cyclic rotations and donor sign so that

$$R_i \sim UW, \qquad R_j^{\varepsilon} \sim UV, \qquad i \neq j, \quad \varepsilon \in \{1, -1\}.$$

Keep the donor and replace the target by $V^{-1}W$. The normal closure is preserved, since

$$(UV)^{-1}(UW) = V^{-1}W, \qquad (UV)(V^{-1}W) = UW.$$

The guaranteed length decrease before free cancellation is $g = 2|U| - |R_j|$. The query finds a maximum positive $g$ over all rotations, signs and distinct slots, or returns no shortening. It is an exact optimization query; failure to find a move is not a knot verdict or a completeness theorem for presentation simplification.

The old query builds a suffix automaton for each oriented doubled donor, then scans eligible doubled targets. The native donor bound $g \le |R_j|$ already avoids donors that cannot improve an incumbent gain. This is effective on repeated relators and on the measured Gordian presentation, but its worst-case operation bound remains $O(s + mL)$.

## 82.2 Capped cyclic windows on a shared suffix-link tree

For a nonempty word of length $d$, the $d$ endpoints from $d - 1$ through $2d - 2$ in its doubled text represent all length-$d$ cyclic windows. Build one suffix automaton over the positive doubled blocks, separated by a symbol distinct from every signed generator. The indexed text has length $2L$. Attach each window to its endpoint state, with a depth cap $d$ and its original slot and endpoint. A cap at a depth inside a suffix-link edge creates an implicit vertex. Thus no marked substring crosses a block separator or wraps around a relator more than once.

Inverse donors are scanned through this completed positive index. At each inverse cyclic-window endpoint, let $h$ be the longest matched suffix. If $h > 0$, attach a donor mark at depth $\min(h, d)$, retaining the original length $d$ for its gain. Every common inverse-donor/positive-target segment of length $u$ is a suffix of a recorded inverse window. The longest match has length at least $u$, so the mark's ancestral suffix path contains it. Conversely, each ancestor denotes a literal suffix of that window. Positive target marks have the same property. This proves both completeness and soundness of the inverse-window representation without inserting inverse blocks into the shared index.

At a capped-tree vertex of depth $h$, any descendant donor and target marks of different original slots give gain $2h - d$, where $d$ is the donor length. Only two donor representatives of distinct slots are needed, ordered first by donor length; retain at most two target representatives of distinct slots as well. If the shortest donor has any eligible target, one of the two target slots differs from it. Otherwise all available targets have that donor's slot, so the second donor is the shortest eligible alternative. If it does not exist, there is no eligible pair. This proves that constant-size summaries recover the maximum gain at each vertex. Duplicate words in different slots remain distinct donors and targets.

Propagate these summaries upward. An optimum common suffix ends at a tree vertex or a cap vertex: between adjacent such vertices there are no new available marks, and increasing the common depth increases gain. Evaluating all these vertices therefore finds a global maximum. Endpoint $e$, depth $h$ and original word length $d$ recover the cyclic start as $(e - h + 1) \bmod d$. The resulting move has exactly the existing version-two relator fields. Deterministic representative ordering need not choose the historical pairwise witness among tied maxima.

There are $O(L)$ automaton states, transitions and window marks, and at most $O(L)$ cap vertices. A depth-first traversal maintains the increasing depths on its suffix-link path. Binary search places each cap; sorting caps along each edge costs $O(L \log(L + 1))$ in total. Constant-size summary propagation is linear. Including the input slots, the shared query takes $O(s + L \log(L + 1))$ integer and dictionary operations and $O(L)$ auxiliary entries. With expected constant-time dictionary operations this is the expected word-time bound. It is not a deterministic constant-time hashing claim for arbitrary Python integers.

## 82.3 A bounded prelude that retains native donor pruning

Building the shared index can cost more than a few successful pairwise scans. Lists with at most four nonempty relators therefore use the unchanged native pairwise query, retaining its exact witness. Since this threshold is fixed, that base case costs $O(s + L)$.

For larger lists, sort donors by decreasing length, breaking equal-length ties by original slot. Targets retain their original order, and donor signs retain the old order. Once the next donor length is at most the best known gain $G$, no remaining donor can improve it. The native target-length bound also remains in force. After a full-donor match, stop the target scan and skip the other sign: no larger gain from that donor is possible. These cutoffs preserve maximum gain, although changing the donor order can change a tied witness.

Charge sorting, donor construction, suffix-link traversal, transition work, clone copying, every target-pair visit, and target scanning to a local allowance

$$Q = 2\max(1, L)\,\mathrm{bitlength}(L + 1).$$

Each local charge first charges the original enclosing group budget. Only exhaustion of the private prelude allowance triggers the complete shared index. A global work limit or external cancellation propagates immediately; it cannot reset the budget or publish the partial incumbent as a completed query. The prelude runs in its own frame, and continuation starts after the exception handler, releasing its traceback and abandoned automaton first. Both phases together retain $O(L)$ auxiliary storage.

If the prelude completes, all omitted candidates have upper bounds at most $G$, so its answer is exact. Otherwise the shared index computes an exact answer independently of the unfinished prelude. Prelude work, including any last charged batch, is $O(L \log(L + 1))$. Thus the adaptive query has $O(s + L \log(L + 1))$ dictionary/word operations and $O(L)$ auxiliary storage. The fixed four-slot and coefficient-two thresholds are part of this bound; allowing a caller to scale them with the input would require charging those parameters explicitly.

The low-level `overlap_move` defaults to `backend='adaptive'`; `'pairwise'` retains the maintained former query, and `'joint'` selects the shared index directly. Optional statistics distinguish fixed-slot queries, completed preludes and shared-index continuations. The native group producer still invokes this search only where relator moves are enabled and its existing control flow calls for an explicit overlap query. Compressed word queries keep their separate representation and search bounds.

## 82.4 Independent replay and the general complexity gap

Every proposed move is checked literally before producer application. The established explicit and compressed certificate replayers independently reconstruct and check the full presentation trace; neither calls the overlap search or its application helper. A valid completed trace still reduces the full source-bound knot-group presentation to one generator and no relators. The new index is a search accelerator, not an additional trusted certificate field. Different valid tied moves may cause later search trajectories to differ, so maximum-gain agreement alone is not a whole-recognition timing or completeness result.

A deterministic balanced-map version would give $O(s + L \log(L + 1))$ comparison operations for the shared index and a conservative $O(s + L \log^2(L + 1))$ bound for this adaptive scheduling policy: the bounded prelude counts dictionary operations, each requiring logarithmically many comparisons in that variant. With $\beta = \lceil \log_2(L + 2) \rceil$ and $b$ bounding generator and slot identifier bit lengths, a conservative adaptive bit bound is $O((s + 1 + L\beta^2)(b + \beta))$. This describes the balanced-map variant, not the implemented Python dictionary as an unconditional deterministic bit-cost theorem.

If a producer begins with $r$ generators and explicit relator length never exceeds $B$, at most $r-1$ eliminations occur. Between eliminations each accepted Whitehead or overlap move strictly decreases integer length, so there are at most $B$ such moves. Including failed final queries, overlap searches contribute at most

$$O(r(B+1)(s+B\log(B+1)))$$

dictionary operations. This charges only those searches. Neither the explicit-length bound $B$ nor successful simplification is established uniformly for all knot diagrams. The new local bound therefore improves one contribution without proving general quasi-polynomial recognition. A shared index whose size is controlled by a compressed grammar, and a complete source-bound search with controlled representation growth, remain separate research problems.

## 82.5 Native reconciliation and measured evidence

The incoming index and overlap module were checked against their hashes in the delivered benchmark record. The native query retains the shared capped index, but its prelude incorporates the maintained full-donor and sign cutoffs. The pinned previous overlap module is executed directly; the group producer, explicit and compressed checker hosts, recognizer and diagram modules are byte-identical to that baseline. The delivered report modules are loaded in their own package for comparison, rather than being treated as the maintained baseline.

Ten focused tests pass in 10.340 seconds. They include all 7,056 ordered pairs of rank-two words of lengths one through three, including unreduced words, against independent enumeration of every rotation and donor sign. Another 350 random multislot cases use that oracle; 600 larger random cases include large integer identifiers and compare maximum gain with the old query. Two thousand additional cases check completed longest-first preludes against historical maximum gain and literal application. Periodic clipping, inverse windows, duplicate slots, empty slots, global cancellation and exact shared work ceilings are covered. A repeated full-donor regression requires exactly one automaton and one target scan, preventing loss of the native early cutoffs. The complete Gordian proof and mutations of its move fields exercise both independent replay backends.

The first full-suite run exposed a historical polling assumption: its pairwise regression requested cancellation on the hundredth skipped-donor callback, but the new sorted prelude had already completed. The original assertions are retained and now explicitly select the unchanged pairwise backend. Adaptive global cancellation and exact resource ceilings remain covered separately. The initial full-suite log is archived; no production algorithm was changed to satisfy that callback-count assumption.

The final full maintained suite passes 958 tests with Regina in 217.815 seconds.

A native nineteen-diagram audit takes 2.754 seconds. The corpus contains fifteen unknot examples and four known nontrivial examples. For each diagram, old, native, report and joint-only group production are followed by both explicit and compressed replay whenever a proof is produced. There are fifteen successful certificates per arm; the other four group searches remain inconclusive. All nineteen native outputs, including absent proofs, match the maintained previous implementation exactly on this corpus. This is an observed compatibility result, not a promise that all tied witnesses are unchanged. Source diagrams, complete proofs, residual queries and policy statistics are in `data/joint-overlap-audit.json`.

The first native experiment retained original donor order inside the bounded prelude. It passed the exact-query checks but regressed on the Gordian residual: old/current median times were 144.083/189.582 ms, with paired ratio 0.765. The delivered longest-first prelude was faster. The final policy (`bbfc40517`) adopts that order and keeps the current early cutoffs. Commit `a2af739a3` records the initial policy; its measurements are in `../fast/results/joint_overlap_initial_20261008.json`.

| Query | old ms | native ms | report ms | joint ms | ratio |
|---|---|---|---|---|---|
| Gordian residual | 239.868 | 128.794 | 105.250 | 579.215 | 1.838 |
| random $8 \times 64$ | 12.979 | 17.161 | 28.181 | 12.333 | 0.757 |
| random $32 \times 64$ | 209.994 | 75.992 | 72.125 | 47.579 | 2.701 |
| random $128 \times 64$ | 4267.267 | 560.816 | 479.754 | 252.181 | 8.978 |
| periodic duplicates | 2.530 | 3.274 | 6.410 | 366.506 | 0.776 |
| full-donor duplicates | 0.376 | 0.753 | 0.962 | 274.179 | 0.687 |
| many empty slots | 0.730 | 1.538 | 0.972 | 0.689 | 0.520 |

The final isolated-query run has seven inputs, six arms, one excluded warmup and five measured rounds: 210 measured queries and 42 warmups. The arms are the actual maintained baseline, an identical baseline control, native adaptive search, an identical native control, the delivered report's adaptive query, and the shared index alone. Input generation is outside the timer; all preparation, failed prelude work, index construction and exact search are inside. Optional query statistics are collected inside these calls. The 81.899-second run retains raw words, moves, work counts, orders, times and source hashes in `../fast/results/joint_overlap_kernels_20261008.json`.

On the Gordian residual, native charged work is 146,504 instead of 304,005; the prelude completes after four oriented automata and four target scans, with exact maximum gain 2,998. The isolated paired time ratio is 1.838. These isolated elapsed measurements are noisy: identical-old ratios across the seven inputs range from 0.880 to 1.262, and identical-native ratios from 0.961 to 1.049. In particular, separate arm medians need not give the median paired ratio. The exact reduction in charged work and the following whole-recognition study provide distinct evidence.

For random presentations with 32 and 128 relators of length 64, the native query enters shared-index continuation. Charged work changes from 514,282 to 122,316 and from 8,076,415 to 524,622, respectively. Their paired elapsed ratios are 2.701 and 8.978. These are synthetic query-capacity examples, not hard knots or evidence of a complete unknot search bound. The eight-relator random case is slower, with paired ratio 0.757; its failed prelude costs more time than the eventual saved pairwise scans.

Repeated-word cases also retain regressions. Periodic duplicates have paired old/native ratio 0.776, and full-donor duplicates have ratio 0.687. Sorting and allowance accounting add work when the old query already stops almost immediately. The native cutoffs nevertheless halve the report's work on the periodic case, from 12,612 to 6,408, and reduce its full-donor work from 3,324 to 2,180. The many-empty-slot case has ratio 0.520; both algorithms scan the original slots, and the wrapper adds another pass and charges those slots explicitly. These cases rule out any universal speedup claim.

| Input | explicit old/new ms | ratio | compressed old/new ms | ratio |
|---|---|---|---|---|
| survivor-00 | 4.886/5.764 | 1.025 | 17.949/18.014 | 0.986 |
| mirror-03 | 6.225/6.725 | 0.923 | 24.963/24.693 | 1.003 |
| mirror-08 | 5.897/5.943 | 0.996 | 20.305/20.080 | 1.000 |
| gordian | 1041.096/911.653 | 1.238 | 1762.236/1694.382 | 1.120 |
| trefoil | 0.058/0.058 | 1.005 | 0.057/0.057 | 0.996 |
| conway | 1.018/1.083 | 0.940 | 1.449/1.123 | 0.988 |
| kinoshita_terasaka | 1.089/1.074 | 1.014 | 1.079/1.083 | 0.997 |

The whole-recognition run uses all nineteen actual diagrams in explicit and compressed search modes, with the same six arms and five measured rounds: 1,140 measured calls and 228 excluded warmups. All 1,140 calls complete with the expected knot verdict. Every timed call starts from fresh PD validation and includes all recognition stages and mandatory certificate replay. The query wrappers collect policy statistics inside the timer; corpus loading, policy binding and result serialization are outside. The 117.363-second run retains every diagram, certificate, stage record

and sample in `../fast/results/joint_overlap_pipeline_20261008.json`. Timeouts would be retained and excluded from completed medians and paired comparisons; none occurred in this run.

For Gordian with explicit search, the old/native medians are 1,041.096/911.653 ms and the median paired ratio is 1.238. Its identical-old and identical-native controls give ratios 1.003 and 1.038. Shared-index-only recognition takes 1,441.976 ms, so the index alone is a whole-call regression on this example. The adaptive prelude is essential to its practical result. With compressed search enabled, Gordian's old/native medians are 1,762.236/1,694.382 ms and its paired ratio is 1.120; the corresponding control ratios are 1.015 and 1.030. This is a measured effect of the explicit query reached by that mode, not a shared fully compressed overlap theorem.

Across the complete corpus, identical-old ratios range from 0.969 to 1.044, and identical-native ratios from 0.965 to 1.076. The small mirror examples and sources which do not reach overlap search supply controls rather than large gains. Mirror 03 with explicit search has paired ratio 0.923, while its identical-native ratio is 1.076; this run does not isolate a small algorithmic regression from timing variation. Mirror 08 is close to unchanged. All raw results, including those unfavorable to the new policy, remain in the archive.

# 83 Adaptive boundary classification of a supplied normal surface

Report 47 supplies a useful refinement of normal component counting: distinguish closed components from components meeting the ambient boundary, and separate each class by orientability. We now integrate this refinement with the maintained finite-manifold validator, edge-point interval model, common-multiplicity reduction and independent certificate replay. The public option `classify_boundary=True` returns these additional counts. It defaults to false, preserving the previous result structure and version-one/version-two proofs.

This is a local normal-surface operation. The input is still a supplied finite triangulation and an admissible integral normal vector. No triangulation-to-diagram provenance or normal-vector search is added. In particular, separating closed and boundary-touching component counts does not give the Euler characteristic of each component or decide whether a disconnected supplied surface contains an essential disc. The existing `compressing_disk` predicate still requires the entire supplied surface to be connected, with Euler characteristic one, one boundary circle and nonzero mod-two boundary class.

## 83.1 Mark boundary edges in the maintained orbit universe

Let $Q$ denote the normal surface after dividing out any common coordinate factor. The maintained surface interval system has one point for each intersection with an ambient edge, ordered in blocks by canonical edge representative. Its orbits are precisely the connected components of $Q$. Let $A$ be the union of blocks belonging to ambient boundary edges. A normal surface component meets the ambient boundary exactly when its orbit meets $A$: every nonempty normal boundary curve contains boundary-edge points, and such a point lies on the boundary of the proper normal surface.

Empty blocks are omitted and adjacent marked blocks are merged. There are $O(t)$ blocks for $t$ tetrahedra, with binary endpoints; no list of normal discs or individual edge intersections is created. For the doubled vector $2Q$, each edge block doubles in length, so a marked interval $[a, b)$ becomes $[2a, 2b)$. This detail depends on the maintained coordinate ordering. Appending a second copy of $A$ shifted by the old total universe length would instead describe a different ordering, and is not the construction used here.

For nonempty $A$, add the existing interval-incidence cone relations: translations identify all points

within each marked interval, and a star joins one representative from every marked interval. These relations merge all original orbits touching $A$ into one, and leave every disjoint orbit unchanged. If $C$ is the original orbit count, $C_A$ the count after coning, and $T$ the number of touched components, then

$$C_A = C - T + 1, \qquad T = C - C_A + 1.$$

This is a counting reduction, not an operation on the actual embedded surface. It adds $O(t)$ interval pairings and retains binary endpoints. The empty marked set has $T = 0$ without a cone query.

### 83.2 Orientation covers refine the boundary partition

Write $C_2$ for the number of components of $2Q$, and $T_2$ for the number of its components meeting the ambient boundary. Since the supported ambient manifold is orientable, doubling normal coordinates gives the normal orientation cover. An orientable component has two lifts; a nonorientable component has one connected orientable lift. Boundary presence is preserved by this covering. Therefore, with subscripts $b$ denoting components with boundary,

$$T = O_b + N_b, \qquad T_2 = 2O_b + N_b, \qquad O_b = T_2 - T, \quad N_b = 2T - T_2.$$

The already verified total counts $O = C_2 - C$ and $N = 2C - C_2$ then give closed counts $O_c = O - O_b$ and $N_c = N - N_b$. The implementation checks nonnegativity and containment in the total orientation classes. If $b$ is the number of boundary circles of $Q$, a nonempty boundary also requires $1 \le T \le \min(C, b)$.

For an original vector $X = gQ$, the sheet-pairing argument of Section 79 applies independently to components with and without boundary. Thus

$$O_b(X) = gO_b + \lfloor g/2 \rfloor N_b, \qquad N_b(X) = (g \bmod 2)N_b.$$

All reported counts concern $X$. The original source fingerprint, Euler characteristic, normal-disc count and boundary-cohomology witness continue to concern $X$ as well.

An even multiplier yields a further useful deduction. If $g$ is even, $X$ consists of $g/2$ copies of the orientable normal surface $2Q$. Consequently

$$O_b(X) = (g/2)T_2, \qquad N_b(X) = 0.$$

Computing $T$ is unnecessary in this case. In particular, for a primitive surface with both orientation types the algorithm can omit the cone query on $Q$, compute only $T_2$, and still determine all six requested original counts. It does not claim to have recovered the unknown partition of $Q$.

### 83.3 Choose queries from already established progress

The three base queries still compute $C$, $C_2$ and $b$. The adaptive policy then uses exact deductions before scheduling any further query:

- If $b = 0$, both touched counts are zero. If $C = 1$ and $b > 0$, they are $T = 1$ and $T_2 = C_2$.
- If the marked union is the entire edge-point universe, every component touches the boundary: $T = C$ and $T_2 = C_2$.
- If $b = 1$, exactly one component touches the boundary, so $T = 1$.
- If $C_2 = 2C$, every component is orientable and $T_2 = 2T$. If $C_2 = C$, every component is nonorientable and $T_2 = T$. Either case needs at most one cone query.
- If both orientation types occur and $g$ is even, compute only the doubled cone count when the preceding deductions do not suffice. Otherwise compute any still-needed counts on $Q$ and $2Q$.

Thus the feature uses zero, one or two additional orbit queries. This is an adaptive reduction in the information that must be computed, rather than a heuristic choice of an unverified result. A private research ablation requests both cones whenever the boundary is nonempty; it is not a new public option.

Each cone system still has $O(t)$ pairings and logarithmic dependence on the edge-point universe size. At most five interval-orbit queries therefore preserve the polynomial local bound in the encoded triangulation and normal coordinates. No quasipolynomial bound for finding a useful normal vector follows from this operation. The gains measured below concern complete calls on supplied vectors, including validation, rather than repeated queries on a prevalidated geometry object.

## 83.4 Proof version three and independent replay

Opting into classification with certificate recording produces `normal-surface-topology-v3`. It binds the full original input, records an exact coordinate divisor $g \geq 1$, retains all three base orbit proofs and the original boundary-cohomology witness, and includes zero to two cone proofs. The claimed topology includes the six new fields: the two boundary-presence totals and their four orientability subclasses. The empty surface uses $g = 1$ and six zero counts. As before, the checker accepts any valid common divisor, not only the producer's gcd.

The checker validates the original source, divides coordinates exactly, reconstructs both interval systems and the marked blocks, and replays every supplied cone proof. It derives omitted counts directly from verified base counts and the reconstructed marked support. It does not call the producer's query planner, orbit search or gcd selection. Redundant valid cone proofs are accepted and checked. Pure orientability permits inference in either direction, so a sound alternate proof choice also verifies. A proof containing only the doubled touched count can suffice for even $g$; changing that divisor to an odd value without supplying the missing information is rejected.

The producer's `max_cycles` allowance is shared across the base and all additional queries. Exhaustion during a cone query returns an inconclusive result with query diagnostics, without partial topology claims or a certificate. The verifier's `max_operations` counts all supplied orbit events before replay, including redundant cone proofs. Cancellation callbacks propagate through both routes. Version-one and version-two formats retain their earlier meanings; the older checker is not claimed to understand version three.

## 83.5 Independent geometry audit and measurements

The pinned baseline is commit `ce180ce1e640`, loaded directly from Git. The native audit checks 548 supplied surfaces against both Regina 7.4 component decomposition and report 47's delivered component implementation. Every new count agrees; all default results and certificates equal the actual maintained baseline exactly, with default proofs replayed in both versions. Direct and adaptive classification agree and independently verify on every input. The audit includes 404 three-query cases, 95 four-query cases and 49 five-query cases.

The corpus reuses 270 layered-solid-torus and boundary-cap inputs, adds all 64 small combinations of an interior vertex sphere, a boundary vertex disc and a Möbius band in a fixed four-tetrahedron subdivision, and includes vertex normal surfaces and compatible combinations in a nine-tetrahedron solid-torus connected sum with $\mathbb{RP}^3$. That final manifold is deliberately not a knot exterior: it supplies a closed one-sided projective plane for validating the general local classifier, without asserting nonexistent knot-diagram provenance. The audit contains 74 inputs with closed orientable components, 15 with closed nonorientable components, 473 with orientable components meeting the boundary and 98 with nonorientable components meeting the boundary. These classes overlap. Raw data: `data/normal-boundary-native-audit.json`.

Eight focused tests additionally exercise 5,001-bit multiplicities, proper common divisors, the unknown primitive partition in the even case, missing and forged cone proofs, mutated topology claims, shared exhaustion and replay limits, caller cancellation and empty inputs. Replay is also tested with the producer's planner, gcd routine and orbit search disabled. The full maintained suite passes 966 tests with Regina in 213.279 seconds.

| Input | queries D/A | direct ms | adaptive ms | count ratio | proof ratio |
|---|---|---|---|---|---|
| meridian8 | 5/3 | 6.151 | 4.970 | 1.222 | 1.238 |
| meridian32 | 5/3 | 75.565 | 69.700 | 1.088 | 1.049 |
| meridian96 | 5/3 | 1279.093 | 1274.040 | 0.999 | 0.971 |
| parallel 5000 | 5/3 | 6.251 | 5.122 | 1.210 | 1.242 |
| closed | 3/3 | 0.698 | 0.696 | 1.003 | 1.030 |
| one boundary | 5/4 | 1.811 | 1.530 | 1.185 | 1.149 |
| pure orientable | 5/4 | 5.637 | 4.236 | 1.338 | 1.279 |
| mixed odd | 5/5 | 5.559 | 5.521 | 0.997 | 1.110 |
| mixed even | 5/4 | 5.550 | 4.435 | 1.259 | 1.243 |
| even 5000 | 5/4 | 7.536 | 7.700 | 1.245 | 1.211 |
| odd 5000 | 5/5 | 5.771 | 5.749 | 0.989 | 0.986 |
| closed one-sided | 5/5 | 13.854 | 14.040 | 0.994 | 0.998 |

The paired benchmark has twelve supplied-vector inputs, eight arms, five measured rounds and one excluded warmup: 1,920 measured complete calls and 384 warmup calls. Some small cases are batched. Direct and adaptive arms answer the same six-field classification question, with common multiplicity reduction enabled on both sides. Separate certified arms include proof production and fresh independent replay. Identical direct and adaptive controls expose timing variation. Two further arms compare the actual old and new default operation, which answers a smaller question and is not used as the classification speedup baseline. Source preparation, manifold validation, coordinate validation and all queries are inside each timer; policy binding, serialization and fixture loading are outside. The 99.926-second run retains every input, shuffled arm order, sample, query statistic, event count and source hash in `../fast/results/normal_boundary_20261008.json`.

For the eight-tetrahedron meridian, the query count falls from five to three and the paired count ratio is 1.222. Production plus replay has ratio 1.238, with median times 10.121 and 8.263 ms. The compact proof falls from 33,609 to 20,613 bytes and from 501 to 278 orbit events. The 32-tetrahedron input has a smaller count ratio, 1.088. At 96 tetrahedra, the three mandatory queries dominate: the count ratio is 0.999 and the certified ratio is 0.971, despite reducing orbit events from 26,591 to 24,038. This run establishes no elapsed improvement for that large case; its identical-control ratios are 0.959 and 0.978.

On the four-tetrahedron mixture of a sphere, a boundary disc and a Möbius band, odd primitive multiplicity needs both cones. Doubling that vector permits the new even shortcut: five queries become four, 55 cycles become 44, and 351 events become 270. The proof decreases from 23,322 to 18,080 bytes. The paired count and certified ratios are 1.259 and 1.243. The purely orientable sphere-plus-two-discs input also needs only one cone, with ratios 1.338 and 1.279. A sphere plus a single Möbius band already has $b = 1$, eliminating the original-surface cone and yielding ratios 1.185 and 1.149.

Large binary multiplicities retain the same query savings after gcd reduction. The even $2^{5000}$ mixed input has paired count ratio 1.245 and certified ratio 1.211. Its separate count medians are 7.536 and 7.700 ms: the median of paired ratios need not equal a ratio of separate medians, particularly with timing shifts between rounds. These raw data are retained without selecting favorable rounds. The corresponding odd $2^{5000} + 1$ case still requires both cones and has ratios 0.989 and 0.986. The closed one-sided mixed input likewise requires both cones and has ratios 0.994 and 0.998. The closed-only input requires no cone under either policy.

Across all inputs the identical direct ratios range from 0.959 to 1.031 and the identical adaptive

ratios from 0.978 to 1.008. The smaller default operation has old/new ratios from 0.986 to 1.023. The primitive mixed-odd case has identical query counts and proof size yet a certified ratio of 1.110; this is timing variation, not evidence of a saved query. Exact query, cycle and proof-size reductions support the adaptive mechanism; the elapsed measurements support modest gains on selected supplied surfaces, not a universal speedup or a general recognition bound.

# 84 A source-bound primitive-power terminal for compressed knot groups

Report 45's arithmetic kernel was previously reproduced but deliberately kept outside the knot-verdict path. The missing step was a terminal contract: the old compressed search succeeded only after reaching one generator and no nonempty relators. A primitive-power relation at rank two does not meet that syntactic condition. We now add a distinct version-five terminal, replay its entire presentation prefix, and independently verify its arithmetic evidence on the reconstructed relator. This adds a positive stopping rule to the existing optional compressed group route.

## 84.1 The minimum-width arithmetic test

For a nonempty positive two-letter word, let $P, Q$ count its letters, $d = \gcd(P, Q)$, and $(p, q) = (P/d, Q/d)$. The prefix-height walk assigns weight $q$ to the first letter and $-p$ to the second. Its total height is zero. Write $\Omega$ for the difference between its maximum and minimum prefix heights, including the empty prefix.

When both counts are positive, each step changes height by $q$ modulo $\ell = p + q$. Coprimality makes the first $\ell$ prefixes visit all residues, so $\Omega \geq \ell - 1$. If equality holds, all heights lie in an interval of exactly $\ell$ consecutive integers. The two possible next heights, $h + q$ and $h - p$, differ by $\ell$, and exactly one remains in that interval. Thus the next letter is forced by the present height. The walk repeats the same length-$\ell$ rotation cycle $d$ times. That cycle is a cyclic conjugate of the Christoffel word with counts $(p, q)$.

Euclidean division of $(p, q)$ expresses its Christoffel word through Nielsen maps $a \mapsto a$, $b \mapsto a^k b$, or $a \mapsto ab^k$, $b \mapsto b$, until one generator remains. Hence the cycle is primitive in $F(a, b)$. Conversely, the rank-two primitive conjugacy classification used in report 45 identifies every cyclically reduced primitive word, after independent generator inversions, with such a cycle. Therefore a nonempty cyclically reduced word is a power of a primitive element exactly when each generator occurs with only one sign and its normalized width is $|u| + |v| - 1$. Here

$$(A, B) = d(u, v), \qquad d = \gcd(|A|, |B|),$$

where $A, B$ are signed exponent sums. The signed-letter weights are $(v, -v, -u, u)$ on $(a, a^{-1}, b, b^{-1})$. A one-letter power has width zero and primitive vector $(\pm 1, 0)$ or $(0, \pm 1)$, so it is included.

For a binary grammar, prefix profiles form a small exact concatenation summary. If $U$ has total, minimum and maximum $(s, l, h)$ and $V$ has $(t, L, H)$, then

$$\text{profile}(UV) = (s + t, \min(l, s + L), \max(h, s + H)).$$

Four signed occurrence counts also add at concatenations. These recurrences use binary integers and never enumerate represented letters. The native query uses the maintained `WordArena` directly, preserving its original generator names and relator slots. It creates no word nodes and calls no equality, longest-prefix or normalization routine.

## 84.2 One shared slope for the intact rank-two state

At a two-generator presentation of a knot group, the relator exponent rows span one rational line: the abelianization is $\mathbb{Z}$. The producer computes a primitive direction $(u, v)$ from the first nonzero row, checks collinearity of every row by $vA - uB = 0$, and performs one shared profile pass over the reachable grammar. It keeps every root slot, including empty and repeated roots. A nonzero, sign-coherent root with the required width supplies the terminal witness. Its exponent and signed vector are derived from its own counts, so inverse-oriented roots are handled correctly.

All sign-incoherent roots are skipped. This makes the positive test safe even if a caller provides a raw grammar: sign coherence itself implies free and cyclic reduction, since inverse letters cannot occur. Absence of a hit on an arbitrary raw root is not a complete primitivity decision; normalizing a general compressed input remains a separate cost. On the cyclically reduced states maintained by the knot search, the report's criterion applies directly. Noncollinear input is outside the shared-slope query's supported family and yields no terminal evidence.

The query runs when the compressed search reaches two live generators, before building a general elimination plan or searching for further Whitehead and overlap moves. This position matters in the actual corpus: a prototype placed after singleton detection had no positive activations, because all tested successful states already contained an eliminable relation. Moving the exact stopping rule before that work permits avoiding the final substitution and normalization across all relator slots. The measurements below determine the resulting whole-call effect.

## 84.3 Why the source-bound terminal proves an unknot

The certificate checker begins with a validated classical one-component planar diagram. It independently reconstructs the full Wirtinger presentation and replays every elimination, Whitehead and relator move in the supplied prefix. Those moves preserve the presented group, and no unaccounted relator slot is discarded. Suppose that the resulting presentation has exactly two live generators and contains a relator $w^d$, with $d > 0$ and $w$ primitive.

A classical knot group is torsion-free, so $w^d = 1$ implies $w = 1$ in that group. Since $w$ belongs to a free basis of the rank-two free group, its normal quotient is infinite cyclic. The full presented group is therefore a quotient of a cyclic group. Its abelianization is still $\mathbb{Z}$, so it must itself be infinite cyclic. The classical knot-group unknot criterion then implies that the original diagram represents the unknot. For exponent one the root-killing step does not require torsion-freeness. These are imported topological facts, as in the report's source-bound argument, not new claims about arbitrary presentations.

Other relator slots need not be individually shown redundant after this terminal is established, but all of them must have been retained in the verified presentation prefix. The distinction is essential: the abstract presentation $\langle a, b \mid a^d \rangle$ for $d > 1$ also has a local primitive-power witness and is not an infinite cyclic knot group. Neither the arithmetic producer nor its arithmetic checker alone returns a knot verdict or accepts a caller's torsion-free flag.

## 84.4 Version five, independent replay and bounded execution

A successful terminal proof retains the usual method, status, original PD and move prefix. Version five replaces `remaining_generator` by `terminal`, containing its kind, original current relation slot, ordered live generator pair, two signed primitive coordinates, positive exponent and width. The outer checker rejects extra or missing fields, unknown versions and mismatched diagrams. Versions one through four continue to require the original rank-one terminal.

After independent prefix replay, the explicit checker walks the selected literal relator and recomputes signed counts and prefix extrema. The compressed checker instead performs a fresh postorder

traversal of that root's grammar, distinct from the producer's sorted reachable-node scan. Both check exactly two live generators, the selected slot, strict integer types, primitive gcd, exponent products, sign coherence and minimum width. The proof's numerical fields are bounded by the reconstructed word length before gcd or profile arithmetic, so forged oversized metadata cannot inflate the replay calculation. Neither route calls the query producer, selects a gcd direction for search, or runs Whitehead discovery.

Every arithmetic pass charges the existing work budget and propagates external cancellation. The kernel allocates no new SLP nodes; its temporary summaries are proportional to the reachable grammar, already bounded by the enclosing node cap. These are cooperative abstract allowances, not hard bounds on Python integer-instruction time or process memory. Search exhaustion remains inconclusive. An unaffordable explicit replay can fail its resource allowance even where compressed replay succeeds; it never silently expands beyond that allowance.

The low-level `compressed_certificate` option `primitive_power` defaults to true and can be disabled for the old search path. The ordinary PD dispatcher still requires the existing compressed-group opt-in. The explicit search and its representation-switching handoff retain their old terminal contract. Internally, the compressed search enables rank-two success only when its caller supplies a distinct terminal-output record; the old call form still means one generator and no relators. Thus an abstract internal success cannot accidentally be encoded with a fabricated remaining generator.

## 84.5 Local cost and what remains unproved

Let $S$ count reachable grammar nodes, $M$ all allocated node identifiers, $m$ current relator slots and $B$ the largest represented-length bit size. The native query uses $O(S\log(S+1)+m)$ structural operations and a constant number of additions/comparisons per profile node, plus gcd and collinearity arithmetic. The sorting term comes from the maintained reachable-node ordering; it is absent from the report's already ordered standalone grammar scan. Prefix heights have $O(B)$ bits, since their absolute value is at most the product of word length and a slope component.

With conventional dictionary access, a conservative arithmetic/identifier bit accounting is

$$O\big(SB + S\log(S+1)\log(M+1) + Sb + m(B^2+\log(M+1))\big),$$

where $b$ bounds generator-label encoding. The quadratic term permits ordinary integer multiplication and gcd bounds. Replay needs only the selected root's postorder summaries and the claimed primitive vector; it does not repeat the all-relator slope search. Python hash tables retain the usual implementation-model qualification; these counts are not an unconditional adversarial dictionary-time theorem.

This polynomial local test removes a potentially long sequence of normalizations or substitutions once a suitable rank-two state exists. It bounds neither the cost of reaching that state nor aggregate grammar growth during the preceding presentation search. Section 85 subsequently supplies the complete independently replayed higher-rank projection contract from report 45. A controlled short-depth schedule remains unproved. No general quasipolynomial unknot-recognition bound follows from the new terminal.

## 84.6 Native audits and whole-recognition evidence

The pinned maintained baseline is `cd77d1bee8fa`. Its actual group host, compressed search and compressed replay are loaded in a separate package. Relevant downstream modules are checked byte-for-byte against that baseline. The algebraic audit compares all 29,540 cyclically reduced words through length nine with the independent literal KMP-root/Whitehead oracle, report

45's delivered arithmetic producer and checker, and native arithmetic replay. All results agree, including 844 primitive-power positives.

The source audit checks 74 actual diagrams: the nineteen-case research corpus and 55 valid single-component braids from a fixed 240-attempt random sample. Disabling the new terminal yields exactly the old producer's certificate or nondecision on every input. Every new positive certificate passes both full source-reconstructing replay paths. Any newly closed state would additionally be checked with the complete independent recognizer; none occurred in this sample. Recorded data do not treat a failed positive group search as proof of knottedness.

Eight focused tests cover short exhaustive words, 300 random parses, large generator identifiers, repeated and empty slots, noncollinear and sign-incoherent inputs, forged numerical fields, schema/source mutations, caller cancellation and shared exhaustion. A nonuniform Fibonacci grammar raised to exponent $2^{5000} + 1$ passes query and replay with expansion, equality and longest-prefix methods disabled. A separate actual-diagram proof has a powered Whitehead prefix of exponent $2^{1024}$; compressed source replay succeeds with query discovery and expansion disabled, while explicit replay correctly exceeds its finite allowance. These are capacity and verifier tests, not difficult-diagram speedup claims. The full maintained suite passes 974 tests with Regina in 210.163 seconds. The diagram audit contains 44 version-five terminal hits, all with exponent one. There are no observed proper-power hits and no newly solved diagrams. The more general torsion-free rule is therefore supported by its proof and algebraic tests, not by a claim of proper-power activity in this corpus. Raw audit results are in `data/primitive-power-native-audit.json`.

| Input | old ms | new ms | ratio | old A/A | new A/A |
|---|---|---|---|---|---|
| survivor-00 | 16.892 | 16.588 | 1.015 | 0.974 | 0.992 |
| survivor-02 | 16.246 | 16.666 | 0.964 | 0.993 | 1.046 |
| mirror-03 | 24.088 | 24.403 | 0.985 | 0.994 | 0.986 |
| mirror-08 | 21.633 | 20.333 | 1.052 | 1.012 | 1.017 |
| survivor-11 | 7.344 | 7.061 | 1.035 | 1.007 | 1.005 |
| gordian | 1500.813 | 1512.295 | 0.998 | 0.979 | 1.006 |
| trefoil | 0.056 | 0.056 | 0.995 | 1.016 | 0.982 |
| conway | 1.017 | 1.118 | 0.979 | 0.980 | 1.046 |
| kinoshita_terasaka | 1.077 | 1.095 | 0.970 | 0.980 | 1.031 |

The complete-recognition benchmark uses nineteen actual diagrams, four shuffled arms, five measured rounds and one excluded warmup: 380 measured calls and 76 warmups. Every measured call completes with the expected verdict. Each starts with fresh PD validation and includes the compressed group search, source reconstruction and independent proof replay when that stage is reached. The old host and its old compressed search/replay modules are pinned together; the comparison does not silently run old search through the new terminal checker. Identical old and new arms measure timing variation. The 43.282-second run retains all inputs, certificates, orders, elapsed samples, charged work and arena statistics in `../fast/results/primitive_power_pipeline_20261008.json`.

The elapsed results are mostly close to unchanged. Survivor 00 has old/new medians 16.892/16.588 ms and paired ratio 1.015. Its charged search work falls from 17,998 to 17,879, but the terminal record makes its compact certificate 1,216 bytes instead of 1,160. Mirror 03 has ratio 0.985, with arena nodes decreasing from 617 to 615 and charged work from 25,558 to 25,431. Mirror 08 has ratio 1.052; this is a small measured improvement, not the large literal-baseline ratio advertised for the standalone delivered arithmetic kernel.

On Gordian, search work decreases from 1,215,916 to 1,215,145, and one longest-prefix call is avoided. The arena still has 9,025 nodes and largest word bit length thirteen. Certificate size increases from 9,607 to 9,663 bytes. Whole-call medians are 1,500.813 and 1,512.295 ms, with paired ratio 0.998; no speedup is established. The expensive prefix leading to rank two dominates this input. The new certificate replaces one final elimination; it does not accelerate that prefix.

The old identical-arm ratios across the corpus range from 0.974 to 1.016; the new controls range from 0.978 to 1.046. Survivor 02 has paired ratio 0.964 and new-control ratio 1.046. Nontrivial-knot examples resolved before the group stage also exhibit small variations. These controls preclude interpreting every few-percent difference as an algorithmic effect. Arena node counts and exact certificate bytes are retained as storage measures; process peak memory was not measured. No result is censored in this run, but the driver retains incomplete calls and excludes them from completed medians and paired ratios.

This integration supplies a sound compressed stopping rule and removes some exact work at the observed terminal states. Its current whole-diagram timing benefit is small or absent. The next research question is whether certified transformations expose primitive-power relators earlier, or support independently replayed higher-rank pair contractions with bounded aggregate grammar growth. Merely repeating a polynomial local query does not establish the required global bound.

# 85 Independently replayed disjoint primitive-pair projections

The version-five terminal of Section 84 uses a primitive-power relation only after the whole presentation reaches rank two. Report 45 also proposes contracting disjoint generator pairs inside a larger presentation, while retaining raw word circuits between rounds. We now implement that operation, including an independent checker for every complete round. This closes the report's whole-round replay gap. It does not supply the missing short-depth producer for arbitrary diagrams. The new search policy is explicitly optional: ordinary-corpus measurements below show overhead rather than a practical speedup.

## 85.1 The quotient map and the invariant preserved by a round

Let a source-reconstructed knot-group presentation have free generators $X$, and let one current relation be $w^d$, where $w$ is primitive in the free factor $F(a,b)$, $d>0$, and its exponent vector $(u,v)$ has both coordinates nonzero. Knot-group torsion-freeness implies that adjoining $w=1$ preserves the presented group. Because $w$ belongs to a free basis of $F(a,b)$, the quotient by its normal closure is infinite cyclic. The primitive character

$$a \longmapsto t^{v}, \qquad b \longmapsto t^{-u}$$

is surjective and kills $w$. It therefore realizes that cyclic quotient: $\gcd(|u|,|v|)=1$, and a surjection between two infinite cyclic groups is an isomorphism. The remaining free generators are unaffected. Substituting this character into *every* relator gives a presentation of the same knot group, with one fewer generator.

The implementation orders $a<b$, reuses the name $a$ for $t$, and orients the character so that the image of the old $a$ is positive:

$$a \longmapsto a^{|v|}, \qquad b \longmapsto a^{-u\,\operatorname{sign}(v)}.$$

Here the $a$ on the right is the new quotient generator; the substitution is simultaneous, so this image is not recursively substituted into itself. Negative letters receive inverse images. The new generator need not be a meridian, and no later step assumes that it is one.

For pairwise disjoint pairs, these operations act on different free factors. The checker validates all donor witnesses against the *pre-round* presentation, constructs all images, and performs one simultaneous substitution across all relator roots. It then deletes the second generator of each pair. A chosen donor becomes a word in its own new cyclic generator with exponent $du|v| - dv\,u\operatorname{sign}(v) = 0$, so that donor's original slot can be replaced by the empty word. All other slots are retained, including empty, repeated and unreduced roots. No claim that other relators are redundant is needed or accepted.

Each round thus preserves the original knot group. Torsion-freeness remains available in subsequent rounds because the group remains isomorphic to the source knot group. For exponent-one donors the local step needs no torsion-freeness hypothesis, but proper-power donors do. The public verdict path obtains this hypothesis from a validated classical knot diagram and complete prefix replay; an arbitrary presentation plus a caller-supplied flag cannot authorize it.

## 85.2 Raw circuits, cached eligibility and the normalization boundary

The producer scans relator slots in their original order and greedily chooses disjoint eligible pairs. Eligibility requires exactly two live generators, one sign per generator, and the minimum-width primitive-power witness established in the preceding section. Sign coherence implies free and cyclic reduction without performing normalization. Pure single-generator powers are excluded from a pair round. A root outside this sufficient raw class is skipped; its reduced representative might still be eligible. The greedy choice is deterministic but is not asserted to minimize depth.

Occurrence summaries are cached by immutable arena node identifier. A terminal contributes one signed occurrence, and a concatenation adds its children's counts. A support with more than two distinct generators is stored as an ineligible marker: later concatenation cannot remove letters from a raw circuit. The traversal stops at cached nodes, so each allocated node's capped support summary is constructed at most once over the search. Positive and negative arithmetic classification results are also cached by root. The selected relation slot is attached to a fresh witness copy; it is not cached with the immutable word, since repeated roots may occupy different slots. Normalization or substitution creates other roots and therefore cannot invalidate these word-content summaries.

A round builds binary power circuits for the images, then maps each reachable original node once using raw concatenation. It does not call compressed substitution helpers that automatically reduce words. Inverse letters and zero-exponent circuits may remain physically present after substitution. On reaching one live generator, an independent postorder sum verifies that every retained root uses that generator and has exponent zero. A word in one generator with zero exponent is the identity; the resulting presentation is exactly infinite cyclic. No expansion, equality query or cancellation search is required at this endpoint. The existing rank-two primitive-power terminal may also close a raw state, because its selected donor is checked for sign coherence independently.

When the next projection round cannot be found and more than one generator remains, the producer records an explicit `normalize_relators` move, cyclically reduces all roots, and retries. If no pair becomes eligible, the earlier singleton, Whitehead and overlap search resumes on the normalized state. This is an adaptive transition between representations, with an explicit proof boundary. The cost of that normalization and subsequent search is separate from the normalization-free round bound below. Even simple stabilized-circle inputs can alternate projections and normalizations under the present slot-order policy; a balanced schedule on an abstract family is not evidence that this policy discovers one on all diagram presentations.

## 85.3 Version six and independent bounded replay

Version six retains the source PD, method, status, complete move prefix and terminal record. A `primitive_projection` move contains exactly its kind and a nonempty list of primitive-power witnesses. Each witness identifies its original donor slot, ordered pair, primitive signed vector, positive exponent and width. Pairs must be disjoint subsets of the current live generators, donor slots must be distinct, and both coordinates must be nonzero. Every witness uses the strict arithmetic checks of version five, including bounds on claimed integers before expensive arithmetic. The checker accepts no extra root-dropping or torsion-freeness fields.

Literal replay constructs the images and substitutes into each literal word; compressed replay independently constructs signed image circuits and uses a fresh postorder mapping of each original root. Neither invokes the producer's planner, application routine, cache or rank-one query. All witnesses are checked before any state mutation. The final `rank_one_exponent_zero` record names exactly the one live generator, and both replay routes check every retained relator. A version-six proof can instead use the previous rank-two terminal after its checked prefix. Versions one through five retain their prior meaning.

The literal checker computes the entire projected letter count and checks its letter and work allowances *before* constructing repeated lists. The compressed checker constructs binary powers under the existing node and work budgets. Both propagate caller cancellation. Cached summaries and other Python allocations are auxiliary memory, so a grammar-node cap is not a hard process-memory bound. Work counters are cooperative structural allowances, not a claim that a Python big-integer operation has unit bit cost.

Both `compressed_certificate` and `group_decide` accept `primitive_projection=True`. The `recognize` API accepts `use_group=True` together with `group_primitive_projection=True`. The CLI flag is `-group-primitive-projection`. At the group host it implies compressed search and mandatory compressed replay; combining it with the older explicit-to-compressed adaptive-search flag is rejected. It defaults to false. Disabling projections reproduces the previous version-five producer exactly in the source audit. Disabling both primitive options retains the earlier rank-one search contract.

## 85.4 A bound for consecutive raw rounds, and its limitations

Let $r_0$ be the initial rank, $m$ the number of retained relator slots, $M_j$ all allocated grammar nodes after round $j$, and $B_j$ the largest bit length of any allocated word length, including image intermediates. Use $B_j \geq 1$ and let $d$ count consecutive raw projection rounds with no intervening normalization or legacy search. Every image exponent is at most a donor's word length. Thus substituting images increases a word's length by at most a factor of the previous maximum length. Binary power construction introduces at most $O(r_0(B_j + 1))$ image nodes, and mapping the old reachable circuit introduces at most one node per old node. Hence

$$B_{j+1} \leq 2B_j + O(1), \qquad M_{j+1} \leq 2M_j + O(r_0(B_j + 1)).$$

Consequently $B_j \leq O((B_0 + 1)2^j)$ and $M_j \leq 2^j O(M_0 + r_0(B_0 + 1)(j + 1))$. Counts and profile heights have $O(B_j)$ bits; generator and node identifiers have their usual encoding costs. There are at most $r_0 - 1$ pair contractions in the full run.

The maintained implementation uses a sorted reachable-node pass for raw application and for some arithmetic queries. It may classify several candidate roots separately, so its bound should not be confused with one shared linear profile scan over all donors. Nevertheless, scanning all $m$ roots, independently classifying each eligible candidate and replaying all chosen witnesses is polynomial in $M_j, B_j, m, r_0$ and identifier bit sizes. Ordinary quadratic integer multiplication/gcd bounds suffice; expected dictionary access is the same implementation-model qualification as in the preceding section. Composing any fixed such polynomial over $d$ rounds yields

$$\mathrm{poly}(M_0, B_0, m, r_0, \text{identifier sizes})\, 2^{O(d)}.$$

The independent raw replay and rank-one endpoint have the same form of bound. This statement does not count a general normalization as a raw round or give an unproved polynomial bound to the existing heuristic search.

Section 92 supersedes the depth-dependent raw-epoch estimate with a polynomial bound using the fixed source's saturated kernel. The short-depth condition below was sufficient in the original

analysis; it is no longer needed for an epoch consisting only of these raw moves. Endpoint discovery and preprocessing remain separate obligations.

If a controlled producer on encoded input size $g$ supplied a polynomial-size initial presentation, reached an accepted endpoint in $O((\log g)^2)$ raw rounds, and had quasipolynomial preprocessing cost, this part of recognition would be quasipolynomial. None of the short-depth, endpoint or preprocessing hypotheses is established for all diagrams. Rank merely decreases by at least one per round, which permits $d$ linear in $r_0$. Success of a greedy batch on a finite family is not a general subexponential algorithm.

## 85.5 Source audits, capacity checks and complete-call measurements

The maintained suite passes 981 tests in 229.282 seconds with Regina. Seven new test methods cover 120 random simultaneous-substitution comparisons, independent literal and compressed replay, cached failed queries and changing root slots, strict schemas and numerical forgeries, source rebinding, all-relator endpoint checks, actual-diagram proof prefixes, API/CLI behavior, cancellation and exhaustion. An initial prototype enabled projections by default and broke the old `primitive_power=False` behavior; the retained exploratory failure log records this. The final implementation keeps projections optional and passes the unchanged earlier terminal tests.

The pinned baseline is `e2cf6faa562b`. The research driver loads its entire Python package into a separate namespace, including the previous recognizer and diagram class. The 79-diagram audit consists of the earlier nineteen-case corpus, 55 valid single-component braids from the fixed 240-attempt sample, and five stabilized circles. Current default certificates and nondecisions equal the pinned baseline exactly. Enabling projections produces 34 version-six certificates containing 158 pair contractions; every positive passes both complete source-reconstructing replay paths. There are no newly solved inputs, lost positives or proper-power donors in that diagram sample. The audit takes 4.695 seconds including baseline setup.

Separate actual five-braid sources have verified powered Whitehead prefixes with exponents $2^4$, $2^{128}$ and $2^{1024}$. They test compressed source replay and the explicit normalization handoff. Separately, abstract balanced presentations use both $(a^{2^b}b)^2$ and $(a^{2^b}b)^3$ for each tree edge. The coprime powers imply the edge word is trivial, and leaf-to-root elimination proves these presentations are $\mathbb{Z}$ without assuming torsion-freeness. They are algebraic capacity fixtures, not diagram inputs. Query and independent replay complete with reduction, cyclic reduction, expansion and equality disabled. At rank 256 and $b = 32$, eight rounds contract 255 pairs; allocated nodes grow from 5,118 to 18,235, all 510 relator slots survive, and 255 remaining roots are nonempty raw circuits with zero exponent. This tests the intended representation contract rather than practical difficulty of an unknot diagram.

| Input | old ms | projection ms | ratio | old A/A | proj. A/A |
|---|---|---|---|---|---|
| survivor-00 | 17.196 | 18.394 | 0.938 | 0.991 | 1.017 |
| mirror-03 | 24.069 | 25.222 | 0.953 | 1.004 | 0.998 |
| survivor-06 | 7.759 | 8.231 | 0.927 | 0.996 | 1.001 |
| survivor-08 | 7.640 | 8.267 | 0.924 | 0.994 | 0.993 |
| gordian | 1989.772 | 1584.861 | 1.002 | 1.024 | 1.007 |
| trefoil | 0.106 | 0.105 | 1.004 | 0.994 | 0.985 |
| conway | 1.777 | 1.770 | 0.998 | 0.995 | 1.014 |

The complete-call benchmark uses nineteen fresh validated diagrams, five measured rounds and one warm-up per arm. Six shuffled arms compare the pinned baseline, current default and enabled projections, with separate A/A controls for all three. The timed call includes diagram validation, recognition, source reconstruction and mandatory independent proof replay. All 570 measured calls complete, with 114 additional warm-ups; the run takes 71.320 seconds. Table ratios are medians

of paired old/projection times, so values below one mean a slower projection call. They need not equal the ratio of the two separately reported time medians. Production, fixture and driver source hashes are frozen before and after execution. The baseline package is pinned independently; no adapter that bypasses a new option check is inserted into the timer.

On the ordinary survivor and mirror unknots, paired ratios range from 0.924 to 0.953. Survivor 06 uses one projection and one normalization; its search work rises from 8,670 to 9,197, nodes from 162 to 166, and certificate bytes from 848 to 989. Gordian uses two projection rounds, three pairs and two normalization handoffs. Its search work rises from 1,215,145 to 1,265,669, nodes from 9,025 to 9,084, and proof bytes from 9,663 to 10,020. Its paired ratio is 1.002 despite separate time medians of 1,989.772 and 1,584.861 milliseconds: substantial within-run variation makes the ratio of these medians misleading. The baseline/current-default paired ratio on this same input is 1.071 despite equivalent search work, further limiting an attribution to projections.

Across all nineteen inputs, old A/A ratios range from 0.991 to 1.024, current-default A/A from 0.980 to 1.018, and projection A/A from 0.985 to 1.017. Current default versus the pinned baseline ranges from 0.983 to 1.071, with the upper endpoint occurring on Gordian. No extra timing run is used to replace these retained observations or turn them into a positive speedup claim.

Projection metadata and unsuccessful eligibility scans can cost more than the work saved by an occasional contraction. These timings justify the opt-in default and give no whole-pipeline speedup claim for this corpus. The useful advance is a complete, independently replayed raw-round implementation on which a better schedule can now be tested. The next asymptotic obligation remains a controlled schedule reaching a decisive endpoint, including the cost of discovering and exposing its donors.

From `../fast`, run `primitive_power_research/projections.py` in `audit` or `benchmark` mode. Complete records are `../fast/results/primitive_projection_audit_20261008.json` and `../fast/results/primitive_projection_pipeline_20261008.json`. The source test log is `data/primitive-projection-full-tests.txt`.

# 86 Contracting unit-coordinate primitive forests in one raw batch

Disjointness is a sufficient condition for simultaneous primitive-pair projection, but it excludes a useful class of overlapping donors. A primitive root whose exponent vector has a unit coordinate solves that generator as an integer power of the other. If the resulting directed dependencies form a forest, all such equations can be composed and substituted in one raw batch. The new version-seven certificate checks this larger operation independently. It is a separate optional mode, so the preceding version-five and version-six search policies remain available without enabling the forest planner.

## 86.1 A scheduling obstruction that matching cannot repair

Fix $q \geq 1$ and consider generators $a, b_1, \ldots, b_k$ with the two relations $(a^q b_i)^2$ and $(a^q b_i)^3$ for each $i$. The coprime powers imply $a^q b_i = 1$, so this is an abstract presentation of $\mathbb{Z}$. Every eligible two-generator donor meets the common generator $a$. A disjoint-pair round can therefore choose at most one pair. Projecting that pair replaces its leaf by $a^{-q}$ and leaves all other donors in the same star form. The surviving cube slot of that leaf becomes a raw one-generator zero-exponent circuit, not a new two-generator donor. Inductively, *every* schedule restricted to these current-relator pair operations needs $k$ rounds to reach rank one, or $k-1$ rounds before the rank-two terminal can finish. Choosing a better matching does not repair this example.

This is an obstruction for the abstract algebraic operation model. It is not a lower bound for all recognition methods, nor a counterexample to a short-depth claim specifically about planar

Wirtinger presentations. The family motivates allowing donors to share an anchor when their exact quotient equations can be composed safely. For this family the forest operation removes all $k$ leaves in one batch.

## 86.2 Why an acyclic unit-coordinate batch preserves the group

Use the source-provenance invariant from Section 85. For a checked donor $w^d$ supported on distinct generators $p, c$, suppose the primitive root has signed exponent coordinates $(u, \epsilon)$, where $\epsilon \in \{1, -1\}$ is its coordinate at the chosen child $c$. Our arithmetic witnesses require coherent signs. Hence the primitive root contains exactly one occurrence of $c^\epsilon$ and is a cyclic conjugate of $p^u c^\epsilon$. Killing $w$ is exactly the substitution

$$c = p^{-u/\epsilon} = p^{-u\epsilon}.$$

Knot-group torsion-freeness justifies replacing $w^d = 1$ by $w = 1$ when $d > 1$. For $d = 1$ this is an ordinary generator elimination. Both primitive coordinates must be nonzero, so no zero image exponent is introduced.

Each selected child has exactly one parent, and selected directed edges must have no cycle. Donors may share parents, and a parent may itself be another edge's child. Following parents eventually reaches an uneliminated anchor $z$. The image of a vertex $x$ is

$$x \longmapsto z^{K_x}, \qquad K_x = \prod_{e \text{ on the path from } x \text{ to } z} k_e, \qquad k_e = -u_e \epsilon_e.$$

The empty path has product one. Leaf-first Tietze elimination proves that the quotient by all selected primitive roots is the free group on the anchors and untouched generators. Substituting these images into every other relator therefore preserves the full presented knot group. Each selected donor has zero exponent in its own anchor after substitution, so precisely those donor slots may be emptied. All other original slots are retained and checked later.

The unit condition is essential. Overlapping arbitrary primitive pairs do not automatically give such a power forest: sharing a non-unit power of a newly introduced cyclic generator can produce an amalgamated group rather than a free cyclic factor. The checker never infers the forest rule from primitivity alone. Cycles are also excluded, even when particular cyclic equations might happen to be consistent. This is a sufficient, independently checkable operation with a clear cost contract.

## 86.3 Selection and shared construction of all quotient images

The planner reuses immutable capped-support and signed-count summaries. It scans original relator slots, excludes noncoherent roots and primitive vectors without a unit coordinate, and uses the existing arithmetic witness for proper powers. When the exponent gcd is one and one signed count is one, the coherent word is immediately a primitive conjugate of $p^u c$; no extra producer profile pass is necessary. The independent checker still recomputes its complete arithmetic profile.

For each eligible donor, the planner chooses an unused unit-coordinate child, preferring the larger identifier when both orientations are available. An undirected union-find structure rejects edges joining vertices already in the same component. Together with at most one parent per child, this maintains a rooted forest. A component's sole vertex without a parent is its root, so an allowed new edge attaches one rooted component to another. The choice is deterministic, but it is not asserted to maximize the number of eliminations over all possible orientations and later states.

At least two selected edges are required for the producer to choose a forest batch. Otherwise it falls back to the existing disjoint-pair planner. After either operation, raw search and the explicit

normalization boundary behave as in the preceding section. A single checked forest edge is valid for replay, even though the producer uses the older operation for that case.

A first implementation materialized every $K_x$ and rebuilt the power $z^{K_x}$ separately. This was polynomial but repeated power-construction loops along a long chain. The final producer follows dependency paths once, then constructs child images from already constructed parent images:

$$I_c = I_p^{k_e}.$$

Both signed parent images are retained, so a negative edge uses the opposite parent image without a normalization step. Binary powering sees only the *local* edge exponent. The result is a shared straight-line program for all images, followed by one raw mapping of the original reachable circuit. This change reduces abstract search work on the 64-generator, 128-bit chain from the prototype's 1,723,368 to 223,340; it is a work-counter comparison, not a timing or whole-recognizer ratio. Exact prototype source snapshots and its initial passing test/audit records are retained.

## 86.4 Independent version-seven replay and resource behavior

A `primitive_forest` move has exactly its kind and an edge list. Each edge contains exactly a child identifier and a primitive-power proof with its original donor slot. The checker requires strict integer types, live distinct pair members, a unit coordinate at the child, a nonzero other coordinate, distinct children and distinct donor slots. It validates every arithmetic donor against the intact pre-batch roots before modifying any root or live-generator set.

The verifier uses a different graph algorithm from the producer. It builds parent-to-child adjacency, starts a queue at every surviving anchor, and propagates along edges. Every live generator must be reached exactly once. An isolated directed cycle cannot be reached from an anchor and is rejected. The traversal also derives signed exponent products for literal replay. Compressed replay constructs signed image circuits in this independently obtained parent-before-child order, then uses a fresh postorder traversal of the original grammar. It calls neither the forest planner nor its application helper. Literal replay independently substitutes repeated letters, checking the complete projected allocation and charged output work before constructing any repeated list.

Version seven permits the earlier prefix operations, explicit normalization handoffs and either the raw rank-one exponent endpoint or the rank-two primitive terminal. Versions one through six retain their existing contracts; a forest move disguised under version six is rejected. The source PD and entire presentation prefix remain mandatory. Neither the local producer nor the local replayer is an arbitrary-presentation knot-verdict API.

Both `compressed_certificate` and `group_decide` accept the option `primitive_forest=True`.

For `recognize`, set `group_primitive_forest=True` and `use_group=True`.

The CLI flag is `-group-primitive-forest`. Forest mode implies the projection route and mandatory compressed proof replay. It is incompatible with the older explicit-to-compressed adaptive-search flag. The mode defaults to false. All graph, grammar and arithmetic passes charge cooperative work and propagate external cancellation. As before, node limits do not cap all Python auxiliary memory, and the work allowance is not unit-cost bit time.

## 86.5 Polynomial cost of a batch, independent of forest depth

Let $r$ be the initial live rank, $m$ the retained slot count, $M$ the allocated SLP-node count and $B \geq 1$ the maximum represented-length bit size. A donor's local exponent has at most $B$ bits. A dependency path has at most $r - 1$ edges, so a path product has $O(rB)$ bits. Word lengths after substitution have $O(rB + B)$ bits. This controls lengths without expanding them.

Shared construction uses $O(B)$ new power nodes per selected edge, not $O(rB)$ per path endpoint. Mapping the original circuit creates at most one new concatenation per original reachable node.

Thus a conservative bound for the new total grammar size is

$$M' \le 2M + O(rB), \qquad B' = O((r+1)(B+1)).$$

Candidate arithmetic checks, union-find selection, path processing, independent graph replay, exact integer products and postorder mapping are all polynomial in the initial encoded parameters. A loose bound may classify all $m$ donors separately over all $M$ nodes; no shared-profile linear-time claim is needed. Standard multiplication/gcd bounds and the usual dictionary-access implementation qualification suffice. The compressed operation has this bound regardless of the depth of the selected forest. Literal replay instead pays for expanded output and may exhaust its budget.

This is stronger than performing one disjoint-pair round per leaf on a star, but it is a local theorem. Section 92 improves the following historical estimate to polynomial cost for a whole fixed-source monomial epoch, including both forests and disjoint pairs. If $d$ raw phases mix these batches with ordinary pair projections, a conservative composition bound is

$$\text{poly}(\text{initial parameters})\, 2^{O(d \log(r_0+1))}.$$

The extra logarithm reflects the possible multiplicative bit-length growth of an entire forest; the preceding $2^{O(d)}$ statement counted disjoint-pair rounds and must not silently be relabeled as a forest-phase bound. Polylogarithmically many raw phases would still give a quasipolynomial bound from polynomial-size input data. No such general producer, endpoint or exposure bound is proved here. Normalization and legacy discovery work remain outside this raw-phase accounting.

### 86.6 Audits and measurements

The audit checks eighty actual diagrams against the complete package at `7bf6f01026a1`. Current default and current disjoint-only certificates are exactly equal to their respective prior versions. Forest mode produces 34 version-seven certificates with 203 forest edges, all of whose complete proofs pass both literal and compressed replay. No input is newly solved or loses a positive result; no donor in this diagram sample is a proper power. Three separate actual five-braid sources carry powered Whitehead prefixes of exponents $2^4$, $2^{128}$ and $2^{1024}$ and pass compressed replay of their version-seven proofs. The small prefix also passes literal replay.

Seven new test methods cover one hundred random signed trees with unrelated retained relations, raw star/chain/balanced families, cycles, duplicate children and slots, non-unit forgeries, strict fields, source rebinding, legacy proofs, independence from producer helpers, allocation/work exhaustion, cancellation and CLI behavior. The full maintained suite passes 988 tests in 207.503 seconds with Regina; the log is `data/primitive-forest-full-tests.txt`.

| Family | rounds pair/forest | search work pair/forest | nodes pair/forest |
|---|---:|---:|---:|
| star 64 | 63/1 | 247,420/111,616 | 514/514 |
| chain 64 | 6/1 | 885,083/223,340 | 57,481/24,508 |
| balanced 64 | 6/1 | 172,215/76,141 | 8,539/3,025 |
| star 256 | 255/1 | 1,120,444/126,688 | 1,090/1,090 |

Capacity families use paired square/cube donors and are abstract presentations of $\mathbb{Z}$, not knot diagrams. Both old and new raw schedules, and their independent replay, run with reduction, cyclic reduction, expansion and equality disabled. Work counts include identical initial grammar construction. At rank 256 the star needs 255 disjoint rounds and one forest batch; search work drops from 1,120,444 to 126,688, with the same 1,090 allocated nodes including the empty node. The 64-generator chain reaches represented-length bit size 8,067; forest construction uses 24,508 nodes versus 57,481 for the older six-round schedule. These exact counters establish reduced work

on the supplied algebraic families without claiming a corresponding speedup on difficult knot inputs.

| Input | old ms | pair ms | forest ms | old/forest | pair/forest |
|---|---|---|---|---|---|
| survivor-00 | 17.392 | 18.444 | 18.772 | 0.930 | 0.978 |
| mirror-03 | 24.764 | 26.112 | 26.855 | 0.918 | 0.952 |
| survivor-06 | 7.398 | 8.050 | 9.173 | 0.806 | 0.908 |
| survivor-08 | 7.500 | 8.046 | 8.219 | 0.920 | 0.985 |
| gordian | 1480.229 | 1514.039 | 1536.808 | 0.964 | 0.964 |
| trefoil | 0.054 | 0.055 | 0.054 | 0.995 | 1.015 |
| conway | 1.023 | 1.015 | 1.033 | 0.991 | 0.991 |

| Crossings | old ms | pair ms | forest ms | old/forest | pair/forest |
|---|---|---|---|---|---|
| 8 | 2.196 | 2.061 | 1.833 | 1.188 | 1.123 |
| 16 | 7.248 | 5.607 | 4.518 | 1.606 | 1.245 |
| 32 | 24.987 | 16.061 | 11.920 | 2.084 | 1.344 |
| 64 | 95.567 | 53.731 | 37.144 | 2.546 | 1.440 |
| 128 | 365.596 | 194.214 | 130.191 | 2.808 | 1.524 |

The complete-recognition experiment uses nineteen fresh validated diagrams, five measured rounds and one warm-up per arm. Eight shuffled arms compare old default, old disjoint projections, current default and forest mode, each with an A/A control. Full source reconstruction and mandatory independent replay are inside the timed recognition call. The separate group-stage experiment uses actual stabilized-circle diagrams with 8 through 128 crossings and six shuffled arms. It bypasses earlier diagram simplification, so its ratios are group-stage measurements, not whole-recognizer speedups. All 760 whole-recognition calls and 150 group-stage calls complete, with 152 and 30 respective warm-ups. The runs take 83.853 and 12.067 seconds. All samples, nondecisions, work counters, proof sizes and frozen source hashes are retained by the driver `primitive_power_research/forests.py` under `../fast`. Table ratios are medians of paired times; a value above one favors forests. They need not equal ratios of separate time medians.

On ordinary survivor/mirror unknots other than survivor 06, old-default/forest ratios range from 0.911 to 0.940. Survivor 06 has ratio 0.806 and forest A/A ratio 1.112, so substantial local timing noise accompanies its observed overhead. Gordian's old-default/forest and old-pair/forest ratios both round to 0.964. Its search work increases from 1,215,145 in the default and 1,265,669 with disjoint projections to 1,305,143 with forests: 141 forest attempts expose just one two-edge batch. Its forest proof is 10,060 bytes, compared with 9,663 and 10,020 bytes respectively. Across the whole corpus, old-default, old-pair, current-default and forest A/A ranges are 0.986–1.050, 0.979–1.019, 0.959–1.051 and 0.986–1.112. Current default versus the pinned default ranges from 0.975 to 1.021. These observations support retaining the optional policy; they do not establish a gain from running the planner on every ordinary search state.

The restricted actual-diagram group stage does benefit. At 128 crossings, old default, disjoint projections and forests take median 365.596, 194.214 and 130.191 milliseconds; paired speed ratios are 2.808 and 1.524 relative to the first two arms. Search work falls from 493,856 to 263,317 to 185,764. Allocated node counts are 24,891, 1,017 and 1,017. Forest mode uses 63 two-edge batches and 63 normalization handoffs rather than 126 disjoint rounds, so even these easy actual inputs do not realize the one-batch abstract-star example. Proof bytes are 8,640, 24,517 and 24,286: a faster checked group stage need not have a shorter proof than the old default. Across the five stage inputs, default A/A ratios are 0.986–1.006, pair A/A 0.980–1.001 and forest A/A 0.967–1.007. Earlier diagram simplification would make these circles easy for the full recognizer; the stage improvement must not be presented as a difficult-unknot or whole-pipeline result.

The new operation addresses one specific scheduling obstruction. It does not establish that useful forests become exposed cheaply, nor that the raw schedule reaches an endpoint for every unknot.

The opt-in policy permits further work on exposure and progress-sensitive dispatch without imposing unproved performance benefits on the ordinary recognition route.

# 87 Sharing structural candidates between primitive planners

The preceding forest operation improves selected algebraic families, but its ordinary-corpus overhead comes largely from looking for donors that are absent. Previously, each search state ran a complete forest planner and, when fewer than two edges were selected, a second disjoint-pair planner. Both rebuilt structural eligibility from the same immutable words. We now prepare one current candidate list, cache word-content classification, and dispatch only the planners whose elementary preconditions can hold. The selected certificate is preserved; this is an implementation improvement rather than a new knot theorem.

## 87.1 What can be cached, and what must remain current

A word arena has immutable rules and never reuses an allocated node identifier. For a root, raw support, signed occurrence counts and sign coherence therefore do not depend on the current relation slot or on the previously applied presentation moves. The planner stores either an ineligible marker or the ordered two-generator support together with its two signed counts. A word with more than two generators, mixed signs for either generator, or fewer than two generators is outside the supported candidate class.

The underlying capped-support metadata still visits each allocated node at most once. The new root descriptor is also computed at most once per distinct root. Repeated roots share the descriptor; empty root zero is ineligible from initialization. Each search state deduplicates the roots whose descriptors are missing, computes only that metadata, and then constructs a candidate list in original slot order. It checks both generator labels against the *current* live set. Neither a donor's slot nor its live membership is stored in the immutable descriptor.

This distinction preserves three important cases. Reordering or duplicating roots changes the selected slot while reusing their content summaries. Removing a live generator makes a descriptor ineligible for the present state without invalidating its stored word content. Normalization or substitution that changes the word creates a different root and receives a new descriptor. If an arena reuses an already interned root, its exact word content is unchanged, so the earlier descriptor is valid. Raw cancellation is never inferred merely from a smaller live set.

The fresh list is a producer-internal snapshot of the current roots and live set. The forest planner and its possible pair fallback consume that same snapshot before any presentation mutation. They still create fresh witness records carrying the actual donor slot. Independent source replay continues to reconstruct the whole presentation and derive the evidence from its own words; it does not trust these caches or snapshots.

## 87.2 An exact dispatch bound and sparse graph setup

The producer uses a forest batch only when it selects at least two edges, with distinct donor slots. Consequently fewer than two structurally eligible current slots make a forest batch impossible. In that case it skips forest setup and tries the pair planner directly. This cannot hide a batch that the preceding forest policy would select. It is an exact upper bound on the possible number of selected edges, not a guess based on the last few failed searches or a timeout that could suppress an available move.

When a forest is possible, its union-find maps are created lazily for the generators encountered by eligible unit-coordinate candidates. Isolated live generators do not require graph entries. The

orientation choice, original-slot scan order and cycle tests are unchanged. The pair fallback consumes the same candidate list instead of scanning all relators again. A negative primitive-power classification remains cached by its exact word root.

For forest candidates, the gcd-normalized signed vector and the result of the unit-coordinate test are cached separately. This avoids repeating big integer gcd work on an unchanged donor. A coherent word whose raw count of one generator is one is already a cyclic conjugate of $a^u b^{\pm 1}$, so the pair planner can construct the same exponent-one primitive witness without running a prefix-profile query. The forest planner already used this theorem. The independent checker still verifies the full arithmetic profile. Proper powers and non-unit primitive vectors continue to use the existing exact classification routine.

On the Gordian source audit, the old forest policy made 141 forest attempts. Only three candidate-slot occurrences exist across those states. The new policy skips graph setup on 140 states and runs the forest planner once, selecting the same two-edge batch as before. All 141 states still rebuild current slot/live membership; the optimization does not claim to eliminate that remaining scan.

## 87.3 Cost, resource accounting and unchanged proof contracts

Let $p$ be the number of queried presentation states, $m$ the retained slot count, $M$ the total allocated grammar nodes, $B$ a maximum length/count bit size and $q_j$ the candidate-slot count at state $j$. Descriptors cover at most $M$ roots and take a constant number of signed count operations each. The underlying grammar summaries also cover at most $M$ nodes. Using ordinary integer arithmetic, a conservative accumulated structural/gcd cost is

$$O\left(p(m+1) + MB^2 + \sum_{j=1}^{p} q_j\right)$$

up to generator/node identifier encoding and dictionary-access qualifications. This is for preparing and filtering candidates. Forest selection additionally pays its usual union-find cost over candidate vertices. Uncached primitive profile queries, applying moves, normalization and the rest of recognition retain their separately stated costs. Lazy forest storage at a state uses only $O(q_j)$ vertex entries, bounded also by the live rank. Persistent summaries remain proportional to the allocated grammar and its encoded count sizes.

The implementation charges both complete slot scans before allocating the fresh-root and current-candidate lists. It additionally charges fresh summaries, donor processing and graph traversal. List-comprehension work is charged in aggregate, while grammar and arithmetic passes continue polling cancellation. These cooperative counters are conservative accounting units, not elapsed-time or per-bit CPU measurements. In fact, the new pair-only path can record slightly *more* charged work because new descriptor accounting is explicit, even where its Python execution gets faster.

No certificate version, schema, checker, word operation, source reconstruction or public option changes. The entire previous verifier/host/word modules are checked byte-for-byte against the pinned baseline in the audit. Default, disjoint-projection and forest modes retain the same choices when both runs complete without exhausting resources. A finite work or wall allowance can change which run finishes because accounting and execution costs change; that is not an unconditional equality claim for all budgeted results. The general bound on finding a short successful schedule remains open.

## 87.4 Equivalence audits and complete-call evidence

The pinned package is `b5b0ae50c4e4`. Eighty actual diagrams are run in all three modes, giving 240 baseline/current comparisons. Every certificate and nondecision is identical. All 159 positive

mode results pass both independent source-reconstructing replay paths. Another 2,000 planner comparisons cover one thousand raw-word states with repeated roots, changed slot orders and varying live sets. Those are planning tests on arbitrary words, not knot verdicts.

Four new test methods exercise cached negative structural/profile results, changing slots and live labels, normalization that exposes a newly eligible root, shared current snapshots, strict resource exhaustion and caller cancellation. A donor of the form $a^{2^{4096}}b^{-1}$ is selected and independently replayed with expansion and the producer's general profile query disabled. A one-candidate search skips the forest planner and returns the same pair witness. The earlier primitive tests also pass unchanged. The full maintained suite passes 992 tests in 234.493 seconds with Regina; the log is `data/primitive-planner-full-tests.txt`.

| Input | pair old/new ms | ratio | forest old/new ms | ratio |
|---|---:|---:|---:|---:|
| survivor-00 | 22.165/28.465 | 1.018 | 19.527/18.530 | 1.036 |
| mirror-03 | 25.315/25.043 | 1.008 | 26.052/25.344 | 1.021 |
| survivor-06 | 8.114/8.232 | 0.996 | 8.499/8.051 | 1.030 |
| survivor-08 | 8.049/8.125 | 0.994 | 8.207/8.165 | 1.010 |
| gordian | 1636.125/1565.059 | 0.999 | 1655.535/1574.392 | 1.026 |
| trefoil | 0.057/0.056 | 0.993 | 0.056/0.056 | 1.000 |
| conway | 1.019/1.039 | 0.998 | 1.028/1.029 | 1.003 |

| Crossings | pair old/new ms | ratio | forest old/new ms | ratio |
|---|---:|---:|---:|---:|
| 8 | 2.069/2.298 | 0.948 | 1.846/2.023 | 0.974 |
| 16 | 5.347/5.503 | 0.970 | 4.377/4.349 | 1.037 |
| 32 | 16.319/15.999 | 1.011 | 12.246/11.705 | 1.069 |
| 64 | 55.563/52.154 | 1.064 | 38.666/34.847 | 1.110 |
| 128 | 200.361/185.550 | 1.072 | 131.456/119.081 | 1.132 |

The whole-recognition benchmark compares previous/current disjoint and forest modes on nineteen fresh validated diagrams, with five measured rounds, one warm-up per arm and an A/A control for each of the four implementations. It includes source reconstruction and mandatory proof replay. The separate checked group-stage experiment uses actual stabilized circles, bypassing earlier simplification; its ratios are not whole-recognition speedups. Both runs preserve all samples and incomplete outcomes, if any, and pin the complete prior Python package. Their tables report medians of paired old/current times, so values above one favor the new planner and need not equal ratios of the displayed separate time medians.

All 760 whole-recognition calls and 200 checked-stage calls complete, with 152 and 40 respective warm-ups. The runs take 96.775 and 11.452 seconds. Whole-call pair-mode ratios range from 0.983 to 1.018, and forest-mode ratios from 0.986 to 1.036. The old/new pair A/A ranges are 0.986–1.023 and 0.942–1.035; the old/new forest A/A ranges are 0.977–1.018 and 0.981–1.034. These controls limit attribution of the small whole-call differences to the planner alone. Gordian's forest ratio is 1.026, while its forest A/A ratio is 1.034. Its old/new pair ratio is 0.999 despite separate medians 1,636.125 and 1,565.059 milliseconds. Survivor 00 similarly has pair ratio 1.018 despite old/new separate medians 22.165 and 28.465 milliseconds. Within-run variation explains why neither ratio of separate medians should replace the retained paired observations.

The larger checked group stages show a clearer effect. At 128 crossings, pair medians are 200.361/185.550 milliseconds and forest medians are 131.456/119.081 milliseconds, with paired ratios 1.072 and 1.132. The forest planner skips 63 unsuccessful graph setups and makes 63 productive attempts instead of 126 attempts. Charged forest work falls from 185,764 to 170,833. Pair work increases slightly from 263,317 to 264,451 under the new accounting, despite its elapsed improvement. Both modes retain 1,017 allocated nodes; their respective proof sizes remain

exactly 24,517 and 24,286 bytes. The words, proofs, round counts and normalization handoffs are unchanged.

This benefit is not universal: the eight-crossing stage has pair/forest ratios 0.948/0.974. At sixteen crossings they are 0.970/1.037, increasing to 1.064/1.110 at sixty-four. Across the stage inputs, old/new pair A/A ranges are 0.980–0.998 and 0.968–1.032; old/new forest A/A ranges are 0.979–1.025 and 1.007–1.037. Forest controls show some systematic variation, but their largest paired shift is smaller than the 128-crossing result. The measured evidence supports cheaper planning on larger exposed stages, not a universal whole-recognition gain. Both modes remain optional.

The driver is `primitive_power_research/planner.py` under `../fast`; its `audit`, `benchmark` and `stages` modes save full proofs, work counters, source hashes and shuffled sample orders. These measurements concern the same certified move choices under cheaper planning, not an enlarged class of solved inputs or a general quasipolynomial recognition result.

# 88 Bounded-run normalization between primitive phases

The raw-phase bounds above deliberately excluded normalization. A useful invariant now includes that boundary for a restricted, explicitly represented class: each relator is a concatenation of a bounded number of powers of individual generators. Monomial substitutions preserve this class, and free and cyclic reduction can be performed by integer addition and endpoint trimming. The implementation uses this observation at the existing `normalize_relators` move. It retains the general reducer when the bounded probe cannot establish its precondition.

## 88.1 Exact summaries and a conservative domain

A run list is a sequence $(g_1, e_1), \ldots, (g_s, e_s)$ with positive generator labels, nonzero signed integer exponents and $g_i \neq g_{i+1}$. It denotes the freely reduced word $g_1^{e_1} \cdots g_s^{e_s}$. The producer evaluates an SLP in postorder, storing either an exact list of at most $q$ runs or an unsupported marker. Terminals have one run. To concatenate two supported children, it folds the right list into a stack containing the left list: equal adjacent generator labels cause exponent addition, and zero sums are removed. The result is stored only if it has at most $q$ runs. The later optimization in Section 88.5 skips subtrees certified uniform by the arena and stops at the first unsupported descendant. It preserves exactly this sufficient domain.

The proof of exactness is induction on the grammar. Each child is freely reduced; only its concatenation boundary can create cancellation. Repeated boundary cancellation and merging give the unique freely reduced word. An unsupported child propagates an unsupported result. In particular, this is a sufficient condition on *all intermediate words*, not a complete test for whether the final word has at most $q$ runs. For example, $(ab)^N((ab)^N)^{-1}$ reduces to the empty word, but its many-run intermediate is deliberately unsupported. It proceeds to the maintained general reducer.

For a supported root, cyclic reduction trims opposite-sign powers of the same generator at its two ends. It subtracts the smaller absolute exponent, removes empty ends, and repeats. Equal-sign endpoint powers are left in place: the implementation preserves the same anchored word as ordinary cyclic reduction, rather than choosing a rotated conjugate. This matters because subsequent proof moves can address word positions.

The production limit is $q = 4$. Only roots of expanded length greater than 64 are probed; short roots keep the existing reducer. If there are no long roots, the probe returns immediately. Before building any new word circuit, it checks every required long root. A failed probe allocates summaries but no new SLP word nodes. A successful probe rebuilds long roots from binary

powers and retains all relation slots, including empty and repeated slots. Short roots require only bounded expanded-word work: with nonempty children a binary parse of at most 64 letters has at most 127 node occurrences. The gate does not conceal an exponentially long cancellation problem.

Summaries are cached by immutable arena identifier and run limit. Their meaning is independent of the current relation slot and live generator set. Producer and replayer have separate caches and implementations. The replayer merges a left list against a right deque, then trims endpoints with a deque; it constructs a left-associated output circuit while the producer balances the run concatenations. Neither imports the other's arithmetic helper. The old proof move, version rules and full source reconstruction remain in force. These routines establish word equality, not an independent knot verdict.

## 88.2 Closure under monomial maps

Suppose the initial relators are given by compact run circuits: each root concatenates at most $q$ single-generator powers, and each power is encoded by binary powering. A monomial map sends each generator to an integer power of one generator, possibly with negative exponent or exponent zero. It replaces each original run by at most one run. Therefore it cannot increase the number of runs before free reduction, and free or cyclic reduction cannot increase that number either.

Every intermediate of the mapped compact circuit also has at most $q$ freely reduced runs. An intermediate inside a power circuit is uniform; an intermediate concatenation covers at most the original $q$ blocks. The same is true of the shared powers used to construct forest images: they are still powers of a single surviving generator. Consequently the bounded probe succeeds throughout any sequence of these substitutions and normalizations, if the starting circuit has this form. Its rebuilt output again has the required compact form. Short roots use only bounded literal work and also admit compact bounded-size circuits.

Both disjoint primitive projections and unit-coordinate forest quotients are monomial maps. Initial Wirtinger relators have at most four letters, so they satisfy the representation condition. This does not imply that the full search stays in the class: nonmonomial Whitehead changes, overlap moves and general eliminations can leave it. Nor can an arbitrary grammar be admitted solely because its *final* normal form happens to have four runs. The intermediate condition is essential to the implemented probe and to the stated closure argument.

## 88.3 Encoded cost including normalization

Let $M$ count historical allocated nodes, $S$ the currently reachable nodes, $m$ the retained root slots, and $B \geq 1$ bound represented-length and exponent bit sizes. Constructing previously absent run summaries takes $O(SqB)$ bit operations for exponent additions, up to identifier and dictionary costs. Stored summaries use $O(MqB)$ bits under the same qualification. Cyclic trimming uses $O(mqB)$ exponent work. Rebuilding compact roots adds $O(mq(B+1))$ nodes; binary-power construction and length bookkeeping have polynomial bit cost, not unit-cost arithmetic. Work counters charge traversal and run processing and poll cancellation; they do not measure individual big-integer instructions or total process memory.

The compact reachable grammar after a successful normalization satisfies

$$S_{j+1} = O\big(mq(B_{j+1}+1)\big),$$

where the fixed short-word gate is absorbed in the constant. Historical nodes are not deleted. A projection or forest batch maps each reachable input node once and adds $O(r_j(B_j+1))$ image nodes. A following compact normalization therefore gives the additive allocation bound

$$M_{j+1} \leq M_j + O\big(S_j + r_j(B_j+1) + mq(B_{j+1}+1)\big).$$

The distinction between reachable and historical nodes prevents an unjustified repeated polynomial blow-up. Without an intervening normalization, the earlier raw-round recurrence still applies.

For disjoint-pair rounds, $B_{j+1} \le 2B_j + O(1)$. Combining raw rounds with any of these supported normalization boundaries yields an encoded time bound

$$\text{poly}(\text{initial encoded parameters})\, 2^{O(d)}$$

for $d$ such rounds, including polynomial-cost candidate checks and replay. For mixed forest phases the corresponding bit bound is $B_{j+1} = O((r_0 + 1)(B_j + 1))$, giving

$$\text{poly}(\text{initial encoded parameters})\, 2^{O(d \log(r_0+1))}.$$

Here $q$ and the short-word gate are fixed, and the phase count includes the monomial batches; redundant normalization alone does not change a normalized state. These extend the earlier raw-phase estimates to this closed normalized class. They do not bound the number of required phases, prove that a terminal is reached, or include arbitrary legacy exposure work. A general quasipolynomial recognition theorem still requires those missing global ingredients.

## 88.4 Correctness audits and measured scope

Six focused tests cover arbitrary binary parses, changed run limits, repeated roots, endpoint signs, huge exponents, conservative failure, monomial closure, independent source replay and resource guards. Conjugates with exponent $2^{4096}$ normalize with expansion, equality, slicing and the general reducer disabled. An actual five-braid source with a supplied powered Whitehead prefix passes complete compressed verification with all producer normalization helpers disabled. The full maintained suite passes 998 tests with Regina in 209.031 seconds.

Against the complete pinned package at `483b7397a108`, eighty validated diagrams give 240 comparisons across default, projection and forest modes. All certificates and nondecisions agree; all 159 positive mode results pass both full source replayers. An independent literal audit checks all 87,381 signed rank-two words through length eight: 66,580 bounded probes succeed and 20,801 decline, including the empty-word short gate. A declined probe makes no negative word or knot claim.

These finite equalities are not a universal statement about search traces. Equivalent rebuilt words may have different grammar shapes, and later heuristics or finite resource allowances can depend on allocated nodes or charged work. Soundness follows from exact normalization and complete source replay, regardless of those possible search differences.

The complete-recognition run uses nineteen actual diagrams, five measured rounds and one warm-up per arm. Eight shuffled arms compare the old and new projection and forest modes, each with an A/A control. All 760 measurements and 152 warm-ups complete; the run takes 88.628 seconds. The following table retains all inputs. Ratios are medians of paired old/new times, so values above one favor the new implementation and need not equal ratios of the separate displayed time medians.

| Input | pair old/new ms | ratio | forest old/new ms | ratio |
|---|---|---|---|---|
| survivor-00 | 18.586/17.619 | 1.055 | 17.893/18.042 | 0.986 |
| survivor-01 | 15.160/15.053 | 1.004 | 15.120/15.052 | 1.000 |
| survivor-02 | 16.720/16.665 | 1.006 | 16.665/16.619 | 1.005 |
| survivor-03 | 11.587/11.672 | 0.989 | 11.646/11.775 | 0.995 |
| mirror-03 | 25.682/26.593 | 1.004 | 25.636/26.613 | 1.007 |
| survivor-04 | 10.680/10.862 | 0.994 | 10.619/10.822 | 1.010 |
| survivor-05 | 20.959/20.010 | 1.047 | 20.375/20.345 | 0.990 |
| survivor-06 | 8.567/8.150 | 1.059 | 8.331/8.232 | 1.012 |
| survivor-07 | 20.513/21.119 | 0.981 | 20.795/21.110 | 0.994 |
| survivor-08 | 8.470/8.319 | 1.018 | 8.160/8.684 | 0.963 |
| mirror-08 | 21.462/23.429 | 0.955 | 21.038/21.568 | 0.982 |
| survivor-09 | 7.897/7.983 | 0.989 | 7.811/7.859 | 0.994 |
| survivor-10 | 7.000/6.955 | 0.989 | 6.830/7.141 | 0.957 |
| survivor-11 | 8.129/7.896 | 1.030 | 8.577/8.157 | 1.031 |
| gordian | 1595.833/1574.728 | 1.004 | 1546.156/1545.253 | 1.001 |
| trefoil | 0.054/0.054 | 0.987 | 0.055/0.055 | 1.019 |
| figure_eight | 0.062/0.062 | 1.000 | 0.062/0.062 | 1.002 |
| conway | 1.041/1.019 | 1.024 | 1.055/1.050 | 0.996 |
| kinoshita_terasaka | 1.045/1.077 | 0.977 | 1.038/1.040 | 0.996 |

Pair-mode ratios range from 0.955 to 1.059, and forest ratios from 0.957 to 1.031. The old/new pair A/A ranges are 0.983–1.069 and 0.963–1.071; the old/new forest controls span 0.967–1.054 and 0.919–1.060. These measurements do not establish an overall recognition speedup. Gordian has pair/forest ratios 1.004/1.001. In both modes its single long-root probe declines, visiting 51 new summaries; the source has no successful bounded normalization. Forest work rises from 1,266,731 to 1,267,256, while its 9,084 nodes and 10,060-byte proof are unchanged. No other input in this corpus activates a long-root producer probe.

| Crossings | pair old/new ms | ratio | forest old/new ms | ratio |
|---|---|---|---|---|
| 8 | 2.060/1.987 | 1.022 | 1.713/1.732 | 0.984 |
| 16 | 5.209/5.266 | 0.988 | 4.151/4.176 | 0.994 |
| 32 | 15.305/15.401 | 0.996 | 10.931/11.187 | 0.994 |
| 64 | 50.333/50.272 | 0.988 | 33.070/33.878 | 1.002 |
| 128 | 186.828/182.397 | 1.026 | 112.551/114.080 | 1.004 |

The separate checked group stages bypass earlier simplification on actual stabilized circles. All 200 measurements and 40 warm-ups complete in 10.578 seconds. Every normalization is below the long-word gate; these rows measure dispatch overhead, not the bounded arithmetic kernel. The 128-crossing stage has pair/forest paired ratios 1.026/1.004. Charged forest work nevertheless rises from 170,833 to 178,960 because the new length preflight is charged on each normalization boundary. Its 1,017 nodes and complete proof are unchanged. Smaller stages include regressions. Neither these easy circles nor bypassing earlier simplification supports a claim about difficult whole-knot recognition.

| Case | bits | old ms | new ms | ratio | old A/A | new A/A |
|---|---|---|---|---|---|---|
| conjugate | 16 | 0.368 | 0.301 | 1.331 | 1.020 | 0.749 |
| conjugate | 64 | 1.215 | 0.998 | 1.245 | 0.985 | 1.006 |
| conjugate | 256 | 4.515 | 3.581 | 1.262 | 0.985 | 0.985 |
| conjugate | 1024 | 18.598 | 16.018 | 1.188 | 1.003 | 0.994 |
| conjugate | 4096 | 81.615 | 74.807 | 1.164 | 1.041 | 1.139 |
| source-replay | 16 | 1.391 | 1.476 | 0.944 | 1.033 | 1.019 |
| source-replay | 64 | 4.743 | 5.045 | 0.943 | 1.016 | 1.008 |
| source-replay | 256 | 18.237 | 18.704 | 0.945 | 0.999 | 0.979 |
| source-replay | 1024 | 74.004 | 82.450 | 0.879 | 0.852 | 1.051 |
| source-replay | 4096 | 319.300 | 338.978 | 0.916 | 0.997 | 1.039 |
| wide-cancellation | 64 | 0.568 | 0.774 | 0.731 | 1.020 | 0.985 |
| wide-cancellation | 1024 | 8.590 | 11.585 | 0.742 | 1.002 | 1.010 |

The final experiment separates twelve controlled cases, with four shuffled arms and five measured rounds. All 240 measurements and 48 warm-ups complete within their five-second/20-million-work allowances; the complete run takes 14.216 seconds. In the table, "bits" is $b$ for an exponent $2^b$. Conjugate rows construct a fresh grammar for $a^{2^b}b^{2^b+3}a^{-2^b}$ and normalize it. Source-replay rows reconstruct an actual source PD and verify a complete supplied proof with a powered prefix; proof discovery is outside the timer. Wide-cancellation rows use $(ab)^{2^b}((ab)^{2^b})^{-1}$ and exercise the unsupported-intermediate fallback. Each result is checked against its known word or full source proof.

The conjugate kernel has paired ratios 1.164–1.331. At $b = 256$, the ratio is 1.262 with both A/A controls 0.985. At $b = 4096$, old/new work is 127,040/110,645 and allocated nodes are 16,396/12,295, but its new A/A ratio 1.139 cautions against attributing the entire 1.164 timing ratio to the implementation. The smallest case is also noisy, with new A/A 0.749.

Complete supplied-proof replay regresses: ratios range from 0.879 to 0.945. At $b = 4096$, old/new separate medians are 319.300/338.978 milliseconds and the paired ratio is 0.916. One bounded replay succeeds, but visiting 16,386 summaries adds enough work that the total rises from 438,720 to 463,311; both versions allocate 20,511 nodes. The earlier powered-prefix operations already populate useful general-reduction caches, so a fresh normalization pass is not automatically a net saving. At $b = 1024$ the old A/A ratio 0.852 also shows substantial timing variation.

Unsupported wide words regress more clearly, with ratios 0.731 and 0.742. At $b = 1024$, both implementations allocate 2,055 nodes, while charged work increases from 15,393 to 21,577. The probe adds no word nodes but still pays to establish that its intermediate summaries are unsupported before calling the general reducer. These retained negative results limit the practical claim: the change supplies a closed-class encoded bound and a faster isolated conjugate kernel, not a universal speed improvement. Projection and forest recognition remain optional.

The reproduction driver is `primitive_power_research/syllables.py` under `../fast`, with `audit`, `benchmark`, `stages` and `kernels` modes. Each result pins the complete prior Python package and records current source hashes before and after execution. Raw shuffled samples, controls, counters, complete certificates and incomplete outcomes are retained. The supplied-prefix timings measure verification of a given proof; they do not measure discovering that proof or recognizing a difficult knot from scratch.

## 88.5 Skipping uniform subtrees and stopping at a rejection witness

The preceding measurements exposed two costs of the first implementation. It descended through every binary-power node even when the arena already knew that the entire subtree repeated one signed letter. It also completed unrelated branches after one unsupported intermediate had made the bounded probe certain to decline. Neither cost is needed to establish the probe's exact

sufficient condition. Both the producer and the separately implemented replayer now avoid them.

Every arena node already has exact raw-word uniformity metadata. A terminal has its signed letter. A nonempty concatenation is uniform precisely when both children have the same nonzero uniform signed letter. The empty word is handled separately. This metadata is maintained by the unchanged arena when a node is created, and reconstructed from source operations in proof replay; no certificate supplies or overrides it.

If a node has uniform signed letter $x$ and expanded length $L$, its run summary is exactly $(|x|, \operatorname{sign}(x)L)$. Moreover, every descendant is uniform in that same signed letter. Skipping those descendants therefore cannot conceal an unsupported intermediate. This is a raw-word assertion: a word that merely *freely reduces* to a power does not receive the shortcut unless the maintained raw metadata proves it. All other nodes retain the independent stack-fold or boundary-deque calculation.

For rejection, define eligibility recursively: terminals are eligible, and a concatenation is eligible exactly when both children are eligible and their freely reduced concatenation has at most $q$ runs. An ineligible descendant makes every ancestor ineligible under this definition, even if later cancellation would produce a small final word. The first such witness therefore suffices to stop the current probe. Its queried root is cached as unsupported at the current run limit, while unvisited siblings receive no cache entry. Encountering a previously cached unsupported descendant uses the same rule. A subsequent query with a different run limit uses its separate cache and re-evaluates the sufficient condition.

Consequently both shortcuts preserve the previous accepted/declined domain exactly. They do not enlarge it or treat a rejection as a negative equality test. All-or-none preflight before word allocation, anchored cyclic trimming, compact output construction and complete source replay remain unchanged. The normalization proof schema and search policy also remain unchanged; finite allowances can still change which calls finish because less work is charged or elapsed.

For one probe, let $F$ count nodes actually encountered before cached or uniform summaries stop descent, or before the first rejection stops the whole query. With fixed out-degree, repeated cached-root hits contribute only the root-slot scan. A conservative summary cost is $O(m + FqB)$, with the same identifier and dictionary qualifications as above. On canonical circuits containing at most $q$ uniform powers per root, $F = O(mq)$: exponent-circuit depth disappears from the summary traversal. Constructing and storing the source grammar still has its original cost, and rebuilding binary powers still has polynomial cost in exponent bit length. Thus this is a query improvement, not a constant-time claim for creating or recognizing an exponentially expanded word.

On unsupported inputs, an early stop can leave unfinished ancestor frames uncached. Different later roots can revisit those frames; no new global "visit every node once" claim is made for rejection paths. The earlier per-phase encoded bounds remain valid. In particular, the normalized monomial class and its conditional depth-dependent bounds are preserved, without supplying the missing global short-depth producer.

Two new tests guard the actual access pattern. A conjugate with exponent $2^{4096}$ forbids access to the descendants of its three uniform powers; both evaluators compute exactly five summaries and three uniform shortcuts. A second test forbids access to an unrelated 1,024-bit-exponent sibling after a five-run left descendant has already failed the four-run limit. It checks repeat queries, an ancestor of the cached rejected root, unchanged word-node allocation, absence of fabricated sibling entries, and successful re-evaluation under a larger run limit. The earlier six normalization tests continue to exercise exact words, arbitrary parses, source independence, fallback and resource guards.

The complete maintained suite passes 1,000 tests with Regina in 235.364 seconds. The audit pins the entire preceding package at `1ea654445325`. All 240 mode results across eighty actual

diagrams are identical, and every one of the 159 positive mode results passes both full source replayers. The exhaustive audit now explicitly compares the previous bounded evaluator as well as independent literal reduction and the new producer/replayer. Its 87,381 words preserve all 66,580 accepted and 20,801 declined probes. The audit takes 17.370 seconds.

| Case | bits | old ms | new ms | ratio | old A/A | new A/A |
|---|---|---|---|---|---|---|
| conjugate | 16 | 0.260 | 0.135 | 1.911 | 0.983 | 0.999 |
| conjugate | 64 | 1.008 | 0.470 | 2.121 | 1.008 | 1.002 |
| conjugate | 256 | 3.885 | 1.830 | 2.127 | 0.998 | 0.996 |
| conjugate | 1024 | 16.253 | 7.470 | 2.171 | 0.979 | 0.971 |
| conjugate | 4096 | 70.150 | 33.540 | 2.104 | 0.982 | 1.021 |
| source-replay | 16 | 2.139 | 1.516 | 1.195 | 1.007 | 1.092 |
| source-replay | 64 | 10.615 | 4.361 | 1.221 | 1.007 | 0.995 |
| source-replay | 256 | 38.597 | 31.080 | 1.244 | 1.015 | 0.995 |
| source-replay | 1024 | 146.655 | 129.474 | 1.133 | 0.912 | 0.994 |
| source-replay | 4096 | 684.159 | 558.246 | 1.243 | 0.969 | 1.029 |
| wide-cancellation | 64 | 1.627 | 1.270 | 1.297 | 0.996 | 1.004 |
| wide-cancellation | 1024 | 12.973 | 9.684 | 1.289 | 0.990 | 0.999 |

The controlled follow-up repeats all twelve kernel/supplied-proof cases, using the preceding *bounded* evaluator as its baseline. It keeps the same five measured rounds, four shuffled arms, one warm-up per arm and five-second/20-million-work limits. All 240 measurements and 48 warm-ups complete in a 21.578-second run. As above, the table reports paired median ratios, and source-replay includes fresh source reconstruction and the entire supplied proof but excludes discovering that proof.

Conjugate ratios range from 1.911 to 2.171, with old/new A/A ranges 0.979–1.008 and 0.971–1.021. At exponent $2^{4096}$ the old/new medians are 70.150/33.540 milliseconds, the paired ratio is 2.104, and work falls from 110,645 to 49,195. Summary nodes fall from 12,295 to five, while allocated grammar nodes remain 12,295 in both versions. The three exact uniform summaries explain the reduced traversal; creating the grammar and rebuilding the normalized power still contribute to the measured time.

Complete supplied-proof replay ratios range from 1.133 to 1.244. At $b = 4096$ the paired ratio is 1.243, with old/new controls 0.969/1.029. Work falls from 463,311 to 365,031, and summaries from 16,386 to seven, including four uniform shortcuts. Both versions still allocate 20,511 nodes and perform the same three compressed equality calls in the whole proof. The source proof's earlier and later operations remain in the timer.

Timing variation remains visible. At $b = 64$, separate old/new medians are 10.615/4.361 milliseconds, but the paired median ratio is only 1.221. At $b = 1024$, the old A/A ratio is 0.912; the smallest supplied-proof case has new A/A 1.092. Raw orders and all measurements are retained. These observations are why the paired samples and exact counters, rather than ratios of independently aggregated medians or comparisons across separate runs, support the reported effect.

Wide-cancellation fallbacks improve by paired ratios 1.297 and 1.289. At $b = 1024$, completed summaries fall from 2,055 to five and work from 21,577 to 16,442, while the 2,055 allocated nodes are unchanged. The five summaries do not mean five traversal steps: the algorithm descends through unfinished ancestors before reaching its rejection witness. Relative to the older *unprobed general reducer* in the preceding experiment, which charged 15,393 work units on this same grammar, a probe still adds overhead. The change reduces that overhead; it does not eliminate it or justify multiplying timing ratios from different runs.

| Input | pair old/new ms | ratio | forest old/new ms | ratio |
|---|---|---|---|---|
| survivor-00 | 19.109/19.282 | 0.991 | 19.327/19.984 | 0.937 |
| survivor-01 | 16.108/16.177 | 0.998 | 16.063/15.840 | 1.007 |
| survivor-02 | 17.418/17.428 | 1.007 | 17.420/17.365 | 1.003 |
| survivor-03 | 12.121/12.103 | 1.003 | 12.159/12.098 | 0.996 |
| mirror-03 | 26.433/26.068 | 1.017 | 26.517/26.143 | 1.017 |
| survivor-04 | 11.052/10.728 | 1.042 | 10.816/10.992 | 1.000 |
| survivor-05 | 20.458/20.482 | 0.987 | 20.630/20.959 | 1.002 |
| survivor-06 | 8.516/8.548 | 0.994 | 8.696/8.440 | 1.025 |
| survivor-07 | 21.294/21.308 | 0.998 | 21.358/21.377 | 0.994 |
| survivor-08 | 8.683/8.676 | 1.006 | 8.692/8.689 | 0.998 |
| mirror-08 | 22.243/22.323 | 1.001 | 22.194/22.422 | 0.990 |
| survivor-09 | 8.426/8.501 | 0.997 | 8.460/8.468 | 1.006 |
| survivor-10 | 7.407/7.271 | 1.000 | 7.476/7.424 | 1.007 |
| survivor-11 | 8.012/7.884 | 1.006 | 7.867/7.951 | 0.996 |
| gordian | 1624.589/1632.275 | 1.006 | 1627.165/1634.281 | 0.989 |
| trefoil | 0.054/0.055 | 0.993 | 0.055/0.054 | 1.006 |
| figure_eight | 0.062/0.062 | 0.997 | 0.061/0.061 | 0.998 |
| conway | 1.017/1.005 | 1.013 | 1.010/0.999 | 1.007 |
| kinoshita_terasaka | 1.045/1.066 | 1.004 | 1.063/1.054 | 1.009 |

The complete-recognition follow-up retains nineteen diagrams, five measured rounds, eight shuffled arms and A/A controls for both old and new pair/forest modes. All 760 measurements and 152 warm-ups complete in 90.061 seconds. Pair-mode ratios range from 0.987 to 1.042, and forest-mode ratios from 0.937 to 1.025. The old/new pair controls span 0.985–1.028 and 0.980–1.020; the old/new forest controls span 0.990–1.034 and 0.987–1.053. The complete-call experiment does not establish a broad recognition gain.

Gordian remains the sole input in this corpus with a long-root producer probe, and its probe still declines. Completed summaries fall from 51 to 40, including seven uniform shortcuts. Pair/forest work falls by only 60 units to 1,267,198/1,267,196. Both modes retain 9,084 nodes and exactly the same proofs, respectively 10,020 and 10,060 bytes. Their paired timing ratios are 1.006 and 0.989, while their new A/A ratios are 0.980 and 1.021. The savings are too small a fraction of the whole search to support a speedup claim from these timings. The supplied huge-prefix proof and isolated word timings establish narrower benefits, with no additional solved diagram or general quasipolynomial discovery theorem. Both recognition modes remain optional.

Run the reproduction driver from `../fast`. The file `primitive_power_research/syllable_frontier.py` supports the `audit`, `benchmark` and `kernels` commands and retains source hashes and baseline package hashes. The prior circle-stage timings all stopped at the unchanged short length gate, so that stage was not rerun as evidence for these deeper traversal shortcuts. No new unknot delivery was present during this check; the `openai/math` head remained `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`.

## 89 Acyclic elimination with nonmonomial word images

Primitive forests replace generators by powers of individual surviving generators. A more general batch can replace a generator by an arbitrary word, provided each defining relator contains that generator exactly once and the chosen definitions have no cyclic dependencies. The resulting implementation compiles a linked word circuit once, rather than expanding definitions or repeatedly normalizing after individual eliminations. This broadens the available raw batch operation. It does not prove that arbitrary knot diagrams admit a short sequence of such batches.

## 89.1 The simultaneous Tietze transformation

Let a selected relator be $Lx^{\varepsilon}R$, where $\varepsilon \in \{1, -1\}$ and neither $L$ nor $R$ contains $x$ or $x^{-1}$. Its equation is equivalent to

$$x = W_x, \qquad W_x = \begin{cases} L^{-1}R^{-1}, & \varepsilon = 1, \\ RL, & \varepsilon = -1. \end{cases}$$

Different chosen generators must use different original relator slots. Draw an edge from $x$ to each other chosen generator occurring in $W_x$. Assume this graph is acyclic. In dependency order, replace every chosen letter in $W_x$ by its already resolved word. This defines a homomorphism $\phi$ from the old free group to the free group on the surviving generators. It fixes every survivor.

Repeated elementary Tietze elimination in that dependency order proves

$$\langle X \mid \mathcal{R} \rangle \cong \langle X \setminus E \mid \phi(r),\ r \text{ in an unselected original slot} \rangle .$$

Each selected equation becomes trivial under $\phi$, which justifies clearing exactly its selected slot. Every other slot, including empty or repeated roots, is retained. A cycle is not allowed: simultaneous equations $x = y$, $y = x$ do not justify deleting both generators. The source-replay checker therefore validates the complete dependency graph before applying the quotient.

This theorem is valid for arbitrary presentations and uses no torsion-freeness hypothesis. A knot verdict still requires the validated source diagram and replay of the entire presentation prefix. The existing rank-one exponent endpoint and rank-two primitive-power endpoint retain their source-dependent interpretation. The local batch helper alone is not a knot-recognition API.

## 89.2 Planning and compiling the raw circuit

The producer caches presence and repetition bit masks at immutable word nodes. At a concatenation, presence is union, and repetition is the union of child repetitions and their common presence. Thus a present but unrepeated generator occurs exactly once, without inspecting expanded letters. Generator bit positions remain fixed throughout a search; current slots and live membership are checked afresh.

Candidates retain the earlier singleton cost ordering: relation length and total generator occurrence count determine the primary score, with original slot and generator order breaking ties. The greedy planner picks at most one definition per generator and one per donor slot. Before adding the outgoing dependencies of a new chosen generator, it checks whether any dependency already reaches that generator. This rejects cycles even when their other edges came from earlier selections. It leaves at least one generator alive. Search uses a batch only when at least two generators are selected and therefore skips planning below rank three. No optimality or general rank-halving property is asserted for this greedy selection.

For each selected donor the producer finds its unique occurrence and constructs $W_x$ by exact substring and inverse operations. These are raw circuits; no free reduction is required. It then links terminal $x$ to the circuit for $W_x$, and terminal $x^{-1}$ to its inverse. Original word concatenations and these new definition links form an acyclic graph. An iterative postorder evaluation maps each encountered source node once, building shared output concatenations. The word circuit may have forward links relative to its old allocation order, so merely scanning old node identifiers would be incorrect. The dependency graph supplies the missing topological justification.

If another batch is available, search continues on raw roots. Otherwise it records the existing explicit normalization boundary before returning to positional overlap and Whitehead moves. A terminal can instead close the raw state. Nonmonomial definitions can create many power runs, so the bounded-run closure theorem of Section 88 does not automatically extend to these normalized phases.

## 89.3 Version eight and independent replay

Section 95 subsequently adds an ordered-witness fast path with scalar summaries and virtual inverses. The graph-based replay described here remains the complete unordered fallback.

A version-eight `elimination_batch` move contains a nonempty `entries` list. Each entry has exactly its original relation slot and eliminated generator. The checker accepts no caller-provided word images, occurrence summaries, dependency graph or topological order. It checks strict integer types, live membership, valid distinct slots, distinct generators and preservation of at least one live generator.

The initial compressed replay independently derives every unique occurrence and its donor support from the reconstructed words. It builds its own definitions and checks their dependency graph with a queue-based topological traversal. It then evaluates the linked word circuit with a second Kahn traversal, using incoming-edge counts and reverse adjacency, rather than the producer's postorder mapping. No producer planner, metadata or application routine is imported. Both evaluators implement the same proved quotient while preserving separate word-processing code.

Literal replay derives the equations directly from lists. Before expanding any substituted definition, it computes all resolved image lengths in dependency order and the total retained-output length. It checks and charges the total storage for positive and negative images and retained words before allocating them. Large compressed certificates can therefore exhaust literal replay while remaining inexpensive to check with word circuits. Failure does not turn into a verdict. Both replay paths propagate external cancellation and retain all existing resource limits. Legacy versions remain accepted, and a version-eight move is rejected in an older wrapper.

## 89.4 Encoded cost of a batch and conditional depth bounds

The following conservative bounds describe the original analysis. Section 92 strengthens the singleton-only bound to linear grammar growth per batch and $2^{O(d)}$ composition, using the disjointness of selected occurrence paths. It separately proves polynomial cost for complete fixed-source monomial epochs.

Let $k$ be the number of chosen definitions, $M$ the historical grammar size, $H \le M$ its maximum reachable height, $r$ the live rank and $B \ge 1$ the maximum represented-length bit size. Each donor image is a substring rotation with a possible inverse. Exact slices add $O(H)$ nodes per donor; shared inverses add at most a constant multiple of the nodes they invert. The linked-circuit evaluation then allocates at most one concatenation per needed source concatenation. A conservative resulting storage bound is

$$M' = O(M + kH + k) = O((k+1)(M+1)).$$

Occurrence scans, dense support masks, greedy cycle checks, slicing, topological replay and integer length arithmetic are polynomial in the encoded input parameters. Separate donor scans may revisit source nodes; the implementation does not claim one linear metadata pass for all replay checks. Identifier encoding and dictionary-access qualifications are the same as for the preceding compressed operations.

If $L$ bounds an old expanded length, a resolved definition along a chain of at most $k$ selected generators has length at most $\max(2, L)^k$. A retained relator has length at most $\max(2, L)^{k+1}$. Hence

$$B' = O((k+1)(B+1)).$$

Together these estimates give polynomial encoded cost for an entire batch, independent of the expanded output size and with polynomial dependence on dependency depth. For $d$ raw phases, allowing both these and the earlier monomial batches, a conservative composition bound is

$$\mathrm{poly}(\text{initial encoded parameters})\, 2^{O(d \log(r_0+1))}.$$

If each successful phase removed a fixed fraction of the remaining rank, there would be $O(\log r_0)$ phases and this part would be quasipolynomial. That rank-reduction hypothesis, arrival at a checked endpoint, and the cost of arbitrary intervening normalization or exposure remain unproved for general diagrams. The new batch closes an encoded-operation gap, not the global discovery gap.

The capacity family is explicitly nonmonomial. With two surviving letters $a, b$, define each successive generator by

$$x_i = (x_{i-1} a x_{i-1}^{-1} b)^{2^s}, \qquad x_0 = a.$$

The raw length recurrence is $\ell_i = 2^s(2\ell_{i-1} + 2)$, $\ell_0 = 1$. Eight levels with $s = 128$ give 1,034-bit output lengths and grow the arena from 1,076 to 3,959 nodes. Sixty-four levels with $s = 64$ grow 4,484 nodes to 17,215 and give 4,162-bit lengths. At sixty-four levels and $s = 1024$, 65,924 nodes grow to 261,055, while the length needs 65,602 bits. These node counts include the empty node. Separately initialized producer and checker arenas give identical structural root digests and exact lengths. These are presentation-capacity checks with two surviving generators, not difficult knot instances or whole-recognition timing claims.

## 89.5 A capped adaptive trial protects the established search

Changing elimination order can make later exposure harder even when the batch itself is cheap and valid. On the Gordian source, direct batch search eliminates 127 generators in two batches but exhausts a two-million-work allowance. The prior search succeeds. Thus soundness and fewer elimination rounds do not imply faster complete recognition.

The first public host treated this new schedule as a bounded trial. Its original policy, measured below, is refined in Section 89.7. With producer allowance $W$, it gives the trial $\min(\lfloor W/4 \rfloor, 50000)$ work units. On a nondecision or local resource exhaustion it reconstructs the original diagram's presentation and invokes the preceding compressed policy. The fallback receives $W$ minus the work charged or reserved for the trial. On an exception the entire trial quota is reserved at minimum, because initial presentation recovery can fail before arena counters are updated. A final aggregate charge can cross the nominal allowance; its reported cost is also deducted. The host retains the trial statistics even if fallback later exhausts its remaining allowance.

Both attempts share the existing wall deadline, and caller cancellation propagates rather than triggering a restart. Mandatory independent replay still checks any returned certificate under its existing verifier allowance. The restart uses the original source, not the modified batch state whose subsequent search stalled. This preserves the useful alternative search order at a bounded cooperative-work cost; it does not promise identical coverage under every finite work or time limit, nor constant bit-time per charged operation.

The `group_decide` API accepts `elimination_batch=True`.

For `recognize`, enable `use_group=True` and `group_elimination_batch=True`.

The CLI flag is `-group-elimination-batch`. The option implies compressed search and replay and is incompatible with the older explicit-to-compressed adaptive-search flag. It defaults to false and can be combined with primitive projection/forest options. The lower-level `compressed_certificate` API also accepts `elimination_batch=True`. It exposes direct batch search with the caller's full allowance; its result is a certificate, not an independently verified verdict.

## 89.6 Validation and measurements

Eight focused tests cover one hundred random dependency DAGs with exact literal homomorphism oracles; positive and negative occurrences; cyclic donor rotations; empty definitions; unrelated

and repeated retained slots; deep nonmonomial circuits; strict fields, cycles, duplicate donors and generators; source rebinding; independent replay; preflight allocation; CLI dispatch; cancellation; and original-source fallback accounting. The full maintained suite passes 1,008 tests with Regina in 213.793 seconds.

Against the full pinned package at `3346396cd135`, eighty actual diagrams preserve all 240 results in the three prior modes. Every positive proof passes both literal and compressed source reconstruction and replay. Direct batch search returns 52 positive certificates, 47 of them version eight; it loses Gordian relative to the old 53-positive set. The capped host returns all 53 prior positives, with no new positive in this sample. The audit, including the capacity checks, takes 13.718 seconds.

| Input | old ms | default ms | portfolio ms | old/portfolio |
|---|---|---|---|---|
| survivor-00 | 16.692 | 16.581 | 14.984 | 1.111 |
| survivor-01 | 15.238 | 14.444 | 29.695 | 0.490 |
| survivor-02 | 15.715 | 15.770 | 10.167 | 1.546 |
| survivor-03 | 10.952 | 11.000 | 42.003 | 0.261 |
| mirror-03 | 23.779 | 23.696 | 6.198 | 3.844 |
| survivor-04 | 9.709 | 9.664 | 31.180 | 0.313 |
| survivor-05 | 19.122 | 19.008 | 16.240 | 1.178 |
| survivor-06 | 7.385 | 7.393 | 5.547 | 1.342 |
| survivor-07 | 19.226 | 19.351 | 20.832 | 0.919 |
| survivor-08 | 8.041 | 7.807 | 7.480 | 1.075 |
| mirror-08 | 19.524 | 19.472 | 16.103 | 1.216 |
| survivor-09 | 7.517 | 7.465 | 5.800 | 1.312 |
| survivor-10 | 6.507 | 6.465 | 5.405 | 1.202 |
| survivor-11 | 7.046 | 7.046 | 5.169 | 1.361 |
| gordian | 1476.434 | 1461.754 | 1490.813 | 0.991 |
| trefoil | 0.058 | 0.056 | 0.060 | 1.011 |
| figure_eight | 0.061 | 0.061 | 0.060 | 1.020 |
| conway | 0.978 | 1.024 | 0.983 | 1.002 |
| kinoshita_terasaka | 1.000 | 1.008 | 0.997 | 1.003 |

The complete-recognition experiment uses nineteen actual diagrams with six shuffled arms: prior default, current default and optional portfolio, each with an A/A control. Five measured rounds and one warm-up per arm give 570 measurements and 114 warm-ups; all complete. The run takes 61.994 seconds. Displayed ratios are medians of paired old/new times, not ratios of the displayed separate medians. Values above one favor the new mode.

Current-default ratios range from 0.983 to 1.043; old/current default A/A ranges are 0.934–1.053 and 0.988–1.029. The new optional portfolio has a much wider range, 0.261–3.844, while its A/A controls span 0.983–1.024. These are real tradeoffs in search order, not a uniform speed improvement. Mirror 03 improves from 23.779 to 6.198 milliseconds, with paired ratio 3.844. It drops from 19 moves, 615 nodes and 25,431 work units to two moves, 212 nodes and 6,352 work units; its proof shrinks from 1,491 to 774 bytes.

The corresponding survivor 03 moves in the opposite direction. Its 10.952/42.003-millisecond medians give paired ratio 0.261. One batch eliminates ten generators, but subsequent search grows the complete proof from 12 to 18 moves, nodes from 267 to 1,145, and work from 11,980 to 36,797. Proof bytes rise from 970 to 1,709. Survivor 01 and survivor 04 also regress, with ratios 0.490 and 0.313. A successful bounded trial can still be slower than the untried baseline, and the work quota is not a theorem of competitive elapsed time.

Gordian takes fallback and retains exactly the old 140-move, 9,663-byte certificate and 9,025-node successful search. The failed trial is charged 50,001 work units, raising total producer work from 1,215,145 to 1,265,146. Complete-call medians are 1,476.434/1,490.813 milliseconds with paired ratio 0.991. The guard prevents the direct batch schedule's loss of this positive, but adds work; no broader coverage claim follows.

| Crossings | old ms | forest ms | direct ms | portfolio ms | old/direct | old/port. |
|---|---|---|---|---|---|---|
| 8 | 2.069 | 1.729 | 1.380 | 1.369 | 1.485 | 1.511 |
| 16 | 6.753 | 4.149 | 2.843 | 2.813 | 2.375 | 2.401 |
| 32 | 24.728 | 11.010 | 6.088 | 5.892 | 4.017 | 4.118 |
| 64 | 92.249 | 33.250 | 12.934 | 12.561 | 7.141 | 7.275 |
| 128 | 351.086 | 110.656 | 30.048 | 365.523 | 11.684 | 0.971 |
| 256 | 1422.297 | 390.203 | 74.867 | 1416.740 | 19.228 | 1.001 |

The separate checked group-stage experiment uses six actual stabilized circles from eight through 256 crossings. Eight shuffled arms compare prior default, prior primitive forest, direct batch search and the capped portfolio, each with an A/A control. All 240 measured calls and 48 warm-ups complete in 53.353 seconds. Every timing includes a fresh validated PD, certificate production and mandatory full source replay. Earlier diagram simplification is deliberately bypassed; these circles are not difficult whole-recognition instances.

Direct batching uses one move on every circle. At 128 crossings, it eliminates 127 generators in one batch, replacing 126 earlier moves. Old/forest/direct medians are 351.086/110.656/30.048 milliseconds; paired direct ratios are 11.684 against the prior default and 3.665 against the primitive forest. Producer work falls from 493,856 to 62,591 and nodes from 24,891 to 1,264. The direct proof has 6,247 bytes versus 8,640 for the old default and 24,286 for the forest.

At 256 crossings the corresponding medians are 1,422.297/390.203/74.867 milliseconds, with paired direct ratios 19.228 and 5.178. Direct work is 157,823 versus 1,905,056, and nodes are 2,544 versus 98,939. The 255-generator batch retains a 12,775-byte proof, compared with 17,600 bytes for 254 old moves. Across the stages, old, forest, direct and portfolio A/A ranges are respectively 0.989–1.017, 0.990–1.012, 0.991–1.043 and 0.993–1.012.

The conservative host cap sacrifices those larger-stage gains. The 128- and 256-crossing direct producers require more than 50,000 work units, so the portfolio abandons them and repeats the prior policy. At 128 crossings its paired ratio is 0.971 and work is 543,857. At 256 crossings its ratio is 1.001 despite increased work of 1,955,056; timing variation does not remove that charged overhead. At 64 crossings the direct producer fits the cap, and the portfolio has paired ratio 7.275 over the old stage. Both the successful smaller trials and the lost larger opportunities are retained. The cap is an engineering policy, not a scale-independent characterization of when batching is advantageous. All new recognition options remain disabled by default.

The driver is `primitive_power_research/elimination.py` in `../fast`, with `audit`, `benchmark` and `stages` modes. Each run pins the complete prior package, checks 138 current source hashes before and after execution, and retains full proofs, shuffled sample orders, controls, work counters and incomplete outcomes. The circle stages bypass earlier simplification and cannot be reported as whole-knot speedups.

## 89.7 Exact reachability and a trial allowance after recovery

Two costs limited the first batch implementation: repeated graph traversal while planning, and an adaptive trial that could exhaust its entire allowance before rebuilding a moderately large source presentation. The follow-up preserves the greedy witness order and independent checker, but caches exact reachability within each planning state and separates the continuation allowance from presentation recovery. This changes resource use and some finite-budget outcomes, not the certified Tietze operation.

**Incremental cycle checks.** At a planning state, a vertex has outgoing edges exactly when its definition has already been selected. Let $R(v)$ be the set of vertices reachable from $v$ by a nonempty directed path. Store these sets as dense integer masks and store reverse adjacency for

the selected edges. Both structures are fresh for each planning state; only the earlier immutable word summaries persist across states.

Suppose the next candidate defines $g$ using the dependency set $P$. The generator $g$ has no previous outgoing edges. Form

$$D = \bigcup_{p \in P} (\{p\} \cup R(p)).$$

Adding the candidate creates a directed cycle if and only if $g \in D$. Reject it in that case. Otherwise set $R(g) = D$, add the reverse edges, and propagate $D$ backwards through the old ancestors of $g$. An ancestor already containing all of $D$ needs no further propagation.

For completeness, every new path must use one of the new edges from $g$. Acyclicity prevents it from revisiting $g$, so its prefix is an old path to $g$ and its suffix ends in $D$. Thus only old ancestors gain reachable vertices, and all their new vertices belong to $D$. Conversely, every old ancestor reaches all of $D$ after the update. If an ancestor $v$ already contains $D$, each old ancestor of $v$ already contains it too, which justifies pruning that propagation branch. Induction on accepted candidates therefore maintains exact reachability. Since scoring, tie-breaking and slot checks are unchanged, completed unconstrained runs select exactly the same greedy batch as the earlier fresh depth-first tests.

Let $r$ be the number of mask positions, $k$ the accepted definitions and $E$ their total dependency edges. If $\sum \delta$ counts dependency entries examined in candidate queries, the query masks take $O(\sum \delta)$ mask operations. During one accepted update each ancestor changes at most once; reverse edges are then visited at most once for each changed endpoint. Across the state, $O(kE)$ propagation visits are a conservative bound. Allowing $O(r)$ bit cost per dense operation gives

$$O((\sum \delta + kE)r)$$

for this reachability work, in addition to the unchanged candidate discovery, sorting and word summaries. The closure requires $O(r^2)$ bits and reverse adjacency $O(E \log(r+1))$ bits, apart from container overhead. In a favorable ordered chain each newly defined vertex has no old ancestors, so propagation vanishes and the number of graph traversal steps becomes linear. The masks can still contain $r$ bits; this is not a linear bit-time claim. Neither an arena-node cap nor a cooperative-work counter alone bounds total process memory or bit operations.

**Recovery-aware adaptive accounting.** The initial source pass reconstructs, charges and reduces the presentation before batch planning starts. Its existing conservative accounting includes block charges for short literal reductions. The old 50,000-work cap can therefore fail before any search on the 256-crossing circle. The follow-up does not change these accounting units or the reconstruction algorithm.

For total producer allowance $W$, the host now gives the trial a total allowance $Q = \lfloor W/4 \rfloor$. After successful source reconstruction and initial root reduction have consumed $R$ work units, the additional search allowance becomes

$$S = \min(50000, Q - R).$$

The total cap remains in force. A marker records $R$ only after all recovery work has been charged to the arena. On subsequent local exhaustion the host deducts the observed trial cost before restarting the established policy from the original source. If recovery fails before this marker exists, it conservatively reserves the whole $Q$, or the reported cost if larger. Aggregate work charges may cross a nominal limit; that reported excess is also deducted. Both attempts retain one wall deadline, external cancellation propagates, and mandatory independent replay keeps its existing separate allowance. Thus the second search receives only the remaining producer work.

The lower-level producer accepts an optional `post_recovery_work` argument. Its default is `None`, preserving the direct caller's full search allowance. The host supplies 50,000. All public batch

flags remain optional and false by default. Increasing the trial's total cap can spend more work before fallback, so this policy is not uniformly faster or uniformly stronger under finite limits.

**Validation and scope.** Twelve focused tests pass in 0.693 seconds; the maintained suite passes 1,012 tests with Regina in 215.420 seconds. New tests exercise late ancestor updates, rejected cycles, changed root order and live sets, both work caps, strict argument types, and recovery-failure versus search-failure accounting. Seven hundred test comparisons and 2,250 audit comparisons agree with an independent literal greedy planner; the audit also compares the pinned old planner on every state.

The baseline is the complete package at `47e87a72c23d`. The eighty source diagrams retain all 240 results in the three earlier modes, and all eighty direct-batch outcomes, including positive certificate hashes, agree with the pinned producer. Every positive audit certificate passes both literal and compressed source replay. Application, presentation recovery and verifier functions are checked for identical source, as are the word arena and independent compressed replay modules. The audit takes 19.573 seconds. Separate 128-, 256- and 512-crossing source trials now finish without fallback at two million total producer work units; these additional cases are not folded into the common eighty-diagram coverage claim.

**Complete recognition.** The controlled follow-up uses nineteen fresh source inputs, eight arms (old and current defaults and batch portfolios, each with an A/A copy), one warm-up and five shuffled measured rounds. All 760 measured calls and 152 warm-ups complete; the run takes 84.716 seconds. Timing includes the entire recognition call and mandatory source replay. Ratios are medians of paired old/new times, not quotients of the marginal time medians.

| Input | default ms | old port. ms | new port. ms | old/new | default/new |
|---|---:|---:|---:|---:|---:|
| survivor-00 | 17.192 | 15.347 | 15.811 | 0.993 | 1.115 |
| survivor-01 | 17.096 | 46.533 | 41.473 | 0.967 | 0.487 |
| survivor-02 | 15.871 | 10.243 | 10.323 | 0.992 | 1.527 |
| survivor-03 | 11.346 | 43.427 | 43.755 | 0.988 | 0.259 |
| mirror-03 | 24.675 | 6.440 | 6.455 | 1.009 | 3.830 |
| survivor-04 | 10.105 | 31.943 | 31.816 | 1.007 | 0.319 |
| survivor-05 | 19.303 | 16.361 | 16.452 | 0.992 | 1.173 |
| survivor-06 | 7.636 | 5.634 | 5.761 | 0.983 | 1.326 |
| survivor-07 | 19.798 | 21.842 | 21.486 | 1.003 | 0.921 |
| survivor-08 | 7.751 | 7.432 | 7.554 | 0.983 | 1.026 |
| mirror-08 | 20.526 | 16.625 | 16.524 | 1.009 | 1.243 |
| survivor-09 | 7.602 | 5.801 | 5.852 | 0.995 | 1.304 |
| survivor-10 | 6.640 | 5.444 | 5.532 | 0.992 | 1.185 |
| survivor-11 | 7.126 | 5.096 | 5.174 | 0.982 | 1.374 |
| gordian | 1518.937 | 1509.908 | 1623.636 | 0.941 | 0.936 |
| trefoil | 0.052 | 0.053 | 0.053 | 1.004 | 0.995 |
| figure_eight | 0.060 | 0.060 | 0.059 | 1.003 | 1.013 |
| conway | 0.990 | 1.006 | 0.988 | 1.018 | 1.002 |
| kinoshita_terasaka | 1.028 | 1.023 | 1.019 | 1.002 | 1.003 |

The default old/current ratios range from 0.980 to 1.023. Old/new portfolio ratios range from 0.941 to 1.018, so this sample establishes no broad whole-input improvement. The four A/A ranges are respectively 0.987–1.016, 0.987–1.031, 0.986–1.016 and 0.967–1.038. Against the current default, the new optional portfolio still ranges from 0.259 on survivor-03 to 3.830 on mirror-03. Those large gains and regressions come from the alternative elimination schedule, already present in the baseline, and should not be attributed to this reachability optimization.

Gordian exposes the price of the larger trial. Recovery costs 39,904 work units, after which the new search spends its continuation allowance and falls back. Its trial charge is 89,905 rather than

50,001; total producer work increases from 1,265,146 to 1,305,050. The fallback proof remains the same 9,663-byte version-five certificate and its own search still uses 9,025 nodes. Whole-call medians are 1,509.908 ms for the old portfolio and 1,623.636 ms for the new one, with a paired ratio of 0.941 and a new-portfolio A/A ratio of 1.038. The extra speculative work is real even though noise affects its timing estimate. Optional status and the existing default are retained.

**Checked group stages on larger circles.** A separate run bypasses earlier simplification and compares the old and new direct producers and portfolios, each with an A/A copy. Every successful call includes source reconstruction, production and mandatory compressed replay. Seven circle diagrams, one warm-up and five shuffled measured rounds give 280 measured calls and 56 warm-ups in 45.563 seconds. Of these, 270 measurements and 54 warm-ups complete. All ten incomplete measurements and both incomplete warm-ups are the old portfolio at 512 crossings, where fallback hits the unchanged 100,000-node cap. They remain in the evidence; no old/new completion-time ratio is reported for that portfolio case.

| Crossings | direct old/new ms | ratio | portfolio old/new ms | ratio |
|---|---|---|---|---|
| 8 | 1.409/1.430 | 0.977 | 1.363/1.382 | 0.986 |
| 16 | 2.920/2.882 | 1.021 | 2.818/2.824 | 1.004 |
| 32 | 6.141/5.797 | 1.060 | 5.915/5.629 | 1.051 |
| 64 | 13.458/12.217 | 1.128 | 13.218/12.003 | 1.075 |
| 128 | 30.760/24.467 | 1.288 | 377.400/23.901 | 15.998 |
| 256 | 74.892/49.422 | 1.515 | 1503.725/50.714 | 30.105 |
| 512 | 227.287/101.014 | 2.326 | –/97.486 | – |

At 256 crossings the direct checked call improves from 74.892 to 49.422 ms, with paired ratio 1.515. Producer work drops from 157,823 to 94,320 while both use 2,544 nodes and the same 12,775-byte certificate. At 512 crossings the direct ratio is 2.326, work drops from 446,591 to 188,528, and both use 5,104 nodes and the same 25,879-byte certificate. The producer removes 255 or 511 generators in one batch. The reduction is entirely in planning: these chains require no propagation through old ancestors.

The revised host recovers the direct batch gains previously forfeited by its trial budget. At 128 crossings its paired ratio against the old host is 15.998; at 256 it is 30.105. At 256 crossings, recovery consumes 72,677 units and subsequent search 21,643, within the new continuation allowance. At 512 crossings, recovery consumes 145,125 units and search 43,403; the new host completes in a median 97.486 ms without fallback. The old host returns no verdict at its node limit, so this last result demonstrates a finite-budget coverage difference, not a measured speedup factor.

The stage A/A ranges are 0.979–1.019 for old direct, 0.830–1.023 for new direct, 0.986–1.031 for completed old portfolios and 0.975–1.030 for new portfolios. In particular the 128-crossing new-direct control is 0.830, so its smaller direct timing gain should be treated cautiously. These controlled stages show where the mechanism helps, while the complete-call corpus above records the cost and absence of a broad improvement. Neither set establishes an asymptotic bound for general knot recognition.

Reproduce the evidence using `primitive_power_research/elimination_reach.py` in `../fast`, with `audit`, `benchmark` and `stages` modes. Each run records 140 current source hashes before and after execution, the pinned baseline, complete certificates, work counters, shuffled orders, A/A controls and all incomplete outcomes. The optimization retains the conditional raw-phase theorem above; neither the global depth bound nor a general quasipolynomial recognizer follows from these measurements.

# 90 Memoized frontiers for exact word inversion and reduction

The compressed word engine already caches exact inverses and freely reduced words. Its earlier traversal nevertheless collected every reachable source node before consulting these caches during evaluation. Repeated queries on slightly extended words could therefore revisit a large cached subgrammar. The new traversal stops at known results. It changes neither the equations for inversion and free reduction nor the independently implemented group certificate moves, and applies to both search and compressed replay.

## 90.1 A cached node is a complete boundary condition

Let an immutable binary word grammar have terminals and concatenation rules $X = YZ$, with children allocated before their parent. For one operation let $K(X)$ store its exact completed result: either $X^{-1}$ or the freely reduced word represented by $X$. Starting from a queried root, visit only nodes absent from $K$. Stop at each cached child, and deduplicate uncached nodes encountered along different paths. Sort the collected nodes by allocation identifier and then evaluate them in that order.

For inversion the rule is

$$K(X) = K(Z)K(Y), \qquad K(a) = a^{-1}.$$

The concatenation is raw; inversion itself performs no cancellation. For free reduction, terminals are already reduced, and the rule is

$$K(X) = \operatorname{red}(K(Y)K(Z)).$$

The existing exact boundary-cancellation operation evaluates this product using compressed longest common prefixes, inversion and slicing. The new traversal does not replace any of those algebraic operations.

When a collected node is evaluated, each child either lay on the cached frontier or was collected earlier in the allocation order. Both therefore have exact results. Induction proves the computed result exact. A node that becomes cached as a side effect of processing another node can also be skipped: the same invariant applies to every published cache entry.

Crucially, a cached node need not have cached descendants. A substring of a known reduced word is reduced, so slicing can certify its new root without individually reducing the intermediate concatenations created by the slice. The bounded-run evaluator can similarly certify a rebuilt reduced root. Only that root's result is needed when reducing an ancestor. Descending below it would compute answers that are unnecessary for the current query.

The discovery pass does not publish cache entries. Evaluation publishes a result only after its exact operation completes. Cancellation, work exhaustion or node exhaustion therefore leaves only completed entries, which remain valid on a later query. The existing resource exceptions and caller cancellation still propagate; no partial traversal is treated as a proof. The old full reachable-node traversal remains available for operations that actually need every source node, such as occurrence counts and cold substitution. Section 91 adds an exact support frontier when substitution metadata is already warm.

## 90.2 Amortized traversal cost and its limits

For completed query $i$, let $u_i$ be the number of previously uncached nodes collected above its frontier. At most $1 + 2u_i$ stack entries are examined, since each collected binary node adds two children. Sorting costs $O(u_i \log(u_i + 1))$ identifier operations. Each collected node has a cache

entry by the end of the successful query, and cache keys are never removed. Thus, for $q$ completed calls and $U$ source nodes first collected over their entire sequence,

$$\sum_i u_i \le U, \qquad T_{\text{traversal}} = O(q + U \log(U+1)).$$

This is a separate bound for each of the inverse and reduction caches. It allows the arena to grow between calls; $U$ can be bounded by its eventual historical node count. Identifier encoding and dictionary-access costs retain the qualifications of the earlier word-engine analysis. With at most $M$ historical nodes, each identifier needs $O(\log(M+1))$ bits. Temporary traversal storage is $O(\max_i u_i)$ such identifiers. The two persistent result caches were already present; this change adds no per-node semantic cache.

An interrupted discovery can be repeated, so the amortized statement excludes repeated aborted attempts. It also excludes the algebraic cost of computing new inverses or reductions, exact big-integer arithmetic, and all other word operations. In particular it does not bound the total grammar generated by an arbitrary knot search. A growing chain with each previous prefix already cached needs one new source concatenation per query, rather than rescanning all earlier prefixes. This removes quadratic traversal on that family without asserting linear bit time for the complete compressed-word computation.

The exact word represented by every result is preserved, but unused cache entries and intermediate allocations need not match the old implementation. Some producer heuristics inspect arena size, and resource budgets count work. Consequently universal equality of finite-budget search outcomes or traces is not claimed. The source-bound audit below checks the maintained modes against the entire pinned prior package.

## 90.3 Validation and measurements

Six focused tests cover two hundred random streams of mixed word operations; cached substrings with uncached descendants; 2,400-deep growing grammars; 2,050-bit represented lengths; interrupted discovery and evaluation; and work/node exhaustion. All focused tests pass in 0.238 seconds. The maintained suite passes 1,018 tests with Regina in 205.119 seconds. The exhaustive word audit checks every signed rank-two word through length seven: 21,845 words and 65,535 inverse, free-reduction and cyclic-reduction comparisons against literal stacks and the pinned prior engine.

Against baseline `4cef49dffbc3`, eighty actual diagrams preserve all 400 outcomes across the default, projection, forest, direct batch and adaptive batch modes, including positive certificate hashes. Each positive is replayed with both the old and current packages, using both literal and compressed reconstruction. Five additional supplied source proofs cover Whitehead and projection prefixes, including an exponent $2^{1024}$; modified source bindings are rejected. All production modules other than the shared word engine are checked for identical source. The audit takes 20.225 seconds.

On Gordian the default producer's work falls from 1,215,145 to 1,154,597 with the same certificate and 9,025 nodes. The adaptive host still falls back and its total work falls from 1,305,050 to 1,244,502. Direct batching still exhausts the two-million-work allowance. These counters isolate saved work; wall-time measurements and their controls are reported separately below.

**Word kernels and supplied-proof replay.** The controlled kernel run has fifteen cases, four shuffled arms (old/current and their A/A copies), five measured rounds and one warm-up. All 300 measured calls and 60 warm-ups complete in 58.503 seconds. Timings for word queries include construction of a fresh arena and its initial word. Bounded literal assertions are performed after the timed call. Source-proof timings instead include fresh diagram validation and full independent

reconstruction/replay of a fixed supplied proof; proof construction is outside that timing. All ratios below are medians of paired old/new times.

| Family | size | old ms | new ms | ratio | old/new A/A |
|---|---|---|---|---|---|
| inverse-chain | 64 | 1.780 | 0.371 | 4.820 | 1.016/1.041 |
| inverse-chain | 256 | 24.051 | 1.454 | 16.772 | 0.981/1.021 |
| inverse-chain | 1024 | 363.424 | 5.850 | 61.990 | 1.003/0.999 |
| inverse-chain | 2048 | 1438.629 | 12.118 | 118.719 | 0.987/0.998 |
| reduce-chain | 64 | 1.726 | 0.337 | 5.180 | 1.002/1.000 |
| reduce-chain | 256 | 23.166 | 1.285 | 17.980 | 0.997/1.005 |
| reduce-chain | 1024 | 356.128 | 5.215 | 68.883 | 0.981/0.998 |
| reduce-chain | 2048 | 1444.585 | 10.489 | 137.726 | 0.998/0.986 |
| inverse-power | 1024 | 99.406 | 4.580 | 21.930 | 1.006/1.006 |
| inverse-power | 4096 | 376.773 | 18.048 | 20.775 | 0.984/1.018 |
| reduce-power | 1024 | 95.824 | 3.933 | 24.523 | 1.006/0.997 |
| reduce-power | 4096 | 363.841 | 14.749 | 24.883 | 0.973/1.001 |
| source-proof | 64 | 3.421 | 3.376 | 1.016 | 0.998/1.001 |
| source-proof | 256 | 12.206 | 12.002 | 1.014 | 0.998/0.999 |
| source-proof | 1024 | 48.693 | 48.560 | 1.005 | 0.994/1.001 |

A chain begins with $ab$, then appends alternating $a, b$ letters and queries the extended root after each append. The table's size is its number of extensions. At 2,048 extensions inverse evaluation improves from 1,438.629 to 12.118 ms, a paired ratio of 118.719; reduction improves from 1,444.585 to 10.489 ms, a ratio of 137.726. Inverse work drops from 4,212,748 to 16,396, with 4,102 nodes in both arms; reduction work drops from 4,212,746 to 16,394, with 2,051 nodes in both. The new inverse traversal collects 2,051 nodes across 2,049 queries, and reduction collects 2,049 across 2,049 queries. These are intentionally simple exact-word kernels that expose the repeated traversal, not difficult knot inputs.

A power case starts at $(ab)^{2^s}$, performs its first query, then makes 128 alternating single-letter extensions and queries. The table's size is $s$, not the expanded word length. At $s = 4096$ the output length has 4,098 bits. Inversion improves 20.775-fold with work 1,094,798 to 29,710; reduction improves 24.883-fold with work 1,094,796 to 29,708. Respective node counts remain 8,454 and 4,227. No expanded power word is allocated. Across all kernel and supplied-proof cases, old A/A ratios range from 0.973 to 1.016 and current A/A ratios from 0.986 to 1.041.

The supplied source proof has a powered Whitehead prefix with exponent $2^s$, followed by a checked projection suffix. Full replay ratios are only 1.016, 1.014 and 1.005 for $s = 64, 256, 1024$, comparable with control variation. At $s = 1024$, work falls from 91,623 to 89,567, while both runs still use 5,151 nodes and three equality calls. The kernel gains therefore do not translate into a comparable gain for these complete supplied proofs.

**Complete recognition.** A second run measures nineteen fresh source inputs with old/current defaults and old/current adaptive batch hosts, each with an A/A control. It includes PD construction, the complete recognition pipeline and mandatory source replay. All 760 measurements and 152 warm-ups complete in 81.088 seconds.

| Input | default old/new ms | ratio | portfolio old/new ms | ratio |
|---|---|---|---|---|
| survivor-00 | 16.972/16.323 | 1.020 | 15.323/14.181 | 1.058 |
| survivor-01 | 14.631/13.997 | 1.034 | 29.927/30.166 | 0.987 |
| survivor-02 | 15.654/14.550 | 1.067 | 10.076/9.189 | 1.095 |
| survivor-03 | 10.968/10.680 | 1.027 | 41.732/38.139 | 1.093 |
| mirror-03 | 23.784/22.661 | 1.050 | 6.198/5.817 | 1.063 |
| survivor-04 | 9.863/9.303 | 1.061 | 31.654/29.299 | 1.073 |
| survivor-05 | 18.738/18.092 | 1.036 | 16.421/15.518 | 1.057 |
| survivor-06 | 7.566/7.205 | 1.020 | 5.656/5.185 | 1.086 |
| survivor-07 | 19.395/18.370 | 1.046 | 21.518/19.711 | 1.083 |
| survivor-08 | 7.549/7.158 | 1.046 | 7.292/6.847 | 1.066 |
| mirror-08 | 19.740/18.715 | 1.047 | 16.013/15.147 | 1.057 |
| survivor-09 | 7.418/7.194 | 1.028 | 5.724/5.456 | 1.047 |
| survivor-10 | 6.463/6.380 | 1.014 | 5.343/5.130 | 1.042 |
| survivor-11 | 6.997/6.849 | 1.024 | 5.024/4.742 | 1.052 |
| gordian | 1462.026/1431.768 | 1.050 | 1498.568/1448.005 | 1.037 |
| trefoil | 0.054/0.054 | 0.987 | 0.054/0.054 | 1.005 |
| figure_eight | 0.059/0.059 | 1.003 | 0.060/0.060 | 0.998 |
| conway | 0.968/1.012 | 0.970 | 0.964/0.974 | 0.989 |
| kinoshita_terasaka | 0.994/1.006 | 0.993 | 0.994/0.996 | 0.994 |

Default old/current paired ratios range from 0.970 to 1.067; portfolio ratios range from 0.987 to 1.095. Several group-search inputs show modest gains. For example survivor-02 improves 1.067-fold in default mode and 1.095-fold with the portfolio; its corresponding current A/A ratios are 1.000 and 0.995. Survivor-03's portfolio ratio is 1.093, but its 38.139 ms current portfolio median remains much slower than its 10.680 ms default median. The word-engine optimization does not remove schedule-dependent regressions.

Gordian default medians are 1,462.026 ms old and 1,431.768 ms current, with paired ratio 1.050. Portfolio medians are 1,498.568 and 1,448.005 ms, with paired ratio 1.037. All four arms return the same 9,663-byte certificate. The paired ratios need not equal quotients of the marginal medians. The default old A/A ratio on Gordian is 0.973, so timing precision remains limited even though its 60,548 saved producer work units are deterministic.

Across the full corpus, the four A/A ranges are 0.936–1.014 for old default, 0.973–1.039 for current default, 0.939–1.015 for old portfolio and 0.969–1.019 for current portfolio. Small regressions and noisy controls are retained. These measurements support modest practical savings on several search inputs, alongside the much larger deliberately isolated traversal gains. They do not establish a uniform end-to-end factor or change the unproved global discovery and grammar-growth requirements.

The driver `compressed_word_research/frontier.py` in `../fast` provides `audit`, `benchmark` and `kernels` modes. Each run pins the complete baseline, verifies 143 source hashes before and after execution, and retains proofs, per-arm outcomes, counters, sample order and A/A controls. Supplied-proof verification and word-kernel improvements do not establish faster general proof discovery or a quasipolynomial recognizer.

## 91 Persistent letter summaries and sparse substitution

The frontier optimization of Section 90 leaves another repeated traversal in singleton discovery and substitution. An unchanged shared subword may already have an exact reduced result, yet a later singleton elimination still walks all its source nodes. The current word engine retains raw letter summaries and uses them to stop a warm substitution at subwords unaffected by the current image map. The operation remains exact simultaneous substitution followed by free reduction.

## 91.1 Immutable presence and repetition

For each raw grammar node $X$, store a pair $(P_X, R_X)$ of bit masks. $P_X$ marks absolute generator labels occurring at least once and $R_X$ marks labels occurring at least twice, counting both signs. A terminal $a$ has $(P_a, R_a) = (\{|a|\}, \varnothing)$; for $X = YZ$,

$$P_X = P_Y \cup P_Z, \qquad R_X = R_Y \cup R_Z \cup (P_Y \cap P_Z).$$

Thus $P_X \setminus R_X$ is exactly its singleton set. A shared child reached twice contributes twice to the expanded word, correctly detected by the intersection term. These are raw occurrences, not exponent sums.

Dense bit positions are assigned once, when a label is first summarized, and appended as further labels appear. This avoids dependence on the numeric magnitude of a label. It also makes earlier masks valid forever, even if a new label is smaller than all previous ones. Returned singleton lists are sorted numerically to preserve the producer's previous tie-breaking behavior. A query follows the unresolved-node frontier, evaluates each newly reached node in allocation order, and publishes only completed exact summaries. The empty root is preseeded. Unlike the reduction cache, this metadata cache is descendant-closed: no other operation inserts a root-only shortcut.

For $U$ distinct nodes first summarized, the completed discovery traversals cost $O(q + U \log(U + 1))$ identifier operations over $q$ root queries, plus mask arithmetic and singleton decoding/sorting. With $a$ historically seen labels and $M$ historical nodes, stored masks use $O(Ma)$ bits, in addition to dictionary, label and node-identifier storage. A bitwise operation is charged for its operand width; it is not constant-time in the bit model. Repeatedly aborted discovery is outside the completed-query amortization. This trades persistent metadata memory for avoided recomputation and does not make a node-count allowance a bound on total process memory.

## 91.2 A fresh substitution map over a cached support frontier

For a call with images $I(a)$, let $C$ contain absolute labels whose supplied signed image is not syntactically that terminal. An absent image is identity. This conservative signed test may include both polarities when only one changes; that causes extra traversal but cannot omit a needed change. Image keys absent from the source terminal table are irrelevant.

Use the support frontier only when every queried root already has a complete summary. If $P_X \cap C = \varnothing$, substitution leaves the raw word $X$ unchanged, so its result is exactly $\mathrm{red}(X)$. The engine calls the existing cached reduction at this boundary. Simply returning a raw root would be incorrect when it contains cancellations. Otherwise descend to the children, deduplicate affected nodes, and evaluate them in allocation order. Terminals use their supplied images once; concatenations combine the two already reduced results through the existing exact cancellation operation. Letters inside images are not recursively substituted.

Induction on this traversal proves equality with the original operation: frontier values are exact reduced unchanged words; affected terminals have exact reduced images; the reduced product of child values is the correct parent value. Each call starts with a new map from source nodes to results, so changing images, root order, duplicate roots and emptied relator slots cannot reuse stale substitution answers. Cancellation and resource exceptions propagate, leaving only exact completed immutable summaries and word results.

If $D$ nodes have potentially changed support, the traversal examines only those nodes and their immediate unaffected frontier, besides the queried root list and supplied image map. Sorting costs $O(D \log(D + 1))$ identifier operations. This accounting excludes mask bit arithmetic and the cost of new frontier reductions and affected reduced products. A cold query retains the old full traversal; it does not pay to discover support solely for this optimization. Ordinary certificate replay generally has cold support, while the producer has already queried singletons. We therefore make no generic replay acceleration claim.

### 91.3 Validation and controlled measurements

Seven focused tests cover 400 random signed substitution fixtures on both warm and cold paths, raw cancellation, irrelevant image keys, new sparse labels, repeated/reordered roots, incomplete metadata, interrupted operations, a 1,500-step growing prefix and a huge prefix with a 2,050-bit length. All 1,025 maintained tests pass with Regina in 223.592 seconds. The pinned package audit against `f9e20ed6f3f1` preserves all 400 outcomes and certificate hashes on eighty diagrams in five modes. Every positive passes both old and current literal and compressed source replay. The exhaustive 21,845 short-word audit and five supplied-proof/source-rebinding checks also pass. Only `compressed_words.py` changes among production modules. The audit takes 19.379 seconds and verifies 145 source hashes.

The default Gordian producer uses 789,805 work units instead of 1,154,597, with 139 warm substitutions. Projection and forest work fall from 1,206,616 and 1,206,614 to 842,292 and 842,290 respectively. Direct batching still exhausts its two-million-work allowance. These are deterministic work counters on the audited inputs, not wall-time ratios or universal equality of budgeted search behavior.

**Repeated word queries.** Twelve kernel cases use four shuffled old/current and A/A arms, five measured rounds and one warm-up. All 240 measurements and 48 warm-ups complete in 118.781 seconds. Timings include fresh arena construction, initial reduction and every singleton query. Ratios are medians of paired old/new times.

| Family | size | old ms | new ms | ratio | old/new A/A |
|---|---|---|---|---|---|
| singleton-chain | 64 | 2.694 | 0.377 | 7.215 | 1.000/0.970 |
| singleton-chain | 256 | 37.673 | 1.436 | 26.188 | 0.988/1.001 |
| singleton-chain | 1024 | 601.293 | 5.830 | 103.056 | 1.003/0.990 |
| singleton-chain | 2048 | 2497.807 | 11.843 | 212.355 | 0.992/0.980 |
| warm-tail | 64 | 24.011 | 1.621 | 14.811 | 1.002/0.999 |
| warm-tail | 256 | 89.129 | 2.371 | 37.422 | 0.997/0.990 |
| warm-tail | 1024 | 382.376 | 5.874 | 66.018 | 0.986/0.995 |
| warm-tail | 4096 | 1608.502 | 20.817 | 79.392 | 1.028/1.001 |
| cold-tail | 64 | 14.468 | 13.746 | 1.061 | 1.072/1.012 |
| cold-tail | 256 | 53.492 | 52.677 | 1.000 | 0.886/0.971 |
| cold-tail | 1024 | 384.189 | 353.264 | 1.088 | 1.002/1.034 |
| cold-tail | 4096 | 1838.591 | 1683.154 | 1.010 | 0.955/1.138 |

The singleton chain appends alternating $a, b$ letters and queries each prefix. At length 2,048 the old/new medians are 2,497.807/11.843 ms, with paired ratio 212.355. Work falls from 6,302,722 to 18,434; both arenas contain 2,049 nodes. Only 2,049 node summaries are built across all the new queries. This isolates the removed quadratic metadata traversal on a growing immutable prefix.

The tail cases start with $(ab)^{2^s}$ followed by a letter $c$, and alternate $c \mapsto d$ and $d \mapsto c$ for 128 substitutions while also retaining the unchanged prefix as a root. Warm cases query singletons before each update; cold cases never populate support metadata. At $s = 4096$, the warm old/new medians are 1,608.502/20.817 ms, with paired ratio 79.392. Work falls from 3,702,287 to 42,910, with 4,103 nodes in both arms. Only two affected nodes are compiled per substitution, and represented length has 4,098 bits. The power word is never expanded.

At the same size the cold arms each perform exactly 2,127,631 work units and have 4,103 nodes. Their paired ratio is 1.010, compared with current A/A 1.138, so no cold-path speedup is supported. Across all kernel cases, old A/A ratios span 0.886–1.072 and current A/A 0.970–1.138. The large warm and growing-prefix gains are structurally explained by the counters; small timing differences lie within substantial control variability.

**Complete source-bound recognition.** Nineteen fresh diagrams use old/current defaults and old/current adaptive batch hosts, each with an A/A copy, five measured rounds and one warm-up. The timed calls include diagram construction, discovery and mandatory source replay. All 760 measurements and 152 warm-ups complete in 114.683 seconds.

| Input | default old/new ms | ratio | portfolio old/new ms | ratio |
|---|---|---|---|---|
| survivor-00 | 30.363/27.030 | 1.176 | 24.289/26.807 | 0.880 |
| survivor-01 | 34.843/19.590 | 2.305 | 46.293/36.448 | 1.017 |
| survivor-02 | 28.345/35.763 | 0.944 | 15.999/16.287 | 0.982 |
| survivor-03 | 18.272/16.435 | 1.112 | 67.068/64.655 | 1.006 |
| mirror-03 | 72.083/40.178 | 1.270 | 10.860/10.988 | 0.979 |
| survivor-04 | 16.244/14.327 | 1.134 | 52.601/71.176 | 0.990 |
| survivor-05 | 30.293/28.974 | 0.731 | 39.424/26.340 | 1.379 |
| survivor-06 | 12.535/11.401 | 1.122 | 9.257/8.913 | 1.057 |
| survivor-07 | 52.252/69.502 | 0.825 | 36.044/35.095 | 1.002 |
| survivor-08 | 24.871/19.020 | 1.129 | 23.752/24.010 | 1.108 |
| mirror-08 | 22.632/20.819 | 1.034 | 16.718/29.291 | 0.647 |
| survivor-09 | 8.007/7.555 | 1.088 | 6.257/6.198 | 0.994 |
| survivor-10 | 11.562/10.078 | 1.145 | 9.342/10.244 | 0.997 |
| survivor-11 | 12.249/10.766 | 1.151 | 8.773/8.473 | 1.008 |
| gordian | 1652.607/1495.833 | 1.172 | 2533.473/1289.029 | 1.330 |
| trefoil | 0.061/0.059 | 1.044 | 0.064/0.061 | 0.997 |
| figure_eight | 0.064/0.065 | 0.982 | 0.064/0.064 | 0.995 |
| conway | 1.058/1.065 | 0.998 | 1.042/1.047 | 0.994 |
| kinoshita_terasaka | 1.087/1.084 | 1.007 | 1.068/1.121 | 0.968 |

Default paired ratios span 0.731–2.305 and adaptive-host ratios 0.647–1.379. The corresponding A/A ranges are 0.799–1.341 for old default, 0.616–1.046 for current default, 0.736–1.321 for old host and 0.739–1.560 for current host. These controls show a noisy shared-machine run; neither its largest apparent gains nor its regressions should be read as stable implementation factors.

On Gordian the default medians are 1,652.607/1,495.833 ms and the paired ratio is 1.172. The host medians are 2,533.473/1,289.029 ms with paired ratio 1.330. Paired medians need not equal ratios of marginal medians; the disparate A/A marginal timings are another reason not to infer precision from these figures. The unchanged audited certificate and deterministic default-work reduction of 364,792 units (31.6 percent) provide firmer evidence of saved computation. We retain the complete sample order, controls and outcomes, and make no uniform end-to-end timing claim from this run.

The retained driver `../fast/compressed_word_research/sparse_substitution.py` has `audit`, `kernels` and `benchmark` modes, pins the complete prior package and freezes source hashes around each run. Evidence is in `data/sparse-substitution-*`. Cold controls, A/A variability and small-case regressions remain visible. The optimization removes repeated work on immutable words; it does not supply a bound on global search, normalization or the size of every generated presentation.

## 92 Stronger local bounds from reports 49–53

The five deliveries in `2c7f4fd68bfd` are now reports 49–53. They address three different costs: algebraic substitution, sparse component observations, and componentwise normal-disc detection. This review strengthens the analysis of the existing algebraic implementation. The new sparse and weighted interfaces were delivered research implementations at this review checkpoint. Section 93 subsequently integrates the sparse observer and its independent checker; Section 94 integrates the weighted census and quadrilateral core. Their tests and audit reruns below are separate from the

earlier placement-only intake. No delivered benchmark ratio is treated as a measurement of the current recognizer.

## 92.1 A fixed source bounds an entire monomial epoch

Report 49 supplies a stronger invariant than the successive length estimates in Sections 85 and 86. Freeze a presentation with $r$ generators, $m$ original relator slots, a shared source grammar of $S_0$ reachable nodes, and maximum raw relator length $L \geq 1$. Write $B = \lceil \log_2(L+1) \rceil$. During the epoch permit only verified disjoint primitive-pair projections and acyclic unit-coordinate forests. Retain every unselected original slot, and perform no normalization, Whitehead move, overlap move or change of source coordinates inside the epoch. Proper-power donors retain the existing source-established torsion-free hypothesis; the lattice calculation does not justify their nonabelian deletion.

Every cumulative substitution has the form

$$x_i \longmapsto z_{\pi(i)}^{k_i}, \qquad k_i \neq 0.$$

Its matrix $K \in \mathbb{Z}^{r \times s}$ has one nonzero entry per row and nonempty disjoint column supports $I_z$. Each column is primitive: initially it is a coordinate vector, and merging two primitive disjoint columns multiplies them by coprime coefficients $|v|$ and $-u\,\mathrm{sign}(v)$. The gcd of the merged column is therefore one.

Let $A$ be the original exponent-sum matrix and $A_J$ its rows selected as donors so far. After $q$ individual contractions, $s = r - q$, $A_J K = 0$, and $\mathrm{rank}_{\mathbb{Q}} A_J = q$. Indeed, the next selected row $a$ has nonzero current image $aK_{\mathrm{old}}$, whereas all previously selected rows annihilate $K_{\mathrm{old}}$. The new row is independent of the previous ones. Disjoint batches linearize immediately. Unit-coordinate forests linearize from leaves toward their surviving roots, so an earlier eliminated leaf cannot alter a still-needed donor.

Dimension and disjoint supports now give

$$\ker_{\mathbb{Q}} A_J = \mathrm{im}_{\mathbb{Q}} K, \qquad \ker_{\mathbb{Z}} A_J = \mathrm{im}_{\mathbb{Z}} K.$$

For the second equality, write an integral vector as $Kc$ with rational $c$. A Bezout combination within each primitive column expresses its coefficient $c_z$ as an integer combination of the vector's integral entries. Thus all coefficients are integral. Restricting to $I_z$ leaves a one-dimensional kernel spanned by that primitive column.

If $t = |I_z| \geq 2$, choose $t-1$ independent rows of the restricted matrix. The signed maximal minors form a nonzero integral kernel vector. Dividing its common content gives the primitive vector $(k_i)_{i \in I_z}$, up to sign. Since every source matrix entry has magnitude at most $L$, Hadamard's bound proves

$$|k_i| \leq (t-1)^{(t-1)/2} L^{t-1}; \qquad t = 1 \implies |k_i| = 1.$$

All cumulative exponents consequently use $O(r(B + \log(r+1)))$ bits throughout the entire epoch. This bound is independent of the number of raw phases.

**Reachable size and historical allocation.** Set $\Lambda = 8(r+1)(B + \lceil \log_2(r+1) \rceil + 2)$, a convenient common bit bound. A raw projection maps each old reachable concatenation to at most one new concatenation. If $P_j$ counts the powering gates introduced in phase $j$, the reachable and historical counts obey different recurrences:

$$S_{j+1} \leq S_j + O(P_j), \qquad N_{j+1} - N_j \leq S_j + O(P_j).$$

There are at most $r-1$ contractions. Each current donor is an image of an original slot, of length at most $L \max_i |k_i|$, so its local exponents also have $O(\Lambda)$ bits. The existing forest updater

shares parent power circuits and charges each edge's exponent rather than re-encoding a full path product. Thus $\sum_j P_j = O(r\Lambda)$, and for $d$ phases,

$$S_j = O(S_0 + r\Lambda), \qquad N_d = N_0 + O(dS_0 + (d+1)r\Lambda) = N_0 + O(rS_0 + r^2\Lambda).$$

Every surviving root remains an image of its fixed source word; length metadata is bounded by $B + O(\Lambda)$. The existing planners, exact primitive checks and raw updates therefore have polynomial total encoded cost on this restricted epoch. The previous exponential recurrences were valid but loose upper bounds. A short-depth assumption is unnecessary here.

The supplied source checker matches the current `primitive_projection.py` Git blob and the complete `apply_projection` syntax tree. Forest applicability follows from the leaf-first linearization and the maintained shared-power update above; the report's standalone lazy implementation only implements disjoint pairs. Its maintenance-only timing claims do not measure native discovery or mandatory source replay. Eagerly exporting its lazy state after every round was slower in the delivered experiment, so adopting a lazy state requires preserving that interface through planning and verification. Section 102 subsequently integrates source-anchored independent replay of mixed projection/forest blocks with one final export; Section 103 then integrates source-anchored producer discovery while retaining the existing greedy move and endpoint priorities.

**The meridional special case.** If an integral character $h$ of the frozen source satisfies $Ah = 0$, then $h = Kc$. On a block where $h$ is nonzero, primitive columns imply

$$k_i = \varepsilon_z h_i / \gcd_{j \in I_z} |h_j|, \qquad \varepsilon_z \in \{1, -1\}.$$

Every entry of $h$ on that block is nonzero. In particular, with consistently oriented Wirtinger meridians $h_i = 1$, all cumulative exponents have magnitude one. Huge-power abstract presentations test a broader supplied-presentation interface; they do not show such growth in this initial meridional epoch. Changes of coordinates can create zero character values or large values, and require their own complexity accounting.

## 92.2 Acyclic singleton batches have linear grammar growth

Report 51 improves the general, nonmonomial batch bound. Let $M$ count the nonempty source nodes reachable from all current roots. A selected donor $U_c c^{\epsilon_c} V_c$ contains its chosen child once in the raw spelling. Follow the unique path $H_c$ to that occurrence in the shared binary grammar. If $H_c$ and $H_d$ shared a node, its word would contain both $c$ and $d$. Deleting either selected occurrence would leave the other, giving dependencies $c \to d$ and $d \to c$. Acyclicity excludes this. Hence

$$H_c \cap H_d = \varnothing \ (c \neq d), \qquad \sum_c |H_c| \leq M.$$

Whole donor grammars can still overlap substantially; only the hole paths are disjoint.

Both complementary substring cuts follow that path. At each internal path node at most one cut needs a new concatenation; completely included siblings are reused. Including the final rotation $V_c U_c$, all donor contexts require at most $M$ new nodes. The source and contexts together have at most $2M$ nodes. In the report's compiler, each has two virtual orientations, and selected terminals redirect to the corresponding signed context. A cycle in this graph would imply a cycle among selected definitions. Topological compilation of at most $4M$ states gives at most $4M$ output nodes, at most $5M$ new persistent nodes including contexts, and at most $6M$ total nodes when the initial arena contains only the reachable source.

At this review checkpoint our producer and independent checker materialized inverses before compilation. Their globally memoized inverse operation introduces at most $2M$ inverse nodes for

the source-context union. The resulting union has at most $4M$ nodes; compiling each once gives the conservative native bounds

$$M_{\text{out}} \le 4M, \qquad N_{\text{after}} - N_{\text{before}} \le 7M.$$

Historical unreachable nodes are included in $N_{\text{before}}$, not silently charged to $M$. These constants do not assert that the delivered virtual-state compiler was already installed. Section 95 subsequently installs its virtual-state checker, with the sharper $5M$ allocation bound. Occurrence/support discovery may still scan a whole donor grammar per selected child, so linear output size does not imply linear running time for the current batch. It does give polynomial bit cost in the actual reachable source: a binary terminal grammar with at most $4M$ nodes has $O(M)$-bit lengths and $O(M^2)$ total length metadata.

For an epoch consisting solely of these singleton batches, $d$ phases give $M_d \le 4^d M_0$ and historical growth $O(M_0 4^d)$. Thus their earlier $2^{O(d \log(r_0+1))}$ composition estimate can be sharpened to $\text{poly}(M_0, r_0, m)2^{O(d)}$. A logarithmic number of phases gives polynomial cost; $O(\log^2 n)$ phases with polynomial initial encoded size give quasipolynomial cost. No such general diagram-to-endpoint phase bound is proved. The polynomial monomial theorem above is stronger on its narrower class and does not extend to arbitrary word-valued singleton definitions.

The report also supplies an ordered witness whose per-node summary is only the maximum selected rank and its multiplicity capped at two. Donor $i$ has summary $(i, 1)$ exactly when its child is unique and all selected dependencies are earlier. Section 95 subsequently adapts this checker to the maintained schema, preserving arbitrary entry order through the existing graph-checking fallback. The report's rank-one-completion guard avoids eager partial-batch regressions on its older baseline; our recovery-aware bounded adaptive trial is a different policy and is retained.

### 92.3 Sparse incidence removes an exponential output interface

Reports 50 and 52 are separate implementations of the same positive sparse reconstruction idea. With $r$ interval-union ports, let $h(T)$ count components meeting exactly the port set $T$. Marked-union orbit queries give the zeta oracle

$$G(U) = \sum_{T \subseteq U} h(T).$$

Subtract already recovered atoms. The residual is nonnegative and monotone. Starting at the full port set, delete a port whenever the residual stays positive. At termination every proper subset has zero residual, so the remaining residual is exactly one new atom's positive weight. Repeating until all total mass is recovered uses $O(1 + sr)$ queries for $s$ nonempty atoms. Balanced block deletion improves this to

$$O\left(1 + s + \sum_{T \neq \varnothing : h(T) > 0} |T| \log\left(1 + \frac{r}{|T|}\right)\right).$$

Every failed block intersects the ultimately recovered support; counting failed blocks at each tree depth proves the bound.

Polynomial support for interval-union ports follows from the classical weighted AHT theorem, rather than being a new sparsity hypothesis. Assign one indicator coordinate per port; signs of the resulting nonnegative orbit weights give the incidence signature. AHT returns polynomially many constant weight blocks in the encoded interval input, hence polynomially many distinct signatures [18, Theorem 16]. The sparse unweighted-query route inherits this bound. It replaces a mandatory $2^r$ output array with positive `[mask,count]` entries; preserving the old dense return type would lose the advantage.

Independent certificates check atom weights, immediate-subset zero witnesses and total mass. On a binary orientation cover, each base component lifts to one or two components with the same port signature. Support discovery can therefore be reused, with signed consistency counts derived from the base and cover multiplicities. This local observation does not retain attachment maps, cyclic orders or slopes needed by arbitrary future hierarchy moves. The reports include a small counterexample to treating incidence equality as a complete equivalence for future attachments.

## 92.4 Weighted components and a smaller normal-disc query

Report 53 develops weighted local-rule replay and a componentwise normal surface census. On a validated connected orientable compact triangulation with exactly one torus boundary, assign integral cell weights: $+1$ per surface vertex, $-1$ per arc and $+1$ per polygon, each charged to a point in its own component. Their orbit sum is its Euler characteristic. Two additional intersection weights evaluate the component's boundary class against an independently checked basis of the torus's mod-two homology. A connected component is a compressing disc exactly when

$$\chi = 1 \quad \text{and} \quad (h_1, h_2) \neq (0, 0).$$

Nonzero boundary class excludes closed components, and surface classification then makes Euler characteristic one equivalent to being a disc. An embedded essential circle on a torus has primitive integral slope and nonzero mod-two class. The predicate must be applied componentwise: totals can cancel across two genuine discs, or combine nondisc components into misleading statistics. Higher-genus boundary requires a different essentiality test.

For an admissible standard vector $x$, subtract at each global vertex the minimum incident triangle coordinate times its vertex-link vector. Call the remainder $Y$. These removed links can be realized disjointly and contribute no compressing discs. If $g$ is the gcd of the quadrilateral coordinates, face matching propagates divisibility by $g$ from a zero triangle around each connected vertex-corner graph. Thus $P = Y/g$ is integral and admissible. If all quadrilaterals vanish, $Y = 0$. Otherwise, with $t$ tetrahedra,

$$\max_i P_i \leq (4t - 1)\frac{Q_{\max}}{g}, \qquad D(x) = gD(P),$$

where $D$ counts compressing-disc components. Two-sided discs scale as parallel copies. One-sided components can produce orientation doubles, but in an orientable ambient manifold these cannot create disc components. Full component counts consequently need not scale by $g$, while $D$ does.

The expensive three-weight orbit query sees only the primitive quadrilateral scale, even if the original vector has enormous additive vertex-link multiplicity and full coordinate gcd one. Reading, validating and binding the original input still costs its original bit size. A positive count becomes an unknot witness only with checked provenance tying this triangulation to the input knot exterior; zero rejects only this supplied vector. The delivered two-weight ambient-homology refinement is a proposal, not executed code.

## 92.5 Reproduced evidence and integration boundary

Before execution, all five original archive integrity checks passed. Their selected suites pass 38, 21, 16, 33 and 103 tests respectively: 211 in total. Report 49's audit exercises 600 signed trees, source-row/cofactor invariants, exact raw-update comparisons, 32,664 signed primitive profiles and the 64,513-bit cumulative-exponent fixture. The copied updater is source-identical to our native operation. Report 51's 54 saved positive certificates pass 108 literal/compressed source replays using its pinned final snapshot. A separate maintained-engine audit checks 400 random singleton DAGs through both our producer and checker, including exact literal results, disjoint hole paths and the $4M/7M$ size bounds.

Report 50 reproduces 512 exhaustive native dense/sparse comparisons and independent replays, huge integer transport and producer-disabled signed checking. Its deliberately gapped larger queries remain censored under the five-second driver allowance, and receive no speedup ratios. Report 52's 1,000-case audit checks both ordinary graphs and binary covers against literal components. Report 53 reproduces the abstract weighted and mutation audits and the frozen Regina corpus: 1,275 normal vectors on 45 triangulations, 5,100 component comparisons and 5,100 certificate replays. These local tests provide evidence for the proposed interfaces. At this review checkpoint the sparse and weighted additions had not run in the maintained suite; Sections 93 and 94 supply the subsequent maintained integrations.

The logs and JSON records are `data/incoming-2c7f-*`; the native grammar check is `../fast/compressed_word_research/incoming_bounds.py`. The delivered files remain unchanged after placement. Checksum-only files omitted by intake are recoverable from the original commit's ZIPs; integrity was checked on fresh original extractions before rerunning any generator. The subsequent integrations in Sections 93 and 94 install the additive sparse incidence API and weighted component replay. Source-bound exterior construction and controlled candidate discovery remain open. Polynomial local epochs and observations reduce specific costs; they leave the general quasipolynomial recognition goal open.

# 93 Native certified sparse component incidence

The sparse observer reviewed in Section 92 is now part of the maintained `fastunknot` package. The implementation adapts report 50's additive interface and independent checker. Report 52 supplies a second implementation and the support-sensitive analysis; its standalone code is not imported. The maintained dense API and its existing certificates remain available for consumers that require a full Boolean array. New sparse callers receive positive `[mask,count]` entries and never allocate that array. This is a supplied interval-relation query, not a knot verdict or a complete hierarchy state.

## 93.1 From coning to positive atoms

Let $C$ be the number of orbits of an interval relation on a binary universe of size $N$, and let $A_1, \dots, A_r$ be interval-union ports. An orbit's signature $T$ consists of all ports it meets; $h(T)$ is its multiplicity. For a nonempty union $A$ of marked intervals, add translations joining consecutive points within each interval and singleton pairings joining the intervals to one hub. All original orbits meeting $A$ then become one, while every disjoint orbit is unchanged. If the augmented orbit count is $C_A$, the number of original orbits avoiding $A$ is $C_A - 1$.

For $U \subseteq [r]$, cone the union of ports outside $U$. This gives

$$G(U) = \sum_{T \subseteq U} h(T) = \begin{cases} C_A - 1, & A = \bigcup_{i \notin U} A_i \neq \varnothing, \\ C, & A = \varnothing. \end{cases}$$

Canonical interval unions share one orbit query and one certificate even when many masks describe the same marked set. Half-open port intervals are validated and merged before use; orbit pairings retain the native inclusive endpoint convention. No step lists the represented points.

Subtract already certified atoms from $G$ to obtain a nonnegative monotone residual. Positive residual survives each accepted deletion. Every rejected single-port deletion remains zero as later deletions shrink the current set. The final positive set is therefore inclusion-minimal and its residual is one exact atom. The empty atom is recovered separately. Repeating until recovered mass equals $C$ proves completeness, including unmarked components. The only algebraic requirement for this peeling argument is nonnegativity; it would fail for arbitrary signed measures because cancellations can hide proper-subset atoms.

The `linear` policy tries ports individually. The `split` policy recursively tries balanced blocks and explores children only when a whole-block deletion fails. It adapts the number of queries to the support sizes without predicting them. Section 92 proves the $O(1 + sr)$ and support-sensitive bounds. The implementation keeps the explicit policy choice; it does not claim a universal faster policy. Residual subtractions, mask arithmetic, canonical unions, and orbit-query costs remain part of the encoded running time. Polynomial support follows from weighted AHT, so this is a polynomial local observer on interval-encoded inputs, not merely a fast special case assuming a small output.

### 93.2 Certificates independently certify completeness

A sparse certificate binds the universe, ordered normalized ports and original pairing relation through native orbit proofs. It records positive atoms in peeling order. For every retained port $i$ in atom $T$, it includes a zero residual witness $Z_i$ with

$$T \setminus \{i\} \subseteq Z_i, \qquad i \notin Z_i.$$

The witness may be larger than the final immediate subset, since it was recorded before later deletions. Monotonicity makes its certified zero valid on the immediate subset too. These witnesses eliminate all unrecovered proper-subset atoms. The certified count at $T$, less previous atoms, fixes its weight. Finally the sum of positive weights must equal the baseline count. No missing positive atom can then remain.

The checker reconstructs the coning relations independently, replays every needed orbit proof, checks each weight and zero witness, and rejects duplicate atoms, invalid masks, missing witnesses, missing proofs and unused union proofs. It imports neither the sparse producer nor its nonnegative-recovery helper nor the orbit-count producer. Source validation and exact local orbit rules provide the nonnegative-measure promise; the checker does not infer it from arbitrary supplied numbers. Consequently disabling all producer calls still permits certificate replay.

The output histogram and the certificate are separate objects in the public result. Callers should derive a trusted histogram from the verified certificate entries, or compare their proposed histogram with those entries; a successful Boolean certificate check does not authenticate a separately modified result wrapper. Hexadecimal integer transport uses the maintained JSON helper and does not change Python's global decimal conversion limit.

### 93.3 Signed relations share the support search

For pairings labeled by a binary parity character, construct the two-sheet cover. A component with consistent parity lifts to two components, and an inconsistent component lifts to one. Lift every port to both sheets. Each lifted component has exactly its base component's signature, so no second support-discovery search is needed. If $h(T)$ and $\widetilde{h}(T)$ are base and cover multiplicities, then

$$h(T) \le \widetilde{h}(T) \le 2h(T), \qquad h_{\text{consistent}}(T) = \widetilde{h}(T) - h(T), \quad h_{\text{inconsistent}}(T) = 2h(T) - \widetilde{h}(T).$$

The cover needs at most one additional coned count per base atom plus its baseline count. The independent checker constructs its own cover pairings, verifies the known-support equations and total mass, then checks the split. Parity consistency means orientability only when the supplied character has independently been shown to encode surface orientation. Interval reflection alone is not that character.

## 93.4 Budgets and API use

One cycle allowance and one orbit-query allowance cover the original baseline, all coned counts, the cover baseline and its counts. The signature allowance bounds the number of recovered atoms. Exhaustion returns `INCONCLUSIVE` with counters and a reason, without an ordinary or signed histogram and without a certificate. It never exposes a partial answer as complete. Callback exceptions propagate through input preparation, recovery, proof construction and verification. Callable objects remain callbacks even when they evaluate as false. Invalid strategy and signature-limit options are rejected before any orbit search, even if an independent work allowance is already zero.

The producer's cycle and query allowances do not meter a subsequent independent replay. Verification has its own cooperative callback and explicit certificate size; a host measuring a complete certified query must include both operations.

These are cooperative allowances, not a hard process-memory or serialized-input sandbox. Bit masks use $r$ bits, counts use $O(\log(N+1))$ bits, and dictionary and integer costs must be charged. Canonical-union and raw-query caches can retain the distinct requests made during discovery. A caller handling hostile serialized certificates must also limit their explicit size. The API does not assert a bound on arbitrary future attachment-map descriptions.

```
from fastunknot.sparse_incidence import analyze_sparse_port_incidence
from fastunknot.sparse_incidence_verify import (
    verify_sparse_port_incidence_certificate,
)
ports = [[(0, 4)], [(2, 6)]]
result = analyze_sparse_port_incidence(
    10, [], ports, strategy='split', record_certificate=True)
assert result['status'] == 'COMPLETE'
assert verify_sparse_port_incidence_certificate(
    10, [], ports, result['certificate'])
# Positive classes: {0: 4, 1: 2, 2: 2, 3: 2}.
```

## 93.5 Native validation and measurements

The delivered 21-test selection is now in the maintained suite. It covers 500 random interval graphs under both policies, 1,000 arbitrary nonnegative measures under both policies, 300 signed interval graphs, 128 disjoint ports, huge binary universes, certificate mutations and producer-disabled replay. Four additional integration tests check validation before resource work, false-valued callable cancellation, exact signed shared-budget boundaries, source rebinding and hexadecimal certificate transport. All 25 focused tests pass in 2.074 seconds.

A separate native audit checks every support subset of the three-port Boolean cube: 256 histograms, 512 dense/sparse comparisons and 768 independent dense or sparse replays. Sixteen large analytic cases cover both sparse policies on disjoint, coincident, nested, genuinely paired and all-signature fixtures. Two $2^{1024}$-point signed identity systems independently check consistent and inconsistent lifts. The dense observer, orbit producer, orbit verifier and integer codec are pinned byte-for-byte to baseline `308d5aad3b70`; the extension adds three modules.

The maintained suite passes all 1,050 tests with Regina in 207.266 seconds. The native audit takes 0.422 seconds. The separate timing run has eight supplied interval systems, six shuffled arms (dense, linear, split and their A/A copies), five measured rounds and one warm-up. All 240 measured calls and 48 warm-ups complete in 7.354 seconds. Each timed call includes input validation, complete discovery, certificate construction and independent source replay. Serialization

and analytic histogram assertions are outside the timing. All runs freeze 139 source hashes; the four dense/orbit dependency modules match the pinned baseline exactly.

| Input | dense ms | linear ms | split ms | dense/linear | dense/split |
|---|---|---|---|---|---|
| disjoint-6 | 7.326 | 1.814 | 1.316 | 4.050 | 5.563 |
| disjoint-8 | 42.230 | 3.144 | 1.978 | 13.153 | 21.600 |
| disjoint-10 | 225.918 | 4.996 | 2.644 | 46.004 | 85.953 |
| coincident-4 | 0.204 | 0.161 | 0.174 | 1.269 | 1.216 |
| coincident-10 | 9.060 | 0.226 | 0.284 | 40.232 | 32.390 |
| nested-10 | 9.504 | 1.310 | 1.556 | 7.269 | 6.123 |
| paired-blocks-8 | 75.687 | 6.477 | 3.926 | 11.544 | 19.357 |
| all-signatures-6 | 36.452 | 41.743 | 42.602 | 0.873 | 0.858 |

Ratios are medians of paired elapsed times. Disjoint and coincident fixtures use blocks of length $L = 2^{256}$. The table's port count is part of the explicit input; it is not the number of expanded points. On ten disjoint ports the dense/linear/split medians are 225.918/4.996/2.644 ms. Paired dense-over-split is 85.953. Orbit queries fall from 1,024 to 26 and cycles from 5,633 to 75. Both sparse policies produce a 26,538-byte certificate, compared with 2,498,706 bytes for dense. The split policy also avoids many counts needed during linear discovery even though the retained minimality witnesses coincide.

The genuinely paired eight-port fixture has both a translation and a reflection on huge blocks. Dense-over-split is 19.357, queries fall from 256 to 21 and certificate bytes from 730,590 to 35,055. Coincident and nested ports demonstrate a different saving: canonical-union caching already gives both dense and sparse the same number of expensive orbit counts. At ten coincident ports both make two counts, and at ten nested ports both make eleven. Sparse still avoids scanning and certifying the entire subset cube. Linear deletion is faster than split on these two fixtures, so split is not a universal scheduling improvement.

The all-signatures fixture uses six ports on 64 singleton components, one component for every signature. All methods make 64 orbit queries and 599 cycles. Sparse adds peeling and zero-witness overhead: dense/linear/split medians are 36.452/41.743/42.602 ms, with dense-over-split ratio 0.858. The sparse certificate grows from the dense 42,130 bytes to 49,505 bytes. This regression motivates retaining the explicit dense API for callers whose output is actually dense. No automatic policy inference from these eight fixtures is installed.

Across the cases, dense A/A ratios range from 0.990 to 1.021, linear A/A from 0.933 to 1.014 and split A/A from 0.978 to 1.053. The large sparse-case gains substantially exceed these controls; the observed all-signatures regression remains visible. The results establish practical savings for these complete supplied-system queries, including verification. They measure neither normal-surface candidate generation nor complete knot recognition. In particular the implementation does not add a new automatic knot-decision stage.

The audit and timing driver is `../fast/incidence_research/sparse.py`; its source manifests, proofs, per-arm query counts, timings and outcomes are retained in `data/sparse-incidence-*`. The original delivered files stay unchanged. Section 94 subsequently integrates the weighted component census; certified knot-exterior provenance remains open. This integration removes an exponential local output interface; it does not prove a quasipolynomial hierarchy or complete general unknot search.

## 94 Native weighted normal components and essential-disc counts

The weighted component operations of report 53 are now integrated into the maintained package. They answer a question unavailable from the earlier aggregate normal-surface statistics: how

many components of this supplied surface are compressing discs? Binary normal coordinates stay compressed. The full component census and the reduced count-only query have independent source-bound certificate replay. They require the existing validated finite, connected, orientable triangulation with one torus boundary. A positive answer concerns that manifold and vector; relating the manifold to an input knot exterior remains a separate provenance obligation.

## 94.1 Transport additive weights along an orbit trace

Assign a vector $w(x) \in \mathbb{Z}^d$ to each point of an interval-pairing universe $[0, N)$. The input is an additive list of half-open intervals with constant vectors, possibly overlapping and with signed entries. Unmarked points have weight zero. Endpoint differences combine the intervals into canonical constant runs. The desired compressed histogram records

$$H(v) = \#\left\{O \text{ an orbit} : \sum_{x \in O} w(x) = v\right\}.$$

Equal vectors share one row with binary multiplicity. Negative Euler weights are ordinary signed integers; the multiplicities remain nonnegative counts.

The producer first obtains a complete trace from the maintained AHT engine. It then transports weights through its local moves. Identity deletion, trimming, periodic merging and transmission alter the description of the same equivalence relation and require no weight change. Truncation removes points after transferring their weights to surviving representatives. Contraction emits the weights of static singleton representatives and closes their gaps. At the end, every original orbit has one emitted representative carrying its complete sum. The total count and every coordinate's total weighted mass are checked against the original source.

For a translation truncation with new endpoint $c$ and period $p$, set $b = c - p$. A removed point $x$ maps to

$$b + ((x - b) \bmod p) \in [b, c).$$

For a constant-weight deleted interval of length $\ell = qp + r$, add $q$ copies of its vector to the whole period and one copy to the cyclic remainder, which has at most two ordinary interval pieces. Keep the preexisting prefix weights as well. All arithmetic uses the binary quotient, remainder and endpoints; no loop has $q$ iterations. For a reflection $x \mapsto a + N - 1 - x$, a removed half-open interval $[u, v)$ contributes its weight to $[a + N - v, a + N - u)$. This reverses the interval, with the endpoint convention accounted for exactly.

The independent weighted verifier first checks the unweighted source trace, then reconstructs the weight transport with separate prefix-event and residue-anchor formulas. It does not call the producer's transport helpers. Its final canonical histogram must match the certificate, including sorted unique vectors, positive multiplicities and total mass. Conservation alone would be insufficient; the local trace replay establishes which original weights belong to each orbit. Tests disable producer entry points during replay and mutate arithmetic, source and trace fields.

This is an implementation of classical weighted AHT's polynomial encoded operation [18, Theorem 16]. Canonical runs prevent an implicit expansion of the point universe. A translation fold maps old suffix breakpoints into one period and introduces only a constant number of boundary points; reflection also transports breakpoints directly. Static gap boundaries are charged to the explicit pairing system. Thus the number of weight runs and emitted histogram blocks remains polynomial in input runs, pairings and trace length. Coordinate magnitudes can grow by summing many original points, but their bit lengths grow only by the logarithm of the represented mass. The dimension $d$, endpoint bits, signed weight bits and exact integer arithmetic are all part of the cost model. Section 101 gives the sharp native bound $R_T \le R_0 + 2T$, explicit mass and output bounds, and the later optimization that restricts vector arithmetic to each receiving interval.

The trace is needed internally even when the caller does not request a certificate. A supplied source-bound trace can instead be checked and reused, allowing several different weight queries on the same interval system. This still performs exact arithmetic replay; it avoids discovering a new orbit-reduction schedule. The cycle allowance bounds orbit discovery. Later weight transport and independent verification are controlled by their cooperative callbacks, not silently charged to that cycle counter.

## 94.2 Three weights suffice on a torus boundary

The validated normal arc system has one orbit per connected surface component. Integral ownership assigns $+1$ per surface vertex, $-1$ per arc and $+1$ per polygon, with every charge placed at an intersection point in that cell's component. The sum is its Euler characteristic. Two more interval weights measure intersections of its boundary with a homology basis of the ambient boundary torus. The producer constructs that basis by a primal tree and a disjoint dual tree. The verifier independently checks that the proposed cycles are closed and independent modulo face boundaries using binary elimination. Generalized triangulations with loops and repeated face edges are supported.

As proved in Section 92, a component is a compressing disc exactly when its Euler characteristic is one and its two boundary parities are not both zero. A nonzero boundary class ensures the component has boundary; classification then forces a disc. On a torus an essential embedded circle has primitive slope and nonzero mod-two class. A closed projective plane also has Euler characteristic one but zero boundary values, so it is excluded without an irreducibility assumption.

Componentwise evaluation is necessary. Two parallel meridian discs have zero total mod-two boundary class but contribute two discs. Conversely a vertex-link disc and a nondisc component can yield misleading aggregate Euler/homology values. The weighted histogram distinguishes their contributions. The external audit includes both phenomena and one-sided components.

The public census has three modes. `disk` uses three weights; `summary` uses five, adding polygon and boundary-point counts; `coordinates` uses $7t$ weights and returns actual connected-component normal coordinate vectors with multiplicities. Boundary-point count is not the number of boundary circles. Full-coordinate mode is useful when an actual disc vector must be supplied to a later geometric operation; the cheaper three-weight query is sufficient when only the count is needed. All modes can reuse the same verified unweighted trace.

## 94.3 The quadrilateral core shrinks the numerical query

For each global vertex, subtract the minimum incident triangle count times its vertex-link vector. These links can be realized disjointly from the remaining surface. Their boundary discs are inessential and do not affect the compressing-disc count. Let $Y$ be the remainder and let $g$ be the gcd of the original quadrilateral coordinates. Matching equations propagate divisibility by $g$ from a zero triangle at each global vertex, so $P = Y/g$ is integral and admissible whenever $g > 0$. If all quadrilaterals vanish, $Y = 0$ and the essential-disc count is zero without any orbit query.

The proof in Section 92 gives

$$D(x) = gD(P), \qquad \max_i P_i \le (4t-1)\frac{Q_{\max}}{g}.$$

This identity concerns discs. It is false for the total number of components: two copies of a one-sided normal surface can form its connected orientation double. The count-only API consequently reports no inferred total component count. The original unreduced census remains available for that question.

For $x = gD + (g+1)L$, where $D$ is a fixed primitive vertex-reduced meridian and $L$ is a vertex link, the full coordinate gcd can be one. Common scaling alone then achieves nothing. The new

operation removes the $g + 1$ links and queries the fixed core $D$, returning exactly $g$ essential discs. The core orbit endpoints have size independent of the bit length of $g$. Reading, validating and fingerprinting $x$, computing its minima/gcd and verifying the outer count still depend on the original input bits. The optimization removes those bits from the expensive component query, not from the entire call.

The producer validates the original vector, constructs the core, and invokes a source-validated three-weight census on it. The count certificate binds the original source, all removed link multiplicities, the divisor, core vector, core certificate and final multiplication. The verifier independently reconstructs the minima and gcd rather than invoking the core producer. A modified original vector is rejected even if it has the same reduced core. This prevents replacing the original source binding with only a small-core fingerprint.

## 94.4 Native contracts and validation

The maintained census, independent census verifier and reduced count query are available through these entry points:

```
normal_surface_components.normal_component_census
normal_component_verify.verify_normal_component_certificate
normal_disk_kernel.normal_compressing_disk_count
```

The last module also supplies the independent count-certificate verifier. The generic weighted APIs are in `weighted_orbits` and `weighted_orbit_verify`. The census can receive a supplied orbit trace; then `max_cycles` must be `None`, because no orbit discovery runs. Invalid mode, cycle and periodic-rule options are rejected before geometry work. Incomplete queries publish no component histogram, disc count or certificate. Cancellation propagates during validation, discovery, transport, interpretation and replay, including for false-valued callables.

The original geometry validator and orbit engine are byte-identical to baseline `04a66388e255`. Six new runtime modules implement the extension. Normal geometry and its exact source encoding are shared trusted reconstruction; weight arithmetic, basis verification, component interpretation and core arithmetic have separate checking paths. This independence is supplemented by an external Regina oracle, which does not use producer geometry or weight transport to obtain expected components.

The report's 30 new test methods and four integration tests pass. They include source/summary mutations, generalized boundary bases, actual component-vector multisets, enormous multiplicities, one-sided scaling and source-bound trace reuse across all three modes with orbit discovery disabled. The native audit repeats 10,000 small weighted-system comparisons across the two periodic rules, 2,500 independent checker cases, 366 mutation checks and huge endpoint tests. Its frozen external corpus contains 1,275 vectors on 45 triangulations, with 3,825 census comparisons and replays plus 1,275 core-count comparisons and replays. Regina explicitly expands only these bounded fixtures; the maintained queries use compressed coordinates. The corpus includes 198 cases with discs whose total boundary class cancels, and cases whose aggregate data suggests a disc that is absent. The native implementations pass all these checks.

## 94.5 Complete supplied-vector measurements

The maintained suite passes all 1,084 tests with Regina in 209.265 seconds; the 34 focused tests pass in 1.566 seconds. The native audit completes in 35.314 seconds. A separate timing run takes 280.161 seconds and completes all 200 measured calls and 40 retained warm-ups. Ten fixtures each have two query methods, an identical A/A copy of each, five shuffled measured rounds and one warm-up per arm. Each timed call includes fresh input validation, certificate production and

independent source replay. Certificate serialization and the expected-count assertion are outside the clock. Both audit and timing freeze the same 153 source, driver, test, fixture and corpus hashes. The four original codec, orbit and geometry dependencies match the baseline exactly.

These are comparisons between the new query interfaces. Full-coordinate census provides more output than a three-weight disc query, and an unreduced census provides more output than the core count-only operation. The table therefore measures the saving from requesting sufficient information for the disc-count task; it is not a comparison against a previously available whole-knot recognizer.

The first family uses primitive meridian discs in layered solid tori with $t = 4, 8, 16, 32, 64, 128$ tetrahedra. Every query reports one compressing disc. The full census carries $7t$ weights and the disc census carries three.

| Tetrahedra | full ms | three ms | full/three | full A/A | three A/A |
|---:|---:|---:|---:|---:|---:|
| 4 | 4.125 | 3.065 | 1.396 | 1.002 | 1.010 |
| 8 | 12.352 | 6.392 | 1.972 | 0.999 | 1.004 |
| 16 | 55.524 | 17.167 | 3.179 | 1.023 | 0.974 |
| 32 | 303.772 | 60.642 | 4.973 | 1.009 | 1.017 |
| 64 | 2254.345 | 310.024 | 7.338 | 1.002 | 1.073 |
| 128 | 16945.105 | 1698.802 | 9.975 | 1.015 | 1.001 |

At $t = 128$, complete full-coordinate and three-weight medians are 16,945.105 and 1,698.802 ms; the median paired ratio is 9.975. Both execute 133 orbit cycles and 21,008 replay events. The saving comes from weight construction and transport, rather than a different orbit schedule: weight dimension falls from 896 to three and maximum constant-weight runs from 385 to eight. The compact serialized certificate falls from 2,504,362 to 1,777,349 bytes; the large common orbit trace remains. Across this family, full A/A paired medians range from 0.999 to 1.023 and three-weight A/A from 0.974 to 1.073. These controls are much smaller than the larger dimension savings. No asymptotic exponent is fitted to these six timing points.

The second family fixes $t = 16$ and supplies $x = 2^b D + (2^b + 1)L$ for $b = 64, 512, 4096, 32768$. Its full coordinate gcd is one, while every query must count exactly $2^b$ essential discs. Both arms already use three weights. The comparison isolates vertex-link removal and quadrilateral-content division, including the original-source checks and outer count proof.

| $b$, $g = 2^b$ | raw ms | core ms | raw/core | raw A/A | core A/A |
|---:|---:|---:|---:|---:|---:|
| 64 | 42.837 | 21.948 | 1.952 | 1.021 | 0.996 |
| 512 | 83.938 | 41.735 | 2.108 | 0.999 | 0.995 |
| 4096 | 116.878 | 43.794 | 2.688 | 1.018 | 0.981 |
| 32768 | 296.942 | 35.789 | 5.620 | 1.041 | 0.967 |

For every scale the core has 11-bit coordinates and 14-bit orbit endpoints. The unreduced query uses 39 cycles, 819 replay events and at most 51 weight runs; the core uses 21 cycles, 400 events and at most eight runs. At $b = 32768$, the original maximum coordinate length is 32,779 bits and its unreduced orbit endpoints have 32,782 bits. The compact proof shrinks from 4,523,182 to 56,757 bytes, while still binding the original source. The reduced orbit work is independent of $b$ for this family; the outer reading, validation, arithmetic and fingerprint work remains sensitive to $b$.

The largest core case has appreciable timing variability. Its five raw/core ratios are 5.187, 4.784, 8.899, 5.620 and 5.977; their median is 5.620. The separate arm medians, 296.942 and 35.789 ms, must not be divided and reported as that paired ratio. The identical core control has a 60.289 ms median, and individual raw/core timings change substantially across rounds, even though their median A/A ratios are 1.041 and 0.967. Across the core family the raw A/A medians range from 0.999 to 1.041 and core A/A from 0.967 to 0.996. Median controls alone do not capture the

observed variation. The raw samples and warm-ups are retained; deterministic operation counts, bit sizes and certificate sizes substantiate the reduction independently of the precise elapsed-time factor.

These measurements neither include finding a normal vector nor establish a uniform saving on arbitrary triangulations. Small already primitive vectors can incur reduction overhead. No automatic knot-decision stage is added by this integration, and a zero count is not evidence that other disc vectors do not exist.

Reproduction uses `../fast/weighted_research/native.py`, with `audit` and `benchmark` modes. It pins the unchanged native dependencies and all runtime, driver, test, fixture and corpus hashes before and after each run. Data are in `data/weighted-components-*`; the delivered report is unchanged. The two-weight ambient-homology refinement remains unimplemented. Section 105 now provides certified diagram-to-exterior construction for a canonical unsimplified triangulation. Complete normal-vector search and hierarchy moves with controlled encoded cost remain global tasks. Efficiently counting discs in one supplied vector does not settle those tasks or establish general quasipolynomial recognition.

# 95 Ordered singleton witnesses and virtual inverse replay

Report 51's ordered-witness idea now accelerates verification of the maintained version-eight singleton batches. The previous checker separately located each donor's singleton and scanned its grammar for dependencies. A source subgrammar shared by many donors was repeatedly visited. The new checker validates a claimed dependency order using one shared summary, then evaluates signed references instead of eagerly constructing inverse words. It retains the existing certificate schema and complete fallback for unordered witnesses. This changes proof processing, not which algebraic moves are justified.

## 95.1 A scalar summary checks the claimed dependency order

A batch selects distinct generators $g_1, \dots, g_k$ and distinct relator slots. The native schema requires at least one surviving generator. The witness lists the selected donors in dependency order: the definition of $g_i$ may mention survivors or $g_j$ with $j < i$, but no later selected generator. Give survivors rank zero and $g_i$ rank $i$. For each source grammar node $X$, record

$$h(X) = \max\{\mathrm{rank}(|a|) : a \text{ occurs in } X\}, \qquad c(X) = \min\{2, \#\text{ occurrences of rank } h(X)\}.$$

The empty node has $(h, c) = (0, 0)$; a letter has its assigned rank and count one. For a concatenation $X = YZ$, set

$$h = \max\{h(Y), h(Z)\}, \qquad c = \min\{2, \mathbf{1}_{h(Y)=h}c(Y) + \mathbf{1}_{h(Z)=h}c(Z)\}.$$

These count literal occurrences, without free or cyclic reduction. A repeated child contributes twice. The donor for $g_i$ is valid in the claimed order exactly when $(h, c) = (i, 1)$. Distinct pivot ranks imply that the unique maximum is $g_i$ itself; every other selected letter has smaller rank. This checks both the singleton property and acyclicity of all definitions.

The checker traverses the union of all current source roots, so a shared node is summarized once. It also rejects any source letter outside the current live set. The summary uses a constant number of integers, with $O(\log(k + 1))$ rank bits per node, rather than an explicit selected-generator support set. The traversal uses the arena's sorted reachable-node list, whose comparisons still cost $O(M \log M)$ for $M$ reachable nodes. Root-list processing, input labels, exact length arithmetic and cooperative work charges remain part of the implementation cost; "one summary per node" is not a linear bit-time claim for the whole verifier.

The producer already has the dependency graph while selecting candidates. It now topologically sorts the selected subgraph before emitting a certificate, in $O(k + E)$ graph operations, where $E$ counts the stored dependencies of selected donors, including survivor edges inspected while filtering. The queue changes neither the candidate score nor the selected donor set. The checker's rank computation imports no producer planning or metadata. The internal planner's default result retains its earlier selection order; the actual search explicitly requests the ordered witness.

## 95.2 Compile both orientations without inverse expansion

Write the selected donor as $U_i g_i^{\epsilon_i} V_i$, with $\epsilon_i \in \{-1, 1\}$. Its raw defining equation is

$$g_i = (V_i U_i)^{-\epsilon_i}.$$

The checker descends the unique maximum-rank path to find the pivot position, then obtains $U_i, V_i$ by source slicing. No replacement word or dependency set supplied by the certificate is trusted. The contexts and their signs are retained as definitions, with no eager inverse construction.

Evaluation states are pairs $(X, \sigma)$, $\sigma \in \{-1, 1\}$. A positive concatenation state visits its children in source order; a negative state visits the children in reverse order with negative signs. A selected signed letter follows its defining context with the product of the requested sign, letter sign and $-\epsilon_i$. A surviving letter remains a signed terminal. A memoized iterative traversal compiles each needed signed state once, with an active-state set as an additional cycle check. It includes every donor definition as a goal, even if all surviving relators omit it. Only after successful evaluation are donor slots zeroed, the remaining roots replaced and eliminated generators removed from the live set.

This is exact noncommutative word substitution. The implementation invokes no occurrence scan, support-mask query, eager inverse, word equality, expansion, free reduction or cyclic normalization during ordered replay. Shared arena slicing and concatenation retain their existing exact contracts. Independent literal replay continues to construct the explicit homomorphism after checking its expansion budget.

Section 92 proves that pivot paths in distinct valid donors are disjoint: if a shared node contained two pivots, the corresponding two definitions would depend on each other. Complementary slices and cyclic contexts therefore add at most $M$ nodes in total. The source-plus-context union has at most $2M$ nodes, and its two orientations give at most $4M$ compiled states. Consequently the new checker satisfies

$$M_{\text{out}} \le 4M, \qquad N_{\text{after}} - N_{\text{before}} \le 5M.$$

Here $N$ counts all persistent arena nodes, including previously unreachable ones; $M$ counts only the current reachable source. The producer and unordered fallback retain their earlier eager-inverse bound of $7M$ added nodes. Bit lengths of represented words and stored lengths remain explicit costs. The result removes repeated local scans and unnecessary inverse grammars; it neither bounds the number of batches nor proves a general successful sequence for every unknot diagram.

## 95.3 Compatibility, budgets and source-bound evidence

The move remains `elimination_batch` with the same original slot and generator records. No new certificate version, optional trust flag or producer ordering promise is introduced. If the rank test fails to establish the claimed order, the checker has not allocated contexts or changed any public state and falls back to the existing occurrence/support and dependency-graph checker. An unordered valid old witness therefore retains its meaning. A cycle, missing pivot or repeated literal pivot still fails independent validation. The rank attempt is charged: old certificates near a finite work cap can require more allowance, even when the unchanged fallback accepts them with sufficient resources. Resource exhaustion never becomes a mathematical rejection.

The full source wrapper still rebuilds every Wirtinger relator from the input diagram, checks the existing terminal and binds the original diagram. New ordered certificates also pass the pinned previous literal and compressed verifiers. The optional batch portfolio retains its existing recovery-aware trial allowance and fallback policy. Its producer pays the additional order construction, and earlier simplification may bypass this backend entirely. End-to-end measurements must therefore accompany the local grammar result.

The native audit checks 400 abstract acyclic systems with two noncommuting survivors, arbitrary contexts, both signs and empty definitions. Each is replayed in forward and reverse witness order by both the pinned old and new compressed engines and by literal substitution: 1,600 compressed and 800 literal checks agree. Allocation bounds are checked on each completed replay. On 95 validated diagrams, default certificates remain identical; the optional batch search gains and loses no completed result in this corpus. All saved positive proofs undergo both literal and compressed source replay under both engines, giving 1,048 successful checks. These comparisons include old unordered proofs and newly ordered proofs, not only newly generated evidence.

Six new tests cover arbitrary labels and noncommuting dependencies, legacy reverse order, shared 1,024-bit powers, literal repeated-pivot forgeries, unreferenced cycles, dead source labels, unchanged candidate selection, producer-disabled replay, public source certificates and interrupted-state publication. The 12 existing batch tests also pass. The new tests disable the producer and the word operations that ordered replay is intended to avoid. The report's original files remain unchanged; this is a native adaptation of its checker rather than application of its older search-policy overlay.

## 95.4 Measured savings, compatibility overhead and whole recognition

The maintained suite passes all 1,090 tests with Regina in 218.217 seconds. The 12 existing batch tests and six new integration tests take 0.556 and 0.045 seconds respectively. The final source-pinned audit takes 8.101 seconds. Three separate benchmark executions contain 180 kernel calls, 120 group-stage calls and 380 whole-recognition calls, plus 36, 24 and 76 retained warm-ups. All 680 measured calls complete. Each fixture has five rounds with shuffled old/new and old/new A/A arms. All four runs preserve the same 424 native Python and corpus hashes; the pinned old package is loaded separately. Kernel, stage and pipeline runs take 39.082, 2.843 and 34.649 seconds respectively.

Kernel timing includes fresh arena construction and complete checked raw substitution. The shared family has $k$ independent definitions $g_i = (ab)^{2^{4096}}$ sharing one source power grammar, with both signs used in retained roots. The tower family has $k$ dependent definitions with noncommuting contexts, each raised to $2^{128}$. The legacy family uses the same towers but reverses their witness order, deliberately exercising the compatibility fallback. No represented long word is expanded. Outputs have matching pinned-old and current structural digests; literal correctness is separately checked by the small-system audit.

| Family | pivots | old ms | new ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|---|
| shared | 8 | 100.833 | 44.068 | 2.266 | 0.988 | 0.996 |
| shared | 32 | 267.890 | 44.679 | 6.007 | 1.000 | 0.994 |
| shared | 128 | 948.063 | 45.743 | 20.552 | 0.993 | 0.984 |
| tower | 8 | 14.265 | 11.313 | 1.258 | 0.977 | 1.003 |
| tower | 32 | 60.742 | 47.640 | 1.284 | 1.008 | 0.988 |
| tower | 128 | 272.603 | 215.662 | 1.243 | 0.978 | 1.046 |
| legacy | 8 | 14.085 | 15.152 | 0.924 | 0.992 | 0.999 |
| legacy | 32 | 60.553 | 67.022 | 0.903 | 0.991 | 0.958 |
| legacy | 128 | 276.179 | 308.654 | 0.895 | 0.953 | 0.990 |

At 128 shared pivots, old/new total medians are 948.063/45.743 ms, with paired ratio 20.552. The

replay work counter falls from 2,699,059 to 55,213. Exactly 4,357 source nodes are summarized once; the represented retained word length has 4,099 bits. Both implementations add 4,100 nodes in this case, so the saving is repeated inspection of shared grammars, not output size. The 128-level tower instead demonstrates virtual inverse savings: new allocations fall from 50,044 to 33,407, and replay work from 391,056 to 221,082. The paired total-time ratio is 1.243, with 16,514-bit output lengths.

The legacy compatibility cost is visible. On reversed 128-level towers, old/new medians are 276.179/308.654 ms and the paired ratio is 0.895. New allocations remain 50,044; replay work rises from 391,056 to 442,138 because the charged rank scan precedes the unchanged fallback. The smaller reversed cases also regress, with ratios 0.924 and 0.903. Kernel old A/A medians range from 0.953 to 1.008 and new A/A from 0.958 to 1.046. The large shared-case gain and the legacy overhead both remain recorded. Newly emitted ordered witnesses avoid this fallback; compatibility does not imply identical finite-budget cost.

Group-stage timing uses validated stabilized-circle diagrams. It includes fresh diagram construction, recovery, the same optional batch portfolio and mandatory independent source replay, while bypassing earlier simplification.

| Crossings | old ms | new ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| 8 | 1.430 | 1.274 | 1.154 | 1.032 | 1.000 |
| 16 | 2.800 | 2.479 | 1.139 | 0.996 | 1.011 |
| 32 | 5.695 | 4.816 | 1.180 | 0.993 | 0.997 |
| 64 | 11.500 | 9.748 | 1.188 | 0.995 | 1.000 |
| 128 | 24.006 | 20.978 | 1.213 | 1.012 | 1.032 |
| 256 | 47.905 | 51.442 | 1.128 | 0.977 | 0.997 |

Paired old/new medians range from 1.128 to 1.213. The largest case shows why paired ratios and separate arm medians must be distinguished: at 256 crossings, old/new medians are 47.905/51.442 ms, but the five paired ratios are 0.931, 1.128, 1.213, 1.084 and 1.208. The last two rounds also have substantially larger absolute times. Median old/new A/A controls across this family range from 0.977 to 1.032 and 0.997 to 1.032, but these do not capture all individual variation. These data suggest modest stage savings, not a uniform factor.

A separate deterministic replay audit checks the saved stage proofs under both verifiers, 24 times in total. At 256 crossings, compressed arena work falls from 31,010 to 18,097 and final arena nodes from 3,049 to 2,038. That counter includes SLP construction and replay after independent diagram reconstruction, but excludes the preceding source-recovery budget. Producer work rises from 94,320 to 95,842 to construct the ordering; the producer still has 2,544 nodes and the certificate remains 12,775 bytes. These counters explain why local verification savings give a smaller whole-stage effect. The additional driver and records are `data/ordered_batch_replay_work.py` and `data/ordered-batch-replay-work.json`.

Whole-recognition timing uses the original 19-diagram corpus and the same optional batch portfolio on both versions. It includes fresh validated inputs, all earlier stages, bounded search and mandatory certificate replay. No incomplete result is omitted; all 380 measured calls complete.

| Input | old ms | new ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| survivor-00 | 14.267 | 14.144 | 1.003 | 1.004 | 0.998 |
| survivor-01 | 27.925 | 28.734 | 0.968 | 1.003 | 0.998 |
| survivor-02 | 9.355 | 9.249 | 1.010 | 0.983 | 0.990 |
| survivor-03 | 38.030 | 38.350 | 0.990 | 1.002 | 1.010 |
| mirror-03 | 5.924 | 5.816 | 1.033 | 1.017 | 1.008 |
| survivor-04 | 28.923 | 28.983 | 1.000 | 1.000 | 1.001 |
| survivor-05 | 15.295 | 15.443 | 0.983 | 1.004 | 1.007 |
| survivor-06 | 5.278 | 5.151 | 1.028 | 1.003 | 0.990 |
| survivor-07 | 19.346 | 19.521 | 0.993 | 1.003 | 1.004 |
| survivor-08 | 7.067 | 6.978 | 0.998 | 0.993 | 0.979 |
| mirror-08 | 15.225 | 15.164 | 0.999 | 0.990 | 0.987 |
| survivor-09 | 5.586 | 5.464 | 1.016 | 1.006 | 1.007 |
| survivor-10 | 5.179 | 5.158 | 1.004 | 1.004 | 1.000 |
| survivor-11 | 4.765 | 4.701 | 1.013 | 1.001 | 0.999 |
| gordian | 1207.402 | 1210.837 | 0.998 | 1.001 | 0.992 |
| trefoil | 0.058 | 0.056 | 1.038 | 1.010 | 0.980 |
| figure_eight | 0.062 | 0.062 | 1.000 | 0.992 | 0.981 |
| conway | 0.974 | 0.975 | 1.002 | 1.001 | 1.003 |
| kinoshita_terasaka | 1.002 | 1.010 | 0.998 | 1.000 | 0.999 |

The ratios range only from 0.968 to 1.038; old A/A medians range from 0.983 to 1.017 and new A/A from 0.979 to 1.010. Gordian medians are 1,207.402 and 1,210.837 ms, with paired ratio 0.998. This workload demonstrates no useful whole-recognizer acceleration from the new checker. Earlier exits and the existing discovery fallback limit its contribution. The strong kernel result therefore concerns compressed ordered proof replay; it is not promoted to a recognition speedup or a general complexity result.

Reproduction uses `../fast/compressed_word_research/ordered_batch.py`; `audit`, `kernels`, `stages` and `pipeline` select separate executions. Baseline `06fcb7685da7` is loaded as an isolated package. The word arena and source wrapper remain byte-identical; the only modified existing runtime modules are the batch planner, batch checker and search call site, plus the new ordered checker. Records and source manifests are retained under `data/ordered-batch-*`. General diagram discovery, controlled hierarchy growth and a complete quasipolynomial recognition bound remain open.

## 96 Adaptive periodic closure with unchanged orbit certificates

The component-certificates delivery in commit `92efcbaed` identifies repeated work in the maintained AHT producer: its periodic-merger scan restarts from the beginning after every successful merge. The new native scheduler remembers successful candidates whose operands have not changed. Its adaptive policy first uses indexed scans, then switches to a heap after spending an all-pairs allowance. This is an implementation improvement to the existing exact periodic rule, with a quadratic per-closure candidate bound and identical completed certificates. It does not change the classical AHT cycle argument or establish general quasipolynomial unknot recognition.

The delivered reference producer is byte-identical to maintained baseline `109dabf896f2`. The integration installs the delivered scheduler, producer change and 11 scheduler tests, plus four native integration tests. The delivery's alternative weighted and normal-profile modules are not needed: our newer weighted census, quadrilateral core and separately implemented arithmetic verifier remain intact. This avoids replacing their source-binding and independent-replay contracts with a different research interface.

### 96.1 Stable identifiers preserve the historical greedy choice

A periodic increasing pairing with period $p$ and inclusive support $[a, d]$ can merge with one of period $q$ when their support overlap contains at least $p + q - \gcd(p, q)$ points. The classical alternative threshold is $p + q$. Both existing exact rules remain unchanged. The old closure tests current list pairs $(i, j)$, $i < j$, in lexicographic order, accepts the first legal merge, replaces row $i$, deletes row $j$, and restarts.

Assign each input row its original position as a permanent identifier. Active identifiers retain their original relative order under deletions. A row's generation starts at zero and increases each time it is replaced. The heap stores successful tests as

$$(i, j, \nu_i, \nu_j, \text{replacement}).$$

It orders first by the two identifiers. An entry is stale if an operand has been deleted or its generation has changed. Initially every pair of periodic rows is tested. After merging $i < j$, delete $j$, increase $i$'s generation, and test precisely the surviving periodic partners of the new row $i$. A failed test between two unchanged operands never needs to be repeated.

The invariant is that every currently legal pair has exactly one nonstale entry, and every nonstale entry contains its exact current replacement. Initialization establishes it. A merge leaves all pairs avoiding $i, j$ unchanged; deletion and the generation increment invalidate the other old entries; the new incident tests restore the missing current entries. Thus induction preserves the invariant. Removing stale minima then exposes the lexicographically first legal pair of active identifiers. Its order is exactly the order of current list positions, so it is the historical greedy choice.

For a recorded merge, the producer counts active identifiers below each operand to recover the old current-list indices. At most $k - 1$ merges occur in a closure with $k$ starting rows, so these rank scans cost $O(k^2)$ operations. The surviving ordered list is identical after every closure. Subsequent carrier selection, transmission, truncation and next-cycle preparation are unchanged deterministic operations. Induction over cycles therefore gives identical orbit counts, begun-cycle counts and complete certificates, for either periodic rule. The orbit and weighted checkers need no new proof rule.

This equality concerns completed mathematical traces. Polling schedules and work statistics can change, so a cooperative wall-clock deadline may stop at a different point. A fixed cycle allowance still counts the same begun AHT cycles. Cancellation propagates, and incomplete runs publish no count or certificate.

### 96.2 Quadratic tests with an adaptive fast path

Pure queue closure on $k \geq 1$ rows performs at most

$$\binom{k}{2} + \sum_{r=1}^{k-1} (r - 1) = (k - 1)^2$$

merger tests. The first term initializes the heap. A merge leaving $r$ rows retests at most $r - 1$ partners of the replacement. Nonperiodic rows only reduce these counts. Each successful test creates one heap entry and each entry is popped at most once, including stale entries. Hence total heap work and worst-case heap storage are quadratic up to the heap's logarithmic factors. For $B$-bit endpoints, a conservative per-closure bit bound is

$$O\Big(k^2\big(G(B) + B + \log^2(k + 1)\big)\Big),$$

where $G(B)$ bounds the merger's gcd and bounded endpoint arithmetic. Heap keys contain $O(\log(k + 1))$-bit identifiers and generations. Working storage is $O(k^2(B + \log(k + 1)))$ bits,

excluding the separately retained full trace. This is a real space tradeoff with the historical linear working list.

An eager heap is wasteful when the first candidate repeatedly succeeds: $k$ identical periodic rows require only $k-1$ old tests, but initializing a heap already requires $\binom{k}{2}$. The adaptive policy therefore begins with the historical candidate order. It finds the first two periodic rows lazily, tests that pair immediately, and restarts immediately on success. After the first failure it collects the remaining periodic indices once, then resumes the same lexicographic sequence without revisiting that failed pair. This also avoids repeated inner scans through nonperiodic rows.

Allow this scan at most $\binom{k}{2}$ candidate tests, where $k$ is the original closure size. A further pending test triggers the queue on the current residual list. It follows that adaptive closure makes at most

$$\binom{k}{2} + \max(k-1,0)^2$$

merger tests. At most $k$ index reconstructions each scan at most $k$ rows. No-merge closures and repeated first-pair successes build no heap; their extra index storage is linear. The threshold is an internal scheduling policy, not a limit that returns an inconclusive mathematical answer. In particular, it is not substituted for the caller's cycle allowance or cancellation hook.

The public selector is `merger_scheduler`, accepting `adaptive` (the default), `legacy` and `queue`. The last two provide controlled comparisons and a linear-scheduler-space option. Existing weighted, sparse-incidence and normal-surface consumers inherit the adaptive default. New statistics record queue runs, switches, peak heap entries and stale pops; the old non-scheduler event counts agree.

### 96.3 A cubic-to-quadratic complete-count witness

The delivery proves that redundant merger scans can dominate a complete run, not merely an isolated phase. For $m \geq 2$, set $P = 10m$, $N = 23m$ and take

$$g_i : [i, P + 2i - 1] \longrightarrow [P + 2i, 2P + 3i - 1].$$

The input is $g_0, \ldots, g_{m-1}$ followed by $m$ copies of $g_m$, hence $2m$ pairings. Under the sharp rule the complete maintained calculation has five cycles and one orbit, with exact test counts

$$T_{\text{old}} = m^3 + \frac{m^2 + 5m}{2} - 4, \qquad T_{\text{queue}} = 5m^2 - 6m + 2.$$

For distinct $i < j$, a merger would require $\gcd(P + i, P + j) \geq 2(j - i)$, impossible because the gcd divides $j - i$. Only duplicate tail rows initially merge. With $r$ duplicates left, the old scan repeats $\binom{m}{2} + mr$ failed tests before the next success, or before its final failed pass. Cycle one consequently has $m^3 + m - 1$ tests. The queue instead has $(7m^2 - 7m + 2)/2$ tests in that cycle.

Truncation removes three points and makes the last row nonperiodic. Cycle two tests $\binom{m}{2}$ pairs. Its carrier transmission creates a period-one row, and another three-point truncation leaves one further nonperiodic row. In cycle three, the period-one row first merges with $g_0$ and then successively absorbs the remaining periodic rows. The old scan uses $2m-3$ tests, while the queue uses $(m-1)^2$. Transmission makes the remaining nonperiodic row an identity. Cycle four deletes that identity and truncates to one point; cycle five contracts that point. Adding the three nonzero phase counts gives the displayed formulas. The native tests check the full counts, cycles and identical independently replayed certificates.

Since there are five cycles, $O(m)$ rows and $O(\log m)$-bit endpoints, the revised complete count on this family has $O(m^2 \operatorname{polylog} m)$ bit cost, replacing the historical cubic merger bottleneck. This is a lower bound for that implementation on explicit interval systems, not a lower bound for orbit counting or a constructed family of hard knot diagrams. On general inputs, the closure bound must still be multiplied by the actual polynomial AHT cycle bound. No improved general degree or quasipolynomial hierarchy theorem is inferred from this family.

### 96.4 Native verification and scope

All 1,105 maintained tests pass with Regina in 210.773 seconds. The 11 delivered scheduler tests pass in 0.537 seconds and four native integration tests in 0.156 seconds. They exercise stale generations and current-index reconstruction, easy cases with no heap, unchanged candidate order, both periodic rules, invalid options before input consumption, false-valued cancellation callbacks, every allowance around the five-cycle endpoint, huge binary inputs, weighted proofs and all three normal-census modes plus the reduced core query. Independent weighted replay succeeds with the schedulers disabled.

The separate native audit takes 32.419 seconds. It compares 1,500 random systems against literal connectivity, using both periodic rules and all three new scheduler choices against the unchanged pinned producer. All 9,000 old/new complete certificates and cycles agree, with 12,000 independent random certificate replays. Four analytic cases include 4,096-bit coordinates and check the exact five-cycle work formulas. The frozen Regina corpus contributes 1,275 normal vectors on 45 triangulations: 3,825 census checks and 1,275 core-count checks, each with independent replay, all passing. These validate the new default throughout the existing weighted pipeline.

### 96.5 Certified native measurements

The interval benchmark retains 420 measured calls and 84 warm-ups over 14 fixtures, with five shuffled rounds and separate A/A copies of the historical, adaptive and pure-queue arms. All complete and return identical certificates across arms. The 101.440-second execution includes fresh pairing construction, complete counting, certificate generation and independent replay in every timed call. Native meridian arc geometry is prepared outside these interval timers. Serialization and cross-arm certificate comparisons are also outside the timers. Table entries are separate median milliseconds; ratios are medians of the five paired old/new ratios, not ratios of those medians.

| Input | old ms | adaptive ms | queue ms | old/adapt. | old/queue |
|---|---|---|---|---|---|
| five-cycle-8 | 1.078 | 0.986 | 1.028 | 1.134 | 1.047 |
| five-cycle-16 | 5.567 | 3.046 | 3.324 | 1.828 | 1.687 |
| five-cycle-32 | 40.857 | 11.152 | 11.539 | 3.721 | 3.380 |
| five-cycle-64 | 303.463 | 42.773 | 45.284 | 7.119 | 6.772 |
| five-cycle-128 | 2405.376 | 167.526 | 179.342 | 14.413 | 13.389 |
| duplicates-16 | 0.292 | 0.308 | 1.113 | 0.969 | 0.257 |
| duplicates-64 | 1.051 | 1.056 | 16.320 | 0.991 | 0.064 |
| duplicates-256 | 4.090 | 4.107 | 291.486 | 1.003 | 0.014 |
| meridian-16 | 9.462 | 9.063 | 9.818 | 1.048 | 0.958 |
| meridian-64 | 241.530 | 192.101 | 203.347 | 1.290 | 1.188 |
| meridian-128 | 1578.288 | 1088.357 | 1163.742 | 1.463 | 1.364 |
| binary-64 | 7.847 | 3.714 | 3.876 | 2.037 | 2.043 |
| binary-1024 | 11.147 | 4.948 | 4.859 | 2.253 | 2.291 |
| binary-4096 | 24.273 | 9.387 | 8.568 | 2.587 | 2.828 |

At $m = 128$, the five-cycle input takes 2,405.376 ms historically and 167.526 ms adaptively, with paired ratio 14.413. Merger tests fall from 2,105,660 to 97,409; the pure queue makes 81,154 tests but takes 179.342 ms. The adaptive heap peaks at 11,907 entries, versus 12,097 for the pure queue. The five cycles and 15,396-byte certificate are unchanged. This exhibits the proved removal of redundant candidate tests while also showing that test counts alone do not predict Python heap overhead.

For 256 identical rows, the old and adaptive engines each make 255 tests; adaptive closure constructs no heap. Their times are 4.090 and 4.107 ms, with paired ratio 1.003. Pure queue

closure makes 65,025 tests, peaks at 48,769 heap entries and takes 291.486 ms. The adaptive first-pair fast path therefore matters. At 16 duplicate rows the adaptive ratio is 0.969, a small regression rather than a universal speedup claim.

For the supplied 128-tetrahedron meridian, the paired adaptive ratio is 1.463 (1,578.288 versus 1,088.357 ms). Both engines make 349,764 candidate tests, and the adaptive engine never enters the queue: this gain comes from indexing the periodic rows instead of rescanning nonperiodic rows inside each inner loop. All 133 cycles and the 1,776,281-byte certificate agree. At fixed $m = 16$, increasing the binary scale to 4,096 bits gives adaptive ratio 2.587 and pure-queue ratio 2.828, with the same deterministic candidate counts as the small-coordinate case.

Across interval fixtures the median A/A ratios range from 0.974 to 1.021 historically, 0.965 to 1.021 adaptively, and 0.956 to 1.013 for the queue. Individual observations can vary more: the smallest five-cycle fixture contains a 4.045-ms historical call alongside roughly 1-ms calls. All rounds are retained; neither those observations nor warm-ups are silently discarded. These are supplied interval systems, not whole-knot recognition timings.

The separate normal-query benchmark takes 539.262 seconds and retains 180 measured calls plus 36 warm-ups across nine fixtures, with historical and adaptive arms and an A/A copy of each. Each timed call includes fresh validation of the original triangulation and vector, geometry construction, the requested census or reduced-core query, certificate production and the independent normal-source verifier. Both engines answer exactly the same query; every call completes and every certificate is identical across arms. No tests, article builds or other benchmark drivers ran concurrently.

| Query | old ms | new ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| disk-16 | 18.453 | 18.267 | 1.020 | 0.981 | 1.038 |
| disk-64 | 301.444 | 242.123 | 1.224 | 0.998 | 0.903 |
| disk-128 | 1691.973 | 1246.086 | 1.361 | 0.989 | 0.999 |
| coordinates-16 | 53.690 | 53.090 | 1.008 | 1.002 | 1.002 |
| coordinates-64 | 2224.639 | 2206.549 | 1.019 | 0.995 | 1.001 |
| coordinates-128 | 16871.366 | 16881.820 | 1.014 | 0.999 | 0.999 |
| core-64 | 21.728 | 21.252 | 1.030 | 0.999 | 1.012 |
| core-4096 | 23.322 | 22.730 | 1.021 | 1.007 | 0.993 |
| core-32768 | 35.037 | 34.245 | 1.025 | 0.993 | 0.999 |

For the 128-tetrahedron disk query, the complete paired ratio is 1.361: 1,691.973 ms historically versus 1,246.086 ms adaptively. The five individual ratios lie between 1.265 and 1.441. There are still 133 orbit cycles, 21,008 weighted replay events and at most eight weight runs; the certificate is still 1,777,349 bytes. The change is confined to finding the same periodic merges more efficiently. All nine normal fixtures retain their merger-test counts and take the indexed-scan path without switching to a heap.

The full-coordinate census retains much more work outside merger scheduling. At 128 tetrahedra it carries 896 coordinates and reaches 385 weight runs; its unchanged certificate has 2,504,362 bytes. Separate old/new medians are 16,871.366 and 16,881.820 ms. Although the paired median ratio is 1.014, the five ratios are 1.014, 1.021, 1.026, 0.629 and 0.733. In the final two rounds the new arm takes 26,811.822 and 23,415.500 ms, well above its earlier roughly 16–17-second calls. The median A/A ratios near one do not explain away this variation. These data establish no reliable full-coordinate speedup. The 64-tetrahedron disk query also has a new-arm A/A ratio of 0.903; its paired ratios range from 0.914 to 1.434, so its median 1.224 gain has appreciable noise.

The core fixtures use $x = gD + (g + 1)L$, $g = 2^b$, on the 16-tetrahedron solid torus, with $D$ its meridian vector and $L$ its vertex-link vector. Both engines already apply the same quadrilateral normalization. Their paired ratios range only from 1.021 to 1.030; at $b = 32,768$ the medians are 35.037 and 34.245 ms. The unchanged reduced query still has 21 cycles, 400 events and a

56,757-byte source-bound certificate. These small timing differences are not a second claim of the much larger normalization gain reported in the preceding weighted-component section. Across normal fixtures the historical median A/A ratios range from 0.981 to 1.007 and the new ratios from 0.903 to 1.038. All 600 measured calls and 120 warm-ups in the two new benchmarks complete, but none is a whole-knot recognition measurement.

The maintained driver is `../fast/merger_research/native.py`, with `audit`, `intervals` and `normal` modes. Records are in `data/adaptive-merger-*`. All runtime dependencies except the orbit producer match baseline `109dabf896f2`; the scheduler is a new module. Each execution freezes 428 Python and corpus hashes. The delivered archive is retained unchanged, and its Git-blob provenance was checked before copying the three scheduler-related files. The two other new deliveries, including the persistent signed-circuit representation for whole raw singleton blocks, remain separate research integrations. The new scheduler improves a supplied component kernel and adds no automatic knot-decision stage.

# 97 Polynomial replay of an entire raw singleton block

The persistent-elimination prototype in `ProveIt_UnknotRecognition_Research_2026-10-09.zip`, delivered in commit `92efcbaed`, gives a stronger representation bound than the per-batch estimate of Section 95. A sequence of raw singleton eliminations can retain all historical word contexts in one signed circuit, with one binding per eliminated generator. Balanced context assembly makes the *whole block* polynomial in its initial encoded parameters. This removes a depth hypothesis for that block; it does not bound unrestricted normalization, relator exposure or complete knot search.

The native integration applies this representation to independent replay of consecutive version-eight `elimination_batch` moves. It changes neither the certificate schema nor producer selection. The new checker reconstructs all contexts from the already independently recovered presentation, imports no producer routines, and exports once at the next different move or terminal. The separately implemented literal verifier is unchanged. At this checker checkpoint the producer still used its existing immutable arena. Section 98 subsequently integrates a separate persistent producer and proves the whole-block bound for its discovery work.

## 97.1 Bindings and raw word semantics

A positive handle refers to a terminal $g$ or an immutable concatenation $(u, v)$ of signed handles; a negative handle denotes the formal inverse. Thus inversion reverses child order and changes both signs without allocating an inverse grammar. A live terminal $g$ may receive exactly one signed binding $g \mapsto w$. Every old occurrence then follows that binding, including occurrences in older images. Concatenations themselves never change.

For a donor $Ug^{\epsilon}V$, where $g$ has exactly one raw occurrence, the relation gives

$$g = (VU)^{-\epsilon}.$$

The checker descends the unique occurrence path through concatenations and bindings. Negative handles reverse the two children and their signs. Siblings before the occurrence are collected in traversal order; siblings after it are collected in reverse nesting order. Reversing the latter list and appending the former gives exactly $VU$. Pairwise balanced assembly creates the image with no expanded-word positions, slicing, free reduction or equality test. Empty images are allowed. In particular, a cancelling pair in a raw donor remains two occurrences rather than disappearing.

Every off-path sibling is $g$-free in the current binding graph. Installing the new binding therefore cannot create a directed cycle. Subsequent eliminations act on all old occurrences through their

new terminal bindings, so induction gives exactly the ordinary raw Tietze substitutions in every retained relator. Only the donor's original slot becomes zero; all other slots are retained. This argument requires the raw-word condition: a reduction oracle that hid occurrences by cancellation would invalidate the path test.

## 97.2 Independent batch validation and legacy witnesses

Allocation order ceases to be a dependency order once terminal bindings can point forward. Each batch therefore starts with Kahn's algorithm over all historical circuit nodes. A repeated child contributes two incidence edges; both edges are removed when that child is processed. Failure to obtain every node rejects the circuit.

Assign ranks $1, \dots, b$ to the batch's selected generators in their claimed dependency order and rank zero to survivors. In dependency order, compute the maximum rank in each current word and its multiplicity capped at two. Bound terminals inherit their target's summary; negative handles have the same summary as their positive counterparts. A donor is valid when its selected generator is its unique largest-rank letter. This simultaneously proves the singleton condition and that all its selected dependencies have strictly smaller rank. Unbound terminals must belong to the current live set.

If the supplied order fails, the legacy path computes a selected-generator support bitset for every node, derives the donor dependency graph, and runs Kahn's algorithm on that graph. It then repeats the rank check in the recovered order. A cyclic graph, missing pivot, repeated pivot, invalid source slot, reused donor or eliminated generator rejects the witness. Thus unordered legacy evidence remains supported without exporting the intermediate grammar. The bitsets add a polynomial factor in the number of selected generators; they are not treated as constant-size machine words.

All donor contexts in one batch are derived against the same pre-batch state. Their bindings are installed together only after validation and construction succeed. The strict rank inequalities prove the resulting binding graph acyclic. Equivalently, install the bindings in increasing dependency rank: each right-hand side can refer only to already installed selected definitions. This has the same raw word values as the old simultaneous substitution. Unused definitions are checked as well as those reached by surviving relators.

The public roots and live set remain unchanged while the entire block is processed privately. The final export walks signed dependencies and emits ordinary immutable arena terminals and concatenations. Each signed state is compiled at most once; a binding becomes an alias. With $S$ circuit nodes, at most $2S$ ordinary nodes are added, often fewer because only retained roots are exported and interning can reuse old nodes. No mutable binding is inserted into the ordinary arena, so its cached inverses, reductions and metadata remain valid. A malformed later batch rejects the whole private block. Cancellation or export exhaustion can leave historical arena allocations, but publish no new relator roots or live-generator set. If the ordinary arena initially retains $N_0$ historical nodes, the combined node count during replay and export is at most $N_0 + 3S$, excluding ancillary arrays, bitsets and integer storage.

## 97.3 Whole-block size and bit bounds

Let $S_0$ be the initial number of nonempty signed-circuit nodes, $D_0$ the maximum concatenation depth with bindings ignored, and $k$ the number of distinct generators eliminated in the block. No move in this block creates a generator, so $k$ is at most the initial live rank minus one. Let $D_i$ be the historical maximum binding-free depth after $i$ eliminations and $H_i$ the maximum full dependency height.

Every dependency path contains at most $i$ binding edges. Between successive binding edges it

follows at most $D_i$ immutable concatenations. Therefore

$$H_i \le (i+1)D_i + i < (i+1)(D_i+1) = A_i.$$

A singleton context contributes at most $H_i$ siblings. Balanced assembly of $q$ siblings adds at most $\max(q-1,0)$ nodes and at most $\lceil \log_2 \max(1,q) \rceil$ concatenation levels above their roots. Hence

$$D_{i+1} \le D_i + \lceil \log_2 \max(2, A_i) \rceil.$$

For simultaneous batches, first fix all sibling lists in the pre-batch graph, then conceptually construct and install images in dependency order. Those lists are no longer than the later historical height, and their original concatenation depths are no larger than the later $D_i$. Thus the same one-generator recurrence upper-bounds the implemented batch construction.

Put $T_i = D_0 + i + 2 \ge 2$. Induction yields

$$D_i \le D_0 + 6i \log_2 T_i.$$

Indeed, this hypothesis implies $A_i \le T_i^2(1 + 6\log_2 T_i) \le T_i^5$: $\log_2 t \le t-1$ and $6t - 5 \le t^3$ for $t \ge 2$. Consequently $\lceil \log_2 \max(2, A_i) \rceil \le 6 \log_2 T_i$, which gives the next induction step because $T_{i+1} \ge T_i$. Summing the bounds on newly created contexts gives the explicit conservative historical-size estimate

$$S_k \le S_0 + \frac{k(k+1)}{2}(D_0+1) + 2k(k-1)(k+1)\log_2(D_0+k+2) = S_*.$$

Only $k$ additional binding entries are needed. Initial import has at most the reachable ordinary source size; signed terminals and interning can reduce it.

A binary circuit of full height $H$ unfolds to at most $2^H$ letters. Thus all exact length integers computed during export need at most

$$B_* = (k+1)\big(D_0 + 6k\log_2(D_0+k+2) + 1\big)$$

bits. Rank summaries themselves need only $O(\log(k+1))$-bit ranks and counts in $\{0,1,2\}$. There are at most $k$ batches; their topological and rank passes cost $O(kS_*)$ graph operations. Context traversal and balanced assembly are bounded by the same historical-height accounting. Unordered legacy batches add selected-support bit operations and dependency-graph work, still polynomial in $S_0, D_0, k$. Source import, sorting, generator identifiers, interning, the retained relator-slot list and $B_*$-bit export arithmetic must also be charged. These observations prove polynomial encoded replay for the whole raw block, rather than composing a separate constant-factor grammar bound once per batch.

The native node allowance charges private circuit nodes and all historical ordinary arena nodes together, including nodes produced during export. All graph scans, context steps and export work use the caller's existing cooperative budget. This is not a hard byte-memory or wall-clock bound; Python containers, integers and temporary arrays have additional costs. The different charged operations can change finite-budget outcomes.

## 97.4 Native correctness and source binding

All 1,112 maintained tests pass with Regina in 210.025 seconds. Seven new tests cover random multibatch agreement with literal substitution, negative handles, forward bindings, empty images, repeated child edges, invalid late batches, shared node/work allowances and false-valued cancellation callbacks. Large binary powers are replayed with expansion, equality, reduction, inversion, slicing and ordinary substitution helpers disabled. A 128-step fixture checks the historical depth/size recurrence and final raw length $2^{128} + 2$. Actual diagram certificates split into consecutive singleton batches pass both source verifiers, and explicit normalization boundaries create separate blocks.

The final independent audit takes 9.257 seconds and freezes 431 Python and corpus hashes. It retains 400 previous single-batch DAG fixtures with 1,600 compressed and 800 literal replays. Another 400 random raw blocks contribute 800 old/new compressed comparisons and 1,151 literal batch steps. All agree as literal words on every retained slot and on the surviving generator set; 135 of the new random blocks exercise legacy dependency-order recovery.

On 79 validated diagrams, both default and optional-batch producer certificates are identical to baseline `63c262ac3c22`; there are no gained or lost positive outcomes. All 920 original source-bound replay checks pass. Splitting 48 distinct actual source certificates into single-entry batches yields another 192 old/current literal/compressed checks, all passing; all 48 exercise the new path, with 49 persistent blocks in total. These split certificates are valid alternative witnesses for the same sources. The maintained producer does not generally choose that serial encoding.

### 97.5 Complete supplied-block measurements

The pinned kernel run takes 350.994 seconds and retains 240 measured calls and 48 warm-ups. Four shuffled arms provide old/new and separate A/A pairs; each fixture has five measured rounds. Timers include fresh binary grammar construction and complete checked replay. All calls complete. Exact output lengths are checked outside the timers; retained structural hashes describe the resulting grammars and are not word-equality proofs when associations differ. Small literal comparisons and the source-bound audit provide separate semantic evidence. Table entries are median milliseconds and median paired old/new ratios, which need not equal ratios of the separate medians.

| Family | pivots | old ms | new ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|---|
| contexts | 8 | 2.715 | 1.212 | 2.301 | 0.988 | 1.009 |
| contexts | 32 | 164.627 | 27.204 | 6.061 | 1.006 | 0.983 |
| contexts | 128 | 12125.140 | 713.619 | 16.988 | 1.008 | 0.993 |
| tower | 8 | 21.830 | 14.760 | 1.474 | 0.994 | 0.996 |
| tower | 32 | 391.609 | 106.792 | 3.722 | 1.010 | 1.012 |
| tower | 128 | 6597.190 | 1095.113 | 6.156 | 1.001 | 0.994 |
| legacy | 8 | 24.184 | 16.445 | 1.485 | 0.951 | 0.995 |
| legacy | 32 | 335.460 | 127.707 | 2.616 | 0.989 | 0.995 |
| legacy | 128 | 5342.976 | 1541.778 | 3.446 | 1.003 | 0.961 |
| shallow | 8 | 0.351 | 0.331 | 1.083 | 0.985 | 0.985 |
| shallow | 32 | 1.700 | 1.635 | 1.111 | 0.970 | 0.822 |
| shallow | 128 | 3.935 | 3.574 | 1.109 | 0.991 | 0.998 |

The context family is the delivery's explicit artificial presentation. With $a = 1$, $b = 2$, $x_i = i + 3$ and $z = k + 4$, its relators are

$$r_0 = x_0^{-1} a x_1 b, \qquad r_i = x_0^{-1} x_{i+1} \ (1 \le i < k), \qquad r_k = x_0^{-1} z.$$

Eliminate $x_i$ from slot $i$ for $i = 0, \dots, k-1$. The retained raw word has length $2^k + 2$. Its padding includes cancelling contexts that an intervening free reduction would remove; this is not a hard-knot construction. At $k = 128$, complete fresh construction/replay takes 12,125.140 ms under the old checker and 713.619 ms persistently, with paired ratio 16.988. Replay work falls from 17,679,028 to 1,366,305. The old checker adds 57,155 ordinary arena nodes; the new checker adds 12,031 at export and retains 6,890 private nodes during replay. Its peak combined node count is 19,184, including the 263-node source arena. The maximum binding-free depth is 855 and the final length has 129 bits. Both engines perform the same 128 raw singleton transformations.

The tower families use $2^{128}$-power contexts and 128 pivots arranged into 32 batches of four. Ordered replay takes 6,597.190 versus 1,095.113 ms, with paired ratio 6.156 and work 9,094,171

versus 1,888,530. Ordinary nodes added are nearly equal, 33,437 versus 33,406: avoiding repeated intermediate traversals is the main saving here. The private graph has 17,026 nodes and the peak combined count is 67,587, more than the old final ordinary count 50,592. The largest output length needs 16,514 bits. Reversing each batch's entry order exercises legacy order recovery 32 times; its paired ratio is 3.446 and work falls from 10,971,306 to 2,978,866. These examples show both the benefit and the additional transient storage of persistent replay.

Two shallow independent batches provide a conversion control. At 128 pivots, the recorded paired ratio is 1.109, but charged replay work rises from 4,605 to 5,057 and the peak combined node count is 781, versus 394 old ordinary nodes. At 32 pivots the new-arm A/A ratio is 0.822, so its small timing gain is especially noisy. Across all kernel fixtures the median A/A ranges are 0.951–1.010 historically and 0.822–1.012 persistently. The result is a whole-block representation improvement with favorable large-fixture timings, not a promise of fewer operations or less memory on every input.

## 97.6 Supplied source proofs and unchanged discovery

The separate source benchmark retains 80 measured calls and 16 warm-ups in 23.549 seconds. Each old/new pair verifies the identical version-eight proof for a stabilized circle diagram, split into consecutive single-entry batches. Every timed call performs fresh diagram validation and the verifier's complete independent presentation reconstruction and replay. Producing and splitting the supplied certificate, serialization and the separate literal cross-check are outside the timers. All calls complete. These are genuine source-bound proofs, but their serial encoding is supplied to the verifier rather than chosen by the current discovery algorithm.

| Crossings | old ms | new ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| 8 | 1.360 | 0.868 | 1.573 | 1.069 | 0.994 |
| 32 | 16.699 | 7.723 | 2.155 | 0.999 | 0.971 |
| 128 | 262.194 | 112.566 | 2.336 | 1.020 | 1.000 |
| 256 | 1064.499 | 449.934 | 2.345 | 0.996 | 0.958 |

At 256 crossings, the unchanged 23,189-byte proof has 255 consecutive raw batches. Complete verification takes 1,064.499 versus 449.934 ms, with paired ratio 2.345. The replay arena's work counter falls from 1,637,855 to 894,796; these counters exclude the preceding independent presentation reconstruction, which is included in the timed calls. Final ordinary nodes fall from 2,800 to 1,785, while 1,273 private circuit nodes make the new combined peak 3,058. Thus the time saving does not imply a smaller peak node count. Across the four fixtures the paired ratios range from 1.573 to 2.345; median A/A ranges are 0.996–1.069 historically and 0.958–1.000 persistently.

The ordinary group-stage benchmark separately runs fresh source validation, optional-batch search and mandatory replay. All 120 measured calls and 24 warm-ups finish in 2.436 seconds. Its producer emits a single batch on these circle fixtures, so the persistent block path does not activate. Paired ratios range from 0.988 to 1.006; historical A/A ratios range from 0.986 to 1.035 and new A/A ratios from 0.992 to 1.026. This gives no useful discovery speedup and confirms why the supplied split-proof result must be labelled separately.

| Crossings | old ms | new ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| 8 | 1.258 | 1.250 | 1.006 | 0.986 | 0.992 |
| 16 | 2.492 | 2.498 | 1.004 | 1.001 | 1.005 |
| 32 | 4.928 | 4.959 | 0.988 | 1.008 | 1.001 |
| 64 | 10.011 | 10.029 | 0.994 | 0.995 | 0.999 |
| 128 | 20.254 | 20.442 | 0.994 | 1.007 | 1.000 |
| 256 | 44.013 | 42.488 | 0.993 | 1.035 | 1.026 |

The whole-recognition comparison uses the same optional group portfolio in both engines on nineteen native diagrams. All 380 measured calls and 76 warm-ups complete in 39.931 seconds, retaining the same certificates across arms. One successful proof, `survivor-10`, has consecutive batches of seven and two generators before normalization. Its paired ratio is 1.047, with old/new medians 5.097 and 4.906 ms; its old/new A/A ratios are 1.007 and 1.002. This is a modest observed gain on an activating native input. Separate deterministic source replay reduces arena work from 1,904 to 1,783 and final ordinary nodes from 165 to 123. The new block has 57 private nodes and peaks at 154 combined nodes before the later normalization. These counters exclude source reconstruction and are retained with an additional literal check in `data/persistent-replay-activation.json`. The Gordian trial still exhausts its producer allowance and uses the unchanged discovery fallback; its final proof cannot benefit from the new block checker. Its paired ratio is 1.001, with separate medians 1,254.964 and 1,273.125 ms. Across the corpus ratios range from 0.979 to 1.047, with old A/A medians 0.987–1.026 and new A/A 0.981–1.015. This establishes no broad whole-recognition improvement.

| Input | old ms | new ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| survivor-00 | 14.047 | 14.072 | 1.003 | 1.009 | 0.997 |
| survivor-01 | 28.759 | 27.750 | 1.034 | 1.026 | 0.997 |
| survivor-02 | 9.234 | 9.132 | 1.007 | 1.009 | 0.988 |
| survivor-03 | 37.838 | 38.481 | 0.995 | 1.004 | 1.002 |
| mirror-03 | 5.672 | 5.745 | 0.987 | 0.999 | 1.007 |
| survivor-04 | 28.870 | 29.141 | 0.995 | 0.996 | 0.997 |
| survivor-05 | 15.128 | 15.316 | 0.988 | 0.998 | 1.002 |
| survivor-06 | 5.085 | 5.145 | 0.994 | 0.998 | 0.991 |
| survivor-07 | 19.294 | 19.317 | 0.998 | 0.987 | 1.005 |
| survivor-08 | 6.857 | 6.897 | 1.001 | 1.003 | 1.002 |
| mirror-08 | 15.080 | 15.078 | 1.003 | 0.999 | 1.001 |
| survivor-09 | 5.402 | 5.420 | 1.003 | 1.005 | 0.993 |
| survivor-10 | 5.097 | 4.906 | 1.047 | 1.007 | 1.002 |
| survivor-11 | 4.681 | 4.668 | 1.000 | 1.008 | 1.000 |
| gordian | 1254.964 | 1273.125 | 1.001 | 1.004 | 0.987 |
| trefoil | 0.057 | 0.057 | 0.979 | 0.995 | 0.981 |
| figure_eight | 0.062 | 0.062 | 0.994 | 0.990 | 1.006 |
| conway | 0.978 | 0.991 | 0.987 | 0.994 | 1.015 |
| kinoshita_terasaka | 1.003 | 1.013 | 0.996 | 0.996 | 0.988 |

In total the four new benchmark modes retain 820 measured calls and 164 warm-ups, all complete. Kernel, supplied-source, group-discovery and full recognition scopes are kept separate. None of the runs overlaps tests, article builds or another benchmark. The driver `../fast/compressed_word_research/persistent_replay.py` provides `audit`, `kernels`, `stages` and `pipeline`; `data/persistent_replay_source.py` runs the supplied-source comparison. The former freezes 431 source hashes and the latter additionally freezes its own script, for 432. The baseline package is isolated from commit `63c262ac3c22`; only its compressed group-replay dispatcher differs among existing runtime modules, plus the new persistent checker. All raw records, proof objects and reproduction metadata are retained under `data/persistent-replay-*`. The three deterministic activation checks are reproduced by `data/persistent_replay_activation.py`; its 433 hashes additionally bind the script and the selected pipeline record.

## 97.7 Remaining integration and global obligations

The new module is `../fast/fastunknot/persistent_elimination_verify.py`; the existing compressed source verifier groups consecutive batches and calls it. An isolated batch keeps the preceding ordered checker. Any different move ends the group before its own semantics are

replayed. No normalization is silently inserted or removed, and all existing certificate versions remain subject to their original source and terminal contracts.

At this checkpoint the strengthening concerns replay of an entire raw block; the producer still materializes intermediate grammars. Section 98 subsequently supplies a compatible persistent planner and its polynomial raw-block discovery bound. Arbitrary alternations with normalization, Whitehead moves and new exposures have not received a comparable global bound. Nor does a supplied block guarantee an unknot terminal. Controlling those remaining operations is necessary for the original general quasipolynomial goal, which remains open.

# 98 Polynomial discovery within a persistent raw singleton block

Section 97 establishes polynomial replay of a supplied raw block. The maintained producer now uses a separately implemented persistent circuit while discovering successive batches. It keeps the existing greedy donor policy, computes its exact scores on the changing raw words, and exports only when that block ends. This closes the representation gap for raw batch discovery. It does not prove that such a block reaches an unknot endpoint or that a bounded number of subsequent normalization and exposure stages suffice.

## 98.1 Exact summaries under changing bindings

The private producer circuit has immutable signed concatenations and one binding per eliminated generator, as in the preceding section. It imports the ordinary arena's reachable source once. Its evaluator uses iterative depth-first dependency traversal, independently of the checker's Kahn implementation. After every batch it discards its private summary cache. A summary is reused only while the bindings and relator roots remain unchanged. The ordinary arena never receives mutable bindings, so its historical caches remain valid.

For every node compute raw length $\ell$, presence mask $P$, and repetition mask $R$. An unbound terminal has length one, its own presence bit, and zero repetition mask. A bound terminal inherits its target's summary, including the all-zero summary for an empty image. Inversion preserves all three values. For a concatenation with children $u, v$,

$$\ell = \ell_u + \ell_v, \qquad P = P_u \vee P_v, \qquad R = R_u \vee R_v \vee (P_u \wedge P_v).$$

Thus $P \wedge \neg R$ gives exactly the raw singleton generators. This calculation performs no cancellation: opposite signed occurrences both count toward repetition.

Global absolute generator counts are obtained in one reverse dependency pass. Assign weight one to each nonempty retained root, summing repeated roots. At a concatenation add its accumulated weight to *both* children; when the children agree, add twice. At a bound terminal pass that weight to its target. At an unbound terminal add it to that generator's count. The number propagated to a node is exactly its occurrence multiplicity in the unfolding of the retained roots. This proves the resulting counts equal the ordinary producer's counts without a separate graph traversal per generator.

For singleton generator $g$ in slot $j$, let $c_g$ be its global count and $\ell_j$ the donor's raw length. The candidate key remains

$$((\ell_j - 2)c_g, \ell_j, j, g).$$

The existing greedy selector receives the same keys and supports. It still requires distinct slots and generators, rejects dependency cycles using its incremental reachability masks, leaves at least one generator, and emits the same stable topological ordering. Selection is shared between the two producer representations; no producer helper is shared with the verifier. Induction on batches proves identical selected entries whenever both representations run the same uninterrupted raw

block to completion without exhausting resources. Different exported associations can affect later heuristics and finite budgets; the equivalence assertion is confined to this raw block.

## 98.2 Whole-block discovery cost and boundaries

The producer constructs each donor's signed context by unique-path descent and balanced sibling assembly. It fixes every context against the pre-batch state, then installs the acyclic batch's bindings together. It uses the same size and height argument as Section 97: with $k$ eliminations, its private historical size is at most $S_*$, its binding-free depth is at most $D_0 + 6k \log_2(D_0 + k + 2)$, and its raw lengths need at most $B_*$ bits as defined there.

Let $r$ be the initial live rank and $m$ the retained slot count. Root occurrence weights and global counts need $O(B_* + \log(m + 1))$ bits: each of the $m$ root unfoldings has binary branching and height bounded by the same historical height, even if some leaves eventually bind to empty words. Candidate scores therefore have polynomial bit length. Each planning pass uses $O(S_* + mr)$ summary and candidate operations before sorting at most $mr$ candidates. Presence and reachability masks have $r$ bits. For the selector, each successful reachability update adds a bit; a node's mask can grow at most $r$ times and each update propagates through at most $r$ predecessor edges. Rejected candidates, sorting, set operations, integer arithmetic and identifier encoding add polynomial costs. There are at most $k$ successful batches and one final unsuccessful planning pass. This proves polynomial encoded cost for discovery and execution of the *entire raw block*, including donor selection. It is stronger than a bound conditional on logarithmically many batches.

The host still selects the first batch in the ordinary arena. If that batch already leaves at most two generators, the existing direct update remains a fast path. Otherwise the private circuit processes successive batches until the rank drops below three or fewer than two donors are selected. A stalled plan is recorded so the host does not repeat that unchanged plan immediately. Primitive projections, normalization, relator moves and terminal checks remain explicit boundaries outside this private block.

Final export traverses reachable absolute nodes in dependency order and compiles both orientations. Bound terminals become aliases, and each oriented concatenation becomes at most one ordinary node. Consequently at most $2S_*$ ordinary nodes are added and the combined historical node count is bounded by $N_0 + 3S_*$. Public relator roots, live generators and the accumulated move list are published only after a complete export. Resource exhaustion may leave ordinary historical allocations and statistics, but no partial public block. Private nodes and ordinary history share the existing node allowance; all passes use the same cooperative work and cancellation mechanism. Integer and container storage require the bit accounting above, not just the node count.

This applies automatically within the existing optional batch search. The bounded speculative trial, source recovery, independent replay, complete fallback, certificate schema and default mode are unchanged. There is no claim that a smaller work counter is a uniform wall-time improvement.

## 98.3 Native validation and measured limits

The comparison pins the preceding maintained producer and persistent checker at commit `4360509f450e` (the full hash is in each data file). Only the existing producer selector and compressed-search dispatcher differ among preexisting runtime modules; the new private producer is additive. Every preexisting verifier, including the complete compressed and literal source checkers, is byte-identical. The driver freezes 434 Python/corpus source hashes before each run and checks them again afterward.

All 1,116 maintained tests pass with Regina available, in 222.967 seconds. The four new methods compare 350 random greedy presentations with ordinary updates and literal batch replay; exercise a sixteen-generator binary-power chain with $2^{128}$ repetitions; test summary-cache invalidation,

cancellation, work/node exhaustion and atomic publication; and check both the terminal fast path and an actual two-batch source certificate. The binary-capacity test disables expansion, equality, reduction and inverse helpers during export and independent replay. It is a supplied transformation schedule, not evidence that greedy discovery chooses that schedule on a knot diagram.

The separate pinned audit checks 84 diagrams, including the maintained corpus, random braid closures and stabilized circles. All default and optional-batch certificates match the predecessor, with no gained or lost positives. There are 840 original-source literal/compressed replays across the two packages, and 24 searches activate persistent production. Activation includes searches that later stall; it does not imply a newly accepted proof. The driver also retains the 400 abstract DAG cases, 1,600 compressed replays and 800 literal replays used to check the unchanged batch interface. The audit takes 8.040 seconds; its source hashes and complete evidence are retained separately from timings.

**Measurement protocol.** All timings run serially after tests and audit completion. Four arms comprise the predecessor, the current producer and an identical A/A control of each. Each case has one excluded warmup and five measured rounds in shuffled order. Every timed call constructs and validates a fresh source diagram and includes the host's mandatory independent replay. The three experiments total 520 measured calls and 104 warmups, all complete. Ratios below are medians of within-round old/new ratios; values above one favor the current producer. They are not ratios of displayed median times. These small, single-environment measurements do not establish a broad recognition speedup.

**An activating source-bound group call.** The actual `survivor-10` source discovers two batches of seven and two generators, then normalizes and reaches the existing terminal. Its unchanged 726-byte certificate is replayed inside the timed group call and additionally checked by the literal verifier outside timing. One private producer block uses 57 nodes, 22 context siblings, maximum concatenation depth three and a combined private/ordinary peak of 158 nodes. Final ordinary producer nodes fall from 177 to 127, while total charged search work, including source recovery, is essentially unchanged: 5,464 versus 5,467. The timing likewise shows no material gain:

| Case | Old ms | New ms | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| survivor-10 | 3.747 | 3.686 | 1.017 | 1.011 | 0.992 |

The private-node peak concerns the raw block; final ordinary-node counts also include subsequent terminal/normalization work. They are not interchangeable memory measurements, and neither includes Python container and integer bytes.

**Terminal fast-path controls.** Stabilized-circle group calls finish in one batch and therefore avoid the private import. Their paired ratios range from 0.981 to 1.012; the current A/A controls range from 0.993 to 1.014. This is an overhead control, not a performance demonstration of persistent production:

| Case | Old ms | New ms | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| circle-8 | 1.253 | 1.253 | 1.000 | 0.992 | 0.998 |
| circle-16 | 2.515 | 2.510 | 1.012 | 0.993 | 1.008 |
| circle-32 | 4.743 | 4.902 | 0.981 | 0.991 | 1.000 |
| circle-64 | 9.555 | 9.578 | 1.005 | 0.988 | 0.998 |
| circle-128 | 19.991 | 20.140 | 0.995 | 0.994 | 1.014 |
| circle-256 | 38.915 | 39.168 | 0.991 | 0.983 | 0.993 |

**Complete native recognition.** The nineteen-diagram corpus runs the same optional batch portfolio in both engines, including the bounded speculative trial and complete fallback. All 380 measured calls and 76 warmups complete. Ratios span 0.944–1.026, with several small regressions. Old A/A ratios span 0.957–1.020 and new A/A ratios span 0.982–1.015. There is no broad ordinary-workload gain:

| Case | Old ms | New ms | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| survivor-00 | 13.957 | 14.511 | 0.958 | 0.996 | 0.999 |
| survivor-01 | 28.418 | 28.902 | 0.973 | 0.957 | 0.982 |
| survivor-02 | 9.260 | 9.563 | 0.965 | 1.002 | 1.003 |
| survivor-03 | 38.839 | 39.286 | 0.993 | 1.000 | 1.011 |
| mirror-03 | 5.741 | 5.758 | 0.995 | 0.997 | 1.010 |
| survivor-04 | 29.682 | 30.025 | 0.992 | 0.993 | 0.999 |
| survivor-05 | 15.225 | 15.677 | 0.969 | 0.978 | 1.000 |
| survivor-06 | 5.098 | 5.172 | 0.986 | 0.998 | 0.997 |
| survivor-07 | 20.408 | 19.805 | 1.026 | 1.018 | 0.995 |
| survivor-08 | 6.867 | 7.166 | 0.958 | 1.008 | 1.003 |
| mirror-08 | 15.169 | 15.522 | 0.979 | 1.005 | 1.001 |
| survivor-09 | 5.493 | 5.823 | 0.944 | 1.017 | 1.015 |
| survivor-10 | 4.937 | 5.023 | 0.984 | 1.002 | 1.006 |
| survivor-11 | 4.665 | 4.663 | 1.000 | 1.002 | 0.996 |
| gordian | 1261.500 | 1253.230 | 1.011 | 1.020 | 0.987 |
| trefoil | 0.054 | 0.055 | 0.990 | 0.996 | 0.996 |
| figure_eight | 0.061 | 0.061 | 0.998 | 1.008 | 1.007 |
| conway | 0.971 | 0.973 | 0.997 | 0.998 | 0.996 |
| kinoshita_terasaka | 1.002 | 1.002 | 0.996 | 0.986 | 1.002 |

For `survivor-10`, the complete pipeline ratio is 0.984, despite the smaller producer grammar. Gordian's trial still exhausts 89,905 charged units, comprising 39,904 recovery units and the 50,000 post-recovery allowance plus the failing charge. Both trials perform two batches eliminating 127 generators and then fall back. Persistent production reduces ordinary trial nodes from 2,788 to 2,122, with 735 private nodes and a raw-block combined peak of 1,982. The final accepted fallback certificate is unchanged; the paired whole-call ratio 1.011 is not evidence that the trial now solves Gordian.

The implementation is `../fast/fastunknot/persistent_elimination.py`; reproduction is `../fast/compressed_word_research/persistent_producer.py`. Complete JSON, logs and table generation are retained as `data/persistent-producer-*` and `data/persistent_producer_tables.py`. The earlier checker kernel ratios concern supplied replay schedules and are not reused as producer speedups. The established improvement here is the whole-block discovery bound and reduced intermediate grammar allocation on the activating sources, with explicit ordinary-workload overhead retained.

## 99 Source-checked annihilation from plain power relations

The delivery `unknot_power_conjugacy_20261008.zip` in commit `b365341b1` studies graphs of power-conjugacy relations and their deletion closure. Its two-generator power-minor argument yields a directly implementable special case. The maintained optional primitive-projection search now tries this case after its existing primitive donors fail. It can delete pairs for which neither individual relation is primitive, while keeping all surviving relator slots and the source-bound cyclic-group endpoint.

## 99.1 An annihilating minor, not just an abelianized relation

Suppose the current full presentation contains relators cyclically spelled $x^a y^b$ and $x^c y^d$, where a zero exponent permits a pure-power relator. Their equations give $x^a = y^{-b}$ and $x^c = y^{-d}$. Taking powers and comparing within a single cyclic subgroup proves

$$x^{ad} = y^{-bd} = x^{cb}, \qquad y^{bc} = x^{-ac} = y^{da}.$$

Consequently $x^\Delta = y^\Delta = 1$ for $\Delta = ad - bc$. If $\Delta \neq 0$ and the ambient group is torsion-free, both $x$ and $y$ are trivial. Substituting the empty word for every occurrence of both generators preserves the presented group. This is not a commutativity assumption, and it does not use a unique-root rule for distinct elements.

The hypothesis is supplied by the existing knot host: every accepted move starts from an independently reconstructed classical knot-group presentation and a checked prefix of group-preserving moves. Internal algebraic routines cannot establish that provenance. A nonzero minor does *not* prove triviality in an arbitrary group; for example $x^2 = y^2 = 1$ permits nontrivial torsion. No caller Boolean replaces the source reconstruction.

The report also proves an unconditional variant. For a group presented solely by plain power rows on $x, y$, let $D$ be the gcd of all absolute two-row minors, with $D = 0$ if all minors vanish. Each minor annihilates both elements, so $D = 1$ implies $x = y = 1$ in any ambient group. Conversely, for the isolated two-generator presentation, if the exponent-row lattice has rank below two, its abelianization has positive rank; at rank two its index is $D$. Thus the isolated group is trivial exactly when $D = 1$. The sufficiency uses the actual power-word shape, not just the exponent matrix. Three rows can have minors $6, 7, 29$, so an unconditional witness may need more than one minor. The native knot-specific rule needs only one *nonzero* minor, using its already established torsion-free premise.

## 99.2 Compressed donor discovery and independent checking

The producer recognizes raw cyclic words with at most two signed-letter runs on distinct absolute generators. It permits a single run as a pure-power supplement. It neither freely reduces a donor nor infers its shape from exponent sums alone. A cyclic cut may split one run into a prefix and suffix, so a linear spelling with three runs can still qualify when its first and last signed letters agree.

For each immutable word node the producer caches at most three run records. Concatenation joins the touching equal signed letters and adds their binary multiplicities. More than three runs is a permanent rejection: concatenation cannot decrease the number of internal runs, and closing the cyclic boundary can remove at most one. Cached content remains valid as substitution creates new immutable nodes. The planner rebuilds slot and live-label associations on every call.

Two-run donors are grouped by their unordered generator pair, with exponents oriented to the sorted labels. At most one pure-power donor per generator is appended to each corresponding group. The first mixed row is nonzero; all rows are collinear exactly when each has zero determinant with that row. Thus a linear scan per group finds a nonzero minor whenever that group has one. Groups are traversed deterministically; each selected pair is disjoint from earlier pairs, and at least one live generator remains. The present collector requires a mixed donor for a group: it does not enumerate pairs consisting solely of two pure-power relators. It is not a complete arbitrary compressed conjugacy or relator-exposure algorithm.

The existing primitive planner has already prepared every coherent two-generator candidate when it stalls. Every mixed cyclic two-run donor belongs to this snapshot. If the snapshot is empty, the host skips the new detector completely: this is an exact eligibility implication, not a heuristic budget cutoff. It avoids repeating a whole-root scan on states with no possible mixed donor, while

preserving the detector's supported discovery class. The first native timing exposed the cost of omitting this check; both runs are retained below.

Version nine adds a `power_pair_delete` move containing the two generator labels and two original source-slot indices per pair. No claimed exponents, determinants or proof-produced word summaries are trusted. The independent compressed checker computes first/last letters, signed occurrence counts and the number of adjacent signed-letter changes, capped at three. Closing the cyclic boundary adds its last/first change. One or two signed labels with distinct absolute generators and at most two cyclic changes are exactly the permitted donor shapes. This checker does not import the producer's run-list representation, planner or updater. The literal checker separately counts changes directly in the supplied words.

Both replayers authenticate all selected donors and nonzero minors before deleting any letters. Their separately implemented substitutions retain every slot, including donors, and remove the two labels everywhere. Public roots and live generators change only after the full substitution succeeds. Exhaustion can leave historical arena nodes but no partial published move. Strict schema, integer, distinct-label, live-membership and version checks reject malformed evidence. Every other certificate move and endpoint keeps its existing rules. Older version-eight certificates cannot authorize this new inference.

### 99.3 A polynomial deletion epoch and a strict local extension

Let $S_0$ be the initial reachable grammar size, $m$ its slot count, $r$ its live rank, and $B$ the largest initial raw-length bit size. One raw deletion maps each old concatenation to at most one new concatenation and maps each terminal to itself or zero. Therefore reachable size and expanded raw length do not increase. With $d \le \lfloor (r-1)/2 \rfloor$ rounds,

$$S_i \le S_0, \qquad N_d \le N_0 + dS_0, \qquad B_i \le B.$$

Each donor pass and substitution costs a polynomial in $S_0, m, r, B$ and identifier size. Exponents have at most $B$ bits and determinant arithmetic has polynomial bit cost. Sorting pair keys and scanning groups are charged; no integer factorization is needed. These bounds prove polynomial encoded cost for a raw epoch consisting solely of these deletions, including the implemented restricted discovery and independent replay. Normalization and changes of generators end that accounting epoch.

For $M \ge 2$, consider the supplied presentation

$$\left\langle t, x_1, y_1, \dots, x_k, y_k \;\middle|\; x_i^M = y_i^{M+1}, \quad x_i^{M+1} = y_i^{M+2} \ (1 \le i \le k) \right\rangle.$$

Each pair has determinant $M(M+2) - (M+1)^2 = -1$, so even the unconditional argument makes all $x_i, y_i$ trivial and leaves the infinite cyclic generator $t$. No displayed donor has a singleton. Neither two-run word is primitive: its one-relator quotient surjects onto $C_M * C_{M+1}$, or respectively $C_{M+1} * C_{M+2}$, which is nonabelian. Its coprime exponent coordinates also exclude a proper power. Thus the existing primitive-forest donor conditions find no move in this spelling, whereas the new detector closes it in one round. This is a strict extension of the local operations, not a hardness claim against the old complete search or a hard family of knot diagrams.

### 99.4 Source exposure and the unintegrated graph proposal

Original consistently oriented meridians all have abelianization charge one. A generator proved trivial must have charge zero. Consequently the new rule cannot activate directly on unmodified meridians; changes of generators are needed to expose its targets. A native source fixture starts

with the closure of the four-strand braid $\sigma_1\sigma_2\sigma_3$. Its independently recovered presentation has relators

$$x_1x_2^{-1}, \quad x_1x_2x_1^{-1}x_3^{-1}, \quad x_3x_1^{-1}.$$

Two explicit Whitehead automorphisms substitute $x_2 \mapsto x_2x_1$ and $x_3 \mapsto x_1x_3$. After checked cyclic normalization the relators are $x_2^{-1}, x_2x_3^{-1}, x_3$. The new move independently authenticates a mixed row and a pure row, deletes $x_2, x_3$, and reaches the existing rank-one endpoint. This proves source-bound integration. These deliberately supplied automorphisms are not claimed to be newly discovered by the automatic search.

The broader delivered graph uses relations $x^a = uy^bu^{-1}$ and classical knot-group balancedness. The primary statement, Lemma 2.3 of Himeno–Motegi–Teragaito, was checked: a nontrivial knot-group element conjugate between two powers has equal absolute exponents. It cannot be replaced by torsion-freeness alone, and inverse conjugacy is not a valid triviality test. That graph algorithm, complete compressed conjugator discovery, modular sieves and its delivered whole-braid audit are not integrated or claimed as native measurements here. This pair implementation uses the elementary plain-power argument above. Section 104 subsequently integrates plain signed-cycle and pure-seed component witnesses; conjugated-power graphs remain unintegrated.

## 99.5 Native evidence, capacity and complete-call measurements

All 54 delivered manifest entries verify before execution, and all 24 delivered test methods pass in 2.989 seconds. The original archive remains unchanged. The retained delivery tests are not the report's larger standalone braid audit or a run of its benchmark. Review of the plain-power theorem, the source contracts and the cited balancedness lemma is separate from those measurements.

The maintained suite passes all 1,121 tests. Five new methods include 400 random signed and cyclically rotated two-power cases with literal output comparison; rejection of words having suitable exponent sums but incorrect run shape; strict schema and version mutations; producer-disabled complete PD replay; binary powers with expansion/reduction/equality disabled; and atomic work exhaustion and false-valued cooperative cancellation. An additional guard assertion ensures the detector is not called when the existing snapshot has no coherent mixed donor.

The comparison freezes the preceding maintained package at `00a33ce6714e` and 438 current Python/corpus file hashes. The final audit uses 88 original diagrams and performs 674 complete literal/compressed source replays, including the separately constructed exposed-prefix certificate. Default, projection and forest certificates all match the predecessor; there are no gained or lost positives. None of these automatic corpus searches discovers a `power_pair_delete` proof. The constructed source proof therefore establishes authentication and integration, not improved automatic coverage on this corpus.

**Binary capacity beyond the previous donor conditions.** For the displayed $P_{k,2^b}$ family, the existing primitive-forest planner finds no eligible donor. The current complete raw search selects one new deletion round, and independent compressed replay reaches the same one-generator presentation. Source construction precedes the reported production counters; production includes its existing forest/projection attempts and the new detector. These counters are structural work charges, not elapsed times. All roots disappear without adding an ordinary arena node:

| Pairs $k$ | Power parameter $b$ | Source nodes | Production work | Replay work |
|---:|---:|---:|---:|---:|
| 1 | 8 | 23 | 404 | 153 |
| 8 | 128 | 2104 | 37778 | 14657 |
| 32 | 256 | 16608 | 298562 | 115969 |
| 4 | 4096 | 32796 | 590282 | 229537 |

The $b = 4096$ row represents exponentially many letters but is supplied directly as a word circuit; its largest exponent has 4,097 bits. It does not measure PD-to-circuit construction, automatic exposure of this family, or the old complete search. The old planner's empty result only establishes the strict separation of local eligibility conditions.

**The initial overhead and its exact remedy.** The initial integration invoked discovery after every unsuccessful primitive round, even when no mixed donor could qualify. A four-arm whole-recognizer comparison found paired old/new ratios 0.935–0.968 on the fourteen survivor and mirror cases. Gordian's ratio was 0.965: charged group work rose from 842,290 to 894,778, with 138 unnecessary detector passes and the same accepted certificate. These consistent small regressions motivated the eligibility guard proved above. The original raw records are preserved as `data/power-pair-initial-benchmark.json`; its noisy tiny negative-control ratios are not treated as gains. The initial audit and test logs are also retained under `data/power-pair-initial-*`. The final audit removes all 758 unsuccessful detector passes in each of the projection and forest modes. Their charged work now matches the predecessor on every one of the 88 diagrams, while the binary capacity fixtures still activate the new rule.

**Final complete-call control.** The final timing runs after the guarded full suite and audit, with no concurrent tests, builds or commits. Both packages use the same optional primitive-forest portfolio on the nineteen native diagrams. Each call constructs a fresh PD, runs complete recognition and includes mandatory independent replay of group certificates. Four shuffled arms contain old/new and an identical A/A control of each. Five measured rounds and one excluded warmup per case give 380 measured calls and 76 warmups, all complete. The initial timing used the same protocol and also completed every call; its results are not silently replaced.

Final paired ratios span 0.976–1.021. Old A/A ratios span 0.973–1.037; new A/A ratios span 0.985–1.028. The consistent initial regression is absent, but no broad speedup is established. Gordian has identical charged group work 842,290 and zero detector passes in both engines; its paired ratio is 0.995. The detector remains available on the binary positive fixtures. Each table ratio is the median of within-round ratios, not the ratio of displayed median times. Absolute times from the separate initial run are not used as a direct before/after comparison:

| Case | Old ms | New ms | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| survivor-00 | 15.282 | 15.273 | 1.010 | 1.001 | 1.006 |
| survivor-01 | 13.756 | 13.006 | 1.021 | 1.023 | 1.000 |
| survivor-02 | 14.175 | 14.291 | 0.976 | 0.984 | 0.985 |
| survivor-03 | 10.049 | 10.283 | 0.987 | 1.016 | 1.017 |
| mirror-03 | 22.859 | 23.708 | 0.987 | 0.997 | 1.012 |
| survivor-04 | 8.756 | 8.920 | 0.991 | 0.999 | 1.028 |
| survivor-05 | 18.507 | 18.011 | 1.003 | 1.015 | 1.008 |
| survivor-06 | 13.310 | 13.285 | 1.002 | 0.973 | 1.010 |
| survivor-07 | 35.088 | 34.764 | 1.007 | 0.993 | 0.995 |
| survivor-08 | 12.943 | 12.869 | 0.993 | 0.993 | 0.986 |
| mirror-08 | 35.289 | 35.294 | 1.005 | 0.981 | 0.998 |
| survivor-09 | 13.386 | 13.356 | 1.002 | 1.003 | 1.005 |
| survivor-10 | 6.100 | 6.050 | 1.008 | 0.996 | 0.993 |
| survivor-11 | 7.071 | 6.646 | 0.981 | 1.037 | 1.010 |
| gordian | 1231.702 | 1227.811 | 0.995 | 0.999 | 1.000 |
| trefoil | 0.056 | 0.055 | 0.989 | 0.985 | 1.013 |
| figure_eight | 0.062 | 0.062 | 0.994 | 1.005 | 1.016 |
| conway | 0.995 | 1.005 | 0.989 | 1.027 | 1.020 |
| kinoshita_terasaka | 1.027 | 1.016 | 1.010 | 1.002 | 0.987 |

**Reproduction and remaining obligations.** The maintained producer and checker are `../fast/fastunknot/power_pair.py` and `../fast/fastunknot/power_pair_verify.py`. The pinned audit and complete-call driver is `../fast/compressed_word_research/power_pairs.py`; its raw evidence, source hashes, logs and table generator are in `data/power-pair-*` and `data/power_pair_tables.py`. The optional projection/forest dispatcher preserves its earlier reductions before trying this new inference; the default recognizer is unchanged. Controlled discovery of useful zero-charge coordinates, general compressed conjugator extraction, and bounds across normalization/exposure remain open. The polynomial deletion epoch and broader local operation do not establish general quasipolynomial unknot recognition.

# 100 Compact coordinates for certified component inventories

The chapter in `ProveIt_UnknotRecognition_Research_2026-10-09.zip` gives a useful bound on the number of distinct connected normal vectors in one supplied surface. Its matching-space argument also identifies redundant coordinates in our weighted census. We now use this observation to reduce the weights transported by the native `coordinates` query, while returning exactly the same full-coordinate inventory. This section separates that local implementation improvement from the unresolved search for useful surfaces.

## 100.1 A linear bound on the number of component types

Let $F$ be an embedded normal surface in a finite compact triangulated 3-manifold with $t$ tetrahedra and $v$ quotient vertices, whose links are connected spheres or discs. Let $Q$ be the set of positive quadrilateral coordinates of $F$, and put $q = |Q| \le t$. A *type* means a distinct full $7t$-coordinate vector of a connected component, with repeated copies stored by a binary multiplicity. The report proves

$$H \le v + 2q, \qquad H \le v + q \quad \text{if all components are two-sided.}$$

These statements concern one simultaneously embedded surface. They do not bound the number of candidate normal surfaces in the triangulation.

Here is a proof, including the role of one-sided components. In the real matching space $V_Q$ supported on $Q$, the kernel of projection to the $q$ quadrilateral coordinates consists exactly of the $v$ vertex-link directions. Indeed, triangle-only face equations identify the corner coefficients around each quotient vertex, whose corner adjacency graph is connected. Thus $\dim V_Q \le v + q$.

Distinct full vectors of disjoint connected two-sided normal surfaces are linearly independent. Otherwise, clearing denominators in an integer-matrix kernel gives an equality between positive integer combinations on disjoint sets of types. Take the required parallel copies in disjoint normal product neighborhoods. Equality of full normal coordinates implies normal isotopy, which preserves the multiset of connected vectors. This contradicts the disjoint sets of types. The coordinate uniqueness used here is the classical normal-surface statement recorded in Burton's Lemma 2.5[6].

Choose one representative of each nonvertex type of $F$. Replace each one-sided representative by the horizontal frontier of a small normal interval neighborhood. Its nontrivial normal line bundle makes this a connected two-sided surface with twice the original vector. For properly embedded surfaces with boundary, the vertical boundary lies in the ambient boundary. Retain two-sided representatives and coalesce equal resulting vectors. These $r$ representatives can be taken mutually disjoint and disjoint from all $v$ small vertex links. Independence gives $r + v \le \dim V_Q \le v + q$, hence $r \le q$. Each resulting vector $z$ has at most two original preimages, $z$ and $z/2$. Adding at most $v$ triangle-only types proves $H \le v + 2q$. When all components are two-sided the replacement is injective and gives the stronger bound.

The factor two is necessary. In the one-tetrahedron layered solid torus, identify face 0 with face

1 by the cycle (0 1 2 3) and leave the other faces on the boundary. In our coordinate order, the vertex-link disc and a Möbius band have vectors

$$\ell = (1, 1, 1, 1; 0, 0, 0), \qquad m = (0, 0, 0, 0; 0, 1, 0).$$

The source $\ell + 3m$ decomposes into the three types $\ell, m, 2m$, attaining $v + 2q = 3$. The source $\ell + 2m$ attains the two-sided bound $v + q = 2$. Our native tests preserve the distinction between the Möbius band and its annular double, including large binary multiplicities.

If source coordinates have at most $B \geq 1$ bits, each component coordinate is bounded by its source entry, and multiplicities are bounded by the total number of normal discs. Since $v \leq 4t$, the coalesced full inventory requires $O(t^2(B + \log(t + 1)))$ bits. AHT already provide polynomial-time component recovery from binary normal coordinates[18, Corollary 17]. The present argument sharpens output accounting; it is not a new proof of polynomial-time normal-surface classification. Equality of the sum of proposed component vectors with the source is insufficient evidence: normal addition can reconnect surfaces. Orbit membership must still be certified.

## 100.2 An injective additive coordinate projection

For each quotient vertex choose a triangle coordinate $a_V$ minimizing its value in the supplied source $x$, with ties broken by flat coordinate index. Retain every positive source quadrilateral coordinate and each anchor with $x_{a_V} > 0$. Let $a$ be the number of retained anchors. The transported dimension is now

$$d = \max(1, q + a), \qquad a \leq v,$$

where the single dummy weight handles the empty source under the existing weighted-orbit API. All omitted source-zero quadrilaterals and anchors are zero on every component, because component vectors are nonnegative and sum to $x$. Selecting the source minimum maximizes the opportunity to omit an anchor: a vertex with any zero triangle needs no anchor weight.

The projection is injective on possible component vectors. Two vectors with the same retained entries have equal quadrilaterals; their difference is triangle-only, so its corner entries are constant around each vertex. Its anchor value is zero at every vertex, forcing the difference to vanish. Consequently equal projected orbit weights mean equal full component vectors. Coalescing by projected weight loses no component types or multiplicities. This is a source-dependent linear projection, not a claim that quadrilateral coordinates alone distinguish vertex links.

For a glued face, matching has the form

$$T_{t,v} + Q_{t,\alpha} = T_{u,p(v)} + Q_{u,\beta}.$$

After assigning quadrilaterals and anchors, propagate $T_{u,p(v)} = T_{t,v} + Q_{t,\alpha} - Q_{u,\beta}$ along a corner spanning forest. The producer constructs this forest once from the face pairings. Each histogram entry then takes $O(t)$ integer additions and comparisons to recover, followed by the existing full-vector matching and topology checks. Values are checked against the original source. With $H = O(t)$, decoding adds $O(t^2)$ arithmetic operations on $O(B + \log(t + 1))$-bit integers; source indexing and sorting have their usual additional polynomial costs.

The orbit relation and its trace are unchanged. The existing disk-corner weight construction now selects only the retained coordinates, reducing both dense vector dimension and the number of initial indicator intervals. The number of selected indicators is at most $q + a$. Weighted transport and its independent prefix-sum arithmetic are unchanged. For a meridian in our $t$-tetrahedron layered fixture, $q = t$ and $a = 0$, so $d$ drops from $7t$ to $t$. In general this improves a local polynomial routine; it does not change the asymptotic complexity class of the complete recognizer.

## 100.3 Certificate compatibility and independent reconstruction

New full-coordinate censuses use `normal-component-census-v2`. The source fingerprint, boundary basis and checked weighted-orbit trace remain mandatory. The selected indices are recomputed from the original source; no claimed coordinate basis or spanning tree is trusted. The summary still contains full $7t$-coordinate vectors. The public `weight_dimension` now reports the smaller transported dimension. Disk and summary queries continue to emit version one with three and five weights respectively. The verifier retains the original dense interpretation of all version-one coordinate certificates.

The checker selects anchors independently. Instead of the producer's face-pair spanning forest, it builds an adjacency index of the validated matching equations and repeatedly solves a row with one unknown triangle. It checks source bounds and then runs the full matching/signature validation on every recovered vector. Its arithmetic decoder neither imports nor calls the producer's coordinate-basis module. Shared finite-manifold geometry validation remains part of the pre-existing trust contract. The weighted checker still performs its separate transport replay.

Tests exercise both schemas, reject changing a dense proof's version or a compact proof's version, disable producer reconstruction during verification, and alter both projected and recovered coordinates. Other cases include empty surfaces, several quotient vertices, relabelings, zero anchors, triangle-only sources, cancellation late in reconstruction, and a multiplicity $2^{5000} + 1$ without sheet expansion. The finite compact torus-boundary contract remains unchanged, as does the separate obligation to certify a knot-diagram-to-triangulation correspondence.

## 100.4 What the bound leaves open

The type bound controls the size of one completed inventory. It does not control how many quadrilateral supports a recognizer must try, how encoded coordinates grow under a geometric hierarchy, or the size of triangulations after cutting and simplification. Nor does a histogram of full component vectors label individual boundary ports for arbitrary future gluing. Those operations still need the incidence information and sufficiency arguments discussed in the sparse-observer sections. Reducing an existing polynomial component query is useful implementation progress, but supplies none of those missing global search or state bounds by itself.

## 100.5 Native validation and paired measurements

The pinned baseline is `aeb891e7159a`. The driver `weighted_research/compact_coordinates.py` records 442 current source and fixture hashes, separately records every baseline package hash, and checks that the source hashes do not change during a run. The dense baseline is loaded in an isolated package namespace from Git; it includes the same adaptive interval-merger implementation. The only changed pre-existing runtime modules are the component geometry, producer and verifier; the compact producer decoder is a new module.

The frozen Regina 7.4 corpus contains 1,275 vectors over 45 triangulations. All three census modes and the normalized disc-count query pass 5,100 oracle comparisons and independent certificate replays. The extra compact audit checks full-summary equality against the dense predecessor on every vector, replays all 1,275 dense certificates with the current verifier, and confirms that every unweighted orbit proof is unchanged. The source-support type bounds pass on every corpus entry, including 252 inputs with nonorientable components. These are finite audit checks accompanying the proof above. The audit takes 35.548 seconds, including the additional predecessor runs; it overlaps the test suite and is not a performance measurement.

The maintained suite passes all 1,125 tests in 220.141 seconds. After tests, audit, commit and publication finished, the benchmark ran serially for 267.924 seconds. It uses six supplied surfaces, four

shuffled arms (dense, compact, and an A/A copy of each), five measured rounds and one retained warm-up round. All 120 measured calls and 24 warm-ups complete successfully. Every timed call includes fresh source validation, coordinate-weight construction, orbit discovery, certificate production and independent replay. Fixture creation, serialization and cross-arm equality checks are outside the timers. Both arms return identical full component summaries. Raw samples, source pins and deduplicated full certificates are retained in `data/compact-coordinates-benchmark.json`.

| Input | Dense ms | Compact ms | Dense/compact | Dense A/A | Compact A/A |
|---|---|---|---|---|---|
| meridian-1 | 1.003 | 0.961 | 1.035 | 0.986 | 1.009 |
| meridian-16 | 54.962 | 22.945 | 2.413 | 0.993 | 0.994 |
| meridian-64 | 2204.769 | 484.808 | 4.548 | 0.988 | 1.042 |
| meridian-128 | 15969.781 | 2744.039 | 5.851 | 0.991 | 0.993 |
| triangle-only | 5.332 | 4.005 | 1.331 | 0.988 | 0.993 |
| binary-1024 | 28.181 | 16.203 | 1.705 | 1.014 | 0.996 |

At 128 tetrahedra, dimension falls from 896 to 128, and the initial and peak piecewise-constant weight-run count falls from 385 to 257. The underlying 133 orbit cycles and 21,008 replay events are unchanged. Median total time falls from 15.970 seconds to 2.744 seconds; the median within-round ratio is 5.851, which need not equal the ratio of separate medians. Certificate size falls from 2,504,362 to 1,867,476 bytes in the recorded compact JSON encoding. This is a complete full-coordinate-query comparison, not a comparison of different output modes.

The corresponding paired gains are 2.413 at 16 tetrahedra and 4.548 at 64. Dense A/A ratios range from 0.986 to 1.014 and compact A/A ratios from 0.993 to 1.042. The one-tetrahedron 1.035 ratio is small enough that it should not be promoted as a meaningful gain. A triangle-only source with two vertex-link types improves by 1.331; an eight-tetrahedron meridian scaled by $2^{1024}$ and supplemented with $2^{1024}+1$ vertex links improves by 1.705. These are six supplied-source cases, not a universal speedup claim or new automatic unknot-recognition coverage.

# 101 A sharp bound and local arithmetic for suffix weight folds

The weighted-transport chapter of the research archive reviewed in Section 100 analyses a full-carrier transfer. It moves all weights from the rightmost carrier range, including points that survive the next truncation, and proves a sharp increase of at most three weight runs. The maintained implementation instead moves only the points actually removed by truncation. We prove a sharp increase of at most two runs for this existing operation, then exploit its small receiving interval to avoid redoing vector arithmetic on the untouched prefix. The complete weighted certificates and the independent verifier are unchanged.

## 101.1 The native two-boundary theorem

Let the current universe be $[0, N)$, with a canonical complete partition into $R$ constant integer-vector runs. Suppose a legal truncation retains $[0, C)$, where $0 < C < N$. Write $J$ for the interior change points, so $R = 1 + |J|$, and split it into $J_- = J \cap (0, C)$ and $J_+ = J \cap (C, N)$. A possible old change at $C$ disappears.

For a positive carrier with period $p$, the native transport sends a removed point $x \geq C$ to

$$\rho(x) = b + ((x - b) \bmod p), \qquad b = C - p.$$

Every image belongs to $[b, C)$. This is iteration of the carrier's inverse until first reaching the retained prefix: for an overlapping carrier all intermediate inverse steps are defined, and for a

disjoint carrier only one step is needed. These are the same partial-isometry facts used in the existing truncation proof. The preexisting prefix weights stay in place.

Extend the removed suffix weights by zero. Their discrete derivative has support only at $C$, $N$ and $J_+$. Summation over residue classes maps these changes into $[b, C)$; the change at $C$ maps to $b$. Extending that residue sum by zero below $b$ can introduce a boundary at $b$, while $C$ is the endpoint of the new universe. Thus every new interior change belongs to

$$J_- \ \cup \ \rho(J_+) \ \cup \ \{b, \rho(N)\},$$

with endpoints outside $(0, C)$ discarded. At most two new boundaries replace the vanished suffix boundaries, and the new run count satisfies

$$R' \leq R + 2.$$

If $b = 0$, or if $p$ divides $N - C$ so the two extra points coincide, the increase is at most one. Simultaneous cancellation of vector jumps only reduces these bounds. The reasoning applies to signed vectors without any positivity assumption.

For a reversing carrier with left domain endpoint $a$, the removed suffix maps through $x \mapsto a + N - 1 - x$ onto $[a, a + N - C) \subseteq [0, C)$. Interior suffix changes move to $a + N - j$ for $j \in J_+$. The only extra potential changes are the receiving interval's two endpoints, so again $R' \leq R + 2$. An endpoint at 0 or $C$ is not an interior change and improves the bound. Other trace operations leave point weights fixed. Static contraction takes an ordered subsequence of values, which cannot increase the number of runs. Consequently, after $T$ native truncations,

$$R_T \leq R_0 + 2T.$$

This sharpens the qualitative polynomial accounting in Section 94; it does not change the orbit schedule.

The report's legal seven-point example distinguishes the two operations. Give every point weight one and take inclusive pairings

$$[1, 2] \longrightarrow [5, 6], \quad [3, 4] \longrightarrow [4, 5], \quad [0, 0] \longrightarrow [3, 3].$$

The maintained scheduler first truncates $N = 7$ to $C = 6$ using the first carrier, of period four. Only point six is removed, and it maps to point two. The native result is

$$(1 \text{ on } [0, 2)), \quad (2 \text{ on } [2, 3)), \quad (1 \text{ on } [3, 6)),$$

so one run becomes three and the $+2$ bound is attained. The report's full range transfer moves both points five and six, leaves point five at zero, and produces four runs, attaining its $+3$ bound. Its claim that full-range transfer is essential to a small run bound must therefore be read as specific to that delivered analysis; our deleted-suffix operation has the stronger bound above. We retain the native operation and its separate checker, rather than adopting the report's shared producer/checker arithmetic.

### 101.2 Encoded-state bounds

For additive source intervals $[l_i, h_i)$ with vectors $v_i$, put

$$A = \sum_i (h_i - l_i) \|v_i\|_\infty.$$

Each original point contribution moves to exactly one representative in its orbit. Therefore every intermediate point-weight coordinate has absolute value at most $A$, even with overlapping

source intervals and signed cancellation. Jump coordinates can be twice as large, which adds only a constant number of bits. If the input has $m$ intervals, maximum coordinate bit length $L$, and universe size $N$, the required coordinate bit length is $O(1 + L + \log(m + 1) + \log(N + 1))$. Quotient/remainder transfers perform no loop proportional to a quotient or to the number of represented points.

The initial sweep has $R_0 \leq 2m + 1$ runs. If the input has $k$ pairings, no trace operation increases their number. A static contraction has at most $2k + 1$ gaps, from the complement of a union of at most $2k$ domain and range intervals. Intersecting these gaps with $R$ weight runs produces at most $R + 4k + 2$ emitted blocks. With $E$ trace events, the total emitted blocks, and hence the number of distinct final weight vectors, are bounded by

$$E(R_0 + 2E + 4k + 2).$$

Multiplicities use at most $O(\log(N + 1))$ bits. This is a coarse general weighted-output bound; the sharper normal-surface type bound of the previous section concerns a special geometric input class.

Together with the checked polynomial orbit trace, these bounds explicitly account for intermediate partitions, integer magnitudes and encoded output. Classical AHT already prove polynomial weighted orbit computation, with a four-boundary transfer bound in Lemma 15 and polynomial accounting in Theorem 16[18]. The present suffix-specific statement improves the maintained observer's accounting. It supplies no bound on the number of geometric hierarchy states or on complete unknot-recognition search.

## 101.3 Restricting vector arithmetic to the receiving interval

Previously, every fold rebuilt a difference map for the entire retained prefix, adding and subtracting every retained run's vector at its endpoints. This repeated dense $d$-coordinate arithmetic even on runs far outside the receiving interval. The new producer preserves the same quotient/remainder range additions for the removed suffix. It canonicalizes only those additions on their receiving interval and merges that small partition with the existing runs there. Untouched prefix and tail runs reuse their immutable vector tuples. Coalescing at the receiving boundaries preserves the exact canonical partition; no approximate or lazily unresolved weight is returned.

Let $S$ be the number of suffix runs and $P$ the number of retained runs intersecting the receiving interval. The new fold uses $O(R)$ traversal and copying work, $O(S \log(S+1))$ endpoint comparisons and $O((S + P)d)$ coordinate arithmetic and equality work, including boundary coalescing. The old prefix reconstruction paid $O(Rd)$ vector work even when $S + P$ was small. Both retain flat canonical run lists, so neither avoids the linear traversal cost. Binary division, multiplication, integer comparison and endpoint bit lengths remain part of the bit model. This is an improvement in work on sparse affected regions, not a new scheduling rule.

Only `weighted_orbits.py` changes among pre-existing runtime modules. The checker continues its separate residue-anchor/discrete-derivative transport and recomputes the complete histogram. Schemas, trace operations, normal-coordinate decoding, multiplicities and statistics are unchanged. Tests cover 2,000 signed translation folds and 2,000 signed reflection folds against direct finite pushforwards, including the exact +2 run bound. The seven-point example is tested on an actual maintained trace. Other checks exercise zero weights, immutable prefix reuse, late cancellation and replay with the producer's new overlay disabled.

## 101.4 Native audit and performance evidence

The driver `weighted_research/suffix_folds.py` compares against `e052c7ef07f3`, the compact-coordinate implementation from the previous section. It records 444 current source/fixture hashes

and checks them before and after each run. The baseline package is loaded from Git in a separate namespace; only its weighted producer differs among pre-existing runtime modules. All independent checker files are byte-for-byte unchanged.

All 1,128 maintained tests pass in 214.506 seconds. The weighted audit passes 10,000 finite systems, split evenly between the classical and Fine–Wilf periodic rules, plus exact huge binary examples. The independent checker passes 2,500 literal cases, 366 mutation checks, producer-disabled replay, and a 20,002-bit endpoint case. The normal audit passes 5,100 comparisons and certificate replays on the 1,275-vector, 45-triangulation Regina corpus. Additional baseline comparisons give identical complete answers, including certificates and statistics, for 2,550 normal queries and 1,000 abstract systems. The successful audit takes 48.158 seconds while overlapping the suite, so its duration is not a performance measurement. The outer record's baseline and source hashes identify this run; a nested kernel record retains its historical baseline metadata.

The first audit-driver attempt used dictionary pairings for the isolated old producer, whose public API requires its own `IntervalPairing` class. That harness error was corrected before the successful pinned audit; the initial log is retained separately. It was not a mathematical or runtime regression. The read-only check of `openai/math` found its head unchanged at `fd4aeeb2ee4f`; no new upstream mathematics was imported in this step.

The complete-query benchmark runs serially after the audit, suite and first publication. It takes 108.073 seconds and retains all 160 measured calls and 32 warm-ups. Eight supplied normal surfaces/modes use five shuffled rounds with an A/A copy of both engines. Every timed call includes fresh geometry, weight construction, orbit discovery, certificate production and independent verification. Input-fixture construction, serialization and equality checks are outside the timer. Full answers, including every certificate field and statistic, agree on every call.

| Query | Old ms | New ms | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| meridian-1 | 0.976 | 0.939 | 1.021 | 0.990 | 0.981 |
| meridian-16 | 21.757 | 21.124 | 1.038 | 1.017 | 1.012 |
| meridian-64 | 449.308 | 409.737 | 1.094 | 0.988 | 0.986 |
| meridian-128 | 2773.559 | 2462.575 | 1.126 | 1.000 | 1.001 |
| triangle-only | 4.151 | 4.172 | 1.000 | 0.999 | 1.007 |
| binary-1024 | 17.024 | 16.230 | 1.050 | 1.000 | 0.996 |
| disk-16 | 17.191 | 17.405 | 1.019 | 0.997 | 1.010 |
| disk-128 | 1213.349 | 1206.785 | 1.008 | 1.000 | 1.000 |

On the 128-tetrahedron full-coordinate query, median total time changes from 2.774 to 2.463 seconds, with a within-round median ratio of 1.126. Production medians are 1.497 and 1.225 seconds. Verification uses identical code and proof; variations in its measured time are not an algorithmic gain. At 64 tetrahedra the total paired ratio is 1.094. Triangle-only queries are unchanged (1.000 after rounding), while the large binary-coordinate fixture gives 1.050. Three-weight disk queries give only 1.019 and 1.008; these small differences should not be advertised as reliable gains. Old A/A ratios range from 0.988 to 1.017 and new A/A ratios from 0.981 to 1.012.

This is an incremental comparison against the compact implementation, not against the earlier dense-coordinate version. Its gains should not be multiplied with ratios from another run to claim a measured combined speedup. The improvement is local to weight production, leaves automatic recognition coverage unchanged, and does not establish a general quasipolynomial bound. All samples and deduplicated complete certificates are retained in `data/suffix-fold-benchmark.json`; the table generator is `data/suffix_fold_tables.py`.

# 102 Source-anchored replay of complete monomial blocks

Report 49's saturated-lattice theorem, reviewed in Section 92, bounds an entire raw monomial epoch by its fixed source. Its delivered lazy implementation authenticates disjoint pair batches on an immutable word grammar. We now adapt that representation to the maintained independent compressed verifier, including mixed blocks of disjoint projections and unit-coordinate forests. Ordinary words are exported once at the end of each maximal consecutive block. Producer discovery is unchanged in this integration.

The report's eager-export experiment is a useful warning: a lazy table does not help if its ordinary circuit is reconstructed after every move. The native checker therefore keeps the table through donor authentication as well as through updates. It falls back to the existing direct replay for an isolated move and exports at every normalization or nonmonomial boundary. No existing certificate schema, trusted hypothesis or terminal condition changes.

## 102.1 An exact source representation

At block entry, retain the immutable source root handles $w_1, \dots, w_m$ and initialize one table entry $(g, 1)$ for each live generator $g$. After some verified moves, an entry $(z, k)$ means that every occurrence of the original generator is replaced by $z^k$, with $k \neq 0$. A negative occurrence uses exponent $-k$. Concatenation order is retained exactly. A separate set of retired donor slots records which roots are now zero; every unselected slot remains the image of its original source root.

A disjoint primitive projection with primitive vector $(u, v)$ on the current labels $a < b$ has the local table

$$a \longmapsto a^{|v|}, \qquad b \longmapsto a^{-u\,\mathrm{sign}(v)}.$$

A unit-coordinate forest solves each child as an integer power of its parent. The existing independent forest checker validates distinct slots and children, unit child coordinates and acyclicity, then propagates powers from surviving roots. If the resulting local table sends $z \mapsto y^e$, composition updates an original entry $(z, k)$ to $(y, ke)$. All donors in a move are authenticated against the pre-move table before any update is installed. This realizes exactly the same simultaneous substitution as the previous checker.

The table is private to the verifier. It is reconstructed from the checked moves and never accepted as a certificate claim. At the end, a memoized postorder traversal exports surviving source roots into the ordinary `WordArena`. Equal signed powers share their binary powering circuits; each visited source concatenation is compiled once. Retired slots export as zero, and all other slot positions are preserved. No word normalization, expanded-letter enumeration, equality oracle or producer helper is needed.

## 102.2 Authenticating a donor without exporting its word

The existing primitive header requires exact integer types, legal slot and label data, positive power exponent, a primitive integer vector, and width $|u| + |v| - 1$. It first bounds claimed integers by the reconstructed donor's raw length. The new checker computes that length lazily on the source grammar: a terminal mapped to $z^k$ has length $|k|$, and a concatenation adds lengths. Only requested donor descendants are visited; the cache is shared within a move and discarded when the image table changes.

For a candidate pair $(a, b)$ with primitive vector $(u, v)$, the signed letters $a, a^{-1}, b, b^{-1}$ have height increments $v, -v, -u, u$. A source terminal mapped to $z^k$ contributes $|k|$ occurrences of the appropriate signed letter. Its prefix-height path is monotone, so its total, minimum and maximum are

$$h = |k|\,\delta, \qquad \min(0, h), \qquad \max(0, h).$$

The checker rejects a mapped label outside the pair. At concatenation, profiles $(h, l, r)$ and $(h', l', r')$ combine to

$$(h + h',\ \min(l, h + l'),\ \max(r, h + r')).$$

Signed occurrence counts add componentwise. Therefore the resulting counts and extrema are exactly those obtained by applying the old checker to the exported word, including all intermediate letters within a binary power. The usual coherence, exponent-vector and width equalities authenticate the donor. This changes representation and evaluation cost, not the primitive word criterion.

Proper-power deletion continues to require the source-established torsion-freeness of the recovered classical knot group. The integer table and lattice theorem do not authorize that deletion in an arbitrary presented group. The helper is internal algebraic replay; a verdict still requires independent diagram validation, presentation recovery, every earlier move, and the final rank-one or rank-two test.

### 102.3 Whole-block encoded cost

Let the frozen source have $r$ live generators, $m$ slots, $S_0$ reachable ordinary grammar nodes, maximum raw length $L \geq 1$, and $B = \lceil \log_2(L + 1) \rceil$. As established in Section 92, the cumulative monomial matrix has primitive columns with disjoint supports. The selected original exponent rows have saturated integer kernel equal to the image of this matrix. On a block of $t$ original generators, cofactors and Hadamard's inequality give

$$|k_i| \leq (t - 1)^{(t-1)/2} L^{t-1} \quad (t \geq 2), \qquad |k_i| = 1 \quad (t = 1).$$

Thus every table exponent has $\Lambda = O(r(B + \log(r + 1)))$ bits throughout the block. This is a bound in the fixed source, not a recurrence that raises the previous bound to a new power at each move. The earlier chained updater was already polynomial on this restricted class; the new representation improves its maintenance cost.

There are at most $r - 1$ authenticated donors in the entire block, since each removes one generator. A donor's length and profile each require at most $O(S_0)$ source-node evaluations. Cumulative image updates cost $O(r^2)$ integer operations over all moves. Lengths have $O(B + \Lambda)$ bits; profile heights and checked primitive metadata have the same asymptotic bit bound. Final export adds at most $O(S_0 + r\Lambda)$ ordinary grammar nodes. In terms of historical allocation $N_0$ at block entry,

$$N_{\text{final}} \leq N_0 + O(S_0 + r\Lambda).$$

The traversal, arithmetic and powering counts are bounded by

$$O(rS_0 + r^2 + m + r\Lambda),$$

with exact integer arithmetic, gcds and dictionary operations costed separately. These counts give polynomial bit complexity; they are not unit-cost claims for unbounded integers. Auxiliary length/profile tables occupy at most one source grammar's worth of entries at a time, alongside the $r$ monomial images and the retired-slot set. Existing historical nodes remain charged to the arena; no collection of unreachable nodes is silently assumed.

The bound ends at a normalization, a Whitehead automorphism, an overlap move, a general singleton batch, a power-pair deletion, or any other operation that changes the source model. The next raw block starts from the resulting ordinary grammar. Neither this argument nor the implementation bounds the number or size of such source resets in general knot recognition.

## 102.4 Compatibility, resource limits and source binding

The compressed dispatcher groups consecutive projection and forest moves, checking version six for projections and version seven for every forest, including forests appearing later inside a mixed block. Isolated moves keep the existing checker. Normalization and all other operations end the group. The literal verifier remains unchanged and is an independent comparison path.

The new state publishes neither roots nor live labels until every move has passed and export has completed. Malformed later evidence and work, node or cancellation failures leave those caller-visible objects unchanged. A failed export may have appended private, unreachable ordinary arena nodes, as with other speculative compressed operations; it cannot yield an accepted state. All source traversals, table updates and output construction use the existing cooperative work and cancellation budget. The ordinary node cap still measures grammar nodes, not total bytes or wall-clock time.

Tests compare 120 signed mixed blocks against both literal and chained replay, retaining unrelated and empty relator slots. Other cases cover general nonunit primitive pairs, binary powering, dead donors, malformed integer fields, atomic work and node exhaustion, cancellation, source rebinding and normalization boundaries. A native 17-strand stabilized circle certificate contains seven two-move raw blocks and one final isolated projection. It passes the new path with producer planning and application disabled; replacing one unit projection by its equivalent forest checks a complete mixed-schema source proof and rejects a version-six downgrade.

## 102.5 Native audit and measured scope

The driver `compressed_word_research/anchored_replay.py` compares against `b41fc4960968`, loading that package in an isolated namespace. It pins 446 current source and fixture files before and after each run. Among pre-existing runtime files only the compressed dispatcher changes; the source-anchored checker is a new module. Producer discovery and the literal checker are byte-for-byte unchanged.

All 1,132 maintained tests pass in 219.025 seconds. A separate audit covers 92 diagrams: the 19-case survivor/control corpus, filtered random closed braids and five stabilized circles. Default, projection and forest discovery return identical certificates in both packages. Every positive is replayed by both the literal and compressed verifiers in both packages, giving 768 source replays. The new checker handles 74 blocks, all in projection mode; the default and forest modes do not activate it in this audit. There is no change in recognition coverage. The final pinned audit takes 8.988 seconds; that duration is not a benchmark.

Three abstract infinite-cyclic presentation families additionally compare exact retained words through run-length representations. These fixtures are algebraic capacity tests, not diagram-derived recognition claims. A balanced 32-generator, 128-bit fixture takes five rounds: final ordinary nodes fall from 8,170 to 3,341 and replay work from 119,154 to 45,156. A 31-round star has 385 final nodes in both engines, but work falls from 86,179 to 27,944. A mixed projection/forest chain with 16 generators and 512-bit local powers has 22,662 versus 22,635 final nodes and 968,191 versus 315,907 replay work. The largest reconstructed cumulative exponents have respectively 641, 129 and 7,681 bits. Exact words, slot positions and rank-one endpoints agree.

The three timing runs execute serially after tests, audit and first code publication. Each uses five shuffled rounds, both engines and an A/A copy of each engine. All 620 measured calls and 124 warm-ups complete; failed calls would have been retained. The tables report median times and medians of within-round ratios, which need not equal ratios of the median times. Raw samples and deduplicated proofs are retained in the three `data/anchored-replay-*.json` benchmark records; their table generator is `data/anchored_replay_tables.py`.

**Supplied algebraic schedules.** Each timed call constructs a fresh grammar, authenticates every move of the supplied schedule and verifies the final rank-one condition. Discovery and exact output comparisons are outside the timer. The four cases give 80 measured calls and 16 warm-ups in 19.049 seconds.

| Case | Old ms | New ms | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| star-4-0 | 0.342 | 0.283 | 1.215 | 1.017 | 1.003 |
| balanced-32-64 | 55.790 | 28.266 | 2.010 | 0.999 | 0.959 |
| star-32-128 | 79.009 | 39.927 | 1.979 | 1.010 | 0.996 |
| chain-16-512-mixed | 907.480 | 308.764 | 2.986 | 0.986 | 0.989 |

The three binary-power families give paired speed ratios of about 2.01, 1.98 and 2.99. Their complete endpoints agree, so these are not ratios against an incomplete baseline. Their supplied schedules and abstract source presentations nevertheless prevent interpreting them as full recognition speedups.

**Complete supplied certificates from diagrams.** Each timed call validates a fresh PD, independently recovers its presentation, and checks the entire supplied certificate through its endpoint. Certificate discovery is excluded. The eight cases give 160 measured calls and 32 warm-ups in 4.390 seconds. The number in each name is the stabilized circle's strand count; it has one fewer crossings.

| Case | Old ms | New ms | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| circle-5-projection | 0.343 | 0.345 | 1.012 | 0.958 | 1.071 |
| circle-5-forest | 0.324 | 0.303 | 1.073 | 1.054 | 0.960 |
| circle-17-projection | 2.213 | 2.100 | 1.069 | 1.001 | 1.024 |
| circle-17-forest | 1.733 | 1.752 | 0.981 | 1.001 | 1.014 |
| circle-65-projection | 21.290 | 19.167 | 1.116 | 0.995 | 1.000 |
| circle-65-forest | 14.043 | 14.087 | 0.988 | 1.001 | 0.998 |
| circle-129-projection | 75.447 | 68.209 | 1.123 | 1.003 | 0.977 |
| circle-129-forest | 47.267 | 47.237 | 0.998 | 1.004 | 0.981 |

The 129-strand projection certificate changes from 75.447 to 68.209 ms, with paired old/new ratio 1.123. Charged work falls from 171,431 to 153,791; ordinary nodes remain 1,017 and maximum word-length bit size remains three. The new route checks 63 blocks comprising 126 moves, visits 378 nodes for lazy length queries, and uses only one-bit cumulative exponents. Both engines make 126 LCP calls and the same 318 frontier queries visiting 698 nodes. This source-bound example gains through reduced repeated maintenance, not through enormous hidden powers. The corresponding forest proofs contain one forest batch and retain the direct checker; their ratios are effectively one. The sub-millisecond five-strand cases are noisy, including A/A variation, and do not support a reliable improvement claim.

**Whole recognition, including discovery.** Both engines enable primitive projections and compressed search on the 19-case corpus. Calls include fresh PD validation, discovery and mandatory independent source replay, with the same 20-million group-work, ten-second group, twelve-second total and 50,000-object limits. All 380 measured calls and 76 warm-ups complete in 34.843 seconds.

| Case | Old ms | New ms | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| survivor-00 | 15.430 | 15.481 | 1.012 | 0.995 | 0.997 |
| survivor-01 | 13.185 | 13.159 | 1.002 | 1.003 | 1.003 |
| survivor-02 | 14.496 | 14.192 | 1.023 | 0.986 | 1.005 |
| survivor-03 | 10.318 | 10.225 | 0.996 | 1.001 | 0.997 |
| mirror-03 | 23.029 | 22.907 | 1.007 | 0.993 | 0.990 |
| survivor-04 | 8.868 | 8.924 | 0.979 | 0.885 | 1.023 |
| survivor-05 | 17.787 | 17.895 | 0.994 | 0.993 | 1.005 |
| survivor-06 | 7.064 | 7.092 | 0.993 | 0.996 | 0.999 |
| survivor-07 | 19.071 | 18.637 | 1.018 | 1.008 | 1.023 |
| survivor-08 | 7.145 | 7.346 | 1.022 | 1.011 | 0.970 |
| mirror-08 | 19.440 | 19.207 | 1.009 | 0.998 | 0.997 |
| survivor-09 | 7.142 | 7.149 | 0.997 | 1.006 | 0.915 |
| survivor-10 | 6.102 | 6.063 | 1.013 | 1.015 | 0.997 |
| survivor-11 | 6.614 | 6.688 | 1.000 | 1.006 | 1.006 |
| gordian | 1240.018 | 1250.434 | 1.007 | 1.013 | 0.988 |
| trefoil | 0.056 | 0.056 | 0.993 | 1.026 | 1.009 |
| figure_eight | 0.062 | 0.061 | 1.013 | 0.993 | 0.993 |
| conway | 1.015 | 0.992 | 1.014 | 1.024 | 1.002 |
| kinoshita_terasaka | 1.049 | 1.017 | 1.003 | 1.023 | 1.002 |

Paired old/new ratios range from 0.979 to 1.023, with no broad improvement. Old A/A ratios range from 0.885 to 1.026 and new A/A ratios from 0.915 to 1.023. The matching direct-discovery audit finds no eligible consecutive monomial blocks on these 19 sources. This corpus therefore supplies a compatibility and overhead control, not evidence for a whole-recognizer speedup. Producer-side lazy discovery was a separate integration task; Section 103 records its subsequent implementation. The native checker improvement proves neither a bound on general source resets nor a general quasipolynomial unknot algorithm.

# 103 Source-anchored discovery with the maintained search priorities

The representation from Section 102 now also supports native producer discovery. Previously, even consecutive raw primitive moves rebuilt ordinary words after each round. The new producer keeps the original roots and a monomial image table through candidate discovery, primitive-width queries and subsequent contractions, then exports once. Its arithmetic is separate from the independent checker. No checker file, certificate schema or source prerequisite changes.

## 103.1 Recomputing eligibility on the fixed source

The producer evaluates signed occurrence counts on the surviving source roots under the current image table. A terminal mapped to $z^k$ contributes $|k|$ positive or negative occurrences according to both the original sign and the sign of $k$. Concatenation adds counts. Each node retains at most two current labels; a larger support is represented by an ineligible marker. A current root is a candidate precisely when it has two labels and neither label occurs with both signs. Slot order remains unchanged.

Crucially, these summaries are recomputed after each image update. A source node with three original labels can become a valid two-label word after two labels acquire the same image. Permanently caching its original ineligibility would lose such donors. The old producer's immutable-node cache was valid because substitution created new ordinary roots; source anchoring changes that cache key.

The maintained projection and forest planners consume this current candidate snapshot. Their only new internal parameter is a primitive-power callback. It evaluates the source word's exact prefix-height extrema under the current signed images, with no intermediate ordinary powering circuits. A coherent word with an occurrence count of one retains the existing immediate primitive case. Other candidates use the same gcd-normalized exponent vector and width criterion as before. Forest component merging, child selection, disjoint-pair selection, donor-slot order and the shared within-round power cache are unchanged. Hence the producer discovers the same greedy move, not merely some algebraically equivalent contraction.

The new producer composes disjoint images directly. For a forest it resolves parent chains with memoized cumulative powers. This differs from the checker's independent forest validation and breadth-first reconstruction. Producer planning proves internal eligibility; only full source-bound replay can turn the resulting trace into an unknot verdict.

## 103.2 Preserving higher-priority operations

The ordinary producer still discovers the first move of an epoch. If that move leaves at most two generators, its existing direct updater is retained. The same direct route is retained when every primitive-vector coordinate in the move has absolute value one. Such a move only relabels signed terminals: it creates no nontrivial binary powering circuit. Complete source timings showed that importing these ordinary cases into lazy discovery added overhead.

Otherwise the lazy block starts from the pre-move ordinary roots and consumes the already discovered move. Thus entry requires a nonunit monomial image and room for further raw discovery. This is an adaptive implementation choice between two exact producers; it does not suspend any donor search, change its order, or introduce a new mathematical hypothesis. A block already entered continues through unit moves as well. The ordinary unit path remains covered by the earlier polynomial raw-epoch theorem. It repeats discovery until no primitive move is available, rank reaches two, or a higher-priority singleton batch becomes available. Returning at rank two preserves the caller's existing primitive terminal test, including the option that disables that test.

When singleton batches are enabled, their priority is evaluated exactly on the lazy state. Every source node carries raw length, a presence bitmask and a repeated-label bitmask. At a terminal with image $z^k$, the repeated bit is set exactly when $|k| > 1$. At concatenation, repeated labels are the union of both children's repeated sets and their support intersection. A reverse pass propagates root multiplicities through source edges, including duplicate children, and accumulates each label's exact global occurrence count weighted by $|k|$. These summaries recover the maintained score $(\ell - 2)\,\mathrm{count}(g)$, length, slot and generator tie-breaks. The existing acyclic singleton selection routine decides whether at least two donors take precedence. If so, the block exports and returns to that ordinary search branch; it does not silently execute a projection first.

A stall also exports before power-pair deletion, normalization, overlap moves or Whitehead search. The caller may repeat a boundary eligibility query on the exported state; this conservative handoff preserves the existing control flow. Ordinary source reconstruction and final endpoint tests remain outside the private image-table interface. Roots, live labels and emitted moves are published only after all discovery and export work succeeds. Limits and cancellation cannot publish a partial trace. As with the checker, speculative ordinary export nodes may remain allocated on failure and stay charged.

## 103.3 Encoded cost and the remaining global problem

Use $r, m, S_0, B, \Lambda$ from Section 102. The fixed-source saturated-kernel theorem again bounds cumulative exponents by $\Lambda = O(r(B + \log(r + 1)))$. Each successful primitive round removes a

label, so a block has at most $r-1$ successful rounds and one final unsuccessful query. Signed counts, lengths and height extrema have $O(B+\Lambda)$ bits. No ordinary nodes are allocated by discovery or image composition; final export gives

$$N_{\text{final}} \le N_0 + O(S_0 + r\Lambda).$$

Unlike supplied-certificate verification, discovery can test unsuccessful donors. The number of successful contractions alone does not bound these queries. A simple safe bound allows $m$ candidate roots in each of $O(r)$ rounds, each visiting at most $S_0$ source nodes. The implementation obtains topological order by sorting reachable ordinary node handles, so one may bound this portion by

$$O\Big(r(m+1)(S_0+1)\log(S_0+2) + rm + r^2 + r\Lambda\Big)$$

node, comparison and arithmetic operations, with integer bit costs separate. This is deliberately a whole-block polynomial bound rather than a claim that every failed profile is charged to a removed generator.

With singleton priority enabled, its source summary passes remain polynomial in the same parameters. Each round supplies at most $mr$ singleton candidates to the existing finite acyclic selector, with bitmasks of at most $r$ bits; its sorting and reachability work must also be counted. There are at most $r$ such calls in a block. These observations establish polynomial encoded maintenance and discovery within this restricted source model. They neither bound the number of nonmonomial source changes nor establish that these moves find an unknot proof for every diagram. The general quasipolynomial objective therefore remains open.

### 103.4 Audit, adaptive correction and complete-call evidence

The native `anchored_producer.py` research driver compares against `339c5359d34b`, which already contains the source-anchored checker. It loads the predecessor in an isolated namespace and pins 449 current source and fixture files before and after each run. Three pre-existing producer modules change: the search dispatcher and the two greedy planners' internal profile callbacks. The source-anchored producer is a new module. All independent checker files, including the compressed dispatcher, remain byte-for-byte equal to the baseline.

The final maintained suite passes all 1,137 tests in 219.992 seconds. Focused tests compare exact greedy projection, forest and singleton-priority choices on 240 randomized signed presentations, including unrelated and empty roots. Other cases cover binary powers, checker-disabled production, producer-disabled source replay, unit-versus-growth dispatch, node/work exhaustion and cancellation without partial public state or trace publication.

The source audit covers 102 diagrams in six modes: default, projection, forest, combined singleton/projection, combined singleton/forest, and projection with the rank-two terminal disabled. Both packages return identical complete certificates or the same absence of a certificate. The two combined Gordian modes exhaust the same five-million-work allowance in both engines; these are recorded incomplete discovery attempts, not successes. Every positive certificate is replayed by both engines in both literal and compressed modes. The adaptive rule leaves these ordinary diagram cases on the direct producer.

Six further cases start from actual five- and seventeen-strand stabilized circles and a supplied legal Whitehead power with exponent $2^b$, for $b = 4, 128, 512$. Subsequent primitive discovery is automatic and activates one two-round lazy block in every case. Complete certificates, including the exposure prefix, agree with the predecessor and pass both compressed source checkers; the $b = 4$ cases also pass both literal checkers. The full audit totals 1,328 source replays and takes 34.060 seconds while overlapping the suite; its duration is not a performance measurement. On the five-strand $b = 512$ case, producer work falls from 54,305 to 39,465 and nodes from 2,591 to

2,078. On the seventeen-strand $b = 4$ case, work instead rises from 5,099 to 5,243: the adaptive criterion is a practical choice, not a proof that every accepted entry is faster. Automatic discovery of the exposure prefix is not claimed.

Abstract infinite-cyclic capacity cases additionally compare every move, terminal claim and retained run-length word. At rank 32 and 128 local bits, the balanced case changes final nodes from 8,170 to 4,250 and total producer work from 160,464 to 93,032. The 32-generator star has 385 nodes in both engines, with work 109,465 versus 56,429. A 16-generator, 512-bit chain in projection mode changes nodes from 38,598 to 29,316 and work from 568,655 to 447,260. The same chain in forest mode ends in one batch and retains exactly the old route: 22,635 nodes and 208,807 work. These are algebraic capacity checks, not new diagram-derived recognition successes.

**Why entry became adaptive.** The first serial timing run used lazy discovery whenever a move left more than two labels. On a 129-strand circle, complete projection discovery and replay changed from 143.994 to 194.017 ms, with paired old/new ratio 0.730; the forest case changed from 92.595 to 142.539 ms, ratio 0.664. Both are real regressions despite passing correctness tests. Signed unit images have no nontrivial powers to amortize the repeated source-summary work. The final entry rule therefore retains the cached ordinary path on those moves. The complete initial audit, tests and all three benchmark runs remain in `data/anchored-producer-before-gate-*`; they are not substituted for measurements of the final implementation.

**Final serial measurements.** After the adaptive correction, the suite, audit and second code publication, the driver reruns all three timing modes serially. Five shuffled rounds retain both engines and an A/A copy of each. All 700 measured calls and 140 warm-ups complete. The tables contain median times and median within-round ratios; these are different estimators, especially when machine timings drift across rounds. No incomplete baseline is used as the denominator of a speedup claim.

For algebraic stages, each timer includes fresh grammar construction, complete greedy discovery, fresh source-grammar reconstruction and independent replay through the rank-one endpoint. Exact output comparisons and proof serialization are outside the timer. All words, moves and terminal claims agree. The five cases take 21.687 seconds for 100 measured calls and 20 warm-ups.

| Case | Old ms | New ms | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| star-4-0 | 0.460 | 0.508 | 0.885 | 1.021 | 1.117 |
| balanced-32-64 | 52.760 | 38.859 | 1.363 | 0.992 | 1.004 |
| star-32-128 | 63.602 | 52.502 | 1.233 | 0.990 | 0.974 |
| chain-16-512 | 474.307 | 413.783 | 1.143 | 0.983 | 1.009 |
| chain-16-512-forest | 286.595 | 284.324 | 0.993 | 0.992 | 0.993 |

The balanced, star and chain binary-growth cases give paired old/new ratios 1.363, 1.233 and 1.143. These are complete discovery-and-replay operations, not the maintenance-only schedules timed in the preceding section. They also use a different baseline, which already has the new checker; ratios from the two sections must not be multiplied into an unmeasured combined speedup. The forest chain ends in one batch and changes neither algorithm nor proof; its ratio 0.993 is a control. The sub-millisecond unit star has a new A/A ratio 1.117, so its 0.885 cross-engine ratio is too noisy to establish a reliable regression or gain.

The source-stage timers include fresh PD validation, discovery and mandatory independent presentation recovery, full proof replay and endpoint verification. Cases named "exposed" additionally construct the stated supplied Whitehead prefix inside the timer. Choosing that prefix is not automatic. The eleven cases take 14.103 seconds for 220 measured calls and 44 warm-ups.

| Case | Old ms | New ms | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| circle-5-projection | 0.687 | 0.694 | 0.996 | 0.947 | 0.979 |
| circle-5-forest | 0.640 | 0.638 | 1.006 | 0.987 | 0.993 |
| circle-17-projection | 4.443 | 4.478 | 1.000 | 1.003 | 1.004 |
| circle-17-forest | 3.715 | 3.661 | 1.014 | 1.022 | 0.989 |
| circle-65-projection | 40.742 | 39.652 | 1.030 | 1.019 | 0.990 |
| circle-65-forest | 28.087 | 27.816 | 1.014 | 1.017 | 1.001 |
| circle-129-projection | 257.665 | 221.376 | 0.994 | 1.011 | 1.009 |
| circle-129-forest | 168.659 | 169.456 | 0.989 | 0.987 | 1.012 |
| circle-5-exposed-128 | 14.690 | 18.173 | 1.021 | 1.003 | 0.990 |
| circle-17-exposed-128 | 29.194 | 28.547 | 1.025 | 1.003 | 0.995 |
| circle-17-exposed-512 | 49.051 | 47.322 | 1.050 | 1.004 | 1.023 |

The 129-strand ordinary projection and forest ratios are now 0.994 and 0.989, respectively. The large regression from unconditional entry is absent; these cases retain ordinary production. The exposed cases give 1.021, 1.025 and 1.050. Their modest effects should be interpreted with the raw samples. In the 129-strand projection run, old-engine times move from about 140 ms in the first round to about 258 ms in later rounds. In the five-strand exposed case, the unpaired median time increases while the median paired ratio is above one. Neither should be presented as a robust source-wide improvement. The seventeen-strand, 512-bit exposed case has ratios above one in all five paired rounds, but its new A/A median is 1.023: the observed benefit remains small compared with the algebraic capacity gains.

Finally, the whole-recognition control uses the nineteen-case corpus with primitive projections and compressed search enabled in both engines. It includes discovery and required source replay, preserving the same 20-million group-work, ten-second group, twelve-second total and 50,000-object limits. All 380 measured calls and 76 warm-ups complete in 35.270 seconds.

| Case | Old ms | New ms | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| survivor-00 | 15.166 | 15.293 | 0.996 | 1.000 | 1.001 |
| survivor-01 | 13.022 | 13.012 | 1.001 | 0.995 | 1.005 |
| survivor-02 | 14.241 | 14.243 | 0.983 | 1.005 | 1.004 |
| survivor-03 | 10.197 | 10.098 | 1.008 | 1.012 | 1.004 |
| mirror-03 | 22.814 | 23.176 | 0.992 | 0.996 | 0.980 |
| survivor-04 | 8.680 | 8.786 | 0.992 | 1.001 | 0.996 |
| survivor-05 | 17.662 | 17.725 | 0.991 | 0.993 | 1.000 |
| survivor-06 | 7.000 | 7.129 | 0.985 | 1.002 | 1.007 |
| survivor-07 | 18.588 | 18.693 | 1.003 | 1.003 | 0.991 |
| survivor-08 | 6.875 | 7.002 | 0.978 | 0.990 | 1.014 |
| mirror-08 | 19.171 | 19.303 | 0.996 | 0.993 | 0.967 |
| survivor-09 | 7.081 | 7.036 | 1.002 | 1.002 | 0.983 |
| survivor-10 | 6.216 | 6.064 | 1.000 | 1.012 | 1.008 |
| survivor-11 | 6.792 | 6.624 | 1.024 | 1.032 | 0.984 |
| gordian | 1311.242 | 1232.129 | 1.064 | 1.076 | 1.001 |
| trefoil | 0.056 | 0.055 | 1.011 | 1.015 | 1.018 |
| figure_eight | 0.061 | 0.060 | 1.017 | 1.007 | 0.949 |
| conway | 0.989 | 0.993 | 0.996 | 1.000 | 1.002 |
| kinoshita_terasaka | 1.030 | 1.028 | 0.985 | 0.998 | 0.998 |

Paired old/new ratios range from 0.978 to 1.064. Old A/A ranges from 0.990 to 1.076 and new A/A from 0.949 to 1.018. In particular the Gordian cross-engine ratio 1.064 is smaller than its old A/A ratio 1.076, so it is not evidence for a recognition improvement. The source audit finds no adaptive lazy entry on the ordinary corpus. Recognition coverage and complete certificates are unchanged. These measurements support a targeted representation choice and polynomial

fixed-source discovery, not a general recognition speedup or a quasipolynomial completeness theorem.

The final raw records and their table generator are:

```
data/anchored-producer-kernels.json
data/anchored-producer-source.json
data/anchored-producer-pipeline.json
data/anchored_producer_tables.py
```

Each record retains every sample, source description, proof reference and source pin. The read-only upstream check finds `openai/math` unchanged at `fd4aeeb2ee4f`; no new external mathematics is imported in this step.

# 104 Source-checked annihilation of plain-power components

The power-conjugacy delivery in commit `b365341b1` contains a graph extension of the pair-minor idea integrated in Section 99. We now implement its plain-power component inference. The maintained producer can detect a contradiction around a cycle involving three or more generators, even when every adjacent pair is individually consistent. A new independent checker authenticates the words, removes tree leaves and multiplies around the remaining cycle. The stronger conjugated-power proposal remains outside this implementation.

## 104.1 What the graph proves in a torsion-free group

An authenticated mixed donor has cyclic spelling $x_u^a x_v^b$, with $u \neq v$ and $ab \neq 0$. It supplies the group equality $x_u^a = x_v^{-b}$. A pure donor $x_u^a$ with $a \neq 0$ supplies $x_u^a = 1$. These are actual word shapes, not merely exponent rows obtained by abelianizing arbitrary relators. Conjugating a whole cyclic spelling does not change its normal closure.

Consider an oriented cycle written as

$$x_0^{a_0} = x_1^{b_0}, \quad x_1^{a_1} = x_2^{b_1}, \quad \ldots, \quad x_{k-1}^{a_{k-1}} = x_0^{b_{k-1}}.$$

Taking integer powers successively gives

$$x_0^A = x_0^B, \qquad A = \prod_{i=0}^{k-1} a_i, \quad B = \prod_{i=0}^{k-1} b_i.$$

Every substitution uses powers of one group element at a time; no commutativity or unique-root property between different elements is assumed. If $A \neq B$, then $x_0^{A-B} = 1$. In the source-established torsion-free knot group, $x_0 = 1$, and each adjacent power equality makes the next generator trivial. Triviality propagates through every attached tree in the same way. A pure-power donor supplies an alternative starting vertex. Thus a pure seed or an inconsistent signed cycle kills its entire connected component.

Signs matter. Plain equalities can give $A = -B \neq 0$, which forces torsion and therefore triviality. A cycle of *conjugated* power equalities need only make $x^A$ conjugate to $x^B$; the same signed conclusion would be invalid. The native rule does not discard conjugator labels or invoke knot-group balancedness. Its only group-specific premise is the already reconstructed classical knot group's torsion-freeness, preserved by the checked move prefix.

There is also an exact completeness statement for this restricted graph abstraction. In a component without pure seeds, assign a nonzero rational potential $h_u$ to one vertex and propagate $ah_u + bh_v = 0$ along a spanning tree. If every remaining edge agrees, multiplying by a common denominator

produces nonzero integers $H_u$. The assignment $x_u = t^{H_u}$ in the infinite cyclic group realizes every graph row with every generator nontrivial. Hence no such generator is forced trivial by those rows alone in the class of torsion-free groups. An inconsistent edge closes an annihilating cycle as above. Other relators of the full presentation may impose further constraints; this graph completeness statement does not make the recognizer complete.

## 104.2 Producer discovery and a different independent proof

The existing power-pair producer's immutable capped run summaries are factored into a shared producer-only row extractor. It retains the same raw cyclic one- or two-run language, including a run split across the cyclic cut. Current slot numbers and live labels are reconstructed on every call. Pair selection keeps its existing order and priority; its certificates are unchanged.

After the primitive and pair rules fail, the component detector is eligible only with at least four live generators, at least three extracted rows and at least two mixed rows. These are necessary conditions for the supported component of at least three labels while leaving one survivor, including a component killed by a pure seed. The checks avoid graph setup when the source cannot provide a supported witness; they do not suspend a possible cycle based on elapsed time or a heuristic score.

The producer traverses each connected component, assigning reduced rational potentials along its spanning tree. A non-tree row that disagrees supplies a cycle witness; a pure row supplies a seed witness. The emitted component contains sorted live labels and sorted source-slot indices for the spanning tree plus one witness. There are exactly as many selected rows as labels. The potential values themselves are neither used as group substitutions nor serialized as trusted certificate data. Distinct selected components have disjoint labels, and at least one live label is retained.

Version ten adds `power_component_delete`. The independent compressed checker reconstructs first and last signed letters, signed counts and capped transition counts directly from the ordinary word grammar. At most two cyclic signed-letter transitions and distinct absolute labels authenticate the permitted power-word shape. The literal checker counts those transitions separately in explicit words. Neither imports the producer's run summaries, graph planner, rational potentials or substitution routine.

For each proposed component, the checker validates strict field types, positive distinct live labels, distinct in-range source slots, and equal numbers of rows and labels. It constructs the authenticated multigraph, treating a pure row as a loop, and verifies connectivity. Repeatedly deleting degree-one vertices leaves its unique cycle. A remaining pure loop is a valid seed. Otherwise the checker walks the remaining cycle and multiplies the signed coefficients on its two sides; the products must differ. Parallel edges are allowed and form a two-edge cycle inside a larger claimed component. This leaf-and-product computation is independent of the producer's rational consistency computation.

Only after every component passes does each replayer substitute the empty word for all its labels in every relator, retaining every slot. Roots or words and the live-label set publish atomically after substitution succeeds. A node or work limit, cancellation, malformed later witness or balanced cycle cannot publish a partial accepted move. Historical speculative arena nodes may remain charged on an interrupted export. Version-nine certificates reject the new inference, and older moves keep their existing requirements.

## 104.3 Binary complexity of discovery, proof and deletion

Let $S$ be the reachable source grammar size, $m$ its relator slots, $r$ its live labels and $B$ its largest raw-length bit size. Every extracted exponent has at most $B$ bits. A reduced rational tree potential is a product of at most $r - 1$ edge ratios, so its numerator and denominator have $O(rB)$ bits. Fractions are a producer data structure, not unit-cost real numbers or group roots. There are

$O(m + r)$ graph visits and rational comparisons, plus sorting and exact gcd/arithmetic costs. The independent cycle products also have $O(rB)$ bits and use at most $r$ factors. No factorization or prime sieve is needed.

A component proof stores $r'$ labels and $r'$ source-slot indices for a component of size $r'$. Across one disjoint batch, its structural size is $O(r(\log(G + 1) + \log(m + 1)))$ bits, where $G$ is the largest live label. It contains no expanded donor word or claimed large determinant. The checker visits source descendants with shared memoization, authenticates their word shape, and checks a linear-sized graph. Its encoded bit cost is polynomial in the actual source and proof sizes.

Deletion only maps an old terminal to itself or zero and an old concatenation to at most one new concatenation. For a deletion-only epoch starting at $S_0, N_0, B_0$, reachable size and raw-length bit size cannot increase:

$$S_i \le S_0, \qquad B_i \le B_0, \qquad N_d \le N_0 + dS_0.$$

Component-only rounds satisfy $d \le \lfloor (r - 1)/3 \rfloor$; when interleaved with the older pair deletions, $d \le \lfloor (r - 1)/2 \rfloor$ remains valid. This accounts for historical allocation without assuming garbage collection. It proves polynomial complete discovery, certificate production and checking within this restricted epoch. Normalization, exposure and other changes of generators end the bound; their number and encoded growth remain uncontrolled for general unknot recognition.

## 104.4 A cycle family beyond all adjacent-pair minors

Choose $n \ge 3$, $b \ge n - 1$, and

$$P = 1 + 2^b, \qquad Q = \frac{P^n - 1}{2^{n-1}}.$$

The quotient is integral because $2^{n-1}$ divides $P - 1$. Consider

$$\left\langle t, x_1, \ldots, x_n \;\middle|\; x_i^P = x_{i+1}^2 \ (1 \le i < n), \quad x_n^P = x_1^Q \right\rangle.$$

The two cycle products differ by exactly one: $P^n - 2^{n-1}Q = 1$. Rotating the cycle gives $x_i = 1$ for every $i$ even without torsion-freeness, so the whole presentation is unconditionally infinite cyclic on $t$. This is a valid algebraic capacity family, not an arbitrary torsion presentation treated as a knot group.

Every edge has coprime exponents greater than one: $P$ is odd and $\gcd(P, Q) = 1$. A positive two-run word with coprime counts $a, b > 1$ has prefix-height width $ab > a + b - 1$, so it fails the maintained primitive criterion. There are no singleton donors or unit primitive coordinates. Each adjacent pair has only one relation and there is no pure seed, so the older pair-minor rule has no witness. The graph rule closes the entire cycle in one move. This strictly enlarges the supported local inference class. It is not a lower bound against every possible search strategy or a hard family of diagrams.

All exponents have $O(nb)$ bits and the ordinary binary word grammar has $O(nb)$ nodes. The new complete operation is polynomial in this encoding although literal expansion has exponential length. The predecessor's configured algebraic search stalls on the measured fixtures; its failure time is not a valid baseline for a speedup ratio.

## 104.5 Native validation and measured scope

The native `power_components.py` research driver pins predecessor `8be43faf3e73` and 453 current source/fixture hashes, checked before and after each run. It loads the predecessor separately and records every proof, source diagram, status and work count. The existing pair-only focused tests

continue to pass after factoring their row extraction. All 1,142 maintained tests pass in 214.775 seconds.

Four hundred random connected signed graphs compare native discovery with an independent rational Gaussian-rank calculation. Valid witnesses also pass literal and compressed checking and retain unrelated raw words and empty slots exactly. Tests reject balanced cycles, disconnected witnesses, arbitrary words having misleading exponent sums, unknown fields, Boolean or duplicate indices, foreign labels and attempts to erase every live generator. Other cases check binary input, producer-disabled replay, version downgrades, source rebinding, cancellation, work exhaustion and node exhaustion without publishing partial roots or labels.

The first node-limit fixture used the word $txt$ as an unrelated retained word. Its cyclic spelling is itself a two-run power row, so it joined the free label to the annihilating component and correctly made that proposed whole-component deletion ineligible. Replacing it by a three-label retained word tests the intended interrupted export. The initial failed fixture log is retained; the final focused and full runs pass.

The ordinary source audit covers 85 diagrams in default, projection and forest modes, with no work-limit failures. Complete certificates and positive coverage are identical to the predecessor. No component detector activates on these ordinary search states. This is consistent with the meridional obstruction: original meridians have nonzero abelianization charge, while a generator proved trivial must have charge zero. Changes of generators are needed to expose eligible targets.

Three additional source fixtures use the closures of the five-, nine- and seventeen-strand stabilized circles. Supplied Whitehead automorphisms send each nonbase generator to a conjugated difference from the base meridian. After checked cyclic normalization, the relators are a pure seed, a chain of adjacent differences and a redundant final pure relator. A direct call to the native component planner yields one component witness. Both full source replayers accept, for a total of 654 source replays across the audit. The predecessor rejects the unsupported version-ten format. These fixtures prove source-bound integration; ordinary search can use its higher-priority primitive or pair rules on the same states, and no newly discovered exposure is claimed. A separate algebraic audit deletes two disjoint product-difference-one cycles in one batch and checks both replayers. The final audit takes 6.436 seconds; it overlaps the suite and is not a timing comparison.

The four binary cycle fixtures have $(n, b) = (3, 64), (8, 64), (16, 128), (32, 128)$. The predecessor's configured algebraic search returns unsuccessful on each; the new search succeeds with exactly one component move. Each complete new trace independently replays against a fresh source grammar. At $(32, 128)$, production uses 146,211 work units and 8,622 ordinary nodes, with no allocation increase during deletion. The baseline's unsuccessful work count is recorded but is not interpreted as a speedup denominator.

**Complete operations and A/A controls.** After testing, audit and first code publication, the three benchmark modes run serially. The new-operation timings use five shuffled rounds of the current engine and an A/A copy. Every stage timer includes construction of the binary source grammar, complete algebraic discovery, rebuilding the source and independent replay through the rank-one endpoint. The four stages retain 40 complete measured calls and eight warm-ups in 2.892 seconds.

| Case | Current ms | A/A copy ms | Paired A/A |
|---|---:|---:|---:|
| cycle-3-64 | 5.482 | 5.483 | 0.993 |
| cycle-8-64 | 14.213 | 14.269 | 0.997 |
| cycle-16-128 | 59.758 | 59.592 | 1.008 |
| cycle-32-128 | 132.175 | 125.257 | 1.055 |

These measurements describe supported operation cost, not speed ratios against an incomplete

predecessor. The 32-generator case has median 132.175 ms and A/A ratio 1.055, so even its absolute timing has visible variability. The polynomial claim follows from the encoded analysis, not from fitting these four points.

Source-operation timers include fresh diagrams, presentation recovery, the supplied Whitehead prefix, direct native graph discovery and complete independent source verification. They do not time automatic selection of the prefix or claim the ordinary search would prefer this move. All 30 measured calls and six warm-ups complete in 0.531 seconds.

| Case | Current ms | A/A copy ms | Paired A/A |
|---|---:|---:|---:|
| exposed-circle-5 | 0.908 | 0.921 | 0.991 |
| exposed-circle-9 | 3.347 | 3.400 | 0.987 |
| exposed-circle-17 | 11.479 | 12.019 | 1.001 |

The nineteen-case whole-recognition control compares both complete engines, with primitive projections and compressed search enabled. Both use the same 20-million group-work, ten-second group, twelve-second total and 50,000-object limits. Five shuffled rounds retain both engines and an A/A copy of each; all 380 measured calls and 76 warm-ups complete in 35.159 seconds.

| Case | Old ms | New ms | Old/new | Old A/A | New A/A |
|---|---:|---:|---:|---:|---:|
| survivor-00 | 16.726 | 16.357 | 1.017 | 1.005 | 0.990 |
| survivor-01 | 14.247 | 13.982 | 1.004 | 0.989 | 0.997 |
| survivor-02 | 16.049 | 15.736 | 1.036 | 1.032 | 1.003 |
| survivor-03 | 11.091 | 10.900 | 1.015 | 1.011 | 1.001 |
| mirror-03 | 24.879 | 24.731 | 0.996 | 1.020 | 1.002 |
| survivor-04 | 9.774 | 9.237 | 1.017 | 1.002 | 1.005 |
| survivor-05 | 18.739 | 18.451 | 1.013 | 1.003 | 0.997 |
| survivor-06 | 7.501 | 7.381 | 0.999 | 0.999 | 1.015 |
| survivor-07 | 19.288 | 19.467 | 1.003 | 0.991 | 1.023 |
| survivor-08 | 7.211 | 7.153 | 1.007 | 0.983 | 1.004 |
| mirror-08 | 20.203 | 19.690 | 1.024 | 0.991 | 0.993 |
| survivor-09 | 7.402 | 7.261 | 1.015 | 1.012 | 0.995 |
| survivor-10 | 6.345 | 6.212 | 1.010 | 1.009 | 0.993 |
| survivor-11 | 6.805 | 6.767 | 1.011 | 0.993 | 1.000 |
| gordian | 1252.345 | 1261.067 | 0.996 | 1.004 | 1.020 |
| trefoil | 0.055 | 0.055 | 0.998 | 0.985 | 0.984 |
| figure_eight | 0.064 | 0.065 | 0.991 | 0.997 | 0.985 |
| conway | 0.958 | 0.966 | 0.992 | 0.996 | 1.007 |
| kinoshita_terasaka | 1.053 | 1.040 | 1.035 | 1.039 | 0.975 |

Paired old/new ratios range from 0.991 to 1.036, old A/A from 0.983 to 1.039, and new A/A from 0.975 to 1.023. The Gordian ratio is 0.996. There is no established broad recognition gain, regression or added ordinary coverage. Across the three modes, all 450 measured calls and 90 warm-ups are retained, including complete proof references and source hashes. The table generator and records are:

```
data/power_component_tables.py
data/power-component-stages.json
data/power-component-source.json
data/power-component-pipeline.json
```

The upstream `openai/math` head remains `fd4aeeb2ee4f`. No new upstream mathematics or incoming archive is imported in this step. The plain component rule extends the maintained local certificate language, while compressed conjugator discovery and a general quasipolynomial recognizer remain open.

# 105 A native, source-checked compact knot exterior

The normal-surface routines of Sections 94 and 100 accept a supplied compact manifold. Their manifold checks alone do not establish that it is the exterior of the input knot. We now supply that missing construction for one canonical, deliberately unsimplified triangulation. The native producer starts from the validated PD, uses four crossing cells per crossing, and replaces each by five finite tetrahedra. A separate checker recovers the same gluing specification from the PD and integer barycentric coordinates. Neither routine imports Regina. Section 106 subsequently adds a restricted native cocycle candidate family. Complete normal-vector search and verified simplification remain further tasks.

## 105.1 Crossing cells and the source theorem

Let $n$ be the number of crossings and put $N = \max(1, n)$. The zero-crossing circle is represented by the positive Reidemeister-I curl with PD $(0, 1, 1, 0)$. Otherwise use the original normalized PD. Its four slots are counterclockwise, slots $0, 2$ belong to the under-strand, and slots $1, 3$ belong to the over-strand. The existing validator checks the spherical rotation system and the single knot component; the bare `Diagram` constructor is not accepted as evidence that those conditions hold.

At crossing $c$, put a tetrahedral cell $C_{c,j}$ in the corner from slot $j$ to slot $j + 1$, with subscripts modulo four. Its vertices are the north pole 0, the south pole 1, and the two ends $2, 3$ of that corner on the knot. Only $2, 3$ are ideal vertices. Write $\tau = (2\ 3)$, fixing $0, 1$, for the vertex permutation on each face gluing. The complete specification is:

1. Across slot $j$, glue $C_{c,j-1}$ to $C_{c,j}$ on face $1 - (j \bmod 2)$ by $\tau$. Thus an under-slot joins faces opposite the south pole, and an over-slot joins faces opposite the north pole.
2. If an edge pairs slots $(c, j)$ and $(d, k)$, glue face 3 of $C_{c,j}$ to face 2 of $C_{d,k-1}$ by $\tau$, and face 3 of $C_{d,k}$ to face 2 of $C_{c,j-1}$ by $\tau$.

All faces occur exactly once. Loops and repeated encounters with a crossing are allowed: this is a generalized triangulation, not an assertion that all vertices of its quotient are distinct.

The topological construction is the Weeks crossing-cell decomposition, documented in the SnapPea kernel.[1] Here is why its essential hypothesis holds. Thicken the knot near the equatorial sphere and retain the two polar points. The crossing and between-crossing meridional cuts partition its exterior into four cells per crossing. Their two bigon faces collapse along successive under- or over-pass segments; their other four faces become the displayed tetrahedral faces. A closed cycle of such bigons would invalidate this collapse. A nonempty one-component diagram encounters both the under- and over-pass of every crossing, so no component consists entirely of either kind. The inserted curl handles the empty case. The polar vertices have sphere links and the knot vertex has a torus link. The resulting mixed ideal triangulation is the knot exterior with its torus boundary collapsed to an ideal vertex. No removal or simplification of the finite poles is required.

The native corner formula uses neither crossing signs nor a chosen strand orientation. Its correspondence is for the unoriented PL exterior; it does not supply a marked meridian–longitude basis. The local face prescription, the bigon-collapse theorem, and the planar source validation are the mathematical basis of the provenance claim. Agreement with an external program is additional evidence, not a replacement for that basis.

[1] Primary source: Jeff Weeks's description of the triangulation algorithm in `link_complement.cpp`, especially the cell decomposition and bigon-collapse sections. The separate Regina implementation is an external comparison, not a runtime dependency or copied implementation.

## 105.2 The first compatible finite subdivision

Give one mixed cell barycentric coordinates

$$(x_0, x_1, x_2, x_3), \qquad x_i \geq 0, \qquad \sum_i x_i = 12.$$

Retain $x_2 \leq 8$ and $x_3 \leq 8$. These disjoint truncations remove only the knot vertices. The remaining convex polyhedron has the original poles and the six directed-edge cut points $E_{vw}$, where $v \in \{2, 3\}$, $w \neq v$, and the two nonzero coordinates of $E_{vw}$ are $x_v = 8, x_w = 4$. Its original faces opposite $0, 1$ are pentagons and those opposite $2, 3$ are quadrilaterals. There are also two triangular truncation faces.

Fan each original face from its interior point $F_f$, whose coordinate $f$ is zero and whose other coordinates are four. Leave each truncation triangle whole. Cone all boundary triangles to $(3, 3, 3, 3)$. This produces

$$5 + 5 + 4 + 4 + 1 + 1 = 20$$

tetrahedra. Convexity and the face fans prove that their interiors are disjoint and cover the truncated cell. As an arithmetic control, the sum of their absolute determinants is $12^3(1 - 2/3^3) = 1600$; none is degenerate. Volume agreement is a test, not the only justification of the subdivision.

The old face maps permute barycentric coordinates by $\tau$, so they carry cut points to cut points and face centres to face centres. The two subdivisions therefore agree on every paired face. The new truncation triangles are precisely the finite boundary. Keeping the finite poles fills their spherical links; no extra spherical boundary components are created.

## 105.3 Pulling from a retained pole: four times fewer tetrahedra

The first implementation unnecessarily inserts a centre in every old face and every crossing cell. Those centres make an arbitrary face permutation easy to accommodate, but every actual gluing fixes both poles. This extra structure permits a smaller compatible subdivision.

Write $P_0, P_1$ for the poles. Pull the polyhedron from $P_0$: triangulate the facets not containing $P_0$ and cone them to $P_0$. The facets in question are the pentagon opposite 0 and the two truncation triangles. Fan the pentagon from $P_1$. The complete tetrahedron list becomes

$$\begin{aligned} &(P_0, P_1, E_{21}, E_{23}), \quad (P_0, P_1, E_{23}, E_{32}), \\ &(P_0, P_1, E_{32}, E_{31}), \quad (P_0, E_{20}, E_{21}, E_{23}), \\ &(P_0, E_{30}, E_{31}, E_{32}). \end{aligned}$$

Convexity proves coverage: a ray from $P_0$ through an interior point exits through exactly one facet not containing $P_0$. Distinct facet triangles give disjoint tetrahedral interiors. On each old face, the induced triangulation is the fan from its smallest numbered finite pole. The map $\tau$ fixes both poles, so these fans agree across every gluing, including the identifications between old faces 2 and 3. The two cut faces remain single triangles. There are now five nondegenerate tetrahedra, twelve external triangles and four internal face pairs per crossing cell. Their absolute determinant sum is again 1600.

**Proposition 105.1.** *For every validated classical one-component PD, the maintained constructor returns a compact triangulation of its unoriented knot exterior with exactly* $20N$ *tetrahedra and* $8N$ *boundary triangles. The original centred variant uses* $80N$ *tetrahedra with the same boundary count. The only boundary component is a torus. Its output format is accepted by the native normal-surface geometry validator.*

*Proof.* Apply the crossing-cell theorem and the pulling subdivision to its $4N$ cells. Truncating the unique ideal knot vertex recovers a regular compact exterior. Each cell contributes five

tetrahedra and two boundary triangles. The previous centred construction instead contributes twenty tetrahedra per cell. All other faces are paired by the local subdivisions or original face identifications. □

The default API now selects `pulling`; `subdivision='centred'` retains the original output byte for byte. The independent checker accepts both, distinguishing them by the tetrahedron count relative to the validated source. No arbitrary retriangulation is accepted. Standard normal-coordinate vectors for the default output have $140N$ entries rather than $560N$. This is a factor-four reduction in explicit geometry and coordinate dimension, not an asymptotic improvement to a complete recognition algorithm.

## 105.4 Independent replay and its scope

The producer labels points symbolically as poles and directed-edge cut points, with centres only in the legacy variant. It pairs equal local facets and transports symbolic labels across old faces. The checker does not import that module. It uses a separately specified list of integer-coordinate polygons, derives the old face from the coordinate that vanishes on all three facet points, and recovers the neighbouring corner directly from PD parity and edge mates. It checks every target tetrahedron and every vertex permutation. Its strict schema rejects booleans masquerading as indices, partial pairings and extra fields. Cancellation callbacks propagate, including false-valued callables.

The checker accepts canonical numbering only. An isomorphic retriangulation requires additional evidence; even a valid one-torus-boundary manifold cannot be substituted for the source exterior. Both paths revalidate the source PD. Once this geometry check succeeds, an independently accepted compressing-disc certificate in the very same triangulation proves that the source knot is unknotted: a knot exterior in $S^3$ is irreducible, and a compression of its torus boundary makes it a solid torus. The construction does not itself produce such a disc. In particular, a boundary vertex-linking disc has inessential boundary and supplies no positive unknot certificate.

After the normalized PD and its edge mates are available, construction and replay perform $O(N)$ fixed-size local combinatorial operations. The explicit output occupies $O(N \log(N+1))$ bits because its face records contain global tetrahedron indices. Source validation is polynomial preprocessing; this is not a claim of linear bit complexity for arbitrary input labels or of cheap normal-surface search on the resulting triangulation. The explicit geometry is still substantially larger than typical simplified external-engine triangulations. The API is optional and no default recognition path changes.

## 105.5 Source controls, compatibility and complete-operation costs

The final native audit uses the nineteen-case maintained corpus, the empty circle and the valid one-component closures retained from 240 seeded random braids. These give 84 source diagrams. Each is tested in its original form, as a mirror, and after reversing the crossing list and rotating each PD row by two slots: 252 diagram variants, with at most 141 crossings.

Every mixed crossing triangulation is isomorphic to Regina's unsimplified diagram complement and has vertex-link Euler characteristics $2, 2, 0$. For each variant, both finite subdivisions pass the native manifold validator and Regina's finite orientable one-torus-boundary checks: 504 comparisons. The current source checker accepts both outputs, and the original checker accepts the retained centred output: 756 successful source replays. All 252 centred outputs exactly equal the first published constructor at commit `8354e8aab`. The external engine is Regina 7.4, distribution 7.4.1. Neither engine's manifold checks alone would establish knot provenance.

The focused tests disable the producer during independent replay, check nondegeneracy and exact

volume for both local subdivisions, reject all 400 single-face mutations of the two one-crossing outputs, and reject a reciprocal but incorrect gluing. They also reject a trefoil exterior offered for a three-crossing unknot source, malformed schemas and unchecked invalid PDs. Cancellation propagates without input mutation. A source-bound boundary vertex-link disc correctly has zero essential-disc count; changing that certificate's count to one is rejected. Seven small external solid-torus controls cover the empty circle, curls of both signs, a reducible three-crossing unknot, a trefoil, a figure-eight and a stabilized circle. The original implementation passed 1,149 maintained tests. The final version adds the retained-subdivision contract test and passes all 1,150 maintained tests in 216.795 seconds. Its eight focused tests pass in 1.353 seconds. The final geometry audit took 56.440 seconds while the suite also ran; that elapsed time is not used as a performance comparison.

There was no newly discovered normal meridian at this construction-only checkpoint; Section 106 subsequently supplies native candidates and source-bound disc proofs. Two exploratory external calls to `nonTrivialSphereOrDisc()`, one on the 80-tetrahedron circle and one on the 20-tetrahedron circle, each reached its 20-second process cap without returning a surface. These are incomplete searches, not negative results or a speed comparison. The existing supplied normal-vector algorithms can now be applied to source-authenticated geometry, but obtaining useful vectors efficiently remains an independent obligation.

### 105.5.1 Measured effect of the pulling subdivision

The final benchmark runs serially after the suite, audit and publication of the runtime change. It compares the pinned first constructor and checker at `8354e8aab` against the final pulling implementation, using the same maintained manifold validator. Each timed call starts afresh, constructs the exterior, independently replays its source geometry and validates the finite manifold. Serialization for the stored output hash is outside the timer. Both variants perform the complete stated operation; they have different canonical subdivisions and therefore different expected output hashes.

The four inputs are stabilized circles with $1, 8, 32, 128$ crossings. There are five rounds and four interleaved arms: old, old repeat, new and new repeat, plus one warmup per arm and input. All eighty measured calls and sixteen warmups complete in a total driver elapsed time of 24.675 seconds. Hashes agree within each subdivision across every repeat. The table gives unpaired time medians and medians of paired ratios.

| Crossings | New tets | Old (ms) | New (ms) | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|---|
| 1 | 20 | 8.444 | 2.269 | 3.748 | 0.996 | 1.043 |
| 8 | 160 | 73.140 | 17.259 | 4.346 | 1.032 | 0.997 |
| 32 | 640 | 285.732 | 67.210 | 4.228 | 1.021 | 0.998 |
| 128 | 2560 | 1217.252 | 284.316 | 4.299 | 0.998 | 0.996 |

The complete-call old/new ratios are 3.748–4.346. Old A/A ratios are 0.996–1.032 and new A/A ratios are 0.996–1.043. At 128 crossings, the median falls from 1.217252 seconds to 0.284316 seconds; tetrahedra fall from 10,240 to 2,560. This is consistent with the exact factor-four reduction in geometry. It measures construction, source replay and manifold validation, not a normal-vector search, a simplified minimal exterior or whole recognition. It supplies no new general quasipolynomial theorem.

Reproduction uses `../fast/normal_orbit_research/exteriors.py`, with `audit` and `benchmark` modes. The driver pins all 457 Python and fixture sources before and after each run, and separately pins the two baseline modules loaded from Git. Final records have the `diagram-exterior-` prefix in `data/`; the first implementation's audit, suite and A/A timing records retain the

`diagram-exterior-centred-` prefix. The coordinate and gluing proofs are in the maintained article, not delegated to the optional external controls.

# 106 Native normal-disc discovery from optimized integral cocycles

Section 105 supplies source-authenticated compact exteriors. We now construct normal-surface candidates in them and return native disc certificates when a candidate succeeds. The arithmetic idea comes from report 31, `unknot_certified_primitives_20261008.zip`: minimize the number of pieces in a cocycle-dual normal surface by changing integer vertex potentials. The maintained implementation adds primitive cocycle discovery, a faster network solver, an independent arithmetic checker, and an optional recognition stage. The local optimization is polynomial. The candidate family is incomplete, with an explicit unknot miss retained below.

## 106.1 Integral cohomology in a maximal-tree gauge

Start with the canonical finite exterior and its validated face pairings. Choose an orientation for each global edge. The native geometry validator already computes signed equivalences of local edges; the new producer uses those signs to recover consistent global endpoints. The old unoriented endpoint dictionary suffices for mod-two boundary tests but does not suffice for this integer computation.

Choose a maximal tree in the connected one-skeleton. Any integral cocycle can be changed by an integral vertex coboundary to vanish on the tree: integrate its values along the unique tree paths. This gauge is unique up to an irrelevant constant vertex potential. Put one variable on each remaining edge and impose the oriented triangular-face equations

$$c([a,b]) + c([b,c]) - c([a,c]) = 0.$$

Local signs transport each term to its global edge orientation. Coincident global edges contribute to the same coefficient; they must not be mistaken for distinct simplicial edges.

Sparse exact rational elimination computes the gauged kernel. The API requires nullity one. Clear the denominators of its nonzero vector and divide by the gcd of all its entries. The result is an integral primitive generator: the integer kernel is saturated in the ambient edge lattice, and a rational line meets that lattice in precisely the multiples of its coordinate-primitive vector. This rank-one argument does not assert that separately normalized rational basis vectors give a saturated basis in higher rank. The implementation explicitly rejects that case.

A knot exterior has $H^1(M;\mathbb{Z}) \cong \mathbb{Z}$, so the canonical source lies in the supported case. There are $O(T)$ variables and equations for $T$ tetrahedra. Each original row has at most three signed contributions. Hadamard bounds on the relevant minors give $O(T)$ bits for a primitive kernel generator. Rational elimination uses polynomial bit cost; its worst-case fill is quadratic and its arithmetic operation count is cubic. The subsequent network improvement therefore does not by itself improve this entire preprocessing bound. Work caps and cancellation also cover elimination and the preceding manifold reconstruction.

## 106.2 A normal surface without expanding its pieces

In tetrahedron $t$, set $h_{t,0} = 0$ and $h_{t,i} = c([0,i])$. Closure makes every other edge value the corresponding height difference. On a glued face, transported heights differ by one integer constant. The union of half-integral affine levels therefore glues to a cooriented normal surface. Its signed

intersections with each oriented global edge are the cocycle values, so it represents the Poincaré dual of the integral class.

If the local heights, ordered by vertex, are $a_0 \le a_1 \le a_2 \le a_3$, the three gaps give a triangle at the lowest vertex, a quadrilateral separating the lower pair from the upper pair, and a triangle at the highest vertex. Zero gaps make ties harmless. There is at most one quadrilateral type, and the number of normal pieces is $a_3 - a_0$. All seven coordinates are stored as binary integers; neither the producer nor the native component query enumerates those pieces.

Let $v_{t,i}$ be the global vertex of local corner $i$. Changing the cocycle by an integer potential $f$ changes the heights to $h_{t,i} + f(v_{t,i})$. The restricted objective is

$$D(f) = \sum_t \left( \max_i (h_{t,i} + f(v_{t,i})) - \min_i (h_{t,i} + f(v_{t,i})) \right).$$

It measures normal pieces. It is not Euler characteristic, genus or Thurston norm.

## 106.3 The network and a linear-size independent witness

Introduce variables $L_t, U_t$ and impose

$$U_t - f(v_{t,i}) \ge h_{t,i}, \qquad f(v_{t,i}) - L_t \ge -h_{t,i}.$$

Minimizing $\sum_t (U_t - L_t)$ gives the span objective. Its dual is a three-layer transshipment with lower nodes $\ell_t$, vertex nodes $v$, and upper nodes $u_t$. Each lower node supplies one unit and each upper node demands one. The arcs $\ell_t \to v_{t,i}$ and $v_{t,i} \to u_t$ have rewards $-h_{t,i}$ and $h_{t,i}$ respectively. Feasibility follows by routing each tetrahedron through one of its own corners. Integral optimal flow chooses one outgoing corner per lower node and one incoming corner per upper node.

At each global vertex, match the chosen incoming and outgoing corners. This gives $T$ quadruples $(t, i, s, j)$ with sources $t$ and targets $s$ each appearing once and $v_{t,i} = v_{s,j}$. For every real potential,

$$D(f) \;\ge\; Q := \sum_{(t,i,s,j)} (h_{s,j} - h_{t,i}),$$

because the common vertex potential cancels in each term and the source and target permutations sum every minimum and maximum once. An integer potential satisfying $D(f) = Q$ proves optimality over both real and integer potentials. The checker needs no network solver.

The maintained checker also reconstructs coordinates independently. With $w_{ab} = |a_a - a_b|$, its triangle count at vertex $v$ is

$$x_v = \frac{1}{2} \min_{a,b \ne v,\ a \ne b} (w_{va} + w_{vb} - w_{ab}).$$

Write $A = w_{01} - x_0 - x_1$, $B = w_{02} - x_0 - x_2$ and $C = w_{03} - x_0 - x_3$. The quadrilateral counts are $(B + C - A, A + C - B, A + B - C)/2$. For integer local heights these are nonnegative integers and agree with the level-set surface. The checker compares the whole vector, checks exact index types and the dual permutation, and requires primal–dual equality. It accepts binary hexadecimal transport without changing the interpreter's decimal-conversion limit.

## 106.4 Reduced-cost augmentation improves the local bound

Report 31 uses Bellman–Ford for each unit augmentation, with a conservative $O(T^3)$ relaxation bound. The native solver uses a DAG initialization and reduced-cost Dijkstra searches. Negate rewards to obtain costs, add a source and sink with unit supply/demand arcs, and give every

original internal arc capacity $T+1$. No internal forward arc can saturate under a total flow of $T$. The network has $O(T)$ vertices and arcs, even when heights have thousands of bits.

Initially the usable arcs follow the order source, lower nodes, vertex nodes, upper nodes, sink. Exact shortest distances in this DAG supply potentials $\pi$ with nonnegative reduced costs $c(u,v)+\pi_u-\pi_v$. In each augmentation run Dijkstra, let $d$ be its reduced distances, and put $\Delta = d(\text{sink})$. Stop as soon as the sink is settled, prioritizing it among equal-distance choices. Update every potential by $\min(d(v),\Delta)$, assigning the increment $\Delta$ to unreachable or unsettled vertices. Every unsettled distance is at least $\Delta$, so this gives the same truncated increments as a full search. This rule matters for disconnected abstract incidence inputs; updating reachable vertices alone need not preserve all inequalities.

For any residual arc of reduced cost $c' \geq 0$, the truncated increments satisfy $a_v \leq a_u + c'$. If $u$ is reachable this follows from the shortest path inequality; otherwise $a_u = \Delta \geq a_v$. Hence all updated reduced costs remain nonnegative. Along the augmenting path both distances are at most $\Delta$ and each shortest-path inequality is tight, so newly introduced reverse arcs have zero reduced cost. The invariant holds inductively.

After $T$ augmentations every original internal arc remains forward-residual, and every used arc also has its reverse. Thus the two reduced-cost inequalities make every used arc tight. Taking $x = -\pi$ gives feasible $L_t = x_{\ell_t}$, $U_t = x_{u_t}$ and $f_v = x_v$. Summing tight equalities over the integral flow proves $\sum_t (U_t - L_t) = Q$. Weak duality and $D(f) \leq \sum_t (U_t - L_t)$ then force $D(f) = Q$. The producer checks the independent witness before returning it.

For $B$-bit heights, every original cost has magnitude at most $H < 2^B$. The source potential stays zero, and the updated sink potential is the original cost of a simple augmenting path, bounded by $O(TH)$. Since all increments are nonnegative and bounded by the sink increment, the other potentials also have magnitude $O(TH)$. Arithmetic therefore needs $O(B + \log(T+1))$ bits. A lazy binary heap gives

$$O(T^2 \log(T+1)) \text{ arithmetic operations}, \qquad O\Big(T^2 \log(T+1)(B + \log(T+1))\Big)$$

bit time for additions and comparisons, apart from input-label indexing and transport. The network uses linear storage. The witness uses $O(T \log(T+1))$ index bits and $O(T(B+\log(T+1)))$ numerical bits. Section 112 subsequently batches zero-cost residual augmentations while retaining these conservative bounds and the same checker. These bounds are for span optimization, not all cohomology preprocessing or complete unknot recognition.

## 106.5 Connectedness is useful, but does not give minimum genus

The report also proves a useful geometric consequence. Call a cooriented normal surface edge coherent if all its crossings with any oriented global edge have the same sign. Local-height surfaces have this property, which survives deletion of component unions. Their signed edge counts form cocycles. Admissible normal coordinates are uniquely determined by unsigned edge counts: the kernel of the six local edge-weight equations is generated by $(1,1,1,1,-1,-1,-1)$, and two distinct admissible vectors differing by such a multiple would force three positive quadrilateral types in one of them.

Suppose a nonempty component union of an optimal surface were zero in $H_2(M, \partial M; \mathbb{Z})$. Its signed cocycle would be an integral coboundary. Deleting it preserves edge coherence and admissibility, and the uniqueness just proved reconstructs the remaining surface from another vertex potential in the same class. Its piece count is strictly smaller, a contradiction. Thus there is no such zero-class union.

If $H^1(M;\mathbb{Z}) = \mathbb{Z}$ and the chosen class is primitive, each component is nonseparating and represents a signed primitive generator. A connected cooriented nonseparating surface is primitive because a

loop can cross it once and return through its connected complement. Oppositely signed components would form a forbidden zero-class union. All signs therefore agree, and a primitive total class permits only one component. The optimized seed is connected under these hypotheses.

The initial native positive certificate retained the independent component-level disc proof. Section 110 now checks rank-one provenance, primitivity and connectedness independently and omits that orbit search for these restricted surfaces. In particular, the primitive cocycle for the closure of the two-braid $(1, 1, -1)$ has a 45-piece genus-one surface; span minimization reduces it to 43 pieces but still gives genus one. That closure is an unknot. Optimizing this restricted family is not a complete unknot test.

## 106.6 Adaptive source integration and the remaining search problem

The initial optional native stage tries the primitive cocycle surface first. If its independent component query certifies a compressing disc, it returns immediately. Otherwise it can run span optimization and test that second surface. The work allowance is shared by source validation, exterior construction, cocycle elimination, optimization, both component queries and positive replay. The later connectedness certificates replace the two component queries while preserving these candidate vectors and the shared allowance. It counts cooperative guards, not bit operations or time. The surrounding recognizer's deadline remains active throughout.

In that version every positive witness contains the exact stage-input PD, the canonical exterior, normal coordinates and the existing normal-disc-count certificate. A separate verifier checks the PD-to-exterior construction, replays the surface proof and requires a strictly positive disc count. Seed discovery, optimization and external topology engines are absent from this replay. The certificate is bound to the PD passed to the stage; portfolio evidence also records any preceding reductions or connected-sum factorization.

The stage follows invariant filters and an enabled group search, precedes the optional external-engine fallback, and is disabled by default. A local cap or an unsuccessful surface returns `INCONCLUSIVE`; the exact recognizer resumes its existing path. No absent disc in one or two seeds proves knottedness. The current successes supply native geometric proofs, while the miss above and recorded budget exhaustions motivate further candidate families and certified simplification. They do not establish general quasipolynomial recognition.

## 106.7 Native audits, restricted successes and retained limits

All 507 entries of report 31's release manifest match, and all fifteen delivered cocycle tests pass in 1.178 seconds. The maintained implementation passes fifteen focused tests in 2.152 seconds and all 1,165 regression tests in 215.211 seconds. A first CLI test mistakenly inspected root help instead of the `recognize` subcommand; its failure log is retained, and the final test also runs an actual CLI query and replays its returned proof.

Independent finite tests enumerate 120 small potential problems and forty five-by-five dual assignments. Additional tests cover repeated vertices, disconnected incidence graphs, ties, negative heights, exact schemas, false-valued cancellation callbacks and mutations of every witness field. Scaling an abstract instance by $2^{20000}$ leaves its network size, heap work and augmentation count unchanged and scales the optimum exactly. The encoded certificate still replays with producer routines disabled. Layered-solid-torus seeds survive arbitrary local vertex and tetrahedron relabelings. A rank-two connected-sum control is rejected; a rank-one control with homological torsion exercises nonunit rational pivots exactly.

The pinned audit compares 200 native optimum values with the delivered Bellman–Ford solver. Every value agrees, and the native independent checker accepts both witnesses. The final solver makes 123,422 residual relaxation attempts across these cases, compared with 632,778 in the

delivery. These counts measure that specific network work, not all bit operations in a source query.

Source testing covers nineteen maintained diagrams, the circle, one named optimized-success input, sixty valid closures retained from 180 seeded random braids, and three larger stabilized circles: 84 cases in total. Seventeen queries return native unknot certificates, all independently replayed, with twelve distinct stored proofs. Sixty-two finish the restricted candidate tests without a certified disc, and five exhaust the shared two-million-guard allowance. No source with a known knotted verdict is reported unknotted. Negative candidate results are never reported knotted.

The named five-crossing success is the closure of $\sigma_1^{-1}\sigma_2\sigma_1\sigma_2^{-1}\sigma_3$ on four strands. Its initial 194-piece surface has no compressing disc; its optimized 84-piece surface has one. The same input also occurs in the random corpus, so the two optimized-positive records represent one distinct example. The three-crossing unknot $(1, 1, -1)$ instead remains genus one after optimization from 45 to 43 pieces. It is an explicit limitation of the objective, not a resource failure. Larger stabilized circles and Gordian also expose the current shared-budget limits; those failures are retained.

For the 74 source cases with at most twelve crossings, both the raw and optimized surfaces are independently expanded by Regina: 148 comparisons of component counts, Euler characteristic and compressing discs all agree. Every optimized surface is connected, as predicted by the primitive rank-one theorem. Fourteen raw and sixteen optimized records contain discs; these are surface controls, not additional independent knot inputs. The maintained implementation itself uses compressed coordinates and component orbits, not Regina or expanded sheets.

The ordinary nineteen-case audit compares the previous release, the new code with the option disabled, and the new code with it enabled, using the established compressed-group portfolio. All verdicts agree. The earlier stages close every case, so there are zero native-seed attempts in that portfolio. This evidence supports compatibility, not broader automatic recognition coverage.

The first published native version explored full Dijkstra regions. Its complete-call timing records showed little ordinary-height improvement. The `cocycle-seed-before-sink-optimize` files retain these measurements. Stopping at the settled sink preserves the proof and reduces source-budget exhaustions from twelve to five with the same seventeen positive results. The final audit took 80.071 seconds while regression tests were also running; its elapsed time is not used as a benchmark. Both suites and audits precede the serial measurements below.

### 106.7.1 Complete seed-optimization calls

The final measurements run serially after the tests, audit and publication of the runtime change. Each experiment has five shuffled paired rounds, old and new A/A controls, and one excluded warmup per arm and input. All 580 measured calls and 116 warmups complete. The source pins are identical across the final audit and all three benchmarks.

The optimization experiment starts from a fresh stabilized-circle diagram, constructs and independently checks its exterior, discovers its primitive cocycle, runs the optimizer, checks the span witness, and validates the resulting normal coordinates. The binary case first adds a global vertex potential of amplitude $2^{4096}$, preserving the primitive class. The baseline is the unchanged delivered solver with a native certificate adapter; its own built-in check is retained, and the adapter also invokes the native independent checker. The new producer already invokes that checker. The common measurement endpoint additionally replays the returned span proof. These are complete adapted seed-optimization calls, not isolated flow times.

| Input | Old (ms) | New (ms) | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| 1 crossing | 8.582 | 7.315 | 1.187 | 1.032 | 0.987 |
| 4 crossings | 72.640 | 51.040 | 1.437 | 1.000 | 1.000 |
| 8 crossings | 238.285 | 152.483 | 1.557 | 0.987 | 0.998 |
| 16 crossings | 895.695 | 499.241 | 1.787 | 1.005 | 0.985 |
| 4 crossings, 4096 bits | 110.527 | 58.919 | 1.906 | 0.991 | 0.999 |

All one hundred measured calls and twenty warmups finish; driver elapsed time is 25.414 seconds. Paired gains range from 1.187 to 1.906, beyond the observed old A/A range 0.987–1.032 and new A/A range 0.985–1.000. At sixteen crossings, the median falls from 895.695 ms to 499.241 ms. The high-bit case falls from 110.527 ms to 58.919 ms. Each implementation's certificate hash is stable across its repeats, and the two optimum values agree. Different optimal potentials may produce different valid witnesses.

### 106.7.2 Full recognition with the native stage deliberately reached

The second experiment disables earlier diagram, braid and invariant shortcuts so that the source reaches the new native stage. Both arms use the same current recognizer; only the span optimizer changes between the delivered implementation and the native one. The raw-circle case exits before either optimizer. The optimized-positive case returns a checked normal-disc proof. The remaining cases continue through the exact Khovanov fallback after the candidate queries fail. All verdicts agree with the known inputs; no inconclusive recognition is treated as completed.

| Input | Old (ms) | New (ms) | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| Raw circle | 376.791 | 374.341 | 1.024 | 1.010 | 0.980 |
| Optimized positive | 299.693 | 288.459 | 1.034 | 0.982 | 0.959 |
| Genus-one miss | 84.344 | 85.157 | 0.986 | 1.003 | 1.002 |
| Trefoil | 92.423 | 85.683 | 1.064 | 1.057 | 0.998 |
| Figure-eight | 161.984 | 145.843 | 1.098 | 1.000 | 0.996 |

All one hundred measured recognitions and twenty warmups finish in 24.132 seconds of driver elapsed time. Paired ratios range from 0.986 to 1.098, much smaller than the optimization-stage gains. The optimized-positive ratio 1.034 is comparable to its new A/A fluctuation (0.959), and the trefoil's 1.064 is close to its old A/A ratio 1.057. The raw-circle control follows the same code in both arms. These results do not justify a broad whole-recognition speedup claim; source preparation, component queries, proof replay and fallback can dominate the saved flow work.

### 106.7.3 The ordinary compressed-group portfolio

The last experiment compares release `535a914a6` with the new release and the native option enabled, using the established compressed-group portfolio on all nineteen ordinary cases. The earlier stages decide every input, so there are zero native-seed attempts in all measured calls. This is an integration and overhead control, not a measurement of native-seed discovery or an expansion of automatic recognition coverage.

| Input | Old (ms) | New (ms) | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| survivor-00 | 15.021 | 15.090 | 0.995 | 0.994 | 0.999 |
| survivor-01 | 12.901 | 12.983 | 0.993 | 1.002 | 1.004 |
| survivor-02 | 14.435 | 14.373 | 1.035 | 1.020 | 1.000 |
| survivor-03 | 10.080 | 10.416 | 0.977 | 1.010 | 1.029 |
| mirror-03 | 22.651 | 22.667 | 1.000 | 0.928 | 0.990 |
| survivor-04 | 8.627 | 8.598 | 0.997 | 0.997 | 0.993 |
| survivor-05 | 17.494 | 17.499 | 1.002 | 1.005 | 0.997 |
| survivor-06 | 7.098 | 7.279 | 0.975 | 1.006 | 1.050 |
| survivor-07 | 18.311 | 18.548 | 0.993 | 1.003 | 1.001 |
| survivor-08 | 6.923 | 6.898 | 0.998 | 1.016 | 1.007 |
| mirror-08 | 18.854 | 19.028 | 0.989 | 1.000 | 1.005 |
| survivor-09 | 7.055 | 6.925 | 0.985 | 1.006 | 1.017 |
| survivor-10 | 6.436 | 6.354 | 1.021 | 1.020 | 0.988 |
| survivor-11 | 6.829 | 6.842 | 1.013 | 1.013 | 0.994 |
| gordian | 1223.718 | 1233.042 | 0.998 | 1.002 | 0.986 |
| Trefoil | 0.063 | 0.064 | 0.984 | 0.987 | 0.985 |
| figure_eight | 0.068 | 0.070 | 0.972 | 0.988 | 1.016 |
| conway | 0.980 | 0.992 | 0.985 | 1.000 | 1.014 |
| kinoshita_terasaka | 1.006 | 1.009 | 0.998 | 0.994 | 1.001 |

All 380 measured calls and 76 warmups finish; driver elapsed time is 34.309 seconds. Old/new paired ratios span 0.972–1.035, old A/A ratios 0.928–1.020, and new A/A ratios 0.985–1.050. Gordian's old and new medians are 1.223718 and 1.233042 seconds, with paired ratio 0.998. No broad ordinary-workload gain or regression is established. The new option remains disabled by default.

Reproduction uses `../fast/normal_orbit_research/seeds.py`, with `audit`, `optimize`, `source` and `pipeline` modes. All 468 runtime, driver, test, fixture and delivery-source pins are checked before and after each run. The pipeline baseline is commit `535a914a6`; the optimization baseline is the unchanged report 31 module, identified separately in the source pins. Data use the `cocycle-seed-` prefix in `data/`. Full source proofs, incomplete queries and the earlier measurements remain available.

# 107 Sparse pivot scheduling for primitive cocycle discovery

Section 106 separates cocycle discovery from span optimization. The former is another significant cost: its initial solver processes triangular-face equations in tetrahedron order and always pivots on the leftmost remaining column. Even sparse input can then create dense intermediate rows and nonunit rational pivots. We now retain the same maximal-tree gauge and all face equations, but change their elimination order.

## 107.1 Exact elimination with live column incidence

Maintain every active row as a sparse dictionary and every column as the set of active rows containing it. Buckets indexed by row support size select a shortest nonzero row; row index breaks ties. Within that row, prefer a coefficient of absolute value one, then a column appearing in fewer active rows, then the column index. These choices are deterministic. They aim to reduce fill and rational arithmetic; they are not a theorem that either quantity remains small on all knot exteriors.

Normalize the chosen row at its pivot $p$ and store it as an equation

$$x_p = -\sum_{i \neq p} a_i x_i.$$

Remove that row from all active column sets. Only rows in the incidence set of $p$ need an elimination update. For each one, subtract its pivot coefficient times the normalized row, maintaining exact rational values, column incidence and support-size buckets. A row that becomes zero is retired. Repeated global edges in a face were already combined before this procedure, so no simplicial independence is assumed.

The invariant is that stored pivot equations and active rows have exactly the original rational nullspace. Normalization and row subtraction are invertible row operations. Every active row has zero entries in previously eliminated columns. Thus a stored pivot equation only refers to variables eliminated later or left free. Reverse *elimination order*, rather than reverse column order, gives valid back substitution. The number of free variables is the exact nullity; the API still requires one.

Clearing denominators and dividing by the coordinate gcd gives the primitive integer generator as before. Require its last nonzero chord coordinate to be positive. This recovers the initial solver's sign convention: in its leftmost-pivot echelon form, the only free coordinate is the last nonzero coordinate of the kernel vector. Since the tree gauge, chord ordering and primitive sign are unchanged, the new method returns the identical heights and normal coordinates, not merely a homologous representative. Consequently the downstream span problem and source-disc certificate remain the same whenever the shared budget lets both versions finish.

There are $O(T)$ rows and columns. Active rows, column incidence and stored pivot equations can require $O(T^2)$ entries. A conservative dense bound is still $O(T^3)$ exact arithmetic updates, with polynomial bit cost controlled by minors of the original sparse integer matrix. Shortest-bucket scans, deterministic row selection and incidence ordering do not improve that worst-case bound. The implementation exposes peak active-row support and nonunit pivot counts to distinguish observed sparsity from a global theorem. Cooperative work guards cover initialization, pivot normalization, incidence updates and back substitution; they remain an operational allowance rather than a bit-complexity measure.

## 107.2 Preserved source witnesses and observed arithmetic savings

An independent dense rational elimination test checks 240 small integer matrices with varied support, rank and coefficients, including a binary scaling case. It compares nullity decisions and rank-one kernels against full reduced row echelon form. Existing geometry tests include relabeled layered solid tori, a higher-rank rejection, and a rank-one manifold with homological torsion that still requires exact nonunit pivots. They also retain source-proof mutations, cancellation, shared-budget endpoints and the genus-one unknot miss. All sixteen focused tests pass in 3.372 seconds; all 1,166 maintained tests pass in 237.520 seconds. The source audit runs concurrently with the latter and takes 60.221 seconds; neither elapsed time is a performance comparison.

The pinned source audit compares the published solver at commit `7a13523cc` with the new implementation on all 84 earlier diagrams. Every vertex label, local height and normal coordinate is identical. All seventeen positive source queries replay independently, and their certificate hashes match the prior stored proofs. Sixty-two queries finish without a disc in the tested family, and the same five reach the shared two-million-guard limit. This scheduling change does not add source coverage. The previous Regina comparisons apply to these identical vectors; they are not counted as new external runs.

Across the 84 preparations, elimination updates fall from 2,286,028 to 286,748 and nonunit pivots from 861 to zero. The largest active support is ten. For Gordian, updates fall from 1,124,538 to 72,167 and preprocessing guards from 1,564,758 to 208,461. Nevertheless its subsequent raw component query still exhausts the shared allowance. The 32-crossing stabilized circle also remains capped during its raw query. These observations locate remaining costs; they do not turn zero nonunit pivots in this corpus into a theorem for arbitrary triangulations.

### 107.3 Serial measurements of complete operations

Both experiments run serially after the tests, source audit and runtime publication. Each uses five shuffled paired rounds, old and new A/A arms, and an excluded warmup per arm and input. All 220 measured calls and 44 warmups finish. Tables report individual-arm medians and medians of within-round ratios; the latter need not equal ratios of the former.

The preparation endpoint starts from a fresh PD, constructs the canonical exterior, independently replays its source geometry, obtains the primitive cocycle and validates all normal coordinates on the reconstructed manifold. Only the cocycle solver changes. This includes construction and validation costs rather than timing an isolated elimination loop.

| Input | Old (ms) | New (ms) | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| circle-1 | 4.901 | 5.005 | 0.969 | 0.998 | 1.006 |
| circle-8 | 56.628 | 41.304 | 1.371 | 1.000 | 1.030 |
| circle-32 | 247.496 | 186.966 | 1.343 | 0.989 | 1.073 |
| circle-64 | 508.502 | 368.708 | 1.372 | 1.012 | 0.996 |
| Survivor 00 | 113.154 | 81.171 | 1.400 | 1.002 | 0.999 |
| Gordian | 2244.886 | 1014.818 | 2.222 | 0.992 | 0.998 |

All 120 measured calls and 24 warmups finish in 59.152 seconds of driver elapsed time. Every old and new repetition produces the same seed hash. The larger circles and survivor input improve by 1.343–1.400; Gordian improves by 2.222, with medians falling from 2.244886 to 1.014818 seconds. Old A/A ratios range from 0.989 to 1.012, and new A/A from 0.996 to 1.073. The one-crossing input has paired ratio 0.969: about three percent overhead, with individual medians 4.901 and 5.005 milliseconds. Maintaining all row buckets and incidence sets has a small-input cost.

The recognition endpoint uses the same current recognizer in both arms and swaps only primitive cocycle discovery. Earlier shortcuts are disabled so that the native stage is reached. A raw seed proves both stabilized circles, the five-crossing example needs span optimization, and the two remaining cases continue to the exact Khovanov fallback. Every verdict matches the known input. Positive native paths include their normal component query and independent source-proof replay.

| Input | Old (ms) | New (ms) | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| circle-8 | 388.982 | 375.429 | 1.038 | 0.999 | 1.035 |
| circle-16 | 1446.584 | 1428.088 | 1.008 | 0.998 | 1.002 |
| Optimized positive | 295.827 | 302.754 | 1.042 | 0.957 | 0.930 |
| Genus-one miss | 170.545 | 168.546 | 0.990 | 1.007 | 1.008 |
| Trefoil | 90.172 | 94.203 | 1.022 | 1.000 | 1.004 |

All 100 measured recognitions and twenty warmups finish in 60.576 seconds. Paired old/new ratios range from 0.990 to 1.042, old A/A from 0.957 to 1.007, and new A/A from 0.930 to 1.035. These results do not establish a broad whole-recognition gain. In particular, the optimized positive's apparent paired gain is smaller than its control fluctuations; its separate arm medians even reverse order. Component queries, span optimization and fallback dominate these complete calls. The native option remains disabled by default, and this experiment makes no new claim about the ordinary portfolio that decides its inputs before reaching this stage.

The reproducible driver is `../fast/normal_orbit_research/cocycle_sparse.py`, with `audit`, `prepare` and `recognize` modes. Audit and measurements retain 470 matching source and fixture hashes, including the prior full source certificate audit. The two archives still in the incoming directory are covered by their earlier integration records. The checked `openai/math` head remains `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`; this change uses exact sparse elimination, not a new external complexity claim.

# 108 Output-sensitive support scans inside the adaptive orbit kernel

The native cocycle stage makes remaining component-query costs visible. A diagnostic profile of the 16-crossing stabilized circle finds 154,215 disjoint-support tests among 154,363 calls to the periodic merger. Such tests cannot succeed. The previous adaptive scheduler avoids repeated cubic scans by switching to a quadratic candidate queue, but still tests every pair on a no-merger pass. The new scheduler adds an interval-specific support scan while retaining that queue as its dense-case fallback.

## 108.1 Pruning failures without changing the proof trace

A periodic pairing has support $[a, d]$ and positive period $p = c - a$. Both maintained merger rules require positive support overlap: the sharp rule requires at least $p + q - \gcd(p, q) \geq 1$ points, and the classical rule requires at least $p + q$. Therefore deleting disjoint pairs from a candidate scan cannot remove a successful merger under either rule.

As before, find the first two periodic rows lazily and try that pair before building any index. Immediate first-pair merger families retain linear eligibility work and construct no support index. After a failure, collect all periodic rows and sort them by support start, with original row index as a tie-breaker. Sweep from left to right, retaining earlier rows in an active set and a heap ordered by support end. Remove a row only when its end is strictly less than the next start: supports meeting at one integer point must remain eligible.

Each active row overlaps the new row, and every overlapping pair appears exactly once, when its later sweep endpoint is visited. Store its two original indices in increasing order and sort the candidate list lexicographically. Resume the merger scan in that order, skipping the already failed first pair. Until a merger succeeds, row values and list indices are fixed. Omitted pairs all fail, so the first success is exactly the old greedy success. Its replacement, survivor position and trace event are unchanged. Induction after each restart proves identical closure output and successful events. The resulting complete AHT certificates and cycle allowances are preserved; scheduling diagnostics and work-guard counts may change. The independent replay checkers are unchanged.

For $k$ input rows, give the closure allowance $A = \binom{k}{2}$. Charge each initial first-pair test and *every enumerated overlap*, including ones later discarded because an earlier candidate merges. If another charge would exceed $A$, abandon that unfinished index and use the existing generation-checked queue on the current rows. No merger from a partial index has been applied. The fallback therefore starts from precisely the same greedy state and cannot change a verdict or trace. Charging index construction, rather than only attempted mergers, prevents repeated dense index builds from escaping the allowance.

A no-merger support pass with $E$ overlaps costs $O(k \log(k+1) + E \log(k+1))$ scalar operations and $O(k + E)$ storage. For disjoint supports $E = 0$, it makes one merger call and takes $O(k \log(k + 1))$ operations rather than testing $\binom{k}{2}$ pairs. There are at most $k$ support builds before closure terminates, and their total charged overlap count is at most $A$. With the fallback, the conservative bound remains $O(k^2 \log(k + 1))$ scheduler operations, $A + \max(k - 1, 0)^2$ merger calls and $O(k^2)$ memory. Binary endpoint comparisons and the exact gcd merger have their ordinary bit costs. This is a bound for one closure; it does not count all AHT cycles or prove a general quasi-polynomial unknot recognizer.

There is also an explicit family on which this changes the cost of the whole orbit computation. On $[0, 2k)$ take the $k$ unit pairings $[2i, 2i] \leftrightarrow [2i + 1, 2i + 1]$. Each cycle truncates the rightmost pair and the next contraction emits its remaining singleton. The algorithm has $k + 1$ cycles, including

the final empty-pairing contraction. The previous no-merger scans make

$$\sum_{j=1}^{k} \binom{j}{2} = \binom{k+1}{3}$$

merger calls; the support scheduler makes exactly $k-1$. Remaining coverage, transmission and independent replay scans give a conservative $O(k^2 \log(k+1))$ scalar-operation bound for the new complete query and replay, compared with the old cubic merger work. The proof trace is identical. This is a restricted interval-system theorem, not a claim that arbitrary normal-surface systems have disjoint supports.

## 108.2 One static-gap scan and an identity contraction

The original producer separately scanned domains and ranges to record certificate gaps, then repeated the sort and coverage scan to perform the contraction. It also rebuilt every immutable pairing even when there was no static gap. The new routine obtains occupied blocks and their complement in one sorted scan and returns the same gap list for the trace event.

If the complement is empty, the order-preserving contraction is the identity, so it reuses the immutable pairing rows. Otherwise a prefix map on occupied blocks gives every endpoint's new rank and reconstructs the pairings. All gaps contain only singleton orbits; their total size is added to the orbit count. This is the same mathematical reduction and the same certificate event as before, with fewer sorts and allocations. Input rows are immutable and the caller already owns its working list; no caller-supplied pairing is mutated. Cooperative cancellation covers both the coverage scan and the nontrivial endpoint map.

## 108.3 Source preservation, bounded queries and independent checks

Five new focused tests pass in 0.158 seconds and all 1,171 maintained tests pass in 210.854 seconds. The new controls compare 400 random closures under both merger rules with the literal restart scheduler, including exact trace order and the per-closure candidate bound. They compare 250 contractions with explicit finite order maps, retain one-point support intersections, exercise the dense fallback and cancellation, and use $2^{20000}$-scaled endpoints without expanding their represented points. Existing integration tests retain exact weighted and normal-component certificates, both merger rules, false-valued callbacks and all relevant cycle allowances.

The pinned audit performs 600 complete orbit comparisons on 300 additional random systems under both rules. Counts, cycles and entire certificates match the published baseline, and every result replays independently. Small random systems do not exhibit the main support-sparsity effect: their total merger calls only fall from 2,625 to 2,618. The maintained finite-graph tests separately compare answers with literal connectivity.

The source audit reruns all 84 previous diagrams with the same shared two-million-guard allowance. Seventeen positive results replay independently and match the earlier source-certificate hashes. Sixty-three cases finish the restricted candidate queries without a disc and four remain capped, compared with sixty-two completed misses and five caps previously. The newly finished case is `survivor-01`: it uses 1,812,276 guards and finds no disc in either the 331-piece raw surface or the 257-piece optimized surface. Its final source status remains inconclusive. No recognition coverage increase is claimed.

An independently reproducible diagnostic wraps the merger to classify every call on the raw 16-crossing stabilized circle. Disjoint calls fall from 154,215 to 467, and overlapping calls remain exactly 148. Both complete source proofs have the same hash. The shared guarded work falls from 1,126,713 to 988,196, a much smaller ratio than merger-call counts because the rest of discovery,

transport and positive replay still runs. The diagnostic and source audit overlap regression tests, so their elapsed times are not used as performance comparisons.

### 108.4 Serial complete-query measurements

The final experiments run serially after tests, audits, diagnostics and the runtime commit. Each has five shuffled paired rounds, old and new A/A controls, and one excluded warmup per arm and input. All 200 measured calls and forty warmups complete. Within each input, all old and new complete certificate hashes agree across repetitions. The tables separate individual arm medians from medians of paired ratios.

The first endpoint counts the complete interval system, constructs its certificate, independently replays that certificate and computes its hash. The disjoint cases use the family proved above. The five-cycle cases retain the earlier merger stress family, once with small and once with 4096-bit endpoints. No represented integer point is expanded.

| Input | Old (ms) | New (ms) | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| disjoint-16 | 2.003 | 1.368 | 1.491 | 1.005 | 1.001 |
| disjoint-64 | 58.093 | 14.533 | 4.128 | 0.991 | 1.026 |
| disjoint-256 | 3253.678 | 211.811 | 15.334 | 1.012 | 0.999 |
| Five-cycle, small | 3.028 | 2.604 | 1.147 | 1.006 | 1.017 |
| Five-cycle, 4096 bits | 10.133 | 7.785 | 1.296 | 0.999 | 1.004 |

All one hundred measured queries and twenty warmups finish; driver elapsed time is 42.799 seconds. The disjoint cases improve by 1.491, 4.128 and 15.334. At $k = 256$, old/new median times are 3.253678 and 0.211811 seconds, and merger calls are exactly 2,796,160 and 255, as the formulas predict. The five-cycle cases improve by 1.147 and 1.296. Old A/A ratios lie in 0.991–1.012 and new A/A in 0.999–1.026. These complete-query measurements include verifier cost; they are not isolated scheduler timings.

The second endpoint is the current full recognizer, with earlier shortcuts disabled and the native stage enabled. Only the orbit producer is swapped between releases. Both arms have no local native work cap, while retaining the same twelve-second global deadline; every measured call completes below that deadline. The three stabilized circles return raw normal-disc proofs, the optimized-positive input needs its second candidate, and the trefoil continues to the exact Khovanov fallback. Common result bookkeeping includes fingerprinting the positive certificate.

| Input | Old (ms) | New (ms) | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| circle-8 | 358.809 | 252.720 | 1.430 | 0.995 | 1.000 |
| circle-16 | 1360.820 | 798.522 | 1.704 | 0.999 | 0.995 |
| circle-32 | 5902.774 | 2803.070 | 2.108 | 0.998 | 0.983 |
| Optimized positive | 285.147 | 215.253 | 1.334 | 1.004 | 1.003 |
| Trefoil | 84.994 | 73.013 | 1.169 | 0.996 | 1.001 |

All one hundred measured recognitions and twenty warmups finish; driver elapsed time is 146.078 seconds. Paired gains range from 1.169 to 2.108, outside old A/A variation 0.995–1.004 and new A/A variation 0.983–1.003. The 32-crossing circle's median falls from 5.902774 to 2.803070 seconds, including native positive replay. The five-crossing optimized success falls from 285.147 to 215.253 milliseconds. These are whole-recognition gains on the explicitly exercised native-stage workload. The 32-crossing source still exhausts the default two-million-guard allowance; its unlimited-local-budget timing is not a new default-budget success. No ordinary-portfolio gain or general recognition complexity bound is inferred from this selected workload.

Reproduction uses `../fast/normal_orbit_research/supports.py`.

Its modes are `audit`, `orbits` and `recognize`. The call-category diagnostic is `data/orbit_support_profile.py`. All 473 runtime, driver, fixture and prior-audit pins are checked across the final experiments. The baseline is commit `7b0746a27`, before either the support scheduler or contraction change. The external `openai/math` head is unchanged at `fd4aeeb2e`, and no new unknot delivery arrived during this integration. The generic restart, hybrid and queue schedulers remain available for comparisons; the specialized support scheduler is the default inside the maintained orbit producer.

# 109 Conservation determines every weight on a certified single orbit

Normal-component queries attach additive integer vectors to interval points: Euler characteristic and boundary intersection numbers for disc detection, or normal-coordinate counters for component extraction. The maintained AHT producer first computes an unweighted orbit certificate, then transports these weights along its reductions. Once that certificate proves that there is exactly one orbit, the transport step has a simpler exact replacement.

## 109.1 A complete invariant under a checked connectivity hypothesis

Let $X = \{0, \dots, N-1\}$ and let $w : X \to \mathbb{Z}^d$ be the additive weight function. If the interval-pairing relation has exactly one orbit, its histogram is forced to be

$$\left\{ \left( \sum_{x \in X} w(x),\ 1 \right) \right\}.$$

For a canonical run partition $(\ell_i, r_i, v_i)_{i=1}^{R}$, using half-open intervals, the weight is computed without expanding points:

$$W = \sum_{i=1}^{R} (r_i - \ell_i) v_i.$$

Signed coordinates, zero runs and overlapping source intervals are allowed; the existing independent canonicalization first recovers their summed piecewise-constant function. This is conservation plus *proved* connectedness, not a connectedness conjecture about primitive normal surfaces.

The producer takes this path only after its complete unweighted AHT run returns one orbit. Public reuse of a supplied trace verifies it before entering the same result routine. The independent weighted checker first reconstructs the source weights, checks their binding to the certificate, and fully replays the unweighted orbit proof. Only then may it replace weight transport by the mass formula. A forged orbit count of one never bypasses the local-rule checker.

The existing `weighted-interval-orbits-v1` format is sufficient. Its canonical weights, unweighted proof and histogram are unchanged, so old and new complete certificates agree exactly. This is shorter computation, not a claim of fewer serialized certificate bytes. The producer uses direct coordinate accumulation; the independent checker uses its own tuple-vector arithmetic and reconstructed weight function. Neither implementation calls the other's transport code to justify the shortcut.

For $R$ runs and dimension $d$, the extra mass calculation uses $O(Rd)$ integer products and additions and $O(d)$ accumulator space. If endpoint lengths have at most $B$ bits and entries at most $C$ bits, intermediate signed sums have $O(B + C + \log(R+1))$ bits. A conservative bit bound is

$$O(Rd\,\mathsf{M}(B + C + \log(R+1)))\,,$$

where $\mathsf{M}$ bounds integer multiplication and dominates addition. This excludes source canonicalization and the mandatory unweighted proof search or replay. It removes weighted work depending on all intermediate transport partitions, but does not make the full orbit query linear.

## 109.2 Why several orbits still require transport

Conservation alone is insufficient when there is more than one orbit. For example, pair 0 with 2 and 1 with 3, and assign weights $1, 2, 3, 4$ respectively. The two orbit weights are 4 and 6. A forged histogram with two weights equal to 5 conserves total mass and has the right orbit count, but is false. The maintained multi-orbit path therefore keeps the complete independent weighted transport and rejects that forgery. An empty universe also retains its existing empty histogram; a one-point universe uses the single-orbit identity.

For a connected normal surface the three-weight disc query now obtains Euler characteristic and both boundary parities directly from total mass, after authenticating connectedness by its interval proof. The existing surface classifier and source-to-exterior verifier still supply the topological and provenance obligations. No new genus assumption, negative knot verdict or source-certificate schema is introduced. The same shortcut works for connected full-coordinate queries and for signed abstract weights.

## 109.3 Compatibility and source audits

Four focused tests pass in 0.004 seconds. They exercise signed overlapping weights on a $2^{20000}$-point connected system with both weight-transport routines disabled, including JSON hexadecimal round trips and supplied-trace reuse. They reject a false multi-orbit histogram with correct total mass, a forged single-orbit count, modified unweighted operations, incorrect source weights and Boolean counts. Empty and singleton universes retain their expected behavior. Cancellation still propagates through verification. The sixteen existing weighted tests also pass in 1.251 seconds. The first full run exposes one obsolete cancellation threshold: the six-tetrahedron fixture's verifier now finishes in 348 callbacks, before the test's hard-coded 600th callback. The test now measures each operation and requires cancellation at its entry, middle and final guards. All seventeen component tests pass in 0.917 seconds; the original failure log is retained. No artificial work is added to preserve a historical callback count. The final full suite passes all 1,175 tests in 242.337 seconds. The final source audit takes 32.626 seconds while that suite also runs; this elapsed time is not a performance comparison.

The pinned audit compares 300 random weighted systems with the published producer and checker. All entire certificates agree. The old checker accepts every new certificate and the new checker accepts every old certificate. Of these systems, 226 have one orbit; only those use the new branch. Total producer weight-replay events fall from 17,415 to 1,706. The latter events remain necessary for the other systems; this count does not include the mandatory unweighted proof operations.

All 84 source queries are repeated at the existing shared two-million-guard allowance. The same seventeen positive proofs replay independently with their previous hashes. Sixty-four restricted misses finish and three cases remain capped, compared with sixty-three misses and four caps before. The newly finished query is `survivor-02`, at 1,998,613 guards; it still finds no disc in either candidate, so there is no new unknot verdict. Gordian and the 32-crossing stabilized circle remain among the capped cases. The 16-crossing circle's shared work falls from 988,196 to 782,613 guards while preserving its certificate. Budget counters measure cooperative work, not bit time.

### 109.4 Complete weighted queries and source recognition

The final measurements run serially after the successful full suite, final audit and runtime publication. Each experiment uses five shuffled paired rounds, old and new A/A controls and one excluded warmup per arm and input. All 180 measured calls and 36 warmups finish. Every repeated old and new certificate hash agrees within its input. Tables report separate-arm median times and medians of within-round ratios.

The weighted endpoint constructs its input, computes the entire histogram and certificate, independently verifies that certificate and computes its hash. Connected cases use the earlier five-cycle interval family with 32 signed weight runs and either three or 128 coordinates. One case uses 4096-bit endpoints. A reflection with 128 two-point orbits is the disconnected control. Both producer and verifier switch together between the baseline and new implementations; the unweighted orbit engine is fixed.

| Input | Old (ms) | New (ms) | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| Connected, 3 weights | 4.840 | 3.930 | 1.188 | 0.947 | 1.055 |
| Connected, 128 weights | 13.064 | 7.648 | 1.729 | 1.003 | 0.977 |
| 128 weights, 4096 bits | 31.456 | 20.706 | 1.542 | 1.004 | 1.023 |
| Disconnected control | 0.824 | 0.822 | 1.005 | 0.979 | 1.010 |

All eighty measured queries and sixteen warmups finish in 1.052 seconds of driver elapsed time. Connected-case paired gains range from 1.188 to 1.729; the 128-coordinate small-endpoint case falls from 13.064 to 7.648 milliseconds. The disconnected ratio is 1.005, within the observed controls. Old A/A ratios range from 0.947 to 1.004 and new A/A from 0.977 to 1.055. These timings include source canonicalization, the unchanged unweighted search and proof replay, so they do not isolate the linear mass summation alone.

The second endpoint is complete recognition with the native stage enabled and earlier shortcuts disabled. Both arms use the current recognizer, switching only weighted production and verification. There is no local native guard cap, and both retain the twelve-second global deadline. The three stabilized circles return raw disc proofs; the five-crossing positive uses span optimization; the trefoil continues to the exact Khovanov fallback. Positive result bookkeeping includes the certificate hash.

| Input | Old (ms) | New (ms) | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| circle-8 | 271.428 | 212.207 | 1.279 | 1.037 | 0.992 |
| circle-16 | 855.262 | 659.084 | 1.313 | 1.001 | 1.009 |
| circle-32 | 2930.303 | 2164.070 | 1.346 | 0.999 | 1.007 |
| Optimized positive | 227.187 | 207.367 | 1.106 | 1.011 | 0.966 |
| Trefoil | 79.395 | 74.039 | 1.081 | 1.022 | 0.995 |

All one hundred measured recognitions and twenty warmups finish, in 101.929 seconds of driver elapsed time. Paired gains range from 1.081 to 1.346, with old A/A ratios 0.999–1.037 and new A/A ratios 0.966–1.009. The 32-crossing circle's medians fall from 2.930303 to 2.164070 seconds, including independent positive replay. Its default two-million-guard run remains capped; this timing does not imply a new default-budget success. The gains concern this explicitly exercised native workload. No ordinary-portfolio speedup or general recognition bound follows.

Reproduction uses `../fast/normal_orbit_research/single_orbit.py`.

Its modes are `audit`, `weighted` and `recognize`. The baseline is commit `8160ec448`. All 475 runtime, driver, fixture and prior-audit source pins agree across the final experiments. The incoming archive inventory and the `openai/math` head `fd4aeeb2e` are unchanged at this intake check. The arithmetic observation here supplements the integrated weighted-component machinery without assuming any new global theorem about unknot recognition.

# 110 Checking cocycle connectedness without an orbit search

The native seed stage can prove more from its construction than it could from arbitrary normal coordinates. Report 31 proves that a minimum-span surface in a primitive rank-one class is connected. Its spanning-tree construction also has this property before optimization. We now check the hypotheses independently and replace the two component searches in this restricted stage by Euler-characteristic calculations. General normal-surface queries retain their orbit machinery. The change improves a sufficient positive test; it does not complete the candidate family.

## 110.1 Two certificates excluding zero-class component unions

Let $M$ be the authenticated compact exterior of the input knot. In particular it is connected and orientable, with $H^1(M;\mathbb{Z}) \cong \mathbb{Z}$ and $H_2(M;\mathbb{Z}) = 0$. An integral cocycle $c$ defines the cooriented, edge-coherent local-height surface $S(c)$ of Section 106. All intersections on one oriented global edge have the same sign. Deleting components preserves this property. Signed intersections of any component union form a cocycle, representing its integral Poincaré-dual relative homology class.

First suppose $c$ vanishes on a spanning tree of the one-skeleton. If a component union $Q$ had zero relative class, its intersection cocycle would be an integral coboundary $\delta g$. No component meets a tree edge, since $S(c)$ has zero unsigned intersections there. Thus $\delta g$ vanishes on the tree, forcing $g$ to be constant on all vertices. The intersection cocycle of $Q$ is zero everywhere. Edge coherence then gives zero unsigned edge intersections, so $Q$ is empty: every normal piece meets edges. This argument excludes every nonempty zero-class union without an optimization witness.

Second suppose a potential $f$ is certified to minimize the total number of pieces in $S(c + \delta f)$. A nonempty zero-class union would have cocycle $\delta g$. Deleting it leaves an admissible edge-coherent normal surface with signed cocycle $c + \delta(f - g)$. Admissible coordinates are uniquely determined by unsigned edge weights: the kernel of the six edge-weight equations in a tetrahedron is generated by $(1, 1, 1, 1, -1, -1, -1)$, and a nonzero displacement would force three positive quadrilateral types in one of the two vectors. Consequently the remaining surface is exactly the local-height surface for $f - g$ in normal coordinates. It has fewer pieces, contradicting optimality. The checked primal–dual span equality supplies this second exclusion certificate.

In either case each connected component is nonseparating: a separating cooriented proper surface represents zero relative class. A nonseparating connected cooriented surface represents a primitive class, because a closed curve can cross it once and return through its connected complement. Rank one therefore makes each component a signed generator. Opposite signs would form a nonempty zero-class union. All signs agree, and a primitive total class permits exactly one component. Coorientation in the orientable ambient manifold also proves orientability.

## 110.2 An independent integral primitivity check

The new checker does not invoke cocycle elimination or trust the producer's reported rank. Rank one follows from the separately authenticated PD-to-exterior construction. It reconstructs global vertices and signed global edge equivalences from the finite face records. Every occurrence of an oriented edge must have the same height difference. Differences around each local triangle telescope to zero, so these consistency checks give one integral cocycle; they also guarantee integral constant height transitions across glued faces.

For a raw witness the graph of zero-valued edges must connect every global vertex. A breadth-first traversal therefore supplies a zero spanning tree, independently of the producer's union-find tree. For an optimized witness the checker instead chooses a tree in the entire edge graph and integrates the cocycle along its paths. If this integral vertex potential is $p$, the gauged value of an edge $a \to b$

is

$$\widehat{c}(a,b) = c(a,b) - p(b) + p(a).$$

The checker requires $\gcd_e \widehat{c}(e) = 1$. Tree-gauged integral cocycles form the integral kernel of the face equations, a saturated lattice. Its rank is one, so this gcd proves that the class generates $H^1(M;\mathbb{Z})$. A zero class has gcd zero and is rejected.

Taking the gcd before gauging would be unsound. Doubling a primitive cocycle and adding a unit vertex coboundary can give raw edge gcd one while the class remains twice a generator. The regression test constructs this example on the canonical exterior and rejects both that class and an exact nonzero coboundary. Local heights may also contain arbitrary tetrahedron constants. Tests transport raw and optimized proofs through 20,000-bit offsets and vertex gauges; the checked class and coordinates remain unchanged, including after hexadecimal JSON serialization.

## 110.3 From a connected surface to an essential disc

The checker validates matching, admissibility and all global edge weights of the supplied normal vector. For a raw proof, these unsigned weights must equal $|c(e)|$, which uniquely authenticates its local-height vector. For an optimized proof, the existing independent span checker reconstructs the adjusted coordinates, checks its corner matching and potential, and requires primal–dual equality. Its vector must equal the supplied vector.

Let $F$ be the number of normal pieces, $A$ the sum of their side counts, and $A_\partial$ the number of normal arcs on unpaired boundary faces. Then

$$\chi = \sum_{e \text{ global}} w_e - \frac{A + A_\partial}{2} + F.$$

The maintained geometry routine equivalently counts one copy of each paired face and every boundary face. These are binary sums over incidence records; no piece expansion or component orbit query is needed.

A connected orientable surface with $\chi = 1$ is a disc. Its relative class is nonzero. Since $H_2(M;\mathbb{Z}) = 0$, the boundary map from $H_2(M, \partial M;\mathbb{Z})$ to $H_1(\partial M;\mathbb{Z})$ is injective, so its boundary is homologically nonzero and hence essential on the boundary torus. This is a compressing disc of the source knot exterior and proves that the knot is trivial. The source authentication is essential to both the rank-one and boundary arguments.

Primitivity and Euler characteristic alone are insufficient. The test suite starts with the raw cocycle on the canonical exterior of the trefoil braid $(1, 1, 1)$ and subtracts two from the potential at global vertex 3. The resulting coherent surface still represents a primitive class and has $\chi = 1$, but an independent component query finds three components and no essential disc. Its zero edges do not span the vertices, and it has no valid optimality witness for those coordinates. Both attempts to use it as a positive certificate are rejected. This concrete knotted example guards the connectedness hypothesis itself.

## 110.4 Implementation, costs and portfolio behavior

The new `diagram-cocycle-disc-v1` format contains the exact stage-input PD, canonical triangulation, local heights, normal coordinates and either a null span witness or the existing primal–dual span certificate. The independent inspector returns certified connected-surface data for valid proofs. The public unknot verifier additionally requires $\chi = 1$. It still accepts the previous `diagram-normal-disc-v1` format by replaying its full normal-component proof. New witnesses have intentionally different hashes; the underlying candidate vectors are preserved.

After geometry authentication, the connectedness check uses $O(T)$ incidence records, breadth-first graph work and gcd operations, plus the existing span replay's $O(T)$ arithmetic after vertex

indexing. For $B$-bit supplied integers (including heights, potentials and coordinates), tree potentials and gauged differences have $O(B + \log(T + 1))$ bits. With $\mathsf{G}(L)$ bounding gcd on $L$-bit integers, a conservative bound for the added graph, primitivity and Euler work is

$$O(T\,(B + \log(T + 1) + \mathsf{G}(B + \log(T + 1)))) .$$

This excludes source/manifold authentication and label indexing; it is not a bound for the whole recognition call. No interval-orbit trace is produced or replayed in this restricted stage. Cohomology discovery still has a conservative cubic arithmetic bound, and optimized discovery still uses the $O(T^2 \log(T + 1))$ span solver.

The shared work allowance and cancellation callback cover source checks, both connectedness/Euler replays and optimization. The legacy `max_cycles` argument is still type-validated; these proofs use zero interval cycles, so a zero allowance no longer prevents success. The stage remains disabled by default, tries the raw surface first and optimizes only on a miss. A non-disc or resource exhaustion yields `INCONCLUSIVE` and returns control to the exact fallback. In particular the unknot $(1, 1, -1)$ still has genus-one raw and optimized seeds. There is no new general subexponential or quasipolynomial bound. Section 111 subsequently adds a bounded family of alternative tree gauges around this raw-then-span schedule, using the same independent certificate verifier.

## 110.5 Source, adversarial and external controls

The 22 focused connectivity, seed and span tests pass in 2.165 seconds; the full suite passes all 1,181 tests in 211.483 seconds. Tests disable cocycle discovery, flow discovery, exterior construction and both orbit production and replay while checking serialized positive witnesses. They also exercise exact schema types, modified height transitions, disconnected Euler-one surfaces, false optima, divisibility hidden by coboundaries and cancellation at the first, middle and final guards. Historical component witnesses, including valid zero-count proofs that must not certify an unknot, remain covered.

The audit repeats all 84 source queries with the existing two-million-guard allowance. It obtains 18 positive records, 65 completed restricted misses and one cap, compared with 17 positives, 64 completed misses and three caps before this change. The eighteen records contain thirteen distinct positive witnesses; the named optimized example also occurs as `random-54`, and repeated curls are retained rather than counted as new constructions. All positives replay independently. The seventeen previous positives have exactly the same normal vectors, and their historical proofs are accepted by both old and new verifiers.

The newly successful input is the 32-crossing stabilized circle: its raw 8,385-piece surface now certifies in 53,673 guards. The 8- and 16-crossing circles use 13,305 and 26,761 guards, compared with 199,317 and 782,613. The previously capped `survivor-00` completes both candidates in 1,295,668 guards, but both remain non-discs. Gordian's raw surface now finishes with Euler characteristic $-125$; its span optimization still exhausts the shared allowance. There is no inferred negative verdict and no newly successful default portfolio case.

Fresh external Regina controls compare both surfaces on every source of at most twelve crossings: 148 controls from 74 source records. Each agrees with the new inspector on connectedness, orientability, Euler characteristic and essential-disc count. These controls supplement, rather than replace, the independently checked source certificates and the proof of the connectivity criterion. The final audit takes 36.108 seconds while the full suite also runs; this elapsed time is not a performance comparison.

On the nineteen ordinary corpus inputs, the previous enabled stage, the new disabled stage and the new enabled stage give identical expected verdicts. Every run completes before attempting this stage. The substantial native-stage gains below therefore do not demonstrate a gain for those ordinary portfolio calls.

### 110.6 Smaller witnesses and complete-call measurements

The final measurements run serially, without tests, audits, builds or commits overlapping them. Each case has one warm-up round and five measured rounds. Each round randomly orders old, old A/A, new and new A/A calls. The tables give each primary arm's median elapsed time and the median of paired ratios; the latter need not equal a ratio of the two medians. All 200 measured and 40 warm-up calls complete. Baseline and new certificate hashes are stable within their own implementations; their coordinate hashes agree across implementations. Different source-proof schemas intentionally produce different complete certificate hashes.

The first experiment times source-bound independent verification of a precomputed positive certificate, together with source parsing, certificate hashing and serialized-byte counting. Producing the certificates is outside the timed sample. It includes all geometry and cohomology checks needed by each respective verifier, with no discovery cache passed into replay. The mode's total elapsed time, including preparation, is 11.136 seconds.

| Input | Old (ms) | New (ms) | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| circle-8 | 55.735 | 21.468 | 2.591 | 0.993 | 1.006 |
| circle-16 | 121.078 | 43.413 | 2.786 | 0.990 | 0.998 |
| circle-32 | 273.556 | 92.205 | 2.982 | 0.973 | 0.992 |
| Optimized positive | 30.206 | 15.744 | 1.937 | 1.001 | 1.000 |

Complete replay improves 1.94–2.98× on these four cases. The canonical compact JSON witnesses for the 8-, 16- and 32-crossing circles shrink from 112,472, 237,105 and 491,293 bytes to 30,077, 60,649 and 121,929 bytes. The optimized positive shrinks from 51,195 to 21,671 bytes, including its height data and dual matching. These sizes include the common source triangulation and PD; they are not sizes of an isolated connectivity subcertificate.

The second experiment starts with a source PD and runs the complete recognizer, including source construction, cohomology discovery, an optional span search, the stage's internal checks, the exact fallback when required, and an additional independent replay of any returned native proof. It also hashes and counts the serialized certificate. Earlier shortcuts are deliberately disabled so that every case reaches the native stage. Both arms have no local seed work cap and retain a twelve-second global deadline. All calls complete before that deadline; the mode takes 53.089 seconds.

| Input | Old (ms) | New (ms) | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| circle-8 | 260.410 | 64.166 | 4.004 | 1.001 | 0.921 |
| circle-16 | 744.828 | 131.303 | 5.756 | 1.002 | 0.998 |
| circle-32 | 2345.215 | 269.486 | 8.702 | 1.005 | 1.002 |
| Optimized positive | 233.967 | 114.397 | 2.050 | 0.999 | 0.999 |
| Genus-one miss | 72.198 | 49.426 | 1.467 | 1.006 | 0.992 |
| Trefoil | 71.814 | 49.088 | 1.459 | 1.001 | 1.008 |

The four native positives improve 2.05–8.70×. The 32-crossing circle's median falls from 2.345 seconds to 0.269 seconds. The genus-one unknot miss and trefoil still use the exact Khovanov fallback, returning `UNKNOT` and `KNOTTED` respectively; their complete calls improve about 1.46× by making the preceding unsuccessful stage cheaper. The new-arm A/A ratio on the smallest circle is 0.921, indicating more variation there than on the other cases; its paired old/new gain is still about four. These selected-workload gains establish neither ordinary default-portfolio acceleration nor a general complexity theorem. The default-budget source audit above separately establishes the newly completed 32-crossing positive.

Reproduction uses `../fast/normal_orbit_research/connectivity.py`, with `audit`, `replay` and `recognize` modes. The baseline is commit `6212f6f69`; its seed producer and source-proof

checker are loaded from the committed source while all unchanged dependencies remain fixed. All 478 runtime, driver, fixture and prior-audit pins agree across the experiments. The incoming archive inventory and the checked `openai/math` head `fd4aeeb2e` are unchanged. This integration uses the existing report 31 theorem and supplies its previously missing independent source-bound hypothesis checks.

# 111 A bounded search through alternative spanning-tree gauges

Section 110 makes raw tree-gauged surfaces cheap to certify. The remaining obstacle is discovery: a primitive surface and its minimum-piece representative can both miss a disc. We now reuse one cohomology solution to construct a bounded family of alternative tree gauges. Five previously unsuccessful source records acquire native disc proofs within the first three deterministic trials. An optional longer search finds a sixth. The family remains incomplete.

## 111.1 Changing the tree without solving cohomology again

Fix an integral cocycle $c$ on the validated finite exterior. For a rooted spanning tree $A$, define $p_A(v)$ by summing signed values of $c$ along the unique tree path from the root to $v$. On every oriented tree edge $u \to v$,

$$p_A(v) - p_A(u) = c(u, v).$$

Consequently

$$c_A = c - \delta p_A$$

vanishes on the tree and represents the same integral cohomology class. If $h_{t,i}$ are the original local heights, the new heights are simply $h_{t,i} - p_A(v_{t,i})$. Adding a common constant is immaterial. A different tree can change the normal surface and its Euler characteristic, while preserving the primitive class.

The independent raw certificate from the preceding section applies unchanged: the zero-valued edges contain a spanning tree, so no nonempty component union represents zero. Rank-one primitivity implies connectedness and orientability. Euler characteristic one therefore supplies an essential disc. The verifier reconstructs the signed cocycle, checks primitivity and the connected zero-edge graph itself; it need not trust or reconstruct the producer's choice of tree.

There is no need to repeat rational cohomology elimination. One verified primitive cocycle supplies every gauge by path integration. Nor does each tree introduce a different span-optimization problem: varying a subsequent vertex potential still ranges over the same set $\{c + \delta f\}$. The scheduler performs at most one span solve, on the original heights. Alternative unoptimized representatives can be valuable even when that common minimum-piece surface is not a disc.

## 111.2 Deterministic discovery and bounded later exploration

The producer recovers global oriented edge endpoints and values from every local height difference, checks consistency, and builds the one-skeleton adjacency lists. Loops cannot belong to a spanning tree and are omitted from those lists; their cocycle values still enter the surface. Parallel edges are retained. Roots are ordered by decreasing nonloop degree, with global vertex label breaking ties. A breadth-first traversal chooses a tree by the fixed adjacency order.

At most four such trials form the default prelude. Their root is the trial index modulo the number of vertices. For explicitly requested trials after the fourth, a private generator with seed 261009471 shuffles each adjacency list before breadth-first traversal. This changes the candidate trees, does not alter global random state, and supplies reproducible experiments in the recorded Python environment. No probabilistic completeness claim depends on these choices.

Normalize every integrated potential at the smallest global vertex. Equality of the resulting tuples proves equality of the gauged cocycles, so the producer skips duplicates, including the original zero potential. The duplicate still consumes its trial allowance and construction work. Each distinct candidate reconstructs binary normal coordinates and checks normal matching before computing Euler characteristic. A positive always passes the full independent source-bound verifier. A negative trial emits no negative knot certificate; its connectedness follows from the previously checked primitive source class and the tree construction.

With $V, E = O(T)$ vertices and edges, graph setup and root ordering use $O(T \log(T+1))$ comparison work after manifold validation. Each of the first four deterministic trials uses $O(T)$ arithmetic and incidence work. If the original heights have $B$ bits, a tree path has at most $V - 1$ edges, so potentials and new heights have $O(B + \log(T + 1))$ bits. Binary coordinate construction and the Euler calculation do not expand normal pieces. Storing the graph and one candidate takes $O(T)$ records; retaining exact duplicate signatures for $R$ requested trials takes $O(RV)$ integers. The optional shuffled trials add the permutation-generation cost of Python's random shuffle; the $O(T)$ traversal bound is conditional on those adjacency orders already being chosen. No worst-case bound for the optional random generator is being used to support a recognition theorem.

### 111.3 Switching between search methods

The optional native stage now follows this bounded schedule:

1. Check the initial primitive cocycle. A success ends the stage before constructing the tree-search graph.
2. Try up to four requested tree gauges, skipping duplicate potentials and stopping immediately on a verified disc.
3. If enabled, minimize the original span and check its surface.
4. On another miss, try any remaining requested tree gauges.

All steps share the existing cooperative work allowance and the enclosing deadline. A cap or unsuccessful search returns `INCONCLUSIVE` to the exact fallback. Thus cheap early successes avoid flow search, and a successful flow avoids every optional later tree. The schedule uses the outcome of completed attempts; it does not infer that a small piece count or improved Euler characteristic predicts future success.

The API options are `tree_trials` for `normal_seed_decide` and `normal_seed_tree_trials` for `recognize`; the CLI uses `-normal-seed-tree-trials`. The default is four. Zero restores the previous raw-then-span search. `-normal-seed-no-optimize` now skips the span solve; set the tree allowance to zero as well to request only the original surface. The stage as a whole remains disabled by default. Boolean, negative and noninteger trial allowances are rejected even when the stage is disabled.

### 111.4 Exploratory alternatives and their limits

Before integration, a standalone experiment tried up to 24 tree gauges on 53 previously unsuccessful raw candidates selected from the existing source corpus. It found six discs; five appeared in deterministic trials two or three, and one in shuffled trial 22. These observations motivated the short default prelude and optional later search. The maintained audit below repeats all 84 source records, checks old-proof preservation and validates the new surfaces independently. Euler changes in the exploratory log are not treated as genus changes: boundary-component counts can vary.

A second experiment minimized the sum of absolute global-edge cocycle values. Each term $|c(e) + f(v) - f(u)|$ can be represented as a two-height span, embedded in the existing four-corner arithmetic API by repeating both endpoints. This objective also strictly decreases on deletion of a nonempty zero-class component union, since every normal piece meets edges. Its rank-one

primitive optima therefore have the same connectedness argument. But the 58 tested old-stage misses of at most twelve crossings produced no Euler-one surface under the two-million-guard optimization allowance; all calls completed. The experiment is retained, without adding another optimization stage or source-proof format to the recognizer.

Neither alternative searches every normal surface. In particular, the three-crossing unknot $(1, 1, -1)$ remains a miss under the default policy. There is still no certified complete search with a general subexponential or quasipolynomial running-time bound.

## 111.5 New source proofs and preserved controls

The 22 focused tree, seed and connectivity tests pass in 3.227 seconds. The full suite passes all 1,187 tests in 229.784 seconds. Tests establish that early successes avoid flow, raw successes avoid tree setup, optimized successes avoid later trials, and the late example appears after the optimization miss. They check exact option types, duplicate skipping, determinism, matching failures, shared-cap exhaustion and cancellation. Positive replay succeeds with tree discovery disabled and uses the unchanged independent verifier.

All 84 source records are run with zero, four and 24 tree trials. The zero policy reproduces the preceding stage's 18 positive records, 65 completed restricted misses and one cap. Four trials produce 23 positives and 60 completed misses; 24 trials produce 24 positives and 59 completed misses. Gordian remains capped in both policies. All eighteen old positive hashes are preserved, and all six new source witnesses replay independently. The following rows list the new successes; trial counts include duplicates.

| Source | $n$ | Trial | Disc pieces | Minimum pieces | Guards |
|---|---|---|---|---|---|
| random-29 | 4 | 3 | 83 | 63 | 11,227 |
| random-79 | 4 | 2 | 102 | 63 | 10,192 |
| random-99 | 3 | 2 | 66 | 43 | 7,785 |
| random-124 | 5 | 3 | 121 | 79 | 14,185 |
| random-150 | 7 | 2 | 153 | 99 | 18,281 |
| random-81 (24 trials) | 5 | 22 | 122 | 71 | 180,578 |

Every displayed disc has more pieces than the span optimizer's returned non-disc. For example, `random-29` changes from an 89-piece raw surface of Euler characteristic zero to an 83-piece disc in trial three; the checked minimum-piece solution has 63 pieces and Euler characteristic $-1$. This proves that a useful disc exists within the same potential family even though the selected optimum is not a disc. It does not prove that every tied minimum-piece solution is a non-disc.

Fresh Regina controls check all 205 distinct generated candidates from the six new source examples and three further controls: trefoil, figure-eight and the genus-one unknot miss. Independent native inspection and Regina agree on connectedness, orientability, Euler characteristic and essential-disc count in every case. The new source certificates do not depend on Regina. The full audit takes 84.487 seconds while the test suite also runs, so that elapsed time is not a performance comparison.

Nineteen ordinary corpus inputs give their expected verdicts with zero, four or 24 trials. All 57 calls finish before reaching the native stage. Thus the additional source coverage is a property of the optional native search, not a demonstrated expansion of the ordinary recognizer's solved corpus or a general running-time improvement.

## 111.6 Complete recognition gains and unsuccessful-search costs

The benchmark compares the published raw-then-span stage with the new four-trial default, except for the explicitly labeled 24-trial case. Each timed endpoint parses the input PD, runs the

complete recognizer and exact fallback when needed, independently replays any returned native proof, and hashes and counts its serialized certificate. Earlier shortcuts are disabled to make the native stage reachable. Both arms use no local seed cap and retain a twelve-second global deadline. The native stage's ordinary disabled default is not being changed by this benchmark.

Runs are serial, without overlapping tests, audits, builds or commits. Every round randomly orders old, old A/A, new and new A/A calls. The initial five-round experiment completes all 160 measured and 32 warm-up calls in 16.851 seconds, but its unchanged raw-success control has a new-arm A/A ratio of 1.237 and an old/new ratio of 0.863. That variation warrants a repeat before interpreting the control. The full initial record remains in `data/cocycle-trees-initial-recognize.json`.

The final nine-round experiment completes all 288 measured and 32 warm-up calls in 28.078 seconds. The table gives median primary-arm times and median paired ratios; these ratios need not equal ratios of median times. All verdicts and each implementation's proof hashes agree across repeated calls and both experiments. Existing positive proofs remain byte-identical across implementations. Newly successful native calls intentionally replace the old inconclusive stage followed by an exact fallback.

| Input | Old (ms) | New (ms) | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| Raw circle | 65.061 | 63.657 | 1.026 | 1.025 | 1.005 |
| Optimized positive | 129.131 | 144.014 | 0.925 | 0.991 | 0.978 |
| Early trial 2 | 50.894 | 38.314 | 1.345 | 1.017 | 0.988 |
| Early trial 3 | 78.248 | 51.757 | 1.516 | 0.995 | 0.991 |
| Five-crossing early | 110.901 | 65.990 | 1.639 | 0.958 | 0.989 |
| Late, 24-trial option | 111.954 | 181.031 | 0.624 | 1.024 | 1.014 |
| Genus-one miss | 51.600 | 59.570 | 0.866 | 1.004 | 1.000 |
| Trefoil | 50.198 | 57.041 | 0.877 | 0.982 | 1.024 |

The three early successes improve complete recognition by 1.35–1.64×. They are `random-99`, `random-29` and `random-124` from the preceding table. The unchanged raw-success control is near parity: its 1.026 paired ratio is comparable to its 1.025 old-arm A/A ratio. There is no measured algorithmic gain to claim for that branch.

The unsuccessful prelude has a visible cost. The previous optimized positive is about eight percent slower by the paired ratio, while the genus-one miss and trefoil cost about 15 and 14 percent more respectively. Both misses still use the exact Khovanov fallback and return their expected `UNKNOT` and `KNOTTED` verdicts. The optional 24-trial example produces a new native disc proof, but its full call is about 60 percent slower than the old stage plus fallback. The initial run also shows this cost, with a paired ratio of 0.540. This is additional native proof coverage, not a speedup for that example; the longer search is therefore an explicit option rather than the default allowance.

These measurements support a bounded cheap prelude with a documented tradeoff. They do not establish a uniform speedup, a solved-corpus gain for the ordinary portfolio, or a global complexity improvement. The zero-trial option retains the earlier candidate schedule when its cost is preferable.

Reproduction uses `../fast/normal_orbit_research/trees.py`, with `audit` and `recognize` modes; the final timing run requests `-rounds 9`. The baseline is `48ef0ccef`. All 481 runtime, driver, fixture and prior-audit pins agree. The earlier exploratory scripts and full logs are retained in `data/tree-gauge-exploration.py` and `data/edge-weight-exploration.py`, with corresponding text outputs. Run those scripts from the `fast` directory with its Python package on `PYTHONPATH`. The incoming archives already have maintained review records, and the checked `openai/math` head remains `fd4aeeb2e`.

# 112 Adaptive blocking flows for cocycle-span minimization

The span network has one supply unit per tetrahedron. The previous solver runs Dijkstra once per unit, even when many units can travel at the same reduced cost. The maintained optimizer now routes batches through zero-reduced-cost residual arcs, using shortest-path searches when the current zero-cost graph is exhausted or batching has been unproductive. Its final primal–dual certificate and independent verifier are unchanged. This improves a polynomial local primitive; it does not complete the normal-surface search or establish a general quasipolynomial recognizer.

## 112.1 Routing a batch while retaining dual feasibility

Recall the $O(T)$-node, $O(T)$-arc transshipment network of Section 106. Each lower node receives at most one unit from the source, each upper node sends at most one unit to the sink, and every internal forward arc has capacity $T + 1$. An acyclic shortest-path initialization supplies integral potentials $\pi$ with nonnegative reduced costs

$$r(u, v) = c(u, v) + \pi(u) - \pi(v)$$

on every usable residual arc.

An augmentation entirely along arcs with $r = 0$ needs no change of potential. Every newly opened reverse arc also has reduced cost zero; every other usable arc retains its previous nonnegative reduced cost. Feasibility of the potentials is therefore preserved after any number of such augmentations. A path has unit bottleneck because its first source arc has capacity one. The numerical size of a height does not determine the number of flow units.

The producer first assigns breadth-first levels in the usable zero-cost graph. Once the sink level is known, it finishes the preceding layer and does not expand deeper vertices. It then traverses only arcs advancing one level. A current-arc pointer at each node skips exhausted or unusable choices. Traversal uses explicit node and edge stacks, so a long residual path does not depend on the interpreter's recursion limit.

When a path reaches the sink, the producer sends one unit and restarts at the source. A failed branch advances its parent's pointer. The resulting block leaves no source-to-sink path in the current level graph, unless all required flow has already been sent. Newly opened reverse arcs go to a lower level and cannot revive an earlier discarded choice in the same block. Further zero-cost paths, if any, are found by another breadth-first level construction. This argument allows zero-cost cycles, parallel arcs and disconnected abstract incidence components.

If no zero-cost path remains, reduced-cost Dijkstra finds the next path. Let $\Delta$ be the sink distance. The existing update

$$\pi(v) \leftarrow \pi(v) + \min(d(v), \Delta)$$

uses increment $\Delta$ for unreachable or unsettled nodes. It preserves nonnegative residual costs, including on arcs entering the reached set from outside, and makes the chosen augmenting path tight. The producer sends one unit along that path, then may try batching again.

After $T$ units have been sent, every original internal forward arc is still residual: its capacity exceeds the total flow. Every used arc also has its reverse residual, so dual feasibility forces equality on it. Extracting vertex potentials and a corner matching therefore gives the same primal–dual optimality argument as before. The existing independent checker reconstructs coordinates, verifies matching balance and requires equality of the primal span and dual reward. It neither imports the blocking-flow helper nor trusts its counters.

## 112.2 A bounded handoff for small batches

Batch construction itself costs a graph scan. When marginal costs are distinct, repeating that scan before every Dijkstra search adds work without routing more flow. The scheduler therefore counts an attempted block as poor when it sends fewer than two units. Two consecutive poor attempts disable batching; a block of at least two units resets that count. The solver then uses ordinary shortest-path augmentations.

A later Dijkstra augmentation with sink distance zero is fresh evidence that units are again moving at the same reduced-cost level. It reenables batching with a reset poor-attempt counter. This restart affects only which valid augmenting paths are tried. Both modes maintain the same residual feasibility and final certificate obligations. Every iteration of the outer loop sends at least one unit, including an iteration whose zero-cost search fails, because that failure is followed by Dijkstra.

The separated-cost control has $T$ independent vertex labels, one for each row, with heights $(0, 0, 0, i)$ in row $i$. All local spans are fixed, and successive marginal costs are distinct. The implementation attempts only two zero-cost searches, sends one unit in its first block and then finishes with $T - 1$ Dijkstra searches. The unbounded prototype instead performed a failed zero-cost scan at almost every remaining unit; this control motivated the handoff before integration.

## 112.3 Worst-case accounting and an improved equal-cost family

Write $N, E = O(T)$ for the network size. One breadth-first level construction uses $O(N+E)$ graph work. Within its block, current-arc pointers advance at most once per arc. A successful path can traverse at most $N - 1$ arcs; if a block sends $q$ units, a conservative cost is $O(N + E + qN)$. Across the whole solve there are at most $T$ attempted blocks and at most $T$ routed units. Thus all blocking work is $O(T^2)$ arithmetic and incidence work. At most $T$ Dijkstra searches remain, preserving the conservative bound

$$O(T^2 \log(T + 1))$$

for arithmetic operations. Arrays, queues, current-arc pointers and explicit stacks take $O(T)$ records. The handoff changes neither bound.

Zero-cost augmentations do not change potentials. The previous bound $O(B + \log(T + 1))$ on arithmetic bit length for $B$-bit heights therefore still applies to both modes. A conservative addition/comparison bit bound remains

$$O\Big(T^2 \log(T + 1)(B + \log(T + 1))\Big),$$

apart from label indexing and input/output. The original certificate size and separate cohomology-preprocessing bounds are unchanged.

There is a direct improvement on an explicit family. Give every row vertices $(0, 1, 2, 3)$ and heights $(0, 0, 0, 0)$. The zero-cost level graph has source, lower, vertex, upper and sink layers. One blocking phase sends all $T$ units along paths of length four, with $O(T)$ scans and no Dijkstra search. The old solver performs $T$ searches, each scanning the $T$ source arcs, so its residual search work is $\Omega(T^2)$ even on this zero-height family. The change is from quadratic to linear residual search work on that family, not a linear bound for arbitrary span inputs or complete unknot recognition. Complete-call measurements below include certificate creation and independent verification as well.

## 112.4 Arithmetic, topology and source audits

The 27 focused blocking, span, seed and tree tests pass in 2.593 seconds. The full suite passes all 1,192 tests in 212.108 seconds. New tests cover linear scan growth on the equal-cost family,

bounded handoff on distinct marginal costs, restarting after zero-distance evidence, a 2,500-node zero-cost path with cycles, cooperative caps and cancellation. Multiplying a nonzero instance by $2^{20000}$ preserves all search counters and scales the objective exactly. Its hexadecimal JSON certificate replays with both flow producer entry points disabled.

Five hundred random arithmetic problems agree with the old optimizer on both optimum value and the entire coordinate vector. In 495 cases the entire certificate is also identical; five use another valid witness. This observed coordinate agreement is not assumed in the correctness proof. Each old and new certificate passes the unchanged independent checker. Aggregate Dijkstra searches fall from 7,928 to 7,398, but total guard work rises from 3,333,719 to 3,765,178. Small arbitrary inputs therefore provide an adverse control, not evidence of a uniform speedup. The complete-call benchmark includes a separate batch of such problems.

The initial audit completed its checks but failed while serializing the 20,000-bit objective with ordinary decimal JSON. The driver now uses the existing exact hexadecimal transport; the interpreter's conversion limit is unchanged. The final audit is repeated after that fix and takes 51.768 seconds. Its elapsed time mixes correctness checks and is not a paired performance comparison. The initial failure log is retained.

All 84 source inputs are also optimized directly with a two-million-guard allowance. Both optimizers finish on 82 cases, preserving every objective and coordinate vector. Six complete source span witnesses are byte-identical; the others can use different valid matchings. The new optimizer also finishes the two previously capped standalone queries:

- Gordian, with 2,820 tetrahedra, uses six Dijkstra searches and routes 2,814 units in blocks. It reduces 3,513 pieces to 2,315 in 1,929,649 guards. An independent source-bound inspection and Regina agree that its surface is connected, orientable and has $\chi = -65$, with no essential disc.
- The 32-crossing circle uses 18 Dijkstra searches for 640 units, reducing 8,385 pieces to 4,235 in 1,019,747 guards. Independent inspection and Regina identify a disc. Its raw surface already supplied a native proof before this change.

These two larger topology checks supplement 74 fresh optimized-surface Regina controls on sources of at most twelve crossings. All 76 comparisons agree on connectedness, orientability, Euler characteristic and disc count.

Complete source-stage queries preserve the four-tree policy's 23 positives, 60 completed restricted misses and one cap, and the 24-tree policy's 24 positives, 59 misses and one cap. Gordian still caps because source construction, cohomology, tree trials and replay share its two-million-guard allowance with optimization. Its newly completed standalone query is not a newly completed full stage. The selected optimized positive and its duplicate `random-54` now have a different valid corner matching, giving one distinct changed positive proof. Other positive hashes are preserved, and every returned source proof replays independently.

The default-stage guards for `survivor-00`, `survivor-01` and `survivor-02` fall from 1,310,623, 1,124,340 and 1,234,216 to 251,349, 208,359 and 172,493 respectively. These remain restricted misses. The optimized five-crossing positive falls from 144,578 to 54,318 guards, and the optional late-tree positive falls from 180,578 to 72,527. Guard counts are cooperative work indicators, not bit-operation counts.

On nineteen ordinary corpus inputs, old and new enabled-stage policies return the same expected verdicts without reaching this native stage. The optimization improvements therefore do not establish an ordinary default-portfolio speedup or new source coverage.

## 112.5 Complete optimization and recognition measurements

The final measurements run serially, without tests, audits, builds or commits overlapping them. Each case has one warm-up round followed by five measured rounds, each randomly ordering old,

old A/A, new and new A/A endpoints. All 260 measured endpoints and 52 warm-ups complete. Tables give median primary-arm times and medians of paired ratios; the latter need not equal ratios of the median times. Each implementation's certificate hashes remain stable across repetitions, and all comparable coordinate hashes agree between implementations. Valid matching choices can make complete certificate hashes differ.

The first experiment includes input construction, span optimization, an additional independent certificate check and hashing. For the two source-derived cases, every timed endpoint also constructs the canonical exterior and solves its primitive cohomology. The binary case scales its four-crossing cocycle heights by $2^{4096}$ before making the arithmetic span query; it is not being used as a primitive source-disc certificate. The random control aggregates 24 complete, independently verified 24-row problems per endpoint to reduce timer noise on small queries. The other optimizer endpoints contain one query each. This mode completes 140 measured endpoints and 28 warm-ups in 46.753 seconds.

| Input | Old (ms) | New (ms) | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| Equal cost, 32 rows | 5.594 | 1.964 | 2.849 | 1.006 | 0.999 |
| Equal cost, 128 rows | 75.118 | 8.017 | 9.064 | 1.012 | 1.001 |
| Equal cost, 512 rows | 1164.347 | 33.169 | 35.563 | 1.007 | 1.025 |
| Distinct costs, 128 rows | 74.723 | 74.804 | 0.985 | 1.004 | 0.999 |
| 24 random 24-row queries | 111.174 | 130.429 | 0.858 | 1.001 | 1.069 |
| Circle, 32 crossings | 1683.749 | 429.510 | 3.914 | 1.000 | 1.015 |
| Circle, 4 crossings, binary | 49.438 | 27.429 | 1.811 | 1.008 | 0.988 |

On the equal-cost family, complete endpoints improve from 2.85× at 32 rows to 35.56× at 512 rows, consistent with the proved residual-search improvement. The largest median falls from 1.164 seconds to 0.033 seconds. The distinct-cost control is close to parity, with a paired ratio of 0.985 after the bounded handoff. The random control is about 17 percent slower by its paired ratio; its new-arm A/A ratio of 1.069 also shows visible variation. This is consistent with the adverse guard-work result above, and prevents a claim of uniformly faster span optimization.

The two source-derived optimization endpoints improve 3.91× and 1.81×. The 32-crossing case includes source construction and cohomology, falling from 1.684 to 0.430 seconds. This is a timing of its optimization query: the recognizer normally succeeds on its raw surface and does not run that optimization.

The second experiment starts from the input PD and times the complete recognizer, including its earlier setup, native discovery, exact fallback when required, an additional independent replay of any returned native proof and certificate/coordinate hashing. Earlier shortcuts are disabled to make the native stage reachable. Both arms use no local seed cap and retain the recognizer's twelve-second global deadline. The late-tree case requests 24 trials; the others use four. This mode completes 120 measured calls and 24 warm-ups in 11.591 seconds.

| Input | Old (ms) | New (ms) | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| Raw circle | 63.035 | 61.919 | 1.034 | 1.032 | 0.995 |
| Optimized positive | 128.244 | 89.515 | 1.459 | 0.956 | 0.974 |
| Early tree positive | 37.066 | 38.425 | 0.999 | 1.000 | 1.005 |
| Late tree positive | 178.053 | 139.721 | 1.280 | 0.964 | 1.049 |
| Genus-one miss | 57.538 | 41.679 | 1.378 | 0.997 | 0.986 |
| Trefoil | 55.427 | 39.291 | 1.404 | 0.976 | 1.000 |

The four cases reaching span optimization improve 1.28–1.46×. They preserve the optimized and late-tree positive proofs' conclusions, and preserve the genus-one unknot miss and trefoil's exact fallback verdicts. The early tree positive has a paired ratio of 0.999 and avoids flow in both arms.

The raw-circle control's 1.034 ratio is comparable to its old-arm A/A ratio of 1.032; no algorithmic gain is claimed for that unchanged branch. These are selected native-stage timings, not evidence of improved ordinary portfolio performance or a general recognition bound.

Reproduction uses `../fast/normal_orbit_research/blocking.py`, with `audit`, `optimize` and `recognize` modes. The baseline is `4fa723b79`; all 483 runtime, driver, fixture and prior-audit source pins agree. The two large topology checks use `data/blocking_capped_sources.py`, run from `fast` with `PYTHONPATH=.` and optional Regina. The incoming archive inventory and the checked `openai/math` head `fd4aeeb2e` remain unchanged.

# 113 Bounded search among tied minimum-span surfaces

The minimum number of normal pieces does not determine the topology of a cocycle surface. Even a fixed optimal dual matching can certify both a disc and a non-disc on the same source exterior. This distinction motivates a new optional bounded fallback: after the selected optimum and all requested tree gauges miss, inspect several other potentials with the same minimum span. The construction uses shortest paths in a difference-constraint graph; it does not run another flow optimization. Every positive retains the existing independent source-bound certificate format.

## 113.1 An exact description of all tied optima

Fix a verified span certificate with optimum $D_*$ and matching rows $(t, i, s, j)$. Let $\ell_t$ be its selected lower corner in tetrahedron $t$, and let $u_t$ be its selected upper corner. Each exists exactly once because the matching uses every lower and upper tetrahedron once. For any potential $f$, write $a_{t,i} = h_{t,i} + f(v_{t,i})$. Cancellation at matched global vertices gives

$$D(f) - D_* = \sum_t (\max_i a_{t,i} - a_{t,u_t}) + \sum_t (a_{t,\ell_t} - \min_i a_{t,i}).$$

Every summand is nonnegative. Consequently $f$ is optimal if and only if all selected lower and upper corners remain extremal. Equivalently, for each tetrahedron and each of its four corners,

$$f(v_{t,\ell_t}) - f(v_{t,i}) \le h_{t,i} - h_{t,\ell_t},$$
$$f(v_{t,i}) - f(v_{t,u_t}) \le h_{t,u_t} - h_{t,i}.$$

These eight inequalities per tetrahedron describe *every* optimum, including those admitting other optimal matchings. Necessity follows from weak-duality equality with this one fixed optimal matching; no uniqueness assumption is involved.

Represent $f(b) - f(a) \le w$ by an arc $a \to b$ of weight $w$. Given the certificate's original potential $f_0$, reweight it as $r(a, b) = w + f_0(a) - f_0(b) \ge 0$. In displacement variables $q = f - f_0$, the inequalities become $q(b) - q(a) \le r(a, b)$. The graph has at most $4T$ vertices and exactly $8T$ arcs, retaining parallel arcs and loops. The weights and the constructed extremal solutions are integral.

Within each connected component of the tetrahedron–vertex incidence structure, this constraint graph is strongly connected. Indeed, fixing $D(f) = D_*$ bounds every local height span by $D_*$. Any two corners of a tetrahedron therefore have potential difference at most $D_* + |h_{t,i} - h_{t,j}|$. Chains of tetrahedra bound every relative potential in that incidence component. If the directed constraint graph were not strongly connected, decreasing all potentials in a sink strongly connected component would preserve all inequalities indefinitely, contradicting that bound. Different incidence components can translate independently.

### 113.2 Two simultaneous extrema from one root

For a root $r$ in a component, let $d(r, v)$ be shortest-path distance in the nonnegative reweighted graph. Normalize the displacement by $q(r) = 0$. Every feasible displacement obeys

$$-d(v, r) \le q(v) \le d(r, v).$$

Both simultaneous bound vectors are feasible:

$$q^+(v) = d(r, v), \qquad q^-(v) = -d(v, r).$$

The upper assertion is the shortest-path triangle inequality along an arc; the lower assertion follows from the same inequality for paths ending at $r$. Thus each vector attains all its corresponding coordinate bounds at once. One Dijkstra search in each direction constructs the pair. Outside the root's incidence component, the producer keeps $q = 0$. A common additive normalization then makes the smallest output potential zero.

If two roots have zero distance in both directions, they are in the same zero-weight strongly connected component. Their forward distance vectors are identical, as are their reverse distance vectors. Testing both roots would therefore produce the same pair. The implementation marks all such roots after the two searches and skips them. It orders the remaining roots by descending total arc incidence, breaking ties by vertex label. It also skips normalized potentials already emitted or equal to the original one. These are exact duplicate rules, not assumptions about surface topology.

Coordinates are reconstructed from the adjusted tetrahedral heights, and every emitted potential replays the unchanged span verifier with the original matching and optimum. At the source stage, the original primitive class and minimum-span connectedness theorem still apply. Negative candidates use normal matching and Euler accounting; a proposed disc must also pass the complete independent source-bound verifier. Failure to find a disc remains inconclusive.

### 113.3 Cost and scheduling limits

Let $R$ be the smaller of the requested root allowance and the number of vertices. After verification and graph construction, at most $2R$ shortest-path searches and $2R$ candidate replays occur. In an indexed-graph model with constant-time table lookup and well-distributed signature hashes, the arithmetic cost is $O((R + 1)T \log(T + 1))$, including label/root sorting. This is a conditional container-cost bound, not an unconditional worst-case bound for Python's hash tables.

The shipped code uses dictionaries for labels and distances and a set of potential tuples for duplicates. Adversarial hash collisions can require $O(T)$ probes per dictionary operation and $O(RT)$ coordinate comparisons per signature lookup. Since $R \le 4T$, a conservative deterministic bound for these container comparisons together with the graph arithmetic is $O((R+1)T^2)$. Storage is $O((R + 1)T)$ integer records; stored duplicate signatures account for the factor $R$. With $B$ the largest input integer bit length, intermediate quantities use $O(B+\log(T+1))$ bits. Charging linear bit cost for integer hashing, addition and comparison gives the corresponding product bound. No loop ranges over a height's numerical value. Both bounds are local polynomial bounds, including when all roots are requested.

The optional source-stage experiment requests four roots, after all previously requested trees and the single span optimization have failed. This placement preserves earlier positive proofs and their guard work. The shared cap and cancellation callback cover the new graph searches, coordinate checks and final replay. Setting `face_roots=0`, or `normal_seed_face_roots=0` in the recognizer, restores the previous search policy. The CLI exposes `-normal-seed-face-roots`. Without span optimization the face fallback is skipped. The face allowance defaults to zero: the measurements below show added coverage but slower complete calls. The entire native normal-surface stage also remains optional in the ordinary portfolio.

Even trying every root does not enumerate all feasible potentials, all vertices of the optimality polytope, or all topology types. The exact inequality description is complete; this extremal candidate search is not. It does not maximize Euler characteristic, prove minimal genus, or complete the normal-surface search required for general quasipolynomial recognition.

## 113.4 Topology changes at an unchanged optimum

On the five-crossing source `optimized-positive`, the original 84-piece optimum is a disc. Another potential certified by the very same matching has 84 pieces and is an annulus. Independent source inspection and Regina agree that both surfaces are connected and orientable; their Euler characteristics are 1 and 0, and their boundary counts are 1 and 2. A dual matching therefore does not fix the normal coordinates or topology. A change in flow tie-breaking must not be justified by assuming that it preserves the selected surface.

The seven-crossing source `random-142` supplies the converse discovery example. Its original optimum has 122 pieces and is an annulus. Four nonequivalent root trials find a 122-piece disc at root 13 in the forward direction, after five distinct non-disc alternatives. An exhaustive pass through root classes also finds a second disc and two three-boundary planar surfaces, all at the same optimum. The retained witnesses share the original matching; they differ in potential and coordinates. The successful bounded source proof is independently replayed with the discovery entry points disabled in a focused test.

With four face roots requested, native positives on the 84-source corpus increase from 23 to 24 under the four-tree policy, or from 24 to 25 when 24 tree trials are requested. There is one distinct new positive proof. The shared-budget Gordian cap remains, leaving 59 and 58 completed restricted misses respectively. The new seven-crossing proof uses 117,848 guards under the four-tree policy; disabling face search leaves a completed miss at 93,011 guards. Every previously positive certificate and its guard work are preserved. These counts describe the explicit four-root option; the default zero-root policy keeps its previous coverage.

## 113.5 Validation and the decision to keep this optional

The 32 focused tests pass in 3.991 seconds. They include forty small arithmetic instances checked on exhaustive potential grids, malformed optimality witnesses, disconnected incidence components, 20,000-bit scaled heights, cancellation, strict option types and exact positive-proof budget boundaries. The final maintained suite passes all 1,197 tests in 232.354 seconds. An earlier full run before changing the default allowance also passed; the final run covers the disabled-by-default policy.

The source audit independently inspects every distinct root-extremal surface on the 74 inputs with at most twelve crossings, together with each original optimum. All 787 surfaces agree with Regina on connectedness, orientability, Euler characteristic and compressing-disc status. There are 713 alternatives, and 21 of the 74 sources exhibit different Euler characteristics among their tied optima. Full witnesses for the two named counterexamples and the new positive are retained. On nineteen ordinary portfolio inputs, both zero-root and four-root settings return the expected verdicts without reaching the native stage.

The first audit reached the final ordinary-corpus loop but failed because its driver used an incorrect corpus-helper attribute. The helper reference was corrected and the failure log retained. The final pinned audit takes 103.135 seconds, overlapping correctness tests; this is not a performance comparison. Its 485 runtime, driver, fixture and prior-audit hashes agree with the final benchmark. The independent span and source-verifier modules are byte-identical to the previous baseline `a8658ec40`.

Two serial five-round experiments compare the complete recognizer with face search explicitly

off or set to four roots. Each case has one warm-up round and then five measured rounds, each randomly ordering off, off A/A, on and on A/A calls. No tests, audits, builds or commits overlap the timed runs. Earlier shortcuts are disabled so that the native stage is reached; both arms have no local seed cap and retain the twelve-second global recognition deadline. All requested tree trials remain the same in both arms, including 24 for the late-tree control. A returned native proof is additionally replayed and hashed inside the timed endpoint.

The first experiment motivated the zero default. The second repeats the comparison with the final source pins. All 280 measured calls and 56 warm-ups across the two experiments finish; per-implementation verdicts and proof hashes are stable across both runs. The final run alone has 140 measured calls and 28 warm-ups and takes 15.003 seconds. Its primary-arm median times and median paired ratios are shown below; a median paired ratio need not equal the ratio of median times.

| Input | Off (ms) | On (ms) | Off/on | Off A/A | On A/A |
|---|---|---|---|---|---|
| Raw circle | 63.764 | 64.729 | 0.986 | 1.003 | 1.011 |
| Optimized positive | 92.861 | 91.446 | 1.015 | 1.002 | 1.003 |
| Early tree positive | 37.904 | 38.139 | 0.992 | 0.951 | 1.020 |
| Late tree positive | 140.885 | 136.867 | 1.028 | 1.065 | 1.007 |
| New face positive | 120.063 | 206.335 | 0.582 | 0.955 | 0.973 |
| Trefoil | 41.078 | 64.424 | 0.638 | 0.984 | 1.025 |
| Genus-one miss | 43.752 | 66.204 | 0.667 | 0.993 | 1.000 |

The new seven-crossing native positive takes about 206 milliseconds with face search versus 120 milliseconds when the existing exact fallback finishes the call. Its paired ratio corresponds to about 72 percent additional time. The two continuing misses cost about 50 and 57 percent more. Earlier positive branches avoid the new search; their small timing differences provide no evidence of algorithmic gains and are accompanied by visible A/A variation. In particular, the late-tree off-arm A/A ratio is 1.065. The experiment therefore supports optional discovery and stronger coverage evidence, not a faster default recognizer. The maintained default continues to request zero face roots.

Reproduction uses `../fast/normal_orbit_research/face.py` with `audit` or `recognize`; the audit requires optional Regina. The existing incoming archive inventory is unchanged. The checked `openai/math` head remains `fd4aeeb2e`, and the arXiv record for Lackenby's hierarchy paper [24] still lists its July 2026 first version. The new optimal-face argument is a local extension of report 31's matching duality, not a resolution of the general recognition-complexity gap.

# 114 Capping primitive annuli without another surface search

The optimal-face experiment exposes a simpler improvement than searching for a tied disc: its seven-crossing source already has a certified annulus in the primitive relative class. That annulus itself proves unknottedness. The native stage now accepts this sufficient witness immediately. The stored surface remains an annulus with zero disc components; a new certificate schema records the separate boundary-capping implication.

## 114.1 Why the primitive annulus gives a compressing disc

Let $M$ be the authenticated compact exterior of a knot in $S^3$, with boundary torus $T$. As used in Section 110,

$$H_2(M;\mathbb{Z}) = 0, \qquad H_2(M,T;\mathbb{Z}) \cong \mathbb{Z}, \qquad H_1(M;\mathbb{Z}) \cong \mathbb{Z}.$$

The inclusion $i_* : H_1(T;\mathbb{Z}) \to H_1(M;\mathbb{Z})$ is onto. For example, Mayer–Vietoris for $S^3 = M \cup N(K)$ identifies $H_1(T;\mathbb{Z})$ with $H_1(M;\mathbb{Z}) \oplus H_1(N(K);\mathbb{Z})$. The exact sequence of the pair therefore makes

$$\partial : H_2(M,T;\mathbb{Z}) \longrightarrow H_1(T;\mathbb{Z})$$

injective, with image the primitive rank-one subgroup $\ker i_*$. Consequently a primitive relative class has nonzero primitive boundary class on the torus. This conclusion uses the source knot exterior; it is not asserted for an arbitrary torus-boundary manifold.

Suppose $S$ is a connected orientable proper surface representing that primitive class, with $\chi(S) = 0$. Classification of compact orientable surfaces leaves a torus or an annulus. A closed torus has zero boundary class, which contradicts the preceding paragraph. Thus $S$ is an annulus.

Its two boundary circles are disjoint simple curves on $T$. If both were inessential, their homology sum would be zero. If both were essential, they would be parallel on the torus; their oriented classes would sum to zero or twice a primitive class. Neither possibility is nonzero primitive. Exactly one boundary circle is therefore essential and the other bounds a disc $D \subset T$. The essential circle lies outside that disc: an essential curve cannot lie inside a disc and cannot cross its boundary.

The interior of the proper annulus is disjoint from $T$, so $S \cup D$ is an embedded disc with boundary the remaining essential circle. Using a boundary collar, push its boundary cap inward and round the corner along the capped circle. The result is a properly embedded compressing disc for $T$. The knot-exterior disc criterion then proves that the source knot is the unknot. No normal vector for this capped disc is needed to verify the implication.

The two essential hypotheses cannot be replaced by Euler characteristic alone. The audit below includes a connected null-class annulus in a trefoil exterior and a primitive disconnected surface there with total Euler characteristic zero. Both are rejected. Coorientation is also essential: the connectedness inspector proves orientability rather than inferring it from Euler characteristic.

## 114.2 A separate witness tag and unchanged surface inspection

The new schema is `diagram-cocycle-annulus-v1`. It contains the same six fields as the previous cocycle disc schema: source PD, canonical finite triangulation, integral heights, normal coordinates, optional span witness, and schema tag. The new independent wrapper translates only the tag for the existing surface inspector and requires Euler characteristic exactly zero. That inspector still checks the source geometry, all matching and coherence conditions, the primitive class after its own tree gauge, and either a zero spanning tree or the minimum-span certificate. Its code and the span optimality verifier are unchanged.

The public seed verifier dispatches the new schema to this wrapper. An annulus relabeled as the old disc schema remains invalid as a positive disc proof; a disc relabeled as an annulus also fails. Legacy cocycle-disc and component-count certificates retain their previous acceptance rules. In particular, the new witness does not redefine a zero disc count as a positive normal-disc count.

Discovery already performs independent surface inspection at the raw and optimized stages. A verified Euler-zero result now stops that search and emits the annulus tag. Tree and face candidates with Euler zero must also pass full independent source inspection before being accepted; the cheap negative-candidate branch cannot emit such a positive. The final stage record has `unknot_witness=annulus-cap` and continues to report `compressing_discs=0` for its actual stored surface.

The new implication adds only constant-size metadata and a constant number of checks after surface inspection. The standalone wrapper repeats the same independent inspection and adds the Euler-zero test. It introduces no new search, no boundary-component enumeration, and no expansion of binary normal coordinates. All previously accounted-for arithmetic, container and

bit costs of source inspection remain applicable. The capping argument is an existence certificate for a disc; it does not yet provide a normal-coordinate disc extraction or a hierarchy-cut operation.

Annulus recognition is enabled within the optional native stage. The seed routine accepts `annulus=False` to recover the old criterion. The recognizer exposes `normal_seed_annulus=False`; the CLI exposes `-normal-seed-no-annulus`. The shared work allowance and cancellation behavior are retained. The separate face search remains disabled by default, and the ordinary recognition portfolio still leaves the whole native stage optional.

### 114.3 Coverage and independent boundary controls

On 84 source inputs with the existing two-million-guard allowance, the four-tree policy now has 24 native positives, 59 completed restricted misses and one cap, compared with 23 positives before annulus recognition. Requesting 24 tree trials now gives 27 positives, 56 misses and one cap, compared with 24 positives before. The capped source remains Gordian. The optional four-face-root policy retains its 24 positives; its newly found disc is now preempted by the earlier optimized annulus.

Three earlier positives, `random-29`, `random-79` and `random-99`, now stop at their raw annuli. Their shared guard work falls from 11,227, 10,192 and 7,785 to 6,561, 6,480 and 4,984 respectively. The seven-crossing `random-142` stops on its original 122-piece optimized annulus at 93,012 guards. Without annulus recognition and without face search it previously completed a miss at 93,011 guards: the extra unit is the final success guard, not another discovery pass. With four face roots enabled, its old disc proof required 117,848 guards.

With the optional 24-tree allowance, `random-81` now succeeds at trial 5 rather than trial 22, reducing guards from 72,527 to 52,641. Two additional four-crossing unknots, `random-1` and `random-127`, succeed at tree trials 10 and 7 respectively. Their annuli have 86 and 111 pieces, and their guard totals are 44,244 and 48,849. These are additional positives for that explicit extended allowance; they remain misses under the default four tree trials.

Seven distinct annulus certificates are retained. Every prior positive certificate still passes the public verifier. Under each of the three compared policies, twenty positive records retain their exact certificate hash; the intentional changes use the new annulus tag and may stop earlier. Cases that do not return an annulus preserve their complete recorded native result, including their work count. On nineteen ordinary corpus inputs, both settings return the expected verdicts without reaching the native stage, so no ordinary-portfolio gain follows from this experiment.

The final audit independently checks all 787 original or root-extremal minimum-span surfaces from the preceding 74-source small-input experiment. Sixteen have Euler characteristic zero, and precisely those pass the new annulus wrapper. Regina agrees on connectedness, orientability, Euler characteristic and boundary count. The seven actual produced annulus witnesses receive separate Regina comparisons, including raw and later-tree surfaces outside that minimum-span experiment. Two trefoil adversaries bring the total to 796 Regina comparisons.

For the sixteen accepted minimum-span annuli and the seven produced witnesses, a separate small-fixture control expands boundary arc pairings, finds each boundary circle and computes its mod-two intersection vector against an independently validated torus homology basis. All 23 comparisons give two circles, exactly one with zero vector. At most 86 boundary points are expanded. This is audit code only; neither recognition nor certificate verification calls it.

The first trefoil adversary is a genuine connected orientable annulus with 20 normal pieces. Set the local height at a corner to one exactly when its canonical global vertex is 3 or 7, and to zero otherwise. This is a pure vertex coboundary. Its two boundary circles both have zero mod-two homology, and the primitive-class checker rejects it. The second adversary changes the primitive trefoil seed by subtracting one at global vertex 3. Its total Euler characteristic is zero, but Regina and the native component census find two components and no compressing disc. The zero-tree

connectedness check rejects this forged annulus witness.

### 114.4 Tests and complete-call measurements

All 34 focused tests pass in 4.809 seconds, and all 1,204 maintained tests pass in 226.376 seconds. Tests cover producer-free annulus replay, raw and optimized early termination, exact budget boundaries, cancellation, strict switch types, schema swaps, malformed fields, wrong sources, the trefoil adversaries, zero and doubled classes, and 20,000-bit gauge transport through hexadecimal JSON. Historical tree and face tests explicitly disable the new criterion when testing their original discovery paths.

The final audit takes 159.502 seconds and pins 488 source files. Its elapsed time is a correctness-run total, not a paired performance result. A preliminary successful audit is retained as a log; the final run follows an update that makes the historical tree, blocking-flow and face research drivers explicitly retain their original disc-only comparison criterion. The cocycle-surface inspector and span verifier remain byte-identical to baseline `b2d13b5f4`.

Measurements run serially without overlapping tests, audits, builds or commits. They start from the input diagram and include the complete recognizer, exact fallback when needed, independent replay of any returned native proof and certificate hashing. Earlier shortcuts are disabled to reach the native stage. Both arms have no local seed cap and retain the twelve-second global deadline. The on/off comparison changes only the annulus criterion. Tree trials are four except for the three explicitly extended-tree cases; face roots are zero except for the labeled prior face-search positive, which requests four.

The initial five-round run has visible A/A variation, including a 1.089 raw-disc on-arm ratio and a 0.876 ratio for the three-crossing raw annulus. It is retained, and a nine-round repeat supplies the final table. Each run has one warm-up round per case, and each round randomizes off, off A/A, on and on A/A endpoints. All 616 measured calls and 88 warm-ups across the two runs finish. Their per-implementation verdicts, witness hashes and native work counts are stable across both runs. The final run has 396 measured calls and 44 warm-ups and takes 37.153 seconds.

| Input | Off (ms) | On (ms) | Off/on | Off A/A | On A/A |
|---|---|---|---|---|---|
| Raw disc control | 64.707 | 65.940 | 0.992 | 0.958 | 0.990 |
| Optimized disc control | 91.996 | 90.158 | 1.012 | 1.008 | 0.995 |
| Raw annulus, 4 crossings A | 50.989 | 31.007 | 1.637 | 1.001 | 0.995 |
| Raw annulus, 4 crossings B | 49.219 | 31.143 | 1.580 | 0.998 | 0.999 |
| Raw annulus, 3 crossings | 36.943 | 23.334 | 1.571 | 0.997 | 1.008 |
| New optimized annulus | 118.295 | 137.411 | 0.865 | 1.005 | 0.991 |
| Prior face-search positive | 206.528 | 137.443 | 1.483 | 0.998 | 0.988 |
| Earlier late-tree positive | 136.703 | 100.128 | 1.349 | 1.031 | 0.970 |
| New tree annulus A | 89.212 | 85.510 | 1.033 | 1.021 | 1.004 |
| New tree annulus B | 92.456 | 85.370 | 1.075 | 0.993 | 0.998 |
| Trefoil miss | 39.753 | 40.171 | 0.992 | 0.998 | 1.001 |

The three early-annulus cases improve 1.57–1.64× by paired ratios. The previous face-search positive improves 1.48×, and the late-tree positive improves 1.35×. The two new extended-tree positives have ratios 1.033 and 1.075; the first is modest relative to its A/A variation, so the main conclusion there is added native coverage. The unchanged raw-disc, optimized-disc and trefoil controls are near parity. Table times are primary-arm medians and ratios are medians of paired ratios; these need not be quotients of the displayed median times.

The newly recognized seven-crossing source without face search is an adverse complete-call control. Its old exact fallback finishes in a median 118 milliseconds; returning and independently replaying the new native proof takes 137 milliseconds, about 16 percent slower by the paired ratio. There

is no claim of a uniform recognition speedup. The criterion adds no surface discovery loop and shortens several existing native searches, but native proof production and replay can cost more than the old fallback on a small diagram.

The reproduction driver is `../fast/normal_orbit_research/annulus.py`. Use its `audit` or `recognize -rounds 9` subcommand; the audit uses optional Regina. The incoming archives and checked `openai/math` head `fd4aeeb2e` are unchanged. This is a stronger polynomial sufficient certificate test within an incomplete candidate search, not a general subexponential or quasipolynomial recognition algorithm.

# 115 Compressed planar capping beyond nonnegative Euler characteristic

The annulus criterion suggests checking genus rather than Euler characteristic alone. A connected orientable surface with Euler characteristic $-5$ can be planar with seven boundary circles. If precisely one of those circles is essential on the exterior torus, the other six can be capped, giving a compressing disc. The new optional native branch recognizes this configuration using a compressed boundary census and independent replay. It introduces no additional surface candidates.

## 115.1 The elementary capping criterion

Let $M$ be an authenticated knot exterior with boundary torus $T$, and let $S$ be a connected orientable properly embedded surface. Write $b$ for its number of boundary circles. Classification gives

$$2g(S) = 2 - \chi(S) - b.$$

Suppose $\chi(S) + b = 2$ and exactly one boundary circle is essential on $T$. Every other boundary circle bounds a disc on $T$. Choose an innermost inessential circle among those remaining. Its boundary disc has interior disjoint from the remaining boundary curves. Attach this disc to $S$ and push the cap slightly into a boundary collar, rounding the corner. Properness supplies a sufficiently small collar in which this operation avoids the rest of the surface. Repeat for all inessential circles, choosing collar depths to preserve disjointness of the caps. This also handles nested boundary circles; the boundary discs need not initially have pairwise disjoint interiors on $T$.

Each cap preserves connectedness and orientability, increases Euler characteristic by one and removes one boundary component. The resulting surface has one boundary component and Euler characteristic $\chi(S) + (b - 1) = 1$. It is therefore a disc, and its remaining boundary is essential on $T$. The source knot is the unknot. The certificate stores $S$ and the boundary census; it does not claim that the stored normal surface contains a disc component or extract a normal vector for the capped disc.

Primitivity is not an additional requirement of this topological capping lemma. The implementation nevertheless retains its established restricted cocycle inspector: authenticated source geometry, coherent primitive class, and either a zero spanning tree or a checked minimum-span witness. These supply connectedness and orientability. Euler characteristic and boundary counts alone cannot replace those hypotheses. In particular, a disconnected surface can satisfy a total Euler identity without being planar and connected.

## 115.2 Counting essential circles with two compressed weights

The boundary-only normal arc system has one point for each intersection of $\partial S$ with a boundary edge of the triangulation. Its interval pairings follow normal arcs in boundary faces. Orbits of this

system are exactly the boundary circles. This point order differs from the full surface edge-point order; the new geometry helper constructs its offsets from the sorted boundary edges themselves.

Choose two cycles forming a basis of $H_1(T;\mathbb{F}_2)$. On the block for a boundary edge, put a two-dimensional integer weight whose entries indicate whether that edge belongs to each basis cycle. The weight of an orbit counts its intersections with the two cycles. Reduction modulo two gives its homological intersection vector. Since the intersection form of a torus is nondegenerate, this vector vanishes exactly when the curve's mod-two homology vanishes. An essential simple curve on a torus represents a primitive integral class, whose reduction modulo two is nonzero. Consequently a boundary orbit is essential exactly when at least one of its two weight sums is odd.

The maintained weighted interval-orbit engine computes a histogram $(w, m_w)$, where $m_w$ counts circles with integer weight vector $w$. The two required counts are

$$b = \sum_{w} m_w, \qquad e = \sum_{w:\, w \bmod 2 \neq (0,0)} m_w.$$

Connectedness and orientability already come from the cocycle witness, so this query needs neither a full-surface orbit census nor its orientation double. Only boundary orbits carry weights. Zero-weight circles remain present in the histogram. Counts and multiplicities stay binary. There is no loop over boundary points, normal pieces or individual circles in discovery or replay.

For $N$ tetrahedra the boundary relation and initial weight intervals have $O(N)$ descriptions, with weight dimension two. They use the already accounted-for weighted-AHT transport bounds of the earlier component sections, including polynomially many run breakpoints in the explicit input and trace. Geometry, integer bit costs, basis construction and independent binary rank checking remain part of the cost. This is a polynomial encoded query on a supplied surface, not a complete search for a genus-zero surface. The candidate family's incompleteness remains the obstacle to a general recognition bound.

## 115.3 Certificate separation and bounded optional discovery

The new schema is `diagram-cocycle-planar-v1`. It contains the six cocycle fields plus a boundary homology basis and a weighted boundary-orbit certificate. Its independent inspector first replays the cocycle-surface checker with its existing acceptance rules. It then builds the boundary relation from that call's validated geometry, validates the supplied basis by binary homology rather than repeating the producer's tree–cotree construction, and invokes the independently implemented weighted-orbit verifier. Only the replayed histogram supplies $b$ and $e$. Acceptance requires $\chi + b = 2$ and $e = 1$.

The wrapper imports no cocycle, flow, orbit or weight-transport discovery. The public seed verifier dispatches the new tag without changing the acceptance rules of disc, annulus or legacy component certificates. Its summary records genus zero and `unknot_witness=planar-cap`; `compressing_discs` still describes the actual stored surface.

The native search tries this query only after a candidate fails the existing disc and enabled annulus tests. Raw, tree, optimized and optional face candidates retain their previous order. A discovered planar proof must pass the full independent source and boundary inspector before a positive verdict is emitted. Negative query results supply no knottedness claim. The first implementation repeated finite-triangulation preparation inside the boundary producer and twice inside each positive checker. Profiling identified this as the largest cost. The revised cocycle inspector has a private detailed return carrying its freshly reconstructed geometry and coordinate analysis; its public return remains the same summary. The planar checker uses that result within the current call. The producer likewise retains validated source geometry and its chosen boundary basis only for the current native search. It reuses coordinate analysis when that candidate has just been fully inspected, and otherwise checks its new coordinate vector against the retained geometry.

No caller-supplied geometry cache is accepted by the certificate checker. Each independent replay still reconstructs and authenticates its own source, and checks all surface and boundary claims. This removes repeated work without replacing positive replay by trust in discovery. All work uses the existing shared guard allowance; the seed API's `max_cycles` additionally bounds the sum of boundary orbit-discovery cycles over all candidates. Incomplete boundary queries can fall through to later candidates, while global work exhaustion and cancellation still terminate the bounded stage.

The new query is explicitly requested with `planar=True` in the seed API. The recognizer exposes `normal_seed_planar=True`; the CLI flag is `-normal-seed-planar`. Its default is false. This preserves the previous search records exactly when the new query is not requested. Unlike the annulus implication, the planar test adds an orbit computation on continuing candidates; the measurements below evaluate that cost.

### 115.4 Coverage and boundary-census controls

On the same 84 source inputs and two-million-guard allowance, the four-tree policy increases from 24 to 25 native positives; it has 58 completed restricted misses and one cap. With 24 tree trials, coverage increases from 27 to 30 positives, with 53 misses and one cap. Gordian remains the capped source. With the planar option off, all recorded results under both policies are identical to the previous annulus audit, including work, certificates, stages and statistics. Every previous positive is independently replayed. Nineteen ordinary-portfolio inputs return the expected answers under both settings without reaching the native stage.

The five-crossing `optimized-positive` and its repeated source `random-54` now succeed at tree trial 2 on a surface with $\chi = -5$ and seven boundary circles. Guard work falls from 54,318 to 29,111. The four-crossing `random-1` succeeds at tree trial 2 with $\chi = -4$ and six boundary circles. Its 20,856 guards replace a four-tree miss costing 36,784, or a later annulus at trial 10 costing 44,244 under the extended policy.

Three previously missed extended-search sources now succeed: `random-25` and `random-176` at trial 12, and `random-162` at trial 18. Their witnesses are three-boundary planar surfaces with $\chi = -1$. The sources `random-25` and `random-176` share a certificate; their work rises from a 43,177-guard miss to a 61,043-guard success. The source `random-162` rises from 75,603 to 157,897 guards. These additional boundary queries are real work, despite adding coverage.

The seven-crossing `random-150` stops on its raw four-boundary planar surface with $\chi = -2$, but its work rises from 18,281 to 25,049. The old route reached a disc at tree trial 2 more cheaply than the raw planar census and replay. The seven affected source labels therefore yield five distinct planar certificates, rather than seven distinct proofs. Each is independently replayed and checked against Regina and an expanded boundary census.

A separate survey regenerates raw seeds, all distinct candidates from 24 tree trials, and the original span optimum for the 74 sources with at most twelve crossings. All 1,798 supplied surfaces are connected and orientable according to Regina. For every surface, the compressed boundary count agrees with Regina, and the entire parity histogram agrees with independently expanded boundary pairings. At most 230 boundary points are expanded in these small-fixture controls. The compressed query uses at most 118 orbit cycles on this corpus.

Exactly 345 surveyed surfaces are planar and have one essential boundary circle. All come from known unknots and all receive independent source-bound planar-certificate replay. Of these, 44 have negative Euler characteristic, across fourteen source labels. Thus the extension is not just a new tag for discs and annuli. The five witnesses actually selected by recognition receive additional Regina and expanded-boundary checks, for 1,803 comparisons of each kind across the survey and source audit. These totals count comparisons, not necessarily distinct surfaces.

Focused tests cover earlier termination, producer-free replay, strict fields, changed sources, false

bases, altered weight sums and multiplicities, positive-genus and null-class trefoil surfaces, and a primitive disconnected trefoil control. A scaled seven-boundary fixture has $7 \cdot 2^{2000}$ boundary circles and $2^{2000}$ essential circles; the compressed census and independent verifier recover those binary counts without expansion. Another test transports a valid source proof through 20,000-bit tetrahedral height offsets and hexadecimal JSON. Exact work and shared cycle boundaries, cancellation in discovery and replay, strict Boolean options and unchanged default results are checked.

### 115.5 Regression checks and the scheduling decision

All 23 focused tests pass in 3.292 seconds. After the geometry-reuse refactoring, all 1,210 maintained tests pass in 220.299 seconds. The final source audit takes 75.389 seconds and the boundary survey takes 57.621 seconds. These are correctness runs, partly overlapping the full test suite; their elapsed times are not performance comparisons. The final audit, survey and benchmark each pin 493 source files.

The initial source and survey records, before geometry reuse, are retained. All surveyed boundary records and all five selected certificates are identical after refactoring. Verdicts under both source policies are unchanged, and the off-arm records still exactly reproduce the preceding annulus audit. The initial timing implementation is preserved as hash-verified source snapshots, together with the earlier audit driver's fixture map. This distinguishes the tested revisions from the final implementation rather than assigning historical measurements its source hashes.

Both timing runs start from source PDs and include complete recognition, any exact fallback, independent replay of the returned native certificate, and certificate hashing. Earlier recognition shortcuts are disabled to reach the native stage; the seed has no local work cap, and the global deadline remains twelve seconds. Each case has a warm-up round and five measured rounds, randomizing off, off A/A, on and on A/A order. Tree trials are four except for the three new extended positives, which request 24; face discovery remains disabled. No tests, audits, builds or commits overlap either timing run.

All 440 measured calls and 88 warm-ups across the initial and revised implementations complete. Per-implementation verdicts and certificate hashes agree across both runs. Native work is stable within each revision; the intentional reductions after geometry reuse are recorded rather than claimed unchanged. The initial run takes 28.735 seconds and the final run 22.443 seconds. The table reports the final revision only: primary-arm median times and medians of paired ratios, which need not equal the ratio of the displayed medians.

| Input | Off (ms) | On (ms) | Off/on | Off A/A | On A/A |
|---|---|---|---|---|---|
| Raw disc control | 61.430 | 64.131 | 0.970 | 1.000 | 0.999 |
| Raw annulus control | 30.756 | 30.544 | 1.007 | 0.989 | 0.999 |
| Earlier seven-boundary proof | 88.652 | 78.822 | 1.121 | 0.995 | 1.006 |
| New four-tree positive | 59.070 | 62.398 | 0.916 | 0.945 | 0.977 |
| Earlier raw planar proof | 88.663 | 89.244 | 0.999 | 1.000 | 0.997 |
| Optimized annulus control | 138.529 | 174.510 | 0.794 | 0.984 | 1.020 |
| New extended positive A | 66.745 | 105.452 | 0.631 | 0.996 | 1.058 |
| New extended positive B | 114.391 | 217.936 | 0.530 | 1.007 | 0.966 |
| New extended positive C | 66.821 | 98.140 | 0.669 | 0.998 | 1.001 |
| Trefoil miss | 41.634 | 45.828 | 0.912 | 0.997 | 0.992 |
| Genus-one miss | 42.727 | 51.690 | 0.830 | 1.002 | 1.003 |

The earlier seven-boundary witness improves the complete endpoint by a paired factor of 1.121: approximately 89 milliseconds becomes 79. The earlier raw planar witness is near parity at 89 milliseconds. The new four-tree positive is modestly slower, with a 0.916 off/on ratio and

noticeable off-arm A/A variation. The three new extended positives are materially slower: their off/on ratios are 0.631, 0.530 and 0.669. The existing optimized-annulus positive also slows because earlier unsuccessful candidates now receive boundary queries; its ratio is 0.794. The trefoil and genus-one continuing misses have ratios 0.912 and 0.831. Raw disc and raw annulus controls bypass boundary discovery and show no algorithmic speedup.

These data support an optional stronger certificate search, not a new universal default. Geometry reuse removes much of the initial overhead, but the extra queries and larger certificates can still cost more than the preceding exact fallback. The maintained default therefore remains `planar=False`. The whole native stage remains optional in the ordinary portfolio, whose nineteen measured inputs never reach it.

Reproduction uses `../fast/normal_orbit_research/planar.py`. Its modes are `audit`, `survey` and `recognize -rounds 5`. Regina and expanded boundary enumeration appear only in the audit controls. The incoming research inventory and checked `openai/math` head `fd4aeeb2e` are unchanged. The new capping proof uses elementary surface and torus topology and does not require a zero-surgery theorem. No complete genus-zero search, explicit normal capped-disc extraction, hierarchy cutting, or general subexponential or quasipolynomial recognition bound is established here.

# 116 Shared discovery geometry and independent positive checks

The preceding planar experiment removed duplicate geometry preparation within its new checker, but the ordinary native search still prepared the same triangulation separately for seed construction, raw-surface replay, tree construction and optimized-surface replay. Profiling a continuing miss identified this repeated validation as a larger cost than boundary weight transport. The new search retains one prepared geometry for candidate construction and defers full source-certificate replay until a candidate could yield a positive answer. Independent verification of positive certificates is unchanged.

## 116.1 A height-based Euler prefilter

For an internally constructed coherent integral cocycle, let $h_{t,i}$ be its four local heights in tetrahedron $t$. At an optimized candidate add the returned vertex potential to the corresponding corners. For each global edge $e$, put $w_e = |h_{t,j} - h_{t,i}|$ using a local occurrence of that edge. Coherence makes this independent of the occurrence. Let $s_f$ be the height span on a global face, counted once, and let $s_t$ be the span of the four tetrahedral heights. Then

$$V(S) = \sum_e w_e, \qquad E(S) = \sum_f s_f, \qquad F(S) = \sum_t s_t, \qquad \chi(S) = \sum_e w_e - \sum_f s_f + \sum_t s_t.$$

Here $F(S)$ counts normal pieces, rather than faces of the ambient triangulation. Every half-integral level between the lowest and highest height cuts one normal polygon from a tetrahedron. On a triangular face it cuts one normal arc; on an edge it cuts one point. Integral endpoint heights therefore give precisely the three counts above. Interior faces are counted once by their gluing pairs, and boundary faces once directly. Tetrahedral constant shifts preserve these counts. Adding a vertex gauge to the corner heights while subtracting that same gauge from the vertex potential leaves the adjusted heights, and hence the surface, unchanged.

The helper scans $O(N)$ local cells and performs $O(N)$ dictionary operations for $N$ tetrahedra. If heights and potentials have at most $B$ bits, its arithmetic and accumulated totals use $O(B+\log(N+1))$ bits. In an indexed or constant-table-cost model its arithmetic work is $O(N(B+\log(N+1)))$, with $O(N)$ stored integer entries. As with the earlier Python container accounting, a conservative

collision-sensitive bound allows $O(N^2(B+\log(N+1)))$ work; no constant-time hashing assumption is needed for the polynomial claim. No normal piece or binary height interval is expanded.

This helper is a discovery prefilter. It does not validate matching, primitivity, connectedness or source provenance. It reads the heights produced by the maintained exact constructor, or those heights plus the potential from the checked span optimizer. Tree and face candidates retain their existing normal-coordinate analysis. If Euler characteristic can support a disc or enabled annulus witness, the full independent cocycle inspector reconstructs the source and checks the entire coordinate vector before any positive verdict. A malformed vector with apparently promising heights therefore cannot bypass verification.

## 116.2 One prepared geometry per candidate search

The rank-one constructor has a private detailed return containing the prepared geometry it already built. Its public function still returns the same seed dictionary, with the same cocycle statistics and allowance semantics. The public tree generator still validates its supplied triangulation; an internal generator accepts the prepared geometry from the current native search. Its integral coherence checks, deterministic root order, gauges, duplicate suppression and candidate vectors are unchanged. Optional face analysis uses this same per-search geometry.

Raw and optimized misses use the height cell count instead of full source-bound certificate replay. The constructor's primitive class and zero-tree construction, gauge invariance and checked minimum-span optimization establish the candidate family's mathematical invariants. These discovery summaries are not negative knot certificates. Candidate failure continues to mean only that the bounded search found no witness.

The existing positive inspectors receive ordinary certificate data and construct their own authenticated source geometry. They accept no geometry cache from discovery. Disc and annulus candidates receive the full cocycle check; optional planar candidates receive the full cocycle and boundary check. The public verifier and its independent arithmetic modules are byte-identical to the preceding revision. Reusing producer geometry and skipping negative replays does not remove any positive-certificate gate.

The candidate schedule, default trial counts and optional planar/face settings do not change. The shared guard allowance covers the new prefilter and retained checks. Guard totals can fall on continuing searches and rise slightly on an immediate positive that now receives the additional cheap prefilter. Exact allowance boundaries are therefore revalidated rather than inferred from old work counts. Global cancellation continues through construction, prefilter scans, discovery and independent positive replay.

## 116.3 Source equivalence and independent cell-count controls

The source audit compares 84 diagrams under four policies: four or 24 tree trials, each with planar discovery disabled or enabled. The old search body is loaded from Git revision `a6947e39a`; its source hash is recorded. It uses the preserved public constructor APIs. Every old record exactly reproduces the preceding planar audit, including its work count. For every completed old case, the new record is identical after removing only the total `work` field. This includes verdicts, certificate hashes, all stages, boundary census records and optimizer statistics. All positive certificates in both arms are independently replayed.

Coverage remains 24 and 27 native positives for the four-tree and extended policies without planar discovery, and 25 and 30 with it. Each policy has the same single capped input, Gordian, at 2,000,001 guards. The comparison does not turn that cap into a completed miss. The nineteen ordinary portfolio sources retain their expected verdicts and do not reach the native stage under either implementation.

Representative work changes illustrate why guard totals should not be confused with timing. The optimized disc fixture falls from 54,318 to 52,090 guards. Its earlier optional planar witness falls from 29,111 to 27,303. The seven-crossing optimized annulus falls from 93,012 to 89,862. The extended-tree `random-81` annulus falls from 52,641 to 47,761, and the trefoil four-tree miss falls from 22,041 to 19,099. In contrast, the immediate `circle-9` disc rises from 13,305 to 13,817 guards because of its new prefilter. That control tests the cost paid when there is no repeated negative replay to remove.

The separate 74-source survey regenerates all 1,798 raw, tree and original minimum-span surfaces from the preceding experiment. For every candidate, the height-based Euler count equals both the independently reconstructed normal-coordinate Euler value and Regina's Euler characteristic. The normal-piece total also agrees with coordinate analysis. Regina reconfirms connectedness, orientability and the preceding boundary counts. Candidate ordering and the number of distinct tree gauges match the earlier survey.

New tests compare the formula under 20,000-bit tetrahedral shifts and 19,000-bit compensated vertex gauges. They check that a continuing trefoil miss performs no positive replay, that every possible disc or planar success requires its independent gate, and that a corrupted producer vector cannot turn a promising height count into a positive answer. Another test corrupts a returned source triangulation and verifies that fresh replay rejects it while the original proof and later independent searches remain valid. Existing exact allowance, cancellation, source, primitivity, schema and binary-coordinate tests remain active.

## 116.4 Tests and complete-call measurements

All 38 focused tests pass in 4.469 seconds, and all 1,214 maintained tests pass in 235.344 seconds. The final source audit takes 135.085 seconds and the cell-count survey takes 20.777 seconds. Those correctness runs partly overlap the test suite; their elapsed times are not paired performance measurements. The audit, survey and both timing runs pin 496 source files. The six existing seed, cocycle, annulus, planar, span and weighted-orbit verifier modules remain byte-identical to the baseline revision.

Timing endpoints start from input diagrams and include complete recognition, any exact fallback, independent replay of returned native certificates and certificate hashing. Earlier shortcuts are disabled to reach the native stage. Both implementations have no local seed cap and retain the twelve-second global recognition deadline. The comparison selects the published or current native search body while retaining the same public construction and verification APIs. No tests, audits, builds or commits overlap the timed runs.

The first five-round experiment shows noticeable variation in the raw-disc old-arm A/A ratio, 1.115, and one later-tree control, 1.074. It is retained, and a nine-round repeat supplies the final table. Each experiment includes one warm-up round per case and randomizes old, old A/A, new and new A/A calls within each round. All 616 measured calls and 88 warm-ups complete. Per-arm verdicts, certificate hashes and native work counts are stable across both runs. The initial run takes 24.056 seconds; the final run, with 396 measured calls and 44 warm-ups, takes 49.367 seconds.

The two later-tree annuli and the later planar cap request 24 tree trials; other cases request four. Only the two labeled planar cases enable planar queries. The face-disc case disables annulus recognition and requests four face roots, so it exercises the actual face-search path. All other cases retain the existing annulus criterion and zero face roots. The table shows primary-arm median times and median paired ratios; these ratios need not equal quotients of the displayed median times.

| Input | Old (ms) | New (ms) | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| Immediate disc | 62.879 | 64.832 | 0.983 | 1.000 | 0.988 |
| Immediate annulus | 31.083 | 31.531 | 0.980 | 1.000 | 1.004 |
| Optimized disc | 90.827 | 70.859 | 1.281 | 1.002 | 1.002 |
| Optimized annulus | 136.156 | 108.234 | 1.262 | 1.001 | 1.006 |
| Later tree annulus A | 105.399 | 67.743 | 1.509 | 0.998 | 0.999 |
| Later tree annulus B | 88.082 | 60.483 | 1.466 | 1.012 | 1.000 |
| Early planar cap | 91.193 | 73.902 | 1.283 | 1.011 | 1.035 |
| Later planar cap | 426.389 | 362.727 | 1.179 | 1.021 | 1.020 |
| Optional face disc | 222.422 | 152.559 | 1.400 | 0.990 | 1.002 |
| Trefoil miss | 42.453 | 20.975 | 2.002 | 0.996 | 0.998 |
| Genus-one miss | 43.638 | 23.442 | 1.887 | 1.010 | 1.005 |

The optimized disc and annulus improve by paired factors 1.281 and 1.262. The two later-tree annuli improve 1.509 and 1.466. Optional planar cases improve 1.283 and 1.179, and the optional face-disc endpoint improves 1.400. The trefoil and genus-one continuing misses improve 2.002 and 1.887, including the same exact fallback used before. These are gains on complete calls through the selected native paths, rather than just on a standalone geometry helper.

The immediate disc and annulus are adverse controls: their paired ratios are 0.983 and 0.980, approximately two percent slower. They pay for the prefilter before their existing independent replay and avoid none of the old negative replay work. There is no claim of a uniform speedup. Absolute late-planar times also vary substantially between runs, while its paired factor remains about 1.18. Comparing raw elapsed levels across experiments would overstate what those measurements establish.

The default candidate schedule is retained because discovery geometry reuse improves continuing searches while preserving every positive gate; no additional search is introduced. The planar and face queries remain opt-in, and the whole native stage remains optional in the ordinary portfolio. Its nineteen control sources still finish before that stage, so this experiment supplies no new ordinary-portfolio speedup claim.

Reproduce using `../fast/normal_orbit_research/reuse.py` in `audit`, `survey` or `recognize -rounds 9` mode. The source audit verifies that the loaded baseline search reproduces the previous records before comparing the new implementation. The checked incoming inventory and `openai/math` head `fd4aeeb2e` are unchanged. This removes redundant polynomial work and supplies a linear cell-count prefilter; it does not complete the surface candidate search, normal capped-disc extraction or hierarchy operations, and establishes no general subexponential or quasipolynomial recognition bound.

## 117 Exact Euler optimization on a minimum-span face

The bounded root searches of Section 113 can find surfaces with different topology at the same minimum piece count, but do not optimize topology over the whole face. The height cell formula in Section 116 leads to a stronger algorithm: when each global edge has at least two face incidences, maximizing Euler characteristic on a certified minimum-span face is an exact, integral, polynomial-size network optimization. The implementation returns the surface in the existing span certificate format and a separate arithmetic primal–dual certificate of its Euler optimum. This solves a restricted optimization problem; the minimum-span face need not contain a disc or annulus even when the knot is trivial.

## 117.1 From cell counts to a nonnegative edge objective

Orient each global edge $e = (a, b)$ and write its coherent cocycle value after an integral vertex potential as $x_e(p) = c_e + p_b - p_a$. Let $d_e$ count its incidences with global triangular faces, with multiplicity: an interior face is included once, a boundary face once, and each of its three local edges contributes an incidence even if some are globally identified. For three real numbers, the span is half the sum of their three absolute pairwise differences. Therefore the normal cell counts give

$$2\chi(S_p) = 2D(p) - \sum_e (d_e - 2)|x_e(p)|, \qquad D(p) = \sum_t \operatorname{span}(h_t + p).$$

This identity also explains why minimizing the piece count alone does not maximize Euler characteristic: the edge contribution can vary among tied minima. It follows from counting cells of the actual coherent level surface, not from dropping normal matching or quadrilateral conditions.

Fix a previously certified minimum $D_*$. On its entire optimal face, maximizing $\chi$ is equivalent to minimizing

$$F(p) = \sum_e w_e |c_e + p_b - p_a|, \qquad w_e = d_e - 2.$$

The nonnegativity hypothesis $d_e \geq 2$ matters. It is checked for every edge; unsupported negative weights cause rejection, rather than silently changing the objective. No universal lower-degree claim about arbitrary finite triangulations is used. The sum $W = \sum_e w_e$ is at most $12N$ for $N$ tetrahedra, because there are at most $4N$ global faces and three incidences per face. Thus weights here have total magnitude linear in the triangulation size, independently of the cocycle heights.

The span matching supplies the exact face constraints. If its selected low and high corners in tetrahedron $t$ are $\ell_t, u_t$, impose, for every corner $i$,

$$p_{v_{\ell_t}} - p_{v_i} \leq h_{t,i} - h_{t,\ell_t}, \qquad p_{v_i} - p_{v_{u_t}} \leq h_{t,u_t} - h_{t,i}.$$

These are the same $8N$ difference inequalities used for bounded root extrema. Writing $z_{t,i} = h_{t,i} + p_{v_i}$, the matching dual gives

$$D(p) - D_* = \sum_t [(\max_i z_{t,i} - z_{t,u_t}) + (z_{t,\ell_t} - \min_i z_{t,i})].$$

The two extremal gaps per tetrahedron are nonnegative and vanish exactly when all its displayed inequalities hold. Consequently these inequalities describe all real and integral potentials attaining $D_*$, including ties and changes in local normal-disc types. There is no additional restriction to the root extrema already sampled by the earlier algorithm.

## 117.2 An independent lower bound and an exact integral solver

For each objective edge choose a signed flow $f_e \in [-w_e, w_e]$. For each constraint $p_v - p_u \leq b_{uv}$ choose a nonnegative flow $\lambda_{uv}$ on $u \to v$. Require the combined flow to balance at every vertex. Then

$$\begin{aligned} F(p) &\geq \sum_e f_e(c_e + p_b - p_a) \\ &= \sum_e f_e c_e - \sum_{uv} \lambda_{uv}(p_v - p_u) \\ &\geq \sum_e f_e c_e - \sum_{uv} \lambda_{uv} b_{uv}. \end{aligned}$$

The checker validates the primal difference inequalities, both kinds of flow bounds, vertex balance and equality of this lower bound with $F(p)$. Equality certifies a global optimum over real potentials, hence also over integer potentials. This proof needs no trust in the flow solver or in a floating-point

optimization result. The arithmetic checker makes no claim about triangulation provenance, connectedness or knottedness.

The solver constructs an integral equality witness using a minimum-cost circulation dual. An objective edge has cost $-c_e$, signed capacity interval $[-w_e, w_e]$, and stored flow $g_e = f_e + w_e \in [0, 2w_e]$. A constraint arc has cost $b_{uv}$ and capacity $W + 1$. Start from the supplied feasible span potential $p^0$. Set $f_e = w_e$ where $x_e(p^0) > 0$, $f_e = -w_e$ where it is negative, and $f_e = 0$ where it vanishes; start all constraint flows at zero. Every usable residual arc then has nonnegative reduced cost with vertex potential $p^0$.

This initial flow need not balance. Its total positive imbalance is at most $W$. A super-source supplies vertices with excess incoming flow and a super-sink receives the deficits. Their initial potentials are respectively the maximum and minimum $p^0$ on the relevant vertices, preserving nonnegative residual costs. Reduced-cost Dijkstra searches and capped potential updates route the imbalance. Each augmentation sends the full integral bottleneck, at least one unit. Feasibility is guaranteed by the balanced signed zero flow: undoing the initial signed edge flow alone gives a feasible repair within the original capacity intervals.

At most $W$ units are sent in total, so no constraint arc of capacity $W + 1$ ever saturates. Its surviving forward residual inequality is exactly the required primal difference constraint for the final potential. For $w_e > 0$, the residual forward and reverse inequalities imply the appropriate sign of $x_e$: an interior signed flow forces $x_e = 0$, while an endpoint flow can have the corresponding nonzero sign. A positive constraint flow also leaves its reverse residual arc and forces the constraint to be tight. These complementary equalities produce equality in the checker's lower bound. All capacities, distances and potentials remain integral; no separate rounding step or integrality assumption is needed.

## 117.3 Cost, binary encoding and verification boundaries

There are $O(N)$ nodes and arcs and at most $W \leq 12N$ shortest-path augmentations. The indexed graph computation uses $O(N^2 \log(N+1))$ arithmetic and comparison operations and $O(N)$ integer storage entries. Neither the number of height levels nor the number of normal pieces is expanded. The private generic weighted-difference solver has an augmentation bound in the *magnitude* of its weight sum; the polynomial assertion here uses the proved linear bound on the geometric weights, not a claim of strong polynomiality for arbitrary binary capacities.

A conservative bit bound is sufficient. Let $B$ bound the input cost and initial-potential bit lengths, $m = O(N)$ be the number of network nodes, $C$ bound original absolute arc costs, and $P$ the largest current absolute potential. A shortest simple residual path has reduced length at most $(m-1)(C+2P)$. The capped update therefore gives $P' \leq P + (m-1)(C+2P)$. Iterating at most $12N$ times bounds all arithmetic lengths by $B + O(N \log(N+1))$ bits. Flows have $O(\log(N+1))$ bits. Together with polynomial-time integer arithmetic, geometry validation and conservative Python-container accounting, this establishes a bit-polynomial bound for the complete restricted stage. The tighter indexed operation count is not presented as a literal Python bit-operation count.

The public optimizer validates the triangulation, signed global coherence and supplied span certificate before constructing this model. It checks its new primal–dual equality, replays the returned span certificate, and compares the optimum edge objective with an independent normal-coordinate Euler count. The surface still uses `normal-cocycle-span-v1`. An unknot claim must pass the existing independent source-bound disc or annulus verifier; those verifiers do not import the new optimizer. A negative Euler optimum is a certificate about this face alone, not a knot certificate. Even maximizing Euler here does not maximize the number or essentiality of boundary circles needed by the separate planar-capping search.

## 117.4 Source controls and a certified limitation

The audit uses the preceding 84-source corpus plus the three-crossing unknot control, with a shared allowance of two million guards per source covering exterior construction, cocycle extraction, span optimization, Euler optimization and independent replays. It completes 83 of 85 cases. Gordian exhausts the allowance during span optimization; `circle-33` exhausts it during Euler optimization. These are incomplete arithmetic experiments, not negative knot verdicts. All completed geometric models have strictly positive edge weights; the smallest observed face incidence count is three.

Ten labels improve their initial Euler value: survivor cases 01 and 03, mirror case 03, Kinoshita–Terasaka, and random cases 0, 19, 30, 34, 142 and 158. Most gain one Euler unit; random case 158 improves from $-9$ to $-7$. Random case 142 improves from an annulus to a disc, preserving its certified 122-piece minimum. On all 74 sources covered by the preceding face survey, the exact optimum agrees with the best Euler value among the 787 previously recorded root candidates, including each original optimum. Agreement on this corpus is an observation; the new global dual is what proves optimality over the whole face.

The 18 nonnegative optima produce 13 distinct source-bound positive certificates, all independently replayed. No additional source becomes recognized relative to the current four-tree native policy with annulus capping. In fact, five already recognized sources—random cases 29, 79, 99, 124 and 150—have a negative optimum on their minimum-span face. Their existing earlier witnesses lie outside that face. The exact optimizer is therefore not a replacement for the broader existing candidate schedule. Regina independently confirms connectedness, orientability, Euler characteristic and the compressing-disc predicate on the 75 sources with at most twelve crossings, including the extra unknot control. The audit retains all 73 distinct indexed arithmetic models and duals for direct replay without running the new optimizer.

For the closure of the two-strand braid $[1, 1, -1]$, cancellation leaves the one-crossing unknot. Nevertheless, the span minimum is 43 normal pieces and the new dual proves that every surface in that face has Euler characteristic at most $-1$. The trefoil control has the same Euler upper bound, at a 45-piece minimum. This gives a concrete obstruction to treating exhaustive minimum-span-face optimization as complete disc or annulus discovery. It also separates this result from a minimum-genus or Thurston-norm algorithm: the fixed piece-count optimum is an additional restriction, and the separate planar-capping condition is not optimized by maximizing Euler characteristic.

The arithmetic tests compare 240 small random constrained problems against exhaustive integer grids, including loops, parallel edges, zero weights and repeated constraints. Exact primal–dual equality additionally certifies the real optimum in each case. Scaling costs and bounds by $2^{20000}$ preserves all flow-operation statistics while scaling the objective, and the resulting certificate survives binary JSON transport. Further checks cover independent replay with the producer disabled, malformed duals, incoherent heights, negative geometric weights, hexadecimal certificate fields, cancellation, source-bound positive verification and exact work-allowance boundaries.

## 117.5 Validation, timings and integration scope

All 14 focused tests pass in 1.753 seconds; the full maintained suite passes all 1,219 tests in 254.202 seconds. The final source audit takes 19.307 seconds. The audit and full suite partly overlap, so those elapsed times are correctness-run observations rather than performance comparisons. The final audit and both timing runs pin the same 501 source files. Existing source-bound disc, annulus, planar, span and weighted-orbit verifiers remain byte-identical to the baseline revision `3c0817c50`.

The performance endpoint starts from a diagram and includes finite exterior construction, primitive cocycle extraction, minimum-span optimization, the selected search and independent source-bound

replay of its selected surface. The old arm searches all distinct root extrema, using the number of global vertices as its root allowance, and retains the highest Euler value found. The new arm solves the whole face and checks its arithmetic dual. This is a complete *candidate-construction* pipeline, not complete knot recognition. Earlier native successes are deliberately not allowed to skip the experiment, and no exact knot-recognition fallback is timed here.

The first five-round run contains 120 measured calls and 24 warm-ups and takes 24.281 seconds. Variation in the original-disc and Kinoshita–Terasaka cases motivates the retained nine-round repeat: 216 measured calls and 24 warm-ups in 40.195 seconds. Each round randomizes roots, roots A/A, exact and exact A/A calls. No tests, audits, builds or commits overlap either timing run. Every measured call completes, and all per-arm Euler values, piece counts, operation counts and certificate hashes are stable within and between both runs. Primary-arm median times and median paired ratios from the final run follow; paired ratios need not equal quotients of the displayed medians.

| Input | Roots (ms) | Exact (ms) | Roots/exact | Roots A/A | Exact A/A |
|---|---|---|---|---|---|
| Trefoil | 38.209 | 37.957 | 1.021 | 0.940 | 0.976 |
| Original disc | 48.704 | 71.274 | 0.690 | 1.004 | 1.003 |
| Tied disc | 115.241 | 115.147 | 0.989 | 0.986 | 0.990 |
| Random 19 | 228.711 | 218.924 | 1.063 | 1.001 | 1.002 |
| Random 158 | 333.819 | 259.215 | 1.283 | 0.996 | 0.994 |
| Kinoshita–Terasaka | 202.822 | 227.087 | 0.825 | 1.027 | 0.995 |

Random case 158 improves by paired factors 1.272 and 1.283 in the two runs; random case 19 improves by 1.054 and 1.062. The original-disc case is slower, with ratios 0.677 and 0.690, and Kinoshita–Terasaka is slower with ratios 0.835 and 0.825. Those final ratios correspond to about 45 and 21 percent additional time. The tied-disc case is near parity. The trefoil's small apparent improvement is not evidence of a reliable gain: its final roots A/A ratio is 0.940. Absolute Kinoshita–Terasaka times also vary substantially between runs. The data support mixed costs, not a uniform speedup.

The new callable `maximize_cocycle_face_euler` API is consequently available for explicit candidate construction and research; it is not added to the default native schedule. All existing candidate and portfolio defaults remain unchanged. Its theoretical gain is an exact optimum over a face described by $8N$ inequalities, together with concrete certified limitations of that face. It supplies neither all normal-surface candidates nor a complete planar-capping search, a hierarchy algorithm, or a general subexponential or quasipolynomial recognition bound.

Reproduce with `normal_orbit_research.euler` in `audit` mode or `benchmark -rounds 9` mode. The indexed models and arithmetic duals are retained in the audit JSON for direct independent replay. The incoming unknot archive inventory and the checked `openai/math` head `fd4aeeb2e` remain unchanged. The new reduction is derived from the maintained height-cell identity and matching-face description above.

## 118 Contracting forced differences before Euler optimization

The exact Euler optimizer of Section 117 still routes flow through vertices whose relative potentials cannot change anywhere on the minimum-span face. The new implementation identifies these forced differences, solves a smaller problem and reconstructs a full certificate on the original indexed model. The independent arithmetic checker and all source-bound surface verifiers remain unchanged. If no vertices can be contracted, an adaptive branch retains the original full-network solver.

## 118.1 Which tight constraints force equality

Start with the verified feasible potential $p^0$. For a difference constraint $p_b - p_a \le d_{ab}$, its slack is $r_{ab} = d_{ab} + p_a^0 - p_b^0 \ge 0$. Put $q_v = p_v - p_v^0$, so the constraint becomes $q_b - q_a \le r_{ab}$. On a directed zero-slack path, the shifted potential cannot increase. If two vertices are mutually reachable using such paths, their shifted potentials must be equal for *every* feasible solution. Thus each strongly connected component of the zero-slack graph is one quotient variable. A one-way tight edge is insufficient: $q_b \le q_a$ still allows strict inequality. The implementation uses iterative Kosaraju traversals, avoiding recursion depth proportional to the graph size.

For a component map $C$, write $p_v = p_v^0 + q_{C(v)}$. An objective term becomes

$$w_e \left| c_e + p_b^0 - p_a^0 + q_{C(b)} - q_{C(a)} \right| .$$

Internal terms are constants and need no flow optimization. Between two components, terms with the same oriented endpoints and shifted offset are combined by adding their weights; reversing an orientation also negates the offset. Among constraints with the same ordered component pair, only the smallest shifted bound is needed. Internal constraints are redundant because their shifted left side is zero and their bounds are nonnegative.

These operations preserve the complete feasible set after the forced substitution and preserve the objective up to its explicitly retained constant terms. They do not choose one representative surface topology or restrict the face to sampled extrema. The zero quotient potential is already feasible, so the existing exact flow solver applies directly to the reduced network. The final potential is expanded and normalized only by an additive constant. Its objective is recomputed on the original edge list.

## 118.2 Lifting the dual without trusting the quotient

The returned certificate still has one flow per original objective edge and per original constraint. An internal objective edge receives the sign-saturating flow $w_e \operatorname{sign}(c_e + p_b^0 - p_a^0)$. A zero-weight edge receives zero. For an aggregated intercomponent term, split its signed flow $F \in [-W_g, W_g]$ across the original intervals $[-w_e, w_e]$: begin at all lower endpoints and distribute $F + W_g$ units in bulk, at most $2w_e$ to each member. Restore each member's original orientation. This uses one pass over the members and never iterates over flow units. Because all members have the same shifted affine expression, the split preserves complementary equality. Each quotient constraint flow is assigned to one original constraint attaining its minimum bound; the discarded parallel constraints receive zero.

The lifted flow now balances on each component as a whole, but may not balance at its individual vertices. Two directed trees inside the zero-slack component repair this. A tree directed toward the root comes from the reverse traversal; a forward traversal supplies a tree directed away from it. In reverse order on the inward tree, send each positive vertex imbalance toward the root, allowing intervening negative imbalances to absorb flow. Every nonroot vertex then has nonpositive imbalance. In reverse order on the outward tree, send the remaining subtree demand from parent to child. Component balance ensures that the root also ends balanced.

All added constraint flows are nonnegative and use original zero-slack arcs. Their constraints stay tight after the quotient potential is expanded, so they preserve complementary equality. Internal constant objective terms, aggregated terms and retained constraint flows therefore give a feasible dual on the original model with the same value as its expanded primal potential. The unchanged checker independently verifies every original flow bound, difference inequality, vertex balance equation and objective equality. It receives neither a component partition nor a compressed certificate whose interpretation would depend on the producer.

### 118.3 Operation bounds and an adaptive no-contraction branch

Let $n, e, m$ be the original counts of vertices, objective terms and constraints; let $k, e', m'$ be their quotient counts, and let $W'$ be the remaining objective weight sum. Construction, expansion and tree routing require $O(n + e + m)$ indexed graph operations, apart from dictionary operations used to combine rows. The reduced shortest-path phase costs

$$O(W'(k + e' + m') \log(k + 1))$$

arithmetic and comparison operations. For the geometric model, $n + e + m = O(N)$ and $W' \leq W \leq 12N$, retaining the previous worst-case polynomial bound. There is no universal asymptotic improvement when the quotient is as large as the original graph. If $k = n$, the implementation bypasses aggregation and lifting and calls the original solver; the additional cost is the linear zero-slack graph analysis. Python hash-collision accounting and binary integer costs remain subject to the conservative polynomial qualifications of the preceding section.

Lifting is independent of the numerical magnitude of its flows. The reduced solver sends at most $W'$ units along simple paths with $O(k)$ arcs, so the sum of its final nonnegative constraint flows is $O(kW')$. Together with the bounded objective flows, the initial positive imbalance to repair is at most $O(W + kW')$. Each tree edge carries at most this amount per traversal. Thus all lifted flows have polynomial magnitude in the geometric input size and $O(\log(N + 1))$ bits. Their sum need not be linear in $N$. The solver's auxiliary capacity $W + 1$ is not a constraint on the original dual: lifted constraint flows need only be nonnegative. Heights and shifted costs remain binary encoded; no level, sheet or flow unit is expanded. Independent arithmetic replay remains linear in the original model size in this model; source and geometry checks are retained.

A concrete family separates the costs. Take $n$ vertices in a directed cycle of zero-slack constraints and one unit-weight objective edge from vertex zero to every other vertex, each with shifted offset one. The old initialization creates $n - 1$ separate unit supplies, so the full solver performs $n - 1$ augmentations. Each shortest-path call scans the source's $n - 1$ arc entries, giving a quadratic scan lower bound for that implementation. The new quotient has one vertex, no variable objective terms and no augmentations; two tree passes lift its dual in linear graph work. The retained operation-count experiment measures this family directly, rather than inferring an asymptotic improvement from wall-clock ratios alone.

### 118.4 Preserved optima and a newly completed large source

The comparison loads the full flow implementation from revision `18e76fe87`. Its complete source hash is retained. Before comparing the new implementation, it exactly reproduces every source record, positive proof, arithmetic dual and Regina result in the preceding audit. All 83 previously completed sources preserve their Euler optimum, minimum piece count and independently checked connected-surface summary under contraction. Their selected potentials need not be the same: 58 source-certificate hashes change. Every resulting source surface and arithmetic dual is checked again.

The new run completes 84 of 85 sources under the same shared two-million-guard allowance. Gordian remains capped during the unchanged span solver. Circle 33, previously capped in Euler optimization, now finishes the entire audited pipeline at 1,197,679 guards; its Euler stage takes 90,106 guards. Its face contracts from 130 variables to 64. This is a newly completed exact-face experiment, not a newly recognized knot: the default native search already has an earlier witness for this source.

All completed sources have a nontrivial contraction. Selected reductions are 14 to 4 variables for the trefoil, 22 to 2 for the original-disc fixture, 30 to 5 for the tied-disc fixture, 46 to 8 for random case 19, 50 to 13 for random case 158, and 46 to 5 for Kinoshita–Terasaka. The corresponding

Euler stage guard counts fall from 23,470 to 5,909; 58,288 to 9,579; 108,324 to 13,618; 292,201 to 21,604; 428,401 to 25,311; and 322,516 to 21,253. Guard counts measure the deterministic allowance, not elapsed time or bit operations.

The audit retains 74 distinct full arithmetic models and lifted duals and 14 distinct positive source proofs, covering 19 positive source labels. Regina reconfirms all 75 small-source controls; their observed boundary counts also agree with the previous optima. Neither that agreement nor equality of Euler characteristic is used to assert equality of all possible surface topologies. The certified negative optimum for the three-crossing unknot remains $-1$. The restriction to the minimum-span face and the incomplete nature of general surface discovery are unchanged.

The forced-cycle family gives the following measured arithmetic guard and arc scan counts. Every quotient has one vertex and no variable objective terms.

| $n$ | Full guards | New guards | Full scans | New augmentations |
|---|---|---|---|---|
| 8 | 503 | 150 | 231 | 0 |
| 16 | 1,851 | 302 | 975 | 0 |
| 32 | 7,043 | 606 | 3,999 | 0 |
| 64 | 27,411 | 1,214 | 16,191 | 0 |
| 128 | 108,083 | 2,430 | 65,151 | 0 |

The new flow stage uses $19n - 2$ measured guards at these sizes; the preceding argument establishes linear indexed graph work without depending on an empirical fit. Additional tests cover 240 random forced-group problems against the full solver, reversed and parallel objective terms, tight one-way constraints, malformed lifted duals, cancellation and the adaptive singleton bypass. A 1,500-vertex cycle with 20,000-bit vertex offsets checks iterative traversal, large cancellation in the dual, binary JSON replay and a linear guard bound. The earlier 240 exhaustive-grid arithmetic controls remain active.

### 118.5 Complete candidate-pipeline measurements

All 18 focused tests pass in 1.905 seconds, and all 1,223 maintained tests pass in 223.627 seconds. The source comparison and cycle-family audit takes 35.539 seconds. The correctness audit and full suite partly overlap; these elapsed values are not paired performance measurements. Both the audit and benchmark pin 505 source files. The arithmetic dual checker, public Euler model construction, native recognizer and existing independent positive verifiers remain byte-identical to the preceding revision.

The isolated timing experiment compares the old and new flow functions within the same complete candidate pipeline: source diagram, finite exterior, primitive cocycle, span optimization, Euler optimization, independent source-bound surface replay and certificate hash. Both arms include the same function-selection wrapper. The baseline module is loaded before timing. There is no local work cap and no early native witness short-circuit; these are candidate-pipeline measurements rather than full knot recognition.

Five rounds randomize old, old A/A, new and new A/A calls for eight sources, following one warm-up round each. All 160 measured calls complete, with 32 warm-ups; the run takes 45.746 seconds. No tests, audits, builds or commits overlap it. Per-arm Euler values, piece counts, operation counts and hashes are stable across rounds. The primary-arm medians and median paired ratios are as follows; the ratios need not equal quotients of the displayed medians.

| Input | Full (ms) | Quotient (ms) | Full/new | Full A/A | New A/A |
|---|---|---|---|---|---|
| Trefoil | 35.807 | 31.535 | 1.128 | 1.001 | 1.006 |
| Original disc | 68.724 | 57.351 | 1.196 | 0.996 | 1.001 |
| Tied disc | 107.483 | 85.417 | 1.253 | 0.999 | 0.984 |
| Random 19 | 204.390 | 135.757 | 1.506 | 1.014 | 0.993 |
| Random 158 | 245.875 | 151.331 | 1.625 | 1.013 | 0.996 |
| Kinoshita–Terasaka | 207.558 | 134.843 | 1.539 | 1.017 | 1.011 |
| Circle 17 | 352.748 | 243.128 | 1.442 | 1.005 | 0.978 |
| Circle 33 | 1072.704 | 643.779 | 1.670 | 0.989 | 1.017 |

All eight selected pipelines improve, by paired factors from 1.128 to 1.670. The A/A medians range from 0.978 to 1.017. The smaller complete-pipeline gains, compared with the reduction in Euler guard work, include unchanged source, cohomology, span and verification costs. This evidence does not establish a uniform speedup on arbitrary indexed models: when no forced equality exists, the new preprocessing is additional linear work before the full solver.

Contraction is now used inside the explicit Euler optimizer, with the singleton fallback described above. The default native recognition schedule remains unchanged, and there is no claim of a default-portfolio speedup. The update improves exact optimization within the existing restricted family and gives an asymptotic improvement on the forced-cycle family; it does not supply a general subexponential or quasipolynomial knot-recognition bound. Reproduce the audit and timings with `normal_orbit_research.quotient`, in `audit` or `benchmark -rounds 5` mode. The checked incoming unknot inventory and `openai/math` head `fd4aeeb2e` remain unchanged.

# 119 Certified boundary shellings before surface discovery

Optimizing a cocycle on one fixed triangulation does not search all useful normal surfaces of the knot exterior. The new optional branch changes that triangulation first: it removes a sequence of embedded boundary tetrahedra, then runs the existing raw, tree-gauge and minimum-span candidate schedule. Every removal preserves the exterior, and every positive answer carries both the removal trace and its connected surface witness. This changes the family of surfaces reached by a bounded search. It does not establish completeness of that family or a general subexponential recognition bound.

## 119.1 A deliberately restricted elementary move

Let $M$ be the verified finite triangulated knot exterior, and let $T$ be a tetrahedron whose four local vertices belong to four distinct global vertex classes. This makes the closed tetrahedron embedded: identification of two of its local edges or faces would identify their endpoint sets, and no such sets coincide. Require that $T$ have $b \in \{1, 2, 3\}$ boundary facets and that its entire intersection with $\partial M$ be exactly their union. In particular, lower-dimensional boundary intersections must also be checked. With one boundary facet opposite vertex $f$, vertex $f$ must be interior. With two boundary facets opposite vertices $f_0, f_1$, the edge $f_0 f_1$ must be interior. With three boundary facets, all vertices and edges already lie in their union, so there is no additional lower-dimensional condition.

The union of the boundary facets is an embedded disc in $\partial T$; the complementary union is another embedded disc. Removing the open tetrahedron and the boundary faces that have no remaining incidence replaces one of these discs by the other. A collar of the attachment disc identifies this operation with removing a boundary ball. It therefore preserves the PL homeomorphism type of $M$. This is the embedded special case of the standard boundary shelling move. Regina's documented move also checks lower-dimensional boundary membership and face identifications; its admissible family is broader than our four-distinct-vertices restriction [51].

Counting boundary facets alone is unsafe. After the first legal shelling of the one-crossing-circle exterior, tetrahedron 3 of the densely renumbered remainder has four distinct global vertices and just one boundary facet, opposite local vertex zero. That vertex nevertheless lies on another boundary face of the exterior. Deleting this tetrahedron fails our replay check. The retained regression constructs the induced face pairing after this forbidden removal and checks that the verifier rejects the trace, rather than relying on a producer's eligibility flag. Generalized tetrahedra with repeated global vertices are also excluded, even when a more general shelling test might permit them.

## 119.2 Stable indices and bounded incidence updates

The trace names tetrahedra by their original indices. Removed rows become tombstones; surviving neighbors have only the appropriate reciprocal face record changed to a boundary face. Dense reindexing is performed once at the end, preserving the original order of survivors. A trace has fewer than $N$ entries for an initial $N$-tetrahedron torus-boundary manifold, and each entry needs $O(\log(N+1))$ bits. There is no sequence of repeatedly copied full triangulations in the certificate.

Initial global vertex and edge labels can be retained during these moves. For a surviving vertex, the move deletes a boundary triangle or a boundary disc from its link, or opens a spherical link into a disc. For a surviving edge it removes an endpoint interval from its interval link, or opens a circle link into an interval. The surviving links stay connected. Classes therefore neither split nor merge; some disappear entirely. This justifies using the initial labels to maintain counts of incident boundary triangles. Both producer and checker update all contributions of one move before using the new boundary status.

A surviving interior class can become boundary, but a boundary class cannot become interior under this monotone deletion process; it can only disappear. Consequently, the boolean boundary status of each original vertex or edge changes at most twice. The producer keeps original incidence lists and reconsiders their tetrahedra only when that status changes. It also rechecks the at most four neighbors whose face records change. Across the whole sequence these lists contain $O(N)$ entries and produce $O(N)$ affected visits, including visits to tombstones. A heap selects the eligible tetrahedron with the most boundary facets, breaking ties by stable index; version numbers discard obsolete heap entries. Sorting affected indices and maintaining the heap cost $O(N\log(N+1))$ indexed comparison operations and $O(N)$ storage, after the initial geometry preparation. Counts, indices and versions have $O(\log(N+1))$ bits. These are bounds on the added shelling stage, not a new bound on all geometry validation or the later cocycle solver. Python hashing retains the conservative collision-cost qualifications used earlier.

The independent replay needs no heap or incidence-to-tetrahedron schedule. It tests the one-vertex or one-edge obstruction directly, exposes actual neighbor faces, and compares every final face record against the prescribed stable renumbering. Its additional work after initial preparation is linear in $N$ in the indexed graph model. The producer instead tests equality of boundary subcomplexes and updates counts in the removed tetrahedron's local labels. These separate implementations are compared against Regina's legal move checker and final triangulation signatures in the retained audit.

## 119.3 Authenticating the simplified exterior and its surface

The new outer schema `diagram-shelling-witness-v1` stores the canonical source triangulation, stable removal indices and an inner surface certificate. The inner certificate uses the existing disc, annulus-cap or planar-cap format on the reduced triangulation. Its source PD still names the supplied original diagram. Verification first independently reconstructs the canonical source relation, then replays all shellings and checks the exact final face pairing. Only after this

authenticated homeomorphism chain does the surface checker accept the reduced exterior as its source.

The connectedness, primitivity, matching, Euler and boundary-capping checks are the same as before. A private checker helper accepts an internally chosen source predicate; public legacy checkers always select the canonical-exterior predicate. The shelling checker supplies the canonical-plus-replay predicate. No public checker accepts a caller's assertion that a triangulation is already authenticated, or accepts a producer's cached geometric analysis. Stripping the outer shelling proof from these new examples makes the old source check fail, as it should. A proof for a different knot also fails source binding.

An essential disc on the reduced exterior transports to an essential disc on the original exterior through the verified homeomorphism. The same holds for the disc obtained by annulus or planar boundary capping. No meridian or longitude coordinate transport is required for this recognition conclusion: essentialness of a boundary curve is a topological property. The certificate need not expand the transported disc into a normal vector on the original triangulation. Its explicit triangulation and trace overhead is $O(N \log(N+1))$ bits in addition to the inner binary surface proof.

The API switches and CLI flag are:

| | |
|---|---|
| Native search | `shellings=True` |
| Recognizer | `normal_seed_shellings=True` |
| CLI | `-normal-seed-shellings` |

This branch shares the existing work allowance and cancellation callback. It defaults to false. A bounded miss continues through the existing recognition fallback and never certifies knottedness. Simplifying the triangulation need not preserve which surfaces the original bounded candidate schedule reaches; any deployment decision must measure that tradeoff as well as tetrahedron counts.

## 119.4 Source coverage and independent topology controls

The retained audit covers the same 84 source diagrams as the geometry-reuse experiment. Every default-off result exactly reproduces its preceding record, including work count, stage summaries and certificate hash. With shellings enabled and the ordinary four-tree candidate budget, the number of native positives increases from 24 to 28. All 24 previous positives survive in this corpus. The four added source labels are random cases 1, 25, 81 and 176; their new witnesses are primitive connected annuli that can be capped. Cases 25 and 176 share their diagram, so source-label counts do not count distinct knots or distinct certificates. Earlier optional extended-tree or planar searches already recognized these examples; the gain here is within the four-tree, planar-off schedule.

Case 1 shrinks from 80 to 40 tetrahedra using 40 shellings. Cases 25 and 176 shrink from 60 to 54, and case 81 from 100 to 94. Their total native work counts change from 32,902 to 8,030; 26,620 to 7,043; 44,320 to 12,059; and 26,620 to 7,043 respectively. The old counts describe completed candidate misses; the new counts include the successful source-bound proof. On the original optimized-disc fixture, only six tetrahedra are removed, from 100 to 94, but a tree witness now preempts optimization and lowers native work from 52,090 to 13,239. Small geometric changes can therefore matter more to the candidate family than their tetrahedron reduction suggests.

Gordian loses just six of 2,820 tetrahedra. Nevertheless the new candidate schedule completes its restricted miss at 1,859,545 guards, with optimized Euler characteristic $-64$, instead of hitting the two-million-guard cap in the old schedule. This is not a recognition of Gordian. Enabling planar queries on the reduced triangulation still exhausts that allowance. Across the corpus, enabling planar queries gives the same 28 positive source labels, although some have earlier or different

proofs. There is no claim that these settings dominate one another on arbitrary inputs.

Regina accepts all 1,061 individual removals and agrees with every final triangulation signature. All 84 reduced inputs remain connected orientable valid manifolds with one torus boundary component. On the 74 sources with at most 12 crossings, the audit constructs all 353 scheduled raw, distinct tree and optimized candidates, even when the native recognizer would stop early. Every candidate is connected and orientable in Regina, and its Euler value agrees with the native coordinate calculation. Another 80 positive-result surface checks across the three audit arms give 433 Regina surface controls in total. Every tested Euler-one surface is independently a compressing disc. The saved corpus contains 47 distinct positive proofs, of which 28 use the shelling envelope; all are independently replayed. The shelling-on, planar-off arm uses 22 distinct proofs for its 28 positive source labels.

The six new test methods exercise all three shelling types and every prefix of the one-crossing-circle reduction; source immutability; repeated-vertex exclusion; the stray-boundary obstruction; malformed traces and face records; all three inner surface schemas; producer-free replay; wrong-source rejection; 20,000-bit harmless height offsets; exact work limits; cancellation; and the recognizer's opt-in path. All 29 focused tests pass in 3.863 seconds, and all 1,229 maintained tests pass in 229.572 seconds. The source audit takes 77.295 seconds and partly overlaps the full suite; these times are validation costs, not isolated performance measurements. Audit and timing datasets each pin 508 source files. On the 19 ordinary pipeline controls, both settings finish through earlier recognizers, with zero native-stage attempts in all 38 runs.

### 119.5 More native witnesses do not imply faster recognition

The isolated benchmark runs complete recognition with the native branch forced, earlier shortcuts disabled, a twelve-second allowance and a 20,000-object fallback allowance. Each timed call includes an additional independent replay and hash of any returned native proof. An inconclusive native attempt continues into reduced Khovanov recognition. Shellings-off and shellings-on each have an A/A control; the four arms run in randomized order after one warm-up round per case. No tests, audits, builds or commits overlap either timing run. All per-arm result signatures, work counts and proof hashes are stable within each run.

The first five-round experiment has 200 measured calls and 40 warm-ups and takes 13.736 seconds. A/A ratios near 1.09 on one case justify a nine-round repeat, which has 360 measured calls and 40 warm-ups and takes 25.931 seconds. All 560 measured calls and 80 warm-ups complete with the expected verdict. The final primary-arm medians and median paired ratios are:

| Source | Off ms | On ms | Off/on | Off A/A | On A/A |
|---|---|---|---|---|---|
| circle-9 | 68.402 | 113.360 | 0.598 | 1.046 | 1.021 |
| random-29 | 32.534 | 58.229 | 0.566 | 1.000 | 0.998 |
| optimized-positive | 73.749 | 72.677 | 1.028 | 1.009 | 1.012 |
| random-142 | 111.458 | 140.961 | 0.800 | 1.008 | 0.997 |
| random-1 | 30.707 | 43.739 | 0.703 | 0.989 | 0.991 |
| random-25 | 24.638 | 46.146 | 0.607 | 1.015 | 1.005 |
| random-81 | 76.981 | 129.804 | 0.593 | 0.998 | 1.000 |
| random-176 | 46.083 | 76.112 | 0.606 | 1.007 | 0.997 |
| trefoil | 20.296 | 24.763 | 0.796 | 1.003 | 0.983 |
| genus-one-miss | 22.174 | 38.669 | 0.567 | 0.997 | 1.003 |

A ratio greater than one favors shellings. Nine of the ten selected pipelines are slower with shellings, with paired off/on ratios from 0.566 to 0.800. The original optimized-disc fixture is near parity at 1.028; its lower native guard count does not translate into a substantial total speedup because the new source proof also authenticates and replays the simplification. Final A/A medians range from 0.983 to 1.046. Absolute timings also drift across cases: the identical diagrams labelled 25 and

176 are timed separately and have different millisecond medians, but similar paired ratios. These measurements support the default decision, not a universal timing law.

All four added corpus positives switch from the reduced-Khovanov fallback to a native annulus proof, yet take longer in this measured full-verification pipeline. The additional three-crossing unknot given by the two-strand braid $[1, 1, -1]$ also gets a native annulus witness after six shellings: tree trial two produces a connected 58-piece surface with two boundary circles. Its full proof is retained separately, and Regina confirms the surface summary. The new total pipeline is slower. Earlier negative results for raw and minimum-span surfaces on that example therefore remain restricted-family statements. The trefoil continues through the fallback and is correctly recognized as knotted. The easy circle and raw-annulus controls incur source-replay overhead on a path that already succeeded before simplification.

Shellings remain an explicit research and proof-construction option. Their independently replayable triangulation changes broaden the native family and can improve bounded-search completion, but the measured corpus does not justify enabling them by default. Future work can examine proof-checking costs and progress-based selection between triangulations while retaining independent source authentication. Neither a short shelling trace nor a polynomial shelling algorithm bounds how many triangulations or normal surface families general unknot recognition would need to explore. The general subexponential and quasipolynomial goals remain open here.

The reproduction driver is `normal_orbit_research.shellings`. Select `audit` mode, or run `recognize -rounds 9` for timings. The implementation is revision `1eb57f251`, published after merging concurrent main-branch research in `77347b0b5`. The incoming unknot inventory and `openai/math` head `fd4aeeb2e` were rechecked and are unchanged.

# 120 Validating vertex links by an incidence census

The shelling measurements expose a cost shared by discovery and independent verification: repeatedly reconstructing and validating finite manifold geometry. The previous validator built four union–find structures, for vertices, signed edges, directed edges and vertex-link edges. The latter two can be removed without dropping a validation condition. Signed edge classes already determine all vertex-link vertices, while triangle-side incidence determines the number of vertex-link edges. The new census retains the existing source checks, rejection rules, matching equations and normal-surface proof formats. It also applies to supplied triangulations whose source is not a knot diagram; no provenance assumption is used to bypass manifold checks.

## 120.1 Oriented ends and the two ends of a loop

A local tetrahedron corner contributes a triangle to the link of its vertex. A vertex of that triangle corresponds to a directed edge leaving the corner. Under a face gluing, these directed edge occurrences are identified exactly when they represent the same end of the same global edge. The signed edge union already records whether each local ascending edge agrees with its chosen representative. Its consistency check rejects an edge identified with its own reverse. Therefore every global edge has precisely two distinct oriented-end classes, each attached to its corresponding global vertex.

This remains true for a loop edge whose two endpoints have the same global vertex. The ends are different vertices of that vertex's link, even though they contribute to the same link. Counting a loop once would undercount its link vertices and invalidate the Euler test. The implementation increments the endpoint census twice for every global edge, including when the endpoint labels coincide. Reversing a representative's orientation only exchanges its two endpoints, so their incidence counts are independent of that choice. The existing endpoint map, which is also returned for later normal geometry, is reused for this computation.

Equivalently, a local directed edge can be labelled by its signed global edge representative and one orientation bit. Reconstructing all such labels with a second union–find computes an equivalence relation already contained in the signed edge structure. The implementation needs only their counts at each vertex, so it does not even materialize the doubled collection of ends.

## 120.2 Counting link edges without reconstructing them

For a global vertex $v$, let $F_v$ be its number of tetrahedral corners, let $V_v$ be the number of incident global edge ends, and let $B_v$ count unpaired triangle-side occurrences in its link. Every link triangle has three sides. An internal link edge pairs exactly two side occurrences, and a boundary link edge has exactly one. Hence, if $I_v$ and $E_v$ denote its numbers of internal and total edges,

$$3F_v = 2I_v + B_v, \qquad E_v = I_v + B_v = \frac{3F_v + B_v}{2}.$$

It follows that

$$2\chi(\mathrm{Lk}\, v) = 2V_v - F_v - B_v.$$

Reciprocity and the prohibition on pairing a face with itself are checked before this identity is used. A link side is paired across its one containing tetrahedral face, so there is no chain that merges three or more side occurrences into one link edge. Pairing different sides of one link triangle is allowed and still contributes two occurrences, as the identity requires. Boundary counts use local incidences with multiplicity: a boundary triangle whose vertices become identified can contribute more than once to one $B_v$.

All earlier manifold prerequisites remain active. Face-connectedness and orientability are checked. The signed edge union rejects reversal, and each global edge link must have zero or two unpaired endpoints; its incidence graph is connected by construction and has degree at most two. These checks make vertex links connected surfaces. The finite-vertex test then requires $2V_v - F_v - B_v = 4$ when $B_v = 0$, and $2V_v - F_v - B_v = 2$ otherwise. These are the same sphere and disc Euler conditions as before, evaluated by a different cell census. The final boundary checks still require exactly one torus. An ideal torus vertex has link Euler characteristic zero and is still rejected. Thus the optimization does not accept ideal or singular vertices by assuming that an authenticated source has already made them harmless.

## 120.3 Operation counts, storage and preserved outputs

Three arrays indexed by the original vertex roots accumulate $F_v$, $B_v$ and $V_v$. They have $4N$ entries each for $N$ tetrahedra. The first counts local corners, the second counts corners of unpaired faces, and the third counts the two endpoints of every global edge. Once the existing global classes and endpoint map have been obtained, this census requires $O(N)$ indexed integer operations and $O(N)$ words. Every counter has $O(\log(N+1))$ bits. No normal sheet or integer height is inspected or expanded. The earlier approach performed additional union–find work on both directed edges and triangle sides, followed by per-link sets of representatives.

Let $P$ be the number of paired tetrahedral face pairs, and let $b$ be the number of unpaired faces. Before the unchanged boundary-component union, the old validator allocates $34N$ union–find nodes, while the new validator allocates $10N$. Including the boundary-component structure gives $34N + b$ and $10N + b$ nodes. Every paired face formerly induced 15 unions: three on vertices, three on link sides, three on signed edges and six on directed edges. It now induces six. If $e_\partial$ is the number of global boundary edges, the complete union counts are therefore $15P + e_\partial$ and $6P + e_\partial$. The measured family below checks these exact counts:

| Tetrahedra | Old nodes | New nodes | Old unions | New unions |
|---|---|---|---|---|
| 20 | 688 | 208 | 552 | 228 |
| 160 | 5,504 | 1,664 | 4,416 | 1,824 |
| 640 | 22,016 | 6,656 | 17,664 | 7,296 |
| 2,560 | 88,064 | 26,624 | 70,656 | 29,184 |

Each union–find node has parent, size and parity entries. Removing the two structures eliminates $72N$ list entries, while the three new census arrays add $12N$. This describes those named structures, not the peak memory of the entire Python process. The old per-link dictionaries and sets also disappear; the endpoint map was already required by the returned geometry. Caching each signed edge's root and parity together avoids finding it again at export.

The complete validator still has the same near-linear indexed union–find bound and linear-size combinatorial output. The usual dictionary and set collision qualifications still apply to its remaining maps. The improvement removes redundant equivalence computations and makes the additional link census linear; it does not change the worst-case complexity class of general unknot recognition. Vertex labels, signed edge labels, matching equations, boundary incidence, endpoint representatives and all other returned fields are preserved exactly. Existing guard callbacks stay at the same locations, so work budgets and certificate-selection behavior remain unchanged even when actual execution becomes faster.

## 120.4 Exact compatibility and independent manifold controls

The audit loads the geometry validator from revision `635465da5` and records its complete source hash. It compares every returned field, rejection message and callback count, not just the final recognition verdict. On each of the 84 source diagrams it compares the current pulling exterior, its shelling reduction and the legacy centred exterior. Nine layered solid tori, from one to 256 tetrahedra, bring the exact geometry comparison to 261 inputs. These include one-vertex triangulations with loop edges and canonical exteriors with interior vertices.

All 252 published native-search records are reproduced exactly: 84 sources each with shellings off, shellings on, and shellings plus planar capping. The 47 distinct positive certificates are unchanged and replay successfully. This includes the previous work-cap outcomes and all stage summaries. Lower CPU cost does not change the deterministic work allowance, whose guard-count schedule is preserved. The 19 ordinary pipeline controls still finish through earlier recognizers, with no native attempt, and their verdicts and methods are unchanged.

The audit also generates 12,000 reciprocal face pairings on one to eight tetrahedra, alternating unrestricted gluing permutations with orientation- compatible ones. The old and new validators agree on all prepared data or rejection messages and on every callback count. Regina independently checks the supported-manifold classification for all 12,000: connected, orientable, valid, finite, and with exactly one torus boundary. Both native implementations accept 684 cases. Among the rejections, 1,985 fail the vertex-link condition, 2,897 are nonorientable, 4,877 are not face-connected, 378 have no finite boundary and 1,179 have a nontorus boundary. Seed, generator code, classification counts and a hash of the ordered generated records are retained for reproduction.

New focused controls explicitly reconstruct local link triangles, directed ends and paired sides by graph traversal. Thirty relabelled fixtures compare that literal census with accepted geometry, including both canonical subdivisions and one-vertex layered solid tori. A fixed two-tetrahedron ideal figure-eight exterior has a torus vertex link and must still fail the finite vertex test. Cancellation and input immutability are also checked. All 42 focused tests pass in 3.655 seconds, and all 1,232 maintained tests pass in 216.855 seconds. The final source and random-pairing audit takes 55.020 seconds and partly overlaps the full suite; these elapsed values are validation costs, not performance comparisons. The audit and timing records pin 510 source files.

### 120.5 Complete recognition and proof-replay measurements

The isolated comparison replaces only the geometry validator, loading the old implementation before timing. Both arms use the same selection wrapper. Each timed call runs complete recognition with earlier shortcuts disabled, the native stage enabled, a twelve-second allowance and a 20,000-object fallback allowance, followed by an independent replay and hash of any native proof. Native misses continue through the reduced-Khovanov fallback. Cases marked "plain" disable shellings; cases marked "shell" enable them. The old/new comparison within each row keeps that choice fixed.

Five randomized rounds of old, old A/A, new and new A/A calls cover eleven pipelines after one warm-up round each: 220 measured calls and 44 warm-ups in 15.901 seconds. A new-arm A/A ratio of 1.053 on the trefoil motivates a nine-round repeat. It completes 396 measured calls and 44 warm-ups in 29.043 seconds. All 616 measured calls and 88 warm-ups complete correctly. Within each case, verdict, method, native work count and certificate hash are identical across all four arms and every round. No tests, audits, builds or commits overlap either timing run.

| Pipeline | Old ms | New ms | Old/new | Old A/A | New A/A |
|---|---|---|---|---|---|
| Circle 9, plain | 123.147 | 93.930 | 1.313 | 1.007 | 1.000 |
| Circle 9, shell | 109.032 | 75.259 | 1.425 | 1.007 | 1.004 |
| Random 29, plain | 31.734 | 23.707 | 1.338 | 0.990 | 0.997 |
| Random 29, shell | 57.294 | 41.883 | 1.389 | 1.020 | 0.986 |
| Original disc, plain | 72.279 | 61.827 | 1.158 | 1.002 | 0.999 |
| Original disc, shell | 69.935 | 50.392 | 1.390 | 1.009 | 1.002 |
| Random 142, shell | 141.855 | 114.957 | 1.238 | 0.993 | 0.995 |
| Random 1, shell | 44.022 | 32.310 | 1.357 | 1.002 | 0.988 |
| Random 81, shell | 68.527 | 49.577 | 1.385 | 1.001 | 1.008 |
| trefoil, shell | 24.406 | 20.977 | 1.164 | 0.999 | 0.995 |
| Three-crossing, shell | 39.935 | 28.772 | 1.393 | 1.004 | 1.007 |

All eleven final paired old/new medians favor the new census, from 1.158 to 1.425. Final A/A medians range from 0.986 to 1.020. Plain raw witnesses improve by factors of 1.313 and 1.338; the plain optimized-disc fixture improves by 1.158. Shelling pipelines improve by 1.238–1.425 on the positive fixtures, and the trefoil's complete negative pipeline improves by 1.164. The benefit comes from repeated geometry construction and verification, while unchanged source authentication, cocycle extraction and optimization limit the total gain. Guard counts stay unchanged because guards do not count every union or array operation.

Absolute timing can drift between cases. In particular, the plain circle row was measured during a slower interval than the shelling-circle row. These rows must not be used as an off/on comparison. A separate randomized experiment measures that choice under the new census, using the preceding shelling driver unchanged. Its 200 measured calls and 40 warm-ups all finish correctly in 10.602 seconds, with A/A medians from 0.989 to 1.048. Nine of ten pipelines still favor shellings off, with paired off/on ratios from 0.577 to 0.857. The original optimized-disc fixture now favors shellings by a factor of 1.245, but this one favorable case does not justify a general default. All four added native corpus positives still take longer with shellings than with their earlier Khovanov fallback in this full-verification experiment. The shelling option therefore remains disabled by default.

The incidence census is enabled for all finite-manifold geometry preparation; there is no new recognition option or certificate schema. Its public validation contract is unchanged, and the ordinary pipeline corpus reaches no native stage, so these measurements do not establish a default-portfolio speedup on that corpus. They establish a reduction in geometry work and measured gains in complete native pipelines, including the independent proof cost that defeated the previous simplification experiment.

Implementation revision `16e918cc7` follows a fresh merge check against main. Reproduce

the validation with the driver `normal_orbit_research.link_census` in `audit` mode, and the paired timing comparison in `recognize -rounds 9` mode. The separate shelling-choice run uses `normal_orbit_research.shellings` in `recognize -rounds 5` mode. All measured source proofs and discovery families are unchanged. In particular, this optimization does not supply the missing general subexponential or quasipolynomial discovery bound. The incoming unknot inventory and `openai/math` revision `fd4aeeb2e` remain unchanged.

# 121 An obstruction to the entire coherent-height family

The tree, minimum-span and Euler-optimization searches vary representatives of a primitive integral cocycle. Their failure on an input previously left two possibilities: the search had missed a useful representative, or the entire coherent-height family was unsuitable. An explicit solid-torus triangulation now separates these possibilities. It has eight tetrahedra and one global vertex. Every primitive integral coherent-height surface on it has the same normal coordinates, describing a connected orientable surface of genus two with one boundary component. A separate admissible vector is a compressing disc. Thus even exhaustive gauge optimization cannot make this restricted family complete on arbitrary solid-torus triangulations.

This is a limitation of the candidate representation, not a negative knot certificate. In particular, it does not establish failure on the canonical diagram exteriors used by the maintained recognizer. Those exteriors have additional structure and vertices. It also does not rule out a method that changes triangulations, compresses surfaces, or searches other normal vectors. The general quasipolynomial recognition question is unchanged.

## 121.1 Certifying a change of triangulation

The new local operation `pachner_23` replaces two distinct tetrahedra joined along a chosen face by three tetrahedra. Number the three corners of the shared face $0, 1, 2$, in ascending local order on the first tetrahedron; call its opposite apex 3 and the second apex 4. These are *formal* labels, not global vertex classes. The new tetrahedra, with the displayed local orders, are

$$(3, 4, 0, 1), \qquad (3, 4, 1, 2), \qquad (3, 4, 2, 0).$$

Pair their repeated triangular facets by the identity on formal labels. The remaining six facets are exactly the boundary facets of the old bipyramid. Transfer every old boundary gluing, including gluings between these six facets, through their formal labels. Surviving old tetrahedra retain their order and are renumbered densely; the three new ones follow.

Both fillings are balls with the same labelled boundary. A PL homeomorphism relative to that boundary descends through its prescribed identifications to the ambient triangulation. The five formal vertices need not be five different global vertices: all coincide in the obstruction below. The selected old tetrahedra must be distinct, and their common face must be paired. This is the usual bipyramid description of a 2–3 Pachner move; Regina documents the more general attaching-face construction [52].

The independent checker imports no replacement producer. It validates both finite manifolds, checks the unchanged outside gluings, and constructs a signature of the six formal boundary facets in each filling. A signature records a free boundary face, a pairing to another formal boundary face, or an attachment to a stable outside tetrahedron, together with the full three-vertex map. It separately verifies the interior face pairings. Equality of the signatures proves the relative-boundary replacement. A valid output manifold of the right size is not by itself sufficient.

For a supplied triangulation of $N$ tetrahedra, face rewriting and signature comparison use $O(N)$ indexed operations and storage, in addition to finite manifold preparation. The certificate records

only the chosen tetrahedron and face, but this experiment retains every intermediate triangulation. A $K$-move record therefore uses $O((KN+K^2)\log(N+K))$ bits for these snapshots. No compressed move-sequence bound or bound on finding a useful sequence is asserted. The operation is an explicit geometry API and is not added to the default recognition search.

## 121.2 A reproducible eight-tetrahedron example

Start with the standard two-tetrahedron layered solid torus returned by `layered_torus(2)`. The certificate includes its complete face pairing; replay requires exact equality with this fixed root. Apply the following six moves, using zero-based tetrahedron and opposite-face indices in each successive canonical output:

$$(1,2),\ (0,1),\ (2,2),\ (3,1),\ (3,2),\ (6,0).$$

Every relative-boundary replacement is checked independently. The final triangulation has one vertex and ten edges. Regina independently recognizes it as a solid torus, with isomorphism signature `iHvrQAkbgfeffhhasssvcaj`. The native argument uses the fixed layered solid-torus root and the six checked moves; no topology-engine verdict is an input to the move checker.

The following table gives local heights $h$, their coherent normal vector $x$, and an independently discovered normal disc vector $d$. Each normal row lists the four triangle coordinates followed by the three quadrilateral coordinates in the maintained convention. The complete face pairing is in `data/coherent-obstruction-certificate.json`.

| $t$ | $h_t$ | $x_t$ | $d_t$ |
|---|---|---|---|
| 0 | $(0,-6,-4,-2)$ | $(2,2,0,0,0,0,2)$ | $(2,2,0,0,0,0,2)$ |
| 1 | $(0,-2,2,-4)$ | $(0,0,2,2,0,2,0)$ | $(0,0,2,2,0,2,0)$ |
| 2 | $(0,-2,-3,2)$ | $(0,0,1,2,0,0,2)$ | $(0,2,3,2,0,0,0)$ |
| 3 | $(0,-3,-1,2)$ | $(0,2,0,2,0,0,1)$ | $(2,2,0,0,0,0,3)$ |
| 4 | $(0,-3,-2,-5)$ | $(2,0,0,2,0,1,0)$ | $(0,0,2,2,0,3,0)$ |
| 5 | $(0,1,-3,-4)$ | $(0,1,0,1,3,0,0)$ | $(0,1,2,3,1,0,0)$ |
| 6 | $(0,1,-4,-2)$ | $(0,1,2,0,2,0,0)$ | $(0,1,2,0,2,0,0)$ |
| 7 | $(0,1,-2,-3)$ | $(0,1,0,1,2,0,0)$ | $(0,1,2,3,0,0,0)$ |

The native interval-orbit certificate proves that $x$ has one component, is orientable, has one boundary component, and has

$$|x|_1 = 41, \qquad \chi(x) = -3, \qquad g(x) = \frac{2-1-(-3)}{2} = 2.$$

The separate weighted normal-disc certificate proves that $d$ contains exactly one compressing-disc component; its total piece count is 51. Regina also checks that this supplied $d$ is itself a compressing disc. These are valid normal coordinates, but the following rank argument excludes them from the coherent family, regardless of the chosen heights.

## 121.3 An exact certificate for all primitive coherent vectors

Let $E$ be the global edge set, with the representative orientations reconstructed by the geometry validator. Build the integral face matrix $A$ with one row per global triangular face. For ascending local face corners $a, b, c$, the equation is

$$c_{ab} + c_{bc} - c_{ac} = 0,$$

with signs changed as required by the chosen global edge orientations. Repeated occurrences of an edge contribute to the same matrix entry. The space of integral cocycles is $\ker(A) \cap \mathbb{Z}^E$.

The checker first verifies that every local height difference agrees with one signed global cochain $c$, that $Ac = 0$, and that $\gcd\{c_e : e \in E\} = 1$. It then verifies a nonzero minor of size $E - 1$. For this example $E = 10$ and the supplied $9 \times 9$ minor has determinant 1. Thus $\operatorname{rank} A \geq 9$, while the nonzero kernel vector gives $\operatorname{rank} A \leq 9$. The rational kernel is exactly $\mathbb{Q}c$. If $qc$ is integral, Bezout's identity for the coordinates of $c$ makes $q$ integral. Hence

$$\ker(A) \cap \mathbb{Z}^{10} = \mathbb{Z}c.$$

This avoids trusting the cocycle producer's elimination or a supplied homology rank. The certificate checker independently reconstructs $A$ and evaluates the selected minor by fraction-free elimination.

There is only one global vertex, so every vertex coboundary is zero. Consequently the only primitive integral cocycles are $c$ and $-c$; there is no nontrivial vertex gauge to optimize. Heights realizing either cochain differ from the displayed heights only by a constant on each tetrahedron and possibly a common sign. Constants leave the normal level gaps unchanged, and negation reverses their order without changing the normal coordinates. Equivalently, their six local edge weights are $|c_e|$, which uniquely determine an admissible local normal vector. The checker verifies these weights against the supplied $x$.

It follows that *every primitive coherent-height surface on this triangulation is normally isotopic to $x$*. None is a disc, an annulus, or a planar surface suitable for the current boundary-capping witness. Allowing nonprimitive integral cocycles does not help here: $kc$ gives the coordinate vector $|k|x$, represented by $|k|$ disjoint parallel copies of the two-sided surface $x$, and zero gives the empty vector. This conclusion concerns the coherent construction, not arbitrary embedded normal surfaces.

### 121.4 Verification cost, scope and retained evidence

There are $O(N)$ global edges and faces. The current family checker builds a dense $O(N^2)$ matrix and evaluates its minor in $O(N^3)$ integer arithmetic operations. Each original row has absolute sum at most three; Hadamard's inequality bounds the bit lengths of its minors by $O(N)$. Together with reading and checking the supplied binary heights and normal coordinates, this gives polynomial bit complexity in the encoded input. This is a certificate for a restricted family on one supplied triangulation, not a certificate of knottedness and not a candidate-discovery algorithm. Its return value makes that scope explicit.

The retained research driver reproduces the complete example without Regina, including the six moves, rank minor, topology proof and disc-count proof. Its replay mode uses the independent checkers and no search. An optional Regina audit confirms both final surfaces and checks 240 additional randomly selected 2–3 moves, arranged as twelve length-20 paths starting from layered solid tori of sizes two, three and eight and a canonical one-crossing diagram exterior. Every native output is isomorphic to Regina's independent replacement. The seed and all selected moves and isomorphism signatures are retained, along with hashes of the native source files.

Focused tests check twenty random relabellings of the entire family, certificate mutations, malformed and same-tetrahedron move sites, nonprimitive cochains, extra global vertices, sign changes and independent tetrahedron offsets of more than 20,000 bits. They also check producer-free replay, exact work limits, cancellation and input immutability. The six new focused tests pass in 0.177 seconds. All 1,238 maintained tests pass in 230.697 seconds, with the complete log retained. The full suite partly overlaps the optional audit and article build; its elapsed time is a validation cost, not a performance comparison.

To reproduce the proof from `../fast`, run

```
python -B -m normal_orbit_research.coherent_obstruction reproduce \
  --record ../synthesis/data/coherent-obstruction-certificate.json
```

```
python -B -m normal_orbit_research.coherent_obstruction replay \
  --record ../synthesis/data/coherent-obstruction-certificate.json
```

Use mode `audit` with the same record and an `-output` path for the optional Regina cross-check. The evidence files are `data/coherent-obstruction-*`. No whole-recognizer speedup is claimed: the maintained candidate schedule and default settings are unchanged. The practical consequence is to direct completeness work toward changes of triangulation, surface compression or a broader normal-vector family, rather than assuming that a better potential optimizer alone will suffice.

# 122 Escaping a coherent family and eliminating vacuous optimization

Section 121 rules out completeness of the integral coherent-height family on arbitrary solid-torus triangulations. It does not make this obstruction invariant under retriangulation. In fact, a single checked 3–2 move on that example changes the primitive coherent surface from genus two to a disc. This motivates testing changes of triangulation, while also identifying an avoidable cost: on a one-vertex triangulation, optimizing vertex potentials cannot change any height span. The maintained span solver now detects this case and constructs a linear arithmetic optimality certificate without a flow network.

These are two separate improvements with different roles. The 3–2 operation extends the explicit, independently checked geometry API. Its greedy search remains an experiment. The one-vertex branch is an automatic optimization of an existing arithmetic API and preserves its proof schema. Neither result supplies a general discovery bound.

## 122.1 An independently verified 3–2 replacement

The input site is a tetrahedron and two distinct local vertices specifying an edge. The producer reconstructs its signed global edge class and requires exactly three local occurrences in three different tetrahedra. All faces containing this edge must be paired within the selected region. Give the edge endpoints formal labels $3, 4$ and the other two corners of the selected tetrahedron labels $0, 1$. Propagating the labels through the interior faces assigns the third belt label 2 and must give exactly the three formal tetrahedral vertex sets

$$\{3, 4, 0, 1\}, \quad \{3, 4, 1, 2\}, \quad \{3, 4, 2, 0\}.$$

The consistency test includes all interior face maps, not merely edge degree. Replace this filling by tetrahedra $(0, 1, 2, 3)$ and $(0, 1, 2, 4)$ joined across their common $(0, 1, 2)$ face, and transfer the six formal boundary gluings. As in the 2–3 direction, globally identified vertices and pairings among formal boundary facets are permitted. An inverse relative-boundary PL homeomorphism gives the topology proof.

The certificate lists the three original tetrahedra and their four formal labels. The independent checker reconstructs both finite manifolds, validates the three distinct indices and formal sets, checks the unchanged exterior, and compares the six boundary signatures. Interior faces are checked separately. It reuses only the independent boundary-signature routine from the 2–3 checker; no move producer is imported. A reordered valid output that does not follow the certificate's prescribed labels is rejected.

After manifold preparation, site discovery, face rewriting and replay take $O(N)$ indexed operations for $N$ tetrahedra. The explicit API performs only one supplied move. A simple repeated scan for an eligible 3–2 edge terminates after fewer than $N$ successful moves because the tetrahedron count strictly decreases. Rebuilding native geometry on every pass gives $O(N^2\alpha(N))$ union–find

and indexed work under the existing word-cost model, excluding the research driver's additional Regina isomorphism audits. This is a bound for finding a local minimum in tetrahedron count, not for finding a disc or for reaching a globally smallest triangulation. A plateau may require temporary increases.

## 122.2 The explicit escape and the coverage experiment

On the eight-tetrahedron obstruction, select tetrahedron 5 and its local edge $\{0,1\}$. The new triangulation has seven tetrahedra and one global vertex. Its primitive coherent surface has 36 normal pieces and is a compressing disc. The retained record includes the entire before/after triangulations, the independently checked replacement and the independent normal-disc-count certificate. Together with the previous six-move record it has a fixed solid-torus source. This does not claim correspondence with a supplied planar diagram.

For broader move validation, four length-100 paths start from layered solid tori with two and eight tetrahedra and canonical one-crossing and trefoil exteriors. At each step, Regina selects an eligible upward or downward site; the native producer performs the supplied move, the independent native checker replays it, and the result is compared with Regina's separate replacement up to triangulation isomorphism. All 400 comparisons pass: 210 are 2–3 moves and 190 are 3–2 moves. Additional tests cover reversed local edge order, relabelled one-vertex inputs, malformed certificates, forged boundary maps, resource limits, cancellation and immutable inputs.

The discovery experiment uses the 82 maintained source diagrams with at most sixteen crossings. It repeatedly chooses the first eligible edge in local incidence order, and then evaluates the raw cocycle and up to four alternative tree gauges. One arm begins at the canonical pulling exterior; the other first performs the existing boundary shellings. Every successful 3–2 replacement is independently replayed and compared with Regina. The arms perform 144 and 73 moves respectively, and yield nonnegative-Euler coherent candidates on 20 and 26 source labels. Regina checks connectivity, orientability and Euler characteristic of these supplied candidates, and checks compressibility when the candidate is a disc.

Neither arm adds a source label beyond the earlier extended-tree and planar capping search. This experiment does not run the minimum-span stage and is not a complete-recognition timing comparison. In particular, its candidate counts must not be compared as though they were full default-pipeline success counts. The negative coverage result does not justify adding this greedy simplifier to the recognition schedule; the explicit move API remains useful for testing more selective geometric strategies.

## 122.3 A direct dual certificate when there is one vertex

The arithmetic objective is

$$D(p) = \sum_{t=1}^{N} \left( \max_i (h_{t,i} + p_{v_{t,i}}) - \min_i (h_{t,i} + p_{v_{t,i}}) \right).$$

If all global vertex labels agree, every local corner receives the same potential $p$. Therefore

$$D(p) = D(0) = \sum_t \big(\max_i h_{t,i} - \min_i h_{t,i}\big) \quad \text{for every real or integral } p.$$

Choose a minimum corner $i_t$ and maximum corner $j_t$ in each tetrahedron. The existing certificate format permits the matching entries $(t, i_t, t, j_t)$. Each tetrahedron is used exactly once on each side and the matched corners share their global vertex, so all potential terms cancel. The matching's dual value is exactly $D(0)$. The unchanged weak-duality checker proves optimality, independently

reconstructing the normal vector from its six local edge weights. This includes tied extrema and constant height rows, where the two corners may coincide.

The new branch returns potential zero, these diagonal matching entries and the unchanged local normal coordinates. It still invokes the independent checker before returning. It allocates no flow network, performs no augmentations or shortest-path searches, and keeps the previous schema and statistics fields. Matching certificates need not be identical to those chosen by the generic solver. The coordinate vector and optimal objective are identical. Multiple-vertex inputs continue through the previous solver. The branch establishes no primitivity, connectedness, genus or knot verdict; those require their existing geometric and source checks.

There are $O(N)$ integer arithmetic operations and $O(N)$ stored integers, including the internal proof replay. If input integers have at most $B$ bits, the work is $O(N(B+\log N))$ in the elementary bit-cost model. No integer height or normal-sheet multiplicity is expanded. The implemented deterministic guard count is exactly $5N+3$.

This improves an actual quadratic case of the old implementation. Take one global vertex and rows $(0,0,0,t)$ for $t=0,\dots,N-1$. The previous flow solver performs $N-1$ shortest-path searches after its initial unit, with a full $2N+3$-node potential update on each search. These updates alone take $\Omega(N^2)$ indexed operations, even though the answer is independent of the potential. The direct primal/dual construction removes that cost rather than merely changing the blocking-flow schedule.

## 122.4 Compatibility, measurements and remaining scope

The arithmetic audit compares the old solver at revision `a6cadc5d6` with the new one on 300 random one-vertex instances and 300 random multiple-vertex instances. In the former, coordinates agree and both certificates pass the unchanged checker; in the latter, the entire result, including all work statistics and certificates, agrees exactly. Binary tests use scaled heights and independent tetrahedron offsets exceeding 20,000 bits without increasing the guard count.

The current native source pipeline also reproduces all 252 saved records on the 84-diagram corpus: shellings off, on, and on with planar capping. All 47 distinct positive proofs are identical and replay successfully. Thus the new arithmetic branch changes no candidate, verdict, deterministic work cap or proof in this maintained diagram workload. That compatibility is not evidence of a recognition speedup on these diagrams. All 18 focused tests pass in 0.772 seconds and all 1,245 maintained tests pass in 240.628 seconds. The final audit takes 47.974 seconds and partly overlaps the full suite; these are validation costs. The audit and benchmark records pin 522 source and evidence files.

The timing run is isolated from tests, audits, article builds and commits. It uses five shuffled paired rounds of old, old-A/A, new and new-A/A calls, after one warmup of each arm. All 280 measured calls and 56 warmups complete. The 190.714-second run checks every result and records all raw samples. Ratios below are medians of paired old/new times, so they need not equal the quotient of the two displayed marginal medians. Larger factors favor the new solver. The A/A paired medians across all rows range from 0.965 to 1.059.

The first table measures the supplied arithmetic problem, including the solver's internal proof replay, another independent replay, and result hashing. These distinct-span rows need not be compatible with any manifold triangulation and are not knot-recognition inputs.

| $N$ | old ms | new ms | paired factor | old guards | new guards |
|---|---|---|---|---|---|
| 16 | 2.971 | 0.747 | 3.975 | 7099 | 83 |
| 64 | 29.619 | 2.616 | 11.339 | 99019 | 323 |
| 256 | 462.164 | 10.286 | 45.128 | 1526539 | 1283 |
| 1024 | 8000.620 | 42.857 | 189.235 | 24194059 | 5123 |

The largest local factor is 189.235, but its scope is this arithmetic call. The guard counts expose the separate asymptotic improvement without relying on wall-clock ratios.

The second table includes two broader workloads. Coherent analysis starts from a supplied triangulation and repeats cocycle extraction, span optimization, normal-surface topology production, both independent proof replays and hashing inside every timed call. Its input construction is outside the timer. Diagram recognition instead runs the complete forced-native recognizer and independently replays any returned native proof. Earlier easy stages are disabled equally in both arms; the established exact fallback remains enabled. The fallback unknot and trefoil controls finish through reduced Khovanov homology.

| Complete measured operation | old ms | new ms | paired factor |
|---|---|---|---|
| Layered 16: coherent analysis | 18.477 | 17.207 | 1.086 |
| Layered 64: coherent analysis | 213.461 | 197.052 | 1.087 |
| Layered 256: coherent analysis | 3427.230 | 2953.287 | 1.158 |
| Genus-two miss: coherent analysis | 6.375 | 5.838 | 1.090 |
| Escaped disc: coherent analysis | 5.525 | 5.023 | 1.103 |
| Canonical 1: coherent analysis | 9.146 | 9.191 | 0.991 |
| Canonical 4: coherent analysis | 58.303 | 58.845 | 0.986 |
| Diagram recognition: native disc | 11.549 | 11.522 | 0.995 |
| Diagram recognition: fallback unknot | 20.452 | 20.235 | 0.999 |
| Diagram recognition: trefoil | 17.656 | 17.595 | 1.004 |

On the five one-vertex supplied-triangulation analyses, paired gains are 1.086–1.158. In particular, the 256-tetrahedron layered torus still takes about 2.95 seconds in the new complete analysis: eliminating the flow does not eliminate topology processing. The two canonical multiple-vertex analysis controls and all three complete-recognition controls show no improvement attributable to the new branch. Their result records, work counts and proof hashes are identical across all arms. All supplied one-vertex coordinate vectors and topology summaries also agree, although their span matchings and work counts change. These distinctions prevent a large arithmetic factor from being presented as a general recognition gain.

The retained driver is `../fast/normal_orbit_research/coherent_escape.py`, with `audit` and `benchmark` modes and a required `-output` path. Benchmark mode accepts `-rounds`. Records, source pins, build and validation logs are `data/coherent-escape-*`. The incoming inventory and `openai/math` HEAD remain unchanged, the latter at revision `fd4aeeb2ee4f`. The remaining discovery problem is to find useful retriangulations or broader surface candidates under a general complexity bound. A local escape and a linear arithmetic special case do not settle it.

## 123 Finite coorientations eliminate a redundant orbit query

The normal-surface topology API previously counted the orbits of a supplied surface, its doubled coordinates, and its boundary separately. Doubling detects orientability, but its orbit computation can repeat much of the first query's work. On the 256-tetrahedron layered meridian, the surface and double each take 261 cycles and record 82,960 local events. The boundary adds 259 cycles and 518 events. A diagnostic profile identifies transmission, contraction and support closure in these queries and their independent replay as the main costs. The profile includes instrumentation overhead; the timings below come from separate isolated measurements.

The implementation now first tries a sufficient finite coorientation certificate. When it succeeds, the double has exactly twice the already verified number of surface components. This removes a complete orbit query and its local trace. Failure retains the previous query and proof format. The test is enabled by default in `normal_surface_topology`; `coorientation=False` restores the old schedule. This API analyzes supplied normal coordinates. The current diagram-recognition stages

use other specialized surface APIs, so these measurements do not assert a speedup of complete knot recognition.

## 123.1 A finite sufficient trivialization

Let $Q$ be the normal vector used by the orbit queries. If multiplicity reduction is enabled, the input is $gQ$, where $g$ is the coordinate gcd; otherwise $Q$ is the full input. Order the global edges as in the normal interval construction and let $w_e$ be their intersection weights. The orbit universe is partitioned into nonempty blocks

$$B_e = [s_e, s_e + w_e) \cap \mathbb{Z}, \qquad s_e = \sum_{e' < e} w_{e'}.$$

Each original normal-arc pairing has its domain in one block $B_u$ and its range in one block $B_v$. Associate to this pairing the bit $\delta = 0$ for an order-preserving map and $\delta = 1$ for a reversing map. Solve the finite graph equations

$$\epsilon_u \oplus \epsilon_v = \delta.$$

A parity traversal either supplies one bit per incident block or finds an inconsistent cycle. The latter is only a failed sufficient test. It does not prove the surface nonorientable, since a valid coorientation might vary within a block.

To prove the positive case, write each point of the doubled universe uniquely as $2i + r$, where $r \in \{0, 1\}$. Doubling an interval pairing $f$ sends it by

$$\widetilde{f}(2i + r) = 2f(i) + (r \oplus \delta).$$

This follows directly from the doubled endpoints: an interval $[a, b]$ becomes $[2a, 2b + 1]$. On block $B_e$, give this point the corrected label $r \oplus \epsilon_e$. The equations imply that every doubled pairing preserves this label. For each label, projection to $i$ is a bijection and conjugates every generating pairing to its base pairing. Thus each label carries exactly the original orbit relation, proving

$$C(2Q) = 2C(Q).$$

Blocks with no incident pairing can be given bit zero: their isolated points have two isolated lifts. The equality also holds for the empty universe. Since the validated ambient manifold is orientable and this double is the normal orientation cover, every component of $Q$ is orientable.

The proof must use the original arc pairings, including reversals on single-point intervals. Such a reversal is the identity on its base point but interchanges the two lifted points. Removing its reversal bit would lose the monodromy and could give an incorrect orientability result. The graph is therefore reconstructed from the original normal geometry, independently of simplifications made during the first orbit query.

Coarse-test failure occurs on both one-sided and two-sided surfaces. The one-tetrahedron Möbius band fails; twice that band is an orientable annulus but can still fail, including when its primitive quotient is used. The one-vertex linking disc also fails on this example. All use the old double query. When the test succeeds on a quotient $Q$, the existing multiplicity-lifting identities recover the topology of $gQ$. No new rule replaces those identities, including the even-multiplicity handling of one-sided components on the fallback path.

## 123.2 An independently checked derived count

Successful proofs use outer schema `normal-surface-topology-v4`. It retains the source fingerprint, the actual surface and boundary orbit proofs, boundary homology, claimed topology and coordinate divisor, and adds an explicit Boolean boundary-classification flag. The double entry

has schema `normal-double-coorientation-v1`, the finite parity certificate, the claimed doubled count and an empty operations list. The coordinate divisor is explicit even when it equals one.

The checker reconstructs the manifold, admissible coordinates, quotient and original arc system. It independently verifies the surface count, reconstructs the finite graph, checks all supplied bits and graph equations, and requires the double's count to equal twice that verified count. It does not invoke the parity search or any orbit producer. Missing or duplicate labels, values other than integer bits, inconsistent equations, extra fields, nonempty derived traces and incorrect counts are rejected. A simultaneous bit reversal is a valid alternative certificate. Existing versions one through three remain accepted.

Boundary classification continues to use its existing component identities and, when needed, cone queries. Both producer and checker still construct the doubled arc system, which those optional queries may need. Only its redundant orbit count is skipped. Failed finite tests produce exactly the old complete result, including its certificate and statistics. Successful queries report zero double cycles and a derived-coorientation statistic. Cycle allowances count actual orbit cycles; replay-operation allowances count recorded local events. Neither allowance purports to bound all geometry or graph work. Cancellation callbacks remain active in that work.

For $N$ tetrahedra there are $O(N)$ edge blocks and original arc pairings. The current helper sorts edge roots, computes binary offsets, and locates each pairing's four endpoints by binary search. It requires $O(N \log N)$ integer operations and comparisons, followed by a linear parity traversal. If the endpoint bit length is bounded by $B$, a safe elementary bit bound is $O(N \log N(B + \log N))$. Storage is $O(N)$ integers; the color witness has $O(N \log N)$ bits. No sheet multiplicity is expanded. These bounds describe the additional test, not the remaining orbit queries or discovery of a useful normal vector.

## 123.3 Validation, costs and scope

The frozen comparison is revision `7d9a6f608`, loading its producer and independent checker separately. The maintained corpus supplies 1,275 normal vectors on 45 triangulations, including compatible mixtures, parallel surfaces, empty vectors, one-sided components and relabelled examples. For each vector the audit tests both multiplicity settings and both boundary-classification settings: 5,100 configurations. Disabling the shortcut reproduces all old records exactly. Enabling it derives 956 double counts, corresponding to 239 input vectors, and leaves all 4,144 fallback records identical. Every topology result agrees with the old result. The audit replays each old proof with both checkers and each new proof with the new checker, for 15,300 comparisons. Live Regina checks component counts, orientability, boundary counts and Euler characteristic on all 1,275 vectors; the frozen component-level records also check the finer boundary classification.

Focused tests verify the pointwise doubled-map conjugacy, relabellings, empty inputs, binary multiplicities exceeding 20,000 bits, exact cycle and replay-event limits, cancellation and proof mutations. Replay is also run with both orbit search and parity search disabled. All 46 focused tests and all 1,250 maintained tests pass. The audit retains 84 representative full proofs, source hashes and deterministic operation counts. The layered meridian family illustrates the saved work. The table counts all surface, double and boundary events; the finite sign witness replaces only the double trace.

| $N$ | old events | new events | sign bits | old bytes | new bytes |
|---|---|---|---|---|---|
| 4 | 118 | 66 | 6 | 8341 | 5038 |
| 16 | 838 | 438 | 18 | 64708 | 34023 |
| 64 | 10918 | 5526 | 66 | 907076 | 458720 |
| 256 | 166438 | 83478 | 258 | 14185011 | 7114254 |

The isolated timing experiment uses five randomized rounds, with old and new implementations

and a same-implementation control for each. All 240 measured calls and 48 warm-up calls finish. Each timed call includes topology production, independent replay and complete proof serialization and hashing. Input construction is outside the timer. The old arm uses the old checker; all arms return the same topology and each arm's proof is deterministic. The binary-scale case multiplies a 64-tetrahedron meridian by $2^{2048}+1$. The empty, Möbius, linking, mixture and genus-two cases also request boundary classification.

| Supplied surface | old ms | new ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| Layered 1 | 1.113 | 0.936 | 1.191 | 1.028 | 1.000 |
| Layered 16 | 18.532 | 11.317 | 1.638 | 1.009 | 1.000 |
| Layered 64 | 227.808 | 121.452 | 1.859 | 0.978 | 1.015 |
| Layered 256 | 3490.381 | 1772.576 | 1.967 | 1.002 | 1.000 |
| Layered 64, binary scale | 221.309 | 122.368 | 1.810 | 1.017 | 1.027 |
| Empty | 0.442 | 0.468 | 0.939 | 0.959 | 1.030 |
| Möbius band | 0.719 | 0.747 | 0.962 | 0.980 | 0.984 |
| Doubled Möbius band | 0.730 | 0.757 | 0.974 | 1.003 | 0.969 |
| Vertex-link disc | 1.548 | 1.570 | 0.983 | 1.006 | 1.003 |
| Sphere/disc/one-sided mixture | 5.509 | 5.534 | 0.999 | 0.985 | 0.996 |
| Sphere plus negative $\chi$ | 5.596 | 5.612 | 0.992 | 0.999 | 0.996 |
| Genus-two coherent | 5.398 | 3.995 | 1.323 | 1.020 | 1.001 |
| Entire 1,275-vector batch | 5025.707 | 4988.370 | 1.003 | 0.992 | 0.996 |

A factor above one favors the new implementation; factors are medians of paired ratios rather than ratios of medians. Layered examples gain up to 1.967×, and the genus-two coherent example gains 1.323×. Several small cases regress, including the empty surface even though its double is derived: the extra graph/proof overhead exceeds the saved query. Same-implementation control medians range from 0.959 to 1.030, so small differences should not be interpreted as stable gains.

A second isolated experiment times the entire 1,275-vector corpus as one batch, with default multiplicity reduction and no boundary classification. Five rounds give 20 measured batches (25,500 surface calls) and four warm-up batches (5,100 calls). Each includes all producers, independent replays and proof serialization/hashing. The aggregate topology digest agrees across all arms. The new arm derives 239 doubles, reduces recorded events from 190,570 to 182,925 and serialized proof bytes from 13,330,351 to 12,861,213. Its paired speed factor is 1.003, with old and new control factors 0.992 and 0.996: effectively neutral at this measurement resolution. The default therefore trades small miss costs for substantial savings on some expensive supplied surfaces; it is not presented as a measured whole-corpus speedup. The explicit off switch preserves the old schedule.

The native research driver is `../fast/normal_orbit_research/coorientation.py`, with `audit` and `benchmark` modes, a required `-output` path and optional `-rounds`. The separate batch driver is `data/coorientation_corpus_benchmark.py`. Records and validation logs are `data/coorientation-*`. These results justify the finite shortcut as a way to remove demonstrably redundant work when a small certificate exists. They do not establish a general speedup on the corpus. The base orbit search, candidate discovery and authentication back to a knot diagram remain separate obligations. General quasipolynomial or subexponential recognition is not proved by this change.

## 124 Certified reflection and interval sweep direction

The preceding coorientation test removes the orientation-double query on some supplied surfaces. The remaining base query can still produce a large trace. Its cost depends on a representation choice: the AHT implementation always transmits through a rightmost carrier and removes a

terminal suffix. The same equivalence relation can be presented in the opposite order, so this convention need not give an efficient trace on a particular input.

The interval API and the supplied-normal-surface API now accept an explicit `sweep_direction` option: `forward`, `reverse`, or `wide`. The default remains `forward`, preserving every old complete result and proof. The last option compares the initial terminal carrier widths and reflects if the left one is wider. This is a scheduling heuristic. It is useful on the layered meridian examples but loses on some other genuine normal surfaces. The implementation therefore exposes it for explicit use without promoting it to the default or claiming an adaptive worst-case improvement.

## 124.1 Reflection preserves the entire equivalence relation

For a universe $\{0, \dots, M-1\}$, let $R(x) = M - 1 - x$. An inclusive pairing $[a, b] \leftrightarrow [c, d]$ becomes

$$[M-1-d, M-1-c] \;\leftrightarrow\; [M-1-b, M-1-a].$$

This order keeps the stored domain to the left of the stored range. The order-reversal flag is unchanged: conjugating an affine isometry by two reflections preserves its slope, and swapping domain and range takes its inverse, which also preserves that slope. Thus $R$ induces a bijection between old and new orbits, including singleton orbits, static gaps and components generated by overlapping reflections. For $M = 0$ the map is the unique empty bijection.

The producer preserves the caller's original ordered pairing list in the certificate. If it reflects the working list, it records one initial `reflect` event and then uses the ordinary AHT reductions. There is no change to transmission, trimming, contraction, periodic merging or truncation. Reflection does not increase the maximum bit length of the universe size and endpoints, and leaves the classical polynomial orbit-counting bound in the number of pairings and $\log(M+1)$ unchanged.

The independent checker applies the displayed endpoint formula to its own current rows. It imports neither the direction selector nor the producer. The event has exactly one field, its operation name; additional fields are rejected. Since the formula is valid on any current universe, replay also permits later reflections and pairs of cancelling reflections. The maintained producer only needs the single initial event.

To preserve the interpretation of old proof formats, reflected traces use version three for the Fine–Wilf merger threshold and version four for the classical AHT threshold. Versions one and two retain their former thresholds and reject reflection events. A new-version proof need not actually use reflection; a legal old reduction remains legal under the corresponding threshold. Unsupported-version mutation tests now use version five. This changes no orbit mathematics in the old versions.

With $k$ interval pairings and a $B$-bit bound on the universe size and endpoints, constructing and checking a reflection takes $O(k)$ integer operations, $O(kB)$ elementary bit work and $O(k)$ stored rows. The trace adds one constant-size event. Neither producer nor checker visits the $M$ individual points. Cancellation is polled while transforming rows. The operation is preparation, not an AHT cycle; a local-event replay allowance nevertheless counts its event. An incomplete computation returns no certificate or orbit count.

## 124.2 A deliberately limited wider-end selector

The selector scans the original pairings, ignores identities, and computes the width remaining after the existing overlapping-reflection trim. Among pairings incident to the smallest left endpoint it takes the widest; among those incident to the largest right endpoint it does the same. It reflects precisely when the left width is larger, keeping the historical direction on ties or when there are no nonidentity pairings.

This describes the initial terminal carriers before periodic mergers. Static-gap contraction does not change the width of an occupied interval. However, the selector does not predict later mergers, transmitted rows, or the suffix actually exposed by a carrier. A large initial carrier can therefore be a poor predictor of the total search. The selector adds a linear scan and makes a single initial choice; it does not switch midway through a reduction and it has no guarantee relative to the better of the two directions. That distinction matters for the proposed introspective recognizer: a future switching policy needs a resource or progress bound that this width comparison does not provide.

The normal-surface wrapper passes the chosen option independently to every actual surface, double, boundary and optional boundary-cone query. Its coorientation shortcut still uses the original geometric arc system. Reflection inside an orbit proof changes neither those geometric parity constraints nor the outer source fingerprint. Coordinate-divisor lifting and boundary classification retain their existing checks and proof formats. The current diagram recognizer's scheduling settings are unchanged.

### 124.3 Compatibility, counterexamples and measurements

The frozen baseline is revision `3310e1e84`. The research driver loads its interval producer, interval checker, normal-surface wrapper and outer checker separately, explicitly reconnecting the old wrapper to the old interval producer and the old outer checker to the old interval checker. Shared immutable pairing objects have the same validated representation. This prevents a comparison from silently running the new search inside the old wrapper.

The audit compares all 1,275 maintained normal vectors in both multiplicity modes and both boundary-classification modes. All 5,100 default-forward records equal the frozen implementation exactly, including proofs, statistics and cycles. Each configuration also runs both explicit reverse and wider-end choices, giving 10,200 comparisons with the same topology and successful independent replay. The frozen component-level oracle checks the expected counts. Another 1,000 random interval systems use literal pointwise components as an independent oracle; both directions under both merger rules give 4,000 matching comparisons, while 2,000 default records reproduce the old kernel exactly.

Focused tests also enumerate all two-pairing systems through four points, exercise random larger systems, transport integers exceeding 20,000 bits, check exact cycle and replay-event caps, test cancellation and malformed proofs, and replay with producer search and direction selection disabled. Two additional reflection events test the involution directly. All 29 focused tests and all 1,255 maintained tests pass. The layered meridian base query shows the effect of changing order. The wider-end option selects the reversed direction on these inputs; event counts include its extra reflection event.

| $N$ | forward events | wide events | fwd. trans. | wide trans. | fwd. bytes | wide bytes |
|---|---|---|---|---|---|---|
| 4 | 52 | 43 | 30 | 20 | 3511 | 2718 |
| 16 | 400 | 151 | 330 | 68 | 30914 | 10152 |
| 64 | 5392 | 583 | 5130 | 260 | 448949 | 49951 |
| 256 | 82960 | 2311 | 81930 | 1028 | 7072861 | 367462 |

For $N = 256$, transmissions fall from 81,930 to 1,028 and serialized base proof bytes from 7,072,861 to 367,462. The base query still performs 259 cycles, compared with 261 before. Thus the large trace reduction comes primarily from doing less work per cycle. The displayed samples do not by themselves prove a linear family bound, and the implementation retains full-list scans within each cycle. A smaller trace is not a proof of linear running time.

The wider-end rule has a concrete contrary example in the corpus entry `finite_figureEight_relabel_1:6`. On this supplied surface, it reduces total cycles from 45 to 41 but increases total local

events from 421 to 621 and proof bytes from 28,581 to 42,682. Both full certificates are retained and independently replayed. This is an ordering counterexample, not a failed topology computation or a knot verdict. It also shows why an adaptive policy cannot infer lower proof-production work merely from fewer AHT cycles.

On the 1,275-vector corpus with default multiplicity reduction and no boundary classification, the wider-end rule shortens 160 event traces, lengthens 359 and leaves 756 unchanged. Aggregate events increase from 182,925 to 189,742, while bytes increase from 12,861,213 to 13,315,621. Always reversing is worse on this metric: 199,448 events and 13,921,942 bytes. These are exact deterministic counts, independent of timing noise.

The isolated timing comparison uses five randomized rounds, with a separate same-implementation control for each arm. Every timed call includes full normal-surface topology production, independent replay, proof serialization and hashing; input construction is outside the timer. The old arm uses the frozen forward producer and checker, while the new arm explicitly requests the wider-end rule. Both arms retain the preceding coorientation shortcut. All timed calls use default multiplicity reduction and omit the optional boundary classification, which is tested separately in the audit. All measured calls finish and their topology digests agree. Per-arm proof digests, counts and sizes are stable across all measured and warm-up calls. There are 240 measured calls and 48 warm-ups; counting the surfaces inside the batch calls gives 25,720 measured and 5,144 warm-up surface analyses.

| Supplied surface | forward ms | wide ms | paired factor | old A/A | new A/A |
|---|---|---|---|---|---|
| Layered 1 | 0.939 | 0.939 | 0.984 | 0.993 | 1.001 |
| Layered 16 | 11.166 | 8.474 | 1.317 | 1.011 | 0.997 |
| Layered 64 | 114.741 | 50.027 | 2.304 | 0.999 | 0.994 |
| Layered 256 | 1767.995 | 524.711 | 3.369 | 0.997 | 0.992 |
| Layered 64, binary scale | 130.815 | 52.799 | 2.240 | 1.012 | 1.003 |
| Empty | 0.645 | 0.688 | 0.939 | 0.934 | 1.041 |
| Möbius band | 1.241 | 1.248 | 0.999 | 1.004 | 1.018 |
| Vertex-link disc | 2.589 | 2.654 | 0.985 | 0.987 | 1.016 |
| Relabelled figure-eight surface | 17.711 | 20.591 | 0.851 | 0.985 | 1.004 |
| Interior trefoil surface | 11.453 | 12.033 | 0.956 | 0.997 | 0.995 |
| Genus-two coherent | 6.943 | 7.454 | 0.937 | 0.985 | 1.021 |
| Entire 1,275-vector batch | 4999.519 | 5147.753 | 0.971 | 0.990 | 1.000 |

The final row times all 1,275 surfaces in one batch. A paired factor greater than one favors the wider-end rule. It is the median of paired ratios, not the ratio of the two timing medians. The layered-256 gain is 3.369×, while the figure-eight counterexample has factor 0.851 (about 17 percent slower). The entire batch has factor 0.971, about 3 percent slower, with control factors 0.990 and 1.000. Across all cases, control medians range from 0.934 to 1.041; the widest variation is on the tiny empty-surface case. These measurements include the selector overhead and the larger proofs on its misses. They support making the alternative available on suitable supplied surfaces, but do not support changing the general default. The explicit `forward` choice continues to reproduce the old behavior exactly.

Reproduce the audit and isolated paired measurements with `../fast/normal_orbit_research/direction.py`, using `audit` or `benchmark`, a required `-output` path and optional `-rounds`. Full records, counterexample certificates, source hashes and validation logs are retained under `data/orbit-direction-*`. The experiment identifies a substantial avoidable trace cost and a counterexample to a simple automatic remedy. It provides a checked alternate execution order, while leaving the broader adaptive scheduling and normal surface discovery problems unresolved. It does not prove a new general subexponential or quasipolynomial unknot-recognition bound.

# 125 A bounded restart race between certified interval sweeps

The width selector of Section 124 chooses an order without observing the ensuing search. Its counterexample shows that fewer cycles need not mean less work. The new optional `sweep_direction='race'` instead interrupts attempts at a measured checkpoint allowance, tries the other direction and increases the allowance geometrically. This gives a precise bound for the scheduling policy. The unit of this bound is a cooperative checkpoint, not an elementary bit operation or elapsed time. That distinction is essential because some uninterrupted blocks sort or scan a whole collection.

The race is available in both `count_orbits` and `normal_surface_topology`. Forward remains the default; the previous forward, reverse and width-selected paths retain their complete records exactly. The race is a new search policy using the existing proof formats and unchanged independent checkers.

## 125.1 The restart policy and its exact accounting

Retain a validated list of the $k$ input pairings on a universe of size $M$, and let $b = \mathrm{bit_length}(M)$. Set the initial allowance to

$$L = 8(k+1)(b+1).$$

The earlier width selector chooses which direction is tried first. At round $r = 0, 1, \ldots$, both directions, in that fixed order, receive allowance $L2^r$. Each attempt starts again from the original immutable pairings. On every checkpoint the caller's cancellation callback runs first; then the scheduler increments the attempt and total work counters. The attempt is interrupted if its counter exceeds the allowance. Consequently, an exhausted attempt costs exactly one more checkpoint than its allowance. A completed attempt immediately returns its answer and ordinary certificate.

The interruption uses a fresh local exception class, so unrelated caller exceptions propagate. Failed attempt frames and their partial certificates are discarded. No intermediate orbit count is promoted to a result. The winning certificate is exactly the one the corresponding fixed direction would have emitted in an unlimited run. Thus independent verification does not need to trust the direction selector, restart decisions, abandoned states or scheduling counters.

The existing `max_cycles` allowance is shared across every attempt. A private hook records each begun AHT cycle, including one interrupted halfway through. Each new attempt receives only the remaining cycle allowance. If it is spent, the race returns incomplete with neither a count nor a certificate. In a normal-surface query, the same accounting continues across all surface, double, boundary and optional cone queries. An empty universe can still finish under a zero-cycle allowance.

Public `cycles` reports all begun cycles. Ordinary structural statistics describe the winning attempt, or the last attempt on an incomplete return. An interrupted helper can leave those partial structural counters behind its actual work; they are not used for the scheduling bound. Additional `race_*` counters separately report attempts, restarts, work exhaustions, initial and final allowances, directional work and cycles, winning work and cycles, and startup work. A final exhaustion at the cycle cap is a work exhaustion even though it causes no further restart. Proof event allowances count only the returned certificate, which contains no abandoned prefixes. These different counters are intentionally not interchangeable.

## 125.2 A competitive checkpoint bound

Fix the input, merger scheduler, merger threshold and certificate-recording setting. Let $W_0, W_1$ be the complete checkpoint counts of the two unlimited fixed-direction runs, excluding the initial width-selection scan, and put $W = \min(W_0, W_1)$. Assume no external cancellation or cycle cap

stops the computation. Let $B = L2^r$ be the least allowance at least $W$. No preceding round can finish either direction, and some direction must finish at allowance $B$. The preceding rounds spend at most

$$2\sum_{j=0}^{r-1}(L2^j + 1) = 2(B - L) + 2r$$

checkpoints. The final round spends at most $2B + 1$: one exhausted attempt and one completed attempt, or a single completed attempt. Therefore the total attempt work $A$ satisfies

$$A \le 4B - 2L + 2r + 1 \le 12\max(L, W).$$

The last inequality is deliberately coarse. For $r = 0$ the first bound is $2L + 1$; for $r > 0$, use $B < 2W$ and $2r + 1 \le 2B$. The initial selection contributes exactly $k + 1$ additional checkpoints on valid input. Thus the policy is competitive in its instrumented work unit, up to the explicitly chosen input-dependent initial allowance. It is not a bound of $12W$ when $W < L$.

The allowance formula is a scheduling choice, not a theorem that an AHT run should finish in $O(kb)$ checkpoints. Its purpose is to let short runs finish without paying for a second direction while retaining a geometric restart bound when that initial allowance is exceeded. The proof does not rely on the width selector being correct: it only changes which attempt is tried first at each scale.

This does not prove constant-factor competitiveness in wall-clock time. Checkpoint spacing is not uniform: input validation, sorting, heap construction, identity filtering and some list operations can perform multiple elementary operations between checkpoints. Big-integer division and gcd also have operand-dependent cost. Each such block still has a polynomial bound in the original pairing count and input bit length, and the ordinary AHT run has a polynomial checkpoint count. Combining those bounds with the displayed inequality preserves polynomial bit complexity. The restart policy runs one search state at a time and retains only the winning trace. None of these statements establishes a sharper general bit-complexity bound for AHT or a bound for discovering a useful normal surface.

## 125.3 Validation and performance consequences

The frozen baseline is revision `cb26c81af`. Its interval producer, interval checker, normal wrapper and outer checker are loaded separately, with the old dependencies explicitly reconnected as in the preceding section. The default compatibility audit covers all 1,275 maintained surfaces, both coordinate-multiplicity settings and both boundary-classification settings: 5,100 entire default records agree. All 5,100 race results have the same topology, agree with the frozen component-level oracle and pass independent replay. The coorientation shortcut and the geometric source bindings remain unchanged.

The interval audit includes 500 random interval systems, 100 random point-pair graphs, four star graphs and a fixed switching example. Both merger thresholds give 1,210 literal component comparisons and checkpoint bound checks. The three existing fixed-direction policies reproduce all 3,630 old complete records exactly. For every race, the audit checks the actual caller checkpoint total, the sharper displayed bound and equality of the winning certificate with its corresponding fixed-direction proof.

The switching example is a 72-point graph with 195 singleton pairings. It deliberately exercises scheduling; ordinary graph union–find is a more appropriate solver for such an expanded small graph. The race first tries forward, exhausts five attempts and completes in reverse on attempt six. The final allowance is 50,176 checkpoints and the successful run uses 50,174. It returns the same reverse certificate, after 175,619 total attempt checkpoints and 196 startup checkpoints. Its 211 begun cycles include 68 in the successful attempt. This supplies a concrete check of switching and accounting, without presenting the synthetic graph as a knot-recognition gain.

Focused tests check exact cycle caps including interrupted attempts, caller cancellation, all three merger schedulers, integers exceeding 20,000 bits, empty universes, normal-surface composition and producer-free replay. They also isolate the scheduler with deterministic checkpoint streams so that second-direction completion and interrupted prefixes do not depend solely on a particular geometric fixture. All 16 focused tests and all 1,261 maintained tests pass. The normal-surface audit exercises 2,972 restarts across its 5,100 configurations. In the default-multiplicity, unclassified corpus batch, 3,586 actual orbit queries use 4,264 attempts, including 678 restarts. The race performs 4,724,655 attempt checkpoints and 65,555 startup checkpoints. Its winning traces use 31,729 cycles, but its total begun-cycle count is 41,464 after including abandoned attempts. Forward uses 31,554 cycles. This is why the public allowance must count the attempted work rather than only the eventual certificate's search.

The race's returned corpus proofs contain 189,333 events and 13,289,274 serialized bytes. That is slightly smaller than the width-only result in the previous section, but remains larger than forward's 182,925 events and 12,861,213 bytes. Competitive checkpoint scheduling is neither a proof-size optimizer nor a guarantee that the first completed result is faster in wall-clock time. The retained star and switching fixtures show the scale of restart overhead directly:

| Interval fixture | $W$ | attempt work $A$ | bound | attempts | total cycles |
|---|---|---|---|---|---|
| Star 16 | 632 | 632 | 1537 | 1 | 16 |
| Star 32 | 2544 | 6130 | 10755 | 3 | 62 |
| Star 64 | 10208 | 34788 | 57349 | 5 | 166 |
| Star 128 | 40896 | 169926 | 276487 | 7 | 403 |
| Switching graph | 50174 | 175619 | 175621 | 6 | 211 |

The isolated benchmark uses five randomized rounds, with a separate control for each implementation. The first comparison uses frozen forward search as its reference. Each timed surface call includes topology production, independent replay, proof serialization and hashing; input construction is outside the timer. Both arms retain coorientation and default multiplicity reduction, and omit optional boundary classification.

| Input | reference ms | race ms | ref./race | ref. A/A | race A/A |
|---|---|---|---|---|---|
| Layered 16 | 10.833 | 9.297 | 1.167 | 0.981 | 0.987 |
| Layered 64 | 110.830 | 58.769 | 1.884 | 0.978 | 0.973 |
| Layered 256 | 1742.291 | 660.816 | 2.630 | 1.007 | 0.998 |
| Empty | 0.363 | 0.436 | 0.833 | 1.000 | 1.036 |
| Möbius band | 0.697 | 0.829 | 0.846 | 0.983 | 1.012 |
| Vertex-link disc | 1.460 | 1.668 | 0.885 | 0.983 | 1.017 |
| Figure-eight surface | 9.407 | 21.093 | 0.447 | 1.003 | 0.984 |
| Interior trefoil surface | 6.679 | 7.805 | 0.855 | 1.017 | 1.000 |
| Entire 1,275-vector batch | 4885.100 | 6928.233 | 0.703 | 0.994 | 0.998 |

The second comparison uses frozen width-selected search as its reference. It isolates the extra cost of checkpoint accounting and restarts beyond the same initial ordering preference. The final two rows time interval production, independent interval replay and proof serialization directly; they are scheduling stress cases, not normal-surface or knot pipelines.

| Input | reference ms | race ms | ref./race | ref. A/A | race A/A |
|---|---|---|---|---|---|
| Layered 256 | 500.632 | 650.542 | 0.762 | 1.005 | 1.001 |
| Figure-eight surface | 10.847 | 20.928 | 0.518 | 1.001 | 0.995 |
| Entire 1,275-vector batch | 4957.851 | 6916.716 | 0.716 | 0.995 | 0.994 |
| Star 64 (interval kernel) | 4.229 | 19.827 | 0.216 | 0.996 | 0.997 |
| Switching graph (kernel) | 24.510 | 106.682 | 0.230 | 1.021 | 0.993 |

All 280 measured calls and 56 warm-ups finish with matching topology or orbit-count digests. Proof and count records are stable within each arm. Across the normal-surface cases there are 51,200 measured and 10,240 warm-up surface analyses, counting every surface within the batch calls; the interval rows add 40 measured and eight warm-up kernel calls. Factors greater than one favor the race, and are medians of paired ratios. Against forward, layered-256 improves by 2.630×, but the corpus factor is 0.703, about 42 percent slower. Against the width-only policy, the corresponding factors are 0.762 and 0.716: the race is about 31 and 40 percent slower. The figure-eight counterexample is over twice as slow as forward, and the two interval stress cases are over four times as slow as their width-selected references. Control medians range from 0.973 to 1.036, much smaller variation than these major regressions. The comparison includes the scheduler's callback overhead and all abandoned searches, while serializing only the winning proof. The optional policy does not replace forward as the default. Avoiding repeated work through resumable searches is a separate implementation candidate; these measurements do not establish its performance.

The driver is `../fast/normal_orbit_research/race.py`, with `audit` and `benchmark` modes, a required `-output` path, and optional `-rounds`. Source pins, full switching certificates, validation logs and timing records are retained as `data/orbit-race-*`. The result adds a rigorously accounted adaptive search option. Promotion to a default requires a favorable practical tradeoff, and a stronger running-time guarantee would require finer accounting or resumable implementations with bounded work between suspension points. The general quasipolynomial recognition goal remains a separate open obligation for this implementation.

# 126 New deliveries and support-sensitive normal-disc discovery

The five archives arriving in revision `0693f0a34` are now preserved, with their single wrapper directories removed, as reports 54–58. A sixth unknot archive from revision `92efcbaed`, overlooked in the earlier catalogue, is report 59. Four further older archives are now reports 60–63. All nine supplied checksum manifests were verified before placement and their pure checksum files were omitted under the intake rule. Report 61 supplies no whole-package checksum manifest; its twelve declared changed Git blobs were verified and its complete delivered content preserved. Delivered text follows the repository's LF normalization. The original archives remain available in their arrival commits; the report directories retain the articles, authored PDFs, code and recorded evidence unchanged. Native integration and fresh validation are separate from this preservation.

The principal integration here is report 57's exact candidate discovery in a supplied compatible quadrilateral sector. Earlier observers test a normal vector already in hand. This module instead enumerates every non-link standard vertex ray in a selected coordinate sector, or uses the smaller Q-ray phase when its positive-Euler criterion suffices. The implementation adds `fastunknot/normal_sector.py` and a separate dense checker, `fastunknot/normal_sector_verify.py`. It uses the existing independently checked essential-disc counts on the resulting candidates. Its public API asserts properties of the supplied triangulation and sector; diagram-bound recognition needs the established exterior provenance in addition.

## 126.1 What each delivery adds, and what its native gate establishes

Report 54 compiles the complete component profile on fixed source atoms. It packs atom counts into a carry-free scalar weight, preserving capacity bounds, and computes full-port cones from the resulting multiplicity-aware incidence graph. A touched type with multiplicity $h$ contributes $h$ times its profile to the merged component, rather than a single copy. If $a$ atoms are active and they occupy $b$ active blocks, the potential $2a - b$ increases on each effective cone and has maximum $2m - 1$ for $m$ fixed atoms. This bounds the number of partition-changing instructions

independently of their binary repetition counts. The sharp example first cones $m$ two-point atoms separately, then joins them in $m-1$ further steps. Compressed threshold descent can use at most $g+1$ state trials for a grammar with $g$ nodes. These results require full-port coning and fixed atoms; they do not represent specified pointwise attachment bijections or refinement of the source atoms. The fresh native gate passes 402 packed/vector comparisons and source-bound replays, rejecting 400 source mutations. All 25 standalone tests pass.

Report 55 compiles a complete orbit trace into a nonnegative integer linear transfer circuit. Scalar observations select an emission, and one transposed evaluation recovers its source profile. Interval prefix sums then recover normal coordinates without transporting every component's dense coordinate vector. All 24 standalone tests pass. A fresh current-tree adapter audit checks 64 supplied normal vectors: 12 positive disc extractions and 52 complete supplied-vector negatives, agreeing with the existing coordinate census. Its positive checker establishes an admissible connected essential disc $y \leq x$. It does not independently establish that $y$ is a component of $x$; a hierarchy consumer requiring that stronger statement must retain the transfer theorem or a stronger certificate. Source provenance and arbitrary cut/glue attachment data remain separate.

Report 56 proves that a frozen raw singleton elimination epoch can be discovered by original-source Horn closure. Rules use exact-once raw occurrence, not exponent sum. The theorem includes all raw elimination orders but excludes intervening free reduction, root extraction and other normalization. Its separate orbit index compiles canonical minima and rank/select queries from a completed trace. A current-tree additive scratch overlay runs these modules and report 58's observers together: 60 test methods pass, including literal graph and huge-integer controls. The group option overlay remains separate from the current dispatcher; these tests do not validate its whole-recognizer hook or normalization search.

Report 58 reconstructs the joint component spectrum of Euler characteristic, actual boundary-circle count and orientability from a boundary transversal and two two-dimensional weighted histograms. If $H = O + A$ is the base histogram and $K(v) = 2O(v) + A(v/2)$ the double, finite-support inversion determines the two parts. At $v = 0$, use $A(0) = 2H(0) - K(0)$ and $O(0) = K(0) - H(0)$; treating zero as an ordinary halving chain would be incorrect. Its core restoration also respects one-sided scaling and peeled vertex-link discs and spheres. The fresh scratch-overlay Regina audit passes 1,395 API comparisons and certificates on 664 bounded vectors from 26 triangulation isomorphism classes. This validates the tested forward-proof vocabulary, with the compatibility restriction below. Abstract disc type alone supplies no boundary-essentiality or slope information.

Report 59 gives an output-sensitive sparse route through the unweighted coning oracle. Its claim that the maintained interface was dense describes its pinned historical snapshot; the current tree already has the sparse incidence APIs of reports 50 and 52. The report still supplies useful independent extraction and certificate checks. All 19 standalone tests pass and a fresh native gate passes 250 genuine maintained-kernel cases, including producer-disabled replay. It is not installed as a competing default implementation merely because that older gap is stated in its README.

The four older deliveries have already informed maintained work but lacked numbered preservation and catalogue rows. Report 60's weighted component observer and persistent elimination prototype overlap the native observers and raw producer/replayer of Sections 94 and 98. Report 61's quadratic merger queue is the source for the adaptive closure in Section 96; the current scheduler additionally avoids disjoint supports. Its cellular disc observer is superseded for that role by the smaller weighted-disc machinery. Report 62's sparse unsigned and signed incidence interface overlaps the integration in Section 93. Report 63 is the preserved power-conjugacy delivery behind Sections 104 and 99. Its abstract graph sharpness does not remove source word or conjugator restrictions. These stale snapshots are preserved as research evidence, rather than applied over the newer implementations. The delivered standalone prototypes and the maintained implementations retain their distinct validation records.

## 126.2 A reflection compatibility defect found at intake

The new interval reflection event postdates these deliveries. Their minima and transversal compilers must not inherit acceptance of a larger proof language without implementing its transformations. On the three-point relation $0 \sim 1$, the actual least representatives are $\{0, 2\}$. The current interval checker correctly accepts a reverse-sweep proof. Report 58's delivered transversal checker then ignores its reflection when lifting selected sets and accepts the false set $\{0, 1\}$, which contains two points from one orbit and none from the other. The cardinality is still two, so a count-only check cannot detect this error.

The retained counterexample includes the complete valid orbit proof and the incorrect accepted transversal certificate. Report 56's index compiler rejects the same valid proof with an unknown-event error. These are different failure modes caused by the same vocabulary extension. Report 58 was written against versions one and two, where its monotone survivor-label argument applies. The fresh forward tests do not justify its checker on reflected traces. Before promotion, the observer needs an explicit supported-version guard or a correctly proved reflection-aware selector. Simply preserving survivor points under reflection would select maxima in the original order, so obtaining canonical minima requires more than reversing endpoint formulas. The sector modules use the established weighted-disc path and are not affected by this delivered transversal checker. Section 127 now integrates the observer after adding the required monotone-event guard; the delivered report remains unchanged.

## 126.3 An exact cone kernel for a supplied sector

Let $t$ be the tetrahedron count and allow one quadrilateral type in each of $k$ selected tetrahedra. At each glued face corner, matching has form

$$T_d - T_c = b_e(q),$$

where $b_e$ involves at most two allowed quadrilateral coordinates. Construct the triangle difference graph and contract every edge with identically zero restricted label. This imposes precisely triangle equality, so it preserves real solutions and the integral lattice. Each allowed quadrilateral meets four faces, so at most $4k$ nonzero-label equations remain. Retain their loops: a loop may constrain $q$ even though its triangle difference vanishes.

At a global vertex with only one contracted class, its triangle value is an independent nonnegative vertex-link factor. At a vertex with $p_v \geq 2$ classes and $m_v$ retained edges, connectivity gives $p_v - 1 \leq m_v$ and $p_v \leq 2m_v$. Hence at most $8k$ triangle-class variables and $k$ quadrilateral variables remain, with at most $4k$ equations. Copying each class value to its original corners and restoring the independent link factors gives mutually inverse nonnegative integer-linear maps. Primitive and extreme rays are preserved. This is an algebraic search kernel; the original triangulation and expansion map are still required for geometry.

Write the reduced matrix as $[B \mid D]$, where $B$ is a graph incidence block. Every square minor of $B$ is zero or a unit: expand on a row with at most one nonzero entry, or observe that when every row contains both opposite entries, the all-ones vector is in the kernel. Each quadrilateral column of $D$ has absolute column sum at most four. Successive determinant expansion on the chosen $D$ columns therefore bounds a mixed minor by the product of their column sums. An extreme ray on its positive coordinate support has a one-dimensional kernel and is given by maximal cofactors, followed by gcd division. If its actual quadrilateral support has size $a$, every triangle coordinate is at most $4^a$, and each positive quadrilateral coordinate is at most $4^{a-1}$. Omitted zero columns give independent coordinate axes and satisfy the same bound. Expansion merely repeats triangle values, so the bounds hold in the original coordinates.

## 126.4 Canonical lifting and adaptive complete enumeration

Integrating the difference labels along spanning trees gives integer potentials $\ell_c(q)$. Non-tree edges give the remaining cycle constraints $R_S q = 0$. The quadrilateral cone is $q \geq 0$ in $E = \ker R_S$, with nullity $d$. Its canonical lift subtracts $\min_{c \text{ at } v} \ell_c(q)$ at each global vertex. It has no vertex-link summand; every other nonnegative lift adds such summands. A lifted Q-extreme ray is standard extreme, because any standard decomposition projects to multiples of its Q-ray and the zero triangle at each vertex forces the link summands to vanish. These canonical conversion facts are part of classical Q-theory [6].

Canonical Euler characteristic is positively homogeneous and subadditive: its linear part is corrected by subtracting each positive vertex-link Euler value times a minimum of linear forms. A positive value somewhere in the quadrilateral cone therefore implies a positive value on some Q-extreme ray. This justifies the preliminary Q-ray screen for positive canonical Euler. An inessential disc, sphere or projective plane may also have positive Euler characteristic. Each positive candidate still undergoes the existing compressed essential-disc count and independent replay.

Complete standard enumeration has two routes. The support route tests every positive variable support of size at most $\operatorname{rank}(M_S) + 1$, retaining precisely one-dimensional strictly positive kernels. It uses at most $2^{9k}$ exact linear systems. The arrangement route adds coordinate hyperplanes and all within-vertex ties $\ell_c(q) = \ell_{c'}(q)$ in $E$. There are at most $32k^2$ distinct nonzero hyperplanes. On each closed chamber the canonical lift is linear. A standard extreme ray must project to an extreme chamber ray, otherwise a nonproportional chamber decomposition would lift to a standard decomposition. Thus it is found among intersections of $d-1$ independent hyperplanes. Some ties concern nonminimal potentials, so candidates must still pass the exact standard-extremality rank test.

The dispatcher compares the actual support-subset count against $\binom{H}{d-1}$ and chooses the smaller. When $d = 0$ there is no non-link ray. For $d > 0$, complete sector search has bound

$$\operatorname{poly}(t,k) \min\{2^{9k}, \binom{H}{d-1}\}.$$

It uses exact rational arithmetic and binary coordinates. This is an adaptive selection between complete enumerators with explicit search-count bounds, rather than a prediction from elapsed progress. Dense support can be easy when nullity is one.

The independent exhaustion checker constructs the original dense matching matrix, eliminates triangle variables with separate arithmetic code, reconstructs the full ray set and compares the transcript, Euler values and negative disc proofs. It imports no sector producer helpers. Negative replay repeats an enumeration and is not claimed polynomial in a short negative transcript's length. The implemented basis allowance counts failed linear systems too. The orbit allowance is per positive candidate; Q and standard phases each receive their named per-sector allowance. Cancellation can impose a shared external limit. Incomplete discovery returns no absence claim. Successful exhaustion excludes only the advertised sector or support range, rather than all normal surfaces in the manifold.

## 126.5 Why support alone cannot give the general target

Searching all supports up to $K$ costs

$$\operatorname{poly}(t,K) \left(1 + \sum_{j=1}^{\min(K,t)} \binom{t}{j} 3^j 2^{9j}\right).$$

This is $t^{O(\log t)}$ for $K = O(\log t)$ under the promise that an essential standard vertex disc exists with that support. It gives no universal small-support promise. In the Fibonacci layered solid tori, an essential meridian meets a selected boundary edge at least $F_{t+3}$ times. A primitive standard vertex disc with support $a$ has edge weight at most $3 \cdot 4^a$ by the height bound. Thus $a \geq (\log_2 F_{t+3} - \log_2 3)/2 = \Omega(t)$. The delivery extends the same conclusion to fundamental meridians using the corresponding $a4^a$ bound. This is a representation obstruction on these triangulations, not a lower bound for unknot recognition.

Their supplied dense meridian sector has nullity one. Its Fibonacci recurrence determines the canonical triangles and quadrilaterals from a single parameter, so the arrangement route tries one system. The real outer task is therefore to discover useful low-complexity sectors or change the geometry under an encoded-volume bound. Neither dense-support examples nor polynomial supplied-vector observers settle that task.

### 126.6 Current-tree validation and measured scope

The native enumeration audit matches all 1,718 frozen Regina sectors in both phases, comparing 2,361 Q-rays and 3,249 standard rays with no caps or mismatches. A separate 481-sector audit verifies 962 certificates with producer functions disabled: 134 positives, 694 negative positive-Euler results, 67 positive-Euler-only results and 67 standard-sector negatives. It rejects 60 mutations and checks resource and callback behavior. Eight maintained integration tests cover methods, positivity traps, source binding, restricted exhaustion, cancellation and immutable inputs. All 1,269 maintained tests pass. The isolated benchmark compares the compact Q-sector producer against the independently implemented dense Q-sector reference. Both arms include candidate discovery, the existing essential-disc certificate, independent witness or exhaustion replay, and proof serialization and hashing. Input construction is outside the timer. This reference is a validation-oriented dense enumerator, not a claim about the performance of optimized Regina or every earlier normal-surface algorithm.

| Supplied sector | dense ms | kernel ms | dense/kernel | dense A/A | kernel A/A |
|---|---|---|---|---|---|
| Layered 4 | 7.077 | 6.465 | 1.109 | 0.997 | 1.030 |
| Layered 8 | 29.596 | 15.713 | 1.893 | 0.999 | 0.994 |
| Layered 10 | 49.844 | 23.216 | 2.176 | 1.010 | 0.996 |
| Empty | 0.279 | 0.270 | 1.037 | 0.995 | 0.963 |
| Annular ray | 0.600 | 0.577 | 1.050 | 1.015 | 0.976 |
| One-sided ray | 0.545 | 0.523 | 1.062 | 0.980 | 0.969 |
| Inessential positive Euler | 3.584 | 3.857 | 0.930 | 0.991 | 0.993 |

Five randomized rounds give 140 measured calls and 28 warm-ups, all complete. Results and coordinates agree across arms; each arm's certificates are stable. The layered cases improve by paired factors 1.109, 1.893 and 2.176. The inessential positive-Euler control regresses by about eight percent. Control medians range from 0.963 to 1.030. Ratios are medians of paired ratios, rather than ratios of the two timing medians. These are supplied-sector measurements. They demonstrate a benefit of the compact construction on the tested family while leaving global sector discovery and complete recognition timing unmeasured.

Reproduction uses the report's preserved scripts on an execution copy with `-fast` pointing to the current tree, and explicit corpus, case and output paths. The native complete-call benchmark driver is `../fast/normal_orbit_research/sectors.py`. Fresh records, delivery placement verification, native gates, the reflection counterexample and publication checks are retained under `data/incoming-sectors-*`. The delivered patches for the other reports remain available for further integration after their specific contracts are reconciled with the current tree. No archive's historical gap claim is treated as current evidence by itself, and none establishes the complete general quasipolynomial goal.

# 127 Component topology from two weights

Report 58's observer is now integrated as `normal_topology_spectrum`, after fixing the trace-vocabulary defect of Section 126. It returns the joint distribution of Euler characteristic, boundary-circle count, orientability and genus or crosscap number, with binary multiplicities. The earlier coordinate census can obtain this information by first recovering every distinct component's full normal vector and then classifying it. The new interface requests only two additive weights. Its initial integration uses one boundary query and two surface queries; Section 130 subsequently derives the double histogram when a coorientation certificate succeeds. This removes the dimension $7t$ coordinate payload when a caller needs abstract component types.

Polynomial encoded computation of supplied normal-surface topology is classical [18]. The integration is a smaller observer with explicit compressed boundary markers, one-sided multiplicity restoration, and independent source-bound replay. The mathematical construction comes from report 58; the maintained adaptation adds the explicit monotone-trace contract and its regression tests. It leaves component coordinates, boundary slopes and essentiality to their existing observers. Those are different outputs, so the timing comparison below states its reference's extra work explicitly.

## 127.1 Least representatives from an order-preserving trace

For an interval relation on $[0, M) \cap \mathbb{Z}$, a least transversal contains precisely the minimum point of each orbit. An output interval $[a, b)$ encodes $b - a$ distinct orbit representatives, not one orbit. Even $2^{20,000}$ isolated points have the one-interval transversal $[0, 2^{20,000})$.

The forward compiler maintains an increasing injection from current points to surviving original labels. Its image is a list of original intervals. Identity deletion, pairing trimming, periodic merging and transmission change the generating relation but preserve its equivalence classes on the same point universe. A suffix truncation discards points equivalent to retained earlier points, so no unfinished orbit loses its least original label. A contraction deletes static intervals: each removed point is then the sole live point of its original orbit, and is emitted. At completion no live points remain, so the emitted union is exactly the least transversal. This proof depends on the increasing injection.

The checker independently replays the original orbit proof and reconstructs the selected set backwards. Undoing a suffix truncation appends unselected points. Undoing a contraction inserts all its static gaps as selected and lifts earlier selected points over them. For half-open gaps $[a_i, b_i)$, write $\ell_i = b_i - a_i$ and $c_i = a_i - \sum_{j<i} \ell_j$. A surviving contracted point $q$ lifts to

$$\lambda(q) = q + \sum_{i : c_i \le q} \ell_i.$$

Lift a selected interval $[u, v)$ to the hull $[\lambda(u), \lambda(v-1)+1)$ and include every inserted gap. The hull's extra points are only those inserted gaps and are already selected, so the union is exact. Prefix sums and binary searches avoid point expansion.

If the trace has $E$ events and $G$ static-gap intervals, reverse insertion creates at most $G$ final representative intervals. Forward deletion leaves at most $1 + G$ live original intervals and at most $1 + 2G$ raw emitted pieces. Each deleted gap can split at most one live interval; suffix deletion adds none. During emission, splitting at the $2g$ gap endpoints adds at most $2g$ pieces, and the live-list sizes telescope over successive contractions. These observations give the stated bounds.

The list implementation takes $O(E(1 + G) + G \log(G + 1))$ endpoint operations for compilation; a conservative reverse-selector bound is $O((E + G)(G + 1) \log(G + 2))$, besides ordinary orbit replay. Endpoint bit lengths are bounded by $\log(M + 1)$. Neither output multiplicity nor a long

interval is expanded. These are polynomial local bounds, rather than an assertion that the list implementation is optimal.

## 127.2 The vocabulary guard and the reflection counterexample

A valid orbit-count proof need not satisfy the monotone-label invariant. On $[0, 3)$ with relation $0 \sim 1$, the reflected count proof is valid, but ignoring reflection during reverse selection yields the false set $\{0, 1\}$ in place of $\{0, 2\}$. Equal cardinality is insufficient.

The maintained reverse selector now explicitly accepts only contraction, suffix truncation, deletion, trimming, merging and transmission. Any other event, including global reflection, rejects the transversal certificate. The forward compiler likewise reports an unsupported monotone-trace request when a source-bound count proof contains reflection. Discovery explicitly requests the forward sweep, so a future default direction change cannot silently change its proof language.

The guard is on the actual operations, not merely the version number. For example, a valid version-three count proof containing only supported monotone events still has the same selector theorem. Two cancelling global reflections remain outside this compiler's contract, even though the net map is the identity. This is an explicit language restriction, not a negative mathematical claim about the source relation. The ordinary forward AHT algorithm is complete for every supported interval input, so the observer remains complete on its supplied-vector domain. Monotone here describes the surviving point labels; reversed arc pairings and nonorientable surfaces remain fully supported.

Tests replay the retained counterexample with the count checker, reject it with the corrected selector, and obtain the correct intervals through forward discovery. Another test places a cancelling reflection pair in a normal surface's boundary proof: count replay accepts the pair, but the outer topology checker rejects that unsupported selector language. Producer and checker continue to propagate caller cancellation, including `ValueError` from a false-valued callable. Independent verification never invokes the forward selector or discovery.

## 127.3 One boundary marker per actual boundary circle

Build the ordinary interval relation for the surface boundary and compute its least transversal. Include those boundary-edge points in the full normal-edge universe, respecting the separate block offsets. Each surface component then receives exactly one marker for each of its boundary circles. An additive marker weight therefore measures the actual count $b$, rather than the number of normal arcs or intersections with a boundary edge.

For the other weight, use the existing normal-cell ownership convention: one per edge point, minus one per global normal arc including boundary arcs, plus one per owned normal disc. Its sum on a component is $\chi$. Thus the full-surface query has integer weight vector $(\chi, b)$, of dimension two independent of $t$.

In the orientable ambient manifold, doubled normal coordinates form the orientation cover. Every original point $i$ has adjacent lifts $2i, 2i + 1$. Replacing each weight interval $[a, b)$ by $[2a, 2b)$ with the same value lifts both observations. Each boundary circle has two separate lifts: its tangent line is orientable and its inward collar line is trivial, so the orientation character restricts trivially there. Consequently the same original boundary markers correctly count boundary circles of the double; no second boundary discovery is required.

## 127.4 Sparse inversion and complete type reconstruction

Let $H(v)$ be the base histogram and $K(v)$ the double histogram for $v = (\chi, b) \in \mathbb{Z}^2$. Split the base into orientable and nonorientable counts $O, A$. The geometric cover gives

$$H(v) = O(v) + A(v), \qquad K(v) = 2O(v) + A(v/2),$$

where a nonintegral half or an absent base key contributes zero. For $v \neq 0$, process present base keys in increasing $|\chi| + b$ and set

$$A(v) = \frac{2H(v) + A(v/2) - K(v)}{2}, \qquad O(v) = H(v) - A(v).$$

An integral nonzero half has smaller norm, so it is already known. Only present keys need processing. At zero the two equations instead give $A(0) = 2H(0) - K(0)$ and $O(0) = K(0) - H(0)$; this distinguishes tori and Klein bottles with the same zero signature.

Reject nonintegral or negative counts and counts exceeding $H(v)$. Then reconstruct the entire base and cover histograms, including keys present only in the cover. This proves existence as well as uniqueness of the computed split. For instance, three Möbius bands and five annuli give $H(0,1) = 3$, $H(0,2) = 5$ and $K(0,2) = 13$. The half-signature contribution is essential to recover zero nonorientable annuli.

An orientable row has genus $(2 - \chi - b)/2$, requiring a nonnegative even defect. A nonorientable row has $2 - \chi - b \geq 1$ crosscaps. The independent algebraic checker reconstructs the complete histograms from proposed rows; it does not run this inversion. It checks classification, canonical ordering, strict integral fields and full-support equality. Integral signatures are required: reducing coordinates modulo two before inversion destroys the unique-halving argument.

For $s$ sparse keys with fixed dimension, the recurrence uses a linear number of integer operations after sorting. With a balanced comparison map and $\ell$-bit fields including temporary sums, the bound is $O(s\ell \log(s+1))$. The Python implementation uses dictionaries, whose integer keys can collide adversarially. Its conservative bound is $O(s^2\ell + s\ell \log(s+1))$. Both are polynomial, but the sharper map bound is not advertised as the deterministic worst-case cost of these Python dictionaries.

## 127.5 Peeling links and restoring one-sided multiples

The default observer uses the existing quadrilateral-content reduction. Peel the minimum triangle coefficient at each global vertex, recording that many vertex links. Matching triangle differences are quadrilateral differences. If all quadrilaterals vanish, the peeled core vanishes too. Otherwise the quadrilateral gcd $g$ divides every remaining triangle, since one triangle at each vertex is zero and differences propagate over the connected corner graph. Query the divided core.

For a core orientable component, multiplying normal coordinates by $g$ gives $g$ copies. For a nonorientable component it gives $\lfloor g/2 \rfloor$ copies of its connected orientation double and one original precisely when $g$ is odd. Double both $\chi$ and $b$ on the orientation-double row. Restore peeled boundary-vertex links as discs and interior-vertex links as spheres, and combine equal types. A separate checker applies these typewise equations without calling the restoration producer.

The outer source-bound certificate binds the original labelled triangulation and coordinate vector, not just the quotient. Replay independently reconstructs the peel and divisibility, the boundary selector, both weighted histograms, the cover equations and the original summary. Euler totals must also equal the original geometry. Reducing core size does not remove input reading, fingerprinting, gcd, division or serialization costs on the original bits. The shared cycle allowance covers all three actual orbit discoveries; compilation and verification retain their cooperative checks. Incomplete discovery publishes no spectrum or completed certificate.

## 127.6 Validation, measurements and composition

The current-tree audit compares all 1,275 vectors in the maintained component corpus under both core settings and both merger thresholds: 5,100 complete spectra and independent certificates agree with the frozen component-level oracle. It retains 56 representative full proofs. A fresh Regina audit checks 1,395 API results and certificates on 664 bounded vectors from 26 triangulation isomorphism classes, including genus, crosscap and boundary multiplicities. The explicit reflection counterexample is rejected by the maintained checker while its original count proof remains valid. All 43 focused tests and all 1,312 maintained tests pass. Focused checks cover source and spectrum mutations, cover-only keys, zero signatures, binary multiplicities, coordinate reduction, callback propagation, exact cycle caps, producer-disabled replay and the newly rejected reflection language.

The isolated benchmark has five randomized rounds, with a separate control for each arm. All 300 measured calls and 60 warm-ups complete. The first comparison uses the existing coordinate census followed by connected topology classification as its reference. It computes the same final abstract spectrum, while additionally producing component normal coordinates. Both arms include independent proof replay and complete certificate serialization and hashing; input construction is outside the timer. All spectra agree, and proof records are deterministic within each arm.

| Input | coord. ms | spectrum ms | coord./new | coord. A/A | new A/A |
|---|---|---|---|---|---|
| Layered 8 | 8.881 | 9.467 | 0.946 | 0.992 | 0.996 |
| Layered 32 | 65.535 | 73.658 | 0.952 | 1.010 | 1.007 |
| Layered 64 | 239.303 | 261.880 | 0.916 | 0.958 | 1.006 |
| Layered 32, binary scale | 120.869 | 69.048 | 1.727 | 0.970 | 0.995 |
| Empty | 0.375 | 0.326 | 1.126 | 1.011 | 0.999 |
| Vertex link | 2.406 | 0.381 | 6.412 | 1.001 | 1.028 |
| Möbius band | 1.350 | 1.137 | 1.188 | 1.010 | 1.004 |
| Even Möbius multiple | 1.393 | 1.131 | 1.231 | 0.991 | 1.003 |
| Mixed components | 11.517 | 2.250 | 5.153 | 1.004 | 1.008 |
| Binary mixture | 15.039 | 2.679 | 5.650 | 1.012 | 0.999 |
| Klein bottle | 2.175 | 1.629 | 1.333 | 0.999 | 0.993 |

This comparison is mixed. Vertex links, the ordinary mixture and the binary mixture improve by factors 6.412, 5.153 and 5.650 respectively. The binary mixture's proof shrinks from 170,120 to 11,530 bytes. The primitive layered examples regress by about 5–9 percent despite the smaller weight dimension. Their connected-coordinate reference benefits from the existing single-orbit shortcut and the finite coorientation-derived double count. The new observer still discovers its full weighted double. Dimension reduction alone does not establish a universal speedup; eliminating that redundant double when a finite coorientation certificate exists is a further integration candidate.

The binary-scaled layered case multiplies the 32-tetrahedron meridian by $2^{2048} + 1$. The binary mixture uses independently large sphere, vertex-disc and one-sided populations, including a coefficient exceeding $2^{4096}$. Its spectrum is restored without expanding those populations.

The second comparison isolates the existing coordinate-core reduction, using the new direct two-weight observer as the reference and the reduced two-weight observer as the new arm. Both return the same original spectrum.

| Input | direct ms | core ms | direct/core | direct A/A | core A/A |
|---|---|---|---|---|---|
| Layered 32, binary scale | 111.527 | 69.691 | 1.618 | 1.007 | 0.998 |
| Vertex link | 2.309 | 0.364 | 6.380 | 0.988 | 1.014 |
| Even Möbius multiple | 1.203 | 1.151 | 1.044 | 1.010 | 0.995 |
| Binary mixture | 15.049 | 2.674 | 5.616 | 1.000 | 0.999 |

Reduction improves the scaled layered case by 1.618 and the binary mixture by 5.616. The binary

mixture's direct proof has 370,470 bytes, compared with 11,530 after reduction. In the pure-link case it removes all expensive orbit queries; on an even Möbius multiple the change is much smaller. Across both comparisons, control medians range from 0.958 to 1.028. Factors are medians of paired ratios, not ratios of the displayed timing medians. These are supplied-normal-vector API measurements, rather than complete knot-recognition timings or a complexity fit.

The runtime lives in the six new `fastunknot` observer modules and has an explicit `census`/`verify` command-line interface through `python -m fastunknot.normal_topology`. The preserved report remains unchanged. The maintained driver `../fast/normal_orbit_research/spectra.py` has `audit` and `benchmark` modes with an output path and optional rounds. Fresh records, proofs, source pins, tests and publication checks are `data/topology-spectra-*`.

If an outer algorithm makes $Q(n)$ queries of encoded size at most $S(n)$, this observer contributes $Q(n)\,\mathrm{poly}(S(n))$ time. Controlling both functions at quasipolynomial size would preserve that class for this stage. The integration does not prove those hypotheses. Abstract component types alone do not supply a useful vector, essential boundary slope, component coordinates, cutting instructions or attachment maps. The sector search and existing diagram-bound positive certificates remain relevant to those separate obligations.

# 128 Mathematical results preserved from reports 64–69

The six deliveries in arrival commit `fe88ae28e` contain mathematical results beyond implementation proposals. Their complete manuscripts, PDFs, proof appendices, source code and experimental records are preserved in `reports/64` through `reports/69`. This section records their mathematical content independently of whether a proposed implementation is promoted. A slower interface, a conditional recognition theorem, or a counterexample remains worth preserving. The next section treats the one runtime change selected from this batch.

All six ZIPs passed CRC validation. Placement removed only the common archive wrapper and redundant pure checksum manifests; text line endings were normalized. The retained placement record hashes every preserved file. Reports 64 and 69 also supplied whole-package SHA256 manifests, checked before normalization. The original arrival commit retains the original archives. There are 33, 33, 45, 649, 652 and 702 preserved files respectively, including historical source snapshots. These snapshots are evidence, not replacements for the maintained runtime. The report catalogue links each master manuscript.

## 128.1 A surface gluing identity and a disc criterion

Reports 64 and 65 work with $r \geq 1$ individually labelled, collared boundary intervals. Distinct intervals are disjoint including endpoints. Every fragment component meets an interval, has nonempty boundary, and retains boundary arcs outside the seam intervals. The quotient must be a surface. Let $P, Q$ partition the interval labels according to their component in the two fragments $S, T$. Form a bipartite multigraph $H$: its vertices are the blocks of $P$ and $Q$, and each label gives its own seam edge. Parallel edges must not be suppressed.

**Proposition 128.1** (Boundary-interval gluing)**.** *Under these hypotheses, the union $U$ satisfies*

$$c(U) = |P \vee Q|, \qquad \chi(U) = \chi(S) + \chi(T) - r,$$
$$\beta_1(U;\mathbb{F}_2) = \beta_1(S;\mathbb{F}_2) + \beta_1(T;\mathbb{F}_2) + r - |P| - |Q| + |P \vee Q|.$$

*In particular $U$ is a disc exactly when all fragment components are discs and $H$ is a tree.*

*Proof.* Connected components are exactly the connected components of $H$, giving the join formula. Each interval has Euler characteristic one, so cellular inclusion–exclusion gives the Euler formula.

A compact surface all of whose components have boundary has $H_2(-;\mathbb{F}_2) = 0$ and hence $\beta_1 = c-\chi$. Substitution proves the third identity; its last term is the cycle rank of $H$ and is nonnegative. If $U$ is a disc, all three nonnegative contributions vanish and $H$ is connected. A connected compact surface with boundary and first Betti number zero is a disc, so the fragment components are discs and $H$ is a tree. Conversely remove leaves of that tree: each step glues a disc along one boundary interval to another disc. The result remains a disc, irrespective of the interval parametrization. □

Thus the defect $\delta = c - \chi$ is additive with the graph cycle correction. Positive defect can be pruned only while the continuation obeys this seam contract. Later capping or cutting changes the problem: capping one boundary of an annulus gives a disc. Cyclic order and twists disappear from the tree criterion because it forbids cycles; they do not disappear from arbitrary surface gluing. A binary run describing $2^B$ parallel seams is not one of these individually labelled intervals. The width parameter is the actual interface, not the number of run records.

## 128.2 Exact exterior completion ranks and original witnesses

Set $n = r - 1$ and delete the root coordinate from the incidence representation of the complete graph on the labels, over $\mathbb{F}_2$. For a partition $P$, choose a spanning tree in each block and wedge its incidence columns to form $f_P \in \Lambda^{d(P)}(\mathbb{F}_2^n)$, where $d(P) = r - |P|$. The construction is independent of the chosen block trees: all give the same minors in characteristic two. A union of incidence columns is independent exactly when its multigraph is a forest. Dependence gives an even-degree subgraph containing a cycle; leaf removal proves the converse. The graph obtained from the two block forests and the seam graph have the same component and cycle counts. Consequently their union is a spanning tree exactly when the gluing graph is a tree. Equivalently, for $d(P)+d(Q) = n$, the top coefficient of $f_P \wedge f_Q$ is one exactly for a disc completion.

**Proposition 128.2** (Sharp completion dimensions)**.** *The disc-completion matrix has rank* $2^{r-1}$ *over* $\mathbb{F}_2$*. Its grade-d block has rank* $\binom{r-1}{d}$*.*

*Proof.* The exterior factorization gives the upper bounds. For each subset $A \subseteq \{1,\dots,n\}$, take the partition with block $\{0\} \cup A$ and all other labels singleton. Its feature is the standard wedge $e_A$. The complementary rooted-star partition has feature $e_{A^c}$; their pairing is nonzero, and pairings with other subsets of the same grade vanish. Thus the complementary-grade pairing is nondegenerate on the full standard basis. Summing the grade dimensions gives $2^n$. □

For a weighted family, process candidates by increasing cost and retain a candidate whenever its feature is independent of earlier retained features. Every dropped feature is a linear combination of features of no greater cost. Pairing with any completing feature gives one for a feasible dropped candidate, so at least one summand is feasible and costs no more. Since the retained family is a subfamily, it preserves the exact minimum for every admitted completion. It retains at most $2^{r-1}$ *original* witnesses. Linear combinations certify domination; they do not describe new embedded surfaces. Hidden geometric conditions must be uniform within each reduction type or proved to respect the feature map.

Rank-based weighted connectivity and matroid representative families are classical. Report 64 explicitly compares its construction with Bodlaender–Cygan–Kratsch–Nederlof and the acyclic representatives of Bergougnoux–Kanté, Theorem 3.8 [53, 54]. The sharp dimensions and their surface contract are the identities reviewed here; this is not a claim that rank-based compression itself is new.

## 128.3 Why full cut strata are larger

Report 64 also proves an exact comparison that need not be used by the runtime to remain mathematically useful. Anchor a cut variable $x_0 = 0$ and work with Boolean square-free polynomials over $\mathbb{F}_2$. The cut indicator of $P$ is

$$C_P(x) = \prod_{uv \text{ in a block forest of } P} (1 + x_u + x_v).$$

Its degree-$d(P)$ homogeneous part is $f_P$: choosing distinct variables from all factors gives the incidence permanent, equal to the determinant in characteristic two. Repeated variables contribute only lower degrees.

For $0 \le d \le r - 2$, the span $W_d$ of grade-$d$ cut functions is exactly the space of square-free polynomials of degree at most $d$. To see the inclusion $W_{d-1} \subseteq W_d$, a grade-$(d-1)$ partition has at least three blocks. Merge three chosen block pairs $ab, ac, bc$. Their indicators sum to the original indicator because the three equality factors sum to one. The rooted-star function $\prod_{i \in A}(1 + x_i)$ has leading term $x_A$. Induction therefore supplies every monomial of degree at most $d$. The reverse inclusion follows from the degree bound. At grade $r - 1$ there is just the one-block partition, with rank one. Hence

$$\operatorname{rank} C_{r,d} = \begin{cases} \sum_{j=0}^{d} \binom{r-1}{j}, & 0 \le d \le r-2, \\ 1, & d = r-1. \end{cases}$$

Grade-separated full cut bases retain $(r-1)2^{r-2} + 1$ rows for $r \ge 2$, whereas the homogeneous exterior bases retain $2^{r-1}$. Both remain single exponential. Projection removes information irrelevant to the complementary grade rather than changing the asymptotic class. A Boolean subset Möbius transform and grade projection convert an existing cut table in $O(r2^{r-1})$ binary scalar operations.

## 128.4 Signed compatibility and the finite-group formula

Report 66 adds orientation consistency to connectivity. A signed partition imposes $x_i + x_j = \epsilon_i + \epsilon_j$ within each block. Anchor $x_0 = 0$ and use the indicator vector of its satisfying assignments. The inner product of two such vectors counts their common assignments modulo two. An inconsistent union has none; a consistent union with $c$ blocks has $2^{c-1}$. This is one exactly when the union is connected. Thus the signed compatibility matrix also has rank $2^{r-1}$: one-block signed states give an identity submatrix. The same cost-ordered original-witness argument preserves minimum cost, including negative costs. It does not preserve the number of feasible witnesses.

The full finite-group result is stronger and is preserved even though no native use is yet selected. Let $G$ have order $q$. A gain partition labels each block by $\alpha_i \in G$ modulo simultaneous *left* multiplication and imposes $x_j = x_i \alpha_i^{-1} \alpha_j$. This convention works for noncommutative groups. There is no zero block. The number of states is

$$D_r(q) = \sum_{k=1}^{r} S(r,k) q^{r-k},$$

where $S(r,k)$ is a Stirling number of the second kind. Let $Z$ be the zeta matrix for extension of these constraints. Define the compatibility kernel $K_G$ to be one for consistent connected unions and zero otherwise.

**Proposition 128.3** (Integer factorization and field-dependent rank)**.** *With $k(s)$ the number of blocks of state $s$,*

$$K_G = Z \operatorname{diag}(g(s)) Z^{\mathsf{T}}, \qquad g(s) = (-q)^{k(s)-1}(k(s)-1)!,$$

$$\det K_G = \prod_{k=1}^{r} \left((-q)^{k-1}(k-1)!\right)^{S(r,k)q^{r-k}}.$$

*Its rank is $D_r(q)$ in characteristic zero. In characteristic $p > 0$ it is*

$$\begin{cases} q^{r-1}, & p \mid q, \\ \sum_{k=1}^{\min(r,p)} S(r,k) q^{r-k}, & p \nmid q. \end{cases}$$

*Proof.* Let $f$ indicate one-block states. A $k$-block state has $q^{k-1}$ one-block extensions. For each fixed extension, its interval is the ordinary partition lattice on those $k$ blocks: the final gauges fix all intermediate labels. Its Möbius value is $(-1)^{k-1}(k-1)!$. Möbius inversion of $f$ thus gives $g(s)$ above. Summing $g$ over common extensions of two states yields one for their consistent connected union and zero for every other union, giving the matrix identity. The zeta matrix is unitriangular over every field. Rank is therefore the number of nonzero diagonal entries, and its determinant is their product. In characteristic $p \mid q$ only $k = 1$ survives; otherwise $(k-1)!$ survives exactly for $k \le p$. □

Already the $r = 2, q = 2$ determinant is $-2$: the rank is two over $\mathbb{F}_2$ and three over $\mathbb{Q}$. For fixed $q$, the rational rank is $2^{\Theta(r \log r)}$, precluding a uniformly single-exponential *rational bilinear representation of this kernel*. This does not rule out other algorithms or more restricted continuation families. The factorization is an incidence-algebra argument in the gain-partition setting; the original manuscript records the classical Dowling and join-matrix context.

For actual orientable fragments, signed seam consistency encodes whether their orientations extend. Euler characteristic adds with the known seam correction. Cost $-\chi$ therefore preserves maximum Euler characteristic among connected orientable completions. A permanent unglued boundary anchor ensures nonempty boundary, where $\chi \le 1$ with equality precisely for a disc. The delivered complete *supplied* triangulated-patch selection algorithm costs $\mathrm{poly}(N, B, b)2^{3b}$ elementary bit operations for frontier width $b$. Whole-site symbolic transfers avoid allocating the temporarily larger exponential space. Reports 64 and 65 give corresponding typed assembly contracts. These are meaningful finite-input algorithms. General knot recognition still needs a source-complete geometric patch producer, controlled width, embeddedness and essential-boundary interfaces. The maintained diagram-to-exterior construction already exists; it does not by itself supply that assembly theorem.

## 128.5 Farkas certificates and a propagation obstruction

Report 67 treats canonical standard coordinates by choosing a zero triangle anchor at every global vertex. In a fixed allowed sector $J$, positive canonical Euler characteristic is rational feasibility of

$$M_J z = 0, \qquad z \ge 0, \qquad c_J^{\mathsf{T}} z \ge 1.$$

The scaling inequality is exact, not a numerical positivity threshold. A negative homogeneous Farkas certificate is a rational $y$ with $M_J^{\mathsf{T}} y \ge c_J$: every feasible nonnegative matching vector then has $c_J^{\mathsf{T}} z \le 0$. The dual alternative gives completeness for this homogeneous linear problem. Anchor enumeration covers the canonical cone. A componentwise objective envelope may prove a negative result, but a positive envelope must be checked against an actual anchor objective.

If forbidding a type makes the positive polyhedron empty, that type is mandatory. Every admissible positive witness must then set its two incompatible peers to zero. Repeated deletions preserve every admissible positive witness; exhaustive propagation has an order-independent terminal closure. A strong mandatory-only backdoor requires this closure to leave compatible allowed masks for every anchor and every assignment on its chosen tetrahedra. This is stronger than finding an admissible basic vector.

For a canonical admissible positive-Euler vector $x$, let $Z(x)$ be its tetrahedra with all three quadrilateral coordinates zero, and put $U(T) = \bigcup_x Z(x)$. Report 67 proves

$$U(T) \subseteq B, \qquad \beta_{\mathrm{LP}}(T) \ge |U(T)| \ge t - |\operatorname{qsupp}(x)|$$

for every strong mandatory-only backdoor $B$. Choose anchors and assignments consistent with $x$. It survives every deletion. At a tetrahedron of $Z(x)$ outside $B$, $x$ witnesses feasibility with each type forbidden, so none can become mandatory. No peer deletion occurs there and all three types remain allowed, contradicting terminal mask compatibility. Taking the union proves the obstruction. Sparse positive witnesses can thus force a large strong parameter, even when they are immediately easy to find.

A policy-dependent deciding set permits early termination on an actual admissible positive vector. It avoids that mask-only obstruction on positive instances, but on instances with no canonical admissible positive-Euler vector it equals the strong parameter. That negative condition is not the same as knottedness. With a polynomial-time exact LP policy, a supplied deciding set of size $b$ gives $A(T)3^b\,\mathrm{poly}(t)$ work; search for an unknown set costs $A(T)\sum_{j\le b}\binom{t}{j}3^j\,\mathrm{poly}(t)$. For one vertex $A(T)=4t$. The delivered Bland simplex has no proved polynomial pivot bound, and no general logarithmic deciding-set theorem is supplied. The manuscript also proves that the whole Fibonacci layered family has one positive canonical admissible meridian ray, full quadrilateral support, and mandatory strong backdoor zero. Large support and easy discovery are compatible; the family does not establish a universal parameter bound. These proofs and negative certificates are preserved without installing the larger LP overlay over the newer runtime.

## 128.6 Exact transport changes and disconnected primitive fibres

Report 68 analyzes a legal relative 2–3 bipyramid move for a transported integral cocycle. Sort the three belt heights as $x \le y \le z$ and the two apex heights as $L \le U$. Set $[a]_+ = \max(a, 0)$ and

$$\begin{aligned} g &= [L - z]_+ + [x - U]_+, \\ \Delta P &= \max(U, y) - \min(L, y) + [\min(U, x) - L]_+ + [U - \max(L, z)]_+. \end{aligned}$$

Then the upward move changes Euler characteristic by $-2g$ and the number of normal pieces by $\Delta P \ge U - L \ge 0$. The downward move has the opposite changes. The proof counts relative cells: the vertex change is $|A - B|$, the internal arc change is $\sum_{C,D,E} \mathrm{range}(A, B, C) - \mathrm{range}(C, D, E)$, and the piece change is the difference of the five tetrahedral height spans. Unchanged boundary cells cancel. Sorting the heights reduces these span expressions to the displayed formulas. A new upward edge difference is at most twice the previous maximum coefficient; downward removal adds none. Thus coefficient-bit growth can be charged to upward moves.

Crucially, preserving a primitive cohomology class does not preserve connectedness of its fibre. The retained source-bound example changes a connected 24-piece essential disc of Euler characteristic one into an 11-piece disc and two four-piece spheres. The total has three components, 19 pieces and Euler characteristic five. Belt/apex heights are $(0, 2, 1, 5, 4)$: $g = 2$ and $\Delta P = 5$ for the reversed upward move. Fresh replay checks both source-bound moves, both archived component certificates and both current topology-spectrum certificates. An independent cell-span calculation checks all $5^5 = 3125$ small height assignments. Primitive *matching-coordinate rays*, used in the next section, have a different connectedness proof. They must not be conflated with primitive cohomology fibres.

There are two further useful results even without native promotion. First, let $m_v$ be the minimum triangle coordinate at global vertex $v$. During a fixed-state scoring pass, the thirteen least corner records per vertex suffice to find each new minimum after a legal 3–2 move: at most twelve old corners are removed, so a surviving minimum occurs among the first thirteen; compare it with the at most eight new corners. This gives linear preparation and constant integer work per candidate. Summaries must be rebuilt after committing a move. Global regauging is outside the lemma's contract. The refined scorer is proved but unimplemented.

The same report proves $m'_v \ge m_v$ and nonincrease of the piece count after vertex-link peeling. Write $a = [\min(U, x) - L]_+$ and $b = [U - \max(L, z)]_+$. The local apical minima rise by $g$ and belt

minima remain unchanged. Taking minima over identified vertices and unchanged exterior corners gives $m'_v \geq m_v$. If $r_v$ is the number of formal apices identified with $v$, then $|C'_v| = |C_v| - 2r_v$, $\sum_v r_v m_v \leq a + b$, and $\Delta P \geq 2(a + b)$. With $\delta_v = m'_v - m_v \geq 0$ the peeled change is

$$-\Delta P + 2\sum_v r_v m_v - \sum_v \delta_v |C'_v| \leq 0.$$

This excludes gcd division and global regauging. It proves neither peeled Euler monotonicity nor bounded essential-disc discovery.

Second, cutting a marked degree-two boundary curve leaves paths with exactly two endpoint *occurrences*. Four additive weights—count, sum of endpoint labels, sum of squared labels, and length—recover a path's endpoints: if $S = x+y$ and $Q = x^2+y^2$, then $(x-y)^2 = 2Q-S^2$. Integral recovery and endpoint adjacency restore the cyclic order. This is a degree-two interface theorem, not reconstruction of arbitrary attachments from moments. The report's broader transport search bounds require explicit coverage assumptions; its surveyed recognition coverage does not increase. The complete theorem statements and negative examples remain in the preserved manuscript.

## 128.7 Support projections and the number of component types

Report 69 fixes the positive support $S$ of a supplied admissible matching vector and its rational matching nullity $d$. Projection to coordinates $J$ is injective on the supported kernel exactly when the complementary columns are independent. A minimum-cost coordinate projection therefore retains exactly $d$ coordinates: maximize the total weight of the complementary column basis by the weighted matroid greedy rule. Here costs are allocated binary slot sizes. A rational decoder and a checked nonsingular modular minor certify exact rank and kernel reconstruction. This optimality is among coordinate projections with a linear decoder on the entire supported space; it is not an optimality claim among all compressed encodings. Under uniform positive scaling, reusing the projection costs within $d$ bits of the new optimum because each slot gains the same scale bit length up to one bit. Capacity-based scalar packing preserves equality of every live point-weight run of this selected-vector observer, not merely its final values. It does not prove fewer orbit runs. The full projection interface has measured regressions and remains optional research code.

**Proposition 128.4** (Unmarked normal component-type inventory)**.** *For the report's compact orientable manifold setting, let $H(x)$ count distinct full normal-coordinate component types and let $a$ count supported vertex-link types. Then $H(x) \leq 2d - a$. If all components are two-sided, $H(x) \leq d$.*

*Proof.* Distinct mutually disjoint two-sided normal types have independent rational coordinate vectors. A dependence can be separated into two positive integer combinations. Both are realized by disjoint parallel copies of the selected two-sided surfaces. Equal coordinates imply normal isotopy and hence equal component-type multisets, contradicting their disjoint type sets.

For each nonlink type choose one component. Replace each one-sided component by its connected two-sided horizontal interval-neighbourhood frontier, whose vector is twice the original. Choose neighbourhoods disjointly and coalesce equal resulting vectors, giving $h$ nonlink two-sided types. Each has at most two original preimages, $z$ and $z/2$. It cannot become a vertex link because its positive quadrilateral survives doubling. Along with the $a$ link vectors, independence gives $h + a \leq d$, while $H - a \leq 2h$. Combining proves $H \leq 2d - a$. □

The bound counts coordinate types, not components or marked attachments. The one-tetrahedron Möbius source $3M$ has $d = 1, a = 0, H = 2$; adding its disjoint vertex link gives $d = 2, a = 1, H = 3$. Both attain the bound. Binary multiplicities remain essential. No global hierarchy branching bound follows without a theorem controlling the markings.

The report also gives safe geometric support blocks: connect positive types using their occurrences in each *original, uncancelled* face equation. Restricting the source to a block preserves matching, and actual face attachments never leave a block. This gives a disjoint surface decomposition and a direct sum of matching spaces. A graph built only after algebraic cancellation can omit geometric attachments and needs a separate proof. Its unit-pivot specialization is the integration in the next section.

### 128.8 Fresh validation and limits of the review

Scratch copies of reports 64–66 pass 18, 36 and 55 test methods respectively. Their fresh audits reproduce: report 64's 44,168 partition pairs, 1,155 incidence minors, 3,336 weighted queries, 72 independent certificates, 2,000 ribbon surfaces, 5,295 projections and 1,800 staged joins; report 65's 813,297 direct matrix entries, 14,014 completions from 1,000 weighted families, 1,000 literal triangulated gluings and 200 grammar comparisons; report 66's 68,581 signed entries, 442 triangulated gluings and all 297 assignments in the finite patch system. The transport audit is described above. These finite checks supplement the manuscript proofs; they cannot establish their infinite statements by exhaustion. No fresh execution of the complete report-67 LP overlay is claimed. Its original evidence is preserved and its polynomial-policy hypothesis remains explicit.

The mathematical review accepts the statements under their stated contracts, without claiming formal proof-assistant verification or originality beyond the manuscripts' attributions. All original proofs remain available even where this synthesis gives only their central argument. The generic LP, geometric transport, signed-patch and packed-coordinate overlays have not been installed over the newer native tree. Neither these mathematical results nor the local speedup below supplies a general quasipolynomial or general subexponential unknot-recognition bound.

## 129 Unit-pivot rays before disc orbit discovery

Report 69 supplies a particularly small sufficient test for a one-dimensional normal matching support. The maintained disc-count API now tries that test after its existing vertex-link removal and quadrilateral-gcd reduction. Success gives a checked connected primitive surface, whose Euler characteristic and boundary homology determine the essential-disc count. Failure uses the complete previous component observer. This is an adaptive choice of proof method based on an inexpensive structural certificate; it does not guess connectedness from coefficient size or from cohomological primitivity. The unit producer and checker are copied unchanged from the delivery; the surrounding disc protocol and native validation are new.

### 129.1 Triangular unit rows certify the integral ray

Let $x \neq 0$ be the validated nonnegative admissible vector, $S$ its positive support, and $M_S$ the matching matrix restricted to that support. Choose a seed $s \in S$. A transcript lists $|S| - 1$ pairs $(\text{row}, \text{pivot})$. At each step the row has exactly one supported coordinate not yet known, its pivot, and its coefficient is $+1$ or $-1$. After the final step all supported coordinates are known. The source already establishes existence; the transcript establishes uniqueness.

**Proposition 129.1** (Integral unit-ray witness)**.** *Such a transcript proves* $\dim_{\mathbb{Q}} \ker M_S = 1$*. There is an integral kernel generator* $u$ *with* $u_s = 1$*, and* $x = x_s u$*. In particular* $\gcd_{i \in S} x_i = x_s$*.*

*Proof.* Set the seed value to one. Each row solves its new coordinate as an integer linear combination of earlier known coordinates, since its pivot is a unit. In discovery order these $|S| - 1$ rows have a triangular minor with diagonal entries $\pm 1$, so their rank is $|S| - 1$. The nonzero matching source gives nullity at least one. Thus every kernel vector is determined by its seed,

and the constructed $u$ equals $x/x_s$, satisfies all remaining equations, and is positive on $S$. Its integral coordinates include one, proving the gcd assertion. No values need to be propagated in the certificate planner: the already validated source and the ordered row pattern suffice. □

The producer chooses the minimum-bit positive coordinate, breaking ties by index. It keeps incident rows, an unknown count and the XOR of unknown column indices for each row. Processing a known column decrements each incident count once. A row reaching one unknown exposes that column if its coefficient is a unit. Each supported column is processed once and each incidence once. The independent checker instead reconstructs the support and examines the chosen original rows in order; it imports no planner. Matching rows have bounded local size, so both use linear sparse-incidence work, besides geometry preparation and source arithmetic. The indexed transcript has $O(|S|\log(t+1))$ bits. This is not a linear-bit-time claim for reading arbitrarily large coordinates.

Failure of this gate does not prove nullity greater than one. For example the rank-one equation $2x_0 - 3x_1 = 0$ with source $(3, 2)$ has no unit pivot. Multiple free variables, an empty support, or a seed that does not expose all columns also miss this sufficient criterion. No repeated seed search or Gaussian elimination is introduced on a miss.

## 129.2 Connectedness and the exact essential-disc count

Write the original source as vertex links plus $mv$, where the existing canonical routine removes maximal complete triangle layers and divides by the quadrilateral gcd $m$. A nonzero reduced core $v$ is primitive: its quadrilateral gcd is one and matching forces the peeled triangle content to divide by the same gcd. A successful unit transcript therefore has seed coordinate exactly one; both producer and checker enforce it.

**Lemma 129.2** (Primitive supported ray is connected)**.** *If an admissible primitive normal vector $v$ spans its supported matching kernel, then its normal surface is connected.*

*Proof.* Each component has a nonnegative integral matching vector supported within $v$, hence equals $\lambda v$ for some positive rational $\lambda$. For a unit seed, $\lambda$ is the component's integral seed value. More generally primitivity and a Bézout combination of the entries also force $\lambda$ to be integral. These positive integers sum to one. There can be only one component. □

In the compact orientable manifold with torus boundary validated by the existing geometry, a connected surface with $\chi(v) = 1$ is a disc when its boundary is nonempty; the closed alternative is a projective plane. The boundary class is computed from two checked torus homology cycles, by intersecting each with the normal boundary modulo two. A nonzero class excludes the closed alternative and proves essential disc boundary. Conversely an essential simple circle on a torus has primitive slope, so its two mod-two coordinates cannot both vanish. Therefore

$$D(mv) = m\,\mathbf{1}\{\chi(v) = 1 \text{ and } [\partial v] \neq 0 \text{ in } H_1(\partial T; \mathbb{F}_2)\}.$$

Here $D$ counts essential-disc components, not all components.

The zero case is also exact: any disc component of $mv$ has vector $kv$ for a positive integer $k$, and $1 = k\chi(v)$ forces $k = 1$ and $\chi(v) = 1$. Its essential boundary supplies the other condition. A disc is two-sided in this ambient manifold, so a positive result consists of $m$ parallel copies. One-sided primitive rays need not have $m$ total components: even scaling gives their connected two-sided frontier. That frontier has Euler characteristic $2\chi(v)$ and cannot create a disc. Removed vertex links are spheres or boundary-parallel discs and contribute zero to $D$. This proves the displayed count for the whole original source.

The argument uses rank and integral coordinates. The disconnected primitive-cohomology example of Section 128 does not satisfy this reasoning merely because its class is primitive.

## 129.3 The whole Fibonacci family has short witnesses

The preserved report proves more than the finite tests. In its actual layered fixture, the three supported coordinates per tetrahedron are $a_i = b_i = F_{t-i+1}$ and $c_i = F_{t-i}$, with $F_1 = F_2 = 1$. The original face equations for $i < t - 1$ include

$$b_i = b_{i+1} + c_{i+1}, \quad c_i = a_{i+1}, \quad a_i = a_{i+1} + c_{i+1}, \quad c_i = b_{i+1},$$

and the terminal triple is equal. For $t = 1$ the chosen seed is $a_0$; otherwise it is $c_{t-2}$. The equalities expose the terminal triple, then the recurrence exposes each previous triple. Every pivot is a unit, so there are exactly $3t - 1$ discovery steps for every $t \geq 1$ and positive scale $m$. The next larger value is at least twice the seed, keeping the minimum-bit/index choice fixed under scaling.

For completeness the source cell counts also prove the topology, rather than assuming a descriptive fixture name. Its global edge weights are $F_2, \ldots, F_{t+3}$, giving $V = F_{t+5} - 2$; its pieces number $P = F_{t+5} - 5$. The total local side count is $6F_{t+3} + 4F_{t+2} - 16$, with $2F_{t+3}$ unpaired boundary arcs, so $E = 2F_{t+5} - 8$ and $\chi = V - E + P = 1$. Boundary weights on the one-vertex torus are $F_{t+1}, F_{t+2}, F_{t+3}$. Adjacent Fibonacci numbers are coprime; their mod-two cocycle is nonzero. A one-vertex triangulation has no nonzero vertex coboundaries, so the boundary class is nonzero. The ray theorem gives exactly $m$ essential discs.

The primitive piece count is exponential in $t$, but the indexed rank witness has $O(t \log(t + 1))$ bits. The full binary source has $\Theta(t^2 + t \log(m + 1))$ bits. This is a family theorem, not a linear-time claim in the tetrahedron count for the entire call. Unlike the delivery's full ray-block API, the native adaptation deliberately retains its existing gcd and division prefix. Only the subsequent unit-witness stage avoids value propagation and elimination. The whole call is still polynomial in its encoded supplied input; its proof and orbit work are substantially smaller on these dense rays.

## 129.4 Protocol, fallback and resource semantics

`normal_compressing_disk_count` defaults to `unit_ray=True`, with strict Boolean option validation. A successful gate returns `normal-disc-count-v2`; its seven outer fields bind the input fingerprint, divisor, vertex links, core coordinates, core certificate and final count. The inner `normal-unit-ray-disc-v1` has exactly six fields: its schema, support transcript, torus homology basis, Euler value, two mod-two boundary coordinates and primitive disc count. The transcript itself has strict support, seed, step and nullity fields.

Replay independently rebuilds the original geometry, source binding, link/gcd reduction, matching support, triangular rank evidence, torus basis, Euler characteristic and boundary parities. It invokes neither the planner, the canonical reduction producer nor orbit discovery. Shared finite geometry and arithmetic helpers remain part of the trusted mathematical boundary; the producer and replay control paths are distinct. An incorrect primitive seed is rejected, rather than used as an asserted multiplicity.

An unsuccessful gate retains the complete component query and its version-one proof. Setting `unit_ray=False` reproduces the old schedule and result exactly. Pure vertex links retain the old zero-core proof. Existing version-one certificates remain accepted. The historical coherent-obstruction driver explicitly disables the gate to preserve its published proof bytes.

The shortcut reports zero orbit cycles, zero weight runs and zero replay events, with its own ray-step count. A `max_cycles=0` allowance bounds orbit discovery, so it permits this algebraic path. The supplied-sector test now correctly finds a certified meridian with zero orbit allowance; a zero basis-enumeration allowance still prevents candidate discovery. Callback cancellation covers preparation, planning and replay and propagates to callers. A missed gate with an exhausted orbit allowance remains inconclusive; it does not certify a negative knot verdict.

The source pipeline still matters. The supplied-sector producer benefits through its disc checker,

but the main cocycle candidate schedule does not use this API and has not acquired new source coverage. This integration is an exact observer improvement for a supplied vector or a supplied sector, not a new complete diagram-recognition implementation.

### 129.5 Validation and complete-call timings

All 1,317 maintained test methods pass in 236.139 seconds. The focused 34-method run passes in 2.896 seconds. New tests cover layered sizes 1, 2, 8, 32, 128 and 256; exact $3t - 1$ transcripts; relabelings; links and binary scalings through $2^{20,000}$; one-sided negatives; nonunit rank-one misses; strict option types; callback cancellation; and field/row/seed mutations with producer-disabled replay.

The corpus audit compares 1,275 vectors under both periodic rules, or 2,550 configurations, against the frozen `66098968e` count producer and checker and the corpus component oracle. Every count agrees. The gate succeeds on 1,878 configurations and uses zero orbit cycles; the 672 misses reproduce the complete baseline result. Disabling the gate reproduces all 2,550 baseline results exactly. The audit retains 44 representative complete proofs and 553 source hashes. Verification of old and new proof languages is checked separately. The corpus contains duplicates by scale and relabeling, so 1,878 is not a count of distinct knot types.

The isolated benchmark uses five rounds and four interleaved arms, including old/old and new/new controls: 220 measured calls and 44 warmups all complete. Each timed call includes production, independent replay and serialization/hash of the complete certificate. The count reference is the frozen producer *and* checker; for supplied-sector calls its disc producer is replaced by that baseline and the maintained outer replay accepts its legacy proof. All arms agree on counts or discovered-coordinate hashes, and proof bytes are deterministic within each arm. Timing ratios below are medians of paired old/new ratios, not ratios of separate medians.

**Supplied-vector disc counts.**

| Input | old ms | new ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| Layered 16 | 12.891 | 3.556 | 3.639 | 0.964 | 1.014 |
| Layered 64 | 229.634 | 26.748 | 8.685 | 1.010 | 0.997 |
| Layered 256 | 1811.970 | 57.238 | 31.386 | 1.003 | 1.012 |
| Layered 64, binary scale | 130.841 | 19.010 | 6.924 | 1.036 | 0.997 |
| Empty | 0.226 | 0.226 | 0.992 | 0.988 | 1.000 |
| One-sided | 0.777 | 0.390 | 1.995 | 0.998 | 0.989 |
| Vertex link | 0.235 | 0.231 | 1.019 | 1.007 | 1.003 |
| Sphere/disc/Mobius mix | 1.898 | 1.010 | 1.901 | 0.994 | 0.994 |
| Sphere/negative Euler mix | 4.298 | 1.616 | 2.654 | 1.001 | 0.995 |

**Complete supplied-sector discovery.**

| Input | old ms | new ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| Layered 16 | 55.591 | 42.830 | 1.303 | 1.001 | 1.002 |
| Layered 64 | 1945.525 | 1802.229 | 1.068 | 1.006 | 1.013 |

The layered-256 query drops from about 1.812 seconds to 0.057 seconds, a paired 31.39-fold gain. Its complete certificate shrinks from 7,098,789 to 36,240 bytes. Layered-64 improves 8.68-fold; its $2^{4096} + 1$ multiple improves 6.92-fold after paying the existing reduction prefix. Empty and pure-link controls are near parity. The mixture cases benefit because link removal leaves a certified ray; this is not promotion of a generic multi-block decomposition.

Complete supplied-sector discovery improves only 1.30-fold at size 16 and 1.07-fold at size 64 because enumeration remains dominant. The A/A medians range from 0.964 to 1.036; the raw

records preserve all rounds and bytes. No full knot-recognition timing is inferred from these measurements. Reproduction is `python -B -m normal_orbit_research.unit_rays audit` and `benchmark -rounds 5`, each with a required `-output` path, from `fast/`. The retained records are `data/unit-ray-audit.json`, `data/unit-ray-benchmark.json` and both test logs. The remaining global problem is controlled complete witness discovery, not the already polynomial inspection of one supplied normal vector. No general quasipolynomial bound follows from this shortcut.

# 130 Deriving a weighted normal double from coorientation

The first native topology-spectrum integration of Section 127 used one boundary query and two weighted surface queries. The ordinary component-count API already had a finite sufficient certificate that the normal double is two copies of its source (Section 123). This section extends that certificate to the entire weighted histogram. A successful finite colouring now removes the second weighted discovery and its replay, rather than merely predicting its number of orbits. An unsuccessful colouring uses the complete double query. The returned topology spectrum remains exact in both cases.

## 130.1 A weight-preserving cover isomorphism

The base normal interval system has points $j$ and the doubled system has adjacent lifts $2j + s$, $s \in \{0, 1\}$. A base arc pairing $\phi$ has reversal bit $r$. Its doubled pairing is

$$2j + s \longmapsto 2\phi(j) + (s \oplus r).$$

For a preserving map this follows by doubling its translation and interval endpoints. For a reversing map, its doubled endpoint is $2d + 1$ rather than $2d$, and substitution reverses the sheet bit. The bit is still necessary when the base interval contains just one point; forgetting it loses the normal orientation cover.

Give each nonempty ambient-edge block a bit $\epsilon_u$. For every base pairing from block $u$ to block $v$, require

$$\epsilon_u \oplus \epsilon_v = r.$$

These are the established block-constant coorientation constraints. Blocks with no incident pairing may take bit zero. Failure is only failure of this sufficient finite representation: an actual coorientation may vary within one large block.

**Proposition 130.1** (Weighted double trivialization)**.** *If these constraints are satisfied, let the base point weight be any vector $w(j) \in \mathbb{Z}^d$ and suppose the doubled weight obeys $w_2(2j + s) = w(j)$. If $H(v)$ is the base orbit histogram and $K(v)$ the doubled histogram, then $K(v) = 2H(v)$ for every weight vector $v$.*

*Proof.* The corrected sheet label $\sigma(j, s) = s \oplus \epsilon_{\mathrm{block}(j)}$ is invariant under every doubled pairing, by the displayed constraint. The map $(j, s) \mapsto (j, \sigma(j, s))$ is a bijection from the doubled point set to two labelled copies of the base set and conjugates each generator to the base generator within a fixed label. Thus each doubled orbit maps bijectively to one base orbit. By the point-weight hypothesis that bijection preserves its sum. Each base orbit has one copy of each label, proving the histogram identity, including zero or negative weight entries and binary multiplicities. □

The multiplicity doubles; the signature does not. A base component of signature $(\chi, b)$ gives two components of signature $(\chi, b)$, not a component of signature $(2\chi, 2b)$. The result is an interval-cover theorem for arbitrary additive weights. The maintained application uses $d = 2$ and does not introduce a general new weighted-proof API.

## 130.2 Both topology weights satisfy the hypothesis

The Euler weight is the sum of point ownership for vertices, minus arc ownership for surface edges, plus corner ownership for discs. Every original half-open ownership interval $[a, b)$ becomes $[2a, 2b)$ at scale two with the same integer weight. Hence both lifts of every point receive its old Euler contribution, even where many ownership intervals overlap.

For the second coordinate, the boundary transversal selects one original point on each actual boundary circle. The embedding into the full point universe preserves edge-block offsets. Its doubled interval selects both adjacent lifts of that point. These are precisely the two lifted boundary markers, so the boundary-marker point weight also pulls back unchanged. No new doubled boundary transversal is required.

In an orientable ambient manifold the normal double is the orientation cover. The colouring therefore also proves every source component orientable. The usual identities $H = O + N$ and $K(v) = 2O(v) + N(v/2)$ then agree with $O = H$, $N = 0$. The maintained sparse inversion and its independent forward checks are still used, avoiding a second special spectrum-restoration implementation. At the zero signature, $N(0) = 2H(0) - K(0) = 0$ and $O(0) = H(0)$, so tori/Klein-bottle ambiguity is resolved by the geometric certificate, not by an Euler-value heuristic. The existing one-sided scaling and vertex-link restoration remain unchanged on fallback inputs.

The finite colouring is never inferred from a primitive cohomology class, from positive Euler characteristic, or from a primitive coordinate vector. The disconnected primitive fibre of Section 128 illustrates why source-specific proofs matter. Even an orientable even multiple of a Möbius band can fail the block-constant test. Such an input receives a complete ordinary double query, not a nonorientability verdict.

## 130.3 Native planning, replay and work accounting

After the base weighted query completes, discovery reconstructs the finite graph from the actual base pairings. A parity search either supplies vertex bits or finds an odd cycle. A zero-parity witness is independently checked before use. Success constructs the doubled histogram by doubling each retained multiplicity and skips construction and weighted discovery of the scale-two system. Failure continues the previous query schedule. The geometry, boundary selector and base weighted observer are unchanged.

There are $O(t)$ nonempty edge blocks and pairings. The graph implementation locates the block of each endpoint by binary search, and the parity producer sorts its vertex keys, giving $O(t \log(t+1))$ indexed arithmetic operations besides source preparation. Endpoint comparisons have the source's binary bit bound. The colouring has $O(t \log(t+1))$ encoded bits, and constructing the histogram takes one checked binary multiplication per retained row. Neither normal points nor components are expanded. On a miss this finite work is extra overhead; it is not silently free in the timings. The entire observer retains its polynomial encoded-input bound. Fewer local queries do not prove a bound on the outer recognition search or hierarchy states.

The option `coorientation=True` is now the default of `normal_topology_spectrum`; it requires a strict Boolean. Setting it false reproduces the old schedule, statistics and certificate exactly. The CLI exposes `-no-coorientation`. Pure vertex-link reduced inputs still require no orbit query and retain their original certificate.

Successful derivation uses outer schema `normal-topology-spectrum-v2`, with the same eight fields as version one. Its query still contains the boundary transversal, base surface proof, double proof and core spectrum. The new inner `normal-topology-double-coorientation-v1` has exactly four fields: schema, dimension two, zero-parity colouring and histogram. It is not an ordinary weighted-orbit certificate and is accepted only through the source-bound outer protocol. The colouring cannot be transplanted to another source graph merely because the claimed histogram

is plausible.

Independent replay first rebuilds source geometry and canonical reduction, checks the boundary selector and fully verifies the base weighted proof. It then reconstructs the graph from that actual base system, checks every colour constraint, and compares the complete proposed double histogram with twice the already certified base histogram. It subsequently checks the spectrum and multiplicity restoration through their existing forward identities. Replay invokes neither the colouring search, orbit discovery, weighted producer, sparse inversion nor canonical-core producer. Finite geometry helpers are shared trusted definitions. Version-one proofs retain their full explicit double replay; a schema swap cannot turn one proof language into the other.

Statistics count actual discoveries. Derivation uses two calls rather than three and records zero double orbit cycles. A shared cycle allowance can therefore admit a complete derived proof where the old schedule is inconclusive. Preparation, colouring and verification remain governed by the cooperative callback, rather than being counted as orbit cycles. An incomplete base query cannot acquire a derived certificate, and a missed colouring cannot bypass an exhausted double allowance.

## 130.4 Validation and performance

The accompanying validation compares the complete maintained observer with a frozen pre-change producer *and* checker, under both periodic rules and both coordinate-reduction modes. It also compares each complete spectrum with the independent component records retained in report 53. The disabled option must reproduce every complete old result, and each failed colouring must do the same. Successful cases must save exactly the old double cycle count and return only orientable components. Retained source-bound proofs are replayed with producer entry points disabled.

Tests independently expand small lifted point systems to check $w_2(2j+s) = w(j)$ and the generator conjugacy, including reversal bits. They also cover relabelings, huge binary scalings, orientable coarse misses, one-sided inputs, strict proof fields, Boolean/integer confusion, altered weights and multiplicities, duplicate rows, colour flips and missing vertices, legacy replay, exact cycle allowances and caller cancellation. A simultaneous colour reversal is accepted because it is a legitimate alternate labelling of the two sheets.

All 1,322 maintained test methods pass in 223.502 seconds; the focused 39-method run passes in 1.582 seconds. The audit covers 1,275 supplied vectors under two periodic rules and two reduction modes, or 5,100 configurations. Every full spectrum and source summary agrees with the frozen `ea366dc9b` producer and checker and with the independent component oracle. All 5,100 disabled-option results reproduce the baseline exactly. Derivation succeeds on 1,088 configurations and saves precisely the old double-query cycle count; the other 4,012 return the complete old result. These counts include scaled and relabelled repeats, not 1,088 distinct knot types. The audit retains 80 paired current/legacy examples and 557 source hashes.

A separate fresh Regina audit rechecks 664 bounded vectors across 26 triangulation isomorphism classes. All 1,395 produced certificates replay and match the freshly computed component types. The controls include orientable genus nine, ten crosscaps and fifteen boundary circles on one component. Runtime sources remain unchanged during each audit.

The isolated five-round benchmark has 300 measured calls and 60 warmups, all complete, with interleaved old/old and new/new arms. Timings include production, independent replay and serialization/hash of the full proof. The native reference is the frozen producer *and* checker. Each pair returns the same spectrum hash, and proof bytes are deterministic within each arm. The following ratios are paired medians.

**Full native topology observer.**

| Input | old ms | new ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| Layered 8 | 9.259 | 4.958 | 1.868 | 1.000 | 0.994 |
| Layered 32 | 68.196 | 32.955 | 2.047 | 0.988 | 0.980 |
| Layered 64 | 251.061 | 114.663 | 2.190 | 1.017 | 1.002 |
| Layered 32, binary scale | 69.058 | 34.356 | 2.004 | 0.997 | 1.003 |
| Empty | 0.305 | 0.307 | 0.993 | 0.952 | 0.990 |
| Vertex link | 0.325 | 0.324 | 1.007 | 1.012 | 0.970 |
| Möbius band | 1.107 | 1.145 | 0.967 | 0.991 | 1.009 |
| Even Möbius multiple | 1.126 | 1.132 | 0.985 | 0.999 | 0.987 |
| Mixed components | 2.218 | 2.228 | 0.995 | 0.994 | 0.978 |
| Binary mixture | 2.641 | 3.496 | 0.861 | 0.775 | 0.997 |
| Klein bottle | 1.635 | 1.651 | 0.990 | 1.004 | 1.008 |
| Layered 256 | 3867.661 | 1704.070 | 2.261 | 0.998 | 0.999 |

The layered-256 call improves from about 3.868 to 1.704 seconds, a paired 2.26-fold gain. Its certificate drops from 14,211,485 to 7,139,727 bytes. The smaller layered gains are 1.87–2.19-fold; the binary-scale case gains 2.00-fold. Pure links and empty reduced inputs keep the identical old proof and are near parity.

**Earlier coordinate-census comparison.**

| Input | coord. ms | new ms | coord./new | coord. A/A | new A/A |
|---|---|---|---|---|---|
| Layered 8 | 8.789 | 5.036 | 1.758 | 0.983 | 0.963 |
| Layered 32 | 62.932 | 32.668 | 1.935 | 0.996 | 0.998 |
| Layered 64 | 238.759 | 115.228 | 2.102 | 1.020 | 0.993 |

The coordinate reference obtains full component normal vectors, then checks their topology with the existing scalar observer; it computes finer embedding data than requested here. This extra work is stated explicitly. The new topology observer is 1.76–2.10-fold faster on these three primitive layered controls, resolving the 5–9 percent regressions recorded for the previous three-query integration. These are supplied-vector calls, not whole knot-recognition measurements.

The original binary-mixture ratio 0.861 comes with a poor old/old control of 0.775. The original records are retained. A justified isolated repeat uses fifteen rounds on that case and two other fallback controls, or 180 measured calls and 12 warmups, all complete. Paired old/new medians are 0.985 for the Möbius band, 0.993 for the small mixture and 0.994 for the binary mixture. Repeat A/A medians range from 0.993 to 1.006. The repeat indicates small extra finite-test costs rather than establishing a fourteen-percent binary-input regression. It does not erase the initial noisy measurement or assert zero miss overhead.

Raw evidence is in `data/weighted-coorientation-*`. From `fast/`, run `python -B -m` with module `normal_orbit_research.coorientation_spectra` with mode `audit` or `benchmark -rounds 5` and a required `-output` path. The fresh oracle command is `python -B -m` with module `topology_research.regina_audit` and mode `audit`, with the retained `topology_research/data/regina_corpus.json`, an output path and `-recheck-oracle`, in the Regina environment. `data/weighted_coorientation_fallback.py` reproduces the longer isolated control. The table and publication-review scripts retain completion, source stability, independent replay and document checks.

This change improves supplied-vector observation and certificate replay. It adds no diagram recognition coverage and no general quasipolynomial theorem. Its relevance to the wider programme is that a proof already used for scalar counts now transports the full additive interface without a redundant expensive orbit pass. Controlled complete witness discovery and geometric continuation remain the outstanding global obligations.

# 131 Cycle overlap, rooted components and local trace search

Arrival `8188525b7` supplies six more manuscripts, now preserved as reports 70–75 with their full PDFs, sources, code and evidence. Placement commit `7dff9ca69` removes only their wrappers and placed archives; existing Git attributes and ignore rules apply, without intake checksum or manual line-ending processing. The subsequent mathematical review and native integration are part of the authorized performance programme. The original report directories remain immutable. Their results are preserved independently of which implementation is selected.

Three themes improve the available theoretical interfaces. Affine profiles bound distinct observed weights without expanding sheets. Cycle envelopes replace total seam count by a smaller two-sided overlap parameter. Rooted completion asks for one useful component rather than requiring an entire surface to be a disc. The geometric reports add exact batch selection and a constructive bounded local trace search, while proving an obstruction to unrestricted active-support parameterization. Report 74 supplies the native minimum-envelope improvement of the next section.

## 131.1 Sharp weighted affine profiles

Report 70 assumes an explicit base graph with uniform fibres of size $W$ and full-fibre maps $x \mapsto \pm x + a \pmod{W}$. A spanning-tree gauge reduces each connected block to its root monodromy. The translation subgroup is described by a divisor $d \mid W$; an optional reflection pairs residues by $r \mapsto c - r \pmod{d}$. This gcd classification is inherited from earlier cover work. The new results concern interval weights and exact attachments.

For $K$ transported signed vector interval records, write their lengths as $\ell_j = q_j d + t_j$, $0 \le t_j < d$, and let $I_j$ be their cyclic residual intervals. A translation orbit has weight

$$F(r) = \sum_j (q_j + \mathbf{1}_{I_j}(r)) z_j.$$

Its endpoint sweep has at most $2K+1$ linear cells. In the reflected case put $G(r) = F(r) + F(c-r)$. A nonfixed residue pair has weight $G(r)$; a fixed residue has $F(r)$, not $2F(r)$. Reflection has at most two fixed residues.

**Proposition 131.1** (Sharp affine profile support)**.** *The histogram has at most $2K+1$ distinct keys without reflection and $2K+3$ with reflection. Across $b$ blocks and $h$ additional pointwise attachments with additive payloads, its support is at most $2K + 3b + h$. The reflected bound is attained for every $K \ge 1$, even with scalar nonnegative weights.*

*Proof.* The indicator intervals give at most $4K$ changing edges for $G$ on the residue cycle. Its reflection quotient is a path, up to endpoint loops. A reflection-fixed edge has exchanged endpoints, so an invariant function cannot change there. Changing edges therefore come in pairs, giving at most $2K$ changes along the quotient and at most $2K+1$ generic values. Correcting the at most two fixed residues adds at most two keys. Small moduli one and two satisfy the bound directly. Block supports add. An attachment touches old orbit vertices; each new affected quotient component contains an attachment, so at most $h$ new keys appear after replacing its old entries.

For sharpness take $d = 4K+4$, $W = 5d$, root maps $x \mapsto x + d$ and $x \mapsto -x$, and weight-one intervals $[d - j, 4d + K + j + 1)$ for $1 \le j \le K$. The two fixed weights are $3K, 4K$. The nonfixed weights are every integer from $6K$ through $8K$, each once. They are disjoint from the fixed weights, giving exactly $2K + 3$ keys. □

The implementation may retain up to $4K + 1$ sweep cells; the sharper theorem is about distinct output keys. Equal weights never identify equal attachment positions. The sparse overlay retains original endpoints and recomputes their actual orbit labels after a bulk change.

During a fixed-base insertion-only epoch, a root translation updates $d$ to $\gcd(d, a)$; a later reflection updates it to $\gcd(d, a - c)$. A first reflection occurs at most once. Every changed divisor is a proper divisor, hence at most half its previous value. There are therefore at most

$$\sum_i \left( \lfloor \log_2 d_i^0 \rfloor + \mathbf{1}_{\text{no initial reflection in } i} \right)$$

bulk rebuild triggers. Redundant maps need no rebuild, although reading their input records still costs time. The forest, blocks, weights and attachment coordinates are fixed during this epoch. Cross-block changes require another contract. Full-fibre structure must be authenticated before applying this kernel to a normal interval system; an arbitrary restricted pairing is not a full-fibre map. The manuscript does not supply that geometric extraction.

## 131.2 Exact ranks from the two-sided cycle envelope

Report 71 assumes candidate partitions refine $\sigma$ and future partitions refine $\rho$ on $r$ individually labelled seam intervals. Form their coarse bipartite multigraph $H$, retaining parallel seams. A disconnected $H$ cannot become a whole-disc completion by vertex splitting and is rejected. For connected $H$, put $s = |\sigma|$, $t = |\rho|$ and

$$\lambda = r + 1 - s - t = \dim Z(H; \mathbb{F}_2).$$

This is two-sided cycle overlap, rather than total interface size.

Choose a fundamental cycle basis $z_1, \dots, z_\lambda$. Splitting a $\sigma$-vertex according to $p$ imposes even-incidence constraints on all but one of its subblocks. Their matrix $A_p$ has $j = |p| - s$ rows; similarly $B_q$ has $k = |q| - t$ rows on the other side. Summing subblock equations gives the omitted coarse equation, so

$$Z(G(p, q)) = \ker \begin{pmatrix} A_p \\ B_q \end{pmatrix} \quad \text{in cycle-basis coordinates.}$$

For $j + k = \lambda$, the split graph has $r$ edges and $r + 1$ vertices. It is a tree exactly when that square stacked matrix is nonsingular. For any other grade sum it cannot be a tree. The separated-interval gluing theorem then gives the disc predicate, under its disc-fragment and geometric contracts.

**Proposition 131.2** (Sharp cycle-envelope dimensions)**.** *The binary completion matrix has rank exactly $2^\lambda$, with grade-$j$ rank $\binom{\lambda}{j}$ for complementary future grade.*

*Proof.* The maximal minors of $A_p$ give a feature supported on $j$-subsets of $[\lambda]$. Laplace expansion of the stacked determinant pairs it with complementary minors of $B_q$. This gives the stated upper bounds. For the lower bounds choose a spanning tree of $H$ and its $\lambda$ chords. Detach the left endpoints of any chord subset $S$ into singleton subblocks, leaving core blocks nonempty because each retains a tree edge. Its constraint rows are the corresponding coordinate units, giving exterior feature $e_S$. Detaching right endpoints of $R$ gives $e_R$. The split graph is a tree exactly when $R = S^c$: each chord is then a pendant edge on the original tree; a chord detached at both ends is a separate component. These states give a complement-permutation submatrix of size $2^\lambda$, and its grade-$j$ restriction has size $\binom{\lambda}{j}$. □

A cost-ordered original-witness basis retains at most $2^\lambda$ candidates and preserves all admitted completion minima. The chord witnesses also force every representative-subset method to retain that many on a universal family. This sharpness is for the specified abstract continuation class, not every embedded knot-region class. A feature row vanishes exactly when $p \vee \rho$ is disconnected, providing a cheap shared viability filter.

The delivered complete finite-grammar compiler derives safe candidate and future envelopes in polynomial time. Its elementary search cost is $\mathrm{poly}(I, B) 2^{3\lambda_{\max}}$, including explicit input size

$I$ and cost-bit bound $B$. A 256-arc example has $\lambda = 3$ and needs eight representatives. A general low-cycle geometric producer remains unproved. Actual seams, fixed-type substitution and essential-boundary/source binding remain obligations; a binary multiplicity is not one physical seam.

### 131.3 One distinguished disc component: a ternary interface

Whole-disc completion can reject a useful disc accompanied by unrelated components. Report 72 distinguishes a root seam, records whether each fragment component is a disc, and permits other components to remain bad. Component-local charges lie in $H = (\mathbb{F}_2)^k$, with $q = |H|$. The root component succeeds exactly when all its fragments are good and its incidence graph is a tree, with the requested total charge.

Its feature coordinates are $(X, A, \gamma)$, where $0 \in X$, $A \subseteq X \setminus \{0\}$, and $\gamma \in H$. A coordinate is one when $X$ is a union of good blocks, $X \setminus A$ omits exactly one point from every block, and those blocks' charges sum to $\gamma$. Complementary exterior coordinates pair the two sides, with charges adding to the target. Saturation under both full partitions ensures that a connected restricted graph containing seam zero must be exactly the actual root component. Thus there is at most one contributing $X$, avoiding an uncontrolled parity sum over several possible components. The feature dimension is

$$q \sum_{X \ni 0} 2^{|X|-1} = q3^{r-1}.$$

**Proposition 131.3** (Exact rooted ranks)**.** *The uncharged binary rank is $3^{r-1}$. Every exact-charge kernel has rank $q3^{r-1}$. For $q \geq 2$, the nonzero-charge kernel has the same rank.*

*Proof.* At each non-root label take three fragment choices: join the good root block, be a good singleton, or be a bad singleton. Their compatibility matrix is

$$M = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad \det M = -1.$$

Labels act independently, giving $M^{\otimes(r-1)}$. Arbitrary root charges add a permutation factor for an exact target. For a nonzero target predicate the charge factor is $J + I$ over $\mathbb{F}_2$; since $q$ is even, $(J + I)^2 = I$. These full-rank submatrices attain the feature upper bound. □

Cost-ordered bases preserve every exact/nonzero completion minimum using original witnesses, even for large signed costs. For exact charges a weighted universal family forces all $q3^{r-1}$ witnesses: use local costs zero, two and one respectively for root/good/bad choices; appropriate complementary choices uniquely expose each candidate as cheapest. There is no claimed matching weighted lower bound for the nonzero-only predicate. Rank alone does not supply such a lower bound. Already with one seam and many root charges, the cheapest two distinct charges suffice for all nonzero-charge queries in that simple family, despite full charge-matrix rank.

Two properly owned peripheral bits suffice for an actual disc on a boundary torus: its simple boundary is essential exactly when its mod-two class is nonzero. Aggregate charge cannot replace component-local charge; two parallel essential discs can sum to zero. A permanent root marker is represented by an algebraic dummy cap, not a physical cap added to the output.

The complete supplied layered-language algorithm costs $\mathrm{poly}(I, B, b)q^3 27^b$ for frontier width $b$ excluding the root marker. Portless non-root components can be forgotten while retaining their costs and witness material; the objective is total assembly cost, not root-only cost. Cyclic rotation of a site order increases frontier width by at most a factor two, giving a root-selection theorem. The delivered executable is a layered-language solver, not a general order compiler. Mesh replay

realizes bad pieces as annuli with formal charges; it does not construct their three-dimensional embedding or actual boundary cocycles.

## 131.4 A tree envelope does not compress every rooted query

The whole-disc cycle bound and the rooted bound cannot be combined by simply replacing $r$ with $\lambda$. A small counterfamily makes the missing root-reachability information explicit.

**Proposition 131.4** (Rooted tree-envelope obstruction)**.** *There are connected coarse envelopes with $\lambda = 0$ whose uncharged rooted compatibility matrix has binary rank $2^{r-1}$. A universal weighted original-witness reducer needs $2^{r-1}$ candidates on a corresponding family, even with nonnegative costs.*

*Proof.* Take $\sigma$ to be one block and $\rho$ to be all singletons. Their graph is a star with $r$ edges and $r+1$ vertices, hence $\lambda = 0$. For every $X \subseteq \{1, \dots, r-1\}$, the left candidate has a good root block $\{0\} \cup X$ and good singleton blocks elsewhere. The right cap is the singleton partition with root zero good and bad annuli at a subset $B$ of the other labels. Its remaining pieces are discs.

The root component is always a star and is a disc exactly when $X \cap B = \varnothing$. The matrix on these two families is the disjointness matrix, a tensor power of $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, whose determinant is $-1$. Its binary rank is $2^{r-1}$. Every other rooted row in this star envelope is zero or depends only on its good root block, so this is also the exact uncharged rank for that envelope.

Give candidate $X$ total cost $(r-1) - |X|$, counting its non-root singleton pieces. Against cap $B = X^c$, the feasible candidates are precisely subsets of $X$. Candidate $X$ is their unique cheapest member. Deleting any candidate therefore changes one completion optimum, proving the weighted lower bound. □

This is an abstract surface-fragment result, realized by discs and annuli with separated seams. It is not a lower bound for all knot algorithms or for one fixed three-dimensional region. It does not force an exponential subfamily for mere feasibility: in this displayed family the root-only candidate succeeds against every cap. The lower bound concerns universal bilinear compatibility and weighted minima. It identifies an additional obligation: control the root-reachable good/bad frontier as well as cycle overlap. Fresh checks compare 1,365 direct/feature entries, 341 literal mesh pairs and 63 uniquely exposed optima. The companion proof and audit are preserved even though the result warns against a tempting performance extrapolation.

## 131.5 Commuting moves and exact peeled batch selection

Report 73 proves that tetrahedron-disjoint formal 3–2 bipyramids commute as relative-boundary replacements, even with shared global vertices or paired facets. Their raw Euler and piece changes add. Every region has only ten formal edge types; any overlapping collapse is centred at one of these, so the deduplicated conflict graph has maximum degree nine. Greedy packing of $M$ candidates therefore supplies at least $\lceil M/10 \rceil$ moves.

Vertex-link-peeled Euler changes are not additive: the retained genuine 32-tetrahedron counterexample has singleton gains one and one but joint gain zero. For a fixed disjoint family $E$, let $r_i = 2g_i$ be raw gain and $c_{vi}$ the reward for omitting move $i$ at global vertex $v$. Coherent minima prove at most two positive endpoints per move and $c_{vi} \le r_i$, with the sharper boundary bound $c_{vi} \le g_i$. The exact omission objective is

$$F(E \setminus R) = F(E) + \sum_{v} \max(\{c_{vi} : i \in R\} \cup \{0\}) - \sum_{i \in R} r_i.$$

Among maximizers choose $R$ of minimum size. Removing any move uniquely maximizing at at most one endpoint loses reward at most its cost, contradicting minimality. Every retained omission

therefore uniquely maximizes at two endpoints, and two omissions cannot share an endpoint. Thus the optimum omission set is a maximum-weight matching with edge weight $c_{ui} + c_{vi} - r_i > 0$. Integer transformation $(|E| + 1)w_i - 1$ selects the fewest omissions among equal weights. This solves the fixed-family problem; it does not optimize the initial packing or authenticate an essential disc.

Every positive edge has an interior endpoint, since two boundary rewards sum to at most $2g_i = r_i$. Hence at most one omission per relevant interior vertex is needed. The maintained pulling source has two interior poles, which form a vertex cover of the reward graph; a linear scan solves its matching problem. It retains at least $\max(0, \lceil M/10 \rceil - 2)$ packed moves with nondecreasing peeled score. The total number of boundary vertices is not two. Geometry preparation and packing costs are charged separately.

Deterministic AVL corner trees retain all records and cache thirteen-key prefixes. A hypothetical singleton deletion removes at most twelve old corners, so its new minimum is among those prefixes and new corners. Committed edits replenish them in logarithmic tree work. The complete endpoint checker reconstructs the full geometry rather than trusting this dynamic index. The delivered large batch/scoring gains are local fixed-workload comparisons, not complete recognition measurements.

## 131.6 Complete active support and a linear root obstruction

For $C = \{x \geq 0 : Ax = 0\}$ and positive objective $c$, report 73 gives one primal–dual active-support certificate:

$$Ax = 0, \quad x \geq 0, \quad c^T x \geq 1, \quad s = A^{\mathsf{T}} y \geq 0, \quad x + s \geq \mathbf{1}.$$

Orthogonality $x^T s = 0$ proves that exactly one of $x_j, s_j$ is positive at every coordinate. A positive $x_j$ witnesses activity; a positive $s_j$ forces that coordinate zero throughout $C$. Conversely summing active primal witnesses and inactive-coordinate Farkas multipliers constructs such a pair when the positive-objective part is nonempty. The negative alternative is the original homogeneous dual $A^{\mathsf{T}} h \geq c$. A lifted homogeneous system with $3N + 2m + 2$ variables realizes the pair, but its exact Bland implementation has no polynomial pivot bound. In the delivered adverse audit it reduces 130 unary calls to 17 lifted calls while time rises from 1.4183 to 22.8305 seconds. It remains an optional interface.

Let $J(C)$ be active coordinates in still-conflicting type groups and define $\rho(C)$ as the rank of their coordinate functionals on $\operatorname{span} C$. Branching on an incompatible active pair $i, j$ into faces $x_i = 0$ and $x_j = 0$ preserves admissible coverage and drops this rank by at least one on each positive child. With a polynomial exact LP policy the binary tree has at most $2^{\rho+1} - 1$ nodes. Complete anchor enumeration is still needed.

For an unrestricted normal root with one zero triangle anchor per global vertex, all quadrilateral types are active. The vector $r$ with triangles zero and all quads one is matching. At any tetrahedron with no chosen anchor, the local vector $w_a$ with four triangle entries $+1$ and three quad entries $-1$ changes every face arc by zero. Both $r$ and $r + w_a$ belong to $C$. Their disjoint quadrilateral images prove

$$\rho(C) \geq t - u \geq t - v,$$

where $u$ is the number of anchored tetrahedra and $v$ the total global vertex count. On pulling sources $t = 20n'$, $v = 4n' + 2$, so $\rho \geq 16n' - 2$. This rules out a polylogarithmic unrestricted-root ambiguity rank on these actual sources; it is not a runtime lower bound or an obstruction to other hierarchy algorithms.

## 131.7 A constructive positive relaxation at every anchor

The same local relation gives a further consequence, made explicit here. Let $p$ count interior vertices of a finite compact manifold triangulation, and let $\chi_{\text{lin}}$ be the usual normal Euler functional

extended linearly to matching vectors without quadrilateral compatibility.

**Proposition 131.5** (Uniform positive anchored relaxation)**.** *For every choice of one zero triangle anchor per global vertex, there is an integral nonnegative matching vector $z$ satisfying all anchors with*

$$\chi_{\mathrm{lin}}(z) = v + p - u \geq p.$$

*In particular every unrestricted anchor root with $p > 0$ has positive Euler LP feasibility. On the pulling diagram sources it has value at least two.*

*Proof.* Take $z = r + \sum_{a \text{ unanchored}} w_a$. In an anchored tetrahedron it has three quads and no triangles; in an unanchored tetrahedron it has four triangles and no quads. Every face has one arc of each type, every global edge has weight two, and every chosen anchor is zero. Thus it is a nonnegative matching vector.

Write $e$ for global edges and $b$ for boundary facets. There are $4t - u$ pieces, $12t$ local sides and $3b$ unpaired boundary arcs, so

$$\chi_{\mathrm{lin}}(z) = 2e - (12t + 3b)/2 + (4t - u).$$

The manifold has $f = 2t + b/2$ faces and $e = v + t + b/2 - \chi(M)$. Its boundary has $v_\partial = v - p$ vertices, $3b/2$ edges and $b$ triangles. Doubling, or mod-two duality, gives $2\chi(M) = \chi(\partial M) = v_\partial - b/2$. Substitution yields $2v - u - v_\partial = v + p - u$. Since $u \leq v$, the bound follows. Pulling sources have $p = 2$. □

This vector is deliberately incompatible: every anchored tetrahedron has all three quad types. It proves LP feasibility and never an embedded normal surface or a knot verdict. Thus an initial negative Euler dual cannot settle these unrestricted pulling roots, whether their diagrams are knotted or unknotted. The formal relative 2–3 and 3–2 moves preserve vertex classes and the external boundary, so this interior-vertex feasibility consequence also persists along those moves. The original linear rank bound is not asserted unchanged after tetrahedra are removed. Type-restricted descendants and other positive-witness tests remain meaningful. Adding a sufficiently small multiple of $r$ also proves all quadrilateral types active in the positive-objective part. Fresh controls check 374 anchor patterns on eleven genuine sources, including both knot and unknot diagrams; every vector matches, has the predicted formal Euler value, and is rejected by the actual admissible-normal validator. The proof is uniform and does not rely on these finite controls.

## 131.8 Commitments, local regions and positive LP witness reuse

Report 75 gives a constant-alphabet encoding for formal Pachner traces with at most $U$ total upward 2–3 moves. A tetrahedron incarnation chooses idle, one of four upward faces, or one of six downward edges. Two different ready committed events cannot compete for an incarnation: its selected port identifies a unique legal event. Disjoint simultaneously enabled replacements commute relative to their boundary and transported cocycle. An intended ready event can be moved ahead of earlier disjoint events, giving a deterministic scheduler with the same endpoint.

If there are $u \leq U$ upward and $d$ downward moves, $d \leq t + u$ and the number of original/new incarnations is $t + 3u + 2d \leq 3t + 5U$. Padding commitments with idle symbols therefore bounds commuting trace classes, including all finite prefixes, by $11^{3t+5U}$. With $U = 0$ the alphabet has seven symbols. Exact sleep-set exploration inherits the bound using persistent incarnation/event identities and the specific geometric commutation relation. This is not a theorem for arbitrary state-dependent independence rules or only for maximal traces.

Protect originals outside an allowed region $F$ of size $i$. Then $d \leq i + u$ and only $3i + 5U$ incarnations need free commitments. The dual graph has degree at most four. A rooted depth-first

tree traversal bounds connected $i$-vertex regions by $t16^{i-1}$; distributing components gives at most $(r+1)(c+1)t^c32^r$ regions of size at most $r$ with at most $c$ components. Canonical-parent reverse search constructs them with polynomial delay rather than enumerating all $t^r$ subsets. The resulting complete bounded search costs

$$t^c 2^{O(r+U)} \operatorname{poly}(t+U+B)$$

when its endpoint predicate is complete and polynomial; otherwise its cost is charged explicitly. The region is a cover of the actual consumed original cells, which need not have only $c$ components. Neither maximum excess size nor upward-burst count can replace total $U$. Upward cocycle transport adds at most one coefficient bit per move; downward removal adds none. Source-complete successful regions with $c = O(\log t)$ and $r + U = O(\log^2 t)$ would give a constructive QP class. No such general coverage theorem is supplied.

The other interface reuses exact positive kernel witnesses for deletion LPs. If a checked witness already has the queried coordinate zero, it certifies feasibility of that deletion without solving again. Local reuse works even with global cache capacity zero; a source/anchor-bound support antichain extends reuse across masks. Only positive deletion queries whose witness content does not affect the fixed propagation policy are skipped, preserving every completed deterministic certificate. The delivered algebraic control has LP calls $3s^2+2s+1$ versus $2s+1$ and Bland pivots $6s^3+3s^2+s$ versus $3s^2+s$. It is not a claimed triangulation realization. The source adapter's four surveyed positives all use zero moves, validating provenance rather than adding move-search coverage. Bounded exhaustion stays inconclusive as a knot question.

### 131.9 Fresh evidence and integration decisions

Scratch validation of reports 70–72 passes 34, 42 and 42 test methods. Fresh audits reproduce the sharp-profile tests; 633,531 cycle-envelope entries over 3,162 matrices, 16,904 weighted queries and 1,412 literal grammar assignments; and the rooted charged/mesh/language audit with 10,000 weighted queries and 3,200 assignments. Their complete original theorem ledgers and proofs remain available. Report 73 passes 51 standard-library tests plus five optional matching tests in an isolated NetworkX 3.7 environment, with independent active and geometric replay. Report 75 passes 28 standard-library tests plus its optional native-topology test, and fresh geometric, region, LP and Regina replays. Detailed fresh counts and reproduction records are retained in `data/incoming-818-*`; the new relaxation check is `data/anchored-relaxation.json` and its companion script.

These finite checks support source-level review, not proof-assistant formalization or an originality claim beyond the reports' attributions. The affine, cycle-envelope, rooted-language, active-face and trace/cache overlays remain separate research interfaces: their geometric admission and complete-source promises are not supplied by their algebraic features. Minimum envelopes are integrated and tested on the maintained native sector API in the next section. All six manuscripts remain preserved, including the structural obstructions and other results without a measured native recognition gain. None establishes a general subexponential or QP recognizer.

## 132 Adaptive standard-ray enumeration from minimum envelopes

Report 74 replaces a particular over-refinement in the native sector enumerator. The older standard phase formed every pairwise equality of triangle potentials, lifted the resulting nonnegative directions, and checked their full standard extremality. A crossing between two potentials above the minimum can create a spurious direction. At quadrilateral matching nullity at most two, the new method enumerates only changes of the actual minimum and the feasible section endpoints. These are exactly the standard rays, so their rank rejection stage disappears.

Three delivered production modules and their independent tests are now integrated unchanged. The native adaptation additionally selects this method automatically at nullity one or two and threads an explicit method option through supplied-sector discovery. Higher nullity retains the complete arrangement/support choice; the quadrilateral phase retains its old behavior. This is a structural adaptive choice with a completeness proof, rather than a heuristic that drops difficult candidates.

## 132.1 Canonical lifting is linear on the right faces

For a compatible allowed quadrilateral sector with $k \le t$ types, let $C = \{q \ge 0 : Aq = 0\}$ be its quadrilateral matching cone. The triangle graph gives linear corner potentials $\ell_c(q)$, grouped by global vertex $v$. Triangle coordinates have the form $\alpha_v + \ell_c(q)$. Their canonical lift is

$$\Psi(q)_c = \ell_c(q) - m_v(q), \qquad m_v(q) = \min_{c \in G_v} \ell_c(q).$$

Every standard sector vector is uniquely $\Psi(q) + \sum_v \beta_v L_v$, $\beta_v \ge 0$, with $L_v$ its vertex-link vector. A non-link extreme ray must have all $\beta_v = 0$, since a nonzero link summand has zero quadrilateral projection and cannot be proportional to the whole vector.

Choose a minimum anchor $b_v$ in each group and define

$$F_b = \{q \in C : \ell_c(q) \ge \ell_{b_v}(q) \text{ for all } c \in G_v\}.$$

On $F_b$ the lift is linear. Projection and lifting are inverse maps between $F_b$ and the full standard coordinate face obtained by setting all selected triangle anchors to zero. This is a face because every standard coordinate is nonnegative. Extreme rays of a face are extreme rays of the whole cone. Conversely every non-link standard extreme ray is canonical and belongs to one such anchored face. Therefore its projected direction is a ray of the *minimum* subdivision, not an arbitrary pairwise-potential arrangement.

This face correspondence holds in every dimension. Within an anchor cell, nonnegative summands of a lifted vector must keep every zero anchor zero; global nonadditivity of $\Psi$ creates no extremality gap. In higher dimension the minimum subdivision is the common lower normal fan of the potential coefficient polytopes, equivalently that of their Minkowski sum, intersected with $C$. This description does not bound arbitrary-dimensional fan size. Canonical conversion itself is classical; the manuscript retains its attribution to normal-surface conversion work.

## 132.2 Complete nullity-two enumeration and its output bound

Let $d = \dim_{\mathbb{Q}} \ker A \le 2$. Normalize by $\sum q_i = 1$, which meets every nonzero nonnegative ray exactly once. The section is empty, a point, or a bounded interval. For a nondegenerate interval choose exact rational $u, w$ with $Au = Aw = 0$, $\sum u_i = 1$, $\sum w_i = 0$, $w \ne 0$, and write $q(s) = u + sw$. Every coordinate inequality gives an exact lower/upper bound or a constant feasibility test, producing $[L, U]$.

For each vertex group put $h_v(s) = \min_c \ell_c(u + sw)$. This is the lower envelope of affine lines. The complete ray list consists of primitive canonical lifts at $L, U$ and every genuine interior slope change of any $h_v$. Between those parameters the same anchor realizes each minimum, so there is a single linear minimum cell and no interior extreme direction. At a genuine change the section is a vertex of an anchored face. An interior zero of a nonnegative affine coordinate must be an identically zero coordinate, so no additional quadrilateral event is missing. Parallel or coincident lines and endpoint-only ties create no extra ray.

Dimension zero gives no non-link ray; dimension one is empty or a point. A two-dimensional matching space may still have an empty or singleton nonnegative section. All these cases are implemented explicitly. The raw linear nullity is not confused with feasible cone dimension.

**Proposition 132.1** (Linear ray inventory)**.** *For a nondegenerate section, let $p$ be retained triangle classes, $g$ their vertex groups and $a \le 4k$ the nonzero-labelled triangle-graph edges. Then*

$$R \le p - g + 2 \le a - k + 4 \le 3k + 4.$$

*Proof.* A line is minimal on an interval, since each comparison with another line defines an interval. Concavity makes successive distinct envelope slopes strictly decrease. A group with $p_v$ classes therefore has at most $p_v - 1$ genuine changes; unioning changes and adding endpoints gives $p - g + 2$. After zero-label contraction, cycle consistency has rank at most $a - p + g$. Its matching nullity two gives rank $k - 2$, so $k - 2 \le a - p + g$ and $p - g + 2 \le a - k + 4$. Each allowed quad labels at most four face-matching edges, hence $a \le 4k$. □

The constant three is an upper bound, not a claimed sharp geometric example. The local $p - g + 2$ bound is often tighter. An abstract affine-potential family can attain the corresponding envelope count, but that does not realize every such potential system as a knot exterior.

The hull implementation sorts slopes, keeps the smallest intercept at equal slope, and pops a previous line when the next intersection is not later than its start. Exact fractions retain all genuine ties. Projected potential intercepts/slopes are computed once and reused across all lifts. With $p = O(k)$ and $R = O(k)$ the post-kernel arithmetic cost is

$$O(pk + p\log(p+1) + R(t + p + k)) = O(tk), \qquad 1 \le k \le t.$$

Kernel construction, source preparation and full coordinate validation are charged separately. Coefficient sizes give polynomial bit time; this is not a quadratic bit-time bound. The growing family has exponentially many pieces but polynomially encoded output, so numeric piece counts cannot be treated as input lengths.

### 132.3 Independent certificates of complete coverage

The new `normal-sector-envelope-v1` binds the exact triangulation, allowed types, matching and section dimensions, rational section vectors, endpoints, active-envelope corner records and the complete primitive ray list. Rationals use reduced numerator/positive-denominator pairs. The coverage checker imports neither the producer, its hull routine nor its contracted kernel builder. It reconstructs an uncontracted corner graph from the actual matching rows, derives its own potentials and independently checks nullity.

Supplied section vectors are checked against every matching equation and normalization; the coordinate bounds certify the entire nonnegative section. For an envelope, active slopes must strictly decrease and successive intersections must occur in order within the section. Each active line comes from an actual source corner. An omitted source line $f$ dominates the proposed concave envelope $h$ exactly when the convex function $f - h$ is nonnegative at its minimum. A supplied slope bracket identifies that minimum at a breakpoint or endpoint, giving one exact dominance comparison per corner rather than testing all corners at all candidate parameters.

These checks certify the entire minimum subdivision and then every lifted ray, with no missing or duplicate output. A positive disc certificate is still a separate native topology/essentiality proof. Coverage of this supplied sector does not authenticate a diagram exterior or prove that no other sector contains a useful disc.

Automatic standard enumeration now dispatches to envelopes for $d = 1, 2$; empty $d = 0$ auto results retain their previous empty schedule. Explicit `method='envelope'` accepts only the standard phase with $d \le 2$. Explicit arrangement/supports remain available; higher-dimensional auto selection still uses the old predicted-work comparison. The quadrilateral phase does not silently become a standard phase.

The envelope's `max_bases` counts begun actual output lifts, not spurious linear systems. A partial iterator raises `SearchLimit` and establishes no absence claim. The complete-coverage producer prechecks `max_rays` against the true planned output and returns inconclusive without a partial certificate. Supplied-sector discovery exposes the method selector and preserves independently checked positive/exhaustion protocols. Its standard exhaustion says only that no vertex disc was found in that sector. Caller cancellation remains cooperative throughout planning and verification.

## 132.4 All-size capped family and its component classification

The supplied fixture attaches one tetrahedral ball to face two of a Fibonacci layered solid torus, allowing the meridional quad in each original tetrahedron and type one in the cap. Put $F_0 = 0$, $F_1 = F_2 = 1$, $a = F_{n+1}$, $b = F_{n+2}$, and for integers $x, y \geq 0$ let $h = \max(0, y - ax)$. Its canonical vector has original rows

$$(F_{n-i+1}x + h, F_{n-i+1}x + h, h, h; 0, 0, F_{n-i}x), \quad 0 \leq i < n,$$

and cap row

$$(ax + h - y, bx + h, 0, h; 0, y, 0).$$

The terminal face gluing forces the last meridional quad to be $x$, the penultimate one to be $x$ when it exists, and the remaining quads to satisfy $q_i = q_{i+1} + q_{i+2}$. Triangle matching makes the two lower triangle entries a common value $\alpha$ and the two upper entries $F_{n-i+1}x + \alpha$. The cap equations give triangle entries $ax + \alpha - y, bx + \alpha, \beta_1, \alpha$. Nonnegativity is exactly $x, y \geq 0$, $\alpha \geq h$ and $\beta_1 \geq 0$. Subtracting the old vertex-link minimum $\beta_0 = \alpha - h$ gives the displayed canonical form. There are only two minimum cells, separated by $y = ax$, so the non-link standard ray parameters are $(1, 0), (1, a), (0, 1)$. The middle ray is absent from the two quadrilateral endpoints, although one endpoint also supplies an essential disc. This demonstrates stronger complete enumeration, not a claim that Q screening fails all useful recognition tests.

The README asserts a component formula; its delivered family script checks finitely many parameters. The following geometric argument records the uniform classification explicitly instead of treating those tests as proof.

**Proposition 132.2** (Every integral vector in the capped sector)**.** *The canonical vector above consists of $x + h$ discs, exactly $x$ of which have essential boundary. Adding nonnegative vertex-link multiplicities $\beta_0, \beta_1$ gives $x + h + \beta_0 + \beta_1$ disc components and the same essential count $x$. These parameters describe all integral vectors of this fixed sector for every $n \geq 1$.*

*Proof.* Restriction to the original torus is $xD_n + hL$, where $D_n$ is the primitive meridian of Section 129 and $L$ its boundary vertex link. It is a disjoint union of $x$ meridian discs and $h$ inessential discs. Each normal piece in a tetrahedron meets a given face in at most one arc. The only type missing that face is its opposite triangle, here cap type two, whose coordinate is zero. Thus every cap piece attaches along exactly one disjoint boundary interval to one old component. The incidence graph for each old component is a star with disc leaves: gluing preserves its disc type and cannot merge two old components or create a separate one.

The cap is the fixture's boundary retriangulation. Its new outer three-face disk maps to the old attaching triangle by a boundary-fixing subdivision homeomorphism. Each added disc's outer boundary arc maps to a path homotopic relative endpoints to its attaching arc. Hence the resulting boundary circles retain their old peripheral classes, preserving the $x$ essential circles. This argument takes place away from identified boundary vertices and respects the formal face gluings.

The two source vertices are boundary vertices. Their added vertex links are inessential disc components. Canonical decomposition separates these links from every standard vector. The

terminal meridional quad has coefficient $F_1 x = x$ and the cap quad has coefficient $y$, so integrality forces integer $x, y$; their canonical triangle values and link minima are then integral. Matching and nonnegativity give exactly the displayed parameters and their nonnegative link additions. □

Consequently primitivity alone does not imply connectedness: $x = 2, y = 1$ has coordinate gcd one but two essential discs. The supported-ray theorem requires its additional nullity-one hypothesis. On this family the previous arrangement method admits $n + 2$ positive directions and rejects $n - 1$; the envelope method emits exactly three. Controls with cap type zero have two rays and are retained separately.

## 132.5 Native validation and isolated timings

All 1,351 maintained test methods pass in 224.903 seconds. The focused 42-method run passes in 3.481 seconds, including the unchanged native unit-ray disc tests. New integration tests forbid calls to the pairwise hyperplane generator and standard-rank filter during automatic envelope enumeration, and independently replay positive and negative sector results. They also exercise actual-ray resource caps, the extra non-Q disc, strict method/phase validation and rejected missing coverage rays.

The native audit selects 9,995 sectors across all 48 frozen triangulations and audits all 9,344 with matching nullity at most two. The declared selection includes every support on the small triangulations and prescribed ray/small/random supports on the larger ones; it is not every possible sector on every larger triangulation. Every emitted set matches the frozen complete standard-ray list filtered by its allowed types. There are 4,853 ray occurrences and 804 interior-ray occurrences across those repeated queries. Automatic and explicit envelope output agree in 9,360 calls, including fifteen capped-family configurations and one extra-disc example. Fifteen fresh Regina full enumerations also match. All 567 captured source hashes remain unchanged during the audit and timings. Fresh family controls check nine ray lists through base size 128 and 97 integral surface vectors against the maintained component observer, supporting the uniform cap proof.

The delivered complete-coverage evidence separately replays sixty eligible independent-audit certificates, six coverage-benchmark certificates and one essential-disc certificate with enumeration imports blocked. Six higher-nullity model-only records remain outside this API. Native publication checks replay those saved coverage proofs with the current independent checker, together with the new capacity certificates. Empty and singleton sections, missing rays, false breakpoints, slope brackets, source changes, noncanonical fractions and callback cancellation have explicit controls.

The isolated benchmark uses five interleaved four-arm rounds and one warmup per arm: 260 measured calls and 52 warmups, all complete. Its reference is the frozen `654c0b2a7` native sector code. Both arms pay full kernel construction. Enumeration times include all emitted coordinates, their serialization and digest; this existing iterator API has no portable coverage proof, so independent coverage replay is not claimed inside that timing. The separate discovery times include its native essential-disc or exhaustion proof, independent replay and full proof serialization. Fixtures are constructed before timing, and ratios are paired medians.

**Complete standard-ray enumeration and output.**

| Input | old ms | new ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| Base 1, cap 1 | 1.068 | 0.714 | 1.496 | 0.959 | 0.985 |
| Base 4, cap 1 | 12.053 | 1.625 | 7.498 | 0.986 | 1.018 |
| Base 8, cap 1 | 79.621 | 3.624 | 21.642 | 0.979 | 1.005 |
| Base 16, cap 1 | 741.848 | 10.808 | 68.440 | 1.016 | 1.013 |
| Base 32, cap 1 | 8449.232 | 43.139 | 197.217 | 1.005 | 0.996 |
| Base 8, cap 0 | 26.903 | 3.304 | 8.209 | 1.002 | 0.995 |
| Base 16, cap 0 | 137.210 | 9.984 | 13.791 | 0.986 | 0.997 |

At base size 32, full enumeration drops from about 8.449 seconds to 0.043 seconds, a paired 197.22-fold gain, returning the same three rays. The old method generates 34 positive directions and rejects 31. Gains are smaller on the smallest source and substantial on the cap-zero controls without a genuine minimum break, where projected hyperplane/rank work is still avoided. These are local enumeration measurements on genuine finite triangulations.

**Complete supplied-sector disc discovery and replay.**

| Input | old ms | new ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| Base 1, cap 1 | 1.743 | 1.484 | 1.166 | 0.985 | 1.015 |
| Base 4, cap 1 | 7.796 | 3.375 | 2.301 | 1.006 | 1.000 |
| Base 8, cap 1 | 28.471 | 6.557 | 4.339 | 0.999 | 1.006 |
| Base 16, cap 1 | 144.934 | 16.111 | 9.002 | 0.993 | 0.986 |
| Base 32, cap 1 | 918.508 | 53.136 | 17.283 | 1.005 | 1.002 |
| Empty sector | 0.626 | 0.627 | 0.996 | 0.997 | 0.999 |

Complete discovery at base size 32 drops from about 0.919 to 0.053 seconds, a paired 17.28-fold gain. The empty-sector control is near parity. These smaller ratios include the remaining kernel, geometry and independent positive-proof costs and are not substituted by the larger enumeration ratios. All complete discovery arms return the same status; individual witnesses are checked, rather than assumed equal from status alone. A/A medians range from 0.959 to 1.018.

Four separately timed new-only calls produce and replay complete coverage certificates at base sizes 1, 8, 32 and 64. Their full elapsed times are approximately 0.002, 0.010, 0.135 and 0.765 seconds, with certificate sizes 501, 1,067, 3,915 and 9,985 bytes. No speedup is inferred from these unpaired capacity observations. Source/coverage replay and asymptotic proofs remain separate evidence from wall-clock measurements.

Reproduction is the maintained `normal_orbit_research.envelopes` driver, with `audit -fresh-regina` or `benchmark -rounds 5` and an explicit `-output` path, from `fast/`. The raw audit, timings, test logs, source hashes and publication checks are retained in `data/envelopes-*`. The complete original report and all its proof appendices remain in `reports/74`.

The global obligation remains selection of a complete, sufficiently small family of useful sectors, with their source and transition costs. Matching nullity of an allowed quad sector is different from full-coordinate nullity on the positive support of a discovered standard ray. The latter being one does not prove that the former is at most two. Report 74 also proves that checked Farkas zero-coordinate implications can restrict a cone to an equivalent smaller face, but supplies no universal reduction to this low-dimensional class. This native local improvement and the preserved cycle/region theorems therefore advance specific operations while leaving general QP recognition unproved.

# 133 Binary patch grammars, boundary observers and causal localization

Reports 76–80 are preserved in full, including their original manuscripts, PDFs, proofs, executable experiments and proposed patches. Placement `eadb83e0e` precedes the separate review recorded here. Executions use scratch copies; the delivered source trees are unchanged. Mathematical results are retained whether or not they improve the native recognizer. The two native integrations are explained in the following sections. None of these reports proves general quasi-polynomial unknot recognition.

## 133.1 A rectangular algebra for whole-disc patch transfer

Report 76 uses actual unions of discs with individually labelled, collared input and output intervals. Every component must meet the external boundary interface. Gluing is zero-rejecting: a sealed component or an incidence cycle permanently prevents the whole assembly from being one disc. This predicate differs from the distinguished-component predicate of Section 131.

Linearize transfers over $\mathbb{F}_2$ and quotient by the radical of all admitted cap tests. Substituting a fixed transfer into a cap gives another admitted cap, so this radical is preserved by left and right composition and tensor product. This substitution argument is needed before quotient multiplication can be used; equality under a single chosen closure would not suffice.

**Proposition 133.1** (Rectangular disk-transfer algebra)**.** *For positive interface sizes $a, b$, the cap quotient satisfies*

$$\mathrm{Hom}(a, b) \cong \mathrm{Mat}_{2^{b-1}\times 2^{a-1}}(\mathbb{F}_2[\varepsilon]/(\varepsilon^2)).$$

*Composition is ordinary rectangular matrix multiplication under this identification, with the closure pairing given by the corresponding trace.*

*Proof.* With one input and one output, the strip is the unit and two separate discs represent $\varepsilon$. Composing the latter with itself seals the middle disc, hence $\varepsilon^2 = 0$. For two ports, actual transfers with partitions $000, 001$ give maps $u_0, u_1 : 1 \to 2$; partitions $010, 000$ give $v_0, v_1 : 2 \to 1$. Direct cap testing gives $v_i u_j = \delta_{ij} I_1$ and $u_0 v_0 + u_1 v_1 = I_2$ in the quotient. The latter sum is an algebraic identity, not a physical glued surface. Iterating the splitting yields $b \cong 1^{\oplus 2^{b-1}}$. The composites of the resulting inclusion and projection maps are matrix units with coefficients in $\mathrm{End}(1)$, proving the formula and its compatibility with composition. The full report supplies the cap tables and explicit maps, including the trace normalization. □

For a homogeneous acyclic binary grammar of width $b$, with $q$ XOR-charge values, the feature bound is $R = q2^{2b-1}$. Cost-ordered elimination retains original candidates, rather than interpreting an XOR basis row as a surface. The elementary all-cap min-cost algorithm has bit cost $\mathrm{poly}(N, b, B)R^4$, including grammar size and signed-cost bits. A binary power of physical length $W$ takes $O(\log W)$ composition stages; each copy may choose independently from its language. Raising one selected witness to the $W$th power would solve a different problem. A witness DAG of size $O(M + SR)$ preserves the selected original word and its costs. For a nonnegative free-star objective, an optimum needs at most $R - 1$ layers. This shortest-word theorem does not permit shortening a prescribed physical span. No general geometric patch exporter or ambient/source binding is supplied by this algebraic compiler.

## 133.2 A small boundary quotient, under a checked simplicity promise

Report 77 concerns a marked closed orientable surface $\Sigma_g$, $g \geq 2$. For a multiple $q$ of $g$, use

$$H_g(q) = (\mathbb{Z}/q)^{2g} \times \mathbb{Z}/g, \quad (x,z)(y,t) = (x+y, z+t+c(x,y)), \quad c(x,y) = \sum_i x_{a_i} y_{b_i}.$$

The inverse is $(-x, -z + c(x,x))$ and a signed power is $(kx, kz + \binom{k}{2} c(x,x))$. The product of the $g$ standard commutators has central coordinate $g = 0$, so the surface relation is respected.

For an independently source-verified simple curve, nonseparating curves have a nonzero first coordinate: their integral homology class is primitive. A separating curve bounding a genus-$h$ side has signature $(0, h)$, up to orientation. Thus zero detects contractibility under this promise, and a nonzero central value recovers the unordered genera $\{h, g-h\}$. The general finite-quotient principle is classical, due to Livingston and Pikaart [55, 56]; this review makes no exhaustive novelty claim.

This small observer is not a general word-problem solver. In particular,

$$w = [[a_1, b_1], a_1[a_1, b_1]a_1^{-1}]$$

vanishes in every class-two observer here but is nontrivial in the surface group. Map $a_1, b_1, a_2, b_2$ to $x, y, y, x$ in a free group and all remaining generators to one. The surface relator maps to $[x,y][y,x] = 1$, whereas $w$ has nonempty free reduction. Agreement among these observers cannot certify simplicity. Ordered boundary words and geometric marking provenance are indispensable; a letter census loses the central coordinate. Ordinary contractibility also does not certify admissibility relative to a punctured or marked boundary pattern.

A power SLP with encoded size $S$, including exponent bits and reference nodes, can be evaluated in $O(Sg \log^2(g+1))$ bit operations using lazy basis-generator handling. Only $2g+1$ modular coordinates are stored, not an exponentially large group table. Safe transport under every oriented marking change requires common coordinate modulus $g$ for odd $g$ and $2g$ for even $g$. Indeed a generator image must retain order dividing $q$, which requires $g \mid \binom{q}{2}$; a twist sends $a_1^g$ to $(a_1 b_1)^g$ with central value $g/2$ when $g$ is even. Thus the narrower even-genus quotient is not invariant under all marking changes. Checking a supplied symplectic image tuple makes subsequent cached transport algebraically safe, at $O(g^2 \log^2 g)$ per value. It does not authenticate that tuple as a geometric change of marking.

## 133.3 Exactly which fixed double-cover banks detect every separator

Identify the character space with the symplectic space $V = \mathbb{F}_2^{2g}$. An essential separating curve determines a proper splitting $V = U \perp W$. In the double cover with nonzero character $\alpha$, either individual lift has nonzero homology exactly when $\alpha$ restricts nontrivially to both sides. When both side covers are connected they have two boundary circles; an individual lifted circle is nonseparating, while the sum of the two classes is zero. Testing only their sum loses detection. Every proper symplectic splitting is realizable by a separating simple curve, by changing a symplectic marking through transvections/Dehn twists.

For a bank $A$, put $X = \langle A \rangle$ and choose a basis $B \subseteq A$. Join each nonbasis character to all basis vectors in its support; also join basis pairs with nonzero symplectic pairing. Its components span the unique finest orthogonal decomposition $X = X_1 \perp \cdots \perp X_s$ in which every bank member lies in one summand. This decomposition is independent of the chosen basis, although its graph presentation can change.

**Proposition 133.2** (Universal fixed banks)**.** *A nonempty bank detects every essential separator through individual-lift homology if and only if this graph is connected and $X$ is coisotropic, $X^\perp \subseteq X$. The criterion costs $O(mg^2 + g^3)$ binary operations for $m$ characters. The minimum bank size is $g+1$.*

*Proof.* Write $d_i = \dim X_i$, $r_i = \dim \operatorname{rad} X_i$. Split each $X_i$ into a nondegenerate part and its radical. The sum of all radicals is isotropic in the complement of the nondegenerate parts. Extend its basis to a symplectic basis there and assign one dual partner per radical vector. This constructs simultaneous orthogonal nondegenerate envelopes $E_i$ of dimensions $d_i + r_i$.

If the interaction graph is disconnected, one envelope and its orthogonal complement form a proper splitting in which every character is pure. If the graph is connected but $X$ is not coisotropic, a single envelope of dimension $d + \dim \operatorname{rad} X < 2g$ gives such a splitting. Conversely any undetected splitting makes every character pure. Graph connectedness forces all of $X$ onto one side. Coisotropy forces a nondegenerate side containing $X$ to have dimension at least $2g$, a contradiction. Coisotropy is tested by $2d - \operatorname{rank}(\omega|_X) = 2g$.

At most $g$ characters either span a noncoisotropic space, or are an independent basis of a $g$-dimensional Lagrangian. In the latter case their interaction graph has $g$ isolated vertices and fails when $g \geq 2$. A Lagrangian basis $\lambda_1, \ldots, \lambda_g$ together with its sum is coisotropic and connected, proving the upper bound. □

In particular, the full $2g$ standard symplectic basis is not universal: its graph has $g$ components. The lower bound applies to fixed degree-two covers and individual-lift homology, not adaptive cover selection, higher-degree covers, or arbitrary curve algorithms.

## 133.4 A quadratic bridge between genus and lifted homology

The same report preserves a separate cellular/Magnus identity. Degree-two signatures over $\mathbb{F}_2$ have multiplication

$$(x, M)(y, N) = (x + y, M + N + xy^{\mathsf{T}}).$$

A positive generator is $(e_i, 0)$, its inverse is $(e_i, e_i e_i^{\mathsf{T}})$, and every signature has fourth power one. Let $J$ have standard symplectic $2 \times 2$ blocks. The surface relator is $(0, J)$; changing a word representative with first coordinate zero can toggle $M$ by $J$.

For a separating simple curve bounding genus $h$,

$$\{\operatorname{rank} M, \operatorname{rank}(M + J)\} = \{2h, 2(g - h)\}.$$

Indeed each commutator contributes $uv^{\mathsf{T}} + vu^{\mathsf{T}}$, giving the bivector of one symplectic summand; congruence preserves its rank and conjugation fixes a zero-first-coordinate signature.

More generally, for any word with zero mod-two first coordinate and a nonzero character $\alpha$, its individual lifted homology satisfies

$$[\widetilde{w}] \neq 0 \quad \Longleftrightarrow \quad M^{\mathsf{T}}\alpha \notin \{0, J\alpha\}.$$

To see this, traverse its letters in the two-vertex cellular cover. The prefix exponent vector determines the sheet of each positive edge; a negative letter adds its own character bit when locating the reversed edge. These contributions are precisely the contraction of the quadratic signature. Since total exponent parity is zero, both sheet chains are the same vector $M^{\mathsf{T}}\alpha$. The two lifted two-cells have the same boundary, the duplication of $J\alpha$, so the boundary image is its one-dimensional span. This proves the formula without expanding the SLP. It also supplies independent cellular and quadratic replay routes.

## 133.5 Causal localization of bounded tetrahedron descent

Report 79 strengthens report 75's local-region theory for a particular objective. Moves are relative 2–3 and 3–2 replacements on marked tetrahedron incarnations, with their source legality and transport checked. Construct the birth DAG of a trace: a directed arc joins a move creating an

incarnation to a later move consuming it, retaining parallel arcs. Two simultaneously legal moves with disjoint consumed incarnations commute. There is also a backward-persistence lemma: if the second move consumes no incarnation born at the first, all its sources survive the first move. Its internal facet, or complete three-tetrahedron edge star, is unchanged, so it was legal before the first move. This justifies swapping those moves.

**Proposition 133.3** (Connected first descent)**.** *If a descending trace uses at most $U$ upward moves, a descending trace exists with $u \le U$ upward moves, $d = u + 1$ downward moves, length $2u + 1$, and a connected original footprint of size at most $3u + 3$.*

*Proof.* Birth-DAG components project to legal traces by repeated swaps using backward persistence, and their tetrahedron losses add. Choose a shortest descending trace within the upward allowance and truncate at its first descent. Then $d = u + 1$. If its birth DAG had several components, one would have positive loss; its legal projection would be a shorter descending trace, a contradiction.

For a connected birth component with $u$ upward and $d$ downward moves, there are $2u + 3d$ consumed source incarnations. At least $u + d - 1$ are born within the component, so at most $u + 2d + 1$ are original. Their original footprint is connected: maintain ancestor blocks of original cells and the bijection between current cut facets and original boundary ports. Each connected source replacement merges adjacent connected ancestor blocks. At the end these blocks are exactly the birth-component footprints. Substitution of $d = u + 1$ gives $3u + 3$. □

Constructive extraction alternates projection to a positive-loss component and truncation at first descent until stable. Each step reduces the number of events; truncation alone need not produce a connected footprint. A six-tetrahedron ball shows the footprint bound is sharp at $U = 1$.

Enumerate connected original regions of radius $3U + 3$ in the bounded-degree dual graph, then the constant-alphabet local traces. With source-bit size $B$, this gives a complete bounded-descent algorithm $2^{O(U)}\,\mathrm{poly}(t + B)$, without a separate locality hypothesis. A negative result exhausts descent under that upward allowance; it is not a knottedness certificate. The report also proves a multiunit loss-$k$ bound $2^{O(U+k)}\,\mathrm{poly}(t + B)$ by decomposing independent birth components and using color coding/perfect hash families. Its deterministic splitter generator is not implemented; the executable packing routine solves one supplied coloring.

A shared cover index avoids restarting a trace in overlapping regions: it filters covers by the actually consumed original footprint. Full predicate completeness requires invariance under legal projection and commutation. Marked-endpoint descent predicates qualify; arbitrary history-dependent or selected-cover callbacks need not. At delivery the preserved implementation remained separate from the maintained trace/source interfaces. Section 148 later reconciles and integrates its bounded search and independent transport checkers, including an optional source-bound diagram adapter. No universal small-upward descent or terminal structure theorem for all unknots follows from this result.

## 133.6 Fresh finite evidence and preserved proof scope

Fresh scratch runs pass all 52, 47, 66 and 36 tests of reports 76–79, respectively. Report 76 checks 44,168 cap entries, 4,964 rectangular products, 41,438 trace pairings, 14,400 weighted queries, 39,926 powered versus explicit cap queries, 855 literal pair meshes and 1,408 complete-word meshes. Its $W = 2^{1024}$ example returns $-3W + 4$ with 1,025 compiled rules and 2,049 reachable witness nodes; this is one compressed-language example. Report 77 compares every one of 32,768 genus-two banks with literal coverage of the ten splittings, exhausts all banks of size at most $g$ for genera two and three, and checks 1,100 random splitting cases. The all-genus claims depend on the proofs, not these finite counts.

Report 78's fresh audit checks 600 flow models against 418,908 boxed potentials, 80 complete bands against 51,824 potentials, and 3,725 strata. Its 100 constructed meshes, 800 legal moves, 1,200 surfaces and 1,360 positive-core budgets pass. All 100 old and new outputs have Euler characteristic one, so this corpus demonstrates no added positive coverage. Report 79 replays 124 saved covered endpoints and eighteen strict descents, in addition to its 36 focused tests. Its much larger delivered regression corpus is preserved, not represented here as freshly rerun. Report 80's scratch reproduction passes 39 tests and seventeen saved coverage proofs. Native audits, integration tests and timings follow below. Fresh logs and structured evidence are retained under `data/incoming-eaad-*`.

# 134 Edge-first coherent surfaces and a reusable span dual

Report 78 changes the optimization order. Earlier selectors minimize normal-piece span $D$, then edge penalty $F$ within that minimum-span face. The new selector minimizes $(F, D)$ lexicographically over all integer vertex potentials in one coherent cocycle class. This reverses two genuine objectives; minimizing $D$ first can exclude the best connected Euler candidate. The native stage is optional because its current measurements show additional cost without additional positive recognition coverage.

## 134.1 Exact span bands and their real parameter

Write $h_{t,i}$ for the four integral corner heights in tetrahedron $t$, and $p_{v(t,i)}$ for global vertex potentials. Then

$$D(p) = \sum_t \bigl(\max_i(h_{t,i} + p_{v(t,i)}) - \min_i(h_{t,i} + p_{v(t,i)})\bigr),$$
$$F(p) = \sum_e w_e |c_e + p_{b(e)} - p_{a(e)}|, \qquad \chi = D - F/2.$$

Here $w_e = \deg_{\text{face}}(e) - 2 \geq 0$ is authenticated from the actual source. The coherent dual is cooriented and hence orientable. Let $D_*$ be the global minimum span with its independent primal/dual certificate.

The minimum-span anchor pairs make the sum of their height differences equal to $D_*$: potential contributions cancel at each global vertex. Consequently $D - D_*$ is the sum of $2T$ nonnegative integral defects from the chosen lower and upper anchors. For radius $k$, pick $j$ positive defects, one of three nonanchor attaining corners for each, and their positive sizes with total at most $k$. This gives at most

$$H(T,k) = \sum_{j=0}^{\min(2T,k)} \binom{2T}{j} 3^j \binom{k}{j} \leq (1+6T)^k$$

constant-span strata. Each has $O(T)$ difference constraints and permits exact min-$F$ network optimization in polynomial bit time. Attainer ties can repeat a potential in several descriptions without losing completeness. When $k = O(\log n)$ and source size and coefficient bits are polynomial, this supplied band can be searched in quasi-polynomial time. There is no theorem that every unknot has a useful coherent disc in such a band. An unrecorded streaming search also does not supply a portable negative certificate for every stratum.

## 134.2 Deleting zero-class coherent components

**Proposition 134.1** (Positive core and excess budgets)**.** *In an authenticated knot exterior, let a coherent surface $S$ represent the positive primitive relative class and have $D = D_* + \delta$ pieces. It*

*contains a positive-class component $C$ with $D(C) \geq D_*$, and*

$$D(S \setminus C) \leq \delta, \qquad \#\pi_0(S) \leq \delta + 1.$$

*If there are $q$ negative-class components, then $2qD_* \leq \delta$ and the total zero-class mass is at most $\delta - 2qD_*$. Also $\chi(C) \geq \chi(S) - \delta$.*

*Proof.* Connected cooriented components in a knot exterior have relative classes $0, +1, -1$. Their sum is $+1$, so there are $q + 1$ positive components. Choose one. The union of all others has zero class. Its signed edge intersection cocycle is exact, say $\delta g$, so deletion replaces the coherent edge cochain by the remaining cochain in the same primitive class. The remaining absolute edge weights uniquely determine its admissible normal coordinates. In one tetrahedron the kernel of the six edge-weight map is generated by $(1, 1, 1, 1, -1, -1, -1)$; two distinct admissible nonnegative vectors cannot differ by a nonzero multiple of it. Thus deletion leaves a coherent representative and the chosen component has at least $D_*$ pieces. Applying this also with orientation reversed to each nonzero component gives $(2q + 1)D_*$ nonzero-class mass. Every component uses a piece, giving the count bound. Finally, nonnegative edge weights give $\chi(S \setminus C) \leq D(S \setminus C)$. □

The report's nine-tetrahedron $S^1$-times-triangle staircase attains the opposite-pair bound for every $q$: $D_* = 3$, with $q + 1$ positive and $q$ negative three-piece components, hence $\delta = 6q$. These are uniform geometric statements, independent of the finite performance corpus.

### 134.3 Why edge-first optimization produces one component

The primary problem is the existing unconstrained integral min-$F$ network. Any independently verified optimal dual $f$ identifies its whole optimal face edge by edge, with $x_e = c_e + p_b - p_a$:

| | |
|---|---|
| $w_e = 0$ | no restriction |
| $\lvert f_e \rvert < w_e$ | $x_e = 0$ |
| $f_e = w_e$ | $x_e \geq 0$ |
| $f_e = -w_e$ | $x_e \leq 0$. |

This is complementary slackness for $w_e|x_e|$, not an extra choice of height chamber. Add upper and lower variables $U_t, L_t$ with constraints $p_{v(t,i)} - U_t \leq -h_{t,i}$ and $L_t - p_{v(t,i)} \leq h_{t,i}$. Minimizing $\sum_t |U_t - L_t|$ on that face is the secondary span network. The two integral networks take $O(T^2 \log T)$ indexed operations in the report's elementary model, with separate polynomial bit costs; they do not expand the height range.

**Proposition 134.2** (Connected lexicographic optimum and success threshold)**.** *Every nonempty primitive coherent lexicographic minimum of $(F, D)$ in a knot exterior is connected and orientable. If $F_* \leq 2D_*$, it is an essential disc or an annulus with exactly one inessential boundary circle, whose cap gives an essential disc. In particular, a coherent essential disc with at most $D_* + 1$ pieces guarantees success.*

*Proof.* Select a positive component as above and coherently delete the zero-class remainder. If its edge penalty is positive, deletion lowers the primary objective; otherwise a nonempty remainder lowers the secondary span. Both contradict optimality. Thus the optimum is connected. Its Euler characteristic is at least $D_* - F_*/2 \geq 0$. Its primitive relative class forces nonempty boundary, so classification of connected orientable surfaces leaves a disc or annulus. Two essential parallel annulus boundaries have total class zero or twice a primitive class, neither the prescribed primitive class; two inessential boundaries have class zero. Exactly one boundary is therefore inessential in the annulus case. Finally a disc with $D \leq D_* + 1$ has $F = 2D - 2 \leq 2D_*$. □

Negative Euler characteristic is a failed candidate, not a knottedness certificate. The primitive/source hypothesis is part of the theorem and cannot be inferred from an optimizer's arithmetic output.

### 134.4 Native proof formats and avoiding the second flow

The arithmetic producer `cocycle_lex.py` emits `normal-cocycle-edge-span-v1`: global vertex identifiers, a primary primal/dual proof, a secondary proof, piece count and Euler characteristic. The independent checker `cocycle_lex_verify.py` rebuilds the whole optimal face from the actual source and verifies both objectives without importing either optimizer. The outer `diagram-cocycle-lex-v1` binds the result to a canonical input diagram, its authenticated exterior and the primitive cochain. Actual normal edge weights, coordinates and Euler characteristic are replayed. Boundary shelling transports this source proof through the existing independently checked shelling record.

The maintained pipeline already has an independently verified global minimum-span dual. If the primary optimum attains that $D_*$, it is automatically also secondary optimal. We retarget the old span proof to the new coordinates and check it again; no second network is needed. Otherwise the producer runs the full secondary network. Both cases have independent replay and the same lexicographic contract.

The opt-in stage runs after the four-tree prelude and optional minimum-span stage, before additional tree trials. It reuses prepared source geometry. Only a successful disc/cappable-annulus candidate receives the outer source proof. Callback cancellation, errors and resource limits propagate through the producer and checker. The existing recognizer fallback remains available. The public switch is `normal_seed_edge_span=True`, or CLI `-normal-seed-edge-span`; the default remains false.

### 134.5 Validation and the reason this stage remains optional

The first native release passes 1,356 tests in 244.385 seconds. Its focused 37-test run includes one optional-dependency skip in the system interpreter; the full native run includes Regina. The audit exactly preserves all 85 legacy default answers and work records when the option is disabled. With it enabled, positives remain 24 of 85. Arithmetic completes on 84 sources, one is capped, 81 reuse the old span dual and three require the second network. Nineteen source-bound positive proofs and 75 fresh Regina connectedness/orientability/Euler/disc controls agree. On two random examples Euler characteristic improves from $-3$ to $-1$, still no disc. Mutation, 20,000-bit gauge, producer-disabled replay, shelling and cancellation tests exercise both secondary formats.

Two serial four-arm benchmarks each complete 220 measured calls and 44 warmups, with full seed/recognition proofs, replay and serialization. Unrelated Lean workers shared the host; both raw runs and A/A controls are retained, and these measurements are not presented as isolated-host timings. Early positive controls stay near parity. Failed seed queries become slower, with old/new ratios about 0.42–0.74; full recognition miss controls have ratios about 0.70–0.74. This is roughly 35–43 percent additional full-recognition time, not a speedup.

**Full recognition controls from the second shared-host run.**

| Complete recognition | default ms | enabled ms | default/enabled |
|---|---|---|---|
| optimized positive | 50.490 | 50.988 | 0.993 |
| genus one miss | 20.867 | 29.597 | 0.721 |
| trefoil | 18.081 | 25.402 | 0.711 |
| figure eight | 32.220 | 43.575 | 0.738 |

No observed coverage gain justifies default activation. The new theorem and independently

replayable candidate remain useful for future adaptive policies or sources where coherent span excess is demonstrably small. The frozen baseline is `5402cd990`; evidence and the maintained `normal_orbit_research.lex` driver are retained under `data/lex-*`.

# 135 Adaptive planar enumeration of complete standard rays

Report 80 extends the minimum-envelope method from a projective interval to a projective polygon. Automatic STANDARD enumeration now selects it when matching nullity is three. The existing nullity-one/two envelope paths remain selected, and higher-dimensional sectors retain the earlier support/arrangement choice. Explicit planar enumeration also accepts lower nullity. Selection is based on an exact geometric parameter rather than elapsed-time guesses. Complete sector discovery still uses its existing independently replayable disc/exhaustion consumer.

## 135.1 The common minimum fan describes actual standard faces

Let $k$ be the number of allowed labelled quadrilateral variables and $d$ the dimension of their matching kernel. Intersect its nonnegative cone with $\sum q_i = 1$. This compact section $P$ has effective dimension at most $d-1$; raw nullity three can still give an empty set, point or segment. For each global vertex the triangle potentials are linear forms, and the canonical lift subtracts their minimum. Choosing an attaining corner therefore defines a convex minimum cell in $P$, by inequalities comparing its form to all the others. Identical projected forms are collapsed.

The common refinement of these minimum fans and the coordinate faces of $P$ describes the canonical standard coordinate faces. A point is an extreme standard ray precisely at a vertex of this refinement. This follows directly from the matching equations and zero-coordinate faces: within a fixed minimum cell the canonical lift is linear; conversely, the zero triangle and quadrilateral coordinates defining a standard face recover the corresponding minimum and coordinate-face constraints. Thus Q-cone extremes alone need not give all standard extremes.

In a two-dimensional section, full-dimensional minimum cells and their faces suffice. A label active only on a boundary face adds no geometry: the difference between it and the adjacent minimum is a nonnegative linear function, whose zero set on a face cannot be an isolated interior point. The implementation clips these convex cells, extracts their internal segments, and overlays segments from different global-vertex groups. The rays are the original polygon vertices, individual fan vertices and intersections between group segments. No standard-rank filter is needed to remove spurious arrangement crossings.

**Proposition 135.1** (Output and construction bounds)**.** *Let $b$ be the number of vertices of $P$, and $m_v$ the number of distinct full-dimensional active minimum cells at global vertex $v$. Then*

$$R \le b + 2\sum_v (m_v - 1) + 9\sum_{v<w} (m_v - 1)(m_w - 1) = O(k^2).$$

*With one nontrivial minimum group the bound is linear. After kernel construction, a deterministic realization has elementary scalar-operation cost $O(k^3 + tk^2)$; rational coefficient-bit costs must also be included.*

*Proof.* A convex subdivision into $m$ minimum cells has at most $2(m-1)$ new vertices and at most $3(m-1)$ internal segments. Euler's planar formula and degree at least three at genuine new vertices give the first bound; the dual planar adjacency bound gives the second, since two convex cells share at most one segment. Summing individual fan vertices and the at most one crossing for each pair of segments from different groups gives the displayed output bound. The source equations provide only $O(k)$ relevant distinct potential forms after the equality contraction. Clipping every cell against its group's dominance inequalities costs $O(k^3)$ in this elementary construction; overlay

costs $O(k^2)$ and lifting at most $O(k^2)$ rays into $7t$ coordinates costs $O(tk^2)$. The exact rational charts and intersections have polynomially bounded bit length in source coefficient bits. A unit-cost rational count alone would not establish the bit-complexity claim. □

### 135.2 Independent area and length coverage replay

The producer is `sector_planar.py`; the independent checker is `sector_planar_verify.py`, using the source-bound `normal-sector-planar-v1` format. Its source model reconstructs the full uncollapsed corner matching graph and checks the chart and bounds. It does not import the producer's polygon clipper. Each submitted convex cell has source-checked vertices and dominance inequalities. Linear dominance at all vertices proves dominance throughout the cell.

Within a group, cells with different projected active forms have disjoint interiors: simultaneous equality on an open set would make the forms identical, contrary to canonical collapse. The checker verifies that their exact areas sum to the area of $P$. A finite union of closed cells cannot leave an open gap of positive area, and closure then covers its boundary. Hence the cells cover $P$. For effective dimension one the corresponding argument uses exact lengths; points and empty sections have separate controls. The checker independently reconstructs the overlay vertices and compares every normalized primitive lifted ray with the submitted list. Missing rays, malformed fractions, incorrect charts, hidden lower forms, source substitutions and cancellation are rejected.

This proves coverage of one supplied compatible sector. A negative disc result also checks every ray's disc predicate independently. It does not prove that the selected sector family contains a disc for every unknot, and its status must not be promoted to a knottedness verdict.

### 135.3 Reusable sparse source preparation, kept explicit

The additive `sector_sparse.py` module validates and prepares immutable source geometry once. Building another sector reuses that prepared source, whose identity is checked before certificate emission. Union by size and path halving contract zero-type corner equalities. Only $4k$ active labelled edges carry dense length-$k$ potential tuples; zero records share constants. Fundamental-cycle coefficients lie in $[-4, 4]$, permitting deterministic radix ordering in $O(k^2)$. Pre-elimination work is $O(t \log t + k^2)$, with subsequent rational elimination accounted separately. This reusable interface is valuable for repeated sector queries. The delivered one-shot measurements are slower than the current dense constructor, so native automatic one-shot enumeration retains that constructor. The planar enumeration benchmark below charges both arms for the same native full kernel preparation.

### 135.4 Seven rays and a uniform integral disc classification

The report gives a geometric family with $n$ Fibonacci base tetrahedra and two boundary caps. Use base type two and cap type one, and put $A = F_{n+1}$, $B = F_{n+2}$, $F_0 = 0$, $F_1 = F_2 = 1$. Every standard matching solution has unique parameters $x, y, z, b, u, v$:

$$\begin{aligned} i < n: &\quad (F_{n-i+1}x + b, F_{n-i+1}x + b, b, b; 0, 0, F_{n-i}x), \\ i = n: &\quad (Ax + b - y, Bx + b, u, b; 0, y, 0), \\ i = n+1: &\quad (Bx + b, Ax + b - z, b, v; 0, z, 0). \end{aligned}$$

Nonnegativity is exactly $x, b, u, v \geq 0$ and $0 \leq y, z \leq Ax + b$. The matching kernel has dimension three. Canonical lifting sets $h = \max(0, y - Ax, z - Ax)$, $b = h$, $u = v = 0$; general solutions add the three boundary vertex links. Substitution into the base equations gives $W_i = W_{i+1} + W_{i+2}$ and hence the Fibonacci rows, while the two attaching face equations determine all cap triangles except $u, v$.

**Proposition 135.2** (All-size double-cap classification)**.** *For every $n \geq 1$, the canonical sector has precisely the seven rays*

$$(1,0,0),\ (1,A,0),\ (1,0,A),\ (1,A,A),\ (0,1,0),\ (0,0,1),\ (0,1,1).$$

*The first four are essential meridian discs; the last three are inessential discs. Every integral canonical vector consists of $x$ meridian discs and $h$ inessential discs. A full integral vector consists of $x$ meridian discs and $b+u+v$ inessential discs.*

*Proof.* The three active forms are $0, Ax-y, Ax-z$. Their minimum fan in the nonnegative projective triangle has three cells and exactly the seven displayed vertices. Coordinate-face extremality proves these are all rays. For components, the base restriction is $x$ parallel meridian discs and $b$ vertex-link discs, identified by uniqueness of normal coordinates. With $u=v=0$, every cap piece meets its attaching face in one arc; the opposite-vertex triangle is absent. Attaching a disc leaf along one interval preserves a disc and cannot merge components. Apply this to the two caps successively. The arcs use distinct global edges even when attaching-face vertices are identified in the quotient.

Essentiality is preserved by replacing each old arc with the corresponding cap-boundary arc. Two disjoint systems in an abstract disc with identical endpoint pairing are related by a PL homeomorphism fixed on the boundary, constructed by cutting outermost arcs. It respects boundary identifications and descends to the quotient boundary torus. Thus the original meridians remain essential and links remain inessential. The free triangles $u, v$ are the new vertex links. Substituting $b=h$ gives the canonical census. The preserved realization proof handles quotient faces with an interior positive collar vanishing on the boundary, rather than assuming embedded face closures. □

Only $(1,0,0)$ among the essential rays is a Q-cone extreme; three further essential standard rays are created by the minimum fan. This uniform theorem explains why the planar enumeration is needed. Primitive integral coordinates alone still do not guarantee a connected surface.

## 135.5 Native validation and complete measurements

All 1,391 maintained tests pass in 250.052 seconds. The focused 72-test run passes in 2.205 seconds. Integration tests disable pairwise hyperplane generation and standard-rank filtering in the planar branch; positive discovery, independent producer-disabled coverage, missing-ray mutations, shared-source identity and actual-ray caps have explicit controls.

The audit selects 9,995 sectors across 48 frozen triangulations and checks every selected one of matching nullity at most three: 9,807 queries, including 463 of nullity three. This selection is exhaustive only on the declared smaller sources; it is not all supports on every large source. All 7,674 ray occurrences match the frozen complete standard lists, including 2,279 non-Q rays. The nullity-three subset contributes 2,821 rays and 1,475 non-Q rays, with 129 essential-disc occurrences, 84 non-Q. Automatic dispatch agrees with direct planar output on every audited query. On all 9,344 lower-nullity queries, the complete output order and statistics exactly equal the frozen incumbent. Eight fresh Regina complete standard enumerations match the frozen lists. The 188 selected sectors of higher raw nullity remain explicitly outside the gate.

Four additional fresh geometric families at base sizes 1, 2, 4 and 8 verify 300 full-coordinate parameter cases, forty independently replayed component censuses, 28 fresh Regina ray classifications and four fresh full standard enumerations. Every family has seven rays, four essential and three inessential. These finite controls support the separate all-size written proof; they do not substitute for it.

The benchmark compares the frozen native sector implementation at `6a34c5c0d`. Both arms pay full native dense kernel construction. Five interleaved four-arm rounds complete 220 measured

calls and 44 warmups on an otherwise idle observed host, with A/A arms retained. Complete enumeration includes every output coordinate, serialization and ray digest. That iterator has no portable coverage proof; such replay is not claimed inside enumeration time. Complete discovery includes its positive or exhaustion proof, independent verification and full proof serialization. Both arms return the same discovery status and their actual witnesses are verified. A smaller witness's differing byte length is recorded.

**Complete standard enumeration and output.**

| Input | old ms | new ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| Base 1, two caps | 4.332 | 3.178 | 1.363 | 0.997 | 1.004 |
| Base 4, two caps | 61.947 | 6.291 | 9.850 | 0.890 | 1.004 |
| Base 8, two caps | 676.910 | 11.143 | 60.054 | 0.999 | 1.011 |
| Base 16, two caps | 11931.837 | 24.230 | 493.136 | 1.008 | 1.001 |
| Base 4, one cap | 1.639 | 1.620 | 1.008 | 1.001 | 0.999 |
| Base 8, one cap | 3.522 | 3.537 | 0.992 | 0.993 | 1.012 |

At base size 16, complete enumeration drops from about 11.932 seconds to 0.024 seconds, a paired 493.14-fold gain, producing the same seven rays. Nullity-two one-cap controls retain the old envelope path and stay near parity. The size-four old A/A ratio is 0.890 in the first run. A separate fifteen-round repeat completes sixty measured calls and four warmups; old and new A/A ratios are 1.011 and 1.001, with paired gain 9.823. Both records are retained, rather than discarding the noisier run.

**Complete supplied-sector disc discovery and replay.**

| Input | old ms | new ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| Base 1, two caps | 3.119 | 7.089 | 0.449 | 0.954 | 1.035 |
| Base 4, two caps | 11.535 | 12.757 | 0.910 | 0.999 | 0.994 |
| Base 8, two caps | 38.402 | 20.807 | 1.854 | 0.989 | 0.997 |
| Base 16, two caps | 191.528 | 41.348 | 4.683 | 1.055 | 1.008 |
| Empty sector | 1.384 | 1.410 | 0.989 | 0.985 | 1.020 |

At size 16, complete discovery drops from about 0.192 to 0.041 seconds, a paired 4.68-fold gain. It improves 1.85-fold at size eight, but is about 2.23 times slower at size one and 10 percent slower at size four. The old iterator can find its first disc before paying full arrangement work, whereas the planar iterator prepares its minimum diagram before yielding. These regressions limit the discovery policy; enumeration gains must not be substituted for them. Callers can still select the old support/arrangement methods explicitly. Full recognition is not timed by these supplied-sector rows. The empty-sector control is near parity.

Four new-only complete coverage calls at base sizes 1, 8, 16 and 32 include production, independent coverage replay and serialization. Times are approximately 0.008, 0.024, 0.054 and 0.182 seconds; proof sizes are 995, 2,048, 3,442 and 6,869 bytes. These are capacity observations without a baseline ratio. Publication review separately replays the thirteen native audit proofs, four family proofs, four capacity proofs and seventeen delivered proofs with the producer disabled.

All 587 captured native source pins stay fixed during audit and timing. They refer to runtime release `cc44aae87`; a subsequent docstring-only edit documents the new dispatch and is checked to preserve its executable syntax tree. Historical edge-first evidence retains its own release pins. Reproduction uses `normal_orbit_research.planar_sectors` with `audit -fresh-regina` or `benchmark -rounds 5`, from `fast/`. Raw evidence, the noisy-row repeat and publication checks are retained under `data/planar-sectors-*`.

# 136 Sharp planar interactions and a Q-corner Euler criterion

Arrival `3e07f86cf` supplies report 81 while the preceding native integration is being completed. Its full delivered manuscript, PDF, alternative implementation and evidence are preserved by placement `6a92806b1`; the committed archive is subsequently retired. The report strengthens the output bound, extends the mathematical admission criterion, and isolates a cheaper Euler-only objective. These results are preserved independently of implementation performance. Its alternative planar modules use different certificate schemas from report 80; their patch is retained for comparison rather than overwriting the maintained producer/checker pair.

## 136.1 The interaction constant improves from nine to one

For the previous section's polygon $P$, set $A_v = m_v - 1$ and $A = \sum_v A_v$. The earlier segment-pair bound is valid but overcounts crossings within convex cell diagrams. Two diagrams with $f_1, f_2$ full-dimensional cells have at most $(f_1 - 1)(f_2 - 1)$ proper transverse crossings of open edges. In general position, Euler's formula gives an overlay cell count $F = f_1 + f_2 - 1 + X$. Each overlay cell is one convex intersection of a pair of cells, so $F \le f_1 f_2$. This proves the crossing bound. For degeneracies, choose a strict interior witness for every retained minimum function. Sufficiently small generic rational perturbations preserve these witnesses and each original proper transverse crossing in a disjoint neighborhood. They do not introduce more retained functions; applying the general-position bound proves the same inequality. Collinear overlaps add only existing endpoints, and crossings at existing vertices are already counted.

**Proposition 136.1** (Sharp abstract interaction bound)**.** *The complete ray count satisfies*

$$R \le b + 2A + \sum_{u<v} A_u A_v \le b + \frac{A(A+3)}{2}.$$

*For every finite prescribed list of nonnegative $A_v$, the first bound is attained by a rational affine-potential system. With normal-sector matching nullity three, the inherited graph count gives $A \le 3k + 3$, so $R \le (9k^2 + 29k + 18)/2$. With one varying group, $R \le 7k + 6$.*

*Proof.* Count original polygon corners, individual diagram vertices and proper cross-group crossings, using the preceding bounds. Since $\sum_v A_v^2 \ge A$, the second inequality follows. For sharpness choose $A_v$ parallel lines per group in pairwise nonparallel rational directions, with small generic offsets so every cross-group intersection lies inside a polygon containing zero. They give distinct boundary endpoints and no triple crossings. If $z = n_v \cdot x$, the forms $h_j = -jz + \sum_{s=1}^{j} \tau_s$ realize the successive strips with breaks at $\tau_j$. The vertex count is exactly the first displayed bound. These systems homogenize to rational potential cones. Realization of the quadratic lower bound by actual manifold triangulations is not proved. □

## 136.2 Certified forced-zero removal admits more feasible cones

For $C = \{q \ge 0 : A_\Sigma q = 0\}$, let $J$ be the coordinates positive in some feasible vector. Removing the other allowed types preserves every feasible standard vector. Summing one positive witness per surviving coordinate gives a vector strictly positive on $J$. Small perturbations in any direction of the rebuilt kernel remain nonnegative there, so that kernel equals the linear span of its feasible cone. Thus feasible-cone dimension at most three, rather than raw nullity at most three, suffices after this exact recompilation.

The following single homogeneous LP identifies $J$:

$$\max \mathbf{1}^T z : \quad A_\Sigma q = 0,\ q \ge 0, \quad 0 \le z \le \mathbf{1}, \quad z \le q.$$

Scaling and summing the individual feasibility witnesses makes every $q_j \geq 1$ on $J$, giving optimum $|J|$ and unique optimal auxiliary $z = \mathbf{1}_J$. The original $q$ need not be unique. Its dual is

$$\min \mathbf{1}^T u : \quad u, v \geq 0, \quad u + v \geq \mathbf{1}, \quad A_\Sigma^T y \geq v.$$

Strong duality supplies independently checkable support witnesses

$$A_\Sigma q = 0, \quad q \geq 0, \quad q_j \geq 1 \ (j \in J), \qquad w = A_\Sigma^T y, \quad w_j = 0 \ (j \in J), \quad w_j \geq 1 \ (j \notin J).$$

Indeed for any feasible $x$, $w^T x = y^T A_\Sigma x = 0$ is a sum of nonnegative terms, forcing every off-$J$ coordinate to vanish; $q$ proves the opposite inclusion. This is a standard maximal-support LP construction specialized to the sector, without a literature-priority claim. Adding $\sum q_j = 1$ or coordinate caps would invalidate its scaling proof. Neither delivered implementation installs this preprocessing. At that delivery all 188 higher-raw-nullity corpus sectors remained outside the native planar gate. Section 146 later implements support removal from completed Q lists and admits 22 of them to planar enumeration; the polynomial LP producer itself remains unimplemented.

### 136.3 Euler maxima need only the quadrilateral corners

**Proposition 136.2** (All-dimensional canonical convexity)**.** *Let $\lambda$ be a linear functional on standard matching vectors with $\lambda(L_v) \geq 0$ for every vertex link. On any nonempty normalized quadrilateral section $P$,*

$$\max_{q \in P} \lambda(\mathrm{lift}(q)) = \max_{a \in \mathrm{Vert}(P)} \lambda(\mathrm{lift}(a)).$$

*Every corner lift spans a non-link standard extreme ray. For compact finite three-manifold triangulations this applies to Euler characteristic; a nonpositive maximum excludes every essential normal disc in that sector.*

*Proof.* Let $U(q)$ be the signed linear triangle-potential matching vector with zero offsets, including singleton vertex groups. Canonical lifting gives

$$\lambda(\mathrm{lift}(q)) = \lambda(U(q)) - \sum_v \lambda(L_v) \min_{c \in G_v} \ell_c(q).$$

Each minimum is concave, so the expression is convex and positively homogeneous. A convex combination of corners has value at most their maximum; the reverse inequality is immediate. A Q-corner is extreme in any minimum cell containing it; the coordinate-face correspondence makes its lift standard-extreme. Euler values of compact manifold vertex links are one or two, meeting the sign hypothesis. An essential normal disc has no vertex-link component and is canonical, by the classical canonical surface properties reviewed by Burton [6]. It has nonzero quadrilateral vector and positive Euler value, contradicting a nonpositive maximum. An empty section contains only link surfaces. □

The maximum is of the normalized value $\chi / \sum q_i$, not the raw Euler characteristic of primitive integer representatives. Positive rescaling preserves sign, but not numerical maxima. A positive Q-corner can be a sphere, projective plane or inessential disc, so the essentiality observer is still required. An essential standard disc need not itself be a Q-corner. This criterion neither replaces complete standard enumeration nor supplies a complete sector family.

A primitive integral vector on an admissible standard extreme ray is connected. Each component is a nonnegative matching vector in the same compatible sector, so extremality makes it a positive scalar multiple of the whole vector. Bezout's identity for the primitive coordinates makes that scalar an integer; it cannot lie strictly between zero and one. This justifies skipping nonpositive-Euler primitive extreme rays during disc discovery. Primitivity without extremality does not give this result.

For nullity at most three, a proposed Euler precheck can decide this sign in $O(1 + t + k^2)$ rational operations after kernel construction. Build the bounded-coefficient linear Euler functional using one representative per physical face and edge in $O(t)$; project the relevant potentials to the fixed-dimensional chart and construct $P$ in $O(k^2)$. Aggregate the linear Euler term and link coefficients in one source pass, then evaluate the group minima at its at most $k$ corners in $O(k^2)$. A positive witness lift costs $O(t + k)$. Dense source replay and compressed disc testing are separate costs. This faster aggregated implementation is not installed; the maintained complete Q phase already provides the logically valid nonpositive-Euler exclusion, with its existing proof and costs. In dimension $d$, active-constraint enumeration tests at most $\binom{k}{d-1}$ corner systems, including collapsed sections, giving $k^{O(d)}$ supplied-sector Euler queries. Logarithmic nullity therefore allows local quasi-polynomial Euler queries, without a global coverage or essentiality theorem.

### 136.4 Arithmetic models, independent evidence and integration limits

The deterministic sort-and-merge realization has arithmetic bound $O(1 + k^3 + (t + k)R)$ and polynomial bit complexity. Both supplied Python implementations use exact sets/dictionaries; adversarial hash collisions are not covered by that sharper cubic bound and can add polynomial work. The geometric bound is not a claim of cubic worst-case Python bit time. The inherited cofactor estimate bounds primitive triangle coordinates by $4^s$ and quadrilaterals by $4^{s-1}$ for actual support $s$: the triangle incidence block is totally unimodular and each quadrilateral column has absolute column sum at most four. Thus output coordinates have $O(k)$ bits, even if elementary pieces are exponentially numerous.

Fresh scratch reproduction passes all thirty relevant tests, checks 304 abstract models with 609 compared rays and 1,409 feasible arrangement candidates, and passes 96 focused certificate/budget/cancellation controls. Its Euler audit compares 607 declared sectors using exact normalized maxima and source standard-ray membership. These fresh results are retained under `data/incoming-eaad-81-*`. The delivered all-48-source Regina audit and 1,344-test historical run are preserved as delivered evidence; they are not called fresh native runs here.

The conditional recognition statement requires a certified polynomial-size diagram exterior, a quasi-polynomially constructed complete family of low-nullity compatible sectors, and exact polynomial-time essential-disc testing per emitted ray. Local enumeration then preserves quasi-polynomial time. The complete-family hypothesis remains the substantive missing theorem. None of the sharpness, forced-zero, convexity or finite audit results proves it.

## 137 Progress-driven discovery before planar refinement

Complete planar enumeration and first-disc discovery have different cost profiles. The preceding release prepares every minimum cell and crossing before yielding a ray. Its small two-cap discovery cases are slower than the former arrangement iterator, even though complete enumeration is much faster. The new discovery policy responds to observed progress: test the original projective Q-corners first, and build the complete minimum overlay only when that prelude has not found a disc. Its complete fallback and proof formats are preserved.

### 137.1 A cheap prefix of the complete ray set

Let $P$ be the normalized nonnegative quadrilateral section. Every corner of $P$ is extreme in any minimum cell containing it. Canonical lifting is a linear isomorphism from that cell's cone to a standard coordinate face, so its corner lift is a standard extreme ray. This is the face theorem of Sections 132 and 136; it does not depend on the Euler convexity hypothesis about vertex links. A Q-corner is therefore a safe first candidate even when the source is not an authenticated knot

exterior. It still needs the existing exact essential-disc certificate before any positive result is returned.

For matching nullity three, construct the source Q-chart and its polygon, segment, point or empty section without constructing minimum cells. Project the retained triangle forms once. There are $O(k)$ forms, so the preparation costs $O(k^2)$ rational operations after kernel construction. Order domain corners by decreasing first allowed quadrilateral coordinate, with exact chart-coordinate ties. This is a deterministic candidate policy, not a theorem predicting essentiality. The first-corner preference is independent of the Fibonacci fixture parameters. Sorting uses constant-size keys in $O(k \log(k+2))$ comparisons. One full canonical lift costs $O(t+k)$ arithmetic operations; the first candidate is thus available in $O(1+t+k^2)$ post-kernel operations, with rational-bit, component-query, independent-replay and output costs separately charged.

**Proposition 137.1** (Complete adaptive discovery stream)**.** *Yielding all domain-corner lifts, then all remaining vertices of the common minimum subdivision, gives each non-link standard extreme ray exactly once. If a consumer stops at a successful corner, no minimum-overlay construction is required. If it resumes after the prelude, the complete stream has the same ray set as full planar enumeration.*

*Proof.* Domain corners are vertices of every subdivision and their lifts are standard-extreme by the coordinate-face correspondence. Build the common minimum subdivision using the already prepared chart and projected forms. Its vertices are exactly the complete standard-ray set. Remove only the exact domain points already yielded; normalization meets each nonzero ray once, so neither duplicate rays nor omissions result. Empty and singleton sections complete at the prelude. A segment resumes through its exact minimum breaks; a polygon resumes through all cells and cross-group intersections. The generator's control flow builds this refinement only after the consumer asks for a candidate beyond the prefix. □

The full geometric arithmetic bound stays $O(1+k^3+(t+k)R)$ under the deterministic realization described earlier. Every Q-corner is among the $R$ output rays and is lifted once. The prefix shares its chart and projected forms with the tail. It does not pay a second full preparation. This is a worst-case preservation result and an early-exit opportunity, not a uniform speedup theorem. Python hash-set collision costs retain the qualifications of the preceding section.

## 137.2 Complete enumeration at logarithmic nullity

The general arrangement fallback also gives a useful local complexity statement beyond the planar gate. This is elementary accounting for the existing implementation, without a novelty claim.

**Proposition 137.2** (Complete low-nullity sector enumeration)**.** *For a supplied compatible sector with $k$ allowed types, $t$ tetrahedra, matching nullity $d$, and encoded source size $B$, the general standard-ray arrangement algorithm has bit cost*

$$\operatorname{poly}(t+k+B)(1+k)^{O(d)}.$$

*The higher-dimensional automatic support/arrangement selector preserves this bound. Hence polynomial source size and logarithmic matching nullity give quasi-polynomial complete enumeration within one supplied sector.*

*Proof.* The contracted matching model has at most $8k$ retained triangle classes. If $p_v$ classes belong to vertex group $v$, the candidate hyperplanes comprise $k$ coordinate forms and at most $\sum_v \binom{p_v}{2} = O(k^2)$ potential differences, projected to the $d$-dimensional matching kernel. Identically zero and duplicate forms can be dropped. An extreme ray of a minimum-cell cone is determined by $d-1$ independent active hyperplanes from this list. This remains true when the feasible cone

has smaller dimension: its forced equalities are among the active constraints. Thus testing all $\binom{O(k^2)}{d-1}$ systems, followed by the exact standard-extremality filter, is complete. Nullity zero and empty sectors have their explicit no-ray branches.

Each candidate requires polynomial-size exact linear algebra, canonical lifting, source-coordinate validation and output. The bounded integral matching coefficients and determinant bounds control elimination bits; the inherited $4^s$ and $4^{s-1}$ coordinate bounds for actual support $s$ give $O(k)$ bits per primitive output coordinate. No elementary pieces are expanded. The automatic support alternative has at most its computed mixed-support count in candidate bases and is selected only when that count is below the arrangement count. Its per-candidate elimination is also polynomial. Computing the two combinatorial bounds uses polynomially many integer operations on polynomial-bit integers. Multiplying the candidate count by these polynomial costs gives the result. Even quadratic worst-case duplicate-table work only squares the candidate count and stays within $(1 + k)^{O(d)}$; no constant-time hashing assumption is needed for this broader complexity class. □

This strengthens the local enumeration accounting beyond the Euler-only logarithmic-nullity observation in report 81. It supplies neither a quasi-polynomial complete sector producer nor a bound for source hierarchy resets and essential-disc testing across all knot inputs. Those additional requirements are still needed for a general recognition theorem.

For the two-cap Fibonacci family, the first-corner behavior is uniform, not only an observation on the timed sizes. Its Q-cone is $(x, y, z) \in \mathbb{R}^3_{\geq 0}$. Only the $x$-corner has nonzero first allowed coordinate $q_0 = F_n x$; the two cap corners have $q_0 = 0$. The decreasing-coordinate policy therefore selects $x = 1, y = z = 0$ first after primitive normalization, since the last base coordinate is $F_1 = 1$. By the all-size component theorem of Section 135, this is exactly one essential meridian disc. A complete exact disc consumer thus succeeds on its first candidate for every $n \geq 1$ without requesting a minimum overlay. Cooperative interruption or an observer resource limit can still produce an inconclusive query. This family theorem does not predict first-corner success for arbitrary triangulations.

### 137.3 One allowance, ordinary positive and negative certificates

The native generator is `sector_planar_discovery_rays`. It uses one actual-ray allowance across the prefix and tail: its counter is not reset when the implementation switches to refinement. The already tested corner points are excluded before any further coordinate lift. A cap produces `INCONCLUSIVE` without an exhaustion certificate. The cooperative callback covers section construction, lifting, resumed refinement and the existing component checks; callable objects with false Boolean value still receive cancellation calls.

Automatic STANDARD discovery selects this policy at matching nullity three. The other automatic dimensions and explicit methods retain their previous behavior. Complete enumeration and planar coverage APIs retain their previous canonical order and statistics; they have no reason to prioritize a potential first disc.

The first candidate is inspected by the maintained compressed disc observer and independently replayed. A success uses the unchanged `normal-sector-witness-v1` source-bound proof. Exhaustion uses the unchanged `normal-sector-exhaustion-v1` with phase STANDARD and every ray's Euler value and required negative disc proof. Its independent dense checker compares a set, so a different traversal order changes no coverage claim. In particular, failing the corner prelude does not become a negative disc result. We deliberately do not install the proposed aggregated Euler-only negative precheck here: this stream keeps full standard-ray coverage and its established portable certificate contract.

### 137.4 Reusing a kernel after a prior Q phase

The bounded-support outer search already tries a complete Q phase before requesting STANDARD discovery on a `POSITIVE_EULER_ONLY` result. Starting a new corner prelude there would repeat work. The search now builds one producer matching kernel per selected sector and passes it through both phases. Its internal STANDARD continuation skips the new prelude because the Q phase has already tested those candidates; it uses the ordinary complete standard iterator. Per-phase candidate allowances keep their previous meaning. The established STANDARD checks still include their own corner entries; this change removes the redundant producer build and avoids adding another discovery prelude. The source/kernel is read-only throughout this query. Independent proof checkers continue to reconstruct geometry from the actual supplied source; cached producer geometry is not substituted for source authentication or verification.

This removes a redundant producer build for each sector requiring a second phase. It does not yet share preparation across different sectors and does not change which supports are enumerated. The outer negative status remains restricted to the declared support allowance, with no unqualified knot verdict. This supplied-triangulation research API is separate from the default diagram-recognition portfolio.

### 137.5 Validation and measurements

All 1,396 maintained tests pass in 227.198 seconds. The focused 48-test run passes in 1.719 seconds. Tests forbid overlay construction on an early disc, compare complete unique adaptive streams on every support of the three-tetrahedron two-cap source, replay a seven-ray negative result, remove each ray from its certificate, stop between stages, and check one shared cap at limits zero, one, three, six and seven. A genuine capped trefoil control verifies one producer kernel per selected support and identity reuse across failed Q and STANDARD phases.

The new native audit exactly reproduces the incumbent's complete canonical ray sequences and geometry statistics on all 9,807 selected eligible sectors. Adaptive complete streams give the same ray sets, without duplicates, on all those queries. On every one of the 463 selected nullity-three sectors, old and new discovery statuses match and the new certificate independently replays. All 44 observed positive queries finish in the corner prelude. All 419 negative proofs replay, with 415 requiring the full overlay and four ending on degenerate sections. Their transcript entry sets exactly equal the incumbent's. Eight fresh Regina full enumerations again match the frozen reference. This is the previously declared selection across 48 triangulations, not all supports on all possible inputs. All 589 captured source pins remain fixed during audit and timing.

The baseline is `89d3e8bee`: both its native sector module and planar module are frozen, with only the sector module's lazy import redirected to the frozen planar module. Other imported dependencies are checked unchanged at the runtime release. This prevents the incumbent timing from silently using the new planner. Five serial interleaved four-arm rounds complete 280 measured calls and 56 warmups. Source fixtures are prepared before timing. Discovery includes full kernel construction, actual disc or exhaustion proof, independent replay and complete serialization. Individual old/new witnesses can differ, so both are checked and their byte lengths are retained. No partial or capped run contributes a ratio.

**Complete supplied-sector discovery and proof replay.**

| Input | old ms | new ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| Base 1 | 6.683 | 2.498 | 2.684 | 1.013 | 1.011 |
| Base 4 | 12.500 | 4.966 | 2.521 | 1.009 | 0.976 |
| Base 8 | 20.785 | 8.848 | 2.363 | 1.012 | 0.991 |
| Base 16 | 42.106 | 23.028 | 1.874 | 0.996 | 1.023 |
| Base 32 | 107.636 | 63.952 | 1.641 | 1.014 | 1.005 |
| Complete negative | 28.560 | 28.914 | 0.994 | 0.995 | 0.997 |
| Empty sector | 1.410 | 1.412 | 1.000 | 0.999 | 0.991 |

The size-one query drops from about 6.683 to 2.498 milliseconds, a paired 2.684-fold gain. At size 32 it drops from about 107.636 to 63.952 milliseconds, a paired 1.641-fold gain. The full negative and empty-sector controls stay near parity. These ratios compare directly with the preceding planar release on the same host; they are not assembled from timings in different releases. The first-corner family proof above explains the successful prefix, while the complete negative control exercises the resumed fallback. There is no claim of a uniform first-corner success rate.

**Unchanged lower-nullity and complete-enumeration controls.**

| Input | old ms | new ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| One cap 4 | 3.440 | 3.452 | 0.996 | 0.999 | 1.002 |
| One cap 8 | 6.634 | 6.611 | 1.003 | 1.005 | 1.008 |
| Enumeration 1 | 3.175 | 3.189 | 0.996 | 0.999 | 1.012 |
| Enumeration 8 | 11.268 | 11.180 | 1.004 | 1.013 | 1.004 |
| Enumeration 16 | 25.141 | 24.671 | 1.018 | 0.986 | 0.973 |

Complete enumeration includes every output coordinate, serialization and ray digest; portable coverage replay is outside that iterator timing. It retains its previous output and costs, with paired ratios near one. Lower-nullity discovery also stays near parity. A/A medians over the whole run range from 0.973 to 1.023.

**Complete bounded-support scans on supplied capped exteriors.**

| Input | old ms | new ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| Capped trefoil | 33.488 | 31.639 | 1.052 | 0.997 | 0.973 |
| Capped figure eight | 64.924 | 63.044 | 1.031 | 1.004 | 1.015 |

These outer API calls exhaust every support of size at most one. Native candidate checks are included; this API does not emit a portable proof of the entire support-family exhaustion, and such replay is not claimed inside these times. Both arms return identical restricted-scope records. The first paired gains are about 1.052 and 1.031. Because these effects are small, a separate fifteen-round four-arm repeat completes 120 measured calls and eight warmups. Its gains are 1.052 and 1.041, with A/A medians between 0.996 and 1.005. Both runs are retained. The structural result is one producer build per support; the small timing estimates do not imply the same gains on every triangulation or in the diagram portfolio.

The maintained driver is `normal_orbit_research.planar_adaptive`, with `audit -fresh-regina` and `benchmark -rounds 5` modes. Raw evidence and the longer repeat are in `data/planar-adaptive-*`. Publication replay checks frozen source versions and saved certificates with the producer routines disabled. It retains the existing thirteen coverage proofs, all 463 discovery certificates and the 44 incumbent positive witnesses. General recognition and complete sector-family production remain separate unproved requirements.

# 138 Signed residuals expose nearby disc sectors

The preceding planar work improves supplied-sector queries. The next step connects those queries to actual diagram exteriors: after coherent candidates fail, search compatible type assignments near one of their supports. This produces a new source-certified disc for a known coherent miss. The search remains optional because nearby-sector coverage is not guaranteed and its cost on misses can exceed the incumbent fallback.

## 138.1 A real disc outside the tested coherent families

The three-crossing closure of the two-strand word $(1, 1, -1)$ is an unknot. Its canonical diagram exterior has sixty tetrahedra. The raw, tree and minimum-span coherent candidates do not supply a disc or cappable annulus. Complete standard enumeration of their actual sectors also misses a disc; positive Euler rays in some of those sectors are inessential.

A radius-one type probe of the raw and optimized supports checks 360 distinct neighbouring assignments without success. At radius two about the optimized support, an unfiltered scan finds a disc on its 4,920th query, after 36.464 seconds. The two edits permit type one at tetrahedron 10 and type two at tetrahedron 12, both previously unoccupied. The resulting sector has matching nullity two. Its first tested standard ray is a primitive disc with every coordinate zero or one; only eight quadrilateral types are actually used, although the allowed sector retains the other seed types. Independent diagram-source and normal-disc replay accepts it. The source-disc proof is independent of the unsuccessful coherent surface.

The two new columns cannot occur separately. Their joint exposure suggests an exact filter based on matching equations rather than attempting every nearby assignment. The filter reduces this example to two single-column candidates and 73 pairs. The standalone filtered probe finds the same disc after 31 queries in about 0.411 seconds, including residual planning. These exploratory measurements motivate the implementation; the paired native measurements below have separate, disclosed scopes.

## 138.2 Eliminating triangles once for all type alternatives

Use the actual standard matching equations, written $B\tau + Dq = 0$. Here $q$ has all $3t$ labelled quadrilateral coordinates; the algebraic system temporarily permits all types so it can describe alternatives. Admissibility is imposed again when a compatible sector is queried. The triangle block is a signed graph-incidence transpose. A spanning forest expresses corner triangles as integral linear potentials in $q$, up to their free link offsets. Every non-tree edge contributes its cycle equation. Pure-Q equations are also retained. The resulting sparse matrix $A$ satisfies

$$\exists \tau \ (B\tau + Dq = 0) \quad \Longleftrightarrow \quad Aq = 0.$$

This is exact linear elimination; no mod-two reduction or expanded normal pieces are used. For nonnegative compatible $q$, subtracting minima and adding link offsets gives the usual nonnegative canonical lift.

Let $S$ be the positive quadrilateral support of an actual validated seed vector. Compute an exact column-space echelon basis for $A_S$, and let $\pi$ be its linear quotient projection. For any nonbase type $j$, define the residual $r_j = \pi(A_j)$. Thus $r_j = 0$ precisely when $A_j$ is in the old column span. Nonzero rational residuals are reduced to a primitive integer direction with positive first entry, retaining a separate sign. Equality of unsigned keys then means proportionality; opposite signs mean a positive cancellation is possible. The scalar magnitudes need not agree.

The forest construction and sparse rational column elimination take polynomial bit work and storage in the actual encoded source. Fundamental cycle entries are bounded integers; elimination

coefficients and reduced residual keys have polynomial bit length by determinant bounds. Dictionary collisions can add polynomial work but do not change this complexity class. The residual matrix is a producer filter, not a certificate that the supplied source is the exterior of a particular diagram.

### 138.3 A complete radius-two filter, including replacements

Represent a compatible type assignment by a word of length $t$ in $\{\varnothing, 0, 1, 2\}$. Distance counts tetrahedra whose assignments differ. Deleting a type only sets its coordinate to zero and therefore restricts to a coordinate face. It cannot create a standard extreme ray absent from the containing cone. Other edits add a type at an empty site or replace the previously allowed type.

**Proposition 138.1** (Signed residual window filter)**.** *The union of non-link standard extreme rays over the full radius-two assignment window about $S$ is preserved by querying only:*

1. *the base sector;*
2. *each single nonbase type with zero residual;*
3. *each pair at distinct tetrahedra with both residuals zero;*
4. *each pair at distinct tetrahedra with nonzero oppositely signed proportional residuals.*

*For a radius-one window, only the first two kinds are needed. Replacements remove their old type before adding the new one, preserving admissibility.*

*Proof.* For any admitted normal vector, let $J$ be its positive nonbase types. Within the window $|J| \le 2$, and projection of its matching equation gives

$$\sum_{j \in J} q_j r_j = 0, \qquad q_j > 0.$$

If $J$ is empty its coordinates use only old types. If $|J| = 1$, its residual must vanish. If $|J| = 2$, either both vanish or they are nonzero opposite multiples. A pair with exactly one zero residual cannot cancel. This proves that every possible positive nonbase support is on the list.

Unused edits need not be kept. If a queried assignment allows a new type whose coordinate on a particular ray is zero, setting all quadrilaterals at that site to zero gives a coordinate face containing the ray. This face also belongs to the retained assignment that restores the old type there. Extremality passes from the original cone to its face and then from that face to the containing retained cone. Pure deletions are handled the same way. Thus every original window ray remains extreme in one of the retained sectors, rather than merely belonging to their feasible union. Conversely every retained assignment is itself within the radius-two window. The ray unions are therefore equal. □

There is no necessity-only positivity shortcut hidden in the actual disc test: each retained sector still receives complete standard-ray enumeration and the established exact disc observer. Opposite residuals alone need not make a replacement feasible after its old columns are removed. For additions at previously empty sites, the seed's strictly positive old support does give sufficiency for feasibility: solve the old-span equation, then add a sufficiently large multiple of that positive seed to remove negative old coordinates. Feasibility still does not establish disc topology or essentiality.

Writing $z$ for the number of zero residual columns and $p_h, m_h$ for the signed bucket sizes in nonzero projective class $h$, the unpruned candidate bound is

$$1 + z + \binom{z}{2} + \sum_h p_h m_h.$$

Same-tetrahedron pairs are removed. This can be far below the naive $1 + 3t + 9\binom{t}{2}$ assignment count, but its worst case remains quadratic. Nonzero residuals with the same sign are not paired; forgetting the sign would waste precisely the positivity information used by this theorem.

### 138.4 The logarithmic-radius conditional accounting

The type-window perspective supplies a precise remaining structural parameter. For a centre with matching nullity $d_0$, an edit deletes at most one old column and adds at most one new column. Deletion cannot increase kernel dimension, and addition increases it by at most one. Every radius-$r$ sector therefore has nullity at most $d_0 + r$. The full assignment window has at most

$$\sum_{j=0}^{r} \binom{t}{j} 3^j$$

members. By Section 137, complete local standard enumeration consequently has total bit cost

$$\mathrm{poly}(t + B)(1 + t)^{O(d_0+r)}.$$

In particular, polynomial source size and $d_0, r = O(\log n)$ permit quasi-polynomial complete window enumeration. The implemented residual planner treats radii one and two; larger radii require positive dependencies of several residuals rather than the pair classification above.

This is a supplied-window theorem. A general recognizer would still need a bounded complete set of centres, a logarithmic-distance coverage theorem for an essential vertex disc, polynomial essentiality testing and controlled source construction. Finding one real disc at radius two proves none of those universal claims. The earlier failed radius-one searches are retained as counterevidence to simply activating the extra queries everywhere.

### 138.5 Native integration and independent source binding

The optional `sector_radius` argument of `normal_seed_decide` accepts zero, one or two. Zero follows the prior code path exactly. After all configured coherent attempts fail, the new stage chooses the candidate with best Euler characteristic and then least piece span. `sector_residual.py` tests its base sector, prepares signed residual groups, then queries single zero columns, opposite pairs and zero pairs. Every stage shares the original work callback. Remaining orbit cycles are shared across all window component queries and any earlier planar capping queries. Budget interruption or incomplete component work cannot become a negative knot claim.

A success uses the existing strict `diagram-normal-disc-v1` schema: canonical input PD, actual exterior triangulation, normal coordinates and the independently checked disc-count proof. The final checker authenticates the diagram exterior and its disc component; it does not run residual planning, coherent extraction or sector discovery. It neither requires nor infers that this new disc is coherent. Boundary-shelling wrappers can now carry this ordinary disc schema as well: the checker first binds the original canonical exterior and every shelling move to the final source. Callback errors remain observable even when inner geometry validators convert malformed source errors to rejection.

The recognition switch is `normal_seed_sector_radius`; its CLI counterpart is `-normal-seed-sector-radius`, used with the native seed stage enabled. A positive result reports `native-normal-sector-window`. The default remains zero, preserving the earlier fast path. This optional source-certified accelerator coexists with the complete recognition fallback; it supplies no negative verdict from a missed local window.

### 138.6 Measured coverage, costs and publication checks

All 1,401 maintained tests pass in 234.317 seconds. The 26-test focused run passes in 4.368 seconds, including actual CLI recognition. Literal small-window ray unions match the residual-filtered union for radii one and two. Independent dense matching confirms the base nullities. Tests cover the new disc, missing source fields, altered coordinates, a different diagram, work interruption,

zero orbit allowance, false-valued cancellation and the ordinary-disc schema transported through genuine boundary shellings.

The native audit uses the same 85-source corpus and verifies exact disabled legacy equality on every input, including work, stages and proof contents. Enabled positives increase from 24 to 31. The seven added source cases have six distinct canonical PD codes, at crossing counts three through six; they are not seven distinct knot types or a random sample of inputs. Every added ordinary disc proof passes independent diagram-source replay and a fresh Regina compressing-disc check. Six have coordinate maximum one and the seventh maximum two. All seven use zero interval-orbit cycles through the existing unit-ray certificate path.

Thirty-four audit cases hit the two-million shared work cap. Twenty finish their requested window without finding a disc; seven stop at a new positive and the 24 incumbent positives return before window construction. These four counts account for all 85 cases. Capped cases do not prove absence, and even complete local-window misses remain inconclusive for the knot. The whole audit takes about 684.44 seconds, illustrating why default activation is not justified by the additional witnesses alone. All 592 captured source pins remain fixed during audit and timing.

The exploratory evidence also retains the 2,340 distinct radius-one queries about selected raw/optimized centres of four diagrams. Every query completes, with matching nullity at most three and at most five emitted rays; none finds a disc. These are genuine failures of that small neighbourhood, not failures of the complete recognition fallback. The radius-two motivating scan is a successful prefix, not an exhaustive negative experiment. Its source, edits, coordinates, original scripts and proof are retained.

Five serial interleaved four-arm rounds complete 200 measured calls and 40 warmups. The seed-query times include all native construction, planning, normal queries, successful source verification and serialized result output. All configured candidate work completes in these uncapped small cases. The separate complete-recognition rows include the established fallback when the native stage misses; every arm reaches the expected final verdict. All A/A arms, deterministic seed-result digests and full evidence byte sizes are retained. These timings compare radius zero and radius two in the same native implementation; the disabled implementation separately matches the frozen `6c35c3e53` source on the complete audit.

**Complete configured native seed queries and output.**

| Input | off ms | on ms | off/on | off A/A | on A/A |
|---|---|---|---|---|---|
| empty circle | 5.772 | 5.696 | 1.017 | 1.013 | 0.992 |
| optimized positive | 57.391 | 67.196 | 0.858 | 1.004 | 1.186 |
| genus one miss | 17.938 | 427.069 | 0.042 | 0.999 | 0.989 |
| trefoil | 15.682 | 673.458 | 0.024 | 0.995 | 1.008 |
| figure eight | 27.397 | 1402.391 | 0.019 | 1.001 | 0.994 |

The genus-one example now produces a native disc, but its native query takes about 427 milliseconds against the incumbent miss's 18 milliseconds. That comparison is a coverage/cost tradeoff: the older native query returns inconclusive and has not recognized the diagram. Early positive controls avoid window construction. The optimized-positive seed row has a noisy first-run ratio 0.858; a separate fifteen-round repeat gives ratio 1.005 with off/on A/A medians 1.010 and 0.989. Both runs are kept.

**Complete forced-portfolio recognition.**

| Input | off ms | on ms | off/on | off A/A | on A/A |
|---|---|---|---|---|---|
| empty circle | 0.035 | 0.033 | 1.048 | 0.931 | 1.262 |
| optimized positive | 50.211 | 51.131 | 1.007 | 1.008 | 1.005 |
| genus one miss | 20.640 | 440.936 | 0.047 | 0.999 | 1.011 |
| trefoil | 18.122 | 715.580 | 0.026 | 1.022 | 1.019 |
| figure eight | 45.882 | 1503.935 | 0.031 | 1.454 | 0.997 |

Complete recognition of the genus-one example takes about 441 milliseconds with the new stage, against about 21 milliseconds with the incumbent's complete fallback. The trefoil and figure-eight controls are likewise much slower with extra windows. This is a regression, not a speedup in complete recognition. A longer noisy-row repeat completes 180 measured calls and twelve warmups. Its empty-circle control is at parity, and the figure-eight off/on ratio is 0.022 with both A/A medians about 1.001. This confirms the large regression rather than discarding it as first-run noise.

The stage therefore stays off by default. Its value is the additional independently replayable normal witnesses and the polynomial residual filter in place of an unfiltered local scan. No performance claim is made for enabling it on every diagram. A future scheduler would need evidence that its predicted window work is worthwhile against the available complete fallback, in addition to any structural completeness theorem.

Reproduction uses `normal_orbit_research.sector_windows`, with `audit -fresh-regina` or `benchmark -rounds 5`, from `fast/`. Raw records, exploratory executions, test logs and the longer control run are in `data/sector-windows-*`. The native runtime revision is `a498521ef`. Publication checks independently replay all 31 enabled positive proofs with coherent/sector/residual/orbit producers disabled, verify frozen source pins and retain rendered-page review. The seven fresh Regina observations support the actual new witnesses; the general window and nullity statements depend on the proofs above.

# 139 Updating window kernels in their small parameter space

The signed-residual filter avoids most nearby type assignments, but the first implementation still rebuilds a matching kernel for each retained one. Its motivating example queries 32 sectors. Each rebuild repeats contraction, cycle construction and rational elimination on nearly the same source. The new implementation retains the exact information computed by the residual planner and updates the kernel in at most $d_0 + 2$ parameters. It also reuses source triangle potentials instead of rebuilding the triangle graph. The independent diagram/disc certificate contract remains the same.

## 139.1 Column decompositions retain more than a residual key

Use the fixed full-Q matrix $A$ of Section 138 and the old support $S$. During exact elimination retain coefficient vectors alongside the column-space basis. For every nonbase column this gives

$$A_j = A_S c_j + r_j.$$

The old planner stored only the projective direction and sign of $r_j$. The new records also retain $c_j$ and the actual rational residual values. The forest's original corner potential matrix $P$ is retained as sparse integer rows. It describes signed matching triangle coordinates $Pq$; subtracting each global vertex minimum gives the canonical nonnegative lift. Neither record expands elementary normal pieces.

Let $K$ be a row basis of $\ker A_S$, already computed by the first sector query. The old kernel vectors, padded with zero new coordinates, remain kernel vectors when types are added algebraically. The following small updates supply all remaining directions.

**Proposition 139.1** (Exact augmented Q kernels)**.** *For a single zero-residual column, append the vector $(-c_j, 1)$ to $K$. For two zero-residual columns, append their two independent vectors. For a nonzero opposite pair $a, b$, choose $\gamma > 0$ with $r_a + \gamma r_b = 0$, and append*

$$(-c_a - \gamma c_b, 1, \gamma).$$

*In each case these vectors and the padded old basis span the full augmented matching kernel and are linearly independent.*

*Proof.* Each displayed vector satisfies the matching equation by its column decomposition. For zero residuals the new coordinates are independent. For a nonzero opposite pair the quotient equation forces the two new coordinates to be proportional with ratio $\gamma$, giving exactly one new direction. Subtracting the appropriate displayed vector from any augmented kernel vector leaves zero new coordinates and hence an old kernel vector. This proves spanning, while independence follows by first examining the new coordinates and then the independent old basis. □

Replacement edits must also set the replaced old coordinates to zero. If $W$ is the augmented basis with $m \leq d_0 + 2$ rows, impose

$$\sum_{a=1}^{m} z_a W_{a,i} = 0$$

for each removed old index $i$. There are at most two such equations. Their small nullspace gives the exact remaining parameter vectors; apply them to $W$ and drop the removed coordinates. This preserves all replacement constraints without solving a fresh wide matching system.

## 139.2 Recovering the incumbent's canonical basis gauge

The basis gauge matters for deterministic projective charts and traversal order. The native wide elimination chooses the earliest possible pivot columns, leaving the complementary free coordinates. Starting from the spanning kernel rows, reduce with columns scanned from right to left. Its pivots are those complementary free coordinates. Reverse the columns back and order the rows by increasing free index. The result is exactly the native nullspace basis, with an identity matrix in those free slots.

This uses thin row elimination on at most $d_0 + 2$ rows. The underlying linear fact is the complementary-basis duality between a matrix's column matroid and its kernel coordinate matroid. Equivalently, the greedy latest independent coordinate functionals on the kernel complement the greedy earliest independent matching columns. The normalized kernel basis is unique once these free coordinates are fixed. Exact source comparisons below test the resulting full basis, not merely its dimension.

## 139.3 Projected triangle geometry without a new contraction

For an updated Q basis $K'$, compute the source corner forms $PK'$ using the retained sparse forest potentials. Equal forms at a global vertex are identified, keeping the first physical corner as representative. This is an exact contraction on the entire updated matching space. It can merge more corners than the old zero-label contraction, because matching constraints can force a nonzero-looking potential difference to vanish. Single-form vertex groups are pure link factors and are omitted in the canonical lift, just as before.

The canonical basis gauge provides free coordinates $F$ with $q_{F_a} = z_a$. A corner form can therefore be expressed directly in those physical Q coordinates, using nonzero coefficients only at the free slots. The normalized envelope and planar producers can reuse their existing interfaces. Potential values can be rational in this representation, but the exact lift at an integral matching Q vector is integral by the original source forest equations. The producer checks this integrality and validates the full normal coordinates against the actual source.

Higher-nullity fallbacks remain complete. The projected kernel supplies linear Q constraints

$$q_j - \sum_a K'_{a,j} q_{F_a} = 0 \qquad (j \notin F)$$

and triangle constraints equating each class to its vertex anchor plus the corresponding potential difference. These describe precisely the updated standard matching space, including one free link offset per nontrivial vertex group. Coordinate-face extremality and the generic support/rank tests therefore continue to apply. Nullity zero means no non-link ray and completes without allocating another geometry kernel.

The change is implemented in `sector_window_basis.py`. It replaces the internal per-window rebuild; the public sparse constructor, standalone enumeration and independent coverage checkers retain their previous paths. The source is read-only throughout the query. Positive source replay still reconstructs geometry independently and never trusts the cached potentials, updated basis, class identifications or planner statistics.

## 139.4 Complexity and retained scope

The residual planner pays a polynomial one-time cost for the actual column decompositions. Per edit the matching update combines at most two new directions with the old kernel, solves at most two equations in at most $d_0 + 2$ parameters, and canonicalizes that thin basis. Triangle projection then uses the retained sparse source rows. Wide kernel elimination and repeated source graph contraction are removed from each candidate query. Exact rational coefficient bits and output costs remain separately charged.

This preserves the local quasi-polynomial accounting for logarithmic nullity and window radius from the preceding section. It neither expands the radius-one/two window nor supplies a universal distance bound. The default window radius remains zero unless the measured full-recognition tradeoff justifies a later policy change. Changes to deterministic guard counts can alter which queries complete under a finite shared work limit; they do not turn capped local searches into absence certificates.

## 139.5 Source comparisons, timings and publication checks

All 1,404 maintained tests pass in 243.872 seconds. The focused sixteen tests pass in 14.118 seconds. For every tested small residual window, the updated full basis exactly equals fresh native reconstruction; the complete ray sets, source-coordinate lifts and extremality tests agree. An additional genuine four-tetrahedron source reaches nullity four and checks both generic arrangement and support enumeration. A real diagram-window test forbids more than one public full basis build and independently replays its resulting ordinary disc proof.

The 85-source audit exactly preserves every disabled legacy result. With radius two enabled, all verdicts equal the preceding release and all 31 native positive witnesses remain available. The seven ordinary window discs again receive fresh Regina confirmation. Shared-work caps fall from 34 to 30; complete window misses rise from twenty to twenty-four. The four additional completed cases are `random-8`, `random-14`, `random-41` and `random-135`. They still do not yield knot verdicts. This is a guard-count improvement, not proof that every query fits the same bit-time or wall-clock allowance.

The updated audit takes about 260.75 seconds, whereas the retained preceding audit took about 684.44 seconds. Those are different execution records, not a paired speedup estimate. The incumbent radius-two corpus observations come from their frozen release data; disabled behavior is checked afresh against the frozen implementation. All 595 captured source pins stay fixed. The first audit attempt reached its final source and failed a research harness tuple/list comparison; the corrected comparison and completed rerun provide the published evidence. No interrupted run supplies final counts.

For fair native timings, the matching-rebuild, seed and recognition modules at `11dde31ee` are frozen. Only their local imports are redirected to the corresponding frozen dependency. No timed mock/patch context is charged to either arm. Five serial interleaved four-arm rounds complete 200 measured calls and forty warmups. Seed times include full construction, planning, actual normal queries, independent successful source replay and serialized output. Both arms complete their configured candidate work. The tested positive certificate digests exactly agree, as do final statuses; implementation statistics and guard counts can differ.

**Complete configured native window queries.**

| Input | rebuild ms | update ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| empty circle | 5.983 | 6.042 | 0.994 | 1.003 | 1.020 |
| optimized positive | 60.032 | 58.637 | 0.992 | 1.011 | 0.964 |
| genus one miss | 417.903 | 286.440 | 1.453 | 0.989 | 0.958 |
| trefoil | 666.928 | 518.970 | 1.259 | 1.014 | 0.986 |
| figure eight | 1400.078 | 967.272 | 1.453 | 1.002 | 0.994 |

The genus-one native query drops from about 417.903 to 286.440 milliseconds, a paired 1.453-fold gain. The trefoil and figure-eight native window queries improve about 1.259 and 1.453 times, respectively. The motivating query now performs one full basis build and 31 updates, retaining its 32 sector queries, 22 emitted rays and exact disc coordinates. Early positive controls return before the update planner and remain near parity.

**Complete recognition with radius-two windows enabled.**

| Input | rebuild ms | update ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| empty circle | 0.020 | 0.019 | 1.036 | 0.961 | 0.948 |
| optimized positive | 50.791 | 50.581 | 1.001 | 1.010 | 1.005 |
| genus one miss | 427.039 | 297.222 | 1.430 | 0.981 | 1.027 |
| trefoil | 696.116 | 557.498 | 1.248 | 0.990 | 1.018 |
| figure eight | 1453.388 | 1020.106 | 1.431 | 1.011 | 1.001 |

Complete enabled recognition improves about 1.430 times on the genus-one case, 1.248 on the trefoil and 1.431 on the figure-eight. These times include the established complete fallback where the native window misses. Every arm reaches the expected final verdict; no partial or capped run supplies a ratio. The improvement is relative to rebuilding the same window kernels, not relative to disabling windows. On these small inputs, radius zero remains substantially faster, as the preceding section records. This optimization does not justify default activation or a universal speedup.

The empty-circle recognition control lasts only about twenty microseconds, so its first-run A/A ratios of 0.961 and 0.948 are not interpreted as a kernel speedup. A separate 31-round four-arm repeat completes 124 measured calls and four warmups; its old/new ratio is 1.015, with A/A ratios 0.993 and 1.008. Both records are retained. The longer control never enters the window planner. Other reported window-search A/A controls are retained in the full tables.

Reproduction uses `normal_orbit_research.window_basis`, with `audit -fresh-regina` or `benchmark -rounds 5`, from `fast/`. Runtime release `b1741c008`, raw records, focused/full tests,

the longer control and publication checks are retained under `data/window-basis-*`. Publication replay checks the frozen and current source pins and all 31 saved positives with matching updates, projected geometry, residual planning, coherent extraction, sector enumeration and orbit producers disabled. Independent source-disc verification remains the authority for positive results.

# 140 Euler screening before window geometry and shared disc observers

The kernel update removes wide matching elimination, but repeated source geometry still costs time. This continuation implements the Euler-corner criterion preserved from report 81 in Section 136. It screens updated windows before building their standard matching geometry. Positive-Euler windows retain their ordinary standard-ray and independently replayed disc queries. A second change shares validated producer geometry with those disc queries, while leaving independent certificate reconstruction in place.

## 140.1 Compile Euler characteristic from the actual source

For each global edge choose one local representative. A normal surface's intersection weight there is

$$w_{\{a,b\}}(x) = x_a + x_b + \sum_{q=0}^{2} x_{4+q} - x_{4+Q(a,b)},$$

where $Q(a,b)$ is the quadrilateral type disjoint from that edge. Matching makes this weight independent of the representative. Count normal arcs on every boundary face and one representative of every paired face; on face $f$ their count is $\sum_{v \neq f} x_v + \sum_q x_{4+q}$. Each elementary normal disc supplies one face of the induced surface cell structure. Thus

$$\chi(x) = \sum_{\text{global edges}} w_e(x) - \sum_{\text{global faces}} \operatorname{arcs}_f(x) + \sum_{t,j} x_{t,j} = \beta \cdot x.$$

The native compiler forms the integer vector $\beta$ by precisely these three sums. It neither expands discs nor assumes that a local edge is a separate global edge. Independent native coordinate analysis remains the reference for the resulting Euler values.

Let $P$ be the retained all-Q source forest potentials of Section 139. The signed standard extension is $U(q) = (Pq, q)$. Compile once the coefficients

$$\gamma_j = \beta_{Q,j} + \sum_c \beta_{\triangle,c} P_{c,j}, \qquad \lambda_v = \sum_{c \in G_v} \beta_{\triangle,c}.$$

The second sum evaluates $\beta$ on the actual vertex link. The compact manifold source has $\lambda_v \in \{1, 2\}$; the compiler checks these values. For an updated basis $K'$ and parameters $z$, the existing projection gives corner forms $\ell_c(z) = (PK')_c z$. Canonical normalization subtracts the minimum in each global vertex group. Consequently

$$\chi(\operatorname{lift}(q)) = \gamma \cdot q - \sum_v \lambda_v \min_{c \in G_v} \ell_c(z), \qquad q = \sum_a z_a K'_a.$$

This evaluates the exact Euler value without allocating the full coordinate vector or reconstructing matching and edge-weight geometry for that vector. It includes every physical corner, even those that the later projected kernel would contract or omit as a pure link factor.

## 140.2 A complete nonpositive screen and a conservative positive branch

The displayed formula is positively homogeneous and convex: the minima of linear forms are concave and all link coefficients are positive. On the compact normalized nonnegative Q section, a convex combination of corners has Euler value at most the same combination of their values. Nonpositive values at every corner therefore exclude positive Euler everywhere in the sector. An essential normal disc is canonical and has nonzero Q coordinates and Euler characteristic one, so it cannot occur in an excluded sector. Positive Euler alone still proves neither connectedness nor essentiality.

The implementation uses the existing exact normalized chart and polygon clipping at matching nullity at most three. Empty, point and segment sections remain possible even when the raw nullity is larger. It evaluates every corner needed for a nonpositive answer; a callback interruption propagates and cannot supply an exclusion. As soon as a positive corner is found, ordinary standard discovery resumes. The projected forms are reused when building the standard kernel, so the screen does not project the source a second time. At nullity above three it continues through the existing complete generic arrangement/support paths without an Euler exclusion.

Nullity one has a cheaper branch. The canonical basis row has a positive unit free coordinate; any negative entry therefore makes its nonnegative cone empty. This check precedes corner projection. Otherwise the sole primitive direction's Euler sign can be evaluated directly from scalar corner forms. No normalized rational rescaling or affine chart is needed. The sign, rather than an unnormalized numerical maximum, is the criterion.

These exclusions are internal search decisions, not portable negative knot certificates. The original radius-one/two window, its candidate order and its positive proof format are retained. A completed nonpositive sector can be skipped, while any positive-Euler sector still receives the full ordinary standard-ray query. Finite shared work and orbit caps remain inconclusive. The radius-zero recognizer path stays unchanged.

## 140.3 Reuse producer geometry without weakening source replay

The public disc-count interface continues to validate the actual source, coordinates and all options. It then delegates to a private observer on that validated geometry. The window already owns a read-only prepared source and has freshly analysed each candidate vector against it; it can call the same private observer directly. This removes an additional source preparation and coordinate analysis from each positive-Euler candidate query.

The canonical disc core is also compared exactly with the analysed input rows. If they agree, its geometry is already available. Otherwise the reduced core is analysed afresh. Equality handles genuine primitive canonical rays without assuming that every supplied normal vector has those properties. Orbit-census fallbacks retain their established behavior.

No certificate verifier uses this private observer. Independent disc replay still reconstructs the source and input geometry, removes links and scale arithmetically, and checks the core proof separately. Successful diagram proofs still authenticate the canonical diagram exterior. The reused producer records are not an accepted substitute for any of these checks.

## 140.4 Cost and scope

The source Euler compilation costs $O(T+\operatorname{nnz} P)$ integer additions after the planner has supplied $P$. For a $d$-row updated basis, projection costs $O(d \operatorname{nnz} P_S)$ arithmetic operations. The implementation scans physical corner forms at each tested Q corner; with $m$ such corners its Euler work is $O(md(T+k))$ plus the exact normalized-chart construction. It does not claim the fully aggregated $O(T+k^2)$ precheck proposed in report 81. Eliminating the remaining repeated physical-corner

scans is a separate possible improvement.

Excluded sectors avoid standard matrix allocation, minimum-envelope/overlay construction, full ray lifts and their topology queries. Positive sectors pay for screening before their unchanged complete standard search. Producer geometry reuse removes repeated validation rather than changing compressed observer mathematics. Bit costs, coefficient growth, output and independent replay remain charged. The preceding local quasi-polynomial bound for logarithmic nullity/radius survives; no universal centre family, distance bound or general quasi-polynomial unknot algorithm follows.

## 140.5 Validation and measured behavior

All 1,407 maintained tests pass in 240.354 seconds; the fifteen focused tests pass in 18.920 seconds. Every tested small source window compares the screen with complete freshly reconstructed standard rays. Both the compiled linear functional and canonical formula agree with independent native coordinate analysis, including scaling by $2^{90} + 1$. A genuine nullity-four source continues through the existing complete fallback. The private prepared observer reproduces all seven genuine fixture queries and their independently replayed disc proofs.

The 85-source audit exactly preserves all disabled legacy outputs, all enabled verdicts and all 31 native positives. Seven ordinary window discs receive fresh Regina confirmation. Work-capped cases fall from thirty to twenty-eight, while complete window misses rise from twenty-four to twenty-six. The newly completed sources are `random-31` and `random-144`, at 1,552,225 and 1,751,788 guard calls. They still yield inconclusive native knot results. The genus-one positive keeps its 32 sector queries and 31 basis updates, emits eighteen rather than twenty-two rays, and prunes twenty sectors, including empty Q sections. Its work drops from 127,635 to 109,610 calls. All 598 captured source pins remain fixed. The enabled incumbent corpus observations are frozen preceding-release data; disabled comparisons and all updated queries are performed afresh. The 234.535-second audit is not a paired timing against earlier audits.

An initial Euler-only pilot completes 120 measured calls and forty warmups. It shows only small gains on two window cases and approximately parity on the trefoil. Its raw evidence and the exact source snapshots needed to recover all 598 original pins are preserved. It does not establish a uniform speedup. Profiling then motivates the prepared-observer integration.

The final comparison freezes five preceding-release modules at `1539f353a`: disc counting, window basis construction, residual queries, seed queries and recognition. Their local imports are redirected only to corresponding frozen dependencies; no timed patch context is added. Five serial interleaved four-arm rounds complete 200 measured calls and forty warmups. Native timings include the whole configured window query, independent successful source replay and serialized output. Both arms finish all configured work. Positive certificate digests agree exactly.

**Complete native queries with radius-two windows.**

| Input | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| empty circle | 6.030 | 5.927 | 0.985 | 0.983 | 1.004 |
| optimized positive | 59.189 | 58.038 | 1.015 | 1.009 | 0.982 |
| genus one miss | 281.501 | 246.270 | 1.143 | 0.989 | 1.067 |
| trefoil | 545.240 | 422.983 | 1.233 | 1.044 | 1.014 |
| figure eight | 994.493 | 754.190 | 1.312 | 1.007 | 0.981 |

The first genus-one seed comparison has a new-arm A/A ratio of 1.067. A separate fifteen-round repeat completes sixty measured calls and four warmups with the same full query, replay and serialization scope. Its paired old/new gain is 1.228, with A/A controls 1.020 and 1.030. Both runs are retained; the noisier first row is not the sole evidence for a genus-one native speedup.

**Complete recognition with radius-two windows enabled.**

| Input | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| empty circle | 0.019 | 0.018 | 1.020 | 0.973 | 0.958 |
| optimized positive | 50.866 | 50.965 | 1.002 | 0.999 | 0.993 |
| genus one miss | 299.243 | 239.963 | 1.230 | 0.993 | 0.983 |
| trefoil | 561.791 | 447.279 | 1.241 | 0.999 | 1.013 |
| figure eight | 1040.929 | 791.445 | 1.318 | 0.990 | 0.999 |

Complete recognition improves about 1.230 times on the genus-one case, 1.241 on the trefoil and 1.318 on the figure-eight. The latter two include the established complete fallback after native misses. These rows have A/A controls within about two percent of parity. Early positive controls remain approximately unchanged. The empty-recognition row is a microsecond control; its ratio is not interpreted as a kernel speedup. These gains are relative to the preceding enabled-window implementation. Radius zero remains much faster on the small controls, so the window stage stays off by default.

Runtime release `f95a86fdf` contains the implementation, regression checks and `normal_orbit_research.window_euler` driver. Use `audit -fresh-regina` or `benchmark -rounds 5` from `fast/`. Publication checks compare 598 frozen/current runtime pins, 595 preceding release evidence pins, all timing outcomes and both saved exploratory/repeat records. They independently replay all 31 saved positive proofs with Euler compilation, prepared observers, matching updates, residual planning, coherent extraction, ray enumeration and orbit producers disabled. The checked `openai/math` head remains `fd4aeeb2ee4f`; no new external revision supplies an additional complexity claim.

# 141 Carrying window updates through cached potential modes

Matching updates use only $d_0 + 2$ parameters, but the preceding version still projects the full source potential matrix onto each updated wide Q basis. This continuation carries the same thin row operations through cached source modes. It computes exactly the same corner forms, then uses the existing Euler screen, standard geometry and independent disc observers.

## 141.1 Corrected columns and augmented potential modes

Retain the decompositions $A_j = A_S c_j + r_j$ of Section 139, with the source potential matrix $P$. For each nonbase type define its algebraic correction

$$D_j = P_j - P_S c_j.$$

The old matching basis contributes $P_S K_0^T$. A zero-residual singleton adds $D_j$; two such columns add their separate corrections. An opposite pair adds $D_a + \gamma D_b$, with the same positive ratio satisfying $r_a + \gamma r_b = 0$ as in the matching update. These are precisely the potentials of the augmented matching modes. A correction with nonzero residual is not a matching vector by itself; it becomes part of the actual matching space only through the cancelling pair and subsequent restrictions. No individual correction is asserted to be an admissible normal surface.

Write the augmented matching rows in full global Q coordinates as $W$, and their potential columns as $\Phi_W = PW^T$. Replacement constraints give a thin row matrix $N$. Sorting the surviving support changes coordinate order but not the full global vector. Canonical gauge reduction gives a thin matrix $G$. Thus, before suppressing zero removed coordinates,

$$K' = GNW, \qquad P(K')^T = \Phi_W N^T G^T.$$

The implementation retains $Z = GN$, then combines the cached columns of $\Phi_W$ with its rows.

**Proposition 141.1** (Exact propagated source potentials)**.** *The propagated corner forms equal direct multiplication of the source potentials by the updated canonical Q basis, at every physical corner. Consequently their equal-form identifications, canonical lifts, Euler screens and complete standard matching cones agree exactly.*

*Proof.* For every augmented mode its potential is the corresponding column of $PW^T$, by the linearity of $P$ and the displayed corrected-column formulas. Replacement and gauge reduction are ordinary linear combinations of these modes, so multiplication commutes with both operations. Removed coordinates are exactly zero after replacement constraints; dropping them therefore changes neither the full global Q vector nor its potential. The displayed matrix equality proves the corner-by-corner identity. Subsequent identification, minimum normalization and matching construction use only these exact forms and the unchanged source, giving the remaining claims. □

## 141.2 Track the actual gauge operations

During the native reversed-column gauge reduction, append an identity matrix to the thin input rows. Only the original matching columns are eligible pivots. Every elimination operation also acts on the appended columns, recording the exact coefficients of the resulting rows in the input mode basis. Reversing the matching columns back and sorting by free index preserves the corresponding transformation rows. Composing this record with the replacement nullspace gives $Z$.

Auxiliary identity columns never become matching variables or pivot choices. They do not change the canonical gauge. The untracked entry point continues to return the same ordinary support/basis pair. The window producer requests an additional private projection plan, which remains outside every portable positive certificate.

## 141.3 Lazy sparse source caches

The old basis potentials are projected at most once and stored by nonzero corner entries. The all-Q potential rows are transposed at most once into sparse columns. Each needed $D_j$ is then accumulated from $P_j$ and the nonzero coefficients of $c_j$, with exact zero cancellations removed. A correction is cached only after its complete construction. An interrupted transpose or column cannot publish an unfinished vector for another query.

Per-window projection combines only the nonzero thin weights and sparse cached potential entries. It reconstructs a full tuple of corner forms for the existing Euler and standard-kernel interfaces. Algebraic zeros are represented exactly; no numerical threshold or hashing surrogate changes form equality. Source caches belong to one read-only source and fixed old support. They are not shared across unrelated centres or triangulations.

Projection remains demand driven. An empty updated kernel or an immediately empty one-dimensional nonnegative cone returns before creating a cache. Unneeded correction columns are never computed. A positive result and all independent source replay retain their previous contract. High-nullity arrangement and support fallbacks use the identical propagated forms.

## 141.4 Arithmetic cost and retained scope

Let $C$ be the number of distinct correction columns actually needed and $m \le d_0 + 2$ the augmented parameter count. Transposition costs $O(\operatorname{nnz} P)$ once. The old basis projection costs $O(\operatorname{nnz} P + d_0 \operatorname{nnz} P_S)$ once. A correction's sparse arithmetic costs

$$O\left(1 + \|c_j\|_0 + \operatorname{nnz} P_j + \sum_{i:(c_j)_i \neq 0} \operatorname{nnz} P_{S_i}\right),$$

in addition to constructing its coefficient record. Its worst case is $O(T(k+1))$, while $C \leq 3T$. The complete correction preparation is therefore polynomial, bounded by $O(CT(k+1))$ arithmetic operations.

After these preparations, each window uses thin matching/gauge operations and sparse combinations of at most $d_0 + 2$ source potential vectors per output basis row. Including dense interface reconstruction, potential propagation costs $O(Td(d_0 + 2))$ in the worst case. It removes repeated scanning of $\operatorname{nnz} P$ and repeated wide source projection from each candidate. The lazy cache can be more expensive on a short query that never reuses a column; the measured tradeoff must include its preparation.

Cached coefficient bits, sparse-vector storage, exact rational arithmetic, output and independent replay remain charged. This changes a polynomial factor in the existing local logarithmic-nullity/radius bound. It does not supply global sector centres, a universal distance bound or a general quasi-polynomial recognition theorem. Shared deterministic guard counts may change separately from wall-clock time.

## 141.5 Validation and measured behavior

All 1,408 maintained tests pass in 241.501 seconds; the twelve focused tests pass in 19.293 seconds. Every tested small literal source window compares propagated forms with direct source projection, full bases with fresh reconstruction, and complete native ray sets and coordinate lifts. A genuine nullity-four source retains both complete generic methods. An interrupted column construction leaves no partial cache entry, and a real diagram window still produces an independently replayed ordinary disc.

The 85-source audit exactly preserves all disabled outputs, all enabled verdicts and all 31 native positives. Seven window discs receive fresh Regina confirmation. The twenty-eight work-capped cases and twenty-six completed window misses remain unchanged. On the genus-one source, thirteen actual projections use one old-basis preparation, one transpose and twenty-four corrected columns. The trefoil and figure-eight use thirty and forty-two projections with fifty-four and seventy-six corrected columns, respectively. Their candidate and emitted-ray counts remain unchanged. Source guard calls increase from 109,610 to 110,981 on the genus-one case, from 178,409 to 180,453 on the trefoil, and from 314,978 to 318,797 on the figure-eight. Cache preparation is genuine work; the improved arithmetic organization is not a reduction in every guard-count allowance. All 599 captured runtime pins stay fixed. The 249.386-second audit is not a paired speedup estimate.

The first dense-mode pilot completes 120 measured calls and forty warmups and is mostly near parity. It exposes the cost of multiplying algebraic zero potential entries. The final version stores sparse modes and accumulates only their nonzero contributions before reconstructing the existing dense corner-form interface. The pilot records and both exact historical source snapshots needed to recover all 599 pins are retained.

The final native comparison freezes four preceding-release modules at `8a6cc03d7`: window basis construction, residual search, seed queries and recognition. Local imports are redirected only to corresponding frozen dependencies. Five serial interleaved four-arm rounds complete 200 measured calls and forty warmups. Native timings include source/cache preparation, all configured queries, independent positive replay and serialized output. Both arms finish all configured work and agree on final statuses and tested positive certificate digests.

**Complete native queries with radius-two windows.**

| Input | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| empty circle | 6.039 | 5.995 | 1.012 | 1.004 | 1.005 |
| optimized positive | 59.252 | 100.487 | 0.927 | 0.965 | 0.989 |
| genus one miss | 240.396 | 234.586 | 1.016 | 1.005 | 0.989 |
| trefoil | 408.116 | 403.060 | 1.015 | 0.982 | 1.006 |
| figure eight | 751.056 | 722.209 | 1.038 | 1.010 | 0.983 |

The optimized-positive native control returns before mode projection, yet its first comparison has a paired old/new ratio of 0.927 and noticeably different raw medians. This is not attributed to the unused cache implementation. A fifteen-round repeat completes sixty control calls with four warmups, a ratio of 0.993 and A/A controls 0.989 and 1.003, resolving that row near parity. Its source evidence confirms that neither arm enters window search.

**Complete recognition with radius-two windows enabled.**

| Input | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| empty circle | 0.018 | 0.018 | 1.011 | 1.000 | 0.952 |
| optimized positive | 51.464 | 51.683 | 0.995 | 0.989 | 1.011 |
| genus one miss | 246.124 | 238.624 | 1.015 | 1.000 | 0.983 |
| trefoil | 444.265 | 428.993 | 1.038 | 1.013 | 1.007 |
| figure eight | 777.020 | 763.385 | 1.019 | 0.983 | 1.004 |

The three window cases have paired complete-recognition ratios of 1.015, 1.038 and 1.019 on the genus-one, trefoil and figure-eight, respectively. These are modest small-case measurements. The same longer repeat performs sixty complete trefoil recognition calls and four warmups, including the established complete fallback after the native miss. Its gain is 1.015, with A/A controls 0.998 and 1.002, smaller than the first row's gain. Both runs remain available. Early full-recognition controls stay near parity; the microsecond empty case is not used to infer a projection speedup. The principal result is the exact amortized projection bound, not a uniform wall-clock gain. Radius zero remains much faster on these small controls, so the window stage continues to default off.

Runtime release `4f181d86c`, all tests, the source audit, dense pilot, final timing run and 120-call control/recognition repeat are retained. From `fast/`, use `normal_orbit_research.window_modes` with `audit -fresh-regina` or `benchmark -rounds 5`. Publication checks verify 599 frozen/current runtime pins, 598 preceding release evidence pins, the recoverable pilot and every final timing outcome. They independently replay all 31 saved positive source proofs with mode projection, corrected-column caches, Euler compilation, prepared observers, matching updates, residual planning, coherent extraction, ray enumeration and orbit producers disabled. Independent source reconstruction remains the authority for positive results.

## 142 Sparse standard geometry in free quadrilateral coordinates

The propagated potentials of Section 141 already have only $d$ parameters. The old window constructor expands each form into $k$ coefficients and every standard constraint into $p + k$ coefficients, where $p$ is the number of retained triangle classes. Low-dimensional producers then multiply those expansions, including their zeros. This continuation keeps the actual coefficients sparse and materializes dense rows only for rank computations that request them.

## 142.1 The unchanged complete matching equations

Let $F_a$ be the free Q coordinates of the canonical updated basis $K'$, so $q_{F_a} = z_a$. Every dependent Q coordinate obeys

$$q_j - \sum_{a=1}^{d} K'_{a,j} q_{F_a} = 0.$$

Each such equation has at most $d+1$ nonzero coefficients. A class potential $\phi_c(z)$ has at most $d$ coefficients, all in the free physical slots. For a vertex-group anchor $b$, the standard equation is

$$x_c - x_b - \sum_{a=1}^{d} (\phi_{c,a} - \phi_{b,a}) q_{F_a} = 0,$$

with at most $d+2$ nonzeros. Together these are the same complete standard matching space as the preceding projected kernel, including its link-offset factors. The representation change neither drops constraints nor replaces standard rays with Q rays.

## 142.2 Exact forms with two access paths

The native `SparseLinearForm` stores the width and the nonzero coefficient dictionary. Its sparse dot product visits only stored entries. Its chart projection visits the same entries for each chart coordinate. Indexing returns the exact coefficient, including omitted zeros; iteration returns the full dense sequence. Coordinate-face restrictions can therefore index the sparse row directly, while the existing rational rank routines can materialize the same dense row by iteration.

**Proposition 142.1** (Representation preserves all kernel queries)**.** *Sparse window constraints and potentials define exactly the previous projected kernel. Their sparse dot and chart projection values equal dense arithmetic; every coordinate-face restriction and dense rank calculation receives exactly the previous coefficient values.*

*Proof.* Only exact zero coefficients are omitted. Summation over nonzero entries is equal to summation over the full row, for every vector or chart form. Indexing and iteration restore zero at each omitted slot and retain every other value. Thus both matching equations and restricted matrices are identical. The basis, normalized potentials, canonical lifts, extrema and rank/nullspace tests consequently remain unchanged. □

The projected lift evaluates its Q constraints and potentials sparsely, then checks integrality and validates the full coordinate vector against the actual source as before. Envelope line construction uses sparse dot products. Planar chart construction uses sparse form projection. Other kernel constructors still use their existing dense tuples. Their fallback arithmetic is retained. The generic arrangement/support paths see the complete restored rows when needed; no low-dimensional promise is applied to a higher-nullity source.

## 142.3 Storage and arithmetic accounting

There are $k-d$ dependent-Q rows and $p-g$ triangle-difference rows, with $g$ retained vertex groups. The stored standard coefficient count is at most

$$(k-d)(d+1) + (p-g)(d+2).$$

Stored potentials use at most $pd$ entries. Thus standard coefficient and potential storage is $O((k+p)(d+1))$, instead of expanded row storage with quadratic dependence on $k+p$. Widths and basis storage remain separately charged. Source class identification still uses its existing

exact form-group dictionary; its hash costs are not replaced by a claimed deterministic linear-time bound.

After the basis and propagated forms are available, filling the sparse coefficient records requires $O((k+p)(d+1))$ scalar operations. Checking a Q vector costs $O(k(d+1))$ sparse coefficient operations. Canonical potential evaluation costs $O(pd)$, followed by the existing minima, physical-corner reconstruction and independent source validation. A one-dimensional direction therefore avoids the old dense $k^2$ cycle-row dot products. Envelope and planar projections likewise avoid multiplying zero physical Q slots; their fan, clipping and actual-ray costs remain charged.

Generic Gaussian elimination still pays for any dense matrix it constructs. Exact coefficient bits, indexing/hash costs, output and certificate replay also remain charged. With logarithmic matching nullity this reduces storage and a polynomial factor of the supplied-window computation. It supplies neither the missing global window family nor a general quasi-polynomial recognition theorem.

## 142.4 Source validation and measured behavior

All 1,411 maintained tests pass in 262.025 seconds; the thirty-eight focused tests pass in 23.105 seconds. Genuine nullity-one, two and three sectors produce the complete expected source rays while dense-form iteration is disabled. Direct projected lifts and coordinate-face extremality tests also work under that restriction. A genuine nullity-four sector retains both complete generic methods. A larger sixteen-layer source verifies the linear stored-coefficient bounds and exact lifts scaled by $2^{90}+1$.

The 85-source audit exactly preserves every disabled output, every enabled verdict, every recorded guard count and all 31 native positives. Seven ordinary window discs receive fresh Regina confirmation. The twenty-eight capped cases and twenty-six completed local misses remain unchanged. All 602 captured runtime pins remain fixed. The 214.396-second audit is not a paired speedup estimate against its predecessor. Its result establishes preservation of the configured source behavior, not a global knot-search completeness theorem.

Native timings freeze seven preceding-release modules at `9abd16fe1`: sector dispatch, envelope and planar production, window basis construction, residual queries, seed queries and recognition. Local imports are redirected only to the matching frozen dependencies, including the generic dispatcher's lazy low-dimensional imports. Five serial interleaved four-arm rounds complete 200 measured calls and forty warmups. The timings include source and cache preparation, complete configured queries, independent positive source replay and serialized output. Every arm completes its configured work. Statuses and tested positive certificate digests agree exactly.

**Complete native queries with radius-two windows.**

| Input | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| empty circle | 5.923 | 6.068 | 0.987 | 0.994 | 1.016 |
| optimized positive | 63.597 | 60.929 | 1.012 | 1.003 | 0.991 |
| genus one miss | 237.021 | 229.417 | 1.024 | 0.995 | 0.976 |
| trefoil | 414.162 | 393.843 | 1.044 | 1.017 | 1.013 |
| figure eight | 722.362 | 679.206 | 1.064 | 0.986 | 1.000 |

The native genus-one, trefoil and figure-eight queries have paired gains of about 1.024, 1.044 and 1.064. Early controls that return before sparse window geometry remain close to parity. These are improvements over the preceding cached-mode implementation, not over disabling the window stage.

**Complete recognition with radius-two windows enabled.**

| Input | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| empty circle | 0.018 | 0.018 | 1.000 | 0.977 | 0.967 |
| optimized positive | 51.249 | 51.234 | 0.996 | 1.006 | 1.007 |
| genus one miss | 244.378 | 230.930 | 1.039 | 1.004 | 0.986 |
| trefoil | 435.224 | 407.538 | 1.063 | 1.008 | 0.975 |
| figure eight | 754.915 | 716.218 | 1.052 | 0.996 | 0.960 |

Complete recognition has paired gains of 1.039, 1.063 and 1.052 on the same three cases. The first figure-eight new-arm A/A control is 0.960, so a separate fifteen-round comparison repeats the two knot cases with full recognition and serialized output. It completes 120 measured calls and eight warmups. The trefoil gain is 1.046, with A/A controls 0.985 and 0.994; the figure-eight gain is 1.052, with controls 0.996 and 1.003. Both repeat rows include the complete fallback after the native miss. The longer records support modest measured gains, rather than a uniform wall-clock theorem. The microsecond empty-recognition control is not used to infer a geometry speedup. Radius zero remains much faster on the small controls, and the window stage stays off by default.

Runtime release `f3772d8a3`, tests, the source audit, final timing run and longer repeat are retained under `data/window-geometry-*`. From `fast/`, use `normal_orbit_research.window_geometry` with `audit -fresh-regina` or `benchmark -rounds 5`. Publication checks compare 602 frozen/current runtime pins, 599 preceding release evidence pins, every timing outcome and all 31 saved positive proofs. The proof replay disables sparse dot/chart evaluation, cached modes, Euler compilation, prepared observers, matching updates, residual planning, coherent extraction, ray enumeration and orbit producers. Independent source geometry remains the authority for positive results.

# 143 Independent disc replay with verifier-owned source geometry

Sparse window geometry reduces producer arithmetic, but positive-Euler candidates still repeatedly reconstruct the same manifold during independent disc-proof replay. This continuation reuses only data independently built by a verifier for one fixed source. Candidate coordinates, reductions, support proofs and component claims continue to be checked for every proof. No producer-prepared geometry is passed to the verifier context.

## 143.1 Bind an owned snapshot of the actual source

The private context copies the supplied triangulation and validates the copy with the ordinary finite-manifold reconstruction. Its prepared geometry is therefore verifier-owned. A proof query also supplies its actual source; the context compares that source with its snapshot recursively, checking container and scalar types as well as values. An integer face index cannot be replaced by a Boolean merely because Python compares them equally. A changed source is rejected rather than silently rebinding the context.

Every query validates its supplied normal coordinates afresh against the owned source geometry. It recomputes the original source/vector fingerprint, then reconstructs link minima, the peeled vector and quadrilateral divisor arithmetically. It compares these values with the certificate and checks the reduced-core proof independently. This is the same reduction logic as fresh public disc replay; it never invokes the producer's canonical-core routine or cached source potentials.

The boundary homology basis depends only on the validated source. The context computes it independently on the first relevant unit-ray proof, then reuses it on later proofs. Boundary weights, parities and the claimed Euler and disc counts still come from the current independently validated

core. A basis is published to the context only after complete construction; a callback interruption cannot leave a partial cached result.

### 143.2 Acceptance equivalence on a fixed source

**Proposition 143.1** (Verifier-owned reuse preserves disc-proof acceptance)**.** *Let $T$ be a supported finite source and let a context be constructed from its owned snapshot. For every query source equal to that snapshot in both structure and types, coordinates $x$ and disc certificate $C$, context replay has the same acceptance result as fresh public replay on $T$. A different query source cannot be accepted by that fixed-source context.*

*Proof.* Owned source validation reconstructs the same deterministic geometry as fresh validation of the identical source. Fresh coordinate validation therefore reconstructs the same rows, matching checks, edge weights and Euler data. The original fingerprint and independently calculated reduction are identical. The component and support-proof checks receive the same actual source and core. An independently computed cached boundary basis is the same deterministic source basis that fresh replay would compute, so the boundary parity comparisons also agree. Every acceptance comparison is consequently unchanged. Type-preserving source comparison rejects a different query source before these checks. □

The scope is one fixed supported source, not arbitrary new triangulations. Public `verify_normal_disk_count_certificate` still reconstructs its supplied source afresh on each call. Its independently validated input analysis can also serve as the core analysis only when the reconstructed core rows equal those input rows exactly. Otherwise the core is analysed afresh. This equality reuse is verifier-owned and makes no assumption that all supplied vectors are primitive or have no vertex links.

The window creates a context lazily, only when a complete disc-count proof needs replay. It supplies the actual source and coordinates on every call. The ordinary final diagram witness retains its existing format and receives fresh canonical diagram/source verification outside this context. No cached acceptance result, producer analysis or producer geometry becomes part of the trusted certificate input.

### 143.3 Cost, ownership and limits

A batch of $M$ relevant proofs previously repeated source preparation and source boundary-basis construction. The context pays these costs once, plus one source snapshot. Each proof still pays for type-preserving source comparison, fresh coordinate validation, original fingerprinting, independent reduction arithmetic, core-proof replay and its current boundary parity. The gain is source-only amortization; certificate-dependent arithmetic is not removed. Generic component proofs retain their established independent reconstruction paths.

Equality checks and snapshot storage are linear in the source encoding. Manifold validation, boundary algebra, coordinate bits, output and actual replay work remain charged. Callback allowances count begun guarded work rather than bit operations, so the measured guard tradeoff is reported separately from paired timings. The context belongs to its verifier and one fixed source; it is not shared across arbitrary diagram exteriors or mutable producer preparations.

These changes preserve the preceding local sector/window complexity scope. They neither authenticate a new diagram-source relationship nor supply a global sector family, hierarchy or quasi-polynomial recognition theorem.

### 143.4 Source comparisons and paired measurements

All 1,415 maintained tests pass in 248.833 seconds; the sixteen focused tests pass in 7.067 seconds. A genuine seven-ray source replays every certificate with producers disabled while constructing one distinct owned source and one independent boundary basis. Fresh and context replay reject tampered counts, reduction/core coordinates, link fields, parity, source fingerprints and input vectors. Both accept genuine proofs scaled by $2^{90} + 1$. Source mutations and Boolean/integer aliases are rejected; interrupted homology construction leaves no cached basis. An additional saved source audit checks eight legacy component-based version-one proofs, including the empty vector, with both fresh and context replay and all producers disabled.

The 85-source audit exactly preserves every disabled legacy result, every enabled verdict and all 31 native positives. Seven window discs receive fresh Regina confirmation. Work-capped cases fall from twenty-eight to twenty-two, while completed local window misses rise from twenty-six to thirty-two. The newly completed sources are `survivor-08`, `survivor-10`, `survivor-11`, `conway`, `random-30` and `random-159`, at 1,896,624, 1,912,330, 1,947,346, 1,943,782, 1,881,685 and 1,969,785 guard calls, respectively. These native searches remain inconclusive for knot recognition. On the motivating genus-one positive, work drops from 110,981 to 106,508 calls; trefoil and figure-eight work drops from 180,453 and 318,797 to 171,166 and 300,945. All 604 captured runtime pins stay fixed. The 196.763-second audit is not a paired throughput comparison against older audit records.

The timing comparison freezes four preceding-release modules at `e1294e757`: disc counting/replay, residual queries, seed queries and recognition. Local imports are redirected only to corresponding frozen dependencies, with no timed patch context. Five serial interleaved four-arm rounds complete 200 measured calls and forty warmups. Native timings include source and cache preparation, all configured queries, independent successful source replay and serialized output. Both arms finish all configured work; statuses and positive certificate digests agree exactly.

**Complete native queries with radius-two windows.**

| Input | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| empty circle | 6.129 | 6.177 | 0.991 | 1.000 | 1.004 |
| optimized positive | 61.325 | 60.445 | 1.011 | 0.992 | 0.990 |
| genus one miss | 228.561 | 202.645 | 1.128 | 0.981 | 1.035 |
| trefoil | 396.591 | 329.329 | 1.253 | 0.929 | 0.999 |
| figure eight | 717.283 | 578.454 | 1.240 | 0.990 | 1.009 |

The native genus-one, trefoil and figure-eight gains are about 1.128, 1.253 and 1.240 in the first run. The native trefoil incumbent A/A ratio of 0.929 prompts a separate fifteen-round repeat. It completes sixty native calls and four warmups, with a paired gain of 1.220 and A/A controls 0.995 and 0.997. Both records are retained; the noisy first row is not the sole evidence for this gain.

**Complete recognition with radius-two windows enabled.**

| Input | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| empty circle | 0.018 | 0.018 | 1.017 | 0.989 | 0.989 |
| optimized positive | 51.985 | 51.746 | 1.007 | 0.997 | 0.991 |
| genus one miss | 238.169 | 206.544 | 1.178 | 0.975 | 0.987 |
| trefoil | 425.219 | 359.686 | 1.174 | 0.992 | 1.027 |
| figure eight | 776.022 | 612.214 | 1.264 | 1.010 | 1.023 |

Complete recognition gains are 1.178, 1.174 and 1.264 on the same three cases. The longer repeat also completes sixty full trefoil recognition calls and four warmups, including complete fallback and serialized output; its gain is 1.225, with A/A controls 1.016 and 0.997. Source-only verifier reuse

gives measured improvements relative to the preceding enabled-window implementation. Early positives remain near parity. The empty-recognition microsecond control is not a replay-cache speedup claim. Radius zero remains much faster on small controls, and windows still default off.

Runtime release `c302eb097`, all tests, the source audit, paired run and 120-call native/full repeat are retained under `data/disc-context-*`. From `fast/`, use `normal_orbit_research.disc_context` with `audit -fresh-regina` or `benchmark -rounds 5`. Publication checks compare 604 frozen/current runtime pins and 602 preceding release evidence pins and check every timing outcome. They independently replay all 31 saved positives with the batch verifier context, prepared producer observers, source-mode caches, Euler compilation, matching updates, residual planning, coherent extraction, ray enumeration and orbit producers disabled. Fresh public source reconstruction remains available and remains the final authority for the ordinary diagram proofs.

# 144 Grouped Euler minima and progress-driven screening

The source Euler compiler of Section 140 evaluates canonical Euler characteristic without constructing a normal vector. Its first implementation still scans all physical corners at every tested Q corner. This continuation groups equal projected forms within each source vertex and compiles one small nonlinear objective for the remaining corner queries. It retains the cheap one-dimensional branch and builds no grouping when the first corner is already positive.

## 144.1 Anchor shifts and constant vertex groups

For the updated basis $K'$ write

$$\alpha_a = \sum_j \gamma_j K'_{a,j}, \qquad F(z) = \alpha \cdot z - \sum_v \lambda_v \min_{c \in G_v} \ell_c(z).$$

Choose one actual form $a_v$ as the anchor for vertex $v$, and let $D_v$ be the distinct nonzero differences $\ell_c - a_v$. Define

$$\bar{\alpha} = \alpha - \sum_v \lambda_v a_v.$$

Then

$$F(z) = \bar{\alpha} \cdot z - \sum_{v : D_v \neq \varnothing} \lambda_v \min\Big(0, \min_{f \in D_v} f \cdot z\Big).$$

The zero anchor difference is supplied explicitly. If a vertex group has only one projected form, it contributes only to $\bar{\alpha}$ and disappears from the nonlinear minima. Adding any common linear offset to a vertex's forms, together with its exact link-weighted change to the signed linear term, cancels under this normalization.

**Proposition 144.1** (Grouped Euler objective is exact)**.** *The grouped functional equals canonical source Euler characteristic at every parameter vector. Duplicated forms and constant vertex groups can be omitted as above without changing any corner value or exclusion result.*

*Proof.* For each vertex, $\min_c \ell_c(z) = a_v \cdot z + \min(0, \min_{f \in D_v} f \cdot z)$. Substitution and collecting the anchor terms give the displayed identity. Duplicates do not alter a minimum. An empty difference set contributes only its anchor linear form. The equality is algebraic and does not require positivity or integral coordinates. Applied to the normalized nonnegative Q section, it therefore retains the preceding complete nonpositive-corner criterion and its independent source interpretation. □

### 144.2 Only refine the objective after a nonpositive first corner

The implementation tests the first Q corner with the existing direct canonical functional. A positive value immediately resumes ordinary standard discovery. A point section also finishes without aggregation. Only resuming to another corner triggers compilation of distinct forms and the anchor-adjusted linear term. Later corners use that one compiled objective. Empty sections and the scalar nullity-one branch keep their previous behavior; higher-nullity generic fallbacks remain available.

The grouping belongs to one updated support, basis and source projection. It is not reused across different sectors. Equality is exact rational value equality, with no numerical tolerance. Compilation completes before a grouped value is used; callback interruptions propagate and cannot give an exclusion from a partial list of corners. Positive corner values still prove neither connectedness nor disc essentiality.

### 144.3 Distinct-form bound and arithmetic cost

The source support contraction of Section 126 removes matching edges with identically zero allowed-Q labels. Corner forms are constant along every such contracted component. Equal-form grouping can merge those components further, including dropping groups that become constant; it cannot split them. Consequently the total number $p$ of distinct forms in nonconstant vertex groups is bounded by the original retained class bound $p \leq 8k$. This bound is about actual source matching geometry, not arbitrary families of affine functions.

Compiling $\alpha$ costs $O(kd)$ arithmetic operations. Scanning the physical corner forms, adjusting anchors and creating distinct differences costs $O(Td + pd)$ coefficient operations, plus exact grouping/index costs. Each later corner evaluates only $O(pd)$ coefficients. Including the unaggregated first corner and compilation, $m$ tested corners cost $O(d(T + k + mp))$ arithmetic operations after source projection. At nullity at most three, the normalized polygon has $O(k)$ vertices; thus this implements the proposed $O(T + k^2)$ post-projection coefficient arithmetic. The normalized chart and polygon costs are still charged.

Python dictionaries implement exact form grouping. Their expected lookup cost and rational hashing/bit costs are distinct from the scalar-operation bound; no deterministic linear-time hash guarantee is asserted. Source preparation, potential projection, coefficient storage, actual ray enumeration, full coordinate output and independent proof replay also remain separately charged. This improves a local polynomial factor. A global centre family, logarithmic distance bound and general quasi-polynomial unknot recognition theorem remain unproved.

### 144.4 Source comparisons and timings

All 1,417 maintained tests pass in 254.597 seconds; the fourteen focused tests pass in 22.808 seconds. Every tested small source window compares the grouped objective, direct canonical formula and fresh source Euler data, including scaling by $2^{90} + 1$. Genuine source tests check anchor-gauge invariance, constant-group elimination and the $8k$ distinct-form bound. A positive first corner is tested with aggregation disabled, proving that no unused objective is compiled. Higher-nullity fallback behavior is retained.

The 85-source audit exactly preserves every disabled output, every enabled verdict and all 31 native positives. Seven ordinary window discs receive fresh Regina confirmation. The twenty-two capped cases and thirty-two completed local misses remain unchanged. The motivating genus-one case builds seven grouped objectives with 127 retained distinct forms in total; trefoil and figure-eight build ten and twelve, with totals 259 and 289. Guard work rises from 106,508 to 108,245, from 171,166 to 173,683 and from 300,945 to 304,918 on these three cases. Grouping preparation is real

work, not a universal guard-count improvement. All 605 captured runtime pins stay fixed. The 185.301-second audit is not a paired throughput comparison.

An initial three-round run completes 120 measured calls and forty warmups. The final five-round comparison completes 200 measured calls and forty warmups, freezing four preceding-release modules at `2c2db6b42`: Euler screening, residual queries, seed queries and recognition. Local imports are redirected only to corresponding frozen dependencies; no timed patch context is charged. Both arms finish their configured native work and agree on statuses and tested positive certificate digests. Native timings include source and grouping preparation, actual queries, independent successful replay and serialized output. The pilot and final records are both retained.

**Complete native queries with radius-two windows.**

| Input | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| empty circle | 5.806 | 5.819 | 0.991 | 0.998 | 1.002 |
| optimized positive | 57.007 | 57.541 | 1.001 | 0.989 | 0.982 |
| genus one miss | 190.226 | 178.944 | 1.064 | 1.014 | 0.987 |
| trefoil | 314.882 | 306.607 | 1.027 | 1.003 | 1.030 |
| figure eight | 540.842 | 522.980 | 1.034 | 0.982 | 1.004 |

The native genus-one, trefoil and figure-eight first-run gains are about 1.064, 1.027 and 1.034. Early controls that return before aggregation remain close to parity. These are small-case measurements over the preceding verifier-context implementation, not comparisons with omitting windows.

**Complete recognition with radius-two windows enabled.**

| Input | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| empty circle | 0.018 | 0.017 | 1.023 | 0.966 | 0.961 |
| optimized positive | 53.087 | 53.316 | 0.996 | 1.012 | 0.997 |
| genus one miss | 199.700 | 193.789 | 1.037 | 1.027 | 1.025 |
| trefoil | 335.007 | 326.341 | 1.042 | 0.992 | 0.991 |
| figure eight | 564.884 | 546.939 | 1.031 | 0.989 | 0.990 |

The first complete-recognition gains are 1.037, 1.042 and 1.031 on the same cases. Because these are modest and the genus-one A/A controls are about 1.027 and 1.025, a separate fifteen-round comparison repeats the genus-one and figure-eight cases. It completes 120 measured calls and eight warmups with complete recognition and serialized output. The old/new ratios are 1.002 and 1.000, approximately parity; A/A controls are 1.008 and 1.068 on the genus-one case and 0.991 and 1.026 on the figure-eight. The repeat therefore does not support a reliable small-case wall-clock gain. Both records are published. No further ratio is inferred from the microsecond empty-recognition control.

The concrete result is the exact grouped objective and improved post-projection coefficient bound, not a uniform practical speedup. Preparation can outweigh saved scans on short or small queries. Radius zero remains much faster on the small controls; the window stage continues to default off. A complete local window still supplies no general negative knot certificate or quasi-polynomial coverage theorem.

Runtime release `71e1e5225`, tests, the source audit, pilot, final timing run and longer repeat are retained under `data/euler-aggregate-*`. From `fast/`, use `normal_orbit_research.euler_aggregate` with `audit -fresh-regina` or `benchmark -rounds 5`. Publication checks verify 605 frozen/current runtime pins, 604 preceding release evidence pins, all timing outcomes and all 31 saved positives with grouping, the batch verifier context, prepared observers, cached modes, Euler compilation, matching updates, residual planning, coherent extraction, ray enumeration and orbit producers disabled. Eight saved generic legacy proofs also receive producer-disabled fresh/context replay.

# 145 All-dimensional Q screening before generic standard discovery

The convexity theorem retained from report 81 applies in every matching nullity, but the first Euler compiler screened only planar sections. This continuation extends the screen to the generic branch. It addresses an actual gap in that integration: the retained source bank contains 177 sectors of nullity four and eleven of nullity five that were outside the planar gate. This is a supplied-sector improvement, with the same missing universal coverage obligations as Section 138.

## 145.1 Q hyperplanes suffice even without a full-dimensional cone

Let $K$ be an injective rational matching basis with $d$ parameter coordinates and $k$ allowed Q coordinates. Write $a_i$ for its coordinate forms, so that $q_i = a_i \cdot z$ and $C = \{z : a_i \cdot z \geq 0\}$. The cone is pointed: a vector and its negative can both be feasible only when all Q coordinates vanish, which by injectivity forces $z = 0$. Discard zero forms and deduplicate the hyperplanes using exact primitive integer normals. Their number $h_Q$ is at most $k$.

**Proposition 145.1** (Complete generic Q-corner screening)**.** *Every nonzero extreme ray of $C$ is generated by a one-dimensional kernel of $d-1$ independent coordinate hyperplanes. Enumerating all such subsets, retaining the feasible orientation and identifying proportional Q vectors therefore enumerates every Q extreme direction. This remains true if the linear span of $C$ has dimension smaller than $d$.*

*Proof.* For a point $z$ in the relative interior of an extreme ray, let $I$ be its zero Q coordinates. If the common kernel of their forms had dimension at least two, choose a direction $u$ in it independent of $z$. The other coordinates of $z$ are strictly positive, so sufficiently small positive $\epsilon$ makes both $z + \epsilon u$ and $z - \epsilon u$ feasible. Their average is $z$, contradicting extremality. The common kernel thus has dimension one, and a basis of its normal space supplies $d-1$ active forms. Conversely, any nonzero feasible direction in such a one-dimensional intersection lies in a one-dimensional face of the pointed cone. This argument includes constraints vanishing on the whole feasible cone; no relative-interior promise in the ambient matching space is required. □

The new internal iterator solves those small rational systems and computes the resulting Q direction. It selects the sign using the first nonzero Q coordinate, rejects a direction containing any negative coordinate, and emits only the lexicographically first independent subset of each ray's active hyperplanes. Column pivots in the transpose of those active normals select that subset by exact elimination. This gives polynomial deterministic deduplication work per candidate without a hash-table assumption or storage of previously emitted Q vectors. The parameter vector itself need not be rescaled to an integral normal surface: canonical Euler characteristic is positively homogeneous, so its sign is unchanged by positive rescaling. The exact grouped objective from Section 144 applies in arbitrary parameter dimension. As before, the first corner uses the direct functional; another corner triggers one grouping compilation.

A nonpositive value at every Q direction proves the Euler exclusion by convexity. No partial scan does so. Source preparation checks the actual compact manifold links, and callback interruption propagates before any exclusion is returned. Positive Euler still requires ordinary standard rays and independently verified disc components.

## 145.2 Two integration points and a shared allowance

Updated radius-one/two windows now use this iterator above nullity three, before allocating their standard matching geometry. Lower-nullity chart and scalar branches retain their previous

behavior. A positive first Q corner avoids grouping; any positive corner resumes the existing complete standard enumeration. Source projections and positive proof formats remain the same. The window stage continues to default off.

The public supplied-sector discovery API has a separate integration above nullity three in automatic mode. Its existing Q-ray producer first lifts and examines Q corners, checking any positive corner for an essential disc. A found disc returns its ordinary independently replayed source witness. A completed nonpositive scan returns the existing Q-phase exhaustion certificate, whose stronger statement excludes positive Euler throughout that sector, even though the caller requested standard discovery. The certificate describes the phase that was actually completed. Explicit ray methods and complete enumeration APIs retain their original behavior.

If the complete Q phase has positive Euler but finds no disc, automatic discovery resumes its complete standard phase. The two phases share the configured candidate-basis allowance: the standard phase receives the initial allowance minus the Q candidates already begun. Thus a cap reached in either phase returns `INCONCLUSIVE`, without an exhaustion proof. The existing independent checker does not import the new producer or Euler compiler. A positive Q phase can add work before the fallback, and a fixed cap can consequently stop sooner than in the preceding standard-only strategy. Neither a Q corner nor positive Euler alone is promoted to a recognition verdict.

## 145.3 What the asymptotic bound improves

The generic internal screen begins at most

$$\binom{h_Q}{d-1}, \qquad h_Q \le k,$$

small rational linear-algebra candidates, with sorting and canonical-active subset selection also charged. Each has polynomial bit cost in the supplied source, basis and coefficient encoding. At most this many feasible directions need Euler evaluation; source compilation and grouping also have polynomial preparation cost. The complete screen therefore costs

$$\mathrm{poly}(T + k + B)(1 + k)^{d-1}$$

in bits, charging coefficient growth through exact rational elimination. By contrast, the standard potential arrangement can have $O(k^2)$ hyperplanes and $\binom{O(k^2)}{d-1}$ candidate intersections. A completed nonpositive screen avoids that arrangement and all its full coordinate outputs and component queries. For a positive sector, both costs remain; no uniformly smaller worst-case bound for complete discovery is asserted.

Logarithmic matching nullity again gives a conditional quasi-polynomial local bound. This is not a general unknot algorithm: it supplies neither logarithmic nullity nor a bounded complete family of admissible sectors for arbitrary knot exteriors. The global recognition bound remains exponential.

## 145.4 Independent source evidence and complete-query measurements

All 1,422 maintained tests pass in 248.449 seconds, and the 23 focused integration tests pass in 11.398 seconds. The tests cover a positive generic corner, interrupted exclusion, a completed Q-only negative proof, a shared cap after positive but nonessential Q rays, and repeat enumeration with one statistics dictionary. An attempted standard arrangement is forbidden in the Q-only exclusion test, so the smaller proof reflects an avoided search rather than a shorter transcript of a search already performed.

The complete higher-nullity audit covers all 188 sectors selected by the retained report 81 selection rule: 177 have nullity four and eleven have nullity five. Every new Q stream equals the frozen previous Q stream in coordinates and order. Direct and grouped Euler values agree with fresh native source coordinate analysis on every Q ray, including scaling by $2^{90} + 1$. The screen's sign decision also agrees with the retained complete standard-coordinate source oracle. Twelve sectors are excluded; each has four Q corners and fifteen standard rays. The other 176 continue their ordinary positive branch. Fresh Regina standard enumerations for all four source triangulations represented by the exclusions reproduce exactly the fifteen compatible rays in every case. All twelve returned Q exclusion proofs pass independent replay.

The complete supplied-sector benchmark uses five randomized paired rounds and four arms per case, including identical-baseline A/A controls. Its six cases complete 120 measured calls and 24 warmups. Each call builds its source kernel, completes automatic discovery, independently replays its returned witness or exhaustion proof, and serializes the result. The previous arm freezes release `e0a2aa2ee`; it does not substitute an earlier delivered implementation. Both arms agree on every status, and all configured work finishes. These are supplied-source queries, not full knot-recognition measurements.

| Source | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| `fibonacci_lst_09-d4` | 182.956 | 62.866 | 2.917 | 0.999 | 1.002 |
| `fibonacci_lst_10-d4` | 228.834 | 83.590 | 2.745 | 1.001 | 0.986 |
| `lst_1_8-d4` | 121.583 | 36.242 | 3.366 | 1.009 | 0.998 |
| `solid_torus_sum_s2xs1-d4` | 138.525 | 31.298 | 4.460 | 1.000 | 1.000 |
| `cap_1_2_3-d4` | 36.996 | 41.811 | 0.900 | 1.000 | 0.988 |
| `cap_3_5_3-d5` | 3502.927 | 3526.416 | 0.993 | 0.995 | 0.990 |

The four nonpositive source cases have paired old/new ratios 2.917, 2.745, 3.366 and 4.460. Their candidate counts fall from 220 to 20 in the first three cases and from 120 to four in the fourth. Each negative proof stores four Q records instead of fifteen standard records. The A/A controls stay within about two percent of parity. The positive nonessential nullity-four control instead has ratio 0.900, about an eleven-percent slowdown, with 120 versus 124 candidates. The nullity-five control is near parity at 0.993, with 12,650 versus 12,655 candidates. Both positive controls finish `NO_VERTEX_DISC_IN_SECTOR`; they demonstrate the cost of a Q phase that must resume the complete standard search. No uniform speedup is claimed.

The 85-diagram native audit completes in 164.808 seconds. Every disabled output, enabled verdict and callback work count agrees exactly with the preceding release. All 31 positives remain; seven source-disc proofs receive fresh Regina confirmation. The twenty-two caps and thirty-two completed local misses also remain unchanged. None of these diagram windows enters the new generic Q arrangement: they exercise the retained lower-nullity branches. Thus the higher-nullity source-bank gains are not attributed to improved coverage or work counts on this diagram corpus.

The diagram timing comparison completes 200 measured calls and forty warmups over five native queries and five complete recognition calls, including complete fallback when the native query is inconclusive. Fresh input records, source and cache preparation, proof replay, configured work and serialized output are charged. Every status and positive certificate agrees across arms; none of these calls enters generic Q screening.

**Complete native queries with radius-two windows.**

| Input | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| empty circle | 5.923 | 6.116 | 0.998 | 0.989 | 1.025 |
| optimized positive | 57.609 | 59.896 | 0.988 | 0.990 | 0.976 |
| genus one miss | 182.137 | 186.831 | 0.970 | 0.997 | 0.961 |
| trefoil | 319.788 | 300.650 | 1.064 | 1.033 | 0.969 |
| figure eight | 538.644 | 535.883 | 1.011 | 1.007 | 0.984 |

**Complete recognition with radius-two windows enabled.**

| Input | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| empty circle | 0.018 | 0.018 | 1.006 | 0.960 | 0.951 |
| optimized positive | 51.192 | 54.268 | 0.937 | 1.000 | 1.071 |
| genus one miss | 195.579 | 211.906 | 0.942 | 1.021 | 1.047 |
| trefoil | 326.238 | 335.769 | 0.998 | 0.996 | 0.990 |
| figure eight | 580.540 | 575.658 | 1.017 | 1.036 | 1.028 |

These measurements do not establish a stable diagram-recognition gain. In particular the genus-one recognition ratio is 0.942 while its new A/A control is 1.047; the figure-eight controls are about 1.036 and 1.028. Variation on these unchanged lower-nullity paths is not attributed to the new generic algorithm, and the source-query gains are not generalized to these diagrams. No reliable wall-clock gain is claimed for these calls.

Runtime release `71c8f8321`, the tests, source-sector audit, complete source timings, 85-diagram audit and complete-recognition timings are retained under `data/generic-euler-*`. From `fast/`, use the module `normal_orbit_research.generic_euler`, with `sector-audit` and `-fresh-regina`, or its `sector-benchmark`, `audit` and `benchmark` modes. Publication review checks all 607 captured runtime pins, 605 preceding release evidence pins, all timing outcomes, twelve source exclusions and all 31 saved positives with the new Q iterator, Euler compiler and producers disabled. Eight legacy generic proofs also receive fresh/context replay. This verifies source exclusions and ordinary disc witnesses, without a new global recognition theorem or a universal sector coverage claim.

# 146 Complete Q support and effective matching dimension

The preceding generic Q phase makes a complete quadrilateral-corner list available when positive Euler characteristic fails to produce an essential disc. Report 81 already isolates forced-zero removal as an admission rule for lower-dimensional geometry (Section 136). This continuation obtains the maximal feasible support from that completed list, recompiles the source matching kernel, and uses the effective dimension for standard enumeration. It also lets the independent checker verify this support information before rebuilding a smaller standard cone.

## 146.1 A completed ray list determines every forced-zero type

For the original allowed support write

$$C = \{q \in \mathbb{R}^k : q \geq 0,\ A_\Sigma q = 0\}.$$

Let $r_1, \ldots, r_m$ be its complete primitive Q extreme directions and set

$$J = \{j : \text{some } (r_i)_j > 0\}.$$

The set $J$ is computed by scanning all coordinates in the completed list. Neither a successful early Q witness nor a capped or interrupted list provides the required completeness premise.

**Proposition 146.1** (Maximal support from complete Q rays)**.** *Every feasible quadrilateral vector vanishes off $J$. Restricting the matching system to the coordinates in $J$ preserves the whole standard normal cone after reinstating those zero coordinates, including all non-link standard rays and their actual source coordinates. The restricted matching kernel equals the linear span of its feasible Q cone.*

*Proof.* The nonnegative matching cone is pointed. Its section $\sum_j q_j = 1$ is compact, so every nonzero feasible vector is a conic combination of its extreme directions. Off-$J$ coordinates therefore

vanish in every feasible vector. Conversely $q_* = \sum_i r_i$ is feasible and strictly positive on every coordinate in $J$. For any vector $u$ in the restricted matching kernel, sufficiently small positive $\epsilon$ makes both $q_* + \epsilon u$ and $q_* - \epsilon u$ nonnegative on $J$; their difference shows that $u$ lies in the feasible cone's linear span. Every feasible standard vector projects to a feasible Q vector, so its removed coordinates were already zero. Conversely reinstating these zeros in a feasible restricted standard vector preserves all original matching and nonnegativity equations. The two standard cones are thus linearly isomorphic. This isomorphism preserves extreme rays, including the vertex links; canonical normalization and the full source coordinates of each non-link ray also agree. No topology or component query is inferred from positive Euler characteristic alone. □

This implements the maximal-support conclusion of the report's LP theorem through a different available witness. It does not install a polynomial LP solver. Obtaining the complete Q list still costs the preceding generic Q search in the *original* matching dimension. Recompilation costs are charged rather than hidden in a dimension parameter.

### 146.2 Adaptive recompilation after a completed positive Q phase

At release `936e83365`, automatic supplied-sector discovery above nullity three retains the Q-first policy described here. The sufficient matching preprocessor of Section 147 later precedes this complete-Q route and can avoid its original-dimensional enumeration. A verified disc returns immediately. A nonpositive complete Q phase returns its ordinary Q exclusion proof. Only `POSITIVE_EULER_ONLY` reaches the support step: all Q corners have been tested, and none supplied an essential disc.

If $J$ is the full allowed support, the already prepared original kernel is reused. Otherwise the implementation rebuilds the supplied source kernel with exactly those allowed types in $J$. The ordinary automatic dispatcher then selects the envelope, planar, or generic standard method using this new kernel's actual nullity. No original coordinate is merely overwritten with a guessed zero; matching geometry is reconstructed from the source. The Q and standard phases retain their shared candidate allowance. Exhaustion in either phase is inconclusive, and interruptions in support scanning, recompilation or proof translation propagate.

Statistics describe the kernel that performed the standard search. A `support_reduction` record retains the requested kernel statistics, original-coordinate indices and number of removed types. A positive witness still contains the actual full triangulation coordinates and the original allowed type list; its existing disc proof suffices independently of the support optimization.

### 146.3 A source proof that still states the original sector

A negative standard result retains the existing `normal-sector-exhaustion-v1` schema, original source digest, original allowed types and standard phase. Its Q records are expanded to the original coordinate width by inserting the removed zeros. The optional `q_support_certificate` is the complete original Q-phase exhaustion proof with status `POSITIVE_EULER_ONLY`. The parent never substitutes a smaller requested sector for the original statement.

The independent checker first validates that the nested proof has the same original source and allowed types, the Q phase and the required status. Nested support annotations inside it are rejected, so there is only one level of support proof. The checker then reconstructs and verifies the complete original Q list by its own dense elimination and reference enumeration. It derives $J$ from that authenticated list; it accepts no producer-supplied retained-index list. Only then does it independently rebuild and enumerate the restricted standard model, restore the original coordinate width, and compare every parent ray, Euler value and negative disc certificate against that model.

A malformed or incomplete nested Q proof cannot license removing any coordinate. An old certificate without the annotation still follows the original full-source checker. Removing a valid annotation also produces a valid original full-sector certificate: its complete parent ray list and source statement are unchanged. The optimization therefore accelerates reconstruction rather than changing the negative statement's meaning.

## 146.4 Raw and effective dimensions have different costs

Let $d$ be the original matching nullity, $s$ the rebuilt nullity and $k_J$ the retained support size. Once the complete Q list is available, finding $J$ costs $O(km)$ scalar inspections, with callbacks and bit access charged. The rebuilt kernel has nullity exactly the feasible Q span dimension $s$. Standard geometry consequently has the preceding $\mathrm{poly}(T+B)(1+k_J)^{O(s)}$ enumeration bound. For $s \le 3$ the native low-dimensional routines apply; in particular the complete planar source ray count is quadratic in retained support size.

The overall *ray geometry* cost still includes the original Q phase, whose upper bound is $\mathrm{poly}(T+k+B)(1+k)^{d-1}$, plus support scanning, source recompilation and the reduced standard search. Independent Q reconstruction and proof reading likewise remain charged. Component queries, essentiality and their proof costs are separate obligations. A small effective dimension does not alone give a polynomial or quasi-polynomial total search when the original Q phase has uncontrolled dimension. The general centre, distance and hierarchy bounds required for unknot recognition remain unresolved.

## 146.5 Source oracles, proof mutations and complete timings

All 1,427 maintained tests pass in 289.263 seconds. The 28 focused tests pass in 13.271 seconds. They check planar admission from raw nullity four, generic reduction from five to four, complete parent records in the original coordinate width, legacy replay without the support annotation, shared-cap boundaries and interruption during source recompilation. Eleven mutations remove or forge nested Q rays, change the source or allowed support, change the nested phase or status, add recursive nesting, or alter parent coverage and Euler data. They are all rejected with the producer, Q iterator and planar producer disabled during independent replay.

The source audit repeats all 188 retained higher-nullity sectors against release `c21e78bcd`. It completes in 53.634 seconds. Every complete status agrees. For every source sector, complete standard rays from the recompiled support equal the retained original-support source oracle in actual full coordinates. Sixty-five sectors have forced-zero types: 22 fall from matching nullity four to three, 41 stay at four, and two fall from five to four. Fresh Regina standard enumerations for all three source triangulations involved reproduce the complete compatible ray lists in all 65 cases. These are actual finite source triangulations, not manufactured post-elimination kernels.

Automatic discovery recompiles 61 of those sectors and returns 61 augmented negative standard proofs. The other four return a verified essential Q witness before recompilation is needed. All 22 raw-nullity-four sectors with feasible nullity three enter the planar standard producer. Every augmented parent proof retains the original allowed type list and coordinate width. The unchanged original full-sector verifier also accepts representative unannotated proofs at effective dimensions three and four. Independent publication review replays all 61 augmented proofs with the producers disabled, and also replays all 31 saved native diagram witnesses from the preceding release.

The paired benchmark freezes both the preceding discovery module and its independent checker. Each call constructs its source, completes standard discovery and all configured work, replays its proof using that arm's checker, and serializes the full result. Five randomized paired rounds with identical-baseline A/A controls complete 100 measured calls and twenty warmups. Both arms agree on every status; every proof is replayed and each arm's certificate digest is stable across its

repetitions.

| Source | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| `cap_3_5_2-4-to-3` | 164.057 | 41.699 | 3.863 | 1.045 | 1.005 |
| `cap_3_5_3-4-to-4` | 78.158 | 78.027 | 0.984 | 1.019 | 1.005 |
| `cap_3_5_3-5-to-4` | 3452.408 | 143.699 | 23.862 | 1.001 | 0.998 |
| `cap_1_2_3-4-to-4` | 40.810 | 40.753 | 1.004 | 1.002 | 1.002 |
| `fibonacci_lst_09-4-to-4` | 60.635 | 61.058 | 0.999 | 0.988 | 0.988 |

The planar-admission case has paired old/new ratio 3.863, with 684 versus nine begun candidates, including its completed Q phase. Its incumbent A/A ratio is 1.045 and the treatment A/A ratio is 1.005; the gain is substantially larger than those fluctuations. The nullity-five-to-four case has ratio 23.862, with 12,655 versus 369 candidates and A/A controls about 1.001 and 0.998. Both the producer and independent replay avoid their respective original larger standard arrangements.

The forced-zero case that stays at dimension four is near parity at 0.984; its candidate count remains 124 in both arms. The full-support positive control and nonpositive Q control have ratios 1.004 and 0.999. Removing coordinates without changing effective dimension does not supply a uniform speedup, and the larger nested proof and source recompilation are included in the treatment cost. No source-query gain is attributed to improved coverage of arbitrary knot diagrams.

The runtime release is `936e83365`. Tests, the complete source audit, fresh Regina controls and paired timings are retained under `data/feasible-span-*`. From `fast/`, use the module `normal_orbit_research.feasible_span` with its `audit` and `benchmark` modes; the audit accepts `-fresh-regina`. Publication checks verify all 609 runtime pins, 607 preceding-release pins, every timing outcome, the complete source-sector verdicts and all producer-disabled proof replays. The ray-geometry improvement and feasible span theorem are local results. A general quasi-polynomial recognizer and an implemented polynomial support producer remain separate unresolved requirements. The retained LP formulation gives a route to the latter; this Q-list integration does not implement that solver.

# 147 Polynomial matching consequences before Q enumeration

The complete-Q support route of Section 146 still pays for ray enumeration in the original matching dimension. This continuation adds a polynomial sufficient preprocessor before that enumeration. It certifies coordinate zeros from exact matching equations, then recompiles the source using only the retained types. Its checker verifies source-row combinations directly, without trusting a producer's forest, reduced matrix or inferred support. The complete Q and standard fallbacks remain available when the inexpensive implications do not find every forced zero.

## 147.1 One-sided equations imply zeros

Suppose earlier verified steps have forced coordinates in $F$ to vanish. For an actual matching consequence $w \cdot q = 0$, assume $w_j \geq 0$ for every allowed coordinate outside $F$. Then every still-active coordinate with $w_j > 0$ is zero in every feasible vector: the equality is a sum of nonnegative terms, and the already forced terms vanish. A row whose active coefficients are all negative is multiplied by $-1$. Rows with both signs are left for the complete search.

**Proposition 147.1** (Sound polynomial sign propagation)**.** *Starting with $F = \varnothing$, repeatedly applying the preceding implication to a fixed rational row family preserves every feasible standard vector. At most $k$ successful steps occur when $k$ types are allowed. Removing the proved-zero*

*coordinates and restoring them as zeros gives a linear isomorphism between the original and restricted standard cones.*

*Proof.* Induct on the steps. Every preceding zero is valid by the induction hypothesis, and the remaining summands in the new matching consequence are nonnegative. A positive coefficient therefore forces its coordinate to zero. Each successful step removes at least one new coordinate, giving the step bound. Every original feasible standard vector already has the removed coordinates zero, while reinstating them in a restricted feasible vector preserves all original matching and nonnegativity equations. The resulting isomorphism preserves the non-link standard rays and actual source coordinates. □

This is the preceding single-row implication procedure. Section 154 strengthens it using two-row spans; neither rule is a complete maximal-support LP solver. For example the two RREF rows

$$(1, 0, -1, 2), \qquad (0, 1, 2, -1)$$

have mixed signs and yield no step. Their sum is $(1, 1, 1, 1)$, so their nonnegative matching cone is actually zero. This counterexample separates soundness from completeness for general row systems. In particular a mixed-sign fixed point does not certify positive feasibility, maximal support or absence of an essential disc.

## 147.2 Recover useful rows cheaply, then compile source provenance

The existing canonical nullspace basis $K$ has one unit free coordinate per row. For each pivot coordinate $p$, its matching RREF equation is

$$q_p - \sum_a K_{a,p} q_{f_a} = 0.$$

Recovering these rows from the already computed basis costs $O(k^2)$ scalar work. Repeated sign propagation on at most $k$ such rows costs $O(k^3)$ scalar inspections; rational sign comparisons and coefficient bits are charged. If no coordinate is forced, no source provenance forest is built.

When useful zeros are found, the compiler reconstructs triangle incidence from the actual source matching equations. A directed edge with equation $t_a - t_b + L(q) = 0$ has label $L$ and a unit source-equation coefficient. A rooted forest stores both its Q potential $P_c$ and the source-row combination $E_c$ satisfying

$$E_c = t_{\text{root}} - t_c + P_c(q).$$

For an edge $a \to b$, the combination $E_a + E_{a\to b} - E_b$ cancels every triangle coefficient and gives $P_a + L - P_b$. Source equations with no triangle terms supply direct Q constraints. Exact elimination pivots only on the allowed Q columns; appended source-row coefficient columns record all the same row operations. The resulting Q RREF rows must equal the rows recovered from the native kernel, and their sign closure must agree. These exact equalities detect a provenance/gauge disagreement before any restricted geometry is used.

The forest, path combinations and augmented elimination have polynomial source-size and bit cost. They are not charged as constant-time preparation: a direct implementation can retain quadratic source path data. Coefficient bit growth is bounded by the usual minor bounds for exact rational elimination. This additional work is compiled only when the cheap recovered rows already show a valid reduction.

### 147.3 Independent linear replay licenses source recompilation

The optional `matching_support_certificate` records the original source digest and allowed type list. Its schema is `normal-sector-matching-support-v1`. Each step contains sorted source-row indices, exact numerator/denominator coefficients and the newly forced coordinates. The independent checker prepares the source anew and forms each stated combination of its actual matching equations. It checks cancellation of *all* triangle coefficients, nonnegative coefficients on every currently unforced allowed Q coordinate, and exact agreement between the claimed new coordinates and those with strictly positive coefficients. Previously forced coefficients need no sign condition because their coordinates have already been authenticated as zero. Nonselected Q types are zero by the original allowed-sector definition.

Malformed source indices, Boolean indices, zero or invalid denominators, wrong source/support identities, repeated indices, unsupported nesting, nonzero triangle remnants and incomplete coordinate claims are rejected. The checker imports neither the producer's forest nor its RREF/closure helpers. Valid coefficients can be represented by the existing large-integer codec; replay uses exact rational arithmetic and no numerical tolerance.

Automatic standard discovery above nullity three runs this preprocessor before a generic Q phase. A proper reduction rebuilds the source kernel and resumes automatic discovery in its smaller matching space, with the same candidate allowance. Low-dimensional progressive corner/envelope/planar branches can therefore run without the original-dimensional Q scan. An ordinary positive witness retains the original allowed list and full source coordinates; its disc proof suffices independently of this preprocessing.

Negative parent certificates also retain the original support and coordinate width. Their checker authenticates the matching implications first and reconstructs the retained model independently. If later complete Q support removes more types, the existing Q support proof concerns that intermediate retained sector; replay checks that precise scope, derives the next support and restores all coordinates to the original parent width. The matching and Q support proofs can therefore compose without narrowing the final statement. Removing all support annotations still leaves an original full-sector proof that the legacy checker can reconstruct directly.

### 147.4 The raw-dimensional Q cost can now disappear

Let $r$ be the matching nullity after the sufficient polynomial preprocessing, and $k'$ its retained support size. The complete ray-geometry search has

$$\mathrm{poly}(T+k+B)+\mathrm{poly}(T+B)(1+k')^{O(r)}$$

bit cost, including the Q prelude, standard fallback, source recompilation and source-equation certificate reading. The old raw nullity need not appear in the enumerative term. Thus logarithmic $r$ gives a conditional quasi-polynomial geometry bound even if the original matching dimension was larger. At $r \le 3$ the native low-dimensional geometry applies directly. Component queries, essentiality and their proof costs remain separate.

The retained matching dimension can exceed the feasible cone's actual span when the sign procedure misses zeros; no maximality is asserted from its fixed point alone. Neither logarithmic $r$ on arbitrary knot exteriors nor a bounded complete family of sectors or centres has been proved. This implements a polynomial sufficient operation toward that goal, without a new general recognition bound or an implementation of the full LP theorem.

### 147.5 Source controls and complete-query measurements

All 1,433 maintained tests pass in 259.100 seconds. The 34 focused tests pass in 13.277 seconds. They cover source-row planar admission before any generic Q scan, exact rational signs, fourteen

source-provenance mutations, legacy full-source replay and composition of a partial matching proof with the complete Q-support fallback. The mixed-row counterexample is tested as an explicit non-completeness boundary. The preceding complete-Q regression route is still exercised with matching preprocessing disabled internally.

The 47.730-second audit repeats all 188 retained higher-nullity source sectors against release `bc98f99a9`. Every complete status agrees. For this entire selected corpus, the polynomial recovered-row closure finds all 126 oracle forced-zero coordinates in 65 sectors; this observation is not promoted to a general maximal-support theorem. Fresh Regina standard enumerations for all three involved source triangulations reproduce every compatible full-coordinate ray list. The full support oracle again gives 22 transitions from dimension four to three, 41 reductions that stay at four and two transitions from five to four.

All 65 affected automatic searches now recompile before their generic Q phase. Four return an ordinary verified essential-disc witness and 61 return negative proofs with the source-equation annotation. The 22 sectors that drop to dimension three enter the progressive planar producer without the original-dimensional Q prelude. A returned ordinary positive witness requires no additional support annotation. Every negative proof retains the original allowed types and full coordinate width. Independent review replays all 61 annotated proofs and all 31 saved diagram witnesses with the new forcing, forest, Q and ray producers disabled. Frozen original full-source replay also accepts representative unannotated proofs at effective dimensions three and four.

The complete-query timing driver freezes both the previous discovery code and its independent checker. Each arm charges source construction, its complete configured search, every component query and proof replay, and full serialized output. Five randomized paired rounds with identical-baseline A/A controls complete 100 measured calls and twenty warmups. Every status agrees across arms, and each arm's certificate digest is stable across its repetitions.

| Source | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| `cap_3_5_2-4-to-3` | 40.904 | 25.583 | 1.614 | 1.002 | 0.982 |
| `cap_3_5_3-4-to-4` | 80.217 | 63.367 | 1.255 | 0.999 | 1.002 |
| `cap_3_5_3-5-to-4` | 142.674 | 122.678 | 1.166 | 1.002 | 0.983 |
| `cap_1_2_3-4-to-4` | 41.597 | 41.056 | 1.023 | 0.990 | 1.000 |
| `fibonacci_lst_09-4-to-4` | 63.082 | 61.222 | 1.040 | 1.027 | 1.013 |

The planar-admission query has paired old/new ratio 1.614 over the already improved preceding release. Its begun candidates fall from nine to five; the original four Q-basis candidates are no longer required. Serialized result size falls from 6,569 to 4,451 bytes. Its A/A controls are about 1.002 and 0.982. The affected same-dimension-four query gains 1.255 with the same 124 candidates; its linear source proof avoids the preceding complete-Q support reconstruction and reduces serialized size from 10,468 to 7,822 bytes. A/A controls are about 0.999 and 1.002.

The dimension-five-to-four query gains 1.166, with 369 versus 368 candidates and serialized size 11,820 versus 8,889 bytes. Its A/A controls are about 1.002 and 0.983. The one-candidate change is not the whole gain: direct linear implication replay replaces authentication of an original-dimensional Q list and its component proofs. Source recompilation and provenance preparation are included rather than treated as cached or zero-cost.

The full-support positive and nonpositive Q controls have ratios 1.023 and 1.040. Their A/A observations include ratios about 0.990 and 1.027; no reliable improvement is inferred on those unchanged paths. Recovering and scanning reduced rows also has a cost when no type is removed. These are host measurements of supplied-source queries, not a general speed guarantee or a claimed complete knot-recognition gain.

Runtime release `4c75b66de`, tests, the complete source audit, fresh Regina controls and paired timings are retained under `data/matching-support-*`.

The driver is the module `normal_orbit_research.matching_support`, run from `fast/` with `audit` or `benchmark`. The audit accepts `-fresh-regina`. Publication review checks 612 runtime pins, 609 preceding-release pins, source oracles, all timing outcomes, and the producer-disabled source and saved-diagram replays. The meaningful asymptotic change is sufficient polynomial preprocessing before enumeration; a complete polynomial support solver, universal logarithmic remaining dimension and general recognition coverage remain unimplemented or unproved.

# 148 Native bounded Pachner descent and diagram-bound disc search

Report 79's bounded-descent theorem was preserved in Section 133, but its marked search implementation remained separate from the maintained source interfaces. This continuation integrates the transport, commitment, region, shared-cover and causal modules, together with their independent source checkers. It also connects a bounded positive search to the exact diagram recognizer. The move primitives, their checkers, primitive cocycle kernel and integer codec are byte-identical to the delivered snapshot; the integration therefore preserves those maintained dependencies. The delivered report and historical evidence remain unchanged.

## 148.1 What the complete parameterized primitive decides

Let $t$ be the initial number of tetrahedra and $U$ the *total* number of upward $2 \to 3$ events allowed in a trace. This is not a bound on maximum excess tetrahedra. The formal relative-boundary replacements preserve the source manifold and transport an integral cochain with independently checked source maps. An incarnation destroyed by a replacement does not reappear, even when later geometry resembles an earlier state.

The first-descent locality theorem of report 79 says that any descending trace with at most $U$ upward events has a connected first-descent witness using $u \leq U$ upward and $d = u + 1$ downward events, and consuming at most $3u + 3$ original tetrahedra. The event birth DAG decomposes traces into components with disjoint original consumption. Project to a component with positive loss and truncate at its first descent, repeating until stable. Relative-boundary persistence permits these projections and commutations; truncation alone need not produce connected consumption. This is a theorem for the formal move model, not an assumed locality property of arbitrary input traces.

The native `find_pachner_descent` searches connected original covers of cardinality at most $\min(t, 3U + 3)$. On complete exhaustion it decides absence of *bounded-upward descent*. A found descent is independently replayed, and its final tetrahedron count must equal $t + u - d < t$. A smaller explicit cover limit gives only a local conclusion. Search, index and work caps remain inconclusive; no such result is a knottedness certificate.

## 148.2 Connected covers and constant-alphabet trace classes

The original dual graph has maximum degree four. A deterministic spanning walk encodes each rooted connected $i$-vertex set with $2(i - 1)$ local ports, so there are at most $t16^{i-1}$ such sets. The native enumerator uses a canonical parent: delete the largest non-root vertex whose deletion leaves a connected set. Every non-singleton connected set has a non-root tree leaf, so this parent exists. Reverse search enumerates each set once with polynomial work per set, using an explicit stack.

Each eligible incarnation can commit to idle, one of four upward face ports, or one of six downward edge ports. Two ready events cannot share an incarnation, because each local port specifies its

unique event. They thus commute. The deterministic scheduler's encoding uses at most

$$i + 3u + 2d \leq i + 5U + 2$$

assigned symbols before first descent. With $i \leq 3U+3$, the complete bounded-descent geometry has $2^{O(U)}\,\mathrm{poly}(t+B)$ bit cost. The native sleep-set engine keeps one representative of each commuting class; naive enumeration remains available as a finite differential oracle. Surviving incarnations preserve commitments and sleeping obligations; source names do not resurrect. Marked geometry and exact cochain values, not a probabilistic hash alone, govern endpoint equality.

Consequently bounded descent is polynomial for $U = O(\log(t+2))$ and quasi-polynomial for $U = O(\log^2(t+2))$. The sufficient connected-cover cardinality and total upward-event bound are both charged. The theorem supplies no universal such allowance for all unknots.

## 148.3 One search over overlapping source regions

Let $\mathcal{R}$ be the permitted original covers and $I(v) = \{R \in \mathcal{R} : v \in R\}$. A frame that has consumed original footprint $F$ keeps

$$C(F) = \bigcap_{v \in F} I(v).$$

The root represents the whole family by a sentinel. Newly consumed original cells intersect this set with their incidence lists. Consuming a born cell adds no new original identity: its ancestors entered the footprint when that incarnation was created. An empty compatible-cover set rejects the branch; any surviving cover supplies a concrete source certificate.

Induction proves that a frame is admitted exactly when some cover contains its whole original consumption. Every prefix then belongs to that same cover, so this is the exact union of the independently restarted search languages. Admitted swaps of independent events retain that property, which makes the existing sleep-set reduction valid on the shared language. Endpoint query deduplication does not discard continuation frames merely because an endpoint was seen: remaining budget, sleeping events and compatible covers can differ.

At that release the complete cover index can have $t2^{O(U)}$ incidences. Section 149 later replaces default eager indexing by exact on-demand footprint completion while retaining the indexed backend. It is explicitly constructed and charged; neither constant memory nor a free index is claimed. Transitions and integral transport are evaluated only when queued branches are visited. Independent replay checks move geometry, cochain transport, original consumption and the advertised cover bound. Constructive birth analysis and disjoint trace composition are also integrated, preserving source labels and additive event counts. The report's deterministic multiunit splitter construction remains a theorem rather than a new runtime implementation here.

## 148.4 A positive diagram adapter with an inexpensive first probe

The optional diagram adapter constructs and independently binds the native knot exterior, optionally applies certified boundary shellings, chooses an integral cocycle and optionally minimizes its span. Its bounded search observes coherent normal surfaces at marked endpoints. The represented fibre may change topology during cochain transport; final positive evidence must be an actual essential normal-disc component, not positive Euler characteristic or an asserted transport of the original embedded fibre.

A bounded-region sleep/naive call now probes the unchanged source endpoint before building any cover index. That endpoint belongs to every permitted cover. A verified source disc returns immediately; an ordinary miss triggers the shared-cover engine. Both calls share the same node and cooperative work allowance. No index is constructed after the node allowance has already

been consumed. An incomplete per-endpoint disc query can still be followed by the full search when another node remains; its later exhaustion retains the engine's inconclusive semantics.

The returned `diagram-transport-disc-v1` proof includes the original PD source, authenticated exterior, optional shellings, initial cochain, every actual move and transport map, final coordinates and component proof. The independent checker imports no source constructor, cocycle solver, move producer, search planner or component search. It replays the source binding and complete move chain, validates the final coherent coordinates, and requires a positive independently verified essential-disc count. A proof for one input cannot substitute the exterior of another diagram.

`recognize(..., use_pachner_seed=True)` inserts this positive stage after the ordinary native seed attempt and before the complete fallback. It defaults off. The CLI exposes `-pachner-seed` and shared upward, node and work allowances. Its local cover limit is $3U + 3$, with $U = 1$ by default. Only a replayed positive returns `UNKNOT`; bounded exhaustion or any local cap proceeds to the maintained complete recognizer. The global deadline remains global, so a caller interruption cannot manufacture a local negative or a positive whose replay did not finish.

### 148.5 Scope of the asymptotic consequence

The complete first-descent primitive now has a maintained implementation with the report's parameterized guarantee. The optional disc adapter searches a bounded marked family and has separately charged component queries and replay. A complete negative family result remains inconclusive for a knot: there is no theorem that every unknot has a disc or a descent in this family. Neither universally small upward allowance nor a recognized terminal structure and globally controlled marked hierarchy is established. Thus this closes an implementation gap in an actual bounded geometric operation, while the six general obligations of Section 5 and the recognizer's exponential worst-case bound remain.

### 148.6 Maintained source checks and complete measurements

All 1,499 maintained tests pass in 258.931 seconds. The 66 focused integration tests pass in 9.620 seconds. They retain the delivered geometric, integral transport, callback, causal projection and composition checks, and add actual recognizer/CLI positive replay, cap-safe complete fallback, unchanged disabled behavior, a source-first index bypass and an exhausted-node index bypass. The report's historical endpoints are read from its unchanged artifacts rather than copied into a new search result.

The native source and diagram audit completes in 41.188 seconds. Ten copied runtime modules and six foundational move/cocycle/codec modules exactly match their delivered source bytes. The diagram adapter is the reconciled part: bounded sleep/naive queries use shared covers after the first source probe, and the recognizer and CLI provide the optional stage.

The two genuine source fixtures attach one or two six-cell descent gadgets to a finite solid torus, giving seven and thirteen initial tetrahedra. Neither has an initial native $3 \to 2$ move. Independent restarted and shared exhaustions at $U = 1$, cover cardinality six, have identical exact cochain-isomorphism endpoint classes: fourteen and twenty-seven. The native shared engine retains twenty-eight and fifty-five marked endpoints, versus 495 and 3,152 restarted frames. All 124 retained source endpoint proofs pass independent replay. The exact class comparisons use full canonical cochain-isomorphism tuples, not digest-only equality; they are distinct from the runtime's marked incarnation keys.

At $U = 0$ both sources completely exhaust bounded descent. At $U = 1$ both return a separately replayed one-up/two-down witness with strict loss one. Fresh Regina checks confirm that each original and returned endpoint is a solid torus. These checks do not replace the source-bound move replay or prove a universal small-upward theorem.

The first-descent comparison uses five randomized paired rounds and four arms per source, including identical-baseline A/A controls. Forty measured calls and eight warmups complete their configured positive query, independent replay and full serialization. Both arms validate and search the same source and declared $U = 1$ connected-cover family. Source/index preparation and the actual transitions are charged in both arms.

| Input | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| descent 1 | 989.355 | 26.204 | 37.721 | 0.991 | 1.014 |
| descent 2 | 3862.141 | 48.123 | 79.691 | 0.997 | 0.995 |

The median paired old/new ratios are 37.721 and 79.691. The first case begins 273 versus six frames and records 210,418 versus 5,263 callback work units; the second begins 630 versus six frames and records 837,402 versus 14,520. Each returned proof gives strict loss one. The A/A controls stay within about two percent of parity. These measurements establish gains for complete *first-descent source queries*, not full family exhaustion or complete knot recognition on these sources.

The diagram audit uses thirteen actual diagrams, including unknots, trefoil and figure-eight controls and hard native-seed survivors. With a 200,000-unit local allowance, all thirteen complete enabled recognition verdicts agree with the frozen preceding recognizer. Disabled statuses, methods and evidence agree exactly. The direct diagram adapter certifies the empty-circle case; all twelve other native queries reach the work cap. The one saved positive also receives fresh Regina confirmation. No added nonempty-diagram coverage is claimed for this cohort. Separate focused tests certify nonempty one-crossing inputs, both zero-move and transported positive proofs, and binding against a different input knot.

Five paired rounds complete 180 measured native/full-recognition calls and 36 warmups. Native query rows compare the delivered independently restarted adapter with the maintained shared adapter, both under the same local work cap. Full recognition rows compare enabling this new stage with omitting it. Preparation, proof replay, serialization and any local cap/fallback are included. The four nonempty native timing controls all cap, so their ratios are bounded-call latency observations rather than completed-family speedups.

**Bounded native diagram queries.**

| Input | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| empty circle | 9.994 | 9.711 | 1.029 | 1.001 | 0.992 |
| optimized positive | 788.913 | 191.477 | 4.120 | 0.995 | 1.002 |
| genus one miss | 883.092 | 200.690 | 4.516 | 1.017 | 0.993 |
| trefoil | 873.501 | 184.554 | 4.733 | 0.998 | 0.966 |
| figure eight | 861.689 | 187.937 | 4.569 | 1.005 | 0.990 |

**Complete recognition, disabled versus enabled Pachner stage.**

| Input | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| empty circle | 0.016 | 0.016 | 1.018 | 0.975 | 0.975 |
| genus one miss | 0.687 | 228.249 | 0.003 | 1.087 | 1.060 |
| trefoil | 0.637 | 211.157 | 0.003 | 0.660 | 0.873 |
| figure eight | 0.900 | 221.463 | 0.004 | 1.026 | 1.012 |

The capped native queries have old/new ratios about 4.12–4.73, while the source-positive empty-circle query is near parity. This does not establish successful search on the capped diagrams. Enabling the stage adds roughly 0.21–0.23 seconds to the three small nonempty recognition controls whose complete fallback takes under one millisecond. Their paired disabled/enabled

ratios are about 0.0028, 0.0031 and 0.0040; the delay dominates despite noisy submillisecond A/A observations. The empty diagram is recognized before the stage in both arms. These actual costs justify retaining the disabled default; no reliable full-recognition speedup is claimed.

Runtime release `fbd3c9a8d`, the tests, exact source audit, paired source and diagram timings are retained under `data/pachner-native-*`. The driver is `causal_research.native`, with `audit`, `source-benchmark` and `diagram-benchmark` modes. Publication review checks 641 runtime pins, all completed timing outcomes, all 124 source endpoint proofs and both strict descents, and the saved source-bound diagram proof with move/search/transport producers disabled. The maintained bounded primitive and optional source-bound disc adapter are concrete integrations of report 79; the global structural closure hypothesis and general quasi-polynomial recognition remain unproved.

# 149 Exact cover existence before ambient indexing

The integrated shared Pachner engine removes restarted trace prefixes but still constructs a complete ambient cover index before visiting the search root. On the actual-diagram audit of Section 148, this preparation consumes the local allowance before useful move exploration. This continuation represents the same existential cover language by exact on-demand queries on consumed footprints. It preserves connected and multicomponent covers, the complete bounded family and independent replay.

## 149.1 A footprint needs a witness, not the whole index

For original dual graph $G$, size limit $R$ and component limit $c$, a footprint $F$ is admitted exactly when

$$\exists S: \quad F \subseteq S, \quad |S| \leq R, \quad \text{components}(G[S]) \leq c.$$

An empty footprint has the empty witness. If $F$ itself satisfies the size and component limits, it is already a witness. Otherwise the oracle searches for additional original cells connecting its components. In particular it must not reject disconnected consumption merely because $c = 1$. A larger connected cover can contain a disconnected footprint.

**Proposition 149.1** (Rooted completion suffices)**.** *If a permitted cover of $F$ exists, one exists with every induced component intersecting $F$. The latter covers can be enumerated without duplication by a rooted reverse search starting from $F$.*

*Proof.* Delete every component of a cover that does not intersect $F$. This retains all required cells, reduces its size and cannot increase its component count. For any rooted set $S \supsetneq F$, choose a spanning forest with one assigned root in $F$ for every tree, for example a deterministic multi-source breadth-first forest. It has a non-root leaf. Deleting that leaf leaves every remaining vertex connected to a root, so at least one non-root deletion preserves the rooted property. Define the canonical parent by deleting the largest non-root vertex with that property. Cardinality decreases until $F$ is reached. The deleted vertex always has a neighbour in its parent, since a component consisting only of that non-root vertex would violate rootedness. Thus adding frontier vertices and accepting exactly children whose canonical parent is the current set generates every rooted extension once. □

The original degree bound is four. For fixed roots $F$, encode any rooted extension by processing a deterministic breadth-first queue initially containing those roots. Each processed vertex gives at most four membership bits for its neighbour ports. Every component meets a root, so the queue reaches the entire extension. At most $R$ vertices are processed, giving at most $\sum_{i=0}^{R} 2^{4i} = 2^{O(R)}$ extensions, independent of an ambient factor raised to $|F|$. Canonical-parent and component tests

cost polynomial work in $R$ and the source encoding. A fixed query therefore has $2^{O(R)}\,\mathrm{poly}(t+B)$ bit cost. No greedy Steiner approximation or numerical relaxation supplies the answer.

## 149.2 The selected witness does not define future eligibility

A search frame retains its actual consumed original set $F$. Adding an event's newly consumed originals $A$ asks about $F \cup A$, using the *original* dual graph. Born cells add no new original identity because their ancestors entered at creation. An admitted frame stores one covering witness solely for the source certificate. The next query uses $F$, not that witness: a later event may require a different cover containing the larger footprint.

For example on the path 0–1–2–3 with $R = 3, c = 1$, consumption $\{0\}$ has witness $\{0\}$. Later consumption $\{0, 2\}$ is still allowed, with witness $\{0, 1, 2\}$. Restricting future moves to the first witness would lose this trace. The new engine does not do so.

The existential predicate is hereditary. If some cover contains a full trace's footprint, it contains every prefix. It therefore admits exactly the union language of the indexed engine. Admitted independent-event swaps have the same final footprint and source marking, and both intermediate footprints lie in that cover. The same sleep-set commutation argument remains valid. Endpoint deduplication still does not delete continuation states with different remaining budgets or obligations.

Witness covers may differ between backends; actual geometry, transported cochain, incarnation histories and parent trace evidence do not depend on which covering witness is chosen. Predicate completeness still requires invariance under marked endpoint/commutation equivalence and independence from an arbitrarily selected witness cover. A history-sensitive callback cannot gain a completeness guarantee from changing representation.

## 149.3 Completed caches, explicit limits and retained interfaces

The oracle caches only a completed existence result for a footprint: one actual witness, or a proved absence. Its global completion-state limit counts begun rooted-extension candidates. Exhausting that limit or a cooperative callback interrupts the query; it never returns or caches a false exclusion. Cheap certified cases, such as a connected footprint already within the limit, need no extension search.

`cover_backend` accepts `auto`, `indexed` and `oracle`. The default uses the on-demand backend when no complete-index cap is requested. An explicit `max_regions` retains its original index-cap meaning and selects the indexed engine in automatic mode. The two explicit backends remain differential oracles. `max_oracle_states` bounds total completion work across a call, not the ambient cover-family count. These limits have distinct meanings; an incompatible explicit oracle/index-cap combination is rejected. The original certificate format and source checker remain unchanged.

Index counters remain zero on an unmaterialized oracle call; they do not assert that no logical covers exist. Separate query, cache-hit and extension state counts describe the actual oracle work. The index implementation and its cap/measurement regression tests remain available explicitly.

For connected bounded first descent, $R \le 3U + 3$. Multiplying the existing $t2^{O(U)}$ geometric trace count by the $2^{O(R)}$ per-footprint completion cost retains $2^{O(U)}\,\mathrm{poly}(t+B)$ overall geometry. This replaces eager preparation with visited-footprint work, rather than improving that parameterized complexity class. It can expose useful geometric transitions within an allowance that previously ended during index construction. For general multicomponent families, no new ambient-independent trace count is asserted; only each supplied-footprint completion has the stated bound.

### 149.4 Actual diagram progress and measured source costs

All 1,506 maintained tests pass in 275.089 seconds. The 27 focused tests pass in 2.332 seconds. Their literal small-graph comparison covers every footprint, size allowance and zero/one/two-component allowance on the selected graphs. Returned witnesses satisfy the exact requested containment and component conditions. Tests include disconnected completion, changing the witness between prefixes, completed negative caches, interrupted/capped queries, exact indexed work/cap boundaries and independent geometric replay. A five-node actual trefoil query performs three completed moves with no ambient index and correctly remains inconclusive.

The 21.817-second audit compares both backends on the preceding genuine seven- and thirteen-tetrahedron solid tori. Each complete exhaustion preserves its full marked endpoint proof sequence: twenty-eight and fifty-five entries agree byte-for-byte after removing only the arbitrary covering-witness field. The search node counts also agree. All 83 current endpoint proofs replay independently. The zero-upward bounded exclusions and one-up/two-down strict descents remain, with fresh Regina confirmations of both original and final solid tori. Both full source exhaustions require no rooted extension states: their visited footprints have immediate permitted witnesses.

Thirteen actual diagrams retain all complete enabled recognition verdicts, and disabled statuses, methods and evidence match the frozen previous recognizer exactly. Native disc successes are unchanged: the empty-circle case is positive and receives fresh Regina confirmation, while all twelve nonempty native calls hit their local allowance. Completed geometric/cochain transitions nevertheless rise from nine to 141 in the same cohort. Eleven nonempty diagrams that formerly performed no moves now do so; the remaining nonempty case rises from nine to thirty-four. This is source exploration progress, not new knot-recognition coverage.

A separate frozen-source index probe establishes the preparation bottleneck on three concrete diagrams. With the same 200,000-unit allowance, indexing fails to finish after 6,092 regions on optimized-positive, 7,079 on genus-one miss and 7,112 on trefoil. The new five-node pilot completes three moves on each without an index. Its complete work totals are 123,529, 46,923 and 42,731. The diagram proof query still returns inconclusive; no positive is inferred from the smaller triangulation or the fact that moves were reached.

The first-descent timings use five randomized paired rounds, forty measured calls and eight warmups, with identical-baseline A/A controls. Both arms charge source preparation, completion/index work, their configured complete positive search, independent strict-descent replay and full serialization. The preceding indexed source is frozen at `ce7fb0737`.

| Input | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| descent 1 | 33.207 | 31.849 | 1.033 | 1.010 | 0.999 |
| descent 2 | 62.264 | 55.380 | 1.125 | 0.991 | 0.988 |

The source ratios are 1.033 and 1.125. Both searches begin six frames. Work falls from 6,563 to 5,496 for the smaller source and from 16,854 to 9,665 for the larger. The larger source's A/A controls are about 0.991 and 0.988; the modest small-source gain is not generalized to a universal speedup. This improves visited-footprint preparation, without changing the bounded first-descent complexity class.

The diagram timings complete 160 measured calls and thirty-two warmups. Both native and complete-recognition rows compare the preceding indexed stage with the new on-demand stage under the same local cap; the stage is enabled in both recognition arms. Source/cochain construction, every actual transition and component query, configured caps, positive replay and full serialized output are charged. They are distinct from comparing the optional stage against omitting it altogether.

**Budgeted native diagram calls, indexed versus on-demand.**

| Input | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| empty circle | 9.535 | 9.476 | 1.003 | 1.005 | 0.998 |
| optimized positive | 189.580 | 637.641 | 0.294 | 1.015 | 1.000 |
| genus one miss | 183.740 | 834.054 | 0.221 | 0.989 | 1.003 |
| trefoil | 183.011 | 827.692 | 0.221 | 0.999 | 1.001 |
| figure eight | 183.139 | 788.638 | 0.232 | 1.003 | 1.005 |

**Complete recognition with the optional stage enabled in both arms.**

| Input | previous ms | current ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| genus one miss | 208.951 | 869.423 | 0.241 | 1.009 | 1.001 |
| trefoil | 203.683 | 864.819 | 0.236 | 0.999 | 0.999 |
| figure eight | 208.323 | 815.071 | 0.255 | 1.011 | 1.009 |

At the same checkpoint allowance, reaching actual moves costs more wall time than processing the old eager index. The nonempty native paired ratios are about 0.29, 0.22, 0.22 and 0.23. Complete enabled recognition ratios are about 0.24, 0.24 and 0.25: old calls take about 0.20–0.21 seconds, while current calls take about 0.82–0.87 seconds. The A/A controls are near parity, so the slower budgeted latency is a real consequence of doing more expensive geometric work, not claimed as a speed gain. The unchanged source-positive control is near parity. No additional disc success is demonstrated, and the optional diagram stage remains off by default.

The useful result is an exact source family that begins exploration without ambient preprocessing and retains complete fallback/replay guarantees. Runtime release `1519b06f7`, all tests, exact source and actual-diagram audits, the supplemental index probe and paired timings are retained under `data/cover-oracle-*`. The driver is `causal_research.cover_oracle` with `audit`, `source-benchmark` and `diagram-benchmark` modes. Publication review checks 643 runtime pins, all timing outcomes, all 83 source endpoint proofs, both strict descents and the saved diagram witness with cover/move/transport producers disabled. General small-budget reach, terminal structure, global hierarchy accounting and quasi-polynomial recognition remain unproved.

# 150 Strict descent epochs reach an actual transported disc

The on-demand cover engine reaches useful moves, but its small single-family search still returns no new nonempty-diagram witness in the preceding audit. This continuation tests two ways to turn that exploration into reach. Exact Euler-based event ordering alone does not succeed on the selected controls. Restarting bounded *strict descents* on successively smaller sources produces an actual new transported disc for the genus-one-miss diagram. The new policy is explicit, with a per-epoch upward allowance and a full original-source proof; it does not change the existing total-upward family.

## 150.1 Failed ordering and monotone pilots

The existing bipyramid compiler scores Euler jumps and normal-disc changes with a fixed number of exact integer operations on five local heights. An ordering-only scratch experiment prefers downward Euler gains and upward moves with smaller Euler loss. All eight calls on four selected actual diagrams remain inconclusive at the two-million-unit allowance. This finite negative observation neither proves such orderings useless in general nor justifies adding a runtime policy with no demonstrated witness.

The maintained monotone `descend_cocycle` pilot instead tries only $3 \to 2$ moves, with exact score selection and independently replayed transport. It is inexpensive on the selected diagrams,

but most have no initial eligible downward move; the optimized-positive control has three without obtaining a disc. These observations motivate using the complete bounded first-descent primitive to unlock a source, then restarting from the smaller endpoint rather than remaining inside one original cover.

## 150.2 One actual source chain through strictly smaller triangulations

An epoch starts at source $(T_i, h_i)$ and calls `find_pachner_descent` with at most $U$ total upward events *in that call*. A successful result is independently replayed and has

$$|T_{i+1}| < |T_i|.$$

The returned integral cochain becomes the next source cochain. All actual move/transport records are concatenated; no new diagram exterior is silently substituted. The adapter checks the coherent surface at the initial source and after each successful epoch. Only an independently verified essential disc component supplies the positive result.

The scope is deliberately different from one bounded family. A chain of $e$ epochs can have up to $eU$ upward events, although each epoch has at most $U$. A stationary result of the bounded descent primitive or a resource cap remains inconclusive for the knot. It is not a terminal recognition theorem for that manifold. The new mode is never described as a witness with a whole-chain upward bound $U$.

**Proposition 150.1** (Strict restart and geometric bit bounds)**.** *From a supplied $t$-tetrahedron marked source, the epoch policy has at most $t$ successful restarts. Its completed first-descent move chain has at most $t(2U+1)$ events. Geometry, cochain transport and bounded descent search have $2^{O(U)}\,\mathrm{poly}(t+B)$ bit cost, with source/proof encoding growth charged; component queries and essentiality replay remain separate costs.*

*Proof.* Tetrahedron count is a nonnegative integer decreasing at every successful epoch. Its first descent has $u \le U$ and $d = u+1$, hence at most $2U+1$ events. Summing gives the restart and event bounds. For the height accounting, let $H$ bound the span of each current integral height row. A formal bipyramid has five vertices. The old edge graph is connected, so differences between its aligned vertex heights are sums of at most four old edge differences, each bounded by $H$ in absolute value. The new tetrahedron heights restrict those aligned values; untouched rows retain their spans. Thus one move has new span at most $4H$. After $m$ moves, bit lengths are at most $B + O(m)$, including the zero-span case. With $m \le t(2U+1)$ this is polynomial in $t, U, B$. The live cell count within an epoch is at most $t+U$. Applying the preceding complete $2^{O(U)}$ bounded-descent geometry bound at most $t+1$ times and charging transport and retained full snapshots preserves the displayed form. No component-query or essentiality cost is removed by this argument. □

This is a concrete polynomial restart measure. It does not prove that a small per-epoch $U$ reaches an essential disc for every unknot, that a stationary source can be recognized cheaply, or that the hierarchy's other pattern/cutting operations have controlled representations. The general obligations of Section 5 therefore remain.

## 150.3 Shared allowances and original-source proof authority

`pachner_epoch_seed_decide` prepares and binds the original exterior, optional boundary shellings and one initial cocycle/span choice. It shares one cooperative work allowance and one geometric-node allowance across initial observation, every epoch and full positive replay. A separate epoch limit can stop the chain, without invalidating already accepted moves or providing a negative knot

verdict. A failed component query also remains inconclusive. Caller exceptions propagate through the same guarded replay interface as ordinary transported witnesses.

A positive result uses the unchanged `diagram-transport-disc-v1` schema: original PD input and source exterior, optional shellings, original height rows, every actual transport step, final coordinates and the terminal disc proof. The independent checker replays the whole concatenated chain from that original input. It uses neither the epoch search nor an asserted intermediate outcome; the final source cannot be replaced by another knot's exterior. Missing or reordered moves, changed cochain values and a false positive count are rejected.

The recognizer and CLI expose this as an explicit optional mode. `pachner_seed_epochs=0` keeps the single-family policy unchanged. A positive value bounds the number of strict descent epochs, with the upward allowance now explicitly *per epoch*. The CLI option is `-pachner-seed-epochs`, used with `-pachner-seed`. Neither the optional stage nor epoch mode becomes a default. A miss continues the maintained complete recognizer, and its global deadline stays global.

## 150.4 Actual diagram witnesses and complete-call measurements

All 1,514 maintained tests pass in 285.175 seconds. The 26 focused tests pass in 13.434 seconds. They exercise the actual 33-epoch nonempty witness, shared epoch/node/work caps, whole-chain producer-disabled replay, input substitution and missing/reordered/altered moves, ordinary complete fallback, unchanged zero-epoch behavior and the explicit CLI mode.

The 112.342-second six-diagram audit compares one bounded family with the explicit epoch policy under a shared two-million-unit allowance. All disabled statuses, methods and evidence match the frozen recognizer, and every zero-local-allowance enabled call reaches the correct complete fallback. The genus-one-miss input additionally runs through the actual enabled epoch recognizer and returns `native-pachner-epochs`, with full independent source replay. Three returned disc proofs, including two initial-source positives, receive fresh Regina confirmation.

For genus-one-miss, the single total-upward-one family remains inconclusive. The epoch policy returns a compressing disc after 33 strict descents from 54 to 21 tetrahedra, using 191 geometric search nodes and 914,561 callback work units. Its original-source proof has 51 actual transport steps: nine upward and forty-two downward events. Thus its per-epoch allowance one plainly does not describe the whole trace's upward count. Every epoch record has strictly decreasing source size and at most one upward event.

The optimized-positive control reaches the cap after 48 descents from 94 to 46 tetrahedra. The trefoil control reaches a complete bounded-stationary source after fourteen descents from 48 to 34, then correctly remains inconclusive; that result is not used as a knottedness certificate. Figure-eight reaches the cap after eighteen descents from 68 to 50. The empty-circle and one-crossing controls return verified source discs without a descent. No universal reach theorem is inferred from these cases.

The failed ordering experiment is separately retained under `data/epoch-order-pilot.*` with its exact driver. Eight capped calls on four selected actual diagrams produce no witness under either lexical or exact local Euler ordering. The monotone-only pilot is also inexpensive but stalls at the initial source on most selected inputs. These observations motivate the epoch route; they do not prove either alternative generally ineffective.

The configured-call comparison completes three randomized paired rounds, 48 measured calls and sixteen warmups, including identical-baseline A/A controls. Both policies share a one-million-unit work allowance and a thousand geometric nodes. The single policy has total upward budget one; the epoch policy has at most 64 strict descents with upward budget one per epoch. Their finite reach is different. Preparation, every component query, configured caps, independent positive replay and full serialization are charged; an old capped result is not represented as a completed family.

**Configured native calls, single family versus epochs.**

| Input | single ms | epoch ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| empty circle | 10.052 | 8.564 | 1.161 | 1.009 | 0.988 |
| genus one miss | 4607.241 | 5195.393 | 0.877 | 1.008 | 0.996 |
| trefoil | 4515.999 | 5421.991 | 0.833 | 0.982 | 0.998 |

**Complete recognition with the two explicit optional policies.**

| Input | single ms | epoch ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| genus one miss | 4896.477 | 5079.321 | 0.974 | 1.013 | 1.002 |

The native genus-one-miss query returns inconclusive in every single-family arm and a replayed unknot certificate in every epoch arm. Its paired old/new ratio is 0.877, with median 4.607 versus 5.195 seconds. Thus the gain is a verified witness within the shared allowance, not reduced latency. The native trefoil control remains inconclusive in both policies, with ratio 0.833 and median 4.516 versus 5.422 seconds. The source-positive empty-circle control has ratio 1.161; no general inference is drawn from that tiny control. A/A observations are included in the table.

Complete genus-one recognition has paired ratio 0.974, median 4.896 versus 5.079 seconds. The single-policy arm reaches the ordinary exact fallback. The epoch arm instead returns its geometric disc proof. Both have the correct unknot verdict. Omitting the optional stage altogether already recognizes this small example in under a millisecond in the preceding comparisons. This alternative original-source geometric proof therefore demonstrates new reach, not a faster default recognizer. The optional stage and epoch mode continue to default off.

Runtime release `82d241319`, the passing tests, original-source proof audit, fresh Regina confirmations and configured-call timings are retained under `data/pachner-epochs-*`. The driver is `causal_research.epochs`, with `audit` and `benchmark` modes. Publication review verifies 646 runtime pins, all six source/complete verdicts, all timing outcomes and all three saved disc proofs with epoch, move, transport and source constructors disabled. The actual 51-step witness, its scope and every strict epoch are checked explicitly. No universal stationary-source recognition theorem or general quasi-polynomial unknot bound is claimed.

# 151 Exact vertex gauges accelerate a strict-descent disc witness

The preceding epoch policy transports one cochain through smaller sources without changing its gauge. Reoptimizing that gauge on the new geometry can expose a simpler coherent fibre. This continuation does so periodically, adds exact independent coboundary replay, and obtains the same actual nonempty-diagram disc in fewer epochs. The optimization remains optional; source, move and final component authority are checked from the original input rather than inferred from a span objective.

## 151.1 A vertex change is an exact integral coboundary

Let $v(t, j)$ be the actual global vertex of corner $j$ in tetrahedron $t$. For a single integral potential $p$ on these global vertices, set

$$h'_{t,j} = h_{t,j} + p(v(t, j)) - (h_{t,0} + p(v(t, 0))).$$

The final subtraction is only row normalization. Along an oriented source edge $a \to b$, the new edge cochain differs by $p(b) - p(a)$. A period around any closed source cycle is consequently

unchanged by telescoping. Local triangle sums and consistency across shared edge occurrences are preserved. The correction is integral, so this is a change of representative within the same integral cohomology class.

**Proposition 151.1** (Source gauge replay is exact)**.** *An independently prepared source, a complete vertex-potential list and the displayed corner equation suffice to authenticate the new edge cochain. No optimizer, claimed minimum, flow transcript or move producer is required.*

*Proof.* Preparation reconstructs the global vertex and oriented edge classes from actual face pairings. Every stated potential must refer to exactly one such vertex. Checking the displayed equation at all corners verifies every new local height difference. Checking all signed edge occurrences confirms that those differences form the same global source cochain plus the stated vertex coboundary. The closed-cycle period identity follows directly. The assertion concerns the cochain class; it does not assert equality of the represented embedded normal fibres. □

The native producer invokes the existing span optimizer on that current source and cochain. It anchors each connected vertex component's irrelevant constant, constructs the new normalized rows and independently verifies the gauge proof before using it. It also requires nonincreasing total disc span. These are producer checks for the chosen policy, not assumptions made by the positive disc verifier.

### 151.2 A mixed move/gauge chain retains the original source

The local `cocycle-vertex-gauge-v1` proof contains the actual source digest, sorted complete vertex/value pairs and new height rows. The checker reconstructs the source, rejects missing/extra/Boolean vertex identities, reads exact integers through the maintained codec, and checks every corner and signed edge equation. Arbitrary common potential offsets cancel under row normalization, including values requiring thousands of encoded bits. There is no expansion proportional to such a value.

Mixed chains use `diagram-transport-disc-v2`. The top-level original PD, exterior, optional shellings, initial heights, final coordinates and terminal disc proof retain their meanings. A move step retains its original triangulation/transport pair and full geometric replay. A gauge step has only its source-bound gauge proof, leaves the current triangulation unchanged and supplies the authenticated next height rows. The independent full-chain checker accepts no gauge step under the old v1 schema. Old v1 proofs still use their original move-only path.

Neither checker imports the gauge optimizer. Every topology-changing Pachner step still has its own source/relative-boundary proof; every gauge step is checked on the exact current source. A lower objective value does not imply an essential disc, and a change in fibre topology is not described as transport of one unchanged embedded surface. The final actual coherent coordinates and essential-component count are independently reconstructed and checked as before.

### 151.3 Periodic use and representation-size accounting

The epoch API exposes `regauge_interval`; zero keeps the previous policy. A positive interval invokes the optimizer after that many strict descents, with the same global work and node allowance. Identity results need no new chain step. The recognizer and CLI expose the interval only with explicit epoch mode, and neither optimization nor epoch search becomes a default. A cap or failure before a complete positive replay remains inconclusive and reaches the complete fallback.

Let $H$ bound each normalized height-row span and $t$ the initial source size. Its total disc span is $W = \sum_t (\max h_t - \min h_t) \le tH$. A verified nonincreasing span optimization has each new row

span at most $W$, hence at most $tH$. In combination with the preceding $4H$ per-move bound, $m$ geometric moves and $g$ gauge calls give height bits bounded by

$$B + O\big(m + g\log(t+1)\big).$$

Strict descent gives $m \le t(2U+1)$ and $g \le t$ for a positive interval, so this growth is polynomial in $t, U, B$.

Proof potentials are normalized per vertex component. Along a spanning path of its source one-skeleton, $p(b) - p(a)$ is the difference between new and old edge cochains. There are at most $4t$ vertices, and each edge correction is bounded by $(t+1)H$ for one gauge call. Anchoring therefore bounds potential magnitudes by $4t(t+1)H$; their bit size is also charged. The independent checker may accept larger harmless offsets supplied by a caller, but reading them costs their actual encoding size.

Source preparation, the span optimizer, every gauge replay and every final component query remain explicit costs. This bound controls representation growth; it does not turn an unaccounted operation into a constant-time one. The bounded geometry/restart argument and the separate essentiality and universal reach obligations retain their preceding scope.

## 151.4 Actual proof reduction and complete configured-call measurements

All 1,521 maintained tests pass in 286.636 seconds. The 28 focused tests pass in 19.532 seconds. They check every actual gauge edge difference as one global vertex coboundary, source and potential forgeries, changed height rows, missing gauge steps, v1 schema downgrade, source substitution, false component counts and producer-disabled full mixed-chain replay. A common $2^{9000}$ potential offset leaves all normalized rows unchanged and still passes replay after the portable integer codec. CLI routing, shared caps, legacy zero-interval behavior and caller exception identity also pass.

The 67.775-second audit compares the frozen ungauged epoch policy with interval-four reoptimization on the same six actual diagrams and shared two-million-unit allowance. Exact disabled recognizer statuses, methods and evidence are preserved, and every zero-local-work enabled call reaches the correct complete fallback. Fresh Regina confirms all three returned disc witnesses. The target nonempty proof also runs through the actual enabled recognizer and full source-bound replay.

For genus-one-miss, both epoch policies certify the unknot. The ungauged chain takes 33 strict descents, 191 nodes and 914,561 callback work units. The gauged chain takes 24 descents, 83 nodes and 622,768 work units, about 32 percent fewer checkpoints, reaching thirty tetrahedra instead of twenty-one. Six optimization calls yield three nonidentity gauge changes. Its v2 proof has four upward and twenty-eight downward geometric events plus three gauge records, 35 steps in total. The earlier disc is not attributed to a changed cohomology class: every intermediate correction is source-replayed as an integral vertex coboundary.

Other controls show the cost of this policy. Optimized-positive still reaches the work cap, after 44 descents rather than 48; eleven optimizer calls produce two actual changes. Trefoil still reaches its bounded-stationary source after fourteen descents; three gauge changes raise its work from 1,343,190 to 1,382,485, about three percent. Figure-eight still caps after eighteen descents with four changes. The empty-circle and one-crossing controls return their initial-source disc without a gauge. None of these nonpositive outcomes is used as a knottedness certificate.

Configured-call timings use three randomized paired rounds, 48 measured calls and sixteen warmups, including identical-baseline A/A controls. Both policies share a one-million-unit allowance and a thousand geometric nodes. Every source preparation, span optimization, gauge check, epoch search, component query, full positive replay and serialized result is charged. The preceding ungauged source and recognizer modules are frozen, rather than disabling just one helper inside the current implementation.

**Configured native epoch queries, ungauged versus periodic gauges.**

| Input | ungauged ms | gauged ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| empty circle | 8.435 | 8.392 | 1.003 | 1.009 | 0.992 |
| genus one miss | 5064.360 | 3222.290 | 1.572 | 1.001 | 1.002 |
| trefoil | 5226.081 | 5066.872 | 1.031 | 0.991 | 0.999 |

**Complete recognition using the two explicit optional epoch policies.**

| Input | ungauged ms | gauged ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| genus one miss | 4807.415 | 3024.992 | 1.589 | 1.002 | 1.002 |

The target native query has paired old/new ratio 1.572, median 5.064 versus 3.222 seconds. Complete optional recognition has ratio 1.589, median 4.807 versus 3.025 seconds. Both arms return an independently replayed unknot certificate, and their A/A controls are essentially at parity. The gauges, optimizer, changed chain schema and all replay costs are included. Thus the gain is measured completed-call work on the demonstrated geometric witness, not an inference from a smaller objective or a capped old result.

The empty-circle control is near parity at 1.003 and reaches no gauge. The trefoil configured native control has ratio 1.031 with both policies capped; its A/A observations are about 0.991 and 0.999. No broad speedup or negative knot conclusion is inferred from that row, and the complete two-million-unit audit already records extra gauge work on its bounded-stationary source. Ordinary recognition with the optional stage omitted still solves the small target far faster. The optional stage, epoch mode and periodic interval therefore retain their disabled defaults.

Runtime release `6fea587df`, the passing tests, six-diagram audit, fresh Regina confirmations and paired timings are retained under `data/epoch-gauge-*`.

The driver is `causal_research.epoch_gauge`, with `audit` and `benchmark` modes. Publication review verifies 650 runtime pins, all completed and capped outcomes and all three original-source disc proofs with gauge optimizer, epoch, move and search producers and source constructors disabled. The three actual gauge records, their exact scope, the 35-step mixed chain and every strict descent are checked explicitly. General cheap support/reach, stationary-source recognition and quasi-polynomial unknot bounds remain unproved.

# 152 Primitive annulus caps on transported sources

The preceding epoch adapter recognizes a positive essential normal-disc component. It can nevertheless continue costly descent from a source already carrying a certifiable primitive annulus. The existing native capping theorem provides a different witness. This continuation authenticates that theorem on a replayed moved source and exposes an explicit terminal annulus option. It preserves all source, connectivity, primitivity and arithmetic requirements; Euler characteristic zero alone never supplies a knot verdict.

## 152.1 Negative pilots identify the useful scope

Radius-one type-window queries at selected smaller endpoints find no extra disc on the sampled controls and consume allowance otherwise used for descent. Radius-two queries likewise find no witness while costing more. Repeated boundary shelling removes additional tetrahedra but supplies no new hard-control success and does not improve the target's checkpoint total. The edge-first/span-second gauge pilot also gives no new witness and raises work on the demonstrated

target. These experiments are retained as finite negative observations, not promoted into new runtime defaults.

The useful distinction is instead topological witness type. The initial optimized-positive source has Euler zero and a independently checked minimum-span primitive connected fibre. Its annulus already admits the existing capping argument, while disc-only epoch search consumes its budget seeking a normal disc. The source query therefore needs an authenticated annulus terminal rather than a larger unproved disc-search family.

## 152.2 The capping theorem and its source hypotheses

**Proposition 152.1** (Primitive annulus cap after source replay)**.** *Let $M$ be a compact knot exterior authenticated from the input diagram and source-replayed relative-boundary replacements. A connected orientable properly embedded primitive relative-class surface with Euler zero is an annulus. Exactly one of its boundary circles is inessential on $\partial M$. Capping that circle on the boundary torus and rounding inward gives an embedded compressing disc. Thus this authenticated surface proves that the input knot is the unknot.*

*Proof.* A connected orientable Euler-zero surface is either an annulus or a closed torus. A knot exterior has $H_2(M) = 0$, so a closed torus cannot represent the nonzero primitive relative class. The boundary of that relative class generates the primitive kernel of $H_1(\partial M) \to H_1(M)$; in particular it is neither zero nor twice an integral class. Two disjoint essential simple circles on a torus are parallel and their signed sum is zero or twice a primitive slope. Two inessential circles have zero sum. Hence the annulus has one essential and one inessential boundary circle. The latter bounds a disk in the torus disjoint from the essential circle. A boundary collar permits the cap to be rounded into $M$ without intersecting the annulus. The remaining boundary is essential, giving a compressing disk. Boundary compressibility of an authenticated knot exterior yields unknotness. This is the existing primitive capping theorem applied after the verified source replacements, not a new literature-priority claim. □

The implementation obtains connectedness and orientability through the retained coherent-surface minimum-span theorem, with an independently verified arithmetic dual witness. Integral primitivity is checked on actual signed source edges after a separate tree gauge. It is not the gcd of ungauged heights and not an assertion from the optimizer. A valid doubled minimum-span certificate still fails this primitive period check; the new regression test exhibits that distinction explicitly.

## 152.3 Bind a terminal surface to the original moved source

The new `diagram-transport-annulus-v1` proof contains the original PD, original exterior, optional initial shellings, initial cochain, the complete actual move/gauge chain and a terminal surface certificate. Source replay is shared privately with the existing disc checker: every geometry and cochain transport is authenticated, and the computed final triangulation is owned by that verification call. No public caller-supplied source cache is accepted.

The terminal `diagram-cocycle-annulus-v1` data contain that final triangulation, reference integral heights, actual normal coordinates and a minimum-span certificate. A closed source predicate binds it to the exact triangulation computed by replay. The existing independent cocycle inspector checks the original input identity, matching geometry, global edge values, primitive period class and the minimum-span primal/dual equality. Only then are its connected surface summary and Euler value used. The required value is exactly zero.

The final annulus is authenticated on the moved source, not asserted to be the unchanged fibre of every earlier cochain. Its reference heights and minimum-span proof are independently checked

on that source. Source rank one and the capping theorem come from the replayed knot exterior. A different source, a missing geometric move, changed gauge data, forged coordinates, invalid dual equality or nonprimitive class cannot substitute for these hypotheses. No normal-coordinate vector for the capped disk is invented.

Move-only and mixed move/gauge disc proofs retain their v1/v2 schemas and full final-component authority. The new annulus format is explicitly different, and the disc checker rejects it. The annulus checker imports no source constructor, optimizer, move producer, epoch planner or normal component search.

### 152.4 A cheap terminal check inside the bounded epoch policy

The optional epoch mode exposes `annulus_caps`. Zero/default false preserves the preceding behavior. At the initial span optimum and every current-source gauge optimum, the adapter retains the minimum-span witness and its reference heights. A geometric move invalidates that witness until another optimum is available. If a retained witness has Euler zero, the full original-source annulus verifier runs before more geometric branching. Only its completed positive replay returns `UNKNOT` with method `native-pachner-annulus`. Other Euler values continue ordinary disc queries and strict descent.

The source preparation, Euler observation, primitive/connectivity replay and all arithmetic checks use the existing shared work allowance. The initial geometric-node allowance is still required; a work or node cap never becomes a capping certificate. A miss or capped query continues the complete recognizer, while the global deadline remains global. The recognizer and CLI require explicit epoch mode for `pachner_seed_annulus=True` or `-pachner-seed-annulus`. No optional geometric policy becomes a default.

This adds a verified terminal witness type to the actual source search. It gives no theorem that every unknot reaches a primitive annulus or disc within small upward/epoch/support bounds. It does not classify a stationary source as knotted or eliminate the hierarchy's missing global obligations.

### 152.5 Actual source proofs and complete configured-call measurements

All 1,529 maintained tests pass in 279.905 seconds. The 36 focused tests pass in 19.897 seconds. They check an actual nonempty-diagram annulus terminal, a nonempty geometric move plus gauge chain, final-source substitution, coherent coordinates, minimum-span dual data, input identity, malformed shellings, caller cancellation and producer-disabled replay. A valid doubled minimum-span certificate still fails capping because its period class is nonprimitive. Existing move-only/mixed disc proofs, explicit CLI routing and complete cap-safe fallback remain covered.

The 52.309-second six-diagram audit compares the frozen disc-only epoch policy with the explicit annulus option. Every disabled recognizer status, method and evidence is preserved, and all zero-local-work enabled calls reach the correct complete fallback. The optimized-positive input now returns `native-pachner-annulus` through the actual enabled recognizer; the preceding policy remains inconclusive at its two-million-unit cap. The current query uses 48,281 checkpoints and no descent epochs.

Fresh Regina verifies the terminal surface as a connected orientable Euler-zero annulus with two boundary components. The original source, primitive class and capping argument are supplied by the independent native proof, not by that surface-shape observation alone. The three other positive proofs remain essential normal discs and receive their usual fresh Regina confirmation. All four complete original-source proofs pass producer-disabled replay.

Genus-one-miss still reaches its disc after 24 epochs; seven annulus observations raise its work from 622,768 to 626,709 without changing the result. Trefoil still stops at its bounded-stationary source

after fourteen descents, and its four capping observations raise work from 1,382,485 to 1,384,737. Figure-eight still caps. These misses are not negative knot certificates. The empty-circle and one-crossing controls return initial discs; one cheap Euler observation raises each work count from 2,037 to 2,098.

The nonempty moved-source regression also authenticates a primitive annulus after an actual upward replacement and a gauge record. Removing the move or substituting the old surface source fails replay. That test checks the source-chain scope rather than claiming additional search coverage. The production optimized-positive witness already exists at the initial span optimum, so no move is required for its improved query.

The four earlier pilot policies are retained under `data/endpoint-negative-pilots.json` with exact drivers and raw logs. They cover radius-one/radius-two endpoint windows, repeated boundary shelling and the edge-first objective. None produces another hard-control witness in the selected calls. These finite observations do not exclude the policies in general; they explain why no new runtime default is attributed to them.

Configured-call timings use three randomized paired rounds, 48 measured calls and sixteen warmups, including identical-baseline A/A controls. Both policies share a one-million-unit allowance and a thousand geometric nodes. Every source, span and cochain preparation, optional capping observation, primitive/connectivity arithmetic, actual configured search/cap, complete original-chain positive replay and serialized result is charged. The older disc-only policy may cap; no such call is described as completed family exhaustion merely because it is timed.

**Configured native calls, disc-only versus annulus caps enabled.**

| Input | disc only ms | cap enabled ms | old/new | old A/A | new A/A |
|---|---:|---:|---:|---:|---:|
| empty circle | 8.456 | 8.760 | 0.964 | 1.001 | 0.995 |
| optimized positive | 4574.050 | 90.172 | 50.726 | 0.994 | 0.961 |
| genus one miss | 3263.042 | 3319.601 | 0.983 | 1.004 | 1.012 |

**Complete recognition with the two explicit optional policies.**

| Input | disc only ms | cap enabled ms | old/new | old A/A | new A/A |
|---|---:|---:|---:|---:|---:|
| optimized positive | 4850.132 | 78.063 | 62.726 | 1.022 | 1.067 |

The optimized-positive native query has paired old/new ratio 50.726, median 4.574 versus 0.090 seconds. The old call reaches its local cap and returns inconclusive; every new call returns the independently replayed annulus-cap witness. Its completed-call latency is therefore compared with a configured old cap, not an old exhausted disc-search theorem. Complete optional recognition has ratio 62.726, median 4.850 versus 0.078 seconds. The old policy proceeds through its cap to exact fallback; the new policy returns the primitive annulus certificate. Both complete calls have the correct unknot verdict.

The new recognition A/A control is 1.067 and the old one 1.022. Those small sample fluctuations are far smaller than the demonstrated order-of-magnitude latency difference, but these remain host observations on one actual input. The native A/A observations are about 0.994 and 0.961. Preparation, arithmetic connectivity and primitivity checks and full source replay are included; none is treated as a cached or free operation.

The empty-circle and genus-one-miss native controls have ratios 0.964 and 0.983. Their tiny or percent-scale overhead is not represented as a speedup; the target disc control still succeeds in both policies. No negative-control or general recognition theorem follows. Omitting the optional stage still solves these small inputs much faster in the maintained complete recognizer. The optional geometric stage, epoch policy and annulus option retain their disabled defaults.

Runtime release `4368395ac`, all tests, exact input-bound proof audit, fresh Regina checks and configured-call timings are retained.

The records are `data/transport-annulus-*` and the pilot bundle named above. The driver is `causal_research.annulus`, with `audit` and `benchmark` modes. Publication review checks 652 runtime pins, six exact and capped complete outcomes, all 48 measured calls and all four source-bound disc and annulus proofs with optimizers and source, move and search producers disabled. The added terminal witness is a precise primitive capping theorem; small universal reach, cheap terminal recognition and the general quasi-polynomial unknot bound remain unproved.

# 153 Binary transport and bit bounds for sector transcripts

The compressed hierarchy needs to operate on binary normal coordinates, not on an expanded list of normal discs. A local verifier must also accept its own exact serialization. The retained sector-exhaustion checker had a mismatch: its disc subproofs accepted signed hexadecimal integer fields, but its ray coordinates and Euler values required Python integers. Thus an otherwise valid serialized transcript could fail without any mathematical change. The nested Q-support reader also used the truth value of a field, rather than its integer value: a string representing zero is nonempty.

The repair decodes only the ray-count and Euler fields with the existing strict integer codec. Integers and explicitly signed hexadecimal strings are accepted; booleans, floats, decimal strings and malformed encodings are rejected. Coordinates must still have the independently computed support width. Decoded rays are compared with the independently rebuilt complete ray set, with the same duplicate, Euler and component-proof checks. The already verified nested Q transcript is decoded before its positive support is extracted. Source identifiers, allowed type indices, phase and status remain subject to their existing schema checks. No producer, claimed complete enumeration or claimed count is trusted.

## 153.1 The size of primitive ray fields

The following determinant argument is the usual vertex-coordinate bound; we include it to make the implementation's bit accounting explicit. Lackenby and Yazdi recall the $2^{7t-1}$ bound and attribute it to Hass and Lagarias[22, Section 2.7]. This is not a new knot-theory theorem.

**Proposition 153.1** (Primitive sector ray size)**.** *For a source triangulation with $t$ tetrahedra, a primitive integral extreme ray of its standard matching cone restricted to an admissible support has every coordinate at most $2^{7t-1}$. Its complete coordinate vector therefore has $O(t^2)$ bits, and its Euler characteristic has $O(t + \log(t+1))$ bits. The ray-coordinate and Euler fields of a transcript with $R$ such rays have $O(Rt^2)$ bits.*

*Proof.* Each original source matching row equates one triangle and one quadrilateral contribution on the two sides of a face. Before cancellation it has two positive and two negative unit terms. The positive triangle and quadrilateral indices are distinct, as are the two negative indices. Identifications can cancel opposite terms but cannot combine equal-sign terms. Thus each row has Euclidean norm at most two. Deleting columns preserves this bound.

Let the ray have $s$ positive coordinates, with $s \leq 7t$. The matching matrix restricted to those columns has rank $s-1$: otherwise its nonnegative cone contains a perturbation of the positive ray in another direction, contradicting extremality. Select $s-1$ independent rows. The vector of signed maximal minors spans their one-dimensional kernel. Hadamard's inequality bounds each minor by $2^{s-1}$. Dividing by their common divisor and choosing the nonnegative sign gives the primitive ray and preserves this bound. For $s = 1$ the primitive entry is one.

There are at most $7t\,2^{7t-1}$ normal discs in the resulting surface. Each has at most four sides and corners. Counting cells gives, for example, $|\chi| \le 9 \cdot 7t\,2^{7t-1}$, sufficient for the stated digit bound. The vector has at most $7t$ entries, each with at most $7t$ binary digits; its quadrilateral subvector and Euler field satisfy the same accounting. □

The bound applies to primitive rays, not arbitrary multiples or arbitrary normal surfaces. The dense reference checker confirms a candidate's one-dimensional supported kernel and compares its primitive direction. The implementation never expands its represented normal discs. Signed hexadecimal transport uses a constant number of characters per four bits, so decoding these fields adds time linear in their encoded size to the existing exact arithmetic work. A maliciously changed 16,385-bit count must still fail the geometric ray equality; a large number is not evidence of a new ray.

This is not a polynomial bound on the entire exhaustion certificate. The number $R$ of rays can be large; negative positive-Euler ray entries also carry independently replayed component proofs, whose size is accounted separately. The checker still reconstructs the reference ray set and can perform a large arrangement enumeration. Small binary coordinate fields alone bound neither search coverage nor hierarchy depth, source growth, restarts, or component-proof generation. In particular no general subexponential recognition bound follows from this repair.

### 153.2 Real-source replay evidence

All 1,534 maintained tests pass in 293.776 seconds; the 24 focused tests pass in 2.474 seconds. Real-source regressions cover all three exhaustion statuses, equivalent signed hexadecimal values, mixed representations, nested Q support containing encoded zeros, and producer-disabled replay. Malformed numbers, altered large counts, duplicate rays, incorrect widths, changed Euler values and encoded metadata indices are rejected. Cancellation continues to propagate.

The retained-source audit replays 134 saved exhaustion certificates: twelve generic-Q exclusions, 61 feasible-span proofs and 61 matching-support proofs. These are proof instances, not 134 distinct triangulations. Every literal certificate passes both the frozen preceding checker and the new checker. Every equivalent hexadecimal transcript fails the frozen checker and passes the new one; every signed-zero variant also passes the new checker. All discovery, kernel, ray and disc-count producers are disabled during replay. The existing dense geometric reconstruction and independent negative component proof checks remain active.

For every retained proof the audit rejects an altered 8,193-bit Euler value and an altered 16,385-bit quadrilateral count after ordinary JSON transport. The latter uses the public binary codec, rather than changing Python's decimal conversion limit. These changed values are mutations, not valid high-coordinate geometric witnesses. The audit checks the primitive quadrilateral digit bound on all retained entries. It does not claim that this finite bank tests every large triangulation or measures a new asymptotic speedup.

Runtime release `b361cd6cd` changes only the existing independent sector checker and adds five regression tests. The audit pins 422 current core-runtime/test sources and checks the preceding 652-source inventory: exactly the verifier changed among those existing sources. Discovery producers, recognition policy and disabled-stage behavior are unchanged.

The reproduction driver is `data/sector_integer_audit.py`.

The records are under `data/sector-integer-*`. The publication review checks exact source hashes, the passing test records and all 134 transport outcomes. The current `openai/math` head is still `fd4aeeb2ee`, checked against its remote during this continuation; no new result is inferred from that catalogue. The general quasi-polynomial search obligations remain open.

# 154 Two-row matching consequences before generic enumeration

The sign preprocessor of Section 147 inspects individual RREF rows. A mixed-sign row may become useful after combination with another row. The actual-source pilot finds fifty additional reductions, including two sectors whose raw matching nullity is four. This continuation adds an exact two-row implication search before the expensive generic-Q phase. The independent source-equation checker and certificate schema are unchanged.

## 154.1 Find every one-sided consequence in a two-row span

Let $F$ be the already authenticated zero coordinates, and let $a, b$ be fixed rational matching consequences. For signs $s_a, s_b \in \{-1, 1\}$, consider

$$w = s_a a + \lambda s_b b, \qquad \lambda \geq 0.$$

For each $j \notin F$, the inequality $w_j \geq 0$ either supplies a rational lower or upper bound on $\lambda$, supplies no bound, or makes the interval empty. Intersecting these constraints and $[0, \infty)$ takes $O(k)$ rational operations. A finite nonempty interval is sampled at its midpoint; an unbounded interval at its lower endpoint plus one. A singleton is sampled at its sole point. Only a strictly positive active coefficient produces a step. Every such coordinate is forced zero by the same nonnegative-sum argument as before.

**Proposition 154.1** (Complete pair-span detection, sufficient global closure)**.** *For any fixed rational row family and authenticated zero set, the four signed interval tests detect a nonzero nonnegative active consequence in any two-row span whenever one exists. Repeated pair steps interleaved with single-row sign propagation preserve the full feasible matching cone and terminate after at most $k$ successful steps. For at most $k$ original rows, the closure costs $O(k^4)$ scalar inspections, with polynomial bit cost. It need not find every forced coordinate in the entire row space.*

*Proof.* Any combination with a nonzero first coefficient can be positively rescaled into one of the stated signed forms. A combination with only one nonzero coefficient is handled by sign propagation; row order and signs also cover this case. The active nonnegativity inequalities describe a closed interval. If it has positive length, each active coefficient is a nonnegative affine function throughout that interval. Such a function is zero at an interior point only if it is identically zero. Thus the midpoint or the selected interior point finds a positive coefficient whenever one is possible. The singleton case is checked directly.

Every step uses an actual matching consequence and previously established zeros, so induction authenticates all newly forced coordinates. Each step adds at least one coordinate. A pair sweep has $O(k^2)$ pairs and four $O(k)$ interval tests per pair; at most $k+1$ sweeps occur. The intervening single-row sweeps have the same polynomial overall bound. Rows are never replaced by generated combinations: each interval uses the original fixed rational rows. Bound endpoints are ratios of their coefficients and sampling adds only constant-factor digit growth. This avoids iterative coefficient blow-up. Source provenance has the existing polynomial rational-elimination cost. Reinstating proved-zero coordinates gives the same full cone isomorphism as in Section 147. □

For example, the preceding mixed rows $(1, 0, -1, 2)$ and $(0, 1, 2, -1)$ now force all four coordinates. But the three rows

$$(1, 0, 0, -2, 1, 1), \quad (0, 1, 0, 1, -2, 1), \quad (0, 0, 1, 1, 1, -2)$$

produce no single or pair step. Their sum $(1, 1, 1, 0, 0, 0)$ forces the first three coordinates zero, while $(0, 0, 0, 1, 1, 1)$ remains feasible. Indeed each pair's active pivot columns require its two

coefficients nonnegative; the two corresponding free columns require their ratio to be both at least two and at most one half. This counterexample concerns general row systems, not a claimed geometric realization. It prevents treating the pair fixed point as a complete maximal-support LP theorem.

### 154.2 Keep source authority and complete discovery intact

The cheap phase uses the existing kernel's canonical RREF equations and first performs the preceding single-row closure. When a pair creates new zeros, single-row propagation resumes before another pair sweep. Source provenance is still compiled only when a reduction exists. The actual source RREF must equal the recovered kernel equations, and replaying the same closure must give the same forced set.

Each step records a rational combination of one or two RREF rows. Their source-equation coefficient vectors are combined exactly, coalesced and serialized as the ordinary sorted source-row combination. No new schema or pair claim is entrusted to the checker. It reconstructs each combination from the original matching equations, requires every triangle term to cancel, and checks its nonnegative active coefficients and exact newly forced set. The checker imports neither the pair search nor its interval calculations.

The native automatic standard branch continues to run preprocessing only above matching nullity three. A successful reduction recompiles the original source with its retained allowed types, then uses the existing adaptive low-dimensional discovery or generic fallback. A completed negative certificate retains the original support and restores omitted coordinates as zeros. Removing its support annotation still permits full original-sector replay. Candidate caps and cancellation remain in force; an unfinished query is inconclusive.

If the retained matching nullity is $r$, the same conditional geometry bound now uses the stronger polynomial preprocessor:

$$\mathrm{poly}(T+k+B)+\mathrm{poly}(T+B)(1+k')^{O(r)}.$$

The quartic sign scan, source recompilation and provenance are charged; component queries and their proofs remain separate. No logarithmic bound on $r$ for arbitrary knot inputs or complete small family of sectors has been proved. More polynomial reductions can remove expensive raw-dimensional work on actual sources without supplying the missing global theorem.

### 154.3 Source pilot, independent replay and complete-query costs

All 1,540 maintained tests pass in 297.188 seconds; the 22 focused tests pass in 6.306 seconds. The six new regressions check exact interval geometry, all fifty additional actual-source implications, original and unannotated sector replay, shared candidate caps, interruption, large rational bounds and the three-row incompleteness counterexample.

The deterministic pilot examines 3,412 full allowed-type assignments on 48 source triangulations, using saved complete standard vertex surfaces as the feasible-support oracle. Pair propagation finds every oracle zero in this finite bank; the preceding one-row closure misses zeros in fifty sectors on fifteen of these sources. There are 176 additional coordinate zeros across those proof instances. Only two of the fifty sectors have raw nullity above three and therefore exercise the maintained automatic preprocessor; the other reductions are tested through its source API. These finite observations are not a proof of universal pair completeness.

The 6.077-second source audit authenticates all fifty implication certificates with pair search, source provenance and kernel producers disabled during replay. All 188 earlier source-control supports remain unchanged. Four complete source queries return the same restricted negative status in the frozen preceding and current implementations. Their full original-sector certificates

and their versions without support annotations independently replay. No exhausted sector is converted into a negative knot verdict.

Both new automatic targets are distinct sectors of the same thirteen- tetrahedron finite figure-eight source. The one-row preprocessor removes one Q type but leaves matching nullity four; the old query then pays for generic Q screening and complete Q-support reduction before reaching nullity three. The new rule removes four types before that screen, leaving nine types and matching nullity three. It goes directly to the existing adaptive planar discovery. Total begun ray candidates fall from 63 to seven. Both final complete queries have nullity three: this is a reduction of work before reaching the same feasible span, not a claim that the old final model still has dimension four.

Fresh Regina enumeration is repeated for each target sector on that source. In each case its seven standard rays agree with the complete independent native transcript, and none is an essential disc. These are two sector checks on one triangulation, not two distinct new knot-exterior witnesses. The source correspondence and general sector coverage obligations remain.

Five paired rounds measure 80 completed calls and sixteen warmups, with identical-baseline A/A arms. Every call includes source validation, kernel construction, the full sufficient sign/pair preprocessing, source-equation provenance, source recompilation, actual ray and component discovery, independent full-sector certificate replay and result serialization. All calls finish the requested sector query; none is a capped result.

| Source sector | single ms | pair ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|
| figure eight sector A | 300.458 | 199.309 | 1.550 | 1.002 | 0.992 |
| figure eight sector B | 302.155 | 195.484 | 1.531 | 0.979 | 0.990 |
| existing reduction | 25.076 | 25.019 | 1.000 | 1.001 | 0.999 |
| unreduced control | 61.305 | 65.907 | 0.954 | 1.001 | 1.043 |

The target paired old/new ratios are 1.550 and 1.531. Median old/new times are approximately 300.458/199.309 milliseconds and 302.155/195.484 milliseconds. Target A/A ratios are near one (old 1.002 and 0.979, new 0.992 and 0.990). These are source-query measurements, not timing claims for full unknot recognition.

The existing single-row reduction keeps its five attempted candidates and is at parity. The unreduced nullity-four Fibonacci control keeps its twenty Q candidates and unchanged stronger `NO_POSITIVE_EULER` result. Its paired ratio is 0.954: added polynomial scanning costs about five percent on these observations. Its new A/A ratio is 1.043, so that small difference is not a precise universal overhead estimate. The automatic stage remains limited to the existing high-nullity branch, and the optional diagram geometry policy still defaults off. No broad recognition speedup is claimed.

The core runtime release is `e78d265ff`; `757cb08fa` adds the informative unreduced measurement control without changing the algorithm. The audit and timings pin 654 source/evidence files and the exact pilot and source-bank hashes. Records are under `data/matching-pairs-*` and `data/matching-pair-pilot.*`.

Reproduce source audits and timings with the driver `normal_orbit_research.matching_pairs`. The independent publication review replays all fifty implications and four complete sector certificates with producers disabled and checks all passing test and timing records. The retained conditional geometry bound improves when this polynomial procedure lowers dimension. A general logarithmic retained dimension, complete bounded sector family and quasi-polynomial unknot-recognition algorithm remain unproved.

# 155 Compressed connectivity of cut-open normal surfaces

The hierarchy needs more than a component census of its cutting surface. It must also locate the connected pieces of the ambient manifold after the cut. Expanding a binary normal vector into individual discs loses the desired complexity bound. This continuation constructs a linear-size interval system for the connectivity of the cut-open manifold and authenticates its count with the existing independent orbit-trace checker. It is a native geometric operation toward compressed cutting, not a full implementation of that hierarchy obligation.

The current source contract remains the existing validated finite connected compact orientable triangulation with one torus boundary. The surface is supplied by admissible nonnegative integral normal coordinates. It need not be connected, primitive, two-sided or incompressible. No correspondence with an input knot diagram is inferred. The result counts the components of $M \setminus \operatorname{int} \nu(F)$, where $\nu(F)$ is a normal regular neighborhood of the surface, including the one-sided case.

## 155.1 Local chambers and affine face regions

In a tetrahedron let $t_v$ be its four triangle counts and let $q$ be its single occupied quadrilateral count, zero when no quadrilateral is present. Cutting the tetrahedron along these normal discs gives

$$1 + q + \sum_{v=0}^{3} t_v$$

connected chambers. Index the $q+1$ successive chambers of the quadrilateral stack first, followed by four arms of $t_v$ chambers each, numbered outward from vertex $v$. The triangle arms meet the appropriate endpoint chamber of the quadrilateral stack. With $q = 0$, all four arms meet the central chamber. This is a description of the chambers; adjacent chambers separated by the surface are never identified by a local tree edge.

Quadrilateral type $j$ separates the vertex pairs $A = \{0, j+1\}$ and $B = \{0,1,2,3\} \setminus A$. Take the chain index to increase from the $A$ side to the $B$ side. On a face, each normal arc cuts off one of its three vertices. The face therefore has a central region and three ordered arms of arc regions. At its corner $v$, the arm has length $t_v + q_{f,v}$, where $q_{f,v}$ is the count of quadrilaterals whose face arc cuts off $v$.

The first $t_v$ face-arm regions map in order to the triangle chambers. Only the face's singleton vertex on one side of the quadrilateral partition has further regions. Its $q$ remaining regions map to chain indices $0, \dots, q-1$ on the $A$ side, and $q, \dots, 1$ on the $B$ side. The central face region maps to the opposite chain endpoint. These are at most two affine pieces per face arm, with slopes $+1$ or $-1$.

For a glued face, normal matching equates the two arc-arm lengths. Its vertex permutation matches regions of the same rank from the corresponding vertices, including its central region. Intersect the two arm partitions at their triangle/quad breakpoints. There are at most three nonempty rank intervals per arm. Composing the two affine chamber maps produces an interval isometry between chamber indices. Add the single central-region pair. Self-face gluings and order reversals are included; boundary faces produce no identification.

**Proposition 155.1** (Binary cut-component reduction)**.** *For $T$ source tetrahedra and maximum coordinate bit length $B$, the described system has*

$$N = T + D, \qquad K \le 20T, \qquad \log(N+1) = O(B + \log(T+1)),$$

*where $D$ is the number of represented normal discs. Its interval orbits are exactly the connected components of the cut-open manifold. Construction, orbit counting and independent trace replay*

*have polynomial bit complexity in the source size and coordinate bit length; replay also charges its supplied trace length. No represented normal disc or chamber family is expanded.*

*Proof.* The normal discs form a quadrilateral stack with four compatible nested triangle stacks near the vertices. Every disc separates one local chamber, so the stated chain and arms account for all $1+q+\sum t_v$ local pieces. Each chamber is connected. Cutting a face along its arcs gives precisely the central region and its three corner arms. The rank descriptions above record which chamber meets every such region. Matching and the face permutation give every actual identification across a glued face, without adding an identification across the cutting surface.

A path in the cut-open manifold can be perturbed to cross source faces through their region interiors, avoiding their edges and vertices. Its sequence of local chambers therefore gives an interval-orbit path. Conversely every interval arrow is a genuine shared face region and joins two connected local chambers. Chaining these paths proves the equivalence. A one-sided surface merely causes its two local sides to be identified by source face gluings; no global coorientation is assumed.

Summing the local counts gives $N = T + D$. Each glued face contributes at most ten interval pairings, and there are at most $2T$ glued face pairs. Also $D \le 7T\,2^B$, proving the digit bound. Only a constant number of breakpoints per face are sorted and composed, so interval construction has linear scalar work in $T$, charged at its endpoint bit sizes. The established Agol–Hass–Thurston interval algorithm gives polynomial orbit work in $K$ and $\log(N+1)$[18]; the retained independent local trace checker charges all supplied events. Finite manifold validation and exact matching validation retain their existing polynomial costs. □

This is an elementary chamber-connectivity reduction using the existing orbit machinery; no new literature-priority claim is made. Regina's `cutAlong` supplies an independent small-input geometric oracle; its API explicitly warns that the expanded triangulation can be very large [57]. The native API supplies the component count and its source-bound proof, not that expanded triangulation.

## 155.2 Independent geometry and binary multiplicity

The producer numbers chain and triangle-arm intervals and composes face maps by evaluating their endpoints. The independent checker derives each face's singleton vertex from its quad partition, constructs its affine segments and intersects their domains. It reconstructs the canonical chamber pairings from the actual validated source and coordinates. It imports neither the producer's chamber constructor nor the orbit planner.

The schema `normal-cut-components-v1` binds the normalized source and coordinates, the cut-component count and the complete ordinary orbit certificate. All pairings, the universe size and the final count must match the independent construction and replay. Signed binary integer transport is retained. Invalid schemas, changed sources or coordinates, changed counts and altered traces do not authenticate a result. Cooperative cancellation propagates. A capped orbit query contains neither a component count nor a certificate, and a replay operation limit means unverified rather than any topological conclusion.

Multiplicity must not be divided out as it is for some surface-census queries. For $m \ge 1$ parallel meridian discs in a solid torus the cut has $m$ pieces; $m$ parallel vertex-link discs give $m+1$. For the retained one-tetrahedron Möbius-band vector, its $m$-fold normal vector gives $\lfloor m/2 \rfloor + 1$ pieces. Odd multiplicity includes a one-sided sheet, whereas even multiplicity consists of annular doubles. The chamber model handles these cases directly on binary counts without producing their copies.

### 155.3 Scope still missing for the hierarchy

The new API is `normal_complement_components`.

Independent replay uses `verify_normal_complement_certificate`.

It is not called by the complete recognizer and supplies no knot verdict or new recognition asymptotic bound. A cut can have exponentially many components; a count does not list them or extract a usable compressed manifold representation.

Post-cut cells with full boundary-pattern transport, component descriptions, regluing maps and provenance for lifting later discs are still absent. The existing one-torus validator also does not validate arbitrary later hierarchy pieces with multiple boundary components. Pattern-sensitive surfaces, bounded rebuilds and logarithmic hierarchy depth remain unproved. Thus this implements a concrete polynomial connectivity operation inside the missing cutting requirement, without claiming that requirement complete.

### 155.4 Geometric cross-checks and component-count measurements

All 1,548 maintained tests pass in 288.894 seconds; the eight focused tests pass in 0.284 seconds. They check actual-source chamber pairings, empty cuts, parallel meridians and vertex links, one-sided sheets and their doubles, mixed link/quad surfaces, source and trace mutations, producer-disabled replay, operation/cycle caps and cancellation. A 16,385-bit coordinate certificate passes ordinary binary JSON transport and independent replay.

The initial geometric pilot matches 1,635 standard-vertex or empty surfaces on 48 source triangulations against Regina's cut-open component counts. The integrated 10.894-second audit repeats those actual cuts with the native source-bound proof and independent chamber reconstruction, and adds 384 compatible mixed-coordinate surfaces. Every one of the 2,019 expanded-cut checks agrees. These are surface instances, not 2,019 distinct triangulations. All native proofs replay with chamber and orbit producers disabled.

Thirteen further binary cases include parallel meridians on one-, four- and sixteen-tetrahedron layered solid tori, with multiplicity one, $2^{128}$ or $2^{500}$; a one-tetrahedron $2^{16384}$ meridian case; and large vertex-link and odd/even Möbius-sheet families. Their counts agree with the geometric family formulas above, and their exact certificates independently replay. No expanded Regina cut is attempted for the enormous multiplicities. The sixteen-tetrahedron meridian family takes 68 orbit cycles for all three scales; the one-tetrahedron meridian family takes four. These observations support the implementation checks, while the general polynomial guarantee comes from the proved chamber reduction and interval algorithm.

The audit retains 157 representative original-source proofs, including all thirteen large cases. The publication review independently replays these, checks the complete oracle records and authenticates 658 source/evidence hashes together with the exact pilot and source-bank hashes.

Five paired timing rounds measure 80 completed component-count calls and sixteen warmups, with identical-baseline A/A arms. Each input is a binary multiple of the primitive meridian of the same one-tetrahedron source. The baseline constructs a Regina cut-open triangulation and counts its components, including its source/surface construction and result serialization. The native arm constructs chamber intervals, counts their orbits, independently replays the source-bound certificate and serializes its result. Thus the comparison has the same requested count but different intermediate artifacts: only the baseline constructs the expanded cut triangulation. It is not a comparison of two complete cut-manifold constructors or two knot recognizers.

| copies | cut $T$ | explicit ms | native ms | old/new | old A/A | new A/A |
|---|---|---|---|---|---|---|
| 1 | 31 | 0.151 | 0.414 | 0.353 | 0.914 | 1.023 |
| 16 | 226 | 0.404 | 0.456 | 0.935 | 1.020 | 1.056 |
| 128 | 1682 | 2.422 | 0.499 | 4.826 | 0.999 | 1.119 |
| 1024 | 13330 | 34.197 | 0.468 | 74.860 | 0.997 | 1.088 |

At 1,024 copies the paired explicit/native ratio is 74.860, with median 34.197 versus 0.468 milliseconds. The explicit cut has 13,330 tetrahedra; the compressed model has 3,073 chamber points, three interval pairings and four orbit cycles. At 128 copies the ratio is 4.826. At one and sixteen copies the ratios are 0.353 and 0.935: native Python validation and independent proof replay have a larger small-input constant than the C++ expanded baseline. These regressions are retained rather than presented as improvements.

The new A/A ratios range from 1.023 to 1.119 and the old ones from 0.914 to 1.020. Submillisecond observations fluctuate; the large ratio is not a precise universal speedup. The important asymptotic distinction is the binary count operation versus constructing every represented chamber. No general recognition speedup or changed recognizer default follows.

Runtime release `cbbd59371`, the passing tests, both source audits and the complete count-task measurements are retained.

The driver is `normal_orbit_research.cut_complement`, with audit and benchmark modes. Records are under `data/cut-complement-*` and `data/complement-chamber-pilot.*`. The operator completes the local cut-connectivity calculation under its existing source contract. It does not extract or validate later hierarchy pieces, transport boundary patterns or prove logarithmic depth. The full compressed-cutting obligation and general quasi-polynomial recognition bound remain open.

# 156 Bounded cut cores and wholly prismatic components

A cut can have exponentially many connected pieces even when its input coordinates are short. The chamber count alone does not distinguish which pieces require further geometric work. We now mark a linear number of local chambers that are not standard normal-disc prisms. Every cut component avoiding these marks is an interval bundle. This yields a source-certified count of a bounded conservative core and the remaining wholly prismatic pieces, including twisted bundles.

## 156.1 What the published cutting algorithms already supply

The earlier wording of Section 5 was too broad about the published record. Lackenby–Yazdi's Theorem 8.6[22] recalls a polynomial algorithm for an orientable normal cutting surface: it constructs the handle core outside the parallelity bundle, determines base-surface genus and boundary count, distinguishes product from twisted bundles, and locates their vertical attachments. The running time is polynomial in $T \log w(F)$. The source algorithm is Lackenby's Theorem 9.3[58]. It distinguishes the handle core from the interval-bundle information instead of expanding every handle.

These are published component algorithms, not absent theorems. Their source-relative realization and integration with the accelerated hierarchy's full patterns and restart bounds remain missing in our implementation. Our classification below is smaller than that parallelity-bundle algorithm: it detects whole components made entirely from prisms, not bundles attached inside a core component. It does not produce vertical attachment locations or a core handle structure.

The output contract matters for arbitrary binary multiplicities. A cut along $m$ parallel meridian discs has $m$ connected pieces. Any ordinary explicit tetrahedron list for the entire cut therefore

contains at least $m$ tetrahedra. Thus no bound such as $200T$ can describe that explicit output for all multiplicities on a fixed $T$-tetrahedron source. The printed sphere/disc retriangulation statements in Theorems 8.2–8.3 require a clarified restriction or compressed-repetition contract before they can be applied literally to such inputs. This is a direct output-cardinality observation, not a criticism of applications to bounded nonparallel cutting families. We do not use an unrestricted linear explicit-output assertion in the native algorithm or its complexity proof.

## 156.2 A linear set of exceptional chambers

In the chamber indexing of Section 155, mark the two endpoints of each occupied quadrilateral chain, coalescing them when $q = 0$. Also mark the innermost chamber of each nonempty triangle arm. There are at most six distinct marks per tetrahedron. Every unmarked chamber lies between two successive parallel normal discs of the same type and is a prism $D^2 \times I$. Its faces inherited from source faces are vertical rectangles; its two horizontal faces lie on the cutting surface.

A corner region at a triangle/quad transition meets a quad-chain endpoint, and a central face region also meets a marked endpoint. Thus gluing an unmarked chamber only to other unmarked chambers matches vertical rectangles between consecutive parallel arcs. The face identifications preserve or reverse their interval direction. No global coorientation is assumed.

**Proposition 156.1** (Conservative cores and normal bundle midsections)**.** *For a normal cut in the validated orientable source with $T$ tetrahedra, at most $6T$ components touch the exceptional chambers. Every remaining component is an interval bundle over a connected embedded normal midsection. The marked components may themselves be interval bundles. The distinction is sufficient rather than maximal.*

*Proof.* Each marked component contains a distinct exceptional chamber, proving the component bound. A component disjoint from the marks is assembled entirely from the stated prisms, glued along their vertical rectangles. Their interval fibres match, possibly with a reversal, giving an interval bundle. This is the usual parallelity construction; no claim is made that our selected components constitute the full maximal parallelity bundle.

Choose compatible middle points of the interval fibres in these prisms. In each tetrahedron they form normal triangles or quadrilaterals of the original occupied type. Their face arcs match because the vertical rectangle gluing identifies the same two bounding normal arcs. Hence they form an embedded normal midsection. Projecting the connected bundle to its base proves that this surface is connected. Reversing fibre transitions can give a nonorientable midsection and a twisted interval bundle; it does not obstruct the construction. □

**Corollary 156.2** (Unpatterned component-type bound)**.** *Let $v$ be the number of source quotient vertices and $q$ the number of positive quadrilateral types in the cutting vector. The cut has at most $6T + v + 2q \leq 12T$ distinct abstract unpatterned component homeomorphism types.*

*Proof.* There are at most $6T$ marked components. The normal midsections of the unmarked components are simultaneously disjoint, use only the original $q$ quadrilateral types, and form one admissible normal surface by union. The component-type theorem of Section 100 therefore bounds their distinct full normal vectors by $v + 2q$. Equal vectors give normally isotopic bases. Since the total manifold is orientable, its interval bundle is the product over an orientable base and the orientable twisted bundle over a nonorientable base. Thus the base determines the unpatterned bundle type. Finally $v \leq 4T$ and $q \leq T$. □

This bounds types, not total multiplicity, marked-component cell complexity or the complexity of embeddings and attachment patterns. It does not construct a compressed inventory or prove that arbitrary labelled attachments have only linearly many types.

### 156.3 Count marked and prismatic pieces without expansion

Let $C$ be the original chamber-orbit count. Add singleton pairings joining all exceptional chamber points to point zero, which is itself marked. If $K$ original orbits touch marks, the new cone merges exactly those $K$ orbits and leaves every unmarked orbit separate. Its orbit count is

$$C_{\text{cone}} = C - K + 1.$$

Therefore the core count is $K = C - C_{\text{cone}} + 1$ and the wholly prismatic count is $C_{\text{cone}} - 1$. The cone adds fewer than $6T$ pairings to the same binary universe, so both queries retain polynomial bit cost.

The opt-in API flag is `classify_prisms`. Default false preserves the original scalar count and its version-one proof. Version two binds both ordinary orbit traces and the conservative classification to the same normalized source/vector fingerprint. The independent checker reconstructs both the face-region system and the exceptional points; it imports neither mark producer nor orbit planner. It verifies both traces, the cone identity and the $6T$ bound. Cycle and replay-operation limits are shared across the two queries. An unfinished classification has no top-level count or proof.

The result field `prismatic_components` includes both product and twisted interval bundles. It is not an assertion that all bases are orientable, nor a maximal interval-bundle decomposition. No full cut geometry, pattern transport, component extraction or knot verdict is added.

### 156.4 Finite geometric and binary proof evidence

All 1,555 maintained tests pass in 287.156 seconds; the 15 focused tests pass in 0.598 seconds. The seven new regressions check scaled actual sources, meridian/link/one-sided families, exact frozen disabled behavior, legacy proof replay, shared cycle and operation caps, malformed cone/classification data and cancellation. The final API uses “prismatic” rather than claiming that every interval bundle is a Cartesian product.

The finite geometric pilot starts from the 1,635 earlier primitive or empty cases and adds parallel multiples two, three and five where the primitive disc count is at most 32. There are 5,241 cut-surface instances on 48 source triangulations. An explicit small chamber graph locates its unmarked components; each component’s prism counts give a normal midsection satisfying the original matching equations. All 7,838 such midsections are connected according to Regina, and their interval-bundle Euler/boundary signatures occur with the required multiplicities among the actual Regina cut pieces. These signature checks are finite geometric consistency checks, not an independent complete 3-manifold homeomorphism test. The interval-bundle conclusion comes from the prism gluing proof above.

The integrated 61.696-second audit matches all 5,241 count classifications and preserves the entire scalar output exactly against the frozen preceding implementation when classification is disabled. Every classification proof replays with mark, chamber and orbit producers disabled. All 157 preceding version-one certificates also replay. Seven large meridian cases on one-, four- and sixteen-tetrahedron sources use scales $2^{128}$, $2^{500}$ or $2^{16384}$. Each has one marked core and $m - 1$ prismatic pieces. The one-tetrahedron family takes eight total orbit cycles, including the 16,385-bit case; the sixteen-tetrahedron family takes 136 cycles at both large scales. The audit retains 55 representative version-two source proofs.

Five paired timing rounds measure 80 completed classification calls and sixteen warmups, with A/A controls. Both arms construct the same valid one-tetrahedron meridian source and compute total, marked and wholly prismatic counts. The explicit reference expands the chamber graph, performs union-find and tests the marked roots. It is an expansion baseline, not an optimized Regina or published parallelity-bundle implementation. The native arm runs both compressed orbit queries, independently replays its complete certificate and serializes the result. All preparation and result serialization are included. Neither arm extracts full core geometry or a component inventory.

| copies | expanded points | explicit ms | native ms | ratio | old A/A | new A/A |
|---|---|---|---|---|---|---|
| 16 | 49 | 0.148 | 0.632 | 0.230 | 0.976 | 1.015 |
| 256 | 769 | 0.410 | 0.660 | 0.611 | 0.976 | 0.995 |
| 4096 | 12289 | 4.720 | 0.677 | 7.062 | 1.013 | 0.980 |
| 65536 | 196609 | 83.157 | 0.748 | 112.287 | 1.018 | 0.907 |

At 65,536 meridians the explicit graph has 196,609 points. Its median classification time is 83.157 milliseconds, versus 0.748 milliseconds for the native method; the paired ratio is 112.287. At 4,096 copies the ratio is 7.062. At sixteen and 256 copies the ratios are 0.230 and 0.611: proof production and independent native replay lose on these small cases. The old A/A ratios range from 0.976 to 1.018 and the new ones from 0.907 to 1.015. These are host observations, not precise universal speedups. The asymptotic benefit is avoiding a graph with one node per represented chamber while returning the same conservative classification.

The default scalar operation, the complete recognizer and the optional geometric recognition defaults remain unchanged. No full-recognition speedup or general quasi-polynomial bound follows from these count-task timings. The existence of linearly many abstract unpatterned types does not extract or authenticate their patterned representations.

Runtime release `eced40c56`, the complete test logs, finite geometric pilot, source-proof audit and timing records are retained.

The driver is `normal_orbit_research.cut_products`. Records are under `data/cut-products-*` and `data/cut-product-pilot.*`. Publication review checks 660 source/evidence hashes, the exact pilot inputs, all timing outcomes and the passing test records. It independently replays all 55 saved version-two source proofs with producers disabled. The next step is a source-authenticated compressed inventory and core geometry, with boundary patterns and attachment data retained; those obligations remain open rather than being replaced by a count or a type bound.

# 157 Compressed inventories of normal bundle midsections

The classifier of Section 156 counts wholly prismatic pieces but does not identify their normal bases. We now recover a binary inventory of those midsections with multiplicities. Marked core components remain counted, without a claimed description of their topology or geometry. This is not the inventory of the full parallelity bundle attached inside a core component.

## 157.1 Additive chamber weights identify actual midsections

Use the source-owned chamber system from Section 155. Give each exceptional chamber weight one in a leading marker coordinate. Give each unmarked triangle-prism chamber one in its original triangle coordinate, and each unmarked quad-prism chamber one in its original quad coordinate. The total dimension is $1 + 7T$. The quad interiors are one interval per occupied type, and triangle-arm interiors are one interval per arm. Along with the at most $6T$ singleton marks, there are at most $11T$ additive indicator intervals. None of these weights expands a represented chamber family.

An orbit's leading sum is positive exactly when its component touches a mark. Such rows contribute only to the core count. A zero-leading row records one normal disc per local prism in that component; the remaining $7T$ entries are its connected normal midsection vector. Equal vectors are grouped with a binary orbit multiplicity. The weighted source and its independent orbit replay determine membership; merely summing proposed vectors to a plausible total would not establish an inventory.

**Proposition 157.1** (Exact binary bundle-base inventory)**.** *The weighted chamber histogram determines the complete normal midsection inventory of the wholly prismatic cut components, and the conservative core count. With maximum source-coordinate bit length $B$, the returned inventory has $O(T^2(B+\log(T+1)))$ bits. Construction and discovery retain the polynomial encoded-input bound of the weighted orbit operation; independent verification also charges its supplied trace length.*

*Proof.* Each chamber receives either a marker or exactly one normal-disc indicator. Therefore a zero-leading orbit consists entirely of prisms. By the preceding midsection proposition, its indicators are the normal triangles and quadrilaterals of one connected midsection. Summing them gives its full normal vector. All orbits with that vector contribute to its multiplicity. Positive-leading orbits are exactly the conservative core components.

There are at most $v+2q \le 6T$ distinct midsection vectors by Section 156. Each entry is bounded by the corresponding cutting coordinate: the total number of prism chambers of a given local type is at most its source disc count. Multiplicities are bounded by the $T+D$ chamber universe, and Euler values have polynomially bounded digits. This proves the stated output size. The weight dimension and indicator count are polynomial in $T$, their coefficients are zero or one, and their endpoints use $O(B+\log(T+1))$ bits. The existing weighted interval operation has its polynomial encoded-input cost[18, Theorem 16], with all dimension, run transport, arithmetic and trace work charged. □

The full normal coordinates are retained rather than divided by their gcd. A connected annular midsection can be the normal double of a one-sided Möbius band. Dividing its vector by two changes the base and its horizontal boundary, even when the two associated total interval bundles happen to share an unpatterned homeomorphism type. For $m$ sheets of the retained Möbius vector, an even $m$ produces one Möbius base and $m/2-1$ annular bases, while odd $m$ produces $(m-1)/2$ annular bases. These are grouped normal vectors, not expanded surface lists.

## 157.2 Independent weights and trace reuse

The native API is `normal_prismatic_inventory`. It validates the actual source and coordinates, constructs the chamber intervals and weights, and applies the existing weighted orbit kernel. Its result contains core and prismatic counts, full normal base coordinates, multiplicities and independently computable Euler values. It does not assert a knot verdict.

The checker reconstructs the chamber graph through the independent face segment model. Its own block-by-block weight construction identifies endpoint marks and prism interiors; it imports neither weight nor inventory producer. After exact weighted replay, it reconstructs every proposed base, checks normal matching, source coordinate bounds and the type bound, and compares the entire ordered inventory. Altered weights, base coordinates, multiplicities, Euler values or source identity do not authenticate a result. The schema is `normal-prismatic-inventory-v1`; signed hexadecimal integer transport is retained.

A caller may supply a previously obtained ordinary cut-orbit certificate. The producer then independently checks its exact fresh geometric pairings and uses it for weighted replay, without another orbit search. A discovery cycle cap cannot be combined with this replay path. Reuse is not a source cache or an assertion that two inputs have the same geometry: only a trace that matches the current owned interval system is accepted, and the inventory proof binds the current source and vector separately.

Cycle limits constrain the underlying orbit search, not all weighted arithmetic. Callbacks apply throughout validation, construction, transport and verification. An incomplete query has no inventory or certificate; a replay operation limit means unverified, never a geometric conclusion.

## 157.3 The remaining core problem

The inventory describes whole cut components made entirely of prisms. It omits the parallelity regions that attach to exceptional geometry inside a marked core component. In particular, connected primitive cuts may have only a core component even while containing extensive interior parallelity. The printed parallelity-bundle algorithms address that richer construction; we have not implemented their full core and attachment output here.

Core geometry, boundary patterns, ordered vertical attachments, arbitrary later hierarchy-piece validation and bounded hierarchy depth remain open. The new standalone operator is not called by recognition and changes neither its defaults nor its general complexity bound. It provides a verified compressed inventory prerequisite without substituting it for full cutting.

## 157.4 Source inventories and complete-query measurements

All 1,564 maintained tests pass in 301.283 seconds; the 24 focused tests pass in 0.734 seconds. The nine new regressions check parallel meridians, vertex links, layered sources, odd/even one-sided families, connected normal doubles, source/weight/base/count mutations, shared trace ownership, caps, operation limits and cancellation. Binary JSON and a 16,385-bit multiplicity pass independent replay without expanding components.

The 68.207-second source audit computes inventories on all 5,241 saved cut-surface instances on 48 triangulations. Every core count, full normal base vector, multiplicity and Euler value agrees with explicit finite chamber recovery using the independent geometric pairing construction. The inventories account for 7,838 midsection instances. This is agreement of complete coordinate inventories, not just agreement of total component counts. Representative bases on all 48 sources receive fresh Regina connectivity and Euler checks. The earlier geometric pilot supplies the full finite midsection/cut-signature context; no unique 3-manifold classification is inferred from those signatures alone.

All native proofs replay with weight, inventory, chamber and orbit producers disabled. Six large cut-orbit traces are then reused with orbit search disabled; their inventories agree exactly with fresh queries. Nine large binary cases include one-, four- and sixteen-tetrahedron meridian families at $2^{128}$ or $2^{500}$ scale, the $2^{16384}$ one-tetrahedron case, and large odd/even Möbius stacks. The meridian families have one base vector with multiplicity $m-1$. Even Möbius stacks retain both the one-sided vector and its connected annular double where present; odd stacks retain the annular vector. The audit preserves 57 representative complete source proofs.

For the large meridian families, maximum observed weight-run counts are six, 24 and 96 for one, four and sixteen source tetrahedra respectively; they do not grow between the tested $2^{128}$ and $2^{500}$ multiplicities. These are finite diagnostic observations, not the proof of the polynomial bound. That bound comes from the source weight construction, normal-type accounting and the existing weighted interval operation.

Five paired timing rounds measure 80 completed inventory calls and sixteen warmups, with identical-baseline A/A arms. Both methods return the same full midsection vectors and multiplicities. The explicit reference expands the finite chamber graph, computes its connected components and accumulates normal-disc counts for each unmarked component. The native arm constructs compressed indicators, runs weighted transport, independently replays the complete source proof and serializes its result. All source construction, validation, reconstruction and serialization are charged. This is an explicit graph baseline, not an optimized published parallelity-bundle constructor.

| copies | explicit ms | native ms | ratio | old A/A | new A/A |
|---|---|---|---|---|---|
| 16 | 0.168 | 0.880 | 0.187 | 0.982 | 1.000 |
| 256 | 0.575 | 0.897 | 0.641 | 0.995 | 1.000 |
| 4096 | 8.060 | 0.928 | 8.647 | 1.011 | 0.989 |
| 65536 | 154.142 | 1.033 | 148.591 | 0.991 | 1.073 |

At 65,536 copies the paired explicit/native ratio is 148.591, with median 154.142 versus 1.033 milliseconds. At 4,096 copies the ratio is 8.647. At sixteen and 256 copies the ratios are 0.187 and 0.641: native proof and weighted reconstruction lose on small inputs. Old A/A ratios range from 0.982 to 1.011 and new ones from 0.989 to 1.073. These remain host observations, not universal speedup factors or full-recognition timings.

Runtime release `a10e8eb83`, all tests, exact source inventories, search-free trace reuse checks and complete-query timings are retained.

The reproduction driver is `normal_orbit_research.prismatic_inventory`. Records are under `data/prismatic-inventory-*`. Publication review checks 664 current source/evidence hashes, the exact pilot input, every timing outcome and the passing test records. It independently replays all 57 saved source proofs with producers disabled. The complete recognizer is unchanged. The next geometric task is parallelity inside the marked cores and its attachment representation; that richer problem is not replaced by this whole-component inventory.

# 158 Source-labelled prism regions attached to cut cores

Whole prismatic cut pieces do not describe the parallelity that remains inside a marked core. A primitive connected cut can have no whole prismatic components and still contain many locally parallel discs. This continuation retains their face-connected prism regions and the exact source faces and ranks where they meet exceptional chambers. The output is a geometric incidence representation and a verified partition of interface ports. It is not yet a full parallelity-bundle or core-handle reconstruction.

## 158.1 Remove exceptional points without losing face locations

Start with the owned chamber model of Section 155. Delete its at most $6T$ exceptional chamber points. On the remaining points use the order-preserving compression

$$\pi(x) = x - \#\{e \text{ exceptional} : e < x\}.$$

For each source face region, retain its original source/target tetrahedra, faces, corner and rank interval. Its two chamber maps are affine with slopes $+1$ or $-1$. Split the rank interval at every preimage of an exceptional point and immediately after it. Each resulting part has one of three roles:

1. Both chambers are prisms: retain the compressed interval pairing and one face-band contribution on its source endpoint family.
2. Exactly one chamber is exceptional: retain a singleton interface port with its source face, corner and rank, core-cell identity, side and compressed prism point.
3. Both chambers are exceptional: retain their source-labelled core-face contact. No prism identification is added.

Because no remaining piece crosses a deleted point, compression preserves its interval isometry. The domain/range normalization used by the orbit kernel does not erase source-side information: port positions are computed from the original affine face maps before normalization. Self-face

gluings are included. A self pairing that changes no connectivity still represents a face band and must contribute to the ribbon profile.

**Proposition 158.1** (Bounded source ports and exact face-region partition)**.** *The construction has at most* $6T$ *exceptional cells and* $24T$ *core/prism face ports. Its interval orbits are exactly the components of the prism face-adjacency graph. Every component touching a core-facing port is identified by the set of those port indices. Components touching no such port are exactly the whole prismatic cut pieces of the preceding classifier. All source locations and multiplicities remain binary and the construction has polynomial source-size and endpoint-bit cost.*

*Proof.* Each local chamber meets at most one region in each source face: the central face region, triangle arms and quad-chain segments have disjoint chamber images. An exceptional chamber therefore supplies at most four glued-face incidences. There are at most $6T$ exceptional chambers, giving the port bound. Each port has exactly one prism endpoint and one exceptional endpoint.

Clipping removes precisely the exceptional vertices and their incident arrows from the original face-adjacency graph. All remaining face arrows are retained through their compressed affine intervals, so their orbits are exactly the stated prism face-components. A component with no port has no original adjacency to an exceptional chamber; it is therefore a whole unmarked cut component. Conversely a whole unmarked component has no such port. The source record keeps the exact original geometric face location of every removed core/prism adjacency.

There are only constantly many exceptional points in each incident tetrahedron and only constantly many affine face pieces per face. Splitting at their preimages adds only a constant number of pieces per face. All indices use $O(B + \log(T + 1))$ bits; no chamber or repeated face band is expanded. □

This identifies prism chains attached within a core even when the cut has only one component. The core cells and their face contacts are local incidence data, not a completed triangulation of the core. Source edge and vertex neighborhoods are not supplied by this representation.

## 158.2 Weighted port ownership and 0/1-handle profiles

Give each compressed prism point one cell-count weight. Give each retained prism/prism face pairing one band-count indicator over its source endpoint interval. Finally give every interface port a distinct singleton indicator at its prism point. The dimension is two plus the number of ports. All work, including overlapping indicators, is charged to the existing weighted orbit operation.

An orbit's port entries are zero or one. A nonzero port-weight vector can occur only once, because that particular port belongs to exactly one orbit. Thus these vectors recover the attached-region port partition without enumerating points or performing a separate search for every port. Zero-port rows coalesce repeated unattached regions with binary multiplicities. For each profile the output records prism-cell count $c$, face-band count $b$ and the ribbon quantity

$$\chi_{01} = c - b.$$

Here "ribbon" refers to the 0/1-handle neighborhood of the prism blocks and their face bands, with corner attachments kept separate. Its Euler value follows from its discs and bands. It is not the Euler characteristic of the full parallelity base: edge handles can fill corner cycles, and vertex data and boundary patterns still require construction. We do not identify touching edge strata as a completed bundle, or infer base genus and boundary counts from $\chi_{01}$.

The number of attached profiles is at most the number of ports. An unattached profile's cell and band counts are linear functions of its full normal midsection vector, so their number of distinct values is bounded by the previous $v + 2q$ type bound. This gives a polynomial output with repeated unattached regions encoded by multiplicity.

### 158.3 Independent source clipping and remaining topology

The API is `normal_prism_face_regions`. Its checker independently constructs affine face segments, walks their exceptional-point events, reconstructs core contacts and ports, and replays the exact weighted source. The producer's splitter, summary and orbit planner are not used. Every source location, port group, cell/band count and ribbon value must match the independently owned construction. The schema is `normal-prism-interfaces-v1`; large integer fields use the existing exact hexadecimal transport. Incomplete queries have no structure or proof. Operation limits mean unverified, and cooperative cancellation propagates.

The source still satisfies the existing finite compact orientable one-torus contract. The result supplies attached face regions and source-labelled interfaces inside cores, not their full parallelity topology. Edge/vertex handles, actual base genus and boundary curves, cyclic ordering of vertical attachments, core geometry and general later hierarchy-piece validation remain unresolved. The complete recognizer is not changed and no general quasi-polynomial recognition bound follows.

### 158.4 Exact source incidence and complete-query evidence

All 1,571 maintained tests pass in 329.583 seconds; the 31 focused tests pass in 0.908 seconds. The seven new regressions exercise regions inside primitive cut cores, whole-piece count preservation, empty prism graphs, self-pairing bands, source/rank/port/partition/weight mutations, shared verification limits, caps and cancellation. A 16,385-bit source case passes binary JSON and producer-disabled replay.

The 28.060-second source audit checks 5,289 complete structures: the 5,241 preceding primitive/scaled cases and 48 compatible mixed surfaces. Independent finite graph expansion agrees with every core-cell record, core/core face contact, core/prism face location, attached-region port group, cell count, face-band count and ribbon Euler value. All preceding unattached counts remain unchanged. Every proof replays with interface, summary and orbit producers disabled.

The distinction from whole-piece inventory is visible on primitive meridians. At $T = 2, 4, 8, 16$, the prism graph has respectively two, seventeen, 204 and 10,893 points, with two, six, fourteen and thirty source-labelled core-facing ports. It finds two, three, three and three attached face regions. Each original cut still has one conservative core and no whole prismatic piece. Thus the new operator records incidence inside that core, rather than renaming an earlier whole-piece count.

The audit retains 58 source proofs: 47 source cases with ports, four primitive inside-core examples and seven large meridian cases. The large cases cover one-, four- and sixteen-tetrahedron sources at $2^{128}$ or $2^{500}$ scale, plus a four-tetrahedron $2^{16384}$ case. Source locations, port ownership and repeated unattached profiles all remain binary. No expanded graph is constructed for these enormous cases.

Five paired timing rounds measure 80 completed structure calls and sixteen warmups with A/A controls. Both arms return byte-identical canonical structures: source core cells and contacts, port locations, region groups and the complete ribbon profiles. The reference expands the independent geometric prism graph and its weights. The native arm charges source validation, geometric clipping, weighted transport, complete independent replay and result serialization. Neither arm returns missing edge/vertex handles or full parallelity topology.

| input | explicit ms | native ms | ratio | old A/A | new A/A |
|---|---|---|---|---|---|
| $T = 4$, copies 1 | 0.864 | 3.037 | 0.286 | 0.999 | 0.996 |
| $T = 16$, copies 1 | 88.150 | 57.157 | 1.299 | 1.047 | 1.018 |
| $T = 4$, copies 256 | 37.006 | 9.453 | 3.896 | 1.003 | 1.005 |
| $T = 4$, copies 4096 | 598.350 | 9.252 | 64.801 | 1.018 | 0.998 |

At $T = 4$ and 4,096 copies, the paired expansion/native ratio is 64.801, with median 598.350 versus 9.252 milliseconds. At 256 copies the ratio is 3.896. The small $T = 4$ primitive case loses at ratio 0.286. The $T = 16$ primitive case has paired ratio 1.299, median 88.150 versus 57.157 milliseconds. That is a source-incidence query gain, not a full recognizer measurement. Its old A/A ratio is 1.047; timing variability and the small sample prevent a universal primitive-source speedup claim.

Old A/A ratios range from 0.999 to 1.047 and new ones from 0.996 to 1.018. All outcomes agree; no capped or unfinished query is counted as a completed structure. The important encoded-input improvement is avoiding chamber expansion while retaining exact source interface locations and repetitions. The complete recognition algorithm and its asymptotic bound remain unchanged.

Runtime release `0367cc5c2`, all tests, source incidence audits and complete-query measurements are retained.

The driver is `normal_orbit_research.prism_interfaces`.

Records are under `data/prism-interfaces-*`.

Publication review checks 668 current source and evidence hashes, every exact structure/timing outcome and the test records, and independently replays all 58 saved source proofs with producers disabled. The next required topology is source edge and vertex handles, actual parallelity-base boundary curves, and their ordered attachment to the core. Those are not supplied by the 0/1-handle ribbon profiles.

## 159 Linear static contraction of weighted interval runs

The binary normal-surface operators depend on weighted interval-orbit transport. Its static contraction step removes intervals of points that have become singleton orbits, accumulates their weights in the histogram, and closes the deleted gaps. The preceding implementation formed a set of every run and gap endpoint and sorted that set at every contraction. Both input families are already ordered. The maintained implementation now merges them directly without collecting or sorting their boundaries.

**Proposition 159.1** (Exact linear boundary sweep)**.** *For a canonical covering of $[0, N)$ by $R$ constant-weight runs and $G$ ordered disjoint inclusive static gaps, contraction requires $O(R + G)$ boundary steps and $O(R)$ output records. It emits exactly the old histogram increments, canonical surviving runs and emission counter, sharing surviving weight tuples. No represented point is enumerated.*

*Proof.* Maintain the current run position, the first gap whose right endpoint has not been passed, and the number $d$ of already deleted points. The next piece ends at the run boundary, at the next gap's left boundary if outside it, or just after that gap's right boundary if inside it. A deleted piece of length $\ell$ adds $\ell$ singleton orbits of its unchanged weight and increases $d$ by $\ell$. A surviving piece $[a, b)$ is sent to $[a - d, b - d)$ with the same weight. Each step passes a run boundary or one gap boundary; each gap pointer advances once. These are precisely the pieces induced by the former sorted boundary set. The same adjacent-equal coalescing restores the same canonical result and the same emission count. Removing gaps from one run cannot increase its number of surviving weight changes, so at most $R$ output runs remain. □

The bound is on boundary processing, not unit-cost manipulation of arbitrary integers or weight vectors. Endpoint arithmetic charges their binary lengths; histogram hashing, integer addition and adjacent-vector comparisons charge the vector dimension and coefficient sizes. Thus the former $O((R + G) \log(R + G + 1))$ endpoint-sort work becomes linear while the existing weighted arithmetic and independent replay accounting remain in force. The independent checker retains

its separate implementation. Certificate schemas and histogram semantics are unchanged, and cooperative cancellation is polled at every processed piece.

Literal signed-vector deletion tests include 2,000 seeded finite systems, complete deletion, no deletion, coincident boundaries and separated runs coalescing after contraction. A 16,385-bit endpoint case checks binary processing without expansion; complete weighted-orbit tests use the existing pointwise oracle and independent checker. This is an improvement to a compressed hierarchy prerequisite. It does not supply core topology, ordered attachments or a bound on hierarchy depth, and it does not establish a general quasi-polynomial unknot recognition algorithm.

### 159.1 Controlled contraction and complete-query measurements

The initial maintained-suite checkpoint passes all 1,574 tests, with eight skips; the final port-encoding checkpoint passes 1,579 tests with the same eight skips. The existing pointwise weighted-orbit oracle and separate checker remain in use. The final independent decoder additionally passes all 21 focused prism tests and the complete source-face audit.

The comparison loads the former contraction function directly from commit `5a5c0ed65`; the full revision and source hash are in the saved record. It changes only that function while retaining current discovery, source geometry and verifier. Every arm forces the same original vector port encoding. All complete output digests agree, including weight statistics and proof data. Five shuffled paired rounds follow one warmup round, with three repetitions per measured arm and an identical sorted-function control.

Local cases include fresh histogram allocation and contraction, excluding prepared inputs and digest comparison. Complete query cases include source validation, clipping, orbit discovery, weighted transport, independent certificate replay and JSON serialization. Ratios are medians of the paired sorted/sweep ratios; values above one favour the sweep. The A/A column uses two identical sorted arms. It exposes measurement variability rather than being an additional algorithm.

| Task | Sorted ms | Sweep ms | Ratio | A/A |
|---|---|---|---|---|
| 1,000 runs, many gaps | 2.219 | 1.859 | 1.176 | 1.008 |
| 1,000 runs, no gaps | 0.683 | 0.423 | 1.609 | 0.979 |
| 10,000 runs, many gaps | 23.049 | 18.808 | 1.241 | 1.020 |
| 10,000 runs, no gaps | 6.953 | 4.020 | 1.729 | 1.050 |
| 100,000 runs, many gaps | 260.508 | 186.340 | 1.400 | 0.991 |
| 100,000 runs, no gaps | 76.732 | 39.084 | 1.891 | 0.939 |
| Complete query, $T = 4$ | 2.888 | 2.847 | 1.021 | 0.993 |
| Complete query, $T = 16$ | 54.195 | 54.183 | 1.007 | 1.003 |
| Complete query, $T = 32$ | 326.235 | 325.675 | 1.006 | 0.997 |

Isolated contraction ratios range from 1.176 to 1.891. Complete query ratios range from 1.006 to 1.021, near their identical-control ratios of 0.993 to 1.003. We therefore claim reduced local boundary cost, not a demonstrated complete-query speedup from this change alone. All outcomes and controls are retained. No universal elapsed-time speedup is inferred from the linear boundary bound, and these are not full-recognizer measurements. Reproduce with `benchmark_weighted_contraction.py -output FILE` from `fast/`. The source-pinned samples are retained in `data/weighted-sweep-benchmark.json`.

## 160 Carry-free encoding of source port ownership

Section 158 assigned each core-facing port its own coordinate in every weight vector. With $P \leq 24T$ ports, each vector had dimension $P + 2$, even when it carried only one port indicator. The new

encoding keeps the cell and band counts and replaces all port coordinates by one nonnegative binary integer. Port $i$ contributes $2^i$ at its unique prism endpoint. The weighted operation still uses ordinary exact integer addition, including when several ports meet the same point.

**Proposition 160.1** (Exact additive ownership bitsets)**.** *Suppose each of $P$ labelled ports supplies exactly one singleton unit indicator in the source universe. For every orbit, the sum of their $2^i$ contributions is the bitset of precisely its port indices. Three-coordinate weighted discovery and independent replay therefore recover exactly the original attached-region partition and all cell/band profiles. Every port-bearing histogram row has multiplicity one.*

*Proof.* Each port belongs to exactly one source point and thus exactly one orbit. For a given orbit let $I$ be the set of its port indices. Its last weight is $\sum_{i\in I} 2^i$. No exponent occurs twice in this sum, so its binary digits are exactly the indicators of $I$, with no carries. Uniqueness of binary expansion makes this representation injective on port subsets. Distinct nonempty subsets are disjoint across orbits and cannot be equal, so a port-bearing row cannot describe several orbits. The first two weight coordinates retain exactly their former definitions. Decoding the subsets and sorting the resulting profiles yields the same geometric structure. □

The restriction to unique unit source indicators is essential. This is not an injective encoding of arbitrary integer vectors: two copies of $2^i$ would be indistinguishable from one copy of $2^{i+1}$. Here source clipping independently reconstructs each labelled port once; weighted replay binds the proof to those exact singleton contributions. Decoding alone is never used to authenticate ownership. The final summary also checks the complete port partition, rejects out-of-range bits and requires multiplicity one on port-bearing rows.

All weights are nonnegative. Exact transport conserves their total mass, whose port coordinate is $2^P - 1$. Therefore every transported port weight has at most $P$ bits. The reduction from $P + 2$ Python integer coordinates to three does not mean constant bit cost: shifts, additions, multiplication, hashing and serialization charge these $O(P)$-bit integers. For $O(T)$ indicator intervals their worst-case total bit footprint remains $O(TP) = O(T^2)$, while the number of coefficient objects and coordinate-loop iterations is reduced. This improves representation and practical costs within the existing polynomial encoded-input bound; it does not alter the global recognition exponent or prove a hierarchy depth bound.

The API's `port_encoding` accepts `vector`, `bitset` and `auto`. The default chooses a bitset when $P > 1$, and otherwise retains the vector whose dimension is no larger. Vector proofs retain `normal-prism-interfaces-v1`; bitset proofs use `normal-prism-interfaces-v2`. Both versions independently reconstruct source-face events and weights before weighted replay. The checker's binary-digit decoder is separate from the producer's least-set-bit decoder. It first bounds the mask length by $P$, then converts it once and visits at most $P$ binary digits per profile. This avoids repeatedly right-shifting a large integer through its absent low-order ports. The geometric structure and ribbon-only interpretation are unchanged.

Tests compare both encodings with pointwise source-face geometry, exercise more than one hundred ports with a 501-bit multiplicity, reject altered source masks and ownership summaries, and replay all 58 saved version-one proofs with producers disabled. General core topology, edge/vertex handles, ordered parallelity boundaries and later hierarchy stages remain unresolved.

## 160.1 Exact source structures and complete-query timing

All 5,289 supplied source structures agree with explicit face recovery, including 48 mixed cases and seven large binary cases. Previous whole-piece counts are preserved. The 58 new proofs and all 58 legacy proofs independently replay with producers disabled. The retained audit authenticates 672 source and evidence hashes. The final full suite passes 1,579 tests with eight skips; all 21 focused

tests also pass after the independent decoder improvement. The test and audit logs remain under `data/prism-port-encoding-*`.

Both measured encodings use the same linear contraction kernel. Each timed call includes fresh source validation, clipping, discovery, weighted transport, independent replay and complete result serialization. Input preparation and structure-digest comparison are outside timing. The output structures agree in all arms, while their certificate encodings and sizes can differ. Five shuffled paired rounds follow one warmup round; an identical vector arm supplies the A/A control. The ratio is the median paired vector/auto ratio, with values above one favouring auto. Prior results are released and garbage is collected before each arm's timer. This prevents small cases from inheriting cleanup of a preceding large workload; those cleanup costs are excluded equally in all arms.

| Source | Vector ms | Auto ms | Ratio | A/A |
|---|---|---|---|---|
| $T = 1$, $P = 0$ | 0.590 | 0.551 | 1.041 | 1.083 |
| $T = 4$, $P = 6$ | 3.054 | 3.033 | 1.008 | 0.992 |
| $T = 16$, $P = 30$ | 56.222 | 33.854 | 1.654 | 1.010 |
| $T = 32$, $P = 62$ | 339.533 | 121.254 | 2.791 | 1.003 |
| $T = 64$, $P = 126$ | 2407.057 | 494.555 | 5.034 | 0.985 |
| $T = 16$, $P = 30$, 501-bit copies | 60.213 | 39.573 | 1.527 | 0.981 |

The primitive $T = 16, 32, 64$ cases have paired vector/auto ratios 1.654, 2.791 and 5.034, with corresponding identical-vector controls 1.010, 1.003 and 0.985. At $T = 64$ the weight dimension drops from 128 to three and the complete serialized result shrinks from 417,245 to 325,587 bytes. The 501-bit multiplicity case has ratio 1.527 with control 0.981. The $T = 4$ case stays near parity, at 1.008 with control 0.992. With no ports, the $T = 1$ auto and vector paths are identical; its observed ratio 1.041 and control 1.083 illustrate small-call timing variability.

These are standalone source-geometry queries. Their gains do not establish a full-recognizer improvement or a quasi-polynomial recognition theorem. Reproduce timings from `fast/` with:

```
benchmark_prism_port_encoding.py -output FILE
```

Saved samples include wire sizes, weight dimensions, run counts, source hashes and all timing controls. Run the source audit from `fast/`:

```
python -B -m normal_orbit_research.prism_interfaces audit -output FILE
```

Run the combined publication check from `synthesis/`:

```
python -B data/weighted_prism_review.py -publication
```

Older dated publication checks retain their original source pins and PDF page numbers; they describe their original releases. The combined current check authenticates both saved proof versions on the updated implementation, rather than rewriting those historical records.

## 161 Full cut-boundary cells and source-certified piece profiles

The hierarchy must retain actual geometric attachments after a normal cut. The preceding chamber and prism operators locate face-connected pieces and their interface sets, but they do not reconstruct the boundary of the cut manifold. We now construct its complete two-dimensional cell structure in binary families, with cyclic polygon attachments, edge endpoints, both sides of normal intersection vertices and a carrier into the original cut chambers. The resulting operator

determines all boundary genera and each cut piece's boundary count and total boundary genus without expanding normal sheets.

This is the boundary of $M \setminus \operatorname{int} \nu(F)$ for the supplied normal surface $F$, including its one-sided components. It is not the base of the full interior parallelity bundle. The ambient source contract remains finite, compact, connected, orientable and with one torus boundary. No source coordinate is divided by a common factor: repeated surfaces change the cut manifold, not just the spelling of its input.

## 161.1 Two horizontal sides and the original cut boundary faces

Give each normal triangle and quadrilateral two local polygon copies. For a triangle near vertex $v$, side zero faces $v$ and side one faces away from it. For quad type $j$, side zero faces $A = \{0, j+1\}$ and side one faces its complement. These are local sides, not an assumed global coorientation. Number the two sides in contiguous blocks for each of the seven disc types.

On a source face $f$, the arc stack near corner $v$ has length $a_v = t_v + q_{f,v}$. Its first $t_v$ arcs are triangles, followed by the applicable quad arcs. Triangle rank increases from $v$. Quad rank increases from $A$ and decreases from the other side. Consequently a face arc's rank maps to its horizontal polygon by at most two affine pieces of slope $+1$ or $-1$. Each arc also has an inward and an outward side relative to its face corner; this specifies which local horizontal copy it meets.

For a glued face, match equal arc ranks and equal corner sides through its vertex permutation. Intersect the two rank partitions and retain both local side maps. This includes self-gluings and the monodromy of one-sided surfaces. For a boundary face, retain its central region and three corner arms. Arc rank $r$ has inward region $r$, and outward region $r+1$, except that the last outward region is the centre. Glue these regions to their corresponding horizontal polygons.

Finally glue the cut segments of each original boundary edge between its two incident boundary-face occurrences. If the edge has $w$ normal points, there are $w+1$ gap segments. The face sends its first corner stack to one arm, its middle gap to its centre, and its last stack to the other arm. Global edge orientation comes from the validated signed edge union, including loop edges. This partitions the gap ranks into three affine pieces on each face. These joins are essential: horizontal-side matching alone would leave the original boundary faces disconnected.

**Proposition 161.1** (Exact compressed cut-boundary complex)**.** *The polygons and joins above are exactly the full boundary of the cut-open manifold. Each edge has two polygon incidences and each vertex link is a circle. Polygon attachments have constant-length cyclic recipes with affine rank parameters. The construction has $O(T)$ cell families and interval joins, with endpoint bit length $O(B + \log(T+1))$ for maximum coordinate bit length $B$. No sheet, edge point or polygon family is expanded.*

*Proof.* Inside each source tetrahedron the frontier of a normal regular neighbourhood has the two stated local copies of every normal disc. Across a glued face, each copy of an arc is attached to the copy on the same corner side; the matching equations identify exactly its rank. This reconstructs the entire horizontal frontier, with both local sides glued even when no consistent global side labelling exists.

The remaining boundary comes from the source boundary faces cut along their normal arcs. Their centre and corner-arm regions are discs. An inward arc copy bounds its rank-$r$ region; an outward copy bounds the next region or the centre. The cut original edge gaps glue the two adjacent source faces. These are exactly the pieces and identifications in the boundary of $M \setminus \operatorname{int} \nu(F)$.

At a normal intersection with a source edge there are two distinct cut vertices, indexed by the side of the globally oriented edge. Rank from a face corner maps to its edge point by a translation or reflection, and its corner side determines the cut-vertex side. Original boundary vertices remain

unchanged. The validated manifold edge links and vertex links imply that the complete boundary neighbourhoods are discs. In particular, these are genuine boundary vertices, not a count inferred from prism 0/1-handles.

Horizontal triangle and quad cycles have three and four arcs. A corner-arm region is triangular at rank zero and quadrilateral at positive rank. A central region has between three and six edges. Cyclic recipes retain their source face, corner, local side, oriented rank and the optional inner arc at positive rank. All edge endpoints are source-labelled affine maps; original gap endpoints split only at the first and last gap. Their cyclic ordering therefore describes full polygon attachments, including repeated vertices and self-identifications.

An internal face contributes at most 18 horizontal joins, a boundary face at most 15 horizontal/original joins, and a boundary edge at most seven gap joins. There are at most $2T$ glued face pairs, $4T$ boundary faces and $6T$ boundary edges, giving at most $138T$ edge-family joins. There are at most $30T$ polygon families and $16T$ vertex families. Every cell count is bounded by a constant multiple of $T + D$, with $D \leq 7T2^B$. All processing uses these binary counts and constantly many events per source face or edge. □

The source orientation fixes opposite signed incidences on a shared edge. The producer uses the standard triangle and quad cycles; independent replay instead builds the crossing-edge polygon and orients it with integer midpoint cross products in the unit tetrahedron. This avoids a shared orientation-table assumption. Small literal cell audits check cyclic closure, signed incidences and connected vertex links independently of orbit discovery.

## 161.2 Owned vertices, edges and exact boundary genera

Each polygon receives one face weight. Assign each normal-arc edge copy to its canonical source horizontal polygon, and each original gap segment to its canonical boundary-face occurrence, with weight minus one. Assign the two cut vertices at a normal edge point through one canonical incident tetrahedron's crossing discs. Assign each original boundary vertex through one incident innermost arm or centre. Each cell is owned exactly once.

Use three coordinates: Euler weight, horizontal-polygon count and original-polygon count. The latter two preserve source provenance and distinguish otherwise equal genus profiles. Weighted orbit sums over polygon adjacency give each connected boundary component's $V - E + F$, and group equal profiles with a binary multiplicity. The closed boundary is orientable, so

$$g = 1 - \chi/2.$$

An odd Euler value or a value greater than two is rejected.

For a global conservation check let $D$ be the normal disc count, $A$ the normal arc count over canonical source faces, $W$ the total normal edge weight, and use subscripts $\partial$ for the original boundary. The cells have counts

$$\begin{aligned} V' &= V_\partial + 2W, \\ E' &= E_\partial + W_\partial + 2A, \\ F' &= F_\partial + A_\partial + 2D. \end{aligned}$$

Every boundary normal arc has two ends, and every boundary edge point is incident to two such arcs; hence $A_\partial = W_\partial$. Because the original boundary is a torus, subtraction gives

$$V' - E' + F' = 2(W - A + D) = 2\chi(F).$$

This checks global mass but does not replace the per-component orbit proof. The independent checker computes horizontal Euler coefficients directly from canonical edge owners and face owners, instead of repeating the producer's individual edge contributions.

## 161.3 Boundary ownership inside cut pieces

Every horizontal polygon has a specific cut chamber on its local side. A triangle's inward copy meets its rank-$r$ triangle-arm chamber. Its outward copy meets the next arm chamber, except at the last triangle, where it meets the appropriate quad-chain endpoint or the centre. A quad's two sides meet chain chambers $r$ and $r+1$. The original boundary regions use the face/chamber maps of Section 155.

These carriers are a partition of all boundary polygons into affine maps to cut chambers. There are at most $42T$ carrier intervals. Join the boundary polygon system to the existing chamber system through these maps. The joint orbits are exactly the connected pieces of the cut manifold: boundary joins lie within a genuine cut piece, and every polygon is attached to its actual chamber. No ownership arrow crosses the cutting surface.

To count boundary components inside a cut piece, first obtain a compressed least transversal of the boundary polygon orbits. Its original-source intervals contain exactly one representative per boundary component. Give these representatives one additional marker coordinate. Joint weighted replay then obtains, for every cut piece, the boundary count $b$, boundary Euler sum $\chi_\partial$, its horizontal and original polygon counts and its number of source chambers. In particular

$$\sum_{i=1}^{b} g_i = b - \chi_\partial/2.$$

This is a total genus, not an asserted distribution of individual genera inside a piece with several boundary components. The complete boundary cells and their carriers retain the underlying attaching data.

**Proposition 161.2** (Polynomial binary boundary and ownership query)**.** *Full cut-boundary construction, its genus inventory and cut-piece boundary counts and total genera have polynomial bit complexity in $T$ and $B$. Independent verification additionally charges the supplied cell records and all supplied trace events. Neither operation enumerates a represented boundary component or a parallel surface copy.*

*Proof.* The boundary interval universe has $N_\partial = 2D + F_\partial + A_\partial \le 6D + 4T$ points and at most $138T$ joins. The chamber universe has $T + D$ points and at most $20T$ joins. Adding at most $42T$ carriers gives at most $200T$ joint joins and a universe of at most $7D + 5T$ points. Their endpoint bit length is $O(B + \log(T+1))$.

Boundary discovery uses the existing monotone Agol–Hass–Thurston trace and compressed least-transversal compiler. Weighted boundary replay reuses that authenticated trace. The joint system requires one further discovery. The fixed weight dimensions are three and five. Initial coefficients are bounded owned-cell counts; intermediate and final totals have polynomial binary lengths. The polynomial interval-orbit, weighted-transport and transversal bounds of the maintained operators apply, with their complete arithmetic and trace accounting[18]. Canonical family sorting and independent cell reconstruction retain polynomial source/bit cost. No multiplicity is used as a loop bound. □

## 161.4 Source replay and the topology that remains

The API is `normal_cut_boundary`. Independent checking uses `verify_normal_cut_boundary`. The schema is `normal-cut-boundary-v1`. It binds the actual source and unscaled coordinates, every face/edge/vertex family and oriented cycle, every chamber carrier, the boundary transversal and both weighted proofs. Rank-event reconstruction and midpoint orientation import no boundary producer. Signed hexadecimal transport supports large binary fields.

Discovery caps share one allowance across boundary discovery and the joint query; weighted trace reuse does not start another discovery. Incomplete queries return no geometry, summary or certificate. The verifier's optional operation cap charges all three supplied traces, including the repeated boundary trace, and means unverified on exhaustion. Cooperative cancellation propagates through source construction, compilation and replay.

Boundary spheres do not certify a ball. The retained solid-torus/$RP^3$ connected-sum source, cut along its standard vertex disc number six, has one sphere boundary but first homology $\mathbb{Z}/2$. Regina's independent cut confirms it is not a ball. The new API therefore returns no ball or unknot verdict. Pattern essentiality and irreducibility would need their own applicable certificates.

The constructor supplies actual boundary cells and their source carriers, not the full compressed three-dimensional interior or an admissible handle structure. It does not compute the interior parallelity base, collapse its handles, transport an arbitrary existing pattern, validate general later hierarchy-piece sources, or establish bounded hierarchy depth. Those remain obligations for the original quasi-polynomial recognition goal. Ordinary recognition is not yet changed.

## 161.5 Source geometry and independent cut evidence

The retained audit checks 1,755 finite cases: all 1,635 supplied pilot surfaces, 48 mixed source cases, 36 analytic examples and 36 relabellings. Every global genus inventory and cut-piece boundary count/total genus agrees with Regina's actual three-dimensional cut. Forty-five cases contain a piece with several boundary components. The literal cell audit checks 340,202 signed edge incidences, closed polygon cycles and connected vertex links. The oracle uses `cutAlong` and each actual boundary component's Euler characteristic[57, 3].

Ten large binary cases include parallel meridians, vertex links and odd/even Möbius multiples. A 16,385-bit multiplicity retains twelve polygon families and 33 edge families in the one-tetrahedron meridian source, with 16 boundary discovery cycles and 20 joint cycles. These numbers describe that supplied family; the general bound remains polynomial in encoded source size. All 110 saved source proofs independently replay with discovery, forward transversal compilation, weighted producers and the cell producer disabled. The source/evidence manifest authenticates 677 hashes. All ten focused test methods pass, including mutations of endpoints, orientations, chamber carriers, representative intervals and topology summaries, and shared caps. The full maintained suite passes all 1,589 tests with Regina available and no skips; its complete log is retained in `data/cut-boundary-tests.txt`.

The data are `data/cut-boundary-audit.json`. Reproduce the source audit from `fast/` with:

```
python -B -m normal_orbit_research.cut_boundary -output FILE
```

The oracle is optional and requires the project's Regina extra; the native constructor and independent checker use only the Python standard library. The retained run uses the Regina Python distribution 7.4.1, whose engine version string is 7.4, on Python 3.13.14.

## 161.6 Complete boundary-profile task measurements

The measured task returns the boundary genus inventory and each cut piece's boundary count and total genus. Each native call validates the source, constructs the complete compressed boundary cells, discovers and transports both orbit systems, independently replays its certificate and serializes its full result. The Regina arm converts the same source, constructs its explicit three-dimensional cut and obtains and serializes the common profiles. Its expanded triangulation is not serialized. These are complete measurements of the common census task; the additional geometric outputs and proof obligations differ.

There are five shuffled paired rounds after one warmup, and an identical Regina control. All 105 measured samples and 21 warmup samples complete and return the same common-profile digest. Input preparation, cleanup of prior results, garbage collection and digest comparison occur outside the timers. Ratios are median paired Regina/native ratios, with values above one favouring native. The A/A column compares the two identical Regina arms.

| Source | Regina ms | Native ms | Ratio | A/A |
|---|---|---|---|---|
| $T = 1$, 0 copies | 0.355 | 1.724 | 0.207 | 1.056 |
| $T = 1$, 1 copy | 0.394 | 8.259 | 0.048 | 1.010 |
| $T = 1$, 16 copies | 1.032 | 14.407 | 0.070 | 1.034 |
| $T = 1$, 128 copies | 6.079 | 15.404 | 0.390 | 1.026 |
| $T = 1$, 1,024 copies | 55.468 | 15.216 | 3.610 | 0.957 |
| $T = 1$, 4,096 copies | 237.651 | 15.235 | 15.599 | 0.963 |
| $T = 16$, 1 copy | 125.832 | 304.516 | 0.414 | 1.007 |

At 1,024 and 4,096 meridian copies the ratios are 3.610 and 15.599, with controls 0.957 and 0.963. The native medians are 15.216 and 15.235 ms; the latter represents the boundary cells and independently replayed proof for a cut with 4,096 sphere boundaries and 4,096 pieces. The small inputs show substantial native overhead: ratios at one and sixteen copies are 0.048 and 0.070. The primitive 16-tetrahedron source also loses, at ratio 0.414, with 304.516 ms native versus 125.832 ms Regina. All these regressions are retained. The binary representation avoids expansion in the multiplicity; it does not promise a universal wall-time improvement.

Reproduce from `fast/` with:

```
python -B benchmark_cut_boundary.py -output FILE
```

The records are `data/cut-boundary-benchmark.json`, including source hashes, completed outcomes, wire sizes and all controls. Publication review replays every saved source proof with producers disabled, authenticates both audit and timing evidence and checks the current article:

```
python -B data/cut_boundary_review.py -publication
```

Run that review from `synthesis/`. These standalone cutting operators retain the original general recognition goal and its unresolved interior, parallelity, pattern and hierarchy complexity requirements.

# 162 Full three-dimensional normal cuts in affine cell families

The boundary constructor in Section 161 retains the complete cut boundary and its chamber carriers. We now also retain the interior cells, giving a full raw-cut CW description with oriented attachments in all four dimensions. The representation uses binary affine families and includes the original source charts and unscaled normal coordinates for reconstruction of local three-ball prototypes. Describing the interior therefore does not require expansion of a large normal vector.

The source remains a validated finite connected compact orientable triangulation with one torus boundary. The normal surface may be empty, disconnected, one-sided or a large multiple. The result describes $M' = M \setminus \operatorname{int} \nu(F)$, with its boundary marked as a subcomplex. It is a raw cut, without a parallelity collapse or an admissible bounded handle core.

## 162.1 All cell dimensions and their source charts

The zero-cells are all original source vertices and the two cut vertices at each normal intersection with a source edge. Normal vertices are boundary vertices; original vertices retain their original boundary status. The global edge orientation distinguishes the two cut sides, including loop edges.

The one-cells are both copies of every normal arc on a canonical source face and all $w+1$ gap segments on every original edge of weight $w$. Interior source edges are included. Gap endpoints split only at ranks zero and $w$; an interior gap joins the upper copy of point $r-1$ to the lower copy of point $r$. A normal-arc copy has affine maps to its cut-vertex endpoints.

The two-cells are both local copies of every normal disc and every region of every canonical source face cut along normal arcs. Canonical faces include one member of each glued pair and every original boundary face. Their centres and corner arms retain the earlier cyclic recipes: triangles, quadrilaterals and three-to-six-sided centres. Horizontal polygons and original boundary-face regions are exactly the boundary two-cells.

The three-cells are all local chambers, with the established quad-chain and triangle-arm numbering. Each is a three-ball in its source tetrahedron. Chain ranks distinguish an empty-stack centre, the two quad endpoints and interior quad prisms. Triangle-arm ranks distinguish the tip and triangle prisms. The source row and rank determine the local prototype without listing its copies.

Source charts are essential. Cellular boundary-matrix coefficients alone need not specify a three-cell's topological attaching map. Every chamber and incidence here also retains its source tetrahedron, local face or normal-disc side and ordered rank. The packet contains the source face permutations and normal vector, copied without aliasing mutable inputs. These identify the abstract local boundary flags before source quotients. The result is a reconstructible source cut, not merely a chain complex with matching Euler characteristic.

## 162.2 Oriented edge/polygon and polygon/chamber maps

A normal-arc edge is incident to a source horizontal polygon, a target horizontal polygon when the face is glued, and its adjacent original-face region. The latter has inward rank $r$ or outward rank $r+1$, with the last outward region replaced by the centre. Source and target triangle/quad transitions retain separate incidence intervals.

A source-edge gap has an incidence for each occurrence on a canonical source face. Its first corner arm, centre gap and last corner arm use at most three affine pieces. Interior edge degree can be large. Keeping each occurrence's own partition avoids a product of degree and common-refinement size, preserving linear encoded storage.

Horizontal polygons attach once to their actual chamber with positive outward incidence. Original boundary-face regions also attach once. An interior original-face region has two incidences: positive on the canonical owner and negative on its paired side. Their triangle/quad breakpoints are retained independently. Two occurrences on the same chamber can cancel as cellular coefficients; their source attaching data are still retained.

The source orientation determines polygon cyclic directions. Independent replay uses the separate integer-midpoint polygon orientation model and direct source-rank events. Singleton maps have a canonical slope-one spelling; source side and permutation data remain separately recorded.

**Proposition 162.1** (Faithful raw-cut cells)**.** *The four cell namespaces and attaching families describe exactly the CW structure induced on $M'$ by the source normal cut. The marked cells are its complete boundary subcomplex, and the signed cellular maps satisfy*

$$\partial_1\partial_2 = 0, \qquad \partial_2\partial_3 = 0.$$

*All source self-identifications and local sides of one-sided surfaces remain.*

*Proof.* Compatible triangle stacks and one quad stack cut each source tetrahedron into the stated chain and arms of balls. Original faces cut into central and corner polygons, original edges into ordered gaps, and every normal disc, arc and edge point has two local frontier copies. These exhaust the actual cut cells.

Normal matching and source permutations identify equal corner ranks and their local edge and vertex flags across glued faces. Horizontal copies meet the chamber on their stated local side. Each original-face region meets its source and paired chambers at that same face rank. Thus all actual attaching incidences, and only those incidences, are retained.

Each polygon traces its disc boundary, so oriented endpoints cancel. Each local chamber has a sphere boundary, so its oriented polygon boundaries cancel. These cancellations survive source identifications, even when two occurrences identify on one chamber. Source charts retain the pre-quotient flags and determine the full attaching maps, beyond their possibly cancelled matrix entries. This proves both chain identities.

Horizontal frontier cells and their normal-arc and cut-vertex cells lie on the new boundary. Original vertices, edge gaps and face regions lie on the boundary exactly when they lie on the original boundary. Their marked attachments reproduce Section 161. Only local corner sides were used, so one-sided monodromy is preserved without a global coorientation. □

## 162.3 Family size and exact Euler accounting

Let $V_M, E_M, F_M, T$ denote source cell counts, $D$ the normal disc count, $A$ the normal arc count on canonical faces and $W$ the total global edge weight. The cut has counts

$$V' = V_M + 2W, \quad E' = E_M + W + 2A, \quad F' = F_M + A + 2D, \quad C' = T + D.$$

Hence $\chi(M') = \chi(M) + \chi(F)$, and its marked boundary has Euler characteristic $2\chi(M')$. These exact binary conservation checks accompany the source attaching-map proof; they do not replace it.

**Proposition 162.2** (Linear encoded raw-cut description)**.** *Construction and independent source verification have polynomial bit complexity in the source tetrahedron count $T$ and maximum coordinate bit length $B$. The explicit integer/record accounting, including source charts, uses $O(T(B + \log(T + 1)))$ bits. No represented cell or chamber is enumerated.*

*Proof.* There are at most $16T$ vertex, $30T$ edge, $26T$ polygon and $5T$ chamber families. Each normal-arc family has at most two source horizontal, two target horizontal and two original-region incidences, giving at most $144T$ records. Source-edge occurrences add at most $36T$ records. Gap endpoints use at most $18T$ pieces. Horizontal polygon/chamber maps use at most $14T$ records; canonical and paired original-face maps add at most $28T$ and $14T$, giving at most $56T$ in total. Cycles have bounded length, and source reconstruction uses $O(T)$ face records and $7T$ coordinates.

All indices, counts and offsets have $O(B + \log(T + 1))$ bits because $D \leq 7T2^B$, $A \leq 4D$ and $W \leq 4D$. Each local occurrence has constantly many rank events. Global sorting charges $O(T \log(T + 1))$ record comparisons. Source manifold and normal matching checks retain their polynomial cost. Independent midpoint orientation uses fixed integer unit-tetrahedron coordinates, with no floating approximation.

Separate incidence partitions avoid quadratic refinement at high-degree edges. Exact arithmetic, source reconstruction and comparison are polynomial in encoded size. The physical count $T + D$ is only a namespace endpoint, never a native loop bound. The expanded finite chain oracle is a separate, explicitly capped reference and is not an encoded-cost homology algorithm. □

## 162.4 Source replay and the remaining hierarchy

The APIs are `normal_cut_cell_complex` and `verify_normal_cut_cell_complex`. The schema is `normal-cut-cells-v1`. It binds the fingerprint, reconstruction charts, unscaled coordinates, every cell family, incidence sign, boundary flag, cyclic recipe and chamber map. Invalid source

geometry raises the existing source error; malformed proofs return false. Cooperative cancellation propagates without a partial result. No orbit discovery is started.

Literal finite checks build the three integral differentials and verify both zero compositions, cyclic polygon closure, boundary-subcomplex closure, opposite boundary-edge incidences and connected boundary vertex links. Independent ranks over $\mathbb{F}_2$ and $\mathbb{F}_3$ are compared with actual Regina cuts. The retained sphere-boundary nonball has Betti vectors $(1, 1, 1, 0)$ over $\mathbb{F}_2$ and $(1, 0, 0, 0)$ over $\mathbb{F}_3$, detecting its interior two-torsion. Homology alone still does not certify a ball.

The first supplied cut now has full raw interior data. Physical cell counts can remain exponential in coordinate bits. General binary-cell homology, parallelity removal, a bounded admissible core, arbitrary pattern transport, general later-piece validation and the hierarchy depth bound are still required. Ordinary recognition is unchanged. This supplies concrete compressed cutting data for the original quasi-polynomial objective without claiming that objective complete.

## 162.5 Complete source and interior-homology evidence

All 1,599 maintained tests pass with Regina available and no skips, and all ten focused cut-cell tests pass. The complete audit covers 1,755 finite cuts: 1,635 pilot surfaces, 48 mixed cases, 36 analytic examples and 36 relabellings. Every marked boundary genus inventory, Euler value and full homology vector over $\mathbb{F}_2$ and $\mathbb{F}_3$ agrees with an actual Regina cut. It checks 233,036 integral zero compositions, 638,563 edge/polygon incidences and 260,842 polygon/chamber incidences. The boundary subcomplex also passes its orientation and connected-vertex-link checks.

Ten large binary cases include meridians, vertex links and odd/even one-sided multiples. In the 16,385-bit one-tetrahedron meridian case, seven vertex, 15 edge, 15 polygon and three chamber families retain the entire raw cut; no represented cell is expanded. All 110 saved source proofs replay with the cell constructor, producer orientation, summaries and orbit discovery disabled. The audit's 682-hash source snapshot is retained at revision `56a70f4da`. The sole subsequently changed snapshot file is the unused benchmark driver; its archived Git content is authenticated separately while all operation sources match the current implementation.

The actual full-cut homology oracle accounts for torsion in finite-field homology using the universal coefficient theorem. If $H_1$ has $p$-torsion rank $t_p$, then

$$b_1(\mathbb{F}_p) = \operatorname{rank} H_1 + t_p, \qquad b_2(\mathbb{F}_p) = \operatorname{rank} H_2 + t_p.$$

Every cut component has nonempty boundary, so $b_3 = 0$. This oracle performs explicit integer homology on expanded cuts; it is not the native constructor's runtime. The retained audit took 1,743.062 seconds, with live sampling locating its cost in Regina's homology calculation.

Reproduce the audit from `fast/` with:

```
python -B -m normal_orbit_research.cut_cells -output FILE
```

Its data and full-suite log are `data/cut-cells-audit.json` and `data/cut-cells-tests.txt`. The native constructor and checker use only the Python standard library; the independent oracle uses the project's Regina extra, with Python distribution 7.4.1 and engine version string 7.4.

## 162.6 Full raw-cut construction and serialization measurements

Both measured arms construct a full representation of the same source cut. Native constructs the binary CW packet, independently verifies its source proof and serializes the entire result, including reconstruction charts and coordinates. Regina constructs its explicit three-dimensional triangulation and serializes its reversible labelled `tightEncoding`. Its exact `tightDecoding` roundtrip is checked outside timing. Isomorphism canonicalization is excluded. The two formats

and geometric refinements differ, while the native source proof authenticates its actual attaching maps.

Five shuffled paired rounds follow one warmup round, with an identical Regina control. All 105 measured samples and 21 warmups finish. Euler values agree in the timed cases; geometric validity comes from the separate source, chain and homology audit, not that scalar timing check. Input preparation, cleanup, garbage collection, decoding and comparison are outside timers. The ratio is the median paired Regina/native ratio; values above one favour native. The A/A column compares the identical Regina arms.

| Source | Regina ms | Native ms | Ratio | A/A |
|---|---|---|---|---|
| $T = 1$, 0 copies | 0.227 | 1.339 | 0.171 | 1.091 |
| $T = 1$, 1 copy | 0.252 | 4.274 | 0.059 | 0.964 |
| $T = 1$, 16 copies | 0.513 | 4.272 | 0.120 | 0.985 |
| $T = 1$, 128 copies | 2.665 | 4.493 | 0.534 | 1.046 |
| $T = 1$, 1,024 copies | 24.968 | 4.559 | 5.407 | 0.962 |
| $T = 1$, 4,096 copies | 117.196 | 4.489 | 25.792 | 0.997 |
| $T = 16$, 1 copy | 101.950 | 52.066 | 1.816 | 1.045 |

The 1,024- and 4,096-copy cases give ratios 5.407 and 25.792 with controls 0.962 and 0.997. The latter has native median 4.489 ms against 117.196 ms Regina, with 41,869 serialized native bytes against 421,039 Regina bytes. The primitive 16-tetrahedron case gives ratio 1.816 with control 1.045; its native and Regina medians are 52.066 and 101.950 ms. Native is much slower on tiny inputs: the one-copy ratio is 0.059. All regressions and controls are retained. Neither the timing ratios nor the binary-family bound imply a general recognition speedup.

Run the timing driver from `fast/`:

```
python -B benchmark_cut_cells.py -output FILE
```

The current source-pinned records are `data/cut-cells-benchmark.json`. The complete publication review replays the 110 saved proofs, verifies the archived source snapshot and current operation/benchmark sources, checks every timing round and reviews the article:

```
python -B data/cut_cells_review.py -publication
```

Run that review from `synthesis/`. Parallelity removal, bounded core geometry, pattern transport and the full hierarchy complexity proof remain the next requirements for the original quasi-polynomial goal.

## 163 Weighted least representatives retain component identity

An orbit histogram records how many components have each additive weight. It does not identify where these components occur in the original interval. This distinction matters when a later geometric query must place one mark on each boundary component, while omitting isolated interior vertices. Selecting one point per orbit and separately computing a weight histogram loses the association needed to make that decision.

The new `weighted_orbit_transversal` returns disjoint half-open runs $[a, b)$ of least original representatives, each tagged by the exact weight of its orbit. A run of length $b - a$ represents that many different orbits with the same weight. Signed additive weights and overlapping input intervals are supported. The run endpoints and weights are binary integers; neither large runs nor their orbits are expanded. Adjacent runs with equal tags are coalesced, but labels separated by a deleted interval remain separated.

### 163.1 Forward transport and its invariant

Discovery uses the monotone forward interval-orbit trace already described in the preceding sections. Its weight transport keeps a piecewise constant function on the current interval. Its label transport keeps the ordered intervals of original points that are still live. At every stage, the weight on a live point is the sum of the original weights already folded into it. The live labels are increasing, and folding removes a point in favour of a smaller point in the same original orbit.

Deletion, trimming, periodic merging and transmission change the description of the equivalence relation without changing these labels or their weights. Translation and reflection truncation add the removed points' weights to their surviving images and retain the appropriate prefix of live labels. A static-gap contraction finishes one orbit per point of the gap. Intersecting the gap, weight runs and live-label intervals emits the corresponding original labels with their final orbit weights. An ordered three-way sweep performs this intersection without visiting individual points.

**Proposition 163.1.** *For a complete accepted monotone trace, the emitted labels are exactly the least original representatives, and every emitted tag is the additive weight of the represented orbit.*

*Proof.* The label invariant is the one used by the unweighted least-representative operator. Weight transport preserves the sum on each equivalence class. At a static contraction, a current point has no remaining identification with another live point, so its weight is final. Its original label is least by monotonicity. The complete trace removes every live point, and therefore emits exactly one representative for each original orbit. □

The implementation additionally checks that the tag histogram equals the weighted histogram, that total weight is conserved, and that the emitted cardinality equals the ordinary orbit count. These checks supplement the transport invariant. Histogram equality alone would accept a permutation of tags among representatives and is therefore insufficient.

### 163.2 An independent backward verifier

The certificate contains the existing weighted interval-orbit certificate and the proposed representative runs. Replay first authenticates the source weight function and the ordinary trace. Its separate weighted construction records emissions in the current coordinates of each contraction and accumulates the histogram in the same pass. It then restores the contractions backwards by inserting their removed gaps. An interval crossing an insertion anchor is split before its labels are lifted. Truncations insert no selected points: the removed suffix contains no least representative. This verifier does not call the producer's forward live-label compiler.

Only monotone forward traces are accepted. A trace using a global reversal may preserve the histogram while changing the least-label invariant. Capped discovery returns an inconclusive result without a selector or certificate; external cancellation propagates. JSON transport retains large integers through the existing exact hexadecimal encoding.

### 163.3 Removing the duplicated weighted pass

The first version called the complete weighted-certificate verifier before running the selector reconstruction. Both performed the same forward weight folds. Profiling the four-tetrahedron primitive example identified this redundancy in the independent replay path. The updated implementation factors the common certificate checks into a private checker that accepts one of two fixed internal replay routines. The public APIs retain their existing proof formats and validation requirements.

Ordinary histogram verification uses its existing weighted replay. Tagged verification uses the forward part of the backward-label reconstruction, which now returns both the selected runs and

the histogram already accumulated at its contractions. The common checker still binds the exact initial weight function, verifies the ordinary interval-orbit trace, compares the sorted histogram, checks the orbit cardinality, and checks signed weight conservation. The tagged caller also compares every proposed original representative and its weight with the independently reconstructed runs. No certificate field chooses which checks or replay routine are used.

**Proposition 163.2.** *For complete monotone traces, the single weighted replay authenticates the same histogram and representative data as the former two-pass construction.*

*Proof.* When the former histogram verifier performs weighted transport, both passes start from the same authenticated canonical weights and pairings. At every trace operation they use the same independent translation, reflection and contraction rules. Induction therefore identifies their current weight functions. A contraction's contribution to the histogram is the sum of the lengths of its intersections with each weight run; those same intersections carry the weighted label emissions. The combined pass retains this histogram and exactly those emissions. Backward gap insertion is unchanged, as are all final histogram and representative comparisons. If the former histogram verifier instead uses its one-orbit shortcut, conservation uniquely determines the same weight that the label replay accumulates. □

The ordinary verifier retains the one-orbit conservation shortcut: after checking the ordinary trace, the total weight determines the unique orbit weight. Tagged verification deliberately does not take that shortcut. It must still authenticate the proposed original label, and hence performs its label reconstruction even for one orbit. Tests explicitly reject a correct weight attached to a wrong representative in this case.

Writing $U$ for source and ordinary-trace authentication, $W$ for forward weighted transport, and $L$ for backward label insertion, the dominant tagged replay work for traces with more than one orbit changes from $U + 2W + L$ to $U + W + L$. The second normalization of the source weights is also removed. This improves the cost of an existing encoded operation; it does not change the general hierarchy complexity class.

The complete decomposition audit was rerun after this change. All 1,755 finite proof records and all 106 saved source certificates are identical to the preceding version. The rerun again checks 119,490 vertex links on 32,437 literal base polygons, as well as the seven large binary cases. The compact audit artifact authenticates the preceding full proof bank and records the new source hashes. Regression coverage includes malformed nested histograms, dimensions and traces, a wrong representative in a single orbit, and cancellation after replay has begun.

The full Regina-enabled suite passes all 1,614 tests with no skips. The three added tests exercise the common certificate checks and the single-orbit label obligation directly.

A paired comparison loads the preceding selector and weighted verifier sources directly from Git revision `42ca57831`. Source preparation, bundle construction, full independent proof replay and serialization are all inside the timer. The current producer and geometric verifier are shared; only the selector replay implementation changes. Every serialized result is byte-identical across the old, old-control and new arms. Five shuffled paired rounds follow a warmup.

| $T$ | Copies | Old (ms) | New (ms) | Old/new |
|---|---|---|---|---|
| 1 | 4096 | 11.10 | 9.92 | 1.128 |
| 4 | 1 | 125.75 | 100.41 | 1.249 |
| 4 | 128 | 294.47 | 241.09 | 1.197 |
| 16 | 1 | 5002.13 | 4167.24 | 1.205 |

Median paired speed ratios range from 1.128 to 1.249, corresponding to about 11–20 percent less time for this complete decomposition task. The identical old-control ratios range from 0.930

to 1.011, so the timings also expose the remaining measurement variability. Every individual measured old/new pair favours the new implementation. These constant-factor savings leave the previous small-case expansion advantage and the unproved general hierarchy bound in place.

### 163.4 Cost and validation scope

Tag emission is linear in the ordered partitions traversed at that contraction. The reference replay deliberately uses simpler backward gap insertion, with polynomial work in the trace length, emitted-run count, gap count and integer bit lengths. These are bounds in terms of the encoded trace and realized weighted partitions; they do not assert a new worst-case bound on every weighted partition generated by the scheduler.

The regression suite compares 750 seeded finite signed-pairing systems with an independent pointwise union model and directly checks the least label and weight of every orbit. It includes overlapping signed weights, empty systems, a forged exchange of tags that preserves the entire histogram, cancellation, cap exhaustion, and a $2^{16384}$-orbit example represented by one tagged run. Producer-disabled replay accepts the large example after exact JSON transport.

## 164 Source parallelity in all handle dimensions and a bounded inherited core

The full raw cut of Section 162 retains the source geometry needed to distinguish product regions from the exceptional pieces of a cut. The new `normal_cut_parallelity` uses that structure to describe the parallelity bundle for the supplied normal surface in the triangulation's dual handle structure. It includes product handles of indices zero, one and two, their complete base surfaces, their fibre twisting, and their interfaces with an inherited complementary core. The number of complementary handles is at most $30T$, independently of normal multiplicity, where $T$ is the number of source tetrahedra.

The input is still a validated finite connected compact orientable triangulation with one torus boundary and an admissible integral standard normal vector. Empty, disconnected and one-sided surfaces are supported. The output is an intermediate decomposition of the supplied cut. An admissible patterned hierarchy and its global running-time bound still require further work.

### 164.1 Selecting the product handles

Use the dual handle decomposition relative to the original boundary: tetrahedron chambers give zero-handles, regions of internal faces give one-handles, gaps on interior edges give two-handles, and interior original vertices give three-handles. Boundary faces and boundary edges do not supply one- or two-handles in this decomposition.

In a tetrahedron, a stack of $m$ identical normal discs has $m-1$ product zero-handles between successive discs. Distinct disc types are separated by exceptional chambers. On a canonical internal face, the arm at corner $v$ contains $A = t_v + q_{fv}$ parallel normal arcs. Its ranks $1, \dots, A-1$ give product one-handles; rank zero and the face centre are exceptional. Finally, on an interior source edge of weight $w$, ranks $1, \dots, w-1$ give product two-handles. End gaps are exceptional. No three-handle is selected.

These selections are local and respect the source face permutations. At an intersection of selected handles the product fibres agree, with an explicit bit recording whether the local fibre parameter reverses. Their union is therefore the source parallelity $I$-bundle. The rank-map slope is retained separately from the fibre bit: an interval consisting of a single point cannot encode a fibre reversal by its point map alone.

## 164.2 Complete base polygons, including contacts with the core

Each selected zero-handle contributes a truncated normal polygon: a hexagon for a triangle and an octagon for a quadrilateral. Its sides alternate between contacts across faces and contacts around source edges. A selected one-handle contributes a rectangle, with two tetrahedron ends and two edge sides. A selected two-handle contributes its edge-link polygon, alternating tetrahedron sectors and internal-face sectors. The source edge link records every occurrence, including repeated tetrahedra and self-face pairings.

All zero–one, zero–two and one–two contacts are glued along their actual base arcs. Each contact has an affine rank map, a permutation of its arc endpoints, and a fibre-reversal bit. Unglued arcs either lie on the original boundary or carry a source-labelled port to a complementary zero- or one-handle. A missing selected lower-index handle leaves a free boundary arc; it is not silently filled by a disc.

This last point is essential. The discussion of boundary handles in the retained July manuscript [24], in the proof of its lemma on cutting along a vertical surface, explains why the naive list of selected zero-, one- and two-handles need not itself be a handle decomposition of the base. Higher-index handles can meet the complementary core. The polygon construction here retains those boundary contacts explicitly. It implements the source-cut parallelity geometry relevant to the cutting theorem, while the later interior parallelity bundle of an admissible patterned structure has different input hypotheses. The published algorithmic precedent is Lackenby's full parallelity construction [58].

A concrete case in the audit is the two-tetrahedron source labelled `fibonacci_lst_02`, with coordinate rows $(0, 0, 1, 1, 1, 0, 0)$ and $(1, 1, 1, 1, 0, 0, 0)$. Its parallelity bundle has one component whose base is a disc, supplied by a single selected two-handle. It has no selected zero- or one-handles and has eight core-facing arcs. A prism-only extraction would miss this component entirely. Its attachment subset is the whole base boundary, so $\chi(A) = 0$, although all eight arcs and their cyclic endpoint identifications must still be retained.

## 164.3 The complementary handles and attachment charts

**Proposition 164.1.** *The unselected handles form an inherited complementary handle structure with at most* $6T$ *zero-handles,* $8T$ *one-handles,* $12T$ *two-handles and* $4T$ *three-handles. Its attaching data and its interfaces with the bundle have* $O(T)$ *source records with binary ranks.*

*Proof.* Each tetrahedron has at most four triangle tips and two endpoints of an occupied quadrilateral chain. With no quadrilateral stack the single centre replaces those two endpoints. Hence there are at most six exceptional chambers. Each internal face has its centre and at most three nonempty arm tips, and there are at most $2T$ paired faces. Each interior edge has at most two end gaps; the total number of source edges is at most $6T$. There are at most $4T$ source vertices. Adding gives the stated $30T$ bound.

A complementary face region has only complementary chamber ends. An end gap around an interior edge similarly meets only complementary chamber and face regions. The source vertex links provide the three-handle charts. Thus the lower attachments of complementary handles remain in the core. The total length of the edge-link sector lists is bounded by the number of local tetrahedron-edge flags; the vertex-link triangles are likewise bounded by local vertex flags. Retaining each flag with its source chart therefore uses linear combinatorial storage. □

The packet contains both ends of every core one-handle, the alternating zero/one attaching sectors of every core two-handle, and the source vertex-link triangles of every core three-handle. It also includes a copied source triangulation and the unscaled normal coordinates. This preserves topological attaching information beyond a cellular boundary matrix.

Every core-facing base arc is at a type transition and has singleton rank. A rectangle end has at most one such transition per face corner and side. An edge-cap tetrahedron sector has at most two, and a face sector has at most one. These yield a linear port bound; the implementation conservatively checks $66T$. Adjacent free cap arcs can face different core handles, even when their integer ranks are consecutive. The constructor splits them at the source triangle/quad transitions before assigning owners.

## 164.4 Classifying the bases without expanding polygons

Four interval-orbit queries suffice. First identify polygon corners across glued arc endpoints. Next add the free-arc endpoint pairings: among these new orbits, those containing free arcs are the boundary loops. The weighted selectors of Section 163 distinguish them from isolated interior-vertex orbits and place exactly one representative mark on each actual loop.

For each connected polygon component accumulate the seven coordinates

$$(\chi, b, h_0, h_1, h_2, e_{\mathrm{old}}, m).$$

Here $b$ is the number of base boundary components, $h_i$ counts selected source handles of index $i$, $e_{\mathrm{old}}$ counts free arcs on the original boundary, and $m$ is the bit mask of core ports. A polygon of degree $d$ initially contributes $1 - d$ to $\chi$. Each glued arc contributes one back, because it was counted twice, and one mark at each quotient vertex completes $V - E + F$. Boundary-loop marks contribute to $b$. Unique port $j$ contributes $2^j$ to $m$, so its encoded bit length is $O(T)$.

The fourth query is the fibre double cover of the polygon system. Let $C(w)$ be the base-component histogram and $D(w)$ its doubled histogram. If $O(w)$ and $N(w)$ count product and twisted components, respectively, then

$$C(w) = O(w) + N(w), \qquad D(w) = 2O(w) + N(w/2),$$

where the second term is zero when any coordinate of $w$ is odd. Process signatures in increasing positive $h_0 + h_1 + h_2$. This recovers both types without confusing an annulus signature with twice a Möbius-band signature. Since the ambient manifold is orientable, the total interval bundle is a product exactly when its base is orientable. The genus or crosscap count then follows from $\chi = 2 - 2g - b$ or $\chi = 2 - k - b$.

## 164.5 Checking the gluing to the core

Let $A \subseteq \partial B$ be the union of core-labelled arcs in all base surfaces, and let $K$ be the complementary core. The bundle meets the core in the interval bundle over $A$. The vertex selector also records whether a vertex is incident to a core port, giving

$$\chi(A) = \#\{\text{vertices incident to core ports}\} - \#\{\text{core ports}\}.$$

The checked identities are

$$\chi(M \setminus \operatorname{int} \nu(F)) = \chi(K) + \chi(B) - \chi(A), \qquad \chi(B) - \chi(A) = h_0 - h_1 + h_2.$$

The second identity explains why counting selected handles alone can give the wrong Euler characteristic of the base. Every port belongs to exactly one base component. A profile containing a port must have multiplicity one; only components with no core contacts can be grouped into repeated profiles. The source polygon packet retains their attachment locations even when profiles alone omit the cyclic order of free arcs.

Independent verification reconstructs the source geometry from the raw cut-cell reference model. It derives contact maps by inverting edge-to-face-to-chamber incidences, rather than calling the

producer's direct rank formulas. It independently assembles all four query systems, checks the two tagged selector proofs and the two weighted proofs, and recomputes the surface classification and gluing identities. A shared discovery-cycle cap covers all four queries; exhausted work has no geometric result. The verification operation cap counts the supplied traces, and cancellation propagates through both paths.

### 164.6 Encoded size, experiments and remaining obligations

There are at most $17T$ polygon families, $44T$ gluing families and $68T$ polygon side slots. If $D$ is the total number of normal discs, the polygon and corner universes have sizes at most $9D$ and $32D$, respectively. These universes are represented by binary interval endpoints. Their storage does not scale with their numerical sizes. The source geometry and core have $O(T)$ records; the attached weights and traces are additional encoded data. As with the existing weighted engine, the implementation's replay accounting is polynomial in the realized trace and weighted partitions. The present work does not turn that conditional accounting into a complete hierarchy complexity theorem.

The finite reference expands the polygons, checks each vertex link is a single interval or circle, checks fibre and base orientation independently, and counts connected components, boundary loops, ports and Euler characteristics. The accompanying audit records its source hashes and case-by-case proof hashes. The regression tests also cover one-sided multiples, interior three-handles, relabellings, forged fibre and endpoint maps, forged core owners, shared caps, cancellation, and exact JSON replay of $2^{16384}$ parallel meridians with a fixed four-zero/three-one core.

The completed audit covers 1,755 finite cases: 1,635 pilot surfaces, 48 compatible mixed surfaces, 36 analytic multiples and 36 relabellings. It checks 32,437 literal base polygons and 119,490 vertex links. All native profiles agree with the finite expansion, and all source certificates replay with producer entry points disabled. The artifact retains 106 complete source proofs and hashes every tested finite proof. Seven binary cases include multipliers $2^{128}$ and $2^{500}$ on one, four and sixteen tetrahedra and $2^{16384}$ copies on one tetrahedron. Their core sizes are respectively 7, 43 and 187 handles, independent of scaling; the corresponding polygon family counts are 5, 29 and 125. The audit completed in 209.58 seconds on this machine. These measurements validate the tested cases and the binary representation; they are not a worst-case recognition theorem.

At the initial publication, the full Regina-enabled regression suite passed all 1,611 tests with no skips, including twelve new selector and parallelity tests.

The following baseline uses the original two-pass selector verifier; the single-pass update is measured in Section 163. The complete supplied-surface task was timed in five shuffled paired rounds after warmup. Both arms validate the source, construct bundle and core charts, classify the base surfaces and serialize the result. The native arm also performs independent source and orbit-certificate replay; the literal arm reconstructs geometry from raw-cell incidences and explicitly checks all polygon vertex links. Their representations and proof obligations differ. Input preparation outside these tasks and garbage collection are excluded. The table gives medians, with the last column the median of paired ratios.

| $T$ | Copies | Native (ms) | Literal (ms) | Literal/native |
|---|---|---|---|---|
| 1 | 1 | 1.65 | 1.14 | 0.691 |
| 1 | 16 | 10.42 | 3.35 | 0.324 |
| 1 | 128 | 10.89 | 17.20 | 1.590 |
| 1 | 1024 | 10.88 | 141.53 | 12.564 |
| 1 | 4096 | 10.79 | 614.32 | 57.221 |
| 4 | 1 | 117.19 | 6.99 | 0.060 |
| 16 | 1 | 4702.48 | 943.69 | 0.202 |

Binary compression wins for large multiplicity: the measured ratio at 4,096 copies is 57.22. The one-tetrahedron native times from 16 to 4,096 copies remain near 11 milliseconds. Small examples pay substantial orbit and proof overhead: the four-tetrahedron primitive case is about 16.7 times slower than expansion, and the sixteen-tetrahedron primitive case about 4.95 times slower. Identical literal-control ratios range from 0.983 to 1.020. These are supplied-surface operator measurements; they establish neither an end-to-end recognizer speedup nor the desired general bound.

There is also a normality obligation at the interface to the hierarchy. The accepted standard coordinates satisfy triangulation matching and quadrilateral constraints. The patterned handle-normality definition in [24] imposes further local restrictions involving the characteristic surface and boundary pattern. The manuscript's nonnegative index lemma assumes that stronger normality and an admissible input handle structure. A standard coordinate vector and a bounded source core alone therefore do not authorize applying that lemma. The patterned construction must supply and check these hypotheses explicitly.

The empty cut on the one-tetrahedron layered solid torus is already a concrete obstruction to assuming admissibility. Its independently replayed core has handle counts $(1, 1, 0, 0)$. The characteristic surface has two zero-handles, the ends of the single manifold one-handle, and no one-handles, since there are no manifold two-handles. With empty pattern, each characteristic zero-handle has index $-4 + 2 \cdot 0 + 0 = -4$. Thus this inherited core fails the first admissibility condition. The saved source certificate in `parallelity-admissibility-obstruction.json` makes the distinction checkable even before any large normal vector is introduced.

This operator supplies the all-dimensional source parallelity extraction missing from the earlier prism-only construction. The next requirements are to carry boundary patterns through the core interfaces, construct and verify an admissible uniform-type structure, and implement the hierarchy operations with their reduction and search bounds. The complete recognizer's general exponential fallback remains. Quasi-polynomial unknot recognition has not yet been established by this implementation.

# 165 Certified empty-pattern handle cancellations and cocore discs

The inherited core of Section 164 need not be admissible. We now implement an initial geometric reduction for the empty boundary pattern, together with a new source-bound positive unknot certificate. The reduction removes eligible two/three-handle pairs and one/two-handle pairs using their actual attaching occurrences. An exposed one-handle cocore is accepted as a compressing disc only after an exact boundary-homology check. The operation is polynomial in the source triangulation size and does not enumerate normal surfaces.

This is one initial stage of the intended hierarchy. Its input is the validated finite compact orientable source with one torus boundary and empty pattern. It does not discard a pattern on a later hierarchy state, and its output does not assert full admissibility. The implementation is `reduce_source_handles`; the recognizer can try it with `normal_seed_handles=True`, or the command-line combination `-normal-seed -normal-seed-handles`.

## 165.1 Source attaching occurrences and geometric cancellations

The source dual structure has $T$ zero-handles, one one-handle for each internal face pair, one two-handle for each interior edge, and one three-handle for each interior vertex. Every two-handle carries its cyclic edge-link sectors and its oriented passages through the one-handles. Repeated occurrences are retained. Its contacts with three-handles are the interior endpoints of its source edge, counted with multiplicity. In particular, an edge looping at an interior vertex has two such contacts.

The two local moves are the empty-suture versions of Procedures 6 and 2, respectively, in Section 6.1 of [58]. A two-handle meeting the remaining three-handles in exactly one disc cancels with that three-handle. After all three-handles are gone, a one-handle meeting the remaining two-handles in exactly one disc cancels with the incident two-handle. These are geometric incidence conditions.

For the first phase, a queue tracks source edges with exactly one remaining interior endpoint. Removing a vertex handle exposes further edges to the boundary. Every interior vertex is reachable from a boundary vertex in the source one-skeleton, so this phase removes all three-handles. This is the source-specialized realization of the published removal of three-handles in a manifold with no closed component.

For the second phase, occurrence counts are maintained at every remaining one-handle. When a count is one, remove that handle and its unique incident two-handle, then decrement the counts along the removed attaching cycle. Each source attaching occurrence is removed at most once. No remaining two-handle uses a cancelled one-handle, so the remaining attaching cycles retain their original source charts. Three-handles have already been removed, avoiding a need to update their attaching data in this phase.

An algebraic coefficient is insufficient. Oppositely oriented occurrences can cancel in a boundary matrix while remaining distinct attaching discs. Also, a coefficient of one in a single relator does not authorize deleting another relator's passage through the same one-handle. The checker accepts exactly one occurrence among *all* remaining two-handles; it neither freely reduces attaching words nor supplies implicit handle slides.

The original source and the move sequence reconstruct the reduced handle presentation geometrically. The result also lists its remaining handle identifiers and characteristic zero-handle indices. For empty pattern, an end of a one-handle with $d$ incident two-handle occurrences has index $-4 + 2d$. At the producer's fixed point, no one-handle has $d = 1$. Handles with $d = 0$ still require attention. For the empty pattern, the second admissibility condition is automatic: any simple closed curve on a spherical zero-handle boundary bounds a disc whose intersection with the pattern is empty. Consequently, a replayed result in which every reported index is nonnegative is admissible for this empty-pattern input. This observation does not extend automatically to later states with nonempty patterns.

## 165.2 Exposed cocores and the essential-boundary obligation

A remaining one-handle disjoint from all two-handles has a properly embedded disc cocore: its boundary lies on the manifold boundary. A graph path joining its two ends without using that one-handle certifies that cutting along the cocore leaves the manifold connected. The graph is the zero/one-handle incidence graph; higher handles do not join its connected components. A forest that processes used one-handles before unused ones supplies candidate unused edges outside the forest and connecting paths for them.

Nonseparation alone does not prove that the disc boundary is essential in a general reducible source. A nonseparating sphere can give a properly embedded nonseparating disc with inessential boundary. The source operator therefore checks the boundary cohomology class of each proposed cocore. Its positive claim does not assume irreducibility.

## 165.3 Transporting a cocore class through the cancellations

Give the final unused one-handle value one and all other remaining one-handles value zero. These are the values of the cocore's dual integral cohomology class on the reduced dual graph. Reverse a one/two cancellation. If the removed one-handle is $f$ and the removed two-handle has oriented

attaching cycle $r$, then $f$ occurs exactly once, with sign $\epsilon \in \{1, -1\}$. Its original value is forced by

$$x_f = -\epsilon \sum_{(g,\sigma)\in r,\ g\neq f} \sigma x_g.$$

This reconstructs the class on all original one-handles. Two/three cancellations do not add one-handle values. Every original two-handle equation is checked afterwards, including those removed with three-handles. Thus the packet contains a source cocycle, rather than only a function on the final graph.

The verifier uses a different characterization. It checks the prescribed values on all remaining one-handles and all original two-handle equations. The cancelled one-handle values are then uniquely determined by the triangular sequence of unit equations. It does not call the producer's reverse-substitution routine.

## 165.4 A source boundary period without normal-disc expansion

For each source tetrahedron $t$ and vertex $v$, construct an integer potential $u_{t,v}$. Across a canonical paired face from $t$ to $s$, with permutation $p$ and dual value $x_f$, impose

$$u_{s,p(v)} - u_{t,v} = x_f \qquad (v \text{ on the paired face}).$$

The components of this corner graph are the source vertex stars. Their links are spheres or discs, and the source edge equations give consistency. The constructor detects any conflicting potential assignment, and the verifier checks every face equation directly.

On an oriented source edge from local vertex $a$ to $b$ in tetrahedron $t$, put $y_e = u_{t,b} - u_{t,a}$. Face equations make this well defined on global oriented edges, and its sum around every source face is zero. Choosing reference corners in the vertex stars identifies this primal cocycle, up to the overall orientation sign, with the transported dual cocore class: a path through a vertex star records the potential difference, and the within-tetrahedron segment contributes no dual-face crossing.

Restrict $y$ to the boundary triangulation. A spanning-tree potential test finds a closed boundary edge walk $\gamma$ with

$$\langle y, \gamma \rangle \neq 0,$$

or establishes that this candidate's restriction is exact. The certificate records the dual values, all corner potentials, the oriented boundary walk and its nonzero period. Global edge orientation is taken from the actual source edge representative, not from an arbitrary final incidence.

**Proposition 165.1.** *An accepted cocore certificate with nonzero boundary period gives a properly embedded disc with essential boundary on the source torus.*

*Proof.* The geometric cancellation trace identifies the final handle presentation with the source manifold. The unused one-handle cocore is a proper disc. Its transported dual cohomology class is determined by the unit cancellation equations. The corner-potential equations identify its restriction to the boundary with $y$, up to sign. Evaluating this restriction on a boundary curve is the algebraic intersection with the disc boundary. If that boundary were inessential on the torus, it would be null-homologous and every such intersection would vanish. The certified nonzero period rules this out. □

For a source known independently to be the exterior of the input knot, this essential disc is an unknot witness. The new `diagram-handle-disc-v1` wrapper checks the original planar diagram, reconstructs its canonical exterior independently, and verifies the entire handle and boundary-period certificate. A reducible torus-boundary manifold is not accepted as a knot exterior merely because it has a compressing disc.

### 165.5 Size, verification and integration

The sum of original two-handle attaching-cycle lengths is at most $6T$. There are $O(T)$ cancellations. The queue phase uses linear incidence updates, with source indexing and ordered traversal contributing at most $O(T \log T)$ combinatorial work. For a candidate cocore, reverse substitution can increase the largest absolute value by at most the length of the removed cycle at each step. Since each such cycle is used once and their lengths sum to $O(T)$, all transported integers have $O(T)$ bits. Corner potentials and the boundary period obey the same bit-length bound.

One candidate uses $O(T)$ arithmetic operations on these integers. Testing at most $O(T)$ candidate cocores therefore takes at most $O(T^3)$ bit operations with elementary integer arithmetic. A positive certificate uses $O(T^2)$ bits. These bounds concern the new cancellation/cohomology kernel; the established source validator and independent raw-cell reconstruction are additional polynomial operations. The intentionally simple independent cancellation checker rescans remaining attaching occurrences at each proposed one/two cancellation.

Source replay reconstructs its handles from the independent raw cut-cell model at zero normal coordinates. It does not call the source-handle constructor, scheduler or cocore producer. It verifies exact occurrence multiplicities, the remaining handle sets, the graph path and every cohomology equation. Move-capped source discovery is inconclusive and emits no partial disc certificate. A valid reduction with no disc is not a negative recognition result. Its checked index list determines whether it is already admissible for the empty pattern; otherwise the unused one-handles remain explicit obligations.

The native normal-seed search can run this stage before extracting a cocycle. On a miss it continues its existing search. Construction and positive replay share the existing work allowance, and public certificate dispatch and the command-line interface accept the new proof type. The stage is optional: the existing hard survivor corpus supplies no cocore hits from these two cancellation rules alone. General boundary-pattern transport, the remaining admissibility reductions, and bounded hierarchy search remain obligations.

An unused bridge has a separating cocore. Its boundary is inessential on the single boundary torus: an essential torus curve is nonseparating and would join the two sides through the boundary complement. In an irreducible knot exterior, such a disc cuts off a ball. This identifies the next empty-pattern reduction to implement, together with its source-chart transport. It cannot be applied to an arbitrary reducible source merely from the Euler characteristic of that side. The original diagram provenance, or another independent irreducibility justification, must remain part of the authority for discarding it.

The source audit checks 193 triangulations: the 48 saved sources, three relabellings of each, and an additional trefoil-exterior connected sum with $S^2 \times S^1$. Independent replay accepts every reduction. Regina's compressing-disc oracle agrees with all 176 positive and 17 negative source outcomes. The reducible solid-torus connected sums remain positive for a compressing disc while not being solid tori. The reducible trefoil control remains negative.

A further 57 canonical diagram exteriors include the hard survivor corpus, knotted controls, the empty circle and seeded small random braids. The stage produces 15 diagram-bound disc certificates; all agree with the independent recognition controls, and Regina confirms the positive source exteriors are solid tori. No hard survivor gains a cocore certificate. Thirteen of the 193 source reductions have all characteristic indices nonnegative and hence already satisfy empty-pattern admissibility. None of the 57 canonical diagram reductions does: unused bridge handles remain to be treated.

The full Regina-enabled suite passes all 1,626 tests with no skips. Twelve new tests cover source reconstruction, geometric multiplicities, exact boundary periods, forged proofs, relabellings, cancellation, shared caps, public recognition and command-line dispatch.

The following comparison measures the complete native positive-certificate stage from its planar

diagram: exterior construction, discovery, internal positive replay, a second public certificate replay and serialization. The existing cocycle path and the optional handle-first path must return the same status. Five shuffled paired rounds follow a warmup, with an identical existing-path control. This does not time the full recognizer portfolio.

| Case | Existing (ms) | Handles (ms) | Existing/handles |
|---|---|---|---|
| Two-crossing circle | 10.87 | 35.14 | 0.310 |
| Eight-crossing circle | 44.66 | 253.40 | 0.177 |
| Span-optimized positive | 53.66 | 120.26 | 0.443 |
| Survivor 06 | 85.27 | 110.48 | 0.791 |
| Trefoil | 15.31 | 20.83 | 0.737 |

The handle stage is slower on every measured case. Its three direct cocore hits cost 2.26–5.66 times the existing stage, and the two misses add about 26–36 percent. Identical control ratios range from 0.980 to 1.008. These measurements justify retaining the optional setting. The current independent checker reconstructs the zero-coordinate raw-cell core and rescans remaining attaching cycles at each cancellation; reducing that replay overhead is a separate optimization opportunity. The present gain is a verified geometric reduction and an additional polynomial certificate construction, not a claimed portfolio speedup.

# 166 Ordered bundle interfaces and certified handlebody cuts

The source decomposition of Section 164 now has a native consumer that proves the topological type of a restricted class of *whole normal cuts*. It recovers the cyclic order of core-facing ports, certifies geometric cancellations in the inherited core, and glues the resulting handlebodies along their actual disk interfaces. Its output is a binary histogram of component genera, including isolated bundles with very large multiplicities. This integrates the interface and core portions of report 85 [1]; its separate sector-block algorithms are not part of this implementation.

The source remains a validated finite, connected, compact, orientable triangulation with one torus boundary, together with an admissible integral standard normal vector. The surface may be empty, disconnected or one-sided. The consumer does not infer a ball from its boundary or homology. Successful certificates prove a handlebody census; unresolved cases carry no negative recognition conclusion. A knot conclusion would additionally require the appropriate connected cutting surface and authenticated diagram exterior.

## 166.1 Two source queries recover the marked boundary

Let $P$ be the number of source ports. The source construction bounds $P \leq 66T$ and supplies the complete polygonal interval bundle, including corner contacts and a separate fibre-reversal bit at every gluing. Number the two ends of port $p$ by $2p$ and $2p+1$.

The first query imposes only corner identifications and assigns the distinct weights $2^j$ to endpoint tokens $j$. Its weighted orbit histogram gives the partition of port endpoints into actual base vertices. At a marked boundary vertex there are one or two such tokens. The second query doubles the corner universe by the interval-fibre orientation. It adds the original-boundary edges but omits all port edges, and assigns distinct weights to the lifted endpoint tokens. Every component with a marked endpoint is then a lift of an interval joining two port ends. Each such interval has two lifts; their endpoint pairing determines the fibre transport bit.

**Proposition 166.1** (Source-bound boundary skeleton)**.** *The two weighted queries determine each ported base boundary circle as a signed cyclic port word, together with its intervening gaps, the*

*presence or absence of original boundary in each gap, and its fibre transport. They distinguish the maximal attaching intervals from whole attaching circles. The returned active skeleton has $P$ gaps and $P$ port edges.*

*Proof.* Remove the interiors of all port edges from the completed boundary circles. A circle containing a port becomes a collection of intervals whose ends are the named tokens. An interval can degenerate to a single common vertex when two ports are consecutive. The second query gives the endpoint matching of these intervals. The first query distinguishes a degenerate interval from a positive original-boundary path: precisely the former has both tokens in the same vertex orbit. These conclusions use the source boundary's one-manifold structure; they do not follow from an arbitrary pairing supplied by a caller.

Alternately cross a port and its matched gap. This traverses each ported boundary circle exactly once, up to orientation and starting point. The implementation starts at the least unused port in its positive local chart. Multiply the fibre bits around the circle; orientability of the three-dimensional total bundle forces trivial boundary holonomy. Consecutive ports separated only by zero gaps form one maximal attaching interval. If a circle has no positive gap, its complete vertical boundary is an annulus. Otherwise the vertical bundle over each maximal closed interval is a disk. Unported boundary circles are intentionally omitted here; their counts and base topology are already in the source profiles. □

The distinction is necessary. Fifteen individual ports in the four-tetrahedron layered meridian example form only three attaching disks. Treating each port as an independent disk changes the gluing graph and its genus. Likewise, a whole attaching circle must not be cut into an arbitrary collection of disks. No extra peripheral curves or inherited pattern words inside an original-boundary gap are preserved by this observer.

## 166.2 Sparse weights retain a polynomial interface

The queries use $2P$ and $4P$ singleton atoms. Under the suffix-only weighted transport of Section 163, every atom is either at one current point or in an already emitted orbit. Thus at most $K$ points carry atoms when the query started with $K$ atoms. Its canonical piecewise constant weight function has at most $2K + 1$ runs, independently of the number of truncations. Repeated atom locations simply combine their weights. With distinct powers of two, every nonzero weight is a disjoint subset mask of at most $K$ bits and each nonzero output mask has multiplicity one.

Consequently the two interface queries have at most $4P + 1$ and $8P + 1$ weight runs. The ordinary interval-orbit trace, binary source ranks and these bounded masks give polynomial encoded work in $T$ and the input coordinate bit lengths. This is a bound for the supplied-cut interface, not a bound on the number of surfaces a complete hierarchy must search. The independent checker reconstructs the polygon geometry from raw cut cells, builds both queries separately, replays their weighted certificates, and traverses the resulting endpoint permutation. It imports none of the interface producer's decoder or query routines.

## 166.3 Geometric singleton pivots in the core

The inherited core has at most $30T$ handles. For this consumer it must have no three-handles. Contract a spanning forest of its zero/one-handle graph; the remaining one-handles have a complete cocore disk system $D_h$. For each two-handle attaching curve $\alpha_i$, retain its actual unsigned intersection count

$$n_{ih} = |\alpha_i \cap \partial D_h|.$$

The alternating source sectors determine these counts, including repeated occurrences. Forest contraction deletes only the tree-handle entries. Signed exponent sums or freely reduced words cannot replace this matrix.

**Lemma 166.2** (Geometric positive pivot)**.** *If $n_{rg} = 1$, attaching the two-handle $r$ reduces the handlebody genus by one and admits a new complete disk system on the other columns with*

$$n'_{ih} = n_{ih} + n_{ig}n_{rh} \qquad (i \neq r,\ h \neq g).$$

*Proof.* Keep $D_g$ fixed. Along $\alpha_r$, choose an intersection with another $\partial D_h$ adjacent to the unique intersection with $\partial D_g$. Slide $D_h$ over a parallel copy of $D_g$ along a thin band following the intervening arc. Its interior misses all disk boundaries and all other attaching curves. The smoothing removes that intersection with $\alpha_r$; for every other curve $\alpha_i$ the replacement adds exactly $n_{ig}$ intersections. The disks remain an embedded complete system. Repeat until $\alpha_r$ misses every disk except $D_g$. Disk $D_h$ has been slid exactly $n_{rh}$ times. The selected one/two-handle pair now cancels by the geometric one-point cancellation theorem [2, Lemma 10.3]. The surviving attaching curves have precisely the displayed incidences. □

This gives a topological existence certificate even when the expanded disk slides would be very long. It does not yet encode the transported position of every later boundary-pattern strand. This distinction separates it from the no-slide source reductions in Section 165.

Only the values zero, one and at least two are needed for subsequent pivots. The map $n \mapsto \min(2, n)$ preserves addition and multiplication into the saturated nonnegative semiring. Hence the independent checker updates

$$c'_{ih} = \min(2, c_{ih} + c_{ig}c_{rh})$$

on constant-sized entries without losing the singleton predicate. A complete schedule has $O(T)$ pivots, $O(T \log(T+1))$ bits of source labels, and at most $O(T^3)$ matrix operations. A connected core with $v$ zero-handles, $e$ one-handles and $f$ two-handles then has genus $e - v + 1 - f$.

## 166.4 A complete source test within the square raw model

For a square remaining incidence matrix, a successful positive-pivot sequence exists exactly when source row-leaf deletion removes every row and column. A leaf is a row with exactly one live nonzero, equal to one. The justification is the permanent-one criterion proved in report 85 and the earlier immutable source closure of report 56. Reversing a successful positive pivot preserves uniqueness of the edge-labelled perfect matching. A nonnegative square matrix with permanent one has a unique matching with unit matched entries; orient every other row incidence toward its column. A directed cycle would produce a second matching, so there is a sink row leaf. Induction removes all rows. Conversely, a row leaf has no fill term in any surviving column, so every source leaf deletion is a valid geometric pivot.

The permanent is a proof device and is never computed. The producer maintains source incidence lists and a heap of eligible row labels, costing $O(I + T \log(T+1))$ structural operations for $I$ source incidences. The checker uses deletion times and scans immutable rows. In the rectangular case the producer first tries this leaf order, then a deterministic saturated pivot schedule. Failure is inconclusive. Even square completeness is only for this raw cancellation model on the selected forest, not recognition of all balls.

## 166.5 Assembling the entire cut

An orientable interval bundle over a connected bordered surface $B$ is a handlebody of genus $1 - \chi(B)$. Indeed a system of $1 - \chi(B)$ proper arcs cuts $B$ into a disk; their inverse images cut

the total bundle into a ball. This includes the orientable bundle over a nonorientable base. Closed bases are excluded by the current consumer.

The full source product-intersection theorem identifies the core/bundle intersection with the vertical bundle over the closed core-labelled base arcs, including endpoint contacts. After adjacent ports are amalgamated, distinct proper runs have separated closures and compatible corner collars. Their vertical bundles are disjoint embedded disks on both sides. This geometric input is indispensable; a cyclic word alone would not prove it.

**Proposition 166.3** (Exact binary handlebody census)**.** *Suppose every core component has a verified cancellation certificate, all bundle bases have nonempty boundary, and every attaching run is proper. Each component of the cut is a handlebody. For a connected component $C$ of the disk-attachment multigraph, its genus is*

$$G_C = \sum_{v \in V(C)} g_v + |E(C)| - |V(C)| + 1.$$

*Unported bundle profiles contribute their binary multiplicities of isolated handlebodies. These contributions determine the complete genus histogram.*

*Proof.* Vertices represent the certified core handlebodies and individual ported bundle components; each maximal disk interface gives one edge. Choose meridian systems in all pieces disjoint from the marked boundary disks. This is possible by a boundary isotopy of the finitely many disks away from a chosen meridian system, extended through a collar. Cut along these meridians and the gluing disks outside a spanning tree. The remainder is a tree of balls joined along disks, hence a ball by successively absorbing leaf balls. The number of cut disks is exactly the displayed genus. Ported profiles have multiplicity one by their distinct source ownership. Every unported component is separate from the core and must be counted with its full multiplicity. No expansion of that multiplicity is needed. □

The producer uses union-find for this graph; the checker reconstructs the components by graph search. Both verify the redundant Euler identity $\chi(K) + \chi(\mathcal{B}) - \chi(K \cap \mathcal{B})$. That identity is a consistency check, not the ball proof. The complete cut is one ball exactly when the histogram is $\{(0, 1)\}$. A union of many balls is distinguished from one ball even when its multiplicity has thousands of bits.

## 166.6 Implementation contracts and the next geometric extension

Three APIs each have an independently implemented verifier:

- `normal_parallelity_boundary_words` recovers the ordered interface;
- `core_handlebody_certificate` certifies the inherited core;
- `normal_cut_handlebodies` assembles the complete cut census.

The low-level core API assumes authenticated geometric charts; the whole-cut wrapper supplies them. The six orbit discoveries share the cycle allowance, and replay counts all six supplied traces against its operation cap. Callback exceptions propagate. Integer transport uses the existing exact hexadecimal codec; the native integration also decodes core component quantities before graph arithmetic, including signed and uppercase hexadecimal forms, while rejecting booleans.

A full attaching circle on a *disk* base can be absorbed as a two-handle, as proved in report 85. Its actual face-sector ports supply an unsigned incidence row and source flags distinguish the ends even for self-pairings. The initial checkpoint below did not implement that extension; Section 166.9 records the subsequent certified implementation. General annular gluing, three-handle removal in cut cores, explicit pattern transport, useful-surface coverage and a global bound on hierarchy descriptions remain separate tasks. The optional initial cocore stage and the full exponential fallback are unchanged; this API does not dispatch a new knot-recognition verdict.

## 166.7 Initial disk-interface audit and evidence boundaries

This subsection preserves the first checkpoint before annulus absorption. Its source pins and timings describe that earlier implementation; the subsequent extension is recorded in Section 166.9.

The integrated audit replays the 106 source cuts supplied with report 85. It produces 30 complete certificates, all accepted by the independent source and genus checkers. Regina constructs the actual cut triangulation and recognizes all its connected components for 99 cases; all 23 completed native cases among those agree exactly in genus and multiplicity. Seven large-coordinate cuts are excluded from literal expansion by the explicit 4,096-normal-disk guard, and their native certificates remain independently replayed. No oracle timeout occurs.

The 76 unresolved native attempts consist of 60 full-circle interfaces, six stalled raw cancellations, and ten cores containing three-handles. These are refusal categories, not assertions that those cuts fail to be handlebodies. In particular, the disk-base annulus absorption theorem identifies further geometric work rather than grounds to silently accept the current refusals.

The 22 focused tests include 1,200 saturated-versus-exact incidence schedules, 750 square matrices checked against exact permanents, literal boundary walks, source relabellings, one-sided bases, forged proofs, shared resource caps, and multiplicity $2^{16384}$. They also disable producers during replay and exercise exact hexadecimal transport before the graph calculation. Fresh audit data and source pins are in `data/marked-cut-audit.json`. These checks concern supplied-cut topology, not completeness of surface discovery or a global knot-recognition bound.

The full Regina-enabled regression suite passes all 1,648 tests with no skips in 501.552 seconds on this host. Its existing expanded-sheet and marked-cover audits also complete. The runtime source pins remain identical to the independent 106-case cut audit. The replay-specific hexadecimal regression accepts valid signed/uppercase encodings and rejects malformed strings and booleans.

A separate algebraic feasibility probe examines the 60 annular refusals. Exactly 38 have only disk-base circle attachments. Appending the proposed unsigned two-handle rows permits complete saturated cancellation in 33 of those 38 cases. This is a prioritization diagnostic, not 33 additional topological certificates: the source-chart absorption constructor and its independent checker are still absent. The probe and its source pins are retained in `data/marked-cut-annulus-probe.py` and the corresponding JSON. Its command is run from the repository root; it never supplies a knot verdict or changes the accepted census.

## 166.8 Complete-cut timings and binary capacity

After the full regression run finished, ten finite cuts were measured in five shuffled paired rounds after warmup. Both arms start with the same face pairings and normal coordinates. The native arm constructs its entire certificate and independently verifies it; the explicit arm builds Regina's cut triangulation and recognizes every connected component. Input parsing and output serialization are outside these timing intervals. Separate identical-explicit controls have casewise median ratios between 0.982 and 1.089. The raw rounds and source pins are retained in `data/marked-cut-benchmark.json`.

Three of the ten finite cases favor the native implementation; seven favor the explicit reference. The 4,096-copy one-tetrahedron example improves by 55.702 in median paired time, while the twelve-tetrahedron single meridian improves by 2.707. The small four-tetrahedron single-meridian example has a ratio of only 0.032: proof construction and replay dominate. These measurements justify neither a universal speedup nor a new default recognition stage. They demonstrate the intended benefit of avoiding literal cut expansion while preserving the complete component query and independent evidence.

The separate capacity runs use multiplicity $2^{16384}$. Production plus replay takes 0.024 seconds for one tetrahedron, 2.406 seconds for four, and 12.652 seconds for eight. Their serialized certificates

| $T$ | Copies | Disks | Native (s) | Explicit (s) | Ratio | A/A |
|---|---|---|---|---|---|---|
| 1 | 1 | 3 | 0.0049 | 0.0010 | 0.194 | 1.089 |
| 1 | 16 | 48 | 0.0106 | 0.0026 | 0.237 | 0.982 |
| 1 | 256 | 768 | 0.0114 | 0.0363 | 3.203 | 0.984 |
| 1 | 4096 | 12288 | 0.0113 | 0.6336 | 55.702 | 1.034 |
| 4 | 1 | 29 | 0.2533 | 0.0081 | 0.032 | 1.025 |
| 4 | 8 | 232 | 0.6152 | 0.0348 | 0.056 | 1.007 |
| 4 | 64 | 1856 | 0.6206 | 0.2602 | 0.417 | 1.013 |
| 8 | 1 | 228 | 2.1979 | 0.2554 | 0.118 | 1.008 |
| 8 | 8 | 1824 | 3.1758 | 2.0417 | 0.658 | 1.007 |
| 12 | 1 | 1592 | 7.0811 | 18.0060 | 2.707 | 1.002 |

Table 2: Whole supplied-cut classification. Native includes proof production and independent replay; explicit constructs the Regina cut and recognizes its components. Ratio is the median paired explicit/native time; A/A compares identical explicit runs.

contain 1,701,021, 30,078,776 and 71,291,795 bytes respectively. At four tetrahedra and multiplicity $2^{1024}$, the corresponding figures are 0.783 seconds and 2,510,984 bytes. These are single-run capacity observations; no expanded reference was attempted. Large binary topology can therefore be certified without enumerating its pieces, but the present proof representation still has a substantial size cost. The eight-tetrahedron case retains 47 ports and three maximal disk attachments rather than expanding its multiplicity.

## 166.9 Certified absorption of disk-base annuli

The next implementation checkpoint removes a specific obstruction left by whole-circle interfaces. A general annulus is still outside the disk-gluing criterion, but a complete vertical annulus of a disk-base bundle determines an actual two-handle. The implementation now constructs its source attaching curve, authenticates it independently, and includes that handle in the geometric cancellation proof. This implements the absorption proposition of report 85; it does not interpret an arbitrary presentation row as an embedded attaching curve.

**Proposition 166.4** (Augmented source core)**.** *Suppose a source-certified bundle component has a disk base and its entire base boundary is attached to the core. It can be absorbed as one two-handle, without adding zero-, one-, or three-handles. Its raw intersection count with one-handle $h$ is the number of its core-one ports owned by $h$, counting all occurrences without signs or cancellation. Any collection of these components can be absorbed simultaneously. The added handles and their total raw occurrences number at most $P$, the number of source ports.*

*Proof.* The bundle over a disk is $D^2 \times I$, and its complete attaching region is $\partial D^2 \times I$. This is the defining attaching region of a three-dimensional two-handle. The two horizontal disks remain free boundary. Take the midpoint of every interval fibre along the attaching annulus. All source transition maps, including fibre reversals, preserve the midpoint, so these local sections define an embedded attaching circle.

A core-zero port contributes an arc within its named zero-handle boundary. A core-one port is a face-sector side of an edge-cap polygon. Its midpoint arc traverses the lateral boundary of the corresponding source one-handle from one end to the other, meeting its cocore once. Thus each such port contributes one unsigned occurrence. Two occurrences with opposite signs still contribute two intersections.

The complete lower attachments of the complementary source handles remain in the core. The product contacts occupy their recorded boundary regions, disjoint from the original two-handle attaching regions after compatible corner rounding. Consequently the new attaching circles

can be placed at the same two-handle stage as the old ones. Distinct bundle components have disjoint attaching annuli. Absorption therefore changes neither the embedded cut manifold nor the interfaces of the remaining bundles. Each absorbed component uses at least one port, and distinct components use disjoint ports, proving both bounds. □

**Face flags authenticate the actual path.** Write the two source ends of one-handle $h$ as $(t_h, f_h)$ and $(u_h, g_h)$. For a core-one port inspect its two cap-edge endpoint flags. Their order must match these two flags or their reverse. Call the resulting local sign $\delta_p \in \{+1, -1\}$. If the canonical boundary word traverses that port with sign $s_p$, the corresponding raw one-handle crossing has sign $s_p\delta_p$. Its initial and terminal zero-handles are read from the appropriate ends of $h$.

Both tetrahedron and face are essential. In a self-paired tetrahedron the one-handle can have equal tetrahedron labels and even equal zero-handle owners at its ends, while its two face flags still differ. Comparing only tetrahedron labels would lose this distinction. Every core-zero port stays at its owner vertex. The producer and independent checker require the end of every oriented port path to equal the beginning of the next, including the final-to-initial join. Deleting the stationary portions then gives the alternating zero/one cycle used by the existing core checker. The signs are retained as chart evidence; the cancellation matrix remains unsigned. A circle with no one-handle crossings contributes a zero row and cannot silently disappear from the required cancellation schedule.

**Eligibility and independent replay.** The adapter requires a single base owner, signature $(\chi, b) = (1, 1)$, orientability, a product bundle, and multiplicity one. The circle must exhaust that component's port set, and no component may be absorbed twice. These facts are obtained from the source-verified base profiles and ordered interface, rather than from the proposed absorption record. Because the base has exactly one boundary component, a complete circle uses its whole vertical boundary. A component with a nondisk base remains unresolved.

The new whole-cut certificate schema is `normal-cut-handlebodies-v2`. Its absorption record names the source attachment, base profile, appended two-handle, and every directed crossing. The independent module `disk_bundle_absorption_verify` reconstructs these data from authenticated polygons and endpoint flags. It imports no producer. Only after exact comparison does it return an augmented geometric core to the existing independent cancellation checker. Every added row must be eliminated; the genus calculation uses the surviving one-handles of this augmented core. Version-one certificates remain replayable, and cuts requiring no absorption continue to use version one.

**The whole-cut census after absorption.** Remove absorbed bases and their annular attachments from the bundle/core graph, replacing the old core by the augmented core. All remaining bases must be bordered and all their attachments must be proper intervals. The disk-gluing theorem then gives the same genus formula

$$g(X) = \sum_{v \in V(X)} g(v) + |E(X)| - |V(X)| + 1.$$

An absorbed disk contributes one to the core Euler characteristic and removes a bundle of Euler characteristic one; its attaching annulus has Euler characteristic zero. Hence the original source-cut Euler identity remains the correct final check. Unported bundle multiplicities are counted as before. This proves the complete histogram only when all augmented core cancellations and all remaining disk gluings succeed.

**Cost and scope.** The adapter adds $O(P)$ records and $O(P \log(T + P))$ bits of finite chart data; it never expands the binary normal multiplicities. Constructing the records is linear in the finite

source data, while the checker also sorts port sets in $O(P \log P)$ comparisons. The augmented core still has $O(T)$ handles and raw occurrences, so the existing polynomial saturated-cancellation bounds apply unchanged in order. The six weighted orbit queries and their shared budgets are unchanged. Failure of a cancellation schedule or rejection of a nondisk annulus carries no negative topology conclusion. This advances the supplied-cut terminal operator, but does not provide useful-surface discovery, arbitrary boundary-pattern transport, or a global quasi-polynomial hierarchy bound.

**Source-cut audit.** The refreshed 106-case audit certifies 63 whole cuts, compared with 30 before absorption. Every new success is among the 33 algebraically eliminable candidates identified by the earlier probe, and all previous successes are preserved. All 63 source certificates replay independently. Regina's actual cut construction and component recognition finish for 99 cases and agree exactly with every one of the 56 native successes within the expansion limit. Seven large-coordinate successes exceed the 4,096-disk literal cap and are independently replayed without expansion. No oracle times out. The source pins and per-case results are retained in `data/disk-absorption-audit.json`.

The 43 remaining refusals comprise 26 nondisk annular cases, seven stalled augmented cancellations (three detected by the square leaf obstruction), and ten cores with three-handles. These categories are not perfectly comparable with the earlier refusal ordering: the boundary and absorption checks now precede cancellation, so four previously stalled core cases are classified first by their nondisk annuli. They are not lost successes.

The focused checks cover actual same-tetrahedron and same-zero-handle self-pairings, source relabellings, signed endpoint charts, malformed and omitted absorption records, an omitted cancellation, disabled producers, exact hexadecimal transport, resource caps, nondisk refusals and stalled augmented cores. Multiplying the two-tetrahedron annular cut by $2^{1024}$ produces and independently verifies exactly $2^{1024} + 1$ solid-torus components without enumerating them. The tests also retain the pre-existing raw cancellation and disk-gluing checks. These are source-bound whole-cut certificates, not an enlargement of the complete useful-surface search.

**Full regression.** The complete Regina-enabled suite passes all 1,656 tests with no skips in 517.462 seconds. It includes 13,846 expanded-sheet topology comparisons and 196,712 independent marked-cover comparisons. The focused absorption, core and cut checks also pass (21 tests followed by the additional conservative refusal test). The full log is `data/disk-absorption-tests.txt`; the runtime source pins remain identical to the 106-cut audit.

**Paired timings and capacity.** After the regression suite finished, eight finite cuts were measured in five shuffled paired rounds after warmup. Four use the newly supported two-tetrahedron annular family and four use a previously supported three-tetrahedron v1 control. All explicit and native histograms agree. Input parsing and output serialization are excluded, while the native timer includes the complete independent replay. The casewise medians of identical explicit-code controls range from 0.976 to 1.073. The driver, full samples, source pins and controls are retained in `data/disk-absorption-benchmark.py` and its companion JSON. Run the driver from the repository root with Regina available.

Within the new annular family, the 4,096-copy cut improves by 6.152 in median paired time: median native production plus replay takes 0.174 seconds, whereas explicit construction and recognition take 1.093 seconds. The latter builds 98,334 tetrahedra and recognizes 4,097 solid-torus components. The three smaller instances favor literal construction; at 16 copies the ratio is only 0.024. In the previously supported control, 256 and 4,096 copies favor the native path, with paired ratios 2.202 and 34.567 respectively. These control gains are not attributed to absorption. There is no universal speedup or new default knot-recognition stage.

| $T$ | Copies | Native (s) | Explicit (s) | Ratio | A/A |
|---|---|---|---|---|---|
| 2 | 1 | 0.0131 | 0.0008 | 0.081 | 1.012 |
| 2 | 16 | 0.1711 | 0.0041 | 0.024 | 1.009 |
| 2 | 256 | 0.1679 | 0.0605 | 0.358 | 0.976 |
| 2 | 4096 | 0.1744 | 1.0935 | 6.152 | 1.073 |
| 3 | 1 | 0.0053 | 0.0010 | 0.199 | 1.019 |
| 3 | 16 | 0.0307 | 0.0049 | 0.159 | 1.004 |
| 3 | 256 | 0.0318 | 0.0699 | 2.202 | 0.994 |
| 3 | 4096 | 0.0337 | 1.2317 | 34.567 | 0.994 |

Table 3: Complete cut queries after disk-base absorption. The $T = 2$ family uses the new v2 annulus certificates; $T = 3$ is a previously supported v1 control. Native includes production and independent replay. Ratio is the median paired explicit/native time, and A/A compares identical explicit runs.

A separate single-run capacity check on the annular family certifies $2^{16384}+1$ solid-torus components in 0.605 seconds, including replay. It absorbs one disk-base component and produces a 15,069,104-byte serialized certificate. The v1 control at the same multiplicity takes 0.088 seconds, uses no absorbed component, and produces 4,388,996 bytes. Neither capacity case attempts literal expansion. They demonstrate binary multiplicity handling while retaining the cost of the present proof representation.

**Next annular target.** A source-pinned profile diagnostic finds that 15 of the 26 remaining annular refusals have fewer fully attached boundary circles than boundary components on every nondisk base involved. There are 19 such base profiles altogether: 13 annuli with one attached circle, four orientable three-holed spheres with two, one orientable five-holed sphere with four, and one nonorientable surface with three boundary components and two attached circles. The other profiles include nine Möbius bands attached along their sole boundary, one annulus attached along both boundaries, and one nonorientable surface attached along both of its boundary components. The diagnostic is retained in `data/annular-profile-diagnostic.json`.

The first group suggests a simultaneous primitive-annulus rule: arcs joining the attached base circles to a remaining boundary component lift to disjoint cocore disks in the interval bundle. Their endpoints must avoid every other marked attaching interval. Turning that observation into an independently checked gluing rule and accounting for all component joins is further work; the 15 candidates are not additional certified cuts in this checkpoint.

## 167 Assessment and recommendations

**What was achieved.** Seven mutually consistent exact recognizers, the seventh now in an accelerated Python version and a Rust version that agree with each other and with nine independently accelerated variants; a verified and cross-validated body of code for the terminal ball-pattern test and for cocycle-dual normal surfaces; a precise, sourced statement of the conditional $L(g+1)^L$ accounting and of everything a $n^{O(\log n)}$ implementation would still need; and one recognizer whose practical reach is well beyond the $2^n$-state cubes.

Reports 64–69 also contribute a preserved mathematical record independent of the performance programme: surface-gluing identities, exact continuation ranks, propagation obstructions and component-type bounds, with their hypotheses and full manuscripts retained in Section 128. The unit-ray integration of Section 129 improves exact supplied-vector disc queries while retaining complete fallback and independent legacy replay. Section 130 then carries finite coorientation to

the complete additive topology histogram, removing a redundant double query when its checked colouring succeeds. Sections 131 and 132 preserve the next six manuscripts, sharpen the relaxation and rooted-context limitations, and integrate complete low-nullity minimum-envelope enumeration with independent coverage and essential-disc replay. Sections 133–136 preserve reports 76–81, including the fixed-cover characterization and bounded-descent locality theorem, and integrate source-checked edge-first candidates and complete planar standard rays. The edge-first stage remains optional after a measured regression; planar enumeration extends the automatic low-nullity branch. The sharper interaction and Q-corner Euler theorems retain their supplied-sector scope. Section 137 then makes discovery progress-driven, testing Q-corners before requesting the complete minimum overlay and sharing a producer kernel across the outer search's two phases. Its local $k^{O(d)}$ complete-enumeration accounting permits logarithmic nullity without proving a complete sector producer for all unknots. Section 138 then uses signed matching residuals to expose nearby admissible sectors and adds seven source-certified native witnesses. Its complete small-window filter and logarithmic-radius accounting are local theorems; the measured cost keeps this stage optional. Section 139 then removes repeated wide matching elimination and triangle-graph reconstruction from those window queries, using exact small-parameter kernel updates and source-projected geometry. All existing window witnesses remain source-certified; measured enabled recognition improves while the default continues to omit costly windows. Section 140 implements the retained Euler-corner criterion before allocating updated standard geometry and shares validated producer geometry with disc observers. Independent source replay remains in place; negative local screens do not supply a general recognition bound. Section 141 carries the thin replacement and gauge operations through lazily cached sparse source potentials, removing repeated wide projection while preserving exact corner geometry and independent disc replay. Its improvement is in amortized projection cost; the small-case recognition timings remain close to parity. Section 142 retains sparse free-coordinate constraints and potentials through low-dimensional ray production, while restoring the same full coefficient sequences for generic rank calculations. Source lifts and independent disc replay remain exact; global coverage is still unproved. Section 143 amortizes source-only work inside an independently constructed verifier context. Actual sources and coordinates remain checked per proof, while final diagram witnesses retain fresh canonical source replay. Section 144 groups equal source corner forms and removes constant nonlinear minima. It refines the objective only after a nonpositive first corner, retaining exact exclusion and the early positive path. Section 145 extends Q screening above the planar gate. Exact active-basis selection avoids hash-dependent deduplication; automatic generic discovery resumes the standard phase only after a positive Q phase fails to find a disc, sharing its candidate allowance across both phases. Section 146 then removes coordinates proved zero by a completed Q list and recompiles the effective matching space. Independent replay authenticates that original Q support before reconstructing the smaller standard cone, while retaining the original full-sector statement. Section 147 then certifies sufficient coordinate zeros by polynomial source-equation implications before generic Q enumeration. Independent linear replay licenses the smaller matching space without assuming maximal support or a universal remaining-dimension bound. Section 148 then integrates report 79's complete bounded Pachner descent, shared source covers and independent cochain transport. Its diagram adapter admits only source-replayed positive discs and keeps the complete fallback. The measured small-diagram cost keeps the new stage optional. Section 149 replaces the ambient cover index by exact visited-footprint completion, preserving the indexed language and source proofs. Actual diagram calls reach more moves, but consume more wall time at the same checkpoint allowance without new successes in the audit. Section 150 then restarts only after source-replayed strict descents and concatenates their actual transport chain. Its explicit per-epoch allowance yields a new nonempty-diagram disc witness, while stationary and capped sources still use the complete fallback. Section 151 then authenticates periodic integral vertex coboundaries on smaller sources. Its mixed move/gauge chain retains original input replay and reduces the actual transported disc witness's epochs and work, without asserting that a span objective proves essentiality. Section 152 then authenticates the retained primitive annulus capping

theorem on a source-replayed endpoint. It checks connectivity, primitive periods and the exact arithmetic witness before accepting an annulus terminal, with the original source and complete fallback preserved. Section 153 repairs signed binary transport of complete sector transcripts and records the primitive ray digit bound; it preserves the independent geometric exhaustion checks. Section 154 adds polynomial pair-span consequences with the existing source-equation authority, removing generic-Q work on two further actual sectors while retaining the incomplete-support fallback. Section 155 implements binary chamber connectivity of a normal-surface cut, with independently reconstructed geometric pairings. It counts cut-open pieces without constructing their full patterned geometry. Section 156 then bounds the conservative marked core and counts wholly prismatic interval-bundle components, preserving the distinction from the published full parallelity-bundle algorithms. Section 157 then recovers source-certified normal midsection vectors and binary multiplicities for whole prismatic pieces. It retains coordinate doubles and independently replayed weighted membership without claiming to extract the parallelity regions inside a core. Section 158 then retains the face-connected prism regions attached inside cores and their exact source-labelled interfaces. Its 0/1-handle profiles keep the missing edge and vertex topology explicit. Section 159 contracts ordered static gaps without sorting all run boundaries again. Section 160 then encodes unique source port ownership in three weighted coordinates, with independent replay of both certificate versions. These are exact compressed-operator improvements; the full core and hierarchy obligations remain in place. Section 161 then constructs the full cut boundary in binary polygon, edge and vertex families, including oriented attaching cycles and carriers into cut chambers. It computes all boundary genera and each cut piece's boundary count and total genus, with independent source-cell reconstruction and trace replay. Section 162 then retains the full three-dimensional raw cut in affine cell families, with reconstruction charts, oriented chamber attachments and the boundary subcomplex. Independent source replay and integral-chain and field-homology checks retain interior information beyond boundary genera. Section 163 binds exact orbit weights to least original representatives. Section 164 then extracts source parallelity handles in all dimensions, classifies their complete bases and fibre covers, and retains a complementary core with at most $30T$ handles and exact source ports. An admissible patterned core, pattern transport and the hierarchy complexity proof remain required. Section 165 then implements initial empty-pattern two/three and one/two cancellations, with exact source incidences and boundary-period certificates for exposed cocore discs. These provide an additional independently checked positive native stage without claiming that every inherited core is now admissible. Section 166 recovers ordered bundle interfaces and certifies complete handlebody genus histograms using geometric core cancellations, source-certified absorption of disk-base annuli, and the remaining disk attachments. The absorption extension raises the supplied-cut audit from 30 to 63 certificates. Nondisk annular gluing and explicit boundary-pattern transport remain geometric implementation tasks.

**Obstacles that remain.** The six items of Section 5 remain obligations for a general complexity theorem. Lackenby's July 2026 paper [24] develops the hierarchy and distinguishes iteration counts from the cost of its encoded operations; the refined acceleration still requires a complete running-time accounting. The local polynomial orbit and compressed-braid algorithms above resolve useful subproblems. They do not establish polynomially bounded geometric state descriptions or a quasi-polynomial complete search for every knot diagram. An implementation must retain this distinction between proved local bounds, restricted-input theorems and the announced general target.

**If work continues.** (1) Track the published record: a full version of the 2021 announcement, or the refined Section 10 algorithm with an explicit cost model, would change the picture. (2) Settle the move-set question of Section 3.4 from Dynnikov's paper, or add shared-corner exchanges to archive 01 and re-run its exhaustive validation. (3) The scanning recognizer now exists in

Rust (Section 9); the Python separator backend now guarantees $O(\sqrt{n})$ frontier width and a subexponential exact scalar query. The next complexity problem is to control intermediate homology multiplicities or supply a different complete decision fallback; order width alone does not settle it. (4) Use the archives' exact recognizers as oracles for any future geometric implementation, as several of their reports suggest.

## References


[1] *Marked parallelity interfaces and certified handlebody cores*, research report 85, 12 October 2026. Preserved manuscript and implementation under `Topology/UnknotRecognition/reports/85/`.

[2] Rustam Sadykov, *Elements of Surgery Theory*, Section 10.1.3, Lemma 10.3 (geometric handle cancellation), author-hosted notes, `https://www.math.ksu.edu/~sadykov/Lecture%20Notes/Surgery%20Theory.pdf`.

[3] Regina project, *Boundary component API*, `BoundaryComponent::eulerChar` and component ownership, `https://regina-normal.github.io/engine-docs/classregina_1_1detail_1_1BoundaryComponentBase.html`.

[4] Nathan J. Fine and Herbert S. Wilf, *Uniqueness theorems for periodic functions*, Proc. Amer. Math. Soc. 16 (1965), 109–114. `https://doi.org/10.1090/S0002-9939-1965-0174934-9`.

[5] Erwin H. Bareiss, *Sylvester's Identity and Multistep Integer-Preserving Gaussian Elimination*, Mathematics of Computation 22 (1968), 565–578. `https://doi.org/10.1090/S0025-5718-1968-0226829-0`.

[6] Benjamin A. Burton, *Converting between quadrilateral and standard solution sets in normal surface theory*, Algebraic & Geometric Topology 9 (2009), 2121–2174. `https://arxiv.org/abs/0901.2629`.

[7] Benjamin A. Burton and Melih Ozlen, *A fast branching algorithm for unknot recognition with experimental polynomial-time behaviour*, arXiv:1211.1079. `https://arxiv.org/abs/1211.1079`.

[8] Regina 7.4.1 Calculation Engine, *Triangulation<3>::isSolidTorus API documentation.* `https://regina-normal.github.io/engine-docs/classregina_1_1Triangulation_3_013_01_4.html`.

[9] Regina 7.4.1 Calculation Engine, *ExampleLink API documentation.* `https://regina-normal.github.io/engine-docs/classregina_1_1ExampleLink.html`.

[10] Juanjo Rué, Dimitrios M. Thilikos and Vasiliki Velona, *Structure and enumeration of $K_4$-minor-free links and link-diagrams*, European Journal of Combinatorics 89 (2020), 103147, Theorem 9. `https://arxiv.org/html/1806.07855v3`.

[11] Hans L. Bodlaender, Benjamin Burton, Fedor V. Fomin and Alexander Grigoriev, *Knot Diagrams of Treewidth Two*, arXiv:1904.03117v2, 2019, Theorems 1–3. `https://arxiv.org/html/1904.03117`.

[12] Avishy Carmi and Eliahu Cohen, *On trivial Jones–Vassiliev polynomials*, arXiv:2606.22410v1, 21 June 2026. `https://arxiv.org/abs/2606.22410`.

[13] Marc Lackenby, *Some fast algorithms for curves in surfaces*, arXiv:2401.16056v2, 20 March 2026, Theorems 1.1–1.3. `https://arxiv.org/html/2401.16056v2`.

[14] Richard J. Lipton and Robert Endre Tarjan, *A separator theorem for planar graphs*, SIAM J. Appl. Math. 36 (1979), 177–189. https://doi.org/10.1137/0136016.

[15] Marc Lackenby, *Unknot recognition is quasi-polynomial time*, YMSC Topology Seminar, 7 November 2024, abstract. https://qzc.tsinghua.edu.cn/en/info/1122/4171.htm.

[16] Gheehyun Nahm, *Basepoints in Khovanov homology and nonorientable surfaces*, arXiv:2605.08457v1, 8 May 2026, Lemma 2.4. https://arxiv.org/abs/2605.08457.

[17] Allen Hatcher, *Algebraic Topology*, Cambridge University Press, 2002, Section 1.3. https://pi.math.cornell.edu/~hatcher/AT/ATpage.html.

[18] Ian Agol, Joel Hass and William P. Thurston, *The computational complexity of knot genus and spanning area*, Trans. Amer. Math. Soc. 358 (2006), 3821–3850. https://arxiv.org/abs/math/0205057.

[19] Tetsuya Abe, *The Rasmussen invariant of a homogeneous knot*, Proc. Amer. Math. Soc. 139 (2011), 2647–2656. https://arxiv.org/abs/1003.5392.

[20] Andrew Lobb, *Computable bounds for Rasmussen's concordance invariant*, Compositio Math. 147 (2011), 661–668. https://arxiv.org/abs/0908.2745.

[21] Report 07, *Unknot recognition: structural certificates and compressed complexes*, delivered 7 October 2026; reports/07/paper/unknot_structural_compression.tex.

[22] Marc Lackenby and Mehdi Yazdi, *The computational complexity of knot genus in a fixed 3-manifold*, 23 May 2022 manuscript, Sections 2.7 and 8. https://people.maths.ox.ac.uk/lackenby/KnotGenusComplexity23May2022.pdf.

[23] Marc Lackenby, *Unknot recognition in quasi-polynomial time*, slides, February 2021, 109 pages, supplied as docs/quasipolynomial-talk.pdf.

[24] Marc Lackenby, *Incompressible surfaces, hierarchies and unknot recognition*, arXiv:2607.23350v1, 25 July 2026, 54 pages. https://arxiv.org/abs/2607.23350.

[25] C. D. Papakyriakopoulos, *On Dehn's lemma and the asphericity of knots*, Annals of Mathematics 66 (1957), 1–26. https://doi.org/10.2307/1970113.

[26] Abdó Roig, Enric Ventura and Pascal Weil, *On the complexity of the Whitehead minimization problem*, International Journal of Algebra and Computation 17 (2007), 1611–1634. https://arxiv.org/abs/math/0608779.

[27] Ilya Kapovich, *Compressed primitivity problem in free groups*, arXiv:2607.21499v1, 23 July 2026. https://arxiv.org/html/2607.21499v1.

[28] The Regina development team, *GroupPresentation::smallCancellation and simplify*, engine documentation, version 7.4.1, accessed 8 October 2026. https://regina-normal.github.io/engine-docs/classregina_1_1GroupPresentation.html.

[29] A. Blumer, J. Blumer, D. Haussler, A. Ehrenfeucht, M. T. Chen and J. Seiferas, *The smallest automaton recognizing the subwords of a text*, Theoretical Computer Science 40 (1985), 31–55. https://doi.org/10.1016/0304-3975(85)90157-4.

[30] Ilya Kapovich, *Small Cancellation Stability and Isomorphism Rigidity for Generic Finitely Presented Groups*, arXiv:2608.17238v1, 18 August 2026. https://arxiv.org/html/2608.17238v1.

[31] Tomohiro I, *Longest Common Extensions with Recompression*, 28th Annual Symposium on Combinatorial Pattern Matching (CPM 2017), LIPIcs 78, 18:1–18:15, Theorem 2. https://doi.org/10.4230/LIPIcs.CPM.2017.18.

[32] Yury Lifshits, *Solving Classical String Problems on Compressed Texts*, arXiv:cs/0604058v1, 2006, Section III. https://arxiv.org/abs/cs/0604058.

[33] Wataru Matsubara, Shunsuke Inenaga, Akira Ishino, Ayumi Shinohara, Tomoyuki Nakamura and Kazuo Hashimoto, *Efficient algorithms to compute compressed longest common substrings and compressed palindromes*, Theoretical Computer Science 410 (2009), 900–913. https://doi.org/10.1016/j.tcs.2008.12.016.

[34] Saul Schleimer, *Polynomial-time word problems*, arXiv:math/0608563. https://arxiv.org/abs/math/0608563.

[35] P. B. Kronheimer and T. S. Mrowka, *Khovanov homology is an unknot-detector*, Publ. Math. IHÉS 113 (2011), 97–208.

[36] Dror Bar-Natan, *Khovanov's homology for tangles and cobordisms*, Geom. Topol. 9 (2005), 1443–1499; and *Fast Khovanov homology computations*, J. Knot Theory Ramifications 16 (2007), 243–255.

[37] Ivan Dynnikov, *Arc-presentations of links: monotonic simplification*, Fund. Math. 190 (2006), 29–76, arXiv:math/0208153.

[38] Harry M. Markowitz, *The elimination form of the inverse and its application to linear programming*, Management Science 3 (1957), 255–269.

[39] Mikhail Khovanov, *A categorification of the Jones polynomial*, Duke Math. J. 101 (2000), 359–426.

[40] Konstantinos Meichanetzidis and Stefanos Kourtis, *Evaluating the Jones polynomial with tensor networks*, Phys. Rev. E 100 (2019), 033303. https://arxiv.org/abs/1807.02119.

[41] Clément Maria, *Parameterized Complexity of Quantum Knot Invariants*, SoCG 2021, LIPIcs 189, 53:1–53:17. https://doi.org/10.4230/LIPIcs.SoCG.2021.53.

[42] Patrick Dehornoy, *Efficient solutions to the braid isotopy problem*, arXiv:math/0703666, 2007. https://arxiv.org/abs/math/0703666.

[43] Marius Crainic, *On the perturbation lemma, and deformations*, arXiv:math/0403266. https://arxiv.org/abs/math/0403266.

[44] Thomas C. Jaeger, *A Remark on Roberts' Totally Twisted Khovanov Homology*, arXiv:1109.1805, Theorem 3.1. https://arxiv.org/abs/1109.1805.

[45] Louis H. Kauffman and Sofia Lambropoulou, *On the classification of rational tangles*, arXiv:math/0311499. https://arxiv.org/abs/math/0311499.

[46] Louis H. Kauffman and Sofia Lambropoulou, *Hard Unknots and Collapsing Tangles*, arXiv:math/0601525, Theorem 5. https://arxiv.org/abs/math/0601525.

[47] João M. Nogueira and António Salgueiro, *On embeddings of 2-string tangles into the unknot, the unlink and split links*, arXiv:2110.15645, Theorem 4.9. https://arxiv.org/abs/2110.15645.

[48] Elwyn R. Berlekamp, *Factoring Polynomials Over Finite Fields*, Bell System Technical Journal 46 (1967), 1853–1859. https://doi.org/10.1002/j.1538-7305.1967.tb03174.x.

[49] Peter B. Kronheimer and Tomasz S. Mrowka, *Knots, sutures and excision*, Journal of Differential Geometry 84 (2010), 301–364, Corollary 7.17. https://people.math.harvard.edu/~kronheim/excision_paper.pdf.

[50] Stefan Kiefer, *Notes on Equivalence and Minimization of Weighted Automata*, arXiv:2009.01217v1, 2020, Section 2, Proposition 2.1. https://arxiv.org/html/2009.01217v1.

[51] The Regina development team, *Regina 7.4.1 Calculation Engine: boundary shelling conditions*, Triangulation::shellBoundary, accessed 9 October 2026. https://regina-normal.github.io/engine-docs/classregina_1_1Triangulation_3_013_01_4.html.

[52] The Regina development team, *Pachner moves on triangulations*, Regina 7.1 Calculation Engine, accessed 9 October 2026. https://regina-normal.github.io/7.1/engine-docs/pachner.html.

[53] Hans L. Bodlaender, Marek Cygan, Stefan Kratsch and Jesper Nederlof, *Deterministic Single Exponential Time Algorithms for Connectivity Problems Parameterized by Treewidth*, Information and Computation 243 (2015), 86–111. https://arxiv.org/abs/1211.1505.

[54] Benjamin Bergougnoux and Mamadou Moustapha Kanté, *Fast Exact Algorithms for Some Connectivity Problems Parameterized by Clique-Width*, Theoretical Computer Science 782 (2019), 30–53, Theorem 3.8. https://arxiv.org/abs/1707.03584.

[55] Charles Livingston, *Maps of surface groups to finite groups with no simple loops in the kernel*, Journal of Knot Theory and Its Ramifications 9 (2000), 1029–1036. https://arxiv.org/abs/math/0002162.

[56] Martin Pikaart, *Large characteristic subgroups of surface groups not containing any simple loops*, Journal of Knot Theory and Its Ramifications 10 (2001), 529–535. https://arxiv.org/abs/math/0005093v2.

[57] Regina 7.4.1 Calculation Engine, *NormalSurface::cutAlong API reference*. https://regina-normal.github.io/engine-docs/classregina_1_1NormalSurface.html.

[58] Marc Lackenby, *The efficient certification of knottedness and Thurston norm*, author manuscript, Section 9.4, Theorem 9.3. https://people.maths.ox.ac.uk/lackenby/knp02719.pdf.

# A Files produced

- `reports/01` – `reports/06`: the extracted archives (flat, one level).
- `synthesis/report.tex`, `synthesis/report.pdf`: this report; `synthesis/make_tables.py` regenerates the tables from `synthesis/data/*.json`; the cross-validation scripts are in `synthesis/data/`.
- `fast/`: the `fastunknot` package (version 0.2 with the later work of Section 10: the default scanner `scan_fast.py` and its geometry `planar.py`), tests and examples; `ablation.py` with `results/ablation.json`; the 0.1 benchmark, kept as `results/benchmark_0.1.json`; and the measuring tools of Section 10, namely `perf.py`, `ab.py` and `crosscheck.py`, with their logs in `results/`.
- `proposals/01` – `proposals/09`: the nine acceleration proposals, with `proposals/README.md` comparing their ideas.
- `rust/`: the Rust crate, `cross_check.py`, `profile.py` and `results/profile.json`.
- `synthesis/data/proposals_bench.py` and its JSON outputs: the same-machine comparison of all engines.